

Research continuation and complete source proofs

Split-Zero Cohomology and Arithmetic Weight Control

13 September 2026

This reader integrates the Split-Zero carrier, finite arithmetic operator comparisons, the historical packet ghost, chain retractions, arithmetic moments, Gaussian resolvents, full kernel layers, symmetric tensor frontiers, measured support branches, the original theta norm-volume identities, quartet derivative costs, gamma-reference determinants, and the complete two-Stieltjes-space and spectral-sum comparisons, the full relative connection, strict normal energy, the conormal derivative with its original residue maps, nested relative-frame transitions and density, the full limiting derivative graph, cyclic arithmetic metric comparisons, every retained conormal depth, exterior trace amplification, the exact equality and coupling maps, positive exterior formulas for arithmetic interpolation volumes, the original source and relation Toda flows with all four loss terms, exact endpoint comparisons, certified theta-seed input with explicit infinite tails, gamma convolution and its full arithmetic coefficient maps, the certified seed coefficients, finite source-to-quotient metric transfer, and the complete complex amplitude in the sum fibres. It retains the original arithmetic functions, measures, scalar actions, support fibres, and complete jets. Part I gives the local continuation with proofs. Part II preserves the complete selected source notes, their exact revisions, and their chronology.

Contents

I	Integrated mathematical continuation	11
1	Exact Split-Zero carriers and the local spectral bridge	11
1.1	Scalar support and independent channel amplitudes	11
1.2	Tesseract sectors and a supported critical fixed locus	13
1.3	Exact sector, holonomy and positive-moment maps	14
1.4	The theta-packet form and its scalar comparison	16
1.5	Projective zeta at all zeros and at its pole	16
1.6	Matrix packets and their full relation module	17
2	The historical packet ghost and its full arithmetic jet quotient	18
3	Historical Hurwitz reconstruction and the complete Laurent jet bridge	22
3.1	Source receipts and exact locators	22
3.2	Every compact Hurwitz input is determined by one simple-zero germ	23
3.3	The complete zero cluster removes the simplicity restriction	26
3.4	The historical period map preserves every finite spectral jet	27
3.5	Exact conjugate-reciprocal involution on the jets and conormal	29
3.6	Period collisions: the exact unlabelled image and its missing fibres	30
3.7	Two prime periods recover the complete finite spectral algebra	31
3.8	Split-Zero support is preserved by the exact maps	33
3.9	All-order Hurwitz motion transported to the period and tesseract coordinates . .	34
3.10	Mathematical scope	35
4	Historical support branches, ideal completion, and measured observations	35
4.1	Exact ordinary and continuous Split-Zero branches	36
4.2	The exact frame comparison retaining all ordinary branches	38
4.3	A fully explicit nontrivial frame with algebraic branches and no frame points . . .	40
4.4	Atomic observation and the exact support data it removes	41
4.5	The exact operator-algebraic branch map	42
5	Reconstruction with changes of support index	44
5.1	The internal quotient and the actual comparison kernel	45
6	The original theta complex and its arithmetic packet	46
6.1	Source spaces and fixed arithmetic factors	46
6.2	The exact operator range, including its inverse	48
6.3	The original extension supplies the boundary coefficient	49
6.4	Actual representative corrections and rank-two control	49
7	Certified moments from the original arithmetic theta source	50
7.1	Source polynomials and the exact reflection map	51
7.2	An absolutely convergent double series for every entry	51

7.3	A complete finite truncation certificate	53
7.4	Exact adjoint, parity, and finite-boundary checks	54
7.5	Reproducible certified finite arithmetic data	54
8	Exact source-metric optimization and aggregate arithmetic control	55
8.1	The fixed arithmetic maps and their boundary	56
8.2	A source-density calculation with its topology retained	57
8.3	Every positive packet metric has an exact finite source lift	57
8.4	Sharp control with all nilpotent jets present	59
8.5	The exact tensor cost of unrestricted metrics	60
8.6	A stronger bound on the actual orthogonal sequence	61
8.7	Original supported maps and the precise result delivered	62
9	The actual theta minimum as a confluent arithmetic kernel	62
9.1	Original functions, maps, coordinates, and measures	63
9.2	Exact constrained minimum over the admitted source	64
9.3	Monic arithmetic polynomials and their exact recurrence	65
9.4	Exact relative weight control from the same kernel	66
9.5	Exact change when one more theta direction is admitted	67
9.6	The original measure supplies a full tail recurrence	67
9.7	Metric dependence and the precise mathematical gain	69
10	The full jet kernel, finite resolvents, and the actual analytic tail	69
10.1	A finite compression in the original Mellin coordinate	70
10.2	Exact quadrature and the full confluent kernel	71
10.3	The infinite multiplication operator and its genuine tail	73
11	Quantitative convergence of the arithmetic Gaussian resolvents	74
11.1	The Gaussian rule inherits the original exponential moments	75
11.2	Exact moments give a quantitative analytic-strip estimate	75
11.3	Every confluent derivative and the original finite algebra	77
12	The exact terminal step and its retained boundary energy	79
12.1	A step written entirely in the original polynomial packet	79
12.2	The complete boundary-energy increment	80
12.3	The exact rational terminal step and its polynomial continuation	81
12.4	The extra jets required to recover the metric from the tail	82
12.5	Actual quartet packets close the energy in two volume ratios	83
12.6	What the recurrence proves about boundary energy	84
12.7	An exact positive-moment counterexample to consecutive energy monotonicity . .	84
13	The exact metric departure and the cost of complete jets	90
13.1	A quantitative constraint on every retained nilpotent chain	91
14	Coherent packets, joint tensor layers, and their exact spectral cost	92

14.1	The arithmetic unit and the coherent one-variable maps	92
14.2	The original reflection and the exact diagonal coefficient	93
14.3	The joint total-degree source and its relation layer	94
14.4	Nested packets on the joint source: the additional projection	95
14.5	Positive layer rank and the complete tensor spectral departure	95
15	Density of the original joint boundaries and determinant dissipation	97
15.1	Polynomial density with the original weighted measures	97
15.2	Strong limits of the actual source representatives	98
15.3	The accumulated determinant cost of the same boundary sequence	100
16	The actual kernel layer, its derivative image, and its quotient	101
16.1	The original arithmetic source and its constrained minimum	101
16.2	The complete next-layer frame and its theta primitives	102
16.3	The exact incidence map and the derivative-image rank	103
16.4	The whole quotient of the next layer by the derivative image	104
16.5	The derivative between consecutive relation quotients	105
16.6	The exact original-theta norm and volume bridge	105
16.7	The complete scalar dictionary for the boundary energy	107
17	Original orthogonal shells and the exact projective-ratio algebra	108
17.1	Two sections of the same polynomial filtration	108
17.2	The full correspondence between the two image spaces	110
17.3	The algebra generated by the original coordinate ratio	111
17.4	A complete finite witness for unequal shell ranks	113
18	The original frontier energy, determinant growth, and the entire border fibre	115
18.1	Original moment coordinates and the relation graph	115
18.2	Exact relative loss and the source derivative energy	117
18.3	The entire small border operator, including its zero extension	118
18.4	A quantified link between the frontier bound and determinant cost	119
19	The signed symmetric summand and its complete spectral cost	120
19.1	The chain projector and its completed and finite targets	120
19.2	Orbit coordinates, their Gram, and all local nilpotents	122
19.3	Complete symmetric spectral sums and their exact departure	123
19.4	The cost in the original symmetric relation layer	125
20	Arithmetic quartet parity and the exact positive frontier operator	126
20.1	The actual two frontier matrices	126
20.2	The original linear reflection separates the two frontiers	127
20.3	The complete positive operator and the sharper arithmetic bound	128
20.4	The same positive frontier on the actual symmetric summand	130
21	Quartet parity in the original derivative energy and volume loss	133

21.1	Original maps, their spaces, and the reflection	133
21.2	The single original boundary block	134
21.3	Complete determinant constraints, including zero cases	135
21.4	Exact energy-volume equality for the original one-variable layer	136
22	The exact two-Stieltjes-space representation of the theta packet	137
22.1	The quadratic algebra and both reflection branches	137
22.2	The original measures and a Hilbert space isomorphism	139
22.3	The two monic families and their original norms	140
22.4	Both full jet kernels and the exact weight operator	141
22.5	The original tesserine coordinates and support lift	142
22.6	The complete rank-one half-line boundary and its derivative cost	143
23	The exact gamma reference and the arithmetic Gram determinants	144
23.1	The original measure, phase, and packet cancellation	144
23.2	Gamma norms from a generating integral	145
23.3	Finite arithmetic determinants in the retained metrics	146
23.4	Full confluent kernel and the original coordinates	147
23.5	The finite map to Romik's gamma parameter	148
23.6	Exact scope of the exponential-weight theorem	149
23.7	Primary source locators	149
24	The symmetric spectral sum and its unequal Stieltjes pair	150
24.1	Original polynomial coordinates and the reflection ideals	150
24.2	The entire defect is the original paired packet	151
24.3	The original unit and all coordinate factorials	153
24.4	The actual matrix weight and its positive half-line pair	154
24.5	Full-jet kernels for the actual constrained minimum	155
24.6	Original primitives and supported maps	156
25	The sum connection on both half-line branches and its exact domain	157
25.1	The actual amplitude, its mass, and the original relative fibre	157
25.2	The complete matrix connection and its actual graph	158
25.3	A branch-preserving quadratic map for the entire matrix weight	159
25.4	The meeting point of the two branches is part of the domain	161
25.5	The even specialization and the unchanged scalar contraction	161
25.6	The actual arithmetic normal contribution is strictly positive	162
25.7	Exact reconstruction from the actual finite normal image	164
26	The original conormal derivative and its finite section commutator	166
26.1	The first thickening and the dual-number map	166
26.2	The original monic section and the exact finite correction	168
26.3	The full arithmetic unit and the original finite representatives	170
26.4	The one-variable residue functional and the original theta relation	171
26.5	Supported algebra and coefficient maps	172

26.6	The arithmetic unit and complete tensor partial residues	173
27	Exact transitions between nested relative frames	175
27.1	Original sources and coefficient inclusions	175
27.2	The original weights and their orthogonal transition	176
27.3	Full source matrices, graph maps and arithmetic observation	177
27.4	Every variable-frame term	179
27.5	Nested chains and both quadratic branches	180
27.6	Exact zero and endpoint scopes	181
27.7	One independent exact original-coordinate calibration	181
28	Density of the actual relative frames and the limiting derivative graph	183
28.1	An exponential bound with the original arithmetic factors	183
28.2	Every centred fibre and its exponential moment	185
28.3	A complete weighted polynomial density proof	185
28.4	The multiplier, literal frames, and their strong limit	186
28.5	A fixed original source and the complete derivative domain	187
28.6	Invariant frames, sparse frames, and all original zero cases	190
28.7	Frame covariance and the exact half-line transport	191
29	The actual normal-image limit, its graph, and its arithmetic observation	192
29.1	Exact contractions of the finite normal images	193
29.2	The algebraic limit and its two exact metric presentations	194
29.3	Original arithmetic jets and the size of reconstruction	194
29.4	The split lift retains the algebraic source of the zero limit	196
30	The cyclic sum at every conormal depth	197
30.1	Original coordinates and the exact collided annihilator	197
30.2	Unscaled invariant bases and a full-unit retraction	199
30.3	The derivative tower, scalar first jets, and the unit obstruction	200
30.4	The finite cyclic derivative and its exact Weyl correction	202
30.5	All first jets and the symmetric conormal kernel	202
30.6	Cyclic relation response and the full product Jacobian	204
30.7	Trace pairings and all four complementary operator blocks	205
30.8	Exact two-factor symmetric presentation at every depth	207
30.9	Split lifts, source provenance and quantitative scope	208
31	The cyclic sum source inside the full connection	209
31.1	Complete polynomial moments and the original finite metric	209
31.2	A full real-exponential family of the same information identity	210
31.3	All cyclic normal energy is retained under frame enlargement	211
31.4	The canonical metric difference has its actual conormal lift	212
32	Exterior trace equality, the spectral projector, and quantitative coupling	213
32.1	1. Original objects and exact domains	213

32.2	2. Exact trace maximum and equality theorem	214
32.3	3. Sharp quantitative trace slack	215
32.4	4. Exact morphism from the projector difference to the coupling	216
32.5	5. Sharp original two-dimensional family	217
32.6	6. Critical generalized blocks at the trace maximum	217
32.7	The original quartet, its critical blocks, and a canonical equality example	219
32.8	7. Integration scope	220
33	The cyclic control as an exact change of arithmetic interpolation volume	220
33.1	The exact volume map and the reflected quartet bound	221
33.2	The positive exterior minor sum and its actual source vector	222
33.3	The exact first admitted degree	224
34	The original source and relation Toda flows with their exact control maps	224
34.1	Original arithmetic objects and the strip of real tilts	225
34.2	The source relation and its Laplace observation	227
34.3	Both monic determinants and their exact determinant-line map	228
34.4	Toda derivatives and the full curvature operators	228
34.5	The rank-one update, trace phase and original rank-two radius	230
34.6	Exact maps to the CV omitted determinant and positive minors	231
34.7	Both exact losses, their equality cases and the telescope	232
34.8	The first admitted degree and the empty quotient	233
34.9	The complete join with the original spectral coupling	235
34.10	Crosswalk and exact continuing arithmetic quantity	236
35	The original theta input and certified infinite-series tails	236
35.1	The original theta function and its Mellin transform	237
35.2	Mellin division, the dagger sign, and the tilted observation	238
35.3	The complete double series and an explicit omitted tail	239
35.4	The second moment and the first tensor coefficient	241
35.5	Certified numerical enclosures and their executable scope	242
36	An exact comparison of equality, volume contraction, and its discarded square	243
37	The original gamma convolution and the exact arithmetic coefficient maps	245
37.1	Source, attribution, and original coordinates	245
37.2	The gamma reference with its actual mass	247
37.3	The sum projection and exact coefficient powers	248
37.4	The supplied analytic generating join, with its branches	249
37.5	The complete density expansion for the original bounded references	251
37.6	Original-coordinate finite Grams and every tilt derivative	252
37.7	The original cyclic quotient and the exact cochain correction	253
37.8	Determinant indices and the original Toda substitution	254
38	Certified theta moments and the original gamma arithmetic coefficients	255

38.1	The theta input and its transform, with the original factors	256
38.2	Every derivative polynomial and the exact higher-moment map	257
38.3	The incomplete-gamma square and a proved omitted tail	258
38.4	The coefficient morphism at the actual gamma reference	259
38.5	Certified infinite quantities and exact rational error transport	261
38.6	Exact tensor coefficient interface through degree six	263
39	The first three original source costs at every tensor degree	264
40	Finite arithmetic Gram transfer with the original minimum quotient metric	266
40.1	The original source, relation and coordinate maps	267
40.2	Full Gamma mass and the finite coefficient evaluation	268
40.3	Finite coefficient errors and polynomial-test bounds	270
40.4	Exact least lifts and the boundary-resolvent remainder	271
40.5	A two-degree map for the actual consecutive ratio	272
40.6	Certified finite envelopes and quotient-dimensional bounds	273
40.7	Exact calibration and a bounded relative-volume counterexample	274
40.8	Verification scope	277
41	The complete complex amplitude in gamma sum descent	277
41.1	Original objects and the question answered	277
41.2	Unitary transport of the actual conditional projection	278
41.3	Strictness for an actual quartet, with its exact constant	279
41.4	Full complex coefficient transform and exact variance norm	279
41.5	The exact induced change of the original finite quotient metric	280
41.6	Support, the original unit, and the domain of arithmetic observation	281
41.7	Exact polynomial calibration with the original gamma mass	283
41.8	Provenance and consequence	284
42	Which completion retains the arithmetic jets?	284
42.1	Logarithmic coordinates and the original measure	284
42.2	Exponential weights and exact jet norms	285
42.3	The energy term carried by the weighted topology	286
42.4	The tesserine coordinate in the actual weight form	287
43	The current arithmetic route and its next calculation	287
43.1	A single chain of exact comparison maps	287
43.2	Topology, duality and the two sides of a weight estimate	288
43.3	Provenance and mathematical status	289
43.4	The original theta norm and the positive arithmetic frontier	291
43.5	The symmetric summand, the actual relation layer, and support	292
43.6	The original derivative cost and both positive-half-line kernels	293
43.7	The gamma determinant and the retained spectral-sum fibre	294
43.8	The sum derivative, its strict normal energy and the full conormal algebra	295
43.9	The full relative limit and the original arithmetic observation	296

43.10	Every relation depth and its full cyclic complement	297
43.11	Exterior amplification and the arithmetic volume to estimate	297
43.12	The original Toda volumes and the complete four-term loss	298
43.13	Certified theta input and a calculated comparison of the two gaps	299
43.14	Gamma convolution, certified higher input, and the complete quotient comparison	299
II	Complete source proof notes and chronology	303
A	PR 8: Finite operator comparison	304
A.1	Exact finite operator quotients of the tau-base theta cohomology	304
B	PR 9: Chain descent	316
B.1	Chain-level arithmetic descent over the tau base	316
C	PR 10: Global retraction	326
C.1	A global retraction of the original theta complex over the tau base	326
D	PR 11: Homotopy control	336
D.1	Homotopy choices and source-preserving arithmetic control	336
E	PR 13: Orthogonal boundary control	342
E.1	Orthogonal theta boundaries and rank-two arithmetic control	342
F	PR 17: Spectral-sum descent and the complete relative arithmetic fibre	348
F.1	Spectral-sum descent with the relative arithmetic fibre retained	348
G	Derived Split-Zero mathematics	354
G.1	SplitZero: reconstruction, internal homology, and balanced finite jets	354
H	PR 14: Coherent arithmetic interpolation and joint tensor boundary control	360
H.1	Coherent arithmetic interpolation and joint tensor boundary control	360
I	Full-jet interpolation kernels and the actual next relation layer	371
I.1	Full-jet interpolation kernels and the actual next relation layer	371
J	Arithmetic frontier estimates and symmetric tensor control	378
J.1	Arithmetic frontier estimates and symmetric tensor control	378
K	Complete spectral-sum descent: relative arithmetic fibres, dual resolution and all nilpotents	388
K.1	Spectral-sum descent with the relative arithmetic fibre retained	388
L	PR 20: Arithmetic sum density, the full relative connection and conormal derivatives	400
L.1	Quantitative sum-fibre connection and the conormal arithmetic trace	400
M	Cyclic arithmetic sum control and the complete contracted relation tower	411
M.1	Sum-generated arithmetic cohomology, scalar control, and exact conormal depth	411

N	Exterior trace amplification with the original arithmetic Gram and full quartet multiplicities	425
N.1	Result and scope	425
N.2	1. GitHub corrections integrated at a pinned revision	426
N.3	2. The original scalar and arithmetic source	427
N.4	3. The existing canonical metric and its rank-two operator	428
N.5	4. Exterior control: the allowance does not multiply by exterior degree	429
N.6	5. The exterior maps live on the original cochain source	430
N.7	6. A determinant line measures the sum of the positive defects	432
N.8	7. Spectral and metric projections are linked explicitly	433
N.9	8. Exact sum counting for a nonreal quartet	434
N.10	9. The quantitative target after this construction	435
N.11	10. Deligne architecture and the boundary retained in this step	436
N.12	11. Verification and reproducibility	437
N.13	References and provenance	437
O	Source and relation volume dynamics for the original arithmetic control	438
O.1	Results and scope	438
O.2	1. Integration of the new formalization report	439
O.3	2. Original base, coefficient square, and arithmetic source	439
O.4	3. One arithmetic function generates both determinant flows	441
O.5	4. The quotient determinant is the source/boundary volume ratio	442
O.6	5. The two classical Toda identities have their actual arithmetic input	443
O.7	6. The phase in the rank-two control is a scalar volume derivative	444
O.8	7. The exact scalar formula and a two-step upper bound	445
O.9	8. What the exterior lower bound forces on these arithmetic volumes	446
O.10	9. The Laplace input can be evaluated from the original theta source	447
O.11	10. Exact finite checks, an essential phase example, and current endpoint	448
O.12	References and source boundary	449
P	Supplied conormal derivative, symmetric restriction and uncompiled tower draft	450
Q	PR 21: conormal tower and cyclic depth formalization with its complete mathematical note	460
Q.1	Original conormal and cyclic relation towers	460
R	The coordinating Zeta contribution: determinant losses and spectral coupling	465
R.1	Exact join: determinant losses and spectral coupling	465
S	Gamma convolution descent and the original arithmetic coefficient correction	467
S.1	1. Scope and the inspected collaboration state	467
S.2	2. The original arithmetic source and its split maps	467
S.3	3. Carry the repository's gamma reference through every tensor power	468
S.4	4. The sum projection and its complete polynomial complement	469
S.5	5. Insert the complete arithmetic multiplier, not a substituted gamma measure	470

S.6	6. Exact arithmetic coefficients at every tensor degree	472
S.7	7. Feed the result into the two original Toda determinants	474
S.8	8. The current upper-control expression, with all reference costs exposed	475
S.9	9. A proved all-degree tail estimate for the actual arithmetic moments	476
S.10	10. What has and has not been concluded	477
S.11	References and exact source roles	477

Part I

Integrated mathematical continuation

1 Exact Split-Zero carriers and the local spectral bridge

Throughout this section, semiring homomorphisms preserve additive zero and multiplicative unit. Let R be a nonzero commutative ring and $\mathbb{B} = (\{0, 1\}, \vee, \wedge)$. Define

$$G(R) = R \sqcup \{\tau\}, \quad e = 0_R \in R \subset G(R).$$

Addition and multiplication of supported elements are those of R , including when their result is zero. The additional element satisfies

$$\tau + r = r + \tau = r, \quad \tau + \tau = \tau, \quad \tau r = r\tau = \tau, \quad \tau^2 = \tau.$$

Thus the semiring zero is τ , and $e \neq \tau$.

1.1 Scalar support and independent channel amplitudes

Proposition 1.1 (Pair carrier and scalar reflection). *There is a semiring isomorphism*

$$\eta_R : G(R) \longrightarrow \{(0, 0)\} \cup (R \times \{1\}) \subset R \times \mathbb{B}, \quad \tau \mapsto (0, 0), \quad r \mapsto (r, 1).$$

The coordinate maps

$$\begin{aligned} p_R : G(R) &\rightarrow R, & p_R(\tau) &= 0, & p_R(r) &= r, \\ \chi_R : G(R) &\rightarrow \mathbb{B}, & \chi_R(\tau) &= 0, & \chi_R(r) &= 1 \end{aligned}$$

are semiring homomorphisms. The map p_R is universal among semiring homomorphisms from $G(R)$ to a ring.

Proof. In the product semiring, two supported pairs have sum $(r + s, 1)$ and product $(rs, 1)$. These identities retain support when the amplitude vanishes. The pair $(0, 0)$ is an additive identity and an absorbing multiplicative zero. This proves closure and all operation identities, and the displayed map is bijective. Its coordinate projections give p_R, χ_R . If $f : G(R) \rightarrow A$ has ring target, then $e + e = e$ gives $f(e) + f(e) = f(e)$, hence $f(e) = 0_A$. Its restriction to the supported copy of R is therefore a unital ring homomorphism $h : R \rightarrow A$. The equality $f = hp_R$ holds on R and on τ , and determines h . Conversely every such h gives hp_R . $\square$

The larger carrier $R \times \mathbb{B}$ contains $(1, 0)$; its exact relation to $G(R)$ is the inclusion above. The pair presentation requires $R \times \{1\}$.

For a nontrivial bounded distributive lattice L , define

$$\begin{aligned} G_L(R) &= \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subset R \times L, \\ (r, \lambda) + (s, \mu) &= (r + s, \lambda \vee \mu), & (r, \lambda)(s, \mu) &= (rs, \lambda \wedge \mu). \end{aligned}$$

Write $z_\lambda = (0, \lambda)$, $\tau_L = z_{0_L}$, and $e_L = z_{1_L}$.

Proposition 1.2 (Lattice reflection with its original coefficient ring). *The displayed carrier is a commutative semiring. The projections*

$$p_L : G_L(R) \rightarrow R, \quad \chi_L : G_L(R) \rightarrow L$$

are respectively the universal ring reflection and the support homomorphism. In particular the amplitude codomain of $G_{\mathbb{B}^2}(\mathbb{C})$ is $\mathbb{C}$.

Proof. A summand with nonzero amplitude has top support, so its sum has top support. A factor with lower support has zero amplitude, so a product of lower support has zero amplitude. These facts prove closure. All laws restrict from the product semiring; distributivity uses that of L . Both projections preserve operations, zero and unit. If $f : G_L(R) \rightarrow A$ has ring target, $e_L + z_\lambda = e_L$ gives $f(z_\lambda) = 0$ by cancellation in A . Restriction to the supported copy of R is a ring homomorphism h and $f = hp_L$. Conversely each h gives this factorization. Surjectivity of p_L gives uniqueness. $\square$

Let $j : \mathbb{C} \rightarrow G(\mathbb{C})$ include the supported copy. It preserves addition, multiplication and unit; it takes ordinary zero to e , whereas the semiring zero is τ . Put

$$M = G(\mathbb{C})^2, \quad p^{(2)} = (p_{\mathbb{C}}, p_{\mathbb{C}}) : M \rightarrow \mathbb{C}^2, \quad \chi^{(2)} = (\chi_{\mathbb{C}}, \chi_{\mathbb{C}}) : M \rightarrow \mathbb{B}^2.$$

For $\epsilon \in \mathbb{B}$ let $u_0 = \tau, u_1 = e$.

Proposition 1.3 (Exact scalar-lattice and two-amplitude embeddings). *There are injective semiring homomorphisms*

$$b : G_{\mathbb{B}^2}(\mathbb{C}) \hookrightarrow M, \quad B : G_{\mathbb{B}^2}(\mathbb{C}^2) \hookrightarrow M$$

defined by

$$\begin{aligned} b(c, (1, 1)) &= (j(c), j(c)), \\ b(0, (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}), \\ B((a, b), (1, 1)) &= (j(a), j(b)), \\ B((0, 0), (\lambda_1, \lambda_2)) &= (u_{\lambda_1}, u_{\lambda_2}). \end{aligned}$$

Their amplitude and support identities are

$$p^{(2)}b = \Delta_{\mathbb{C}}p_{\mathbb{B}^2}, \quad \chi^{(2)}b = \chi_{\mathbb{B}^2}, \quad p^{(2)}B = p_{\mathbb{B}^2}, \quad \chi^{(2)}B = \chi_{\mathbb{B}^2},$$

where $\Delta_{\mathbb{C}}(c) = (c, c)$.

Proof. The pair presentation identifies M with

$$\{((a, b), (\lambda_1, \lambda_2)) \in \mathbb{C}^2 \times \mathbb{B}^2 : \lambda_1 = 0 \Rightarrow a = 0, \quad \lambda_2 = 0 \Rightarrow b = 0\},$$

with componentwise operations. The map b restricts the product semiring map $(c, \lambda) \mapsto ((c, c), \lambda)$. Lower support in its source forces $c = 0$, so it lands in the displayed subset. It retains the amplitude and both support bits and is injective. The map B restricts the identity coordinate map on $\mathbb{C}^2 \times \mathbb{B}^2$; lower support in its source forces both amplitudes to vanish, so it too lands in M and is injective. The prescriptions agree at the top-support zero. Both maps preserve zero and unit. Projection proves all four identities. $\square$

The element $(j(1), \tau) \in M$ is outside both images. There is no function $r : M \rightarrow G_{\mathbb{B}^2}(\mathbb{C}^2)$ satisfying both $p_{\mathbb{B}^2}r = p^{(2)}$ and $\chi_{\mathbb{B}^2}r = \chi^{(2)}$: it would send that element to $((1, 0), (1, 0))$, outside its target carrier. This identifies the precise obstruction to a reverse map preserving both projections.

The synchronized carrier has the exact map

$$G(\mathbb{C}^2) \cong G(\mathbb{C}) \times_{\mathbb{B}} G(\mathbb{C}) \hookrightarrow M.$$

It takes τ to (τ, τ) and a supported (a, b) to $(j(a), j(b))$. Equality of the two support characters leaves exactly those cases, proving bijectivity onto the fibre product. Operations agree componentwise. Its image has masks $(0, 0), (1, 1)$, while the other masks in M carry independent mixed amplitudes.

The map $p^{(2)}$ is the universal ring reflection of M . Indeed, (e, e) is additively idempotent, so every ring-target map kills it. Adding (e, e) to $x \in M$ gives the fully supported element of amplitude $p^{(2)}x$. The map is consequently determined by a ring homomorphism from the supported copy of $\mathbb{C}^2$, proving the universal property. A semiring isomorphism $G_{\mathbb{B}^2}(\mathbb{C}) \cong M$ would induce a ring isomorphism $\mathbb{C} \cong \mathbb{C}^2$. This is impossible because $(1, 0)$ is a nontrivial idempotent of $\mathbb{C}^2$, and $\mathbb{C}$ has only 0, 1. The positive relation between the original carriers is b , with diagonal amplitude projection.

1.2 Tesserine sectors and a supported critical fixed locus

Reserve τ for split absence and use κ for conjugation. Let

$$\sigma(s) = 1 - s, \quad \kappa(s) = \bar{s}, \quad K_4 = \{1, \sigma, \kappa, \sigma\kappa\}.$$

The real tesserine algebra and its complexification are

$$A_{\mathbb{R}} = \mathbb{R}[K_4] = \mathbb{R}[\sigma, \kappa]/(\sigma^2 - 1, \kappa^2 - 1), \quad A_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} A_{\mathbb{R}}.$$

The polynomial algebra in this presentation is commutative.

Proposition 1.4 (Character algebra and quartet coordinates). *For $\epsilon, \delta \in \{1, -1\}$ let*

$$E_{\epsilon\delta} = \frac{1}{4}(1 + \epsilon\sigma)(1 + \delta\kappa).$$

Character evaluation gives an algebra isomorphism

$$\mathcal{F} : A_{\mathbb{C}} \rightarrow \mathbb{C}^4, \quad x \mapsto (\text{ev}_{\epsilon,\delta}(x))_{\epsilon,\delta},$$

with inverse $(d_{\epsilon\delta}) \mapsto \sum d_{\epsilon\delta} E_{\epsilon\delta}$. For the exact quartet element

$$Q(s) = s \cdot 1 + (1 - s)\sigma + \bar{s}\kappa + (1 - \bar{s})\sigma\kappa, \quad s = \beta + i\gamma,$$

the four character values are

$$c_{++} = 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1), \quad c_{--} = 4i\gamma.$$

Projection to the last two specified characters is a unital algebra map $\pi : A_{\mathbb{C}} \rightarrow \mathbb{C}^2$. The composite πQ is a real affine bijection

$$\mathbb{C} \rightarrow \mathbb{R} \times i\mathbb{R}, \quad s \mapsto (a, b) = (c_{-+}(s), c_{--}(s)), \quad s = \frac{1}{2} + \frac{1}{4}(a + b).$$

Proof. Evaluation of $E_{\epsilon\delta}$ at (ϵ', δ') is $\frac{1}{4}(1 + \epsilon\epsilon')(1 + \delta\delta')$. It is one when both signs agree and zero otherwise. This proves the two inverse formulas and all orthogonal idempotent identities, retaining the factors of four. Character evaluation preserves the algebra operations. Directly,

$$\begin{aligned} c_{++} &= s + (1 - s) + \bar{s} + (1 - \bar{s}) = 2, \\ c_{+-} &= s + (1 - s) - \bar{s} - (1 - \bar{s}) = 0, \\ c_{-+} &= s - (1 - s) + \bar{s} - (1 - \bar{s}) = 2(s + \bar{s}) - 2 = 4\beta - 2, \\ c_{--} &= s - (1 - s) - \bar{s} + (1 - \bar{s}) = 2(s - \bar{s}) = 4i\gamma. \end{aligned}$$

The proposed inverse recovers $\beta + i\gamma$. Conversely every real a and purely imaginary b is recovered by substituting that inverse into the two character expressions. $\square$

The quartet image has real affine dimension two. Its real linear span has dimension three, spanned by E_{++}, E_{-+}, iE_{--} , retaining the constant character coordinate. The real tesserine algebra has dimension four. The maps $Q, \mathcal{F}, \pi$ and the displayed affine inverse specify the exact relationship of these objects.

Proposition 1.5 (Supported sector lift and fixed locus). *Define*

$$\Psi : \mathbb{C} \rightarrow M, \quad \Psi(s) = (j(c_{-+}(s)), j(c_{--}(s))).$$

It has $\chi^{(2)}\Psi(s) = (1, 1)$ for all s . The $G(\mathbb{C})$ -linear automorphisms

$$T_{\sigma}(x, y) = ((-1)x, (-1)y), \quad T_{\kappa}(x, y) = (x, (-1)y)$$

of the free semimodule M give

$$\Psi(\sigma s) = T_{\sigma}\Psi(s), \quad \Psi(\kappa s) = T_{\kappa}\Psi(s).$$

Here $-1 = j(-1)$ is a supported unit. Their product satisfies

$$\text{Fix}(T_{\sigma}T_{\kappa}) = \{\tau, e\} \times G(\mathbb{C}), \quad \Psi^{-1} \text{Fix}(T_{\sigma}T_{\kappa}) = \{s : \Re s = \frac{1}{2}\}.$$

Proof. The sector formulas give $(a, b)(1 - s) = (-a, -b)$ and $(a, b)(\bar{s}) = (a, -b)$. Multiplication by $j(-1)$ negates supported amplitudes and fixes τ . It preserves addition, commutes with scalar multiplication and squares to identity. This proves the automorphism and equivariance statements. Its fixed points are τ and supported $j(c)$ with $c = -c$. Since $2 \neq 0$ in $\mathbb{C}$, the latter is precisely e . The product action is $((-1)x, y)$, proving its fixed locus. On the image of Ψ the first coordinate is supported and is fixed exactly when $4\beta - 2 = 0$. $\square$

The scalar-lattice channel atoms have images $b(z_{10}) = (e, \tau)$ and $b(z_{01}) = (\tau, e)$. The separate support swap has fixed masks $(0, 0), (1, 1)$. Since $\chi^{(2)}\Psi$ is constantly $(1, 1)$, it cannot detect the critical line by that swap. The support action induced by $T_\sigma T_\kappa$ is identity; its amplitude action is $(-a, b)$. The preceding proposition supplies the exact fixed-locus test with the first-channel supported zero retained.

1.3 Exact sector, holonomy and positive-moment maps

Fix a prime p , $\ell_p = \log p > 0$, and $u \in S^1$. For an arithmetic packet, take $u = \psi(\text{Frob}_p)$ with $L/\mathbb{Q}$ finite abelian, p unramified, and $\psi : \text{Gal}(L/\mathbb{Q}) \rightarrow S^1$ a character. All powers use $p^w = \exp(w\ell_p)$ with the positive real logarithm.

Proposition 1.6 (Sector-to-holonomy map with constants retained). *The analytic function*

$$H_{p,u} : \mathbb{C}^2 \rightarrow \mathbb{C}^\times, \quad H_{p,u}(a, b) = u \exp\left(\frac{\ell_p}{4}(a + b)\right)$$

gives a function $H_{p,u}p^{(2)}$ on M . On the quartet image,

$$H_{p,u}(c_{-+}(s), c_{--}(s)) = up^{s-1/2} =: z_{p,u}(s),$$

$$\frac{\log |z_{p,u}(s)|}{\ell_p} = \frac{c_{-+}(s)}{4} = \Re s - \frac{1}{2}, \quad |z_{p,u}(s)|^2 - 1 = \exp\left(\frac{\ell_p}{2}c_{-+}(s)\right) - 1.$$

Proof. The exact inverse for the sector coordinates gives $s - \frac{1}{2} = (a + b)/4$. Substitution proves the exponential identity. On that image a is real and b is purely imaginary. Since $|u| = 1$, taking absolute values yields $\exp(\ell_p a/4)$. Taking its real logarithm and, separately, its square proves the two remaining identities. $\square$

The radial exponent is $c_{-+}/4$; the exponent $c_{-+}/2$ belongs to the squared modulus. The map $H_{p,u}$ is used with its analytic-function type, without any assertion that it preserves semiring addition.

Proposition 1.7 (Rank-one packet). *For $z \in \mathbb{C}$ define*

$$v_2(z) = \begin{pmatrix} 1 \\ z \end{pmatrix}, \quad T_2(z) = v_2(z)v_2(z)^* = \begin{pmatrix} 1 & \bar{z} \\ z & |z|^2 \end{pmatrix}, \quad P_z(w) = w - \bar{z}.$$

Then $T_2(z)$ is Hermitian positive semidefinite of rank one and

$$\ker T_2(z) = \mathbb{C}(-\bar{z}, 1)^T.$$

It is Toeplitz exactly when $|z| = 1$, equivalently when the unique root of P_z lies on S^1 . For $z = z_{p,u}(s)$ this is equivalent to $\Re s = \frac{1}{2}$.

Proof. For $x \in \mathbb{C}^2$, $x^*T_2(z)x = |v_2(z)^*x|^2 \geq 0$. The vector $v_2(z)$ is nonzero and hence its outer product has rank one. The adjoint kernel equation is $x_0 + \bar{z}x_1 = 0$, giving the displayed space. A two-by-two matrix is Toeplitz exactly when its diagonal entries agree, here $1 = |z|^2$. The root $\bar{z}$ has modulus $|z|$. The final equivalence follows from the preceding proposition and $\ell_p > 0$. $\square$

Theorem 1.8 (Positive finite packets and their exact diagonal defect). *Let $r \geq 1$, $\alpha_j > 0$, and $z_j \in \mathbb{C}$. For $n \geq 1$ put*

$$v_n(z) = (1, z, \dots, z^{n-1})^\top, \quad M_n = \sum_{j=1}^r \alpha_j v_n(z_j) v_n(z_j)^*.$$

The matrix M_n is Hermitian positive semidefinite. For $n \geq 3$, it is Toeplitz exactly when every $|z_j| = 1$. With matrix indices starting at zero,

$$\mathcal{D}(M_n) := (M_n)_{22} - 2(M_n)_{11} + (M_n)_{00} = \sum_{j=1}^r \alpha_j (|z_j|^2 - 1)^2.$$

For these packets and $n \geq 3$, vanishing of $\mathcal{D}$ already characterizes Toeplitzness. If the z_j are pairwise distinct and $r \leq n$, the rank is r . If $r < n$, coefficient vectors identify the kernel with

$$\left(\prod_{j=1}^r (w - \bar{z}_j) \right) \mathbb{C}[w]_{\leq n-1-r} \subset \mathbb{C}[w]_{\leq n-1}.$$

In particular $r = n - 1$ gives corank one and exactly the conjugated roots $\bar{z}_1, \dots, \bar{z}_{n-1}$.

Proof. The identity

$$x^* M_n x = \sum_j \alpha_j |v_n(z_j)^* x|^2$$

proves positivity and shows that its kernel is the simultaneous kernel of all $v_n(z_j)^*$. The first three diagonal entries are $\sum \alpha_j$, $\sum \alpha_j |z_j|^2$, and $\sum \alpha_j |z_j|^4$. Their displayed linear combination gives the defect without rescaling any weight. Toeplitzness equates these entries, forcing zero defect. Every weight is positive and every square nonnegative, hence all $|z_j| = 1$. Conversely that condition gives $(M_n)_{k\ell} = \sum_j \alpha_j z_j^{k-\ell}$, a Toeplitz matrix. The same argument proves the zero-defect characterization.

Let V have columns $v_n(z_j)$. Its first r rows have Vandermonde determinant $\prod_{i < j} (z_j - z_i) \neq 0$, so V has rank r . The established kernel identity is $\ker M_n = \ker V^*$, proving the rank. For $P_x(w) = \sum_{k=0}^{n-1} x_k w^k$, the kernel equations say $P_x(\bar{z}_j) = 0$. Distinct linear factors divide P_x exactly when these equations hold: division by one factor preserves the other root equations since that factor is nonzero at those roots. This proves the product and degree bound. At $r = n - 1$, the remaining factor is a scalar; for a nonzero kernel vector it is nonzero, so there are no further roots. $\square$

The size bound is sharp for general positive packets. At $n = 2$, the data $(\alpha_1, \alpha_2) = (4, 1)$ and $(z_1, z_2) = (1/2, 2)$ give

$$M_2 = 4 \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1/4 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}.$$

This is positive definite and Toeplitz, while neither atom is on S^1 . At size three the diagonal is $(5, 5, 65/4)$ and the defect is $45/4$. One atom retains the two-by-two criterion. Zero-weight atoms are invisible; repeated atoms combine their weights before applying the distinct-column rank count.

The exact sector defect is therefore

$$\mathcal{D}(M_n) = \sum_{j=1}^r \alpha_j \left[\exp \left(\frac{\log p_j}{2} c_{-+}(s_j) \right) - 1 \right]^2.$$

The function chain

$$M \xrightarrow{p^{(2)}} \mathbb{C}^2 \xrightarrow{H_{p,u}} \mathbb{C}^\times \xrightarrow{z \mapsto v_n(z) v_n(z)^*} \text{Herm}_{\geq 0}(n)$$

has explicit maps at every step. Positive finite summation supplies the preceding theorem.

1.4 The theta-packet form and its scalar comparison

Proposition 1.9 (Eigenpair form quotient and sector channel). *For every complex d -by- d matrix A , Hermitian matrix G , and nonzero eigenvector v with $Av = \rho v$, let $W = A^*G + GA - G$. Then*

$$v^*Wv = (2\Re\rho - 1)v^*Gv.$$

*On the domain of these data where $v^*Gv \neq 0$, the quotient*

$$\omega(A, G, v, \rho) = \frac{v^*Wv}{v^*Gv} = 2\Re\rho - 1 = \frac{c_{-+}(\rho)}{2}$$

is real. Its associated holonomy satisfies

$$|z_{p,u}(\rho)|^2 = \exp(\ell_p\omega), \quad z_{p,u}(\rho) = u \exp\left(\frac{\ell_p}{4}(c_{-+}(\rho) + c_{--}(\rho))\right).$$

Proof. Since $(Av)^* = \bar{\rho}v^*$,

$$v^*(A^*G + GA - G)v = (Av)^*Gv + v^*G(Av) - v^*Gv = (\bar{\rho} + \rho - 1)v^*Gv.$$

This proves the unquotiented identity even when the right factor is zero. Division on the stated domain gives ω , and $c_{-+}(\rho) = 4\Re\rho - 2$ gives the factor $1/2$. The holonomy identities follow from the exact sector exponential above. $\square$

This is the typed map from an eigenpair of the form packet to its Split-Zero sector and local moment matrix. The moment matrix is positive for every parameter, so that positivity cannot supply a further zero-defect conclusion. For $\rho = 1, u = 1$,

$$z = \sqrt{p}, \quad T_2(z) = \begin{pmatrix} 1 & \sqrt{p} \\ \sqrt{p} & p \end{pmatrix} \geq 0,$$

and its diagonal entries differ. On the form side $A = (1), G = (1), v = (1)$ give $W = (1)$ and $\omega = 1$; the exact maps take these data to the same off-critical packet. Any arithmetic identity forcing the actual family of W to vanish on its eigenvectors must therefore be established from that family's construction. These maps retain that requirement rather than replacing it by Gram positivity.

1.5 Projective zeta at all zeros and at its pole

A pair in $G(\mathbb{C})^2$ is unimodular exactly when at least one amplitude is nonzero. Such a coordinate is a supported unit. If both amplitudes are zero, every linear combination has zero amplitude and cannot be the unit. Let $\mathbb{P}_{\text{um}}^1(G(\mathbb{C}))$ denote unimodular pairs modulo common unit multiplication.

Proposition 1.10 (Projective amplitude section and reciprocal). *The map*

$$p_* : \mathbb{P}_{\text{um}}^1(G(\mathbb{C})) \rightarrow \mathbb{P}^1(\mathbb{C}), \quad [x : y] \mapsto [p(x) : p(y)]$$

has an injective section

$$\iota : \mathbb{P}^1(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad [a : b] \mapsto [j(a) : j(b)],$$

whose image is the full-support component. With

$$\mathcal{R} : G(\mathbb{C}) \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad x \mapsto [1 : x], \quad r : \mathbb{C} \rightarrow \mathbb{P}^1(\mathbb{C}), \quad c \mapsto [1 : c],$$

one has $p_\mathcal{R} = rp_{\mathbb{C}}$ and $\mathcal{R}j = \iota r$.*

Proof. Unimodularity ensures the projected pair is nonzero. A common supported unit projects to a nonzero complex scalar, so p_* is well defined. A nonzero ordinary pair becomes a unimodular supported pair; changing its representative lifts to common supported unit multiplication. Thus ι is well defined, and $p_*\iota$ is identity. Every full-support pair is the supported lift of its amplitude pair. The pair $(1, x)$ is always unimodular, and projection of its coordinates proves both remaining identities. $\square$

In particular $\mathcal{R}(e) = [1 : e] = \infty_e$ and $\mathcal{R}(\tau) = [1 : \tau] = \infty_\tau$. Both project to $[1 : 0]$, while their Boolean projective supports are $[1 : 1]$ and $[1 : 0]$. Similarly $[e : 1]$ and $[\tau : 1]$ project to the ordinary zero while retaining different supports.

Proposition 1.11 (Two exact charts for the supported zeta map). *For the meromorphic zeta function with its simple pole of residue one at 1, the full-support lift is*

$$Z_{\text{sp}} : \mathbb{C} \rightarrow \mathbb{P}_{\text{um}}^1(G(\mathbb{C})), \quad Z_{\text{sp}}(s) = \begin{cases} [j(\zeta(s)) : 1], & s \neq 1, \\ [1 : e], & s = 1. \end{cases}$$

It equals $\iota\hat{\zeta}$. At every zeta zero ρ its value is $[e : 1]$. Near the pole it has the chart

$$Z_{\text{sp}}(s) = [1 : j(h(s))], \quad h(s) = 1/\zeta(s), \quad h(1) = 0, \quad h'(1) = 1.$$

The reciprocal is a finite-amplitude coordinate away from the zeta zeros, including its extension at 1. At a zero of order m , it has a pole of order m .

Proof. Put $t = s - 1$ and write $\zeta(1+t) = t^{-1} + g(t)$ with g holomorphic near zero. Then $h(1+t) = t/(1+tg(t))$. Its denominator has value one at zero, so this is holomorphic there with value zero and derivative one. On the overlap where $\zeta \neq 0$, common multiplication by $j(\zeta(s))$ gives $[1 : j(h(s))] = [j(\zeta(s)) : 1]$. This proves the transition and pole value. The first chart directly gives $[e : 1]$ at each zero. At a zero of order m , write $\zeta(\rho+t) = t^m u(t)$ with $u(0) \neq 0$. Its reciprocal is $t^{-m} u(t)^{-1}$, proving its pole order; the first chart retains the original order m . $\square$

The swap $J_*[x : y] = [y : x]$ is induced by an invertible permutation matrix and commutes with ι, p_* . Hence J_*Z_{sp} is the global reciprocal projective map, sending zeta zeros to ∞_e and the pole to $[e : 1]$. Analytic statements use the complex structure transported to the full-support component by ι ; no topology on the extra support components is needed.

1.6 Matrix packets and their full relation module

Theorem 1.12 (Block packet with explicit hypotheses). *Let V be a complex Hilbert space of finite dimension $d > 0$ with fixed positive-definite Hermitian inner product. Let $g \in \text{GL}(V)$ and $m \geq 1$. Use that metric and its orthogonal direct sums for all adjoints. Define*

$$W_m(g) : V \rightarrow V^{m+1}, \quad v \mapsto (v, gv, \dots, g^m v), \quad \mathbb{T}_m(g) = W_m(g)W_m(g)^*,$$

$$D_m(g) : V^m \rightarrow V^{m+1}, \quad \begin{cases} (D_m(g)y)_0 = -g^* y_1, \\ (D_m(g)y)_k = y_k - g^* y_{k+1} \quad (1 \leq k \leq m-1), \\ (D_m(g)y)_m = y_m. \end{cases}$$

The matrix $\mathbb{T}_m(g)$ is positive semidefinite and

$$\ker \mathbb{T}_m(g) = \text{im } D_m(g), \quad \text{rank } \mathbb{T}_m(g) = d, \quad \text{corank } \mathbb{T}_m(g) = md.$$

It is block Toeplitz exactly when g is unitary for the fixed metric.

Proof. The Gram factorization gives positivity and $\ker \mathbb{T}_m = \ker W_m^*$. The first component of $W_m v$ is v , so W_m is injective and the kernel dimension is $(m+1)d - d = md$. The adjoint is

$$W_m^*(v_0, \dots, v_m) = \sum_{k=0}^m (g^*)^k v_k.$$

Substitution of $D_m(y)$ gives

$$-g^* y_1 + \sum_{k=1}^{m-1} ((g^*)^k y_k - (g^*)^{k+1} y_{k+1}) + (g^*)^m y_m = 0.$$

Thus its image lies in the kernel. If $D_m(y) = 0$, the last component gives $y_m = 0$, and backward substitution gives each preceding $y_k = 0$. Its image dimension is md , proving equality with the kernel. The block of indices j, k is $g^j (g^*)^k$. For unitary g this is g^{j-k} , hence block Toeplitz. Conversely block Toeplitzness equates blocks $(0, 0), (1, 1)$, giving $I = gg^*$. Invertibility yields $g^* = g^{-1}$ and unitarity. $\square$

At $m = 0$ the only block is I , so the bound $m \geq 1$ is essential. At $m = 1$, the full kernel is $\{(-g^* y, y) : y \in V\}$; its vector polynomials are $(wI - g^*)y$. Taking determinant gives the exact scalar map

$$wI - g^* \mapsto \det(wI - g^*) = \overline{\det(\bar{w}I - g)}.$$

The equality follows by conjugating the determinant expansion and transposing. It retains conjugate eigenvalues and multiplicities. For the standard metric and $J_t = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ with $t > 0$, the polynomial is $(w - 1)^2$ but

$$J_t J_t^* = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix} \neq I.$$

Its block packet consequently fails Toeplitzness, although the determinant roots lie on the unit circle. The kernel remains $\{(-J_t^* y, y) : y \in \mathbb{C}^2\}$, retaining the matrix information that determinant forgets.

2 The historical packet ghost and its full arithmetic jet quotient

The source of the packet construction in this section is the response dated 24 June 2026, 11:13:04 AM in the SSD export *Globalization Semiring Analysis*, whose exact path, hash and response-line locators are recorded in `logbook/historical_split_transcripts_20260912.md`. The source proves the permutation-packet calculation. We give that calculation completely, then construct its comparison with the full local algebras of the arithmetic packet. In particular, we retain the nilpotent jet operators and distinguish the source's complex-linear permutation from arithmetic conjugation by explicit maps.

Let Z be a finite set of complex numbers stable under $c(\rho) = \bar{\rho}$ and $\sigma(\rho) = 1 - \rho$. Assign a positive integer m_ρ constant on each orbit. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad A[f] = [sf].$$

We use the original coordinate s , without replacing its multiplicative operator by its semisimple part. Put

$$L_Z = \bigoplus_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad e_{\rho,k} = t_\rho^k \quad (0 \leq k < m_\rho)$$

in the indicated direct summand. The coefficient inner product makes all $e_{\rho,k}$ orthonormal; it is an explicitly chosen reference inner product, not the theta-source metric.

Proposition 2.1 (Original-coordinate identification with all jets). *The map*

$$T : E_h \longrightarrow L_Z, \quad T[f]_\rho = \sum_{k=0}^{m_\rho-1} \frac{f^{(k)}(\rho)}{k!} t_\rho^k$$

is a complex-algebra isomorphism. In these coordinates

$$TAT^{-1} = A_{ss} + N, \quad A_{ss}e_{\rho,k} = \rho e_{\rho,k}, \quad Ne_{\rho,k} = \begin{cases} e_{\rho,k+1} & k+1 < m_\rho, \\ 0 & k+1 = m_\rho. \end{cases}$$

Proof. Taylor multiplication modulo $t_\rho^{m_\rho}$ proves multiplicativity. Its kernel consists exactly of polynomials divisible by every $(s - \rho)^{m_\rho}$ and hence by their product h , since the factors are pairwise coprime. For completeness an inverse is explicit. Set $h_\rho = h/(s - \rho)^{m_\rho}$, and let a_ρ be the Taylor polynomial of $1/h_\rho(s)$ at ρ through degree $m_\rho - 1$. Then $E_\rho = [a_\rho h_\rho]$ has residue 1 at the ρ factor and 0 at every other factor. Thus

$$T^{-1}e_{\rho,k} = E_\rho[(s - \rho)^k]$$

gives the inverse. Multiplication by $s = \rho + t_\rho$ gives the displayed operator, including its last nilpotent column. $\square$

In every subsequent operator formula on L_Z , the symbol A denotes the transported operator $TAT^{-1} : L_Z \rightarrow L_Z$ of Proposition 2.1. The polynomial-coordinate quotient at the end is equipped with the original multiplication-by- s operator.

Define complex-linear permutations C_0, S_0 and a complex-linear parity operator H on this complete basis by

$$C_0e_{\rho,k} = e_{\bar{\rho},k}, \quad S_0e_{\rho,k} = e_{1-\rho,k}, \quad He_{\rho,k} = (-1)^k e_{\rho,k}.$$

Also let K be the conjugate-linear map $K(\sum a_{\rho,k}e_{\rho,k}) = \sum \bar{a}_{\rho,k}e_{\rho,k}$. All four maps commute, square to the identity, and preserve products in L_Z , with K conjugate-linear on scalars.

Proposition 2.2 (Typed arithmetic involutions). *Under T , ordinary conjugation of polynomial coefficients is the conjugate-linear map $\mathcal{C} = C_0K$, and pullback by $s \mapsto 1 - s$ is the complex-linear map $\mathcal{S} = S_0H$. In particular*

$$C_0 = \mathcal{C}K, \quad S_0 = \mathcal{S}H, \quad \mathcal{C}A = A\mathcal{C}, \quad \mathcal{S}AS = I - A.$$

Here the equality involving $\mathcal{C}$ is an equality of conjugate-linear maps. Moreover $C_0N = NC_0$, $S_0N = NS_0$, and $HN = -NH$.

Proof. If $f^\#(s) = \overline{f(\bar{s})}$, then $(f^\#)^{(k)}(\rho)/k! = \overline{f^{(k)}(\bar{\rho})/k!}$, giving $\mathcal{C} = C_0K$. The k th derivative of $f(1 - s)$ at ρ is $(-1)^k f^{(k)}(1 - \rho)$, giving $\mathcal{S} = S_0H$. The ideal (h) is preserved by both operations: conjugation fixes the monic polynomial h , and $h(1 - s) = (-1)^{\deg h} h(s)$. Conjugating coefficients fixes the indeterminate s , whereas substituting $1 - s$ replaces multiplication by s by multiplication by $1 - s$. This proves the two identities for A . The assertions about N follow on each basis vector, including the vector of largest jet degree, which both sides send to zero. $\square$

Theorem 2.3 (Historical ghost with complete multiplicities). *Set*

$$R = C_0S_0, \quad \Gamma = C_0 - S_0, \quad P_- = \frac{I - R}{2}, \quad P_+ = \frac{I + R}{2}.$$

These satisfy

$$\Gamma^* = \Gamma, \quad \Gamma^2 = 4P_-, \quad \Gamma e_{\rho,k} = 0 \iff \Re \rho = \frac{1}{2}.$$

On a nonreal off-axis quartet of common multiplicity m , Γ has eigenvalues -2 and 2 each with multiplicity m , and 0 with multiplicity $2m$. Furthermore

$$\text{rank } P_- = 2m, \quad \text{Tr}(\Gamma^*\Gamma) = 8m.$$

On a critical-line orbit $\Gamma = P_- = 0$ at every jet degree.

Proof. The permutations C_0, S_0 are commuting selfadjoint involutions in the declared reference inner product. Consequently $(C_0 - S_0)^2 = 2I - 2C_0S_0$. The equality $C_0e_{\rho,k} = S_0e_{\rho,k}$ says exactly that the two labelled basis vectors have the same root, namely $\bar{\rho} = 1 - \rho$, which is equivalent to $2\Re\rho = 1$.

On an off-axis nonreal quartet order the roots as $(\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho})$. At every fixed jet degree the four joint (C_0, S_0) eigenspaces have characters $(1, 1), (1, -1), (-1, 1), (-1, -1)$. Explicit joint eigenvectors are

$$e_{\rho,k} + \epsilon e_{\bar{\rho},k} + \delta e_{1-\rho,k} + \epsilon \delta e_{1-\bar{\rho},k}, \quad \epsilon, \delta \in \{1, -1\}.$$

They form a basis: their pairwise inner products are zero and each has squared norm 4. The eigenvalue of Γ is $\epsilon - \delta$. Thus its squared eigenvalues are 0, 4, 4, 0, giving the rank and trace with every one of the m jet degrees included. On a critical orbit the two permutations agree on every basis vector. $\square$

For real off-axis orbits the same proof gives rank m and squared trace $4m$, since the orbit has two elements. Actual nontrivial zeta zeros in $0 < \Re s < 1$ are nonreal: for real $0 < s < 1$ the alternating series $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s}$ is strictly positive by grouping successive pairs, and $\zeta(s) = \eta(s)/(1 - 2^{1-s}) < 0$. For completeness the paired terms satisfy

$$(2n-1)^{-s} - (2n)^{-s} = s \int_{2n-1}^{2n} x^{-s-1} dx.$$

They converge uniformly on each compact subset of $\Re s > 0$, because on such a subset $|s| \leq M$, $\Re s \geq \delta > 0$, and the absolute value is at most $M(2n-1)^{-1-\delta}$. Thus η is analytic there. The product identity $\eta(s) = (1 - 2^{1-s})\zeta(s)$ follows first in $\Re s > 1$ by splitting even terms and then by analytic continuation. Division is used here only on the real interval $0 < s < 1$, where the denominator is strictly negative. Hence the quartet formula applies to off-axis arithmetic packets.

Proposition 2.4 (The exact group-algebra quotient). *For $W = \langle c, \sigma \mid c^2 = \sigma^2 = 1, c\sigma = \sigma c \rangle$ the homomorphism*

$$q : \mathbb{C}[W] \longrightarrow \mathbb{C}[u]/(u^2 - 1), \quad c \longmapsto u, \quad \sigma \longmapsto u$$

is surjective, with

$$I_q := \ker q = (c - \sigma)\mathbb{C}[W] = \mathbb{C}(c - \sigma) \oplus \mathbb{C}(1 - c\sigma).$$

Under the representation $c \mapsto C_0, \sigma \mapsto S_0, I_q L_Z = P_- L_Z$. Thus $L_Z/P_- L_Z$ is the largest quotient of the W -representation on which $C_0 = S_0$.

Proof. In the basis $(1, c, \sigma, c\sigma)$, q sends $a + bc + d\sigma + ec\sigma$ to $(a + e) + (b + d)u$. Its kernel is therefore spanned by $c - \sigma$ and $1 - c\sigma$. Multiplication by c sends $c - \sigma$ to $1 - c\sigma$, and multiplication by any generator stays in their span. This proves the ideal formula. Since $P_- = \Gamma^2/4$ and $\Gamma = \Gamma P_-$, their images agree and equal $I_q L_Z$. A quotient kills I_q precisely when it kills this image; this proves its universal property without any invariance assumption about the arithmetic generator. $\square$

The generator acts on the complete local rings. The next calculation constructs exactly the extra kernel needed to descend that action.

Theorem 2.5 (Arithmetic saturation of the historical ghost quotient). *Put*

$$Be_{\rho,k} = (\Re\rho - \tfrac{1}{2})e_{\rho,k}, \quad Z_{\text{crit}} = \{\rho \in Z : \Re\rho = \tfrac{1}{2}\}, \quad L_{\text{off}} = \bigoplus_{\rho \notin Z_{\text{crit}}} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}).$$

The smallest A -invariant subspace containing $I_q L_Z$ is exactly

$$I_q L_Z + A(I_q L_Z) = L_{\text{off}}.$$

On each off-axis orbit this sum is direct. The largest quotient on which both the original generator A descends and $C_0 = S_0$ is therefore the exact algebra and W -module quotient

$$L_Z \longrightarrow L_{Z_{\text{crit}}}, \quad (f_\rho)_\rho \longmapsto (f_\rho)_{\rho \in Z_{\text{crit}}}.$$

Its kernel is L_{off} , including every local multiplicity. In polynomial coordinates this is the quotient

$$\mathbb{C}[s]/(h) \longrightarrow \mathbb{C}[s]/(h_{\text{crit}}), \quad [f]_h \longmapsto [f]_{h_{\text{crit}}}, \quad h_{\text{crit}} = \prod_{\rho \in Z_{\text{crit}}} (s - \rho)^{m_\rho},$$

whose kernel is $(h_{\text{crit}})/(h)$; the target is zero when $h_{\text{crit}} = 1$.

Proof. The permutation R sends ρ to $1 - \bar{\rho}$, preserving its imaginary part and reversing $\Re \rho - 1/2$. It preserves the jet index and commutes with N . Accordingly $RB = -BR$ and

$$A = \frac{1}{2}I + B + iY + N, \quad Ye_{\rho,k} = \Im \rho e_{\rho,k}, \quad RY = YR, \quad RN = NR.$$

This is an exact identity for A , with N retained. If $v \in P_-L_Z$, then $(\frac{1}{2}I + iY + N)v \in P_-L_Z$ whereas $Bv \in P_+L_Z$. Therefore

$$P_+Av = Bv \quad (v \in P_-L_Z).$$

On each off-axis orbit all diagonal entries of B are nonzero, so $B : P_-L_Z \rightarrow P_+L_Z$ is an isomorphism on that orbit, its inverse being the restriction of the original diagonal operator B^{-1} . For $w = w_- + w_+$ on that orbit, take $v = B^{-1}w_+$; then $P_+Av = w_+$ and $w - Av \in P_-L_Z$. Thus every w lies in $P_-L_Z + A(P_-L_Z)$. If $Av \in P_-L_Z$ with $v \in P_-L_Z$, then $Bv = P_+Av = 0$ and $v = 0$, proving directness.

On a critical orbit $P_- = 0$. Both A and all permutations preserve the direct sum of off-axis factors, so the subspace just calculated is invariant and is precisely the smallest A -invariant subspace containing P_-L_Z . If a linear map F intertwines A with an operator on its target and satisfies $FC_0 = FS_0$, its kernel contains I_qL_Z and is A -invariant. It consequently contains L_{off} . Conversely the displayed coordinate projection intertwines A , preserves products and has $C_0 = S_0$ on its target. This proves the universal quotient assertion, also for targets carrying all the stated structures. The polynomial quotient and its kernel follow from $h_{\text{crit}} \mid h$ and Proposition 2.1. $\square$

Proposition 2.6 (Exact displacement and nilpotent contribution). *In the declared reference coefficient metric, let $W_{\text{ref}} = A^* + A - I$. Then*

$$W_{\text{ref}} = 2B + N + N^*, \quad \frac{W_{\text{ref}} - RW_{\text{ref}}R}{2} = 2B, \quad \frac{W_{\text{ref}} + RW_{\text{ref}}R}{2} = N + N^*.$$

For the original tessarine coordinate $c_{-+}(\rho) = 4(\Re \rho - 1/2)$, the first displayed component equals multiplication by $c_{-+}/2$.

Proof. The adjoint of A_{ss} is diagonal with entries $\bar{\rho}$, so $A_{\text{ss}}^* + A_{\text{ss}} - I = 2B$. Because R is a self-adjoint permutation commuting with N , it also commutes with N^* . Conjugation by R reverses B . Adding and subtracting the two resulting identities proves both formulas, including the entire nilpotent term. The tessarine identification uses $c_{-+} = 2(2\Re \rho - 1) = 4B$ with its original factor 4. $\square$

The precise map from this reference calculation to a theta-source metric retains the full metric matrix. Let G_E be that positive Hermitian matrix in the original polynomial basis of E_h ; then in the declared coefficient basis of L_Z it is

$$G_L = (T^{-1})^* G_E T^{-1}.$$

Writing the original polynomial-coordinate generator as A_E and the transported one as $A_L = TA_ET^{-1}$ for this formula, one obtains

$$\begin{aligned} W_E &= A_E^* G_E + G_E A_E - G_E, \\ W_L &= A_L^* G_L + G_L A_L - G_L = (T^{-1})^* W_E T^{-1}, \\ W_L &= G_L W_{\text{ref}} + [A_L^*, G_L], \quad [A_L^*, G_L] = A_L^* G_L - G_L A_L^*. \end{aligned}$$

The first two identities follow by inserting $A_L = TA_ET^{-1}$ and cancelling adjacent factors $T^{-1}T$ and $T^*(T^{-1})^*$. The last identity follows by expanding $G_L(A_L^* + A_L - I) + (A_L^* G_L - G_L A_L^*)$. Thus both nilpotent and metric-commutator contributions are present in the exact arithmetic weight; no zero-location estimate is supplied by discarding either term.

Finally form the split-zero semiring $G(\text{End}_{\mathbb{C}} L_Z)$, using the associative endomorphism ring. The original ghost has supported lift $\Gamma^\bullet$; when the comparison cancels it is $e = 0^\bullet$, whereas the absent comparison is τ . Conjugation by T extends to a semiring isomorphism

$$G(\text{End}_{\mathbb{C}} E_h) \longrightarrow G(\text{End}_{\mathbb{C}} L_Z), \quad X^\bullet \longmapsto (TXT^{-1})^\bullet, \quad \tau \longmapsto \tau.$$

It preserves addition, composition, the unit, e , and τ directly. Thus the complete jet comparison preserves supported cancellation. Theorem 2.5 states exactly which arithmetic factors are removed if that cancellation is imposed as a generator-compatible quotient. It does not assert that those factors are absent from the original arithmetic cohomology.

The quotient itself induces an endomorphism map on an exact specified domain. Define

$$\mathcal{D} = \{X \in \text{End}_{\mathbb{C}} L_Z : X(L_{\text{off}}) \subseteq L_{\text{off}}\}.$$

It is a unital subalgebra, and the induced quotient action is a surjective unital homomorphism

$$d : \mathcal{D} \longrightarrow \text{End}_{\mathbb{C}} L_{Z_{\text{crit}}}, \quad \ker d = \text{Hom}_{\mathbb{C}}(L_Z, L_{\text{off}}).$$

Indeed invariance of L_{off} is exactly what makes the quotient action well defined; the kernel condition says exactly that every value of X lies in L_{off} . Any endomorphism of the critical direct summand lifts by making it zero on the off-axis direct summand, proving surjectivity. Thus

$$G(d) : G(\mathcal{D}) \longrightarrow G(\text{End}_{\mathbb{C}} L_{Z_{\text{crit}}})$$

is the explicit split-zero descent map. Both A and Γ belong to $\mathcal{D}$; it sends $A^\bullet$ to the supported critical generator, $\Gamma^\bullet$ to the supported zero e , and τ to τ . This statement includes the zero target algebra when the critical summand is absent.

3 Historical Hurwitz reconstruction and the complete Laurent jet bridge

This section reconstructs a compact probability measure from the complete local Hurwitz zero cluster at any nontrivial zeta zero. It also proves the complete finite-jet relation with the marked Laurent period, including the involution coefficients and the exact collision quotient. Two distinct prime periods recover the entire finite spectral algebra with every multiplicity retained.

3.1 Source receipts and exact locators

1. The original *Split-Zero CUE Hurwitz phase space* source has SHA256 95CCA3275F2012E32F9E506EC0211273D5E7DA1FD72AD90F7167AA37094E94CC. Its exact excerpts accompany this edition in `sources/historical/hurwitz_laurent/`: `compact_hurwitz_and_residues.tex`

reproduces source lines 722–856; `zero_motion_and_bernstein.tex` reproduces lines 1110–1257; and `all_moments_claim.tex` reproduces lines 1907–1912. The statements below retain the original coordinate s , displacement t , support coordinate y , support bound L , and moments $m_n(\mu)$.

2. The *Mathematical Functor Construction* transcript has SHA256 8F94D3A8172F8585837A4E4F6060B5F8DAC38CF627C9689335F13792B7710D85. Its complete mathematical proof block, source lines 14142–15894, is retained as `sources/historical/hurwitz_laurent/marked_period_laurent_conormal_proofs.md`. Theorems 3, 7, 11 and 13 give the marked Laurent augmentation and period maps; Theorems 15–18 give the packet and Laurent involution; Theorems 20–25 give the marked conormal chart. Within that source, lines 15143–15227 prove $J_u^* = J_{-u}$, lines 15397–15502 give the $d \log |a_\infty|$ chart, and lines 15504–15631 give the augmentation-conormal diagram.
3. The corresponding standalone *Terzo-CC-CVS-JT deformation intertwiner* witness has SHA256 BAF7DB6506710CA9AA5B7705C7D5687B4E66536C08A9FE77549ABBC041BC9044. Its Laurent deformation appears at source lines 490–542 and its conormal calculation at lines 638–760. The complete proofs needed here are reproduced below and in the accompanying transcript excerpt.

The historical source gives the forward zero-motion recursion and the marked conormal map. The proofs below give the inverse moment recursion, the complete multiple-zero version, the full nilpotent Laurent isomorphism, and its collision quotient. Source line numbers and hashes identify the particular source editions; these comparisons make no claim of global mathematical priority.

3.2 Every compact Hurwitz input is determined by one simple-zero germ

Keep the original family

$$Z_{\mu,t}(s) = \int_{[0,L]} \zeta(s, 1+ty) d\mu(y), \quad m_n(\mu) = \int_{[0,L]} y^n d\mu(y), \quad m_0(\mu) = 1.$$

Here μ is a Borel probability measure, L is any finite support bound, and t is the original displacement; no rescaling of either variable is made. For $|t|L < 1$, put

$$b_n(s) = \frac{(-1)^n}{n!} (s)_n \zeta(s+n), \quad Z_{\mu,t}(s) = \sum_{n \geq 0} m_n(\mu) b_n(s) t^n. \quad (\text{H1})$$

The expansion is the source's compact Taylor formula. To verify its analytic scope directly, for $\text{Re } s > 1$ expand each $(k+ty)^{-s}$ in its binomial Taylor series. On compact s -sets and $|t|L \leq r < 1$, the ratios $|ty|/k \leq r$ give normally convergent double sums. Integration is termwise valid. The identity extends meromorphically in s ; near a nontrivial zero ρ all displayed coefficients are holomorphic. Joint holomorphicity and normal convergence in the strip can be checked directly: for a compact set K in $0 < \text{Re } s < 1$, the product

$$\left| \frac{(s)_n}{n!} \right| = \prod_{j=1}^n \left| 1 + \frac{s-1}{j} \right| \leq \prod_{j=1}^n \left(1 + \frac{C_K}{j} \right) \leq e^{C_K} (n+1)^{C_K}$$

is polynomially bounded uniformly in $s \in K$. The factor $\zeta(s+n)$ is uniformly bounded for $n \geq 1$ by the absolutely convergent Dirichlet series at $1 + \min_{s \in K} \text{Re } s$. Since $m_n \leq L^n$, the terms of (H1) are bounded by a constant times $(n+1)^{C_K} (L|t|)^n$. Thus the series is normally convergent when $L|t| < 1$, including on a surrounding compact annulus for any contour used below. Cauchy estimates give the same local derivative control. For $n \geq 1$, $\text{Re}(\rho+n) > 1$ and none of $\rho, \rho+1, \dots, \rho+n-1$ is zero, so

$$b_n(\rho) \neq 0. \quad (\text{H2})$$

The zeta factor in (H2) is nonzero because on $\operatorname{Re} z > 1$ the logarithmic Euler-product series

$$\sum_{p \text{ prime}} \sum_{k \geq 1} \frac{p^{-kz}}{k}$$

converges absolutely, and its exponential is $\zeta(z)$ there. This conclusion uses the actual zero's strip $0 < \operatorname{Re} \rho < 1$; it does not assume RH.

Fix a simple nontrivial zero ρ , so $\zeta(\rho) = 0$ and $\zeta'(\rho) \neq 0$. The analytic implicit function theorem produces the unique local holomorphic germ

$$\rho_\mu(t) = \rho + w_\mu(t), \quad w_\mu(t) = \sum_{j \geq 1} c_j(\mu) t^j, \quad Z_{\mu,t}(\rho_\mu(t)) = 0. \quad (\text{H3})$$

This is the existing source theorem. Its simplicity hypothesis is retained explicitly, rather than silently extending a scalar branch through a multiple zero.

The inverse recurrence. For each integer $N \geq 1$, the input moment is recovered by the exact finite formula

$$m_N(\mu) = -\frac{1}{b_N(\rho)} \sum_{n=0}^{N-1} m_n(\mu) [t^{N-n}] b_n \left(\rho + \sum_{j=1}^N c_j(\mu) t^j \right). \quad (\text{H4})$$

Thus $(c_1, \dots, c_N)$ determines $(m_1, \dots, m_N)$ and conversely. In particular, two compact probability inputs have the same zero-trajectory jet through order N exactly when their first N moments agree.

Proof. Substitute (H3) in (H1). Its coefficient of t^N is

$$0 = \sum_{n=0}^N m_n(\mu) [t^{N-n}] b_n(\rho + w_\mu(t)).$$

The $n = N$ summand is exactly $m_N(\mu) b_N(\rho)$. It contains no positive-order coefficient of w . Every other summand uses only $m_0, \dots, m_{N-1}$ and $c_1, \dots, c_N$. Division is legitimate by (H2), proving (H4). In the same coefficient equation the only occurrence of c_N is $\zeta'(\rho) c_N$, from the $n = 0$ summand, because every other Taylor term has either a positive power of t already or at least two positive-order factors of w . This proves the forward triangular recurrence and both directions of the finite-jet assertion by induction. $\square$

The first two inverse formulas, with every sign retained, are

$$m_1 = \frac{\zeta'(\rho) c_1}{\rho \zeta(\rho + 1)}, \quad (\text{H5})$$

$$m_2 = -\frac{2}{\rho(\rho+1)\zeta(\rho+2)} \left[\zeta'(\rho) c_2 + \frac{\zeta''(\rho)}{2} c_1^2 - m_1 (\zeta(\rho+1) + \rho \zeta'(\rho+1)) c_1 \right]. \quad (\text{H6})$$

These follow by $b_1(s) = -s\zeta(s+1)$ and $b_2(s) = s(s+1)\zeta(s+2)/2$, not by changing the Hurwitz convention.

Exact measure reconstruction on the original interval. Assume $L > 0$. For every integer $M \geq 1$ and $0 \leq j \leq M$, set

$$\lambda_{M,j} = \binom{M}{j} \sum_{k=0}^{M-j} (-1)^k \binom{M-j}{k} \frac{m_{j+k}(\mu)}{L^{j+k}}, \quad \mu_M = \sum_{j=0}^M \lambda_{M,j} \delta_{Lj/M}. \quad (\text{H7})$$

Then every $\lambda_{M,j} \geq 0$, their sum is one, and μ_M converges weakly to μ . Thus (H4) and (H7) give an explicit reconstruction of the entire input from the zero germ, with the original support coordinate and bound present in every formula.

Proof. Expanding $(1 - y/L)^{M-j}$ proves

$$\lambda_{M,j} = \int_{[0,L]} \binom{M}{j} (y/L)^j (1 - y/L)^{M-j} d\mu(y).$$

This proves positivity, and the binomial theorem proves the sum is one. At fixed original y , let J have binomial law with M trials and success probability y/L . Its original-coordinate variable LJ/M has mean y and variance

$$\mathbb{E}[(LJ/M - y)^2] = \frac{y(L - y)}{M} \leq \frac{L^2}{4M}. \quad (\text{H8})$$

For continuous f on $[0, L]$ and any $\delta > 0$, split the expectation according to $|LJ/M - y| \leq \delta$. Uniform continuity and Chebyshev's inequality give

$$|\mathbb{E}f(LJ/M) - f(y)| \leq \omega_f(\delta) + 2\|f\|_\infty \frac{L^2}{4M\delta^2},$$

uniformly in y . First take $M \rightarrow \infty$, then $\delta \rightarrow 0$, and integrate against μ . This is weak convergence. For Lipschitz f , Cauchy-Schwarz gives the explicit bound $\text{Lip}(f)L/(2\sqrt{M})$. If $L = 0$, the input is exactly δ_0 , all positive moments vanish, and the original zero branch is constant; this exceptional fibre is retained separately. $\square$

Injectivity without a prespecified common support bound. If two compactly supported probability measures have the same zero germ at the chosen simple ρ , (H4) gives equal moments. Choose any common finite support bound for the two measures. Their measures (H7) are identical at every M , and their weak limits are therefore identical. Hence the map

$$\left\{ \begin{array}{c} \text{compact probability measures} \\ \text{on } [0, \infty) \end{array} \right\} \longrightarrow \left\{ \begin{array}{c} \text{holomorphic zero germs} \\ \text{at the fixed simple } \rho \end{array} \right\},$$

$$\mu \longmapsto \rho_\mu$$

is injective onto its actual image. This statement asserts no surjectivity onto arbitrary holomorphic germs.

The exact support endpoint is also recovered:

$$\sup \text{supp } \mu = \lim_{n \rightarrow \infty} m_n(\mu)^{1/n}. \quad (\text{H9})$$

Indeed, if the endpoint is L_μ , then $m_n \leq L_\mu^n$. For any $0 < a < L_\mu$, positive mass lies in $(a, L_\mu]$, so $m_n \geq a^n \mu((a, L_\mu])$. Taking n th roots and then $a \uparrow L_\mu$ proves the claim. The zero endpoint gives δ_0 and both sides zero.

The completed residue comparison has its own exact factorization. Define

$$\Lambda_{\mu,t}(s) = \pi^{-s/2} \Gamma(s/2) Z_{\mu,t}(s).$$

Then, for the same displacement domain $|t|L < 1$,

$$\text{Res}_{s=1} \Lambda_{\mu,t} = 1, \quad \text{Res}_{s=0} \Lambda_{\mu,t} = -1 - 2tm_1(\mu), \quad (\text{HRes1})$$

and consequently

$$\mathfrak{B}(\mu, t) := \text{Res}_{s=0} \Lambda_{\mu,t} + \text{Res}_{s=1} \Lambda_{\mu,t} = -2tm_1(\mu). \quad (\text{HRes2})$$

Proof. For $\operatorname{Re} \mathfrak{a} > 0$ and $\operatorname{Re} s > 1$, termwise integration of the absolutely convergent geometric series gives

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \int_0^\infty x^{s-1} \frac{e^{-\mathfrak{a}x}}{1 - e^{-x}} dx.$$

At zero the integrand's second factor is $x^{-1} + (1/2 - \mathfrak{a}) + O(x)$, uniformly on compact subsets of the indicated parameter half-plane. Subtract these two terms on $(0, 1)$. The remaining integrals define a function $H(s, \mathfrak{a})$ holomorphic for $\operatorname{Re} s > -1$, and

$$\Gamma(s)\zeta(s, \mathfrak{a}) = \frac{1}{s-1} + \frac{1/2 - \mathfrak{a}}{s} + H(s, \mathfrak{a}).$$

The tail integral is holomorphic there because of exponential decay. Since $\Gamma(1) = 1$ and $\Gamma(s) = s^{-1} + O(1)$, the formula proves $\operatorname{Res}_{s=1} \zeta(s, \mathfrak{a}) = 1$ and $\zeta(0, \mathfrak{a}) = 1/2 - \mathfrak{a}$. For $\mathfrak{a} = 1 + ty$, the parameters stay in a compact subset of that half-plane as y ranges over the original support. Uniformity therefore permits integration of both coefficients against the probability measure. It gives residue one at $s = 1$ and $Z_{\mu,t}(0) = -1/2 - tm_1(\mu)$. Finally $\pi^{-1/2}\Gamma(1/2) = 1$, $\Gamma(s/2) = 2/s + O(1)$, and $\pi^{-s/2} = 1 + O(s)$ give (HRes1) and their sum (HRes2). $\square$

Thus the residue map factors through the explicitly recovered first moment:

$$\rho_\mu \mapsto \frac{\zeta'(\rho)c_1(\mu)}{\rho\zeta(\rho+1)} = m_1(\mu) \mapsto -2tm_1(\mu).$$

The full germ retains every higher moment through (H4), with measure reconstruction given by (H7).

3.3 The complete zero cluster removes the simplicity restriction

The preceding source theorem used a simple scalar branch. The full cluster gives an inverse for **every** nontrivial zero, with its original multiplicity retained. Fix a nontrivial zero ρ of exact order $r \geq 1$. Choose a closed disc about ρ entirely inside $0 < \operatorname{Re} s < 1$, whose boundary Γ contains no zero and whose interior contains no zero of ζ other than ρ . Thus the entire disc also excludes the pole at 1. Shrink the complex t -disc so $Z_{\mu,t}$ is nonzero on Γ and has exactly r zeros inside, with multiplicity. The normal-convergence estimate following (H1) supplies joint holomorphicity and uniform convergence to ζ on Γ ; Rouché's theorem then proves the exact zero count.

For $1 \leq j \leq r$ define the holomorphic power-sum germs

$$q_j(t) = \frac{1}{2\pi i} \oint_\Gamma (s - \rho)^j \frac{\partial_s Z_{\mu,t}(s)}{Z_{\mu,t}(s)} ds.$$

They are exactly the sums of the j th powers of the displacements of the r zeros, counted with multiplicity, by the residue theorem. Set $e_0(t) = 1$ and recursively

$$je_j(t) = \sum_{i=1}^j (-1)^{i-1} e_{j-i}(t) q_i(t).$$

Newton's identities show that the polynomial

$$P_\mu(s, t) = \sum_{j=0}^r (-1)^j e_j(t) (s - \rho)^{r-j} = (s - \rho)^r + \sum_{N \geq 1} P_N(s) t^N \quad (\text{H10})$$

has precisely the full zero cluster as its divisor, and every $P_N(s)$ has degree at most $r - 1$ in the original displacement $s - \rho$. The original point ρ and monic leading coefficient one are both retained. The polynomial is independent of any enumeration or Puiseux branch choices.

There is a jointly holomorphic unit $U_\mu(s, t)$ on a smaller disc such that

$$Z_{\mu,t}(s) = U_\mu(s, t)P_\mu(s, t), \quad U_\mu(s, t) = \sum_{N \geq 0} U_N(s)t^N, \quad U_0(s) = \frac{\zeta(s)}{(s - \rho)^r}. \quad (\text{H11})$$

For completeness, joint holomorphicity does not need to be assumed from the root parametrization. Choose an inner circle still enclosing all zeros for small t ; on that circle the quotient Z/P has nonzero denominator. Its Cauchy integral against $1/(\zeta - s)$ defines a jointly holomorphic function of (s, t) inside, since the integrand is jointly holomorphic on the fixed contour. For each fixed t , the zeros of Z and P agree with their exact multiplicities, so the quotient has removable singularities at the roots and the Cauchy integral equals Z/P . The quotient has no zero, because the two divisors already coincide. This proves (H11). The contour variable ζ here denotes a complex integration variable, not the Riemann zeta function.

Now the input moments have the exact recursive recovery

$$m_N(\mu) = \frac{1}{b_N(\rho)} \sum_{k=1}^N U_{N-k}(\rho)P_k(\rho), \quad (\text{H12})$$

followed by the exact analytic division

$$U_N(s) = \frac{m_N(\mu)b_N(s) - \sum_{k=1}^N U_{N-k}(s)P_k(s)}{(s - \rho)^r}. \quad (\text{H13})$$

In (H13) the numerator vanishes to order at least r ; its holomorphic extension at ρ has value equal to its r th derivative there divided by $r!$. Thus the formula retains every derivative needed by a multiple zero, rather than dividing a nonvanishing germ by a zero.

Proof. Comparing the coefficient of t^N in (H1) and (H11) gives

$$m_N(\mu)b_N(s) = U_N(s)(s - \rho)^r + \sum_{k=1}^N U_{N-k}(s)P_k(s).$$

Evaluation at $s = \rho$ and the nonvanishing (H2) prove (H12). Rearranging the same analytic equality proves (H13), including the divisibility and the value at the centre. Starting from the completely known U_0 , the pair (H12)–(H13) is triangular and recovers every moment from the polynomial-cluster germ.

Two measures with the same polynomial-cluster jet through order N therefore have the same moments through N , by induction. Conversely, equal moments through N give $Z_{\mu,t} - Z_{\nu,t} = O(t^{N+1})$ as holomorphic functions on the fixed circle. Since both denominators remain nonzero there, their logarithmic derivatives, contour power sums, Newton coefficients and hence polynomials (H10) agree through order N . This proves the converse without choosing scalar zero branches. Finally (H7) reconstructs the measure, so the cluster germ at any nontrivial zero determines the entire compact probability input. $\square$

For $r = 1$, $P_N(s) = -c_N$, so (H12)–(H13) reduce to the inverse branch recurrence. For $r > 1$, the contour construction keeps all r zeros and their multiplicities even when individual zeros have fractional-power trajectories. This is a fully proved extension of the historical source's simple-zero statement.

3.4 The historical period map preserves every finite spectral jet

The transcript's marked Frobenius character is

$$z_{p,\omega}(s) = \omega p^{s-1/2}, \quad p > 1, \quad \omega \in S^1, \quad a = \log p > 0. \quad (\text{J1})$$

For the original prime-circle construction, p is a rational prime and $\omega = \psi(\text{Frob}_p)$; the proof below only uses the displayed positive a and unit-modulus ω . It retains the mark and phase.

Fix $\rho \in \mathbb{C}$ and an integer multiplicity $m \geq 1$, and write

$$z = \omega p^{\rho-1/2}, \quad E_{\rho,m} = \mathbb{C}[X]/(X - \rho)^m, \quad L_{z,m} = \mathbb{C}[U, U^{-1}]/(U - z)^m.$$

The original spectral coordinate remains X , and the original Laurent period generator remains U .

Exact jet isomorphism. There is a complex-algebra isomorphism

$$\Phi_{\rho,m} : L_{z,m} \xrightarrow{\sim} E_{\rho,m}, \quad [U] \mapsto z \sum_{k=0}^{m-1} \frac{a^k (X - \rho)^k}{k!}. \quad (\text{J2})$$

Its inverse is

$$[X] \mapsto \rho + \frac{1}{a} \sum_{k=1}^{m-1} \frac{(-1)^{k+1}}{k} \left(\frac{U - z}{z} \right)^k. \quad (\text{J3})$$

For every $0 \leq k < m$, (J2) induces on the k th associated conormal grade the exact multiplier

$$[(U - z)^k] \mapsto (za)^k [(X - \rho)^k]. \quad (\text{J4})$$

Proof. The target of (J2) is a unit: it is z times a polynomial with constant term one and nilpotent remaining part. Its difference from z lies in $(X - \rho)$, so its m th power vanishes. The Laurent universal property and the quotient therefore define (J2). In $\mathbb{C}[t]/t^m$, truncated $\exp(t)$ and $\log(1 + t)$ are inverse operations. This identity can be proved without convergence: formal differentiation shows $(\log(\exp t))' = 1$ and its constant term is zero; likewise $\exp(\log(1 + t))/(1 + t)$ has derivative zero and constant term one. Reducing the resulting formal identities modulo t^m gives (J3) and both composites. Finally

$$\Phi(U - z) = za(X - \rho) + (X - \rho)^2 \left(z \sum_{j=2}^{m-1} \frac{a^j (X - \rho)^{j-2}}{j!} \right),$$

so its k th power modulo $(X - \rho)^{k+1}$ is exactly $(za)^k (X - \rho)^k$. For $m = 1$, the sums in (J3) are empty and both rings are the specified scalar fibres, as required. $\square$

For a finite primary spectral block $A = \rho I + N$ with $N^m = 0$, this means that its actual period operator is exactly

$$\omega \exp(a(A - \tfrac{1}{2}I)) = z \sum_{k=0}^{m-1} \frac{a^k N^k}{k!}. \quad (\text{J5})$$

There is no deletion of nilpotent multiplicity in this map. Indeed, $\exp(aN) - I = N(aI + NB(N))$; its second factor is an invertible polynomial in N , so its powers have exactly the same kernels as the corresponding powers of N . This proves preservation of all Jordan block lengths locally.

3.5 Exact conjugate-reciprocal involution on the jets and conormal

Define the original spectral involution

$$\rho^\dagger = 1 - \bar{\rho}, \quad z^\dagger = \omega p^{\rho^\dagger - 1/2} = 1/\bar{z}. \quad (\text{J6})$$

On polynomial classes define the conjugate-linear algebra isomorphism

$$\mathcal{S}_E : E_{\rho,m} \longrightarrow E_{\rho^\dagger,m}, \quad [f(X)] \longmapsto [\bar{f}(1 - X)]. \quad (\text{J7})$$

On Laurent classes define

$$\mathcal{S}_L : L_{z,m} \longrightarrow L_{z^\dagger,m}, \quad \left[\sum c_n U^n \right] \longmapsto \left[\sum \bar{c}_n U^{-n} \right]. \quad (\text{J8})$$

Both have square the identity after returning to the original labelled block, and the full finite-jet diagram commutes:

$$\boxed{\mathcal{S}_E \Phi_{\rho,m} = \Phi_{\rho^\dagger,m} \mathcal{S}_L.} \quad (\text{J9})$$

Proof. The polynomial generator transforms by

$$X - \rho \longmapsto 1 - X - \bar{\rho} = -(X - \rho^\dagger).$$

The Laurent generator transforms by the exact identity

$$U - z \longmapsto U^{-1} - \bar{z} = -\bar{z} U^{-1} (U - z^\dagger). \quad (\text{J10})$$

The factor $-\bar{z}U^{-1}$ is a unit, so the respective m th-power ideals map to one another and both maps are defined. Applying each twice fixes constants and the original generator. To check (J9), it suffices to check the Laurent generator and constants. Its left image is

$$\bar{z} \sum_{k=0}^{m-1} \frac{(-a)^k (X - \rho^\dagger)^k}{k!},$$

which is the inverse of $z^\dagger \exp(a(X - \rho^\dagger))$ modulo $(X - \rho^\dagger)^m$ and hence the right image. $\square$

For clarity, the entire nonlinear mixing of Laurent Taylor coefficients under (J8) is explicit. With $t' = U - z^\dagger$, for $1 \leq k < m$ one has

$$\boxed{\mathcal{S}_L((U - z)^k) = \sum_{\ell=k}^{m-1} (-1)^\ell \binom{\ell-1}{\ell-k} \bar{z}^{k+\ell} (t')^\ell.} \quad (\text{J11})$$

Proof. By (J10), the expression is

$$(-\bar{z})^k U^{-k} (t')^k = (-1)^k \bar{z}^{2k} (t')^k (1 + \bar{z}t')^{-k}.$$

The formal binomial expansion $(1 + v)^{-k} = \sum_{j \geq 0} (-1)^j \binom{k+j-1}{j} v^j$, truncated when the total degree reaches m , gives (J11). The constant basis vector maps to one separately. $\square$

In particular, the conormal maps are

$$[X - \rho] \longmapsto -[X - \rho^\dagger], \quad [U - z] \longmapsto -\bar{z}^2 [U - z^\dagger], \quad (\text{J12})$$

and their compatibility with (J4) is an exact equality of coefficients:

$$-\bar{z}^2 (z^\dagger a) = -\bar{z}a = \bar{z}\bar{a}(-1).$$

The higher grade k has factors $(-1)^k$ and $(-1)^k \bar{z}^{2k}$ respectively, with the corresponding $(za)^k$ comparison. This supplies the all-order anti-involution/conormal morphism requested by the historical construction; it does not erase the higher terms in (J11).

On the real analytic radial chart $s = \beta + i\gamma$, the same exact map gives

$$\nu = \log |z| = (\beta - \tfrac{1}{2}) \log p, \quad d\nu = (\log p) d\beta, \quad c_{-+} = 4(\beta - \tfrac{1}{2}), \quad \nu = \frac{\log p}{4} c_{-+}. \quad (\text{J13})$$

The source's marked transverse chart $j_S(\nu) = [((e^\nu, 1, \dots, 1), 0)]$ pulls its displayed form $\alpha = d \log |a_\infty|$ back to $d\nu$, since $\log(e^\nu) = \nu$. Thus the composite character→radial chart has pullback $\alpha = (\log p) d\beta$, with the original prime and tessarine factor retained. The local identity requires no conjecture about zeros. A global identification with any separately defined Weil quadratic form is not asserted by this local chart calculation; the exact algebraic and chart maps proved here are the morphisms actually available from this source block.

3.6 Period collisions: the exact unlabelled image and its missing fibres

Let Z be a finite set of distinct spectral points ρ , with original multiplicities $m_\rho \geq 1$, and

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[X]/h(X) \cong \prod_{\rho \in Z} E_{\rho, m_\rho}.$$

The last map is the CRT map $f \mapsto (f \bmod (X - \rho)^{m_\rho})$; pairwise coprimality gives its inverse by the Bezout identity. No multiplicity is discarded.

Set $z_\rho = \omega p^{\rho-1/2}$. The equality of two period values is exactly

$$z_\rho = z_\sigma \iff \rho - \sigma \in \frac{2\pi i}{\log p} \mathbb{Z}. \quad (\text{J14})$$

The proof is the exact kernel $2\pi i \mathbb{Z}$ of the complex exponential. In particular, a fixed prime's character cannot automatically be treated as globally injective on a zero packet. No noncollision claim for actual zeta zeros is inserted here.

For each distinct period value z , put $M_z = \max_{\rho: z_\rho=z} m_\rho$ and

$$P(U) = \prod_z (U - z)^{M_z}, \quad L_P = \mathbb{C}[U, U^{-1}]/P(U).$$

There is an injective algebra map

$$\Phi_Z : L_P \hookrightarrow E_h \quad (\text{J15})$$

whose ρ th component is Laurent reduction modulo $(U - z_\rho)^{m_\rho}$ followed by (J2). Its image consists exactly of the tuples which, after (J3) on each labelled primary block, have the same Laurent Taylor coefficients at a common period value through every order visible in both blocks. More explicitly, if $a_\rho \in E_{\rho, m_\rho}$, write

$$\Phi_{\rho, m_\rho}^{-1}(a_\rho) = \sum_{k=0}^{m_\rho-1} b_{\rho, k} (U - z_\rho)^k.$$

Then (a_ρ) lies in the image exactly when

$$b_{\rho,k} = b_{\sigma,k} \quad \text{whenever } z_\rho = z_\sigma \text{ and } 0 \leq k < \min(m_\rho, m_\sigma). \quad (\text{J16})$$

The exact dimension and cokernel dimension as complex vector spaces are

$$\dim \operatorname{im} \Phi_Z = \sum_z M_z, \quad \dim \operatorname{coker} \Phi_Z = \sum_\rho m_\rho - \sum_z M_z. \quad (\text{J17})$$

Proof. Use CRT also on P . For a fixed common value z , a Laurent polynomial vanishes on every corresponding spectral block exactly when it is divisible by $(U - z)^{m_\rho}$ for every such ρ , which is equivalent to divisibility by $(U - z)^{M_z}$. Therefore the kernel of the original Laurent evaluation map is exactly (P) , giving (J15). Reduction of one degree- $< M_z$ polynomial to each smaller truncation gives (J16). Conversely choose any one block of maximal multiplicity M_z ; its Laurent coefficients define the common polynomial, and (J16) forces all other components to be its truncations. These constructions prove the image equality, while the dimensions follow from the explicit polynomial bases. $\square$

Keeping the labels restores a full exact isomorphism:

$$\prod_{\rho \in Z} L_{z_\rho, m_\rho} \xrightarrow{\Pi \Phi_{\rho, m_\rho}} E_h. \quad (\text{J18})$$

Thus the historical period map and the current finite arithmetic jet object have a proved relation at both the labelled and unlabelled levels. The collision compatibility space is exactly the image (J16). An explicit quotient records every lost independent coefficient: choose one maximal block ρ_z over each z , with $m_{\rho_z} = M_z$, and define the complex-linear map

$$\Delta_Z : E_h \longrightarrow \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho}, \quad (a_\rho) \longmapsto (b_{\rho,k} - b_{\rho_z,k})_{\substack{z, \rho \neq \rho_z \\ 0 \leq k < m_\rho}}. \quad (\text{J19})$$

It is surjective: set all maximal-block coefficients to zero and prescribe each remaining block's coefficients to be the desired output. Its kernel is exactly (J16). Consequently

$$0 \longrightarrow L_P \xrightarrow{\Phi_Z} E_h \xrightarrow{\Delta_Z} \bigoplus_z \bigoplus_{\substack{\rho: z_\rho = z \\ \rho \neq \rho_z}} \mathbb{C}^{m_\rho} \longrightarrow 0 \quad (\text{J20})$$

is an exact sequence of complex vector spaces. The first map is an algebra homomorphism; the last map is explicitly linear and is not incorrectly promoted to a ring homomorphism. Its target has the cokernel dimension in (J17).

If Z is stable under $\rho \mapsto 1 - \bar{\rho}$ with the same multiplicities, the maps (J7) and (J8), together with permutation of the labelled factors, give conjugate-linear involutions on both sides of (J18). Formula (J9) proves that (J18) intertwines them. The unlabelled construction (J15) is also stable: $z \mapsto 1/\bar{z}$ permutes the common period values and preserves M_z ; (J10) carries the defining ideal of one value to that of the paired value.

3.7 Two prime periods recover the complete finite spectral algebra

Retain a finite nonempty set Z of distinct spectral points with the original orders $m_\rho \geq 1$, and $E_h = \mathbb{C}[X]/h(X)$, where $h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}$. Choose distinct rational primes p, q and fixed marks $\omega_p, \omega_q \in S^1$. Put

$$a = \log p, \quad b = \log q, \quad u_\rho = \omega_p e^{a(\rho-1/2)}, \quad v_\rho = \omega_q e^{b(\rho-1/2)}.$$

Define the units $\mathfrak{A}_p, \mathfrak{A}_q \in E_h$ by their complete original spectral CRT components:

$$\begin{aligned} (\mathfrak{A}_p)_\rho &= u_\rho \sum_{k=0}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \\ (\mathfrak{A}_q)_\rho &= v_\rho \sum_{k=0}^{m_\rho-1} \frac{b^k (X - \rho)^k}{k!}. \end{aligned} \tag{P1}$$

The period-algebra map therefore has the exact type

$$\mathcal{P}_{p,q} : \mathbb{C}[U^{\pm 1}, V^{\pm 1}] \longrightarrow E_h, \quad U \longmapsto \mathfrak{A}_p, \quad V \longmapsto \mathfrak{A}_q. \tag{P2}$$

This map uses the actual periods of the original generator; the notation $\mathcal{P}_{p,q}$ leaves the theta map Θ unchanged.

Theorem 3.1 (Complete two-prime Laurent presentation). *The map (P2) is surjective, with every local order retained. More precisely, in $\mathcal{L} = \mathbb{C}[U^{\pm 1}, V^{\pm 1}]$, put*

$$\begin{aligned} B_\rho(U) &= v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j, \\ I_\rho &= ((U - u_\rho)^{m_\rho}, V - B_\rho(U)). \end{aligned} \tag{P3}$$

Here $\binom{b/a}{0} = 1$, and for $j \geq 1$ the coefficient is $(b/a)(b/a - 1) \cdots (b/a - j + 1)/j!$. The ideals I_ρ are pairwise comaximal, and

$$\begin{aligned} \ker \mathcal{P}_{p,q} &= \bigcap_{\rho \in Z} I_\rho = \prod_{\rho \in Z} I_\rho, \\ \mathcal{L} / \bigcap_{\rho \in Z} I_\rho &\xrightarrow{\sim} E_h. \end{aligned} \tag{P4}$$

The map on each primary quotient is the original finite exponential map (J2), with the second generator determined by (P3).

Proof. First the two marked periods separate distinct complex points. If their values at ρ, σ agree, there exist integers k, l with

$$a(\rho - \sigma) = 2\pi i k, \quad b(\rho - \sigma) = 2\pi i l.$$

Eliminating $\rho - \sigma$ gives $k \log q = l \log p$, hence $q^k = p^l$. Prime valuations in the positive rationals give $k = l = 0$, including negative integer exponents. Thus $\rho = \sigma$. Each fixed phase cancels only within its own prime character; neither mark was changed.

For a fixed ρ , set $y = (U - u_\rho)/u_\rho$. The quotient $\mathcal{L}/I_\rho$ is exactly $\mathbb{C}[y]/y^{m_\rho}$: its generators satisfy

$$U = u_\rho(1 + y), \quad V = v_\rho \sum_{j=0}^{m_\rho-1} \binom{b/a}{j} y^j.$$

Both expressions are units, since their constant terms are nonzero and their remaining parts are nilpotent. The Laurent inverses therefore impose no extra relation. These expressions give a map from the Laurent quotient to $\mathbb{C}[y]/y^{m_\rho}$, whose inverse sends y to $(U - u_\rho)/u_\rho$; their composites fix the displayed generators.

The map to E_{ρ, m_ρ} is

$$y \longmapsto \sum_{k=1}^{m_\rho-1} \frac{a^k (X - \rho)^k}{k!}, \quad X \longmapsto \rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} y^j \tag{P5}$$

for it and its inverse, respectively. The finite logarithm and exponential identities proved in (J2)–(J3) verify both composites. The identity $(1+y)^{b/a} = \exp((b/a) \log(1+y))$, reduced modulo y^{m_ρ} , verifies the prescribed V -image. For completeness its binomial coefficients are fixed by the formal differential equation $(1+y)F' = (b/a)F$, $F(0) = 1$: $(j+1)[y^{j+1}]F = (b/a-j)[y^j]F$. This recurrence gives exactly the coefficients in (P3). Thus I_ρ is precisely the kernel of the ρ -component of (P2), including order one.

For two different labels, either $u_\rho \neq u_\sigma$, or their v -values differ by the separation already proved. In the first case the powers of the two distinct linear U -factors have a polynomial Bezout identity, proving $I_\rho + I_\sigma = \mathcal{L}$. In the second case, $(V - v_\rho)^{m_\rho} \in I_\rho$: modulo I_ρ , $V - v_\rho = B_\rho(U) - v_\rho$ is a multiple of $U - u_\rho$. The corresponding powers of the two distinct linear V -factors again give a Bezout identity. Hence all primary kernels are pairwise comaximal.

The Chinese-remainder map now gives

$$\mathcal{L} / \bigcap_{\rho} I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} \mathcal{L} / I_{\rho} \xrightarrow{\sim} \prod_{\rho \in Z} E_{\rho, m_{\rho}} \xleftarrow{\sim} E_h. \quad (\text{P6})$$

For clarity, the first map is onto because Bezout identities give classes ϵ_ρ equal to one in factor ρ and zero in every other factor; summing ϵ_ρ times prescribed representatives recovers any tuple. Its kernel is the intersection by definition. Pairwise comaximality also gives intersection equal to product, first for two ideals by multiplying a Bezout identity by an element of the intersection, and then by induction. These arguments prove every assertion in (P4).

The original coordinate itself has the explicit inverse image

$$[X] \mapsto \sum_{\rho \in Z} \epsilon_\rho \left(\rho + \frac{1}{a} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} \left(\frac{U - u_\rho}{u_\rho} \right)^j \right). \quad (\text{P7})$$

Each term is exactly (P5) in its own primary factor; the other factors vanish through ϵ_ρ . Thus the inverse retains the original generator and all of its nilpotent jets. $\square$

This presentation is an exact statement for each finite packet. The single-prime collision quotient (J20) and the two-prime map (P6) specify their respective information. The factorwise split lifts below retain their individual support labels.

3.8 Split-Zero support is preserved by the exact maps

For any ring R , keep the original $G(R) = \{\tau\} \sqcup R^\bullet$, global zero τ , and supported zero $e_R = 0_R^\bullet$. Every ring map $f : R \rightarrow S$ gives

$$G(f)(\tau) = \tau, \quad G(f)(r^\bullet) = f(r)^\bullet. \quad (\text{S1})$$

This preserves addition and multiplication: if one input is τ , use its defining identity/absorber laws, and if both are supported, use the ring homomorphism identities with the supported tag retained. In particular $G(f)(e_R) = e_S \neq \tau$, even if f has a nontrivial kernel.

Apply (S1) separately to each map (J2), and then take the product of these lifted semiring isomorphisms. The resulting map is

$$\prod_{\rho \in Z} G(L_{z_\rho, m_\rho}) \xrightarrow{\sim} \prod_{\rho \in Z} G(E_{\rho, m_\rho}). \quad (\text{S2})$$

Every individual supported-zero coordinate is retained. The support map on both sides is the full vector of presence indicators in $\{0, 1\}^Z$; under (S2) it is unchanged.

For a finite nonempty index set $\mathcal{I}$, the comparison with applying G once to a product ring is the map

$$\begin{aligned} \iota_R : G \left(\prod_{i \in \mathcal{I}} R_i \right) &\longrightarrow \prod_{i \in \mathcal{I}} G(R_i), \\ \tau &\longmapsto (\tau)_i, \quad (r_i)_i^\bullet \longmapsto (r_i^\bullet)_i. \end{aligned} \quad (\text{S4})$$

It is an injective unital semiring homomorphism. For two supported inputs, the operations agree coordinate by coordinate by the ring laws. When a τ occurs, its additive-identity and multiplicative-absorber laws also agree coordinate by coordinate. The two sorts of image are disjoint because $\mathcal{I}$ is nonempty; equality of supported images recovers every r_i . Its support image is exactly the empty and full support vectors. For ring maps $f_i : R_i \rightarrow S_i$, direct evaluation proves

$$\left(\prod_i G(f_i) \right) \iota_R = \iota_S G \left(\prod_i f_i \right). \quad (\text{S5})$$

This is the exact natural comparison with the factorwise map (S2). If $\mathcal{I}$ is empty, the same comparison has the two-element semiring $G(0)$ as source and the one-element empty product as target, so it is not injective; its unique map identifies e and τ . The product map (S2) itself remains the identity between empty products in that case.

For the unlabelled map, a source factor $G(L_{z,M_z})$ maps to all spectral factors over z , using (J2) after the quotient $L_{z,M_z} \twoheadrightarrow L_{z,m_\rho}$. This is injective because one target block has multiplicity M_z . Its support image consists exactly of the vectors which are constant on every collision class $q^{-1}(z)$, where $q : Z \rightarrow \{z_\rho\}$ is $\rho \mapsto z_\rho$. A source τ gives τ in every such factor, while a supported element gives a supported element in every factor, including factors in which its amplitude becomes zero. The support embedding is exactly

$$q^{-1} : \mathcal{P}(\{z_\rho\}) \hookrightarrow \mathcal{P}(Z), \quad (\text{S3})$$

which preserves empty set, unions and intersections. The direct-image map $q_* : \mathcal{P}(Z) \rightarrow \mathcal{P}(\{z_\rho\})$ preserves unions and has $q_* q^{-1} = \text{id}$, but need not preserve intersections. For two distinct labels ρ, σ in the same collision class, $\{\rho\} \cap \{\sigma\} = \emptyset$ while $q_* \{\rho\} \cap q_* \{\sigma\} = \{z\}$. Thus (S3), rather than an unsupported lattice quotient isomorphism, is the exact support relation; the amplitude compatibility is (J16).

For a same-centre truncation $m' \leq m$, the quotient

$$L_{z,m} \twoheadrightarrow L_{z,m'}, \quad E_{\rho,m} \twoheadrightarrow E_{\rho,m'}$$

commutes with (J2) because the finite power series simply reduce to the corresponding lower nilpotence order. Its lifted map sends the supported nonzero class $(U - z)^{m'}$ to e when $m' < m$, and never to τ . This supplies a concrete example of the original distinction between a vanished jet amplitude and absent support, inside the current arithmetic finite-jet packet.

3.9 All-order Hurwitz motion transported to the period and tesserine coordinates

For the actual zero branch (H3), the exact period germ is

$$z_\mu(t) = \omega p^{\rho_\mu(t)-1/2} = z_\rho \exp \left((\log p) \sum_{n \geq 1} c_n(\mu) t^n \right). \quad (\text{C1})$$

Its N th coefficient is

$$[t^N]z_\mu(t) = z_\rho \sum_{k=1}^N \frac{(\log p)^k}{k!} \sum_{\substack{n_1+\dots+n_k=N \\ n_j \geq 1}} \prod_{j=1}^k c_{n_j}(\mu) \quad (N \geq 1). \quad (\text{C2})$$

The empty coefficient is z_ρ . Conversely, using the unique logarithm germ vanishing at $z_\mu/z_\rho = 1$,

$$c_N(\mu) = \frac{1}{\log p} [t^N] \log(z_\mu(t)/z_\rho). \quad (\text{C3})$$

Equations (C2)–(C3) are inverse because the same formal exponential/logarithm calculation used in (J2)–(J3) applies at every finite order. Combining (C3) with (H4) proves that the marked local period germ also determines the entire compact Hurwitz input.

For real t in the germ's domain,

$$\begin{aligned} \log |z_\mu(t)| &= (\log p) \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right), \\ c_{-+}(\rho_\mu(t)) &= 4 \left(\operatorname{Re} \rho - \frac{1}{2} + \sum_{n \geq 1} \operatorname{Re} c_n(\mu) t^n \right). \end{aligned} \quad (\text{C4})$$

All coefficients, including tangential ones, are retained in (C1)–(C3). The real projection in (C4) is explicitly the radial/tessierine observable; its loss of imaginary coefficients is a specified map. For $t \neq 0$, the Hurwitz family generally lacks the original functional equation, so $1 - \rho_\mu(t)$ is not asserted to be another zero of that family. The tessierine coordinate formula is the proved coordinate map evaluated at the actual branch, with this spectral distinction preserved.

3.10 Mathematical scope

The maps (J2)–(J20) identify the full finite arithmetic algebra with its labelled Laurent jets and calculate the exact quotient of unlabelled single-prime observations. Equations (H12)–(H13) recover all moments from the actual zero cluster, and (H7) reconstructs the entire measure. These statements retain the original multiplicities, support coordinates and marked periods.

The constructions give exact algebraic and analytic comparison maps. They do not supply the uniform estimate on the arithmetic moment matrices required by the weight-control calculation elsewhere in this reader.

The accompanying source records retain the historical locators and independent proof reviews. The proofs in this section specify their own hypotheses and maps in full.

4 Historical support branches, ideal completion, and measured observations

This chapter gives the complete branch, ideal-completion, atomic-kernel, and operator-algebra proofs extracted and continued from the historical Split-Zero support program. Its mathematical report and independent audit are, respectively, `logbook/historical_next_responses_20260912.md` and `logbook/frame_branch_independent_audit_20260912.md`. The equation identifiers B, I, M, A and O are local to this chapter.

Source locators and comparison. Exact complete response bodies are staged at `sources/historical/next_responses/preboolean_prime_branches.md` (source lines 23137–23956; response

heading 23104; exported timestamp 7/6/2026, 12:23:35 PM), sources/historical/next_response/preboolean_arithmetic_branch_maps.md (source lines 25162–25715; heading 25058; timestamp 7/6/2026, 7:58:41 PM), and sources/historical/next_responses/pointfree_frame_arithmetic_claim.md (source lines 25754–26221; heading 25721; timestamp 7/6/2026, 10:01:27 PM). Their shared original source has SHA-256 8d1780a24df3bb31db981b5e437558de31e6d94bf21bb54103a0a336d6a73091. The respective excerpt SHA-256 values are f5f11dd6b26a3a9d6a4c77cf2f7b86898759a19fa740c993d5749c8d47c02618, e84873955876b2be5f7738f746b4e133fc479cfe726bcb85d477f9aec7d6d10e, and 57c6662732b723e451a495f0224507f61e3a969956178018af7633ae06797ac0. For each excerpt, its first line is the first indicated original source line, so every original locator has the exact excerpt locator obtained by subtracting that first source line and adding one.

The core corpus comparison is split_zero_projective_monads_surcomplex/split_support_geometry_arithmetic_curve_v11.tex, SHA-256 5fa6a55aa9d1a2e57599e206087d3d2466b25dd99fa1a90d5f86367aafab1b4d. Its lines 1772–2054 already prove the ordinary prime-filter branches and their arithmetic maps; lines 2001–2018 explicitly impose arbitrary-join preservation for locale points. The support and measured realizations were also compared at lines 5006–5096 and 5346–5378. Accordingly the ordinary branch classification is a surviving historical construction, while the ideal quotient, explicit descent witnesses, atomic kernel and full L^∞ correspondence below supply the additional proved maps.

The final historical response, source lines 25982–26020 (excerpt lines 229–267), equates ordinary amplitude-fixed semiring branches with frame points without imposing arbitrary-join preservation. The exact correction is the classification (B1)–(B5), the frame quotient (I2), and the point-descent criterion (I3)–(I10). The measure-algebra construction then exhibits the kernel pairs that ordinary branches can distinguish. No external KMS classification or Morita equivalence is used as a theorem of this chapter.

4.1 Exact ordinary and continuous Split-Zero branches

The results below are stated in classical set theory. The ultrafilter construction explicitly uses Zorn’s lemma. No claim about existence or nonexistence of points in a constructive foundation is inferred from that use of choice; such an internal comparison would also have to specify its truth-value object and its available joins.

Let R be a nonzero commutative ring and let L be a nontrivial bounded distributive lattice. Retain the source carrier and all its exceptional coordinates:

$$G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(r, 1_L) : r \in R\} \subseteq R \times L,$$

$$(r, \lambda) + (s, \mu) = (r + s, \lambda \vee \mu), \quad (r, \lambda)(s, \mu) = (rs, \lambda \wedge \mu),$$

$$\tau_L = (0, 0_L), \quad e_L = (0, 1_L), \quad 1 = (1_R, 1_L).$$

Write $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $e = 0_R^\bullet$ distinct from global zero τ . The indicated subset is the carrier throughout.

Here $\mathbb{B} = \{0, 1\}$ has join and meet as its semiring operations. The supported copy in $G(R)$ has $r^\bullet + s^\bullet = (r + s)^\bullet$ and $r^\bullet s^\bullet = (rs)^\bullet$; its global zero satisfies $\tau + x = x + \tau = x$ and $\tau x = x\tau = \tau$. The map $G_{\mathbb{B}}(R) \rightarrow G(R)$ defined by $(0, 0) \mapsto \tau$, $(r, 1) \mapsto r^\bullet$ is a bijection and preserves these operations in each support case, hence is a unital semiring isomorphism. To verify the general carrier, the displayed operations are the restrictions of the product semiring $R \times L$. This subset is closed: a sum of nonzero amplitude has top support because at least one input has nonzero amplitude; a product of nonzero amplitude has two nonzero input amplitudes and hence top support. Every zero-amplitude

output belongs to the zero fibre. The zero and unit belong to the subset. Thus associativity, commutativity and distributivity restrict from the product, including all supported cancellations. All homomorphisms below preserve zero and unit. Write $z_\lambda = (0, \lambda)$.

Theorem B1 — all ordinary amplitude-fixed branches. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \xrightarrow{\sim} \{\beta : G_L(R) \rightarrow G(R) \text{ unital semiring map} : \beta(r, 1_L) = r^\bullet\}. \quad (\text{B1})$$

It sends ϵ to

$$\beta_\epsilon(r, 1_L) = r^\bullet, \quad \beta_\epsilon(0, \lambda) = \begin{cases} \tau & \epsilon(\lambda) = 0, \\ e & \epsilon(\lambda) = 1. \end{cases} \quad (\text{B2})$$

The map is surjective as a semiring map; its exact kernel congruence is equality of amplitudes together with equality of selected support values:

$$(r, \lambda) \sim_\epsilon (s, \mu) \iff r = s \text{ and } \epsilon(\lambda) = \epsilon(\mu). \quad (\text{B3})$$

Proof. The source formulas agree at $(0, 1_L)$ since $\epsilon(1_L) = 1$. They preserve zero and unit. On the zero fibre, preservation of addition and multiplication is precisely preservation of join and meet. If one argument has nonzero amplitude, its lattice coordinate is 1_L ; adding a zero-fibre argument leaves it unchanged and multiplying by that argument gives $(0, \lambda)$, matching respectively addition by τ or e , and multiplication by τ or e in $G(R)$. With both arguments at full support, even if their sum or product cancels to amplitude zero, both outputs are supported and ring identities prove preservation. This verifies all exceptional products and cancellations.

Conversely, $(0, \lambda)$ is additively idempotent. The only additive idempotents of $G(R)$ are τ and e : a supported $r^\bullet + r^\bullet = r^\bullet$ implies $2r = r$ in the ring, hence $r = 0$. Therefore every amplitude-fixed β sends the zero fibre into $\{\tau, e\}$. Identifying this two-element subsemiring with $\mathbb{B}$ gives a map ϵ preserving bottom, top, meet and join. Formula (B2) reconstructs β ; both constructions are inverse. Every amplitude and τ occurs in the image, and inspecting (B2) gives exactly (B3). $\square$

Now let L be a frame: all joins exist and finite meets distribute over arbitrary joins. Its zero fibre has specified infinitary joins

$$\bigvee_{i \in I}^{\text{fib}} (0, \lambda_i) = (0, \bigvee_{i \in I} \lambda_i).$$

Theorem B2 — exactly the continuous branches. Under (B1), the subset of branches preserving all these zero-fibre joins is exactly

$$\text{Hom}_{\text{Frm}}(L, \mathbb{B}). \quad (\text{B4})$$

Indeed the corresponding equality for (B2) says exactly $\epsilon(\bigvee_i \lambda_i) = \bigvee_i \epsilon(\lambda_i)$, and finite meets are already preserved. This is the precise additional structure required by the transcript's final frame formulation. Its ordinary algebraic maps retain a larger, explicitly identified category.

The classification extends to every map between original lattice-split carriers. For commutative unital rings R, A and nontrivial bounded distributive lattices L, M ,

$$\text{Hom}_{\text{Semi}}(G_L(R), G_M(A)) \cong \text{Hom}_{\text{Ring}}(R, A) \times \text{Hom}_{\text{DLat}_{0,1}}(L, M). \quad (\text{B5})$$

The pair (h, f) gives $(r, \lambda) \mapsto (h(r), f(\lambda))$. Conversely an arbitrary semiring map sends each additively idempotent z_λ to amplitude zero, say $z_{f(\lambda)}$; this preserves finite joins, meets and bottom. The identity $e_L = (1_R, 1_L) + (-1_R, 1_L)$ makes its image have top support, because one summand is the target unit; additive idempotence makes its amplitude zero. Thus $e_L \mapsto e_M$ and f preserves

top. Now $(r, 1_L) + e_L = (r, 1_L)$ forces its image to have top support. Write that image $(h(r), 1_M)$. The supported operations give a unital ring map h , and the formulas recover every source element, consistently at e_L . Coordinate composition proves naturality and both inverse constructions. In particular the entire two-factor datum is recoverable from the semiring map; scalar amplitude and support are not freely interchanged.

4.2 The exact frame comparison retaining all ordinary branches

For the underlying bounded distributive lattice of a frame L , let $\text{Idl}(L)$ be the set of lattice ideals. An ideal is a nonempty lower set closed under finite joins. Its bottom is $\{0_L\}$, its top is L , its finite meet is intersection, and the join of a family of ideals is

$$\bigvee_{\alpha} I_{\alpha} = \{x \in L : x \leq y_1 \vee \cdots \vee y_n \text{ for some finite } n \geq 0, y_j \in \bigcup_{\alpha} I_{\alpha}\}. \quad (\text{I1})$$

The empty finite join is 0_L ; this retains the bottom ideal in every formula.

Theorem I1 — frame quotient and principal-ideal section. The maps

$$q : \text{Idl}(L) \rightarrow L, \quad I \mapsto \bigvee I, \quad \eta : L \rightarrow \text{Idl}(L), \quad \lambda \mapsto \downarrow \lambda \quad (\text{I2})$$

have the following exact properties:

1. $\text{Idl}(L)$ is a frame and q is a surjective frame homomorphism.
2. η is an injective bounded-lattice homomorphism, $q\eta = \text{id}_L$, and $q(I) \leq \lambda \iff I \subseteq \eta(\lambda)$.
3. The kernel congruence of q is exactly $I \sim J \iff \bigvee I = \bigvee J$. No claim that η preserves arbitrary joins is made.

Proof. Formula (I1) is an ideal and is the least ideal containing every I_{α} ; intersections give infima. To prove frame distributivity, let $x \in I \cap \bigvee_{\alpha} J_{\alpha}$. Choose $x \leq y_1 \vee \cdots \vee y_n$ with $y_j \in J_{\alpha_j}$. In the distributive lattice,

$$x = x \wedge (y_1 \vee \cdots \vee y_n) = (x \wedge y_1) \vee \cdots \vee (x \wedge y_n).$$

Each $x \wedge y_j$ belongs to $I \cap J_{\alpha_j}$, proving $I \cap \bigvee_{\alpha} J_{\alpha} \subseteq \bigvee_{\alpha} (I \cap J_{\alpha})$; the reverse inclusion follows from containment. Thus $\text{Idl}(L)$ is a frame.

The supremum of the right side of (I1) equals $\bigvee_{\alpha} \bigvee I_{\alpha}$, proving arbitrary-join preservation by q , including the empty family. Since L is a frame,

$$(\bigvee I) \wedge (\bigvee J) = \bigvee_{i \in I, j \in J} (i \wedge j) = \bigvee (I \cap J).$$

The last equality holds because each $i \wedge j$ is in the intersection, and each $x \in I \cap J$ is the term $x \wedge x$. Hence q preserves finite meets and the top. Principal ideals satisfy $\downarrow \lambda \cap \downarrow \mu = \downarrow (\lambda \wedge \mu)$ and $\downarrow \lambda \vee \downarrow \mu = \downarrow (\lambda \vee \mu)$; they also preserve bottom/top. Their suprema are the original elements, proving $q\eta = \text{id}_L$, injectivity and surjectivity. The adjunction equivalence and the kernel description follow immediately from the definition of supremum. $\square$

Theorem I2 — all algebraic branches become frame points before the quotient. There are mutually inverse bijections

$$\text{Hom}_{\text{DLat}_{0,1}}(L, \mathbb{B}) \cong \text{Hom}_{\text{Frm}}(\text{Idl}(L), \mathbb{B}), \quad (\text{I3})$$

given by

$$\widehat{\epsilon}(I) = \bigvee_{\lambda \in I} \epsilon(\lambda) \quad \text{and} \quad \epsilon = \widehat{\epsilon}\eta. \quad (I4)$$

A frame point $\widehat{\epsilon}$ in (I3) factors through q if and only if its original ϵ is a frame point of L . Thus the exact inclusion of point sets is

$$q^* : \text{Pt}(L) \hookrightarrow \text{Pt}(\text{Idl}(L)), \quad \epsilon \mapsto \epsilon q, \quad (I5)$$

and its image is the points constant on the kernel congruence in Theorem I1.

Proof. In (I4), the value is 1 precisely when the ideal meets the prime filter $\epsilon^{-1}(1)$. For a join of ideals, a value 1 is witnessed by an element below a finite join of source elements; finite primeness then gives an element of value 1 in one of those source ideals. The reverse implication follows by containment, proving arbitrary-join preservation. For intersections, if I and J each have a value 1 element i, j , their meet $i \wedge j \in I \cap J$ has value 1; the reverse implication is immediate. Top and bottom follow from $\epsilon(1) = 1, \epsilon(0) = 0$. Therefore $\widehat{\epsilon}$ is a frame homomorphism.

Every ideal is the join in $\text{Idl}(L)$ of its principal ideals, so an arbitrary frame point is completely determined by its restriction along η , giving both inverse identities. If ϵ preserves arbitrary joins, $\widehat{\epsilon}(I) = \epsilon(\bigvee I) = (\epsilon q)(I)$. Conversely if $\widehat{\epsilon} = \chi q$ for a frame map χ , then $\epsilon = \widehat{\epsilon}\eta = \chi q\eta = \chi$, so ϵ is a frame map. A set map constant on the fibres of surjective q factors uniquely; if that map is a frame map, the induced factor preserves arbitrary joins and finite meets by lifting families through q . This proves the final image claim. $\square$

Theorem I3 — original Split-Zero comparison and exact congruence. Every bounded-lattice map $f : L \rightarrow M$ induces

$$G_f(R) : G_L(R) \rightarrow G_M(R), \quad (r, \lambda) \mapsto (r, f(\lambda)). \quad (I6)$$

For the nontrivial frames in Theorem I1 this gives a split pair of unital semiring maps

$$G_L(R) \xrightarrow{G_\eta(R)} G_{\text{Idl}(L)}(R) \xrightarrow{G_q(R)} G_L(R), \quad G_q G_\eta = \text{id}. \quad (I7)$$

They preserve τ and supported e as distinct elements. The exact kernel of $G_q(R)$ consists of equal amplitudes and equal suprema of zero-fibre ideals:

$$(r, I) \sim (s, J) \iff r = s \text{ and } \bigvee I = \bigvee J. \quad (I8)$$

Proof. A nonzero amplitude in $G_L(R)$ has full support and maps to full support because $f(1_L) = 1_M$; hence (I6) lands in the exact target carrier. Finite join and meet preservation establish the semiring identities coordinate by coordinate, including zero-amplitude cancellation. Formula (I7) follows from $q\eta = \text{id}_L$, and (I8) is equality of the two coordinates of the image. The bottom and top of the nontrivial source and target lattices differ; their images under these bounded maps are again bottom and top, proving the exact zero statements. $\square$

Thus an ordinary branch which fails zero-fibre join continuity is a fully explicit frame point **upstairs**, and the arithmetic map fails to descend through a proved frame quotient **downstairs**. The original support object and all its algebraic branches remain related by the maps (I2)–(I8).

The two available lifts have an exact comparison, which prevents a split algebra map from being mistaken for a split frame map. Given an ordinary $\epsilon : L \rightarrow \mathbb{B}$, both

$$\epsilon q : \text{Idl}(L) \rightarrow \mathbb{B}, \quad \widehat{\epsilon} : \text{Idl}(L) \rightarrow \mathbb{B} \quad (I9)$$

are bounded-lattice maps and agree on every principal ideal. The first descends through q by definition; the second preserves arbitrary joins by Theorem I2. They coincide on all ideals exactly when

ϵ preserves arbitrary joins. The forward implication follows because $\epsilon q = \widehat{\epsilon}$ and surjectivity of the frame map q force its factor ϵ to preserve joins; the converse is (I4). This also yields two amplitude-fixed Split-Zero branch maps out of $G_{\text{Idl}(L)}(R)$, with exactly the same criterion for equality.

There is a further exact localic structure. The map $j = \eta q$ is a nucleus on $\text{Idl}(L)$: it is monotone, $I \subseteq j(I)$ because each $i \in I$ lies below $\bigvee I$, it satisfies $j^2 = j$ because $q\eta = \text{id}_L$, and it preserves finite meets because both q and η do. Its fixed ideals are exactly the principal ideals. Their inherited meet and nucleus-applied join

$$\bigvee_{\alpha}^{\text{fix}} \downarrow \lambda_{\alpha} = j \left(\bigvee_{\alpha}^{\text{Idl}(L)} \downarrow \lambda_{\alpha} \right) = \downarrow \left(\bigvee_{\alpha}^L \lambda_{\alpha} \right) \quad (\text{I10})$$

identify the fixed-point frame exactly with the original L . This proves the quotient/sub-locale correspondence at the level of operations, while retaining the precise failure of η to preserve the ambient ideal-frame joins.

4.3 A fully explicit nontrivial frame with algebraic branches and no frame points

Let L be the Lebesgue probability measure algebra of $[0, 1]$: measurable subsets modulo null symmetric difference. Write $[A]$ for the class of A , keep the original measure $\mu(A)$, and use union, intersection and complement for finite Boolean operations. These operations are well defined modulo null sets, since changing their inputs changes any output only within the union of those null differences. The exact order is $[A] \leq [C]$ if and only if $\mu(A \setminus C) = 0$.

Lemma M1 — completeness and atomlessness. This L is a complete Boolean algebra, hence a frame, and has no atoms.

Proof. For an arbitrary family $([A_i])_{i \in I}$, let

$$b = \sup \{ \mu(A_{i_1} \cup \dots \cup A_{i_n}) : n \geq 0 \}.$$

Choose finite unions F_n with measures tending to b , and put $B = \bigcup_n F_n$. Replacing F_n by the union of the first n such finite unions makes them increasing without changing their defining kind, so $\mu(B) = b$. For each original A_i , the finite-union supremum bounds $\mu(F_n \cup A_i) \leq b$, and continuity from below gives $\mu(B \cup A_i) \leq b = \mu(B)$. Thus $A_i \setminus B$ is null. If $[C]$ is another upper bound, every one of the countably many constituent sets in B is contained in C modulo a null set; hence $B \setminus C$ is null. This proves that $[B]$ is the supremum. Infima follow by complements.

In a complete Boolean algebra finite meets distribute over arbitrary joins: if $v = \bigvee_i (a \wedge b_i)$, then every $b_i \leq \neg a \vee v$; taking joins and intersecting with a gives $a \wedge \bigvee_i b_i \leq v$. The reverse inequality is immediate. Thus L is a frame.

For a nonzero $[A]$, the function $F(t) = \mu(A \cap [0, t])$ is continuous because $|F(t) - F(u)| \leq |t - u|$. It takes the values 0 and $\mu(A) > 0$ at the endpoints. There is therefore t with $F(t) = \mu(A)/2$. The class of $A \cap [0, t]$ is strictly between 0 and $[A]$, so no nonzero class is an atom. $\square$

Theorem M2 — exact frame points of a complete Boolean algebra. If B is any complete Boolean algebra, its frame points correspond bijectively to its atoms:

$$\text{Hom}_{\text{Frm}}(B, \mathbb{B}) \cong \text{At}(B), \quad a \mapsto \epsilon_a(b) = \mathbf{1}_{\{a \leq b\}}. \quad (\text{M1})$$

Proof. An atom below $\bigvee_i b_i$ lies below some b_i : otherwise every $a \wedge b_i = 0$, while frame distributivity gives $a = a \wedge \bigvee_i b_i = \bigvee_i (a \wedge b_i) = 0$, a contradiction. Thus ϵ_a preserves arbitrary joins. It preserves finite meets, top and bottom directly.

Conversely, a frame map $\epsilon : B \rightarrow \mathbb{B}$ into the two-element Boolean algebra preserves complements, since complements are characterized by their finite meet 0 and join 1. By De Morgan it therefore also

preserves arbitrary meets. Put $a = \bigwedge \{b : \epsilon(b) = 1\}$. Its image is 1, so $a \neq 0$. If $0 < c \leq a$ and $\epsilon(c) = 0$, then $\epsilon(\neg c) = 1$, forcing $a \leq \neg c$ and hence $c = 0$, a contradiction. Consequently $\epsilon(c) = 1$, which by definition of a implies $a \leq c$. Thus a is an atom. Every element of the filter contains a ; conversely if $a \leq b$, then $\epsilon(b) = 1$. This proves the inverse formulas and their uniqueness. $\square$

In particular the measure algebra of Lemma M1 has **no frame points**.

It nevertheless has ordinary lattice characters. Here is an explicit witnessed failure of the required continuity. Let $D_n = [0, 2^{-n}]$ as original measurable subsets of $[0, 1]$; their classes are nonzero and any finite intersection is the smallest selected D_n . The filter generated by these classes is proper. Order proper filters containing it by inclusion; the union of every chain is proper, so Zorn's lemma gives a maximal proper filter F . In a Boolean algebra, maximal proper filters contain exactly one of $b, \neg b$: if neither lay in F , adjoining b would produce 0, so some $f \in F$ satisfies $f \wedge b = 0$ and hence $f \leq \neg b$, which already puts $\neg b \in F$. Its characteristic function is therefore a bounded-lattice homomorphism $\epsilon_F : L \rightarrow \mathbb{B}$.

Every $\epsilon_F([D_n]) = 1$, while every $\epsilon_F([D_n^c]) = 0$. But

$$\bigvee_{n \geq 1} [D_n^c] = [[0, 1] \setminus \{0\}] = 1_L, \quad \epsilon_F(1_L) = 1 \neq \bigvee_{n \geq 1} 0 = 0. \quad (\text{M2})$$

Thus (B2) gives an actual surjective semiring branch $\beta_{\epsilon_F} : G_L(R) \rightarrow G(R)$ preserving amplitudes, even though the frame has no point. Formula (M2) gives its exact failed zero-fibre join equation. The point $\hat{\epsilon}_F$ of $\text{Idl}(L)$ does not descend through q . More explicitly, set

$$J = \bigvee_{n \geq 1}^{\text{Idl}(L)} \downarrow [D_n^c].$$

Then $q(J) = 1_L = q(L)$, but $\hat{\epsilon}_F(J) = 0$ and $\hat{\epsilon}_F(L) = 1$. This is an explicit pair in the kernel congruence of q distinguished by the upstairs point; it proves the obstruction to descent at an exact source/target pair.

In fact **every** ordinary branch on this measure algebra fails already countable-join preservation. For its ultrafilter F , start at $a_0 = 1$. Split a representative of a_n into two equal-measure pieces by the continuous cumulative-measure construction in Lemma M1. Exactly one of their classes belongs to F ; call it a_{n+1} . The resulting decreasing sequence satisfies $a_n \in F$ and $\mu(a_n) = 2^{-n}$. Its infimum is 0 by continuity of measure. Consequently $b_n = \neg a_n$ increases to 1 and each $\epsilon_F(b_n) = 0$, again contradicting countable-join preservation. The ideal $I = \bigcup_n \downarrow b_n$ then satisfies $q(I) = q(L) = 1$, $\hat{\epsilon}_F(I) = 0$, and $\hat{\epsilon}_F(L) = 1$. This gives a kernel-congruence witness for each individual algebraic branch, not merely for the displayed dyadic choice.

4.4 Atomic observation and the exact support data it removes

For a complete Boolean algebra B , put

$$b_{\text{at}} = \bigvee_{a \in \text{At}(B)} a, \quad b_{\text{na}} = \neg b_{\text{at}}.$$

The frame-point evaluation map is

$$\text{ev} : B \rightarrow \mathcal{P}(\text{At}(B)), \quad x \mapsto \{a : a \leq x\}. \quad (\text{A1})$$

Theorem A1 — exact atomic quotient. This is a surjective complete Boolean-algebra homomorphism. Its kernel congruence is

$$\text{ev}(x) = \text{ev}(y) \iff x \wedge b_{\text{at}} = y \wedge b_{\text{at}}, \quad (\text{A2})$$

and its inverse image of the empty set is the full principal ideal $\downarrow b_{\text{na}}$. The complete decomposition retaining that ideal is

$$B \xrightarrow{\sim} (\downarrow b_{\text{at}}) \times (\downarrow b_{\text{na}}), \quad x \mapsto (x \wedge b_{\text{at}}, x \wedge b_{\text{na}}), \quad (\text{A3})$$

with inverse $(u, v) \mapsto u \vee v$ and respective local units $b_{\text{at}}, b_{\text{na}}$.

Proof. Each coordinate of (A1) is the frame point in (M1), so it preserves arbitrary joins and finite meets. Complements are preserved because an atom lies below exactly one of $x, \neg x$. For any subset S of atoms, the element $\bigvee_{a \in S} a$ maps exactly to S , by pairwise disjointness of distinct atoms and the arbitrary-join argument in (M1). Thus evaluation is surjective. Distributivity gives $x \wedge b_{\text{at}} = \bigvee_{a \in \text{At}(B)} (x \wedge a) = \bigvee_{a \leq x} a$, which proves (A2) and the empty-fibre claim. The two complementary components in (A3) reconstruct x by distributivity, and disjointness of $b_{\text{at}}, b_{\text{na}}$ proves both inverse identities and preservation of complete Boolean operations. $\square$

The complement on either local Boolean factor $\downarrow b$ in (A3) is explicitly $u \mapsto b \wedge \neg u$; its local top is b , not the global top of B unless $b = 1$.

This supplies an exact relation to the source's adelic measure-algebra support. In a finite measure-algebra realization, frame-point evaluation detects precisely its measure atoms; (A3) retains the entire nonatomic component rather than treating lack of points as lack of support. For a discrete countable measure with positive measure at each singleton, the singleton classes are the atoms and $b_{\text{at}} = 1$; evaluation is an isomorphism to the power set of the original places. For the Lebesgue example, $b_{\text{at}} = 0, b_{\text{na}} = 1$, and every frame-point evaluation is absent while the entire nontrivial Boolean carrier persists.

When applying $G_{\text{ev}}(R)$, retain the exceptional case: if there are no atoms, $\mathcal{P}(\text{At}(B))$ is the one-element lattice, so the exact carrier $G_{\mathcal{P}(\emptyset)}(R)$ is isomorphic to R , and this quotient identifies τ_B and e_B with ordinary zero. It is the explicit ring-reflection map on this observation, not evidence that $\tau_B = e_B$ in the original carrier. The split comparison (I7) has nontrivial lattices on both sides and preserves them as distinct; it remains available when the point-observation quotient collapses them.

Concretely, $\Psi : G_{\{\ast\}}(R) \rightarrow R, (r, \ast) \mapsto r$, is bijective and preserves both operations because the unique support coordinate is fixed. In the atomless case its composition is the exact identity

$$\Psi \circ G_{\text{ev}}(R) = p_B : G_B(R) \rightarrow R, \quad p_B(r, \lambda) = r. \quad (\text{A4})$$

4.5 The exact operator-algebraic branch map

The core manuscript's L^∞ realization permits an additional precise bridge to state spaces. Work with the same finite probability measure space, complete measure algebra B , and the complex algebra $M = L^\infty(\Omega, \mu; \mathbb{C})$ equipped with essential-supremum norm and complex conjugation. The original measure and every atomic mass are retained.

Theorem O1 — all Boolean branches extend exactly to multiplicative states. There is a natural bijection

$$\text{Hom}_{\text{DLat}_{0,1}}(B, \mathbb{B}) \cong \{\chi : M \rightarrow \mathbb{C} : \chi \text{ is a unital complex-linear } \ast\text{-homomorphism}\}. \quad (\text{O1})$$

The correspondence is specified on every projection by

$$\chi \epsilon(\mathbf{1}_A) = \epsilon([A]), \quad (\text{O2})$$

and on arbitrary $f \in M$ by the uniform extension of this finite formula:

$$\chi_\epsilon \left(\sum_{j=1}^n c_j \mathbf{1}_{A_j} \right) = \sum_{j=1}^n c_j \epsilon([A_j]), \quad (O3)$$

where the A_j form the specified measurable partition of Ω modulo null sets. Every such χ is positive and has norm 1.

Proof. Exactly one member of a finite partition has ϵ -value 1: finite-join preservation gives at least one, and pairwise disjointness with meet preservation gives at most one. Thus (O3) selects one of the original values c_j . Refining two partitions by all intersections shows the formula is independent of the simple-function presentation. The unique selected intersection gives additivity and multiplicativity for two simple functions, while conjugating their values gives star preservation. The selected value has modulus at most the essential supremum of the simple function, so the map is contractive.

Every bounded measurable complex function has simple functions tending to it in essential-supremum norm: partition a square containing its essential range into finitely many squares of diameter at most 2^{-n} and use one original value in each nonempty square. The resulting evaluations are Cauchy by contractivity. Their limit defines a unique continuous extension independent of the approximating sequence. Addition, multiplication and conjugation commute with these uniform limits, proving the required homomorphism identities. Positivity follows from $f = g^*g$ with $g = \sqrt{f}$ for every nonnegative measurable f ; hence $\chi(f) = |\chi(g)|^2 \geq 0$. Contractivity and $\chi(1) = 1$ prove norm 1.

Conversely a unital star homomorphism sends projections to real idempotents 0 or 1. Since $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$ and $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_{A \cap B}$, its projection values form the required Boolean lattice map. It is contractive because $\|f\|_\infty^2 \mathbf{1} - f^*f$ is the square of a bounded measurable function, so positivity gives $|\chi(f)|^2 \leq \|f\|_\infty^2$. It agrees with (O3) on simple functions, and continuity forces equality on all of M . This proves both inverse constructions. $\square$

Theorem O2 — normal branches, exact atomic masses, and failure of normality. For the correspondence (O1), the following are equivalent:

1. χ preserves suprema of bounded increasing nets of nonnegative elements of M (normality).
2. ϵ is a frame point of the complete measure algebra.
3. There is a measure atom a such that

$$\chi(f) = \frac{1}{\mu(a)} \int_a f d\mu \quad (f \in M). \quad (O4)$$

In particular the atomless Lebesgue example has multiplicative states for all of its ordinary branches, and none is normal; its exact countable failure is exhibited by (M2).

All bounded increasing nets used here have a supremum in the specified M . An explicit proof for nonnegative nets retains the finite measure: put $b = \sup_i \int_\Omega f_i d\mu$. Choose indices with integrals tending to b , and use directedness to replace them by an increasing dominating sequence. Choose representatives outside one common null set for the countably many sequence comparisons, so this selected sequence is pointwise increasing there. Its pointwise supremum g has integral b . For every fixed original index i , each finite maximum with a sequence member lies below a net member, so $\int_\Omega \max(g, f_i) d\mu \leq b = \int_\Omega g d\mu$ by monotone convergence. Thus $f_i \leq g$ almost everywhere. Every upper bound of the original net bounds the selected sequence and hence g , proving that g is the required least upper bound.

Proof. The supremum in M of the finite-union projections from an arbitrary family $([A_i])$ is the projection $\mathbf{1}_{\bigvee_i [A_i]}$: Lemma M1's countable essential union is an upper bound, and its increasing finite selected unions show it is the least upper bound also among arbitrary measurable nonnegative functions. Normality applied to this increasing net therefore gives preservation of its full projection

join. Together with (O2), this is precisely frame continuity of ϵ , proving $1 \Rightarrow 2$. By Theorem M2, a frame point corresponds to a unique atom a .

Every bounded measurable f is essentially constant on a . To see this, use the finite measurable partitions of its essential range from the proof of Theorem O1. Each partition has exactly one cell whose intersection with a is nonzero, and by atomhood that intersection equals a modulo null sets. Its diameter tends to 0, so the simple approximants restrict to constants c_n on a , with $|c_n - c_m|$ bounded by their uniform difference. Their limit c is the almost-everywhere value of f on a . Formula (O3) selects exactly these constants and gives $\chi(f) = c$; integration over the original atom gives precisely (O4), with the factor $1/\mu(a)$ retained. Thus $2 \Rightarrow 3$.

Finally let a bounded increasing net $f_i \geq 0$ have supremum f in M , and let c_i be its respective constants on a . These constants increase; put $c = \sup_i c_i$. Every f_i is bounded above by the function equal to c on a and to f off a ; hence the least-upper-bound property of f forces its constant on a to be at most c . Conversely each $f_i \leq f$ forces $c_i \leq f|_a$, and taking their supremum gives the reverse inequality. Thus $\chi(f) = \sup_i \chi(f_i)$. This proves $3 \Rightarrow 1$, including every original coefficient and atomic mass. $\square$

Equations (B1),(O1),(O2) form the promised typed connection between algebraic Split-Zero branches and operator-algebraic states. They identify exactly which branches can be normal observations of the measured support.

5 Reconstruction with changes of support index

The web formalization constructs reverse reconstruction, its fibrewise quotients, and an inverse on each actual semimodule. This section gives the explicit categorical comparison, including changes of support index, used to connect those objects to the arithmetic chart. No assumption of injective transition maps or zero bottom fibre is made.

Fix a commutative ring R and its split semiring $G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}$. A diagram object consists of a join-semilattice L with bottom and R -modules V_l , with linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying identity and composition. A morphism from (L, V) to (L', W) is a pair (α, f) , where $\alpha : L \rightarrow L'$ preserves bottom and joins with R -linear fibre maps

$$f_l : V_l \rightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l. \quad (\text{D1})$$

Composition is $(\beta, g)(\alpha, f) = (\beta\alpha, (g_{\alpha(l)} f_l)_l)$. These are the literal carriers and maps of the diagram category.

Theorem 5.1 (Full support-index reconstruction). *The diagram category just defined is equivalent to the category of $G(R)$ -semimodules and their $G(R)$ -linear maps. The forward functor has carrier*

$$\mathcal{T}(L, V) = \coprod_{l \in L} V_l$$

and operations

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx). \quad (\text{D2})$$

Its global zero is $(\perp, 0)$. On morphisms it is $\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$.

The inverse sends a semimodule M to

$$\begin{aligned} L_M &= eM, \quad l \vee k = l + k, \quad \perp = 0_M, \\ V_l &= \{m \in M : em = l\}, \quad \rho_{lk}(m) = m + k \quad (l + k = k). \end{aligned} \quad (\text{D3})$$

The natural comparison maps are

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \mapsto m, \quad m \mapsto (em, m), \quad (\text{D4})$$

and the label-and-fibre isomorphism

$$L \longrightarrow e\mathcal{T}(L, V), \quad l \mapsto (l, 0), \quad x \in V_l \mapsto (l, x). \quad (\text{D5})$$

Proof. For (D2), threefold addition transports both parenthesizations to $l \vee k \vee j$. Linearity and composition of the ρ 's make both amplitudes $\rho_l x + \rho_k y + \rho_j z$ in that same fibre. The bottom zero and commutativity follow likewise. For supported scalars the module laws follow from those of each V_l . Their addition law at a vector in fibre l is the ordinary R -module addition in V_l ; in particular $0_R^\bullet$ gives $(l, 0)$. The τ cases use the global bottom zero. Distributivity over vectors follows from linearity of each transition. These checks prove every semimodule axiom. Equation (D1) proves that $\mathcal{T}(\alpha, f)$ respects addition; its scalar and zero laws are immediate. Identity and composition hold on each pair.

For the reverse direction, $e^2 = e$, $e + e = e$, and $1 + e = 1$ in $G(R)$. Thus every $l \in eM$ is additively idempotent; eM is closed under addition, which supplies a semilattice with the specified bottom. For $m \in V_l$,

$$m + l = (1 + e)m = m, \quad m + (-1_R)^\bullet m = em = l.$$

The inherited addition in V_l , with zero l and inverse $(-1_R)^\bullet m$, is therefore an abelian group. For any $r \in R$, $e(r^\bullet m) = em = l$, since $er^\bullet = e$. Its scalar action satisfies every R -module law inherited from the semimodule, and $0_R m = em = l$ is its local zero. This proves that V_l is the specified module, without cancellation assumptions on all of M .

If $l + k = k$, then $e(m + k) = l + k = k$, so ρ_{lk} has the correct codomain. Its zero law is $l + k = k$; its additivity uses $k + k = k$. Also $r^\bullet k = k$, proving R -linearity. Identity follows from $m + l = m$, and composition from $k + j = j$. A split-linear map $F : M \rightarrow N$ takes eM to eN , with $\alpha(l) = F(l)$. It preserves joins and bottom, restricts to R -linear maps on each fibre, and $F(m + k) = F(m) + F(k)$ proves (D1). This defines the inverse functor.

Both arrows in (D4) are inverse as functions. Addition on the reconstructed carrier has amplitude

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

because $e(m + n) = l \vee k$. Thus (D4) is additive; its scalar identities follow from (D2), including τ . It is a split-linear isomorphism. In the reconstructed total, $e(l, x) = (l, 0)$, so (D5) is an isomorphism of semilattices and of the original fibres. Its transition map is exactly $(l, x) + (k, 0) = (k, \rho_{lk}x)$, recovering the given ρ . For a morphism (α, f) , (D5) takes $(l, 0)$ to $(\alpha(l), 0)$; for F , (D4) sends a representative to $F(m)$. These equalities prove naturality of both inverse comparisons, including support-index changes. Hence they are a categorical equivalence. $\square$

5.1 The internal quotient and the actual comparison kernel

Let $B_l \subseteq V_l$ be transition-stable submodules. The quotient diagram has fibres V_l/B_l with their induced transitions. The map

$$q : \mathcal{T}(L, V) \rightarrow \mathcal{T}(L, V/B), \quad (l, x) \mapsto (l, [x]) \quad (\text{D6})$$

is onto and split-linear. It is the coequalizer of the two maps $\mathcal{T}(L, B) \rightrightarrows \mathcal{T}(L, V)$ given by inclusion and $(l, b) \mapsto (l, 0)$.

To prove the universal property against any split-linear target N , let f equalize these maps. If $x - y \in B_l$,

$$f(l, x) = f(l, y) + f(l, x - y) = f(l, y) + f(l, 0) = f(l, y).$$

The last equality is the image of $(l, y) + (l, 0) = (l, y)$; no subtraction or cancellation in N was used. Thus f descends as a function through (D6). Lifting representatives proves additivity, scalar laws and uniqueness. Different labels remain distinct under (D6). In particular a boundary is carried to its original support-labelled zero.

For a cochain map $f : C \rightarrow D$ of R -module complexes at a fixed degree, put $Z_C = \ker d_C$, $B_C = \text{im } d_C^{\text{prev}}$ inside Z_C , and $K_{\text{rep}} = \{z \in Z_C : f(z) \in B_D\}$. Then the map

$$K_{\text{rep}}/B_C \xrightarrow{\sim} \ker H(f), \quad [z] \mapsto [z] \quad (\text{D7})$$

is a linear isomorphism. It is well-defined since f maps boundaries to boundaries. The prequotient map $K_{\text{rep}} \rightarrow \ker H(f)$, $z \mapsto [z]$, has kernel exactly B_C ; every kernel class has a cycle representative, proving surjectivity. If $fa = bf$ for chain endomorphisms, $aK_{\text{rep}} \subseteq K_{\text{rep}}$ and $aB_C \subseteq B_C$; (D7) therefore intertwines the actual induced actions. If f is a morphism of diagrams of complexes, these constructions commute with the transitions and reconstruct by (D2). For the equivariant assertion across supports, a, b are diagram endomorphisms as well. The equation $d^2 = 0$ then becomes the map $(l, x) \mapsto (l, 0)$ from its original source degree to its original target degree. It is different from constant absence when $l \neq \perp$, and (D6)–(D7) are the exact comparison maps.

For completeness, the full semimodule category retains two precisely related kernel constructions. For $F = \mathcal{T}(\alpha, f) : M = \mathcal{T}(L, V) \rightarrow N = \mathcal{T}(L', W)$, put $L_0 = \alpha^{-1}(\perp')$. Then

$$\begin{aligned} \ker_{G(R)} F &= \mathcal{T}(L_0, (\ker f_l)_{l \in L_0}), \\ F^{-1}(eN) &= \mathcal{T}(L, (\ker f_l)_{l \in L}) = M \times_N eN. \end{aligned} \quad (\text{D8})$$

The inclusion of the first into the second is induced by $L_0 \hookrightarrow L$. Indeed L_0 contains bottom and is closed under joins. Naturality in (D1) takes each fibre kernel into the next one. The formula $F(l, x) = (\alpha(l), f_l x)$ equals the global zero exactly when $\alpha(l) = \perp'$ and $f_l x = 0$; it belongs to eN exactly when $f_l x = 0$. These pointwise descriptions prove both universal properties: any split-linear map satisfying the respective condition has its unique factorization through the displayed subsemimodule. Thus reconstruction of the all-label fibre kernels in (D7) gives the second object. If α is the identity, the categorical kernel is its bottom fibre, with the inclusion in (D8) retaining the exact comparison.

6 The original theta complex and its arithmetic packet

The September 12 source chain has advanced from a support dictionary to an actual characteristic-zero cochain construction. We retain its complete coefficient square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

In particular $p_{\mathbb{Z}}$ has infinite target. The absolute pointed base is $\{\tau, 1\}$, with τ absorbing for multiplication; it is not obtained by imposing $e = 1$. The map p_R sends both τ and e_R to 0_R , while the support map distinguishes them. All of the linear maps below lift on each given support fibre; their ordinary zero values lift to that fibre's supported zero.

For clarity, the fixed section called R_0 in the web notes is denoted R_{ref} here. The corrected sequence keeps its name R_m , including its genuinely different level-zero member. This is an explicit notation correspondence between the same maps.

6.1 Source spaces and fixed arithmetic factors

Let

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+) : p_{b,k}(F) = \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \end{aligned}$$

Use the original Fourier, theta and Mellin transforms

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \quad (\text{A1})$$

The seminorms $p_{b,k}$ define the strong Schwartz topology. Logarithmic coordinates give completeness by uniform convergence of every derivative on compact intervals and the weighted bounds. The conditions defining V are closed in the even Schwartz space.

Proposition 6.1 (The fixed multiplier and injective theta map). *The map $\Theta : V \rightarrow \mathcal{B}$ is continuous and injective, $D\Theta = \Theta D$, and*

$$\Theta\widehat{\phi}(x) = x^{-1}\Theta\phi(1/x).$$

On $0 < \operatorname{Re} s < 1$, define

$$H_\phi(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)} \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

Then

$$\mathcal{M}\Theta\phi = gH_\phi, \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s). \quad (\text{A2})$$

For the exact original source $\phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$,

$$H_{\phi_*} = 1, \quad H_{P(D)\phi_*} = P, \quad \mathcal{M}\Theta(P(D)\phi_*) = gP. \quad (\text{A3})$$

Proof. Schwartz estimates give uniform convergence of each differentiated theta series on $x \geq \delta > 0$ and all bounds at infinity. Poisson summation and the two source moments give the displayed Fourier identity; applying the same bounds to $\widehat{\phi}$ gives all bounds at zero, with continuous dependence on Schwartz seminorms. Differentiating the series proves intertwining. For $x > 0$, the absolutely convergent divisor rearrangement gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{k \geq 1} \left(\sum_{n|k} \mu(n) \right) \phi(kx) = \phi(x).$$

Absolute convergence follows from $\sum_k d(k) |\phi(kx)| < \infty$. This proves injectivity.

In a right half-plane absolute integration gives $\mathcal{M}\Theta\phi = 2\zeta M_+\phi$. Both sides continue to the open strip, with the source's zero moments removing the relevant apparent singularities, which proves (A2). More explicitly $M_+\phi$ is holomorphic for $\operatorname{Re} s > -2$, since evenness and $\phi(0) = 0$ give $\phi(x) = O(x^2)$ at zero; the theta side is entire. The multiplier defining H_ϕ is holomorphic and nonzero on the open strip. The Gaussian integral gives

$$M_+\phi_*(s) = \pi^{-s/2} \{2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)\} = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

The source moments follow from its value at zero and this formula at $s = 1$. Integration by parts proves $H_{D\phi} = sH_\phi$; all boundary terms vanish. These calculations prove (A3), with both factors of two retained. $\square$

The actual one-leg complex, in degrees zero and one, is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$. Let $Q = \mathcal{B}/\Theta V$ and $q : \mathcal{B} \rightarrow Q$. The two-leg complex is

$$C_{+-} = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(v, w) = \Theta(v - \widehat{w}). \quad (\text{A4})$$

Its source operator is $(D, 1 - D)$. Its degree-zero cycles are exactly $(\widehat{w}, w)$; its degree-one homology is Q . This follows from injectivity of theta and $\operatorname{im} d = \Theta V$. Thus there is a concrete inclusion of C_+ into C_{+-} , with the extra degree-zero cycle retained through the map $w \mapsto (\widehat{w}, w)$.

6.2 The exact operator range, including its inverse

Proposition 6.2 (Source and target Euler inverses). *For $\rho \in \mathbb{C}$, on the indicated kernel define*

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

Then

$$(D - \rho)\mathcal{B} = \ker(\mathcal{M}(\cdot)(\rho)), \quad (D - \rho)S_\rho = 1, \quad \mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho}. \quad (\text{A5})$$

The operator $D - \rho$ is injective. For $0 < \operatorname{Re} \rho < 1$, the same formula, extended evenly, gives

$$(D - \rho)V = \ker(H(\cdot)(\rho)). \quad (\text{A6})$$

These inverses are continuous for the displayed source topologies.

Proof. Differentiation proves the right-inverse identity. For F with zero Mellin value the upper tail equals the negative lower tail. In $x = e^y$ coordinates, with $\sigma = \operatorname{Re} \rho$, the upper bound $C_N e^{-Ny}$ becomes $C_N e^{-Ny}/(N - \sigma)$ for $N > \sigma$, and the lower bound $C'_N e^{Ny}$ becomes $C'_N e^{Ny}/(N + \sigma)$ for $N + \sigma > 0$. Differentiating uses $\partial_y S_\rho F(e^y) = -\rho S_\rho F(e^y) - F(e^y)$. These estimates prove every required seminorm bound and continuity. The homogeneous solution $cx^{-\rho}$ fails the target decay at one end unless $c = 0$, proving injectivity and uniqueness. Integration by parts proves the Mellin identity, with its removable value at ρ filled in.

For the source, $H_\phi(\rho) = 0$ implies $M_+\phi(\rho) = 0$, since its coefficient is nonzero in the open strip. Near zero the same inverse is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Dominated differentiation gives smooth even extension and value zero. The upper tail gives Schwartz decay. Integration of $(D - \rho)S_\rho \phi = \phi$ on $\mathbb{R}$ gives $(1 - \rho) \int S_\rho \phi = \int \phi = 0$. Thus the second moment is preserved. The same estimates, with derivatives under the integral at zero, prove continuity on V . $\square$

For any polynomial $h(t) = \prod_i (t - \rho_i)^{m_i}$ with distinct centres in the open strip, iteration of (A5)–(A6) gives

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H), \quad E_h = \mathbb{C}[t]/(h). \quad (\text{A7})$$

Here j_h retains the full Taylor jets through the Chinese-remainder isomorphism. Order of inversion does not affect the answer, by uniqueness for the injective $h(D)$. The source jet map is onto by (A3). The target jet map is onto: the finitely many functions $y^k e^{\rho_i y}/k!$ are independent on any real interval; their Gram against a smooth nonnegative bump positive on an interval is invertible. Its inverse produces compactly supported smooth log functions with any prescribed vector of moments. Consequently the actual quotient square is

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\Theta} & \mathcal{B}/h(D)\mathcal{B} \\ j_h H \downarrow \wr & & \wr \downarrow j_h \mathcal{M} \\ E_h & \xrightarrow{\times j_h g} & E_h. \end{array} \quad (\text{A8})$$

Theorem 6.3 (Finite arithmetic kernel, quotient and packet section). *The actual theta sequence gives the exact sequence*

$$0 \longrightarrow Q[h(D)] \longrightarrow E_h \xrightarrow{\times j_h g} E_h \longrightarrow Q/h(D)Q \longrightarrow 0. \quad (\text{A9})$$

For a nonempty finite packet Z of actual nontrivial zeros with their full orders, put

$$h = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = E_h, \quad v = g/h, \quad \varepsilon = j_h(h/g).$$

The element ε is a unit. Let $R_Z(u)$ be the unique polynomial of degree $< d$ representing εu , and set

$$R_{\text{ref}}u = S_h^{\mathcal{B}}\Theta(R_Z(u)(D)\phi_*), \quad J_Z F = j_h \mathcal{M}F, \quad \sigma = qR_{\text{ref}}.$$

Then

$$\mathcal{M}R_{\text{ref}}u = (g/h)R_Z(u), \quad J_Z R_{\text{ref}} = 1, \quad J_Z \Theta = 0, \quad Q = Q[h(D)] \oplus h(D)Q. \quad (\text{A10})$$

The projector is $\sigma \bar{J}_Z$; its image is $Q[h(D)]$, and its kernel is $h(D)Q$.

Proof. If $h(D)[F] = 0$, write the unique $h(D)F = \Theta\phi$ and send $[F]$ to $j_h H\phi$. Changing F by a theta relation changes ϕ by $h(D)V$, so this is well-defined. If its value is zero, (A7) gives $\phi = h(D)\psi$, and injectivity yields $F = \Theta\psi$. Its image lies in $\ker(j_h g)$ by (A8). Conversely a polynomial representative P of a vector in that kernel gives the class $[S_h \Theta(P(D)\phi_*)]$. It is defined by (A7), and changing P by hT changes the lift by $\Theta(T(D)\phi_*)$. These are inverse maps. The cokernel description in (A9) follows by quotienting the right side of (A8), proving exactness with its original arrows.

For the packet, $j_h g = 0$, and v is nonvanishing at every chosen zero because the complete order was retained. The Mellin formula proves $J_Z R_{\text{ref}} = j_h v \varepsilon = 1$. The kernel-to-quotient comparison of (A9) is multiplication by $j_h v$, an isomorphism. Its inverse specifies σ . Thus every class has a unique sum of a class in $Q[h(D)]$ and one in $h(D)Q$, proving (A10) and the projector assertions. $\square$

6.3 The original extension supplies the boundary coefficient

With $A = M_t$ on E , define $\ell(u) = [t^{d-1}]R_Z(u)$. Polynomial division, without changing R_Z or its unit, gives

$$DR_{\text{ref}} - R_{\text{ref}}A = f_0 \ell, \quad f_0 = \Theta\phi_*. \quad (\text{A11})$$

Indeed $tR_Z(u) - R_Z(tu)$ is divisible by the monic h , and its degree is at most d ; its quotient is precisely $\ell(u)$. Apply (A3) and Mellin injectivity, or multiply by the injective $h(D)$, to obtain (A11). For one simple zero this coefficient is $1/g'(\rho) = 1/(2\xi'(\rho))$.

The invariant subspace $\mathcal{B}_Z = \Theta V + R_{\text{ref}}E$ fits into the actual extension

$$0 \rightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J_Z} E \rightarrow 0. \quad (\text{A12})$$

It splits complex-linearly through R_{ref} ; in these coordinates its operator is

$$\begin{pmatrix} D_V & \phi_* \ell \\ 0 & A \end{pmatrix}.$$

The extension class, using the resolution $\mathbb{C}[t] \xrightarrow{h} \mathbb{C}[t]$, is the class of $R_Z(1)(D)\phi_*$ in $V/h(D)V$. Under (A7) it is exactly $\varepsilon \neq 0$. This follows by applying $h(D)$ to the chosen lift $R_{\text{ref}}(1)$. It has no equivariant section into $\mathcal{B}$, since every equivariant image of E would be killed by $h(D)$, which is injective there. Its exact replacement is the equivariant roof

$$E[-1] \xleftarrow{\sim} [V \rightarrow \mathcal{B}_Z] \longrightarrow [V \rightarrow \mathcal{B}]. \quad (\text{A13})$$

The left arrow is J_Z in degree one and zero in degree zero; exactness of (A12) proves it a quasi-isomorphism. Thus no relation or homotopy is suppressed by the ordinary direct-sum notation.

6.4 Actual representative corrections and rank-two control

Throughout this subsection $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$, with inner product conjugate-linear in the first slot. Coefficient spaces carry the Euclidean inner products in the retained bases; every adjoint below

uses these specified metrics. Put $\phi_j = D^j \phi_*$, $f_j = D^j f_0$, and $\Phi_m(c) = \sum_{j=0}^m c_j \phi_j$. For $F_m = \Theta \Phi_m : \mathbb{C}^{m+1} \rightarrow \mathcal{H}$, define

$$\begin{aligned} H_m &= F_m^* F_m, & C_m &= F_m^* R_{\text{ref}}, & B_m &= -H_m^{-1} C_m, \\ \Pi_m &= F_m H_m^{-1} F_m^*, & R_m &= R_{\text{ref}} + F_m B_m = (1 - \Pi_m) R_{\text{ref}}. \end{aligned} \quad (\text{A14})$$

The columns f_j are independent: Mellin would turn a dependence into $gP = 0$. Thus $H_m > 0$. The correction is an original theta boundary, giving $qR_m = \sigma$, $J_Z R_m = 1$, and $F_m^* R_m = 0$. For every original coefficient matrix $B : E \rightarrow \mathbb{C}^{m+1}$,

$$\begin{aligned} (R_{\text{ref}} + F_m B)^* (R_{\text{ref}} + F_m B) &= G_m + (B - B_m)^* H_m (B - B_m), \\ G_m &= R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \end{aligned} \quad (\text{A15})$$

Expansion proves the identity; injectivity from $J_Z R_m = 1$ proves its strict positivity.

Let $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$, both with codomain $\mathbb{C}^{m+2}$. In the retained source coordinates,

$$K_m = e_0 \ell + S_m B_m - I_m B_m A, \quad D R_m - R_m A = F_{m+1} K_m. \quad (\text{A16})$$

Integration by parts in dx , with the source boundary term $[x \bar{F} H]_0^\infty = 0$, gives

$$W_m = A^* G_m + G_m A - G_m = -(R_m^* F_{m+1} K_m + K_m^* F_{m+1}^* R_m).$$

All but the last source column pair to zero. With $z_m = R_m^* f_{m+1}$ and $b_m = e_m^* B_m$ this is exactly

$$W_m = -z_m b_m - b_m^* z_m^*. \quad (\text{A17})$$

In particular rank $W_m \leq 2$. Factoring $G_m^{-1} W_m$ through the two columns gives its nonzero spectrum as that of

$$-\begin{pmatrix} c_m & v_m \\ u_m & \bar{c}_m \end{pmatrix}, \quad u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m.$$

Cauchy-Schwarz gives $u_m v_m \geq |c_m|^2$. Hence its exact two-sided allowance is

$$\epsilon_m = |\operatorname{Re} c_m| + \sqrt{u_m v_m - (\operatorname{Im} c_m)^2}, \quad -\epsilon_m G_m \leq W_m \leq \epsilon_m G_m. \quad (\text{A18})$$

This includes ranks zero and one. For a reflection-stable packet, $\operatorname{Tr}(G_m^{-1} W_m) = 2 \operatorname{Re} \operatorname{Tr} A - d = 0$. Every eigenvector retains the equality

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{A19})$$

The next sections compute the actual arithmetic moment problem behind G_m , sharpen its constraints, and determine the topology of the original jet observation. They use (A11)–(A19) rather than an assigned metric in place of the theta representative.

7 Certified moments from the original arithmetic theta source

The moment matrices in the arithmetic kernel formulas can be calculated directly from the specified theta source. This section gives an exact incomplete-gamma series, proves an explicit bound for the entire omitted double tail, and applies directed interval arithmetic to the retained terms. The construction retains the original factor $g = 2\xi$, the measure dx , the Euler operator $D = -x\partial_x$, and every polynomial coefficient. No information about the location of a zeta zero enters this calculation.

7.1 Source polynomials and the exact reflection map

Keep

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad f_j = D^j \Theta \phi_*, \quad \Theta \phi(x) = 2 \sum_{n \geq 1} \phi(nx). \quad (\text{AM.1})$$

Introduce polynomials with their original integer coefficients by

$$P_0(z) = 4z^2 - 6z, \quad P_{j+1}(z) = 2z(P_j(z) - P'_j(z)), \quad Q_j(z) = \sum_{k=0}^j (-1)^k \binom{j}{k} P_k(z). \quad (\text{AM.2})$$

Write p_{ja} and q_{ja} for the coefficients of z^a . Both polynomials have degree $d_j = j + 2$, with leading coefficients 2^{j+2} and $(-1)^j 2^{j+2}$, respectively, and both have zero constant coefficient. The formula defining Q_j is an explicit change of polynomial coordinates; neither family is discarded.

Define the linear involution on the actual Hilbert observation by

$$\mathcal{T} : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}_{>0}, dx), \quad (\mathcal{T}F)(x) = x^{-1}F(1/x). \quad (\text{AM.3})$$

It is unitary and self-adjoint, satisfies $\mathcal{T}^2 = 1$, and, on the strong Schwartz space $\mathcal{B}$,

$$D\mathcal{T} = \mathcal{T}(1 - D), \quad \mathcal{T}f_0 = f_0, \quad \mathcal{T}f_j = (1 - D)^j f_0. \quad (\text{AM.4})$$

Consequently, the complete source series and its reflected series are

$$f_j(x) = 2 \sum_{n \geq 1} P_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \tilde{f}_j(x) := (1 - D)^j f_0(x) = 2 \sum_{n \geq 1} Q_j(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad (\text{AM.5})$$

and $f_j(1/x) = x \tilde{f}_j(x)$.

Proof. For $z = \pi x^2$, applying $-x\partial_x$ to $P(z)e^{-z}$ gives $2z(P - P')e^{-z}$. This proves the recurrence and (AM.5). The leading coefficients and zero constant coefficients follow by induction. Under the Fourier convention $\hat{\phi}(\xi) = \int \phi(x)e^{-2\pi i x \xi} dx$, differentiating the Fourier transform of the Gaussian gives

$$\widehat{x^2 e^{-\pi x^2}}(\xi) = \left(\frac{1}{2\pi} - \xi^2 \right) e^{-\pi \xi^2}, \quad \widehat{x^4 e^{-\pi x^2}}(\xi) = \left(\xi^4 - \frac{3\xi^2}{\pi} + \frac{3}{4\pi^2} \right) e^{-\pi \xi^2}.$$

Multiplication by the exact coefficients $4\pi^2, -6\pi$ therefore gives $\hat{\phi}_* = \phi_*$. Its value at zero is zero, and its integral is $\hat{\phi}_*(0) = 0$. The theta-Fourier identity already proved for the source V then gives $\mathcal{T}f_0 = f_0$.

Substitution $u = 1/x$ gives $\int |x^{-1}F(1/x)|^2 dx = \int |F(u)|^2 du$, so $\mathcal{T}$ is unitary. It is an involution and hence self-adjoint. Direct differentiation proves $D\mathcal{T} = \mathcal{T}(1 - D)$, including the constant 1 arising from the original factor x^{-1} . Induction proves the other identities in (AM.4). Local uniform convergence of each differentiated Gaussian series follows on $x \geq \delta > 0$ by a polynomial times Gaussian majorant. The series gives decay of all Euler derivatives at infinity, and its exact reflected expression gives the corresponding decay at zero. Thus these operations apply to the original functions in $\mathcal{B}$. $\square$

7.2 An absolutely convergent double series for every entry

For integers $i, j \geq 0$, retain the raw moment entry

$$H_{ij} = \langle D^i f_0, D^j f_0 \rangle_{L^2(dx)} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j |2\xi(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{AM.6})$$

The equality is the original Mellin Plancherel identity. In particular H_{ij} is real because the source functions are real. For $L \geq 0$ and $\lambda > 0$, set

$$I_L(\lambda) = \int_1^\infty x^{2L} e^{-\lambda x^2} dx = \frac{\Gamma(L + 1/2, \lambda)}{2\lambda^{L+1/2}}, \quad \Gamma(a, \lambda) = \int_\lambda^\infty u^{a-1} e^{-u} du. \quad (\text{AM.7})$$

Theorem 7.1 (Exact original arithmetic moment series). *With $\lambda_{nm} = \pi(n^2 + m^2)$, the following double series converges absolutely, including after the finite coefficient sums are replaced by their absolute values:*

$$H_{ij} = 2 \sum_{n,m \geq 1} \sum_{a=1}^{d_i} \sum_{b=1}^{d_j} (p_{ia} p_{jb} + q_{ia} q_{jb}) \pi^{a+b} n^{2a} m^{2b} \cdot \frac{\Gamma(a+b+1/2, \lambda_{nm})}{\lambda_{nm}^{a+b+1/2}}. \quad (\text{AM.8})$$

Equivalently, it is the Gram integral on the positive half-domain

$$H_{ij} = \int_1^\infty (f_i(x) f_j(x) + \tilde{f}_i(x) \tilde{f}_j(x)) dx. \quad (\text{AM.9})$$

Thus the precise map to the positive Gram presentation is

$$\mathbb{C}[s] \longrightarrow L^2([1, \infty), dx)^{\oplus 2}, \quad P \longmapsto (P(D)f_0, P(1-D)f_0)|_{[1, \infty)}. \quad (\text{AM.10})$$

It preserves the original polynomial inner product (AM.6) and is injective. The signs in (AM.8) are retained; the positivity is the Gram positivity of this specified map.

Proof. Split the integral in (AM.6) at $x = 1$. On $(0, 1)$, set $x = 1/u$. The two factors u supplied by (AM.5) cancel the Jacobian u^{-2} exactly. This gives (AM.9). For absolute convergence, write

$$C_j = \sum_{a=1}^{d_j} |p_{ja}| \pi^a, \quad E_j = \sum_{a=1}^{d_j} |q_{ja}| \pi^a. \quad (\text{AM.11})$$

For $n, x \geq 1$, the polynomial bounds are

$$|P_j(\pi n^2 x^2)| \leq C_j n^{2d_j} x^{2d_j}, \quad |Q_j(\pi n^2 x^2)| \leq E_j n^{2d_j} x^{2d_j}.$$

Since $n^2 + m^2 \geq 2$ and $x^2 \geq 1$,

$$(n^2 + m^2)x^2 \geq \frac{1}{2}(n^2 + m^2) + x^2.$$

The absolute integrand series is therefore bounded by

$$4(C_i C_j + E_i E_j) x^{2(d_i + d_j)} e^{-\pi x^2} \left(\sum_{n \geq 1} n^{2d_i} e^{-\pi n^2/2} \right) \left(\sum_{m \geq 1} m^{2d_j} e^{-\pi m^2/2} \right),$$

which is integrable on $[1, \infty)$. This proves all required sum-integral interchanges. Substitution $u = \lambda x^2$ gives (AM.7). The two theta factors contribute 4, and (AM.7) contributes $1/2$, leaving exactly the factor 2 in (AM.8).

For complex polynomial coefficients, conjugate the coefficients of the first factor in the Gram integral. Equation (AM.9) then proves the inner-product assertion in (AM.10). If a nonzero polynomial P were in its kernel, its first component would vanish on $[1, \infty)$. Put $U(z) = \sum_j P_j(z) c_j$ when $P(s) = \sum_j c_j s^j$. The distinct degrees and nonzero leading coefficients of the P_j imply $U \neq 0$. Multiplying the vanishing theta series by $e^{\pi x^2}/2$ gives

$$U(\pi x^2) = - \sum_{n \geq 2} U(\pi n^2 x^2) e^{-\pi(n^2-1)x^2}.$$

The right side decays exponentially times a fixed polynomial as $x \rightarrow \infty$, by the same Gaussian bound. A nonzero polynomial on the left cannot have that decay. This contradiction proves injectivity. $\square$

7.3 A complete finite truncation certificate

Let $H_{ij}^{[N]}$ be (AM.8) with both theta indices restricted to $1 \leq n, m \leq N$, keeping all coefficients p_{ia}, q_{ia} . For the calculation of this entry choose an integer $N \geq \max(d_i, d_j)$. Define, for $k = i, j$,

$$\begin{aligned} r_{k,N} &= \left(\frac{N+2}{N+1} \right)^{2d_k} e^{-\pi(2N+3)}, & T_{k,N} &= \frac{(N+1)^{2d_k}}{1 - r_{k,N}}, \\ A_{k,N} &= \sum_{n=1}^N n^{2d_k} e^{-\pi(n^2-1)} + T_{k,N} e^{-\pi((N+1)^2-1)}. \end{aligned} \quad (\text{AM.12})$$

All these numbers are specified positive real constants. The choice of N ensures $r_{k,N} < 1$, since

$$\log r_{k,N} < \frac{2d_k}{N+1} - \pi(2N+3) < 2 - 5\pi < 0.$$

Theorem 7.2 (Rigorous complete double-tail bound). *For every $i, j \geq 0$ and $N \geq \max(i, j) + 2$,*

$$\boxed{|H_{ij} - H_{ij}^{[N]}| \leq \mathcal{E}_{ij,N} := 4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) \cdot I_{d_i+d_j}(\pi((N+1)^2 + 1))}. \quad (\text{AM.13})$$

In particular the bound tends to zero as $N \rightarrow \infty$, for each fixed pair i, j , and provides an explicit finite certificate for every raw arithmetic moment.

Proof. For $x \geq 1$, the ratio of successive terms of $n^{2d_k} e^{-\pi n^2 x^2}$ is at most

$$\left(\frac{n+1}{n} \right)^{2d_k} e^{-\pi(2n+1)}.$$

The logarithm of this expression has derivative $-2d_k/(n(n+1)) - 2\pi < 0$ for positive real n . For $n \geq N+1$ every ratio is consequently at most $r_{k,N}$. Summing the geometric majorant yields

$$\sum_{n \geq N} n^{2d_k} e^{-\pi n^2 x^2} \leq T_{k,N} e^{-\pi(N+1)^2 x^2}, \quad \sum_{n \geq 1} n^{2d_k} e^{-\pi n^2 x^2} \leq A_{k,N} e^{-\pi x^2}. \quad (\text{AM.14})$$

For the second inequality, divide by $e^{-\pi x^2}$, bound each retained factor $e^{-\pi(n^2-1)x^2}$ at $x = 1$, and apply the first inequality to the tail.

Write $F_{k,N}$ for the finite P_k -series and $\tilde{F}_{k,N}$ for the finite Q_k -series. Equations (AM.11) and (AM.14) give

$$\begin{aligned} |f_k(x)|, |F_{k,N}(x)| &\leq 2C_k x^{2d_k} A_{k,N} e^{-\pi x^2}, \\ |f_k(x) - F_{k,N}(x)| &\leq 2C_k x^{2d_k} T_{k,N} e^{-\pi(N+1)^2 x^2}. \end{aligned}$$

The reflected inequalities have exactly E_k in place of C_k . In each product of (AM.9), use the exact decomposition $ab - a_N b_N = (a - a_N)b + a_N(b - b_N)$. The resulting bound on the absolute difference of integrands is

$$4(C_i C_j + E_i E_j)(T_{i,N} A_{j,N} + A_{i,N} T_{j,N}) x^{2(d_i+d_j)} e^{-\pi((N+1)^2+1)x^2}.$$

Its integral is exactly the right side of (AM.13). Finally, the $A_{k,N}$ remain bounded for fixed k , the $T_{k,N}$ grow at most polynomially, and the integral decays exponentially in $(N+1)^2$. For completeness, with $L = d_i + d_j$ fixed and $\lambda \geq 2L$, the inequality $\log x \leq (x^2 - 1)/2$ for $x \geq 1$ gives

$$I_L(\lambda) \leq e^{-\lambda} \int_1^\infty e^{-(\lambda-L)(x^2-1)} dx \leq \frac{e^{-\lambda}}{2(\lambda-L)}.$$

Here the last step uses $x^2 - 1 \geq 2(x-1)$. This proves the asserted limit with an additional elementary upper bound when desired. $\square$

The finite matrix $(H_{ij}^{[N]})_{0 \leq i, j \leq m}$ is itself positive definite for every $N \geq 1$. Indeed it is the Gram matrix of $(F_{j,N}, \tilde{F}_{j,N})$ on the same half-domain; the injectivity proof in (AM.10) applies verbatim to the finite sum. This fact does not assign an ordering to its difference from the full matrix. The relation to the full matrix is the entrywise enclosure (AM.13), whose signs and error bounds are explicit.

7.4 Exact adjoint, parity, and finite-boundary checks

On $\mathcal{B}$ in $L^2(dx)$, integration by parts gives $\langle DF, H \rangle = \langle F, (1 - D)H \rangle$. Therefore the complete moments satisfy

$$H_{i+1,j} + H_{i,j+1} = H_{ij}, \quad H_{ij} = \sum_{k=0}^i (-1)^k \binom{i}{k} H_{0,j+k}, \quad H_{01} = \frac{1}{2} H_{00}. \quad (\text{AM.15})$$

All boundary terms vanish because the original functions have rapid decay at both ends. More explicitly,

$$(Df_i)f_j + f_i(Df_j) - f_i f_j = -\partial_x(x f_i f_j),$$

whose integral on $(0, \infty)$ is zero. Moving each of the i copies of D in the first argument to the second gives the middle identity; symmetry gives the last one.

The finite sums retain an exact boundary correction at $x = 1$:

$$H_{i+1,j}^{[N]} + H_{i,j+1}^{[N]} - H_{ij}^{[N]} = F_{i,N}(1)F_{j,N}(1) - \tilde{F}_{i,N}(1)\tilde{F}_{j,N}(1). \quad (\text{AM.16})$$

To prove this, use $DF_{j,N} = F_{j+1,N}$ for the first component and $D\tilde{F}_{j,N} = \tilde{F}_{j,N} - \tilde{F}_{j+1,N}$ for the second. Applying the preceding integration-by-parts calculation on $[1, \infty)$ gives the first displayed endpoint product for the first component and its negative for the second. The endpoints at infinity vanish term by term. In particular the exact finite identity $H_{01}^{[N]} = H_{00}^{[N]}/2$ follows even before the omitted tail is enclosed, since $F_{0,N} = \tilde{F}_{0,N}$.

One may also retain the exact triangular coordinate map

$$B_{ja} = \begin{cases} \binom{j}{a} (-1/2)^{j-a}, & a \leq j, \\ 0, & a > j, \end{cases} \quad C = BHB^T. \quad (\text{AM.17})$$

This is the Gram matrix of $u_j = (D - 1/2)^j f_0$, with inverse coordinate matrix $(B^{-1})_{ja} = \binom{j}{a} (1/2)^{j-a}$. No original moment is lost. Since $\mathcal{T}u_j = (-1)^j u_j$, unitarity proves $C_{ij} = 0$ for odd $i + j$. The operator $D - 1/2$ is skew-symmetric on this displayed strong Schwartz domain, so for $i + j = 2q$,

$$C_{ij} = (-1)^{i+q} \|u_q\|^2, \quad \|u_q\|^2 = \int_{\mathbb{R}} t^{2q} |2\xi(1/2 + it)|^2 \frac{dt}{2\pi} > 0. \quad (\text{AM.18})$$

For the first equality move all i powers to the second argument, and compare with moving the q powers in $\langle u_q, u_q \rangle$. Mellin Plancherel gives the second equality. Strict positivity follows from the nonzero original function and the polynomial injectivity already proved.

7.5 Reproducible certified finite arithmetic data

The companion program `scripts/check_arithmetic_moments.py` implements the exact integer recurrence (AM.2), evaluates every retained term of (AM.8) by Arb directed ball arithmetic, and enlarges its radius by an upward-rounded value of (AM.13). Thus the final balls enclose the infinite-source entries, including rounding error and both omitted theta tails. The default run uses $0 \leq i, j \leq 5$, $N = 7$, and 110 decimal working digits; its machine-readable output is `checks/arithmetic_moments.json`.

Let $\Delta_k = \det(H_{ij})_{0 \leq i,j < k}$ and $\Delta_0 = 1$. The squared norm h_k of the monic degree- k polynomial for the original measure μ is

$$h_k = \frac{\Delta_{k+1}}{\Delta_k}. \quad (\text{AM.19})$$

Indeed the coefficients below the leading term solve the Gram system making that monic polynomial orthogonal to degrees below k . Substitution of that unique solution into its squared norm gives the Schur complement of the leading k -by- k Gram block. The block determinant identity gives (AM.19). The interval calculation certifies strict positivity of all six leading principal determinants recorded in the companion certificate and all six corresponding h_k ; those are the actual arithmetic norms used by the orthogonal recurrence, through the finite degrees stated.

Here are explicit certified intervals from that run. Each row means the entire interval $[c - r, c + r]$, with the printed decimal centre c and radius r interpreted exactly. Rounding the stored Arb balls to the displayed decimals was performed outwards.

raw moment	c	r
H_{00}	1.279007247846485140480	$4.67 \cdot 10^{-22}$
H_{01}	0.6395036239232425702398	$3.33 \cdot 10^{-23}$
H_{02}	-12.73579749060893715027	$2.81 \cdot 10^{-21}$
H_{11}	13.37530111453217972051	$2.57 \cdot 10^{-21}$
H_{12}	6.687650557266089860256	$2.85 \cdot 10^{-22}$
H_{22}	389.8820543216399270934	$1.65 \cdot 10^{-20}$

Symmetry supplies the other entries of this original three-by-three moment block. The six original monic squared norms are enclosed by

k	c for h_k	r
0	1.279007247846485140480	$4.67 \cdot 10^{-22}$
1	13.05554930257055843539	$2.69 \cdot 10^{-21}$
2	250.0089768190499145175	$4.88 \cdot 10^{-20}$
3	6694.666685525627506773	$4.76 \cdot 10^{-19}$
4	224298.1953314520556463	$9.95 \cdot 10^{-18}$
5	9383436.471794727053469	$2.54 \cdot 10^{-16}$

The analytic double-tail bound is less than $2.961369 \cdot 10^{-63}$ for every entry of the six-by-six matrix, and is less than $1.150030 \cdot 10^{-83}$ for H_{00} . These bounds account for omitted source terms; the displayed intervals additionally include finite-term arithmetic and decimal conversion errors.

The checker also evaluates (AM.15) and the independent exact finite-boundary identity (AM.16), and detects removal of either the reflected half-domain, the theta factor 2, or the adjoint constant 1. Its checks execute identically under ordinary Python and Python's optimization flag. Finite interval certificates do not assert an infinite-degree weight estimate; (AM.8)–(AM.14) give the exact arithmetic input and its complete error control for calculating further degrees.

8 Exact source-metric optimization and aggregate arithmetic control

This section continues the actual theta complexes in the September 12 chain-descent, global-comparison, Deligne-control, and orthogonal-boundary notes. It proves an aggregate bound on the distinguished orthogonal representatives and computes exactly the optimization over all finite polynomial theta corrections. Every matrix below is recorded in the original polynomial coordinates of the packet. Auxiliary comparison maps and metrics are written explicitly.

8.1 The fixed arithmetic maps and their boundary

Retain

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad q : \mathcal{B} \longrightarrow Q, \quad \mathcal{H} = L^2(\mathbb{R}_{>0}, dx). \end{aligned}$$

The Hilbert inner product is conjugate-linear in its first variable. The original transform conventions are

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \hat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx.$$

The original source vector and its iterates are

$$\phi_* = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \phi_j = D^j \phi_*, \quad f_j = \Theta \phi_j, \quad \mathcal{M}f_j(s) = s^j g(s), \quad g = 2\xi.$$

Fix a nonempty finite set Z of actual nontrivial zeros with their full multiplicities m_ρ , and retain

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h_Z, \quad E = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

The coordinate basis of E is $1, t, \dots, t^{d-1}$. Let j_h denote the full Taylor-jet/Chinese-remainder map, let $\varepsilon_Z = j_h(h_Z/g)$, and write $R_Z(u)$ for the unique polynomial of degree less than d representing $\varepsilon_Z u$. The finite source construction gives

$$\begin{aligned} R_{\text{ref}} u &= S_{h_Z}^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*), \quad JF = j_h \mathcal{M}F, \quad \sigma_Z = q R_{\text{ref}}, \\ JR_{\text{ref}} &= 1_E, \quad J\Theta = 0, \quad DR_{\text{ref}} - R_{\text{ref}} A = f_0 \ell_Z, \quad \ell_Z(u) = [t^{d-1}] R_Z(u). \end{aligned} \quad (\text{TC1})$$

Here $S_\rho^{\mathcal{B}} F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy$, on its zero-Mellin-moment domain, and its iterated inverse defines $S_{h_Z}^{\mathcal{B}}$. Thus all local units in g/h_Z and the factor two in $g = 2\xi$ remain present.

For $L \geq 0$ define the actual finite source maps

$$\Phi_L : \mathbb{C}^{L+1} \longrightarrow V, \quad c \longmapsto \sum_{j=0}^L c_j \phi_j, \quad F_L = \Theta \Phi_L : \mathbb{C}^{L+1} \longrightarrow \mathcal{B}.$$

Every allowed representative in the family considered here has the form

$$R = R_{\text{ref}} + F_L C, \quad C : E \longrightarrow \mathbb{C}^{L+1}. \quad (\text{TC2})$$

In particular $qR = \sigma_Z$ and $JR = 1_E$. Let $I_L, S_L : \mathbb{C}^{L+1} \rightarrow \mathbb{C}^{L+2}$ send the standard coordinate vector e_j respectively to e_j, e_{j+1} . Directly,

$$DR - RA = \Theta K_R, \quad K_R = \phi_* \ell_Z + \Phi_{L+1}(S_L C - I_L C A) : E \longrightarrow V. \quad (\text{TC3})$$

This formula keeps the entire source boundary. Its class in Q is zero through $q\Theta = 0$.

Put $G_R = R^* R$, a positive definite Hermitian matrix in the fixed packet coordinates. For $F, H \in \mathcal{B}$, integration by parts gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

because $[x \overline{F(x)} H(x)]_0^\infty = 0$. Substitution of (TC3) therefore proves the exact identity

$$W_R := A^* G_R + G_R A - G_R = -(R^* \Theta K_R + (\Theta K_R)^* R). \quad (\text{TC4})$$

The adjoints here are adjoints of maps from finite-dimensional coefficient spaces to $\mathcal{H}$; no Hilbert extension of Θ is presumed.

8.2 A source-density calculation with its topology retained

We give the density argument used below, so the realization theorem has its full analytic input available. The map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dy), \quad (\mathcal{U}F)(y) = e^{y/2} F(e^y)$$

is unitary by $x = e^y$. Fourier transformation with kernel e^{ity} and measure $dt/(2\pi)$ sends $\mathcal{U}f_j$ to $(1/2 + it)^j g(1/2 + it)$.

The Gaussian theta series defining $f_0(e^z)$ converges holomorphically when $|\operatorname{Im} z| < \pi/4$. On every closed narrower strip, $\operatorname{Re}(e^{2z}) \geq c_a e^{2\operatorname{Re} z}$, with $c_a = \cos(2a) > 0$; its polynomial Gaussian summands consequently have superexponential decay as $\operatorname{Re} z \rightarrow +\infty$. The source ϕ_* is Fourier-fixed: the Gaussian is Fourier-fixed and the transform sends $D(D-1)$ to $(1-D)(-D) = D(D-1)$. Poisson summation and its two zero moments give $f_0(x) = x^{-1} f_0(1/x)$. Thus the holomorphic function $k(z) = e^{z/2} f_0(e^z)$ obeys $k(z) = k(-z)$, first on the real axis and then throughout the strip. The same decay holds at the negative end. Moving the Fourier contour to $\operatorname{Im} z = a$ sign t has vanishing vertical edges and proves, for every $0 < a < \pi/4$,

$$|g(1/2 + it)| \leq C_a e^{-a|t|}. \quad (\text{TC5})$$

Let $d\mu(t) = |g(1/2 + it)|^2 dt/(2\pi)$. This finite positive measure has a positive exponential moment. If $v \in L^2(\mu)$ is orthogonal to every polynomial, the finite complex measure $d\nu = \bar{v} d\mu$ has a smaller positive exponential moment, by Cauchy-Schwarz. Its Fourier transform is holomorphic on a strip and all derivatives at zero vanish, by the polynomial orthogonality. Its power series and analytic continuation make it identically zero. Uniqueness of the Fourier transform of finite measures gives $\nu = 0$, hence $v = 0$ in $L^2(\mu)$. Polynomials are therefore dense in $L^2(\mu)$. Multiplication by $g(1/2 + it)$ is a unitary map from $L^2(\mu)$ onto $L^2(dt/(2\pi))$: its norm equality is the definition of μ ; surjectivity follows by division off the measure-zero zero set of the nonzero entire function g . It follows that

$$\overline{\bigcup_{L \geq 0} F_L \mathbb{C}^{L+1}}^{\mathcal{H}} = \mathcal{H}. \quad (\text{TC6})$$

This is a statement in the displayed Hilbert topology. The identities $JR = 1$ below continue to hold in the original test-function space.

8.3 Every positive packet metric has an exact finite source lift

Define

$$H_m = F_m^* F_m, \quad C_m = F_m^* R_{\text{ref}}, \quad B_m = -H_m^{-1} C_m, \quad \Pi_m = F_m H_m^{-1} F_m^*, \quad R_m = R_{\text{ref}} + F_m B_m. \quad (\text{TC7})$$

The columns of F_m are linearly independent, because a relation has Mellin transform $g(s)P(s) = 0$ and hence $P = 0$. Thus $H_m > 0$, Π_m is the orthogonal projection onto the stated finite theta space, and

$$F_m^* R_m = 0, \quad G_m := R_m^* R_m = R_{\text{ref}}^* R_{\text{ref}} - C_m^* H_m^{-1} C_m > 0. \quad (\text{TC8})$$

Its strict positivity follows from $JR_m = 1$. Equation (TC6) proves $\Pi_m \rightarrow 1$ strongly. Since E is finite dimensional, $R_m \rightarrow 0$ in $\operatorname{Hom}(E, \mathcal{H})$, and $G_m \rightarrow 0$ in matrix norm.

Exact fixed-level image. At every specified level m , the entire image of the source-Gram map is

$$\{(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) : C : E \rightarrow \mathbb{C}^{m+1}\} = \{G_m + \Delta : \Delta \succeq 0, \operatorname{rank} \Delta \leq m+1\}. \quad (\text{TC8a})$$

Indeed completing the square, using $F_m^* R_m = 0$, gives

$$(R_{\text{ref}} + F_m C)^*(R_{\text{ref}} + F_m C) = G_m + (C - B_m)^* H_m (C - B_m). \quad (\text{TC8b})$$

The extra term is positive semidefinite and has the stated rank bound. Conversely, a positive semidefinite Δ of rank at most $m + 1$ has a factorization $\Delta = T^*T$ with $T : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$: diagonalize the finite Hermitian matrix Δ , retain its nonnegative eigenvalues, and insert their square roots into $m + 1$ rows. Taking $C = B_m + H_m^{-1/2}T$ proves the converse. The auxiliary factorization records a coefficient map; it changes none of the original packet data.

Exact realization theorem. For every specified positive definite Hermitian matrix G in the original polynomial coordinates of E , there exist a finite L and a coefficient map $C : E \rightarrow \mathbb{C}^{L+1}$ for which

$$R = R_{\text{ref}} + F_L C, \quad JR = 1, \quad qR = \sigma_Z, \quad R^*R = G. \quad (\text{TC9})$$

The construction is as follows. Choose $m \geq d - 1$ with $G_m \prec G$, set $L = m$, and let $I_{d-1,m} : \mathbb{C}^d \rightarrow \mathbb{C}^{m+1}$ pad a vector with zeros. The principal matrix $I_{d-1,m}^* H_m I_{d-1,m}$ is exactly H_{d-1} . Put

$$\Delta = G - G_m > 0, \quad C = B_m + I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}. \quad (\text{TC10})$$

Both square roots are the uniquely specified positive Hermitian roots of finite positive matrices on the displayed coefficient spaces, using their fixed standard Hermitian forms and the fixed polynomial-coordinate Hermitian form on E . The resulting map remains expressed in the original packet coordinates. Indeed

$$R = R_m + F_m I_{d-1,m} H_{d-1}^{-1/2} \Delta^{1/2}, \quad R_m^* F_m = 0,$$

so its Gram is

$$R^*R = G_m + \Delta^{1/2} H_{d-1}^{-1/2} H_{d-1} H_{d-1}^{-1/2} \Delta^{1/2} = G.$$

All added columns are finite complex linear combinations of the original $D^j \phi_*$ followed by Θ . This proves (TC9) inside the declared source family, with a finite coefficient degree.

The extension class has the literal type $\text{Ext}_{\mathbb{C}[t]}^1(E, V) \simeq V/h_Z(D)V$; its arithmetic coordinate ε_Z has type E . The exact map between these types is

$$\iota_h : E \longrightarrow V/h_Z(D)V, \quad [p(t)] \longmapsto [p(D)\phi_*]. \quad (\text{TC10a})$$

It is well-defined and $\mathbb{C}[t]$ -linear because replacing p by $p + h_Z a$ adds $h_Z(D)a(D)\phi_*$. It is an isomorphism, with the explicit inverse

$$\kappa_h : V/h_Z(D)V \longrightarrow E, \quad [\psi] \longmapsto j_h \left(\frac{\mathcal{M}\psi}{\mathcal{M}\phi_*} \right). \quad (\text{TC10b})$$

The denominator is

$$\mathcal{M}\phi_*(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2),$$

which is holomorphic and nonzero near every $\rho \in Z$. Every $\psi \in V$ has $\psi(x) = O(x^2)$ at zero, so its Mellin transform is holomorphic for $\text{Re } s > -2$. Mellin integration by parts gives $\mathcal{M}(p(D)\psi) = p(s)\mathcal{M}\psi$ near Z ; this proves well-definedness of κ_h and $\kappa_h \iota_h = 1_E$. For the other inverse law, let $\psi \in V$ have vanishing Mellin jets along h_Z . For the first factor $t - \rho$, its source inverse on $x > 0$ is

$$S_\rho \psi(x) = x^{-\rho} \int_x^\infty \psi(y) y^{\rho-1} dy = - \int_0^1 \psi(tx) t^{\rho-1} dt.$$

The second expression and $0 < \text{Re } \rho < 1$ prove smooth even extension to zero, with value zero; differentiating in every order is justified by the integrable factor $t^{\text{Re } \rho - 1}$. The first expression gives Schwartz decay at infinity. Integration of $(D - \rho)S_\rho \psi = \psi$ on the full real line gives $(1 - \rho) \int S_\rho \psi = \int \psi = 0$, so $S_\rho \psi \in V$. Its Mellin transform is $\mathcal{M}\psi(s)/(s - \rho)$, so the remaining zero jets permit iteration through every factor of h_Z . Consequently $\ker(j_h \mathcal{M}|_V) = h_Z(D)V$. Apply this to $\psi - p(D)\phi_*$,

where $p = j_h(\mathcal{M}\psi/\mathcal{M}\phi_*)$, to obtain $\iota_h\kappa_h = 1$. Thus the class of the actual arithmetic extension is precisely $\iota_h(\varepsilon_Z)$, which is nonzero. For every source change (TC2), this class is unchanged. Explicitly,

$$h_Z(D)R(1) = \Theta(R_Z(1)(D)\phi_* + h_Z(D)\Phi_L C(1)),$$

and the second term is zero in $V/h_Z(D)V$. There is still no strictly D -equivariant section: an intertwiner $T : E \rightarrow \mathcal{B}$ has image killed by $h_Z(D)$, whereas $h_Z(D)$ is injective on $\mathcal{B}$. To see this injectivity, $(D - \rho)F = 0$ implies $F(x) = cx^{-\rho}$; the bounds defining $\mathcal{B}$ force $c = 0$, and one iterates over the factors of h_Z . The exact map relating any section to the original arithmetic action is (TC3), with its retained nonzero extension class.

8.4 Sharp control with all nilpotent jets present

For a positive definite G , set

$$W_A(G) = A^*G + GA - G, \quad \epsilon_A(G) = \sup_{u \neq 0} \frac{|u^*W_A(G)u|}{u^*Gu}, \quad \eta_Z = \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|. \quad (\text{TC11})$$

For every eigenvector $Au = \rho u$,

$$u^*W_A(G)u = (2 \operatorname{Re} \rho - 1)u^*Gu. \quad (\text{TC12})$$

Consequently $\epsilon_A(G) \geq \eta_Z$, for every positive G . We now construct metrics arbitrarily close to that bound without removing a single nilpotent jet.

Fix $\epsilon > \eta_Z$, and fix any $G_0 > 0$, both in the original coordinates. Define

$$\begin{aligned} F(t) &= e^{-t}e^{tA^*}G_0e^{tA}, \\ G_\epsilon &= \int_{-\infty}^{\infty} e^{-\epsilon|t|}F(t)dt, \\ W_A(G_\epsilon) &= \epsilon \int_{-\infty}^{\infty} \operatorname{sign}(t)e^{-\epsilon|t|}F(t)dt. \end{aligned} \quad (\text{TC13})$$

The last identity is a conclusion, proved below. The factor e^{-t} records the original weight-one character; the action is still $e^{tA} = U_Z(e^t)$.

Let $e_\rho \in E$ be the exact Chinese-remainder idempotent and write $P_\rho = M_{e_\rho}$ and $N_\rho = (A - \rho I)P_\rho$. On $P_\rho E$, $N_\rho^{m_\rho} = 0$. Directly,

$$e^{tA} = \sum_{\rho \in Z} e^{t\rho} \sum_{j=0}^{m_\rho-1} \frac{t^j}{j!} N_\rho^j P_\rho. \quad (\text{TC14})$$

Thus every matrix entry in the integral is a finite sum of polynomial-exponential functions. The real part of each exponent $\bar{\rho} + \sigma - 1$ has absolute value at most η_Z . This proves absolute convergence of (TC13), including its derivative, and proves the endpoint limits needed for integration by parts. In particular $G_\epsilon > 0$. Since $F'(t) = A^*F(t) + F(t)A - F(t)$, integration by parts on the negative and positive half-lines gives (TC13). The two values at $t = 0$ cancel exactly. Both half-line integrals are strictly positive matrices, so

$$-\epsilon G_\epsilon \prec W_A(G_\epsilon) \prec \epsilon G_\epsilon. \quad (\text{TC15})$$

For completeness the construction requires no unevaluated analytic integral once the packet data and G_0 are specified. Put

$$I_k(z, \epsilon) = k! \left((\epsilon - z)^{-k-1} + (-1)^k (\epsilon + z)^{-k-1} \right). \quad (\text{TC16})$$

The substitutions $t \mapsto -t$ on the negative half-line and $\int_0^\infty t^k e^{-at} dt = k! a^{-k-1}$ show that

$$G_\epsilon = \sum_{\rho, \sigma \in Z} \sum_{i=0}^{m_\rho-1} \sum_{j=0}^{m_\sigma-1} \frac{I_{i+j}(\bar{\rho} + \sigma - 1, \epsilon)}{i!j!} \cdot (N_\rho^i P_\rho)^* G_0(N_\sigma^j P_\sigma). \quad (\text{TC17})$$

All local Jordan powers and every cross term remain in this formula. Combining (TC12), (TC15), and the actual realization (TC9) proves

$$\inf_{G>0} \epsilon_A(G) = \inf_{\substack{L \geq 0 \\ C: E \rightarrow \mathbb{C}^{L+1}}} \epsilon_A((R_{\text{ref}} + F_L C)^*(R_{\text{ref}} + F_L C)) = \eta_Z. \quad (\text{TC18})$$

The equality concerns all finite polynomial corrections. It does not assert that the distinguished orthogonal sequence (TC7) attains the same infimum.

Exact attainment criterion. The infimum in (TC18) is achieved precisely when every eigenvalue on either boundary $|2 \operatorname{Re} \rho - 1| = \eta_Z$ is semisimple. For $E = \mathbb{C}[t]/h_Z$, this means $m_\rho = 1$ for those zeros.

To prove necessity, a bound $\epsilon_A(G) \leq \eta_Z$ integrates to

$$\|e^{tA} u\|_G \leq e^{(1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0),$$

and applying the other inequality backward gives

$$\|e^{-tA} u\|_G \leq e^{(-1+\eta_Z)t/2} \|u\|_G \quad (t \geq 0).$$

On a generalized eigenspace on the upper boundary the first estimate bounds $e^{t(A-\rho I)}$ uniformly for $t \geq 0$, up to a unit-modulus scalar. A nonzero nilpotent block has a vector whose exponential contains a nonzero highest power of t , hence is unbounded. The lower boundary is treated by the backward estimate. This proves necessity.

For sufficiency, use the unchanged CRT direct-sum map $u \mapsto (P_\rho u)_\rho$. Make its summands mutually orthogonal by taking the pullback sum of chosen positive metrics on them. On a semisimple boundary summand, $A = \rho I$, so the control form is exactly $(2 \operatorname{Re} \rho - 1)$ times that summand's metric. For each interior summand choose a number strictly between $|2 \operatorname{Re} \rho - 1|$ and η_Z , and use (TC13) on that summand. Its control is strictly smaller than η_Z . The sum therefore has the required bound. When $\eta_Z = 0$ there are only boundary summands, and the same construction gives zero control exactly. The source realization (TC9) then supplies an actual finite corrected representative for this metric. Its boundary (TC3) persists; zero metric control only states that its Hermitian cross-pairing in (TC4) is zero.

8.5 The exact tensor cost of unrestricted metrics

For the original ordered tensor space $E^{\otimes n}$, retain

$$A_n = \sum_{j=1}^n I^{\otimes(j-1)} \otimes A \otimes I^{\otimes(n-j)}, \quad W_{A_n, n}(G) = A_n^* G + G A_n - nG. \quad (\text{TC19})$$

Its eigenvalues, with algebraic multiplicity, are all ordered sums $\rho_1 + \cdots + \rho_n$. This follows by tensoring complete invariant flags, or directly by the CRT decomposition and the nilpotent sum on each ordered product block. Hence

$$\max_{\lambda \in \operatorname{Spec} A_n} |2 \operatorname{Re} \lambda - n| = n\eta_Z. \quad (\text{TC20})$$

The upper bound is the triangle inequality; equality uses the ordered repeated tuple $(\rho, \dots, \rho)$ for a zero attaining η_Z . Repeating (TC12)–(TC17) with the weight factor e^{-nt} proves the exact value

$$\inf_{G>0} \sup_{v \neq 0} \frac{|v^* W_{A,n}(G)v|}{v^* G v} = n\eta_Z. \quad (\text{TC21})$$

The nilpotent sum on a product block is retained throughout this argument. Thus an unrestricted choice of a product-image metric, even a non-product metric, has a precisely calculated linear spectral cost. Establishing sublinear excess for the actual arithmetic image requires mathematical information about its spectrum or a further comparison which constrains that spectrum. Equations (TC18) and (TC21) are calculations of the optimization; they supply no missing estimate by choosing representatives.

8.6 A stronger bound on the actual orthogonal sequence

Return to the specific R_m in (TC7), and take Z stable under $\rho \mapsto 1 - \bar{\rho}$, including multiplicities. Write

$$b_m = e_m^* B_m : E \rightarrow \mathbb{C}, \quad z_m = R_m^* f_{m+1} : \mathbb{C} \rightarrow E.$$

The source boundary of R_m lies in the span of $f_0, \dots, f_{m+1}$. Orthogonality to the first $m+1$ vectors gives directly from (TC3)–(TC4)

$$W_m = A^* G_m + G_m A - G_m = -z_m b_m - b_m^* z_m^*, \quad \text{rank } W_m \leq 2. \quad (\text{TC22})$$

Reflection stability gives

$$\text{Tr}(G_m^{-1} W_m) = 2 \text{Re Tr } A - d = 0. \quad (\text{TC23})$$

Let $\epsilon_m = \epsilon_A(G_m)$. The G_m -self-adjoint map $G_m^{-1} W_m$ therefore has at most two nonzero eigenvalues, namely $\epsilon_m, -\epsilon_m$; if the rank is zero both are zero.

Aggregate displacement theorem. For every finite m in this original orthogonal source sequence,

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \max(2 \text{Re } \rho - 1, 0) = \sum_{\rho \in Z} m_\rho |\text{Re } \rho - \tfrac{1}{2}|. \quad (\text{TC24})$$

Here the second equality uses the given reflection involution and its preservation of multiplicities.

For a full proof of the first inequality, take a complete A -invariant flag, keeping all generalized eigenvectors. Choose a G_m -orthonormal frame adapted to that flag and denote its explicit coordinate map by $T : \mathbb{C}^d \rightarrow E$. It satisfies $T^* G_m T = I$; it is only a map used in the proof, while the recorded A, G_m, W_m remain in the original coordinates. The matrix $T^{-1} A T$ is upper triangular with diagonal entries the zeros ρ , repeated m_ρ times. Therefore the Hermitian matrix $H = T^* W_m T$ has diagonal entries $\delta_j = 2 \text{Re } \rho_j - 1$. By (TC22)–(TC23) its positive part H_+ has trace ϵ_m . Let P be the orthogonal coordinate projection onto the indices with $\delta_j > 0$. As $0 \leq P \leq I$ and $H \leq H_+$,

$$\sum_{\delta_j > 0} \delta_j = \text{Tr}(PH) \leq \text{Tr}(PH_+) \leq \text{Tr}(H_+) = \epsilon_m. \quad (\text{TC25})$$

Each trace inequality follows by inserting the positive square root of P or H_+ , so no sign is inferred from a product of noncommuting matrices. This proves (TC24).

More generally, whenever the positive rank of a control form is r_+ , the same argument gives $\sum_j \max(\delta_j, 0) \leq r_+ \epsilon_A(G)$. In the present source construction $r_+ \leq 1$, which is why the aggregate bound has no dimension loss.

The source's exact scalar certificate is retained:

$$u_m = z_m^* G_m^{-1} z_m, \quad v_m = b_m G_m^{-1} b_m^*, \quad c_m = b_m G_m^{-1} z_m, \quad \epsilon_m^2 = u_m v_m - (\operatorname{Im} c_m)^2. \quad (\text{TC26})$$

Indeed the two nonzero eigenvalues are those of $-\begin{pmatrix} c_m & v_m \\ u_m & c_m \end{pmatrix}$, and (TC23) gives $\operatorname{Re} c_m = 0$. If $a_{m+1} = (1 - \Pi_m)f_{m+1}$ and $\eta_{m+1} = \|a_{m+1}\|^2$, the exact rank-one Gram update gives

$$G_{m+1} = G_m - \eta_{m+1}^{-1} z_m z_m^*, \quad u_m = \eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right). \quad (\text{TC27})$$

The determinant identity is the rank-one determinant lemma, obtained by choosing a basis whose first vector spans $G_m^{-1} z_m$, or by expanding the determinant multilinearly. Thus a fully arithmetic version of (TC24) is

$$\eta_{m+1} \left(1 - \frac{\det G_{m+1}}{\det G_m} \right) v_m - (\operatorname{Im} c_m)^2 \geq \left(\sum_{\rho \in Z} m_\rho |\operatorname{Re} \rho - \tfrac{1}{2}| \right)^2. \quad (\text{TC28})$$

Every entry on the left is an actual theta Gram or cross-Gram entry with the original source representative. This identifies a stronger target for the arithmetic moment estimates than a bound for a single eigenvector.

8.7 Original supported maps and the precise result delivered

The passage from a finite source correction C to its representative, Gram, and boundary is the explicit chain of maps

$$C \mapsto R_{\text{ref}} + \Theta \Phi_L C \mapsto (qR, JR, R^*R, K_R) = (\sigma_Z, 1_E, G_R, K_R). \quad (\text{TC29})$$

For every active support label λ , the linear maps lift as $(\lambda, u) \mapsto (\lambda, Ru)$ and $(\lambda, u) \mapsto (\lambda, K_R u)$. The quotient of the boundary is $(\lambda, \Theta K_R u) \mapsto (\lambda, 0_Q)$; external absence remains the external zero τ . For the two-leg differential $d(\phi, \psi) = \Theta(\phi - \widehat{\psi})$, the two primitives are $(K_R u, 0)$ and $(0, -\widehat{K_R u})$. Their difference $(K_R u, \widehat{K_R u})$ is the retained degree-zero cycle. Thus all source changes in the optimization have actual typed cochain lifts with the same support synchronization as the input programme.

The new conclusions are the exact fixed-level Gram image (TC8a), the finite source realization (TC9), the sharp source-metric infimum with its attainment criterion (TC18), the exact tensor infimum (TC21), and the aggregate arithmetic bound (TC24)–(TC28). The metric realization leaves the extension unit, the residue/Jacobian pairing, the quotient classes and the generator unchanged. The original distinguished orthogonal metrics remain a smaller, arithmetically specified family. Their relative estimates, including all shrinking denominators in (TC26), are the next concrete arithmetic calculation.

9 The actual theta minimum as a confluent arithmetic kernel

This section continues the orthogonal-boundary construction of the 12 September 2026 tau workbench, through PR 13. Its new calculation identifies the permitted finite source corrections with a constrained polynomial minimum, evaluates that minimum by a complete jet kernel, and proves the exact two-vector boundary formula for that kernel. The arithmetic packet, Euler operator, local units, and multiplicities are retained throughout.

9.1 Original functions, maps, coordinates, and measures

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad D = -x\partial_x.$$

Let $\mathcal{B}$ be the space of smooth functions on $\mathbb{R}_{>0}$ such that $\sup_{x>0} x^b |D^k F(x)| < \infty$ for every integer b and every $k \geq 0$. The original maps and function are

$$\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*, \quad g(s) = 2\xi(s), \quad \mathcal{M}f_0 = g.$$

Write $\phi_j = D^j \phi_*$ and $f_j = D^j f_0 = \Theta\phi_j$. The Hilbert observation is $\mathcal{H} = L^2(\mathbb{R}_{>0}, dx)$; the first argument of every inner product is conjugate-linear. The actual Mellin-line map

$$\mathcal{U} : \mathcal{H} \longrightarrow L^2(\mathbb{R}, dt/(2\pi)), \quad \mathcal{U}F(t) = \mathcal{M}F(1/2 + it)$$

is unitary. Indeed $x = e^y$ gives the unitary map $F \mapsto e^{y/2} F(e^y)$ to $L^2(dy)$, and the displayed Mellin transform is its Fourier transform with the positive exponential e^{ity} . Plancherel has measure $dt/(2\pi)$ in this convention. Integration by parts on $\mathcal{B}$ gives $\mathcal{U}DF = (1/2 + it)\mathcal{U}F$.

Fix a nonempty finite packet Z of actual nontrivial zeros of g , including the full multiplicity m_ρ of each selected zero. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[s]/(h), \quad A = M_s : E \longrightarrow E.$$

The basis of E remains $1, s, \dots, s^{d-1}$. Let j_h denote polynomial remainder together with its equivalent complete Taylor-jet description. Set

$$v(s) = \frac{g(s)}{h(s)}, \quad \varepsilon = (j_h v)^{-1} \in E^\times, \quad U_\varepsilon = M_\varepsilon.$$

The function v is entire and has nonzero value at each selected centre after the corresponding cancellation. The matrix U_ε retains all derivatives of g/h , including the factor 2 in $g = 2\xi$. It commutes with A . For $u \in E$, let $R_Z(u)$ be the unique polynomial of degree below d representing εu . Write $R_{\text{ref}} : E \rightarrow \mathcal{B}$ for the source note's fixed finite section R_0 . This alias records its identity and avoids colliding with the $m = 0$ corrected member of the sequence R_m . The fixed section is characterized by

$$\mathcal{M}R_{\text{ref}}u = vR_Z(u), \quad J_Z R_{\text{ref}} = I_E, \quad J_Z \Theta = 0. \quad (\text{TB.1})$$

These are the actual finite section and jets of the tau construction; the proofs below do not replace them with a finite model.

Introduce the two specified measures on the same Mellin line

$$d\mu(t) = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad d\nu_Z(t) = |v(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{TB.2})$$

Both have every polynomial moment. For μ this follows from the Gaussian-series Mellin decay proved in PR 13; division by the fixed polynomial h preserves that decay outside a compact set, and the removable values of v are bounded on the compact set. The densities are positive except on discrete sets, because g and v are nonzero entire functions. Consequently the squared norm of every nonzero polynomial in either measure is strictly positive. No positive arithmetic residue or Weil pairing has been inserted in (TB.2).

9.2 Exact constrained minimum over the admitted source

For $m \geq 0$, the original permitted changes of representative are

$$R_B u = R_{\text{ref}} u + \sum_{j=0}^m f_j(Bu)_j, \quad B : E \longrightarrow \mathbb{C}^{m+1}. \quad (\text{TB.3})$$

Set $N = d + m$. Denote by $\mathcal{P}_N$ the polynomials of degree at most N , with their coefficient columns in the monomial basis, and define the surjection

$$C_N : \mathcal{P}_N \longrightarrow E, \quad C_N P = j_h P. \quad (\text{TB.4})$$

Let H_N^ν be its actual moment Gram matrix:

$$(H_N^\nu)_{ij} = \int_{\mathbb{R}} \overline{(1/2 + it)^i} (1/2 + it)^j d\nu_Z(t), \quad 0 \leq i, j \leq N. \quad (\text{TB.5})$$

It is positive definite by (TB.2). Define

$$K_N = C_N (H_N^\nu)^{-1} C_N^*. \quad (\text{TB.6})$$

This is positive definite: C_N is onto, so C_N^* is injective, and its nonzero vectors have positive quadratic form under $(H_N^\nu)^{-1}$.

Theorem 9.1 (The actual source minimum). *The orthogonally corrected representative R_m of PR 13 has the following explicit numerator and Gram matrix:*

$$p_{N,u} = (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad \mathcal{M} R_m u = v p_{N,u}, \quad (\text{TB.7})$$

$$G_m = R_m^* R_m = U_\varepsilon^* K_N^{-1} U_\varepsilon. \quad (\text{TB.8})$$

The polynomial correction

$$B_m(u)(s) = \frac{p_{N,u}(s) - R_Z(u)(s)}{h(s)} \quad (\text{TB.9})$$

has degree at most m . Its coefficients give exactly the map $B_m : E \rightarrow \mathbb{C}^{m+1}$ in the original source family (TB.3). Thus (TB.7) specifies the original theta boundary itself.

Proof. Taking Mellin transforms in (TB.3) gives

$$\mathcal{M} R_B u = v(s) \left(R_Z(u)(s) + h(s) \sum_{j=0}^m (Bu)_j s^j \right).$$

The expressions in parentheses range over precisely

$$\{P \in \mathcal{P}_N : C_N P = U_\varepsilon u\}. \quad (\text{TB.10})$$

Indeed the displayed expressions have the required residue. Conversely the difference between a polynomial in (TB.10) and $R_Z(u)$ is divisible by h , and the degree bound makes its quotient a polynomial of degree at most m . Mellin Plancherel identifies its squared Hilbert norm with $P^* H_N^\nu P$; this equality includes the measure $dt/(2\pi)$.

The vector $p = p_{N,u}$ in (TB.7) obeys $C_N p = U_\varepsilon u$. For $z \in \ker C_N$,

$$z^* H_N^\nu p = z^* C_N^* K_N^{-1} U_\varepsilon u = 0.$$

Therefore every competitor $p + z$ has the exact squared norm

$$(p + z)^* H_N^\nu (p + z) = u^* U_\varepsilon^* K_N^{-1} U_\varepsilon u + z^* H_N^\nu z. \quad (\text{TB.11})$$

Positivity proves uniqueness of the minimum. The original orthogonal projection construction is the unique minimum over the same affine source family, so its representative is (TB.7). Polarization proves (TB.8), while (TB.10) proves (TB.9). Every equality has been proved before application of the quotient; after applying q and J_Z one still has $q R_m = \sigma_Z$ and $J_Z R_m = I_E$. $\square$

The formula also extends to $m = -1$, $N = d - 1$, with no allowed source correction: C_{d-1} is the identity in the retained polynomial coordinates, and (TB.7) is precisely $R_Z(u)$.

9.3 Monic arithmetic polynomials and their exact recurrence

For this subsection use ν_Z , with no symmetry assumption on the selected packet. Let q_n be its monic polynomial of degree n orthogonal to all smaller degrees, and let $\kappa_n = \|q_n\|_{\nu_Z}^2 > 0$. The polynomials are explicitly the result of subtracting the projection of s^n onto smaller monomials. This is a specified unitriangular change of auxiliary polynomial basis; no coordinate of E , no arithmetic operator, and no polynomial coefficient has been replaced by an orthonormal coordinate.

Lemma 9.2 (The retained vertical-line recurrence). *There are $b_n \in \mathbb{C}$, with $b_n + \bar{b}_n = 1$, such that*

$$sq_0 = q_1 + b_0 q_0, \quad sq_n = q_{n+1} + b_n q_n - \frac{\kappa_n}{\kappa_{n-1}} q_{n-1} \quad (n \geq 1). \quad (\text{TB.12})$$

Proof. On the original line $s = 1/2 + it$, one has $\bar{s} = 1 - s$. Thus multiplication by s has adjoint multiplication by $1 - s$ on polynomial arguments. The coefficient of q_{n+1} in sq_n is one because both are monic. If $j \leq n - 2$, then

$$\langle q_j, sq_n \rangle_{\nu_Z} = \langle (1 - s)q_j, q_n \rangle_{\nu_Z} = 0,$$

since the first polynomial has degree at most $n - 1$. For $j = n - 1$ the same identity is $-\kappa_n$; division by κ_{n-1} gives the displayed negative coefficient. Finally

$$(b_n + \bar{b}_n)\kappa_n = \langle q_n, sq_n \rangle + \langle sq_n, q_n \rangle = \kappa_n.$$

This proves all coefficients, including their signs. No assertion that $b_n = 1/2$ is needed for a general packet. $\square$

Define the retained complete residue columns

$$v_n = j_h q_n \in E. \quad (\text{TB.13})$$

Changing the auxiliary monomial basis in (TB.6) to the stated monic basis and retaining its diagonal Gram matrix gives

$$K_N = \sum_{n=0}^N \frac{v_n v_n^*}{\kappa_n}. \quad (\text{TB.14})$$

Equivalently, this is the complete jet of the polynomial kernel

$$\mathcal{K}_N(z, w) = \sum_{n=0}^N \frac{q_n(z) \overline{q_n(w)}}{\kappa_n}. \quad (\text{TB.15})$$

In Taylor-jet coordinates at each centre, v_n has entries $q_n^{(k)}(\rho)/k!$ for $0 \leq k < m_\rho$. The map from these entries to the retained basis of E is the inverse confluent Vandermonde/Chinese-remainder map. In particular the factorials and all repeated-root derivatives remain in the construction.

Theorem 9.3 (Confluent kernel boundary formula). *For every $N \geq d - 1$,*

$$\boxed{AK_N + K_N A^* - K_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N}}. \quad (\text{TB.16})$$

Before taking any jets the corresponding polynomial identity is

$$(z + \bar{w} - 1)\mathcal{K}_N(z, w) = \frac{q_{N+1}(z) \overline{q_N(w)} + q_N(z) \overline{q_{N+1}(w)}}{\kappa_N}. \quad (\text{TB.17})$$

Proof. Apply j_h to (TB.12). The result replaces its left side by Av_n , because $j_h(sq_n) = A(j_hq_n)$. Substitute these relations into

$$AK_N + K_N A^* - K_N = \sum_{n=0}^N \frac{(Av_n)v_n^* + v_n(Av_n)^* - v_n v_n^*}{\kappa_n}.$$

The diagonal terms vanish by $b_n + \bar{b}_n = 1$. The negative contribution involving v_{n-1} has coefficient $-1/\kappa_{n-1}$, and cancels exactly the preceding positive contribution. The sole uncanceled term is the $n = N$ upper edge, giving (TB.16). The identical substitution before applying j_h proves (TB.17) for all complex z, w . This is a polynomial identity; no division by $z + \bar{w} - 1$ is performed at its zero set, and differentiating it proves the full confluent version. $\square$

9.4 Exact relative weight control from the same kernel

The original integration-by-parts form remains

$$W_m = A^* G_m + G_m A - G_m = -\{R_m^*(DR_m - R_m A) + (DR_m - R_m A)^* R_m\}. \quad (\text{TB.18})$$

The boundary term $[x\bar{F}H]_0^\infty$ vanishes for $F, H \in \mathcal{B}$. Put

$$L_N = AK_N + K_N A^* - K_N.$$

Commutation of A with U_ε in (TB.8) proves the exact comparison

$$\boxed{W_m = U_\varepsilon^* K_N^{-1} L_N K_N^{-1} U_\varepsilon.} \quad (\text{TB.19})$$

For clarity, multiplication gives

$$K_N^{-1} L_N K_N^{-1} = K_N^{-1} A + A^* K_N^{-1} - K_N^{-1},$$

which is exactly the middle factor in (TB.18). This proves the morphism between the inverse-kernel and original Hilbert metrics.

Define the actual real and complex numbers

$$\alpha_N = v_{N+1}^* K_N^{-1} v_{N+1}, \quad \beta_N = v_N^* K_N^{-1} v_N, \quad \gamma_N = v_N^* K_N^{-1} v_{N+1}. \quad (\text{TB.20})$$

Then the least symmetric allowance for the actual pair (G_m, W_m) is

$$\boxed{\epsilon_m = \frac{|\operatorname{Re} \gamma_N| + \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.} \quad (\text{TB.21})$$

Here $-\epsilon_m G_m \preceq W_m \preceq \epsilon_m G_m$. To prove minimality, the invertible change $y = K_N^{-1} U_\varepsilon u$ turns its Rayleigh quotient into $y^* L_N y / (y^* K_N y)$. By (TB.16) the nonzero eigenvalues of this Hermitian comparison are the eigenvalues of the two-by-two matrix

$$\frac{1}{\kappa_N} \begin{pmatrix} \gamma_N & \beta_N \\ \alpha_N & \bar{\gamma}_N \end{pmatrix}.$$

Its trace and determinant give

$$\lambda_\pm = \frac{\operatorname{Re} \gamma_N \pm \sqrt{\alpha_N \beta_N - (\operatorname{Im} \gamma_N)^2}}{\kappa_N}.$$

Cauchy-Schwarz in K_N^{-1} gives $|\gamma_N|^2 \leq \alpha_N \beta_N$, so the radicand is nonnegative; its square root is at least $|\operatorname{Re} \gamma_N|$. The maximum absolute eigenvalue is therefore (TB.21), including rank-one and zero cases.

For a packet closed under $\rho \mapsto 1 - \bar{\rho}$, with the multiplicities retained, $2 \operatorname{Re} \operatorname{tr} A = d$. Taking the trace in (TB.16) after multiplying by K_N^{-1} gives

$$0 = \operatorname{tr}(K_N^{-1} L_N) = \frac{2 \operatorname{Re} \gamma_N}{\kappa_N}. \quad (\text{TB.22})$$

Thus its real part is zero without any coordinate choice. Every eigenvector $Au = \rho u$ still satisfies

$$\frac{u^* W_m u}{u^* G_m u} = 2 \operatorname{Re} \rho - 1. \quad (\text{TB.23})$$

In particular (TB.21) has the intrinsic lower bound $\epsilon_m \geq \max_{\rho \in Z} |2 \operatorname{Re} \rho - 1|$. All of its quantities are now specified by arithmetic moments, complete jets, and the original local unit.

9.5 Exact change when one more theta direction is admitted

The step $m \mapsto m + 1$ means $N \mapsto N + 1$. Set $w = v_{N+1}$, $k = \kappa_{N+1}$, and $\theta = w^* K_N^{-1} w \geq 0$. Formula (TB.14) gives

$$K_{N+1} = K_N + \frac{w w^*}{k}. \quad (\text{TB.24})$$

Multiplication directly verifies the inverse identity

$$K_{N+1}^{-1} = K_N^{-1} - \frac{K_N^{-1} w w^* K_N^{-1}}{k + \theta}.$$

Consequently

$$G_{m+1} = G_m - \frac{U_\varepsilon^* K_N^{-1} w w^* K_N^{-1} U_\varepsilon}{k + \theta}, \quad (\text{TB.25})$$

$$\frac{\det G_{m+1}}{\det G_m} = \frac{k}{k + \theta} > 0. \quad (\text{TB.26})$$

The inverse formula follows by multiplying by (TB.24); If $w = 0$, both updates are identities and the determinant ratio is one. For $w \neq 0$, the determinant formula follows by extending $K_N^{-1} w$ to a basis, in which the rank-one map $K_N^{-1} w w^* / k$ has one possible nonzero eigenvalue θ/k . Hence the determinant factor of K_{N+1} relative to K_N is $1 + \theta/k$.

These are the same Hilbert metrics at adjacent relation levels. Subtract the two polynomial numerators in (TB.7) and divide their difference by h , as in (TB.9). This gives a source polynomial P_u of degree at most $m + 1$. The actual representative difference is $R_{m+1}u - R_m u = \Theta(P_u(D)\phi_*)$, an element of ΘW_{m+1} , where $W_{m+1} = \operatorname{span}(\phi_*, D\phi_*, \dots, D^{m+1}\phi_*)$. The quotient map $\mathcal{B}/\Theta W_m \rightarrow \mathcal{B}/\Theta W_{m+1}$ has kernel $\Theta W_{m+1}/\Theta W_m$; injectivity of Θ identifies this kernel with the original one-dimensional space W_{m+1}/W_m . The scalar e sends a class to the zero in its same support fibre, while τ sends it to external absence. None of (TB.24)–(TB.26) changes those maps.

9.6 The original measure supplies a full tail recurrence

There is a second exact description which uses the unchanged measure μ of (TB.2). Let Q_n denote its monic orthogonal polynomials and $h_n = \|Q_n\|_\mu^2$. The reality and functional equation of g make this measure invariant under $s \mapsto 1 - s$; uniqueness of the monic orthogonal polynomial then proves $Q_n(1 - s) = (-1)^n Q_n(s)$. The recurrence proof above therefore gives

$$sQ_n = Q_{n+1} + \frac{1}{2}Q_n - \frac{h_n}{h_{n-1}}Q_{n-1} \quad (n \geq 1). \quad (\text{TB.27})$$

For $n = 0$, omit the last term.

Define the rational function-valued map and coefficient rows

$$r(u)(s) = \frac{R_Z(u)(s)}{h(s)}, \quad d_n u = \frac{1}{h_n} \int_{\mathbb{R}} \overline{Q_n(1/2 + it)} r(u)(1/2 + it) d\mu(t). \quad (\text{TB.28})$$

At an actual line zero a rational pole in this expression is interpreted in $L^2(\mu)$; its product with g is the entire function $vR_Z(u)$. In particular its norm and every displayed pairing are finite. Polynomials are dense in $L^2(\mu)$: the positive exponential moment from PR 13 makes the Fourier transform of an orthogonal annihilator analytic in a strip, and all its derivatives vanish at zero, so Fourier uniqueness annihilates it. Therefore Parseval gives

$$\mathcal{U}R_m u = g \sum_{n>m} Q_n d_n u, \quad \boxed{G_m = \sum_{n>m} h_n d_n^* d_n.} \quad (\text{TB.29})$$

The series is a norm-convergent positive matrix series.

Let $\ell_Z(u) = [s^{d-1}]R_Z(u)$ be the original rank-one extension coefficient. Polynomial division gives the exact identity $sr - rA = \ell_Z$. Pair it with Q_n , and use the adjoint relation $s^* = 1 - s$ and (TB.27). With $d_{-1} = 0$, the result is

$$\boxed{\frac{h_{n+1}}{h_n} d_{n+1} = d_{n-1} + d_n(\tfrac{1}{2}I - A) - \delta_{n0}\ell_Z.} \quad (\text{TB.30})$$

This computation uses only finite pairings of the rational function and polynomials; it does not differentiate an unproved termwise infinite expansion.

The polynomial correction in (TB.3) has last row $b_m = -d_m$, since its polynomial is $-\sum_{j=0}^m Q_j d_j$ and Q_m is monic. Furthermore $R_m^* f_{m+1} = h_{m+1} d_{m+1}^*$. Substitution in the exact PR 13 boundary contraction gives

$$\boxed{W_m = h_{m+1}(d_{m+1}^* d_m + d_m^* d_{m+1}).} \quad (\text{TB.31})$$

Thus the moment-kernel description (TB.16)–(TB.21) and the full tail description (TB.29)–(TB.31) calculate the same operator and metric through explicitly proved maps.

Theorem 9.4 (The full source defect and its observation of the packet). *The entire original derivative boundary, including its orthogonal lower source term, is*

$$\boxed{DR_m - R_m A = -\frac{h_{m+1}}{h_m} Q_m(D) f_0 d_{m+1} - Q_{m+1}(D) f_0 d_m.} \quad (\text{TB.35})$$

An exact primitive in V is obtained by replacing f_0 with ϕ_* in this formula. The finite observation

$$\mathcal{O}_m : E \longrightarrow \mathbb{C}^{2d}, \quad u \longmapsto (d_m A^k u, d_{m+1} A^k u)_{k=0}^{d-1} \quad (\text{TB.36})$$

is injective. In particular, its two retained source rows, together with the unchanged arithmetic action, distinguish every vector and every nilpotent jet in the original packet.

Proof. Use the finite expression $r_m = r - \sum_{j=0}^m Q_j d_j$ and compute

$$sr_m - r_m A = \ell_Z - \sum_{j=0}^m (sQ_j d_j - Q_j d_j A).$$

Insert (TB.27) and group coefficients of Q_j . For $0 \leq j < m$ the coefficient is zero by (TB.30); the $j = 0$ equation includes precisely the term ℓ_Z . The coefficient at Q_m is $-h_{m+1} d_{m+1}/h_m$, and that at Q_{m+1} is $-d_m$. Multiplying by g and applying the inverse Mellin map proves (TB.35). Its displayed

primitive belongs to V , since every $D^j \phi_*$ does, and its theta image is the entire right side. Thus the term $Q_m(D)f_0$ remains an actual supported boundary even though its pairing with R_m vanishes.

Suppose that $\mathcal{O}_m u = 0$. Since $h(A) = 0$, polynomial division by h proves $d_m A^k u = d_{m+1} A^k u = 0$ for every $k \geq 0$. Equation (TB.30) for $n = m + 1$, applied to each $A^k u$, gives $d_{m+2} A^k u = 0$ for every k . Induction with the same recurrence gives $d_n A^k u = 0$ for all $n > m$ and $k \geq 0$. In particular every term of the positive series in (TB.29) vanishes on u , so $u^* G_m u = 0$. The exact left inverse $J_Z R_m = I_E$ makes G_m positive definite, hence $u = 0$. This proves injectivity without assuming distinct roots or removing a Jordan direction. $\square$

9.7 Metric dependence and the precise mathematical gain

Positivity in (TB.8) is the positivity of the original $L^2(dx)$ norm on an injective representative map. For the same-packet residue and dual-metric comparisons (TB.32)–(TB.33), assume explicitly that $Z = Z^\dagger$ under $\rho \mapsto 1 - \bar{\rho}$, with every full order retained. The preceding Christoffel-kernel and source-boundary results allow arbitrary packets. Reflection on a general packet is the conjugate-linear map $E_Z \rightarrow E_{Z^\dagger}$ defined in Section 4.3. The original arithmetic residue matrix S , satisfying $S^* = -S$ and $A^* S + S A = S$, is still related to its dual metric by

$$G_m^D = S^* G_m^{-1} S, \quad W_m^D = -S^* G_m^{-1} W_m G_m^{-1} S. \quad (\text{TB.32})$$

The original arithmetic Weil form, with $J_g = M_{j_h g'}$, is

$$u^* S J_g v = \langle u, (G_m^{-1} S J_g) v \rangle_{G_m}. \quad (\text{TB.33})$$

Thus the map relating the positive source metric to the arithmetic form is explicitly $G_m^{-1} S J_g$. A positive matrix G_m does not supply its positivity.

For each fixed packet, the original polynomial theta span is dense in $\mathcal{H}$, as proved in PR 13. Hence $G_m \rightarrow 0$. Equations (TB.8) and (TB.26) then imply

$$\lambda_{\min}(K_{d+m}) \rightarrow +\infty, \quad \prod_{m=m_0}^M \frac{\kappa_{d+m+1}}{\kappa_{d+m+1} + v_{d+m+1}^* K_{d+m}^{-1} v_{d+m+1}} \rightarrow 0. \quad (\text{TB.34})$$

Indeed $K^{-1} = U_\varepsilon^{-*} G_m U_\varepsilon^{-1} \rightarrow 0$, and the product telescopes to the positive determinant ratio. Absolute decay of the Gram matrix thus has a specified arithmetic-kernel counterpart. The relative quantity remains the exact expression (TB.21), with the spectral lower bound (TB.23).

The new result is a source-constrained, multiplicity-preserving arithmetic kernel formula and its exact finite relative control. It makes the next calculation depend on the concrete ν_Z -moments and the two retained complete jet vectors, rather than on freely selected positive matrices. No estimate that makes (TB.21) tend to zero for all actual packets has been proved here. A claim of spectral purity would require proving that estimate on these arithmetic quantities; (TB.23) identifies exactly what such a result would establish.

For standard background on orthogonal polynomial recurrence and kernel identities, see NIST DLMF, Section 18.2, <https://dlmf.nist.gov/18.2>. All identities used above, including the vertical-line signs, full jet maps, and metric comparisons, have been proved here for the retained objects.

10 The full jet kernel, finite resolvents, and the actual analytic tail

Retain the arithmetic measure, coordinates and maps of Section 9:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = |g(s)/h(s)|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

In particular, every selected zero has its complete multiplicity in h , and c is the coordinate column of the unit in the retained monomial basis of E . Write q_k for the monic orthogonal polynomials of ν_Z , $\kappa_k = \|q_k\|_{\nu_Z}^2$, and $a_k = \kappa_k/\kappa_{k-1} > 0$ for $k \geq 1$. The exact recurrence is

$$sq_k = q_{k+1} + b_k q_k - a_k q_{k-1}, \quad b_k + \bar{b}_k = 1, \quad (\text{KR.1})$$

where the last term is omitted at $k = 0$.

10.1 A finite compression in the original Mellin coordinate

Fix $n \geq d$, so $N = n - 1$ is the degree bound in the preceding kernel construction. On the coefficient space with basis $q_0, \dots, q_{n-1}$, put

$$H_n = \text{diag}(\kappa_0, \dots, \kappa_{n-1}), \quad (J_n)_{k,k} = b_k, \quad (J_n)_{k+1,k} = 1, \quad (J_n)_{k,k+1} = -a_{k+1}. \quad (\text{KR.2})$$

All unspecified entries are zero, and indices start at zero. Multiplication of the displayed matrices proves

$$J_n^* H_n + H_n J_n = H_n. \quad (\text{KR.3})$$

The isometry from this coefficient space to its polynomial image is $u \mapsto \sum_k q_k u_k$. For a standard coefficient Hermitian form, the corresponding isometry is

$$\mathcal{I}_n u = \sum_{k=0}^{n-1} \frac{q_k}{\sqrt{\kappa_k}} u_k, \quad B_n = H_n^{1/2} J_n H_n^{-1/2}. \quad (\text{KR.4})$$

This explicitly stated auxiliary coordinate map leaves A , the packet coordinates, q_k , and κ_k unchanged. The matrix B_n has lower off-diagonal entry $\sqrt{a_{k+1}}$, upper off-diagonal entry $-\sqrt{a_{k+1}}$, and satisfies $B_n^* + B_n = I$.

Lemma 10.1 (Nodes and endpoint resolvents). *The characteristic polynomial of J_n , and hence of B_n , is $q_n(z)$. Its roots $\lambda_{n,1}, \dots, \lambda_{n,n}$ are distinct and lie on $\text{Re } z = 1/2$. There are positive numbers $\omega_{n,j}$ and $\sigma_{n,j}$ such that*

$$\begin{aligned} m_n(z) &= \kappa_0 e_0^* (zI - B_n)^{-1} e_0 = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \omega_{n,j} &= \kappa_0, \\ F_n(z) &= e_{n-1}^* (zI - B_n)^{-1} e_{n-1} = \frac{q_{n-1}(z)}{q_n(z)} = \sum_{j=1}^n \frac{\sigma_{n,j}}{z - \lambda_{n,j}}, \\ \sum_j \sigma_{n,j} &= 1. \end{aligned} \quad (\text{KR.5})$$

Here m_n and F_n are respectively the initial and terminal observations of the same finite compression.

Proof. Expanding the tridiagonal determinant along its last row gives $D_{k+1}(z) = (z - b_k)D_k(z) + a_k D_{k-1}(z)$, with $D_0 = 1$, $D_1 = z - b_0$. This is (KR.1), proving $D_n = q_n$. The matrix $(B_n - \frac{1}{2}I)/i$ is Hermitian. If an eigenvector has first coordinate zero, its first row equation forces its second coordinate to be zero; recurrence through the rows forces every coordinate to vanish. Thus each eigenspace has dimension at most one. Hermitian diagonalization gives a complete eigenbasis, hence all eigenvalues are simple. The same argument beginning at the last row shows every eigenvector has nonzero last coordinate. If P_j are its rank-one orthogonal eigenprojections, set $\omega_{n,j} = \kappa_0 e_0^* P_j e_0 > 0$ and $\sigma_{n,j} = e_{n-1}^* P_j e_{n-1} > 0$. Their sums and the two spectral expansions follow from $\sum_j P_j = I$. The last diagonal cofactor of $zI - B_n$ is $q_{n-1}(z)$, proving the ratio in (KR.5). $\square$

Define $\delta(z) = |\operatorname{Re} z - 1/2|$. For $\delta(z) > 0$, (KR.3)–(KR.4) give $\|(zI - B_n)^{-1}\| \leq \delta(z)^{-1}$: for a vector u , the real part of $u^*(zI - B_n)u$ is $(\operatorname{Re} z - 1/2)\|u\|^2$, and Cauchy–Schwarz then gives the lower singular-value bound. In particular $|F_n(z)| \leq \delta(z)^{-1}$, and

$$\operatorname{Re} F_n(z) = (\operatorname{Re} z - \tfrac{1}{2}) \sum_j \frac{\sigma_{n,j}}{|z - \lambda_{n,j}|^2}. \quad (\text{KR.6})$$

10.2 Exact quadrature and the full confluent kernel

Theorem 10.2 (A polynomial formula valid at every packet centre). *Put*

$$L_{n,j}(z) = \frac{q_n(z)}{(z - \lambda_{n,j})q'_n(\lambda_{n,j})}. \quad (\text{KR.7})$$

This denotes the polynomial obtained by cancelling the displayed linear factor. For every polynomial P of degree at most $2n - 1$,

$$\int P(s) d\nu_Z = \sum_j \omega_{n,j} P(\lambda_{n,j}). \quad (\text{KR.8})$$

Consequently the complete arithmetic kernel is

$$K_{n-1} = \sum_{j=1}^n \frac{L_{n,j}(A) c c^* L_{n,j}(A)^*}{\omega_{n,j}}. \quad (\text{KR.9})$$

Formula (KR.9) includes all Taylor derivatives at all selected zeros, even when a selected critical zero equals a quadrature node.

Proof. For degree below n , the vector $P(J_n)e_0$ records exactly the coefficients of P in the monic orthogonal basis: in the successive multiplications no term of degree n occurs. Its initial coefficient is $\kappa_0^{-1} \int P d\nu_Z$. Thus the spectral expansion in (KR.5) proves (KR.8) in that degree range. For a general P in the asserted range, polynomial division gives $P = q_n U + R$, where $\deg U \leq n - 1$ and $\deg R < n$. On the line, the conjugate of $U(s)$ is again a polynomial in s , namely $\overline{U(1 - \bar{s})}$. Orthogonality therefore gives $\int q_n U d\nu_Z = 0$. The same multiple vanishes at all the nodes, proving (KR.8).

The polynomials $L_{n,j}$ form the interpolation basis of $\mathcal{P}_{n-1}$. Apply (KR.8) to $\overline{L_{n,j}(s)} L_{n,k}(s)$, interpreted as a polynomial on the line; its degree is at most $2n - 2$. Their Gram matrix is $\operatorname{diag}(\omega_{n,j})$. In this basis the residue map has column $j_h L_{n,j} = L_{n,j}(A)c$. The definition $K = CH^{-1}C^*$ now proves (KR.9). This is a basis change in the polynomial source only. Polynomial evaluation at A retains its nilpotent powers, so the formula gives the complete jets without any distinct-centre or simple-zero assumption. $\square$

To identify the quadrature endpoints exactly, one has

$$\sigma_{n,j} = \frac{\omega_{n,j} |q_{n-1}(\lambda_{n,j})|^2}{\kappa_{n-1}} = \frac{\kappa_{n-1}}{\omega_{n,j} |q'_n(\lambda_{n,j})|^2}. \quad (\text{KR.10})$$

Indeed an eigenvector of B_n at λ has coordinates proportional to $\overline{q_k(\lambda)}/\sqrt{\kappa_k}$: substitution into the recurrence proves this assertion, including the endpoint row. The squared ratio of its last and first coordinates proves the first equality. The preceding polynomial kernel formula gives $\mathcal{K}_{n-1}(\lambda_{n,j}, \lambda_{n,j}) = 1/\omega_{n,j}$. In the original Christoffel–Darboux polynomial identity, set $w = \lambda_{n,j}$, divide its common polynomial factor $z - \lambda_{n,j}$, then evaluate at that point. This gives

$$\frac{1}{\omega_{n,j}} = \frac{q'_n(\lambda_{n,j}) \overline{q_{n-1}(\lambda_{n,j})}}{\kappa_{n-1}}.$$

Its squared modulus gives the second equality in (KR.10).

Theorem 10.3 (The exact resolvent metric for the packet). *Suppose, for this formula, that no selected centre equals one of the specified n quadrature nodes. Define maps on the original space E by*

$$Q_n = q_n(A), \quad X_{n,j} = (A - \lambda_{n,j}I)^{-1}c, \quad S_n = \sum_j \sigma_{n,j} X_{n,j} X_{n,j}^*, \quad r_n = F_n(A)c. \quad (\text{KR.11})$$

Then Q_n is invertible, $S_n > 0$, and

$$\boxed{K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*}. \quad (\text{KR.12})$$

In particular, the original relative-control quantity at $m = n - 1 - d$ is exactly

$$\epsilon_m = \sup_{y \neq 0} \frac{|y^*(cr_n^* + r_n c^*)y|}{y^* S_n y}. \quad (\text{KR.13})$$

The identity is valid whenever that index belongs to the original source sequence; $m = -1$ is its already specified fixed-section extension. Every off-line packet obeys the nonintersection condition for every n . At an intersecting critical centre, (KR.9) gives the exact polynomial continuation, with no deletion of that centre or its jets.

Proof. The finite algebra E has an inverse for a polynomial precisely when the polynomial is nonzero at every centre: the Euclidean algorithm gives an inverse modulo h , or, equivalently, the finite Taylor inverse at each centre gives the inverse under the complete Chinese-remainder map. This proves the asserted invertibilities. Substitution of (KR.7) and (KR.10) into (KR.9) proves the first identity in (KR.12). It proves $S_n > 0$, since the residue map is onto at $n \geq d$, hence $K_{n-1} > 0$. The identity $AX_{n,j} = \lambda_{n,j} X_{n,j} + c$, together with $\lambda_{n,j} + \bar{\lambda}_{n,j} = 1$, proves

$$AS_n + S_n A^* - S_n = c \left(\sum_j \sigma_{n,j} X_{n,j} \right)^* + \left(\sum_j \sigma_{n,j} X_{n,j} \right) c^*.$$

The partial fractions in (KR.5), evaluated in the finite algebra, identify the sum with $F_n(A)c$. Because Q_n commutes with A , both kernel and its weight form are congruent, with the same factor κ_{n-1}^{-1} , to S_n and its displayed weight form. The invertible change $y = Q_n^* x$ proves equality of their Rayleigh suprema. The previous exact map from K_{n-1} to G_m then proves (KR.13). All of these maps are stated in the original quotient coordinates; the Taylor expansion of each rational factor includes every power through $m_\rho - 1$. $\square$

The scalar form of (KR.13), with every inverse taken on the same retained packet coordinates, is

$$\alpha_n = r_n^* S_n^{-1} r_n, \quad \beta_n = c^* S_n^{-1} c, \quad \gamma_n = c^* S_n^{-1} r_n,$$

$$\boxed{\epsilon_m = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}}. \quad (\text{KR.13a})$$

Indeed the nonzero eigenvalues of $S_n^{-1/2}(cr_n^* + r_n c^*)S_n^{-1/2}$ coincide, with multiplicities, with the nonzero eigenvalues of $\begin{pmatrix} \gamma_n & \beta_n \\ \alpha_n & \bar{\gamma}_n \end{pmatrix}$: factor the former through its two displayed vectors and interchange the two rectangular factors. Its characteristic polynomial gives $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Cauchy-Schwarz gives $|\gamma_n|^2 \leq \alpha_n \beta_n$, so their greatest absolute value is (KR.13a), including zero or one-dimensional rank. The full congruence (KR.12) specifies exactly how the original q_n , κ_{n-1} and local unit enter the metric whose relative form is evaluated here.

10.3 The infinite multiplication operator and its genuine tail

Let $\mathcal{L} = L^2(\nu_Z)$, and let M be multiplication by s , with domain $\{f : sf \in \mathcal{L}\}$. It is a closed operator and $M^* = I - M$. These statements follow directly from the multiplication formula; its resolvent off the line is multiplication by $(z - s)^{-1}$, of norm at most $\delta(z)^{-1}$. The auxiliary vectors $\psi_k = q_k / \sqrt{\kappa_k}$ form a complete orthonormal basis. Completeness follows from the positive exponential moment of the actual arithmetic measure by the Fourier-annihilator proof already given for the theta measures.

Let P_n project onto $\psi_0, \dots, \psi_{n-1}$, and put $Q_n^{\text{tail}} = I - P_n$; this projection is distinct from the polynomial map $Q_n = q_n(A)$ in (KR.11). Define

$$M_n^{\text{tail}} = Q_n^{\text{tail}} M Q_n^{\text{tail}} \quad \text{on } Q_n^{\text{tail}} \text{Dom } M, \quad \zeta_n(z) = \langle \psi_n, (z - M_n^{\text{tail}})^{-1} \psi_n \rangle. \quad (\text{KR.14})$$

This tail resolvent exists at every off-line z , with norm at most $\delta(z)^{-1}$. Here is the domain justification. Every polynomial belongs to $\text{Dom } M$, so P_n preserves that domain and $Q_n^{\text{tail}} \text{Dom } M = \text{Dom } M \cap Q_n^{\text{tail}} \mathcal{L}$. The operator $(M - \frac{1}{2}I)/i$ is self-adjoint multiplication by t . Its off-diagonal blocks for $P_n \oplus Q_n^{\text{tail}}$ extend to a bounded, finite-rank self-adjoint operator, by (KR.1). Subtracting this operator preserves self-adjointness: for a self-adjoint T and bounded self-adjoint C , the definition of the adjoint gives $(T - C)^* = T^* - C$ on exactly $\text{Dom } T^*$. The resulting diagonal operator reduces the two projections. Its tail block is therefore self-adjoint on the displayed tail domain. Multiplication by i and addition of $1/2$ gives the claimed tail resolvent.

Theorem 10.4 (Exact finite-section, terminal, and tail comparison). *For $\delta(z) > 0$, let*

$$m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t).$$

Then

$$H_n^{\text{obs}}(z) = \frac{\zeta_n(z)}{1 + a_n F_n(z) \zeta_n(z)}, \quad (\text{KR.15})$$

$$m(z) - m_n(z) = \frac{(-1)^n \kappa_n \zeta_n(z)}{q_n(z)^2 [1 + a_n F_n(z) \zeta_n(z)]} = \frac{(-1)^n}{q_n(z)^2} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z(t). \quad (\text{KR.16})$$

Every denominator in these formulas is nonzero off the original Mellin line. The terminal finite observations additionally obey the exact Schur recurrence

$$F_1(z) = \frac{1}{z - b_0}, \quad F_{n+1}(z) = \frac{1}{z - b_n + a_n F_n(z)}. \quad (\text{KR.17})$$

Proof. Under $\mathcal{I}_n \oplus I$, the block operator M is

$$\begin{pmatrix} B_n & -\sqrt{a_n} e_{n-1} \langle \psi_n, \cdot \rangle \\ \sqrt{a_n} \psi_n e_{n-1}^* & M_n^{\text{tail}} \end{pmatrix}. \quad (\text{KR.18})$$

The signs follow directly from (KR.1); both coupling maps are bounded on the displayed summands. Eliminate the tail component in $z - M$. The finite Schur complement is $zI - B_n + a_n \zeta_n(z) e_{n-1} e_{n-1}^*$. For $\text{Re } z > 1/2$, the real part of its quadratic form is at least $\delta(z) \|u\|^2$, because $\text{Re } \zeta_n(z) > 0$; on the other side all signs reverse. The sign of this scalar real part follows either from the self-adjoint tail resolvent or from the resolvent identity $\text{Re} \langle v, (z - M_n^{\text{tail}})^{-1} v \rangle = (\text{Re } z - 1/2) \|(z - M_n^{\text{tail}})^{-1} v\|^2$. Thus the Schur complement is invertible and $1 + a_n \zeta_n F_n \neq 0$. Direct multiplication verifies the rank-one inverse formula

$$R_n - \frac{a_n \zeta_n}{1 + a_n \zeta_n F_n} R_n e_{n-1} e_{n-1}^* R_n, \quad R_n = (zI - B_n)^{-1}. \quad (\text{KR.19})$$

The analogous bottom Schur calculation, paired against ψ_n , gives (KR.15), since $\langle \psi_n, (z - M)^{-1} \psi_n \rangle = H_n^{\text{obs}}(z)$.

The two endpoint cofactors of $zI - B_n$ give

$$e_0^* R_n e_{n-1} = \frac{(-1)^{n-1} \sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}, \quad e_{n-1}^* R_n e_0 = \frac{\sqrt{\kappa_{n-1}/\kappa_0}}{q_n(z)}. \quad (\text{KR.20})$$

These cofactors are the products of the successive upper and lower entries, respectively; the formula also covers $n = 1$. Pair (KR.19) against $\sqrt{\kappa_0} e_0$, the coefficient vector of the unchanged function 1. Substitute (KR.20) and $a_n \kappa_{n-1} = \kappa_n$. This proves the first equality in (KR.16); (KR.15) proves the second. Finally (KR.17) is (KR.1) divided by q_{n+1} , equivalently the last-coordinate finite Schur complement. $\square$

There is also a direct polynomial check on every factor in (KR.16). The polynomial

$$p_n(z) = \int \frac{q_n(z) - q_n(s)}{z - s} d\nu_Z(t)$$

has degree below n . At a quadrature node its ratio residue is $p_n(\lambda_{n,j})/q'_n(\lambda_{n,j}) = \int L_{n,j} d\nu_Z = \omega_{n,j}$, so $m_n = p_n/q_n$. Orthogonality to the polynomial $(q_n(z) - q_n(s))/(z - s)$ gives

$$m - m_n = \frac{1}{q_n(z)^2} \int \frac{q_n(s)^2}{z - s} d\nu_Z.$$

The polynomial $(-1)^n \overline{q_n(1 - \bar{s})}$ is monic and has the same orthogonality conditions as q_n ; uniqueness makes it equal to q_n . On the line this says $q_n(s)^2 = (-1)^n |q_n(s)|^2$, giving (KR.16) again. In particular the explicit error inequality is

$$|m(z) - m_n(z)| \leq \frac{\kappa_n}{\delta(z) |q_n(z)|^2}. \quad (\text{KR.21})$$

The initial and terminal observations have now been related by the actual block maps (KR.18)–(KR.20), the tail (KR.14), and the exact remainder (KR.16). The relative form uses the terminal observation through (KR.11)–(KR.13). An estimate for the initial observation can be propagated through these equalities while retaining the factors q_n , κ_n , a_n , and the tail denominator. None of those arithmetic quantities is discarded in passing from one observation to the other.

11 Quantitative convergence of the arithmetic Gaussian resolvents

Retain the actual measure and every coordinate of Section 10:

$$s = \frac{1}{2} + it, \quad d\nu_Z(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad E = \mathbb{C}[s]/(h), \quad A = M_s, \quad c = j_h 1.$$

The polynomial h contains the complete multiplicities of its selected actual zeros. The quotient g/h is therefore entire, including at the selected critical zeros. Use the initial Gaussian observation of (KR.5) and (KR.8),

$$m_n(z) = \sum_{j=1}^n \frac{\omega_{n,j}}{z - \lambda_{n,j}}, \quad m(z) = \int \frac{d\nu_Z(t)}{z - s}, \quad \mathcal{E}_n(z) = m(z) - m_n(z), \quad \delta(z) = |\operatorname{Re} z - \tfrac{1}{2}|. \quad (\text{GC.1})$$

The Gaussian weights are positive and the distinct nodes lie on the original Mellin line, as proved in Section 10. Its exact quadrature (KR.8) and exact error (KR.16) are the inputs below. No asymptotic for q_n or κ_n is assumed.

For each fixed $0 < a < \pi/2$, the number

$$E_a = \max \left\{ \int e^{at} d\nu_Z(t), \int e^{-at} d\nu_Z(t) \right\} \quad (\text{GC.2})$$

is finite. Here is the connection to the proved arithmetic decay. The actual theta calculation in Section 42 gives $|g(1/2 + it)| \leq C_b e^{-b|t|}$ for every $0 < b < \pi/4$. Choose $a/2 < b < \pi/4$. Outside a sufficiently large compact interval, the monic polynomial h of degree d satisfies $|h(1/2 + it)| \geq c_h (1 + |t|)^d$. Thus the density multiplied by $e^{a|t|}$ is bounded there by a constant times $(1 + |t|)^{-2d} e^{-(2b-a)|t|}$. On the compact interval the entire quotient is bounded. Both parts are integrable, proving (GC.2) with the original factor $dt/(2\pi)$ retained.

11.1 The Gaussian rule inherits the original exponential moments

Lemma 11.1 (Exponential moment domination). *For every real v with $|v| \leq a$, write $\lambda_{n,j} = 1/2 + it_{n,j}$. Then*

$$\sum_j \omega_{n,j} e^{vt_{n,j}} \leq \int e^{vt} d\nu_Z(t) \leq E_a. \quad (\text{GC.3})$$

Proof. Let H be the real Hermite interpolation polynomial of degree at most $2n - 1$ whose values and first derivatives at the distinct real nodes $t_{n,j}$ equal those of $f(t) = e^{vt}$. Its existence and uniqueness follow from the invertibility of the interpolation map: a polynomial in its kernel has a double zero at each of the n nodes, hence is zero in degree below $2n$; domain and target both have dimension $2n$. For a real t outside the nodes the interpolation remainder is

$$f(t) - H(t) = \frac{f^{(2n)}(\xi)}{(2n)!} \prod_j (t - t_{n,j})^2 \geq 0 \quad (\text{GC.4})$$

for some real ξ between t and the nodes. To prove the remainder, subtract $C \prod_j (x - t_{n,j})^2$ from $f(x) - H(x)$, choosing C to produce an additional zero at $x = t$. The resulting function has the n double zeros and the additional zero. Repeated Rolle's theorem gives a zero of its $2n$ -th derivative; solving at that zero gives (GC.4). The last sign follows from $f^{(2n)}(x) = v^{2n} e^{vx} \geq 0$. At a node the remainder is zero.

Integrate the pointwise inequality. The exponential and every polynomial are integrable by (GC.2). Since H is a polynomial in $t = (s - 1/2)/i$ of degree at most $2n - 1$, (KR.8) yields

$$\int H(t) d\nu_Z(t) = \sum_j \omega_{n,j} H(t_{n,j}) = \sum_j \omega_{n,j} e^{vt_{n,j}}.$$

This proves the first inequality in (GC.3). Pointwise convexity of e^{vt} in $v \in [-a, a]$, followed by integration, proves the second. No bound on the largest quadrature node has been inserted. $\square$

11.2 Exact moments give a quantitative analytic-strip estimate

Define

$$\Phi(u) = \int e^{iut} d\nu_Z(t), \quad \Phi_n(u) = \sum_j \omega_{n,j} e^{iut_{n,j}}, \quad \Delta_n(u) = \Phi(u) - \Phi_n(u). \quad (\text{GC.5})$$

These functions are holomorphic on $|\operatorname{Im} u| < a$. Indeed, on each smaller closed strip, every differentiated integrand is dominated by a constant times $e^{at} + e^{-at}$, because $|t|^k e^{b|t|} \leq C_{k,b,a} (e^{at} + e^{-at})$ when $b < a$. Dominated convergence proves all the derivatives of Φ ; the finite sum is entire. Equation (GC.3), applied at $v = -\operatorname{Im} u$, gives

$$|\Delta_n(u)| \leq 2E_a \quad (|\operatorname{Im} u| < a), \quad \Delta_n^{(k)}(0) = 0 \quad (0 \leq k < 2n). \quad (\text{GC.6})$$

The derivative identities are exactly the moments in (KR.8).

Lemma 11.2 (The complete strip bound). *For $|\operatorname{Im} u| < a$,*

$$|\Delta_n(u)| \leq 2E_a \left| \tanh\left(\frac{\pi u}{4a}\right) \right|^{2n}. \quad (\text{GC.7})$$

Proof. The explicit holomorphic coordinate maps

$$w = \tanh\left(\frac{\pi u}{4a}\right), \quad u = \frac{2a}{\pi} \operatorname{Log} \frac{1+w}{1-w}$$

are inverse bijections between the strip and the unit disk. For $|w| < 1$, the fraction lies in the right half-plane, so its principal logarithm has imaginary part in $(-\pi/2, \pi/2)$, which proves the asserted strip range. Compose $\Delta_n/(2E_a)$ with this inverse. By (GC.6), the resulting disk function is bounded by one and has a zero of order at least $2n$ at zero. Divide by w^{2n} , filling in its removable value at zero. On each circle $|w| = r < 1$, its modulus is at most r^{-2n} ; the maximum principle gives this bound inside the circle. Letting r increase to one proves the required bound by $|w|^{2n}$, hence (GC.7). $\square$

Theorem 11.3 (Direct off-line convergence with an explicit rate). *For every $0 < a < \pi/2$, every $n \geq 1$, and every z off the original Mellin line,*

$$|m(z) - m_n(z)| \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\delta(z)}{\pi}\right) (4n)^{-2a\delta(z)/\pi}. \quad (\text{GC.8})$$

On a compact set K disjoint from that line the same bound holds uniformly after replacing $\delta(z)$ everywhere by $\delta_K = \min_{z \in K} \delta(z) > 0$.

Proof. Put $w = z - 1/2$. On $\operatorname{Re} w > 0$, the elementary Laplace identity and Fubini's theorem give

$$\mathcal{E}_n(z) = \int_0^\infty e^{-wu} \Delta_n(u) du. \quad (\text{GC.9})$$

For example the absolute integral for the original measure before subtraction is at most $\nu_Z(\mathbb{R})/\operatorname{Re} w$, so the interchange is justified. On $\operatorname{Re} w < 0$, the exact orientation is

$$\mathcal{E}_n(z) = - \int_0^\infty e^{wu} \Delta_n(-u) du. \quad (\text{GC.10})$$

These follow respectively from integrating $e^{-(w-it)u}$ and $-e^{(w-it)u}$. Set $b = \pi/(4a)$. Equation (GC.7) gives

$$|\mathcal{E}_n(z)| \leq 2E_a I_n(\delta(z)), \quad I_n(\delta) = \int_0^\infty e^{-\delta u} \tanh(bu)^{2n} du. \quad (\text{GC.11})$$

For $0 < x < 1$, direct integration gives

$$\log \frac{1-x}{1+x} = -2 \int_0^x \frac{dv}{1-v^2} \leq -2x.$$

Taking $x = e^{-2bu}$ yields $\tanh(bu)^{2n} \leq e^{-4ne^{-2bu}}$. With $\beta = \delta/(2b) = 2a\delta/\pi$, the same variable change gives

$$\begin{aligned} I_n(\delta) &\leq \frac{1}{2b} \int_0^1 x^{\beta-1} e^{-4nx} dx \\ &\leq \frac{1}{2b} (4n)^{-\beta} \Gamma(\beta), \quad \Gamma(\beta) = \int_0^\infty y^{\beta-1} e^{-y} dy. \end{aligned} \quad (\text{GC.12})$$

Substitution in (GC.11) proves (GC.8). On the compact set one uses $e^{-\delta(z)u} \leq e^{-\delta_K u}$ in (GC.11) and then (GC.12) at δ_K ; no monotonicity of the gamma factor is assumed. This proves locally uniform convergence on both components of the off-line domain. $\square$

11.3 Every confluent derivative and the original finite algebra

Write $F^{[k]} = F^{(k)}/k!$. The exact error (KR.16), with its vertical-line sign, gives for all $k \geq 0$

$$\mathcal{E}_n^{[k]}(z) = (-1)^n \sum_{j=0}^k (q_n^{-2})^{[k-j]}(z) (-1)^j \int \frac{|q_n(s)|^2}{(z-s)^{j+1}} d\nu_Z(t). \quad (\text{GC.13})$$

Every derivative under this integral is justified on a compact off-line neighborhood by its positive distance from the line and the finite measure $|q_n|^2 \nu_Z$. The product rule for Taylor coefficients has no binomial factors; those are precisely cancelled by the convention $F^{[k]} = F^{(k)}/k!$. For completeness its polynomial factor is explicitly

$$(q_n^{-2})^{[r]}(z) = \frac{(-1)^r}{q_n(z)^2} \sum_{r_1+\dots+r_n=r} \prod_{j=1}^n \frac{r_j+1}{(z-\lambda_{n,j})^{r_j}}. \quad (\text{GC.14})$$

This follows by multiplying the Taylor series of the n original factors $(z - \lambda_{n,j})^{-2}$, retaining every cross term.

Differentiating (GC.9)–(GC.10) gives the independent estimate

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{2E_a}{k!} \int_0^\infty u^k e^{-\delta(z)u} \tanh(bu)^{2n} du. \quad (\text{GC.15})$$

The integrable majorant $u^k e^{-\delta_K u}$ on a smaller off-line compact neighborhood justifies this differentiation. If $0 < \sigma < \delta$, the exponential series gives $u^k e^{-(\delta-\sigma)u} \leq k!/(\delta-\sigma)^k$. Equations (GC.11)–(GC.12) prove

$$|\mathcal{E}_n^{[k]}(z)| \leq \frac{4aE_a}{\pi(\delta-\sigma)^k} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi}, \quad 0 < \sigma < \delta \leq \delta(z). \quad (\text{GC.16})$$

Taking $\delta = \delta_K$ makes this a simultaneous compact-uniform bound on every fixed finite list of derivatives.

Here is the complete algebra transport, including the part of the original packet outside this resolvent domain. Write

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{d_\rho}, \quad H_\rho(s) = \frac{h(s)}{(s - \rho)^{d_\rho}}, \quad U_\rho(s) = \sum_{k=0}^{d_\rho-1} (H_\rho^{-1})^{[k]}(\rho) (s - \rho)^k,$$

and define the actual polynomial remainder

$$e_\rho = j_h(H_\rho U_\rho) \in E. \quad (\text{GC.17})$$

Its remainder is one modulo $(s - \rho)^{d_\rho}$ and zero modulo every other factor of h . The distinct factors are coprime, and a polynomial divisible by each is divisible by their product: successive use of the Bezout identity proves this last statement. It follows directly that

$$e_\rho e_\eta = 0 \ (\rho \neq \eta), \quad e_\rho^2 = e_\rho, \quad \sum_{\rho \in Z} e_\rho = 1. \quad (\text{GC.18})$$

Restriction of a polynomial to its truncated Taylor coefficients therefore gives the exact algebra isomorphism

$$E \longrightarrow \bigoplus_{\rho \in Z} \mathbb{C}[\epsilon_\rho]/(\epsilon_\rho^{d_\rho}), \quad j_h P \longmapsto \left(\sum_{k < d_\rho} P^{[k]}(\rho) \epsilon_\rho^k \right)_\rho. \quad (\text{GC.19})$$

Its inverse sends the coordinate ϵ_ρ^k to $e_\rho(s-\rho)^k$; composition is the identity by the just proved congruences. Multiplication is preserved because the Taylor coefficient of a product is the exact convolution of the two coefficient lists, truncated only by the displayed ideal.

Let $Z_{\text{off}} = \{\rho \in Z : \text{Re } \rho \neq 1/2\}$ and $Z_{\text{line}} = Z \setminus Z_{\text{off}}$. Define

$$e_{\text{off}} = \sum_{\rho \in Z_{\text{off}}} e_\rho, \quad e_{\text{line}} = 1 - e_{\text{off}}, \quad E_{\text{off}} = e_{\text{off}}E, \quad E_{\text{line}} = e_{\text{line}}E. \quad (\text{GC.20})$$

Then $E = E_{\text{off}} \oplus E_{\text{line}}$ as algebras with their respective idempotent units. The projection $u \mapsto e_{\text{off}}u$ has exactly kernel E_{line} and commutes with the original multiplication operator A . Under (GC.19), its restriction at every retained centre is $A_\rho = \rho I + N_\rho$, where multiplication by ϵ_ρ gives $N_\rho^{d_\rho} = 0$ and retains all lower powers. This constructs the precise map from the full packet to the off-line resolvent domain, together with its full kernel and the operator on that kernel. An empty off-line set gives the zero summand, with the full original packet in the kernel.

For a function f holomorphic near Z_{off} , define

$$f(A_{\text{off}}) = \sum_{\rho \in Z_{\text{off}}} \sum_{k=0}^{d_\rho-1} f^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \quad (\text{GC.21})$$

Here $e_\rho(A)$ means multiplication by the explicitly constructed remainder (GC.17). Equation (GC.19) proves that (GC.21) is a unital algebra homomorphism on the off-line summand. No diagonalizability of A has entered this construction.

In particular, finite geometric expansion in each nilpotent block proves

$$\begin{aligned} m(A_{\text{off}}) &= \int (A_{\text{off}} - sI)^{-1} d\nu_Z(t), \\ m_n(A_{\text{off}}) &= \sum_j \omega_{n,j} (A_{\text{off}} - \lambda_{n,j}I)^{-1}, \\ \mathcal{E}_n(A_{\text{off}}) &= \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \mathcal{E}_n^{[k]}(\rho)(A - \rho I)^k e_\rho(A)|_{E_{\text{off}}}. \end{aligned} \quad (\text{GC.22})$$

For the first equality the expansion is

$$(A - sI)^{-1} e_\rho(A) = \sum_{k < d_\rho} \frac{(-1)^k (A - \rho I)^k e_\rho(A)}{(\rho - s)^{k+1}}.$$

Each coefficient is integrable since $\delta(\rho) > 0$, and its integral is exactly $m^{[k]}(\rho)$. This proves the integral identity, rather than assigning a formal meaning to it.

For a concrete bound keep any fixed positive Hermitian form G_* on the original coefficient space and restrict it to E_{off} . When that summand is nonzero, put

$$\delta_* = \min_{\rho \in Z_{\text{off}}} \delta(\rho) > 0, \quad C_{\rho,k} = \left\| (A - \rho I)^k e_\rho(A)|_{E_{\text{off}}} \right\|_{G_*}.$$

The triangle inequality and (GC.16) give, for every $0 < \sigma < \delta_*$,

$$\left\| m(A_{\text{off}}) - m_n(A_{\text{off}}) \right\|_{G_*} \leq \frac{4aE_a}{\pi} \Gamma\left(\frac{2a\sigma}{\pi}\right) (4n)^{-2a\sigma/\pi} \sum_{\rho \in Z_{\text{off}}} \sum_{k < d_\rho} \frac{C_{\rho,k}}{(\delta_* - \sigma)^k}. \quad (\text{GC.23})$$

Thus the full initial resolvent converges in the original finite coefficient space, with every nilpotent contribution controlled. Applying the displayed operators to the retained vector $e_{\text{off}}c$ gives the corresponding vector statement with the additional factor $\|e_{\text{off}}c\|_{G_*}$.

The polynomial kernel (KR.9) continues to act on all of E , including E_{line} and collisions with quadrature nodes. The off-line estimates above concern the initial observation m_n . Its exact map to the terminal observation F_n is the block and tail comparison (KR.14)–(KR.20), with the factors q_n, κ_n, a_n and the tail denominator still present. The estimate (GC.23) may be inserted into those identities; it does not remove their arithmetic factors or the critical-line summand of (GC.20).

12 The exact terminal step and its retained boundary energy

All measures, packet coordinates and maps in this section are the original arithmetic ones. Thus $s = 1/2 + it$, $d\nu_Z = |g(s)/h(s)|^2 dt/(2\pi)$, $E = \mathbb{C}[s]/(h)$, $A = M_s$, and $c = j_h 1$. Every selected zero occurs with its complete multiplicity in h . Retain the monic q_n , their actual norms κ_n , and

$$a_n = \frac{\kappa_n}{\kappa_{n-1}} > 0, \quad q_{n+1}(s) = (s - b_n)q_n(s) + a_n q_{n-1}(s), \quad b_n + \bar{b}_n = 1. \quad (\text{KT.1})$$

No recurrence coefficient is replaced by a limiting value.

12.1 A step written entirely in the original polynomial packet

For $n \geq d = \deg h$, put

$$v_j = q_j(A)c, \quad K = K_{n-1} = \sum_{j=0}^{n-1} \frac{v_j v_j^*}{\kappa_j} > 0, \quad M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}. \quad (\text{KT.2})$$

Use the positive form $\langle x, y \rangle_n = x^* M_n y$, conjugate-linear in its first argument, and define

$$u_n = v_n, \quad w_n = v_{n-1}, \quad \alpha_n = \langle w_n, w_n \rangle_n, \quad \beta_n = \langle u_n, u_n \rangle_n, \quad \gamma_n = \langle u_n, w_n \rangle_n. \quad (\text{KT.3})$$

The arithmetic relative allowance at source level $m = n - d - 1$ will be denoted by $\epsilon^{(n)}$. The fixed-section extension at $m = -1$ has already been specified; the original corrected sequence has $n \geq d + 1$. The exact kernel boundary identity gives

$$\epsilon^{(n)} = |\operatorname{Re} \gamma_n| + \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}, \quad \operatorname{Re} \gamma_n = \operatorname{Re} \operatorname{tr} A - \frac{d}{2} =: g_Z. \quad (\text{KT.4})$$

Indeed the kernel weight form is $(u_n w_n^* + w_n u_n^*)/\kappa_{n-1}$. Its possible nonzero generalized eigenvalues are obtained by reversing the two rectangular factors, giving $\operatorname{Re} \gamma_n \pm \sqrt{\alpha_n \beta_n - (\operatorname{Im} \gamma_n)^2}$. Taking the trace after multiplication by K^{-1} gives the second identity. Cauchy–Schwarz proves the nonnegativity of the radicand and justifies the maximum absolute value in (KT.4).

Theorem 12.1 (Universal scalar and metric step). *Write $a = a_n$, and retain the exact next vector*

$$t_n = (A - b_n I)u_n + a_n w_n = v_{n+1}. \quad (\text{KT.5})$$

Then

$$M_{n+1} = \frac{1}{a} \left(M_n - \frac{M_n u_n u_n^* M_n}{a + \beta_n} \right), \quad (\text{KT.6})$$

$$\begin{aligned} \alpha_{n+1} &= \frac{\beta_n}{a + \beta_n}, \\ \gamma_{n+1} &= \frac{\langle t_n, u_n \rangle_n}{a + \beta_n}, \\ \beta_{n+1} &= \frac{1}{a} \left(\langle t_n, t_n \rangle_n - \frac{|\langle u_n, t_n \rangle_n|^2}{a + \beta_n} \right). \end{aligned} \quad (\text{KT.7})$$

Every denominator is strictly positive. These identities hold also when any selected critical centre equals a quadrature node, and when $q_n(A)$ is singular.

The exact determinant changes of the kernel and original theta metric are

$$\boxed{\frac{\det K_n}{\det K_{n-1}} = 1 + \frac{\beta_n}{a_n}, \quad \frac{\det G_{m+1}}{\det G_m} = \frac{a_n}{a_n + \beta_n} = 1 - \alpha_{n+1}.} \quad (\text{KT.8})$$

Proof. The unchanged kernel update is $K_n = K_{n-1} + u_n u_n^* / \kappa_n$. Its inverse is verified by multiplication:

$$K_n^{-1} = K_{n-1}^{-1} - \frac{K_{n-1}^{-1} u_n u_n^* K_{n-1}^{-1}}{\kappa_n + u_n^* K_{n-1}^{-1} u_n}.$$

Insert $\kappa_n = a \kappa_{n-1}$ and the definition of M_n ; this gives (KT.6). At the next step the ordered vectors in (KT.3) are $u_{n+1} = t_n$, $w_{n+1} = u_n$. Pair each ordered pair with (KT.6). The three resulting expressions are precisely (KT.7), including their conjugations. The denominator is positive because $a > 0$, $\beta_n \geq 0$. The determinant of the rank-one update is $1 + u_n^* K_{n-1}^{-1} u_n / \kappa_n = 1 + \beta_n / a$; this follows directly by expanding the determinant multilinearly in its columns. The original metric is $G_m = U_\varepsilon^* K_{n-1}^{-1} U_\varepsilon$, with the same fixed invertible local unit at adjacent levels. Its determinant ratio is therefore the inverse kernel ratio. Equation (KT.7) gives the last equality in (KT.8). All calculations use polynomial residues at A ; no inverse of $q_n(A)$ has been used. $\square$

12.2 The complete boundary-energy increment

Suppose first that $\beta_n > 0$. The exact orthogonal projection for $\langle \cdot, \cdot \rangle_n$ is

$$\begin{aligned} P_n x &= x - u_n \frac{\langle u_n, x \rangle_n}{\beta_n}, & p_n &= P_n w_n, & z_n &= P_n A u_n, \\ \Delta_n &= \alpha_n \beta_n - |\gamma_n|^2 = \beta_n \|p_n\|_n^2, & \Omega_n &= \beta_n \|z_n\|_n^2. \end{aligned} \quad (\text{KT.9})$$

The projection records a specified map inside the original packet; $A u_n$, its removed coefficient, and its remaining vector z_n are all retained in (KT.9).

Theorem 12.2 (Exact transverse energy law). *At every such step,*

$$\boxed{\Delta_{n+1} = \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}}. \quad (\text{KT.10})$$

Consequently

$$\boxed{\Delta_{n+1} - \Delta_n = \frac{\beta_n}{a_n(a_n + \beta_n)} (\|z_n\|_n^2 + 2a_n \operatorname{Re} \langle z_n, p_n \rangle_n - a_n \beta_n \|p_n\|_n^2)}. \quad (\text{KT.11})$$

The exact energy law gives the following two-sided bounds by the sharp triangle inequalities:

$$\frac{|a_n \sqrt{\Delta_n} - \sqrt{\Omega_n}|}{\sqrt{a_n(a_n + \beta_n)}} \leq \sqrt{\Delta_{n+1}} \leq \frac{a_n \sqrt{\Delta_n} + \sqrt{\Omega_n}}{\sqrt{a_n(a_n + \beta_n)}}. \quad (\text{KT.12})$$

The actual relative allowance is recovered exactly by

$$\boxed{\epsilon^{(n)} = |g_Z| + \sqrt{g_Z^2 + \Delta_n}}. \quad (\text{KT.13})$$

For a reflection-stable packet with its multiplicities, $g_Z = 0$, so the energy law calculates the square of the actual allowance without a trace term.

Proof. Let $T = \langle t_n, t_n \rangle_n$ and $v = \langle u_n, t_n \rangle_n$. Substitution of (KT.7) gives

$$\alpha_{n+1}\beta_{n+1} - |\gamma_{n+1}|^2 = \frac{\beta_n T - |v|^2}{a_n(a_n + \beta_n)} = \frac{\beta_n \|P_n t_n\|_n^2}{a_n(a_n + \beta_n)}.$$

Since $P_n u_n = 0$, the exact recurrence (KT.5) gives $P_n t_n = P_n A u_n + a_n P_n w_n = z_n + a_n p_n$. This proves (KT.10). Expand the squared norm and subtract $\Delta_n = \beta_n \|p_n\|_n^2$ to prove (KT.11). The triangle inequality and its reverse, applied to these two specific vectors, prove (KT.12). Finally (KT.4) gives (KT.13). Its real trace term is unchanged with n ; equivalently the same trace computation can be performed at each adjacent positive kernel. Reflection permutes all roots and multiplicities, so $2 \operatorname{Re} \operatorname{tr} A = d$ in that case. $\square$

If $\beta_n = 0$, positivity of M_n forces $u_n = v_n = 0$. Then $\gamma_n = 0$, the current kernel weight is zero, $K_n = K_{n-1}$, and (KT.5)–(KT.7) give the entire exceptional step:

$$t_n = a_n w_n, \quad \alpha_{n+1} = \gamma_{n+1} = 0, \quad \beta_{n+1} = a_n \alpha_n, \quad \Delta_n = \Delta_{n+1} = 0, \quad \epsilon^{(n)} = \epsilon^{(n+1)} = 0. \quad (\text{KT.14})$$

Moreover $v_n = 0$ is exactly the divisibility $h \mid q_n$. Since every root of q_n is simple and lies on the Mellin line, this particular event forces the selected packet to consist of simple line roots. It does not remove the resulting packet or the next vectors in (KT.14).

12.3 The exact rational terminal step and its polynomial continuation

Where $q_n(A)$ and $q_{n+1}(A)$ are invertible, use the actual terminal maps and Gram from Section 10:

$$T_n = F_n(A), \quad r_n = T_n c, \quad Q_n = q_n(A), \quad K_{n-1} = \kappa_{n-1}^{-1} Q_n S_n Q_n^*.$$

The map Q_n is the exact isometry

$$Q_n : (E, S_n^{-1}) \longrightarrow (E, M_n), \quad Q_n c = u_n, \quad Q_n r_n = w_n. \quad (\text{KT.15})$$

Indeed $Q_n^* M_n Q_n = S_n^{-1}$ follows by inverting the displayed Gram congruence, and the vector identity uses $F_n = q_{n-1}/q_n$. This proves the precise relation between the rational and polynomial forms of (KT.3)–(KT.13).

The terminal recurrence gives

$$T_{n+1} = (A - b_n I + a_n T_n)^{-1}, \quad \boxed{S_{n+1} = T_{n+1} (c c^* + a_n S_n) T_{n+1}^*}. \quad (\text{KT.16})$$

For a direct full-jet proof, the scalar kernel of a terminal observation is

$$H_{F_n}(z, w) = \frac{F_n(z) + \overline{F_n(w)}}{z + \overline{w} - 1}.$$

The quotient denotes its removable polynomial/rational continuation where numerator and denominator both vanish. It is identically the positive resolvent Gram $\sum_j \sigma_{n,j} / [(z - \lambda_{n,j})(\overline{w} - \overline{\lambda}_{n,j})]$. Substitution of $F_{n+1} = (z - b_n + a_n F_n)^{-1}$ gives

$$H_{F_{n+1}}(z, w) = F_{n+1}(z) \overline{F_{n+1}(w)} [1 + a_n H_{F_n}(z, w)]. \quad (\text{KT.17})$$

This identity holds first away from its removable sets and hence through them by continuation. Taking the complete residue on both variables proves (KT.16). It also follows from the original kernel update and $Q_{n+1}^{-1} Q_n = T_{n+1}$. Thus a quadrature collision is handled by the polynomial formulas (KT.2)–(KT.14), which retain all coordinates; (KT.15) states exactly the isometry on the open set where the rational presentation is defined.

In this presentation the determinant still retains the entire polynomial factor:

$$\frac{\det S_{n+1}}{\det S_n} = \frac{a_n^{d-1}(a_n + \beta_n)}{|\det(A - b_n I + a_n T_n)|^2}, \quad \det(A - b_n I + a_n T_n) = \prod_{\rho \in Z} \left(\frac{q_{n+1}(\rho)}{q_n(\rho)} \right)^{m_\rho}. \quad (\text{KT.18})$$

The determinant of $a_n S_n + cc^*$ gives the first formula. The second follows by evaluating the polynomial ratio on each full triangular Taylor block $\rho I + N_\rho$. The determinant has the stated multiplicity even when the block is not semisimple.

12.4 The extra jets required to recover the metric from the tail

There is an exact map from the scalar tail identity to the matrix energy, with a specified number of Taylor coefficients. For (KT.19)–(KT.22), take centres at which q_n is nonzero, so F_n is holomorphic there. Every off-line centre meets this condition. At a critical quadrature intersection the universal polynomial formulas (KT.2)–(KT.14) retain the complete packet. The terminal function satisfies $\overline{F_n(1 - \bar{z})} = -F_n(z)$, by its positive spectral sum on the line. Thus

$$H_{F_n}(z, w) = \frac{F_n(z) - F_n(1 - \bar{w})}{z - (1 - \bar{w})}. \quad (\text{KT.19})$$

For centres ρ, σ , and Taylor orders $0 \leq k < m_\rho, 0 \leq l < m_\sigma$, let $S_n^{\text{jet}}(\rho, k; \sigma, l)$ denote the coefficient of $(z - \rho)^k (\bar{w} - \bar{\sigma})^l$ in this kernel. In particular, at a reflection pair,

$$\boxed{S_n^{\text{jet}}(\rho, k; \sigma, l) = (-1)^l F_n^{[k+l+1]}(\rho) \quad \text{if } \rho = 1 - \bar{\sigma}.} \quad (\text{KT.20})$$

Here $F^{[j]} = F^{(j)}/j!$. To prove the formula, put $z = \rho + x, 1 - \bar{w} = \rho - y$ in (KT.19), and expand

$$\frac{x^r - (-y)^r}{x + y} = \sum_{j=0}^{r-1} x^{r-1-j} (-y)^j.$$

The coefficient of $x^k y^l$ is supplied exactly by $r = k + l + 1, j = l$, proving (KT.20) without dividing by zero.

For $D_{\rho\sigma} = \rho + \bar{\sigma} - 1 \neq 0$, the corresponding complete coefficient is

$$\begin{aligned} S_n^{\text{jet}}(\rho, k; \sigma, l) &= \sum_{i=0}^k F_n^{[i]}(\rho) \frac{(-1)^{k-i+l} \binom{k-i+l}{k-i}}{D_{\rho\sigma}^{k-i+l+1}} \\ &\quad + \sum_{j=0}^l \overline{F_n^{[j]}(\sigma)} \frac{(-1)^{k+l-j} \binom{k+l-j}{k}}{D_{\rho\sigma}^{k+l-j+1}}. \end{aligned} \quad (\text{KT.21})$$

This is the full product of the numerator Taylor series and the geometric expansion of $(D_{\rho\sigma} + x + y)^{-1}$. Both constant terms in the numerator are retained. If $V_Z : E \rightarrow \bigoplus_\rho \mathbb{C}^{m_\rho}$ is the fixed complete Taylor-coordinate map, then

$$S_n^{\text{jet}} = V_Z S_n V_Z^*, \quad S_n = V_Z^{-1} S_n^{\text{jet}} (V_Z^{-1})^*. \quad (\text{KT.22})$$

This follows by applying V_Z to each resolvent column in the positive Gram representation of S_n .

To specify the domain of the analytic extraction, let $h_{\text{off}} = \prod_{\text{Re } \rho \neq 1/2} (s - \rho)^{m_\rho}$. The full original algebra projects onto $E_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}})$; the kernel is the retained critical-line CRT summand, with its original action of A . Above the off-line summand use the doubled algebra $\widehat{E}_{\text{off}} = \mathbb{C}[s]/(h_{\text{off}}^2)$. Its exact reduction sequence of $\mathbb{C}[s]$ -modules is

$$0 \longrightarrow E_{\text{off}} \xrightarrow{[p] \mapsto [h_{\text{off}} p]} \widehat{E}_{\text{off}} \xrightarrow{[p] \mapsto [p]} E_{\text{off}} \longrightarrow 0. \quad (\text{KT.23})$$

The first map is well-defined and injective because $h_{\text{off}}^2 \mid h_{\text{off}}p$ holds exactly when $h_{\text{off}} \mid p$; its image is precisely the kernel of the reduction. All maps commute with multiplication by s . The reduction is also a unital algebra map. The image of the injection is the square-zero ideal $(h_{\text{off}})/(h_{\text{off}}^2)$: the product of the images of $[p]$ and $[q]$ is $[h_{\text{off}}^2 pq] = 0$. This records its complete inherited multiplication as well as its module isomorphism with E_{off} . The nilpotent lengths in the middle term are exactly $2m_\rho$, and reduction retains the lower m_ρ coefficients. At reflected actual zeros, the full multiplicities are equal, by the functional equation and conjugation of g . Thus (KT.20) uses at most order $2m_\rho - 1$, exactly a coefficient of this doubled algebra. Formula (KT.21) uses only the stated lower coefficients. Equations (KT.20)–(KT.23) prove the precise passage from terminal Taylor data to the full metric.

Write $\mathcal{E}_n = m - m_n$ for the initial observation error, and retain the genuine tail ζ_n from (KR.14). For every off-line point its real part has the sign of $\text{Re } z - 1/2$, so $\zeta_n \neq 0$. The exact identity (KR.16) also gives

$$\mathcal{E}_n(z) = \frac{(-1)^n \kappa_n}{q_n(z)^2} H_n^{\text{obs}}(z), \quad H_n^{\text{obs}}(z) = \frac{1}{\kappa_n} \int \frac{|q_n(s)|^2}{z - s} d\nu_Z.$$

The real part of H_n^{obs} has that same strict sign: its defining measure is positive of total mass one. Hence $\mathcal{E}_n(z) \neq 0$ off the line as well. Solving the actual tail equation, before taking any residues, therefore gives the exact extraction

$$F_n(z) = \frac{1}{a_n} \left(\frac{(-1)^n \kappa_n}{q_n(z)^2 \mathcal{E}_n(z)} - \frac{1}{\zeta_n(z)} \right). \quad (\text{KT.24})$$

It holds in the complete doubled algebra at the off-line centres. Every reciprocal there is explicit: if $f(\rho + x) = \sum_{j=0}^{2m_\rho-1} f_j x^j$ modulo x^{2m_ρ} , with $f_0 \neq 0$, the inverse coefficients are

$$b_0 = f_0^{-1}, \quad b_k = -f_0^{-1} \sum_{j=1}^k f_j b_{k-j} \quad (1 \leq k < 2m_\rho). \quad (\text{KT.25})$$

Multiplication of the two finite series proves all coefficients. The already proved bounds (GC.13)–(GC.16) apply to every one of the error coefficients required here. Equations (KT.24)–(KT.25), then (KT.20)–(KT.22), give a complete finite extraction of the terminal metric on E_{off} from the initial error and genuine tail data. In the retained monomial quotient coordinates, set $A_{\text{off}} = M_s$, $\pi_{\text{off}}([p]_h) = [p]_{h_{\text{off}}}$, and $c_{\text{off}} = [1] = \pi_{\text{off}}c$. Explicitly, retain the same ν_Z , q_n , nodes and terminal weights, and put

$$S_n^{\text{off}} = \sum_j \sigma_{n,j} (A_{\text{off}} - \lambda_{n,j} I)^{-1} c_{\text{off}} c_{\text{off}}^* (A_{\text{off}}^* - \bar{\lambda}_{n,j} I)^{-1}. \quad (\text{KT.26})$$

For the original quotient map $\pi_{\text{off}} : E \rightarrow E_{\text{off}}$, the resolvent intertwining identity gives $S_n^{\text{off}} = \pi_{\text{off}} S_n \pi_{\text{off}}^*$ whenever the full rational Gram is defined. The right side of (KT.26) remains defined at every off-line centre even when the full packet has a critical quadrature collision. No replacement of the measure by $|g/h_{\text{off}}|^2 dt / (2\pi)$ occurs in this projection. The full polynomial quadrature formula, applied to these projected residue columns, proves even at such a collision that

$$\pi_{\text{off}} K_{n-1} \pi_{\text{off}}^* = \kappa_{n-1}^{-1} q_n (A_{\text{off}}) S_n^{\text{off}} q_n (A_{\text{off}})^*. \quad (\text{KT.27})$$

These extraction formulas retain the inverse error, the inverse tail, q_n , κ_n , and a_n , including their cancellations. The initial error bound alone is not used to replace any of those terms. The polynomial metric step (KT.6) continues to include the critical-line summand, where the off-line resolvent domain ends.

12.5 Actual quartet packets close the energy in two volume ratios

Assume here that the selected actual packet is closed under both $\rho \mapsto \bar{\rho}$ and $\rho \mapsto 1 - \bar{\rho}$, with the complete multiplicities. These are exact invariances of the specified divisor; no value of $\text{Re } \rho$ has been imposed.

Theorem 12.3 (Two consecutive arithmetic volume contractions). *Let G_{-1} be the fixed-section metric already identified with $U_\varepsilon^* K_{d-1}^{-1} U_\varepsilon$, and define the actual positive determinant ratios*

$$r_m = \frac{\det G_m}{\det G_{m-1}} \quad (m \geq 0).$$

Then $0 < r_m \leq 1$, and for every $m \geq 0$, with $n = d + m + 1$,

$$\epsilon_m^2 = \frac{\kappa_{d+m+1}}{\kappa_{d+m}} \frac{(1 - r_m)(1 - r_{m+1})}{r_{m+1}}. \quad (\text{KT.28})$$

This is the original relative allowance and the original theta Gram sequence, including their local unit and all multiplicities.

Proof. Conjugation invariance of the divisor makes every coefficient of the monic h real. The entire function g obeys $g(\bar{s}) = \overline{g(s)}$, so the unchanged density $|g(1/2 + it)/h(1/2 + it)|^2$ is even in t . For a monic orthogonal polynomial q , form $q^c(s) = \overline{q(\bar{s})}$. Substitution $t \mapsto -t$ and conjugation of each orthogonality equation show that q^c is monic with the same degree and the same lower-degree orthogonality. Positivity of the polynomial Gram proves uniqueness, so $q^c = q$. Thus every q_n has real coefficients in the original variable s . The coefficient of q_n in sq_n is therefore real as well; (KT.1) proves its exact value $b_n = 1/2$. This is a consequence of the given packet symmetries, with every other recurrence coefficient retained.

The original companion matrix A , vector c , residue columns v_j , and Gram matrices K_{n-1}, M_n are real. Therefore the scalar γ_n in (KT.3) is real. The second specified symmetry gives $g_Z = 0$ in (KT.4), so $\gamma_n = 0$ exactly. Hence $\epsilon_m^2 = \alpha_n \beta_n$. At the preceding and following steps (KT.7)–(KT.8) give

$$\alpha_n = 1 - r_m, \quad \alpha_{n+1} = 1 - r_{m+1}, \quad \beta_n = \frac{a_n \alpha_{n+1}}{1 - \alpha_{n+1}}.$$

All these denominators are positive: $G_m > 0$ and $a_n/(a_n + \beta_n) > 0$. Substitution proves (KT.28), including the exceptional zero-column steps. The inequality $0 < r_m \leq 1$ follows directly from the same determinant formula. At $m = 0$ the preceding metric is precisely the stated G_{-1} , so no earlier undefined source level is used. $\square$

12.6 What the recurrence proves about boundary energy

The unchanged arithmetic source has supplied the explicit energy increment (KT.11). Its three terms are the squared transverse A -image, its real pairing with the retained terminal vector, and the specified negative term $-a_n \beta_n \|p_n\|_n^2$. Thus the source recurrence produces an exactly computable boundary energy, including its sign-determining cross term. No sign for their sum has been assumed. Every term is a polynomial residue, an actual moment Gram inverse, or a retained recurrence coefficient. The rational route to those same quantities is (KT.15)–(KT.25). The original theta representative and its boundary are recovered from the preceding kernel-minimum formula and its explicit division by h . Consequently all updates are maps between the original source representatives, with the same quotient class and supported zero. For each support label the displayed linear maps lift by $(\lambda, x) \mapsto (\lambda, Tx)$; external absence remains τ , and a zero vector in that fibre remains its supported zero throughout.

12.7 An exact positive-moment counterexample to consecutive energy monotonicity

This section concerns the moment-to-packet construction for the explicitly specified positive measure below. It proves that positivity, the vertical orthogonal-polynomial recurrence, reflection stability of the packet, and the terminal rank-two identity together allow an increase of the relative weight

energy at consecutive degrees. The measure is specified by its nine atoms; no assertion identifies it with the arithmetic measure $d\nu_Z(t) = |g(\frac{1}{2} + it)/h(\frac{1}{2} + it)|^2 dt/(2\pi)$. The exact correspondence with that construction will be the same polynomial residue map, multiplication operator, kernel sum and Hermitian defect, with the measure retained as part of the data throughout.

12.7.1 The measure and every moment needed below

Keep the original coordinate $s = \frac{1}{2} + it$. Define a probability measure ν on this line by the following table. In a row containing two signs, the displayed weight is assigned to each of the two atoms.

t	weight at each atom	
0	115/288	
$\pm 1/4$	29/120	
$\pm 1/2$	13/240	
$\pm 3/4$	11/2520	
± 1	1/6720	(TEC.1)

All nine weights are strictly positive, and

$$\frac{115}{288} + \frac{29}{60} + \frac{13}{120} + \frac{11}{1260} + \frac{1}{3360} = 1. \quad (\text{TEC.2})$$

Write $\mu_k = \int t^k d\nu$ and $v = 1/16$. The paired atoms give $\mu_k = 0$ for every odd integer $k \geq 1$, including $k = 1, 3, 5, 7, 9$. For positive integers r , the even-moment formula, with all original atom weights retained, is

$$\mu_{2r} = 4^{-2r} \left(\frac{29}{60} 1^{2r} + \frac{13}{120} 2^{2r} + \frac{11}{1260} 3^{2r} + \frac{1}{3360} 4^{2r} \right). \quad (\text{TEC.3})$$

Here the factor 4^{-2r} comes from the explicitly displayed atom locations in (TEC.1). The four contributions and their sums are

r	$\frac{29}{60} 1^{2r}$	$\frac{13}{120} 2^{2r}$	$\frac{11}{1260} 3^{2r}$	$\frac{1}{3360} 4^{2r}$	sum	
1	29/60	13/30	11/140	1/210	1	
2	29/60	26/15	99/140	8/105	3	
3	29/60	104/15	891/140	128/105	15	
4	29/60	416/15	8019/140	2048/105	105	
5	29/60	1664/15	72171/140	32768/105	939	(TEC.4)

Consequently

$$(\mu_0, \mu_2, \mu_4, \mu_6, \mu_8, \mu_{10}) = (1, v, 3v^2, 15v^3, 105v^4, 939v^5). \quad (\text{TEC.5})$$

Equations (TEC.2)–(TEC.5), together with cancellation of each odd moment between the two atoms, verify all moments through degree ten.

12.7.2 The actual monic orthogonal polynomials of this measure

Use the inner product

$$\langle P, Q \rangle_\nu = \int \overline{P(\frac{1}{2} + it)} Q(\frac{1}{2} + it) d\nu(t), \quad x = s - \frac{1}{2}. \quad (\text{TEC.6})$$

The following table retains the monic polynomials in the original s -coordinate and gives their actual squared norms.

k	$q_k(s)$	$\kappa_k = \langle q_k, q_k \rangle_\nu$
0	1	1
1	x	v
2	$x^2 + v$	$2v^2$
3	$x^3 + 3vx$	$6v^3$
4	$x^4 + 6vx^2 + 3v^2$	$24v^4$
5	$x^5 + 10vx^3 + 15v^2x$	$114v^5$

(TEC.7)

Proof. For an explicit verification, define real polynomials

$$\begin{aligned}
 P_0(t) &= 1, & P_1(t) &= t, & P_2(t) &= t^2 - v, \\
 P_3(t) &= t^3 - 3vt, & P_4(t) &= t^4 - 6vt^2 + 3v^2, \\
 P_5(t) &= t^5 - 10vt^3 + 15v^2t.
 \end{aligned}
 \tag{TEC.8}$$

Direct substitution gives $q_k(\frac{1}{2} + it) = i^k P_k(t)$. Thus

$$\langle q_j, q_k \rangle_\nu = (-i)^j i^k \int P_j(t) P_k(t) d\nu(t). \tag{TEC.9}$$

If j, k have different parity, the integrand on the right is odd and its integral is zero. The remaining six off-diagonal integrals, using (TEC.5), are precisely

$$\begin{aligned}
 \int P_0 P_2 d\nu &= v(1 - 1) = 0, \\
 \int P_0 P_4 d\nu &= v^2(3 - 6 + 3) = 0, \\
 \int P_2 P_4 d\nu &= v^3(15 - 21 + 9 - 3) = 0, \\
 \int P_1 P_3 d\nu &= v^2(3 - 3) = 0, \\
 \int P_1 P_5 d\nu &= v^3(15 - 30 + 15) = 0, \\
 \int P_3 P_5 d\nu &= v^4(105 - 195 + 135 - 45) = 0.
 \end{aligned}
 \tag{TEC.10}$$

For the diagonal integrals, direct polynomial multiplication gives

$$\begin{aligned}
 \int P_0^2 d\nu &= 1, & \int P_1^2 d\nu &= v, \\
 \int P_2^2 d\nu &= v^2(3 - 2 + 1) = 2v^2, \\
 \int P_3^2 d\nu &= v^3(15 - 18 + 9) = 6v^3, \\
 \int P_4^2 d\nu &= v^4(105 - 180 + 126 - 36 + 9) = 24v^4, \\
 \int P_5^2 d\nu &= v^5(939 - 2100 + 1950 - 900 + 225) = 114v^5.
 \end{aligned}
 \tag{TEC.11}$$

In (TEC.9), the diagonal phase is $(-i)^k i^k = 1$. These equations therefore prove every entry of the Gram matrix asserted in (TEC.7). Each polynomial in that table is monic. A nonzero polynomial of degree at most five has at most five distinct roots, so it cannot vanish at all nine atoms in (TEC.1). Positivity of the weights makes the inner product positive definite on polynomials of degree at most five.

If two monic polynomials of the same degree were both orthogonal to every lower degree polynomial, their difference would have lower degree and zero squared norm. Positive definiteness forces that difference to vanish. Thus (TEC.7) consists of the actual monic orthogonal polynomials and norms of the stated measure, including the fifth norm. $\square$

Polynomial multiplication in (TEC.7) also proves

$$sq_k = q_{k+1} + \frac{1}{2}q_k - kvq_{k-1} \quad (0 \leq k \leq 4), \quad (\text{TEC.12})$$

where the last term is omitted for $k = 0$. For $1 \leq k \leq 4$, the displayed coefficient is exactly $a_k = \kappa_k/\kappa_{k-1} = kv = k/16 > 0$. Thus (TEC.12) has the required vertical sign and $b_k + \overline{b_k} = 1$, with $b_k = 1/2$. The fifth norm in (TEC.7) is $\kappa_5 = 114/16^5 = 57/524288$; its value was derived from the actual tenth moment $939/16^5$.

12.7.3 The retained packet and its exact evaluation isomorphism

Set

$$h(s) = (s - \frac{3}{4})(s - \frac{1}{4}) = s^2 - s + \frac{3}{16}, \quad E = \mathbb{C}[s]/(h), \quad j_h : \mathbb{C}[s] \longrightarrow E. \quad (\text{TEC.13})$$

Both packet centres lie in $0 < \operatorname{Re} s < 1$, and the reflection $\rho \mapsto 1 - \overline{\rho}$ exchanges them. They are distinct, and each has its complete multiplicity one. In the retained monomial basis $(1, s)$ of E , the unit and multiplication operator are

$$c = j_h 1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad A = M_s = \begin{pmatrix} 0 & -3/16 \\ 1 & 1 \end{pmatrix}. \quad (\text{TEC.14})$$

Indeed $s^2 = s - 3/16$ in this quotient, which proves both columns of the multiplication matrix.

The complete evaluation map is

$$\begin{aligned} T : E &\longrightarrow \mathbb{C} \oplus \mathbb{C}, & [p] &\longmapsto (p(3/4), p(1/4)), \\ T &= \begin{pmatrix} 1 & 3/4 \\ 1 & 1/4 \end{pmatrix}, & T^{-1} &= \begin{pmatrix} -1/2 & 3/2 \\ 2 & -2 \end{pmatrix}. \end{aligned} \quad (\text{TEC.15})$$

Both evaluations vanish on the ideal (h) , so the map is well defined. Evaluations preserve addition, multiplication and the unit. For any pair (u_+, u_-) , the polynomial

$$2u_+(s - \frac{1}{4}) + 2u_-(\frac{3}{4} - s) \quad (\text{TEC.16})$$

evaluates to that pair. It has the coefficients given by T^{-1} , and multiplication of the two matrices in (TEC.15) gives the identity. This proves the typed algebra isomorphism, including its inverse. Evaluation of sp also proves

$$TA = \tilde{A}T, \quad \tilde{A} = \operatorname{diag}(3/4, 1/4), \quad Tc = \begin{pmatrix} 1 \\ 1 \end{pmatrix}. \quad (\text{TEC.17})$$

In the original quotient E one has $x^2 = v$. Reducing the explicit polynomials of (TEC.7), with no deletion of their supported residue classes, gives

k	$v_k := j_h q_k$ in the monomial basis $(1, s)$	
0	$(1, 0)^T$	
1	$(-1/2, 1)^T$	
2	$(1/8, 0)^T$	
3	$(-1/8, 1/4)^T$	
4	$(5/128, 0)^T$	
5	$(-13/256, 13/128)^T$	(TEC.18)

For example $q_3 = 4vx = x/4$, $q_4 = 10v^2 = 5/128$, and $q_5 = 26v^2x = (13/128)x$ in E . These calculations establish every entry of (TEC.18) directly from the original ideal relation.

12.7.4 The original kernels, terminal defects and exact energies

For $1 \leq N \leq 4$, define

$$\begin{aligned} K_N &= \sum_{k=0}^N \frac{v_k v_k^*}{\kappa_k}, \\ L_N &= AK_N + K_N A^* - K_N, \\ \epsilon_N &= \left\| K_N^{-1/2} L_N K_N^{-1/2} \right\|. \end{aligned} \tag{TEC.19}$$

All adjoints in (TEC.19) are conjugate transposes in the explicitly retained monomial coordinates. The columns v_0, v_1 are linearly independent. If a nonzero column u had $u^* K_N u = 0$, the positive sum in (TEC.19) would force both $v_0^* u = 0$ and $v_1^* u = 0$, a contradiction. Thus each K_N is positive definite, so its positive inverse square root and the displayed Hermitian operator norm are well defined.

Using (TEC.7) and (TEC.18), the successive kernels are

$$\begin{aligned} K_1 &= \begin{pmatrix} 5 & -8 \\ -8 & 16 \end{pmatrix}, & K_2 &= \begin{pmatrix} 7 & -8 \\ -8 & 16 \end{pmatrix}, \\ K_3 &= \begin{pmatrix} 53/3 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}, & K_4 &= \begin{pmatrix} 131/6 & -88/3 \\ -88/3 & 176/3 \end{pmatrix}. \end{aligned} \tag{TEC.20}$$

In particular the last step is the exact positive rank-one update

$$K_4 - K_3 = \frac{v_4 v_4^*}{\kappa_4} = \frac{25}{6} cc^*. \tag{TEC.21}$$

Multiplication by the matrix in (TEC.14) gives

$$L_3 = \begin{pmatrix} -20/3 & 20/3 \\ 20/3 & 0 \end{pmatrix}, \quad L_4 = \begin{pmatrix} -65/6 & 65/6 \\ 65/6 & 0 \end{pmatrix}. \tag{TEC.22}$$

These defects satisfy the original terminal identity, with its exact coefficients and signs:

$$L_N = \frac{v_{N+1} v_N^* + v_N v_{N+1}^*}{\kappa_N} \quad (N = 3, 4). \tag{TEC.23}$$

For $N = 3$, the numerator calculated from (TEC.18) is

$$\begin{pmatrix} -5/512 & 5/512 \\ 5/512 & 0 \end{pmatrix}, \quad \kappa_3 = 3/2048.$$

For $N = 4$, it is

$$\begin{pmatrix} -65/16384 & 65/16384 \\ 65/16384 & 0 \end{pmatrix}, \quad \kappa_4 = 3/8192.$$

Division gives exactly the two matrices in (TEC.22), proving (TEC.23) without an appeal to a numerical rank calculation.

Theorem 12.4 (Strict increase at two consecutive permitted levels). *For the positive measure (TEC.1) and the complete reflection-stable packet (TEC.13), the energies in (TEC.19) satisfy*

$$\epsilon_3^2 = \frac{25}{99}, \quad \epsilon_4^2 = \frac{4225}{15136}, \quad \epsilon_4^2 - \epsilon_3^2 = \frac{39875}{1498464} > 0. \tag{TEC.24}$$

In the indexing $N = d + m$, with $d = \dim_{\mathbb{C}} E = 2$, these are the consecutive levels $m = 1$ and $m = 2$, both with $m \geq 0$.

Proof. Transport the two forms by the proved isomorphism (TEC.15):

$$\begin{aligned}\tilde{K}_3 &= TK_3T^* = \begin{pmatrix} 20/3 & -2/3 \\ -2/3 & 20/3 \end{pmatrix}, \\ \tilde{K}_4 &= TK_4T^* = \begin{pmatrix} 65/6 & 7/2 \\ 7/2 & 65/6 \end{pmatrix}.\end{aligned}\tag{TEC.25}$$

By (TEC.17), the transported defect is

$$\tilde{L}_N = TL_NT^* = \tilde{A}\tilde{K}_N + \tilde{K}_N\tilde{A}^* - \tilde{K}_N.\tag{TEC.26}$$

For either matrix $\tilde{K} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ in (TEC.25), one has

$$\tilde{L} = \text{diag}(a/2, -a/2), \quad \det(\tilde{L} - \lambda\tilde{K}) = \lambda^2(a^2 - b^2) - a^2/4.\tag{TEC.27}$$

To check that this calculation gives the norm defined in the original coordinates, set $H = K^{-1/2}LK^{-1/2}$. Then

$$\det(H - \lambda I) = \det(K)^{-1} \det(L - \lambda K), \quad \det(T(L - \lambda K)T^*) = |\det T|^2 \det(L - \lambda K).\tag{TEC.28}$$

The two determinant factors are nonzero and independent of λ . Consequently the eigenvalues of the Hermitian matrix H equal the roots of (TEC.27), with their multiplicities. The roots are the opposite real numbers $\pm a/(2\sqrt{a^2 - b^2})$. A Hermitian matrix has operator norm equal to the greatest absolute value of its eigenvalues, so

$$\epsilon^2 = \frac{a^2}{4(a^2 - b^2)}.\tag{TEC.29}$$

For $N = 3$, substitution of $a = 20/3, b = -2/3$ gives $\epsilon_3^2 = (400/9)/(4 \cdot 396/9) = 25/99$. For $N = 4$, substitution of $a = 65/6, b = 7/2$ gives $\epsilon_4^2 = (4225/36)/(4 \cdot 3784/36) = 4225/15136$. Their difference, retaining the two denominators, is

$$\frac{4225 \cdot 99 - 25 \cdot 15136}{15136 \cdot 99} = \frac{418275 - 378400}{1498464} = \frac{39875}{1498464} > 0.\tag{TEC.30}$$

Both energies are nonnegative, and their squared values have strictly positive difference, so $\epsilon_4 > \epsilon_3$. The original coordinate characteristic polynomials provide the same calculation:

$$\begin{aligned}\det(L_3 - \lambda K_3) &= 176\lambda^2 - 400/9, \\ \det(L_4 - \lambda K_4) &= (3784/9)\lambda^2 - 4225/36.\end{aligned}\tag{TEC.31}$$

Finally $N = 3 = 2 + 1$ and $N = 4 = 2 + 2$, proving the stated degree range without using the lower degree $N = 1$. $\square$

12.7.5 The exact scope of the obstruction

For this measure the construction is the composite of the polynomial source with the original residue quotient

$$\mathcal{P}_N \hookrightarrow \mathbb{C}[s] \xrightarrow{j_h} E, \quad E \xrightarrow{T} \mathbb{C} \oplus \mathbb{C},\tag{TEC.32}$$

and multiplication is intertwined exactly by (TEC.17). Its source inner product is (TEC.6); its monic source Gram matrix is $\text{diag}(\kappa_0, \dots, \kappa_N)$; its kernel is the full sum (TEC.19). Thus it has precisely the vertical recurrence, positive kernel update, complete packet quotient and terminal defect maps appearing in the arithmetic moment construction. Equations (TEC.24) and (TEC.30) prove that these data alone do not imply nonincreasing consecutive energy. An argument for that property for ν_Z must therefore establish and use additional identities or estimates of its stated arithmetic density.

The present example establishes the indicated obstruction to a universal recurrence-based argument; it asserts no consecutive increase for ν_Z .

The companion exact checker verifies 88 rational and complex-rational identities in both ordinary Python and optimized Python. It checks the atom weights and relevant moments, source Gram matrix, packet isomorphism, residue columns, kernel updates, terminal identities and generalized characteristic polynomials. Those finite checks supplement the complete computations above.

13 The exact metric departure and the cost of complete jets

Retain the original packet $E = \mathbb{C}[s]/h$, its polynomial coordinates and $A = M_s$. Write its eigenvalues with their algebraic multiplicities as $\rho_1, \dots, \rho_d$, and put $\delta_j = \operatorname{Re} \rho_j - 1/2$. No eigenvalue or repeated factor is discarded. For a positive Hermitian matrix G in these coordinates define

$$\begin{aligned} W_G &= A^*G + GA - G, \quad S_G = G^{-1/2}W_GG^{-1/2}, \quad \epsilon_G = \|S_G\|_{\text{op}}, \\ \mathfrak{d}_G(A) &= \operatorname{Tr}(G^{-1}A^*GA) - \sum_{j=1}^d |\rho_j|^2. \end{aligned} \tag{MD1}$$

The positive square root is used only for the explicit isometry $x \mapsto G^{1/2}x$ from the specified metric space (E, G) to the Euclidean coordinate space. The original A, G , their entries and their maps remain those in (MD1).

Theorem 13.1 (Exact departure identity). *The number $\mathfrak{d}_G(A)$ is real and nonnegative, and*

$$\operatorname{Tr} S_G = 2 \sum_j \delta_j, \quad \operatorname{Tr}(S_G^2) = 4 \sum_j \delta_j^2 + 2\mathfrak{d}_G(A). \tag{MD2}$$

*It vanishes precisely when A is normal for the specified metric, that is, when $A(G^{-1}A^*G) = (G^{-1}A^*G)A$. For the original orthogonal representatives R_m , their matrices G_m and rank-two forms W_m , one has exactly*

$$\epsilon_m = \left| \sum_j \delta_j \right| + \sqrt{2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j \right)^2}. \tag{MD3}$$

For a reflection-stable packet this becomes

$$\boxed{\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A)}. \tag{MD4}$$

Proof. Let $C = G^{1/2}AG^{-1/2}$. Then $S_G = C + C^* - I$ and $\operatorname{Tr}(G^{-1}A^*GA) = \operatorname{Tr}(C^*C)$. For completeness, choose a unit eigenvector of C , extend it to an orthonormal basis, and repeat on the lower-right compression. Induction gives a unitary matrix U with upper-triangular $T = U^*CU$ whose diagonal lists the same eigenvalues with their algebraic multiplicities. Unitary invariance of the trace gives

$$\mathfrak{d}_G(A) = \sum_{i < j} |T_{ij}|^2 \geq 0. \tag{MD5}$$

If it is zero, T is diagonal and C is normal. Conversely, an upper-triangular normal matrix is diagonal: its first row norm equals its first column norm, eliminating the strict first-row entries, and induction eliminates every remaining such entry. Thus normality of C is equivalent to the stated equality for A , through the specified isometry.

Expanding $(C + C^* - I)^2$ under the trace gives

$$\mathrm{Tr}(S_G^2) = 2 \mathrm{Tr}(C^* C) + 2 \mathrm{Re} \mathrm{Tr}(C^2) - 4 \mathrm{Re} \mathrm{Tr} C + d.$$

The traces of C and C^2 are respectively $\sum_j \rho_j$ and $\sum_j \rho_j^2$, even in the presence of nilpotents. Since $2|\rho|^2 + 2 \mathrm{Re} \rho^2 - 4 \mathrm{Re} \rho + 1 = 4(\mathrm{Re} \rho - 1/2)^2$, this proves (MD2). The trace formula follows directly from $S_G = C + C^* - I$.

For G_m , Section 6 proved $\mathrm{rank} S_{G_m} \leq 2$. List its two possibly nonzero real eigenvalues a, b , appending zero if necessary (also when $d = 1$). Equations (MD2) give

$$a + b = 2 \sum_j \delta_j, \quad (a - b)^2/4 = 2 \sum_j \delta_j^2 + \mathfrak{d}_{G_m}(A) - \left(\sum_j \delta_j\right)^2.$$

Thus the radicand is nonnegative, and $\max(|a|, |b|) = |(a + b)/2| + |a - b|/2$, proving (MD3). Reflection-stability pairs δ with $-\delta$, with the same multiplicities, proving (MD4). $\square$

The actual arithmetic kernel of Section 9 also gives this quantity without changing the local unit. Indeed $G_m = U_\varepsilon^* K_N^{-1} U_\varepsilon$, $N = d + m$, and $U_\varepsilon A = A U_\varepsilon$. Cyclicity of trace gives

$$\mathfrak{d}_{G_m}(A) = \mathrm{Tr}(K_N A^* K_N^{-1} A) - \sum_j |\rho_j|^2. \quad (\text{MD6})$$

To verify the cancellation, substitute $G_m^{-1} = U_\varepsilon^{-1} K_N (U_\varepsilon^*)^{-1}$ in (MD1), commute A with U_ε and A^* with U_ε^* , and cancel the outer similar factors inside the trace. This calculation removes no source representative: their exact recovery remains (TB.7)–(TB.9).

13.1 A quantitative constraint on every retained nilpotent chain

For a selected root ρ of full order r , let

$$I_\rho : \mathbb{C}[z]/z^r \longrightarrow E$$

be the original Chinese-remainder inclusion: the ρ -component is the given Taylor polynomial in $z = s - \rho$, and every other component is zero. Its Gram matrix is $G_\rho = I_\rho^* G I_\rho$, in the basis $1, z, \dots, z^{r-1}$. Put $N_\rho z^k = z^{k+1}$, with $z^r = 0$. The exact intertwining relation is $A I_\rho = I_\rho(\rho I + N_\rho)$.

Theorem 13.2 (Full-jet metric inequalities). *For every positive original metric G , write $\delta = \mathrm{Re} \rho - 1/2$. Then $\epsilon_G \geq 2|\delta|$, and for all $0 \leq k < k + \ell < r$,*

$$\frac{(G_\rho)_{k+\ell, k+\ell}}{(G_\rho)_{k, k}} \leq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^\ell. \quad (\text{MD7})$$

In particular, for $r \geq 2$,

$$\kappa_2(G_\rho) \geq \left[\frac{r}{2} (\epsilon_G^2 - 4\delta^2) \right]^{-(r-1)}. \quad (\text{MD8})$$

The bracket is strictly positive when $r \geq 2$. Here κ_2 is the ratio of the largest and smallest eigenvalues in the specified Taylor coordinates.

Proof. The definition of ϵ_G is the two-sided form inequality $-\epsilon_G G \leq W_G \leq \epsilon_G G$. On an eigenvector it gives $\epsilon_G \geq 2|\delta|$. Pull this same inequality back by I_ρ . Let $L = G_\rho^{1/2} N_\rho G_\rho^{-1/2}$ and $T = 2\delta I + L + L^*$. The pulled-back inequality gives $\|T\| \leq \epsilon_G$. Nilpotence retains $\mathrm{Tr} L = \mathrm{Tr}(L^2) = 0$, whence

$$\mathrm{Tr} T = 2r\delta, \quad \|L\|_F^2 = \frac{1}{2} \mathrm{Tr}(L + L^*)^2 = \frac{1}{2} \mathrm{Tr} T^2 - 2r\delta^2 \leq \frac{r}{2} (\epsilon_G^2 - 4\delta^2). \quad (\text{MD9})$$

The first equality for the Frobenius norm follows by expanding the square and using the two zero traces. The final inequality uses the r real eigenvalues of T , each bounded in modulus by ϵ_G . For the

original metric norm, $\|N_\rho\|_{G_\rho} = \|L\| \leq \|L\|_F$. Apply its ℓ -th power to the specified basis vector z^k . Since $N_\rho^\ell z^k = z^{k+\ell}$, the resulting squared norm inequality is exactly (MD7).

When $r \geq 2$, $N_\rho \neq 0$, so (MD9) makes the bracket strictly positive. Put $k = 0, \ell = r - 1$. The Rayleigh inequalities give $\kappa_2(G_\rho) \geq (G_\rho)_{0,0}/(G_\rho)_{r-1,r-1}$; combine with (MD7) to obtain (MD8). $\square$

All entries in (MD7) for the original sequence are explicitly $I_\rho^* U_\varepsilon^* K_N^{-1} U_\varepsilon I_\rho$. Thus this theorem supplies constraints on the actual moment calculation, with its original Taylor factors. At a critical root of order $r \geq 2$, a sequence with relative control approaching zero necessarily has the stated diverging Taylor-metric condition number. The surviving jets have not been erased; the inequalities calculate their relative metric scales. This is compatible with the complete graph and weighted-jet maps in Section 42, which retain those same coordinates.

14 Coherent packets, joint tensor layers, and their exact spectral cost

This section integrates the complete arithmetic interpolation construction of the 12 September coherent-interpolation delivery, whose source revision is 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, and proves its comparison with the preceding kernel and departure calculations. All zeros in a packet retain their complete multiplicities. A reflected packet is required only in statements explicitly using its reflection. The original spaces $V, \mathcal{B}$, map Θ , generator $D = -x\partial_x$, and Hilbert observation $L^2(dx)$ remain fixed.

14.1 The arithmetic unit and the coherent one-variable maps

For $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, write

$$d = \deg h, \quad E_h = \mathbb{C}[s]/h, \quad A_h = M_s, \quad v_h = g/h, \quad v_h = j_h v_h, \quad \varepsilon_h = v_h^{-1}, \quad g = 2\xi. \quad (\text{CT.1})$$

Complete multiplicities make v_h invertible. Let $F_h \in \mathcal{B}$ be the original inverse satisfying $h(D)F_h = f_0$ and $\mathcal{M}F_h = g/h$, and put $\mathcal{T}_h P = P(D)F_h$. The integral construction of F_h , its estimates at both ends, and injectivity of the Mellin transform are proved in Section 6. They give

$$\mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (\text{CT.2})$$

Let $C_N : \mathcal{P}_N \rightarrow E_h$ be the full remainder map in the original monomial coordinates, let $M_{h,N}$ be the Gram matrix of $\mathcal{P}_N$ for $d\nu_h = |g(1/2 + it)/h(1/2 + it)|^2 dt/(2\pi)$, and use U_a for multiplication by a on E_h . The two kernel matrices occurring in the constructions have the exact types

$$K_N = C_N M_{h,N}^{-1} C_N^*, \quad J_{h,N} = U_{v_h} C_N, \quad \mathcal{C}_{h,N} = U_{v_h} K_N U_{v_h}^*. \quad (\text{CT.3})$$

Consequently their actual representative and metric coincide:

$$\begin{aligned} M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1} &= M_{h,N}^{-1} C_N^* K_N^{-1} U_{\varepsilon_h}, \\ G_{h,n} &= \mathcal{C}_{h,n+d}^{-1} = U_{\varepsilon_h}^* K_{n+d}^{-1} U_{\varepsilon_h}. \end{aligned} \quad (\text{CT.4})$$

Indeed invert the congruence in (CT.3) and cancel the adjacent inverse factors. The first line is exactly the polynomial numerator in (TB.7)–(TB.9). Its difference from $\text{rem}_h(\varepsilon_h u)$ is divisible by h and has quotient of degree at most n . Applying (CT.2) gives the same original theta correction. Thus the arithmetic unit remains in every coefficient map and every source representative.

For completeness, the minimum assertion follows directly without an existence assumption. For any positive Gram matrix M and surjective J , set $G = (JM^{-1}J^*)^{-1}$ and $L = M^{-1}J^*G$. Then $JL = I$, $L^*ML = G$, and for $Jz = 0$, $z^*ML = z^*J^*G = 0$. Every feasible vector is $Lu + z$, so

$$(Lu + z)^* M (Lu + z) = u^* Gu + z^* Mz. \quad (\text{CT.5})$$

Positivity makes this the unique minimum. In (CT.3), positivity holds because a nonzero polynomial cannot vanish almost everywhere on the original integration line; surjectivity follows from complete polynomial remainders. The kernel is exactly $h\mathcal{P}_n$ when $N = n + d$. Hence $R_{h,n} = (I - P_{L_n})s_h$, where $s_h = \mathcal{T}_h \text{rem}_h(\varepsilon_h \cdot)$ and $L_n = \Theta \text{span}(\phi_*, \dots, D^n \phi_*)$.

If $h \mid H$ are full packets, put $m = H/h$. The factors h, m are coprime. The original CRT injection $\iota : E_h \rightarrow E_H$ inserts zero jets at the additional centres, and satisfies $A_H \iota = \iota A_h$. Multiplication $P \mapsto mP$ is an isometry of the declared polynomial images, since $(g/H)mP = (g/h)P$; it satisfies

$$\mathcal{T}_H(mP) = \mathcal{T}_h P, \quad J_H \mathcal{T}_H(mP) = \iota J_h \mathcal{T}_h P. \quad (\text{CT.6})$$

At the added centres the second identity follows from the full zeros of g/h ; at the old centres it is the identity of entire functions. Both numerators for $s_H \iota = \mathcal{T}_H(m \text{rem}_h(\varepsilon_h \cdot))$ have degree below $\deg H$ and the same complete jets, proving equality. Since L_n is independent of the packet,

$$R_{H,n} \iota = R_{h,n}, \quad \iota^* G_{H,n} \iota = G_{h,n}, \quad \iota^* W_{H,n} \iota = W_{h,n}. \quad (\text{CT.7})$$

These are congruences, with no interchange of inversion and compression. Their split maps at a retained support λ are $(\lambda, u) \mapsto (\lambda, \iota u)$; all added zero jets are represented zeros at that support.

14.2 The original reflection and the exact diagonal coefficient

The original arithmetic density $|g(1/2 + it)|^2$ is even in t , since the real Taylor coefficients of g give $g(1/2 - it) = \overline{g(1/2 + it)}$. If Q_j is the monic orthogonal polynomial for that density in the variable s , the polynomial $(-1)^j Q_j(1 - s)$ is monic and orthogonal to all lower degrees, by the change $t \mapsto -t$. Uniqueness therefore gives

$$Q_j(1 - s) = (-1)^j Q_j(s), \quad \int t |Q_j(1/2 + it)|^2 |g(1/2 + it)|^2 \frac{dt}{2\pi} = 0. \quad (\text{CT.8})$$

The diagonal recurrence coefficient is thus exactly $\langle Q_j, sQ_j \rangle / \|Q_j\|^2 = 1/2$. For lower degrees the identity $D^* + D = I$ eliminates every coefficient except the adjacent one and gives its sign and actual norm ratio. More explicitly, sQ_j is orthogonal to Q_i for $i \leq j - 2$, because its inner product is the inner product of Q_j with $(1 - s)Q_i$. For $i = j - 1$ the leading term gives coefficient $-\|Q_j\|^2 / \|Q_{j-1}\|^2$. This proves, with the original vectors and norms,

$$Du_j = u_{j+1} + \frac{1}{2}u_j - \frac{\mathfrak{h}_j}{\mathfrak{h}_{j-1}}u_{j-1}, \quad u_j = Q_j(D)f_0, \quad \mathfrak{h}_j = \|u_j\|^2. \quad (\text{CT.9})$$

The final term is omitted at $j = 0$. The even density in (CT.8) is the additional reason the imaginary part of the diagonal coefficient vanishes; the adjoint identity by itself fixes only its real part.

For a packet stable under $\rho \mapsto 1 - \bar{\rho}$, including the same complete orders, the original maps are

$$\mathfrak{j}_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}, \quad \mathfrak{j}_{\mathcal{B}} F(x) = x^{-1} \overline{F(1/\bar{x})}.$$

The second is antiunitary for dx , and $D\mathfrak{j}_{\mathcal{B}} = \mathfrak{j}_{\mathcal{B}}(I - D)$. The identities $h^\dagger = (-1)^d h$ and $g^\dagger = g$ imply $(g/h)^\dagger = (-1)^d g/h$; they give $\mathfrak{j}_{\mathcal{B}} s_h = s_h \mathfrak{j}_h$ by the same complete jets and degree bounds as above. Equation (CT.9), or invariance of the polynomial space under $s \mapsto 1 - s$, shows $\mathfrak{j}_{\mathcal{B}} L_n = L_n$. The unique minimum (CT.5) then commutes with these maps. If $\mathfrak{j}_h u = C_h \bar{u}$,

$$C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}. \quad (\text{CT.10})$$

Substitution into a quadratic form proves that an upper bound and its reflected lower bound use this same $G_{h,n}$.

14.3 The joint total-degree source and its relation layer

Fix $k \geq 1$, $M \geq k(d-1)$, and set

$$E^{(k)} = E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{P}_M^{(k)} = \mathbb{C}[s_1, \dots, s_k]_{\leq M}, \quad \mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}.$$

Here $\leq M$ means total degree. The full jet map is reduction by each $h(s_j)$, followed by multiplication by $v_h^{\otimes k}$, and its kernel is

$$\mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)). \quad (\text{CT.11})$$

Successive monic division in the declared order $s_1, \dots, s_k$ gives $P = \sum_j h(s_j) P_j$ in this kernel, with $\deg P_j \leq M - d$. Division lowers total degree at every replacement, so this bound holds at every division step. In the original complex $[V \rightarrow \mathcal{B}]$, placed in degrees 0, 1, the primitive for the j -th term uses $(-1)^{j-1} \phi_*$ in the j -th factor and F_h in the others, with the polynomial $P_j(D_1, \dots, D_k)$ applied to the whole tensor. The preceding $j-1$ factors have degree one, so the tensor differential supplies the further factor $(-1)^{j-1}$. Its boundary is exactly $+\mathcal{T}_h^{(k)}(h(s_j) P_j)$.

Write $\mathcal{B}_M = \mathcal{T}_h^{(k)} \mathcal{I}_{h,k,M}$ inside $\mathcal{H}_k = L^2(d^k x)$. The arithmetic moment matrix in the retained monomials is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s^{\alpha_j}} s^{\beta_j} d\nu_h, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (\text{CT.12})$$

Product integration proves the first formula. A polynomial vanishing almost everywhere for this product measure is zero, by successive one-variable uniqueness, proving positivity. The complete remainder basis has total degree at most $k(d-1)$, proving surjectivity. Thus (CT.5) proves

$$R_M = (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad J^{(k)} R_M = I, \quad q^{(k)} R_M = \sigma_h^{\otimes k}. \quad (\text{CT.13})$$

The kernel dimension is $\binom{M+k}{k} - d^k$. Since $\mathcal{T}_h^{(k)}$ is injective by its entire Mellin multiplier, the orthogonal relation layer $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ has exactly

$$\dim \mathcal{E}_M = \binom{M+k}{k-1}. \quad (\text{CT.14})$$

Define all of the following maps on the original $E^{(k)}$:

$$B_M = D^{(k)} R_M - R_M A^{(k)}, \quad Y_M = P_{\mathcal{E}_M} R_M, \quad C_M = (I - P_{\mathcal{B}_M}) B_M. \quad (\text{CT.15})$$

The complete jet of B_M is zero and its numerator has degree at most $M+1$, so $B_M(E^{(k)}) \subseteq \mathcal{B}_{M+1}$. Both Y_M, C_M therefore have target $\mathcal{E}_M$. Integration by parts on these rapidly decreasing functions gives $(D^{(k)})^* + D^{(k)} = kI$. Substitution into $R_M^*(D^{(k)} + (D^{(k)})^*) R_M = kG_M$, followed by orthogonality to $\mathcal{B}_M$, proves

$$\begin{aligned} W_M &= (A^{(k)})^* G_M + G_M A^{(k)} - kG_M = -(Y_M^* C_M + C_M^* Y_M), \\ \Delta_M &= G_M - G_{M+1} = Y_M^* Y_M \succeq 0. \end{aligned} \quad (\text{CT.16})$$

For the second identity use the orthogonal decomposition of $\mathcal{B}_{M+1}$ and $R_{M+1} = R_M - Y_M$. The actual quotient map $[z] \mapsto (I - P_{\mathcal{B}_M})z$ identifies $\mathcal{B}_{M+1}/\mathcal{B}_M$ with $\mathcal{E}_M$, with inverse the inclusion followed by quotient. A transported boundary is zero at its next support and remains represented there; this map sends no supported input to external τ .

For a reflection-stable packet, put $C_{h,k} = C_h^{\otimes k}$. The original product reflection acts on a numerator by $(-1)^{kd} \overline{P(1 - \overline{s_1}, \dots, 1 - \overline{s_k})}$. This retains total degree and preserves the ideal in (CT.11), with

its displayed factor. Its action on $\mathcal{H}_k$ is antiunitary. The unique minimum (CT.5) therefore commutes with reflection and gives

$$C_{h,k}^* G_M C_{h,k} = \overline{G_M}, \quad A^{(k)} C_{h,k} = C_{h,k} (kI - \overline{A^{(k)}}), \quad C_{h,k}^* W_M C_{h,k} = -\overline{W_M}. \quad (\text{CT.16a})$$

If $W_M u = \lambda G_M u$, conjugation followed by $(C_{h,k}^*)^{-1}$ in the last identity gives $W_M C_{h,k} \bar{u} = -\lambda G_M C_{h,k} \bar{u}$. All generalized eigenvalues are real because $G_M > 0$. This invertible antilinear map pairs their full multiplicities. The source-layer maps themselves retain the reflection types. If $j_E = j_h^{\otimes k}$ and j_E is the restriction of the original function reflection to $\mathcal{E}_M$, then

$$j_E Y_M = Y_M j_E, \quad j_E C_M = -C_M j_E. \quad (\text{CT.16b})$$

Both boundary projections commute with reflection, proving the first identity. For the second use $D^{(k)} j_H = j_H (kI - D^{(k)})$, $A^{(k)} j_E = j_E (kI - A^{(k)})$, and $j_H R_M = R_M j_E$ in (CT.15). Expanding its two terms gives $j_H B_M = -B_M j_E$; the projection then proves (CT.16b).

14.4 Nested packets on the joint source: the additional projection

Let $h \mid H$, $q = \deg(H/h)$, $N = M + kq$, and $\iota_k = \iota_{hH}^{\otimes k}$. Multiplication of a numerator by $\prod_{j=1}^k (H/h)(s_j)$ gives a degree-preserving-with-shift map $\mathcal{P}_M^{(k)} \rightarrow \mathcal{P}_N^{(k)}$, with

$$\mathcal{T}_H^{(k)} (\prod_j (H/h)(s_j) P) = \mathcal{T}_h^{(k)} P, \quad J_H^{(k)} \mathcal{T}_h^{(k)} P = \iota_k J_h^{(k)} \mathcal{T}_h^{(k)} P. \quad (\text{CT.17})$$

The second equality follows on pure monomials from (CT.6) and then by linearity. In particular it maps the smaller relation space into the larger one in the same $\mathcal{H}_k$. Let

$$\mathcal{F} = \mathcal{B}_{H,k,N} \cap \mathcal{B}_{h,k,M}^\perp, \quad Z = P_{\mathcal{F}} R_{h,k,M}.$$

Since $s_H^{\otimes k} \iota_k = s_h^{\otimes k}$, the exact comparison is

$$\begin{aligned} R_{H,k,N} \iota_k &= R_{h,k,M} - Z, \\ \iota_k^* G_{H,k,N} \iota_k &= G_{h,k,M} - Z^* Z, \\ \iota_k^* W_{H,k,N} \iota_k &= W_{h,k,M} - ((A_h^{(k)})^* Z^* Z + Z^* Z A_h^{(k)} - k Z^* Z). \end{aligned} \quad (\text{CT.18})$$

Indeed nested orthogonal projections commute, and their difference is $P_{\mathcal{F}}$; apply them to the common section to prove the first line. Pythagoras proves the second; intertwining of $A^{(k)}$ and direct expansion prove the third. Also

$$\dim \mathcal{F} = \binom{N+k}{k} - (\deg H)^k - \binom{M+k}{k} + d^k. \quad (\text{CT.19})$$

This is the difference of the exact kernel dimensions. For $k = 1$, $M = n + d$, both boundary spaces are the original L_n , so $\mathcal{F} = 0$ and (CT.18) recovers (CT.7). For joint tensors (CT.18) retains every additional allowed boundary and calculates its effect on the control form.

14.5 Positive layer rank and the complete tensor spectral departure

List $\rho_1, \dots, \rho_d$ with all algebraic multiplicities, put $\delta_i = \text{Re } \rho_i - 1/2$, $\sigma_1 = \sum_i \delta_i$, $\sigma_2 = \sum_i \delta_i^2$, and define

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2}, \quad \epsilon_M = \|S_M\|_{\text{op}}, \quad r_M = \text{rank } \Delta_M, \\ \mathfrak{d}_M^{(k)} &= \text{Tr}(G_M^{-1} (A^{(k)})^* G_M A^{(k)}) - \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \rho_{i_j} \right|^2, \\ \mathcal{D}_k &= \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k \delta_{i_j} \right|. \end{aligned} \quad (\text{CT.20})$$

All sums are ordered and retain repeated entries. The isometry $x \mapsto G_M^{1/2}x$ in this calculation leaves the original jet coordinates, source functions, metric and layer maps unchanged. For a map into $\mathcal{E}_M$, the Frobenius symbol below denotes its Hilbert–Schmidt norm for the layer’s inherited $L^2(d^kx)$ inner product.

Theorem 14.1 (Exact tensor departure and relative layer cost). *One has $\mathfrak{d}_M^{(k)} \geq 0$, and exactly*

$$\begin{aligned} \mathrm{Tr} S_M &= 2kd^{k-1}\sigma_1, \\ \mathrm{Tr}(S_M^2) &= 4\{kd^{k-1}\sigma_2 + k(k-1)d^{k-2}\sigma_1^2\} + 2\mathfrak{d}_M^{(k)}. \end{aligned} \quad (\text{CT.21})$$

The positive and negative inertias of W_M each obey

$$n_+(W_M), n_-(W_M) \leq \min(\mathrm{rank} Y_M, \mathrm{rank} C_M), \quad \mathrm{rank} Y_M = r_M \leq \binom{M+k}{k-1}. \quad (\text{CT.22})$$

The complete ordered spectral cost satisfies

$$\boxed{\mathcal{D}_k \leq \frac{1}{2}\|S_M\|_1 \leq \|Y_M G_M^{-1/2}\|_F \|C_M G_M^{-1/2}\|_F}, \quad \mathcal{D}_k \leq r_M \epsilon_M. \quad (\text{CT.23})$$

In particular, the actual source matrices satisfy

$$\mathcal{D}_k^2 \leq \mathrm{Tr}(G_M^{-1} \Delta_M) \mathrm{Tr}(G_M^{-1} C_M^* C_M). \quad (\text{CT.24})$$

If the packet is reflection-stable, put $p_M = n_+(W_M) = n_-(W_M)$. Then

$$p_M \leq \min\{r_M, \mathrm{rank} C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k \leq p_M \epsilon_M. \quad (\text{CT.24a})$$

In this case (CT.21) becomes $\mathrm{Tr}(S_M^2) = 4kd^{k-1}\sigma_2 + 2\mathfrak{d}_M^{(k)}$. When $p_M > 0$, it implies

$$\epsilon_M^2 \geq \frac{2kd^{k-1}\sigma_2 + \mathfrak{d}_M^{(k)}}{p_M}. \quad (\text{CT.25})$$

Proof. Triangularize A_h over $\mathbb{C}$. The tensor product of this basis makes $A^{(k)}$ triangular, with diagonal entries $\sum_j \rho_{ij}$. Thus these are all its eigenvalues with algebraic multiplicity; its nilpotent entries have not been discarded. Now apply the unitary triangularization argument of (MD5) to $T = G_M^{1/2} A^{(k)} G_M^{-1/2}$. The sum of squared absolute values of its strictly upper triangular entries is exactly $\mathfrak{d}_M^{(k)}$, proving nonnegativity. Since $S_M = T + T^* - kI$, expansion of its squared trace gives

$$\mathrm{Tr}(S_M^2) = 4 \sum_{i_1, \dots, i_k} \left(\sum_j \delta_{i_j} \right)^2 + 2\mathfrak{d}_M^{(k)}.$$

The k diagonal terms in this ordered sum contribute $kd^{k-1}\sigma_2$; the $k(k-1)$ ordered cross terms contribute $k(k-1)d^{k-2}\sigma_1^2$. The linear trace is computed in the same way. This proves (CT.21), including $k = 1$.

Write $\hat{Y} = Y_M G_M^{-1/2}$ and $\hat{C} = C_M G_M^{-1/2}$. Then $S_M = -(\hat{Y}^* \hat{C} + \hat{C}^* \hat{Y})$. Its form is zero on $\ker \hat{Y}$. A positive definite subspace of dimension greater than $\mathrm{rank} \hat{Y}$ intersects this kernel nontrivially, a contradiction; the same argument holds for a negative definite subspace and with $\hat{C}$ in place of $\hat{Y}$. Sylvester inertia under the displayed invertible congruence proves the first part of (CT.22). The identity $\Delta_M = Y_M^* Y_M$ gives equality of their kernels, hence their ranks, proving the rest.

In the unitary triangular basis of T , the diagonal of S_M consists of $2 \sum_j \delta_{i_j}$. For any Hermitian matrix S , its spectral decomposition gives $|S_{ii}| \leq \sum_a |\lambda_a| |U_{ia}|^2$. Sum over i to obtain $\sum_i |S_{ii}| \leq \sum_a |\lambda_a| = \|S\|_1$. This proves the first inequality in (CT.23). For finite matrices B, C with a common row space, $\|B^* C\|_1 \leq \|B\|_F \|C\|_F$: choose a singular-value decomposition of $B^* C$, use its left and

right singular vectors to express its trace norm as $|\operatorname{Tr}(UB^*C)|$ for a unitary U , and apply Cauchy-Schwarz to the entries of BU^* and C . The triangle inequality now proves the second inequality of (CT.23). Finally (CT.22) bounds the number of nonzero eigenvalues of S_M by $2r_M$, so $\|S_M\|_1 \leq 2r_M\epsilon_M$. Squaring the middle bound and writing its two Frobenius norms as traces proves (CT.24).

For a reflected packet $\sigma_1 = 0$, with every multiplicity paired. The exact reflection (CT.16a) pairs the nonzero eigenvalues of S_M , giving equal positive and negative counts p_M . Equation (CT.22) proves the bounds on p_M , and there are exactly $2p_M$ nonzero eigenvalues of S_M , each of square at most ϵ_M^2 . Their trace norm is at most $2p_M\epsilon_M$, proving (CT.24a); their squared trace proves (CT.25). If $p_M = 0$, the pairing implies $W_M = 0$ directly, and (CT.21) states that all the nonnegative terms vanish; no division by p_M is made. $\square$

Every quantity in (CT.24) is attached to the same arithmetic moment matrix and its next original relation layer. In particular, $0 \preceq \Delta_M \prec G_M$, because $G_{M+1} > 0$, so all eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$ lie in $[0, 1)$, and its trace is strictly smaller than r_M when $r_M > 0$. Its actual value is retained in (CT.24). Equations (CT.21)–(CT.25) calculate necessary relative layer costs and the full nilpotent departure for the constructed tensor family. They assert no unproved upper bound on these arithmetic quantities.

There is an exact determinant refinement of the same source loss. For $r_M > 0$, put

$$\pi_M = \frac{\det G_{M+1}}{\det G_M}, \quad \tau_M = \operatorname{Tr}(G_M^{-1}\Delta_M), \quad \chi_M = \operatorname{Tr}(G_M^{-1}C_M^*C_M).$$

Then

$$1 - \pi_M \leq \tau_M \leq r_M(1 - \pi_M^{1/r_M}) < r_M, \quad \boxed{\mathcal{D}_k^2 \leq \tau_M \chi_M \leq r_M(1 - \pi_M^{1/r_M})\chi_M}. \quad (\text{CT.26})$$

To prove every factor, let $t_1, \dots, t_{r_M} \in (0, 1)$ be the positive eigenvalues of $G_M^{-1/2}\Delta_M G_M^{-1/2}$. The other eigenvalues are zero, so $\tau_M = \sum_i t_i$ and $\pi_M = \prod_i (1 - t_i)$. Induction on the number of factors proves $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$. For positive numbers with a fixed sum, their product is at most the product with all entries equal: a maximum exists on the closed nonnegative simplex, lies in its interior when the sum is positive, and any unequal pair can be replaced by its average to increase the product, since $((a+b)/2)^2 - ab = (a-b)^2/4$. Apply this fact to $1 - t_i$ to obtain $\pi_M \leq (1 - \tau_M/r_M)^{r_M}$, proving both bounds. The last two inequalities in (CT.26) now follow from (CT.24). For $r_M = 1$, $\tau_M = 1 - \pi_M$ exactly; for $r_M = 0$, $\Delta_M = 0$, $\pi_M = 1$, and $W_M = 0$, so this case requires no exponent $1/r_M$.

15 Density of the original joint boundaries and determinant dissipation

Retain the complete packet h , its degree $d \geq 1$, the arithmetic unit v_h , and every original map in Section 14. In particular

$$s_j = \frac{1}{2} + it_j, \quad v_h = g/h, \quad g = 2\xi, \quad d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad E^{(k)} = E_h^{\otimes k}, \quad \mathcal{H}_k = L^2(d^k x). \quad (\text{JD.1})$$

The one-variable cyclic-density argument in the coherent-interpolation source has a full joint-source extension. We prove it here, including the exact relation between the disappearing Hilbert representatives and their unchanged finite jets.

15.1 Polynomial density with the original weighted measures

Lemma 15.1. *Let μ be a finite positive Borel measure on $\mathbb{R}^k$ such that $\int e^{a \sum_j |t_j|} d\mu < \infty$ for some $a > 0$. The complex polynomials in $t_1, \dots, t_k$ are dense in $L^2(\mu)$.*

Proof. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial. Define the finite complex measure $d\eta = \bar{f} d\mu$. For $0 < b < a/2$, Cauchy–Schwarz gives

$$\int e^{b \sum_j |t_j|} d|\eta| \leq \|f\|_{L^2(\mu)} \left(\int e^{2b \sum_j |t_j|} d\mu \right)^{1/2} < \infty. \quad (\text{JD.2})$$

Thus $\Phi(z) = \int e^{i \sum_j z_j t_j} d\eta(t)$ is holomorphic in the connected tube $|\text{Im } z_j| < b$. On every smaller closed tube, each differentiated integrand is dominated by the integrable function in (JD.2), multiplied by a constant; the strict exponent margin absorbs each monomial. This justifies every derivative and the convergent Taylor series on small polydiscs. All derivatives at zero vanish by polynomial orthogonality. The Taylor series therefore vanishes near zero, and successive overlapping polydiscs along any path in the tube show $\Phi = 0$ throughout it. In particular the Fourier transform of η vanishes on $\mathbb{R}^k$.

For clarity, uniqueness here follows from an explicit approximate identity. Put

$$\gamma_\epsilon(x) = (4\pi\epsilon)^{-k/2} e^{-|x|^2/(4\epsilon)} = (2\pi)^{-k} \int e^{-\epsilon|u|^2} e^{iu \cdot x} du. \quad (\text{JD.3})$$

The second equality is the product Gaussian integral, obtained in one variable by differentiating its Fourier integral in x , integrating by parts in u , and evaluating its value at zero. The Gaussian integrability and finiteness of $|\eta|$ justify Fubini. Consequently $\eta * \gamma_\epsilon = 0$. For any continuous compactly supported φ , integration and Fubini give $\int (\varphi * \gamma_\epsilon) d\eta = 0$. These convolutions converge uniformly to φ : split the integral at a fixed small radius, use uniform continuity within it, and use the Gaussian tail and boundedness outside it. Hence $\int \varphi d\eta = 0$. Uniqueness of finite Borel measures from their continuous compactly supported integrals gives $\eta = 0$, and therefore $f = 0$ almost everywhere. The orthogonal complement of the polynomial closure is zero, proving the lemma. $\square$

The actual measure ν_h has every exponential moment of order less than $\pi/2$, with its full factor $dt/(2\pi)$, as proved in (GC.2). Multiplying it by any fixed polynomial in t preserves the same open range of exponential moments: choose a strictly larger exponent still below $\pi/2$ and absorb the polynomial into the difference. Product integration therefore verifies the lemma for both measures

$$\nu_h^{\otimes k}, \quad d\mu_{h,k}(t) = |h(s_1)|^2 \prod_{j=1}^k d\nu_h(t_j). \quad (\text{JD.4})$$

No factor h has been discarded. Its finitely many zeros on the first integration line have measure zero. The multiplication map

$$U_h : L^2(\mu_{h,k}) \longrightarrow L^2(\nu_h^{\otimes k}), \quad U_h P = h(s_1)P \quad (\text{JD.5})$$

is a unitary surjection: its norm identity follows directly from (JD.4), and its inverse is division by $h(s_1)$ away from that null set, with an arbitrary value on the null set. Polynomials in $s_j = 1/2 + it_j$ are precisely polynomials in t_j , since these coordinate changes are invertible. Thus (JD.5) and the lemma prove density of $h(s_1)\mathbb{C}[s_1, \dots, s_k]$ in $L^2(\nu_h^{\otimes k})$. This dense space is contained in the original joint ideal $(h(s_1), \dots, h(s_k))$.

15.2 Strong limits of the actual source representatives

The Mellin Plancherel map, with its original sign and measure, is

$$(\mathcal{M}_k F)(t) = \int_{(0,\infty)^k} F(x) \prod_{j=1}^k x_j^{-1/2+it_j} d^k x, \quad \mathcal{M}_k : \mathcal{H}_k \longrightarrow L^2 \left(\frac{d^k t}{(2\pi)^k} \right). \quad (\text{JD.6})$$

Indeed the substitutions $x_j = e^{u_j}$ send F to $e^{\sum_j u_j/2} F(e^{u_1}, \dots, e^{u_k})$ isometrically in $L^2(d^k u)$, and (JD.6) becomes its Fourier transform with kernel $e^{it \cdot u}$. The inverse has kernel $e^{-it \cdot u}$ and factor $(2\pi)^{-k}$. Multiplication by $\prod_j v_h(s_j)$ is a unitary surjection from $L^2(\nu_h^{\otimes k})$ to the target in (JD.6). For surjectivity divide by this product almost everywhere: the entire nonzero function v_h has only isolated zeros on the line, and the product zero set is null. Composing with the inverse of (JD.6) gives the unitary extension

$$\mathcal{U}_{h,k} : L^2(\nu_h^{\otimes k}) \longrightarrow \mathcal{H}_k, \quad \mathcal{U}_{h,k} P = \mathcal{T}_h^{(k)} P \quad \text{for every polynomial } P. \quad (\text{JD.7})$$

The polynomial identity follows from the actual Mellin transform of F_h and D_j , so this is an extension of the original source map, not a change of representative.

Theorem 15.2 (Joint boundary density and exact finite-jet graph closure). *For each $k \geq 1$, the original total-degree boundary spaces $\mathcal{B}_M$ satisfy*

$$\overline{\bigcup_{M \geq k(d-1)} \mathcal{B}_M} = \mathcal{H}_k, \quad P_{\mathcal{B}_M} \longrightarrow I \quad \text{strongly.} \quad (\text{JD.8})$$

For the actual coherent minima of (CT.13), in the original fixed finite monomial tensor coordinates,

$$\|R_M\|_{E^{(k)} \rightarrow \mathcal{H}_k} \longrightarrow 0, \quad G_M \downarrow 0, \quad \det G_M \longrightarrow 0. \quad (\text{JD.9})$$

Here $G_M > 0$ at every finite M , and the middle limit is both decreasing in the positive-semidefinite order and convergent in matrix operator norm. Every finite jet is still retained: $J^{(k)} R_M = I$ for every M .

Define J_{pol} on the dense original source $\mathcal{D}_{h,k} = \mathcal{T}_h^{(k)} \mathbb{C}[s_1, \dots, s_k]$ by its complete arithmetic jets, including multiplication by $v_h^{\otimes k}$. Its graph has exactly the closure

$$\overline{\text{Graph}(J_{\text{pol}})} = \mathcal{H}_k \oplus E^{(k)}. \quad (\text{JD.10})$$

As a closed linear relation this has the split exact sequence

$$0 \longrightarrow E^{(k)} \xrightarrow{u \mapsto (0,u)} \overline{\text{Graph}(J_{\text{pol}})} \xrightarrow{(F,u) \mapsto F} \mathcal{H}_k \longrightarrow 0, \quad F \longmapsto (F, 0). \quad (\text{JD.11})$$

Proof. Every polynomial in $h(s_1) \mathbb{C}[s_1, \dots, s_k]$ belongs to one finite total-degree ideal $\mathcal{I}_{h,k,M}$. Its density proved in (JD.5), followed by the unitary map (JD.7), proves the first assertion in (JD.8). Given $F \in \mathcal{H}_k$ and $\epsilon > 0$, choose $b \in \mathcal{B}_N$ with $\|F - b\| < \epsilon$. For every $M \geq N$, the distance from F to $\mathcal{B}_M$ is at most that norm, so $\|(I - P_{\mathcal{B}_M})F\| < \epsilon$. This proves strong convergence.

Apply it to each of the d^k images of the fixed monomial basis under $s_h^{\otimes k}$. The identity $R_M = (I - P_{\mathcal{B}_M})s_h^{\otimes k}$ makes each column tend to zero. For a vector with coordinate norm one, Cauchy-Schwarz bounds its image by the square root of the sum of the squared column norms, proving the first limit in (JD.9). Since $G_M = R_M^* R_M$ and $G_M - G_{M+1} = Y_M^* Y_M$, the norm limit and monotonicity follow. The determinant limit follows by continuity of the determinant in these same finite coordinates. Positivity and $J^{(k)} R_M = I$ are the proved finite interpolation identities; none of these finite algebraic equations is lost in taking the Hilbert norm limit.

The kernel of J_{pol} is exactly the union of the $\mathcal{B}_M$, so its graph closure contains $\mathcal{H}_k \oplus \{0\}$ by (JD.8). For any $u \in E^{(k)}$, the vectors $(R_M u, u)$ belong to its graph and converge to $(0, u)$ by (JD.9). The closure is a linear subspace, and therefore contains every (F, u) ; the reverse inclusion is its declared ambient space. This proves (JD.10), and the maps, exactness and section in (JD.11) follow immediately by evaluating their two coordinates. In particular the closure is a relation with vertical fibre $E^{(k)}$, and it cannot be the graph of a single-valued operator because $d^k > 0$. $\square$

There are corresponding exact limits for the original weight form and both interpolation kernels. Write $U = U_{v_h^{\otimes k}}$, let K_M be the remainder-only polynomial kernel on the joint source, and let $\mathcal{C}_M = G_M^{-1} = UK_M U^*$ be the complete arithmetic interpolation matrix. In the same fixed coordinates,

$$\begin{aligned} \|W_M\| &\leq (2\|A^{(k)}\| + k)\|G_M\| \longrightarrow 0, \\ \lambda_{\min}(\mathcal{C}_M) &= \|G_M\|^{-1} \longrightarrow \infty, \\ \lambda_{\min}(K_M) &\geq \frac{1}{\|U\|^2\|G_M\|} \longrightarrow \infty. \end{aligned} \tag{JD.9a}$$

The first line follows by the triangle inequality applied to the definition of W_M . Positivity of G_M proves the second. For the third, use $K_M = U^{-1}\mathcal{C}_M U^{-*}$ and $\|U^{-*}z\| \geq \|z\|/\|U\|$. Thus these coercive limits include every retained local jet, and the arithmetic unit remains in the comparison. The relative weight is related to its absolute form by the exact congruence

$$W_M = G_M^{1/2} S_M G_M^{1/2}, \quad \lambda_{\min}(G_M)\epsilon_M \leq \|W_M\| \leq \|G_M\|\epsilon_M. \tag{JD.9b}$$

The upper bound is submultiplicativity; applying it instead to $S_M = G_M^{-1/2} W_M G_M^{-1/2}$ gives the lower bound. Equations (JD.9a)–(JD.9b) retain the metric factors needed to pass from the proved absolute limits to the relative quantity.

The complementary quotient of the same closed relation is also explicit:

$$\frac{\overline{\text{Graph}(J_{\text{pol}})}}{\mathcal{H}_k \oplus \{0\}} \xrightarrow{\sim} E^{(k)}, \quad [(F, u)] \mapsto u, \quad u \mapsto [(0, u)]. \tag{JD.11a}$$

The kernel, well-definedness and both inverse identities follow by subtracting the two displayed pairs. With the product Hilbert norm the quotient norm is exactly $\|u\|$, since $\inf_{H \in \mathcal{H}_k} (\|F - H\|^2 + \|u\|^2)^{1/2} = \|u\|$, attained at $H = F$. Thus every completed-boundary class retains its original finite jet through an explicit isometric quotient.

This limit has an exact Split-Zero morphism. For a complex vector space X , use its one-support lift $\hat{X} = \{\tau_X\} \sqcup X^\bullet$, with τ_X the additive identity, vector addition within $X^\bullet$, and the original $G(\mathbb{C})$ -scalar action. In particular $0_X^\bullet \neq \tau_X$. Any linear map $T : X \rightarrow Y$ lifts to $\hat{T}(\tau_X) = \tau_Y$ and $\hat{T}(u^\bullet) = (Tu)^\bullet$; direct substitution checks addition and every scalar, including the supported scalar zero and external scalar zero. Give the supported fibre its norm topology and τ_X its isolated topology. Equation (JD.9) then says exactly

$$\hat{R}_M \longrightarrow \hat{0}, \quad \hat{0}(\tau_E) = \tau_{\mathcal{H}}, \quad \hat{0}(u^\bullet) = 0_{\mathcal{H}}^\bullet \quad (u \in E^{(k)}). \tag{JD.12}$$

Thus the limiting morphism retains the support of every input. Equation (JD.11) identifies all finite jets in the vertical fibre of the closed relation; it supplies the full typed relation between these supported Hilbert limits and the unchanged algebraic jets.

15.3 The accumulated determinant cost of the same boundary sequence

Continue to use the original ordered spectral sum and layer energy

$$\mathcal{D}_k = \sum_{i_1, \dots, i_k} \left| \sum_{j=1}^k (\text{Re } \rho_{i_j} - 1/2) \right|, \quad \chi_M = \text{Tr}(G_M^{-1} C_M^* C_M), \quad \tau_M = \text{Tr}(G_M^{-1} (G_M - G_{M+1})). \tag{JD.13}$$

The list $\rho_1, \dots, \rho_d$ includes every algebraic multiplicity. Define $\pi_M = \det G_{M+1} / \det G_M$. For integers $k(d-1) \leq a < b$, the following exact finite inequality holds whenever $\mathcal{D}_k > 0$:

$$\boxed{\mathcal{D}_k^2 \sum_{M=a}^{b-1} \frac{1}{\chi_M} \leq \sum_{M=a}^{b-1} \tau_M \leq \log \frac{\det G_a}{\det G_b}.} \tag{JD.14}$$

All denominators in this statement are proved positive. Indeed (CT.24) gives $\mathcal{D}_k^2 \leq \tau_M \chi_M$, with both factors nonnegative; strict positivity of the left side forces strict positivity of each factor. Let $t_{M,j} \in [0, 1)$ be the full eigenvalue list of $G_M^{-1/2}(G_M - G_{M+1})G_M^{-1/2}$, including zeros. The finite original matrices give exactly

$$\tau_M = \sum_j t_{M,j}, \quad \pi_M = \prod_j (1 - t_{M,j}), \quad -\log \pi_M - \tau_M = \sum_j \int_0^{t_{M,j}} \frac{u}{1-u} du \geq 0. \quad (\text{JD.15})$$

The first two identities are congruence and determinant identities; the last follows by integrating the derivative of $-\log(1-t) - t$. Sum (CT.24) divided by χ_M , then sum (JD.15). The product of the determinant ratios telescopes without a change of coordinates, proving (JD.14).

If $\mathcal{D}_k = 0$, every repeated tuple $(i, \dots, i)$ has zero summand, hence $\text{Re } \rho_i = 1/2$ for every retained zero. Conversely these equalities give $\mathcal{D}_k = 0$. The second inequality in (JD.14) and the exact remainder (JD.15) hold also in this case; no quotient $0/0$ is introduced. Equation (JD.9) proves that the right-hand side tends to infinity as $b \rightarrow \infty$ for each fixed a . Equations (JD.13)–(JD.15) retain the actual rate through χ_M , the complete relative Gram losses and their explicit logarithmic remainder. They impose no unproved upper estimate on that rate.

The same sequence supplies two further exact energy identities:

$$G_a = \sum_{M=a}^{\infty} Y_M^* Y_M, \quad \sum_{M=a}^{\infty} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M=a}^{\infty} \tau_M = \infty. \quad (\text{JD.16})$$

The first series converges in matrix operator norm, by telescoping $G_M - G_{M+1} = Y_M^* Y_M$ and the proved limit $G_M \rightarrow 0$. Taking traces of its finite partial sums proves the second identity; all summands are nonnegative. To prove the third, suppose instead that $\sum_M \tau_M$ were finite. Then beyond some index all $\tau_M \leq 1/2$, so every $0 \leq t_{M,j} \leq 1/2$. Integrating $(1-u)^{-1} \leq 2$ over $[0, t_{M,j}]$ gives $-\log(1-t_{M,j}) \leq 2t_{M,j}$. Hence the tail sum of $-\log \pi_M$ would be bounded by $2 \sum_M \tau_M < \infty$. The finite initial terms are finite because $G_M > 0$. Telescoping would therefore bound $\log(\det G_a / \det G_b)$ uniformly in b , contradicting $\det G_b \rightarrow 0$. This proves the divergence, including packets entirely on the critical line and every retained nilpotent chain. The equality $\tau_M = \|Y_M G_M^{-1/2}\|_{\text{HS}}^2$ identifies the relative energy in this divergence with the same layer map appearing in the convergent absolute energy series.

16 The actual kernel layer, its derivative image, and its quotient

This section integrates the complete kernel-layer delivery of 12 September 2026. Its archive hash and member hashes are recorded in `sources/web_kernel_layer_delivery/LOCAL_STAGING_PROVENANCE.json`. The source note's formulas (9)–(28) identify the interpolation kernel with an explicit basis of the next original theta relation layer. We first prove those maps in the retained coordinates and then calculate the exact derivative image and the entire quotient of that layer. The new quotient calculation retains the part of the layer not reached by the summed Euler derivative.

16.1 The original arithmetic source and its constrained minimum

Keep $g = 2\xi$, a nonempty full packet $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$, and $d = \deg h$. Fix $k \geq 1$, and set

$$E = E_h^{\otimes k}, \quad E_h = \mathbb{C}[s]/(h), \quad A = \sum_{j=1}^k A_{h,j}, \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times. \quad (\text{KL.1})$$

All tensor coordinates use the original monomial remainders. The function F_h , with Mellin transform g/h , is the original test function satisfying $h(D)F_h = f_0$. The exact source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{M}_k \mathcal{T}(P) = P(s_1, \dots, s_k) \prod_{j=1}^k v_h(s_j), \quad s_j = \frac{1}{2} + it_j. \quad (\text{KL.2})$$

The Mellin multiplier proves injectivity on polynomials. Its Hilbert norm is the polynomial norm for the unchanged product measure

$$d\nu_h(t) = |v_h(\frac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad \nu_h^{\otimes k}. \quad (\text{KL.3})$$

Let $\mathcal{P}_N$ be the polynomials of total degree at most N , and let $\mathcal{J}_N : \mathcal{P}_N \rightarrow E$ be full reduction by $h(s_1), \dots, h(s_k)$, followed by multiplication by the original unit $v_h^{\otimes k}$. For $N \geq k(d-1)$ this is onto, since the monomial remainder basis has at most that total degree. Fix an integer $N \geq k(d-1)$ for the remainder of this section; all subsequent levels $N+1, N+2$ satisfy the same threshold. Its kernel and its original theta image are

$$\mathcal{I}_N = \mathcal{P}_N \cap (h(s_1), \dots, h(s_k)), \quad \mathcal{L}_N = \mathcal{T}\mathcal{I}_N. \quad (\text{KL.4})$$

Let $p_j(s)$ be the actual monic orthogonal polynomial of ν_h , and retain $\kappa_j = \|p_j\|_{\nu_h}^2 > 0$. All moments exist by the proved arithmetic decay. The Gram is strictly positive because the density is positive except at a discrete set; this proves existence and uniqueness by finite orthogonal projection. For multi-indices set

$$p_\alpha = \prod_j p_{\alpha_j}, \quad \kappa_\alpha = \prod_j \kappa_{\alpha_j}, \quad a_\alpha = \bigotimes_j (v_h[p_{\alpha_j}]_h). \quad (\text{KL.5})$$

The monic triangular change preserves the full space of degree at most N , and product integration makes its Gram diagonal. It follows, by direct constrained minimization, that the original canonical representative and its metric are

$$\mathcal{K}_N = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha} > 0, \quad G_N = \mathcal{K}_N^{-1}, \quad R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (\text{KL.6})$$

Indeed the full jet of this vector is $\mathcal{K}_N G_N u = u$. If a coefficient vector has zero full jet, its inner product with the displayed coefficient vector is zero. Every other representative of that same jet is its sum with such a vector; its squared norm is therefore $u^* G_N u$ plus the squared norm of the added relation. This proves the unique minimum, its metric, and $R_N \perp \mathcal{L}_N$. These are the original maps $J^{(k)} R_N = I$ and $q^{(k)} R_N = \sigma$. The notation $\mathcal{K}_N$ here includes the local unit. To specify the comparison in the full tensor dimension, put

$$r_\alpha = \bigotimes_j [p_{\alpha_j}]_h, \quad K_N^{\text{rem}} = \sum_{|\alpha| \leq N} \frac{r_\alpha r_\alpha^*}{\kappa_\alpha}.$$

Then $\mathcal{K}_N = U_{v_h^{\otimes k}} K_N^{\text{rem}} U_{v_h^{\otimes k}}^*$, as an equality of operators on $E_h^{\otimes k}$. When $k = 1$, K_N^{rem} is exactly the earlier remainder-only kernel K_N .

16.2 The complete next-layer frame and its theta primitives

Write $\Sigma_n = \{\alpha \in \mathbb{N}^k : |\alpha| = n\}$ and $\mathbf{H}_n = \mathbb{C}^{\Sigma_n}$. The coordinate identification sends $x \in \mathbf{H}_n$ to the homogeneous polynomial $\sum_\alpha x_\alpha z^\alpha$; these z_j are coefficient variables and are not substituted for the original Mellin variables s_j . For $\beta \in \Sigma_{N+1}$, define the actual functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \mathcal{E}_N x = \sum_\beta x_\beta e_\beta, \quad \mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp. \quad (\text{KL.7})$$

The full jet of e_β is zero, hence it is in $\mathcal{L}_{N+1}$. The new-degree polynomial is orthogonal to all old polynomials and $R_N a_\beta \perp \mathcal{L}_N$, so $e_\beta \in \mathcal{E}_N$. The leading total-degree terms of the e_β are the distinct monomials s^β , proving independence by the injectivity of $\mathcal{T}$. For any element of $\mathcal{E}_N$, subtract its full degree- $N+1$ combination of the e_β . The remainder has zero jet and degree at most N , hence is in $\mathcal{L}_N$. It is also orthogonal to that space, so it is zero. This proves the exact isomorphism

$$\mathcal{E}_N : \mathbf{H}_{N+1} \xrightarrow{\sim} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1}/\mathcal{L}_N. \quad (\text{KL.8})$$

The inverse of the second arrow is $[f] \mapsto (I - P_{\mathcal{L}_N})f$.

Every vector in (KL.7) has a primitive in the original tensor complex. Its numerator polynomial is

$$p_\beta - \sum_{|\alpha| \leq N} p_\alpha \frac{a_\alpha^* G_N a_\beta}{\kappa_\alpha}. \quad (\text{KL.9})$$

Successive monic division in the ordered variables $s_1, \dots, s_k$ expresses this zero-jet polynomial as $\sum_i h(s_i) Q_{i,\beta}$, where $\deg Q_{i,\beta} \leq N+1-d$. At each replacement the leading power of the divided variable decreases and total degree does not increase, proving the bound. After the last division, the remainder has degree below d in every variable; its full jet is zero, so uniqueness of the complete remainder forces it to vanish. In the i -th tensor factor replace F_h by $(-1)^{i-1} \phi_*$, and apply $Q_{i,\beta}(D_1, \dots, D_k)$ to the whole tensor. The preceding $i-1$ degree-one factors give the differential sign $(-1)^{i-1}$. Together with $h(D)F_h = \Theta \phi_*$, the boundary of this primitive is exactly $+\mathcal{T}(h(s_i)Q_{i,\beta})$. Summing proves the asserted primitive with all ordered signs retained.

Put

$$\mathbf{U}_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad \mathbf{D}_{N+1} = \text{diag}(\kappa_\beta), \quad \mathbf{H}_N^{\text{lay}} = \mathcal{E}_N^* \mathcal{E}_N.$$

Expanding (KL.7) and using old/new polynomial orthogonality proves

$$\mathbf{H}_N^{\text{lay}} = \mathbf{D}_{N+1} + \mathbf{U}_N^* G_N \mathbf{U}_N > 0, \quad \mathcal{E}_N^* R_N = -\mathbf{U}_N^* G_N. \quad (\text{KL.10})$$

Define the projection loss $Y_N = P_{\mathcal{E}_N} R_N$. In particular its exact formula is $Y_N = -\mathcal{E}_N (\mathbf{H}_N^{\text{lay}})^{-1} \mathbf{U}_N^* G_N$. Its rank is exactly $\text{rank } \mathbf{U}_N$, because every other displayed factor is injective or invertible on its declared space.

16.3 The exact incidence map and the derivative-image rank

Define the actual old top-degree columns and their original norms by

$$\mathbf{A}_N = (a_\alpha)_{\alpha \in \Sigma_N} : \mathbf{H}_N \rightarrow E, \quad \mathbf{D}_N = \text{diag}(\kappa_\alpha)_{\Sigma_N}. \quad (\text{KL.11})$$

Define the coordinate incidence map by

$$\mathbf{T}_n : \mathbf{H}_n \rightarrow \mathbf{H}_{n+1}, \quad (\mathbf{T}_n x)_\beta = \sum_{i: \beta_i > 0} x_{\beta - \mathbf{e}_i}. \quad (\text{KL.12})$$

On the specified homogeneous coefficient polynomials this is multiplication by $\ell(z) = z_1 + \dots + z_k$. A product of two nonzero polynomials over $\mathbb{C}$ is nonzero: choose a monomial order compatible with products and multiply the two leading terms. Since $\ell \neq 0$, $\mathbf{T}_n$ is injective.

The full unprojected derivative defect and its projected layer map are

$$B_N = D^{(k)} R_N - R_N A, \quad C_N = (I - P_{\mathcal{L}_N}) B_N : E \rightarrow \mathcal{E}_N.$$

They have the exact coordinate factorization

$$\boxed{C_N = \mathcal{E}_N \mathbf{T}_N \mathbf{D}_N^{-1} \mathbf{A}_N^* G_N.} \quad (\text{KL.13})$$

To prove it, multiply the finite numerator in (KL.6) by $s_1 + \cdots + s_k$. Monicity makes its coefficient at the new top polynomial p_β equal to $\sum_{i:\beta_i>0} a_{\beta-\mathbf{e}_i}^* G_N u / \kappa_{\beta-\mathbf{e}_i}$, precisely the row on the right side of (KL.13). The numerator of $R_N A u$ has degree at most N . Subtract the right side of (KL.13) from $B_N u$: its numerator has degree at most N and its full jet is zero, so it is an original relation in $\mathcal{L}_N$. Orthogonal projection proves (KL.13), including the original quotient statement $[B_N u] = [C_N u]$ in $\mathcal{B}^{(k)} / \mathcal{L}_N$.

Theorem 16.1 (Exact derivative rank and its full kernel). *With every original jet column retained,*

$$\boxed{\text{rank } C_N = \text{rank } A_N, \quad \ker C_N = \mathcal{K}_N \ker A_N^*} \quad (\text{KL.14})$$

Define the original weight form $W_N = A^* G_N + G_N A - k G_N$. It obeys the refined bound

$$\text{rank } W_N \leq 2 \min\{\text{rank } A_N, \text{rank } U_N\}. \quad (\text{KL.15})$$

Proof. Both $\mathcal{E}_N$ and $\mathbf{T}_N$ are injective; D_N and G_N are invertible. Thus (KL.13) has rank $\text{rank } A_N^*$, which equals the rank of its conjugate transpose. Its kernel is $G_N^{-1} \ker A_N^* = \mathcal{K}_N \ker A_N^*$, with the displayed invertible coordinate map retained. The original Green identity and orthogonality give $W_N = -(Y_N^* C_N + C_N^* Y_N)$. Each summand has rank at most the lesser of the two map ranks; the rank of a sum is at most the sum of its ranks. Use (KL.10) and (KL.14) to obtain (KL.15). No nonzero homogeneous-leading-term assertion has been substituted for an actual column a_α . $\square$

16.4 The whole quotient of the next layer by the derivative image

For $k \geq 2$, let H'_n be the homogeneous polynomials of degree n in $z_1, \dots, z_{k-1}$. There is a split exact sequence of vector spaces

$$0 \longrightarrow H_n \xrightarrow{\mathbf{T}_n} H_{n+1} \xrightarrow{\rho_n} H'_{n+1} \longrightarrow 0, \quad \rho_n f = f(z_1, \dots, z_{k-1}, -\sum_{i < k} z_i). \quad (\text{KL.16})$$

For a complete proof and inverse, divide f by the monic linear polynomial $\ell = z_k + \sum_{i < k} z_i$ in the last variable. Every replacement retains total degree, giving uniquely

$$f = \ell q + \iota r, \quad q \in H_n, \quad r = \rho_n f \in H'_{n+1}, \quad (\text{KL.17})$$

where ι includes a polynomial independent of z_k . For uniqueness set $z_k = -\sum_{i < k} z_i$ first to obtain $r = 0$, then use injectivity of multiplication by ℓ . This also proves surjectivity, exactness and the specified section ι . The ordered last coordinate and the signs in the substitution remain part of the map.

Theorem 16.2 (Exact unused-layer quotient). *Set $Z_N = \text{im}(D_N^{-1} A_N^*) \subseteq H_N$. Then the actual quotient is*

$$\boxed{\mathcal{E}_N / \text{im } C_N \simeq (H_N / Z_N) \oplus H'_{N+1}} \quad (\text{KL.18})$$

For an original layer representative $e = \mathcal{E}_N f$, divide $f = \ell q + \iota r$ by (KL.17); the map sends $[e]$ to $([q], r)$. Its inverse sends $([q], r)$ to $[\mathcal{E}_N(\ell q + \iota r)]$.

Proof. The invertibility of G_N in (KL.13) gives $\text{im } C_N = \mathcal{E}_N \mathbf{T}_N Z_N$. Changing e by such an element changes exactly q by an element of Z_N , while leaving r unchanged. Thus the first map is well-defined. Changing the representative of $[q]$ has the same effect on the inverse, so the inverse is well-defined. The unique division (KL.17) proves both inverse identities. For $k = 1$, $\mathbf{T}_N$ is an isomorphism between the one-dimensional shell spaces and the same statement holds with $H'_{N+1} = 0$. $\square$

In particular $\dim H'_{N+1} = \binom{N+k-1}{k-2}$ for $k \geq 2$, by counting monomials; the first summand has dimension $\binom{N+k-1}{k-1} - \text{rank } A_N$. Their sum is exactly the original layer dimension minus $\text{rank } C_N$. The quotient therefore retains both the full-jet rank loss and the explicitly complementary relation directions. For every coordinate vector, (KL.9) supplies its original theta primitive. The quotient is of actual test-function relations, with those primitives, rather than only a quotient of dimensions.

16.5 The derivative between consecutive relation quotients

The original Euler sum has $D^{(k)}\mathcal{L}_N \subseteq \mathcal{L}_{N+1}$, because it multiplies the numerator polynomial by $s_1 + \cdots + s_k$ and preserves the ideal in (KL.4). Hence it induces the well-typed map

$$\partial_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [D^{(k)}f]. \quad (\text{KL.19})$$

Under the exact layer coordinates of (KL.8), its matrix is

$$\partial_N = \mathcal{E}_{N+1} \mathsf{T}_{N+1} \mathcal{E}_N^{-1}, \quad (\text{KL.20})$$

where each $\mathcal{E}$ is followed or preceded by its quotient isomorphism as in (KL.8). Write $e_\beta^{(N)}$ for (KL.7) at level N , and $e_{\beta+\mathbf{e}_i}^{(N+1)}$ for the next layer. The actual recurrence $sp_j = p_{j+1} + b_j p_j - (\kappa_j/\kappa_{j-1})p_{j-1}$ gives the complete retained difference

$$\begin{aligned} D^{(k)}e_\beta^{(N)} - \sum_i e_{\beta+\mathbf{e}_i}^{(N+1)} = & \mathcal{T} \left[\left(\sum_i b_{\beta_i} \right) p_\beta - \sum_{i:\beta_i>0} \frac{\kappa_{\beta_i}}{\kappa_{\beta_i-1}} p_{\beta-\mathbf{e}_i} \right] \\ & - D^{(k)}R_N a_\beta + R_{N+1} \sum_i a_{\beta+\mathbf{e}_i}. \end{aligned}$$

The final term in the one-variable recurrence is omitted at $j = 0$. Every term on the right has numerator degree at most $N + 1$. Each term on the left has zero full jet, so their difference is in $\mathcal{L}_{N+1}$. This proves (KL.20) with all lower polynomial coefficients retained. Applying the explicit ordered division in (KL.9) to this displayed numerator supplies its original theta primitive.

It follows that ∂_N is injective and its cokernel is exactly H'_{N+2} , with maps (KL.16)–(KL.17) at index $N + 1$. Thus the original derivative connects consecutive relation quotient types by a proved injection; it has not been made an endomorphism of a fixed quotient. The ordinary relation-admission map has the same declared source and target:

$$\text{adm}_N : \mathcal{L}_{N+1}/\mathcal{L}_N \longrightarrow \mathcal{L}_{N+2}/\mathcal{L}_{N+1}, \quad [f] \longmapsto [f] = 0.$$

Equation (KL.19) instead sends the representative to its next derivative class. These are explicitly different typed maps of the same original representative, related by (KL.20) and the inclusion of their actual numerator relations.

All maps here have their original supported lifts. At a fixed support label λ , a linear map T acts by $(\lambda, x) \mapsto (\lambda, Tx)$. The relation-admission map therefore sends a killed class to the receiving fibre's supported zero. The external zero τ remains external absence. Each displayed kernel and quotient records its full supported vector space and each derivative retains its consecutive source and target; no passage to the Hilbert closure is used.

16.6 The exact original-theta norm and volume bridge

Specialize to $k = 1$. We now distinguish the degree-indexed G_N of (KL.6) from the correction-indexed original metric by writing

$$\mathbf{G}_m = G_{d+m} \quad (m \geq -1), \quad \mathbf{R}_m = R_{d+m}, \quad r_m = \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{m-1}} \quad (m \geq 0). \quad (\text{KL.21})$$

At $m = -1$, $\mathcal{P}_{d-1}$ contains no nonzero multiple of h . Thus the original representative is exactly the fixed section

$$s_h u = \mathbf{R}_{-1} u = \mathcal{T}_h \text{rem}_h(\varepsilon_h u), \quad \varepsilon_h = v_h^{-1}, \quad \mathbf{G}_{-1} = s_h^* s_h.$$

This fixes the initial determinant in (KL.21), without introducing a negative-degree theta polynomial. Put $L_{-1} = 0$ and

$$f_j = D^j f_0, \quad L_j = \text{span}(f_0, \dots, f_j), \quad u_j = (I - P_{L_{j-1}})f_j, \quad \mathfrak{h}_j = \|u_j\|^2 > 0 \quad (j \geq 0). \quad (\text{KL.22})$$

These $u_j = Q_j(D)f_0$ are the monic orthogonal theta vectors for the original measure $|g(1/2 + it)|^2 dt/(2\pi)$. Their definition does not depend on h . The monic p_j and their norms κ_j continue to use the original packet-dependent measure $|g/h|^2 dt/(2\pi)$. Multiplication by the unchanged monic polynomial h is the exact isometry connecting their source polynomial spaces:

$$P \mapsto hP, \quad \|hP\|_{\nu_h}^2 = \|P\|_{|g|^2 dt/(2\pi)}^2, \quad \mathcal{T}_h(hP) = P(D)f_0. \quad (\text{KL.23})$$

This follows by cancellation of the pointwise factor $|h|^2$ in the integral; the identities extend across the discrete zeros by their entire expressions. It is an isometry onto the specified subspace of multiples of h , with its degree shifted by the full degree $d = \sum_\rho m_\rho$.

Theorem 16.3 (Original norm as an exact relative Gram volume). *For every $j \geq 0$, including the initial theta vector,*

$$\boxed{\kappa_{d+j} = \mathfrak{h}_j r_j}. \quad (\text{KL.24})$$

Consequently, for $m \geq 0$,

$$\boxed{\frac{\kappa_{d+m+1}}{\kappa_{d+m}} = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} \frac{r_{m+1}}{r_m}, \quad \frac{\det \mathbf{G}_m}{\det \mathbf{G}_{-1}} = \prod_{j=0}^m \frac{\kappa_{d+j}}{\mathfrak{h}_j}}. \quad (\text{KL.25})$$

All ratios have positive denominators and all packet multiplicities are retained.

Proof. At the degree level $N = d + j - 1$, define $e_{d+j} = \mathcal{T}_h p_{d+j} - \mathbf{R}_{j-1} a_{d+j}$. Its numerator has zero full jet, is monic of degree $d + j$, and is divisible by the original monic h . By ordinary monic division its quotient is monic of degree j . Under (KL.23) this is f_j plus a vector in L_{j-1} . By (KL.7) it is orthogonal to that space. The defining orthogonal projection in (KL.22) therefore proves

$$e_{d+j} = u_j, \quad \mathfrak{h}_j = \kappa_{d+j} + a_{d+j}^* \mathbf{G}_{j-1} a_{d+j}. \quad (\text{KL.26})$$

When $j = 0$, the quotient of the numerator by h is the constant 1, so the same argument explicitly gives $e_d = f_0 = u_0$; no previous theta layer is required.

For the full unit cancellation put $v_l = [p_l]_h = p_l(A_h)[1]_h$, and $K_{l-1} = \sum_{i=0}^{l-1} v_i v_i^* / \kappa_i$, using the original monomial remainder coordinates. At $l = d + j$,

$$a_l = U_{v_h} v_l, \quad \mathbf{G}_{j-1} = U_{\varepsilon_h}^* K_{l-1}^{-1} U_{\varepsilon_h}, \quad a_l^* \mathbf{G}_{j-1} a_l = v_l^* K_{l-1}^{-1} v_l. \quad (\text{KL.27})$$

Both cancellations use $U_{\varepsilon_h} U_{v_h} = I$; hence they preserve every derivative of the original local unit at every full jet. The degree update $K_l = K_{l-1} + v_l v_l^* / \kappa_l$ has determinant ratio $1 + v_l^* K_{l-1}^{-1} v_l / \kappa_l$. To verify this scalar identity directly, expand $\det(I + xy^*)$ multilinearly in its columns: all terms with two updated columns vanish since those columns are multiples of x , leaving $1 + y^* x$. The fixed-unit congruences in (KL.27) then give

$$r_j = \frac{\kappa_l}{\kappa_l + v_l^* K_{l-1}^{-1} v_l} = \frac{\kappa_{d+j}}{\mathfrak{h}_j}.$$

This also proves $0 < r_j \leq 1$, including the case $v_l = 0$. Divide two consecutive identities and multiply the first $m + 1$ identities to obtain (KL.25). $\square$

For two full packets h, H , their original norm sequences are therefore connected at the same theta degree j by the exact scalar equality

$$\frac{\kappa_{\deg h+j}^{(h)}}{r_j^{(h)}} = \mathfrak{h}_j = \frac{\kappa_{\deg H+j}^{(H)}}{r_j^{(H)}}. \quad (\text{KL.28})$$

When $h \mid H$, the corresponding source isometry is $P \mapsto (H/h)P$, since $(g/H)(H/h)P = (g/h)P$. At the old centres this equality preserves every jet, and at each added centre g/h has the complete zero order, so it realizes exactly the CRT insertion of zero jets. The reference numerator has degree below $\deg H$ and is its unique full remainder. Both corrected representatives then subtract the same projection onto the packet-independent L_j , proving the coherent equality $\mathbf{R}_{H,j} \iota_{hH} = \mathbf{R}_{h,j}$, including $j = -1$. Thus (KL.28) connects precisely these original coherent source maps and their fixed-section metrics.

16.7 The complete scalar dictionary for the boundary energy

For $m \geq 0$, set $n = d + m + 1$ and retain the terminal notation

$$M_n = \kappa_{n-1}^{-1} K_{n-1}^{-1}, \quad \alpha_n = v_{n-1}^* M_n v_{n-1}, \quad \beta_n = v_n^* M_n v_n, \quad \gamma_n = v_n^* M_n v_{n-1}.$$

Define the original theta projection rows by

$$d_j = \mathfrak{h}_j^{-1} u_j^* s_h : E_h \rightarrow \mathbb{C}, \quad \mathbf{R}_m = s_h - \sum_{j=0}^m u_j d_j, \quad c_m = d_m \mathbf{G}_m^{-1} d_{m+1}^*. \quad (\text{KL.29})$$

Here an expression $u_j^* F$ denotes the Hilbert inner product $\langle u_j, F \rangle$, conjugate-linear in its first argument. The row d_j therefore retains the original unnormalized vector and the exact factor $\mathfrak{h}_j^{-1}$.

The complete row and scalar comparison is

$$\boxed{\begin{aligned} d_m &= -\frac{v_{n-1}^* K_{n-1}^{-1} U_{\varepsilon_h}}{\kappa_{n-1}}, \\ d_{m+1} &= -\frac{v_n^* K_{n-1}^{-1} U_{\varepsilon_h}}{\mathfrak{h}_{m+1}}, \\ d_m \mathbf{G}_m^{-1} d_m^* &= \frac{\alpha_n}{\kappa_{n-1}}, \\ d_{m+1} \mathbf{G}_m^{-1} d_{m+1}^* &= \frac{\kappa_{n-1} \beta_n}{\mathfrak{h}_{m+1}^2}, \\ \mathfrak{h}_{m+1} c_m &= \overline{\gamma_n}. \end{aligned}} \quad (\text{KL.30})$$

To prove the first row, the polynomial numerator of s_h has degree below d . In (KL.29), each u_j has numerator hQ_j , monic of degree $d+j$. Thus the coefficient of degree $d+m = n-1$ in the numerator of $\mathbf{R}_m$ is $-d_m$. The finite expansion (KL.6) gives the same coefficient as $a_{n-1}^* \mathbf{G}_m / \kappa_{n-1}$. Cancelling the same units as in (KL.27) proves the first row, also at $m = 0$. For the second row use $u_{m+1} = e_n$, so (KL.10) gives $u_{m+1}^* \mathbf{R}_m = -a_n^* \mathbf{G}_m$. The orthogonality of u_{m+1} to L_m also gives $u_{m+1}^* \mathbf{R}_m = u_{m+1}^* s_h = \mathfrak{h}_{m+1} d_{m+1}$. This proves its formula. Finally $\mathbf{G}_m^{-1} = U_{v_h} K_{n-1} U_{v_h}^*$; insert both rows, cancel both unit pairs and the adjacent kernel inverse. The three remaining products are exactly those in (KL.30), including the conjugation in the last line.

Theorem 16.4 (Exact original-norm formula for the reflected packet). *Suppose the original full packet is stable under $\rho \mapsto 1 - \bar{\rho}$. For its actual relative allowance ϵ_m ,*

$$\boxed{\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2.} \quad (\text{KL.31})$$

If the packet is also stable under ordinary conjugation, then $c_m = 0$, so its complete formula becomes

$$\epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{KL.32})$$

Proof. The terminal identity (KT.4), with the original full multiplicities, gives $\text{Re } \gamma_n = \text{Re tr } A_h - d/2 = 0$: reflected centres contribute complementary real parts with equal orders. Its two possible nonzero generalized eigenvalues then yield $\epsilon_m^2 = \alpha_n \beta_n - |\gamma_n|^2$. The preceding and succeeding rank-one kernel updates, also proved directly in (KL.27), give

$$\alpha_n = 1 - r_m, \quad \beta_n = \frac{\kappa_n}{\kappa_{n-1}} \frac{1 - r_{m+1}}{r_{m+1}}.$$

Insert (KL.25) into the product $\alpha_n \beta_n$ and the last line of (KL.30) into $|\gamma_n|^2$. Every positive ratio is retained, and this is (KL.31).

For the final assertion, conjugation stability makes h real in the original monomial coordinates. The real Taylor coefficients of g give $(g/h)(1/2 - it) = \overline{(g/h)(1/2 + it)}$, so the actual ν_h is even in t . Its monomial Gram matrix is real: the integrand for entry $s^i s^j$ at $-t$ is the complex conjugate of its integrand at t . Therefore the unique solution of the real Gram linear equations for the monic p_l has real coefficients. Polynomial division by the real monic h makes each v_l real, and the finite sum for K_{n-1} is real. Its inverse and M_n are real as well. Consequently γ_n is real. The already proved reflection trace identity makes its real part zero, hence $\gamma_n = 0$. Equation (KL.30), with $\mathfrak{h}_{m+1} > 0$, proves $c_m = 0$. No division by a jet column, an off-line displacement, or a boundary energy has been used; all zero-column cases remain included. $\square$

The source PR15 denotes the relative determinant at correction m by $\delta_m = \det \mathbf{G}_{m+1} / \det \mathbf{G}_m$. Its exact index map is therefore $\delta_m = r_{m+1}$, with $\delta_{-1} = r_0$. Its cross-term row product is the c_m of (KL.29). Equations (KL.24)–(KL.31) prove the full map from its original-theta scalar formula to the terminal-kernel formula, including both monic norm families, the unit congruences, the conjugation, the fixed initial section, and the complete orders. These equalities calculate the same actual boundary energy at every finite stage. They do not supply an estimate tending to zero; that requires further information about these now explicitly related arithmetic quantities.

The packet-independent norms in (KL.24)–(KL.32) also have the complete positive-half-line formulas (SP.18)–(SP.21) and the gamma-reference determinant formulas (TG.16)–(TG.19). Those sections prove the original affine and quadratic coordinate maps, their inverses, both measures and every phase. Consequently these formulas can be substituted directly into the identities here; the actual unit, packet determinant ratios, full zero orders and source degree indices remain unchanged.

17 Original orthogonal shells and the exact projective-ratio algebra

The source's degree shell consists of its original orthogonal product polynomials. This section proves its full comparison with the homogeneous monomial shell, including every lower-degree coefficient. It also calculates the algebra generated by the ratio of the two original tensor coordinates. The arithmetic local unit and all multiplicities remain in the resulting linear maps.

17.1 Two sections of the same polynomial filtration

Retain a complete nonempty arithmetic packet

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(g/h), \quad g = 2\xi. \quad (\text{KPC.1})$$

The complete orders make v_h a unit. For $k \geq 1$, write $E^{(k)} = E_h^{\otimes k}$ in its original monomial tensor coordinates and use the original full-jet map

$$J(P) = U_{v_h^{\otimes k}}[P]_{(h(s_1), \dots, h(s_k))}, \quad J : \mathbb{C}[s_1, \dots, s_k] \longrightarrow E^{(k)}. \quad (\text{KPC.2})$$

Here U_a is multiplication by the stated element a . The integration variables remain $s_j = 1/2 + it_j$, with

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}. \quad (\text{KPC.3})$$

Let $p_j(s) = \sum_{r=0}^j p_{j,r} s^r$, with $p_{j,j} = 1$, be the original monic orthogonal polynomials for this measure. Retain $\kappa_j = \int |p_j(s)|^2 d\nu_h > 0$. For a multi-index α , put

$$p_\alpha = \prod_{j=1}^k p_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad a_\alpha = J(p_\alpha). \quad (\text{KPC.4})$$

All moment integrals in these definitions use the original density.

For $N \geq 0$, let $\mathcal{P}_{\leq N}$ be the polynomials of total degree at most N , let $\mathcal{P}_{\leq -1} = 0$, and let $\mathcal{H}_N$ be the homogeneous polynomials of degree N . The highest-degree map $\ell_N : \mathcal{P}_{\leq N} \rightarrow \mathcal{H}_N$ has kernel $\mathcal{P}_{\leq N-1}$. It therefore identifies $\mathcal{H}_N$ with the associated-graded space $\mathcal{P}_{\leq N}/\mathcal{P}_{\leq N-1}$. There are two specified sections:

$$\begin{aligned} i_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha s^\alpha, \\ O_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) &= \sum_{|\alpha|=N} c_\alpha p_\alpha. \end{aligned} \quad (\text{KPC.5})$$

Theorem 17.1 (The exact lower-degree correction). *Both maps in (KPC.5) are linear sections of ℓ_N . Their difference $L_N = O_N - i_N$ has target $\mathcal{P}_{\leq N-1}$ and is exactly*

$$L_N \left(\sum_{|\alpha|=N} c_\alpha s^\alpha \right) = \sum_{|\gamma| < N} \left(\sum_{\substack{|\alpha|=N \\ \gamma_j \leq \alpha_j \ (1 \leq j \leq k)}} c_\alpha \prod_{j=1}^k p_{\alpha_j, \gamma_j} \right) s^\gamma. \quad (\text{KPC.6})$$

The image of O_N is the orthogonal complement of $\mathcal{P}_{\leq N-1}$ in $\mathcal{P}_{\leq N}$ for the original product measure, and its basis in (KPC.5) has Gram $\text{diag}(\kappa_\alpha)_{|\alpha|=N}$. Define the two full residue maps on the same $\mathcal{H}_N$ by

$$A_N^{\text{hom}} = J i_N, \quad A_N^{\text{orth}} = J O_N.$$

Then, with $J_{<N} = J|_{\mathcal{P}_{\leq N-1}}$,

$$A_N^{\text{orth}} = A_N^{\text{hom}} + J_{<N} L_N. \quad (\text{KPC.7})$$

In particular the original orthogonal-shell columns of A_N^{orth} are exactly a_α .

Proof. Expand each factor in p_α . Its coefficient of s^γ is $\prod_j p_{\alpha_j, \gamma_j}$ when $\gamma_j \leq \alpha_j$ for all j , and is zero otherwise. If $|\alpha| = |\gamma| = N$, these inequalities force $\gamma = \alpha$, whose coefficient is one. This proves both section identities and the complete expression (KPC.6). For a monomial s^γ of degree less than $N = |\alpha|$, at least one $\gamma_j < \alpha_j$. In the product integral for $\langle s^\gamma, p_\alpha \rangle$, that factor is zero by one-variable orthogonality. All product integrals are justified by their finite moments and Cauchy–Schwarz. Distinct product polynomials are mutually orthogonal, and their norms multiply to κ_α . Their leading monomials make them independent. Their number is $\dim \mathcal{P}_{\leq N} - \dim \mathcal{P}_{\leq N-1}$, so they exhaust the indicated orthogonal complement. Applying the unchanged linear map J to $O_N = i_N + L_N$ proves (KPC.7). This application includes the fixed multiplication by $v_h^{\otimes k}$. For $N = 0$ the sum in (KPC.6) is empty and $L_0 = 0$, so the same proof includes the initial shell. $\square$

The associated-graded target is explicit as well. Put $F_N E^{(k)} = J(\mathcal{P}_{\leq N})$ and $F_{-1} E^{(k)} = 0$. Projection onto $F_N E^{(k)} / F_{N-1} E^{(k)}$, composed with either map in (KPC.7), gives the same surjection

$$\overline{A}_N : \mathcal{H}_N \twoheadrightarrow F_N E^{(k)} / F_{N-1} E^{(k)}. \quad (\text{KPC.8})$$

Indeed their difference lands in $F_{N-1} E^{(k)}$, and $\mathcal{P}_{\leq N} = i_N \mathcal{H}_N + \mathcal{P}_{\leq N-1}$ proves surjectivity. If $Z_N = \ker \overline{A}_N$ and $K_N^a = \ker A_N^a$ for $a \in \{\text{hom}, \text{orth}\}$, then $K_N^a \subseteq Z_N$ and the exact sequence is

$$0 \longrightarrow Z_N / K_N^a \longrightarrow \mathcal{H}_N / K_N^a \longrightarrow \mathcal{H}_N / Z_N \longrightarrow 0. \quad (\text{KPC.9})$$

The middle space is identified with $\text{im } A_N^a$ by $[c] \mapsto A_N^a c$, the last space with the target of (KPC.8), and the first space with $\text{im } A_N^a \cap F_{N-1} E^{(k)}$. These identifications follow directly from their kernels and are bijective: for the first image, membership in F_{N-1} is exactly the condition $c \in Z_N$. Thus equality of the graded maps retains, rather than removes, the two potentially different lower-filtration image spaces. In particular, once $N-1 \geq k(d-1)$, complete monomial remainders show $F_{N-1} E^{(k)} = E^{(k)}$, so (KPC.8) has zero target; the full maps (KPC.7) still have their stated images and kernels.

17.2 The full correspondence between the two image spaces

Theorem 17.2 (Common quotient and exact graph correspondence). *Write $I_h = \text{im } A_N^{\text{hom}}$, $I_o = \text{im } A_N^{\text{orth}}$, $K_h = \ker A_N^{\text{hom}}$, and $K_o = \ker A_N^{\text{orth}}$. Define*

$$Q_N = \mathcal{H}_N / (K_h + K_o), \quad \Gamma_N = \{(A_N^{\text{hom}} c, A_N^{\text{orth}} c) : c \in \mathcal{H}_N\}. \quad (\text{KPC.10})$$

The maps $q_h : I_h \rightarrow Q_N$ and $q_o : I_o \rightarrow Q_N$, given respectively by $A_N^{\text{hom}} c \mapsto [c]$ and $A_N^{\text{orth}} c \mapsto [c]$, are well-defined surjections, with exact sequences

$$\begin{aligned} 0 \longrightarrow A_N^{\text{hom}}(K_o) \longrightarrow I_h \xrightarrow{q_h} Q_N \longrightarrow 0, \\ 0 \longrightarrow A_N^{\text{orth}}(K_h) \longrightarrow I_o \xrightarrow{q_o} Q_N \longrightarrow 0. \end{aligned} \quad (\text{KPC.11})$$

One has the exact vector-space identifications

$$\boxed{\Gamma_N = I_h \times_{Q_N} I_o \cong \mathcal{H}_N / (K_h \cap K_o).} \quad (\text{KPC.12})$$

The projection $\Gamma_N \rightarrow I_h$ has kernel $\{0\} \oplus A_N^{\text{orth}}(K_h)$; the projection onto I_o has kernel $A_N^{\text{hom}}(K_o) \oplus \{0\}$.

Proof. Changing a representative of an element of I_h changes c by K_h , hence leaves its class modulo $K_h + K_o$ unchanged. The same argument applies to I_o . The formulas give surjectivity. If $q_h(A_N^{\text{hom}} c) = 0$, write $c = c_h + c_o$ with $c_h \in K_h, c_o \in K_o$. Then $A_N^{\text{hom}} c = A_N^{\text{hom}} c_o$. Conversely every such vector maps to zero. This proves the first kernel formula in (KPC.11); interchanging the two maps proves the second.

The paired map from $\mathcal{H}_N$ onto Γ_N has kernel exactly $K_h \cap K_o$, proving the last identification in (KPC.12). Every paired image belongs to the fibre product. Conversely, suppose $(A_N^{\text{hom}} c, A_N^{\text{orth}} b)$ belongs to the fibre product. Then $c - b = c_h + c_o$ for some $c_h \in K_h, c_o \in K_o$. Put $e = c - c_h = b + c_o$. It has the two desired images, so the pair belongs to Γ_N . This proves equality with the full fibre product, not merely an inclusion. Taking the kernel of each coordinate projection gives the asserted remaining formulas. $\square$

The quotient in (KPC.10) also has its precise universal property. If linear maps $f_h : I_h \rightarrow Z$ and $f_o : I_o \rightarrow Z$ satisfy $f_h A_N^{\text{hom}} = f_o A_N^{\text{orth}}$, their common composite on $\mathcal{H}_N$ vanishes on both K_h and K_o , hence on their sum. It therefore factors uniquely through Q_N , and that factor gives both original maps after the surjections in (KPC.11). Likewise, a linear map $T : I_h \rightarrow I_o$ satisfying $T A_N^{\text{hom}} = A_N^{\text{orth}}$ exists exactly when $K_h \subseteq K_o$. Necessity follows by evaluating on K_h ; under that inclusion the formula $T(A_N^{\text{hom}} c) = A_N^{\text{orth}} c$ is well-defined and proves sufficiency. It is an isomorphism exactly

when $K_h = K_o$. Thus (KPC.11)–(KPC.12) give the full typed correspondence even when no such isomorphism exists.

There is also an explicit polynomial deformation of the sections. For a complex parameter z , define, without division by z ,

$$p_j^{(z)}(s) = \sum_{r=0}^j p_{j,r} z^{j-r} s^r, \quad O_{N,z}(c) = \sum_{|\alpha|=N} c_\alpha \prod_j p_{\alpha_j}^{(z)}(s_j). \quad (\text{KPC.13})$$

For $|\gamma| < N$, its coefficient of s^γ is exactly the coefficient in (KPC.6) multiplied by $z^{N-|\gamma|}$; at degree N it is c_γ . Hence $O_{N,0} = i_N$, $O_{N,1} = O_N$, and $\ell_N O_{N,z} = I$ for every z . The maps $JO_{N,z}$ are matrices polynomial in the parameter, and (KPC.8) is unchanged throughout. A rank condition $\text{rank } JO_{N,z} \leq r$ is exactly the vanishing of every $(r+1)$ -minor. No constancy of their rank follows from this deformation. In particular (KPC.13) retains every original coefficient at $z = 1$; it does not replace the original measure or its orthogonal minimum.

17.3 The algebra generated by the original coordinate ratio

Now take $k = 2$ and retain $h(0) \neq 0$. Every original nontrivial zero in the arithmetic packet has $0 < \text{Re } \rho < 1$, so this domain includes that packet. Work in the original algebra

$$E^{(2)} = \mathbb{C}[s_1, s_2]/(h(s_1), h(s_2)), \quad \zeta = s_1 s_2^{-1}. \quad (\text{KPC.14})$$

The inverse is defined because s and $h(s)$ are coprime; the Euclidean identity for these polynomials supplies an inverse modulo h . All ordered pairs of centres and their multiplicities are retained.

Theorem 17.3 (Complete ratio algebra and homogeneous-shell rank). *For every distinct ratio $r = \rho/\sigma$, put*

$$M_r = \max_{\rho/\sigma=r} (m_\rho + m_\sigma - 1), \quad b = \sum_r M_r, \quad \mu_\zeta(Z) = \prod_r (Z - r)^{M_r}. \quad (\text{KPC.15})$$

Then evaluation at ζ has exactly kernel (μ_ζ) , and gives the unital algebra embedding

$$\mathbb{C}[Z]/(\mu_\zeta) \hookrightarrow E^{(2)}, \quad [P] \mapsto P(\zeta). \quad (\text{KPC.16})$$

Its image is the subalgebra $\mathbb{C}[\zeta]$. For every $N \geq 0$, the full arithmetic homogeneous-shell map is

$$\sum_{j=0}^N c_j s_1^j s_2^{N-j} \mapsto v_h^{\otimes 2} s_2^N \sum_{j=0}^N c_j \zeta^j. \quad (\text{KPC.17})$$

Its kernel consists exactly of the coefficient polynomials $\sum_{j=0}^N c_j Z^j$ divisible by μ_ζ , and

$$\boxed{\text{rank } A_N^{\text{hom}} = \min\{N+1, b\}.} \quad (\text{KPC.18})$$

The arithmetic unit and s_2^N in (KPC.17) are retained invertible multiplication maps. Formula (KPC.18) does not replace the orthogonal-shell map (KPC.7).

Proof. The factors $(s - \rho)^{m_\rho}$ of h are pairwise coprime. For each factor, a Euclidean identity with the product of the others gives a polynomial equal to one modulo that factor and zero modulo all others. Their residue classes are mutually orthogonal idempotents and sum to one: each of these assertions holds modulo every factor, hence modulo their product. They give the full Chinese-remainder isomorphism, with inverse obtained by multiplying each component representative by its idempotent and adding. Tensoring these explicit maps gives

$$E^{(2)} \cong \prod_{(\rho, \sigma)} \mathbb{C}[x, y]/(x^{m_\rho}, y^{m_\sigma}), \quad x = s_1 - \rho, \quad y = s_2 - \sigma. \quad (\text{KPC.19})$$

In each component the basis is the full set $x^a y^c$, $0 \leq a < m_\rho, 0 \leq c < m_\sigma$. The inverse of $\sigma + y$ is explicitly $\sum_{j=0}^{m_\sigma-1} (-1)^j y^j / \sigma^{j+1}$, as multiplication and cancellation of consecutive powers verify.

Write $m = m_\rho, n = m_\sigma$. In this component the ratio retains the exact identity

$$\zeta - \frac{\rho}{\sigma} = \frac{\sigma x - \rho y}{\sigma(\sigma + y)}. \quad (\text{KPC.20})$$

Every monomial of total degree $m + n - 1$ vanishes, since one exponent must reach m or n . Thus the $(m + n - 1)$ -st power of (KPC.20) is zero. Its preceding power has the exact value

$$\left(\zeta - \frac{\rho}{\sigma}\right)^{m+n-2} = \binom{m+n-2}{m-1} \frac{\sigma^{m-1}(-\rho)^{n-1}}{\sigma^{2(m+n-2)}} x^{m-1} y^{n-1} \neq 0. \quad (\text{KPC.21})$$

Indeed only the displayed monomial survives in the numerator power; in the inverse-denominator power only its constant coefficient can multiply that monomial without making it zero. The displayed coefficient is nonzero over $\mathbb{C}$ because $\rho, \sigma \neq 0$. For $m = n = 1$ the preceding power is the zeroth power one, and the first power is zero, so this case is included. The local nilpotent index is therefore exactly $m + n - 1$.

For any nonzero polynomial $P(Z)$, factor at the local ratio r as $P(Z) = (Z - r)^e B(Z)$ with $B(r) \neq 0$. The element $B(\zeta)$ is invertible: after division by its nonzero constant $B(r)$, it is one plus a nilpotent polynomial in $\zeta - r$, whose finite geometric series is an inverse. Hence $P(\zeta) = 0$ in that component exactly when $e \geq m + n - 1$. Across all ordered components, the condition at each coincident ratio is its maximum exponent, and conditions at distinct ratios are coprime. This proves that the global evaluation kernel is precisely (μ_ζ) , including the zero polynomial. It proves (KPC.16) and its stated image.

Each homogeneous monomial satisfies $s_1^j s_2^{N-j} = s_2^N \zeta^j$ inside the original algebra. Applying the original arithmetic unit proves (KPC.17). Both multiplication factors there are invertible, so its kernel is exactly the asserted restricted polynomial kernel. For $N < b$, no nonzero polynomial of degree at most N is divisible by the monic μ_ζ . For $N \geq b$, all such multiples are $\mu_\zeta \mathbb{C}[Z]_{\leq N-b}$, which has dimension $N - b + 1$, since multiplication by the nonzero polynomial is injective. Subtracting this dimension from $N + 1$ gives (KPC.18) in both cases. $\square$

In the retained monomial tensor coordinates, set $S_1 = A_h \otimes I$, $S_2 = I \otimes A_h$, $c = [1] \otimes [1]$, and $Z_{\text{mat}} = S_1 S_2^{-1}$. Then (KPC.17) is exactly

$$(c_0, \dots, c_N) \mapsto U_{v_h^{\otimes 2}} S_2^N \sum_{j=0}^N c_j Z_{\text{mat}}^j c. \quad (\text{KPC.22})$$

This retains each complete Taylor block under (KPC.19) rather than replacing either coordinate by its eigenvalue. Multiplication by $v_h^{\otimes 2} s_2^N$ is a linear isomorphism from $\mathbb{C}[\zeta]$ to its displayed image, with inverse multiplication by $s_2^{-N} (v_h^{\otimes 2})^{-1}$ on that image. The image after this multiplication is not asserted to be a unital subalgebra.

The original summed generator also has an exact map between adjacent shells. Write $A^{(2)} = M_{s_1+s_2}$ on the retained algebra and $U = v_h^{\otimes 2}$. Then

$$A^{(2)}(U s_2^N f(\zeta)) = U s_2^{N+1} (1 + \zeta) f(\zeta). \quad (\text{KPC.22a})$$

This follows from $s_1 + s_2 = s_2(1 + \zeta)$ and commutativity of the original multiplication maps. Multiplication by $1 + Z$ sends degree at most N to degree at most $N + 1$, and sends a multiple of μ_ζ to another such multiple. Therefore (KPC.22a) is well-defined on the complete source quotients in (KPC.17). It is an adjacent-shell action; it does not assert that $\mathbb{C}[\zeta]$ alone is invariant under $A^{(2)}$. For example, if the packet has two distinct centres ρ, σ , the diagonal pairs (ρ, ρ) and (σ, σ) both have ratio one, while the constant terms of their summed generators are 2ρ and 2σ . A polynomial in the ratio has the same constant term at both pairs, so it cannot equal that summed coordinate. Formula (KPC.22a) retains the missing second-coordinate multiplication explicitly.

Theorem 17.4 (Exact algebra-retraction criterion). *In the local algebra $R_{\rho,\sigma} = \mathbb{C}[x, y]/(x^m, y^n)$, with $\rho, \sigma \neq 0$, the ratio-subalgebra inclusion $\mathbb{C}[\zeta] \hookrightarrow R_{\rho,\sigma}$ admits a unital complex-algebra retraction exactly when $\min(m, n) = 1$. In the full original tensor algebra $E_h^{\otimes 2}$, the inclusion (KPC.16) admits a unital complex-algebra retraction exactly when h is squarefree.*

Proof. First suppose $m = 1$, so $x = 0$ and $\zeta = \rho/(\sigma + y)$. This element is a unit, and $y = \rho\zeta^{-1} - \sigma$. Its inverse belongs to the ratio subalgebra: $\zeta = r + w$, with $r = \rho/\sigma \neq 0$ and $w^n = 0$, has inverse $\sum_{j=0}^{n-1} (-1)^j w^j / r^{j+1}$. Thus the ratio subalgebra contains y and is the whole local algebra. If $n = 1$, then $y = 0$ and $x = \sigma\zeta - \rho$, giving the same conclusion. In both cases the inclusion is an isomorphism and its inverse is the asserted retraction.

Now suppose $m, n \geq 2$, and put $L = m + n - 1$. The exact local index proved above identifies the ratio subalgebra with $B = \mathbb{C}[t]/(t^L)$, by $t \mapsto \zeta - \rho/\sigma$. In any putative retraction $f : R_{\rho,\sigma} \rightarrow B$, the images of x and y have zero constant term because their powers vanish. If the coefficient of t in $f(x)$ were a nonzero number a , the coefficient of t^m in $f(x)^m$ would be $a^m \neq 0$. Since $m < L$, that term survives in B , contradicting $x^m = 0$. The same argument using $n < L$ shows that $f(y)$ also has zero linear coefficient. The exact expression (KPC.20) then makes $f(\zeta - \rho/\sigma)$ have zero linear coefficient: its numerator lies in (t^2) and its denominator is a unit. A retraction would send that element to t . This contradiction proves the local converse.

If h is squarefree, (KPC.19) identifies the full tensor algebra with $\mathbb{C}^{\mathbb{Z} \times \mathbb{Z}}$. The ratio subalgebra is the algebra of functions constant on every fibre of $(\rho, \sigma) \mapsto \rho/\sigma$. Indeed (KPC.15) has exponent one at each ratio, and ordinary polynomial interpolation gives every assignment of values on the distinct ratios. Choose one ordered pair in each such fibre. Restriction of a function to those chosen pairs, then assigning that value to the entire corresponding fibre, is a unital algebra map back to this subalgebra. It is the identity on the subalgebra, proving existence of a retraction.

Conversely, let $M = \max_{\rho} m_{\rho} \geq 2$. The ratio-one component in (KPC.15) is

$$B_1 = \mathbb{C}[t]/(t^{2M-1}), \quad \zeta \mapsto 1 + t. \quad (\text{KPC.22b})$$

This is because $\rho/\sigma = 1$ for nonzero centres exactly when $\rho = \sigma$, and the maximum of $2m_{\rho} - 1$ is $2M - 1$. Compose a proposed global retraction with projection onto B_1 . A unital map from the product (KPC.19) into B_1 selects exactly one local factor. To justify that statement, an idempotent of B_1 has constant term zero or one. If it is zero, the idempotent lies in the nilpotent ideal (t) , and an idempotent which is nilpotent is zero, since all its positive powers equal itself. If the constant term is one, apply this argument to one minus the idempotent. Hence the only idempotents are zero and one. The mutually orthogonal source-factor idempotents sum to one, so exactly one maps to one and all others map to zero, proving the selection statement.

Suppose the selected factor is indexed by (ρ, σ) . Images of its nilpotent coordinates have zero constant term, so the constant term of its ratio image is ρ/σ . Equation (KPC.22b) forces this ratio to be one; thus $\rho = \sigma$, with common multiplicity $m = m_{\rho} \leq M < 2M - 1$. The relation $x^m = 0$ again forces the coefficient of t in the image of x to vanish, since a nonzero coefficient would produce a nonzero t^m term in the target. The same holds for y . Their ratio minus one therefore has zero linear coefficient by (KPC.20), contradicting its required image t in (KPC.22b). This proves the global converse while retaining all source-factor idempotents and the original ratio embedding. $\square$

17.4 A complete finite witness for unequal shell ranks

The following positive polynomial model tests the proposed passage from homogeneous rank to the original orthogonal rank. It is not a claim that $1/2$ is an arithmetic zero. Use the specified measure

$$\nu_{\text{ex}} = \delta_{1/2-i} + \delta_{1/2} + \delta_{1/2+i} \quad \text{on the variable } s, \quad h_{\text{ex}}(s) = s - \frac{1}{2}, \quad v_{\text{ex}} = 1. \quad (\text{KPC.23})$$

All three masses are exactly one. Through degree two the monic orthogonal polynomials and norms are

$$\begin{array}{c|c|c} j & p_j(s) & \kappa_j \\ \hline 0 & 1 & 3 \\ 1 & s - \frac{1}{2} & 2 \\ 2 & (s - \frac{1}{2})^2 + \frac{2}{3} & \frac{2}{3}. \end{array} \quad (\text{KPC.24})$$

To verify the complete Gram, their three value columns are

$$(1, 1, 1)^T, \quad (-i, 0, i)^T, \quad (-1/3, 2/3, -1/3)^T.$$

Their pairwise Hermitian inner products are zero, and their squared norms are respectively 3, $1 + 0 + 1 = 2$, and $1/9 + 4/9 + 1/9 = 2/3$. Thus the moment form through degree two is positive definite, as required for the source and next layer used below.

Take $k = 2$, $N = 1$. Here $d = 1$ and the threshold $N \geq k(d - 1) = 0$ holds. The original tensor residue algebra is $E_{\text{ex}}^{(2)} = \mathbb{C}$, with $s_1 = s_2 = 1/2$, and its full jet map is evaluation at that ordered pair. For $c = c_{10}s_1 + c_{01}s_2$, the two section and image maps are exactly

$$\begin{aligned} i_1(c) &= c_{10}s_1 + c_{01}s_2, \\ O_1(c) &= c_{10}(s_1 - \tfrac{1}{2}) + c_{01}(s_2 - \tfrac{1}{2}), \\ L_1(c) &= -\tfrac{1}{2}(c_{10} + c_{01}), \\ A_1^{\text{hom}}(c) &= \tfrac{1}{2}(c_{10} + c_{01}), \quad A_1^{\text{orth}}(c) = 0. \end{aligned} \quad (\text{KPC.25})$$

Thus the lower coefficient in (KPC.7) cancels the entire homogeneous image. The ranks are one and zero. The kernels are $K_h = \{(c_{10}, c_{01}) : c_{10} + c_{01} = 0\}$ and $K_o = \mathbb{C}^2$, so the common quotient in (KPC.10) is zero and the full correspondence is $\mathbb{C} \oplus \{0\}$. The ratio is $\zeta = 1$, its minimal polynomial is $Z - 1$, and (KPC.18) gives the correct homogeneous rank $\min(2, 1) = 1$. Equation (KPC.25) proves explicitly why it gives no equality with the orthogonal rank.

The actual finite minimum and next layer in this model can also be retained. In the source with product measure $\nu_{\text{ex}}^{\otimes 2}$, the degree-at-most-one orthogonal basis is p_0p_0, p_1p_0, p_0p_1 , with norms 9, 6, 6 and jets 1, 0, 0. Hence the full interpolation kernel is $K_1 = 1/9$, the metric is $G_1 = 9$, and its minimum is the constant polynomial $R_1u = u$. Multiplication by the original sum $s_1 + s_2$ acts on the packet by $A = 1$. Consequently

$$B_1u = (s_1 + s_2 - 1)u \in \mathcal{L}_1, \quad C_1 = (I - P_{\mathcal{L}_1})B_1 = 0, \quad W_1 = A^*G_1 + G_1A - 2G_1 = 0. \quad (\text{KPC.26})$$

Indeed $\mathcal{L}_1$ is spanned by $s_1 - 1/2, s_2 - 1/2$, the complete degree-one relation kernel. Thus the derivative rank zero is verified directly without presupposing the shell-rank formula.

At the next shell, ordered as $(2, 0), (1, 1), (0, 2)$, the original jets and norm diagonal are

$$U = \begin{pmatrix} 2/3 & 0 & 2/3 \end{pmatrix}, \quad D_{\text{next}} = \text{diag}(2, 4, 2).$$

The actual layer vectors are $e_{20} = (s_1 - 1/2)^2$, $e_{11} = (s_1 - 1/2)(s_2 - 1/2)$, and $e_{02} = (s_2 - 1/2)^2$. Each has zero jet and is orthogonal to $\mathcal{L}_1$, either by its product moments or by the layer construction. Their Gram is exactly

$$H = D_{\text{next}} + U^*G_1U = \begin{pmatrix} 6 & 0 & 4 \\ 0 & 4 & 0 \\ 4 & 0 & 6 \end{pmatrix}. \quad (\text{KPC.27})$$

For example the diagonal squared norm of e_{20} is $(1 + 0 + 1)(1 + 1 + 1) = 6$, and its pairing with e_{02} is $(-1 + 0 - 1)^2 = 4$; the odd cross terms vanish. The eigenvalues of the outer two-coordinate block are 10, 2, and the middle eigenvalue is 4, proving positivity. The next full kernel and metric therefore are

$$K_2 = \frac{1}{9} + \frac{(2/3)^2}{2} + \frac{(2/3)^2}{2} = \frac{5}{9}, \quad G_2 = \frac{9}{5}. \quad (\text{KPC.28})$$

The representative itself is the explicit polynomial

$$R_2 u = \left[1 + \frac{3}{5} \left((s_1 - \frac{1}{2})^2 + (s_2 - \frac{1}{2})^2 \right) \right] u. \quad (\text{KPC.28a})$$

For a direct verification, it has jet u . Its values for u equal to one are $-1/5$ at the four product nodes where both coordinates differ from $1/2$, $2/5$ at the four nodes where exactly one differs, and one at the central node. Its squared norm is therefore $4/25 + 16/25 + 1 = 9/5$. It is orthogonal to the five relation basis vectors $s_1 - 1/2, s_2 - 1/2, e_{20}, e_{11}, e_{02}$. The first two and e_{11} have an odd coordinate, so their pairings vanish by the explicit sign symmetry of the nine-node measure. For e_{20} , only the two noncentral first-coordinate values contribute, and for each of them the sum of the three values of $R_2(1)$ is $-1/5 + 2/5 - 1/5 = 0$. Its pairing is thus zero; the same computation with the coordinates interchanged treats e_{02} . These five independent zero-jet polynomials span the kernel in the six-dimensional degree-at-most-two source, whose jet map is onto the one-dimensional target. Every other feasible polynomial differs by a vector in this kernel. Pythagoras and positive definiteness therefore prove the unique-minimum assertion directly, including its actual norm. Thus the nonzero original relation layer and its positive metric loss are retained alongside the zero derivative layer. This complete finite calculation rejects only the proposed equality between homogeneous and orthogonal-shell ranks. The original arithmetic shell continues to be given by (KPC.4)–(KPC.7), and the exact graph and quotient maps relating its two presentations are (KPC.9)–(KPC.12).

Every linear map $T : V \rightarrow W$ in these comparisons has a support-preserving Split-Zero lift on the following specified carriers at a fixed support λ :

$$\widehat{V}_\lambda = \{\tau_V\} \sqcup (\{\lambda\} \times V), \quad \widehat{W}_\lambda = \{\tau_W\} \sqcup (\{\lambda\} \times W).$$

Addition has external τ as identity and uses vector addition inside the supported fibre. The scalar action is $r^\bullet(\lambda, v) = (\lambda, rv)$, while the external scalar zero sends every input to external absence and every scalar sends external absence to external absence. The lift maps τ_V to τ_W and (λ, v) to (λ, Tv) . Its addition identity follows from $T(v + w) = Tv + Tw$; its supported-scalar identity follows from $T(rv) = rTv$, including $r = 0$. The external cases follow from the definitions. Thus every displayed kernel vector maps to the represented zero at λ , and the graph, quotient and lower-degree maps retain this support even when the two ranks differ.

18 The original frontier energy, determinant growth, and the entire border fibre

The symmetric-frontier delivery of 12 September computes coordinates on the actual next relation layer. We retain these coordinates and prove their exact relation to the accumulated source energy of Section 15. The calculation also identifies the entire zero fibre of the delivery's smaller border operator, including its nilpotent extension.

18.1 Original moment coordinates and the relation graph

Keep a nonempty full-order packet h , its degree $d, g = 2\xi, E = \mathbb{C}[s]/h, v = j_h(g/h)$, and $A = M_s$. The positive measure remains

$$d\nu_h(t) = |g(1/2 + it)/h(1/2 + it)|^2 \frac{dt}{2\pi}.$$

Let p_n be the monic orthogonal polynomial obtained from $1, s, \dots, s^n$ without dividing by its norm, and let $\omega_n = \|p_n\|_{\nu_h}^2 > 0$. The original generator has the exact recurrence

$$sp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad a_n = \omega_n / \omega_{n-1} \ (n \geq 1), \quad a_0 = 0, \quad b_n + \bar{b}_n = 1. \quad (\text{FE.1})$$

Here $p_{-1} = 0$. To verify the coefficients, multiplication by s has adjoint multiplication by $1 - s$ on the original line. Its pairing against degrees below $n - 1$ vanishes, the leading coefficient is one, and the pairing against p_{n-1} is $-\omega_n$. Its diagonal is $\int s|p_n|^2 d\nu_h/\omega_n$, which proves the last identity and retains its possibly nonzero imaginary part.

Fix $k \geq 1$ and $M \geq k(d - 1)$, and use $A_k = \sum_j A_j$ on the unchanged $E_k = E^{\otimes k}$. For $\alpha \in \mathbb{N}^k$, put

$$p_\alpha = \prod_j p_{\alpha_j}(s_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = v^{\otimes k}[p_\alpha] \in E_k.$$

All remainder coordinates, including every nilpotent jet and the entire local arithmetic unit, are retained in z_α . The original source map $\mathcal{T}^{(k)}$ is the map in (CT.11)–(CT.13). Its orthogonal polynomial images have norms ω_α . The constrained minimum gives exactly

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}, \quad R_M u = \sum_{|\alpha| \leq M} \mathcal{T}^{(k)} p_\alpha \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (\text{FE.2})$$

This is (CT.5) in the explicitly declared orthogonal source basis; no moment mass or polynomial norm has been changed.

Order the indices β of total degree $M + 1$, and put

$$r = \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\omega_\beta), \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \quad T_+ = [\mathcal{T}^{(k)} p_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k.$$

The two source maps have the same column ordering. Since the image of T_+ is orthogonal to the entire degree- M image,

$$b = T_+ - R_M F : \mathbb{C}^r \rightarrow \mathcal{E}_M, \quad H = b^* b = \Omega + F^* G_M F, \quad b^* R_M = -F^* G_M. \quad (\text{FE.3})$$

Here $\mathcal{E}_M = \mathcal{B}_{M+1} \cap \mathcal{B}_M^\perp$ is the original relation layer. The complete jets of b vanish; both of its terms are orthogonal to $\mathcal{B}_M$. Taking the highest orthogonal degree of bu recovers $T_+ u$, proving injectivity. For any boundary of degree $M + 1$, subtract the image under b of its highest coefficient. The difference lies in $\mathcal{B}_M$. This gives the inverse on the orthogonal layer and proves surjectivity. It also gives an original primitive: divide the numerator of each column of b in the fixed order by $h(s_1), \dots, h(s_k)$. The primitive in factor j has coefficient $(-1)^{j-1}$; its tensor differential contributes the same sign, giving the original positive boundary.

Define the second frontier matrix by

$$E_+ = [w_\beta], \quad w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}. \quad (\text{FE.4})$$

The notation E_+ denotes a matrix $\mathbb{C}^r \rightarrow E_k$. From (FE.2), the highest coefficient of $D^{(k)} R_M - R_M A_k$ is $\Omega^{-1} E_+^* G_M$. Projection onto the graph (FE.3) therefore proves, with the original types and signs,

$$Y_M = -b H^{-1} F^* G_M, \quad C_M = b \Omega^{-1} E_+^* G_M, \quad \Delta_M = Y_M^* Y_M = G_M F H^{-1} F^* G_M. \quad (\text{FE.5})$$

The notation C_M is the original derivative-layer map in (CT.15), not the inverse interpolation matrix $\mathcal{C}_M$.

Applying (FE.1) to (FE.2) cancels every internal neighbouring pair, because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonals give exactly $k C_M$. The remaining pairs have top degree $M + 1$, proving

$$\mathcal{Z}_M = A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{FE.6})$$

Alternatively (FE.5) inserted into $W_M = -(Y_M^* C_M + C_M^* Y_M)$ gives exactly (FE.6). Thus both the polynomial cancellation and its actual boundary primitive yield the same original form.

18.2 Exact relative loss and the source derivative energy

For the next formulas only, use the displayed finite isometries from the specified metric coordinate spaces and define

$$U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}, \quad P = U U^*, \quad K_U = U^* U. \quad (\text{FE.7})$$

The original F, E_+, Ω, G_M are still given by (FE.1)–(FE.5), and can be recovered by the inverse coordinate maps in (FE.7). In particular no polynomial is assigned norm one and no arithmetic coefficient is suppressed.

Theorem 18.1 (Frontier energy and every relative Gram eigenvalue). *In these explicitly related coordinates,*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = U V^* + V U^*, \\ T_M &= G_M^{-1/2} \Delta_M G_M^{-1/2} = U (I + K_U)^{-1} U^* = P (I + P)^{-1}, \\ G_M^{-1/2} G_{M+1} G_M^{-1/2} &= (I + P)^{-1}. \end{aligned} \quad (\text{FE.8})$$

If $\lambda_1, \dots, \lambda_q > 0$ are all positive eigenvalues of P , including multiplicity, then

$$\begin{aligned} q &= \text{rank } F = \text{rank } Y_M = \text{rank } \Delta_M, \\ \tau_M &= \text{Tr}(G_M^{-1} \Delta_M) = \sum_{i=1}^q \frac{\lambda_i}{1 + \lambda_i}, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M} = \prod_{i=1}^q (1 + \lambda_i)^{-1}, \\ \chi_M &= \text{Tr}(G_M^{-1} C_M^* C_M) = \|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{FE.9})$$

The empty product equals one if $q = 0$. In addition $\text{rank } C_M = \text{rank } E_+$.

Proof. Equation (FE.6) proves the first line of (FE.8). The identity $H = \Omega^{1/2}(I + U^* U) \Omega^{1/2}$ substituted into (FE.5) proves the second line's first expression. The equality $(I + U U^*) U = U (I + U^* U)$, multiplied by the indicated invertible factors, identifies it with $P (I + P)^{-1}$. Subtract it from I to get the third line. The positive matrices $I + K_U$ and $I + P$ are invertible directly from their quadratic forms.

The graph map b is injective, and H, Ω, G_M are invertible, so (FE.5) proves both asserted ranks. Equivalently, the formula for T_M identifies its positive eigenvalues with $\lambda_i / (1 + \lambda_i)$, retaining zeros on $\ker P$. The trace and determinant formulas now follow from the finite spectral decomposition in these same coordinates. Finally

$$C_M G_M^{-1/2} = b \Omega^{-1/2} V^*, \quad (b \Omega^{-1/2})^* (b \Omega^{-1/2}) = I + U^* U.$$

Its squared Hilbert–Schmidt norm is $\text{Tr}(V(I + U^* U)V^*)$, which expands to the last line of (FE.9). This is the energy of the original source-layer map for its inherited $L^2(d^k x)$ inner product. $\square$

The exact remainder in the accumulated determinant calculation is therefore also a frontier formula:

$$-\log \pi_M - \tau_M = \sum_{i=1}^q \left(\log(1 + \lambda_i) - \frac{\lambda_i}{1 + \lambda_i} \right) = \sum_{i=1}^q \int_0^{\lambda_i} \frac{x}{(1+x)^2} dx. \quad (\text{FE.10})$$

Differentiate each displayed scalar expression and evaluate at zero to prove the equality, including every eigenvalue and its sign. Thus (JD.14) uses the exact original derivative cost $\|V\|_{\text{HS}}^2 + \|V U^*\|_{\text{HS}}^2$, and (JD.16) is precisely the pair of identities

$$\sum_{M \geq a} \|Y_M\|_{\text{HS}}^2 = \text{Tr } G_a, \quad \sum_{M \geq a} \sum_i \frac{\lambda_{M,i}}{1 + \lambda_{M,i}} = \infty. \quad (\text{FE.11})$$

Both use the already proved actual boundary density; the second retains its full inverse-interpolation cost.

18.3 The entire small border operator, including its zero extension

Let $n = d^k$, and define

$$B = [U, V] : \mathbb{C}^{2r} \rightarrow \mathbb{C}^n, \quad K = B^*B, \quad J_0 = \begin{pmatrix} 0 & I_r \\ I_r & 0 \end{pmatrix}, \quad A_0 = J_0 K, \quad H_0 = K^{1/2} J_0 K^{1/2}. \quad (\text{FE.12})$$

The symbol A_0 here is a finite border operator, not the arithmetic generator A or A_k . This distinction is accompanied by the exact maps below. The source's unscaled border operator is

$$J[F, E_+]^* G_M[F, E_+], \quad J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

With $D_\Omega = \text{diag}(\Omega^{1/2}, \Omega^{1/2})$, it is exactly $D_\Omega^{-1} A_0 D_\Omega$: substitution of (FE.7) gives

$$[F, E_+]^* G_M[F, E_+] = D_\Omega K D_\Omega.$$

This states the full coordinate similarity, including zero.

Put $N = \ker K = \ker B$ and $\mathcal{R} = \text{ran } K$. They are orthogonal complements since K is Hermitian, and $K^{1/2}$ restricts to an invertible positive map on $\mathcal{R}$. There is an exact sequence of spaces with their specified operators:

$$0 \longrightarrow (N, 0) \longrightarrow (\mathbb{C}^{2r}, A_0) \xrightarrow{K^{1/2}} (\mathcal{R}, H_0|_{\mathcal{R}}) \longrightarrow 0. \quad (\text{FE.13})$$

Indeed $K^{1/2} A_0 = H_0 K^{1/2}$, its kernel is N , and its image is $\mathcal{R}$. On N , $A_0 = 0$. The map

$$Q|_{\mathcal{R}} = B(K^{1/2}|_{\mathcal{R}})^{-1}, \quad Q|_N = 0 \quad (\text{FE.14})$$

is a partial isometry onto $\text{ran } B$: for $x, y \in \mathcal{R}$, substitution of $B^*B = K$ gives $\langle Qx, Qy \rangle = \langle x, y \rangle$. It satisfies $B = QK^{1/2}$ and

$$S_M = B J_0 B^* = Q H_0 Q^*, \quad S_M|_{(\text{ran } B)^\perp} = 0. \quad (\text{FE.15})$$

Thus the restriction of H_0 to $\mathcal{R}$ and that of S_M to $\text{ran } B$ are unitarily intertwined by an explicitly proved map; both remaining orthogonal complements and their zero actions are specified.

Theorem 18.2 (Exact zero fibre of the border comparison). *Let $Z_0 = \ker(H_0|_{\mathcal{R}})$ and $q_0 = \dim Z_0$. The complete generalized zero eigenspace of A_0 is*

$$\mathcal{N}_0 = (K^{1/2})^{-1}(Z_0), \quad A_0^2|_{\mathcal{N}_0} = 0. \quad (\text{FE.16})$$

Define $L = (K^{1/2}|_{\mathcal{R}})^{-1} : \mathcal{R} \rightarrow \mathcal{R}$. For its restriction to $z \in Z_0$, set

$$Lz = (K^{1/2}|_{\mathcal{R}})^{-1}z, \quad N_0 z = J_0 K^{1/2} z \in N. \quad (\text{FE.17})$$

The map $N_0 : Z_0 \rightarrow N$ is injective, and the exact vector space decomposition $\mathcal{N}_0 = N \oplus LZ_0$ gives

$$A_0(n + Lz) = N_0 z. \quad (\text{FE.18})$$

Consequently its zero Jordan data consist of exactly q_0 blocks of size two and $\dim N - q_0$ blocks of size one. Its nonzero Jordan blocks have size one and coincide, with full multiplicities, with the nonzero eigenvalues of S_M .

Proof. If $A_0^m x = 0$, (FE.13) gives $H_0^m K^{1/2} x = 0$. Hermitian spectral decomposition of $H_0|_{\mathcal{R}}$ identifies its generalized zero space with Z_0 , proving that x belongs to the preimage in (FE.16). Conversely, if $K^{1/2} x \in Z_0$, then $K^{1/2} A_0 x = 0$, so $A_0 x \in N$ and $A_0^2 x = 0$. This proves (FE.16) exactly.

For $z \in Z_0$, one has $K^{1/2}J_0K^{1/2}z = H_0z = 0$, proving $N_0z \in N$. Since J_0 is invertible and $K^{1/2}$ is injective on $\mathcal{R}$, N_0 is injective. Equation $K^{1/2}Lz = z$ gives the direct decomposition, and $A_0Lz = J_0KLz = J_0K^{1/2}z$ proves (FE.18). Choose a basis of Z_0 , use its injective image as part of a basis of N , and extend it in N . Each pair Lz, N_0z is a size-two chain, and the remaining basis vectors of N are size-one chains. This also proves $q_0 \leq \dim N$ without a rank assumption.

For a nonzero eigenvalue λ , take an eigenvector $z \in \mathcal{R}$ of H_0 . If $x = Lz$, then $n = A_0x - \lambda x \in N$, so $x + n/\lambda$ is an eigenvector of A_0 lifting z . The lift is unique because A_0 vanishes on N and $\lambda \neq 0$. This supplies every eigenvector of the Hermitian quotient and leaves no nontrivial Jordan chain at a nonzero eigenvalue: applying (FE.13) to such a chain would give one for the Hermitian quotient, and a residual chain contained in N is impossible at nonzero λ . Finally (FE.15) identifies those eigenvalues with the actual relative form. All claims include their algebraic multiplicities. $\square$

In particular,

$$\dim \ker S_M = n - \text{rank } B + q_0, \quad \dim \ker A_0 = 2r - \text{rank } B, \quad \text{mult}_0(A_0) = 2r - \text{rank } B + q_0. \quad (\text{FE.19})$$

The first equality is (FE.15), the second uses the invertibility of J_0 , and the third counts (FE.18). Thus replacing the source's border calculation by a Hermitian one has an exact kernel, quotient, complement and zero extension, all explicitly retained here. At any fixed retained support λ , each displayed linear map has its Split-Zero lift $(\lambda, x) \mapsto (\lambda, Tx)$, with external $\tau \mapsto \tau$. A size-two chain in (FE.18) is therefore the exact supported chain

$$(\lambda, Lz) \mapsto (\lambda, N_0z) \mapsto (\lambda, 0). \quad (\text{FE.19a})$$

The last value is the represented zero in that fibre. For $z \neq 0$, injectivity of N_0 proves that the intermediate value has nonzero amplitude; neither map sends it to external absence.

18.4 A quantified link between the frontier bound and determinant cost

Let $\lambda_M = \|UU^*\|_{\text{op}}$, and retain the original positive recurrence quantity

$$\Gamma_M = \max_{|\alpha|=M} \sum_{j=1}^k \{a_{\alpha_j+1} + (k-1)a_{\alpha_j}\} > 0. \quad (\text{FE.20})$$

The strict inequality follows from $a_{\alpha_j+1} > 0$. For clarity, its upper estimate uses an exact matrix comparison. Let $Z = [z_\alpha]_{|\alpha|=M}$, $\Omega_M = \text{diag}(\omega_\alpha)$, and let $L_{\alpha\beta} = a_{\beta_j}$ for $\alpha = \beta - e_j$, summed over those j . Then $E_+ = ZL$, and $\Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is the displayed sum in (FE.20): $\omega_\alpha a_{\alpha_j+1} / \omega_{\alpha+e_j} = 1$ for each outgoing edge, and the other predecessors contribute $\sum_{l \neq j} a_{\alpha_l}$. For a matrix with nonnegative entries, an eigenvector component of largest absolute value bounds the eigenvalue's absolute value by its largest row sum. The matrix is similar to the positive Hermitian matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$, so

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_M Z \Omega_M^{-1} Z^* \preceq \Gamma_M \mathcal{C}_M.$$

It follows that $VV^* \preceq \Gamma_M I$. For any vector x , Cauchy-Schwarz applied to (FE.8) gives $|x^* S_M x| \leq 2\|U^* x\| \|V^* x\|$. Thus

$$\epsilon_M \leq 2\sqrt{\Gamma_M \lambda_M}. \quad (\text{FE.21})$$

Every actual eigenline survives in the full tensor family. If $\delta_* = \max_{\rho \in Z} |\text{Re } \rho - 1/2|$, its repeated tensor gives $\epsilon_M \geq 2k\delta_*$. Combining this with (FE.9) and (FE.21) proves, at every finite degree,

$$\lambda_M \geq \frac{k^2 \delta_*^2}{\Gamma_M}, \quad \frac{\det G_M}{\det G_{M+1}} \geq 1 + \frac{k^2 \delta_*^2}{\Gamma_M}. \quad (\text{FE.22})$$

Indeed the determinant inverse is $\prod_i (1 + \lambda_i)$, which is at least $1 + \lambda_M$. For integers $k(d-1) \leq a < b$, multiplication, followed by the increasing logarithm, gives the explicit source constraint

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{k^2 \delta_*^2}{\Gamma_M} \right). \quad (\text{FE.23})$$

When $\delta_* = 0$, this remains a valid inequality with zero right side and no division by a zero spectral quantity. It is an exact necessary growth relation for the constructed arithmetic matrices, not an assumed upper estimate on that growth.

The original derivative energy supplies a second exact growth bound without the recurrence maximum. By (FE.8), $\|Y_M G_M^{-1/2}\|_{\text{op}}^2 = \lambda_M / (1 + \lambda_M)$. By (FE.9), $\|C_M G_M^{-1/2}\|_{\text{op}}^2 \leq \chi_M$. The original cross-pairing formula, the operator triangle inequality, and submultiplicativity therefore give

$$\epsilon_M^2 \leq 4 \frac{\lambda_M}{1 + \lambda_M} \chi_M. \quad (\text{FE.24})$$

If $\delta_* > 0$, combine it with the retained eigenline $\epsilon_M \geq 2k\delta_*$. The finite number $\lambda_M / (1 + \lambda_M)$ is strictly below one, so

$$\chi_M > k^2 \delta_*^2, \quad \lambda_M \geq \frac{k^2 \delta_*^2}{\chi_M - k^2 \delta_*^2}, \quad \log \frac{\det G_M}{\det G_{M+1}} \geq -\log \left(1 - \frac{k^2 \delta_*^2}{\chi_M} \right). \quad (\text{FE.25})$$

Every denominator and logarithm in (FE.25) has just been proved valid. The middle inequality follows by rearranging (FE.24); the last uses $\det G_M / \det G_{M+1} \geq 1 + \lambda_M$. Summing it over any $k(d-1) \leq a < b$ gives the corresponding finite accumulated determinant inequality. For $\delta_* = 0$, (FE.24) remains valid, and (FE.25) is left un-divided; no 0/0 value is assigned to an original zero-energy layer.

For the actual conjugation-and-reflection packets, the further exact original-layer calculation in Section 21 proves $\epsilon_M^2 \leq [\lambda_M / (1 + \lambda_M)] \zeta_M \leq [\lambda_M / (1 + \lambda_M)] \chi_M$. Here ζ_M is the operator-norm square of the same derivative map whose trace cost is χ_M . Its proof uses the two actual reflection parities in that layer. For $k = 1$, it proves the exact equality $\epsilon_M^2 = (1 - \pi_M) \chi_M$, retaining both zero frontier cases. Thus (FE.24)–(FE.25) retain their general hypotheses, while (QD.8)–(QD.15) give the stronger arithmetic conclusions and all their finite accumulated determinant bounds.

19 The signed symmetric summand and its complete spectral cost

This section proves the cochain, coordinate, and spectral comparisons for the symmetric summand in the arithmetic frontier delivery. The original packet is $E = \mathbb{C}[t]/h$, of dimension $d \geq 1$, and its operator is $A = M_t$. The tensor degree is $k \geq 1$. All orders in $h = \prod_\rho (t - \rho)^{m_\rho}$ remain present. The original joint representative, metric, and layer maps of (CT.12)–(CT.16) are denoted by R_M, G_M, Y_M, C_M , with $M \geq k(d-1)$. The source's symmetric construction is used through the maps proved below; its orbit sums keep their literal factorials.

19.1 The chain projector and its completed and finite targets

The completion in this subsection is specifically the completed projective tensor product $\widehat{\otimes}_\pi$ of the strong-Schwartz locally convex spaces. The global-retraction source, equations (16), (19), and (24), constructs continuous maps

$$\Lambda : \mathcal{B} \longrightarrow V, \quad \Lambda \Theta = 1, \quad s : Q \longrightarrow \mathcal{B}, \quad s[F] = (1 - \Theta \Lambda) F, \quad qs = 1. \quad (\text{SS.1})$$

Its specified cutoffs and moment projection occur in $\Lambda = P_V (1 - A_\alpha B_\beta)^{-1} Y_{\alpha, \beta}$. The convergence and continuity of this formula are proved in that source; no Hilbert-space replacement of these topologies

is used here. Put $C = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, $P = sq$ as a cochain map with degree-zero part zero, and $H^1 = \Lambda$. Then in the two degrees respectively, $\Lambda\Theta = 1$ and $\Theta\Lambda = 1 - sq$ prove

$$dH + Hd = 1 - P. \quad (\text{SS.2})$$

For homogeneous inputs define the chain permutation by

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i||x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (\text{SS.3})$$

Here the degrees in the exponent belong to the original inputs. An adjacent interchange contributes $(-1)^{ab}$ for degrees a, b . Interchanging twice contributes $(-1)^{2ab} = 1$; disjoint interchanges commute; and the two three-interchange braid words both contribute $(-1)^{ab+ac+bc}$. Thus (SS.3) is a representation of S_k . On two adjacent factors, $d(x \otimes y) = dx \otimes y + (-1)^a x \otimes dy$. After interchange the coefficients of $dy \otimes x$ and $y \otimes dx$ are respectively $(-1)^{ab}$ and $(-1)^{ab+b}$, on either side of the cochain equation. This proves $dT_\pi = T_\pi d$, first for generators and then for every permutation. Each map is continuous and therefore extends to $C^{\widehat{\otimes}_\pi k}$.

The signed average and its complementary map are

$$S_k = \frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi, \quad 1 - S_k. \quad (\text{SS.4})$$

For each γ , exactly $k!$ pairs (π, σ) satisfy $\pi\sigma = \gamma$, and their signed coefficient is $\text{sgn}(\gamma)/(k!)^2$. Hence $S_k^2 = S_k$. In top degree every factor has degree one, so $T_\pi = \text{sgn}(\pi)P_\pi$ and S_k is the ordinary symmetric projection there. The unsigned chain average has top part $k!^{-1} \sum_{\pi} \text{sgn}(\pi)P_\pi$; this exact equation specifies its alternating target.

To verify the completed cohomology directly, extend the finite sum

$$H_k = \sum_{j=1}^k P^{\widehat{\otimes}_\pi(j-1)} \widehat{\otimes}_\pi H \widehat{\otimes}_\pi 1^{\widehat{\otimes}_\pi(k-j)}. \quad (\text{SS.5})$$

The degree -1 factor contributes the Koszul sign $(-1)^{\sum_{i < j} |x_i|}$. Expanding $dH_k + H_k d$, the terms in which the differential and homotopy occupy different slots cancel in pairs. The term in slot j is $P^{\widehat{\otimes}(j-1)} \otimes (1 - P) \otimes 1^{\widehat{\otimes}(k-j)}$ by (SS.2). Summing telescopes to $1 - P^{\widehat{\otimes} k}$. Continuity extends this identity from the dense algebraic tensor product. Replace H_k by the explicit equivariant homotopy

$$\overline{H}_k = \frac{1}{k!} \sum_{\pi} T_\pi H_k T_\pi^{-1}. \quad (\text{SS.6})$$

The operator $P^{\widehat{\otimes} k}$ commutes with every permutation, so (SS.6) has the same homotopy identity and commutes with S_k . The maps $q^{\widehat{\otimes}_\pi k}$ and $s^{\widehat{\otimes}_\pi k}$, restricted to the images of (SS.4), now give continuous inverse cohomology maps. Thus

$$H^r(\text{im } S_k) = \begin{cases} (Q^{\widehat{\otimes}_\pi k})^{S_k}, & r = k, \\ 0, & r \neq k. \end{cases} \quad (\text{SS.7})$$

The complementary cohomology is the image of $1 - \frac{1}{k!} \sum_{\pi} P_\pi$ on the same completed Q -tensor. The full cochain isomorphism and its inverse are $x \mapsto (S_k x, (1 - S_k)x)$ and $(u, v) \mapsto u + v$. Equations (SS.5)–(SS.7) prove the completion statement using the supplied continuous contraction; they require no exactness assertion for arbitrary completed tensor products.

Here is the finite-rank cochain map that supports the finite traces. The full packet jet $J_h : \mathcal{B} \rightarrow E$ annihilates ΘV , and descends to a continuous $j_h : Q \rightarrow E$. The original packet inclusion $\sigma_h : E \rightarrow Q$ satisfies $j_h \sigma_h = 1$, as $J_h R_M = 1$ and $q R_M = \sigma_h$ in one variable. Set $\iota_h = s \sigma_h : E \rightarrow \mathcal{B}$ and define

$$\Pi_h^0 = 0, \quad \Pi_h^1 = \iota_h J_h, \quad J_h \iota_h = 1. \quad (\text{SS.8})$$

This is a continuous finite-rank cochain projector: $\Pi_h^1 \Theta = 0$ and $(\Pi_h^1)^2 = \Pi_h^1$. It commutes with permutations after taking identical tensor powers. Consequently $\Pi_h^{\otimes k} S_k$ and $\Pi_h^{\otimes k} (1 - S_k)$ are the finite cochain projectors with ranks $\binom{k+d-1}{d-1}$ and $d^k - \binom{k+d-1}{d-1}$. The projector S_k on the full infinite complex itself has not been assigned either finite rank. The finite inclusion and retraction on cohomology are precisely $\sigma_h^{\otimes k}$ and $j_h^{\otimes k}$, restricted to the respective permutation summands. Their composite on the packet is the identity.

For $U(a) = e^{(\log a)A}$, $a > 0$, the arithmetic action obeys $J_h U_{\mathcal{B}}(a) = U(a) J_h$. Thus $J_h U_{\mathcal{B}}(a) \iota_h = U(a)$. If $L : F \rightarrow X$, $R : X \rightarrow F$, and F is finite dimensional, the trace of the finite-rank map LR is $\text{Tr}_F(RL)$: expanding LRx in a basis of F expresses both as the sum of the same coefficient functionals evaluated on the same image vectors. Apply this identity to (SS.8) and its tensor powers. Expanding a permutation trace in a tensor basis contracts one matrix index around each cycle, proving

$$\text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|}). \quad (\text{SS.9})$$

This proof uses all entries of $U(a)$ and applies to its nonsemisimple action. The cochain supertrace carries $(-1)^k$. The complementary finite supertrace is exactly $(-1)^k$ times $(\text{Tr } U(a))^k$ minus the trace in (SS.9). No infinite trace is assigned by these equations.

At a fixed permutation-invariant support fibre, every displayed linear map lifts by $(\lambda, x) \mapsto (\lambda, Tx)$. For arbitrary masks the support of (SS.4) is the specified orbit $\text{join } c(\lambda) = \bigvee_{\pi} \pi \lambda$, and the amplitude maps include the original transports into that fibre. The two projection maps then decompose the transported vector in the fibre over $c(\lambda)$; their sum is that transport. Their sum has not been declared the identity on a different original fibre. The upstream kernel is exactly the kernel of its specified transport $V_{\lambda} \rightarrow V_{c(\lambda)}$. At invariant masks the transport is the identity and the decomposition above applies. Every vanishing projected amplitude remains the supported zero in its target fibre; external absence is sent to external absence.

19.2 Orbit coordinates, their Gram, and all local nilpotents

Let $e_1, \dots, e_d$ denote the original power basis with indices shifted by one. For $n \in \mathbb{N}^d$, $|n| = k$, let b_n be the sum of the distinct basis tensors having occupancy n . Define

$$N = \binom{k+d-1}{d-1}, \quad I : \mathbb{C}^N \longrightarrow E^{\otimes k}, \quad I e_n = b_n, \quad D_o = I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, \quad L = D_o^{-1} I^*. \quad (\text{SS.10})$$

Distinct tensor basis vectors are orthonormal for this coordinate calculation; hence D_o records exactly their orbit sizes. It is separate from the arithmetic Gram. One has $LI = 1$ and $IL = P_{\text{sym}} = k!^{-1} \sum P_{\pi}$, since either map replaces every coefficient on an orbit by its average. Here $\text{Sym}^k E$ denotes this invariant image. Its comparison with the symmetric tensor quotient is the quotient map restricted to that image: it sends b_n to $(k! / \prod_i n_i!) e_1^{n_1} \cdots e_d^{n_d}$. The inverse sends this quotient monomial to $(\prod_i n_i! / k!) b_n$. These formulas prove the isomorphism and retain the exact diagonal factors between the two presentations.

Permutation invariance of the arithmetic constraints and measure makes the unique constrained minimum satisfy $R_M P_{\pi} = P_{\pi} R_M$; applying the inverse permutation to a feasible minimizer gives a feasible vector of the same norm, and uniqueness identifies it. Thus G_M and G_M^{-1} commute with P_{sym} . Retain the matrices

$$G_M^s = I^* G_M I, \quad W_M^s = I^* W_M I, \quad A^s = L A^{(k)} I, \quad A^{(k)} I = I A^s, \quad (G_M^s)^{-1} = L G_M^{-1} L^*. \quad (\text{SS.11})$$

The last equality includes both D_o^{-1} factors in L . Indeed its product with $I^* G_M I$ is

$$L G_M^{-1} (L^* I^*) G_M I = L G_M^{-1} P_{\text{sym}} G_M I = LI = 1.$$

This proves the inverse formula without exchanging arbitrary compression and inversion.

In these unscaled coordinates, if $Ae_j = \sum_i A_{ij}e_i$, the full generator is

$$A^{(k)}b_n = \left(\sum_i n_i A_{ii} \right) b_n + \sum_{\substack{i \neq j \\ n_j > 0}} (n_i + 1) A_{ij} b_{n+e_i-e_j}. \quad (\text{SS.12})$$

For a specified target tensor of occupancy $n + e_i - e_j$, each of its $n_i + 1$ entries equal to e_i can have come from an e_j in a source tensor, and these are exactly the contributing preimages. The diagonal term counts its n_i unchanged positions. This proves every coefficient in (SS.12). Using the original quotient's complete CRT isomorphism, set $E_\rho = \mathbb{C}[x_\rho]/x_\rho^{m_\rho}$, $t = \rho + x_\rho$. The explicit CRT map is reduction modulo each $(t - \rho)^{m_\rho}$; its inverse is the unique degree $< d$ polynomial with the prescribed full jets. Applying its tensor power and restriction through I, L is the exact change from the original orbit coordinates to the following direct sum:

$$\text{Sym}^k E \simeq \bigoplus_{\substack{\ell_\rho \geq 0 \\ \sum_\rho \ell_\rho = k}} \bigotimes_\rho \text{Sym}^{\ell_\rho} E_\rho, \quad w_\ell = \prod_\rho \binom{\ell_\rho + m_\rho - 1}{m_\rho - 1}. \quad (\text{SS.13})$$

The isomorphism groups the unscaled tensor orbit sums by the number of factors in each CRT summand; these grouped basis vectors exhaust the full basis, which proves the direct sum and its dimensions.

On the summand ℓ , the generator is $\lambda_\ell + N_\ell$, where $\lambda_\ell = \sum_\rho \ell_\rho \rho$ and N_ℓ is the sum of the multiplications by the x_ρ in their original slots. Put $h_\ell = \sum_\rho \ell_\rho (m_\rho - 1)$. Every application raises the total local exponent by one, so $N_\ell^{h_\ell+1} = 0$. On the tensor of local constants,

$$N_\ell^{h_\ell} \left(\bigotimes_\rho 1^{\otimes \ell_\rho} \right) = \frac{h_\ell!}{\prod_\rho ((m_\rho - 1)!)^{\ell_\rho}} \bigotimes_\rho (x_\rho^{m_\rho-1})^{\otimes \ell_\rho} \neq 0. \quad (\text{SS.14})$$

The coefficient counts the ordered choices of increments in each slot, each of which must be incremented exactly $m_\rho - 1$ times. Therefore the nilpotent index is exactly $h_\ell + 1$. Even when different ℓ have the same λ_ℓ , their direct summands in (SS.13) are retained separately. Traces may kill individual nilpotent diagonal contributions, while the complete operators (SS.12)–(SS.14) retain them.

19.3 Complete symmetric spectral sums and their exact departure

List $\rho_1, \dots, \rho_d$ with every algebraic multiplicity and put

$$\delta_i = \text{Re } \rho_i - \frac{1}{2}, \quad \sigma_1 = \sum_i \delta_i, \quad \sigma_2 = \sum_i \delta_i^2, \quad a_k = \binom{k+d-1}{d}, \quad b_k = \binom{k+d-1}{d+1}. \quad (\text{SS.15})$$

Binomial coefficients with lower index exceeding the upper index are zero. Thus these formulas include $k = 1$ and $d = 1$. Triangularize A . The induced ordered orbit basis is also triangular under (SS.12), with diagonal $\lambda_n = \sum_i n_i \rho_i$, once for each $n \in \mathbb{N}^d$, $|n| = k$. For example, order the monomials compatibly with the triangular basis's index weight; each off-diagonal substitution in (SS.12) decreases that weight. Consequently

$$\det(z - A^s) = \prod_{|n|=k} (z - \lambda_n), \quad \sum_{k \geq 0} q^k \text{Tr}(U(a)^{\otimes k} | \text{Sym}^k E) = \prod_{i=1}^d (1 - qa^{\rho_i})^{-1}. \quad (\text{SS.16})$$

The second identity is an identity of formal power series: expand each geometric series and select its occupancy. On grouping repeated roots it is $\prod_\rho (1 - qa^\rho)^{-m_\rho}$, and the coefficient for grouped

occupancy ℓ is precisely w_ℓ from (SS.13). This gives the exact multiplicity dictionary between the cycle trace (SS.9) and the full local jets.

Define the symmetric absolute spectral cost and metric departure by

$$\begin{aligned}\mathcal{D}_k^s &= \sum_{|n|=k} \left| \sum_i n_i \delta_i \right| = \sum_{|\ell|=k} w_\ell \left| \sum_\rho \ell_\rho (\operatorname{Re} \rho - \tfrac{1}{2}) \right|, \\ \mathfrak{d}_M^s &= \operatorname{Tr}((G_M^s)^{-1} (A^s)^* G_M^s A^s) - \sum_{|n|=k} |\lambda_n|^2, \\ S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2}, \quad \epsilon_M^s = \|S_M^s\|_{\text{op}}.\end{aligned}\tag{SS.17}$$

The square root is used only as the displayed isometry for computing the original Hermitian pencil; it changes none of the orbit vectors, source functions, arithmetic constants, or entries of G_M^s .

Theorem 19.1 (Exact occupancy moments and symmetric departure). *The full unweighted occupancies satisfy*

$$\begin{aligned}\sum_{|n|=k} \sum_i n_i \delta_i &= a_k \sigma_1, \\ \sum_{|n|=k} \left(\sum_i n_i \delta_i \right)^2 &= (a_k + b_k) \sigma_2 + b_k \sigma_1^2.\end{aligned}\tag{SS.18}$$

In the original arithmetic orbit Gram,

$$\begin{aligned}\mathfrak{d}_M^s &\geq 0, \quad \operatorname{Tr} S_M^s = 2a_k \sigma_1, \\ \operatorname{Tr}((S_M^s)^2) &= 4\{(a_k + b_k) \sigma_2 + b_k \sigma_1^2\} + 2\mathfrak{d}_M^s.\end{aligned}\tag{SS.19}$$

In particular, if $\bar{\delta} = \sigma_1/d$, the exact mean-square spread of the occupancy offsets is

$$\sum_{|n|=k} \left(\sum_i n_i \delta_i - k\bar{\delta} \right)^2 = \frac{Nk(k+d)}{d(d+1)} \left(\sigma_2 - \frac{\sigma_1^2}{d} \right).\tag{SS.20}$$

Proof. The product $\prod_i (1 - q_i)^{-1}$ enumerates every occupancy. Apply $q_i \partial_{q_i}$ and then set all $q_i = q$. Its coefficient of q^k gives $\sum n_i = [q^k] q(1 - q)^{-d-1} = a_k$. For two distinct indices, applying the two derivatives gives $\sum n_i n_j = [q^k] q^2 (1 - q)^{-d-2} = b_k$. Applying $q_i^2 \partial_{q_i}^2$ gives $\sum n_i (n_i - 1) = 2b_k$. These coefficient identities follow by multiplying geometric series, or by induction from $(1 - q)^{-r} = \sum_{m \geq 0} \binom{m+r-1}{r-1} q^m$. Therefore $\sum n_i^2 = a_k + 2b_k$. Expanding the square, the cross terms have coefficient b_k ; collecting the diagonal terms gives (SS.18). The identities $a_k = Nk/d$ and $a_k + b_k = Nk(k+d)/(d(d+1))$ follow directly from the factorials, including the stated edge cases. Expanding the centered square proves (SS.20).

Let $T = (G_M^s)^{1/2} A^s (G_M^s)^{-1/2}$. A unitary upper triangularization exists by induction: choose a unit eigenvector as the first basis vector, complete it to an orthonormal basis, and repeat on the remaining diagonal block. If its strictly upper triangular part is N_T , its Hilbert–Schmidt squared norm is exactly $\mathfrak{d}_M^s$; the diagonal entries are the complete λ_n . This proves nonnegativity while retaining every nonzero off-diagonal entry. Since $S_M^s = T + T^* - kI$, its diagonal is $2 \sum_i n_i \delta_i$; each strictly upper triangular entry and its adjoint contribute twice their squared absolute value to the squared trace. Thus $\operatorname{Tr}((S_M^s)^2) = 4 \sum_n (n \cdot \delta)^2 + 2\mathfrak{d}_M^s$. Equation (SS.18) proves (SS.19). $\square$

For comparison with (CT.20)–(CT.21), the ordered tensor has an occupancy n exactly $k! / \prod_i n_i!$ times. The permutation complement therefore has that multiplicity minus one. Its characteristic

polynomial, trace, and absolute spectral cost are obtained with these nonnegative integer multiplicities. In particular

$$\mathcal{D}_k = \mathcal{D}_k^s + \sum_{|n|=k} \left(\frac{k!}{\prod_i n_i!} - 1 \right) |n \cdot \delta|. \quad (\text{SS.21})$$

This equality retains every contribution omitted by the symmetric inclusion. Moreover the permutation projection is orthogonal also for G_M , because it commutes with that Gram. Its image and kernel are invariant under $A^{(k)}$. Thus the full metric operator is the orthogonal direct sum of its symmetric and complementary restrictions; the squared Hilbert–Schmidt norms and the eigenvalue-square sums add separately. The full departure in (CT.20) is exactly $\mathfrak{d}_M^s$ plus the corresponding complementary departure, each nonnegative. This proves the full operator comparison as well as the scalar identity (SS.21).

19.4 The cost in the original symmetric relation layer

Restrict the original source maps without changing their Hilbert targets:

$$Y_M^s = Y_M I, \quad C_M^s = C_M I, \quad \Delta_M^s = G_M^s - G_{M+1}^s = (Y_M^s)^* Y_M^s, \quad W_M^s = -((Y_M^s)^* C_M^s + (C_M^s)^* Y_M^s). \quad (\text{SS.22})$$

Their images lie in the invariant original relation layer, whose dimension is $r_{\text{lay}} = p_{\leq k}(M+1)$. Indeed an unscaled symmetric polynomial basis in exact degree $M+1$ is indexed by its partitions into at most k parts. The invariant full-jet map is onto at every $M \geq k(d-1)$: average a complete tensor remainder to lift each invariant jet. The difference of the two kernel dimensions is therefore the number of the new degree $M+1$ basis vectors. The injective polynomial realization and orthogonal layer quotient transport this dimension to the actual source layer, proving the assertion with its original relation ideal. Put

$$r_M^s = \text{rank } Y_M^s, \quad \tau_M^s = \text{Tr}((G_M^s)^{-1} \Delta_M^s), \quad \chi_M^s = \text{Tr}((G_M^s)^{-1} (C_M^s)^* C_M^s). \quad (\text{SS.23})$$

Then the precise relative source costs are

$$\begin{aligned} n_{\pm}(W_M^s) &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}\}, \\ \mathcal{D}_k^s &\leq \frac{1}{2} \|S_M^s\|_1 \leq \sqrt{\tau_M^s \chi_M^s}, \quad \mathcal{D}_k^s \leq r_M^s \epsilon_M^s. \end{aligned} \quad (\text{SS.24})$$

To prove the inertia bound, the Hermitian form in (SS.22) vanishes on $\ker Y_M^s$. Any strictly positive or strictly negative subspace of dimension greater than $\text{rank } Y_M^s$ intersects that kernel nontrivially, a contradiction. Apply the same argument to C_M^s . In the triangular basis used in (SS.19), the absolute diagonal sum of S_M^s is $2\mathcal{D}_k^s$. Its spectral expansion gives $|S_{ii}| \leq \sum_j |\lambda_j| |U_{ij}|^2$; summing over i bounds this sum by the trace norm. Finally, for any two finite-domain Hilbert-space maps B, C , their singular-vector decomposition and entrywise Cauchy–Schwarz prove $\|B^* C\|_1 \leq \|B\|_{\text{HS}} \|C\|_{\text{HS}}$. Apply it to $B = Y_M^s (G_M^s)^{-1/2}$ and $C = C_M^s (G_M^s)^{-1/2}$ in (SS.22). The squared norms are τ_M^s, χ_M^s . The inertia bound gives at most $2r_M^s$ nonzero eigenvalues, proving the last assertion.

If the packet is stable, with full orders, under $\rho \mapsto 1 - \bar{\rho}$, its original antiunitary reflection commutes with permutations. Restricting (CT.16a) to the invariant space pairs every nonzero eigenvalue of the Hermitian pencil with its negative. Thus $\sigma_1 = 0$, and with $p_M^s = n_+(W_M^s) = n_-(W_M^s)$,

$$\begin{aligned} p_M^s &\leq \min\{r_M^s, \text{rank } C_M^s, r_{\text{lay}}, \lfloor N/2 \rfloor\}, \\ \mathcal{D}_k^s &\leq p_M^s \epsilon_M^s, \\ p_M^s (\epsilon_M^s)^2 &\geq \frac{2Nk(k+d)}{d(d+1)} \sigma_2 + \mathfrak{d}_M^s. \end{aligned} \quad (\text{SS.25})$$

There are $2p_M^s$ nonzero eigenvalues, each of absolute value at most ϵ_M^s . Bounding their absolute and squared sums and using (SS.19) proves both inequalities. They involve no division by p_M^s ; when it vanishes, (SS.19) forces every displayed nonnegative term to vanish.

Let $\delta_* = \max_i |\delta_i|$. If it is positive, $|n \cdot \delta| \leq k\delta_*$ yields the additional explicit lower cost

$$\mathcal{D}_k^s \geq \frac{(a_k + b_k)\sigma_2 + b_k\sigma_1^2}{k\delta_*}. \quad (\text{SS.26})$$

Indeed square each occupancy offset, bound it by $k\delta_*|n \cdot \delta|$, and sum using (SS.18). For a reflected packet this becomes $\mathcal{D}_k^s \geq N(k + d)\sigma_2/(d(d + 1)\delta_*)$. Each repeated eigenvector $v^{\otimes k}$ is nonzero and invariant, and its exact original Rayleigh quotient is $2k\delta_i$. Hence $\epsilon_M^s \geq 2k\delta_*$. Equations (SS.25)–(SS.26) additionally measure the cumulative cost of every occupancy, with all multiplicities and original layer ranks. If any $m_\rho > 1$, (SS.14) gives a nonzero nilpotent on the all- ρ summand. An operator with that Jordan block cannot be unitarily diagonal, so the sum of squares defining $\mathfrak{d}_M^s$ is strictly positive for every positive G_M^s . This retains a further finite-metric contribution even when all $\delta_i = 0$.

Finally the density proved in (JD.8) passes to the invariant source by the bounded permutation projection. The canonical representatives are the restrictions $R_M I$; hence $G_M^s \downarrow 0$ and $\det G_M^s \rightarrow 0$, while the full jets remain the identity at every finite M . With $\pi_M^s = \det G_{M+1}^s / \det G_M^s$, the positive eigenvalues t_j of $(G_M^s)^{-1/2} \Delta_M^s (G_M^s)^{-1/2}$ lie in $(0, 1)$, their number is r_M^s , and $\tau_M^s = \sum t_j$, $\pi_M^s = \prod (1 - t_j)$. The arithmetic mean–geometric mean inequality therefore gives

$$(\mathcal{D}_k^s)^2 \leq \tau_M^s \chi_M^s \leq r_M^s (1 - (\pi_M^s)^{1/r_M^s}) \chi_M^s \quad (r_M^s > 0). \quad (\text{SS.27})$$

For $r_M^s = 0$, (SS.22) gives $\Delta_M^s = W_M^s = 0$, without using an exponent $1/r_M^s$. Since $-\log(1 - t) - t = \int_0^t u/(1 - u) du$, the same finite source sequence also obeys, for integers $k(d - 1) \leq a < b$ and $\mathcal{D}_k^s > 0$,

$$(\mathcal{D}_k^s)^2 \sum_{M=a}^{b-1} (\chi_M^s)^{-1} \leq \sum_{M=a}^{b-1} \tau_M^s \leq \log \frac{\det G_a^s}{\det G_b^s}. \quad (\text{SS.28})$$

Positivity of every denominator follows from (SS.24). Telescoping proves the last inequality. Moreover $\sum_{M=a}^\infty \tau_M^s = \infty$: otherwise the tail has $t_j \leq \tau_M^s \leq 1/2$, and $-\log(1 - t_j) \leq 2t_j$ would keep the telescoped logarithm bounded, contrary to $\det G_M^s \rightarrow 0$. These identities transfer the entire determinant and relative-energy calculation to the source's smaller invariant family through its exact inclusion; they supply no assumed arithmetic upper estimate.

20 Arithmetic quartet parity and the exact positive frontier operator

Let h be a nonempty packet of actual zeros with their full orders, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Their composition also preserves the packet. Retain $d = \deg h$, $g = 2\xi$, $v_h = g/h$, $v = j_h v_h$, and the actual measure $d\nu_h = |v_h(1/2 + it)|^2 dt/(2\pi)$. Throughout this section $k \geq 1$, $M \geq k(d - 1)$, and

$$E_k = (\mathbb{C}[s]/h)^{\otimes k}, \quad A_k = \sum_{j=1}^k A_{h,j}, \quad \mathcal{C}_M = G_M^{-1}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{AP.1})$$

The metric G_M is the actual total-degree minimum of (CT.12)–(CT.13). All coordinates below remain its original monomial tensor coordinates.

20.1 The actual two frontier matrices

Let q_j be the original monic orthogonal polynomials for ν_h , and $\kappa_j = \int |q_j|^2 d\nu_h$. Thus κ_0 retains the mass of the original measure. For a multi-index α , put

$$q_\alpha = \prod_{j=1}^k q_{\alpha_j}(s_j), \quad \kappa_\alpha = \prod_{j=1}^k \kappa_{\alpha_j}, \quad z_\alpha = U_{v^{\otimes k}}[q_\alpha] \in E_k. \quad (\text{AP.2})$$

The source's notation p_j, ω_j denotes these same q_j, κ_j : their monic leading terms and identical lower-degree orthogonality determine the polynomial uniquely, and then determine its norm. No polynomial or measure is divided by its norm. The triangular polynomial change of basis and product integration in (CT.12) give exactly

$$\mathcal{C}_M = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\kappa_\alpha}. \quad (\text{AP.3})$$

Indeed this basis has Gram diagonal κ_α , and its full interpolation matrix has columns z_α ; substitution in $JM^{-1}J^*$ proves (AP.3), retaining $v^{\otimes k}$ in every column.

Write $a_j = \kappa_j / \kappa_{j-1}$ for $j \geq 1$, $a_0 = 0$, and retain the ordered frontier $|\beta| = M + 1$. Its size is $r = \binom{M+k}{k-1}$. Define maps with the same declared source $\mathbb{C}^r$:

$$\begin{aligned} \Omega &= \text{diag}(\kappa_\beta)_{|\beta|=M+1}, & F &= [z_\beta]_{|\beta|=M+1}, \\ w_\beta &= \sum_{j:\beta_j > 0} a_{\beta_j} z_{\beta-e_j}, & E_+ &= [w_\beta]_{|\beta|=M+1}. \end{aligned} \quad (\text{AP.4})$$

The original recurrence $sq_j = q_{j+1} + b_j q_j - a_j q_{j-1}$, with $b_j + \bar{b}_j = 1$, applied to (AP.3) gives

$$\mathcal{Z}_M := A_k \mathcal{C}_M + \mathcal{C}_M A_k^* - k \mathcal{C}_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*, \quad W_M = G_M \mathcal{Z}_M G_M. \quad (\text{AP.5})$$

To check the cancellation completely, an adjacent pair $\alpha, \alpha + e_j$ both of degree at most M has coefficients κ_α^{-1} and $-a_{\alpha_j+1} \kappa_{\alpha+e_j}^{-1}$, whose sum is zero. Its transposed conjugate cancels in the same way. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, which cancels the last term. The remaining frontier pair has $\beta = \alpha + e_j$ and coefficient $\kappa_\alpha^{-1} = a_{\beta_j} / \kappa_\beta$; summing these pairs is exactly (AP.4)–(AP.5). The same sum gives the full inverse-metric update

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (\text{AP.6})$$

20.2 The original linear reflection separates the two frontiers

Conjugation closure makes h real, and reflection closure then gives $h(1-s) = (-1)^d h(s)$. The original functional equation gives

$$v_h(1-s) = (-1)^d v_h(s), \quad q_j(1-s) = (-1)^j q_j(s). \quad (\text{AP.7})$$

For the second identity, the actual density is even in t . The polynomial $(-1)^j q_j(1-s)$ is monic, and substitution $t \mapsto -t$ verifies its orthogonality to every lower degree. The difference from q_j has degree less than j and is orthogonal to itself, so positivity proves it is zero. This proves (AP.7) with the original leading coefficient.

Let $\mathfrak{c} : E_k \rightarrow E_k$ be the complex-linear map substituting $s_j \mapsto 1 - s_j$ in every factor. It is well-defined because the ideal $(h(s_1), \dots, h(s_k))$ is preserved, and $\mathfrak{c}^2 = I$. Let C denote its matrix in the retained original basis. The original function map

$$(\mathfrak{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}) \quad (\text{AP.8})$$

is a linear unitary involution for $d^k x$, by the change of variables $y_j = x_j^{-1}$. Its Mellin transform replaces s_j by $1 - s_j$. Consequently, on an original numerator,

$$\mathfrak{t}_k \mathcal{T}_h^{(k)} P = (-1)^{kd} \mathcal{T}_h^{(k)} P(1 - s_1, \dots, 1 - s_k). \quad (\text{AP.9})$$

Equations (AP.7) and (AP.2) show that its full jet is $\mathfrak{c}\mathcal{J}P$, with exactly the displayed unit sign. Total degree and the original relation ideal are preserved. Applying $\mathfrak{t}_k$ to the unique minimum for a prescribed jet therefore gives the unique minimum for its $\mathfrak{c}$ -image. Uniqueness follows from the orthogonal decomposition (CT.5). Hence

$$\mathfrak{t}_k R_M = R_M C, \quad C^* G_M C = G_M, \quad C z_\alpha = (-1)^{kd+|\alpha|} z_\alpha. \quad (\text{AP.10})$$

In particular, with $\eta = (-1)^{kd+M+1}$,

$$CF = \eta F, \quad CE_+ = -\eta E_+. \quad (\text{AP.11})$$

Every term defining w_β has degree M , which proves the second identity without suppressing any coefficient.

Declare the following maps between the original positive spaces and their coordinate Hilbert spaces:

$$P = G_M^{1/2} C G_M^{-1/2}, \quad U = G_M^{1/2} F \Omega^{-1/2}, \quad V = G_M^{1/2} E_+ \Omega^{-1/2}. \quad (\text{AP.12})$$

Here $G_M^{1/2}$ and $\Omega^{1/2}$ are the explicit isometries from the declared original metric spaces to Euclidean coordinate spaces, with their displayed inverses. Every expression in (AP.12) retains both original matrices. From (AP.10), P is unitary and $P^2 = I$; hence $P^* = P$. Equations (AP.11) imply $PU = \eta U$, $PV = -\eta V$, and thus

$$U^* V = (PU)^*(PV) = -U^* V = 0. \quad (\text{AP.13})$$

20.3 The complete positive operator and the sharper arithmetic bound

Put $A = U^* U$, $B = V^* V$, retaining their source type $\mathbb{C}^r$. Equivalently,

$$A = \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, \quad B = \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}. \quad (\text{AP.14})$$

These positive-semidefinite matrices are built from the two original frontiers; A is distinct from the arithmetic generator A_k .

Theorem 20.1 (Exact positive frontier operator). *The actual relative weight satisfies*

$$\begin{aligned} S_M &= G_M^{-1/2} W_M G_M^{-1/2} = UV^* + VU^*, \\ \boxed{\epsilon_{k,M} &= \|UV^*\|, \quad \epsilon_{k,M}^2 = \lambda_{\max}(A^{1/2} B A^{1/2}),} \quad p_M = \text{rank}(A^{1/2} B^{1/2}), \\ \text{Tr}(S_M^2) &= 2 \text{Tr}(AB). \end{aligned} \quad (\text{AP.15})$$

Its positive and negative inertias both equal p_M . Define the original incidence bound and volume ratio by

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad \lambda_{k,M} = \|U\|^2, \quad \pi_M = \frac{\det G_{M+1}}{\det G_M}. \quad (\text{AP.16})$$

Then

$$\boxed{\epsilon_{k,M}^2 \leq \Gamma_{k,M} \lambda_{k,M} \leq \Gamma_{k,M} (\pi_M^{-1} - 1).} \quad (\text{AP.17})$$

All formulas include the zero-rank cases and every full arithmetic jet. No growth estimate for $\Gamma_{k,M}$ or π_M is assumed.

Proof. Equation (AP.5) proves the first equality in (AP.15). Decompose the retained coordinate Hilbert space into the two orthogonal eigenspaces of the explicit Hermitian involution P . Relative to the order $(\eta, -\eta)$, the operator has blocks

$$S_M = \begin{pmatrix} 0 & T \\ T^* & 0 \end{pmatrix}, \quad T = UV^*|_{\ker(P+\eta I)}. \quad (\text{AP.18})$$

Indeed U^* vanishes on the opposite eigenspace and V^* vanishes on the first, by (AP.11). Squaring gives diagonal blocks TT^*, T^*T . For each nonzero singular value σ of T , choose unit singular vectors x, y with $Ty = \sigma x$, $T^*x = \sigma y$; then (x, y) and $(x, -y)$ have eigenvalues $\sigma, -\sigma$. The remaining eigenspaces are $\ker T^* \oplus \ker T$, with eigenvalue zero. These vectors form the full orthogonal decomposition by the spectral theorem for T^*T , including all its zero vectors. It follows that $\epsilon = \|T\| = \|UV^*\|$ and both inertias equal $\text{rank}(UV^*)$.

Now $(UV^*)(UV^*)^* = UBU^*$. Its nonzero eigenvalues and rank are those of $B^{1/2}AB^{1/2}$: the maps $UB^{1/2}$ and its adjoint transfer each nonzero eigenvector, with inverse divided by that eigenvalue. The latter matrix is $T_0^*T_0$, with $T_0 = A^{1/2}B^{1/2}$, while $T_0T_0^* = A^{1/2}BA^{1/2}$. Their full positive spectra agree by the same maps, and their ranks equal $\text{rank } T_0$. This proves the norm and rank formulas, including rank zero. Finally (AP.13) kills $(UV^*)^2$ and $(VU^*)^2$. Taking the trace of the remaining two products gives the last line of (AP.15).

For the incidence estimate, let $Z = [z_\alpha]_{|\alpha|=M}$ and $\Omega_M = \text{diag}(\kappa_\alpha)$. Define $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, and zero otherwise; summation applies if more than one specified incidence contributes. Then $E_+ = ZL$. The matrix $N = \Omega_M L \Omega^{-1} L^*$ has nonnegative entries. Its row sum at α is exactly

$$\sum_{j=1}^k \left(a_{\alpha_j+1} + \sum_{l \neq j} a_{\alpha_l} \right) = \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (\text{AP.19})$$

To verify each factor, its j -th contributing frontier is $\beta = \alpha + e_j$, and $\kappa_\alpha a_{\beta_j} / \kappa_\beta = 1$. The predecessors of this β supply $\sum_l a_{\beta_l}$, which is precisely the displayed summand. Every eigenvalue of N has modulus at most the largest row sum: in an eigenvector choose a coordinate of largest absolute value and apply that row equation. The matrix N is similar, by $\Omega_M^{1/2}$, to the Hermitian positive matrix $\Omega_M^{1/2} L \Omega^{-1} L^* \Omega_M^{1/2}$. Its eigenvalues therefore lie in $[0, \Gamma_{k,M}]$. Undoing the congruence and using (AP.3) proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z \Omega_M^{-1} Z^* \preceq \Gamma_{k,M} \mathcal{C}_M, \quad VV^* \preceq \Gamma_{k,M} I. \quad (\text{AP.20})$$

Consequently $\epsilon^2 = \|UV^*\|^2 \leq \|U\|^2 \|V\|^2 \leq \Gamma_{k,M} \lambda_{k,M}$. Finally (AP.6) gives

$$\pi_M^{-1} = \det(I + UU^*) = \prod_j (1 + \lambda_j(UU^*)) \geq 1 + \lambda_{k,M}. \quad (\text{AP.21})$$

All eigenvalues here are nonnegative, proving the second bound in (AP.17). If $U = 0$ or $V = 0$, (AP.15) gives zero weight directly; all determinant denominators remain positive. $\square$

The arithmetic departure has an exact expression in these same two frontier Grams. List the original ρ_i with every algebraic multiplicity and put $\delta_i = \text{Re } \rho_i - 1/2$. Comparing (AP.15) with the fully proved tensor trace (CT.21) gives

$$\mathfrak{d}_M^{(k)} = \text{Tr}(AB) - 2 \sum_{i_1, \dots, i_k} (\delta_{i_1} + \dots + \delta_{i_k})^2. \quad (\text{AP.22})$$

This retains the complete nonsemisimple generator in the left-hand definition of (CT.20). For $k = 1$, a four-distinct-point tesserine orbit of common full order μ has the further exact chain, with $M = d + m$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_{1,M}^2 = \lambda_{\max}(A^{1/2}BA^{1/2}) \leq \Gamma_{1,M}(\pi_M^{-1} - 1). \quad (\text{AP.23})$$

The first inequality, original character, multiplicity and critical supported-zero case are proved in (R9); (AP.15)–(AP.17) prove the rest. Thus the original character observation is related to an explicit positive frontier operator through the original metric, full interpolation unit and declared isometries.

20.4 The same positive frontier on the actual symmetric summand

The cochain projector used here is exactly $S_k = (k!)^{-1} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi$, with the original Koszul permutations T_π . Its top-degree action is the ordinary permutation projection, because T_π there equals $\text{sgn}(\pi) P_\pi$. Equations (SS.1)–(SS.9) prove its continuous cochain maps, completed projective-tensor cohomology, and finite packet projector, including the complementary summand. We now use its finite invariant image with the original total-degree representatives R_M .

Write $e_1, \dots, e_d$ for the original power basis of E , with indices shifted by one, and retain each occupancy $n \in \mathbb{N}^d$, $|n| = k$. Let b_n be the literal sum of the distinct basis tensors of occupancy n . Define the maps and dimensions by

$$\begin{aligned} N_s &= \binom{k+d-1}{d-1}, & I : \mathbb{C}^{N_s} &\longrightarrow E_k, & Ie_n &= b_n, \\ D_o &= I^* I = \text{diag}_{|n|=k} \frac{k!}{\prod_i n_i!}, & L &= D_o^{-1} I^*, & P_s &= IL = \frac{1}{k!} \sum_{\pi} P_\pi, & LI &= I_{N_s}. \end{aligned} \quad (\text{AP.24})$$

Indeed the distinct original tensor basis vectors are orthonormal for this coordinate calculation; hence their disjoint orbit sums have precisely the displayed diagonal Gram D_o . The map L averages coefficients on each orbit, proving all the compositions in (AP.24). In particular $P_s^* = P_s$ and $P_s^2 = P_s$. The factorials in D_o are retained separately from the arithmetic Gram.

Permutation invariance of the original product measure, relation ideal, total degree and full interpolation unit gives $R_M P_\pi = P_\pi R_M$, by applying the unitary inverse permutation to each constrained minimum. Thus G_M and G_M^{-1} commute with P_s . Put

$$\begin{aligned} R_M^s &= R_M I, & G_M^s &= I^* G_M I, & W_M^s &= I^* W_M I, \\ A_k^s &= L A_k I, & A_k I &= I A_k^s, & W_M^s &= (A_k^s)^* G_M^s + G_M^s A_k^s - k G_M^s, \\ I^* G_M &= G_M^s L, & G_M I &= L^* G_M^s, & \mathcal{K}_M^s &:= (G_M^s)^{-1} = L G_M^{-1} L^*. \end{aligned} \quad (\text{AP.25})$$

These are the same representatives and matrices as (SS.10)–(SS.11). For completeness, $I^* G_M = I^* P_s G_M = I^* G_M P_s = I^* G_M IL$, proving the first rectangular identity and then its adjoint. The generator commutes with every factor permutation, so it preserves the image of I , proving its intertwining identity and the displayed weight formula. Finally

$$(L G_M^{-1} L^*)(I^* G_M I) = L G_M^{-1} P_s G_M I = LI = I_{N_s},$$

which proves the inverse formula with both factors D_o^{-1} present. Positivity follows since I is injective and $G_M \succ 0$. The notation $\mathcal{K}_M^s$ refers only to this inverse kernel; it will not denote a reflection.

Keep the complete, possibly redundant frontier of (AP.4), with its unchanged domain $\mathbb{C}^r$ and norm matrix Ω . Its symmetric target maps are

$$\begin{aligned} F_s &= L F : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, & E_s &= L E_+ : \mathbb{C}^r \longrightarrow \mathbb{C}^{N_s}, \\ \mathcal{Z}_M^s &= F_s \Omega^{-1} E_s^* + E_s \Omega^{-1} F_s^*, & W_M^s &= G_M^s \mathcal{Z}_M^s G_M^s, \\ \mathcal{K}_{M+1}^s &= \mathcal{K}_M^s + F_s \Omega^{-1} F_s^*. \end{aligned} \quad (\text{AP.26})$$

The rectangular identities (AP.25), applied to $I^* G_M \mathcal{Z}_M G_M I$, prove the weight formula. Applying L and L^* to (AP.6) proves the inverse update. It uses the same orbit maps at both source degrees. Thus the update concerns the actual next symmetric minimum, not a newly selected positive form.

The original linear reflection C commutes with every factor permutation, since it substitutes the same $1 - s_j$ in each factor. Its invariant-coordinate matrix is

$$\begin{aligned} C_s &= L C I, & C I &= I C_s, & C_s L &= L C, & C_s^2 &= I_{N_s}, \\ C_s^* G_M^s C_s &= G_M^s, & C_s F_s &= \eta F_s, & C_s E_s &= -\eta E_s, & \eta &= (-1)^{kd+M+1}. \end{aligned} \quad (\text{AP.27})$$

For example $CI = P_s CI = ILCI$, and $C_s L = LCP_s = LP_s C = LC$; these prove the first intertwiners and then the involution. The metric equation follows by substituting $IC_s = CI$ into $I^* G_M I$ and using (AP.10). Applying L to (AP.11) proves both parities with the full arithmetic unit and its original sign.

Define the explicit isometries and the two retained frontier Grams by

$$\begin{aligned} P_M^s &= (G_M^s)^{1/2} C_s (G_M^s)^{-1/2}, \\ U_s &= (G_M^s)^{1/2} F_s \Omega^{-1/2}, \quad V_s = (G_M^s)^{1/2} E_s \Omega^{-1/2}, \\ A_s &= U_s^* U_s, \quad B_s = V_s^* V_s. \end{aligned} \tag{AP.28}$$

The superscript on P_M^s distinguishes this transported reflection from the fixed permutation projection P_s in (AP.24). Equation (AP.27) proves that P_M^s is a Hermitian involution and that U_s, V_s have opposite eigenvalues $\eta, -\eta$. Their ranges are consequently orthogonal, so $U_s^* V_s = 0$.

Theorem 20.2 (Exact symmetric arithmetic frontier). *For the actual signed symmetric cohomology summand, the original Hermitian pencil has*

$$\begin{aligned} S_M^s &= (G_M^s)^{-1/2} W_M^s (G_M^s)^{-1/2} = U_s V_s^* + V_s U_s^*, \\ \boxed{\epsilon_M^s &= \|U_s V_s^*\|, \quad (\epsilon_M^s)^2 = \lambda_{\max}(A_s^{1/2} B_s A_s^{1/2}),} \\ p_M^s &= \text{rank}(A_s^{1/2} B_s^{1/2}), \quad n_+(W_M^s) = n_-(W_M^s) = p_M^s, \\ \text{Tr}((S_M^s)^2) &= 2 \text{Tr}(A_s B_s). \end{aligned} \tag{AP.29}$$

With the original $\Gamma_{k,M}$ of (AP.16), put

$$\lambda_M^s = \|U_s\|^2, \quad \pi_M^s = \frac{\det G_{M+1}^s}{\det G_M^s}.$$

Then the same source sequence satisfies

$$\boxed{(\epsilon_M^s)^2 \leq \Gamma_{k,M} \lambda_M^s \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), \quad \lambda_M^s \leq \lambda_{k,M}.} \tag{AP.30}$$

All formulas retain every occupancy and nilpotent direction in (SS.12)–(SS.14), as well as the ungrouped frontier multiplicities.

Proof. Equation (AP.26) gives the first line of (AP.29). Decomposing $\mathbb{C}^{N_s}$ into the two orthogonal eigenspaces of P_M^s gives exactly the block operator (AP.18), now with $T_s = U_s V_s^*$ between those two spaces. The singular-vector maps and their inverses proved after (AP.18) apply to these declared matrices without any rank assumption. They give opposite eigenvalues for each positive singular value, $\text{rank rank}(A_s^{1/2} B_s^{1/2})$, and all remaining eigenvalues zero. The same two explicit composites $U_s B_s^{1/2}$ and its adjoint identify the positive eigenvalues with those of $A_s^{1/2} B_s A_s^{1/2}$. Finally $U_s^* V_s = 0$ kills the two square cross factors; the traces of the other two products both equal $\text{Tr}(A_s B_s)$. This proves every assertion in (AP.29).

Applying L and L^* to the original inequality (AP.20), and using (AP.25), proves

$$E_s \Omega^{-1} E_s^* \preceq \Gamma_{k,M} \mathcal{K}_M^s, \quad V_s V_s^* \preceq \Gamma_{k,M} I_{N_s}. \tag{AP.31}$$

Thus $\|U_s V_s^*\|^2 \leq \|U_s\|^2 \|V_s\|^2 \leq \Gamma_{k,M} \lambda_M^s$. Taking determinants of the exact inverse update in (AP.26) gives

$$(\pi_M^s)^{-1} = \det(I_{N_s} + U_s U_s^*) = \det(I_r + A_s) \geq 1 + \lambda_M^s, \quad 0 < \pi_M^s \leq 1. \tag{AP.32}$$

The determinant equality follows either by the two rectangular composites and their nonzero spectra, or by multiplying the two block factors for $\det(I + XY) = \det(I + YX)$. The inequality multiplies the nonnegative factors $1 + \lambda_j(A_s)$, exactly as in (AP.21). This proves the first two bounds in (AP.30).

For the final bound, the literal orbit maps give

$$\begin{aligned} L^* G_M^s L &= P_s G_M P_s \preceq G_M, \\ A_s &= \Omega^{-1/2} F^* P_s G_M P_s F \Omega^{-1/2} \preceq \Omega^{-1/2} F^* G_M F \Omega^{-1/2} = A. \end{aligned} \quad (\text{AP.33})$$

Indeed G_M commutes with the orthogonal projection P_s , so their omitted cross terms vanish and $G_M - P_s G_M P_s = (I - P_s) G_M (I - P_s) \succeq 0$. Taking the largest Rayleigh quotient of the two positive matrices in (AP.33) proves $\lambda_M^s \leq \lambda_{k,M}$. No comparison of arbitrary compressed inverse matrices is used. $\square$

There is also an exact comparison of the two relative operators. The map

$$Q_M = G_M^{1/2} I (G_M^s)^{-1/2} : \mathbb{C}^{N_s} \longrightarrow \mathbb{C}^{d^k}$$

has $Q_M^* Q_M = I_{N_s}$, and the inverse formula in (AP.25) gives $Q_M Q_M^* = P_s$. Direct multiplication therefore gives

$$S_M^s = Q_M^* S_M Q_M, \quad U_s = Q_M^* U, \quad V_s = Q_M^* V, \quad \epsilon_M^s \leq \epsilon_{k,M}. \quad (\text{AP.34})$$

Here $G_M^{1/2}$ commutes with P_s by its spectral decomposition. The full S_M also commutes with P_s , since both A_k and G_M do. Consequently (AP.34) is its invariant orthogonal restriction; the complementary operator remains on $\ker P_s$. The norm inequality follows directly by applying S_M to the unit vector $Q_M u$ and then the contraction Q_M^* .

If desired, the source frontier can be grouped with every factorial retained. For an orbit $\mathcal{O}$ of degree- $M + 1$ multi-indices, let $m_j(\beta) = |\{l : \beta_l = j\}|$ and

$$n_{\mathcal{O}} = \frac{k!}{\prod_{j \geq 0} m_j(\beta)!}, \quad \hat{f}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lz_{\beta}, \quad \hat{e}_{\mathcal{O}} = \sum_{\beta \in \mathcal{O}} Lw_{\beta}, \quad \hat{\omega}_{\mathcal{O}} = n_{\mathcal{O}} \kappa_{\beta}. \quad (\text{AP.35})$$

The product norm κ_{β} is constant on an orbit. Equivariance of the full unit, polynomials and incidence sums implies that Lz_{β} and Lw_{β} are individually constant on the same orbit. Hence, for example,

$$\sum_{\beta \in \mathcal{O}} \frac{(Lz_{\beta})(Lw_{\beta})^*}{\kappa_{\beta}} = \frac{\hat{f}_{\mathcal{O}} \hat{e}_{\mathcal{O}}^*}{\hat{\omega}_{\mathcal{O}}}. \quad (\text{AP.36})$$

The same calculation with both columns of the first or second type proves the corresponding kernel and incidence identities. Equations (AP.35)–(AP.36) are the exact map to the source's orbit-factorial presentation. The ungrouped matrices in (AP.26)–(AP.34) continue to retain all their columns.

All zero-rank cases follow without inversion of A_s or B_s . If their square-root product vanishes, then $p_M^s = 0$, $W_M^s = 0$, and $\epsilon_M^s = 0$, even if the kernel update is nonzero. If $\pi_M^s = 1$, (AP.32) forces $U_s = 0$, with the same conclusion. The dimension N_s is always at least one; if it is one, a Hermitian involution has only one nonzero eigenspace, so one of U_s, V_s vanishes and the weight is zero. This argument also covers a zero-dimensional reflection eigenspace in a larger target. It neither removes the retained quotient nor asserts existence of a particular actual packet. For $k = 1$, every orbit cardinality is one and $I = L = I_d$; all the displayed formulas recover (AP.15)–(AP.21) exactly.

Finally the symmetric source is still $R_M I$, and its two actual boundary maps are $Y_M I$ and $C_M I$ in (SS.22). The original relation-graph map, followed by the bounded permutation projection on its Hilbert target, realizes the redundant frontier used here. Grouping it by (AP.35) gives the invariant relation layer of dimension $p_{\leq k}(M + 1)$, as proved in (SS.22)–(SS.24). Its primitive is the original fixed-order division primitive followed by S_k ; the cochain identity in (SS.3)–(SS.4) preserves its boundary and all signs. At invariant support fibres the same maps lift with that support; vanishing amplitudes remain the corresponding supported zero. For other masks the original orbit-joined transport and its kernel remain as stated after (SS.9). Thus (AP.30) is a proved bound for the same

finite arithmetic cohomology summand, with no assumed asymptotic estimate for either $\Gamma_{k,M}$ or π_M^s .

Section 21 carries these same two parities through the original graph frame into the actual derivative layer. The equality $-\widehat{Y}^*\widehat{C} = UV^*$ proved there supplies the exact original derivative cost for the positive operator here. Sections 22 and 24 calculate its one-factor and symmetric-square counterparts through explicit quadratic algebra maps, with all local nilpotents and both reflection components.

21 Quartet parity in the original derivative energy and volume loss

This continuation retains the original maps of (FE.3)–(FE.9) and the actual quartet reflection of (AP.7)–(AP.13). Their combination proves an exact block identity for the two maps into the original theta relation layer. It yields a stronger derivative energy estimate than (FE.24), including an operator-norm version and the resulting finite accumulated determinant bounds.

21.1 Original maps, their spaces, and the reflection

Fix a nonempty packet of actual zeros, with every selected zero's full multiplicity, closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$. Keep

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad g = 2\xi, \quad v_h = g/h, \quad v = j_h v_h, \quad E_k = (\mathbb{C}[s]/h)^{\otimes k}.$$

Here $k \geq 1$, $M \geq k(d-1)$, and the original generator is $A_k = \sum_{j=1}^k A_{h,j}$. Let $G = G_M > 0$ be the actual degree- M minimum metric in the original monomial remainder basis, and let $R = R_M : E_k \rightarrow \mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$ be its original representative, so $R^*R = G$.

For the unchanged measure $d\nu_h(t) = |v_h(1/2 + it)|^2 dt / (2\pi)$, let q_j be the monic orthogonal polynomials and $\kappa_j = \|q_j\|_{\nu_h}^2$. Write $q_\beta = \prod_j q_{\beta_j}$, $\kappa_\beta = \prod_j \kappa_{\beta_j}$, and $z_\beta = U_{v \otimes k}[q_\beta]$. With the original ordering of $|\beta| = M+1$, retain

$$\begin{aligned} r &= \binom{M+k}{k-1}, \quad \Omega = \text{diag}(\kappa_\beta) > 0, \quad F = [z_\beta] : \mathbb{C}^r \rightarrow E_k, \\ E_+ &= [w_\beta] : \mathbb{C}^r \rightarrow E_k, \quad w_\beta = \sum_{j:\beta_j>0} \frac{\kappa_{\beta_j}}{\kappa_{\beta_j-1}} z_{\beta-e_j}, \\ T_+ &= [\mathcal{T}_h^{(k)} q_\beta] : \mathbb{C}^r \rightarrow \mathcal{H}_k, \quad b = T_+ - RF : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M, \\ H &= b^*b = \Omega + F^*GF > 0. \end{aligned} \tag{QD.1}$$

The space $\mathcal{E}_M$ is the actual next theta relation layer with its inherited L^2 norm. The column Gram of T_+ is Ω ; its image is perpendicular to the old polynomial space containing $R(E_k)$. Thus expanding its Gram proves H in (QD.1). The leading degree map gives the inverse of b : each layer vector's highest polynomial coefficient determines it, and subtracting that graph image leaves a vector both in the old relation space and perpendicular to it. In particular b is a bijection onto the original layer, with no arbitrary Hilbert replacement.

The two original maps into that same layer and their weight are

$$\begin{aligned} Y &= -bH^{-1}F^*G : E_k \rightarrow \mathcal{E}_M, \quad C_D = b\Omega^{-1}E_+^*G : E_k \rightarrow \mathcal{E}_M, \\ \Delta &= Y^*Y = G - G_{M+1}, \quad W = A_k^*G + GA_k - kG = -(Y^*C_D + C_D^*Y). \end{aligned} \tag{QD.2}$$

The notation C_D is explicitly the map C_M of (FE.5), and does not denote the inverse interpolation matrix. Its formula records the top polynomial coefficient of the original Euler defect; the graph

projection and the original theta primitive in (FE.3)–(FE.5) supply its actual source realization. The weight identity also follows directly by inserting the two displayed maps and cancelling the middle $b^*b = H$.

Let $C : E_k \rightarrow E_k$ replace each residue variable s_j by $1 - s_j$, and let

$$(\mathbf{t}_k f)(x_1, \dots, x_k) = \left(\prod_j x_j^{-1} \right) f(x_1^{-1}, \dots, x_k^{-1}).$$

The original packet symmetries and functional equation give $h(1 - s) = (-1)^d h(s)$, $v_h(1 - s) = (-1)^d v_h(s)$, and $q_j(1 - s) = (-1)^j q_j(s)$. For the last identity, the reflected polynomial with its sign restored is monic and has the same lower-degree orthogonality for the even density; uniqueness of the monic orthogonal polynomial proves equality. The variable change $x_j \mapsto x_j^{-1}$ proves that $\mathbf{t}_k$ is a unitary involution for $d^k x$, with Mellin action $s_j \mapsto 1 - s_j$. Thus it preserves total degree, the relation ideal and the unique constrained minimum. Writing $\eta = (-1)^{kd+M+1}$, one obtains

$$\mathbf{t}_k R = RC, \quad C^2 = I, \quad C^*GC = G, \quad CF = \eta F, \quad CE_+ = -\eta E_+, \quad \mathbf{t}_k b = \eta b. \quad (\text{QD.3})$$

The last equality uses both $\mathbf{t}_k T_+ = \eta T_+$ and $CF = \eta F$. It states the exact target reflection on the whole original relation layer, since b is onto.

For explicit coordinate isometries put

$$\begin{aligned} P &= G^{1/2}CG^{-1/2}, \quad P^* = P, \quad P^2 = I, \quad U = G^{1/2}F\Omega^{-1/2}, \quad V = G^{1/2}E_+\Omega^{-1/2}, \\ \hat{Y} &= YG^{-1/2}, \quad \hat{C} = C_D G^{-1/2} : \mathbb{C}^{d^k} \rightarrow \mathcal{E}_M, \quad J_b = bH^{-1/2} : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M. \end{aligned} \quad (\text{QD.4})$$

The matrix $G^{1/2}$ is the isometry from (E_k, G) to its Euclidean coordinate space, with inverse $G^{-1/2}$. Similarly $\Omega^{1/2}$ records the original polynomial norms, and J_b is a unitary map with inverse $H^{-1/2}b^*$. No commutation of H with Ω is assumed. In these exact coordinates,

$$\hat{Y} = -J_b H^{-1/2} F^* G^{1/2}, \quad \hat{C} = J_b H^{1/2} \Omega^{-1} E_+^* G^{1/2}. \quad (\text{QD.5})$$

21.2 The single original boundary block

Theorem 21.1 (Exact quartet derivative block and its energy bound). *The two original domain parities and their entire block relation are*

$$YC = \eta Y, \quad C_D C = -\eta C_D, \quad \hat{Y}P = \eta \hat{Y}, \quad \hat{C}P = -\eta \hat{C}, \quad -\hat{Y}^* \hat{C} = UV^*. \quad (\text{QD.6})$$

Let

$$\begin{aligned} S &= G^{-1/2}WG^{-1/2}, \quad \epsilon = \|S\|_{\text{op}}, \quad \lambda = \|UU^*\|_{\text{op}}, \quad \pi = \frac{\det G_{M+1}}{\det G}, \\ \zeta &= \|\hat{C}\|_{\text{op}}^2, \quad \chi = \|\hat{C}\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.7})$$

Then all these actual finite quantities satisfy

$$\boxed{\epsilon = \|\hat{Y}^* \hat{C}\|_{\text{op}} = \|UV^*\|_{\text{op}}, \quad \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq \frac{\lambda}{1 + \lambda} \chi.} \quad (\text{QD.8})$$

The operator-norm cost and the trace cost are exactly

$$\begin{aligned} \hat{C}^* \hat{C} &= V(I + U^*U)V^*, \quad \zeta = \lambda_{\max}(V(I + U^*U)V^*), \\ \chi &= \text{Tr}(V(I + U^*U)V^*) = \|V\|_{\text{HS}}^2 + \|VU^*\|_{\text{HS}}^2. \end{aligned} \quad (\text{QD.9})$$

In particular the operator-norm replacement of χ in (QD.8) is valid without a rank assumption.

Proof. From $C^*GC = G$ and $C^2 = I$ one gets $GC = C^*G$. Thus $F^*GC = (CF)^*G = \eta F^*G$, and similarly $E_+^*GC = -\eta E_+^*G$. Substitution in (QD.2) proves the first two parities, and the coordinate change proves the next two. Equations (QD.2) and $b^*b = H$ give, in the order shown,

$$\hat{Y}^*\hat{C} = -G^{1/2}FH^{-1}H\Omega^{-1}E_+^*G^{1/2} = -UV^*.$$

Decompose $\mathbb{C}^{d^k} = \ker(P - \eta I) \oplus \ker(P + \eta I)$, orthogonally. The first map in (QD.6) annihilates the second summand and the second map annihilates the first. Consequently $X = \hat{Y}^*\hat{C}$ maps the second summand into the first and vanishes on the first. The actual weight is

$$S = -(X + X^*) = -\begin{pmatrix} 0 & X_{+-} \\ X_{+-}^* & 0 \end{pmatrix}.$$

Squaring gives diagonal blocks $X_{+-}X_{+-}^*$ and $X_{+-}^*X_{+-}$. Their largest eigenvalues both equal $\|X_{+-}\|^2$ when this map is nonzero, by its singular value decomposition; if it is zero both blocks vanish. Hence $\|S\| = \|X\|$, including empty parity summands and all zero ranks.

The retained Gram identity in (QD.1) is $H = \Omega^{1/2}(I + U^*U)\Omega^{1/2}$. Substitution gives

$$\hat{Y}^*\hat{Y} = U(I + U^*U)^{-1}U^* = UU^*(I + UU^*)^{-1}.$$

The second equality follows by multiplying $(I + UU^*)U = U(I + U^*U)$ by its positive invertible factors. Thus $\|\hat{Y}\|^2 = \lambda/(1 + \lambda)$, also when $\lambda = 0$. Submultiplicativity in the exact identity $\epsilon = \|\hat{Y}^*\hat{C}\|$ proves the first inequality of (QD.8). On the other hand $\hat{C} = b\Omega^{-1/2}V^*$, so its Gram is exactly $V\Omega^{-1/2}H\Omega^{-1/2}V^* = V(I + U^*U)V^*$. This proves (QD.9). A positive finite matrix has largest eigenvalue at most its trace, proving $\zeta \leq \chi$ and the second inequality. $\square$

This is an actual-source strengthening of (FE.24): its general factor 4 is replaced by 1 under the proved original quartet parity. The retained derivative map still lands in the original $\mathcal{E}_M$; only the displayed exact isometries were used to calculate its norm.

21.3 Complete determinant constraints, including zero cases

Set $\delta_* = \max_{\rho \in Z} |\operatorname{Re} \rho - 1/2|$ and $q_* = 4k^2\delta_*^2$. The name q_* denotes this nonnegative scalar, not an orthogonal polynomial. All original packet multiplicities remain in E_k, A_k, G_M .

Theorem 21.2 (Quartet derivative cost and accumulated volume). *At every admitted degree, including $\delta_* = 0$,*

$$q_* \leq \epsilon^2 \leq \frac{\lambda}{1 + \lambda} \zeta \leq (1 - \pi) \zeta \leq (1 - \pi) \chi. \quad (\text{QD.10})$$

If $\delta_ > 0$, the exact source quantities necessarily obey*

$$\begin{aligned} \zeta > q_*, \quad \chi \geq \zeta > q_*, \quad \lambda \geq \frac{q_*}{\zeta - q_*} \geq \frac{q_*}{\chi - q_*}, \\ \log \frac{\det G_M}{\det G_{M+1}} \geq -\log \left(1 - \frac{q_*}{\zeta_M} \right) \geq -\log \left(1 - \frac{q_*}{\chi_M} \right). \end{aligned} \quad (\text{QD.11})$$

For $\delta_ > 0$ and any integers $k(d-1) \leq a < b$, this gives*

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\zeta_M} \right) \geq \sum_{M=a}^{b-1} -\log \left(1 - \frac{4k^2\delta_*^2}{\chi_M} \right). \quad (\text{QD.12})$$

Every denominator in (QD.11)–(QD.12) is positive, and every logarithm has a strictly positive argument, by the preceding inequalities.

Proof. For every selected centre ρ , the original finite algebra has a nonzero eigenvector of A_h with eigenvalue ρ : in its local factor use $(s - \rho)^{m_\rho - 1}$ and put zero in the other full CRT factors. Its k -fold tensor is nonzero and has eigenvalue $k\rho$ for A_k . The quotient of $W = A_k^*G + GA_k - kG$ on that original vector is exactly $2k \operatorname{Re} \rho - k$. The operator norm of its Hermitian relative matrix is at least the absolute value of this Rayleigh quotient. Taking the largest displacement and squaring proves $q_* \leq \epsilon^2$, with no semisimplicity assumption.

The inverse update is $G_{M+1}^{-1} = G_M^{-1} + F\Omega^{-1}F^*$. Conjugating by the original metric isometry gives

$$\pi = \det(I + UU^*)^{-1} = \prod_i (1 + \lambda_i)^{-1},$$

where all positive eigenvalues of UU^* occur with their multiplicities; the empty product is one. Since the product is at least $1 + \lambda$, $\pi \leq (1 + \lambda)^{-1}$, and therefore $\lambda/(1 + \lambda) \leq 1 - \pi$. Insert (QD.8) and $\zeta \leq \chi$ to prove (QD.10).

When $q_* > 0$, (QD.10) implies $\lambda > 0$ and $\zeta > 0$. Because $\lambda/(1 + \lambda) < 1$, it then implies $\zeta > q_*$. Rearranging its first cost inequality gives $\lambda(\zeta - q_*) \geq q_*$, proving the lower bound for λ . The function $x \mapsto q_*/(x - q_*)$ decreases for $x > q_*$, proving its comparison with the bound using χ . Finally $\pi^{-1} \geq 1 + \lambda \geq \zeta/(\zeta - q_*)$. Apply the increasing logarithm, and use $\zeta \leq \chi$, to obtain (QD.11). Adding the finite inequalities telescopes their original determinants and proves (QD.12). $\square$

At $\delta_* = 0$, (QD.10) remains the complete undivided statement. No value is assigned to $0/0$ when a derivative cost is zero. If $F = 0$, then $U = Y = 0$, $\lambda = 0$, $\pi = 1$ and $W = 0$; the derivative map C_D and its cost can still be nonzero. Conversely $\zeta = 0$ means $C_D = 0$, and the injectivity of $b\Omega^{-1/2}$ gives $V = E_+ = 0$; then $W = 0$ while the metric can still contract through F . These are retained source cases, and neither is excluded by division.

The same proof also gives the stagewise estimate with q_* replaced by the actual ϵ_M^2 whenever $\epsilon_M > 0$, since then $\zeta_M > \epsilon_M^2$. Thus a positive boundary cost remains detected even when every selected centre has real part $1/2$; no nilpotent contribution has been removed by using a zero spectral displacement.

For comparison with (FE.22)–(FE.23), retain the actual positive recurrence quantity

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), \quad a_j = \kappa_j/\kappa_{j-1}, \quad a_0 = 0.$$

The proved arithmetic incidence bound (AP.20) is $VV^* \preceq \Gamma_{k,M}I$, and therefore $\epsilon_M^2 = \|UV^*\|^2 \leq \Gamma_{k,M}\lambda_M$. It yields, for all $\delta_* \geq 0$, the finite consequence

$$\log \frac{\det G_a}{\det G_b} \geq \sum_{M=a}^{b-1} \log \left(1 + \frac{4k^2\delta_*^2}{\Gamma_{k,M}} \right). \quad (\text{QD.13})$$

Here $\Gamma_{k,M} > 0$ because every $a_{\alpha_j+1} > 0$. This is the explicit accumulated determinant consequence of the already proved quartet operator estimate (AP.17), whereas (QD.8)–(QD.12) additionally use the original derivative energy and its stronger operator-norm cost. Neither chain asserts an upper asymptotic estimate for those arithmetic quantities.

21.4 Exact energy–volume equality for the original one-variable layer

Theorem 21.3 (One-variable arithmetic equality). *For $k = 1$, at every admitted original degree M ,*

$$\boxed{\zeta_M = \chi_M, \quad \epsilon_M^2 = \frac{\lambda_M}{1 + \lambda_M} \chi_M = (1 - \pi_M) \chi_M.} \quad (\text{QD.14})$$

These equalities include $F = 0$, $E_+ = 0$, and $\delta_* = 0$. Whenever $\chi_M > 0$, they give the exact original determinant identity

$$\log \frac{\det G_M}{\det G_{M+1}} = -\log \left(1 - \frac{\epsilon_M^2}{\chi_M} \right), \quad 0 \leq \epsilon_M^2 < \chi_M. \quad (\text{QD.15})$$

Proof. For $k = 1$, (QD.1) has $r = 1$ at every M . The actual graph isometry $J_b : \mathbb{C} \rightarrow \mathcal{E}_M$ therefore proves that this original relation layer is one-dimensional. Let $e = J_b 1$, so $\|e\| = 1$. There are unique row maps $y, c : \mathbb{C}^d \rightarrow \mathbb{C}$ with $\hat{Y} = ey$ and $\hat{C} = ec$. Their actual cross-pairing is $\hat{Y}^* \hat{C} = y^* c$. Its norm is $\|y\| \|c\|$: the upper bound is submultiplicativity, and if both rows are nonzero, evaluating on the unit vector $c^*/\|c\|$ attains it. If either row is zero the claimed equality is zero on both sides. The positive matrix $c^* c$ has its only possible positive eigenvalue $\|c\|^2$; thus its largest eigenvalue and trace agree, including $c = 0$. Equations (QD.8)–(QD.9) now prove $\epsilon_M^2 = \|\hat{Y}\|^2 \chi_M = \lambda_M \chi_M / (1 + \lambda_M)$ and $\zeta_M = \chi_M$.

The matrix U has one column, so UU^* has rank at most one. Its determinant product consequently gives $\pi_M = (1 + \lambda_M)^{-1}$, including $U = 0$. This proves the last equality in (QD.14). If $\chi_M > 0$, division gives $1 - \epsilon_M^2/\chi_M = \pi_M > 0$, and applying the logarithm proves (QD.15). If $\chi_M = 0$, one retains (QD.14); it sets $\epsilon_M = 0$ and imposes no additional value on π_M . In particular the possible contraction from F remains present in the zero derivative-cost case. $\square$

The equality (QD.14) is proved for the actual one-variable arithmetic source and its original one-dimensional relation layer. Thus the coefficient one in (QD.8) is an exact source identity throughout this family, at all nonzero or zero frontier ranks; no auxiliary model is required to test it.

22 The exact two-Stieltjes-space representation of the theta packet

We retain the actual function $g(s) = 2\xi(s)$, a nonempty finite packet of its nontrivial zeros with their complete orders, and the original monic packet polynomial h . The packet is closed under conjugation and under $\rho \mapsto 1 - \bar{\rho}$, hence also under $\rho \mapsto 1 - \rho$. No zero in this packet is real: the argument with the alternating eta series in Section 2 applies to the whole real interval $0 < s < 1$. In particular $g(1/2) \neq 0$. Throughout this section all polynomial degrees, powers of $1/2$, and local nilpotent orders are retained.

22.1 The quadratic algebra and both reflection branches

Put

$$z = s - \frac{1}{2}, \quad x = -z^2, \quad d = \deg h = 2D. \quad (\text{SP.1})$$

The even degree follows by partitioning the packet into reflection pairs $\{\rho, 1 - \rho\}$; each pair has the same order at its two members and neither member is $1/2$. For each such pair choose one member, write $\alpha_\rho = \rho - 1/2$, and put

$$r_\rho = -\alpha_\rho^2, \quad H(x) = \prod_{\{\rho, 1-\rho\}} (x - r_\rho)^{m_\rho}. \quad (\text{SP.2})$$

The product is independent of the chosen member. Distinct reflection pairs have distinct r_ρ , because equal squares imply equal or opposite α_ρ . Thus H is monic of degree D , $H(0) \neq 0$, and every original order occurs unchanged in H . Multiplication of the two factors in each pair proves

$$h(s) = (-1)^D H\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.3})$$

Conjugation permutes the factors of H , so H has real coefficients. Define $A_H = \mathbb{C}[x]/(H)$, let X denote multiplication by x in its basis $1, x, \dots, x^{D-1}$, and let c_H be the column of its unit.

Theorem 22.1 (A full algebra isomorphism with its original coordinates). *There is a unital complex algebra isomorphism*

$$\begin{aligned} \mathcal{T} : \mathbb{C}[s]/(h) &\longrightarrow \mathbb{C}[x, z]/(H(x), z^2 + x) = A_H \oplus zA_H, \\ [s] &\longmapsto \frac{1}{2} + z. \end{aligned} \quad (\text{SP.4})$$

The inverse sends x to $-(s - 1/2)^2$ and z to $s - 1/2$. The multiplication in the displayed direct sum is

$$(a, b)(c, e) = (ac - xbe, ae + bc). \quad (\text{SP.5})$$

Here the direct sum describes the underlying module, with the specified multiplication. In the ordered target basis $1, x, \dots, x^{D-1}, z, zx, \dots, zx^{D-1}$, the matrix T of $\mathcal{T}$ has column n , $0 \leq n < 2D$, equal to the coefficients of

$$\begin{aligned} a_n(x) &= \sum_{2j \leq n} \binom{n}{2j} 2^{-(n-2j)} (-x)^j, \\ b_n(x) &= \sum_{2j+1 \leq n} \binom{n}{2j+1} 2^{-(n-2j-1)} (-x)^j. \end{aligned} \quad (\text{SP.6})$$

Its determinant and the original multiplication matrix are exactly

$$\det T = (-1)^{D(D-1)/2}, \quad TAT^{-1} = \frac{1}{2}I_{2D} + \begin{pmatrix} 0 & -X \\ I_D & 0 \end{pmatrix}, \quad A = M_s. \quad (\text{SP.7})$$

Proof. The substitutions in (SP.4) and its inverse send the two defining relations into zero by (SP.3) and $z^2 = -x$. They compose to the identity on the generators and therefore on both algebras. Reduction first by the monic polynomial $z^2 + x$ and then by the monic polynomial $H(x)$ gives a unique expression $a + zb$ with $\deg a, \deg b < D$. Multiplication and the relation $z^2 = -x$ give (SP.5). Expanding $(1/2 + z)^n$ and separating even and odd powers proves (SP.6). In the interlaced basis $1, z, x, zx, \dots, x^{D-1}, zx^{D-1}$, this change of basis is triangular with diagonal entry $(-1)^{\lfloor n/2 \rfloor}$. Their product is $(-1)^{D(D-1)} = 1$. Moving all the odd basis members after the even members has $D(D-1)/2$ inversions, proving the stated determinant. The columns of the inverse are explicitly the original polynomials $-(s - 1/2)^2$ and $(s - 1/2)(-s - 1/2)^2$. Finally multiplication of $a + zb$ by $1/2 + z$ gives $a/2 - xb + z(a + b/2)$, proving (SP.7). $\square$

Here is the complete local meaning of the two branches, including all derivatives. In the factor belonging to $r = -\alpha^2$, of order m , write $\eta = x - r$ and

$$B_r = \mathbb{C}[\eta]/(\eta^m), \quad Z_r = \alpha \sum_{j=0}^{m-1} \binom{1/2}{j} \left(-\frac{\eta}{\alpha^2}\right)^j. \quad (\text{SP.8})$$

The formal binomial identity gives $Z_r^2 = \alpha^2 - \eta = -x$ in B_r . Its constant term is $\alpha \neq 0$, so Z_r is invertible. There are inverse algebra maps

$$\begin{aligned} B_r[z]/(z^2 + x) &\longrightarrow B_r \times B_r, & a + zb &\longmapsto (a + Z_rb, a - Z_rb), \\ (u, v) &\longmapsto \frac{u+v}{2} + z \frac{u-v}{2Z_r}. \end{aligned} \quad (\text{SP.9})$$

The two idempotents are $(1 \pm z/Z_r)/2$. Each component has length m , and corresponds respectively to the original zero $s = 1/2 + \alpha$ or $s = 1/2 - \alpha$. To check the identification on the full local algebras, in the first original factor put $w = s - (1/2 + \alpha)$. The map is $\eta = -2\alpha w - w^2$, with inverse $w = Z_r(\eta) - \alpha$. Its linear coefficient -2α is nonzero, and the two compositions are identities modulo the respective m -th powers. For the second factor replace α by $-\alpha$. This proves the full local isomorphisms without replacing a local algebra by evaluation at its centre. Taking their products gives (SP.4).

The entire functional equation $g(1-s) = g(s)$ also has an exact form in this variable. There is an entire function

$$\Phi(x) = \sum_{n=0}^{\infty} \frac{(-1)^n g^{(2n)}(1/2)}{(2n)!} x^n, \quad g(s) = \Phi\left(-\left(s - \frac{1}{2}\right)^2\right). \quad (\text{SP.10})$$

Indeed the Taylor series of the entire even function $g(1/2 + z)$ contains only even powers, and converges for every z ; its even-power series therefore has infinite radius in $x = -z^2$. By the local isomorphisms above, Φ vanishes to exactly order m_ρ at each r_ρ . Consequently

$$V(x) = (-1)^D \frac{\Phi(x)}{H(x)} \text{ is entire,} \quad v_h(s) := \frac{g(s)}{h(s)} = V(x), \quad v = j_h v_h, \quad \mathcal{T}(v) = (V \bmod H, 0). \quad (\text{SP.11})$$

The element $V \bmod H$ is a unit in A_H , because the removed orders were complete. We write U_V for its multiplication matrix, with all its Taylor coefficients modulo H retained.

22.2 The original measures and a Hilbert space isomorphism

The two original measures are

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad d\nu_h(t) = |v_h(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}. \quad (\text{SP.12})$$

Both densities are even. They have all polynomial moments, by the actual theta decay already established for the kernel construction. They are positive except at a discrete set, hence give a positive norm to every nonzero polynomial. Their pushforwards under $t \mapsto t^2$ are, on $x > 0$, exactly

$$\begin{aligned} d\lambda_0(x) &= \frac{|\Phi(x)|^2}{2\pi\sqrt{x}} dx, \\ d\lambda_h(x) &= \frac{|V(x)|^2}{2\pi\sqrt{x}} dx, \quad d\lambda_0(x) = |H(x)|^2 d\lambda_h(x). \end{aligned} \quad (\text{SP.13})$$

For example the two half-lines contribute $2 dt = dx/\sqrt{x}$; this proves the constants in (SP.13). The formulas use the entire quotients at removable zeros.

Theorem 22.2 (Both Hilbert components with an explicit inverse). *For either pair $(\nu, \lambda) = (\nu_h, \lambda_h)$ or (μ, λ_0) , the map*

$$\begin{aligned} \mathcal{U} : L^2(\lambda) \oplus L^2(x\lambda) &\longrightarrow L^2(\nu), \\ (a, b) &\longmapsto [t \mapsto a(t^2) + itb(t^2)] \end{aligned} \quad (\text{SP.14})$$

is unitary. Its inverse, defined for almost every $x > 0$, is

$$a(x) = \frac{f(\sqrt{x}) + f(-\sqrt{x})}{2}, \quad b(x) = \frac{f(\sqrt{x}) - f(-\sqrt{x})}{2i\sqrt{x}}. \quad (\text{SP.15})$$

Multiplication by the original coordinate $s = 1/2 + it$ is carried to $(a, b) \mapsto (a/2 - xb, a + b/2)$. Its exact domain is

$$a \in L^2(\lambda) \cap L^2(x\lambda), \quad b \in L^2(x\lambda) \cap L^2(x^2\lambda). \quad (\text{SP.16})$$

Reflection $f(t) \mapsto f(-t)$ becomes $(a, b) \mapsto (a, -b)$.

Proof. The two parity components are orthogonal. More explicitly their sum of squared absolute values at t and $-t$ is $2|a(t^2)|^2 + 2t^2|b(t^2)|^2$; integrating proves $\|\mathcal{U}(a, b)\|_\nu^2 = \|a\|_\lambda^2 + \|b\|_{x\lambda}^2$. The same identity applied to the even and odd parts of an arbitrary $f \in L^2(\nu)$ proves that (SP.15) belongs to the stated direct sum and is its inverse. There is no atom at $t = 0$, so no extra value is required there. Multiplication gives $(1/2 + it)(a + itb) = a/2 - xb + it(a + b/2)$. On the original space its domain is $f \in L^2(\nu)$ with $tf \in L^2(\nu)$, since $|1/2 + it|^2 = 1/4 + t^2$. Applying the same parity norm identity to tf gives exactly (SP.16), including both conditions on each component. The reflection formula follows by substitution. $\square$

Multiplication by the original h is also preserved: from $L^2(\mu)$ to $L^2(\nu_h)$ it is an isometry, and in (SP.14) it is $(a, b) \mapsto ((-1)^D H a, (-1)^D H b)$. It is onto as an L^2 map, since division by h off its finitely many real-line zeros has precisely the inverse norm identity; those points have measure zero. Its action on polynomial degree, however, remains the actual multiplication by the monic polynomial of degree $2D$. Thus no finite polynomial subspace is identified with one of a different degree without this explicit multiplication map.

22.3 The two monic families and their original norms

Let P_n be the monic orthogonal polynomial of degree n for λ_h , and let R_n be the monic polynomial for $x\lambda_h$. Set

$$p_n = \int |P_n|^2 d\lambda_h, \quad u_n = \int x |R_n|^2 d\lambda_h. \quad (\text{SP.17})$$

All these numbers are strictly positive; the zeroth ones retain the two masses. Orthogonal projection of x^n onto lower powers in the positive moment matrix proves existence and uniqueness. Since that matrix is real, all coefficients are real.

Theorem 22.3 (Monicity, the Christoffel transform, and exact norm ratios). *The original vertical-line monic polynomials are*

$$q_{2n}(s) = (-1)^n P_n(x), \quad q_{2n+1}(s) = (-1)^n z R_n(x), \quad \kappa_{2n} = p_n, \quad \kappa_{2n+1} = u_n. \quad (\text{SP.18})$$

For every $n \geq 0$, $P_n(0) \neq 0$, and

$$c_n = \frac{P_{n+1}(0)}{P_n(0)} < 0, \quad R_n(x) = \frac{P_{n+1}(x) - c_n P_n(x)}{x}, \quad u_n = -c_n p_n. \quad (\text{SP.19})$$

Consequently the original recurrence coefficients obey

$$a_{2n+1} = -c_n \quad (n \geq 0), \quad a_{2n} = \frac{p_n}{-c_{n-1} p_{n-1}} \quad (n \geq 1). \quad (\text{SP.20})$$

The same statements with $\lambda_0, P_n^0, R_n^0, p_n^0, u_n^0$ give the original theta norms $\mathfrak{h}_j$ exactly:

$$\mathfrak{h}_{2n} = p_n^0, \quad \mathfrak{h}_{2n+1} = u_n^0 = -c_n^0 p_n^0, \quad \frac{\mathfrak{h}_{2n+1}}{\mathfrak{h}_{2n}} = -c_n^0, \quad \frac{\mathfrak{h}_{2n+2}}{\mathfrak{h}_{2n+1}} = \frac{p_{n+1}^0}{-c_n^0 p_n^0}. \quad (\text{SP.21})$$

Proof. Since $x = -z^2$, the leading coefficient of each polynomial in (SP.18) in the original variable s is one. Lower-degree even polynomials correspond to lower-degree polynomials in x , and the odd ones correspond to z times lower-degree polynomials in x . Their mixed inner products vanish by parity. Equation (SP.14) therefore proves all the required orthogonality and norm identities, and uniqueness identifies the original q_j .

If $n > 0$ and $P_n(0) = 0$, write $P_n = xQ$, with Q real, monic and of degree $n - 1$. Orthogonality to Q would give $0 = \int x |Q|^2 d\lambda_h > 0$, a contradiction. For $n = 0$, $P_0(0) = 1$. The numerator in (SP.19) vanishes at zero; its quotient by x is monic of degree n . For each polynomial L of degree below n , its inner product with L for $x\lambda_h$ is $\int (P_{n+1} - c_n P_n) \bar{L} d\lambda_h = 0$. This proves the asserted polynomial formula. Since $R_n - P_n$ has lower degree, $\int R_n P_n d\lambda_h = p_n$, while $\int R_n P_{n+1} d\lambda_h = 0$. Multiplying its numerator by R_n and integrating proves $u_n = -c_n p_n$, and positivity proves $c_n < 0$. Taking successive quotients of (SP.18) gives (SP.20). The proof uses only positivity and the actual measure identity, so its application to λ_0 proves (SP.21). $\square$

22.4 Both full jet kernels and the exact weight operator

Fix an original source degree $N \geq 2D - 1$, and put $n_e = \lfloor N/2 \rfloor$, $n_o = \lfloor (N - 1)/2 \rfloor$. Use the actual residue map $p \mapsto v_j h p$ from degree at most N , with its norm from ν_h , onto the original $E_h = \mathbb{C}[s]/h$. Write G_N^{src} for its minimum metric in the original monomial coordinates. Thus at $N = d + m$ it is precisely the earlier G_m , including the fixed-section endpoint $m = -1$. Define the full columns

$$f_n = U_V P_n(X) c_H, \quad e_n = U_V R_n(X) c_H, \quad K_e = \sum_{n=0}^{n_e} \frac{f_n f_n^*}{p_n}, \quad K_o = \sum_{n=0}^{n_o} \frac{e_n e_n^*}{u_n}. \quad (\text{SP.22})$$

No column is replaced by scalar evaluation: $P_n(X)$, $R_n(X)$, and U_V act on the whole length- D algebra. Since $n_e, n_o \geq D - 1$, the first D monic polynomials in each family give a basis modulo H . The unit U_V is invertible, so $K_e, K_o > 0$. Set $G_e = K_e^{-1}$, $G_o = K_o^{-1}$.

Theorem 22.4 (The complete metric and a positive half-line comparison). *In the coordinates (SP.4), the actual minimum metric is*

$$T^{-*} G_N^{\text{src}} T^{-1} = \begin{pmatrix} G_e & 0 \\ 0 & G_o \end{pmatrix}, \quad \det G_N^{\text{src}} = \det G_e \det G_o. \quad (\text{SP.23})$$

For $W_N = A^* G_N^{\text{src}} + G_N^{\text{src}} A - G_N^{\text{src}}$, its full transported form and original relative allowance are

$$T^{-*} W_N T^{-1} = \begin{pmatrix} 0 & G_o - G_e X \\ G_o - X^* G_e & 0 \end{pmatrix}, \quad \epsilon_{N-d} = \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|_{\text{op}}. \quad (\text{SP.24})$$

All zero eigenvalues and both signs of the relative form are retained: they are the positive and negative singular values of the displayed D -by- D matrix, with the remaining zero eigenvalues of total multiplicity twice its nullity.

Proof. For completeness, if a surjective linear map from a coefficient space with positive Gram matrix M to a target has matrix J , its minimum metric is $(JM^{-1}J^*)^{-1}$. Indeed the vector $M^{-1}J^*(JM^{-1}J^*)^{-1}y$ has image y ; it is orthogonal in M to $\ker J$, since its inner product with w in that kernel is $y^*(JM^{-1}J^*)^{-1}Jw = 0$. All other preimages differ by such a w , so Pythagoras proves the formula. The polynomial basis (SP.18) has diagonal source Gram matrix κ_j . Under T , its residue columns are $((-1)^n f_n, 0)$ or $(0, (-1)^n e_n)$. Thus the inverse minimum metric is exactly $\text{diag}(K_e, K_o)$, proving the first identity. The second follows from $|\det T| = 1$, proved in (SP.7).

Substitute (SP.7) and (SP.23) into the definition of W_N . The diagonal $1/2$ terms cancel its subtracted metric; multiplying the remaining two block matrices gives the first formula in (SP.24). Taking the congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ gives a Hermitian matrix $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$. If $Bv = \sigma u$ and $B^*u = \sigma v$, then (u, v) and $(u, -v)$ have eigenvalues σ and $-\sigma$. Singular value decomposition supplies all these vectors; the two kernels supply the remaining vectors. Its operator norm is consequently $\|B\|$. An invertible coordinate change preserves the quotient $|y^* W y|/(y^* G y)$; this proves the original relative allowance, including its zeros. $\square$

The positive comparison in (SP.24) can also be written entirely inside one copy of A_H . Define

$$R = G_e^{-1/2} G_o G_e^{-1/2} > 0, \quad L = G_e^{1/2} X G_e^{-1/2}. \quad (\text{SP.25})$$

Then

$$\epsilon_{N-d}^2 = \left\| (R - L) R^{-1} (R - L)^* \right\|_{\text{op}}. \quad (\text{SP.26})$$

To verify this without changing a metric implicitly, observe that the matrix B in (SP.24) is $(R - L) G_e^{1/2} G_o^{-1/2}$. The product of the last factor with its adjoint is R^{-1} , so $BB^* = (R - L) R^{-1} (R - L)^*$.

This proves (SP.26). In particular $\epsilon_{N-d} = 0$ holds precisely when $G_o = G_e X$. In that case X is similar to the positive Hermitian matrix R , hence is diagonalizable with positive real spectrum. The minimal polynomial of multiplication by x on A_H is H : a polynomial killing this operator kills the unit, and is therefore divisible by H . Diagonalizability then implies that H is squarefree. Thus a repeated original zero forces a strictly positive finite allowance for every N in this theorem. The complete operator in (SP.26) describes this effect of the retained nilpotents; no bound on its limiting size has been assumed.

Both finite updates are explicit. When N is even, passing to $N + 1$ leaves K_e fixed and adds $e_{n_o+1}e_{n_o+1}^*/u_{n_o+1}$ to K_o ; when N is odd it leaves K_o fixed and adds $f_{n_e+1}f_{n_e+1}^*/p_{n_e+1}$ to K_e . In either case, writing K for the changing block, v for the full added column and k for its positive norm, the determinant ratio is

$$\frac{\det G_{N+1}^{\text{src}}}{\det G_N^{\text{src}}} = \left(1 + \frac{v^* K^{-1} v}{k}\right)^{-1}. \quad (\text{SP.27})$$

Indeed $K + vv^*/k = K^{1/2}(I + ww^*)K^{1/2}$ with $w = K^{-1/2}v/\sqrt{k}$; the middle matrix is identity on $w^\perp$ and has eigenvalue $1 + \|w\|^2$ on its span. This argument also proves (SP.27) when $v = 0$, with ratio one.

22.5 The original tessierine coordinates and support lift

For an original zero $\rho = \beta + i\gamma$, let $\delta = \beta - 1/2$. The coordinate introduced here is exactly

$$r_\rho = \gamma^2 - \delta^2 - 2i\delta\gamma. \quad (\text{SP.28})$$

Retain the group-algebra element and its original characters

$$\begin{aligned} Q(\rho) &= \rho 1 + (1 - \rho)\sigma + \bar{\rho}\kappa + (1 - \bar{\rho})\sigma\kappa \quad \text{in } \mathbb{C}[K_4], \\ c_{++} &= 2, \quad c_{+-} = 0, \quad c_{-+} = 2(2\beta - 1) = 4\delta, \quad c_{--} = 4i\gamma. \end{aligned} \quad (\text{SP.29})$$

Direct multiplication, with all constants unchanged, gives

$$r_\rho - \bar{r}_\rho = -4i\delta\gamma = -\frac{c_{-+}c_{--}}{4}. \quad (\text{SP.30})$$

Every actual γ here is nonzero. It follows that r_ρ is real if and only if $\beta = 1/2$; in that event $r_\rho = \gamma^2 > 0$. Thus the selected actual packet lies on the critical line exactly when the roots of its complete H are positive real numbers. Their multiplicities are still exactly the original ones. On an eigenvector of A at ρ , the relative Rayleigh quotient of W_N is $2\beta - 1$, because $u^* A^* G u + u^* G A u - u^* G u = (2\beta - 1)u^* G u$. Hence

$$|c_{-+}|/2 = 2|\delta| \leq \epsilon_{N-d}. \quad (\text{SP.31})$$

This is the one-factor spectral bound; the tensor and full-orbit multiplicity bounds proved earlier remain in their original scope.

Finally each algebra isomorphism (SP.4) and (SP.9) induces its exact split-zero isomorphism: send τ to τ and a supported amplitude a to the supported image of a . The semiring laws follow directly from the algebra homomorphism laws and absorption by τ ; the inverse is the same lift of the displayed inverse. In particular the target of the lift of (SP.9) is $G(B_r \times B_r)$, whose common external support is retained. The map to $G(B_r) \times G(B_r)$ sends τ to (τ, τ) and a supported pair (u, v) to its two supported coordinates. It is an injective semiring homomorphism, since both original amplitudes and the external point are recovered; its image consists exactly of the two coordinates both absent or both supported. The two mixed-support fibres remain explicit in the product target. Character evaluation $G(\mathbb{C}[K_4]) \rightarrow G(\mathbb{C})$ likewise sends the supported $Q(\rho)$ to supported c_{-+} . At a critical zero this image is the supported zero e , and the common external τ remains its separate absent fibre throughout these exact maps.

22.6 The complete rank-one half-line boundary and its derivative cost

The two positive kernels in (SP.22) have a further exact compatibility identity. Their finite discrepancy retains the full local algebra, even though its matrix rank is at most one. Use the kernels at the specified original degree N , and keep every column f_n, e_n defined in (SP.22). For the two possible degree parities one has

$$K_e - XK_o = \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \quad (\text{SP.32})$$

Here the matrices K_e, K_o in the first row have upper indices $n, n-1$, and those in the second row have upper indices n, n . In particular these are different finite kernels at the two parities.

Proof. The full residue recurrence and its conjugate give the one-factor inverse-metric boundary identity

$$AC_N + C_N A^* - C_N = \frac{z_{N+1}(a_{N+1} z_N)^* + (a_{N+1} z_N) z_{N+1}^*}{\kappa_{N+1}}, \quad C_N = (G_N^{\text{src}})^{-1}, \quad z_j = U_v[q_j]_h.$$

Indeed substitute $C_N = \sum_{j=0}^N z_j z_j^* / \kappa_j$ and $Az_j = z_{j+1} + b_j z_j - a_j z_{j-1}$. The diagonal cancels because $b_j + \bar{b}_j = 1$; each interior adjacent term cancels because $a_{j+1} / \kappa_{j+1} = 1 / \kappa_j$. Only the displayed last pair remains. Transform this identity by T on the left and T^* on the right. By (SP.7) and (SP.23), its upper right block is $K_e - XK_o$.

If $N = 2n$, the two transformed columns are $Tz_N = ((-1)^n f_n, 0)$ and $Tz_{N+1} = (0, (-1)^n e_n)$. Their signs cancel, and $a_{2n+1} / \kappa_{2n+1} = 1 / p_n$, giving the first row. If $N = 2n + 1$, they are $Tz_N = (0, (-1)^n e_n)$ and $Tz_{N+1} = ((-1)^{n+1} f_{n+1}, 0)$. Their product has the minus sign, and $a_{2n+2} / \kappa_{2n+2} = 1 / u_n$, giving the second row. No column has been assumed nonzero. This proves (SP.32) on the whole length- D algebra at both degree parities. $\square$

For the following scalar contractions use the same current metrics G_e, G_o . Equation (SP.32), multiplied on the left by G_e and on the right by G_o , gives the complete outer-product formula for $G_o - G_e X$. Its exact norm is

$$\epsilon_{N-d}^2 = \begin{cases} (f_n^* G_e f_n)(e_n^* G_o e_n) / p_n^2, & N = 2n, \\ (f_{n+1}^* G_e f_{n+1})(e_n^* G_o e_n) / u_n^2, & N = 2n + 1. \end{cases} \quad (\text{SP.33})$$

To prove this, the matrix in (SP.24) becomes the outer product of $G_e^{1/2} f$ and $G_o^{1/2} e$, divided by the displayed positive norm, with the displayed parity sign. For vectors a, b , the norm of ab^* is $\|a\| \|b\|$: Cauchy-Schwarz gives the upper bound, and the vector $b / \|b\|$ attains it when $b \neq 0$. If either vector is zero the equality remains true. This proves (SP.33). Thus the full relative weight has at most the two nonzero eigenvalues $\epsilon_{N-d}, -\epsilon_{N-d}$, and all its other eigenvalues are zero. The full packet and its nilpotents still enter each of the four quadratic contractions in (SP.33).

Let χ_N denote the trace cost of the original one-factor derivative map at source degree N , as in (QD.7), and put $\pi_N = \det G_{N+1}^{\text{src}} / \det G_N^{\text{src}}$. Define

$$\begin{array}{c|c|c} & \lambda_N & \chi_N \\ \hline N = 2n & \frac{e_n^* G_o e_n}{p_n} & \frac{u_n + e_n^* G_o e_n}{p_n^2} (f_n^* G_e f_n) \\ N = 2n + 1 & \frac{f_{n+1}^* G_e f_{n+1}}{p_{n+1}} & \frac{p_{n+1} + f_{n+1}^* G_e f_{n+1}}{u_n^2} (e_n^* G_o e_n). \end{array} \quad (\text{SP.34})$$

These formulas compute the actual cost, rather than choosing one. In the original one-column graph frame the next-layer Gram is $H_{\text{lay}} = \kappa_{N+1} + z_{N+1}^* G_N^{\text{src}} z_{N+1}$, and the derivative row has coefficient $a_{N+1} z_N^* G_N^{\text{src}} / \kappa_{N+1}$. Taking its squared Hilbert–Schmidt norm against the original metric gives $\chi_N = H_{\text{lay}}(z_N^* G_N^{\text{src}} z_N) / \kappa_N^2$. Inserting the two transformed columns just used proves (SP.34), including every original norm and sign. The determinant update (SP.27) now gives, without any division by a zero column,

$$\pi_N = (1 + \lambda_N)^{-1}, \quad \epsilon_{N-d}^2 = (1 - \pi_N) \chi_N. \quad (\text{SP.35})$$

Substitution from the two rows of (SP.34) verifies the second identity against (SP.33). If the new column vanishes then $\lambda_N = 0, \pi_N = 1$ while the derivative cost may persist; if the predecessor column vanishes then $\chi_N = 0$ while the new column may still contract the metric. These are the two original zero-frontier cases, carried through the full half-line kernel formulas.

23 The exact gamma reference and the arithmetic Gram determinants

The norm sequence in (KL.24) is attached to the actual measure

$$d\mu(t) = |g(\tfrac{1}{2} + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi. \quad (\text{TG.1})$$

We construct an explicit gamma reference for this measure, prove its polynomial norms, and retain the full arithmetic multiplier in finite Gram determinants. This provides another exact formula for the original boundary allowance; no asymptotic estimate for an arithmetic determinant is assumed.

23.1 The original measure, phase, and packet cancellation

The defining identity for the completed function gives, on the original line $s = 1/2 + it$,

$$\begin{aligned} g(\tfrac{1}{2} + it) &= \Gamma(\tfrac{1}{4} + it/2) a(t), \\ a(t) &= -\pi^{-1/4} (t^2 + \tfrac{1}{4}) e^{-it \log \pi/2} \zeta(\tfrac{1}{2} + it). \end{aligned} \quad (\text{TG.2})$$

Indeed $s(s-1) = -(t^2 + 1/4)$ and $\pi^{-s/2} = \pi^{-1/4} e^{-it \log \pi/2}$. Both the minus sign and this phase remain in the multiplier.

For $\lambda > 0$, define

$$d\sigma_\lambda(t) = |\Gamma(\lambda + it/2)|^2 \frac{dt}{2\pi}. \quad (\text{TG.3})$$

In particular

$$d\mu = |a|^2 d\sigma_{1/4}, \quad d\nu_h = |a_h|^2 d\sigma_{1/4}, \quad a_h(t) = \frac{g(1/2 + it)/h(1/2 + it)}{\Gamma(1/4 + it/2)}. \quad (\text{TG.4})$$

The last expression defines the values at every zero of the original full packet h ; away from those points it equals $a(t)/h(1/2 + it)$. The function g/h is entire because the complete order of each selected zero is removed. Gamma has no zero and has no pole on the real t axis. Thus (TG.4) retains the actual cancellation at each critical packet centre, without introducing a pole into the measure.

Multiplication gives surjective Hilbert isometries

$$\begin{aligned} U_a : L^2(\mu) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto aF, \\ U_{a_h} : L^2(\nu_h) &\longrightarrow L^2(\sigma_{1/4}), \quad F \longmapsto a_h F. \end{aligned} \quad (\text{TG.5})$$

The integral identity proves the isometry. The zero set of each multiplier on the real axis is discrete, since it comes from a nonzero analytic function. Given $H \in L^2(\sigma_{1/4})$, define $F = H/a$, respectively

H/a_h , off this null set and give F arbitrary finite values on it. Then $\int |F|^2 d\mu = \int |H|^2 d\sigma_{1/4}$, respectively the packet identity, proving surjectivity and the stated inverse. These are maps between the indicated Hilbert spaces. Their images of polynomial subspaces are exactly $a\mathbb{C}[t]$ and $a_h\mathbb{C}[t]$, with the displayed multipliers retained. Moreover the packet isometry from (KL.23) satisfies the exact diagram

$$U_{a_h}(h(1/2 + it)P(t)) = U_a P(t). \quad (\text{TG.6})$$

23.2 Gamma norms from a generating integral

We give the integral derivation to fix all constants independently of different conventions for Meixner–Pollaczek polynomials. The beta integral, with $r = e^x$, gives

$$\int_{\mathbb{R}} \frac{e^{\lambda x}}{(1 + e^x)^{2\lambda}} e^{iux} dx = \frac{\Gamma(\lambda + iu)\Gamma(\lambda - iu)}{\Gamma(2\lambda)}. \quad (\text{TG.7})$$

For completeness the beta integral follows by applying the change of variables $v = r_1 + r_2$, $y = r_1/(r_1 + r_2)$ to the product of the two absolutely convergent gamma integrals; the Jacobian is v . The conditions $\text{Re}(\lambda \pm iu) > 0$ hold for real u . The function on the left before Fourier transformation and its second derivative are integrable: at each end they are bounded by a constant times $e^{-\lambda|x|}$. Two integrations by parts therefore make its Fourier transform integrable. Fourier inversion applies. One direct justification is to multiply the transform by $e^{-\varepsilon u^2}$, apply Fubini, and use the Gaussian approximate identity; dominated convergence removes that factor because the transform is integrable. Using its evenness and then $t = 2u$, we obtain

$$\int_{\mathbb{R}} e^{iyt} d\sigma_{\lambda}(t) = \frac{2\Gamma(2\lambda)}{(2 \cosh y)^{2\lambda}} = 2^{1-2\lambda} \Gamma(2\lambda) (\cosh y)^{-2\lambda}. \quad (\text{TG.8})$$

The factor 2 in the numerator is the Jacobian of the original change $t = 2u$.

The following integral proves the required exponential integrability.

The function $(2 \cosh(x/2))^{-2\lambda}$ extends holomorphically to $|\text{Im } x| < \pi$, using the branch positive on the real axis. On each closed narrower strip it is bounded by $Ce^{-\lambda|\text{Re } x|}$. In (TG.7), for $u > 0$, shift the contour to $\mathbb{R} + ic$, where $0 < c < \pi$. The vertical sides of the truncated rectangle tend to zero by that bound. For $u < 0$, shift down instead. It follows that

$$|\Gamma(\lambda + iu)|^2 \leq C_{\lambda,c} e^{-c|u|}.$$

Consequently $\int e^{v|t|} d\sigma_{\lambda}(t) < \infty$ for every $0 < v < \pi/2$, by choosing $2v < c < \pi$. This proves holomorphic continuation of both sides of (TG.8) to $|\text{Im } y| < \pi/2$, where they agree by the identity theorem and use the branch inherited from the real axis.

For z near zero use the logarithms that vanish at $z = 0$ and set

$$F_{\lambda}(z, t) = (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} = \sum_{n \geq 0} P_n^{(\lambda)}(t/2; \pi/2) z^n. \quad (\text{TG.9})$$

For real z near zero this is $(1 + z^2)^{-\lambda} \exp(t \arctan z)$. Apply (TG.8) at $y = -i(\arctan z + \arctan w)$, where z, w are real and sufficiently small. The identity

$$\cos(\arctan z + \arctan w) = \frac{1 - zw}{\sqrt{1 + z^2} \sqrt{1 + w^2}}$$

then proves

$$\int_{\mathbb{R}} F_{\lambda}(z, t) F_{\lambda}(w, t) d\sigma_{\lambda}(t) = 2^{1-2\lambda} \Gamma(2\lambda) (1 - zw)^{-2\lambda}. \quad (\text{TG.10})$$

All derivatives in z, w at zero can be passed through the integral: on a smaller closed bidisc their integrands are bounded by a polynomial in $|t|$ times $e^{c|t|}$, with $c < \pi/2$. Thus comparison of the Taylor coefficients in (TG.10) is justified, not formal.

Define the monic polynomial and its unchanged squared norm by

$$b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2), \quad \gamma_n^{(\lambda)} = 2^{1-2\lambda}n!\Gamma(n+2\lambda). \quad (\text{TG.11})$$

The coefficient of t^n in (TG.9) is $1/n!$, since $\arctan z = z + O(z^3)$. Consequently $b_n^{(\lambda)}$ is monic. Equation (TG.10) gives the full orthogonality identity

$$\int_{\mathbb{R}} b_n^{(\lambda)}(t)b_j^{(\lambda)}(t)d\sigma_\lambda(t) = \gamma_n^{(\lambda)}\delta_{nj}. \quad (\text{TG.12})$$

All these polynomials have real coefficients, as is also clear from the real form of the generating function. Differentiating (TG.9) gives $(1+z^2)\partial_z F_\lambda = (t-2\lambda z)F_\lambda$. Its coefficient of z^n proves

$$\begin{aligned} b_0^{(\lambda)} &= 1, & b_1^{(\lambda)} &= t, \\ b_{n+1}^{(\lambda)} &= tb_n^{(\lambda)} - n(n+2\lambda-1)b_{n-1}^{(\lambda)}, & n &\geq 1, \\ \gamma_n^{(\lambda)}/\gamma_{n-1}^{(\lambda)} &= n(n+2\lambda-1), & n &\geq 1. \end{aligned} \quad (\text{TG.13})$$

For the actual reference $\lambda = 1/4$, abbreviate $b_n = b_n^{(1/4)}$ and $\gamma_n = \gamma_n^{(1/4)}$. Thus

$$\boxed{\gamma_n = \sqrt{2}n!\Gamma(n+\tfrac{1}{2}), \quad \gamma_{n+1}/\gamma_n = (n+1)(n+\tfrac{1}{2}).} \quad (\text{TG.14})$$

These formulas agree, with the displayed Jacobian and monic factors, with NIST DLMF §18.19, equations (18.19.8)–(18.19.9), and §18.22, equation (18.22.8).

23.3 Finite arithmetic determinants in the retained metrics

For $N \geq 0$ let $N = \text{diag}(\gamma_0, \dots, \gamma_N)$, and keep both matrices

$$\begin{aligned} \mathbf{B}_N &= \left(\int_{\mathbb{R}} b_i(t)b_j(t)|a(t)|^2 d\sigma_{1/4}(t) \right)_{0 \leq i,j \leq N}, \\ \mathbf{J}_N &= {}_N^{-1/2} \mathbf{B}_N {}_N^{-1/2}, \quad D_N = \det \mathbf{J}_N, \quad D_{-1} = 1. \end{aligned} \quad (\text{TG.15})$$

The diagonal factors are explicit invertible coordinate maps; $\mathbf{B}_N = {}_N^{1/2} \mathbf{J}_N {}_N^{1/2}$ recovers every original inner product. The integrals converge with no unproved growth estimate. Indeed, integration of the step function $\lfloor x \rfloor$ gives, initially for $\text{Re } s > 1$, and then by analytic continuation for $\text{Re } s > 0, s \neq 1$,

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx.$$

The initial equality follows by integrating each constant step on $[n, n+1)$, or by summation by parts of the absolutely convergent Dirichlet series. The last integral converges absolutely and locally uniformly in the asserted half-plane since $0 \leq \{x\} < 1$. On $s = 1/2 + it$, it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4}$, because $|s/(s-1)| = 1$. Equation (TG.2) therefore bounds $|a(t)|$ by a polynomial of degree three. The exponential integrability just proved gives all its polynomial moments. The packet multiplier has the same property on the tail because h is monic, and is bounded on each compact interval by its cancelled analytic expression in (TG.4). Each matrix is positive definite: the integral of the squared modulus of a nonzero polynomial is positive, because a vanishes only at isolated real points.

Theorem 23.1 (The actual theta norms as arithmetic determinants). *For all $n \geq 0$, with the initial value included,*

$$\mathfrak{h}_n = \gamma_n \frac{D_n}{D_{n-1}}, \quad \frac{\mathfrak{h}_{n+1}}{\mathfrak{h}_n} = (n+1)(n+\frac{1}{2}) \frac{D_{n+1}D_{n-1}}{D_n^2}. \quad (\text{TG.16})$$

Proof. Let $Q_n(t)$ be the monic polynomial orthogonal to degrees below n in $L^2(\mu)$. Its existence and uniqueness follow from the positive definite Gram matrix. Expressing $Q_0, \dots, Q_N$ in $b_0, \dots, b_N$ gives an upper triangular matrix with diagonal entries one. The determinant of the Gram matrix is therefore unchanged, and orthogonality gives $\det \mathbf{B}_N = \prod_{j=0}^N \|Q_j\|_\mu^2$.

The algebra isomorphism $\Psi : \mathbb{C}[s] \rightarrow \mathbb{C}[t]$, $(\Psi p)(t) = p(1/2 + it)$, preserves the original integral inner product. On the affine space of degree- n monic polynomials the map $p \mapsto i^{-n} \Psi p$ is its monic version and preserves the norm because $|i^{-n}| = 1$. It takes all lower-degree polynomial subspaces onto themselves. Thus the original monic theta residual has squared norm $\mathfrak{h}_n = \|Q_n\|_\mu^2$. The degree-dependent phase is used on this affine space; it is not asserted to define a linear map on the entire polynomial algebra.

Finally $D_N = \det \mathbf{B}_N / \prod_{j=0}^N \gamma_j$. Divide this equality at $N = n$ by the equality at $N = n-1$, taking an empty product to be one. This proves the first formula. Dividing consecutive formulas and using (TG.14) proves the second. $\square$

Define $\mathbf{B}_N^{(h)}, \mathbf{J}_N^{(h)}, D_N^{(h)}$ by (TG.15) with the entire-cancelled multiplier a_h of (TG.4). The same proof, with no restriction on whether the packet centres are critical, gives

$$\kappa_n^{(h)} = \gamma_n \frac{D_n^{(h)}}{D_{n-1}^{(h)}}. \quad (\text{TG.17})$$

Combining (TG.17) and (TG.16) with (KL.24) yields the exact packet-to-reference determinant map, for every $m \geq 0$,

$$r_m^{(h)} = \frac{(d+m)! \Gamma(d+m+1/2)}{m! \Gamma(m+1/2)} \frac{D_{d+m}^{(h)}}{D_{d+m-1}^{(h)}} \frac{D_{m-1}}{D_m}. \quad (\text{TG.18})$$

Every determinant is strictly positive and the full degree $d = \sum_\rho m_\rho$ is retained.

For a packet stable under both ordinary conjugation and $\rho \mapsto 1 - \bar{\rho}$, and for every $m \geq 0$, (KL.32) becomes

$$\epsilon_m^2 = (m+1)(m+\frac{1}{2}) \frac{D_{m+1}D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{TG.19})$$

This is the exact same finite-stage allowance. Its factor $(m+1)(m+1/2)$ is explicit; the arithmetic determinant quotient remains in the formula.

23.4 Full confluent kernel and the original coordinates

Set $\hat{b}_j(s) = i^j b_j((s-1/2)/i)$. These are monic polynomials in the original variable s . With $\hat{\mathbf{r}}_N = \text{diag}(1, i, \dots, i^N)$, their actual packet Gram matrix is

$$\mathbf{B}_{N,s}^{(h)} = \hat{\mathbf{r}}_N^* \mathbf{B}_N^{(h)} \hat{\mathbf{r}}_N. \quad (\text{TG.20})$$

Indeed $\hat{b}_j(1/2 + it) = i^j b_j(t)$; taking the conjugate in the first argument gives exactly the displayed congruence, including every phase.

Let $\mathbf{C}_N$ have columns $[\hat{b}_j]_h \in E_h = \mathbb{C}[s]/(h)$, in the original monomial remainder coordinates. For $N \geq d-1$,

$$K_N = \mathbf{C}_N (\mathbf{B}_{N,s}^{(h)})^{-1} \mathbf{C}_N^*, \quad \mathbf{G}_{N-d} = U_{\varepsilon_h}^* K_N^{-1} U_{\varepsilon_h}. \quad (\text{TG.21})$$

Here $\mathbf{G}_{-1}$ is the fixed initial-section metric, and $\varepsilon_h = (j_h(g/h))^{-1}$ is the same full local unit as before. To prove the first identity, choose the actual monic orthogonal polynomial basis through degree N . If T is its invertible coefficient matrix in the $\hat{b}_j$ basis, then

$$T^* \mathbf{B}_{N,s}^{(h)} T = \text{diag}(\kappa_0, \dots, \kappa_N).$$

Inverting gives $(\mathbf{B}_{N,s}^{(h)})^{-1} = T \text{diag}(\kappa_j^{-1}) T^*$. The columns of $\mathbf{C}_N T$ are the full remainders $[p_j]_h$. Substitution proves exactly $K_N = \sum_{j=0}^N [p_j]_h [p_j]_h^* / \kappa_j$, the original interpolation kernel. The first d monic polynomials span all remainders, so $\mathbf{C}_N$ is onto and K_N is positive definite. The second identity is the fixed-unit minimum-lift formula already proved in (KL.27), with the exact index $N = d + m$. No single remainder column is required to be nonzero or invertible.

For a centre ρ , put $t_\rho = (\rho - 1/2)/i$ and write $f^{[\ell]} = f^{(\ell)}/\ell!$. The full Taylor coordinates of these columns are explicitly

$$\hat{b}_j^{[\ell]}(\rho) = i^{j-\ell} b_j^{[\ell]}(t_\rho), \quad 0 \leq \ell < m_\rho. \quad (\text{TG.22})$$

This follows by differentiating the affine argument ℓ times. CRT converts these full Taylor coordinates to the monomial remainder coordinates, and multiplication by U_{v_h} retains all derivatives of g/h . Thus (TG.21) computes the entire confluent kernel, including nilpotent orders and critical centres.

23.5 The finite map to Romik's gamma parameter

Romik's expansion uses $P_n^{(3/4)}(t/2; \pi/2)$, with the unchanged variable $t/2$; see his §3 and Appendix A.2. The exact finite change to the parameter in (TG.3) is

$$\begin{aligned} b_n^{(1/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{1/2}{k} b_{n-2k}^{(3/4)}(t), \\ b_n^{(3/4)}(t) &= \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{n!}{(n-2k)!} \binom{-1/2}{k} b_{n-2k}^{(1/4)}(t). \end{aligned} \quad (\text{TG.23})$$

To prove these formulas, divide the two generating functions: $F_{1/4}(z, t) = (1 + z^2)^{1/2} F_{3/4}(z, t)$. The binomial series at zero and coefficient comparison prove the first formula. Multiplication by $(1 + z^2)^{-1/2}$ proves the inverse. Each finite coefficient matrix is triangular with diagonal one, preserves parity, and is invertible. Applied to the same measure, it transports Gram matrices by its full congruence and transports every Taylor column by the same coefficient matrix. Consequently the kernel expression (TG.21) is invariant under this change of polynomial basis.

The two reference measures themselves have the exact Hilbert-space map

$$L^2(\sigma_{1/4}) \longrightarrow L^2(\sigma_{3/4}), \quad F(t) \longmapsto \frac{\Gamma(1/4 + it/2)}{\Gamma(3/4 + it/2)} F(t). \quad (\text{TG.24})$$

The gamma quotient has neither zeros nor poles for real t . Cancellation of its squared modulus proves the isometry and its inverse; this isometry has a multiplier, whereas (TG.23) is the finite polynomial coordinate map. Their respective domains and metrics have both been given explicitly.

Romik's coefficient asymptotics concern the expansion of Ξ in a fixed gamma-orthogonal basis. Inoue's §2 similarly treats expansion coefficients and convergence. In the present construction the arithmetic norms are the determinants (TG.16). Equations (TG.20)–(TG.24) are the exact maps connecting those polynomial presentations to these determinants; no assertion that their coefficient asymptotics already estimate D_n is used.

23.6 Exact scope of the exponential-weight theorem

The original theorem of Lubinsky–Mhaskar–Saff does include tail order one. In their 1988 paper, Definition 2.1 requires $W = e^{-Q}$, with Q even and continuous, Q' existing on $(0, \infty)$, $xQ'(x)$ bounded at zero, and, eventually,

$$Q' > 0, \quad x^2|Q'''|/Q' \leq C, \quad 1 + xQ''/Q' \rightarrow \alpha > 0. \quad (\text{TG.25})$$

Theorem 2.3 admits a finite generalized Jacobi factor, a lower-degree polynomial exponential, and a bounded nonnegative factor $\Psi(t) \rightarrow 1$. Its Theorem 2.4 replaces the polynomial exponential by V , but requires $\log V(t)/Q(t) \rightarrow 0$, as well as bounded reciprocal approximation on the expanding interval by polynomials of degree $o(n)$. The precise statements are on printed pages 67–70; their proof is on pages 80–81. These hypotheses concern the full weight, not its gamma factor alone.

We prove a concrete obstruction to inserting the actual arithmetic multiplier into those displayed hypotheses. Hardy’s theorem supplies real numbers $T_j \rightarrow \infty$ with $g(1/2 + iT_j) = 0$. Removing a fixed full polynomial packet leaves an infinite subsequence of these zeros. At each remaining T_j , the multiplier a_h of (TG.4) has the local form

$$a_h(t) = (t - T_j)^{\ell_j} v_j(t), \quad \ell_j \geq 1, \quad v_j(T_j) \neq 0, \quad (\text{TG.26})$$

where v_j is analytic in a neighbourhood of T_j . This follows from the Taylor series at an isolated zero, with its entire order ℓ_j retained.

Let $B(t) > 0$ be any continuous comparison amplitude near all sufficiently large T_j , and suppose it has no zeros there. For every j and every $\eta > 0$, continuity and (TG.26) give a nonempty interval $I_{j,\eta}$ about T_j on which

$$|a_h(t)|/B(t) < \eta. \quad (\text{TG.27})$$

Thus a decomposition with this ratio as the bounded multiplier Ψ cannot satisfy $\Psi \rightarrow 1$. The obstruction holds on intervals of positive measure, so changing the value only at each zero does not repair it. A finite generalized Jacobi factor has only finitely many real zeros or singularities; choosing the subsequence beyond them leaves (TG.27) unchanged.

For the logarithmic hypothesis choose any eventually positive continuous Q . At a remaining T_j , shrink the neighbourhood so that $Q(t) \leq Q(T_j) + 1$, and choose $\eta = \exp(-j(Q(T_j) + 1))$ in (TG.27). On the resulting positive-measure interval,

$$\frac{\log(|a_h(t)|/B(t))}{Q(t)} < -j \quad (t \neq T_j). \quad (\text{TG.28})$$

Hence this ratio cannot be V with $\log V/Q \rightarrow 0$. An additional factor $\Psi \rightarrow 1$ is bounded away from zero and infinity on the tail and does not remove the contradiction. This disproves the indicated application of Theorems 2.3–2.4, including almost-everywhere modifications of the weight. It makes no claim that other asymptotic methods are unavailable. The exact positive Gram matrices (TG.15)–(TG.21) retain the arithmetic information at all those zeros and are available for such further analysis.

23.7 Primary source locators

The content-reading route and source hashes are recorded in `work/theta_norm_literature_route_20260912.json` and `work/theta_norm_literature_sources/download_receipt.json`. The principal public locators are:

- D. S. Lubinsky, H. N. Mhaskar and E. B. Saff, *A proof of Freud’s conjecture for exponential weights*, Constructive Approximation 4 (1988), 65–83, Definition 2.1, Theorems 2.3–2.5 and §4: <https://math.vanderbilt.edu/saffeb/texts/81.pdf>.

- Dan Romik, *Orthogonal polynomial expansions for the Riemann xi function in the Hermite, Meixner–Pollaczek, and continuous Hahn bases*, §3.1–3.3 and Appendix A.2: <https://www.math.ucdavis.edu/~romik/data/uploads/papers/riemannxi-final.pdf>.
- Hiroto Inoue, *Expansion of Dirichlet L-function on the critical line in Meixner–Pollaczek polynomials*, arXiv:1412.1220, §2.1–2.2: <https://arxiv.org/abs/1412.1220>.
- NIST DLMF, equations (18.19.8)–(18.19.9), (18.22.8): <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22>.
- G. H. Hardy, *Sur les zéros de la fonction $\zeta(s)$ de Riemann*, Comptes rendus 158 (1914), 1012–1014, §1–3. The exact page images and transcriptions are linked in the machine-readable source route.

24 The symmetric spectral sum and its unequal Stieltjes pair

The complete PR 17 note, *Spectral-sum descent with the relative arithmetic fibre retained*, supplies the invariant relation ideal and the original matrix weight used below. We prove their further reflection decomposition, its full paired-packet defect, and the minimum-metric and weight formulas. The coordinate x in this section belongs to the spectral sum. We write x_h for the one-factor coordinate of Section 22; their exact relation on the defect will be proved.

24.1 Original polynomial coordinates and the reflection ideals

Keep $g = 2\xi$, the same nonempty quartet-closed packet h with every full order, $d = 2D$, and

$$h(s) = (-1)^D H(-(s - \tfrac{1}{2})^2), \quad A_H = \mathbb{C}[x_h]/(H), \quad v_h = g/h, \quad v = j_h v_h. \quad (\text{SSP.1})$$

In particular $D \geq 1$ and $H(0) \neq 0$, as proved in (SP.1)–(SP.3). Write

$$\begin{aligned} S &= s_1 + s_2, & r &= s_1 - s_2, & \Delta &= r^2, & Z &= S - 1, & x &= -Z^2, \\ s_1 &= \tfrac{1}{2} + (Z + r)/2, & s_2 &= \tfrac{1}{2} + (Z - r)/2. \end{aligned} \quad (\text{SSP.2})$$

The swap fixes S and negates r . Thus substitution gives a filtered algebra isomorphism $\mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{S_2}$, with $\deg S = 1$, $\deg \Delta = 2$: expansion in r keeps precisely its even powers, and the inverse coordinate map in (SSP.2) keeps the factors $1/2$.

Write $h(1/2 + y) = \sum_{j=0}^D b_j y^{2j}$, with $b_D = 1$. Define the following entire finite polynomials:

$$\begin{aligned} A_0(x, \Delta) &= \sum_{j=0}^D b_j 4^{-j} \sum_{\ell=0}^j \binom{2j}{2\ell} (-x)^{j-\ell} \Delta^\ell, \\ A_1(x, \Delta) &= \sum_{j=1}^D b_j 4^{-j} \sum_{\ell=0}^{j-1} \binom{2j}{2\ell+1} (-x)^{j-\ell-1} \Delta^\ell, \quad Q_\Delta = \Delta A_1. \end{aligned} \quad (\text{SSP.3})$$

Binomial expansion, with $x = -Z^2$, proves exactly

$$\begin{aligned} h(s_1) &= A_0 + rZ A_1, & h(s_2) &= A_0 - rZ A_1, \\ A_0(0, \Delta) &= (-1)^D H(-\Delta/4), & A_1(0, \Delta) &= \frac{(-1)^{D+1}}{2} H'(-\Delta/4). \end{aligned} \quad (\text{SSP.4})$$

For the last identity one can also differentiate the difference $h(1/2 + (Z + r)/2) - h(1/2 + (Z - r)/2)$ at $Z = 0$, and divide by $2r$; both sides are polynomials, so the identity includes $r = 0$. The polynomial $A_0(0, \Delta)$ has degree D and leading coefficient 4^{-D} . No localization at r , Δ , Z , or x has been made.

Theorem 24.1 (The original symmetric algebra with both parity modules). *Let $\mathcal{E} = (E_h \otimes E_h)^{S_2}$, $E_h = \mathbb{C}[s]/(h)$. There is an exact algebra isomorphism*

$$\mathcal{E} \cong \mathbb{C}[x, Z, \Delta]/(Z^2 + x, A_0, ZQ_\Delta). \quad (\text{SSP.5})$$

Put

$$B_e = \mathbb{C}[x, \Delta]/(A_0, xQ_\Delta), \quad B_o = \mathbb{C}[x, \Delta]/(A_0, Q_\Delta). \quad (\text{SSP.6})$$

The underlying module of (SSP.5) is $B_e \oplus ZB_o$. The two maps and the product are

$$\begin{aligned} \beta : B_e &\rightarrow B_o, & [a] &\mapsto [a], & \alpha : B_o &\rightarrow B_e, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \alpha(be), \beta(a)e + \beta(c)b), \\ \alpha\beta &= xI_{B_e}, & \beta\alpha &= xI_{B_o}. \end{aligned} \quad (\text{SSP.7})$$

The map β is a unital algebra homomorphism; α is a homomorphism of $\mathbb{C}[x, \Delta]$ -modules. Its full multiplicative relation is $\alpha(b)\alpha(c) = x\alpha(bc)$, and $\alpha(1) = x$. Reflection in both original variables acts as $(a, b) \mapsto (a, -b)$. The dimensions are

$$\dim B_e = D(D + 1), \quad \dim B_o = D^2. \quad (\text{SSP.8})$$

Proof. In the polynomial algebra $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) , where $H_0 = A_0(-Z^2, \Delta)$ and $H_1 = ZA_1(-Z^2, \Delta)$. The factors $1/2$ in the sum and difference of the original generators give both ideal containments. An invariant polynomial in this ideal can be written $H_0a + rH_1b$. Average this expression under $r \mapsto -r$. The result is $H_0a_e + \Delta H_1b_o$, where a_e and b_o are polynomials in S, Δ ; the odd part of b is rb_o . Conversely both invariant generators lie in the original ideal. Every invariant class in the quotient has an invariant lift by this same averaging. Therefore

$$\mathcal{E} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1). \quad (\text{SSP.9})$$

Replace S by $1 + Z$ and adjoin $x = -Z^2$; this proves (SSP.5) with inverse given by the original substitutions.

Before imposing A_0, ZQ_Δ , division by $Z^2 + x$ gives the free module $\mathbb{C}[x, \Delta] \oplus Z\mathbb{C}[x, \Delta]$. Multiplying the two remaining generators by arbitrary elements gives

$$A_0(a + Zb) + ZQ_\Delta(c + Ze) = A_0a - xQ_\Delta e + Z(A_0b + Q_\Delta c). \quad (\text{SSP.10})$$

Its even and odd coefficient ideals are precisely those in (SSP.6). This proves the module decomposition, including both directions of its defining relations. Multiplication by Z is $(a, b) \mapsto (-\alpha b, \beta a)$, so direct multiplication proves (SSP.7) and the two compositions. All operations are well-defined by the displayed ideals.

Reflection sends Z, r to $-Z, -r$, and fixes x, Δ . The one-factor decomposition (SP.4) has reflection spaces of dimensions D, D . On the invariant tensor square the positive space is the sum of their two symmetric squares, with dimension $2D(D + 1)/2$. The negative space consists of their mixed symmetrized tensors, with dimension D^2 . They are exactly the even and odd coefficient modules in (SSP.10), proving (SSP.8), including $D = 1$. $\square$

24.2 The entire defect is the original paired packet

Theorem 24.2 (Two exact sequences with every original local order). *The map α is injective and β is surjective. Regard A_H as a $\mathbb{C}[x, \Delta]$ -module through $x \mapsto 0, \Delta \mapsto -4x_h$. The two exact sequences of modules are*

$$\begin{aligned} 0 &\longrightarrow B_o \xrightarrow{\alpha} B_e \xrightarrow{\rho_0} A_H \longrightarrow 0, & \rho_0[a] &= [a(0, -4x_h)], \\ 0 &\longrightarrow A_H \xrightarrow{\iota_0} B_e \xrightarrow{\beta} B_o \longrightarrow 0, & \iota_0[q] &= [Q_\Delta q(-\Delta/4)]. \end{aligned} \quad (\text{SSP.11})$$

In particular both defects retain the original polynomial H and every order in it. On the full algebra $\mathcal{E}$, multiplication by the original sum is

$$A_S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \quad (\text{SSP.12})$$

Its full kernel and cokernel at $S = 1$ are

$$\ker(A_S - I) = \{(\iota_0 q, 0) : q \in A_H\}, \quad \text{coker}(A_S - I) \cong A_H, \quad [(a, b)] \mapsto \rho_0 a. \quad (\text{SSP.13})$$

Proof. Let $\mathcal{R} = \mathbb{C}[x, \Delta]$. Since $A_0(0, \Delta) \neq 0$, the prime polynomial x does not divide A_0 . If $xb = A_0c + xQ_\Delta e$, then $x(b - Q_\Delta e) = A_0c$ forces $c = xc_1$, whence $b = A_0c_1 + Q_\Delta e$. This proves injectivity of α without dividing by x in either quotient. Surjectivity of β follows from its quotient definition. The image of α is xB_e , because every polynomial representative of an even class is also an odd coefficient representative. Therefore

$$\text{coker } \alpha \cong \mathcal{R}/(A_0, x) \cong \mathbb{C}[\Delta]/((-1)^D H(-\Delta/4)) \cong A_H. \quad (\text{SSP.14})$$

The last isomorphism sends Δ to $-4x_h$; its inverse sends x_h to $-\Delta/4$. This proves the first sequence, with all constants.

We justify the cancellation required for the second sequence. The quotient $\mathcal{R}/(A_0, Q_\Delta) = B_o$ has finite dimension D^2 , already proved in (SSP.8). If a nonconstant irreducible polynomial divided both A_0, Q_Δ , then it would divide every polynomial annihilator of x and every polynomial annihilator of Δ in this quotient. Such nonzero univariate annihilators exist by linear dependence of the powers in the finite vector space. Over $\mathbb{C}$, a common irreducible divisor of a nonzero polynomial in x alone and one in Δ alone is impossible: their irreducible factors are respectively $x - a$ and $\Delta - b$. Thus $\gcd(A_0, Q_\Delta) = 1$ in $\mathcal{R}$.

The kernel of β consists exactly of the classes $[Q_\Delta p]$ in B_e . Multiplication by Q_Δ gives a surjection

$$\mathcal{R}/(A_0, x) \longrightarrow \ker \beta, \quad [p] \longmapsto [Q_\Delta p]. \quad (\text{SSP.15})$$

It is well-defined since $A_0 Q_\Delta = x Q_\Delta = 0$ in B_e . If $Q_\Delta p = A_0 c + x Q_\Delta e$, the coprimality just proved forces $A_0 \mid p - xe$. Hence $[p] = 0$ in $\mathcal{R}/(A_0, x)$, proving injectivity. Compose (SSP.15) with the inverse last map of (SSP.14) to obtain exactly ι_0 in (SSP.11). Finally multiplication by Z is the off-diagonal part of (SSP.12). Its kernel has odd component zero by injectivity of α , and even component $\ker \beta$. Its cokernel has zero odd quotient by surjectivity of β , and even quotient $\text{coker } \alpha$. This proves (SSP.13). $\square$

The full multiplication on the second defect is also explicit. Put

$$\mathfrak{c}_H = 2(-1)^D x_h H'(x_h) \pmod{H}.$$

Then, for every $q, t \in A_H$,

$$\iota_0(q)\iota_0(t) = \iota_0(\mathfrak{c}_H qt). \quad (\text{SSP.15a})$$

Indeed $xQ_\Delta = 0$ gives $Q_\Delta^2 = Q_\Delta Q_\Delta(0, \Delta)$ in B_e . Equation (SSP.4), followed by $\Delta = -4x_h$, identifies $Q_\Delta(0, -4x_h)$ with $\mathfrak{c}_H$. Substitution in the two products proves (SSP.15a). Thus the module isomorphism ι_0 transports the inherited ideal multiplication to $q \star t = \mathfrak{c}_H qt$, retaining the derivative and its nilpotent values.

The generator in (SSP.11) is the full class $Q_\Delta = \Delta A_1$ modulo (A_0, xQ_Δ) . Reducing its coefficient at $x = 0$ before forming this class would change the map. In particular (SSP.4) can vanish on repeated local fibres through H' ; the proved injective map ι_0 still retains those fibres. Equation (SSP.15) specifies the complete mixed nilpotent class that carries them.

24.3 The original unit and all coordinate factorials

The original full unit on $\mathcal{E}$ is

$$v^{(2)} = [v_h(s_1)v_h(s_2)] \in \mathcal{E}^\times. \quad (\text{SSP.16})$$

This bracket denotes its complete double Taylor class modulo $(h(s_1), h(s_2))$, followed by the invariant identification; the values of all admitted derivatives are retained. Since $d = 2D$, $v_h(1-s) = v_h(s)$. Thus this unit has the form $(u_e, 0)$ under (SSP.5), and its two multiplication maps are

$$U_e = M_{u_e} : B_e \rightarrow B_e, \quad U_o = M_{\beta(u_e)} : B_o \rightarrow B_o. \quad (\text{SSP.17})$$

Both are invertible, by multiplication with the components of the original inverse unit. Their exact intertwiners and defect actions are

$$\begin{aligned} U_e \alpha &= \alpha U_o, & \beta U_e &= U_o \beta, & \rho_0 U_e &= M_{V(x_h)^2} \rho_0, \\ U_e \iota_0 &= \iota_0 M_{V(x_h)^2}, & V(x_h) &= (-1)^D \Phi(x_h)/H(x_h). \end{aligned} \quad (\text{SSP.18})$$

Indeed the first two equations are the module multiplication laws in (SSP.7). The algebra homomorphism $\mathcal{E} \rightarrow A_H$ given by $(a, b) \mapsto \rho_0 a$ is the original specialization $s_2 = 1 - s_1$, followed by its even quotient: $Z = 0$, $\Delta = 4(s_1 - 1/2)^2 = -4x_h$. It sends (SSP.16) to $v_h(s_1)v_h(1-s_1) = V(x_h)^2$, proving the third equation with all jets. For the last equation, $xQ_\Delta = 0$ implies

$$u_e Q_\Delta q(-\Delta/4) = Q_\Delta u_e(0, \Delta) q(-\Delta/4). \quad (\text{SSP.19})$$

The same specialization proves (SSP.18). In particular the factor $g = 2\xi$ occurs in each original unit factor and is present in the squared defect unit.

Here are explicit finite coordinate maps to the unscaled symmetric basis used earlier. Let $T : E_h \rightarrow A_H \oplus z_h A_H$ be the exact matrix (SP.6), with $z_h = s - 1/2$. Use the bases $a_i = x_h^i$ and $b_i = z_h x_h^i$, $0 \leq i < D$, in that target. The new symmetric basis is ordered in these three groups:

$$\begin{aligned} a_i \odot a_j \ (i \leq j), & \quad b_i \odot b_j \ (i \leq j), & \quad a_i \odot b_j \ (0 \leq i, j < D), \\ u \odot v = u \otimes v + v \otimes u \ (u \neq v), & \quad u \odot u = u \otimes u. \end{aligned} \quad (\text{SSP.20})$$

The first two groups are the basis of B_e , and the last, after removal of its displayed Z , is the basis of B_o . These identifications follow by the inverse substitutions in (SSP.2) and (SP.4): swap invariance leaves even powers of r , and reflection leaves even powers of Z in the first two groups and odd powers in the last. Replace $Z^{2j} r^{2a}$ by $(-x)^j \Delta^a$, and $Z^{2j+1} r^{2a}$ by $Z(-x)^j \Delta^a$. This finite coefficient prescription gives each basis polynomial explicitly, without choosing values at roots.

Let I_{old} and I_{new} insert the literal orbit sums in, respectively, the original monomial tensor basis and the basis (SSP.20) before passage to its invariant space. Put

$$\begin{aligned} D_{\text{old}} &= I_{\text{old}}^* I_{\text{old}}, & L_{\text{old}} &= D_{\text{old}}^{-1} I_{\text{old}}^*, \\ D_{\text{new}} &= I_{\text{new}}^* I_{\text{new}}, & L_{\text{new}} &= D_{\text{new}}^{-1} I_{\text{new}}^*, \\ T_s &= L_{\text{new}}(T \otimes T) I_{\text{old}}, & T_s^{-1} &= L_{\text{old}}(T^{-1} \otimes T^{-1}) I_{\text{new}}. \end{aligned} \quad (\text{SSP.21})$$

Each diagonal entry of D_{old} , D_{new} is one for a repeated basis vector and two for two distinct vectors. These are the original orbit cardinalities. The products IL are both the permutation average $(I + \text{swap})/2$; it commutes with $T \otimes T$. Hence the two displayed matrices compose to identity. This proves all factors in the coordinate maps. Moreover

$$|\det T_s| = |\det T|^{d+1} = 1. \quad (\text{SSP.22})$$

To verify the exponent, triangularize T over $\mathbb{C}$; on symmetric tensors its diagonal entries are the products of its diagonal entries indexed by $i \leq j$. Each original diagonal entry occurs $d + 1$ times. The similarity inducing that triangularization and the final permutation into (SSP.20) do not change the absolute determinant. Equation (SP.7) then proves (SSP.22). No tensor basis vector has been divided by its length.

24.4 The actual matrix weight and its positive half-line pair

Retain the original density

$$w_h(t) = |v_h(1/2 + it)|^2 / (2\pi). \quad (\text{SSP.23})$$

It is even and has the exponential and polynomial integrability proved for the theta source. For a fixed source degree M , put $L_M = \lfloor M/2 \rfloor$ and keep every relative direction $0 \leq a \leq L_M$. The PR 17 matrix weight is exactly

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (\text{SSP.24})$$

For each u , this finite matrix is real Hermitian and strictly positive. Indeed its quadratic form on a nonzero vector c is one half of the integral of $|\sum_a c_a (-v^2)^a|^2$ against the two positive densities. Their zeros are discrete, while that polynomial is nonzero, proving strict positivity. The actual decay gives finite integrals. Evenness of w_h gives $W(-u) = W(u)$, with every matrix entry retained.

For a symmetric original numerator

$$P(S, \Delta) = \sum_{a=0}^{L_M} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

the original coordinate change $u = t_1 + t_2$, $v = t_1 - t_2$ has $dt_1 dt_2 = du dv / 2$, and $S = 1 + iu$, $\Delta = -v^2$. Expansion and Fubini, justified by the actual moment bounds, give

$$\|\mathcal{T}_h^{(2)} P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* W(u) f(1 + iu) du. \quad (\text{SSP.25})$$

Here $\mathcal{T}_h^{(2)} P = P(D_1 + D_2, (D_1 - D_2)^2) F_h^{\otimes 2}$; its Mellin multiplier is $P(S, \Delta) v_h(s_1) v_h(s_2)$. Thus (SSP.25) includes both factors $1/(2\pi)$, the real Jacobian $1/2$, and every relative-coordinate sign.

Define the matrix-valued measure on $x > 0$ by

$$d\Lambda(x) = W(\sqrt{x}) \frac{dx}{\sqrt{x}}. \quad (\text{SSP.26})$$

For a matrix measure $d\Lambda = W_\Lambda dx$, its squared L^2 norm is $\int f^* W_\Lambda f dx$; every component has the declared dimension $L_M + 1$. The full unitary map and inverse are

$$\begin{aligned} \mathcal{U}_\Sigma : L^2(\Lambda) \oplus L^2(x\Lambda) &\longrightarrow L^2(\mathbb{R}, W(u) du), \\ (A, B) &\longmapsto f(1 + iu) = A(u^2) + iuB(u^2), \\ A(x) &= \frac{f(1 + i\sqrt{x}) + f(1 - i\sqrt{x})}{2}, \\ B(x) &= \frac{f(1 + i\sqrt{x}) - f(1 - i\sqrt{x})}{2i\sqrt{x}}. \end{aligned} \quad (\text{SSP.27})$$

To prove the norm identity, add the two quadratic forms at u and $-u$. The cross terms cancel and give $2A^*WA + 2u^2B^*WB$. The substitution $x = u^2$ proves the direct-sum norm with exactly (SSP.26). The same calculation on the parity parts of an arbitrary f proves that the displayed inverse belongs to both asserted spaces. There is no atom at zero, so the almost-everywhere inverse requires no value there. Multiplication by $S = 1 + iu$ is

$$(A, B) \longmapsto (A - xB, A + B), \quad \begin{cases} A \in L^2(\Lambda) \cap L^2(x\Lambda), \\ B \in L^2(x\Lambda) \cap L^2(x^2\Lambda). \end{cases} \quad (\text{SSP.28})$$

Its exact domain follows by adding $\|uf\|^2$ to $\|f\|^2$, since $|1 + iu|^2 = 1 + u^2$; the same parity calculation gives the two additional weighted norms.

The finite source image in (SSP.27) is precisely

$$\deg A_a \leq \left\lfloor \frac{M-2a}{2} \right\rfloor, \quad \deg B_a \leq \left\lfloor \frac{M-2a-1}{2} \right\rfloor \quad (0 \leq a \leq L_M). \quad (\text{SSP.29})$$

A negative degree bound means the zero polynomial. In particular the last odd relative direction can have a zero allowed polynomial space while its ambient Hilbert component is still retained. The exact coefficient maps are obtained by expanding each original $S^j = (1+Z)^j$:

$$(1+Z)^j = \sum_{2\ell \leq j} \binom{j}{2\ell} (-x)^\ell + Z \sum_{2\ell+1 \leq j} \binom{j}{2\ell+1} (-x)^\ell. \quad (\text{SSP.30})$$

The inverse replaces x by $-(S-1)^2$, Z by $S-1$ in $\sum_a \Delta^a(A_a(x) + ZB_a(x))$. This proves both (SSP.29) and preservation of the original weighted total-degree filtration.

24.5 Full-jet kernels for the actual constrained minimum

Fix $M \geq 2(d-1) = 4D-2$. Let $\mathcal{I}_e = \{(a,j) \in \mathbb{N}^2 : 2a+2j \leq M\}$, and $\mathcal{I}_o = \{(a,j) \in \mathbb{N}^2 : 2a+2j+1 \leq M\}$. The positive source Gram matrices and full residue columns are

$$\begin{aligned} (M_e)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell} d\Lambda_{ac}(x), & F_{a,j} &= u_e[\Delta^a x^j]_{B_e}, \\ (M_o)_{(a,j),(c,\ell)} &= \int_0^\infty x^{j+\ell+1} d\Lambda_{ac}(x), & E_{a,j} &= \beta(u_e)[\Delta^a x^j]_{B_o}. \end{aligned} \quad (\text{SSP.31})$$

Use the exact bases (SSP.20) for these columns and let J_e, J_o be the corresponding column matrices. Each Gram is strictly positive: a nonzero coefficient vector gives a nonzero polynomial vector in (SSP.29), whose squared norm is positive for the strictly positive matrix density. All its moments are finite by (SSP.24)–(SSP.26).

The two jet maps are onto at the stated degree. Indeed the original symmetric orbit basis $s_1^i s_2^j + s_1^j s_2^i$ for $i < j < d$, and the diagonal basis $s_1^i s_2^i$, has degree at most $2(d-1)$. Its full residue map is onto $\mathcal{E}$, and multiplication by the actual unit is invertible. Reflection preserves the degree filtration and this unit, so parity projection proves surjectivity onto each of B_e, B_o separately. Equivalently the concrete basis (SSP.20) has maximum degrees $4D-4, 4D-2$, and $4D-3$ in its three groups. This verifies the endpoint $M = 4D-2$, including $D=1, M=2$.

Define the actual kernels and their inverse metrics by

$$K_e = J_e M_e^{-1} J_e^*, \quad K_o = J_o M_o^{-1} J_o^*, \quad G_e = K_e^{-1} > 0, \quad G_o = K_o^{-1} > 0. \quad (\text{SSP.32})$$

All columns here are full polynomial remainders with the full unit. In particular their matrix sizes are $D(D+1)$ and D^2 , respectively.

Theorem 24.3 (The same original minimum and its unequal relative blocks). *Let R_M^s be the original symmetric canonical representative in the unscaled monomial orbit coordinates, $G_M^s = (R_M^s)^* R_M^s$, and A_M^s its original sum generator. Then*

$$\begin{aligned} T_s^{-*} G_M^s T_s^{-1} &= \text{diag}(G_e, G_o), & \det G_M^s &= \det G_e \det G_o, \\ T_s A_M^s T_s^{-1} &= I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}. \end{aligned} \quad (\text{SSP.33})$$

The original weight and its exact allowance satisfy

$$\begin{aligned} W_M^s &= (A_M^s)^* G_M^s + G_M^s A_M^s - 2G_M^s, \\ T_s^{-*} W_M^s T_s^{-1} &= \begin{pmatrix} 0 & \beta^* G_o - G_e \alpha \\ G_o \beta - \alpha^* G_e & 0 \end{pmatrix}, \end{aligned} \quad (\text{SSP.34})$$

$$\epsilon_M^s = \left\| G_e^{-1/2} (\beta^* G_o - G_e \alpha) G_o^{-1/2} \right\|_{\text{op}}.$$

If the displayed rectangular block has rank p , the relative weight has p positive and p negative eigenvalues, given by its positive singular values and their negatives. Its zero multiplicity is

$$D(D+1) + D^2 - 2p = D + 2(D^2 - p). \quad (\text{SSP.35})$$

Thus the dimension difference, all additional zero directions, and all nonzero multiplicities are retained.

Proof. The actual product source has the permutation-invariant Hilbert norm and full invariant jet map specified before (SSP.25). For a symmetric target, apply swap to its unique full minimum. The image and norm are unchanged, so uniqueness makes that minimum symmetric. Therefore minimizing over precisely the invariant degree- M polynomials gives the original R_M^s , not a new family of positive forms.

The exact coefficient transformation (SSP.30) has source Gram $\text{diag}(M_e, M_o)$; mixed terms vanish by the proved parity isometry. Its full jet map in coordinates (SSP.21) is $\text{diag}(J_e, J_o)$. For a positive source Gram M and surjective map J , the vector

$$M^{-1}J^*(JM^{-1}J^*)^{-1}y$$

has image y and is M -orthogonal to $\ker J$. Every other preimage differs by a kernel vector, and its squared norm is the squared norm of this vector plus the positive squared norm of that difference. Thus its minimum metric is exactly $(JM^{-1}J^*)^{-1}$. Apply this calculation to the two declared blocks to prove the metric congruence in (SSP.33). The coordinate representative is $R_M^s T_s^{-1}$, and its two source coefficient blocks are explicitly $M_e^{-1}J_e^*G_e$ and $M_o^{-1}J_o^*G_o$. Hence both its source function and full jets agree with the original minimum. Equation (SSP.22) proves the determinant equality. Equation (SSP.12) proves the generator similarity.

Multiplying the two generator blocks with the metric in (SSP.33) cancels the two identity terms against $2G_M^s$. This gives exactly the middle line of (SSP.34). The relative congruence by $\text{diag}(G_e^{-1/2}, G_o^{-1/2})$ has the form $\begin{pmatrix} 0 & B \\ B^* & 0 \end{pmatrix}$ with the rectangular block displayed there. For each positive singular value σ and corresponding unit singular vectors u, v , its vectors (u, v) and $(u, -v)$ have eigenvalues $\sigma, -\sigma$. The remaining kernel is $\ker B^* \oplus \ker B$, with dimensions $D(D+1) - p$ and $D^2 - p$. This proves its norm, all signs, and (SSP.35). The coordinate map

$$\text{diag}(G_e^{1/2}, G_o^{1/2})T_s(G_M^s)^{-1/2}$$

is unitary by (SSP.33) and intertwines the two relative Hermitian matrices. Thus these are the complete relative eigenvalues of the original arithmetic weight. No inverse of α, β, B , or either frontier rank has been used. $\square$

In particular zero allowance is exactly the finite equation $\beta^*G_o = G_e\alpha$, with both rectangular maps retained. It does not follow from positivity of the two source Gram matrices. The map from those positive matrices to this equation is the fully proved difference block (SSP.34). Every original zero ρ , of any order m_ρ , has an eigenvector represented in its local factor by $(s - \rho)^{m_\rho - 1}$. Its square tensor is symmetric, nonzero, and has sum eigenvalue 2ρ . Its original Rayleigh quotient is therefore

$$\frac{u^*W_M^s u}{u^*G_M^s u} = 4 \operatorname{Re} \rho - 2, \quad 4 \left| \operatorname{Re} \rho - \frac{1}{2} \right| \leq \epsilon_M^s. \quad (\text{SSP.36})$$

This retains the exact tensor factor and every original local direction in the full metric; it makes no assumption about the limiting size of that metric difference.

24.6 Original primitives and supported maps

For any invariant relation written $H_0A + \Delta H_1B$, the original identity is

$$H_0A + \Delta H_1B = h(s_1) \frac{A + rB}{2} + h(s_2) \frac{A - rB}{2}. \quad (\text{SSP.37})$$

The two original cochain primitives apply the first coefficient to $\phi_* \otimes F_h$ and the negative of the second coefficient to $F_h \otimes \phi_*$. The second tensor differential contributes its own minus sign, so their total boundary is precisely (SSP.37). For an even relation $A_0a + xQ_\Delta b$, take $A = a$, $B = -Zb$; for an odd relation $Z(A_0a + Q_\Delta b)$, take $A = Za$, $B = b$. Substitution of $Z^2 = -x$ proves both assertions with their full signs. Thus the two coefficient ideals in (SSP.6) have the original theta primitives, including the factor xQ_Δ in the even ideal. At a finite admitted degree the polynomial identity is the same original relation. For completeness, divide a relation P of total degree at most M first by the monic $h(s_1)$, then divide its remainder by the monic $h(s_2)$. The final remainder is zero because P lies in the original ideal. This gives $P = h(s_1)C_1 + h(s_2)C_2$, with $\deg C_1, \deg C_2 \leq M - d$. Average under swap and put $B_1 = (C_1 + \text{swap}(C_2))/2$, $B_2 = \text{swap}(B_1)$. Then $A = B_1 + B_2$ is even in r , and $B = (B_1 - B_2)/r$ is a polynomial even in r . Their degrees satisfy $\deg A \leq M - d$ and $\deg B \leq M - d - 1$, and their coefficients in (SSP.37) are exactly B_1, B_2 . This proves the original finite-degree primitive bounds, including a zero coefficient when its degree bound is negative.

Every algebra isomorphism in (SSP.5), (SSP.9), and (SSP.14) has the exact G -lift: it fixes external τ and sends a supported amplitude to its supported algebra image. The algebra-homomorphism laws and the absorbing rule for τ prove the semiring laws; the inverse is the lift of the displayed inverse. The algebra maps β, ρ_0 have the same functorial lift. The module maps α, ι_0 and all Hilbert, jet, kernel and representative maps above have their fixed-label linear lifts $(\lambda, v) \mapsto (\lambda, Tv)$, fixing external absence. Their exact module actions are the ones proved in (SSP.7), (SSP.11), and (SSP.18). A class killed in either exact sequence goes to the zero in its receiving supported fibre; the represented kernel and cokernel A_H retain every local order and the unit V^2 . The direct sum in (SSP.6) is equipped with the multiplication (SSP.7); its parity components have not been assigned independent external supports by this map.

25 The sum connection on both half-line branches and its exact domain

This section integrates the complete sum-connection delivery, whose source note is retained in the appendices at its declared PR 20 revision 2144c358c2b41aa235b15cf0fa472289aec8673f. We prove its mass-dependent derivative identity in the original source and then carry the full relative connection into the quadratic coordinate. Both branches, all relative coefficients, and the condition where the two branches meet are retained. Only the stated dagger stability is needed until evenness is explicitly imposed.

25.1 The actual amplitude, its mass, and the original relative fibre

Keep $g = 2\xi$, $v_h = g/h$, the entire quotient at the full selected zero orders, and the original $F_h \in \mathcal{B}$ with Mellin transform v_h . Let the packet be stable under $\rho \mapsto 1 - \bar{\rho}$. The empty polynomial $h = 1$ is also allowed for this analytic calculation; no nonzero finite jet quotient is inferred in that case. Set

$$a_h^{\text{SC}}(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h^{\text{SC}}(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (\text{FC.1})$$

The superscript distinguishes the delivered spectral amplitude from the gamma-reference multiplier in (TG.4). Their exact relation is $a_h^{\text{SC}}(t) = \Gamma(1/4 + it/2) a_h^{\text{TG}}(t) / \sqrt{2\pi}$. Taking its squared modulus gives precisely the two original measure formulas; neither mass nor phase is replaced.

For $k \geq 2$ use

$$u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = - \sum_{i < k} y_i, \quad t_i = u/k + y_i, \quad \Psi(u, \mathbf{y}) = \prod_i a_h^{\text{SC}}(t_i). \quad (\text{FC.2})$$

The Jacobian has absolute value one. One way to check its constant is to use coordinates $(u, t_1, \dots, t_{k-1})$, whose inverse sets $t_k = u - \sum_{i < k} t_i$ and has determinant of absolute value one. The replacement $t_i = y_i + u/k$, $i < k$, is triangular with diagonal one. For $k = 2$, $v = t_1 - t_2 = 2y_1$, so the earlier measure $du dv/2$ is unchanged.

The test-space estimates imply that $e^{r/2} F_h(e^r)$ and all its derivatives and polynomial multiples are Schwartz on $\mathbb{R}$. Indeed arbitrary powers of e^r bound both tails, and the logarithmic derivatives are the stated D -derivatives of F_h . Fourier differentiation and integration by parts therefore show that a_h^{SC} and all its derivatives are Schwartz. These facts justify the product integrals and differentiated integrals below.

Dagger stability gives $\overline{a_h^{\text{SC}}} = (-1)^d a_h^{\text{SC}}$. Thus this amplitude and its derivative have the same constant real or imaginary phase. Off the discrete zero set, $w'_h = 2\overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'$, with the product real, and consequently

$$I_h := \int \frac{|w'_h|^2}{w_h} = 4 \int |(a_h^{\text{SC}})'|^2 = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx, \quad \int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})' = 0. \quad (\text{FC.3})$$

Values assigned to the quotient on its discrete zero set do not change these integrals. The second equality is the original Mellin Plancherel identity; the last integral is $\frac{1}{2} \int w'_h = 0$ by Schwartz decay. In particular every number in (FC.3) is finite.

Put $m(u) = \|\Psi(u, \cdot)\|^2 = w_h^{*k}(u) > 0$. Positivity holds at each u : the factors have only discrete zeros, whose inverse images in a relative fibre are a countable union of null hyperplanes. The integral of m is exactly μ_h^k . Let $\Pi_{0,u}$ be orthogonal projection onto the line spanned by $\Psi(u, \cdot)$, and let $n(u) = (1 - \Pi_{0,u})\partial_u \Psi$. Constant phase gives $\langle \Psi, \partial_u \Psi \rangle = m'/2$, so

$$n = \partial_u \Psi - \frac{m'}{2m} \Psi, \quad \boxed{\int \frac{|m'|^2}{m} + 4 \int \|n\|^2 = \frac{\mu_h^{k-1}}{k} I_h}. \quad (\text{FC.4})$$

To prove the equality, differentiate the product in (FC.2). Its derivative is

$$k^{-1} \sum_i (a_h^{\text{SC}})'(t_i) \prod_{j \neq i} a_h^{\text{SC}}(t_j).$$

Before the common factor k^{-2} , each of the k diagonal terms in its squared norm has integral $\mu_h^{k-1} \|(a_h^{\text{SC}})'\|^2$, and every distinct-index term contains $|\int \overline{a_h^{\text{SC}}}(a_h^{\text{SC}})'|^2 = 0$. Thus its integral is $\mu_h^{k-1} I_h/(4k)$. The orthogonal line and complementary components have squared norms $|m'|^2/(4m)$ and $\|n\|^2$; integration proves (FC.4).

25.2 The complete matrix connection and its actual graph

Choose a fixed finite independent homogeneous relative polynomial family θ_α , of degrees ℓ_α , in the $k-1$ independent variables $z_i = s_i - S/k$. At source degree M , retain exactly

$$P(\mathbf{s}) = \sum_\alpha \theta_\alpha(\mathbf{z}) f_\alpha(S), \quad \deg f_\alpha \leq M - \ell_\alpha, \quad S = k/2 + iu, \quad c_P(u) = (f_\alpha(k/2 + iu))_\alpha. \quad (\text{FC.5})$$

The family can be the whole admitted relative monomial family or the specified invariant family. Its cardinality r and all degree bounds remain part of the source; it is not held fixed while M changes. In $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, define

$$\begin{aligned} j_u c &= \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha(i\mathbf{y}) c_\alpha, & W &= j_u^* j_u > 0, & B &= j_u^* j'_u, & T &= (j'_u)^* j'_u, \\ \Pi &= j_u W^{-1} j_u^*, & \Gamma &= W^{-1} B, & N &= (1 - \Pi) j'_u, & \mathcal{N} &= N^* N = T - B^* W^{-1} B. \end{aligned} \quad (\text{FC.6})$$

Positivity follows from independence of the relative polynomials and the almost-everywhere non-vanishing just used. Schwartz estimates on compact u -intervals prove smoothness of these finite matrices and of j_u in operator norm. Direct differentiation gives

$$\begin{aligned} \partial_u(j_u c) &= j_u(c' + \Gamma c) + Nc, & j_u^* N &= 0, \\ \Gamma^* W + W \Gamma &= W', & \mathcal{N} &\succeq 0, \\ \|\partial_u(j_u c)\|^2 &= (c' + \Gamma c)^* W (c' + \Gamma c) + c^* \mathcal{N} c. \end{aligned} \quad (\text{FC.7})$$

For example $j_u \Gamma = \Pi j'_u$; adding the complementary term proves the first identity. Differentiating $W = j_u^* j_u$ proves metric compatibility, and the projection identity proves the normal Gram and positivity. In this original fixed frame the common phase makes $B = B^*$, hence $B = W'/2$. After a variable change of frame $j^C = jC(u)$, one instead keeps the full formulas

$$\begin{aligned} W^C &= C^* W C, & B^C &= C^* B C + C^* W C', \\ \Gamma^C &= C^{-1} \Gamma C + C^{-1} C', & N^C &= N C, & \mathcal{N}^C &= C^* \mathcal{N} C. \end{aligned} \quad (\text{FC.8})$$

They follow by the product rule and the unchanged projection image. For instance $C = e^{iu} I$ gives $B^C = B + iW$, so the fixed-frame identity $B = W'/2$ is not silently imposed on a changed frame.

The actual derivative graph is the image of $P \mapsto (j_u(c'_P + \Gamma c_P), N c_P)$. Addition is an isometric injection on that graph by (FC.7); its image is precisely $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P$. Here $\mathcal{U}_k$ is the original Mellin-Fourier isometry followed by (FC.2). On the test space it obeys

$$\mathcal{U}_k D^{(k)} = (k/2 + iu) \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \quad (\text{FC.9})$$

The first two formulas follow by integration by parts and differentiation of the Mellin integral. Applying $-x_i \partial_{x_i}$ to $\log x_i$ gives the last formula, including $1/k$. Therefore addition of both graph components, followed by $-i \mathcal{U}_k^{-1}$ and the original full jet, recovers $J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P$. The bare derivative lands in the image of graph addition; to land in the pair graph itself one uses the actual map $v \mapsto (\Pi_u \partial_u v, (1 - \Pi_u) \partial_u v)$.

25.3 A branch-preserving quadratic map for the entire matrix weight

Put $x = u^2 > 0$, $v = \sqrt{x}$, and retain the ordered branches $u = v$ and $u = -v$. These letters denote the spectral sum coordinate, not the original positive Mellin variables in (FC.9). For any coefficient function define

$$c(v) = a(x) + ivb(x), \quad c(-v) = a(x) - ivb(x), \quad \mathbf{M}(x) = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}. \quad (\text{FC.10})$$

Its inverse is $a = (c(v) + c(-v))/2$, $b = (c(v) - c(-v))/(2iv)$. No relation between $W(v)$ and $W(-v)$ is assumed. Write $W_{\pm} = W(\pm v)$. The full positive paired density is

$$\begin{aligned} \mathbf{H}(x) &= \mathbf{M}^* \frac{\text{diag}(W_+, W_-)}{2v} \mathbf{M} \\ &= \frac{1}{2\sqrt{x}} \begin{pmatrix} W_+ + W_- & i\sqrt{x}(W_+ - W_-) \\ -i\sqrt{x}(W_+ - W_-) & x(W_+ + W_-) \end{pmatrix} > 0. \end{aligned} \quad (\text{FC.11})$$

The inverse and the change of variable on each half-line prove the unitary equality

$$\int_{\mathbb{R}} c(u)^* W(u) c(u) du = \int_0^\infty \begin{pmatrix} a \\ b \end{pmatrix}^* \mathbf{H}(x) \begin{pmatrix} a \\ b \end{pmatrix} dx. \quad (\text{FC.12})$$

Here $\begin{pmatrix} a \\ b \end{pmatrix}^*$ means the adjoint of the two-block column. Positivity follows from the invertibility of M at every $x > 0$; surjectivity on these L^2 spaces follows from its pointwise inverse and the same norm identity. Both branch weights and their off-diagonal difference remain in (FC.11).

For the precise polynomial image in (FC.5), write $f_\alpha(S) = \sum_n f_{\alpha,n} S^n$. Then

$$\begin{aligned} a_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j \leq n} \binom{n}{2j} (k/2)^{n-2j} (-x)^j, \\ b_\alpha(x) &= \sum_n f_{\alpha,n} \sum_{2j+1 \leq n} \binom{n}{2j+1} (k/2)^{n-2j-1} (-x)^j. \end{aligned} \quad (\text{FC.13})$$

Their degree bounds are respectively $\lfloor (M - \ell_\alpha)/2 \rfloor$ and $\lfloor (M - \ell_\alpha - 1)/2 \rfloor$, with a negative bound meaning a zero component. The inverse is exactly $f_\alpha(S) = a_\alpha(-(S - k/2)^2) + (S - k/2)b_\alpha(-(S - k/2)^2)$. Expanding the original powers proves both formulas and their inverse, including every $k/2$ coefficient. Thus (FC.12) retains the full filtered source.

Theorem 25.1 (The complete paired connection and normal energy). *Define*

$$\begin{aligned} \Gamma_b(x) &= \text{diag} \left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v} \right), \\ A(x) &= M^{-1} \Gamma_b M + M^{-1} M' = M^{-1} \Gamma_b M + \text{diag}(0, (2x)^{-1} I), \\ Q(x) &= M^* \frac{\text{diag}(\mathcal{N}(v), \mathcal{N}(-v))}{2v} M \succeq 0, \quad \nabla_x = \partial_x + A. \end{aligned} \quad (\text{FC.14})$$

For $q = (a, b)^T$, the exact differentiated norm is

$$\int_{\mathbb{R}} \|\partial_u(j_u c(u))\|^2 du = 4 \int_0^\infty x (\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx. \quad (\text{FC.15})$$

The transformed coefficient derivative and its metric identities are

$$\begin{aligned} J(x) \nabla_x, \quad J &= \begin{pmatrix} 0 & 2ixI \\ -2iI & 0 \end{pmatrix}, \quad J^2 = 4xI, \quad J^* H J = 4xH, \\ A^* H + H A &= H' + \frac{1}{2x} H. \end{aligned} \quad (\text{FC.16})$$

Consequently $A - (4x)^{-1} I$ is the connection matrix compatible with the full density H . The explicit Jacobian term in (FC.16) remains in the transported original derivative.

Proof. Let $F = (c(v), c(-v))^T = Mq$ and $K = \text{diag}(2vI, -2vI)$. The two original covariant derivative values are exactly $K(F' + \Gamma_b F) = KM \nabla_x q$. The minus sign on the second branch is from $d(-v)/dx = -1/(2v)$. Multiplying the explicit inverse of M gives $M^{-1}KM = J$ and $M^{-1}M' = \text{diag}(0, 1/(2x)I)$. The block-scalar matrix K commutes with the two-block weight and its squared modulus is $4xI$. This proves the first energy term and the first line of (FC.16). Applying (FC.12) to the original normal Gram proves the second energy term; (FC.7) then proves (FC.15).

For metric compatibility set $D = \text{diag}(W_+, W_-)/(2v)$. The original equality $\Gamma^* W + W \Gamma = W'$, with the two opposite derivatives of v , gives $\Gamma_b^* D + D \Gamma_b = D' + D/(2x)$. The additional term is the derivative of the factor $1/(2v)$. Substitution of $H = M^* D M$ and the two terms defining A proves the last line of (FC.16). Subtracting $1/(4x)I$ from the matrix and its adjoint removes exactly that extra term, proving the last assertion. $\square$

Multiplication by the original spectral sum has the exact paired matrix

$$S = k/2 I + \begin{pmatrix} 0 & -xI \\ I & 0 \end{pmatrix}. \quad (\text{FC.17})$$

This follows by multiplying $a + iub$ by $k/2 + iu$. On smooth compactly supported sections in $x > 0$, the commutator is $[J\nabla_x, S] = iI$. To prove it with all variable-frame terms present, conjugate to the two branch coordinates used in the proof: multiplication is $\text{diag}(k/2 + iv, k/2 - iv)$, the connection part commutes with this block-scalar matrix, and multiplication of its derivative by K gives iI on both blocks. Conjugating back proves the claim. Through (FC.9) this is the same original pair $D^{(k)}, \mathcal{L}_k$, with its original sign.

25.4 The meeting point of the two branches is part of the domain

For the L^2 identity (FC.12) no value at zero is required. For the derivative domain it is essential. Define the original domain to consist of coefficient functions c that are locally absolutely continuous on $\mathbb{R}$, have finite norm in (FC.12), and have finite derivative energy in (FC.7). Equivalently these are the coefficient functions for which $j_u c(u)$ has an L^2 weak derivative: on each compact interval the smooth left inverse $W^{-1}j_u^*$ and strict positivity of W prove both directions by the product rule. Its exact image under (FC.10) consists of locally absolutely continuous q on $(0, \infty)$ with the two finite integrals in (FC.12), (FC.15), and the joining condition

$$\lim_{x \downarrow 0} (a(x) + i\sqrt{x} b(x)) = \lim_{x \downarrow 0} (a(x) - i\sqrt{x} b(x)). \quad (\text{FC.18})$$

These finite one-sided limits exist under the stated integrability conditions. Indeed W is bounded above and below by positive constants near zero and Γ is bounded there. The two finite integrals imply that each branch coefficient and its ordinary u -derivative are square-integrable near zero. The fundamental theorem for locally absolutely continuous functions and Cauchy–Schwarz then give their finite endpoint limits. Their union has a weak derivative in L^2 precisely when those limits agree; integration by parts against a compactly supported test function shows that a mismatch contributes their nonzero jump times a delta distribution. This proves (FC.18) and both directions of the domain correspondence. For the original polynomial image (FC.13), the condition holds automatically. In general, finite energy on the two open branches alone does not impose it: a function constant with different values on the two sides near zero has zero ordinary derivative there, while its joined distributional derivative has that nonzero jump. The matching condition preserves the original domain without discarding either branch.

25.5 The even specialization and the unchanged scalar contraction

If the original packet also has conjugation closure, w_h is even. For the symmetric square with relative frame $1, \Delta, \Delta^2, \dots$, the actual j_u is even in u , since its two amplitude factors are interchanged by $u \mapsto -u$ and amplitude evenness. Thus W and $\mathcal{N}$ are even and Γ is odd. Set $L(x) = W(\sqrt{x})/\sqrt{x}$, $L_N(x) = \mathcal{N}(\sqrt{x})/\sqrt{x}$, and $\Gamma_x = \Gamma(\sqrt{x})/(2\sqrt{x})$. Equations (FC.14)–(FC.16) reduce, by direct block substitution, to

$$\begin{aligned} H &= \text{diag}(L, xL), & Q &= \text{diag}(L_N, xL_N), & A &= \text{diag}(\Gamma_x, \Gamma_x + (2x)^{-1}I), \\ \int \|\partial_u(j_u c)\|^2 du &= \int_0^\infty [4x(a' + \Gamma_x a)^* L(a' + \Gamma_x a) + (b + 2x(b' + \Gamma_x b))^* L(b + 2x(b' + \Gamma_x b)) \\ &\quad + a^* L_N a + x b^* L_N b] dx. \end{aligned} \quad (\text{FC.19})$$

These are exactly the two measures and the complete relative family of the preceding symmetric-square section. Its unequal finite quotient components are obtained from this same filtered source, using its already proved full jet and actual minimum.

For any $k \geq 2$ under the same quartet assumption, the scalar sum density m is even. Define the actual half-line density $\rho_k(x) = m(\sqrt{x})/\sqrt{x}$. It retains total mass $\int_0^\infty \rho_k = \mu_h^k$. The source

contraction (FC.4) has the precise radial form

$$4 \int_0^\infty x \frac{|\rho'_k(x) + \rho_k(x)/(2x)|^2}{\rho_k(x)} dx + 4 \int_0^\infty \frac{\|n(\sqrt{x})\|^2}{\sqrt{x}} dx = \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.20})$$

To verify it, $m(u) = u\rho_k(u^2)$ for $u > 0$, so $m'(u) = \rho_k(x) + 2x\rho'_k(x)$. Changing variables in $2 \int_0^\infty m'^2/m du$ gives its first term. Reflection of all relative variables is a unitary map of each fibre and carries $n(u)$ to $-n(-u)$, so its norm is even; changing variables gives the second term. This proves (FC.20) with the stated drift and the entire relative contribution. It is the transported source estimate, with its mass and domain. The complete matrix energy remains (FC.15), on the actual graph and with its normal Gram, rather than being assigned the scalar constant-column bound.

All linear maps on the displayed filtered sources and graph images have the original fixed-support lift: they preserve the support label, send a supported zero to the corresponding supported zero, and send external absence to external absence. Their additive and scalar identities follow from the proved linear maps. In particular the arithmetic observation is taken after graph addition and the specified inverse Mellin isometry; no arithmetic jet is assigned to an arbitrary independent measurable normal section.

25.6 The actual arithmetic normal contribution is strictly positive

We now use the actual zero set of the amplitude in (FC.1). Hardy's theorem that infinitely many zeros of the Riemann zeta function lie on the critical line, with the original source and reading record retained in Section 23, implies that

$$\mathcal{Z}_h = \{T \in \mathbb{R} : a_h^{\text{SC}}(T) = 0\} \text{ is infinite and discrete.} \quad (\text{FC.21})$$

Indeed a finite polynomial h removes only its selected finite set of complete zero orders. Every remaining critical-line zero is still a zero of v_h , and the nonzero entire function v_h has a discrete zero set. This also applies to $h = 1$. At any $T \in \mathcal{Z}_h$, keep its entire order $\ell_T \geq 1$ and its original leading coefficient

$$a_h^{\text{SC}}(T + \delta) = c_T \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad (a_h^{\text{SC}})'(T + \delta) = \ell_T c_T \delta^{\ell_T-1} + O(\delta^{\ell_T}), \quad c_T = \frac{(a_h^{\text{SC}})^{(\ell_T)}(T)}{\ell_T!} \neq 0. \quad (\text{FC.22})$$

These are expansions of the original amplitude, including its fixed phase and its factor $1/\sqrt{2\pi}$.

Theorem 25.2 (Strict scalar residual and every finite normal direction). *For the actual amplitude and every integer $k \geq 2$,*

$$\int_{\mathbb{R}} \|n(u)\|^2 du > 0, \quad \int_{\mathbb{R}} \frac{|m'(u)|^2}{m(u)} du < \frac{\mu_h^{k-1}}{k} I_h. \quad (\text{FC.23})$$

For $k \geq 3$, $\|n(u)\| > 0$ for every real u . For any nonempty finite independent relative polynomial frame of (FC.5), its full normal Gram satisfies

$$\begin{aligned} \mathcal{N}(u) &> 0 \quad \text{for every } u \in \mathbb{R}, & k &\geq 3, \\ \mathcal{N}(u) &> 0 \quad \text{for every } u \notin \mathcal{Z}_h + \mathcal{Z}_h, & k &= 2. \end{aligned} \quad (\text{FC.24})$$

In particular $\mathcal{N}(u) > 0$ almost everywhere for every $k \geq 2$, since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable. No lower bound uniform in u , the relative degree, or the packet is asserted by these strict inequalities.

Proof. First fix $T \in \mathcal{Z}_h$ and choose $b_2, \dots, b_k$ outside $\mathcal{Z}_h$. Put $u_0 = T + \sum_{i=2}^k b_i$ and $A_b = \prod_{i=2}^k a_h^{\text{SC}}(b_i) \neq 0$. Within that fixed original sum fibre consider the transverse path

$$t_1 = T + \delta, \quad t_i = b_i \quad (2 \leq i < k), \quad t_k = b_k - \delta.$$

It changes the relative variables while retaining $u = u_0$. The derivative ∂_u is still the original derivative at fixed relative variables in (FC.2), and its product formula gives

$$\begin{aligned}\Psi(u_0, \mathbf{y}(\delta)) &= c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \\ \partial_u \Psi(u_0, \mathbf{y}(\delta)) &= \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\end{aligned}\tag{FC.25}$$

Only the derivative of the first amplitude has order $\ell_T - 1$; every other product term retains the full factor $a_h^{\text{SC}}(T + \delta)$ and has order at least ℓ_T . Since $m(u_0) > 0$ and m is smooth, subtracting $m'(u_0)\Psi/(2m(u_0))$ does not change that nonzero leading coefficient. Thus $n(u_0, \cdot)$ is nonzero at nearby points of this path and, by continuity, on a nonempty open subset of the entire relative fibre. Its squared fibre norm is strictly positive. Differentiated Schwartz estimates show that Ψ and $\partial_u \Psi$ are continuous as functions of u with values in $\mathcal{H}$. Smoothness and positivity of m therefore make n continuous in that Hilbert norm. Its squared norm stays positive on some interval around u_0 , proving positive integrated residual. Equation (FC.4) proves (FC.23).

For $k \geq 3$, the same construction works at every prescribed u_0 . Choose $b_3, \dots, b_{k-1}$ outside the zero set if that list is nonempty. Choose b_2 outside both $\mathcal{Z}_h$ and the translate-reflection of that set which would make $b_k = u_0 - T - \sum_{i=2}^{k-1} b_i$ a zero. These two discrete sets cannot exhaust the real line. This proves the pointwise scalar statement.

To prove the full matrix assertion, fix u and suppose $N_u c = 0$. The exact projection formula gives $j'_u c = j_u d$, where $d = \Gamma_u c$ is a finite coefficient vector. Write $\vartheta_c(\mathbf{y}) = \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}$, and similarly ϑ_d . Equality in $\mathcal{H}$ says

$$(\partial_u \Psi) \vartheta_c = \Psi \vartheta_d.$$

Both sides have smooth representatives in the relative variables, so their almost-everywhere equality is equality everywhere: a nonzero continuous difference at a point would remain nonzero on a set of positive measure.

Suppose $k \geq 3$. For each $T \in \mathcal{Z}_h$, restrict to the affine hyperplane $t_1 = u/k + y_1 = T$, of dimension $k - 2 \geq 1$. Each of the other original coordinates is a nonconstant affine function on this hyperplane. The points where one of their amplitude factors vanishes are consequently a countable union of proper affine hyperplanes of measure zero. Their complement is dense and is open near each of its points. At any such point p , apply (FC.25). In the asserted smooth identity the left side has leading coefficient $\ell_T c_T A_b \vartheta_c(p)/k$ in degree $\ell_T - 1$, whereas the right side has order at least ℓ_T . Hence $\vartheta_c(p) = 0$. Continuity proves its vanishing on the entire real hyperplane. Division in y_1 then shows that the polynomial ϑ_c has factor $y_1 - (T - u/k)$: the remainder polynomial in the other variables vanishes on all their real values, and therefore is identically zero. There are infinitely many distinct such T , while the degree in y_1 is finite. Thus $\vartheta_c = 0$, and independence of the original relative polynomials gives $c = 0$. This proves injectivity of N_u and therefore $\mathcal{N}(u) > 0$ on the whole finite coefficient space.

For $k = 2$, take $u \notin \mathcal{Z}_h + \mathcal{Z}_h$. At every $T \in \mathcal{Z}_h$, the other amplitude $a_h^{\text{SC}}(u - T)$ is nonzero. The same expansion forces the univariate polynomial ϑ_c to vanish at $y_1 = T - u/2$. These are infinitely many distinct points, so again $\vartheta_c = 0$ and $c = 0$. The discrete subset $\mathcal{Z}_h$ of $\mathbb{R}$ is countable, hence so is its set of pairwise sums. This proves every assertion in (FC.24). $\square$

The consequences hold for the actual full derivative graph and both quadratic branches. For every nonzero admitted source polynomial P ,

$$0 < \int_{\mathbb{R}} c_P(u)^* \mathcal{N}(u) c_P(u) du < \infty,\tag{FC.26}$$

$$\mathcal{Q}(x) > 0 \quad \text{for all } x > 0 \text{ when } k \geq 3, \quad \mathcal{Q}(x) > 0 \quad \text{for almost every } x > 0 \text{ when } k = 2.$$

Indeed a nonzero vector polynomial c_P is nonzero outside a finite set on the real line, and (FC.24) supplies strict positivity almost everywhere. The integral is finite because $\|N_u c_P\| \leq \|j'_u c_P\|$, and

the latter is a polynomial multiple of differentiated Schwartz columns. The assertion about Q follows from its invertible congruence (FC.14). For $k = 2$, the exceptional positive x for which either $\sqrt{x}$ or $-\sqrt{x}$ belongs to $\mathcal{Z}_h + \mathcal{Z}_h$ form a countable set. Under every declared smooth invertible frame, $N_u^C c_P^C(u) = N_u c_P(u)$ and $\mathcal{N}^C = C^* \mathcal{N} C$. Thus the full normal energy and strictness are preserved by the exact coefficient isomorphism $c_P^C = C^{-1} c_P$, including its original filtered image.

The excluded two-factor fibre can actually vanish. For the quartet packet, the original amplitude is even, so

$$\partial_u \Psi(0, y) = \frac{1}{2} ((a_h^{\text{SC}})'(y) a_h^{\text{SC}}(-y) + a_h^{\text{SC}}(y) (a_h^{\text{SC}})'(-y)) = 0. \quad (\text{FC.27})$$

Since the relative frame is independent of u , this gives $j'_0 = 0$, $N_0 = 0$, and $\mathcal{N}(0) = 0$. Also $m'(0) = 0$, hence $n(0) = 0$ exactly. The positive total residual and the almost-everywhere full matrix assertion therefore retain this original zero fibre. They prove that every nonzero actual finite polynomial direction has a positive normal contribution while leaving its complete size in (FC.15) and (FC.26).

25.7 Exact reconstruction from the actual finite normal image

Fix $k \geq 2$, the same actual packet, and one of the finite relative families in (FC.5). Retain its original degree bound M and the literal polynomial basis

$$\begin{aligned} \mathcal{I}_M &= \{(\alpha, n) : 0 \leq n \leq M - \ell_\alpha\}, \quad q_M = |\mathcal{I}_M|, \quad P_{\alpha, n}(\mathbf{s}) = \theta_\alpha(\mathbf{z}) S^n, \\ P_p &= \sum_{(\alpha, n) \in \mathcal{I}_M} p_{\alpha, n} P_{\alpha, n}, \quad c_{P_p}(u) = C_M(u) p, \quad (C_M(u))_{\beta, (\alpha, n)} = \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{FC.28})$$

Thus C_M is a specified rectangular polynomial matrix, not a change of relative frame. Each of its columns retains the original power of S . Its derivative is $(C'_M)_{\beta, (\alpha, n)} = \delta_{\alpha\beta} i n (k/2 + iu)^{n-1}$ for $n \geq 1$, and zero for $n = 0$. Independence of the relative family over $\mathbb{C}$, followed by comparison of the powers of S , proves uniqueness of these coefficients. The source is the exact space $\mathcal{P}_M^\theta = \text{span}\{P_{\alpha, n}\}$; it is the full original admitted source when that family is the complete admitted relative basis, and the declared summand otherwise. If $\mathcal{I}_M$ is empty, all the finite source and image spaces below are zero and their maps are the unique maps of zero spaces. We write the nonempty case explicitly.

Let $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$, with the original measures $du d\mathbf{y}$. Define the normal and tangential maps by their actual amplitudes,

$$\begin{aligned} \mathbf{N}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{N}_M p)(u) &= N_u C_M(u) p, \\ \mathbf{T}_M &: \mathbb{C}^{q_M} \longrightarrow \mathcal{K}, & (\mathbf{T}_M p)(u) &= j_u (C'_M(u) + \Gamma_u C_M(u)) p, \\ H_M^N &= \mathbf{N}_M^* \mathbf{N}_M = \int_{\mathbb{R}} C_M(u)^* \mathcal{N}(u) C_M(u) du > 0, & \mathcal{R}_M^N &= \text{ran } \mathbf{N}_M. \end{aligned} \quad (\text{FC.29})$$

The coefficient adjoints in this subsection use the displayed literal basis and its ordinary Euclidean coordinate inner product. The computed metric pulled back from the normal image is H_M^N . The original source-amplitude metric and their exact comparison operator remain

$$\begin{aligned} H_M^0 &= \int_{\mathbb{R}} C_M(u)^* W(u) C_M(u) du > 0, & K_M^{N:0} &= (H_M^0)^{-1} H_M^N, \\ (K_M^{N:0})^* H_M^0 &= H_M^0 K_M^{N:0} = H_M^N, & p^* H_M^N q &= p^* H_M^0 K_M^{N:0} q. \end{aligned} \quad (\text{FC.29a})$$

Indeed $W > 0$ and a nonzero vector polynomial $C_M p$ give $H_M^0 > 0$, with finite entries by the original moment bounds. Direct multiplication proves the remaining identities, so $K_M^{N:0}$ is self-adjoint and strictly positive in the H_M^0 inner product. This is the exact operator between the two retained forms; no identity of their norms is assumed. All columns of $\mathbf{N}_M$ and $\mathbf{T}_M$ belong to $\mathcal{K}$. For the first map, projection contraction gives $\|N_u C_M p\| \leq \|j'_u C_M p\|$, whose square is integrable by the polynomially weighted Schwartz estimates. For the second, it is the projection by Π_u of $\partial_u(j_u C_M p)$, and the same

estimates apply. Equation (FC.26) proves strict positivity of H_M^N for every nonzero coefficient vector. In particular the normal map is injective on the whole original finite source, including nonzero polynomials whose original arithmetic class is a supported zero.

Theorem 25.3 (The actual normal image recovers the source and both graph parts). *The inverse on $\mathcal{R}_M^N$, and its precise ambient formula, are*

$$\begin{aligned} L_M^N &= (H_M^N)^{-1} \mathbf{N}_M^* : \mathcal{K} \rightarrow \mathbb{C}^{q_M}, & \mathbf{N}_M^* r &= \int_{\mathbb{R}} C_M(u)^* N_u^* r(u) du, \\ L_M^N \mathbf{N}_M &= I_{q_M}, & \Pi_M^N &= \mathbf{N}_M L_M^N \text{ is the orthogonal projection onto } \mathcal{R}_M^N. \end{aligned} \quad (\text{FC.30})$$

The full original derivative graph is $\mathcal{G}_M = \text{ran}(\mathbf{T}_M, \mathbf{N}_M)$. It is recovered from its normal image by the exact linear bijection

$$\begin{aligned} \text{Rec}_M : \mathcal{R}_M^N &\longrightarrow \mathcal{G}_M, & r &\longmapsto (\mathbf{T}_M L_M^N r, r), \\ \text{Rec}_M^{-1}(t, r) &= r, & p &= L_M^N r, & P &= P_{L_M^N r}. \end{aligned} \quad (\text{FC.31})$$

Its complete metric identity retains both components:

$$\begin{aligned} H_M^\partial &= \mathbf{T}_M^* \mathbf{T}_M + H_M^N, & \|(\mathbf{T}_M p, \mathbf{N}_M p)\|^2 &= p^* H_M^\partial p, \\ \|\text{Rec}_M r\|^2 &= \|r\|^2 + \|\mathbf{T}_M L_M^N r\|^2 = (L_M^N r)^* H_M^\partial (L_M^N r). \end{aligned} \quad (\text{FC.32})$$

In particular reconstruction is not asserted to preserve the normal norm alone; (FC.32) specifies its full metric.

Proof. Every entry of the integral in (FC.30) converges by Cauchy–Schwarz, since it pairs r with one of the finitely many L^2 columns of $\mathbf{N}_M$. Pairing against an arbitrary coefficient vector proves that this integral is its Hilbert adjoint. Therefore $L_M^N \mathbf{N}_M = (H_M^N)^{-1} H_M^N = I$. Multiplication shows that Π_M^N is self-adjoint and idempotent, that its image is $\mathcal{R}_M^N$, and that it fixes every vector of that image. This proves the projection and inverse claims. The image is finite dimensional and hence closed; all displayed maps are bounded. If $r = \mathbf{N}_M p$, (FC.30) makes its reconstructed pair exactly $(\mathbf{T}_M p, \mathbf{N}_M p)$. Conversely every pair in $\mathcal{G}_M$ has this form. Injectivity of $\mathbf{N}_M$ gives both uniqueness of p and the inverse in (FC.31). The two graph components are pointwise orthogonal by $j_u^* N_u = 0$; their squared norms give (FC.32). Their addition consequently is the original derivative amplitude and has the same full graph norm, as already proved in (FC.7). $\square$

The full arithmetic derivative observation can now be specified on this actual normal image. Put

$$\begin{aligned} I &= (h(s_1), \dots, h(s_k)), & E &= \mathbb{C}[\mathbf{s}]/I = E_h^{\otimes k}, & U(\mathbf{s}) &= \prod_i v_h(s_i), \\ v^{(k)} &= j_I U \in E^\times, & \partial_S^{\text{rel}} &= \frac{1}{k} \sum_i \partial_{s_i}, & \beta_h &= (v^{(k)})^{-1} j_I (\partial_S^{\text{rel}} U). \end{aligned} \quad (\text{FC.33})$$

Here j_I means the complete polynomial Hermite class in the original product remainder coordinates. It retains every order in each factor. The entire function U is a unit at the finite packet; its inverse and every derivative used in β_h are taken in those full local algebras. Thus this formula uses no global division across a zero of v_h . If $h = 1$, the target E is the zero algebra and all arithmetic observations below are its unique zero map; the preceding analytic reconstruction continues unchanged.

For a nonempty packet define the finite matrix $J_M^{\log} : \mathbb{C}^{q_M} \rightarrow E$ by its original source columns. Mellin differentiation and (FC.9) give their exact formulas

$$\begin{aligned} (J_M^{\log})_{\alpha, n} &= J^{(k)} \mathcal{L}_k T_h^{(k)} P_{\alpha, n} = j_I (\partial_S^{\text{rel}} (U P_{\alpha, n})) \\ &= n v^{(k)} [\theta_\alpha S^{n-1}]_I + \beta_h v^{(k)} [\theta_\alpha S^n]_I. \end{aligned} \quad (\text{FC.34})$$

For $n = 0$ the first term is defined to be zero, and no negative power is used. To verify the last equality, the original coordinate map gives $\partial_S^{\text{rel}} S = 1$ and $\partial_S^{\text{rel}} z_i = 0$. Hence the relative polynomial does not acquire a derivative term, while differentiating S^n gives precisely its original factor n . The other product-rule term is the full β_h just defined. Equivalently $\beta_h = k^{-1} \sum_i j_I(v'_h/v_h)(s_i)$, with the quotient taken only in its actual unit neighbourhoods. Both factors $g = 2\xi$ and every Taylor coefficient remain in these columns.

The exact observation and the complete commuting source path are

$$\begin{aligned} \text{Obs}_M^{\log} : \mathcal{R}_M^N &\rightarrow E, & \text{Obs}_M^{\log}(r) &= \mathbf{J}_M^{\log}(H_M^N)^{-1} \mathbf{N}_M^* r, \\ \text{Obs}_M^{\log}(\mathbf{N}_M p) &= J^{(k)} \mathcal{L}_k \mathcal{T}_h^{(k)} P_p, \\ \mathcal{R}_M^N &\xrightarrow{\text{Rec}_M} \mathcal{G}_M \xrightarrow{\text{Add}} i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta, \\ i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta &\xrightarrow{-i\mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_h^{(k)} \mathcal{P}_M^\theta \xrightarrow{J^{(k)}} E. \end{aligned} \tag{FC.35}$$

Substitute $r = \mathbf{N}_M p$ in (FC.30)–(FC.31). Graph addition gives $\partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P_p = i\mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P_p$, and the last two arrows give exactly the columns in (FC.34). This proves equality of the two descriptions of the observation. All functions in the penultimate space are actual finite tensor test functions, since logarithmic multiplication preserves $\mathcal{B}$. Consequently this is an arithmetic observation on the specified finite normal image with a proved source inverse. The ambient formula for L_M^N also defines an orthogonal projection, but no arithmetic-jet interpretation of an arbitrary unrelated measurable normal section is required by (FC.35).

The reconstruction is unchanged under a declared smooth invertible relative frame. In that frame its coefficient matrix is $C_M^C = C(u)^{-1} C_M$, and the exact identities in (FC.8) give

$$\begin{aligned} N^C C_M^C &= N C_M, & j^C((C_M^C)' + \Gamma^C C_M^C) &= j(C_M' + \Gamma C_M), \\ (H_M^N)^C &= H_M^N, & (\mathbf{N}_M)^C &= \mathbf{N}_M, & (\mathbf{T}_M)^C &= \mathbf{T}_M. \end{aligned} \tag{FC.36}$$

The derivative of C^{-1} cancels exactly against the $C^{-1}C'$ connection term. Thus the original normal-image inverse, the full reconstructed graph and the arithmetic observation are the same maps on the same source after its explicit coefficient transport. Their fixed-support linear lifts retain the original label and all supported zeros. In particular a nonzero polynomial relation can have a nonzero normal image whose reconstructed arithmetic derivative is the full conormal response (FC.34); the source relation is retained before either quotient observation.

26 The original conormal derivative and its finite section commutator

The sum-connection source constructs differentiation on the first thickening of the original packet ideal. We prove its algebra map to dual numbers, then compute the exact finite endomorphism obtained from the original monic remainder section and from its full arithmetic unit. The conormal row and the section correction have different domains; their explicit factorization, complete ranks, and residue comparison are retained throughout.

26.1 The first thickening and the dual-number map

Let $k \geq 1$, and retain a nonempty packet of actual zeros with all complete orders,

$$h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho}, \quad d = \deg h \geq 1, \quad r_0 = |\mathcal{Z}|, \quad g = 2\xi, \quad v_h = g/h. \tag{CW.1}$$

Write $h_i = h(s_i)$, and put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad I = (h_1, \dots, h_k), \quad E = P/I, \quad E^{[2]} = P/I^2, \quad N = I/I^2, \\ \pi : E^{[2]} &\rightarrow E, \quad \partial_\Sigma = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad \delta : E^{[2]} \rightarrow E, \quad [p] \mapsto [\partial_\Sigma p]. \end{aligned} \quad (\text{CW.2})$$

The derivative maps I^2 into I by the product rule, so δ has exactly the stated type. For each i , the class of h_i maps to zero under π , while $\delta[h_i] = h'_i/k \neq 0$ in the retained remainder basis. Thus factoring δ through π would fail on this explicit class. The full algebra map (CW.6) below retains both its quotient value and derivative. Monic division in each original variable gives a unique expansion

$$p = \sum_{\mathbf{a} \in \mathbb{N}^k} h_1^{a_1} \cdots h_k^{a_k} p_{\mathbf{a}}, \quad \deg_{s_i} p_{\mathbf{a}} < d. \quad (\text{CW.3})$$

Only finitely many terms occur. Existence follows by repeated one-variable division, which commutes for distinct variables. For uniqueness, the leading monomials $\prod_i s_i^{da_i+b_i}$, $0 \leq b_i < d$, of the corresponding products are all distinct; comparison in a monomial order, followed by subtraction, proves independence. This also proves that the h_i are algebraically independent and that P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the original remainder basis.

Consequently the original conormal coefficient map is the E -module isomorphism

$$\iota_N : E^k \xrightarrow{\sim} N, \quad (a_i) \mapsto \left[\sum_i h_i \tilde{a}_i \right]_{I^2}, \quad \dim E = d^k, \quad \dim E^{[2]} = (k+1)d^k. \quad (\text{CW.4})$$

Changing a lift $\tilde{a}_i$ adds an element of I^2 . Modulo I^2 , (CW.3) has exactly its constant coefficient and its k linear h_i -coefficients, proving every assertion in (CW.4). In particular $N^2 = 0$ in $E^{[2]}$. The derivative obeys

$$\delta(ab) = \pi(a)\delta(b) + \pi(b)\delta(a), \quad \delta\iota_N(a_i) = \frac{1}{k} \sum_i h'_i a_i. \quad (\text{CW.5})$$

These identities follow before taking the indicated quotients; the terms still containing h_i vanish only in E .

Theorem 26.1 (A full algebra bridge for the coefficient derivative). *Let $B = E[\epsilon]/(\epsilon^2)$, where ϵ is a formal infinitesimal, and let $\mathfrak{j} = (h'_1, \dots, h'_k)$ be the indicated ideal of E . There is a unital algebra map*

$$\Phi : E^{[2]} \rightarrow B, \quad a \mapsto \pi(a) + \epsilon\delta(a). \quad (\text{CW.6})$$

The derivative induces $\bar{\partial}_\Sigma : E \rightarrow E/\mathfrak{j}$, and its image and kernel are exactly

$$\begin{aligned} \text{im } \Phi &= \{q + \epsilon t : [t]_{\mathfrak{j}} = \bar{\partial}_\Sigma q\}, \\ \ker \Phi &= \iota_N \ker \left(E^k \xrightarrow{(a_i) \mapsto k^{-1} \sum_i h'_i a_i} E \right). \end{aligned} \quad (\text{CW.7})$$

There is the exact sequence of complex vector spaces

$$0 \longrightarrow \ker(\delta|_N) \longrightarrow E^{[2]} \xrightarrow{\Phi} B \xrightarrow{\Psi} E/\mathfrak{j} \longrightarrow 0, \quad \Psi(q + \epsilon t) = [t]_{\mathfrak{j}} - \bar{\partial}_\Sigma q. \quad (\text{CW.8})$$

The map Φ is an algebra homomorphism; Ψ is given its displayed complex-linear type. The complete ranks and dimensions are

$$\begin{aligned} \text{rank}(\delta|_N) &= d^k - (d - r_0)^k, \\ \dim \ker \Phi &= (k-1)d^k + (d - r_0)^k, \\ \dim \text{im } \Phi &= 2d^k - (d - r_0)^k. \end{aligned} \quad (\text{CW.9})$$

Proof. Equation (CW.5) gives

$$\Phi(ab) = \pi(a)\pi(b) + \epsilon\{\pi(a)\delta(b) + \pi(b)\delta(a)\} = \Phi(a)\Phi(b),$$

and $\delta(1) = 0$, proving the unital algebra claim. Changing a representative of $q \in E$ by $\sum_i h_i p_i$ changes its derivative modulo I by $k^{-1} \sum_i h'_i [p_i] \in \mathfrak{j}$. Thus $\bar{\partial}_\Sigma$ is well-defined, with its product rule along $E \rightarrow E/\mathfrak{j}$. Every $\Phi(a)$ consequently satisfies the condition in (CW.7).

Conversely let that condition hold. Take the unique original monic remainder $\sigma(q) \in P$, of degree below d in every variable. Then

$$t - \delta[\sigma(q)]_{I^2} = \frac{1}{k} \sum_i h'_i a_i$$

for some $a_i \in E$, by the definition of $\mathfrak{j}$. The element $[\sigma(q)]_{I^2} + \iota_N(a_i)$ has image $q + \epsilon t$. This proves the image equality. Its kernel requires $\pi(a) = 0$, hence $a \in N$, and then exactly the vanishing of the row in (CW.5). This proves the kernel statement. The map Ψ is onto, since $\Psi(\epsilon t) = [t]$, and its kernel is the already computed image of Φ . This proves (CW.8) with its exact map types.

For the ranks, retain an ordered local tensor block at centres ρ_i , of full lengths m_i . In its coordinates

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_1}, \dots, z_k^{m_k}), \quad z_i = s_i - \rho_i,$$

write $h(s_i) = z_i^{m_i} c_i(z_i)$, $c_i(0) \neq 0$. Modulo the original local ideal,

$$h'_i = m_i c_i(0) z_i^{m_i-1}, \quad h'_i a_i = m_i c_i(0) z_i^{m_i-1} a_i|_{z_i=0}. \quad (\text{CW.10})$$

Thus the original input constant coefficient in the i -th variable is sent into that variable's top nilpotent direction; all remaining variables and the factor $m_i c_i(0)$ are retained. The row image is the ideal spanned by monomials having at least one exponent $m_i - 1$. Its dimension is $\prod_i m_i - \prod_i (m_i - 1)$. The excluded monomials remain in the complementary vector space and in $E/\mathfrak{j}$. Sum over every ordered tuple of centres. The two sums are

$$\sum_{\rho_1, \dots, \rho_k} \prod_i m_{\rho_i} = d^k, \quad \sum_{\rho_1, \dots, \rho_k} \prod_i (m_{\rho_i} - 1) = (d - r_0)^k. \quad (\text{CW.11})$$

This proves the first line of (CW.9). Subtract it from $\dim N = kd^k$ for the kernel, then subtract that kernel from $\dim E^{[2]}$ for the image, proving the other two. These counts include every simple and repeated local factor. $\square$

When the packet has the original reflection $a^\dagger(\mathbf{s}) = \overline{a(1 - \mathbf{s})}$, its ideals are stable because $h^\dagger = (-1)^d h$. The derivative satisfies $\delta(a^\dagger) = -(\delta a)^\dagger$. Set $(q + \epsilon t)^\dagger = q^\dagger - \epsilon t^\dagger$. Then $\Phi(a^\dagger) = \Phi(a)^\dagger$, by (CW.6). This retains the sign of the infinitesimal and the original conjugate-linear reflection.

26.2 The original monic section and the exact finite correction

Let $\sigma : E \rightarrow P$ denote the multivariate monic remainder section of (CW.3), and set $j = [\sigma]_{I^2} : E \rightarrow E^{[2]}$. It is complex-linear, with $\pi j = I_E$. Let $S = \sum_i s_i$, and write Z and $\widehat{Z}$ for multiplication by $S - k/2$ on E and $E^{[2]}$, respectively. The original product rule gives the rectangular operator identity

$$Z\delta - \delta\widehat{Z} = -\pi, \quad D_0 = \delta j : E \rightarrow E. \quad (\text{CW.12})$$

The original finite arithmetic generator is $A_k = Z + (k/2)I_E$.

For each i , let $\ell_i : E \rightarrow E_{\widehat{i}} := \bigotimes_{j \neq i} E_h$ extract the coefficient of s_i^{d-1} in $\sigma(u)$. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th variable. If $k = 1$, the target of ℓ_1 is $\mathbb{C}$, and c_1 is its unit column. These maps give the exact section defect

$$\begin{aligned} \mathbf{t}_Z &:= \widehat{Z}j - jZ = \iota_N(c_1 \ell_1, \dots, c_k \ell_k) : E \rightarrow N, \\ [Z, D_0] &= -I_E + \mathbf{J}, \quad \mathbf{J} = \delta \mathbf{t}_Z = \frac{1}{k} \sum_i M_{h'_i} c_i \ell_i. \end{aligned} \quad (\text{CW.13})$$

Indeed multiplying the original remainder by s_i creates only its top s_i^d term outside the admitted coordinate range. Monic division subtracts exactly $h_i \sigma_i(\ell_i u)$. Adding these terms for $S - k/2$ proves the first formula. Compose (CW.12) with j and use $\widehat{Z}j = jZ + \mathbf{t}_Z$; this proves the commutator with its minus sign. Applying (CW.5) proves its last formula. In particular this is a specified endomorphism of the original finite vector space, obtained by a declared section of π .

Theorem 26.2 (All projectors, eigenvalues and ranks of the section correction). *The original coordinate endomorphisms*

$$P_i = \frac{1}{d} M_{h'_i} c_i \ell_i, \quad P_i^2 = P_i, \quad P_i P_j = P_j P_i, \quad \text{rank } P_i = d^{k-1} \quad (\text{CW.14})$$

are commuting idempotents. For $T \subset \{1, \dots, k\}$, put

$$P_T = \prod_{i \in T} P_i \prod_{i \notin T} (I - P_i). \quad (\text{CW.15})$$

Their pairwise products are zero when their labels differ, and $\sum_T P_T = I_E$. On their ranges the full correction has

$$J = \frac{d}{k} \sum_i P_i, \quad J|_{\text{ran } P_T} = \frac{d|T|}{k} I. \quad (\text{CW.16})$$

For $d > 1$, its eigenvalue dj/k has multiplicity $\binom{k}{j} (d-1)^{k-j}$, for every $0 \leq j \leq k$. For all $d \geq 1$,

$$\text{Tr } J = d^k, \quad \text{rank } J = d^k - (d-1)^k, \quad \ker J = \bigcap_i \ker \ell_i. \quad (\text{CW.17})$$

For $d = 1$, every $P_i = I$, $J = I$, and only the $j = k$ eigenspace is nonzero, of dimension one.

Proof. The polynomial h'_i has degree $d-1$ and leading coefficient d . Therefore

$$\ell_i M_{h'_i} c_i = d I_{E_i}. \quad (\text{CW.18})$$

Substitution proves $P_i^2 = P_i$. It also proves that $M_{h'_i} c_i$ is injective, and hence that the rank is d^{k-1} . Operators in distinct variables are tensor products of their respective one-variable maps, so they commute. The one-variable space is the direct sum of $\mathbb{C}h'$ and the polynomials of degree at most $d-2$: the former has nonzero top coefficient d , and every polynomial splits by subtracting that top coefficient times h'/d . These are the image of P_i and the kernel of ℓ_i , of dimensions one and $d-1$.

Taking tensor products of these direct decompositions proves all the identities for P_T and gives $\dim \text{ran } P_T = (d-1)^{k-|T|}$ when $d > 1$. On this range exactly the P_i for $i \in T$ act as identity, proving (CW.16). Summing over $|T| = j$ gives the asserted multiplicity. Its zero space is the all-complement range, of dimension $(d-1)^k$, proving the rank and kernel in (CW.17). Finally

$$\text{Tr } J = \frac{d}{k} \sum_i \text{rank } P_i = \frac{d}{k} k d^{k-1} = d^k. \quad (\text{CW.19})$$

For $d = 1$, the one-variable complement is zero and its image is the entire one-dimensional space. Only the all-image tensor survives. This proves the stated endpoint directly, without assigning a positive multiplicity to a zero-dimensional space. $\square$

These are algebraic idempotents in the retained coordinates. For any actual positive minimum metric G_M , their exact metric adjoints and coordinate transports are

$$P_i^{\dagger G_M} = G_M^{-1} P_i^* G_M, \quad \tilde{P}_i = G_M^{1/2} P_i G_M^{-1/2}, \quad \tilde{J} = G_M^{1/2} J G_M^{-1/2}. \quad (\text{CW.20})$$

The transports retain the products, ranks and algebraic spectrum. For each i , the individual $\tilde{P}_i$ is Hermitian precisely when $P_i^* G_M = G_M P_i$, whereas $\tilde{J}$ is Hermitian precisely when $J^* G_M = G_M J$. None of these metric equalities is assumed. The same isometry carries the original weight to

$$G_M^{-1/2} W_M G_M^{-1/2} = \tilde{Z}^* + \tilde{Z}, \quad \tilde{Z} = G_M^{1/2} Z G_M^{-1/2}, \quad W_M = A_k^* G_M + G_M A_k - k G_M. \quad (\text{CW.21})$$

It carries (CW.13) to the identical commutator with all three operators transported. Thus the algebraic correction and the original Hermitian weight have an explicit common coordinate map, with every adjoint still attached to its actual metric.

The full conormal row rank in (CW.9) and the section rank in (CW.17) are connected by exactly

$$E \xrightarrow{(c_i \ell_i)_i} E^k \xrightarrow{\iota_N} N \xrightarrow{\delta|_N} E. \quad (\text{CW.22})$$

Its composite is J . The first map retains the specific highest-coordinate coefficients of the monic section; the full conormal row accepts arbitrary k input coefficients. This proves the relationship between the two ranks without replacing one domain by the other.

26.3 The full arithmetic unit and the original finite representatives

Keep the complete Taylor classes

$$\hat{U} = \left[\prod_i v_h(s_i) \right]_{I^2} \in (E^{[2]})^\times, \quad U = \pi(\hat{U}) \in E^\times, \quad \beta_h = U^{-1} \delta \hat{U} \in E. \quad (\text{CW.23})$$

Finite multivariate Hermite interpolation supplies these classes from the entire original functions near every ordered packet point. Their values are nonzero at those points because the removed orders were complete. One can also prove invertibility in the thickening directly: lift U^{-1} to b ; then $\hat{U}b = 1 + n$, $n \in N$, and $b(1 - n)$ is the inverse since $N^2 = 0$. Thus this thickening retains the original unit and its first relation derivatives.

Theorem 26.3 (The unit-conjugated finite Weyl identity). *Define the actual unit section and its derivative by*

$$j_U = \hat{U} j U^{-1} : E \rightarrow E^{[2]}, \quad \pi j_U = I_E, \quad D_U = \delta j_U = M_{\beta_h} + M_U D_0 M_U^{-1}. \quad (\text{CW.24})$$

Then, with all multiplication operators in the original basis,

$$\begin{aligned} t_U &:= \hat{Z} j_U - j_U Z = M_{\hat{U}} t_Z M_U^{-1} : E \rightarrow N, \\ [Z, D_U] &= -I_E + J_U, \quad J_U = \delta t_U = M_U J M_U^{-1}. \end{aligned} \quad (\text{CW.25})$$

The complete spectrum, multiplicities, ranks and trace in (CW.16)–(CW.19) therefore hold for J_U as well.

Proof. The section equation follows from $\pi j = I$ and $\pi(\hat{U}) = U$. Apply (CW.5) to $\hat{U} j(U^{-1}u)$. Its two terms are $(\delta \hat{U})U^{-1}u$ and $U D_0(U^{-1}u)$, which prove (CW.24). The multiplication maps $\hat{Z}$, $M_{\hat{U}}$ commute, as do Z , M_U^{-1} . This proves the formula for t_U . Its image lies in N , so the product-rule term $(\delta \hat{U})\pi(t_Z M_U^{-1}u)$ vanishes exactly. The other term is $U J M_U^{-1}u$, proving the last equality in (CW.25). Finally M_{β_h} commutes with multiplication by Z . Conjugating (CW.13) by M_U proves the commutator. Similarity gives the entire stated spectrum and its zero space, without making an orthogonality claim. $\square$

These sections have an original analytic realization. Let

$$\mathcal{T}_h^{(k)} p = p(D_1, \dots, D_k) F_h^{\otimes k}, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad R_{\text{ref}}^{(k)} u = \mathcal{T}_h^{(k)} \sigma(U^{-1}u). \quad (\text{CW.26})$$

The original Mellin transform gives $\mathcal{M}_k \mathcal{T}_h^{(k)} p = (\prod_i v_h(s_i))p$. Its thickened Taylor class is $\widehat{U}[p]_{I^2}$, so the thickened class of $R_{\text{ref}}^{(k)} u$ is exactly $j_U u$. Multiplication by $\mathcal{L}_k$ differentiates this Mellin transform by ∂_Σ . Hence

$$J^{(k)} \mathcal{L}_k R_{\text{ref}}^{(k)} = \mathbf{D}_U. \quad (\text{CW.27})$$

This is the finite endomorphism of the specified original section, with its actual logarithmic source observable.

For the actual degree- M minimum, $M \geq k(d-1)$, retain its original coefficient Gram $\mathcal{M}_M > 0$ and full jet matrix $\mathcal{J}_M$, including multiplication by U . Its numerator and source function are exactly

$$p_{M,u} = \mathcal{M}_M^{-1} \mathcal{J}_M^* G_M u, \quad G_M = (\mathcal{J}_M \mathcal{M}_M^{-1} \mathcal{J}_M^*)^{-1}, \quad R_M u = \mathcal{T}_h^{(k)} p_{M,u}. \quad (\text{CW.28})$$

Indeed its jet is u , and its coefficient vector is $\mathcal{M}_M$ -orthogonal to $\ker \mathcal{J}_M$. Every other preimage differs by such a kernel vector; Pythagoras proves this same original constrained minimum. Define the concrete thickened section and admitted relation by

$$j_M u = \widehat{U}[p_{M,u}]_{I^2}, \quad a_M = j_M - j_U : E \rightarrow N, \quad \mathbf{D}_M = \delta j_M = \mathbf{D}_U + \delta a_M. \quad (\text{CW.29})$$

The image statement follows from $\pi j_M = \pi j_U = I_E$. Its numerator is the original allowed relation $p_{M,u} - \sigma(U^{-1}u) \in I$, not an independently chosen vector in the conormal layer. The corresponding exact identities are

$$\begin{aligned} J^{(k)} \mathcal{L}_k R_M &= \mathbf{D}_M, & [Z, \mathbf{D}_M] &= -I_E + \mathbf{J}_M^{\text{sec}}, \\ \mathbf{J}_M^{\text{sec}} &:= \delta(\widehat{Z} j_M - j_M Z) = \mathbf{J}_U + [Z, \delta a_M], & \text{Tr } \mathbf{J}_M^{\text{sec}} &= d^k. \end{aligned} \quad (\text{CW.30})$$

The first follows by the same original Mellin differentiation. For the second, compose (CW.12) with j_M . For the third, subtract (CW.25) and use $\delta \widehat{Z} a_M = Z a_M^\delta$, where $a_M^\delta = \delta a_M$, since $\pi a_M = 0$. This gives exactly the displayed commutator difference. Its trace is zero by the cyclic identity for finite matrices; (CW.19) gives the final trace. Only the specified fixed-section correction is assigned the ranks (CW.17). Every change to the actual minimum is retained through the computed map a_M and its commutator.

26.4 The one-variable residue functional and the original theta relation

Take $k = 1$, write $\ell : E_h \rightarrow \mathbb{C}$ for the coefficient of s^{d-1} , and let $U = j_h v_h$. The product derivative $g' = h' v_h + h v_h'$ gives

$$\mathbf{J}_U = M_{j_h g'} c_1 \ell M_U^{-1}, \quad \mathbf{J}_U u = (j_h g') \ell(U^{-1}u), \quad \text{rank } \mathbf{J}_U = 1, \quad \text{Tr } \mathbf{J}_U = d. \quad (\text{CW.31})$$

The rank follows either from (CW.25) or from the nonzero column $j_h g' = U h'$ and the nonzero functional. The trace of this rank-one map is $\ell(U^{-1} j_h g') = \ell(h') = d$.

For the original reflection-stable packet, put $f^\dagger(s) = \overline{f(1-\bar{s})}$. Its residue pairing is the well-defined form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{\widetilde{f}^\dagger(s) \widetilde{u}(s)}{g(s)} ds = \ell(U^{-1} f^\dagger u). \quad (\text{CW.32})$$

Changing either polynomial representative by h changes the integrand by a function holomorphic at all selected centres: $h^\dagger = (-1)^d h$, and $g = h v_h$, with v_h nonvanishing there. To prove the last equality, take $p = \text{rem}_h(U^{-1} f^\dagger u)$, of degree below d . The residues of the displayed quotient equal those of p/h . In the partial-fraction expansion of p/h , the coefficient of $1/s$ at infinity is the sum of its coefficients at the simple-pole terms; terms of pole order at least two have no $1/s$ coefficient. Since h is monic of degree d , that coefficient is exactly the original coefficient $\ell(p)$. This proves (CW.32),

keeping every higher local pole before the residue is taken. Consequently the finite correction is precisely

$$J_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \text{Tr } J_U = \mathcal{R}_Z(1, j_h g') = \sum_{\rho \in Z} m_\rho = d. \quad (\text{CW.33})$$

The source map producing the same Jacobian is the original logarithmic theta observation:

$$J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi), \quad \mathcal{M} \Theta \phi = g H_\phi. \quad (\text{CW.34})$$

Differentiate the Mellin product before taking jets. Its $g H'_\phi$ term vanishes modulo the original h ; its other term is exactly (CW.34). For $\phi = P(D)\phi_*$, one has $H_\phi = P$, so the input ranges through all original finite coefficients. Thus this map's finite image is the full ideal $(j_h g')$, of dimension r_0 by (CW.10), rather than a selected one-dimensional input line. The exact connection to (CW.31) is its precomposition with

$$E_h \xrightarrow{\ell M_U^{-1}} \mathbb{C} \xrightarrow{c_1} E_h. \quad (\text{CW.35})$$

The rank-one correction and the rank- r_0 multiplication map therefore use the same Jacobian with their distinct input maps explicitly specified. Applying (CW.32) to (CW.34) gives, with the full conjugation,

$$\mathcal{R}_Z(f, j_h(g') j_h P) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho). \quad (\text{CW.36})$$

Indeed the residue of g'/g at a zero of order m_ρ is m_ρ , by differentiating its local product.

There is also an exact global source map

$$\nu : V \rightarrow Q = \mathcal{B}/\Theta V, \quad \nu(\phi) = q(\log x \Theta \phi). \quad (\text{CW.37})$$

Logarithmic multiplication preserves $\mathcal{B}$, by its defining two-sided polynomial bounds: $|\log x| \leq x + x^{-1}$ and $D^j((\log x)F) = (\log x)D^j F - jD^{j-1}F$ for $j \geq 1$ give every required bound from the original ones. Thus this map is defined. For the original $D = -x\partial_x$, $[D, \log x] = -I$ and $D\Theta = \Theta D$. Therefore

$$D_Q \nu - \nu D_V = q[D, \log x]\Theta = -q\Theta = 0. \quad (\text{CW.38})$$

For $a > 0$, let $\Lambda_a F(x) = F(ax)$. The identities $\Theta \Lambda_a = \Lambda_a \Theta$ and $(\log x)\Lambda_a - \Lambda_a(\log x) = -\log(a)\Lambda_a$ similarly give $\nu \Lambda_a = \Lambda_a \nu$. This proves scaling equivariance on the exact source V and target Q . Since $J_h \Theta = 0$, there is the induced full jet map $\bar{J}_h : Q \rightarrow E_h$ with $\bar{J}_h q = J_h$; its value on ν is exactly (CW.34).

26.5 Supported algebra and coefficient maps

Write $\text{pr}_0, \text{pr}_\epsilon : B \rightarrow E$ for the constant and infinitesimal coefficient maps. The first is a unital algebra homomorphism and the second is complex-linear, with the exact product rule

$$\text{pr}_\epsilon(bc) = \text{pr}_0(b) \text{pr}_\epsilon(c) + \text{pr}_0(c) \text{pr}_\epsilon(b). \quad (\text{CW.39})$$

The strongest typed lift of (CW.6) is the semiring map

$$G(E^{[2]}) \xrightarrow{G(\Phi)} G(B), \quad \pi^\tau = G(\text{pr}_0)G(\Phi), \quad \delta^\tau = (\text{pr}_\epsilon)^\tau G(\Phi). \quad (\text{CW.40})$$

Here $G(\Phi)$ and $G(\text{pr}_0)$ fix external τ and send supported amplitudes to their supported algebra images. The map $(\text{pr}_\epsilon)^\tau$ is the stated split-linear coefficient map; (CW.39) specifies its precise relationship to multiplication. Its value on the supported unit of B is the supported zero e_E .

Every vector in the algebraic kernel in (CW.7) maps under $G(\Phi)$ to supported zero, while external absence maps to external absence. A supported conormal element n maps to $(\epsilon \delta n)^\bullet$, retaining its

coefficient derivative; constant projection gives e_E . The sections, coefficient maps, exact sequences, and original representatives above lift at each fixed support by $(\lambda, u) \mapsto (\lambda, Tu)$. Thus the first thickening, its full kernel and quotient, and the source-selected endomorphism retain their distinct receiving fibres and the common external τ .

If $h = 1$, then $I = P$, $E = E^{[2]} = N = 0$, and all finite maps in (CW.2) have zero source and target. The source function is still the original theta seed $F_h = \Theta\phi_*$, whose map to this arithmetic quotient is zero. This states that case explicitly; the projector formulas dividing by d above have the declared domain $d \geq 1$.

26.6 The arithmetic unit and complete tensor partial residues

Return to the nonempty original packet of (CW.1), with $d \geq 1$ and $k \geq 1$. The unit itself has the following exact image under the algebra bridge already proved above.

Corollary 26.4 (The full unit and its first relation derivative). *For the original $\widehat{U}, U, \beta_h$ of (CW.23),*

$$\begin{aligned}\Phi(\widehat{U}) &= U + \epsilon\delta\widehat{U} = U(1 + \epsilon\beta_h), \\ \Phi(\widehat{U}^{-1}) &= \Phi(\widehat{U})^{-1} = U^{-1}(1 - \epsilon\beta_h).\end{aligned}\tag{CW.41}$$

Proof. The first identity is (CW.6) evaluated on the full thickened unit. The definition $\beta_h = U^{-1}\delta\widehat{U}$ gives its displayed factorization. Since $\epsilon^2 = 0$, $(1 + \epsilon\beta_h)(1 - \epsilon\beta_h) = 1$. The map Φ is unital and multiplicative, so applying it to $\widehat{U}\widehat{U}^{-1} = 1$ proves the second identity with its minus sign. In particular $\Phi(\widehat{U}a) = \Phi(\widehat{U})\Phi(a)$ transports the original arithmetic product through this same algebra map. $\square$

Retain all the original complete orders, $g = 2\xi$ and $v_h = g/h$, and write

$$\begin{aligned}E_h &= \mathbb{C}[s]/(h), & E &= E_h^{\otimes k}, & E_{\widehat{i}} &= \bigotimes_{j \neq i} E_h, \\ v_i &= j_h(v_h) \text{ in the } i\text{-th factor}, & \gamma_i &= j_h(g') \text{ in the } i\text{-th factor}, & U &= \prod_i v_i.\end{aligned}$$

Every v_i is the original complete invertible jet. Let $\ell_h : E_h \rightarrow \mathbb{C}$ extract the coefficient of s^{d-1} from the unique one-variable monic remainder. Let $c_i : E_{\widehat{i}} \rightarrow E$ insert the constant polynomial one in the i -th factor, and let $\ell_i : E \rightarrow E_{\widehat{i}}$ apply ℓ_h in that factor and identity in every other factor. Thus ℓ_i extracts the original coefficient of s_i^{d-1} , retaining every other-variable coordinate. For $k = 1$, $E_{\widehat{i}} = \mathbb{C}$, and c_i is the unit column. Let M_a denote multiplication by the displayed element in its declared algebra.

Define the coefficient-valued partial residue map

$$\mathcal{R}_i : E \longrightarrow E_{\widehat{i}}, \quad \mathcal{R}_i(u) = \sum_{\rho \in \mathbb{Z}} \text{Res}_{s_i=\rho} \frac{\widetilde{u}(s_1, \dots, s_k)}{g(s_i)} ds_i.\tag{CW.42}$$

The coefficients in the other original variables are taken modulo their original ideals after the indicated residue. Each residue uses the usual positively oriented complex s_i -coordinate; the residue notation carries no additional $2\pi i$ factor. Equivalently, each residue equals $1/(2\pi i)$ times the corresponding positively oriented contour integral.

This map is well-defined on E . A change by $h(s_i)p$ in its own variable changes the integrand by $p/v_h(s_i)$, which is holomorphic at every selected centre because the removed zero orders were complete. A change by $h(s_j)p$, $j \neq i$, produces that same factor $h(s_j)$ in the remaining coefficient algebra, hence zero in $E_{\widehat{i}}$. Linearity then covers the full original ideal. The finite sum of coefficient residues is complex-linear and $E_{\widehat{i}}$ -linear.

The exact coefficient formula is

$$\boxed{\mathcal{R}_i = \ell_i M_{v_i}^{-1}}.\tag{CW.43}$$

To prove it, let p_i be the degree-below- d remainder in s_i of $v_i^{-1}u$, retaining the full coefficient algebra $E_{\widehat{\mathcal{C}}}$. The residue in (CW.42) equals the corresponding residue of $p_i/h(s_i)$: the difference in the numerator vanishes to the complete selected orders, so division by h is locally holomorphic. Every coefficient of p_i/h has an ordinary partial-fraction expansion. The coefficient of s_i^{-1} at infinity is the sum of the finite simple-pole coefficients; higher-pole terms contribute no s_i^{-1} coefficient. Since h is monic and $\deg_{s_i} p_i < d$, that coefficient is precisely $\ell_i(p_i)$. This proof applies coefficientwise in the finite algebra $E_{\widehat{\mathcal{C}}}$ and gives (CW.43) with every other-variable nilpotent coefficient retained.

Define the original partial correction endomorphisms

$$C_i = M_{\gamma_i} c_i \mathcal{R}_i : E \longrightarrow E. \quad (\text{CW.44})$$

They satisfy

$$\begin{array}{l} C_i^2 = dC_i, \quad C_i C_j = C_j C_i, \\ \text{rank } C_i = d^{k-1}, \quad \text{Tr } C_i = d^k. \end{array} \quad (\text{CW.45})$$

Indeed $g' = h'v_h + hv'_h$ gives $\gamma_i = v_i h'_i$ in the original i -th quotient, so

$$\mathcal{R}_i M_{\gamma_i} c_i = \ell_i M_{v_i}^{-1} M_{\gamma_i} c_i = \ell_i M_{h'_i} c_i = dI_{E_{\widehat{\mathcal{C}}}}. \quad (\text{CW.46})$$

The last equality uses the literal degree $d - 1$ and leading coefficient d of the original derivative h' . This proves the first formula in (CW.45). It also proves that $M_{\gamma_i} c_i$ is injective and $\mathcal{R}_i$ is onto. The factorization (CW.44) therefore has rank $\dim E_{\widehat{\mathcal{C}}} = d^{k-1}$. Alternatively a specified right inverse for $\mathcal{R}_i$ is $a \mapsto v_i s_i^{d-1} c_i(a)$, using (CW.43). The two rectangular factors in (CW.44) have traces $\text{Tr}_E(AB) = \text{Tr}_{E_{\widehat{\mathcal{C}}}}(BA)$; apply (CW.46) to obtain $d d^{k-1} = d^k$. Operators belonging to distinct variables are tensor products of their respective one-variable maps with identity on the other factors. Their compositions therefore commute, proving the remaining identity.

These operators give precisely the correction of the original full-unit section:

$$\mathbf{J}_U = \frac{1}{k} \sum_{i=1}^k C_i. \quad (\text{CW.47})$$

For proof, retain the complete thickened unit $\widehat{U}$, its residue $U = \prod_i v_i$, and its full first derivative $\beta_h = U^{-1} \delta \widehat{U}$. The original source-selected operator is still

$$\begin{aligned} \mathbf{D}_U &= M_{\beta_h} + M_U \mathbf{D}_0 M_U^{-1}, & [Z, \mathbf{D}_U] &= -I_E + \mathbf{J}_U, \\ \mathbf{J}_U &= M_U \left(\frac{1}{k} \sum_i M_{h'_i} c_i \ell_i \right) M_U^{-1}. \end{aligned} \quad (\text{CW.48})$$

Write $U = v_i U_{\widehat{\mathcal{C}}}$, with $U_{\widehat{\mathcal{C}}} = \prod_{j \neq i} v_j$. The maps c_i, ℓ_i are $E_{\widehat{\mathcal{C}}}$ -linear, so the operator $M_{U_{\widehat{\mathcal{C}}}}$ commutes through their indicated source and target actions, and its inverse then cancels with exactly that same full other-variable unit. Thus each summand in (CW.48) is

$$M_U M_{h'_i} c_i \ell_i M_U^{-1} = M_{v_i h'_i} c_i \ell_i M_{v_i}^{-1} = M_{\gamma_i} c_i \mathcal{R}_i = C_i. \quad (\text{CW.49})$$

This proves (CW.47). Every v_i retains its original value, and M_{β_h} remains in $\mathbf{D}_U$. It commutes with the original multiplication operator $Z = M_{\sum_i s_i - k/2}$, which is why it contributes zero to that specified commutator.

For completeness, retaining all k residue variables gives

$$\begin{aligned} \mathcal{R}_{\text{all}} : E &\longrightarrow \mathbb{C}, \\ \mathcal{R}_{\text{all}}(u) &= \sum_{\rho_1, \dots, \rho_k \in \mathcal{Z}} \text{Res}_{s_k = \rho_k} \cdots \text{Res}_{s_1 = \rho_1} \frac{\widetilde{u}(\mathbf{s})}{\prod_i g(s_i)} ds_1 \cdots ds_k. \end{aligned} \quad (\text{CW.50})$$

Here the formula means the ordered iterated coefficient residues, with each original variable positively oriented. Each variable operates on a different tensor factor, so swapping their evaluation order gives the identical scalar coefficient and introduces no sign. Formula (CW.43) in each factor proves

$$\mathcal{R}_{\text{all}} = \ell_h^{\otimes k} M_U^{-1}, \quad \boxed{C_1 \cdots C_k(u) = \left(\prod_i \gamma_i \right) \mathcal{R}_{\text{all}}(u).} \quad (\text{CW.51})$$

The map $\ell_h^{\otimes k} : E_h^{\otimes k} \rightarrow \mathbb{C}$ is the tensor product of the original one-variable highest-coefficient functionals. Tensoring the one-variable factorization (CW.44) proves the second equality directly, so it keeps all repeated-root and mixed nilpotent data before the final scalar functional is evaluated. The column and functional are both nonzero: their contraction is

$$\begin{aligned} \mathcal{R}_{\text{all}} \left(\prod_i \gamma_i \right) &= \prod_{i=1}^k \left(\sum_{\rho \in \mathcal{Z}} \text{Res}_{s_i=\rho} \frac{g'(s_i)}{g(s_i)} ds_i \right) \\ &= \prod_{i=1}^k \left(\sum_{\rho} m_{\rho} \right) = d^k. \end{aligned} \quad (\text{CW.52})$$

This is the rank-one full-product correction, with its exact nonzero eigenvalue and trace d^k . Its square is d^k times itself by (CW.51)–(CW.52). The residue of g'/g is m_{ρ} because $g(s) = (s - \rho)^{m_{\rho}} u_{\rho}(s)$ with $u_{\rho}(\rho) \neq 0$; hence $g'/g = m_{\rho}/(s - \rho) + u'_{\rho}/u_{\rho}$. This gives the complete multiplicity proof used in (CW.52).

At $d = 1$, (CW.46) gives $C_i = I_E$ on the one-dimensional E ; consequently $J_U = I_E$, and (CW.51) is the same identity. At $k = 1$, every formula reduces exactly to (CW.31)–(CW.33), including the original residue functional $\mathcal{R}_{\mathcal{Z}}(1, u)$, rank one and trace d . For $h = 1$, the original arithmetic quotient is zero, as already stated above; the positive-degree residue coefficient formulas have the declared domain $d \geq 1$. All repeated fibres, complete arithmetic units, and their original source and target maps have been retained.

27 Exact transitions between nested relative frames

This continuation starts with the original amplitude, relative coordinates, finite source powers, graph and arithmetic observation of FC.1–36. It proves the maps produced when the relative frame is enlarged. Every finite weight, original source coefficient and normal component is retained. The published source is used unchanged.

27.1 Original sources and coefficient inclusions

Fix the original packet polynomial $h, g = 2\xi$, the full entire quotient $v_h = g/h$, and the integer $k \geq 2$. Keep

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad u = \sum_{i=1}^k t_i, \quad t_i = u/k + y_i, \quad y_k = -\sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(u/k + y_i). \quad (\text{NT.1})$$

The ambient fibre is $\mathcal{H} = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$, and the full ambient space is $\mathcal{K} = L^2(\mathbb{R}, du; \mathcal{H})$. Thus the original Jacobian $du d\mathbf{y}$ and the amplitude mass are unchanged. The original relative variables are $z_i = s_i - S/k$, $S = \sum_i s_i$, so $z_i = iy_i$ on this fibre.

Let the finite independent homogeneous relative families at levels $M \leq L$ have column counts r_M, r_L and degrees $\ell_{\alpha}^M, \ell_{\beta}^L$. Their inclusion is the literal constant matrix $E = E_{LM} : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$

satisfying

$$\theta_\alpha^M(\mathbf{z}) = \sum_\beta \theta_\beta^L(\mathbf{z}) E_{\beta\alpha}, \quad j_M(u) = j_L(u)E, \quad j_a(u)c = \Psi(u, \mathbf{y}) \sum_\alpha \theta_\alpha^a(i\mathbf{y})c_\alpha. \quad (\text{NT.2})$$

Comparison of homogeneous components gives $E_{\beta\alpha} = 0$ unless $\ell_\beta^L = \ell_\alpha^M$. Independence of the smaller family makes E injective. For an actual nested monomial family, E is its original zero-extension matrix; no scaling of columns is introduced.

For $a = M, L$ retain the complete admitted source index set and its literal coefficient matrix,

$$\begin{aligned} \mathcal{I}_a &= \{(\alpha, n) : 0 \leq n \leq a - \ell_\alpha^a\}, & q_a &= |\mathcal{I}_a|, \\ P_p^a &= \sum_{(\alpha, n) \in \mathcal{I}_a} p_{\alpha, n} \theta_\alpha^a(\mathbf{z}) S^n, & (C_a(u))_{\beta, (\alpha, n)} &= \delta_{\alpha\beta} (k/2 + iu)^n. \end{aligned} \quad (\text{NT.3})$$

Define the source inclusion $F = F_{LM} : \mathbb{C}^{q_M} \rightarrow \mathbb{C}^{q_L}$ by $F_{(\beta, m), (\alpha, n)} = E_{\beta\alpha} \delta_{mn}$. Every nonzero entry is admitted since $L \geq M$ and the relative degrees agree. Direct expansion gives the exact identities

$$P_{Fp}^L = P_p^M, \quad C_L F = E C_M, \quad C'_L F = E C'_M. \quad (\text{NT.4})$$

Uniqueness of the relative-polynomial expansion and of each power of S proves that F is injective. Every factor $(k/2 + iu)^n$ and its derivative $in(k/2 + iu)^{n-1}$ is retained.

27.2 The original weights and their orthogonal transition

At every real u define, with the ambient Hilbert adjoint,

$$\begin{aligned} W_a &= j_a^* j_a > 0, & B_a &= j_a^* j'_a, & \Gamma_a &= W_a^{-1} B_a, & \Pi_a &= j_a W_a^{-1} j_a^*, \\ N_a &= (I - \Pi_a) j'_a, & \mathcal{N}_a &= N_a^* N_a, & a &= M, L. \end{aligned} \quad (\text{NT.5})$$

These are the original FC matrices. Their positivity and smoothness follow from independence of the relative family and the original Schwartz amplitude: a nonzero relative polynomial cannot vanish almost everywhere where Ψ is nonzero. The inverse matrices therefore exist at every real u , including $u = 0$.

The weighted coefficient retraction and coefficient projection are

$$R = W_M^{-1} E^* W_L : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_M}, \quad Q = ER : \mathbb{C}^{r_L} \rightarrow \mathbb{C}^{r_L}. \quad (\text{NT.6})$$

The symbol Q here is a coefficient projection; Π_a remains the ambient orthogonal projection. From (NT.2),

$$\begin{aligned} W_M &= E^* W_L E, & B_M &= E^* B_L E, & RE &= I, & Q^2 &= Q, & Q^* W_L &= W_L Q, \\ \Gamma_M &= R \Gamma_L E, & \Pi_L \Pi_M &= \Pi_M \Pi_L = \Pi_M. \end{aligned} \quad (\text{NT.7})$$

The first identities follow by multiplication of $j_M = j_L E$ and $j'_M = j'_L E$. For the projection identities, Π_a is self-adjoint, idempotent and has image $V_a = \text{ran } j_a$; $V_M \subseteq V_L$ implies that Π_L fixes $\text{ran } \Pi_M$, and taking adjoints proves the other order.

Theorem 27.1 (The complete connection transition). *Define the coefficient transition and its ambient image by*

$$K = K_{LM} = \Gamma_L E - E \Gamma_M = (I - Q) \Gamma_L E, \quad A = A_{LM} = j_L K. \quad (\text{NT.8})$$

Then $K : \mathbb{C}^{r_M} \rightarrow \mathbb{C}^{r_L}$ has image in the W_L -orthogonal complement of $\text{ran } E$, and

$$\begin{aligned} E^* W_L K &= 0, & A &= (\Pi_L - \Pi_M) j'_M, & N_M &= A + N_L E, \\ \text{ran } A &\subseteq V_L \cap V_M^\perp, & \text{ran } (N_L E) &\subseteq V_L^\perp, & A^* N_L E &= 0. \end{aligned} \quad (\text{NT.9})$$

Consequently the exact positive transition form is

$$\boxed{\mathcal{N}_M = E^* \mathcal{N}_L E + \Delta_{LM}}, \quad \begin{aligned} \Delta_{LM} &= A^* A = K^* W_L K \succeq 0 \\ &= E^* \Gamma_L^* W_L \Gamma_L E - \Gamma_M^* W_M \Gamma_M. \end{aligned} \quad (\text{NT.10})$$

For each $c \in \mathbb{C}^{r_M}$ the following equivalences and rank bound are exact:

$$c^* \Delta_{LM} c = 0 \iff Kc = 0 \iff Ac = 0, \quad \text{rank } \Delta_{LM} = \text{rank } K \leq \min(r_M, r_L - r_M). \quad (\text{NT.11})$$

Proof. Equation (NT.7) gives $E^* W_L \Gamma_L E = W_M \Gamma_M$; subtracting $E^* W_L E \Gamma_M$ proves $E^* W_L K = 0$. The original identity $j_a \Gamma_a = \Pi_a j'_a$ gives $j_L K = \Pi_L j'_L E - \Pi_M j'_M = (\Pi_L - \Pi_M) j'_M$. Splitting $I - \Pi_M = (\Pi_L - \Pi_M) + (I - \Pi_L)$ and using $j'_M = j'_L E$ proves $N_M = A + N_L E$. The commuting projection identities show that $\Pi_L - \Pi_M$ is the orthogonal projection onto $V_L \cap V_M^\perp$. This proves the two range statements and their orthogonality. Taking the Gram of this orthogonal sum proves the first line of (NT.10). Expanding $K = \Gamma_L E - E \Gamma_M$, using $E^* W_L \Gamma_L E = W_M \Gamma_M$ and its adjoint, gives the second line. Positivity of W_L implies that the quadratic form of $K^* W_L K$ vanishes exactly on $\ker K$. Injectivity of j_L gives $\ker A = \ker K$. The image of K lies in a space of dimension $r_L - r_M$; rank-nullity now proves (NT.11). $\square$

For a locally absolutely continuous original coefficient $c(u)$, write $\nabla_a = \partial_u + \Gamma_a$. The full derivative has the three mutually orthogonal components

$$\begin{aligned} \nabla_L(Ec) &= E \nabla_M c + Kc, \\ \partial_u(j_M c) &= j_M \nabla_M c + Ac + N_L Ec, \\ \|\partial_u(j_M c)\|^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c + c^* E^* \mathcal{N}_L E c, \\ \|\nabla_L(Ec)\|_{W_L}^2 &= \|\nabla_M c\|_{W_M}^2 + c^* \Delta_{LM} c. \end{aligned} \quad (\text{NT.12})$$

The first equality is the product rule with constant E and (NT.8). The derivative formula follows from (NT.9) and the original connection decomposition. The image of its first term lies in V_M , the second in $V_L \cap V_M^\perp$, and the third in $V_L^\perp$. Their orthogonality proves every energy identity without discarding any coefficient derivative.

27.3 Full source matrices, graph maps and arithmetic observation

Define the actual maps on the original coefficient bases by

$$\begin{aligned} \mathbf{N}_a p &= N_a C_a p, & \mathbf{T}_a p &= j_a(C'_a + \Gamma_a C_a) p, & \mathbf{A}_{LM} p &= A_{LM} C_M p, \\ H_a^0 &= \int C_a^* W_a C_a du, & H_a^N &= \mathbf{N}_a^* \mathbf{N}_a, & H_a^\partial &= \mathbf{T}_a^* \mathbf{T}_a + H_a^N, \\ H_{LM}^{\text{tr}} &= \mathbf{A}_{LM}^* \mathbf{A}_{LM} = \int C_M^* \Delta_{LM} C_M du. \end{aligned} \quad (\text{NT.13})$$

Here the displayed equalities of functions take place in $\mathcal{K}$. All integrals are finite: each term is an orthogonal projection of an original polynomially weighted derivative amplitude, and the FC Schwartz estimates apply before any projection. Equations (NT.4), (NT.9), and (NT.12) give

$$\begin{aligned} \mathbf{N}_M &= \mathbf{A}_{LM} + \mathbf{N}_L F, & \mathbf{T}_L F &= \mathbf{T}_M + \mathbf{A}_{LM}, \\ \mathbf{T}_M^* \mathbf{A}_{LM} &= 0, & \mathbf{A}_{LM}^* \mathbf{N}_L F &= 0, & \mathbf{T}_M^* \mathbf{N}_L F &= 0, \\ H_M^0 &= F^* H_L^0 F, & H_M^N &= F^* H_L^N F + H_{LM}^{\text{tr}}, & F^* \mathbf{T}_L^* \mathbf{T}_L F &= \mathbf{T}_M^* \mathbf{T}_M + H_{LM}^{\text{tr}}, & H_M^\partial &= F^* H_L^\partial F. \end{aligned} \quad (\text{NT.14})$$

These identities follow first pointwise, then by integration. In particular, the positive amount transferred into the larger tangential image is precisely the amount removed from the normal form for the same original polynomial.

Let $\widehat{\Pi}_a$ denote the bounded orthogonal projection on $\mathcal{K}$ obtained by applying $\Pi_a(u)$ pointwise. Measurability follows from smoothness, and contraction follows by integrating the pointwise contraction. Define

$$\begin{aligned}\mathcal{U}_{LM}(t, n) &= (t + \widehat{\Pi}_L n, (I - \widehat{\Pi}_L)n), \quad t \in \text{ran } \widehat{\Pi}_M, \quad n \in \ker \widehat{\Pi}_M, \\ \mathcal{U}_{LM}^{-1}(t_L, n_L) &= (\widehat{\Pi}_M t_L, (I - \widehat{\Pi}_M)t_L + n_L).\end{aligned}\tag{NT.15}$$

The domain and codomain are respectively the orthogonal decompositions of $\mathcal{K}$ at levels M and L . The inverse formula is verified by the nested projection identities. Orthogonality of $t, \widehat{\Pi}_L n$ and $(I - \widehat{\Pi}_L)n$ shows that $\mathcal{U}_{LM}$ is unitary between these sum spaces. On the actual finite derivative graph it is exactly

$$\mathcal{U}_{LM}(\mathbf{T}_M p, \mathbf{N}_M p) = (\mathbf{T}_L F p, \mathbf{N}_L F p).\tag{NT.16}$$

Graph addition on either side is the same original function $\partial_u(j_M C_M p) = \partial_u(j_L C_L F p)$, so (NT.16) also supplies the exact map between the two presentations.

For completeness the finite normal maps are injective on nonempty actual polynomial sources. The FC.21–27 proof uses the infinite discrete set $\mathcal{Z}_h$ of unremoved critical-line zeros, retaining each original order ℓ_T and coefficient $c_T = a^{(\ell_T)}(T)/\ell_T!$. If $N_a(u)c = 0$, then $\Psi' \vartheta_c = \Psi \vartheta_d$ with finite relative polynomials and $d = \Gamma_a c$. On a fibre through $t_1 = T$ where all other amplitude factors are nonzero, the leading terms in the original transverse coordinate δ are

$$\Psi = c_T A_b \delta^{\ell_T} + O(\delta^{\ell_T+1}), \quad \Psi' = \frac{\ell_T}{k} c_T A_b \delta^{\ell_T-1} + O(\delta^{\ell_T}).\tag{NT.17}$$

Thus ϑ_c vanishes on that hyperplane. When $k \geq 3$ the other nonzero factors can be chosen in a dense subset of each hyperplane, so polynomial continuity gives vanishing on every such hyperplane; infinitely many distinct T force $\vartheta_c = 0$. For $k = 2$ and $u \notin \mathcal{Z}_h + \mathcal{Z}_h$, every remaining second factor $a(u - T)$ is nonzero; infinitely many distinct univariate roots give the same conclusion. Since $\mathcal{Z}_h + \mathcal{Z}_h$ is countable, $\mathcal{N}_a > 0$ almost everywhere for every finite nonempty frame. A nonzero vector polynomial $C_a p$ is nonzero outside a finite set: one nonzero component polynomial suffices to bound its common real zero set. Integrating proves $H_a^N > 0$. This argument also includes nonzero source relations whose arithmetic class is zero.

Set $\mathcal{R}_M^N = \text{ran } \mathbf{N}_M$ and $\mathcal{R}_{L|M}^N = \text{ran}(\mathbf{N}_L F)$. The concrete normal transition and its inverse on these specified images are

$$\begin{aligned}\mathcal{Q}_{LM} : \mathcal{R}_M^N &\longrightarrow \mathcal{R}_{L|M}^N, & \mathcal{Q}_{LM} r &= (I - \widehat{\Pi}_L)r, & \mathcal{Q}_{LM} \mathbf{N}_M p &= \mathbf{N}_L F p, \\ H_{L|M}^N &= F^* H_L^N F > 0, & \mathcal{Q}_{LM}^{-1} r_L &= \mathbf{N}_M (H_{L|M}^N)^{-1} F^* \mathbf{N}_L^* r_L, \\ \|r\|^2 - \|\mathcal{Q}_{LM} r\|^2 &= \|\mathbf{A}_{LM} p\|^2 \quad (r = \mathbf{N}_M p).\end{aligned}\tag{NT.18}$$

Positivity of $H_{L|M}^N$ follows from injectivity of F and of $\mathbf{N}_L$. Substitution of $r_L = \mathbf{N}_L F p$ proves the inverse formula in both orders. Thus $\mathcal{Q}_{LM}$ is a contraction and a linear isomorphism between these finite images, with its complete norm loss specified. No inverse bound independent of the frame level is inserted.

The bounded ambient extension obtained from the exact smaller normal image inverse is

$$\begin{aligned}V_{LM} &= \mathbf{N}_L F (H_M^N)^{-1} \mathbf{N}_M^* = (I - \widehat{\Pi}_L) \Pi_M^N : \mathcal{K} \rightarrow \mathcal{K}, \\ \Pi_M^N &= \mathbf{N}_M (H_M^N)^{-1} \mathbf{N}_M^*, & \|V_{LM}\| &\leq 1, & V_{LM}|_{\mathcal{R}_M^N} &= \mathcal{Q}_{LM}.\end{aligned}\tag{NT.18a}$$

The equality follows by multiplying the first line of (NT.18) by $(H_M^N)^{-1} \mathbf{N}_M^*$. The two factors on the right are orthogonal projections and hence contractions. The smaller normal image is fixed by Π_M^N , proving the restriction statement. For $L = M$ this ambient extension is Π_M^N and restricts to the identity precisely on its declared image.

Retain the original arithmetic target, full quotient unit and derivative,

$$\begin{aligned}\mathfrak{A}_h &= \mathbb{C}[s_1, \dots, s_k] / (h(s_1), \dots, h(s_k)), \\ U &= \prod_i v_h(s_i), \quad v = [U], \quad \delta = \frac{1}{k} \sum_i \partial_{s_i}, \quad \beta = v^{-1}[\delta U], \\ (j_a^{\log})_{\alpha, n} &= nv[\theta_\alpha^a S^{n-1}] + \beta v[\theta_\alpha^a S^n].\end{aligned}\tag{NT.19}$$

At $n = 0$ the first term is zero; no negative power is taken. The full Hermite class, multiplicities, $g = 2\xi$ and unit derivatives are retained. The identity $\delta S = 1, \delta z_i = 0$ proves these columns by the product rule applied before taking the quotient. Equation (NT.4) then proves $j_L^{\log} F = j_M^{\log}$ entry by entry. On the actual normal image put $\text{Obs}_a^{\log} r = j_a^{\log}(H_a^N)^{-1} \mathbf{N}_a^* r$. The exact arithmetic transition is

$$\boxed{\text{Obs}_L^{\log} \mathcal{Q}_{LM} = \text{Obs}_M^{\log} \quad \text{on } \mathcal{R}_M^N.}\tag{NT.20}$$

Indeed both sides on $r = \mathbf{N}_M p$ are $j_M^{\log} p$ by (NT.18)–(NT.19). Their original source path is graph addition, followed by $-i\mathcal{U}_k^{-1}$ and the complete original arithmetic jet; (NT.16) preserves that path exactly. For $h = 1$ the arithmetic target is the zero algebra and the observation is its unique zero map; every analytic identity above continues unchanged.

To retain any declared target metric as well, choose its fixed original quotient basis and let $G_M^{\text{tar}}, G_L^{\text{tar}} > 0$ be the two declared Hermitian matrices in that same basis. The coordinate identity $\tau_{LM} : \mathfrak{A}_h \rightarrow \mathfrak{A}_h$ has the exact comparison operator

$$\begin{aligned}D_{LM}^{\text{tar}} &= (G_M^{\text{tar}})^{-1} G_L^{\text{tar}}, \quad (D_{LM}^{\text{tar}})^* G_M^{\text{tar}} = G_M^{\text{tar}} D_{LM}^{\text{tar}} = G_L^{\text{tar}}, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 &= p^* (j_M^{\log})^* G_L^{\text{tar}} j_M^{\log} p, \\ \|\text{Obs}_L^{\log} \mathcal{Q}_{LM} \mathbf{N}_M p\|_{G_L^{\text{tar}}}^2 - \|\text{Obs}_M^{\log} \mathbf{N}_M p\|_{G_M^{\text{tar}}}^2 &= p^* (j_M^{\log})^* (G_L^{\text{tar}} - G_M^{\text{tar}}) j_M^{\log} p.\end{aligned}\tag{NT.20a}$$

Direct multiplication proves the comparison identities, and (NT.20) proves the two observed-form identities. This specifies both target forms through the actual coordinate map without assigning a sign to their difference. In the zero target all these are the unique zero forms and maps, with no positive-dimensional inverse needed.

27.4 Every variable-frame term

Let $G_M(u), G_L(u)$ be smooth invertible matrices of their declared sizes. The changed frame and coefficients are $j_a^G = j_a G_a, c_a^G = G_a^{-1} c_a$. The same inclusion now has the generally variable matrix

$$E^G = G_L^{-1} E G_M, \quad R^G = G_M^{-1} R G_L, \quad \Gamma_a^G = G_a^{-1} \Gamma_a G_a + G_a^{-1} G'_a.\tag{NT.21}$$

For any such variable inclusion the exact formulas are

$$\begin{aligned}\Gamma_M^G &= R^G (\Gamma_L^G E^G + (E^G)'), \quad K^G = (E^G)' + \Gamma_L^G E^G - E^G \Gamma_M^G, \\ K^G &= G_L^{-1} K G_M, \quad A^G = A G_M, \quad N_a^G = N_a G_a, \quad \Delta_{LM}^G = G_M^* \Delta_{LM} G_M.\end{aligned}\tag{NT.22}$$

To prove the first equality, differentiate $j_M^G = j_L^G E^G$ before taking its mixed Gram: $B_M^G = (E^G)^* B_L^G E^G + (E^G)^* W_L^G (E^G)'$. Multiplication by $(W_M^G)^{-1}$ gives it. In the second line, differentiate $G_L^{-1} E G_M$ and substitute both transformed connections. The terms containing G'_L and G'_M cancel separately, leaving $G_L^{-1} K G_M$. The original projection images are unchanged; applying $I - \Pi_a$ to $(j_a G_a)'$ therefore gives $N_a G_a$. This proves the remaining statements and every variable-inclusion version of (NT.9)–(NT.12). In particular the term $(E^G)'$ cannot be deleted from the transition.

For the same original polynomial source, $C_a^G = G_a^{-1} C_a$ and $C_L^G F = E^G C_M^G$. Its tangential and normal amplitudes obey

$$\mathbf{N}_a^G = \mathbf{N}_a, \quad \mathbf{T}_a^G = \mathbf{T}_a, \quad \mathbf{A}_{LM}^G = \mathbf{A}_{LM}.\tag{NT.23}$$

The first and third follow from (NT.22); in the second, the derivative of G_a^{-1} cancels the connection term $G_a^{-1}G'_a$. Thus the literal source matrices in (NT.14), the normal-image transition, its inverse and the arithmetic observation are all the same maps after their explicit coefficient transport. The transformed coefficient functions are exactly this image of the original filtered source; their degree bounds are not replaced by a new polynomial frame convention.

27.5 Nested chains and both quadratic branches

For a third level $T \geq L$ retain $E_{TM} = E_{TL}E_{LM}$ and $F_{TM} = F_{TL}F_{LM}$. Expansion of (NT.8) proves

$$\begin{aligned} K_{TM} &= K_{TL}E_{LM} + E_{TL}K_{LM}, & A_{TM} &= A_{TL}E_{LM} + A_{LM}, \\ \Delta_{TM} &= \Delta_{LM} + E_{LM}^* \Delta_{TL} E_{LM}, & H_{TM}^{\text{tr}} &= H_{LM}^{\text{tr}} + F_{LM}^* H_{TL}^{\text{tr}} F_{LM}. \end{aligned} \quad (\text{NT.24})$$

The two ambient summands of A_{TM} lie in $V_T \cap V_L^\perp$ and $V_L \cap V_M^\perp$; they are orthogonal. Taking their Gram proves the third identity, and integration with (NT.4) proves the fourth. Induction gives the same mutually orthogonal sum and positive telescoping formula along every finite nested chain. The actual graph maps and normal maps compose:

$$\mathcal{U}_{TL}\mathcal{U}_{LM} = \mathcal{U}_{TM}, \quad \mathcal{Q}_{TL}\mathcal{Q}_{LM} = \mathcal{Q}_{TM} \quad \text{on the corresponding original source image.} \quad (\text{NT.25})$$

For the normal maps this follows directly from $(I - \hat{\Pi}_T)(I - \hat{\Pi}_L) = I - \hat{\Pi}_T$. For the graph maps, both compositions are the two orthogonal projections of the same graph-added vector. These proofs retain the intermediate positive transition forms. The ambient normal extensions also compose on all of $\mathcal{K}$: $V_{TL}V_{LM} = V_{TM}$. Indeed their matrix product contains $(H_L^N)^{-1}\mathbf{N}_L^*\mathbf{N}_L = I_{q_L}$, leaving $\mathbf{N}_T F_{TL} F_{LM} (H_M^N)^{-1}\mathbf{N}_M^*$.

Now put $x = u^2 > 0$, $v = \sqrt{x}$ and, at each level, keep

$$M_a = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H_a = M_a^* \frac{\text{diag}(W_a(v), W_a(-v))}{2v} M_a, \quad Q_a^n = M_a^* \frac{\text{diag}(\mathcal{N}_a(v), \mathcal{N}_a(-v))}{2v} M_a. \quad (\text{NT.26})$$

Let $\mathcal{A}_a = M_a^{-1} \text{diag}(\Gamma_a(v)/(2v), -\Gamma_a(-v)/(2v)) M_a + M_a^{-1} M'_a$ be the complete paired connection. Its derivative includes the original $M_a^{-1} M'_a = \text{diag}(0, I/(2x))$. Write $\tilde{E} = \text{diag}(E, E)$ and

$$\begin{aligned} \mathcal{K}_{LM} &= \mathcal{A}_L \tilde{E} - \tilde{E} \mathcal{A}_M = M_L^{-1} \text{diag}\left(\frac{K(v)}{2v}, -\frac{K(-v)}{2v}\right) M_M, \\ H_M &= \tilde{E}^* H_L \tilde{E}, \quad \tilde{E}^* H_L \mathcal{K}_{LM} = 0, \\ \boxed{Q_M^n &= \tilde{E}^* Q_L^n \tilde{E} + 4x \mathcal{K}_{LM}^* H_L \mathcal{K}_{LM}.} \end{aligned} \quad (\text{NT.27})$$

The identity $M_L \tilde{E} = \text{diag}(E, E) M_M$ proves the first two transport assertions by substitution. Applying (NT.10) on both branches gives the normal identity: multiplication by $4x = 4v^2$ cancels precisely the squares of $1/(2v)$ in $\mathcal{K}_{LM}$, leaving $M_M^* \text{diag}(\Delta_{LM}(v), \Delta_{LM}(-v)) M_M/(2v)$. Thus both original signs and both Jacobian factors are retained. For $\boldsymbol{\eta}_M = (a_M, b_M)^t$ the paired inclusion is $\boldsymbol{\eta}_L = \tilde{E} \boldsymbol{\eta}_M$ and

$$(\partial_x + \mathcal{A}_L) \tilde{E} \boldsymbol{\eta}_M = \tilde{E} (\partial_x + \mathcal{A}_M) \boldsymbol{\eta}_M + \mathcal{K}_{LM} \boldsymbol{\eta}_M. \quad (\text{NT.28})$$

Weighted orthogonality in (NT.27) shows that its tangential energy gain is exactly the normal energy loss in that equation. It follows pointwise, before integration, that the complete FC.15 energy is the same on the two included presentations.

In variable frames the paired inclusion is instead $\mathcal{E}(x) = M_L^{-1} \text{diag}(E^G(v), E^G(-v)) M_M$. The corresponding transition is exactly $\mathcal{E}' + \mathcal{A}_L^G \mathcal{E} - \mathcal{E} \mathcal{A}_M^G$; applying (NT.22) on both branches and the product rule proves this formula, including its derivative. The transported original density retains $(\mathcal{A}_a)^* H_a + H_a \mathcal{A}_a = H'_a + H_a/(2x)$ at both levels, since this identity is obtained by differentiating the same two original branch densities. No evenness assumption is used in (NT.26)–(NT.28).

27.6 Exact zero and endpoint scopes

The identities above hold at every real u . If a coefficient vector or admitted polynomial is zero, each displayed linear map sends it to its corresponding zero. If the smaller frame is empty, its coefficient and source spaces are zero and each inclusion, transition and inverse-on-zero-image is the unique such map. For equal frames with an invertible constant E , the weighted projection Q is the identity, so $K = A = \Delta_{LM} = 0$; the normal map is transported exactly by E . For unequal frames, (NT.11) is the precise equality criterion; strict positivity of the actual finite $\mathcal{N}_L$ does not force strict positivity of Δ_{LM} . The maps in (NT.9) and their Gram identity give the exact relation in every such case.

For a quartet packet and $k = 2$, the original amplitude is even. At $u = 0$ its product derivative is $\frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0$. Every fixed relative frame therefore has $j'_a(0) = \Gamma_a(0) = N_a(0) = 0$ and $K_{LM}(0) = A_{LM}(0) = \Delta_{LM}(0) = 0$. Under a smooth variable frame, $\Gamma_a^G(0)$ can have the retained term $G_a(0)^{-1}G'_a(0)$; (NT.22) still gives $K^G(0) = N_a^G(0) = 0$. The integrated finite-source normal inverse in (NT.18) remains valid because its positivity is an almost-everywhere and integrated assertion, proved in (NT.17).

Let $\mathcal{D}_a$ denote the coefficient functions locally absolutely continuous on $\mathbb{R}$ for which $\int c^* W_a c$ and $\int \|\partial_u(j_a c)\|^2$ are finite. For the constant inclusion, the exact domain correspondence is

$$E\mathcal{D}_M = \mathcal{D}_L \cap \{c_L : c_L(u) \in \text{ran } E \text{ for almost every } u\}. \quad (\text{NT.29})$$

The forward inclusion follows from $j_L E c = j_M c$ and constant E . Conversely, use any constant left inverse of E , for example $(E^* E)^{-1} E^*$, to obtain a locally absolutely continuous c_M from c_L . The closed fixed subspace condition and continuity give $c_L = E c_M$ everywhere. The same amplitude and derivative identities then give both required finite integrals. This proves both inclusions without identifying distinct coefficient metrics.

On the quadratic branches the original domain has the exact trace condition

$$\lim_{x \downarrow 0} (a_M(x) + i\sqrt{x} b_M(x)) = \lim_{x \downarrow 0} (a_M(x) - i\sqrt{x} b_M(x)). \quad (\text{NT.30})$$

The analogous condition after $\tilde{E}$ is equivalent because E is injective. The finite norm and energy on each open branch give finite traces: near zero, W_M has positive upper and lower bounds and Γ_M is bounded, so the coefficient and its ordinary derivative are both locally square-integrable. The fundamental theorem and Cauchy-Schwarz give the one-sided limits. Integration by parts then shows that unequal limits give their jump times δ_0 , whereas equality joins them into the original weak-derivative domain. This proves the trace assertion. For smooth invertible gauges through zero the matching condition is transported by their common value at zero. For gauges specified only on the punctured branches one must retain the original condition with $G_M(v)c_M^G(v)$ and $G_M(-v)c_M^G(-v)$; it is exactly (NT.30) applied to the original coefficient. No new endpoint value is assigned by the half-line formulas alone.

Finally every displayed linear source, graph and observation map has the original fixed-support lift: on a fixed support label it acts by the proved linear map, preserves a supported zero as a supported zero, and sends external absence to external absence. Its additivity and scalar laws follow directly from linearity. In particular a nonzero polynomial relation is carried by $P_{Fp}^L = P_p^M$ before any full Hermite quotient, and its arithmetic derivative is the unchanged value (NT.19)–(NT.20).

27.7 One independent exact original-coordinate calibration

This finite calibration has ambient space $\mathbb{C}^5$ with its standard Hermitian form. It checks the proved transition using literal polynomial matrices, without identifying this finite model with the arithmetic

amplitude (NT.1). Keep

$$\begin{aligned}
 p(u) = 1 + u, \quad q(u) = 1 + u + u^2, \quad j_L(u) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & 2p & -p \\ 2q & -q & q \end{pmatrix}, \quad E = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \\
 j_M(u) = j_L(u)E &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & p \\ 3q & 0 \end{pmatrix}.
 \end{aligned} \tag{NT.31}$$

The first three rows make j_L injective, and E is injective. For $a = (1, 2, -1)^t$ and $b = (2, -1, 1)^t$, direct multiplication gives

$$\begin{aligned}
 W_L &= I_3 + p^2 aa^t + q^2 bb^t > 0, \quad W_M = \begin{pmatrix} 2 + 9q^2 & 1 \\ 1 & 2 + p^2 \end{pmatrix} > 0, \\
 B_L &= pp'aa^t + qq'bb^t = W'_L/2, \quad B_M = \text{diag}(9qq', pp') = W'_M/2, \\
 W_M(1) - W_M(-1) &= \begin{pmatrix} 72 & 0 \\ 0 & 4 \end{pmatrix}, \quad [W_M(0), W_M(1)] = \begin{pmatrix} 0 & -69 \\ 69 & 0 \end{pmatrix}, \\
 W_L(1) - W_L(-1) &= 4aa^t + 8bb^t, \quad [W_L(0), W_L(1)] = \begin{pmatrix} 0 & 25 & -15 \\ -25 & 0 & -5 \\ 15 & 5 & 0 \end{pmatrix}.
 \end{aligned} \tag{NT.32}$$

Positivity follows from the squared norm of the injective original maps. Every displayed difference is nonzero, so both original weights are non-even and have noncommuting values.

At $u = 1$, put $J = j_L(1)$ and $D = j'_M(1)$. Their matrices and the actual smaller connection are

$$\begin{aligned}
 J &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & 4 & -2 \\ 6 & -3 & 3 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 9 & 0 \end{pmatrix}, \\
 W_L(1) &= \begin{pmatrix} 41 & -10 & 14 \\ -10 & 26 & -17 \\ 14 & -17 & 14 \end{pmatrix}, \quad W_M(1) = \begin{pmatrix} 83 & 1 \\ 1 & 6 \end{pmatrix}, \quad \Gamma_M(1) = \frac{1}{497} \begin{pmatrix} 486 & -2 \\ -81 & 166 \end{pmatrix}.
 \end{aligned} \tag{NT.33}$$

An independent construction of the transition projection starts with

$$v = (171, -277, -603, 440, 48)^t.$$

Multiplication verifies $v = J(171, -277, -603)^t$, $(JE)^*v = 0$, and $v^*v = 665483$. The larger and smaller column spaces have dimensions three and two. Their orthogonal difference is therefore exactly the line spanned by v , so

$$\begin{aligned}
 \Pi_L(1) - \Pi_M(1) &= \frac{vv^*}{665483}, \quad A(1) = \frac{8}{665483}v \begin{pmatrix} 54 & 55 \end{pmatrix}, \\
 N_L(1)E &= \frac{1}{1339} \begin{pmatrix} -1458 & -146 \\ 459 & -202 \\ -567 & 92 \\ 54 & 55 \\ 225 & 6 \end{pmatrix}.
 \end{aligned} \tag{NT.34}$$

For the last matrix define $T = 1339^{-1} \begin{pmatrix} 1458 & 146 \\ -459 & 202 \\ 567 & -92 \end{pmatrix}$. Direct multiplication gives $W_L(1)T = J^*D = \begin{pmatrix} 54 & 2 \\ -27 & 4 \\ 27 & -2 \end{pmatrix}$. Thus $D - JT$ is exactly the displayed matrix $N_L(1)E$. It is orthogonal to every column of J and hence to v . This verifies both normal components independently of the proposed connection difference. Their full Grams are

$$A(1)^*A(1) = \frac{1}{665483} \begin{pmatrix} 186624 & 190080 \\ 190080 & 193600 \end{pmatrix}, \quad E^*\mathcal{N}_L(1)E = \frac{1}{1339} \begin{pmatrix} 2025 & 54 \\ 54 & 55 \end{pmatrix}. \quad (\text{NT.35})$$

Take the original coefficient curve $c(u) = (1 + iu, u^2 - i)^t$. Its directly differentiated vector at one is

$$w = Dc(1) + JEc'(1) = (i, 2, 2 + i, 5 - i, 9 + 18i)^t, \quad w^*w = 441. \quad (\text{NT.36})$$

The three components in (NT.12), with their original values, are

$$t = \frac{1}{497} \begin{pmatrix} 484 + 985i \\ 1079 - 247i \\ 1563 + 738i \\ 2158 - 494i \\ 4356 + 8865i \end{pmatrix}, \quad N_L(1)Ec(1) = \frac{1}{1339} \begin{pmatrix} -1604 - 1312i \\ 257 + 661i \\ -475 - 659i \\ 109 - i \\ 231 + 219i \end{pmatrix}, \quad (\text{NT.37})$$

$$A(1)c(1) = \frac{8(109 - i)}{665483}v,$$

$$w = t + A(1)c(1) + N_L(1)Ec(1), \quad \boxed{441 = \frac{217065}{497} + \frac{58496}{51191} + \frac{320}{103}}.$$

Multiplication of the explicit vectors proves their sum and their squared norms; their orthogonality was proved from the original column spaces above. All three energies are strictly positive. The number 441 is this pointwise original energy. Multiplying c by any smooth compactly supported scalar function equal to one near $u = 1$ retains the same pointwise value and derivative and makes the corresponding integrated energy finite.

The companion exact script `nested_relative_frame_fixture_20260912.py` retains these original matrices and verifies 24 rational-function identities and nonzero witnesses, with identical successful records in ordinary and optimized Python. Its full matrix receipts and independent sample review are pinned in `nested_relative_frame_fixture_20260912_manifest.json`. The full general proof is (NT.1)–(NT.30); the single concrete family above supplies an independent nonzero calculation of the transition and the surviving normal component.

28 Density of the actual relative frames and the limiting derivative graph

This working continuation proves the nested-frame limit for the original arithmetic amplitude. The input is the sum-connection construction (FC.1)–(FC.36), retained at the source hash recorded in the accompanying receipt. All statements below use the original measure, full packet, relative coordinates, polynomial source, and derivative. The proof uses an exponential moment that is established directly from the arithmetic amplitude. No recurrence or Christoffel asymptotic is assumed.

28.1 An exponential bound with the original arithmetic factors

Let

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad h(s) = \prod_{\rho \in \mathcal{Z}} (s - \rho)^{m_\rho},$$

$$d = \deg h, \quad v_h(s) = g(s)/h(s), \quad a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w(t) = |a(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w(t) dt. \quad (\text{ND.1})$$

Here the set of selected zeros is finite and each selected multiplicity is the full multiplicity in g . The quotient is its entire extension at those zeros. The empty product $h = 1$ is allowed. The packet is stable under $\rho \mapsto 1 - \bar{\rho}$, as in FC.1. For a monic polynomial this says $h(1 - \bar{s}) = (-1)^d \bar{h}(s)$; the corresponding identity for g has factor one. Hence on the original line

$$\overline{a(t)} = (-1)^d a(t), \quad \overline{a^{(j)}(t)} = (-1)^d a^{(j)}(t). \quad (\text{ND.2})$$

There is no change of phase or division by the mass μ_h .

Theorem 28.1 (Actual amplitude and all derivative tails). *For every integer $j \geq 0$ and every real b with $0 < b < \pi/4$, there is a finite constant $C_{h,j,b}$ such that*

$$|a^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|} \quad (t \in \mathbb{R}). \quad (\text{ND.3})$$

The amplitude is a nonzero entire function of t . In particular $0 < \mu_h < \infty$, all polynomially weighted derivative integrals used below exist, and its real zero set is discrete.

Proof. We give the tail argument, including the gamma factor. For $\lambda > 0$, $\tau \in \mathbb{R}$, and $0 < \phi < \pi/2$, rotate Euler's gamma integral in the sector with angle $\theta = \text{sgn}(\tau)\phi$, taking $\theta = 0$ when $\tau = 0$. The principal power is holomorphic on the open sector. On the circle of radius R the exponential has modulus at most $e^{-R \cos \phi}$; its product with the power and arc length tends to zero as $R \rightarrow \infty$. The small circular contribution is bounded by a constant times ε^λ and tends to zero. Cauchy's theorem therefore gives

$$\begin{aligned} \Gamma(\lambda + i\tau) &= e^{i\theta(\lambda+i\tau)} \int_0^\infty r^{\lambda+i\tau-1} e^{-e^{i\theta}r} dr, \\ |\Gamma(\lambda + i\tau)| &\leq e^{-\phi|\tau|} \Gamma(\lambda) (\cos \phi)^{-\lambda}. \end{aligned} \quad (\text{ND.4})$$

At $\tau = 0$ the second bound follows directly from Euler's integral. Its constant is bounded when λ ranges over a compact subinterval of $(0, \infty)$.

For completeness, summing the counting function $[x]$ on $\Re s > 1$ gives $\zeta(s) = s \int_1^\infty [x] x^{-s-1} dx$. Splitting $[x] = x - \{x\}$ yields

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx. \quad (\text{ND.5})$$

The last integral is absolutely and locally uniformly convergent on $\Re s > 0$; its differentiated integrals converge on compact subsets because $\int_1^\infty x^{-1-\sigma} (\log x)^n dx$ is finite for every $\sigma > 0$. Thus (ND.5) continues the same zeta function to that half-plane away from $s = 1$. For $s = \sigma + it$, $1/4 \leq \sigma \leq 3/4$, we have $|s/(s-1)| \leq 5$, $|s|/\sigma \leq 1 + 4|t|$, and consequently $|\zeta(s)| \leq 6 + 4|t|$.

Write $z = t + iv$ with $|v| \leq 1/4$, so that $s = 1/2 + iz = (1/2 - v) + it$. In (ND.4) take $\lambda = (1/2 - v)/2 \in [1/8, 3/8]$ and $\tau = t/2$. The modulus of $\pi^{-s/2}$ is exactly $\pi^{-(1/2-v)/2}$. The two polynomial factors in g , followed by (ND.5), therefore give the bound

$$|g(1/2 + iz)| \leq C_\phi (1 + |t|)^3 e^{-\phi|t|/2} \quad (|\Im z| \leq 1/4). \quad (\text{ND.6})$$

For $|t| \geq \max(1, 2 \max_{\rho \in \mathcal{Z}} |\Im \rho|)$, each factor $|s - \rho|$ is at least $|t|/2$. Thus $|h(s)| \geq 2^{-d} |t|^d$; for the empty product the lower bound is one and the maximum over roots is unnecessary. Division by this original polynomial, without removing its remaining unit, bounds v_h on the unbounded part of the strip. The complementary closed rectangle is compact and the entire quotient is bounded there, including every cancelled root. Choose $2b < \phi < \pi/2$. The polynomial factor times $e^{-(\phi/2-b)|t|}$ is bounded. It follows that $|a(z)| \leq C_{h,b} e^{-b|\Re z|}$ on this strip, with the original factor $1/\sqrt{2\pi}$ included in $C_{h,b}$. The Cauchy estimate on the circle of radius $1/8$ about a real t now gives

$$|a^{(j)}(t)| \leq j! 8^j C_{h,b} e^{b/8} e^{-b|t|}.$$

This proves (ND.3). Entirety follows from cancellation at the specified finite roots. The function cannot vanish identically: the formula for g is nonzero on the real interval $s > 1$, and division by a nonzero polynomial preserves nonidentity. The identity theorem makes the real zeros isolated. Positivity and finiteness of the mass follow from nonidentity and (ND.3). $\square$

28.2 Every centred fibre and its exponential moment

Fix $k \geq 2$, put $r = k - 1$, and retain the real coordinates

$$u = \sum_{i=1}^k t_i, \quad y_i = t_i - u/k \quad (1 \leq i \leq r), \quad y_k = -\sum_{i=1}^r y_i, \quad t_i = u/k + y_i,$$

$$S = k/2 + iu, \quad z_i = s_i - S/k = iy_i, \quad \Psi_u(y) = \prod_{i=1}^k a(u/k + y_i), \quad d\mu_u(y) = |\Psi_u(y)|^2 dy. \quad (\text{ND.7})$$

The independent relative variables are $y_1, \dots, y_r$. The determinant of the map with ordered variables $(u, y_1, \dots, y_r) \mapsto (t_1, \dots, t_k)$ is $(-1)^r$. Indeed moving the first column to the last contributes $(-1)^r$, and the resulting block determinant is $\det \begin{pmatrix} I_r & \mathbf{1}/k \\ -\mathbf{1}^T & 1/k \end{pmatrix} = 1/k + r/k = 1$. In particular the original product measure is exactly $du dy$. For $k = 2$, $v = t_1 - t_2 = 2y_1$ gives $du dv/2$. Neither the centre $k/2$ nor this Jacobian is changed.

For every fixed u , each of the k affine functions $y \mapsto u/k + y_i$ is nonconstant. The inverse image of a real zero of a is a proper affine hyperplane. There are at most countably many such hyperplanes, each of Lebesgue measure zero. Consequently $\Psi_u \neq 0$ almost everywhere for every u , including any fibre on which a particular normal derivative vanishes. The orders of the zeros remain the orders of the original product; the measure assertion does not modify the product on a hyperplane.

For every $0 < \eta < \pi/2$, choose b so that $\eta < 2b < \pi/2$. Since $\sum_{i=1}^k |t_i| \geq \|y\|_1 - r|u|/k$, (ND.3) proves the explicit bound

$$\begin{aligned} \int_{\mathbb{R}^r} e^{\eta\|y\|_1} d\mu_u(y) &\leq C_{h,0,b}^{2k} e^{2br|u|/k} \int_{\mathbb{R}^r} e^{-(2b-\eta)\|y\|_1} dy \\ &= C_{h,0,b}^{2k} e^{2br|u|/k} \left(\frac{2}{2b-\eta} \right)^r < \infty. \end{aligned} \quad (\text{ND.8})$$

This bound is uniform on compact u -intervals. It proves both finite moments and $0 < m_k(u) := \mu_u(\mathbb{R}^r) < \infty$ for every u . The coordinate change and Tonelli's theorem retain the full mass identity

$$\int_{\mathbb{R}} m_k(u) du = \mu_h^k. \quad (\text{ND.9})$$

At fixed relative coordinate the exact derivative is

$$\Psi'_u(y) = \frac{1}{k} \sum_{i=1}^k a'(u/k + y_i) \prod_{j \neq i} a(u/k + y_j). \quad (\text{ND.10})$$

Every derivative of this product, multiplied by any polynomial in (u, y) , is square integrable in $du dy$. To verify this claim directly, every derivative is a finite sum of products of derivatives of the original $a(t_i)$, with the powers of $1/k$ from (ND.7) retained. Each product is bounded by a constant times $e^{-b \sum |t_i|}$ by (ND.3). A polynomial in (u, y) becomes a polynomial in $t_1, \dots, t_k$ under the displayed invertible linear map; each of its monomials is integrable against $e^{-2b \sum |t_i|}$. The same bound on a compact u -interval proves continuity, and all orders of differentiation, of the relevant functions with values in $L^2(dy)$, by dominated convergence.

28.3 A complete weighted polynomial density proof

Theorem 28.2 (Exponential moments imply the required density). *Let μ be a finite positive Borel measure on $\mathbb{R}^r$ such that $\int e^{\eta\|y\|_1} d\mu(y) < \infty$ for some $\eta > 0$. Complex polynomials in all r coordinates are dense in $L^2(\mu)$.*

Proof. All polynomials belong to this space by the exponential moment. Suppose $f \in L^2(\mu)$ is orthogonal to every polynomial, using inner products conjugate linear in the first entry, and define the finite complex measure $d\nu = \bar{f} d\mu$. For $0 \leq 2\tau < \eta$, Cauchy-Schwarz gives

$$\int e^{\tau\|y\|_1} d|\nu|(y) \leq \|f\|_{L^2(\mu)} \left(\int e^{2\tau\|y\|_1} d\mu(y) \right)^{1/2} < \infty. \quad (\text{ND.11})$$

Every monomial moment of ν is zero. Fix $\xi \in \mathbb{R}^r \setminus \{0\}$. The function

$$F_\xi(z) = \int e^{iz\xi \cdot y} d\nu(y)$$

is holomorphic on the connected strip $|\Im z| \|\xi\|_\infty < \eta/2$. To see the differentiability, restrict to a compact substrip and choose $\tau < \eta/2$ strictly larger than its bound on $|\Im z| \|\xi\|_\infty$. Every polynomial factor from differentiation is bounded by the remaining exponential slack in (ND.11), which justifies differentiation under the integral. In a disc about zero, the exponential series is absolutely integrable by the same bound. Its n -th coefficient is $i^n \int (\xi \cdot y)^n d\nu(y)/n! = 0$, since this is a polynomial moment. Thus F_ξ vanishes near zero and throughout the strip by the holomorphic identity theorem. In particular $\hat{\nu}(\xi) = F_\xi(1) = 0$. The zero moment also gives $\hat{\nu}(0) = 0$.

Here is the Fourier uniqueness argument needed for this finite complex measure. For $\varepsilon > 0$ put

$$G_\varepsilon(x) = (4\pi\varepsilon)^{-r/2} e^{-|x|^2/(4\varepsilon)} = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} d\xi. \quad (\text{ND.12})$$

The equality follows in one variable by differentiating the Gaussian Fourier integral and integrating by parts: its derivative is $-x/(2\varepsilon)$ times itself and its value at zero is $\sqrt{\pi/\varepsilon}$. The latter Gaussian integral follows by squaring it and using polar coordinates. Taking products gives (ND.12). Since ν is finite and the frequency Gaussian is integrable, Fubini's theorem gives, with the same positive Fourier sign,

$$(G_\varepsilon * \nu)(x) = (2\pi)^{-r} \int_{\mathbb{R}^r} e^{-\varepsilon|\xi|^2} e^{i\xi \cdot x} \hat{\nu}(-\xi) d\xi = 0.$$

For every bounded uniformly continuous function b , another absolutely convergent Fubini integral yields $\int (b * G_\varepsilon) d\nu = 0$. Moreover $b * G_\varepsilon \rightarrow b$ uniformly: in its difference from b , the integration over $|x| < \delta$ is bounded by the uniform modulus of continuity at δ , while the complementary integral is at most $2\|b\|_\infty \int_{|x| \geq \delta} G_\varepsilon$, which tends to zero after the substitution $x = \sqrt{\varepsilon}z$. Thus $\int b d\nu = 0$ for all such b .

For each nonempty closed set $F \subset \mathbb{R}^r$, the bounded Lipschitz functions $b_n(x) = \max(0, 1 - n \text{dist}(x, F))$ converge pointwise to 1_F . Dominated convergence against $|\nu|$ proves $\nu(F) = 0$; this also holds for the empty closed set. In particular $\nu(\mathbb{R}^r) = 0$. The Borel sets with zero ν -measure form a Dynkin system: complements have measure zero by this total mass identity, and countable disjoint unions have measure zero by countable additivity. Closed sets form a pi-system generating the Borel sigma-algebra. The pi-lambda theorem therefore gives $\nu(A) = 0$ for every Borel set A . Applying this to the positive and negative level sets of $\Re f$ and $\Im f$ shows $f = 0$ almost everywhere. Finally, the orthogonal complement of the closed polynomial span is zero, so the Hilbert orthogonal decomposition proves that span is all of $L^2(\mu)$. $\square$

28.4 The multiplier, literal frames, and their strong limit

Let $\mathcal{H} = L^2(\mathbb{R}^r, dy)$. For each fixed u , the exact multiplication map is

$$\begin{aligned} \mathcal{M}_u : L^2(\mu_u) &\longrightarrow \mathcal{H}, & q &\longmapsto \Psi_u q, \\ \mathcal{M}_u^{-1} F(y) &= \begin{cases} F(y)/\Psi_u(y), & \Psi_u(y) \neq 0, \\ 0, & \Psi_u(y) = 0. \end{cases} \end{aligned} \quad (\text{ND.13})$$

It is a unitary map: its squared norm is exactly $\int |q|^2 |\Psi_u|^2 dy$, and the inverse has weighted squared norm $\int |F|^2 dy$ because the zero set is null. These are maps between the displayed weighted and unweighted spaces. No claim of bounded multiplication by $1/\Psi_u$ on unweighted $\mathcal{H}$ is made or used. Assigning zero on null hyperplanes only selects a representative of a Hilbert class; every analytic amplitude and all of its pointwise zero orders remain those of (ND.7).

For $m \geq 0$ use the full monomial family

$$\begin{aligned} \mathcal{A}_m &= \{\alpha \in \mathbb{N}^r : |\alpha| \leq m\}, \quad \theta_\alpha(z) = z^\alpha, \quad \theta_\alpha(iy) = i^{|\alpha|} y^\alpha, \\ j_m(u) : \mathbb{C}^{\mathcal{A}_m} &\rightarrow \mathcal{H}, \quad j_m(u)c = \Psi_u \sum_{\alpha \in \mathcal{A}_m} i^{|\alpha|} y^\alpha c_\alpha, \end{aligned} \quad (\text{ND.14})$$

$$V_m(u) = \text{ran } j_m(u), \quad W_m(u) = j_m(u)^* j_m(u), \quad \Pi_m(u) = j_m(u) W_m(u)^{-1} j_m(u)^*.$$

All moments defining these matrices are finite by (ND.8). If $c \neq 0$, its displayed polynomial is nonzero. By continuity it is nonzero in a nonempty open set, which has positive μ_u -measure because the density is positive almost everywhere. Thus $W_m(u) > 0$ for every u . Multiplication proves that Π_m is self-adjoint, idempotent, and has range V_m , so it is precisely the orthogonal projection onto that range. The inverse here is the positive source Gram W_m^{-1} ; no inverse of a possibly zero normal Gram is used.

For $n \geq m$, let E_{nm} be literal zero-padding of the coefficient array. The complete inclusion and Gram maps are

$$\begin{aligned} j_n E_{nm} &= j_m, \quad W_m = E_{nm}^* W_n E_{nm}, \quad E_{\ell n} E_{nm} = E_{\ell m}, \\ V_m &\subset V_n, \quad \Pi_n \Pi_m = \Pi_m \Pi_n = \Pi_m. \end{aligned} \quad (\text{ND.15})$$

The first line follows term by term in the original polynomial sum. The second line follows from the nested ranges and orthogonal complements. In particular $\Pi_n - \Pi_m$ is the orthogonal projection onto $V_n \cap V_m^\perp$.

Theorem 28.3 (Full relative frames on every actual fibre). *For every $u \in \mathbb{R}$ and every $F \in \mathcal{H}$,*

$$\lim_{m \rightarrow \infty} \|F - \Pi_m(u)F\|_{\mathcal{H}} = 0. \quad (\text{ND.16})$$

For every finite m , however, $\|I - \Pi_m(u)\|_{\mathcal{H} \rightarrow \mathcal{H}} = 1$.

Proof. The density theorem applies to μ_u by (ND.8), separately for each real u . The phases $i^{|\alpha|}$ are nonzero constants and hence the literal monomials in (ND.14) have precisely the full polynomial span. Applying the onto isometry (ND.13) proves $\bigcup_m V_m(u) = \mathcal{H}$. Given F and $\varepsilon > 0$, choose $v \in V_M(u)$ with $\|F - v\| < \varepsilon$. For $m \geq M$, orthogonal projection minimizes distance, so $\|F - \Pi_m F\| \leq \|F - v\| < \varepsilon$. This proves (ND.16). The Hilbert space $L^2(\mathbb{R}^r)$ is infinite dimensional: indicators of infinitely many disjoint finite positive-measure balls are linearly independent. Each V_m has finite dimension. Its orthogonal complement therefore contains a unit vector, on which $I - \Pi_m$ has norm one. Orthogonal projection contraction gives the opposite inequality, proving the asserted operator norm exactly. $\square$

28.5 A fixed original source and the complete derivative domain

Fix one original polynomial $P(s_1, \dots, s_k)$ of total degree D . In the exact linear coordinates $S = \sum s_i$, $z_i = s_i - S/k$, $\sum z_i = 0$, it has the unique expansion

$$P = \sum_{|\alpha| \leq D} z^\alpha f_\alpha(S), \quad \deg f_\alpha \leq D - |\alpha|. \quad (\text{ND.17})$$

Uniqueness is coefficient comparison in the independent variables $(S, z_1, \dots, z_r)$; substitution by an invertible linear map preserves total degree. For $m \geq D$, retain an admitted source degree $M_m \geq$

$\max(D, m)$ and the full relative family $\mathcal{A}_m$. The fixed source has coefficient column

$$c_{P,m}(u)_\alpha = \begin{cases} f_\alpha(k/2 + iu), & |\alpha| \leq D, \\ 0, & D < |\alpha| \leq m. \end{cases} \quad c_{P,n} = E_{nm} c_{P,m}. \quad (\text{ND.18})$$

Every original constraint $\deg f_\alpha \leq M_m - |\alpha|$ is respected; neither the fixed polynomial nor its coefficient degree is changed as the frame grows. One may in particular take $M_m = m$ for all $m \geq D$, which is the full degree- m source of FC.28.

Put

$$\begin{aligned} p_u(y) &= P(k/2 + iu, iy_1, \dots, iy_r), \quad A_P(u, y) = \Psi_u(y) p_u(y) = j_m(u) c_{P,m}(u), \\ \partial_S^{\text{rel}} &= \frac{1}{k} \sum_{i=1}^k \partial_{s_i}, \quad F_P(u, y) = \Psi'_u(y) p_u(y), \\ \partial_u A_P &= F_P + i \Psi_u(\partial_S^{\text{rel}} P)_u. \end{aligned} \quad (\text{ND.19})$$

The notation $P(S, z)$ in these formulas denotes the exact substitution $s_i = S/k + z_i$; $z_k = -\sum_{i < k} z_i$. At fixed relative coordinates $dS/du = i$, so the displayed derivative includes exactly the factor i/k on the sum of the original coordinate derivatives. By (ND.3) and (ND.10), $A_P, F_P, \partial_u A_P$ belong to $\mathcal{K} = L^2(\mathbb{R}_u; \mathcal{H})$. In fact all their polynomially weighted derivatives belong to $\mathcal{K}$. In particular A_P lies in the actual derivative domain

$$\mathcal{D}_u = \{A \in \mathcal{K} : \partial_u A \in \mathcal{K} \text{ in the distributional sense}\}. \quad (\text{ND.20})$$

The smooth derivative in (ND.19) equals the distributional derivative, as follows by integration by parts against compactly supported test functions and the preceding integrability bounds.

We also retain its original Mellin realization and sign. If $\mathcal{M}f(s) = \int_0^\infty f(x) x^{s-1} dx$, the original test vector F_h has $\mathcal{M}F_h = v_h$, and the polynomial source is $\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k) F_h^{\otimes k}$, where $D_i = -x_i \partial_{x_i}$. The normalized critical-line transform has factor $(2\pi)^{-k/2}$, followed by (ND.7). Integration by parts gives $\mathcal{M}(D_i f) = s_i \mathcal{M}f$, so its image is exactly A_P , with the original product of a 's. Writing $x_i = e^{q_i}$, the Mellin integral becomes the Fourier transform with kernel $e^{it_i q_i}$ of $e^{(q_1 + \dots + q_k)/2} f(e^{q_1}, \dots, e^{q_k})$. The change of variable has squared norm exactly $\int |f|^2 d\mathbf{x}$; Fourier Plancherel and the Jacobian one in (ND.7) give the isometry $\mathcal{U}_k$ in FC.9. On this original test domain,

$$\begin{aligned} \mathcal{U}_k \mathcal{T}_h^{(k)} P &= A_P, \quad \mathcal{U}_k D^{(k)} = S \mathcal{U}_k, \quad \partial_u \mathcal{U}_k = i \mathcal{U}_k \mathcal{L}_k, \\ D^{(k)} &= \sum_i D_i, \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -I. \end{aligned} \quad (\text{ND.21})$$

The second derivative identity follows by differentiating the Fourier integral with $\partial t_i / \partial u = 1/k$; the commutator follows from $-x_i \partial_{x_i} \log x_i = -1$, summed with the displayed coefficient $1/k$. All these operations are on the original test vectors and their logarithmic derivatives. Their spectral images have the established L^2 derivatives, so no new domain or arbitrary assignment of an arithmetic jet is introduced.

In the degree- m frame define the original matrices and maps

$$\begin{aligned} B_m &= j_m^* j'_m, \quad \Gamma_m = W_m^{-1} B_m, \quad N_m = (I - \Pi_m) j'_m, \quad \mathcal{N}_m = N_m^* N_m, \\ t_m(P) &= j_m(c'_{P,m} + \Gamma_m c_{P,m}), \quad r_m(P) = N_m c_{P,m}. \end{aligned} \quad (\text{ND.22})$$

Every term is the FC.6 term in the literal enlarged frame. Since $j_m \Gamma_m = \Pi_m j'_m$, the product rule gives

$$\begin{aligned} t_m(P) &= \Pi_m \partial_u A_P, \quad r_m(P) = (I - \Pi_m) \partial_u A_P = (I - \Pi_m) F_P, \\ \partial_u A_P &= t_m(P) + r_m(P), \quad \langle t_m(P, u), r_m(P, u) \rangle_{\mathcal{H}} = 0, \\ \|\partial_u A_P\|_{\mathcal{H}}^2 &= \|t_m(P)\|_{\mathcal{H}}^2 + \|r_m(P)\|_{\mathcal{H}}^2. \end{aligned} \quad (\text{ND.23})$$

For the second equality, the coefficient derivative in (ND.19) belongs to $V_m(u)$, so its orthogonal complement is zero. This equality identifies a single fixed function F_P , independent of frame degree, which controls every residual.

Theorem 28.4 (Vanishing of the fixed-source actual normal residual). *For every fixed original polynomial P and every real u ,*

$$\begin{aligned} \|r_m(P, u)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P, u) - \partial_u A_P(u)\|_{\mathcal{H}} &\longrightarrow 0, \\ \|r_m(P)\|_{\mathcal{H}} &\longrightarrow 0, & \|t_m(P) - \partial_u A_P\|_{\mathcal{H}} &\longrightarrow 0. \end{aligned} \quad (\text{ND.24})$$

The fibre convergence of both norms is uniform on compact u -intervals. The derivative graph converges, in the original $\mathcal{H} \oplus \mathcal{H}$ norm, to $(\partial_u A_P, 0)$, while its sum is exactly $\partial_u A_P$ at every degree.

Proof. For each u , $F_P(u) \in \mathcal{H}$, and (ND.16) applied to this vector gives the first limit through (ND.23). Furthermore,

$$0 \leq \|r_m(P, u)\|^2 \leq \|F_P(u)\|^2, \quad \int_{\mathbb{R}} \|F_P(u)\|^2 du < \infty. \quad (\text{ND.25})$$

All functions here are measurable; in fact j_m, j'_m, W_m are smooth in u by the compact-interval bounds after (ND.10), and W_m^{-1} is smooth since its determinant is everywhere positive. The dominated convergence theorem proves the integrated residual limit. Since $t_m - \partial_u A_P = -r_m$, it proves the two tangential limits and the asserted graph convergence as well.

For $n \geq m$, (ND.15) gives the exact orthogonal decomposition

$$\begin{aligned} r_m(P, u) &= r_n(P, u) + (\Pi_n - \Pi_m)F_P(u), \\ \|r_m(P, u)\|^2 - \|r_n(P, u)\|^2 &= \|(\Pi_n - \Pi_m)F_P(u)\|^2 \geq 0. \end{aligned} \quad (\text{ND.26})$$

The two terms are orthogonal because the first lies in $V_n^\perp$ and the second in V_n . Each squared residual is continuous by the same smoothness argument. On a compact set K the open sets $\{u : \|r_m(P, u)\|^2 < \varepsilon\}$ are increasing and cover K . A finite subcover, followed by taking its largest index, makes the squared residual less than ε everywhere on K . This proves compact uniform convergence without a global uniform-in-source assertion. $\square$

For a fixed finite source core $\text{span}\{P_1, \dots, P_q\}$ let $R_m c = \sum_j c_j r_m(P_j)$. Cauchy-Schwarz gives

$$\|R_m\|^2 \leq \sum_{j=1}^q \|r_m(P_j)\|_{\mathcal{H}}^2 \rightarrow 0, \quad \|R_m^* R_m\| \rightarrow 0, \quad R_m^* R_m - R_n^* R_n \succeq 0 \ (n \geq m). \quad (\text{ND.27})$$

The last inequality is (ND.26) integrated for each linear combination of the fixed sources. These are exactly the normal Gram matrices restricted by literal inclusion of that source core. The core and its coefficient norm are fixed in this statement. The full degree-varying source dimension, its minimum arithmetic norm, and the size of its normal-image inverse are not replaced by this finite-core limit.

The exact translation to the original source-amplitude metric is also available on this same fixed core. For an independent core put $(G_0)_{ij} = \langle A_{P_i}, A_{P_j} \rangle_{\mathcal{H}}$. It is independent of m and strictly positive: a nonzero linear combination of the original polynomials is nonzero on an open subset of the real critical-line coordinates, and the original product amplitude is nonzero almost everywhere there. Its squared integral is therefore positive. With $H_m = R_m^* R_m$, define

$$K_m = G_0^{-1} H_m, \quad K_m^* G_0 = G_0 K_m = H_m, \quad \|R_m\|_{G_0 \rightarrow \mathcal{H}}^2 \leq \lambda_{\min}(G_0)^{-1} \sum_j \|r_m(P_j)\|_{\mathcal{H}}^2 \longrightarrow 0.$$

The first equalities follow by matrix multiplication, and the bound follows from $c^* G_0 c \geq \lambda_{\min}(G_0) \|c\|^2$ and the same column Cauchy-Schwarz inequality. Multiplication by the positive

square root $G_0^{1/2}$ is the isometry from this original coefficient metric to the Euclidean metric. It carries K_m to the positive matrix $G_0^{-1/2} H_m G_0^{-1/2}$; its largest eigenvalue is exactly $\|R_m\|_{G_0 \rightarrow \mathcal{H}}^2$. Thus K_m also tends to zero in its original metric operator norm. These explicit congruence and comparison maps preserve both the source-amplitude metric and the normal metric.

The arithmetic maps themselves are retained by the same inclusions. For a nonempty packet set $I = (h(s_1), \dots, h(s_k))$, let j_I denote the full finite jet map, and put $U(\mathbf{s}) = \prod_i v_h(s_i)$. At every selected root $v_h(\rho) \neq 0$, since its full zero order has been divided out; hence $v = j_I U$ is a unit in the full tensor jet algebra. The original source and logarithmic derivative observations are

$$J(\mathcal{T}_h^{(k)} P) = v[P], \quad J\mathcal{L}_k \mathcal{T}_h^{(k)} P = j_I(\partial_S^{\text{rel}}(UP)). \quad (\text{ND.28})$$

Indeed the Mellin transform of $\mathcal{T}_h^{(k)} P$ is UP ; insertion of $k^{-1} \sum \log x_i$ differentiates it by ∂_S^{rel} . At a selected tuple the algebra of germs modulo the full powers is finite; a germ with nonzero constant term is invertible by its finite geometric expansion in the nilpotent maximal ideal. This proves the asserted unit property including all mixed jet orders. Since (ND.18) is the identity on P , both quantities in (ND.28) are identical at every degree. No passage of this jet map through a Hilbert-space limit is required for, or asserted by, the density theorem. If $h = 1$, the quotient by $I = (1)$ is the zero algebra and both finite arithmetic targets are zero.

28.6 Invariant frames, sparse frames, and all original zero cases

Let G be any fixed subgroup of the factor permutation group S_k . Its action on $\sum y_i = 0$, expressed in the first r coordinates, is real linear with determinant of absolute value one. To check the determinant, the action on the full t variables is a permutation matrix with determinant ± 1 ; conjugating by (ND.7) leaves u fixed and acts linearly on y , so the remaining block has determinant ± 1 . The product Ψ_u is unchanged by this action. Therefore

$$(\mathbf{U}_g F)(y) = F(g^{-1}y), \quad \mathbf{R}_G = \frac{1}{|G|} \sum_{g \in G} \mathbf{U}_g \quad (\text{ND.29})$$

defines unitary group actions on both $\mathcal{H}$ and $L^2(\mu_u)$; they are intertwined by $\mathcal{M}_u$. The adjoint of $\mathbf{U}_g$ is $\mathbf{U}_{g^{-1}}$, so $\mathbf{R}_G^* = \mathbf{R}_G$. Counting the $|G|$ products $gh = \ell$ for each ℓ gives $\mathbf{R}_G^2 = \mathbf{R}_G$. Its image is the invariant subspace: its image is fixed under every group element, and its value on a fixed vector is that vector. Thus it is exactly the invariant orthogonal projection.

Let $V_m^G(u)$ be Ψ_u times all G -invariant relative polynomials of degree at most m , using any full independent basis of that space with its original coefficients. The degree is preserved by the linear action and by its finite average. Applying $\mathbf{R}_G$ to polynomial approximants in the density theorem proves density in the invariant weighted subspace. Intertwining by (ND.13) therefore gives

$$\overline{\bigcup_m V_m^G(u)} = \mathcal{H}^G, \quad \Pi_m^G(u) \xrightarrow{\text{strongly}} \mathbf{R}_G. \quad (\text{ND.30})$$

To verify the second formula for an arbitrary F , decompose it as $\mathbf{R}_G F + (I - \mathbf{R}_G)F$. The second part is orthogonal to every V_m^G , and the first is approximated by their dense union by the same best-approximation argument as (ND.16). If the original polynomial P is G -invariant, then $A_P, F_P, \partial_u A_P$ are invariant, since the group fixes u and Ψ_u . Its expansion in a full homogeneous invariant basis retains the bound $\deg f_\alpha \leq D - \ell_\alpha$: apply coefficient comparison in powers of S and the homogeneous relative grading to (ND.17). Consequently (ND.23)–(ND.27) hold for these actual invariant frames, with Π_m^G in place of Π_m . In particular their fixed-source residual also tends to zero pointwise, on compact u -intervals, and in $\mathcal{H}$. This includes full symmetric orbit frames without changing the unscaled orbit coefficients.

For an arbitrary nested family of finite polynomial spaces E_m , let $V_m^E(u) = \mathcal{M}_u E_m$ and $V_\infty^E(u) = \overline{\bigcup_m V_m^E(u)}$. Its exact strong projection limit is

$$\Pi_m^E(u) \xrightarrow{\text{strongly}} Q_u^E := \text{proj}_{V_\infty^E(u)}. \quad (\text{ND.31})$$

Here is a proof requiring no density assumption. For $n \geq m$, orthogonality of the nested projections gives $\|\Pi_n^E F - \Pi_m^E F\|^2 = \|\Pi_n^E F\|^2 - \|\Pi_m^E F\|^2$. The squared norms increase and are bounded by $\|F\|^2$. Thus the projections applied to F are Cauchy. Their limit belongs to V_∞^E , and its difference from F is orthogonal to each V_m^E , hence to their closure. This proves (ND.31) and identifies the exact projection map. A fixed source admitted from some level onward is a sum of fixed members of that relative polynomial space times polynomials in S . Its coefficient derivative replaces each such polynomial by i times its S -derivative and therefore belongs to the same relative space. The residual limit is consequently precisely $(I - Q_u^E)\partial_u A_P = (I - Q_u^E)F_P$. This convergence is also in $\mathcal{K}$, by pointwise strong convergence and the integrable bound $4\|F_P(u)\|^2$ for the squared difference from its limiting residual. The limit field is measurable as a pointwise norm limit of measurable projection fields.

The projection in (ND.31) need not be the identity. A concrete actual example uses $E_m = \mathbb{C}$ at every level. Set $b_u = m_k(u)^{-1} \int y_1 d\mu_u(y)$, which is finite by (ND.8). The nonzero vector $F = \Psi_u(y)(y_1 - b_u)$ has zero inner product with Ψ_u , so every $\Pi_m^E F = 0$. It is nonzero because μ_u has positive density almost everywhere and the affine hyperplane $y_1 = b_u$ is null. This gives the exact projection and an explicit failed approximation, without asserting that the sparse space is unrelated to the full one.

All zero and empty cases have concrete maps. The zero source has $A_P = F_P = t_m = r_m = 0$ at every degree. A nonzero source whose arithmetic class $[P]$ or $v[P]$ vanishes is still the same nonzero polynomial in (ND.17), with the same amplitude and convergence; (ND.28) retains its full arithmetic class. The packet $h = 1$ gives the actual $2\xi/\sqrt{2\pi}$ amplitude and satisfies every analytic estimate above. No actual fibre has zero mass for $k \geq 2$, by the hyperplane argument following (ND.7). One may append an initial empty frame $V_{-1} = \{0\}$ with $\Pi_{-1} = 0$; the full nonempty sequence starts with the constant frame at degree zero. Empty source arrays have the unique zero maps.

The original exceptional two-factor fibre is retained. If the packet makes a even, as for the quartet in FC.27, then a' is odd and (ND.10) gives

$$\Psi'_0(y) = \frac{1}{2}(a'(y)a(-y) + a(y)a'(-y)) = 0, \quad j'_m(0) = 0, \quad N_m(0) = 0, \quad r_m(P, 0) = 0. \quad (\text{ND.32})$$

The derivative of the coefficient polynomial may be nonzero; (ND.23) keeps that term in the tangential component. The density proof and the limiting derivative graph require no positive inverse for $\mathcal{N}_m(0)$. Their only fibre inverse is W_m^{-1} , which remains strictly positive there. No conclusion about one-factor relative frames is hidden in the assumption $k \geq 2$: for $k = 1$ the relative coordinate space has one point, the multiplication image is one-dimensional when $a(u) \neq 0$ and is zero when $a(u) = 0$. At such a zero it is not onto the scalar ambient Hilbert space, and the pointwise density-to-identity assertion (ND.16) was explicitly proved only for $k \geq 2$.

28.7 Frame covariance and the exact half-line transport

Let $C_m(u)$ be any smooth invertible change of relative frame. Use $\tilde{j}_m = j_m C_m$ and $\tilde{c}_{P,m} = C_m^{-1} c_{P,m}$. The product and projection remain the same, while the complete connection and normal formulas are

$$\begin{aligned} \tilde{W}_m &= C_m^* W_m C_m, & \tilde{B}_m &= C_m^* B_m C_m + C_m^* W_m C'_m, \\ \tilde{\Gamma}_m &= C_m^{-1} \Gamma_m C_m + C_m^{-1} C'_m, & \tilde{N}_m &= N_m C_m, & \tilde{N}_m \tilde{c}_{P,m} &= r_m(P), \\ \tilde{j}_m(\tilde{c}'_{P,m} + \tilde{\Gamma}_m \tilde{c}_{P,m}) &= t_m(P), & \tilde{E}_{nm} &= C_n^{-1} E_{nm} C_m. \end{aligned} \quad (\text{ND.33})$$

The first two lines follow by differentiating $j_m C_m$ and using the unchanged projection image; its complementary projection annihilates $j_m C'_m$. Differentiating $C_m^{-1} c_{P,m}$ cancels the $C_m^{-1} C'_m$ term and

proves the tangential identity. Finally $\tilde{j}_n \tilde{E}_{nm} = \tilde{j}_m$, so these are the exact typed inclusion maps for the same nested images. Arbitrary growth of C_m is not used to assign an independent coefficient domain: the admitted columns are precisely the pullbacks of (ND.18), and the actual amplitudes and graph norms remain unchanged.

For clarity the residual limit can also be transported into the original two-branch half-line representation without an evenness assumption. For $R \in \mathcal{K}$, let $x = u^2 > 0$, $v = \sqrt{x}$, and set

$$\begin{aligned} R_+(x) &= R(v), \quad R_-(x) = R(-v), \quad e_R(x) = \frac{1}{2}(R_+(x) + R_-(x)), \quad o_R(x) = \frac{R_+(x) - R_-(x)}{2i\sqrt{x}}, \\ R_+ &= e_R + i\sqrt{x} o_R, \quad R_- = e_R - i\sqrt{x} o_R, \\ \|R\|_{\mathcal{K}}^2 &= \int_0^\infty \frac{\|R_+\|^2 + \|R_-\|^2}{2\sqrt{x}} dx = \int_0^\infty \left(\frac{\|e_R\|^2}{\sqrt{x}} + \sqrt{x} \|o_R\|^2 \right) dx. \end{aligned} \tag{ND.34}$$

The first equality is the positive Jacobian on each branch; the second is the parallelogram identity after the displayed exact inverse. These formulas define an onto isometry between the full-line space and the displayed two weighted half-line spaces. Thus (ND.24) carries the actual residual to zero in the sum of both original weighted norms, and carries the full derivative graph to the same transported derivative. Pointwise fibre convergence holds at every $x > 0$ on both branches. At the meeting point $u = 0$, the original smooth amplitudes have their original trace; no new independent half-line boundary values are introduced. The proof does not assert convergence of $\partial_u r_m$, which would require control of a different derivative. It proves exactly the fixed-source convergence of the normal and tangential parts of the original first derivative, with their sum and arithmetic source fixed at every degree.

29 The actual normal-image limit, its graph, and its arithmetic observation

Keep the finite dagger-stable packet, its entire quotient $v_h = g/h$, $g = 2\xi$, and an integer $k \geq 2$. The empty packet $h = 1$ is included. We retain the original amplitude and coordinates

$$a(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad t_i = u/k + y_i, \quad y_k = -\sum_{i < k} y_i, \quad \Psi(u, \mathbf{y}) = \prod_{i=1}^k a(t_i), \quad S = k/2 + iu. \tag{NN.1}$$

The original measures are $du d\mathbf{y}$, with the unit Jacobian proved in (FC.2). No mass, phase, residue unit or source coefficient is replaced. Write $\mathcal{H} = L^2(d\mathbf{y})$ and $\mathcal{K} = L^2(du; \mathcal{H})$.

For each integer $m \geq 0$, let $\mathcal{P}_m$ be the complete original polynomial source of total degree at most m . Its relative frame consists of every monomial $\theta_\alpha(\mathbf{z})$ of degree $\ell_\alpha \leq m$ in the $k-1$ independent variables $z_i = s_i - S/k$, $i < k$, with $z_k = -\sum_{i < k} z_i$ retained as the dependent coordinate. Its literal source basis consists of $\theta_\alpha S^n$, $0 \leq n \leq m - \ell_\alpha$, as in (FC.28). Keep its coefficient map $C_m(u)$, frame $j_m(u)$, orthogonal projection $\Pi_m(u)$, connection $\Gamma_m(u)$, and normal map $N_m(u)$ of (FC.6). Thus the actual amplitude, normal and tangential source maps are

$$\begin{aligned} A_m p &= j_m C_m p = \Psi P_p, \\ N_m p &= N_m C_m p = (I - \Pi_m) \partial_u A_m p, \\ T_m p &= j_m (C'_m + \Gamma_m C_m) p = \Pi_m \partial_u A_m p. \end{aligned} \tag{NN.2}$$

All three maps have values in the same original space $\mathcal{K}$. The last two equalities include the derivative of every original power of S ; its factor is inS^{n-1} , not nS^{n-1} .

Let $Q_{\ell m}$ be the literal coefficient inclusion for $\mathcal{P}_m \subseteq \mathcal{P}_\ell$, $\ell \geq m$, and $E_{\ell m}$ its relative-frame inclusion. The coefficient identities and amplitude identity are

$$C_\ell Q_{\ell m} = E_{\ell m} C_m, \quad j_\ell E_{\ell m} = j_m, \quad A_\ell Q_{\ell m} = A_m. \tag{NN.3}$$

They follow by comparing each literal column $\theta_\alpha S^n$. In particular these are inclusions of the original sources, and not an identification of coefficient spaces of different dimensions.

The weighted-density theorem for the amplitude (NN.1) proves $\Pi_m(u) \rightarrow I$ strongly for every u , and proves for every fixed original polynomial P that

$$\begin{aligned} r_m(P) &:= (I - \Pi_m)\partial_u(\Psi P) \longrightarrow 0 \quad \text{in } \mathcal{K}, \\ t_m(P) &:= \Pi_m\partial_u(\Psi P) \longrightarrow \partial_u(\Psi P) \quad \text{in } \mathcal{K}. \end{aligned} \quad (\text{NN.4})$$

Here and below $m \geq \deg P$. The full density proof, including its original exponential estimate, the division across the selected full zero orders, all fibre zero sets and the invariant-source variant, is given in the preceding relative-density section. Equivalently, the first statement of (NN.4) is obtained by applying those actual fibre projections to the one fixed function $\partial_u(\Psi P) \in \mathcal{K}$, then applying dominated convergence to the bound $\|(I - \Pi_m)\partial_u(\Psi P)\| \leq \|\partial_u(\Psi P)\|$. Thus no unproved estimate about a sequence of changing source polynomials is used here.

29.1 Exact contractions of the finite normal images

Put $\mathcal{R}_m^N = \text{ran } N_m$, and retain $H_m^N = N_m^* N_m > 0$ and $L_m^N = (H_m^N)^{-1} N_m^*$ of (FC.29)–(FC.30). Their positivity and the exact source inverse hold for the actual arithmetic amplitude, including $h = 1$. All these source spaces contain the constant monomial and are nonzero. Define the bounded orthogonal projection on $\mathcal{K}$

$$(\mathbb{P}_m f)(u) = \Pi_m(u) f(u). \quad (\text{NN.5})$$

The finite-frame entries are smooth, so this is a measurable contraction; self-adjointness and idempotence hold after integrating their pointwise identities. The relative inclusions give $\mathbb{P}_\ell \mathbb{P}_m = \mathbb{P}_m = \mathbb{P}_m \mathbb{P}_\ell$ for $\ell \geq m$.

The exact transition from one normal image to the next is

$$\begin{aligned} V_{\ell m} &: \mathcal{R}_m^N \longrightarrow \mathcal{R}_\ell^N, \\ V_{\ell m} r &= (I - \mathbb{P}_\ell) r = N_\ell Q_{\ell m} L_m^N r. \end{aligned} \quad (\text{NN.6})$$

To prove both its codomain and its second expression, write uniquely $r = N_m p = (I - \mathbb{P}_m)\partial_u A_m p$. The nested projection identity and (NN.3) give

$$(I - \mathbb{P}_\ell) r = (I - \mathbb{P}_\ell)\partial_u A_m p = N_\ell Q_{\ell m} p.$$

Injectivity follows from injectivity of N_ℓ and of the literal inclusion. Moreover

$$\|V_{\ell m}\| \leq 1, \quad V_{mm} = I, \quad V_{nm} = V_{n\ell} V_{\ell m} \quad (n \geq \ell \geq m). \quad (\text{NN.7})$$

Projection contraction proves the first assertion; the source formula and $L_\ell^N N_\ell = I$ prove the others. The source and transition square commutes exactly:

$$L_\ell^N V_{\ell m} = Q_{\ell m} L_m^N. \quad (\text{NN.8})$$

In particular these changing normal images have a specified algebraic relation even though their Hilbert subspaces are not asserted to be nested by the ambient identity map.

Their complete energy difference on the fixed source is

$$\begin{aligned} r_m(P) &= (\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P) + r_\ell(P), \\ \|r_m(P)\|^2 - \|r_\ell(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2, \\ \|t_\ell(P)\|^2 - \|t_m(P)\|^2 &= \|(\mathbb{P}_\ell - \mathbb{P}_m)\partial_u(\Psi P)\|^2. \end{aligned} \quad (\text{NN.9})$$

Indeed $\mathbb{P}_\ell - \mathbb{P}_m$, $\mathbb{P}_m$, and $I - \mathbb{P}_\ell$ are mutually orthogonal projections. Apply their orthogonal decomposition to the unchanged derivative amplitude. The full nested-frame connection identities in the preceding section identify this middle component in the original coefficients and also under each stated variable frame change.

29.2 The algebraic limit and its two exact metric presentations

Consider the algebraic direct limit of the injective transitions $V_{\ell m}$. A class represented by $r \in \mathcal{R}_m^N$ is sent to the original polynomial with coefficients $L_m^N r$. Equation (NN.8) proves that the map is well-defined. It is onto $\mathcal{P}_\infty = \mathbb{C}[s_1, \dots, s_k]$, since every polynomial lies in a finite source. If its polynomial is zero then $r = N_m L_m^N r = 0$, so it is injective. We have proved the exact complex-linear isomorphism

$$\operatorname{colim}_{m, V_{\ell m}} \mathcal{R}_m^N \xrightarrow{\sim} \mathcal{P}_\infty. \quad (\text{NN.10})$$

Because (NN.7) is contractive, the limit seminorm is well-defined by

$$\|[r]\|_{N, \infty} = \lim_{\ell \rightarrow \infty} \|V_{\ell m} r\| = \inf_{\ell \geq m} \|V_{\ell m} r\|. \quad (\text{NN.11})$$

Representing the same class at a later index leaves the limit unchanged; representing two classes at a common index proves homogeneity and the triangle inequality. For the fixed polynomial given by (NN.10), (NN.4) proves

$$\|[r]\|_{N, \infty} = 0 \quad \text{for every class.} \quad (\text{NN.12})$$

Thus its Hausdorff seminorm quotient, and its completion, are the zero vector space. This conclusion retains the nonzero algebraic space in (NN.10), together with the exact zero quotient map.

For comparison retain the complete derivative graph $\mathcal{G}_m = \operatorname{ran}(\mathbf{T}_m, \mathbf{N}_m)$ with its original sum norm. Its transitions are

$$W_{\ell m}(\mathbf{T}_m p, \mathbf{N}_m p) = (\mathbf{T}_\ell Q_{\ell m} p, \mathbf{N}_\ell Q_{\ell m} p). \quad (\text{NN.13})$$

Uniqueness of the source proves that these maps are well-defined, injective and compose. Orthogonality and (NN.3) show that both sides have squared norm $\|\partial_u A_m p\|^2$, so every transition is isometric. Addition of the two components commutes with transitions and gives the exact isometry

$$\operatorname{colim}_{m, W_{\ell m}} \mathcal{G}_m \xrightarrow{\text{Add}} \{\partial_u(\Psi P) : P \in \mathcal{P}_\infty\} \subset \mathcal{H}. \quad (\text{NN.14})$$

Its extension identifies the Hilbert completion with the closure of this specified derivative image in the original $\mathcal{H}$. The derivative image is injective in its polynomial source. Indeed $\Psi P \in L^2(du; \mathcal{H})$ and a zero weak derivative in u make every scalar pairing with an $\mathcal{H}$ vector a constant distribution in u . Such a constant belongs to $L^2(du)$ only when zero. Hence $\Psi P = 0$. The nonzero entire one-factor amplitude has a discrete real zero set, so Ψ is nonzero almost everywhere in the original coordinates; a polynomial vanishing almost everywhere there is the zero polynomial.

The algebraic reconstruction maps of (FC.31) give the commuting identity

$$W_{\ell m} \operatorname{Rec}_m = \operatorname{Rec}_\ell V_{\ell m}. \quad (\text{NN.15})$$

It follows by substituting $r = N_m p$. Thus the two algebraic limits are explicitly isomorphic through the original source, while (NN.12) and (NN.14) compute their two distinct retained metrics and their exact completion maps.

29.3 Original arithmetic jets and the size of reconstruction

For a nonempty packet retain the original finite algebra $E = \mathbb{C}[\mathbf{s}]/(h(s_1), \dots, h(s_k))$, its complete Hermite coordinates, and $U = \prod_i v_h(s_i)$. Its jet is a unit. The fixed polynomial observation is the actual derivative jet

$$\mathcal{J}^{\log} P = j_I \left(\frac{1}{k} \sum_i \partial_{s_i}(UP) \right) \in E. \quad (\text{NN.16})$$

It includes the entire product derivative of U at every retained nilpotent order. Formula (FC.34) gives its exact finite matrices $J_m^{\log}$, including the original Mellin factors and the $-i\mathcal{U}_k^{-1}$ arrow from the derivative amplitude. Their coefficient inclusions give $J_\ell^{\log} Q_{\ell m} = J_m^{\log}$, whence

$$\text{Obs}_\ell^{\log} V_{\ell m} = \text{Obs}_m^{\log}, \quad \text{Obs}_m^{\log} = J_m^{\log} L_m^N. \quad (\text{NN.17})$$

These equalities follow from (NN.8), so they define on (NN.10) exactly the original algebraic map $\mathcal{J}^{\log}$.

This map is nonzero, with an explicit original source witness. Choose any retained tuple $\rho = (\rho_1, \dots, \rho_k)$, set $\sigma = \sum_i \rho_i$, and take $P_\rho = S - \sigma$. The complete jet evaluation at the constant coordinate of that tuple is

$$(\mathcal{J}^{\log} P_\rho)(\rho) = U(\rho) \neq 0. \quad (\text{NN.18})$$

The product-rule term multiplying P_ρ vanishes at that tuple; the other term is $U \partial_S^{\text{rel}} S = U$, with its full $1/k$ retained. The nonvanishing follows from taking every selected zero to its full order in h .

There is also a permutation-invariant witness that is itself an original theta relation. Take $P_{\text{rel}} = \sum_i h(s_i) \in I$. Its original full jet is zero, while its retained derivative is

$$\mathcal{J}^{\log} P_{\text{rel}} = \frac{1}{k} U_k \sum_i [h'(s_i)]_I \neq 0. \quad (\text{NN.18a})$$

The term differentiating U still has the factor P_{rel} and is zero in E . If $d = 1$, the sum of derivatives is exactly k . If $d \geq 2$, its degree- $d - 1$ standard remainder terms are the distinct monomials ds_i^{d-1} ; their nonzero coefficients cannot cancel. Multiplication by the full unit U_k proves the nonvanishing in both cases, including all confluent orders. Each $h(s_i)$ term has the original theta primitive with the two matching Koszul signs, so this witness retains the actual relation before its supported-zero quotient. It is available in the full invariant source as well as in the full tensor source.

Fix a genuine original target metric as follows. Choose a degree $m_* \geq k(d - 1)$. The original jet map $J_* p = j_I(U P_p)$ on $\mathcal{P}_{m_*}$ is onto: full monic remainders have total degree at most $k(d - 1)$, and multiplication by $j_I U$ is invertible. With its original amplitude metric $H_{m_*}^0 = A_{m_*}^* A_{m_*}$, the retained minimum-source metric is

$$G_* = (J_*(H_{m_*}^0)^{-1} J_*^*)^{-1} > 0. \quad (\text{NN.19})$$

Indeed for a target vector q , the source $(H_{m_*}^0)^{-1} J_*^* G_* q$ maps to q ; it is orthogonal in the original source metric to $\ker J_*$. Its squared norm is $q^* G_* q$, and adding any kernel vector adds its nonnegative squared norm. This proves the stated actual minimum and its constants.

The exact sizes of the finite inverse, graph reconstruction and arithmetic observation are

$$\begin{aligned} \|L_m^N\|_{N \rightarrow 0}^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^0 (H_m^N)^{-1/2}), \\ \|\text{Rec}_m\|^2 &= \lambda_{\max}((H_m^N)^{-1/2} H_m^\partial (H_m^N)^{-1/2}), \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_* J_m^{\log} (H_m^N)^{-1/2}). \end{aligned} \quad (\text{NN.20})$$

Here H_m^0 , H_m^N , and H_m^∂ are the three original forms in (FC.29)–(FC.32). For a nonzero coefficient vector p , the denominator of each norm ratio is $p^* H_m^N p$; the respective numerators are $p^* H_m^0 p$, $p^* H_m^\partial p$, and $p^* (J_m^{\log})^* G_* J_m^{\log} p$. The change of variable $b = (H_m^N)^{1/2} p$, with its exact inverse, proves the displayed eigenvalue formulas without altering any of these original forms.

All three norms tend to infinity as $m \rightarrow \infty$. For the first two use any fixed nonzero polynomial: its original amplitude and its derivative amplitude both have fixed positive norm, whereas (NN.4) makes its normal norm tend to zero and (FC.26) keeps that norm strictly positive at every finite degree. For the third use the single fixed polynomial P_ρ in (NN.18); its G_* target norm is a fixed positive number. Each operator norm is bounded below by this specified ratio for every sufficiently large m , proving divergence. Equivalently, (NN.17) is a nonzero algebraic observation on the limit, and cannot descend continuously through its zero Hausdorff quotient.

Every changing actual minimum metric is retained as well. Define G_m by the same minimum-source construction at degree $m \geq m_*$. Its original-source inclusion makes $G_m \preceq G_*$: the earlier minimum lift is an allowed later lift with exactly the same amplitude norm. Set $B_m = G_*^{-1/2} G_m G_*^{-1/2} > 0$. The precise comparison is

$$\begin{aligned} q^* G_m q &= (G_*^{1/2} q)^* B_m (G_*^{1/2} q), \quad 0 < B_m \preceq I, \\ \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 &= \lambda_{\max}((H_m^N)^{-1/2} (J_m^{\log})^* G_m J_m^{\log} (H_m^N)^{-1/2}), \\ \lambda_{\min}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2 &\leq \|\text{Obs}_m^{\log}\|_{N \rightarrow G_m}^2 \leq \lambda_{\max}(B_m) \|\text{Obs}_m^{\log}\|_{N \rightarrow G_*}^2. \end{aligned} \quad (\text{NN.21})$$

The first equality is direct multiplication. Its minimum and maximum eigenvalue inequalities applied to $q = J_m^{\log} p$, followed by the same Rayleigh supremum, prove every remaining assertion. Thus the divergence proved for the fixed actual metric is transported to the changing actual metric by an explicit positive operator; its displayed eigenvalues have not been assigned an unproved asymptotic.

29.4 The split lift retains the algebraic source of the zero limit

For a complex vector space V , use the split semimodule $G(V) = \{\tau\} \sqcup V$ over $G(\mathbb{C})$. The external τ is the additive identity, addition of supported vectors is their original vector addition, supported scalar multiplication is the original one, and multiplication involving external τ is external τ . Its supported zero is $e_V = 0_V \neq \tau$. For a linear map $f : V \rightarrow W$, the map $G(f)$ sends τ to τ and a supported vector v to the supported vector $f(v)$. Checking separately external and supported inputs proves additivity and compatibility with every scalar. Composition is the original composition on supported inputs and fixes τ .

Consequently (NN.10), (NN.15), and (NN.17) give exact split-semimodule maps. In particular the seminorm quotient in (NN.12) has the split lift

$$G(\mathcal{P}_\infty) \longrightarrow G(0) = \{\tau, e_0\}, \quad \tau \longmapsto \tau, \quad P \longmapsto e_0 \quad (P \in \mathcal{P}_\infty). \quad (\text{NN.22})$$

Its two fibres are exactly $\{\tau\}$ and the entire supported polynomial source. The algebraic arithmetic map is simultaneously $G(\mathcal{J}^{\log}) : G(\mathcal{P}_\infty) \rightarrow G(E)$. For a nonempty packet it cannot factor through (NN.22): evaluating a proposed factorization on the supported zero polynomial forces the image of e_0 to be e_E , whereas (NN.18) supplies a polynomial whose supported image has nonzero amplitude. This is an exact factorization obstruction between the explicitly constructed maps, with the original source, supported zeros and external absence retained.

For $h = 1$, the algebraic target E is zero, so its arithmetic observation is the zero map and its split lift does factor through (NN.22). The amplitude, density theorem, injective finite normal maps, inverse/graph divergence and the complete graph limit remain unchanged. The arithmetic norm formulas involving a nonzero G_* target are then replaced by their unique zero-target maps. For a full invariant source under a fixed finite permutation group, all constructions restrict through the exact Reynolds maps of the density section; the derivative and source are invariant, $P_\rho = S - \sigma$ is invariant, and the arithmetic target is the corresponding invariant subspace with the restriction of its original target form. Thus the same proofs retain that invariant source and its original metric rather than adding noninvariant vectors. To verify the asserted minimum metric on this subspace, permutations act unitarily on the original tensor amplitude space because they preserve its product measure and its identical one-factor amplitudes. The full jet map is equivariant. Averaging an arbitrary lift of an invariant target with the literal factor $1/|\mathfrak{G}|$ gives an invariant lift of the same target, so the invariant jet map is onto. Uniqueness of the full minimum shows that every permutation fixes that minimum lift. It is consequently the minimum in the invariant source as well, with exactly the restriction of G_m or G_* .

30 The cyclic sum at every conormal depth

This section continues the complete cyclic-sum source, especially its Sections 1–3 and 7–8, and the separately supplied conormal-tower proof. It retains the original packet, arithmetic unit, tensor variables and reflection. The new constructions are the retraction and invariant dimensions at every depth, the exact scalar first-jet image, its full conormal comparison, and the finite cyclic Weyl correction with its complementary map.

30.1 Original coordinates and the exact collided annihilator

Let $k \geq 1$, and let the nonempty original monic packet be

$$h(s) = \prod_{\nu=1}^a (s - \rho_\nu)^{m_\nu}, \quad d = \sum_{\nu=1}^a m_\nu \geq 1. \quad (\text{CC.1})$$

The ρ_ν are distinct, and every m_ν is the full order retained by the packet. In the arithmetic application $g = 2\xi$, $v_h = g/h$, and the chosen packet consists of zeros of g with those full orders. Thus $v_h(\rho_\nu) \neq 0$. For the original arithmetic reflection and pairing, retain the delivered packet hypothesis: the map $\rho \mapsto 1 - \bar{\rho}$ permutes the packet and preserves each full multiplicity. This is the hypothesis used by the involution in (CC.42)–(CC.48). The polynomial-algebra statements below apply to the displayed original roots without any assumption that they lie on the critical line. Put

$$\begin{aligned} P &= \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k), \quad S = \sum_{i=1}^k s_i, \\ B_r &= P/I^r, \quad A_r = B_r^{\mathfrak{S}_k} \quad (r \geq 1), \\ U_r &= \left[\prod_{i=1}^k v_h(s_i) \right]_{I^r} \in A_r^\times. \end{aligned} \quad (\text{CC.2})$$

An entire function in (CC.2) means its full finite Taylor class on each local factor. These classes are compatible under every quotient map. Their invertibility follows from their nonzero constant terms: in a finite local algebra a class $u_0 + n$, with $u_0 \neq 0$ and $n^N = 0$, has inverse $u_0^{-1} \sum_{j=0}^{N-1} (-n/u_0)^j$. This fixes U_r without replacing any of its Taylor coefficients.

For an ordered tuple $\mathbf{t} = (\nu_1, \dots, \nu_k)$, write

$$\begin{aligned} \lambda_{\mathbf{t}} &= \sum_i \rho_{\nu_i}, & D_{\mathbf{t}} &= \sum_i (m_{\nu_i} - 1), & M_{\mathbf{t}} &= \max_i m_{\nu_i}, \\ L_{\mathbf{t},r} &= 1 + D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}, & \ell_{\lambda,r} &= \max_{\lambda_{\mathbf{t}}=\lambda} L_{\mathbf{t},r}, \\ \Lambda_k &= \{\lambda_{\mathbf{t}}\}, & \chi_r(T) &= \prod_{\lambda \in \Lambda_k} (T - \lambda)^{\ell_{\lambda,r}}, & q_r &= \deg \chi_r. \end{aligned} \quad (\text{CC.3})$$

The letter T is the polynomial indeterminate in χ_r ; its evaluation in the original algebra is $T = S$. Equal sums are grouped before taking the maximum.

Theorem 30.1 (Full local depth and the exact scalar contraction). *There are algebra isomorphisms in the original shifted coordinates*

$$B_r \longrightarrow \prod_{\mathbf{t} \in \{1, \dots, a\}^k} \mathbb{C}[z_1, \dots, z_k] / (z_1^{m_{\nu_1}}, \dots, z_k^{m_{\nu_k}})^r, \quad s_i \longmapsto \rho_{\nu_i} + z_i. \quad (\text{CC.4})$$

In the $\mathbf{t}$ -factor an exact basis consists of the monomials

$$z^\alpha, \quad \sum_i \left\lfloor \frac{\alpha_i}{m_{\nu_i}} \right\rfloor \leq r-1. \quad (\text{CC.5})$$

The nilpotency order of $S - \lambda_{\mathbf{t}} = \sum_i z_i$ on that factor is exactly $L_{\mathbf{t},r}$. Consequently

$$I^r \cap \mathbb{C}[S] = (\chi_r(S)), \quad C_r := \mathbb{C}[T]/(\chi_r) \xrightarrow{\alpha_r} A_r, \quad [F] \mapsto [F(S)] \quad (\text{CC.6})$$

is an injective unital algebra map. The original arithmetic coefficient map on the cyclic source is $\eta_r = M_{U_r, \alpha_r} : C_r \rightarrow A_r$. It is injective and intertwines multiplication by T and S .

Proof. For $r = 1$, separate-variable Chinese remainder division gives $I = \bigcap_{\mathbf{t}} J_{\mathbf{t}}$, where $J_{\mathbf{t}} = ((s_i - \rho_{\nu_i})^{m_{\nu_i}})_i$. Two different tuples give comaximal ideals: in a coordinate where their roots differ the two one-variable powers are coprime, and their Bezout identity already gives 1 in the sum of the two ideals. For finitely many pairwise comaximal ideals, intersection equals product. Therefore $I^r = (\prod J_{\mathbf{t}})^r = \prod J_{\mathbf{t}}^r = \bigcap J_{\mathbf{t}}^r$. The same Bezout identity, or its power expansion, proves that the powers remain comaximal. This proves (CC.4).

The local ideal is monomial. A monomial belongs to its r -th power precisely when some non-negative integers b_i , with $\sum b_i = r$, satisfy $m_{\nu_i} b_i \leq \alpha_i$ for every i . Such b_i exist precisely when the sum of the displayed floors is at least r . This proves (CC.5), including independence. Write $\alpha_i = m_{\nu_i} b_i + c_i$, $0 \leq c_i < m_{\nu_i}$. The greatest surviving total degree is $D_{\mathbf{t}} + (r-1)M_{\mathbf{t}}$: put every allowed block b_i in one coordinate attaining $M_{\mathbf{t}}$, and put $c_i = m_{\nu_i} - 1$ in every coordinate. All monomials of larger degree vanish. In $(\sum z_i)^{D_{\mathbf{t}}+(r-1)M_{\mathbf{t}}}$ that chosen monomial has the nonzero coefficient $(D_{\mathbf{t}} + (r-1)M_{\mathbf{t}})! / \prod \alpha_i!$. Distinct monomials cannot cancel. The asserted exact order follows. The annihilator in $\mathbb{C}[T]$ of evaluation on this factor is therefore $((T - \lambda_{\mathbf{t}})^{L_{\mathbf{t},r}})$. Intersecting these ideals takes the maximum at each repeated λ , proving (CC.6). The class of $F(S)$ is invariant, and multiplication by the unit U_r preserves injectivity and the stated intertwining. $\square$

Proposition 30.2 (Degree, scalar powers and all endpoints). *For the nonempty packet,*

$$\begin{aligned} q_r &\geq k(d-1) + (r-1)d + 1, \\ \chi_r \mid \chi_1^r, \quad \chi_r = \chi_1^r &\iff (r=1) \text{ or } (k=1) \text{ or } (m_{\nu} = 1 \ \forall \nu). \end{aligned} \quad (\text{CC.7})$$

These are assertions about the complete collided polynomials.

Proof. Repeated monic division in the unchanged variables gives the unique finite expansion

$$F = \sum_{\mathbf{b} \in \mathbb{N}^k, 0 \leq c_i < d} a_{\mathbf{b}, \mathbf{c}} \prod_i h_i^{b_i} s_i^{c_i}. \quad (\text{CC.8})$$

The leading monomials have the pairwise distinct exponent vectors $d\mathbf{b} + \mathbf{c}$, so a finite dependence is impossible. Equivalently P is free over $\mathbb{C}[h_1, \dots, h_k]$ with the displayed remainder basis. Membership in I^r is exactly the vanishing of the coefficients with $\sum b_i < r$. It follows that the highest homogeneous part of every element of I^r belongs to $(s_1^d, \dots, s_k^d)^r$. Indeed each highest-degree basis term has that property, and its lower terms have smaller total degree.

Put $D = k(d-1) + (r-1)d$. The monomial with exponents $(rd-1, d-1, \dots, d-1)$ survives the corresponding monomial ideal and has degree D . For each $0 \leq j \leq D$, choose a divisor of this monomial of degree j . Its floor sum is still at most $r-1$, and its coefficient in S^j is a nonzero multinomial coefficient. Thus the leading homogeneous part S^j of a monic polynomial of degree j in S cannot belong to that monomial ideal. Such a polynomial cannot lie in I^r . Apply this to $\chi_r(S)$ to obtain the lower bound.

For every tuple, $M_{\mathbf{t}} \leq 1 + D_{\mathbf{t}} = L_{\mathbf{t},1}$, so $L_{\mathbf{t},r} \leq rL_{\mathbf{t},1}$. Taking maxima proves divisibility. Each of the three stated equality cases follows directly from (CC.3). Conversely suppose $r, k > 1$, and put $M = \max_{\nu} m_{\nu} > 1$. Choose a root ρ of order M and its pure k -tuple. At $\lambda = k\rho$, every tuple has $D_{\mathbf{t}} \leq k(M-1)$ and $M_{\mathbf{t}} \leq M$; the pure tuple attains both bounds. Therefore

$$r\ell_{\lambda,1} - \ell_{\lambda,r} = (r-1)(k-1)(M-1) > 0. \quad (\text{CC.9})$$

This proves strict divisibility even when other tuples collide there. $\square$

For one root of order m , (CC.3) reads $\chi_r(T) = (T - k\rho)^{1+k(m-1)+(r-1)m}$. For $k = 1$, $\chi_r = h^r$ and α_r is an isomorphism. For $d = 1$, $\chi_r = (T - k\rho)^r$, but the invariant algebra can have a complement to the cyclic image: for $k \geq 2$, $r = 3$, the two symmetric degree-two polynomials $\sum z_i^2$ and $\sum_{i < j} z_i z_j$ are independent. The empty packet $h = 1$ is treated separately: $I = P$, $B_r = A_r = C_r = 0$, $\chi_r = 1$, and every displayed finite linear map is the zero map. No formula involving a chosen root or q_r^{-1} is applied in that case. The arithmetic source itself need not be zero merely because its finite packet projection is zero.

The attaining tuple can change with depth. As a declared algebraic example, take roots $1/2 + it$, $t = 0, 1, 2, 3$, of orders $6, 4, 4, 1$, and $k = 2$. At $\lambda = 1 + 3i$ the only unordered pairs are $(0, 3)$ and $(1, 2)$; they give respectively $6r$ and $4r + 3$. Thus $\ell_{\lambda,r} = \max(6r, 4r + 3)$. This example asserts no location or multiplicity of an arithmetic zero.

30.2 Unscaled invariant bases and a full-unit retraction

For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_\nu = k$, put $\lambda_{\mathbf{n}} = \sum n_\nu \rho_\nu$. In the factor for a fixed ordered tuple of that occupation, its stabilizer is $\prod_\nu \mathfrak{S}_{n_\nu}$. For each orbit of the surviving monomials (CC.5), take the sum of the distinct monomials in that orbit, with coefficient one. These are the original unscaled invariant basis elements. Their exact number is

$$a_{\mathbf{n},r} = \sum_{p=0}^{r-1} [t^p x_1^{n_1} \cdots x_a^{n_a}] \prod_{\nu=1}^a \prod_{j=0}^{r-1} (1 - x_\nu t^j)^{-m_\nu}, \quad a_{\lambda,r} = \sum_{\lambda_{\mathbf{n}}=\lambda} a_{\mathbf{n},r}. \quad (\text{CC.10})$$

Coefficient extraction here is finite formal algebra. Each variable in a group of order m_ν has a pair of labels (j, c) , $j \geq 0$, $0 \leq c < m_\nu$, representing the original exponent $m_\nu j + c$. An orbit is a multiset of these labels, and the surviving condition is that their j 's sum to at most $r - 1$. The geometric series for each label proves (CC.10). An invariant in the full product (CC.4) is determined by its entry at one ordered tuple per occupation; the other entries are its variable-permuted translates. The stabilizer condition is precisely the one just imposed. This proves both the claimed basis and

$$\begin{aligned} \dim B_r &= d^k \binom{r+k-1}{k}, \\ \dim A_r &= \sum_{\lambda} a_{\lambda,r} = \sum_{p=0}^{r-1} [t^p x^k] \prod_{j=0}^{r-1} (1 - xt^j)^{-d}, \\ a_{\mathbf{n},1} &= \prod_{\nu} \binom{n_\nu + m_\nu - 1}{m_\nu - 1}. \end{aligned} \quad (\text{CC.11})$$

For the first equality use (CC.8): there are $\binom{r+k-1}{k}$ vectors $\mathbf{b}$ with $\sum b_i < r$, and d^k possible remainders. For the second alternative expression, permutation acts on that same basis by permuting its k pairs (b_i, c_i) ; its unscaled orbit sums give the stated series.

Theorem 30.3 (Constructive retraction at every depth). *There is an explicit $\mathbb{C}[T]$ -linear retraction $\Pi_r : A_r \rightarrow C_r$ of η_r , with the complete arithmetic unit retained. Put $K_r = \ker \Pi_r$ and $Q_r = 1 - \eta_r \Pi_r$. Then*

$$\begin{aligned} C_r \oplus K_r &\rightarrow A_r, & (c, v) &\mapsto \eta_r(c) + v, \\ A_r &\rightarrow C_r \oplus K_r, & w &\mapsto (\Pi_r w, Q_r w) \end{aligned} \quad (\text{CC.12})$$

are inverse $\mathbb{C}[T]$ -module isomorphisms. In particular $A_r/\eta_r C_r \rightarrow K_r$, $[w] \mapsto Q_r w$ is an isomorphism, inverse to inclusion followed by quotient.

Proof. For each λ , use the exact CRT idempotent $e_\lambda(T)$ in C_r , and let $A_{\lambda,r} = e_\lambda(S)A_r$. Write $N = S - \lambda$ on this summand. Choose an ordered tuple attaining $\ell = \ell_{\lambda,r}$. Choose a coordinate of maximal multiplicity in it, put all $r - 1$ allowed blocks in that coordinate, and put every remainder at its

maximum $m_{\nu_i} - 1$. Call the resulting exponent α^* . Thus $|\alpha^*| = \ell - 1$. Let $\theta : A_{\lambda,r} \rightarrow \mathbb{C}$ extract the coefficient of z^{α^*} in that specific local factor, in the basis (CC.5). Define, with $X = T - \lambda$,

$$\begin{aligned}\Pi_\lambda^0(v) &= \sum_{j=0}^{\ell-1} \theta(N^{\ell-1-j}v) X^j \pmod{X^\ell}, \\ w_\lambda &= e_\lambda(S)U_r, \quad p_\lambda(X) = \Pi_\lambda^0(w_\lambda), \\ \Pi_\lambda(v) &= p_\lambda(X)^{-1} \Pi_\lambda^0(v).\end{aligned}\tag{CC.13}$$

The equality $N^\ell = 0$ gives $\Pi_\lambda^0(Nv) = X\Pi_\lambda^0(v)$, coefficient by coefficient: the constant term on the left is zero and the other coefficients shift by one. Also

$$p_\lambda(0) = \left(\prod_i v_h(\rho_{\nu_i}) \right) \frac{(\ell-1)!}{\prod_i \alpha_i^*!} \neq 0.\tag{CC.14}$$

To verify (CC.14), the selected term in $N^{\ell-1}$ has that multinomial coefficient; multiplying by a positive-degree term of the unit would exceed the greatest surviving total degree. The idempotent $e_\lambda(S)$ is exactly one on this local factor. Set $t = p_\lambda/p_\lambda(0) - 1$. Its exact inverse is $p_\lambda(0)^{-1} \sum_{j=0}^{\ell-1} (-t)^j$. This retains every coefficient of p_λ . The intertwining now gives $\Pi_\lambda(e_\lambda(S)U_r F(S)) = F(\lambda + X) \pmod{X^\ell}$. Take the product of these maps and use scalar CRT to define Π_r . It is the asserted retraction. Its kernel is S -stable. The identities $Q_r \eta_r = 0$, $\Pi_r Q_r = 0$, and $Q_r|_{K_r} = 1$ prove every map and inverse in (CC.12). $\square$

For any polynomial or entire f , multiplication by $f(S)$ has on $A_{\lambda,r}$ the form $f(\lambda)1$ plus a polynomial without constant term in the nilpotent N . Powers of a nilpotent map have trace zero, as is seen in a basis adapted to its kernel filtration. The same statement holds on the two invariant summands in (CC.12). Thus

$$\begin{aligned}\mathrm{Tr}_{A_r} f(S) &= \sum_\lambda a_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{C_r} f(T) &= \sum_\lambda \ell_{\lambda,r} f(\lambda), \\ \mathrm{Tr}_{K_r} f(S) &= \sum_\lambda (a_{\lambda,r} - \ell_{\lambda,r}) f(\lambda).\end{aligned}\tag{CC.15}$$

This also proves $a_{\lambda,r} \geq \ell_{\lambda,r}$. All nilpotent coordinates remain in (CC.4), (CC.10) and (CC.13); the trace formula follows from those actual operators.

30.3 The derivative tower, scalar first jets, and the unit obstruction

Denote the quotient maps by $\pi_r : B_{r+1} \rightarrow B_r$ and $\rho_r : C_{r+1} \rightarrow C_r$. The original derivative is

$$\delta_r : B_{r+1} \longrightarrow B_r, \quad [F] \longmapsto \left[\frac{1}{k} \sum_i \partial_{s_i} F \right], \quad \partial_r : C_{r+1} \longrightarrow C_r, \quad [F] \longmapsto [F'].\tag{CC.16}$$

Each is a complex linear map. Differentiating a product of $r+1$ elements of I leaves at least r such factors, so $\partial_S I^{r+1} \subset I^r$. Formula (CC.3) gives $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Thus the derivative of a multiple of χ_{r+1} is a multiple of χ_r , proving the second descent as well. On representatives,

$$\begin{aligned}\delta_r \alpha_{r+1} &= \alpha_r \partial_r, & \pi_r \alpha_{r+1} &= \alpha_r \rho_r, \\ \delta_r(xy) &= \pi_r(x) \delta_r(y) + \pi_r(y) \delta_r(x), & \pi_r \delta_{r+1} &= \delta_r \pi_{r+1}.\end{aligned}\tag{CC.17}$$

The last equality is a map $B_{r+2} \rightarrow B_r$. The symmetric subspaces are preserved because the derivative averages all k coordinates with the literal factor $1/k$. Every δ_r is onto: the invertible coordinates

$S, z_1, \dots, z_{k-1}$, where $z_i = s_i - S/k$, have $\partial_S S = 1$ and $\partial_S z_i = 0$. Integrating powers of S gives a polynomial primitive of any polynomial representative. Averaging such a primitive proves surjectivity on the invariant subspaces too.

The same monic expansion (CC.8) identifies

$$I^j/I^{j+1} \cong (P/I)[\zeta_1, \dots, \zeta_k]_j, \quad \zeta_i \mapsto [h_i], \quad \text{gr } \delta = \frac{1}{k} \sum_i [h'_i] \partial_{\zeta_i}. \quad (\text{CC.18})$$

Indeed differentiating $h^{\mathbf{b}} r_{\mathbf{b}}$ in degree $j = |\mathbf{b}|$ gives $b_i h^{\mathbf{b}-e_i} h'_i r_{\mathbf{b}}/k$ in degree $j - 1$; differentiating $r_{\mathbf{b}}$ stays in degree j . Both the integer b_i and the original polynomial h'_i survive.

Introduce a formal square-zero parameter ϵ , with its ordinary coefficient algebra meaning. The two maps

$$\begin{aligned} \Phi_r : B_{r+1} &\longrightarrow B_r[\epsilon]/(\epsilon^2), & b &\longmapsto \pi_r b + \epsilon \delta_r b, \\ \phi_r : C_{r+1} &\longrightarrow C_r[\epsilon]/(\epsilon^2), & F &\longmapsto \rho_r F + \epsilon \partial_r F \end{aligned} \quad (\text{CC.19})$$

are unital algebra homomorphisms by (CC.17) and its one-variable product rule. Extending α_r coefficientwise to α_r^ϵ gives the exact square $\Phi_r \alpha_{r+1} = \alpha_r^\epsilon \phi_r$. The coefficient projections of these algebra maps are precisely the quotient and derivative maps in (CC.16).

Put $R(T) = \prod_{\lambda \in \Lambda_k} (T - \lambda)$ and $\psi_r = \chi_r R$. Then

$$\begin{aligned} \ker \phi_r &= (\psi_r)/(\chi_{r+1}), & \dim \text{im } \phi_r &= q_r + |\Lambda_k|, \\ \ker \rho_r &= (\chi_r)/(\chi_{r+1}) \cong \mathbb{C}[T]/(\chi_{r+1}/\chi_r), & \partial_r(\ker \rho_r) &= \chi'_r C_r. \end{aligned} \quad (\text{CC.20})$$

For the first assertion, F and F' vanish modulo $(T - \lambda)^{\ell_{\lambda,r}}$ exactly when F has order at least $\ell_{\lambda,r} + 1$ there. The order implication in (CC.16) gives $\psi_r \mid \chi_{r+1}$, so the asserted kernel is well defined and has dimension $q_{r+1} - q_r - |\Lambda_k|$. This proves the image dimension and the injective factorization $\mathbb{C}[T]/(\psi_r) \hookrightarrow C_r[\epsilon]/(\epsilon^2)$. Multiplication by χ_r gives the displayed isomorphism onto $\ker \rho_r$; its kernel is exactly (χ_{r+1}/χ_r) . Differentiating $\chi_r F$ modulo χ_r gives $\chi'_r F$. At each λ , multiplication by χ'_r maps the input constant to a nonzero multiple of $(T - \lambda)^{\ell_{\lambda,r}-1}$. Its image has one dimension per λ , proving all image and kernel claims even at repeated collided roots.

Write

$$\beta_r = U_r^{-1} \delta_r U_{r+1} \in A_r, \quad \Phi_r(U_{r+1}) = U_r(1 + \epsilon \beta_r), \quad \Phi_r(U_{r+1}^{-1}) = U_r^{-1}(1 - \epsilon \beta_r). \quad (\text{CC.21})$$

The first formula is the definition of the full logarithmic derivative class, the second follows directly from (CC.19), and the third follows by multiplication and $\epsilon^2 = 0$. In particular,

$$\delta_r \eta_{r+1} = \eta_r \partial_r + M_{\delta_r U_{r+1}} \alpha_r \rho_r. \quad (\text{CC.22})$$

This formula retains the full derivative of the arithmetic unit.

There is a complete decomposition of its last term using (CC.12):

$$\begin{aligned} b_r &= \Pi_r(\delta_r U_{r+1}) \in C_r, & n_r &= Q_r(\delta_r U_{r+1}) \in K_r, \\ \mathcal{N}_r : C_r &\longrightarrow K_r, & \mathcal{N}_r(F) &= \alpha_r(F) n_r, \\ \delta_r \eta_{r+1} &= \eta_r(\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r. \end{aligned} \quad (\text{CC.23})$$

Here K_r is a $\mathbb{C}[T]$ -submodule, so multiplication by $\alpha_r(F)$ preserves it. Both Π_r and Q_r intertwine this multiplication; applying them to the last term of (CC.22) proves (CC.23). To state the obstruction independently of the chosen retraction, let $\mathbf{q}_r : A_r \rightarrow A_r/\eta_r C_r$ be the module quotient and put

$$\Omega_r = \mathbf{q}_r M_{\delta_r U_{r+1}} \alpha_r : C_r \longrightarrow A_r/\eta_r C_r. \quad \mathbf{q}_r \delta_r \eta_{r+1} = \Omega_r \rho_r, \quad \Omega_r = 0 \iff \beta_r \in \alpha_r(C_r). \quad (\text{CC.24})$$

To prove the equivalence, evaluate Ω_r at 1. Its vanishing says $\delta_r U_{r+1} = U_r \alpha_r(\gamma_r)$ for a unique $\gamma_r \in C_r$, precisely the condition on β_r . Conversely that equality makes every value of Ω_r zero by closure of the subalgebra $\alpha_r(C_r)$. Under (CC.12), Ω_r corresponds exactly to $\mathcal{N}_r$. When it vanishes, $b_r = \gamma_r$, and the exact pulled-back derivative is $\partial_r + M_{\gamma_r} \rho_r$. The formulas construct both the obstruction and its strongest scalar derivative correspondence without deleting the complementary directions.

30.4 The finite cyclic derivative and its exact Weyl correction

The quotient derivative has two different depths in its domain and codomain. To compute its finite endomorphism on the prescribed scalar coordinates, let $s_r : C_r \rightarrow C_{r+1}$ send a class to the same polynomial of degree less than q_r given by monic division by χ_r . This is a complex-linear section of ρ_r . Put

$$\begin{aligned} L_r(F) &= [T^{q_r-1}]F_0(T), & d_r &= \partial_r s_r : C_r \rightarrow C_r, \\ D_r^{\text{red}} &= d_r + M_{b_r}, & A_r^C &= M_T : C_r \rightarrow C_r, \end{aligned} \quad (\text{CC.25})$$

where F_0 is that unchanged monic remainder. The scalar $L_r(F)$ retains its original coefficient; it is not a functional defined by dividing by a norm. The complete differentiated arithmetic map on this section is

$$\delta_r \eta_{r+1} s_r = \eta_r D_r^{\text{red}} + \mathcal{N}_r : C_r \longrightarrow A_r. \quad (\text{CC.26})$$

Thus the reduced derivative and its complementary component have explicit maps into the original arithmetic algebra.

Theorem 30.4 (Finite sum-coordinate Weyl relation). *For the original monic remainder coordinates,*

$$\begin{aligned} [A_r^C - k/2, D_r^{\text{red}}] &= -I_{C_r} + J_r, & J_r(F) &= [\chi_r'] L_r(F), \\ \text{rank } J_r &= 1, & J_r^2 &= q_r J_r, & \text{Tr } J_r &= q_r. \end{aligned} \quad (\text{CC.27})$$

The full arithmetic correction is the image of the original retained relation:

$$\begin{aligned} \rho_r([\chi_r]_{\chi_{r+1}} L_r(F)) &= 0, \\ \delta_r(U_{r+1} \alpha_{r+1}([\chi_r] L_r(F))) &= \eta_r([\chi_r'] L_r(F)). \end{aligned} \quad (\text{CC.28})$$

No complementary derivative is suppressed in (CC.26).

Proof. Since TF_0 has degree at most q_r , its exact monic remainder is $TF_0 - \chi_r L_r(F)$. Differentiating it and reducing gives

$$d_r A_r^C F = [F_0 + TF_0' - \chi_r' L_r(F)], \quad A_r^C d_r F = [TF_0'].$$

Their difference proves (CC.27), because multiplication by b_r commutes with multiplication by T . The vector $[\chi_r']$ has degree $q_r - 1$ and leading coefficient $q_r \neq 0$. Thus $L_r([\chi_r']) = q_r$; the operator is rank one, its square is q_r times itself, and its trace is that same scalar, either in the remainder basis or by the rank-one matrix formula. For (CC.28), the quotient kills χ_r , and the product rule leaves $U_r \alpha_r(\chi_r') L_r(F)$; the term with $\delta_r U_{r+1}$ still contains $\alpha_r(\chi_r) = 0$. The original relation itself remains in B_{r+1} before this application. Finally $\mathcal{N}_r$ is $\mathbb{C}[T]$ -linear by (CC.23). Consequently the full relation obtained by applying S to (CC.26) and subtracting (CC.26) applied to TF is exactly $-\eta_r + \eta_r J_r$; this proves the intertwining including the complementary map. $\square$

At $r = 1, k = 1, C_1 = \mathbb{C}[s]/(h)$, L_1 is the coefficient of s^{d-1} , and (CC.27) gives $[M_{s-1/2}, d_1 + M_{b_1}] = -I + [h'] L_1$. In the original weighted coordinates $u = v_h F$, its conjugated correction is $u \mapsto v_h h' L_1(v_h^{-1} u)$. Multiplication by v_h and by v_h^{-1} are the explicit inverse coordinate maps; the expression is the one-variable full-unit correction in the preceding finite Weyl calculation. Formula (CC.28) gives its all-depth extension using the actual relation layer rather than differentiating a quotient zero.

30.5 All first jets and the symmetric conormal kernel

Set $E = B_1, E^{[2]} = B_2, A_h = \mathbb{C}[s]/(h)$, and write $c_i = [h'(s_i)] \in E$. Let a continue to denote the number of distinct roots and put $b = d - a = \deg \gcd(h, h')$. The conormal coordinates are the actual

isomorphism

$$\begin{aligned} \iota : E^k &\longrightarrow I/I^2, & (v_i) &\longmapsto \left[\sum_i h_i \tilde{v}_i \right], \\ \partial_N : E^k &\longrightarrow E, & (v_i) &\longmapsto \frac{1}{k} \sum_i c_i v_i. \end{aligned} \quad (\text{CC.29})$$

Here $\tilde{v}_i$ is its original separate-variable monic remainder. Changing a lift changes the displayed sum by I^2 . Equation (CC.8) proves injectivity, surjectivity and the inverse coefficient extraction. Applying the product rule gives exactly $\delta_1 \iota = \partial_N$. The full conormal differential and the specified contraction are the explicit maps

$$E^k \longrightarrow \bigoplus_i E \, ds_i \longrightarrow E, \quad (v_i) \longmapsto \sum_i c_i v_i \, ds_i \longmapsto \frac{1}{k} \sum_i c_i v_i. \quad (\text{CC.30})$$

Thus the intermediate cotangent module and every coefficient of the sum direction are retained.

Theorem 30.5 (Exact ranks and complete first-jet observations). *Put $T_h = \mathbb{C}[s]/(h, h')$. Then*

$$\begin{aligned} \text{im } \partial_N &= (c_1, \dots, c_k), & \text{coker } \partial_N &\cong T_h^{\otimes k}, \\ \text{rank}_{\mathbb{C}} \partial_N &= d^k - b^k, & \dim \ker \partial_N &= (k-1)d^k + b^k. \end{aligned} \quad (\text{CC.31})$$

Equip $E \oplus E^k$ with multiplication $(x, v)(y, w) = (xy, xw + yv)$, unit $(1, 0)$. The maps

$$\begin{aligned} J_{\text{all}} : B_2 &\longrightarrow E \oplus E^k, & [F] &\longmapsto ([F], ([\partial_i F])_i), \\ J_S : B_2 &\longrightarrow E[\epsilon]/(\epsilon^2), & [F] &\longmapsto [F] + \epsilon \delta_1[F], \\ \mathcal{L} : E \oplus E^k &\longrightarrow E[\epsilon]/(\epsilon^2), & (x, v) &\longmapsto x + \frac{\epsilon}{k} \sum_i v_i \end{aligned} \quad (\text{CC.32})$$

are unital algebra homomorphisms and satisfy $J_S = \mathcal{L} J_{\text{all}}$. Their kernels and ranks are

$$\begin{aligned} \ker J_S &= \iota(\ker \partial_N), & \dim \text{im } J_S &= 2d^k - b^k, \\ \ker J_{\text{all}} &= \iota \left(\bigoplus_i \text{ann}_E(c_i) \right), & \dim \ker J_{\text{all}} &= kb d^{k-1}, \\ \dim \text{im } J_{\text{all}} &= (k+1)d^k - kb d^{k-1}. \end{aligned} \quad (\text{CC.33})$$

For squarefree h , J_{all} is an isomorphism onto the displayed square-zero algebra.

Proof. The row in (CC.29) has precisely the ideal of its entries as image. Quotienting by these entries in the separate tensor factors gives $T_h^{\otimes k}$, with the map induced by the original polynomial quotients. The Euclidean algorithm gives $(h, h') = (\gcd(h, h'))$, so $\dim T_h = b$. This proves the cokernel and rank in (CC.31), and rank-nullity on the $k d^k$ -dimensional source proves its kernel dimension.

Each coordinate derivative maps I^2 into I ; hence the maps in (CC.32) are well defined. The product rule for each derivative gives exactly the specified square-zero multiplication. The factor $1/k$ gives $J_S = \mathcal{L} J_{\text{all}}$. An element in either jet kernel has zero ordinary quotient, so it is a conormal class. Applying (CC.29) gives its sum derivative, while applying each coordinate derivative gives $(c_i v_i)_i$. This proves the asserted kernels. Multiplication by h' on A_h has cokernel T_h , hence has rank $d - b$ and kernel dimension b . Tensoring the kernel and image with the other $k - 1$ vector-space factors shows that $\dim \text{ann}_E(c_i) = b d^{k-1}$. The dimension of B_2 is $(k+1)d^k$ by (CC.11), which proves both image dimensions. If $b = 0$, the full map has zero kernel and equal source and target dimensions; its inverse is multiplicative because it is the inverse of a bijective algebra homomorphism. $\square$

The images also have explicit coordinates. Let $j : E \rightarrow B_2$ choose the original remainder with $\deg_{s_i} < d$. Set $d_0(x) = \delta_1 j(x)$ and $D_0(x) = ([\partial_i j(x)])_i$. Directly from (CC.29),

$$\begin{aligned} \text{im } J_S &= \{x + \epsilon(d_0(x) + v) : x \in E, v \in (c_1, \dots, c_k)\}, \\ \text{im } J_{\text{all}} &= \{(x, D_0(x) + (c_i v_i)_i) : x \in E, (v_i) \in E^k\}. \end{aligned} \quad (\text{CC.34})$$

Every lift differs from $j(x)$ by exactly one conormal vector, which proves both inclusions of each image equality. In particular $\ker J_S = J_{\text{all}}^{-1}(\ker \mathcal{L})$, where $\ker \mathcal{L} = \{(0, v) : \sum_i v_i = 0\}$. This identifies precisely the relative directions removed by the sum contraction after all first derivatives have been retained.

The same calculation restricts to the permutation invariants. On E^k the simultaneous action is $(\sigma v)_i = \sigma(v_{\sigma^{-1}i})$; all the maps above intertwine this action. Evaluation at the first component gives the explicit vector-space isomorphism

$$(E^k)^{\mathfrak{S}_k} \longrightarrow A_h \otimes (A_h^{\otimes(k-1)})^{\mathfrak{S}_{k-1}}. \quad (\text{CC.35})$$

For its inverse, start with a tensor fixed by the stabilizer of index one. For each i , choose a permutation carrying one to i and use its translate as component i . Two choices differ by the stabilizer, so the component is independent of that choice. The resulting tuple is invariant, proving the inverse in full. When $k = 1$, the tensor with zero factors is $\mathbb{C}$ and both maps are the identity on A_h .

For a vector space of dimension n , the invariant tensors in its k -fold tensor power have a basis of unscaled sums over words with prescribed occupation numbers in a fixed basis. There are $\binom{n+k-1}{k}$ such occupations for $n \geq 1$, and zero for $n = 0, k \geq 1$. This proves the dimensions used here directly. Taking invariants in the exact cokernel sequence of (CC.31) remains exact: average a lift by $k!^{-1} \sum_{\sigma} \sigma$ to lift any invariant image. Consequently

$$\begin{aligned} \text{coker } \partial_N^{\text{sym}} &\cong (T_h^{\otimes k})^{\mathfrak{S}_k}, \\ \dim(E^k)^{\mathfrak{S}_k} &= d \binom{d+k-2}{k-1}, \\ \text{rank } \partial_N^{\text{sym}} &= \binom{d+k-1}{k} - \binom{b+k-1}{k}, \\ \dim \ker \partial_N^{\text{sym}} &= d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}. \end{aligned} \quad (\text{CC.36})$$

The final binomial is defined to be zero when $b = 0$. For the declared polynomial fixture $h = s^2, k = 2$, the row is $(v_1, v_2) \mapsto s_1 v_1 + s_2 v_2$; its image is the span of $s_1, s_2, s_1 s_2$, of dimension three, and its kernel has dimension five. On invariants the domain has dimension four, target dimension three, image span $s_1 + s_2, s_1 s_2$, and kernel dimension two. These are exact quotient computations in the stated algebraic fixture, without an assertion about arithmetic roots.

In the original cochain convention a permutation of k factors of cochain degree one contributes $\text{sgn}(\sigma)$. Thus the top-degree action of $k!^{-1} \sum_{\sigma} \text{sgn}(\sigma) T_{\sigma}$ is $k!^{-1} \sum_{\sigma} \sigma$ on these tensor coefficients: the two signs multiply to one. This is the explicit identification with the prescribed symmetric cochain summand. Its top-degree supertrace multiplies each ordinary trace by the retained factor $(-1)^k$.

30.6 Cyclic relation response and the full product Jacobian

Fix the monic-division identity $\chi_1(S) = \sum_i h_i B_i(\mathbf{s})$ in the original polynomial ring; it exists by (CC.6) and (CC.8). Define

$$\begin{aligned} \mathcal{B} : C_1 &\longrightarrow E^k, & F &\longmapsto ([B_i]_I \alpha_1(F))_i, \\ \partial_N \mathcal{B}(F) &= \alpha_1([\chi'_1]F), & \delta_1[U \chi_1(S) F(S)]_{I^2} &= U_1 \alpha_1([\chi'_1]F). \end{aligned} \quad (\text{CC.37})$$

The first map is well defined and $\mathbb{C}[T]$ -linear because each coefficient multiplies the well-defined class $\alpha_1(F)$. Differentiate the displayed division identity with $k^{-1} \sum_i \partial_i$. Terms still containing h_i vanish modulo I ; the derivative of S is exactly one. This proves the first identity. Differentiating the full-unit relation proves the second, since terms with an undifferentiated χ_1 vanish only at the final reduction. These maps give the exact conormal-to-cyclic Jacobian comparison, including $\ker(\partial_N \mathcal{B}) = \ker M_{\chi_1'}$, since α_1 is injective. Its rank is $|\Lambda_k|$ by (CC.20), and its kernel has dimension $q_1 - |\Lambda_k|$.

The full product Jacobian has a different explicit factorization through the same original conormal row. Define

$$\begin{aligned} J_{\text{prod}} &= \prod_i c_i \in E, \\ \mathcal{P} : E &\longrightarrow E^k, & v &\longmapsto \left(v \prod_{j \neq i} c_j \right)_i, \\ \partial_N \mathcal{P} &= M_{J_{\text{prod}}} : E \rightarrow E. \end{aligned} \tag{CC.38}$$

Indeed each of the k summands in the row is $J_{\text{prod}} v / k$. This proves the factorization with the original factor $1/k$. It is permutation equivariant. Its exact kernel is $\mathcal{P}^{-1}(\ker \partial_N) = \text{ann}_E(J_{\text{prod}})$. Also $\ker \mathcal{P} = \bigcap_i \text{ann}_E(\prod_{j \neq i} c_j)$; these equalities follow component by component and retain the intermediate directions rather than declaring the two observations interchangeable.

In a local factor $\mathbb{C}[z]/(z^m)$, the literal factorization $h(\rho + z) = z^m u_\rho(z)$ gives $h'(\rho + z) = m u_\rho(0) z^{m-1}$ modulo z^m . Thus in an ordered product factor, multiplication by J_{prod} sends a constant to $\prod_i m_i u_{\rho_i}(0) \prod_i z_i^{m_i-1}$, with nonzero coefficient, and annihilates the maximal ideal. There is one image dimension for each of the a^k ordered root tuples. Therefore

$$\text{rank } M_{J_{\text{prod}}} = a^k, \quad \ker M_{J_{\text{prod}}} = \sqrt{0_E}, \quad \dim \sqrt{0_E} = d^k - a^k. \tag{CC.39}$$

To check the kernel assertion globally, a class is nilpotent precisely when its constant coefficient vanishes at each local factor: the local maximal ideals are nilpotent and the product is finite. This is exactly the computed annihilator. Under permutation invariants, one image dimension remains for each occupation of the distinct roots, giving rank $\binom{a+k-1}{k}$. The invariant kernel is $(\sqrt{0_E})^{\mathfrak{S}_k}$, either by restriction or by averaging a lift of each invariant image. The factorization (CC.38) continues to hold on these invariant spaces.

The arithmetic unit remains on every observed conormal vector:

$$\delta_1(U_2 \iota(v)) = U_1 \partial_N(v), \quad U_1 \partial_N \mathcal{P} = M_{U_1 J_{\text{prod}}}. \tag{CC.40}$$

The first formula follows from (CC.17), since $\pi_1 \iota(v) = 0$. Multiplication by U_1^{-1} is its explicit inverse change on the target, so the kernels and ranks remain those proved above while all actual Taylor coefficients stay in the map.

30.7 Trace pairings and all four complementary operator blocks

For a monic one-variable polynomial p , all finite residue loops below have positive orientation. The functional $\lambda_p(F) = \sum_{\zeta} \text{Res}_{T=\zeta} F(T) p(T)^{-1} dT$ is well defined on $\mathbb{C}[T]/(p)$: adding pQ adds an entire polynomial differential at every loop. At a root of order m , write $p = (T - \zeta)^m c(T)$, where $c(\zeta) \neq 0$. Then $p'/p = m/(T - \zeta) + c'/c$. Hence

$$\lambda_p(p'F) = \sum_{\zeta} m_{\zeta} F(\zeta) = \text{Tr}_{\mathbb{C}[T]/(p)} M_F. \tag{CC.41}$$

For the last equality, in each basis $1, T - \zeta, \dots, (T - \zeta)^{m-1}$, the multiplication matrix has diagonal $F(\zeta)$ repeated m times; positive nilpotent powers have zero diagonal. This proves the trace identity with all original multiplicities.

The packet reflection gives the conjugate-linear algebra involution

$$F^\dagger(\mathbf{s}) = \overline{F(1 - \overline{s}_1, \dots, 1 - \overline{s}_k)}$$

on each B_r , because it permutes the root factors of h with their full orders. On the scalar algebra its restriction is $F^{\dagger k}(T) = \overline{F(k - \overline{T})}$. The sum set and all the exponents in (CC.3) are permuted by $\lambda \mapsto k - \overline{\lambda}$; hence $\chi_r^{\dagger k} = (-1)^{q_r} \chi_r$. This proves well-definedness on each quotient and $\alpha_r(F)^\dagger = \alpha_r(F^{\dagger k})$. Consequently the cyclic pairing

$$R_{\chi_r}(F, G) = \lambda_{\chi_r}(F^{\dagger k} G) \quad \text{satisfies} \quad R_{\chi_r}(F, \chi'_r G) = \sum_{\lambda} \ell_{\lambda, r} \overline{F(k - \overline{\lambda})} G(\lambda). \quad (\text{CC.42})$$

This is (CC.41) applied to the same full quotient, with its specified involution.

For the original one-factor arithmetic pairing, put $v_h = [v_h]_h$, $\varepsilon_h = v_h^{-1}$, and $\mathcal{R}_Z(f, u) = \lambda_h(\varepsilon_h f^\dagger u)$. The identity $g' = h'v_h + hv'_h$ gives $j_h(g') = h'v_h$. Thus the complete unit calculation is

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) = \text{Tr}_{A_h} M_{f^\dagger P}. \end{aligned} \quad (\text{CC.43})$$

The product functional $\lambda_h^{\otimes k}$ consequently satisfies $\lambda_h^{\otimes k}(J_{\text{prod}} F) = \text{Tr}_E M_F$. To verify it without a tensor restriction on F , expand F in the original separate-variable monomial basis and apply (CC.41) factor by factor; both sides are linear in every coefficient. The product $U_1^{-1} U_1$ gives the same exact unit cancellation for its original weighted arithmetic version. This describes the full product trace and (CC.38) supplies its map through the sum-direction conormal observation.

Let $\chi_{A, r}(T) = \prod_{\lambda} (T - \lambda)^{a_{\lambda, r}}$. Equation (CC.15) gives the genuine quotient trace of every entire $f(S)$ on $T_r = A_r / \eta_r C_r$:

$$\text{Tr}_{T_r} f(S) = \sum_{\lambda} \text{Res}_{T=\lambda} f(T) \left(\frac{\chi'_{A, r}}{\chi_{A, r}} - \frac{\chi'_r}{\chi_r} \right) dT. \quad (\text{CC.44})$$

Its proof is the local logarithmic derivative calculation (CC.41) with exponents $a_{\lambda, r} - \ell_{\lambda, r}$. It retains the module T_r ; its formula is a trace of its induced operator, not a replacement of its nilpotent coordinates.

For an arbitrary linear endomorphism $M : A_r \rightarrow A_r$, use the fixed full-unit splitting (CC.12). Write $\iota_K : K_r \rightarrow A_r$ for inclusion and regard Q_r as a map with codomain K_r . Its full transformed matrix is

$$\begin{pmatrix} \Pi_r M \eta_r & \Pi_r M \iota_K \\ Q_r M \eta_r & Q_r M \iota_K \end{pmatrix} : C_r \oplus K_r \longrightarrow C_r \oplus K_r. \quad (\text{CC.45})$$

Applying the two inverse maps of (CC.12) on either side of M proves every block. In these original coordinates,

$$\text{Tr}_{A_r} M = \text{Tr}_{C_r}(\Pi_r M \eta_r) + \text{Tr}_{K_r}(Q_r M \iota_K). \quad (\text{CC.46})$$

The cross maps in (CC.45) remain part of the transformed operator. For completeness, invariance of trace under these inverse changes follows from $\text{Tr}(XY) = \text{Tr}(YX)$, proved by expanding $\sum_{i,j} X_{ij} Y_{ji}$. The quotient T_r has the induced map $[v] \mapsto [Mv]$ exactly when $M \eta_r C_r \subseteq \eta_r C_r$, equivalently $Q_r M \eta_r = 0$. Necessity follows by applying a well-defined quotient map to $[\eta_r c] = 0$; sufficiency follows by comparing any two representatives. In that case (CC.12) identifies its map with $Q_r M \iota_K$. This proves the exact quotient criterion and the always-defined complementary compression when it fails.

For a multiplier $M = M_w$ with $w \in A_r$, this criterion is

$$M_w \eta_r C_r \subseteq \eta_r C_r \iff w \in \alpha_r(C_r). \quad (\text{CC.47})$$

Indeed its value on $\eta_r(1) = U_r$ gives necessity after multiplication by U_r^{-1} . Conversely subalgebra closure proves sufficiency. If $w = f(S)$, both cross maps in (CC.45) are zero by $\mathbb{C}[T]$ -linearity, proving (CC.44) directly. The full weighted vector pairing is instead the explicitly typed trace

$$\mathrm{Tr}_{A_r} M_{(\eta_r F)^\dagger \eta_r G} = \mathrm{Tr}_{A_r} M_{U_r^\dagger U_r \alpha_r (F^\dagger_k G)}. \quad (\text{CC.48})$$

Multiplicativity of the involution proves this identity before any trace calculation. Insert that unchanged multiplier into all four blocks of (CC.45) to compute its full decomposition. In particular $U_r^\dagger U_r$ has an induced quotient multiplier precisely when it satisfies (CC.47); no membership is assumed from invariance under permutations. If an original positive Hermitian form is represented by H on A_r , the same coordinates give its full pullback

$$\begin{pmatrix} \eta_r^* H \eta_r & \eta_r^* H \iota_K \\ \iota_K^* H \eta_r & \iota_K^* H \iota_K \end{pmatrix}. \quad (\text{CC.49})$$

This follows by evaluating the form on $\eta_r c + \iota_K v$ and $\eta_r c' + \iota_K v'$; all four terms appear. The algebraic retraction entails no orthogonality assertion for that original metric.

30.8 Exact two-factor symmetric presentation at every depth

For $k = 2$, retain $S = s_1 + s_2$, $R = s_1 - s_2$ and $\Delta = R^2$, with inverse $s_1 = (S + R)/2$, $s_2 = (S - R)/2$. Write the original polynomial as $h(s) = \sum_{j=0}^d c_j s^j$, $c_d = 1$, and define

$$\begin{aligned} H_0(S, \Delta) &= \sum_{j=0}^d c_j 2^{-j} \sum_{2a \leq j} \binom{j}{2a} S^{j-2a} \Delta^a, \\ H_1(S, \Delta) &= \sum_{j=1}^d c_j 2^{-j} \sum_{2a+1 \leq j} \binom{j}{2a+1} S^{j-2a-1} \Delta^a. \end{aligned} \quad (\text{CC.50})$$

Binomial expansion gives exactly $h(s_1) = H_0 + R H_1$, $h(s_2) = H_0 - R H_1$, including $R = 0$. Therefore the linear change of generators with coefficients $1/2$ gives $I = (H_0, R H_1)$ in $\mathbb{C}[S, R]$. For every $r \geq 1$ the complete invariant relation ideal is

$$\begin{aligned} J_r &= \left(H_0^{r-j} \Delta^{\lceil j/2 \rceil} H_1^j : 0 \leq j \leq r \right) \subset \mathbb{C}[S, \Delta], \\ A_r &\cong \mathbb{C}[S, \Delta]/J_r, \quad C_r \hookrightarrow \mathbb{C}[S, \Delta]/J_r, \quad [F(T)] \mapsto [F(S)]. \end{aligned} \quad (\text{CC.51})$$

To prove the ideal identity, express any element of I^r as a sum of the generators $H_0^{r-j} R^j H_1^j$ times arbitrary polynomials in S, R . Average under $R \mapsto -R$. For even j , the coefficient's even part is a polynomial in S, Δ , giving $\Delta^{j/2}$. For odd j , its odd part is R times such a polynomial, giving $\Delta^{(j+1)/2}$. This proves containment in J_r . Each displayed generator is conversely an invariant member of I^r , proving equality. Every invariant quotient class has an invariant lift by the same average with coefficient $1/2$. Thus the induced algebra map is onto and has exactly this kernel. The injection in (CC.51) is precisely (CC.6) in those coordinates; its kernel before taking C_r is (χ_r) .

At depth one this recovers the literal presentation $\mathbb{C}[S, \Delta]/(H_0, \Delta H_1)$, with the relative factor Δ intact. Under the earlier quartet assumptions, its H_0, H_1 are exactly those of (SSP.9): both equal the even and odd coefficients of the same unchanged polynomial $h((S + R)/2)$. The further substitution $Z = S - 1$, $x = -Z^2$ is its original (SSP.2) map, so the cyclic injection sends $F(T)$ to $F(1 + Z)$ in the complete SSP algebra. Its weighted module injection sends it to $[v_h((S + R)/2) v_h((S - R)/2)] F(S)$; that product is invariant, and its finite Taylor class is exactly U_r . The inverse SSP coordinate formulas, and (CC.12) for the cyclic summand, retain the full relative and parity modules at every step.

The two parity modules themselves have exact formulas at every depth. Under those same quartet assumptions, put $\mathcal{R} = \mathbb{C}[x, \Delta]$ and use the unchanged polynomials A_0, A_1 of (SSP.3), so that $H_0 =$

$A_0(-Z^2, \Delta)$, $H_1 = ZA_1(-Z^2, \Delta)$. For all indices within $0 \leq j \leq r$, define

$$\begin{aligned} E_{2a} &= A_0^{r-2a} \Delta^a (-x)^a A_1^{2a}, \\ O_{2a+1} &= A_0^{r-2a-1} \Delta^{a+1} (-x)^a A_1^{2a+1}, \\ \mathfrak{a}_{e,r} &= (E_{2a}, xO_{2a+1}) \subset \mathcal{R}, & \mathfrak{a}_{o,r} &= (E_{2a}, O_{2a+1}) \subset \mathcal{R}, \\ B_{e,r} &= \mathcal{R}/\mathfrak{a}_{e,r}, & B_{o,r} &= \mathcal{R}/\mathfrak{a}_{o,r}. \end{aligned} \tag{CC.51a}$$

Then $A_r = B_{e,r} \oplus ZB_{o,r}$ as a module in these specified coordinates. Its exact maps and product are

$$\begin{aligned} \beta_r^{\text{par}} : B_{e,r} &\rightarrow B_{o,r}, & [a] &\mapsto [a], \\ \gamma_r^{\text{par}} : B_{o,r} &\rightarrow B_{e,r}, & [b] &\mapsto [xb], \\ (a, b)(c, e) &= (ac - \gamma_r^{\text{par}}(be), \beta_r^{\text{par}}(a)e + \beta_r^{\text{par}}(c)b), \\ \gamma_r^{\text{par}} \beta_r^{\text{par}} &= xI_{B_{e,r}}, & \beta_r^{\text{par}} \gamma_r^{\text{par}} &= xI_{B_{o,r}}. \end{aligned} \tag{CC.51b}$$

The superscript distinguishes these parity maps from the full logarithmic unit class β_r defined in (CC.21). For a direct proof, substitute $Z^2 = -x$ into each generator of (CC.51). Even $j = 2a$ gives E_{2a} , and odd $j = 2a + 1$ gives ZO_{2a+1} . Before imposing them the module is freely $\mathcal{R} \oplus Z\mathcal{R}$, by monic division in Z . Multiplication by $c + Zd$ shows that their even coefficient ideal is exactly (E_{2a}, xO_{2a+1}) , and their odd coefficient ideal is exactly (E_{2a}, O_{2a+1}) . This proves both directions of the claimed decomposition. The containments between those ideals and their multiples by x prove well-definedness of $\beta_r^{\text{par}}, \gamma_r^{\text{par}}$; multiplication in $\mathcal{R}[Z]/(Z^2 + x)$ proves every sign, product and composition in (CC.51b). At $r = 1$ these are exactly (SSP.6)–(SSP.7), because $E_0 = A_0$ and $O_1 = \Delta A_1 = Q_\Delta$. The original unit is still the full class of the product in (CC.2), carried through these inverse coefficient maps.

30.9 Split lifts, source provenance and quantitative scope

For a complex vector space V , retain $G(V) = \{\tau\} \sqcup \{v^\bullet : v \in V\}$, with $e_V = 0_V^\bullet$, τ the additive identity, and $v^\bullet + w^\bullet = (v + w)^\bullet$. The scalar action is $c^\bullet v^\bullet = (cv)^\bullet$, and an absent scalar or vector produces τ . For a complex-linear map $L : V \rightarrow W$, its exact lift sends $\tau \mapsto \tau$, $v^\bullet \mapsto (Lv)^\bullet$. It preserves addition and scalar action: on two supported inputs this is the linearity of L , and cases with τ follow from the given identity and absorbing rules. For an algebra homomorphism, the same formula also preserves supported products and units. These formulas apply to every map above with its proved type; a derivative uses the linear lift, while the maps (CC.19) and (CC.32) use the multiplicative lift.

In particular, for a conormal vector $n = \iota(v)$,

$$G(\pi_1)(n^\bullet) = e_E, \quad G(J_S)(n^\bullet) = (\epsilon \partial_N v)^\bullet, \quad G(\delta_1)(n^\bullet) = (\partial_N v)^\bullet. \tag{CC.52}$$

The common source is the retained thickening. Every arrow fixes external absence, and every supported kernel vector maps to the receiving supported zero. Equations (CC.37)–(CC.40) give the actual original-unit morphisms connecting cyclic, full conormal and product-Jacobian observations before this quotient.

The exact source map accompanying this section records hashes and line locators for the complete delivered cyclic note, the complete conormal pasted source, and the earlier proof fragments. The cyclic note's depth-one retraction and Sections 7–8 are proved here at every original depth, with complete invariant dimensions, the scalar-power equality classification, the higher-degree lower bound, and the full-unit complement in (CC.24). Equations (CC.45)–(CC.48) provide the precise operator interpretation for the general weighted trace. Equations (CC.29)–(CC.36) prove the first-jet and invariant rank calculations supplied in the pasted source. The uncompiled status of its embedded Lean text describes that historical delivered draft. The separately recorded PR 21 successor has its own checked CI provenance; it does not make the new written proofs of this chapter kernel-checked. No new Lean execution, zero-location calculation, or claimed arithmetic asymptotic is part

of these proofs. The formulas give the exact finite algebra, retained relation derivatives and trace maps used by the adjoining cyclic metric calculation. The quantitative interpolation metric remains the original one, with its degree cost from (CC.7) and all its cross terms from (CC.49).

31 The cyclic sum source inside the full connection

Keep the original nonempty dagger-stable packet h , its full orders, $g = 2\xi$, $v_h = g/h$, and $k \geq 2$. Use exactly (NN.1), $S = k/2 + iu$, and $m_k(u) = w_h^{*k}(u) > 0$, with $w_h = |a_h|^2$ and $a_h(t) = v_h(1/2 + it)/\sqrt{2\pi}$. The complete cyclic-sum source note is retained in the source appendices. Its annihilator is

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_{\lambda}}, \quad \ell_{\lambda} = \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right), \quad C_{\chi} = \mathbb{C}[S]/(\chi). \quad (\text{CM.1})$$

Keep the full algebra $E = E_h^{\otimes k}$, the original unit $U_k = v_h^{\otimes k}$, the injective algebra map $\alpha : C_{\chi} \rightarrow E^{S_k}$, and the separate module injection $\eta = M_{U_k} \alpha$. Their exact construction, retraction, complement, nilpotent orders and trace comparison are proved in the cyclic conormal section and the complete delivered source. In particular the original analytic source is the injective map

$$\begin{aligned} \mathcal{V}_k(P) &= P(D_1 + \cdots + D_k) F_h^{\otimes k}, \\ \mathcal{U}_k \mathcal{V}_k(P) &= \Psi_{c_P}, \quad c_P(u) = P(k/2 + iu), \\ J^{(k)} \mathcal{V}_k(P) &= \eta[P]_{\chi}. \end{aligned} \quad (\text{CM.2})$$

Here $\mathcal{U}_k$ is the original Mellin–Fourier isometry and the change to the original sum and relative coordinates. Its norm is $\int |c_P|^2 m_k du$, with its original mass μ_h^k .

31.1 Complete polynomial moments and the original finite metric

Put $\mu_j = \int t^j |a_h|^2 dt$, $\nu_j = \int t^j |a'_h|^2 dt$, and $\theta_j = \int t^j \overline{a_h} a'_h dt$. The fixed arithmetic phase makes $\overline{a_h} a'_h = w'_h/2$; integration by parts with the original decay gives

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (\text{CM.3})$$

Define the formal series $\mathcal{M}(z) = \sum \mu_j z^j / j!$ and $\mathcal{N}(z) = \sum \nu_j z^j / j!$. The entire finite moment arrays are

$$\begin{aligned} M_{j,k} &= \int u^j m_k(u) du = j! [z^j] \mathcal{M}(z)^k, \\ B_{j,k} &= \int u^j \|\partial_u \Psi(u, \cdot)\|^2 du \\ &= j! [z^j] \left(\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right). \end{aligned} \quad (\text{CM.4})$$

For completeness, the multinomial expansion of $(t_1 + \cdots + t_k)^j$ and Fubini give the first identity. The derivative is $k^{-1} \sum_i a'_h(t_i) \prod_{r \neq i} a_h(t_r)$. There are k diagonal terms, each carrying k^{-2} , and $k(k-1)$ distinct-index terms with that same factor. Their formal series are respectively $\mathcal{N} \mathcal{M}^{k-1} / k$ and $(k-1) \Theta(z)^2 \mathcal{M}^{k-2} / k$, where $\Theta(z) = -z \mathcal{M}(z) / 2$ by (CM.3). This proves the second identity with every cross term and the original k^{-2} retained.

In the literal source coordinates $1, S, \dots, S^N$, write $c_j(u) = (k/2 + iu)^j$. Its original amplitude

and derivative Gram matrices are, entry by entry,

$$\begin{aligned} H_{ij}^0 &= \int \bar{c}_i c_j m_k du, \\ H_{ij}^\partial &= \int \bar{c}_i c_j \|\Psi'\|^2 du + \int \bar{c}_i c'_j m_k du \\ &\quad + \frac{1}{2} \int \bar{c}_i c'_j m'_k du + \frac{1}{2} \int \bar{c}'_i c_j m'_k du, \quad c'_j = ij(k/2 + iu)^{j-1} \quad (j \geq 1), \quad c'_0 = 0. \end{aligned} \quad (\text{CM.5})$$

Each integrand coefficient is obtained by multiplying the displayed original polynomials. Integrating its u^r coefficient uses $M_{r,k}$, $B_{r,k}$, or $\int u^r m'_k = -rM_{r-1,k}$, with zero for $r = 0$. This is an explicit finite matrix construction from the original one-factor arrays; it keeps the two separate cross terms. Both forms are positive definite. The amplitude map is injective by $\Psi \neq 0$ almost everywhere. A zero derivative would give an $L^2(du; \mathcal{H})$ function constant in u , hence zero, and the same argument then gives the zero polynomial.

Let p_j be the original monic orthogonal polynomials for m_k , with full norms $\omega_j = \int |p_j(k/2 + iu)|^2 m_k(u) du$, and $b_j = [p_j]_\chi$ in the fixed remainder coordinates. For $N \geq \deg \chi - 1$, put

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j, \quad G_N = K_N^{-1}, \quad T_N u = \sum_{j=0}^N \text{coeff}(p_j) \frac{b_j^* G_N u}{\omega_j}. \quad (\text{CM.6})$$

Thus T_N is the actual coefficient matrix of the canonical cyclic representative, and its two complete forms are

$$\mathcal{R}_N = \mathcal{V}_k T_N, \quad \mathcal{R}_N^* \mathcal{R}_N = T_N^* H^0 T_N = G_N, \quad (\partial_u \mathcal{U}_k \mathcal{R}_N)^* (\partial_u \mathcal{U}_k \mathcal{R}_N) = T_N^* H^\partial T_N. \quad (\text{CM.7})$$

The first equality on forms follows by multiplying (CM.6) in the orthogonal polynomial coordinates; the second is the definition of the original derivative Gram (CM.5). The latter has not been replaced by the constant-column Fisher estimate.

31.2 A full real-exponential family of the same information identity

The original-amplitude density theorem proves $|a_h^{(j)}(t)| \leq C_{h,j,b} e^{-b|t|}$ for every $b < \pi/4$. It follows, by choosing $2b > |\theta|$, that for each real $|\theta| < \pi/2$ the following are finite actual integrals:

$$M(\theta) = \int e^{\theta t} |a_h(t)|^2 dt > 0, \quad N(\theta) = \int e^{\theta t} |a'_h(t)|^2 dt. \quad (\text{CM.8})$$

The same estimates with a slightly larger real exponential on compact subsets prove differentiation under the integral for every polynomial power. Hence M, N are holomorphic for $|\Re z| < \pi/2$, and their Taylor coefficients at zero are the original arrays in (CM.4). Using Fubini with this actual exponential, rather than only formal coefficients, gives

$$\begin{aligned} \int e^{\theta u} m_k(u) du &= M(\theta)^k, \\ B_k(\theta) &:= \int e^{\theta u} \|\Psi'(u)\|^2 du = \frac{1}{k} N(\theta) M(\theta)^{k-1} + \frac{k-1}{4k} \theta^2 M(\theta)^k. \end{aligned} \quad (\text{CM.9})$$

In this calculation $\int e^{\theta t} \bar{a}_h a'_h = -\theta M(\theta)/2$. The diagonal and distinct-index product terms are therefore exactly those used in (CM.4).

The exact Hilbert-space multiplier is

$$L^2(e^{\theta t} dt) \xrightarrow{f \mapsto e^{\theta t/2} f} L^2(dt), \quad f \mapsto e^{\theta t/2} f, \quad \text{inverse } q \mapsto e^{-\theta t/2} q. \quad (\text{CM.10})$$

It is unitary by direct integration. For the original amplitude set $a_\theta = e^{\theta t/2} a_h$. Its mass and derivative energy are

$$\|a_\theta\|^2 = M(\theta), \quad \|a'_\theta\|^2 = N(\theta) - \frac{\theta^2}{4} M(\theta) > 0. \quad (\text{CM.11})$$

Expansion of a'_θ , followed by the cross integral above, proves the equality. If its derivative norm were zero, the nonzero L^2 function a_θ would be constant, which is impossible; this proves strict positivity. The product amplitude is exactly $\Psi_\theta = e^{\theta u/2} \Psi$, its density is $m_{k,\theta} = e^{\theta u} m_k$, and its relative line projection is the same Π_0 , since the multiplier is a nonzero scalar on each fibre. For $n_k = (I - \Pi_0) \Psi'$, its normal component is consequently $n_{k,\theta} = e^{\theta u/2} n_k$. Applying the complete product-rule proof of (FC.4) to this specific amplitude, including its mass, proves

$$\int e^{\theta u} \frac{(m'_k + \theta m_k)^2}{m_k} du + 4 \int e^{\theta u} \|n_k(u)\|^2 du = \frac{4}{k} M(\theta)^{k-1} \left(N(\theta) - \frac{\theta^2}{4} M(\theta) \right). \quad (\text{CM.12})$$

All square quotients are taken away from the already stated fibre zero sets; $m_k > 0$ itself everywhere. The multiplier keeps the same zeros of the original one-factor amplitude. Its normal component is nonzero on the same fibres as that in (FC.21)–(FC.27), so the second integral is strictly positive. No mass has been assigned the value one. The arithmetic maps (CM.1)–(CM.2) still use the original U_k ; (CM.10) specifies the exact weighted Hilbert comparison used solely to evaluate the displayed original exponential integrals.

31.3 All cyclic normal energy is retained under frame enlargement

The cyclic source uses the single relative coefficient 1 at every scalar degree N . Its line projection is Π_0 . The complete relative frame of total degree m has projection Π_m . For any fixed cyclic polynomial P , $m \geq \deg P$, the actual derivative amplitude has the three mutually orthogonal components

$$\begin{aligned} d(P) &= \partial_u(\Psi c_P) = t_0(P) + b_m(P) + r_m(P), \\ t_0(P) &= \Pi_0 d(P) = \Psi \left(i c_{P'} + \frac{m'_k}{2m_k} c_P \right), \\ b_m(P) &= (\Pi_m - \Pi_0) d(P), \quad r_m(P) = (I - \Pi_m) d(P), \\ n_k c_P &= b_m(P) + r_m(P). \end{aligned} \quad (\text{CM.13})$$

Here $c_{P'}(u) = P'(k/2 + iu)$. The derivative of the cyclic coefficient is $i c_{P'}$; multiplication by c_P commutes with the fibre projections. Hence projecting its derivative gives precisely the first two formulas, and subtracting proves the last one. The nesting $\Pi_0 \leq \Pi_m$ proves orthogonality of all three components. With the original global norms this gives

$$\|n_k c_P\|^2 = \|b_m(P)\|^2 + \|r_m(P)\|^2, \quad \|d(P)\|^2 = \|t_0(P)\|^2 + \|b_m(P)\|^2 + \|r_m(P)\|^2. \quad (\text{CM.14})$$

The proved density limit (NN.4) now yields the exact convergence

$$r_m(P) \rightarrow 0, \quad b_m(P) \rightarrow n_k c_P \quad \text{in } \mathcal{K}. \quad (\text{CM.15})$$

Thus the complete fixed cyclic normal vector is the limit of the added relative tangential vectors, with its entire positive norm retained by (CM.14). This is the explicit map between the constant relative frame of the new cyclic construction and the enlarging full relative frames. It makes no substitution of one frame for another. For a fixed finite collection of original cyclic polynomials, the corresponding restricted Gram matrices converge as well: each entry is a pairing of finitely many convergent $\mathcal{K}$ vectors. No statement for a sequence of changing canonical polynomials is inferred from this fixed-source convergence.

31.4 The canonical metric difference has its actual conormal lift

For $N \geq \deg \chi - 1$, let R_N^{full} be the original full-source minimum on the whole arithmetic quotient E , and let $\mathcal{R}_N$ be (CM.7). This full minimum exists at this degree: $\deg \chi - 1 \geq k(d - 1)$. Here is a direct proof with the original monic polynomial retained. For every $0 \leq q \leq k(d - 1)$, choose $0 \leq \alpha_i < d$ with $\sum_i \alpha_i = q$. The monomial $\mathbf{s}^\alpha$ in S^q has coefficient $q! / \prod_i \alpha_i! \neq 0$. It is already a standard remainder monomial. Every term with some exponent at least d , when reduced by the original monic $h(s_i)$, strictly lowers total degree. Thus the remainder of S^q has total degree exactly q , with that same nonzero coefficient. Remainders of lower powers have strictly lower degree. A nonzero polynomial in S of degree at most $k(d - 1)$ therefore cannot annihilate S in the full algebra. Since χ is its exact annihilator, the asserted degree inequality follows. Every monic multivariate remainder now belongs to the original full source at degree N , so the full jet map is onto. Set

$$\Delta_N = \mathcal{R}_N - R_N^{\text{full}}\eta, \quad E_N = \Delta_N^* \Delta_N \succeq 0. \quad (\text{CM.16})$$

Both representatives have the same full original jets. Their difference is therefore an actual polynomial-source boundary. The full minimum is orthogonal to that boundary, proving the complete Gram and control identities

$$\begin{aligned} G_N &= \eta^* G_N^{\text{full}} \eta + E_N, \\ W_N - \eta^* W_N^{\text{full}} \eta &= A_\chi^* E_N + E_N A_\chi - k E_N, \quad A_\chi = M_S. \end{aligned} \quad (\text{CM.17})$$

The second equality follows by substituting the first and using $A_k \eta = \eta A_\chi$. For an actual eigenvector $A_\chi x = \lambda x$, its exact contribution is

$$x^*(W_N - \eta^* W_N^{\text{full}} \eta)x = (2\Re \lambda - k) \|\Delta_N x\|^2. \quad (\text{CM.18})$$

This is the full effect of the positive Gram difference on that original eigenline, including either sign of $2\Re \lambda - k$.

Write $\Delta_N x = \mathcal{T}_h^{(k)} Q_x$ in the original polynomial coordinates. Since its full jet is zero and U_k is a unit, $Q_x \in I = (h(s_1), \dots, h(s_k))$. Fixed-order monic division supplies $Q_x = \sum_i h(s_i) Q_{i,x}$, linearly in x . Denote the fixed conormal map of this ideal by $\partial_I : I/I^2 \rightarrow E$, $[Q] \mapsto [k^{-1} \sum_i \partial_{s_i} Q]_I$. It is independent of the source cutoff N . The retained conormal map and its derivative observation are

$$\begin{aligned} \kappa_N : C_\chi &\longrightarrow I/I^2, \quad x \longmapsto [Q_x]_{I^2}, \\ \partial_I \kappa_N(x) &= \frac{1}{k} \sum_i h'(s_i) [Q_{i,x}]_I, \\ J^{(k)} \mathcal{L}_k \Delta_N x &= U_k \partial_I \kappa_N(x), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \end{aligned} \quad (\text{CM.19})$$

The first expression is independent of the displayed division because it is the actual class of the uniquely specified polynomial Q_x . The derivative of I^2 lies in I ; its product rule gives the second expression and proves independence of every conormal coordinate choice. For the last equality differentiate the full $U Q_x$: the term $(\partial_S^{\text{rel}} U) Q_x$ lies in I , whereas the other term is the stated U_k times the complete conormal row. Mellin differentiation supplies exactly $\mathcal{L}_k$, as in (FC.34). Thus the actual canonical-minimum difference has an explicit map into the original conormal layer and then to its full arithmetic derivative, with every selected order and unit retained.

The original quotient sends the supported class of Q_x to its supported zero. The conormal lift in (CM.19) is retained before that quotient, so its possibly nonzero differentiated observation is obtained from the stated original relation, not from differentiating the scalar supported zero. For $Q_x \in I^2$ the conormal class and this derivative vanish; for general $Q_x \in I$ their full kernels are related exactly by $\ker \kappa_N = \{x : Q_x \in I^2\}$ and $\ker(J^{(k)} \mathcal{L}_k \Delta_N) = \kappa_N^{-1}(\ker \partial_I)$, since U_k is invertible. The original conormal kernel is computed in the conormal section. The external τ maps to τ along every split lift.

32 Exterior trace equality, the spectral projector, and quantitative coupling

Date: 2026-09-12. This is an independent finite-dimensional proof for the exterior-amplification integration. All norms and adjoints below use the original Hermitian form. No arithmetic asymptotic estimate, Lean execution, or external-source verification is asserted.

32.1 1. Original objects and exact domains

Let $\mathcal{C}$ be a finite-dimensional complex vector space with the positive-definite Hermitian form

$$\langle x, y \rangle_G = x^* G y$$

(conjugate-linear in the first variable). Let $k \in \mathbb{R}$, let $A \in \text{End}_{\mathbb{C}}(\mathcal{C})$, and set

$$A^\sharp = G^{-1} A^* G, \quad H = A^\sharp + A - k I_{\mathcal{C}}.$$

Suppose the actual operator H has rank at most two and trace zero. Its eigenvalues, with zero eigenvalues retained, are $+\epsilon, -\epsilon, 0, \dots, 0$ for a uniquely determined $\epsilon \geq 0$.

Let $V = \mathcal{C}_{>}$ be the direct sum of the **full generalized eigenspaces** of A for the eigenvalues satisfying $\text{Re } \lambda > k/2$. Let $W = V^{\perp_G}$, and let $P : \mathcal{C} \rightarrow \mathcal{C}$ be the G -orthogonal projection with image V and kernel W . Let Q be the spectral projector with image V and kernel the direct sum of all the other full generalized eigenspaces of A .

For specificity, this last projector is an exact polynomial in the original operator. Factor the minimal polynomial as $m_A = m_+ m_-$, where m_+ contains exactly the selected eigenvalues with their original minimal-polynomial exponents and m_- contains all the remaining eigenvalues with their original exponents. The two factors are coprime. Choose polynomials a, b with

$$a(z)m_+(z) + b(z)m_-(z) = 1.$$

Then

$$Q = b(A)m_-(A).$$

Indeed, the polynomial $b m_-$ is congruent to one modulo m_+ and to zero modulo m_- ; applying it to each full generalized eigenspace gives the asserted identity and zero maps. This also proves $Q^2 = Q$, $AQ = QA$, and independence from the choice of the Bezout polynomials.

All block maps below use the original orthogonal direct sum $\mathcal{C} = V \oplus W$, without a change in either restricted Hermitian form. Since $A(V) \subseteq V$, define

$$B = A|_V : V \rightarrow V, \quad D = (1 - P)A|_W : W \rightarrow W, \quad T = PA|_W : W \rightarrow V.$$

Then the exact block identities are

$$A = \begin{pmatrix} B & T \\ 0 & D \end{pmatrix}, \quad A^\sharp = \begin{pmatrix} B^\sharp & 0 \\ T^\sharp & D^\sharp \end{pmatrix}, \quad H = \begin{pmatrix} B^\sharp + B - k I_V & T \\ T^\sharp & D^\sharp + D - k I_W \end{pmatrix}. \quad (\text{E1})$$

In particular, as maps $W \rightarrow V$,

$$PH|_W = PA|_W = T. \quad (\text{E2})$$

This equality includes the proof of the vanishing term: $PA^\sharp|_W = ((1 - P)AP)^\sharp|_W = 0$, because $(1 - P)AP = 0$.

Write

$$L = \text{Tr}_{\mathcal{C}}(PH) = \text{Tr}_V(B^\sharp + B - k I_V) = \sum_{\text{Re } \lambda > k/2} m_A^{\text{alg}}(\lambda)(2 \text{Re } \lambda - k), \quad (\text{E3})$$

where $m_A^{\text{alg}}(\lambda)$ is the original algebraic multiplicity of λ in A . The final equality follows by taking the trace of B , whose characteristic polynomial consists exactly of the selected full generalized blocks, and by $\text{Tr } B^\sharp = \overline{\text{Tr } B}$. Thus $L > 0$ exactly when $V \neq 0$.

If $\epsilon = 0$, then $H = 0$. For every eigenvector $x \neq 0$ of A ,

$$0 = \langle x, Hx \rangle_G = (2 \operatorname{Re} \lambda - k) \langle x, x \rangle_G,$$

so all eigenvalues have real part $k/2$, and $V = 0$. Hence the case $L > 0$ automatically has $\epsilon > 0$. Also $V \neq \mathcal{C}$ in that case, since $P = I$ would give $L = \text{Tr } H = 0$.

32.2 2. Exact trace maximum and equality theorem

Theorem E4. With these definitions and $L > 0$,

$$0 < L \leq \epsilon,$$

and the following five statements are equivalent:

1. $L = \epsilon$.
2. $P = Q$.
3. V reduces A : both V and $V^\perp_G$ are A -invariant.
4. $PA = AP$.
5. $T = 0$.

Moreover, each statement is equivalent to the positive H -eigenline being contained in V and the negative H -eigenline being contained in W .

Proof. Choose vectors u_+, u_- on those two eigenlines with

$$\langle u_+, u_+ \rangle_G = \langle u_-, u_- \rangle_G = 1, \quad \langle u_+, u_- \rangle_G = 0.$$

This records the eigenline projectors using the existing form and does not replace that form. The full operator, including its zero eigenspace, is

$$Hx = \epsilon u_+ \langle u_+, x \rangle_G - \epsilon u_- \langle u_-, x \rangle_G. \quad (\text{E4a})$$

Put $a = \|Pu_+\|_G^2$, $b = \|Pu_-\|_G^2$. Then

$$L = \epsilon(a - b), \quad 0 \leq a, b \leq 1. \quad (\text{E4b})$$

This proves $L \leq \epsilon$. Equality holds precisely when $a = 1$ and $b = 0$. Orthogonality of P identifies these equalities with $u_+ \in V$ and $u_- \in W$. In that case (E4a) gives $PH(1 - P) = 0$, and (E2) gives $T = 0$.

By (E1), $T = 0$, reduction of A , and $PA = AP$ are equivalent. If $P = Q$, the identity $AQ = QA$ already proved implies $PA = AP$. Conversely, if $T = 0$, the decomposition in (E1) is a block diagonal decomposition of the original A . The spectrum of B , with multiplicities, is the selected spectrum, while the characteristic polynomial of D contains exactly the remaining factors: the full characteristic polynomial is the product of the two block characteristic polynomials. Applying the defining projector polynomial to these blocks gives $Q = I_V \oplus 0_W = P$.

Finally suppose $T = 0$. Equation (E1) shows that P commutes with H . The two nonzero eigenspaces of H are one-dimensional. The restriction of an idempotent commuting with H to either such eigenspace is therefore either identity or zero. It follows that $a, b \in \{0, 1\}$. Since $\epsilon(a - b) = L > 0$, necessarily $a = 1, b = 0$, and hence $L = \epsilon$. This proves every stated implication. $\square$

The hypothesis $L > 0$ matters for the converse from reduction to $L = \epsilon$. For example, an empty selected subspace has $P = Q = 0$ and reduces A , while $L = 0$ may hold with $\epsilon > 0$. The theorem concerns exactly the nonempty right-half-plane subspace that appears in (E3).

32.3 3. Sharp quantitative trace slack

For a map $F : W \rightarrow V$, set $\|F\|_{\text{HS},G}^2 = \text{Tr}_W(F^\sharp F)$. This is the Hilbert-Schmidt norm with the original restricted metrics on both its domain and codomain.

Keep u_+, u_- as above and set

$$c = \langle Pu_+, Pu_- \rangle_G, \quad M = \begin{pmatrix} a & c \\ \bar{c} & b \end{pmatrix}. \quad (\text{E5})$$

The matrix M is the compression of the original orthogonal projection P to $E = \text{span}(u_+, u_-)$, expressed in that specified eigenbasis. For $z = (z_+, z_-)$,

$$z^* M z = \|P(z_+ u_+ + z_- u_-)\|_G^2, \quad z^* (I_2 - M) z = \|(1 - P)(z_+ u_+ + z_- u_-)\|_G^2. \quad (\text{E6})$$

Both matrices are positive semidefinite. In particular,

$$ab - |c|^2 \geq 0, \quad (1 - a)(1 - b) - |c|^2 \geq 0. \quad (\text{E7})$$

Theorem E8. The exact identity is

$$\epsilon^2 - L^2 - \|T\|_{\text{HS},G}^2 = \epsilon^2 (\det M + \det(I_2 - M)). \quad (\text{E8})$$

Consequently,

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2, \quad \epsilon - L \geq \frac{\|T\|_{\text{HS},G}^2}{\epsilon + L}. \quad (\text{E9})$$

Proof. By (E2) and selfadjointness of H ,

$$\begin{aligned} \|T\|_{\text{HS},G}^2 &= \text{Tr}_V(PH(1 - P)HP|_V) \\ &= \text{Tr}_\mathcal{C}(PH^2P) - \text{Tr}_\mathcal{C}((PHP)^2). \end{aligned} \quad (\text{E10})$$

Formula (E4a) gives $H^2x = \epsilon^2 u_+ \langle u_+, x \rangle_G + \epsilon^2 u_- \langle u_-, x \rangle_G$. Thus the first trace on the right is $\epsilon^2(a + b)$. The compressed operator is the difference of the two rank-one maps generated by Pu_+ and Pu_- , with coefficient ϵ . Squaring these maps and taking their traces gives

$$\text{Tr}_\mathcal{C}((PHP)^2) = \epsilon^2(a^2 + b^2 - 2|c|^2).$$

For clarity, the cross trace used here is $\text{Tr}((v \langle v, \cdot \rangle_G)(w \langle w, \cdot \rangle_G)) = |\langle v, w \rangle_G|^2$, directly by composition of the two rank-one maps. Therefore

$$\|T\|_{\text{HS},G}^2 = \epsilon^2(a + b - a^2 - b^2 + 2|c|^2). \quad (\text{E11})$$

Subtract this expression and $L^2 = \epsilon^2(a - b)^2$ from ϵ^2 . The result is

$$\epsilon^2(1 - a - b + 2ab - 2|c|^2) = \epsilon^2((ab - |c|^2) + ((1 - a)(1 - b) - |c|^2)),$$

which is exactly (E8). Inequalities (E7) prove its nonnegativity, and division by the positive number $\epsilon + L$ proves the second inequality in (E9). $\square$

Equality in the first inequality of (E9) holds precisely when M has eigenvalues zero and one. Indeed, its eigenvalues μ_1, μ_2 lie in $[0, 1]$, and the two nonnegative determinants both vanish precisely when $\mu_1 \mu_2 = 0$ and $(1 - \mu_1)(1 - \mu_2) = 0$. This is equivalent to the two eigenvalues being zero and one. It also means that E reduces P and that $P|_E$ has rank one: a compression eigenvector with eigenvalue zero satisfies $\|Px\|^2 = 0$, while a compression eigenvector with eigenvalue one satisfies $\|(1 - P)x\|^2 = 0$, by (E6). The reverse implication follows by restricting a rank-one orthogonal projection on E .

The same inequality can be expressed by exact commutators on $\mathcal{C}$:

$$[P, A] = \begin{pmatrix} 0 & T \\ 0 & 0 \end{pmatrix}, \quad [P, H] = \begin{pmatrix} 0 & T \\ -T^\sharp & 0 \end{pmatrix},$$

so

$$\epsilon^2 - L^2 \geq \|[P, A]\|_{\text{HS},G}^2 = \frac{1}{2} \|[P, H]\|_{\text{HS},G}^2. \quad (\text{E12})$$

32.4 4. Exact morphism from the projector difference to the coupling

Because Q has image V and restricts to identity on V , there is a unique linear map $X : W \rightarrow V$ such that

$$Q = \begin{pmatrix} I_V & X \\ 0 & 0 \end{pmatrix}, \quad Q - P = \begin{pmatrix} 0 & X \\ 0 & 0 \end{pmatrix}. \quad (\text{E13})$$

The equation $AQ = QA$, with (E1), yields the **exact Sylvester morphism**

$$\mathcal{S}_{B,D} : \text{Hom}_{\mathbb{C}}(W, V) \longrightarrow \text{Hom}_{\mathbb{C}}(W, V), \quad \mathcal{S}_{B,D}(X) = BX - XD = T. \quad (\text{E14})$$

It is an isomorphism. Here is an explicit proof and inverse retaining the original metrics. Set $\beta = \min_{\lambda \in \sigma(B)} \text{Re } \lambda$ and $\delta = \max_{\mu \in \sigma(D)} \text{Re } \mu$. By the full spectral selection, $\beta > k/2 \geq \delta$. The original operator exponentials have polynomial-times-exponential bounds because each nilpotent part in each generalized eigenspace has finite order. More explicitly, with the spectral projectors $\Pi_{\lambda}^B, \Pi_{\mu}^D$ and their nilpotency orders $\nu_{\lambda}^B, \nu_{\mu}^D$, the exact formulas are

$$\begin{aligned} e^{-tB} &= \sum_{\lambda \in \sigma(B)} e^{-t\lambda} \sum_{j=0}^{\nu_{\lambda}^B-1} \frac{(-t)^j}{j!} (B - \lambda I)^j \Pi_{\lambda}^B, \\ e^{tD} &= \sum_{\mu \in \sigma(D)} e^{t\mu} \sum_{\ell=0}^{\nu_{\mu}^D-1} \frac{t^{\ell}}{\ell!} (D - \mu I)^{\ell} \Pi_{\mu}^D. \end{aligned} \quad (\text{E15})$$

These formulas follow by expanding the exponential of scalar plus nilpotent on every generalized eigenspace; every sum in the nilpotent part terminates at the displayed order. Consequently the following integral converges in the original operator norm, for every $Y : W \rightarrow V$:

$$\mathcal{S}_{B,D}^{-1}(Y) = \int_0^{\infty} e^{-tB} Y e^{tD} dt. \quad (\text{E16})$$

Differentiating the integrand and integrating its derivative from zero to infinity proves

$$B \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) - \left(\int_0^{\infty} e^{-tB} Y e^{tD} dt \right) D = Y.$$

There is no upper-end boundary term, by $\beta - \delta > 0$ and the finite polynomial factors in (E15). If $BZ - ZD = 0$, differentiation shows that $e^{-tB} Z e^{tD}$ is constantly Z , while the same decay shows its limit is zero. Thus $Z = 0$, proving uniqueness and the asserted isomorphism.

Equip $\text{Hom}(W, V)$ with its original Hilbert-Schmidt norm, and define its actual positive smallest singular value

$$s_{B,D} = \min_{\|Y\|_{\text{HS},G}=1} \|BY - YD\|_{\text{HS},G} > 0. \quad (\text{E17})$$

The minimum exists because the unit sphere is compact, and is positive because (E14) is injective. The identity (E13) gives $\|X\|_{\text{HS},G} = \|Q - P\|_{\text{HS},G}$. Combining (E9), (E14), and (E17) proves

$$\epsilon^2 - L^2 \geq \|T\|_{\text{HS},G}^2 \geq s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E18})$$

Thus the difference between the algebraic spectral projector and the orthogonal projection is related to the trace defect by a proved, invertible, typed map.

An explicit bound on the inverse is also available with all original operator norms retained. Write

$$\kappa_{B,D} = \int_0^{\infty} \|e^{-tB}\|_{\text{op},G} \|e^{tD}\|_{\text{op},G} dt < \infty.$$

The ideal inequality $\|RYS\|_{\text{HS}} \leq \|R\|_{\text{op}}\|Y\|_{\text{HS}}\|S\|_{\text{op}}$ follows by diagonalizing the positive operators $R^\sharp R$ and $SS^\sharp$ in their respective existing Hermitian spaces and bounding their largest eigenvalues. Applying it under (E16) yields

$$\|Q - P\|_{\text{HS},G} \leq \kappa_{B,D}\|T\|_{\text{HS},G} \leq \kappa_{B,D}\sqrt{\epsilon^2 - L^2}. \quad (\text{E19})$$

If desired, (E15) gives the fully specified finite upper bound

$$\kappa_{B,D} \leq \sum_{\lambda,\mu} \sum_{j=0}^{\nu_\lambda^B-1} \sum_{\ell=0}^{\nu_\mu^D-1} \frac{(j+\ell)! \|(B-\lambda I)^j \Pi_\lambda^B\|_{\text{op},G} \|(D-\mu I)^\ell \Pi_\mu^D\|_{\text{op},G}}{j! \ell! (\text{Re } \lambda - \text{Re } \mu)^{j+\ell+1}}. \quad (\text{E20})$$

The scalar integral used is $\int_0^\infty t^n e^{-at} dt = n!/a^{n+1}$ for $a > 0$, proved by the substitution $v = at$ and n integrations by parts beginning with $\int_0^\infty e^{-v} dv = 1$. Every denominator in (E20) is positive. No bound on nonnormal operator norms has been replaced by a bound on eigenvalue real parts.

32.5 5. Sharp original two-dimensional family

Fix $k \in \mathbb{R}$, $\ell > 0$, and $t \in \mathbb{C}$. On $\mathbb{C}^2$ with its specified form $G = I_2$, define

$$A = \begin{pmatrix} k/2 + \ell/2 & t \\ 0 & k/2 - \ell/2 \end{pmatrix}, \quad H = A^* + A - kI_2 = \begin{pmatrix} \ell & t \\ \bar{t} & -\ell \end{pmatrix}. \quad (\text{E21})$$

Here $H^2 = (\ell^2 + |t|^2)I_2$, so $\epsilon = \sqrt{\ell^2 + |t|^2}$. The selected full generalized spectral subspace is exactly $V = \mathbb{C}(1, 0)$, since the two eigenvalues $k/2 \pm \ell/2$ are distinct. The original orthogonal projection is $P = \text{diag}(1, 0)$. Hence

$$L = \ell, \quad T = t : \mathbb{C} \rightarrow \mathbb{C}, \quad Q = \begin{pmatrix} 1 & t/\ell \\ 0 & 0 \end{pmatrix}, \quad s_{B,D} = \ell. \quad (\text{E22})$$

The formula for Q is also obtained directly from the polynomial $(A - (k/2 - \ell/2)I_2)/\ell$; thus its stated kernel is the full other eigenspace, with no orthogonality assumption. Direct calculation gives

$$\epsilon^2 - L^2 = |t|^2 = \|T\|_{\text{HS},G}^2 = s_{B,D}^2 \|Q - P\|_{\text{HS},G}^2. \quad (\text{E23})$$

Both quantitative constants in (E18) are attained by this family, including every $t \neq 0$. The equality theorem gives $L = \epsilon$ precisely when $t = 0$, exactly when $P = Q$ and the selected subspace reduces A .

32.6 6. Critical generalized blocks at the trace maximum

The equality theorem also constrains every full critical generalized block, including its original nilpotent structure.

Theorem E24. Suppose $L > 0$ and $L = \epsilon$. For every eigenvalue λ of the original operator A with $\text{Re } \lambda = k/2$, let

$$Z_\lambda = \ker(A - \lambda I)^{\nu_\lambda}$$

be its full generalized eigenspace, where ν_λ is its original exponent in the minimal polynomial. Then

$$HZ_\lambda = 0, \quad A|_{Z_\lambda} = \lambda I_{Z_\lambda}, \quad A^\sharp|_{Z_\lambda} = (k - \lambda)I_{Z_\lambda} = \bar{\lambda}I_{Z_\lambda}. \quad (\text{E24})$$

In particular, Z_λ reduces A , its spectral projector equals its G -orthogonal projection, and the original eigenvalue λ is semisimple. Consequently, the presence of even one critical-line Jordan chain of length greater than one implies the strict inequality

$$\epsilon > L \quad (\text{E25})$$

whenever $L > 0$.

Proof. At equality, Theorem E4 gives $P = Q$, $u_+ \in V$, and $u_- \in W$. Hence the restriction of the original Hermitian defect to the original orthogonal complement is

$$H|_W : w \mapsto -\epsilon u_- \langle u_-, w \rangle_G. \quad (\text{E26})$$

It is negative semidefinite and has rank one. Every critical generalized eigenspace is in $\ker Q = W$, by the defining polynomial for Q . This conclusion uses the proved equality $P = Q$; it makes no claim that the original unselected spectral subspace was orthogonal before that equality was proved.

Fix such a λ , and let R_λ be the original G -orthogonal projection onto Z_λ . The subspace Z_λ is A -invariant. With $A_\lambda = A|_{Z_\lambda}$, its compressed Hermitian form is therefore

$$R_\lambda H|_{Z_\lambda} = A_\lambda^\sharp + A_\lambda - kI_{Z_\lambda}. \quad (\text{E27})$$

The adjoint on the right uses the original form restricted to Z_λ ; (E27) follows from the defining adjoint identity for two vectors in that subspace. The trace is

$$\text{Tr}_{Z_\lambda}(R_\lambda H|_{Z_\lambda}) = 2 \text{Re Tr } A_\lambda - k \dim Z_\lambda = (2 \text{Re } \lambda - k) \dim Z_\lambda = 0. \quad (\text{E28})$$

Here $A_\lambda = \lambda I + N_\lambda$ with N_λ nilpotent, so $\text{Tr } A_\lambda = \lambda \dim Z_\lambda$. To justify the nilpotent trace directly, choose a basis compatible with the filtration $\ker N_\lambda^j$; the map N_λ lowers that filtration and its matrix has zero diagonal.

Choose any G -orthonormal basis $z_1, \dots, z_r$ of Z_λ , solely to evaluate the trace in its existing form. Equations (E26) and (E28) give

$$0 = \sum_{j=1}^r \langle z_j, H z_j \rangle_G = -\epsilon \sum_{j=1}^r |\langle u_-, z_j \rangle_G|^2. \quad (\text{E29})$$

Since $\epsilon > 0$, every displayed square is zero. Formula (E26) now gives $H z_j = 0$ for every basis vector, and hence $H Z_\lambda = 0$.

The defining equation $H = A^\sharp + A - kI$ implies, for every $z \in Z_\lambda$,

$$A^\sharp z = kz - Az \in Z_\lambda. \quad (\text{E30})$$

Thus Z_λ is invariant under both A and $A^\sharp$. Its orthogonal complement is A -invariant: if $w \perp_G Z_\lambda$ and $z \in Z_\lambda$, then $\langle z, Aw \rangle_G = \langle A^\sharp z, w \rangle_G = 0$. It is also $A^\sharp$ -invariant by the corresponding identity with A and $A^\sharp$ exchanged. This proves the asserted reduction and shows that the restricted adjoint in (E27) is the actual restriction of $A^\sharp$.

On Z_λ put $N = A_\lambda - \lambda I$. It is nilpotent by the definition of this full generalized eigenspace, and (E30) gives

$$N^\sharp = kI - A_\lambda - \bar{\lambda}I = \lambda I - A_\lambda = -N, \quad (\text{E31})$$

because $\lambda + \bar{\lambda} = k$. A nilpotent skew-adjoint map on a positive-definite Hermitian space is zero, as the following calculation proves. If $N \neq 0$, take its nilpotency order $r \geq 2$, choose v with $N^{r-1}v \neq 0$, and set $y = N^{r-1}v$, $z = N^{r-2}v$. Then $Nz = y \neq 0$ and $Ny = 0$, whereas

$$\|y\|_G^2 = \langle Nz, y \rangle_G = \langle z, N^\sharp y \rangle_G = -\langle z, Ny \rangle_G = 0, \quad (\text{E32})$$

contradicting positive definiteness. Consequently $N = 0$, proving (E24).

Because Z_λ reduces A , the original orthogonal direct sum $\mathcal{C} = Z_\lambda \oplus Z_\lambda^{\perp_G}$ gives a block diagonal matrix for A . The complement contains exactly the other generalized eigenspaces: its characteristic polynomial is the quotient of the full characteristic polynomial by that of the full λ -generalized block, so it has no eigenvalue λ . The polynomial spectral projector for λ is therefore identity on Z_λ and zero on its orthogonal complement. It equals R_λ , as claimed. Finally, (E4) already proves $L \leq \epsilon$; (E24)

rules out equality whenever a critical Jordan chain of length greater than one is present. This gives (E25). $\square$

Distinct critical eigenspaces are also mutually orthogonal in the original form. For $x \in Z_\lambda, y \in Z_\mu$, (E24) gives

$$\bar{\lambda}\langle x, y \rangle_G = \langle Ax, y \rangle_G = \langle x, A^\sharp y \rangle_G = \bar{\mu}\langle x, y \rangle_G.$$

If $\lambda \neq \mu$, the inner product vanishes. Therefore the full critical subspace $Z = \bigoplus_{\operatorname{Re} \lambda = k/2} Z_\lambda$ is an original orthogonal sum, reduces A , and has the exact diagonal action $\sum_\lambda \lambda R_\lambda$, with $H|_Z = 0$. The shifted operator $A|_Z - (k/2)I_Z$ is skew-adjoint. This describes the full critical block and its projectors, retaining all its original eigenvalues and multiplicities.

32.7 The original quartet, its critical blocks, and a canonical equality example

Apply the preceding maps to the actual cyclic module and canonical Gram G_N of the complete exterior source note. For the exact packet $h(s) = \prod_{\varepsilon, \eta = \pm 1} (s - (\frac{1}{2} + \varepsilon\delta + i\eta\gamma))^m$, $\delta, \gamma > 0$, its distinct sum roots and full lengths are

$$\lambda_{a,b} = k/2 + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k, \quad \ell_k = 1 + k(m - 1), \quad q = \ell_k(k + 1)^2. \quad (\text{E33})$$

Indeed the four occupation counts $(t, a - t, b - t, k - a - b + t)$ exist for every integer $\max(0, a + b - k) \leq t \leq \min(a, b)$; conversely each tuple has these two margins. Equality of sums forces both margins to agree because δ, γ are nonzero real numbers. The top surviving local monomial has degree $k(m - 1)$ and coefficient $(k(m - 1))!/((m - 1)!)^k$, proving the stated full nilpotent length. Summing positive real defects over these actual cyclic blocks gives

$$L = 2\delta\ell_k(k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq q - 1. \quad (\text{E34})$$

For $k = 2n$ the sum of $2a - k$ over $a > k/2$ is $n(n + 1)$; for $k = 2n + 1$ it is $(n + 1)^2$, proving the floor and its constant. If k is even and $m > 1$, each $a = k/2$ block is critical and has length $\ell_k > 1$. The full cyclic local algebra is a single Jordan block in powers of $S - \lambda_{a,b}$, so (E24)–(E25) prove the strict inequality $\epsilon_{h,k,N}^{\text{cyc}} > L$ at every admitted N . For a larger packet containing this quartet, the same sums survive as CRT factors and their maximal nilpotent lengths are at least ℓ_k . Thus (E34) remains a lower bound; its admission threshold is the larger packet's actual $N \geq \deg \chi_{h,k} - 1$.

Equality can occur for a specified positive canonical moment measure. Here is a complete calibration, with no claim about zeta zeros. Take $k = 1, \mu, \sigma > 0$ and

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad S = \frac{1}{2} + iu, \quad \chi = (S - \frac{1}{2})^2 - \sigma^2, \quad N = 1. \quad (\text{E35})$$

The Gaussian integral and one integration by parts give mass μ , mean zero and second moment $\mu\sigma^2$; odd moments vanish by the original reflection $u \mapsto -u$. Consequently the original monic orthogonal polynomials through degree two are $p_0 = 1, p_1 = S - \frac{1}{2}, p_2 = (S - \frac{1}{2})^2 + \sigma^2$, with $\omega_0 = \mu, \omega_1 = \mu\sigma^2$. In the fixed remainder coordinates $(1, S)$ the column map $B = [b_0, b_1] = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}$ is invertible, and the actual canonical Gram is $G = B^{-*} \operatorname{diag}(\mu, \mu\sigma^2) B^{-1}$. Reduction gives $b_2 = 2\sigma^2 b_0$, whence

$$a = \mu\sigma^2, \quad d = 4\mu\sigma^4, \quad c = 0, \quad \epsilon = 2\sigma = L. \quad (\text{E36})$$

The formula follows directly from $\epsilon^2 = (ad - |c|^2)/\omega_1^2$ and the two simple roots $1/2 \pm \sigma$. All original masses and coordinates are retained. For $0 < \sigma < 1/2$ both roots lie in the open critical strip, but they are declared calibration roots. Thus positive measure and canonical interpolation alone cannot supply universal strictness; (E14)–(E18) identify the exact projector and coupling maps governing equality for each original arithmetic measure.

The full exterior cochain idempotent also has its complementary cochain idempotent. For an idempotent e commuting with d , the maps $x \mapsto (ex, (1-e)x)$ and $(u, v) \mapsto u + v$ are inverse cochain maps, since $e(1-e) = 0$ and both summands are d -stable. They induce the same inverse maps on cycles modulo boundaries. Ordinary traces here are taken on the retained finite arithmetic submodules: the signed tensor permutation preserves the finite packet, so restriction of e and $1-e$ gives finite invariant summands and inclusion followed by projection is the stated idempotent. In a basis of these two summands, a commuting finite operator is block diagonal, proving additivity of its ordinary trace; its top cochain sign is the original $(-1)^{kp}$. The splitting of the entire possibly infinite-dimensional theta cohomology itself supplies these maps but no ordinary finite trace.

32.8 7. Integration scope

Equations (E3), (E4), (E8), (E18), and (E24) may be applied directly whenever the exterior-amplification construction supplies its actual finite Hermitian form, its original multiplication operator, and its proved rank-two trace-zero Hermitian defect. All constants $G, k, \epsilon, L, s_{B,D}$ remain attached to that specific construction and degree. The proof establishes an equality classification, an exact projector morphism, a sharp quantitative slack identity, and strictness in the presence of a retained critical Jordan chain. It supplies no uniform estimate for $s_{B,D}$, $\kappa_{B,D}$, or ϵ as arithmetic parameters change; such an estimate would require a further calculation on those original varying objects.

Independent algebra check: the determinant-slack calculation (E8)–(E11), its equality classification, and the sharp family (E21)–(E23) were separately recomputed by the bounded `determinant_slack_check` agent. That agent then independently checked all steps of (E24)–(E32), including the passage from zero compressed trace to ambient annihilation and the use of full primary spaces in identifying the projectors. Both derivations agreed with the formulas above. The projector conclusion applies to each full primary space, including every Jordan block at its eigenvalue; it asserts nothing about arbitrary coordinate projectors onto individually selected copies of a repeated eigenspace. These are second mathematical derivations, not Lean or numerical certificates.

33 The cyclic control as an exact change of arithmetic interpolation volume

Retain a nonempty reflection-stable packet with every original order, the tensor degree $k \geq 1$, the cyclic algebra $C = \mathbb{C}[S]/(\chi)$ of dimension $q \geq 1$, and its fixed monic remainder coordinates. Let $p_j(S)$ be the original monic orthogonal polynomials on $S = k/2 + iu$ for the measure $m_{h,k}(u) du$, and put

$$b_j = [p_j]_\chi, \quad \omega_j = \int_{\mathbb{R}} |p_j(k/2 + iu)|^2 m_{h,k}(u) du > 0, \quad K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G_N = K_N^{-1}. \quad (\text{CV.1})$$

Here $N \geq q - 1$; the original $\omega_0 = \mu_h^k$ remains. All determinants below are taken in these same coordinates. Set $K_{-1} = 0$ and

$$\begin{aligned} D_j &= \det K_j, & \tilde{K}_N &= K_{N-1} + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}, & \tilde{D}_N &= \det \tilde{K}_N, \\ a &= b_N^* G_N b_N, & d &= b_{N+1}^* G_N b_{N+1}, & c &= b_N^* G_N b_{N+1}. \end{aligned} \quad (\text{CV.2})$$

The matrix $\tilde{K}_N$ records the actual polynomial subspace spanned by $p_0, \dots, p_{N-1}, p_{N+1}$, with its inherited norm: the degree- N orthogonal direction is omitted and the degree- $(N+1)$ direction is retained. It can be singular. No inverse for it is used.

Proposition 33.1 (Exact adjacent-volume identities). *The following equalities hold, including $N = q - 1$:*

$$\begin{aligned} 1 - \frac{D_{N-1}}{D_N} &= \frac{a}{\omega_N}, \\ \frac{D_{N+1}}{D_N} - 1 &= \frac{d}{\omega_{N+1}}, \\ \frac{\tilde{D}_N}{D_N} &= \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d}{\omega_{N+1}}\right) + \frac{|c|^2}{\omega_N \omega_{N+1}}. \end{aligned} \quad (\text{CV.3})$$

Consequently the original rank-two control allowance satisfies

$$\boxed{(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \tilde{D}_N}{D_N}}. \quad (\text{CV.4})$$

Equivalently, retaining the entire complex cross term,

$$(\epsilon_{h,k,N}^{\text{cyc}})^2 = \frac{\omega_{N+1}}{\omega_N} \left[\left(1 - \frac{D_{N-1}}{D_N}\right) \left(\frac{D_{N+1}}{D_N} - 1\right) - \frac{|c|^2}{\omega_N \omega_{N+1}} \right]. \quad (\text{CV.5})$$

Proof. The first q monic polynomials give a triangular basis modulo χ , so K_N is positive definite in the stated range. For an invertible K and columns x, y , direct block elimination of $\begin{pmatrix} K & x \\ -y^* & 1 \end{pmatrix}$ in its two possible orders gives $\det(K + xy^*) = \det K(1 + y^* K^{-1}x)$. Applying this with $K = K_N$ and the displayed signed increments gives the first two equations, even if the matrix after subtraction is singular. For completeness the same block elimination with two columns proves $\det(K + UV^*) = \det K \det(I_2 + V^* K^{-1}U)$. Take

$$U = \left(-\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right), \quad V = \left(\frac{b_N}{\sqrt{\omega_N}}, \frac{b_{N+1}}{\sqrt{\omega_{N+1}}}\right).$$

The positive square roots only specify this auxiliary factorization; the original columns, coordinates and norms in (CV.1) are unchanged. The resulting two-by-two determinant is

$$\det \begin{pmatrix} 1 - a/\omega_N & c/\sqrt{\omega_N \omega_{N+1}} \\ -\bar{c}/\sqrt{\omega_N \omega_{N+1}} & 1 + d/\omega_{N+1} \end{pmatrix},$$

which proves the third equation with its plus sign. Subtract that equation from the sum of the first two determinant ratios as ordered in (CV.4). The resulting value is $(ad - |c|^2)/(\omega_N \omega_{N+1})$. The original control formula is $\epsilon^2 = (ad - |c|^2)/\omega_N^2$, proving (CV.4)–(CV.5). In particular its numerator is nonnegative by the Gram determinant of the two original vectors b_N, b_{N+1} in G_N . For $N = q - 1$, K_{N-1} has rank $q - 1$, so $D_{N-1} = 0$ and $a = \omega_N$; every equation still holds. For $q = 1$ the two-vector Gram determinant is zero, consistently with the trace-zero one-dimensional control. There is no positive-dimensional determinant or Gram inverse to apply when $h = 1$; the coefficient quotient is then zero and its unique maps have the split lift $\{\tau, e\}$. $\square$

33.1 The exact volume map and the reflected quartet bound

Let $\mathcal{P}_N$ be the span of $p_0, \dots, p_N$ with its original integral norm, and let $T_N : \mathcal{P}_N \rightarrow C$ be reduction modulo χ . In the displayed polynomial basis its matrix is $[b_0 \cdots b_N]$, and its source Gram is $\text{diag}(\omega_0, \dots, \omega_N)$. The adjoint is therefore the explicit map

$$T_N^* x = \sum_{j=0}^N p_j \frac{b_j^* x}{\omega_j}, \quad T_N T_N^* = K_N, \quad (\text{CV.6})$$

where the star uses the fixed Euclidean pairing on the written target coordinates, solely to specify this typed adjoint. The canonical right inverse is $T_N^* G_N$, whose original source Gram is G_N . Taking its q th exterior power maps the fixed unscaled coordinate wedge to a vector of squared norm $\det G_N = D_N^{-1}$ in $\bigwedge^q \mathcal{P}_N$. The exact exterior source map is

$$\bigwedge^q \mathcal{V}_{h,k} : \bigwedge^q \mathcal{P}_N \longrightarrow \bigwedge^q (\mathcal{B}^{\widehat{\otimes} k}).$$

It preserves the determinant norm D_N^{-1} , because the original source map preserves every pairwise integral inner product. To pass to the original tensor cochain source, use instead $\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q$, where $\text{Alt}_q(v_1 \wedge \cdots \wedge v_q) = \sum_{\pi \in S_q} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(q)}$. The exterior quotient satisfies $\text{wed Alt}_q = q! I$: every permuted summand contributes its permutation sign twice. Expanding the tensor inner product gives $q!$ copies of its determinant, so the literal tensor-source squared norm is $q! D_N^{-1}$. Its target is $\mathcal{B}^{\widehat{\otimes} kq}$ in top cochain degree kq . This proves the asserted interpolation-volume interpretation without changing its target metric or source mass. Replacing T_N by its restriction to the omitted-direction subspace gives exactly $\widetilde{K}_N$; the construction needs no assertion that this restricted map is onto.

Now take the packet polynomial to be exactly

$$h(s) = \prod_{\varepsilon, \eta \in \{+1, -1\}} \left(s - \left(\frac{1}{2} + \varepsilon \delta + i \eta \gamma \right) \right)^m$$

for a complete actual nonreal quartet $\frac{1}{2} \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, of common order m , and set $\ell_k = 1 + k(m-1)$. Its cyclic annihilator has $q = \ell_k(k+1)^2$. The complete exterior trace calculation gives $L_k \leq \epsilon$, where

$$L_k = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \quad (\text{CV.7})$$

Substitution in (CV.4), with all quantities nonnegative, proves the exact necessary arithmetic-volume inequality

$$\boxed{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N \geq \frac{\omega_N}{\omega_{N+1}} D_N L_k^2, \quad N \geq \ell_k(k+1)^2 - 1.} \quad (\text{CV.8})$$

The cross term has been transferred to the actual omitted-direction determinant by (CV.3); it has not been discarded. Thus (CV.4) supplies an exact expression for the still unestimated quantity

$$\frac{\epsilon^2}{k^6} = \frac{\omega_{N+1}}{k^6 \omega_N} \frac{D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N}{D_N}. \quad (\text{CV.9})$$

Every term comes from the specified arithmetic convolution measure and the complete quotient. No decay rate for (CV.9) is asserted here. The formula identifies which signed determinant combination, and which orthogonal-norm ratio at growing degree, the next analytic calculation has to estimate together. For a larger reflection-stable packet containing this quartet, the quartet sums remain present and their full nilpotent lengths are at least ℓ_k . Indeed the same ordered quartet factors embed as CRT summands of the larger tensor algebra, and the largest nilpotent length at a fixed sum is a maximum over all such summands. Their positive trace contribution is therefore at least (CV.7). Inequality (CV.8) remains valid, with every K_j, D_j, ω_j formed from that larger packet and the admission threshold replaced by its actual $N \geq \deg \chi_{h,k} - 1$. No equality for that larger degree is inferred from the quartet subpacket.

33.2 The positive exterior minor sum and its actual source vector

Write $\Delta_N = D_{N+1} + D_{N-1} - D_N - \widetilde{D}_N$. For an increasing finite index set J , let B_J be the matrix whose columns are the original b_j , in that increasing order, and put $\Omega_J = \prod_{j \in J} \omega_j$, with the empty

product equal to one. For every $q \geq 2$ and $N \geq q - 1$,

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i}. \quad (\text{CV.10})$$

For $q = 2$ the index set I is empty and the sum has its single displayed two-column term. For $q = 1$ set this sum to zero; then $\Delta_N = 0$ exactly. Thus the signed combination in (CV.4) is an explicit sum of positive exterior contributions, retaining every original column and squared polynomial norm.

Proof. First take any finite set L of column indices and form $K_L = \sum_{j \in L} b_j b_j^* / \omega_j$. Multilinearity of its q columns expands its determinant as

$$\det K_L = \sum_{(j_1, \dots, j_q) \in L^q} \det[b_{j_1}, \dots, b_{j_q}] \prod_{r=1}^q \frac{\overline{(b_{j_r})_r}}{\omega_{j_r}}.$$

If two indices coincide, the determinant in that summand is zero. For a fixed increasing set $J = \{i_1 < \dots < i_q\}$, the remaining terms are indexed by its permutations. Their sum is

$$\frac{\det B_J}{\Omega_J} \sum_{\sigma \in \mathfrak{S}_q} \text{sgn}(\sigma) \prod_{r=1}^q \overline{(b_{i_{\sigma(r)}})_r} = \frac{|\det B_J|^2}{\Omega_J}.$$

This proves directly, including singular K_L , $\det K_L = \sum_{J \subseteq L, |J|=q} |\det B_J|^2 / \Omega_J$. Apply it to the four index sets

$$\{0, \dots, N+1\}, \quad \{0, \dots, N-1\}, \quad \{0, \dots, N\}, \quad \{0, \dots, N-1, N+1\}.$$

For each q -element subset of the first set, the coefficients in the signed combination defining Δ_N are respectively $1 + 1 - 1 - 1 = 0$ if it contains neither N nor $N+1$, $1 - 1 = 0$ if it contains exactly one of them, and 1 if it contains both. The surviving subsets have exactly the form $I \cup \{N, N+1\}$ in (CV.10). If $q = 1$, no subset contains both indices and the result is zero. No inverse for a smaller or omitted-direction Gram matrix has occurred, so this proof also applies at $N = q - 1$. $\square$

There is a precise source map for the positive sum. Let $e_C = [1] \wedge [S] \wedge \dots \wedge [S^{q-1}]$ be the fixed coordinate wedge in $\bigwedge^q C$, of squared norm one for the coordinate Euclidean pairing used in (CV.6). For $q \geq 2$ define

$$\mathcal{E}_N = \left(\bigwedge_{i=1}^{q-2} \mathcal{P}_{N-1} \right) \wedge p_N \wedge p_{N+1} \subseteq \bigwedge^q \mathcal{P}_{N+1}, \quad \Lambda_N = \left(\bigwedge^q T_{N+1} \right) |_{\mathcal{E}_N}.$$

The increasing vectors $e_I = (\bigwedge_{i \in I} p_i) \wedge p_N \wedge p_{N+1}$ form an orthogonal basis of $\mathcal{E}_N$ with their original squared norms $\Omega_{I \cup \{N, N+1\}}$. They satisfy $\Lambda_N e_I = \det B_{I \cup \{N, N+1\}} e_C$. Consequently the adjoint in these explicitly specified source and target pairings gives the original, unscaled vector

$$\zeta_N := \Lambda_N^* e_C = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\Omega_{I \cup \{N, N+1\}}} e_I, \quad (\text{CV.10a})$$

$$\|\zeta_N\|^2 = \Delta_N, \quad \Lambda_N \zeta_N = \Delta_N e_C, \\ \|\mathcal{V}_{h,k}^{\otimes q} \text{Alt}_q \zeta_N\|^2 = q! \Delta_N.$$

The first two equalities follow by inserting the displayed orthogonal basis and its full norms. The last equality follows from the pairwise isometry of $\mathcal{V}_{h,k}$ and the unscaled alternating-map calculation following (CV.6). Its target is the original tensor source $\mathcal{B}^{\widehat{\otimes} kq}$ in degree kq . For $q = 1$ take $\mathcal{E}_N = \{0\}$, $\Lambda_N = 0$, and $\zeta_N = 0$, so that all three zero identities remain exact. Thus (CV.10) comes with an explicitly typed original arithmetic source vector and the literal tensor factorial. Combining it with (CV.4) gives $\epsilon^2 = (\omega_{N+1} / \omega_N) \|\zeta_N\|^2 / D_N$; no estimate as k or N grows is inferred from this equality.

33.3 The exact first admitted degree

At $N = q - 1$ expand the original monic annihilator in the original orthogonal basis:

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\int \overline{p_j(k/2 + iu)} \chi(k/2 + iu) m_{h,k}(u) du}{\omega_j}. \quad (\text{CV.11})$$

This is a finite polynomial identity: both leading coefficients are one, and orthogonality gives every displayed coefficient, without a change of measure. The reduction map T_{q-1} is a linear isomorphism, since the q original monic remainders are a basis. Its inverse is therefore its canonical minimum section. Consequently $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $0 \leq i, j < q$. Reduction of (CV.11) gives $b_q = -\sum_{j < q} \gamma_j b_j$ and hence

$$a = \omega_{q-1}, \quad c = -\gamma_{q-1} \omega_{q-1}, \quad d = \sum_{j < q} |\gamma_j|^2 \omega_j, \quad (\text{CV.12})$$

$$\boxed{\epsilon_{h,k,q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\|\chi\|^2 - \omega_q - |\gamma_{q-1}|^2 \omega_{q-1}}{\omega_{q-1}}}.$$

The first expression follows by substituting the preceding three exact Gram entries into (CV.4); the second follows by Pythagoras applied to (CV.11). Reflection makes c purely imaginary, so the same is true of γ_{q-1} ; it has not been set to zero. Let $\mathcal{P}_{q-2}$ be the original integral orthogonal projection onto $\mathcal{P}_{q-2}$, with $\mathcal{P}_{-1} = 0$ for $q = 1$. Then the numerator in (CV.12) is exactly $\|\mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi\|^2$ by the original source isometry. Meanwhile the full $\mathcal{V}_{h,k} \chi$ has zero cyclic and full jets, since $\chi(S) \in I$, and the original fixed-order polynomial division gives its theta primitive as in (CM.19). These are different specified source maps: projecting that boundary is an integral orthogonal operation and has the cyclic reduction $T_{q-2}(\mathcal{P}_{q-2} \chi) = \sum_{j < q-1} \gamma_j b_j$, where T_{q-2} is the restriction of T_{q-1} to $\mathcal{P}_{q-2}$; its actual full arithmetic jet is $J^{(k)} \mathcal{V}_{h,k} \mathcal{P}_{q-2} \chi = \eta(\sum_{j < q-1} \gamma_j b_j)$. For $q = 1$ the first map is the unique map from $\mathcal{P}_{-1} = 0$. This exact formula retains its possibly nonzero observation. For $q = 1$ the sum is empty and both sides are zero. The degree- q annihilator and its projection have thus given a complete original-source formula at the first admitted degree, with no large-degree estimate added.

34 The original source and relation Toda flows with their exact control maps

This is a standalone continuation of the identities CV.1–CV.12. It retains the original polynomial quotient, arithmetic measure, monic coordinates, arithmetic unit, source relations and complex phase. The delivered `Tau_Toda_Volume_Control/NOTE.tex` was read in full from its original archive; its SHA256 is

027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375.

The calculations below prove the exact maps underlying the determinant quotient, its first and second tilt derivatives, the omitted-direction determinant, the positive minor sum and the finite upper bound. No asymptotic arithmetic estimate is asserted.

34.1 Original arithmetic objects and the strip of real tilts

The original spaces and transforms are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ is even, } \phi(0) = \int_{\mathbb{R}} \phi(x) dx = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for every } b \in \mathbb{Z}, j \geq 0\}, \\ D &= -x\partial_x, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{M}F(s) &= \int_0^\infty F(x) x^{s-1} dx. \end{aligned}$$

The series and its Euler derivatives converge uniformly at large x with every power of decay. For small x , Poisson summation with Fourier kernel $e^{-2\pi ixy}$ gives $\Theta\phi(x) = x^{-1}\Theta\hat{\phi}(1/x)$; both zero terms vanish because $\phi(0) = \hat{\phi}(0) = 0$. The Fourier transform preserves the stated two zero constraints and evenness. This formula proves every small- x seminorm after Euler differentiation, hence $\Theta : V \rightarrow \mathcal{B}$. It also commutes with D on these spaces by termwise differentiation. Mellin transforms of elements of $\mathcal{B}$ are entire, and integration by parts gives $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$; both endpoint terms vanish with their original coefficients.

Let h be a monic, nonempty packet of actual nontrivial zeros of

$$g(s) = 2\zeta(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

with their complete orders. Assume the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Set $d = \deg h$ and retain

$$v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = [v_h]_h \in E_h^\times, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}. \quad (\text{TVB.1})$$

Here the jet of an entire function f means its unique polynomial Hermite remainder of degree less than d : in the local factor $\mathbb{C}[t]/(t^{m_\rho})$, it is $\sum_{j < m_\rho} f^{(j)}(\rho) t^j / j!$, with $t = s - \rho$. The coprime factors of h give the isomorphism to their product by monic division and Bezout's identity, which supplies the unique global remainder. The constant $v_h(\rho)$ is nonzero at each selected zero because its full order was divided out. If the local jet is $\sum a_j t^j$, its inverse coefficients are $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$ for $n < m_\rho$. Multiplication verifies this inverse in every local factor, and the product isomorphism gives the inverse of v_h .

The original F_h has a construction within $\mathcal{B}$. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, define

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy.$$

The equality is the vanishing Mellin value. For any desired power bound at infinity choose $A > \Re \rho$ with $|F(y)| \leq C_A y^{-A}$ for $y \geq 1$; the first integral is bounded by $C_A x^{-A}/(A - \Re \rho)$ there. At zero choose B with $B + \Re \rho > 0$ and $|F(y)| \leq C_B y^B$ for $0 < y \leq 1$; the second integral is bounded by $C_B x^B/(B + \Re \rho)$. The exact identity $(D - \rho)S_\rho F = F$ now gives all Euler derivative bounds inductively, so $S_\rho F \in \mathcal{B}$. Mellin integration by parts proves $\mathcal{M}(S_\rho F) = \mathcal{M}F/(s - \rho)$, with its removable value at $s = \rho$ supplied by the derivative of the numerator. For the original seed $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2} \in V$, direct substitution $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2), \quad \Re s > -2.$$

For $\Re s > 1$, absolute convergence permits the two-sided theta sum to be integrated termwise, yielding $\mathcal{M}(\Theta\phi_0) = 2\zeta(s)\mathcal{M}\phi_0(s) = g(s)$; both sides continue to the same entire function. Apply S_ρ successively to $\Theta\phi_0$ over the full list of roots of h , including each repeated occurrence. At every stage the next vanishing Mellin value holds by its remaining order. This constructs $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = g/h$

and $h(D)F_h = \Theta\phi_0$; the latter also follows by Mellin injectivity, which is Fourier injectivity after $x = e^y$. Fix $k \geq 1$, set $c = k/2$, $S = s_1 + \cdots + s_k$, and put $I = (h(s_1), \dots, h(s_k)) \subset \mathbb{C}[s_1, \dots, s_k]$. Let χ be the original monic generator of $\{P \in \mathbb{C}[S] : P(S) \in I\}$. Its degree is $q \geq 1$ and

$$C = \mathbb{C}[S]/(\chi), \quad A = M_S : C \longrightarrow C. \quad (\text{TVB.2})$$

The coordinate basis of C is always $1, S, \dots, S^{q-1}$. The polynomial exists because multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$ has an annihilating polynomial. Division by its least-degree monic annihilator proves that its entire annihilating ideal is (χ) . In particular the injection

$$\eta : C \longrightarrow (E_h^{\otimes k})^{\mathfrak{S}_k}, \quad [P] \longmapsto v_h^{\otimes k}[P(S)]_I \quad (\text{TVB.3})$$

has exactly zero kernel: multiplication by the displayed unit is invertible and the preceding annihilator ideal describes the other factor's kernel. The original tensor source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}, \quad D = -x\partial_x, \quad \mathcal{M}F_h = v_h.$$

On the original Mellin line its scalar observation has squared norm

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c + iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad \int m_k = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{TVB.4})$$

Indeed the tensor Mellin–Plancherel density is $\prod_i w_h(t_i) d^k t$. In coordinates $u = \sum_i t_i$, $y = (t_1, \dots, t_{k-1})$, the inverse is $(y, u) \mapsto (y_1, \dots, y_{k-1}, u - \sum y_i)$. With (y, u) in this order its determinant is 1; with (u, y) its determinant is $(-1)^{k-1}$. Integrating the original density over y gives (TVB.4), with no scalar divisor.

For every compact substrip $|\Re\theta| \leq r < \pi/2$, all integrals

$$M_h(\theta) = \int e^{\theta t} w_h(t) dt, \quad Z(\theta) = M_h(\theta)^k = \int e^{\theta u} m_k(u) du \quad (\text{TVB.5})$$

and all their polynomially weighted derivatives converge locally uniformly. Here is an elementary sufficient bound. Summation by parts for the alternating Dirichlet series, whose partial sums have modulus at most one, gives $|\eta(s)| \leq |s|/\Re s$ when $\Re s > 0$. On $\Re s = 1/2$, the denominator in $\zeta(s) = \eta(s)/(1 - 2^{1-s})$ has modulus at least $\sqrt{2} - 1$. For $0 < \varphi < \pi/2$, rotate the defining gamma integral to the ray with argument $\text{sgn}(t)\varphi$. The large arc vanishes by $e^{-R\cos\varphi}$ and the small arc by the positive real part $1/4$ of its exponent. Its absolute value is bounded by

$$|\Gamma(1/4 + it/2)| \leq \Gamma(1/4)(\cos\varphi)^{-1/4} e^{-\varphi|t|/2}.$$

For sufficiently large $|t|$, the monic polynomial h satisfies $|h(1/2 + it)| \geq C_h(1 + |t|)^d$ with $C_h > 0$. The two factors $s, s - 1$, the preceding zeta bound and the gamma bound consequently give

$$w_h(t) \leq C_{h,\varphi}(1 + |t|)^{6-2d} e^{-\varphi|t|}$$

outside a compact interval. On that interval v_h is entire, since the selected orders are complete. Choose $r < \varphi < \pi/2$ and absorb every fixed polynomial in the remaining exponential decay. This proves the asserted domination and differentiation under the integral. Fubini gives the exact product M_h^k in (TVB.5). The nonzero entire function v_h has a discrete real-line zero set; therefore $w_h > 0$ almost everywhere. Its convolutions are positive almost everywhere (in fact at every u for $k \geq 2$), since their fibre integrands are positive off a finite union of measure-zero inverse images of this discrete set. Thus the norm of every nonzero polynomial is positive.

In what follows θ is real, $|\theta| < \pi/2$, and the source pairing, conjugate-linear in its first argument, is exactly

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (\text{TVB.6})$$

The map $f(u) \mapsto e^{\theta u/2} f(u)$ is an onto isometry from $L^2(e^{\theta u} m_k du)$ to $L^2(m_k du)$, with inverse $f \mapsto e^{-\theta u/2} f$. It commutes with multiplication by $c + iu$. This specifies the exact Hilbert comparison for the tilt. The polynomial algebra, its reduction and η are unchanged. When the target of $\mathcal{V}$ is observed at this tilt, its declared tensor norm is

$$\|F\|_{k,\theta}^2 = \int_{\mathbb{R}^k} e^{\theta \sum_i t_i} |\mathcal{M}_{\text{unit}} F(t_1, \dots, t_k)|^2 d^k t,$$

where $\mathcal{M}_{\text{unit}}$ is the original Mellin–Plancherel map with a factor $(2\pi)^{-1/2}$ in each coordinate. The map $f(t) \mapsto e^{\theta \sum_i t_i/2} f(t)$ is an onto isometry from this weighted Mellin Hilbert space to the original one, with the opposite exponential as inverse. The restriction to $\mathcal{V}P$ is exactly (TVB.6), by the same sum-fibre integral as (TVB.4). Thus every tilted exterior norm below has a specified target norm; at $\theta = 0$ it is the original tensor-source norm.

34.2 The source relation and its Laplace observation

Write $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, with $\mathcal{P}_{-1} = 0$. For $N \geq q - 1$, put $n = N + 1$, $m = N - q + 1$. Literal monic division gives the exact sequence

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{T_N} C \longrightarrow 0, \quad M_\chi Q = \chi Q, \quad T_N P = [P]_\chi. \quad (\text{TVB.7})$$

Its kernel is precisely the indicated image, and its first q monomial columns prove surjectivity. Denote that kernel by $\mathcal{D}_N = \chi \mathcal{P}_{m-1}$.

This sequence has an explicit map to the original theta boundaries. To see every sign, choose the quotients Q_i from successive monic division, in the order $s_1, \dots, s_k$, so that

$$\chi(S) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{TVB.8})$$

The remainder is zero by the definition of χ and uniqueness of the tensor remainder basis. Put $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$. It is even, vanishes at zero and has integral zero. Direct gamma integration of its two terms gives $\mathcal{M}(\Theta \phi_0) = g$, hence $h(D)F_h = \Theta \phi_0$ by Mellin injectivity on the original rapidly decreasing source. For every polynomial P , an actual primitive of $\mathcal{V}(\chi P)$ in the tensor complex $(V \xrightarrow{\Theta} \mathcal{B})^{\otimes k}$ is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{TVB.9})$$

The preceding $i - 1$ factors have cochain degree one. The tensor differential therefore contributes $(-1)^{i-1}$ at the i th factor, cancelling the written sign and yielding (TVB.8). Euler differentiation preserves the source conditions and commutes with Θ , so all terms have the indicated domains. Conversely the full jet of $\mathcal{V}P$ is $\eta T_N P$; injectivity in (TVB.3) shows that the admitted polynomial source has no further zero-jet elements. In particular every boundary vector in (TVB.7) retains its positive source norm before the quotient sends it to represented zero.

For any polynomial P , define its Laplace observation by

$$\mathcal{L}(P)(\theta) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Differentiation under (TVB.5) proves the typed intertwining $\mathcal{L}(SP) = (c + i\partial_\theta) \mathcal{L}(P)$. Coefficient-wise conjugation gives the exact Gram observation

$$Z_\chi = \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0. \quad (\text{TVB.10})$$

Thus the two signs and both polynomial factors arise from $\langle M_\chi 1, M_\chi 1 \rangle_\theta$. This is a map on source functions, with no replacement of the higher contracted ideal by a power of χ .

34.3 Both monic determinants and their exact determinant-line map

Set

$$\mathfrak{D}_j = \det[Z^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{B}_j = \det[Z_\chi^{(a+b)}]_{a,b=0}^{j-1}, \quad \mathfrak{D}_0 = \mathfrak{B}_0 = 1. \quad (\text{TVB.11})$$

All nonempty determinants are positive by polynomial positivity. The coefficient map from $1, S, \dots, S^{j-1}$ to $1, u, \dots, u^{j-1}$ is explicitly

$$S^a = (c + iu)^a = \sum_{b=0}^a \binom{a}{b} c^{a-b} i^b u^b. \quad (\text{TVB.12})$$

Its determinant is $i^{j(j-1)/2}$ and its Gram determinant multiplier is exactly one. Therefore $\mathfrak{D}_j$ is also the original S -monomial Gram determinant. The same calculation applies to the relation-weight pairing $\langle P, Q \rangle_{\chi, \theta} = \langle \chi P, \chi Q \rangle_\theta$ and $\mathfrak{B}_j$. This proves the coordinate correspondence without changing the original monic polynomial.

Let $p_j(S; \theta)$ be the monic source orthogonal polynomials, and $r_j(S; \theta)$ the monic polynomials for this relation-weight pairing. Retain their original squared norms

$$\omega_j = \|p_j\|_\theta^2 = \mathfrak{D}_{j+1}/\mathfrak{D}_j, \quad \nu_j = \|\chi r_j\|_\theta^2 = \mathfrak{B}_{j+1}/\mathfrak{B}_j. \quad (\text{TVB.13})$$

The equations follow by triangular monic Gram elimination, whose diagonal change is one. In particular $\omega_0 = M_h(\theta)^k$. The full mass has not been divided out.

Give C its written coordinate Euclidean pairing only when specifying adjoints. Let M_N be the original monomial source Gram, and J_N its reduction matrix. The minimum lift is

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = M_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N. \quad (\text{TVB.14})$$

Here the last expression specifies polynomial coefficients. It satisfies $T_N R_N = I$, $\text{im } R_N \perp \mathcal{D}_N$ and $R_N^* R_N = G_N$ in the original source pairing. Indeed the first equality is matrix multiplication, and pairing with any vector in $\ker J_N$ gives zero. Every other lift is $R_N x + B$ for $B \in \mathcal{D}_N$, so its squared norm is $x^* G_N x + \|B\|^2$. This proves the minimum assertion and uniqueness.

Use the source basis $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$. Its change from the n source monomials is triangular with diagonal one. Replacing its first q columns by $R_N[1], \dots, R_N[S^{q-1}]$ adds boundary columns and again has determinant one. The resulting Gram is the direct sum of G_N and the relation monomial Gram. Consequently

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_m}, \quad D_N^{\text{CV}} := \det K_N = V_N^{-1}. \quad (\text{TVB.15})$$

More precisely the map $\det C \otimes \det \mathcal{P}_{m-1} \rightarrow \det \mathcal{P}_N$ sends the written coordinate wedges to the wedge of these quotient lifts followed by M_χ of the relation monomials; its coordinate determinant is 1. Putting the kernel wedge first instead contributes the explicit sign $(-1)^{qm}$. Its norm identity is $\det G_N \mathfrak{B}_m = \mathfrak{D}_n$. Composing the wedge with $\mathcal{V}^{\otimes n} \text{Alt}_n$ multiplies its squared norm by exactly $n!$: expanding the unscaled alternating tensor sum gives $n!$ copies of the Gram determinant. The original tensor target is $\mathcal{B}^{\widehat{\otimes} kn}$ in cochain degree kn .

34.4 Toda derivatives and the full curvature operators

For either positive real u -weight in (TVB.11), write $Q_j(u)$ for its real monic orthogonal polynomials and h_j for their norms. Orthogonality and leading coefficients prove

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1} \ (j \geq 1), \quad a_0 = 0.$$

For $i \leq j-2$, differentiating $\langle Q_i, Q_j \rangle = 0$ shows that $\partial_\theta Q_j$ is orthogonal to Q_i , because uQ_i and $\partial_\theta Q_i$ have degree at most $j-1$. For $i = j-1$ the same identity gives its coefficient $-a_j$. Thus

$\partial_\theta Q_j = -a_j Q_{j-1}$ and $h'_j = b_j h_j$. To compute b'_j , differentiate $h_j b_j = \langle Q_j, u Q_j \rangle$. The two polynomial derivative terms contribute $-2a_j h_j$. The derivative of the measure contributes $\|u Q_j\|^2 = h_j(a_{j+1} + b_j^2 + a_j)$. After subtracting $h'_j b_j$ this gives $b'_j = a_{j+1} - a_j$. The monic Gram determinant is $\prod_{j < n} h_j$, hence

$$(\log \mathfrak{D}_n)'' = \frac{\omega_n}{\omega_{n-1}} \quad (n \geq 1), \quad (\log \mathfrak{B}_m)'' = \frac{\nu_m}{\nu_{m-1}} \quad (m \geq 1). \quad (\text{TVB.16})$$

Both 0th logarithms have second derivative zero. Multiplication by the square of the determinant proves each Hankel-Toda equation with coefficient one. Set

$$\alpha_{N+1} = \omega_{N+1}/\omega_N, \quad \beta_m = \nu_m/\nu_{m-1} \quad (m \geq 1), \quad \beta_0 = 0, \quad \ell_N = \log V_N.$$

Then $\ell''_N = \alpha_{N+1} - \beta_m$.

There is a finer exact source-map identity. Let $X = (S - c)/i$, which is multiplication by the real variable u in (TVB.6). Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ to $\mathcal{D}_N$ in that pairing. Differentiating $T_N R_N = I$ gives $R'_N x \in \mathcal{D}_N$. Differentiating orthogonality against each fixed boundary polynomial gives

$$R'_N = -P_{\mathcal{D}_N} X R_N =: -I_N, \quad O_N := (1 - P_N) X R_N. \quad (\text{TVB.17})$$

Every derivative is an actual boundary with the primitive (TVB.9). Differentiating its norm, including the derivative of the measure, gives

$$G'_N = R_N^* X R_N, \quad G''_N = R_N^* X^2 R_N - 2R_N^* X P_{\mathcal{D}_N} X R_N.$$

The projection to $\text{im } R_N$ is $R_N G_N^{-1} R_N^*$ and, together with $P_{\mathcal{D}_N}$, sums to P_N . Subtracting the square of the first derivative therefore proves

$$G''_N - G'_N G_N^{-1} G'_N = O_N^* O_N - I_N^* I_N. \quad (\text{TVB.18})$$

This is an equality of maps $C \rightarrow C^{\text{anti}*}$, where $C^{\text{anti}*}$ consists of conjugate-linear functionals on C . For example $O_N^* O_N$ sends x to the functional $y \mapsto \langle O_N y, O_N x \rangle_\theta$; this assignment is complex-linear in x . The fixed coordinate Euclidean pairing identifies that antidual with the written matrix coordinates.

Write $b_j = [p_j]_\chi$, and $a = b_N^* G_N b_N$, $d_+ = b_{N+1}^* G_N b_{N+1}$. The canonical next boundary has the particularly exact factorization

$$\mathfrak{b}_N := p_{N+1} - R_N b_{N+1} = \chi r_m, \quad \nu_m = \omega_{N+1} + d_+. \quad (\text{TVB.19})$$

To prove it, χr_m is monic of degree $N + 1$ and orthogonal to $\mathcal{D}_N$. Subtract p_{N+1} ; the difference lies in $\mathcal{P}_N \cap \mathcal{D}_N^\perp$ and has reduction $-b_{N+1}$. Uniqueness of R_N proves the first equality; the two terms in its definition are orthogonal, giving the second. The coefficient of p_N in $R_N x$ is $b_N^* G_N x / \omega_N$. The leading coefficient of X as an S -polynomial is $1/i = -i$. Consequently the full maps are

$$O_N x = -i p_{N+1} \frac{b_N^* G_N x}{\omega_N}, \quad I_N x = -i \chi r_{m-1} \frac{b_{N+1}^* G_N x}{\nu_{m-1}} \quad (m \geq 1), \quad I_N = 0 \quad (m = 0). \quad (\text{TVB.20})$$

For the second equation, $X \chi r_j$ is an old boundary if $j \leq m - 2$. For $j = m - 1$ its coefficient of χr_m is $-i$. Moving X across the pairing conjugates this coefficient to i , and $\langle \chi r_m, R_N x \rangle = -b_{N+1}^* G_N x$ by (TVB.19). Division by the original norm ν_{m-1} gives exactly the second sign in (TVB.20).

Both maps have rank at most one. For $N \geq q$, put $\delta_N = V_N/V_{N-1}$. The rank-one calculation in the next section gives $a/\omega_N = 1 - \delta_N$ and $\delta_N = \omega_N/\nu_{m-1}$. Taking traces in (TVB.20) gives the separate, positive source costs

$$\begin{aligned} \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1} (1 - \delta_N), \\ \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N = \frac{\nu_m - \omega_{N+1}}{\nu_{m-1}}. \end{aligned} \quad (\text{TVB.21})$$

Their difference is $\alpha_{N+1} - \beta_m$ as required. Thus the individual positive map costs each retain the common quantity $\alpha_{N+1}\delta_N$; assigning them separately the two Toda coefficients would lose this quantity. At $N = q - 1$, the outer cost is α_q and the inner cost is zero. If a single notation for the cost identities is wanted at this endpoint, define only the scalar convention $\delta_{q-1}^{\text{end}} := \det K_{q-2} / \det K_{q-1} = 0$; the included column has leverage $a/\omega_{q-1} = 1$. Together with $\beta_0 = 0$ the cost identities are $\alpha_q(1 - \delta_{q-1}^{\text{end}}) = \alpha_q$ and $\beta_0 - \alpha_q\delta_{q-1}^{\text{end}} = 0$. This convention is not a quotient involving V_{q-2} and does not introduce an inverse for K_{q-2} or a norm ν_{-1} . All statements concern the original polynomial source and its original boundary subspace.

34.5 The rank-one update, trace phase and original rank-two radius

The source-orthogonal basis gives

$$K_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad K_{N+1} = K_N + \frac{b_{N+1} b_{N+1}^*}{\omega_{N+1}}. \quad (\text{TVB.22})$$

The first q monic remainder columns are a triangular basis; K_N is positive definite precisely in the admitted range. Block elimination of $\begin{pmatrix} K & U \\ -V^* & I \end{pmatrix}$ in either order proves $\det(K + UV^*) = \det K \det(I + V^* K^{-1} U)$. Applying it to the positive update and the subtraction of b_N from K_N yields, for $N \geq q$,

$$\frac{a}{\omega_N} = 1 - \delta_N, \quad \frac{d_+}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1, \quad \delta_N = \frac{\omega_N}{\nu_{m-1}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_m}. \quad (\text{TVB.23})$$

The last two identities also follow directly from (TVB.15). Each δ is positive and at most one, since increasing the source space weakly decreases every minimum squared norm. The exact inverse update is

$$G_{N+1} = G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + d_+}. \quad (\text{TVB.24})$$

Multiplying this expression by K_{N+1} proves it directly; the denominator is $\nu_m > 0$ by (TVB.19).

Reflection of the tensor algebra sends S to $k - \bar{S}$ and preserves its annihilator ideal. Monicity gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic. The exact algebra isomorphism

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P(S)] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q(u)] = [Q((S - c)/i)] \quad (\text{TVB.25})$$

transports the original ideal by its leading unit i^q . Its coefficient matrix E is (TVB.12) in dimension q . Thus A transports to $cI + iT$, where $T = M_u$ in the real quotient. Its Gram transports to $E^{-*} G_N E^{-1}$, its kernel to $E K_N E^*$, and $\det G_N$ is unchanged because $|\det E| = 1$. Write

$$\sigma = \text{Tr } T = \frac{\text{Tr } A - cq}{i} \in \mathbb{R}, \quad \psi_N = \sigma - \ell'_N.$$

Reality of $\hat{\chi}$, the weight and its monic orthogonal polynomials makes all the u -coordinate source and quotient Grams real. Let $Q_j(u) = i^{-j} p_j(c + iu)$ and $d_j = [Q_j]$. Then $E b_j = i^j d_j$ and both source norms are ω_j .

The leading coefficient of the u -polynomial minimum lift of x is $d_N^* G_N x / \omega_N$, where in this calculation G_N denotes the transported real-coordinate Gram. The difference $X R_N - R_N T$ has zero reduction, and subtracting $(Q_{N+1} - R_N d_{N+1}) d_N^* G_N / \omega_N$ cancels its highest coefficient. Its remainder is an old boundary. Pair with R_N ; old boundaries vanish and $R_N^* (Q_{N+1} - R_N d_{N+1}) = -G_N d_{N+1}$. Taking the trace after multiplication by G_N^{-1} proves

$$\ell'_N = \sigma - \frac{d_N^* G_N d_{N+1}}{\omega_N}, \quad \frac{b_N^* G_N b_{N+1}}{\omega_N} = i \psi_N. \quad (\text{TVB.26})$$

The second equation is back in the fixed original S coordinates: the phase is $\overline{i^N} i^{N+1} = i$. Both signs are retained.

For completeness, the u recurrence proved above transports to $Sp_j = p_{j+1} + (c + ib_j^{\text{rec}})p_j - \alpha_j p_{j-1}$ in the original source coordinates. Inserting this in (TVB.22) telescopes all interior terms and gives

$$AK_N + K_N A^* - kK_N = \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}.$$

For $W_N = A^*G_N + G_N A - kG_N$, multiplication on both sides by G_N gives the corresponding identity for W_N . Its Hermitian relative operator is $H_N = G_N^{-1/2}W_N G_N^{-1/2}$. It has rank at most two and trace $2\Re \text{Tr } A - kq = 0$. The sum of the squares of its eigenvalues is $2(ad_+ - |z_N|^2)/\omega_N^2$, where $z_N = b_N^* G_N b_{N+1} = i\omega_N \psi_N$. This last assertion follows by expanding the square of $(xy^* + yx^*)/\omega_N$ for $x = G_N^{1/2}b_N, y = G_N^{1/2}b_{N+1}$; its trace is $[2ad_+ + 2\Re(z_N^2)]/\omega_N^2$. Its nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \frac{ad_+ - |z_N|^2}{\omega_N^2} = \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2. \quad (\text{TVB.27})$$

The first expression is valid already at $N = q - 1$.

34.6 Exact maps to the CV omitted determinant and positive minors

Retain $D_j^{\text{CV}} = \det K_j$, put $K_{-1} = 0$, and form

$$\tilde{K}_N = K_{N-1} + \frac{b_{N+1}b_{N+1}^*}{\omega_{N+1}}, \quad \tilde{D}_N = \det \tilde{K}_N, \quad \Delta_N = D_{N+1}^{\text{CV}} + D_{N-1}^{\text{CV}} - D_N^{\text{CV}} - \tilde{D}_N.$$

This is the actual quotient kernel of the source with p_N omitted and p_{N+1} included. No inverse of it is used. The two-column determinant formula preceding (TVB.23), with one negative and one positive update, proves

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} = \left(1 - \frac{a}{\omega_N}\right) \left(1 + \frac{d_+}{\omega_{N+1}}\right) + \frac{|z_N|^2}{\omega_N \omega_{N+1}}, \quad \epsilon_N^2 = \alpha_{N+1} \frac{\Delta_N}{D_N^{\text{CV}}}. \quad (\text{TVB.28})$$

In particular, for $N \geq q$, the complete phase map is the exact nonnegative difference

$$\frac{\tilde{D}_N}{D_N^{\text{CV}}} - \frac{\delta_N}{\delta_{N+1}} = \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.29})$$

The sign of ψ_N is retained by the first derivative in (TVB.26); the omitted determinant observes its square.

For any finite column set L , determinant multilinearity gives

$$\det \left(\sum_{j \in L} \frac{b_j b_j^*}{\omega_j} \right) = \sum_{\substack{J \subseteq L \\ |J|=q}} \frac{|\det B_J|^2}{\prod_{j \in J} \omega_j}, \quad B_J = [b_j]_{j \in J \text{ increasing}}.$$

Terms with repeated indices vanish. The $q!$ remaining permutations of a fixed increasing index set yield its determinant times its conjugate, which proves the displayed identity including singular kernels. Apply it to the four column sets defining Δ_N . A set containing neither of the two last indices has coefficient $1 + 1 - 1 - 1 = 0$; one containing exactly one has coefficient $1 - 1 = 0$; one containing both has coefficient one. Hence

$$\Delta_N = \sum_{\substack{I \subseteq \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det B_{I \cup \{N, N+1\}}|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \quad (\text{TVB.30})$$

For $q = 1$ it is zero. This proof applies at $N = q - 1$. For $q \geq 2$ define the actual source subspace $\mathcal{E}_N = (\bigwedge^{q-2} \mathcal{P}_{N-1}) \wedge p_N \wedge p_{N+1}$ and $\Lambda_N = (\bigwedge^q T_{N+1})|_{\mathcal{E}_N}$. For the fixed coordinate wedge $e_C = 1 \wedge S \wedge \cdots \wedge S^{q-1}$, whose coordinate Euclidean squared norm is one, the adjoint is

$$\zeta_N := \Lambda_N^* e_C = \sum_{|I|=q-2} \frac{\overline{\det B_{I \cup \{N, N+1\}}}}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \left(\bigwedge_{i \in I} p_i \right) \wedge p_N \wedge p_{N+1}. \quad (\text{TVB.31})$$

The displayed source wedges are orthogonal with the written full product norms. Therefore $\|\zeta_N\|_\theta^2 = \Delta_N$ and $\Lambda_N \zeta_N = \Delta_N e_C$. At $\theta = 0$ its literal original tensor-source image has squared norm $q! \Delta_N$ under $\mathcal{V}^{\otimes q} \text{Alt}_q$. At other real tilts use the onto Hilbert comparison after (TVB.6) on each scalar source factor; the same determinant norm and factorial hold in that specified tilted observation. For $q = 1$ set $\mathcal{E}_N = 0$, $\Lambda_N = 0$, $\zeta_N = 0$.

34.7 Both exact losses, their equality cases and the telescope

For $N \geq q$ define

$$v_N = -\log \delta_N, \quad s_N = (v_N + v_{N+1})/2, \quad t_N = (v_{N+1} - v_N)/2, \quad \mathcal{R}_N = e^{2s_N} = \frac{V_{N-1}}{V_{N+1}} \geq 1.$$

Expanding $(1 - e^{-v_N})(e^{v_{N+1}} - 1)$ gives $2e^{t_N} \cosh s_N - e^{2t_N} - 1$. Completing this scalar square, with both terms retained, yields

$$\boxed{\epsilon_N^2 = \alpha_{N+1} \sinh^2 s_N - \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 - \psi_N^2.} \quad (\text{TVB.32})$$

Thus the exact upper bound and its equality conditions are

$$\epsilon_N \leq \sqrt{\alpha_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}, \quad \text{equality} \iff \psi_N = 0 \text{ and } \delta_N(2 - \delta_{N+1}) = 1. \quad (\text{TVB.33})$$

Indeed $\alpha_{N+1} > 0$ and both lost squares in (TVB.32) are nonnegative. Multiplication of $e^{t_N} = \cosh s_N$ by $2\sqrt{\delta_N \delta_{N+1}}$ gives precisely the displayed equation. This includes the case $\mathcal{R}_N = 1$, when both contractions equal one and $\epsilon_N = \psi_N = 0$ by (TVB.27). The corresponding equality with the full positive minor source is

$$\frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N} - \frac{\|\zeta_N\|_\theta^2}{D_N^{\text{CV}}} = (e^{t_N} - \cosh s_N)^2 + \frac{\psi_N^2}{\alpha_{N+1}}. \quad (\text{TVB.34})$$

It follows from (TVB.28), (TVB.31) and (TVB.32), so its minor equality has the same two necessary and sufficient conditions. In particular equality is a fully specified source-volume equation, not a consequence of positivity alone.

There is a direct formulation in the actual volumes, without logarithmic variables. Put $U_N = \sqrt{\alpha_{N+1}}(\mathcal{R}_N - 1)/(2\sqrt{\mathcal{R}_N})$. Since

$$e^{t_N} = \frac{V_N}{\sqrt{V_{N-1} V_{N+1}}}, \quad \cosh s_N = \frac{V_{N-1} + V_{N+1}}{2\sqrt{V_{N-1} V_{N+1}}},$$

equation (TVB.32) gives the exact two losses

$$U_N^2 - \epsilon_N^2 = \alpha_{N+1} \frac{(2V_N - V_{N-1} - V_{N+1})^2}{4V_{N-1} V_{N+1}} + \frac{|z_N|^2}{\omega_N^2}. \quad (\text{TVB.34a})$$

Thus the volume equality is the arithmetic midpoint $2V_N = V_{N-1} + V_{N+1}$ together with $z_N = 0$. Equal consecutive logarithmic contractions alone do not give this midpoint: their extra equation $V_N^2 = V_{N-1} V_{N+1}$ implies it only if all three positive volumes are equal, by the square $(\sqrt{V_{N-1}} -$

$\sqrt{V_{N+1}})^2$. In the original inverse-volume coordinates $D_- = D_{N-1}^{\text{cv}}$, $D_0 = D_N^{\text{cv}}$, $D_+ = D_{N+1}^{\text{cv}}$ and $E_N = \tilde{D}_N$, the same identities become

$$\begin{aligned} E_N D_0 - D_- D_+ &= \frac{D_0^2 |z_N|^2}{\omega_N \omega_{N+1}}, \\ \frac{D_0(D_+ - D_-)^2}{4D_- D_+} - \Delta_N &= \frac{(2D_- D_+ - D_0(D_- + D_+))^2}{4D_0 D_- D_+} + E_N - \frac{D_- D_+}{D_0}. \end{aligned} \quad (\text{TVB.34b})$$

Substitute $D_j = V_j^{-1}$ in (TVB.34a), or multiply out its two squares, to verify every denominator. Both terms on the right are nonnegative by the first equation. Equality is equivalently $D_0 = 2D_- D_+ / (D_- + D_+)$ and $E_N = (D_- + D_+)/2$; it then gives $\Delta_N = E_N - D_0$. In particular phase zero at these degrees records a positive omitted-direction determinant $E_N = D_- D_+ / D_0$, with its exact map specified in (TVB.28), rather than a zero determinant.

The algebraic exterior lower bound applies at each real tilt as well as at zero. Here is a direct proof. In the Hermitian matrix H_N above let P be the orthogonal projection to the image, under $G_N^{1/2}$, of the sum of all full generalized eigenspaces of A with $\Re \lambda > c$. Invariance of that subspace makes the transformed A upper triangular in its orthogonal direct sum with the complement, without asserting that the spectral projection itself is orthogonal. Taking the trace of its first diagonal block gives

$$\text{Tr}(PH_N) = L := \sum_{\Re \lambda > c} \ell_\lambda (2\Re \lambda - k).$$

Since H_N has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros, $\text{Tr}(PH_N) \leq \epsilon_N$: its positive eigenline contributes at most ϵ_N and its negative eigenline contributes a nonpositive number. All algebraic multiplicities ℓ_λ are retained. Thus $L \leq \epsilon_N$ for the same tilted minimum metric, without choosing another quotient metric.

Combining this with (TVB.33), using monotonicity of $\sinh$ on $[0, \infty)$ and its inverse, proves

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.35})$$

For $N = N_0 + 2j$, $0 \leq j < r$, with $N_0 \geq q$, every denominator in one logarithm is the numerator in the next. Summing the exact logs and then their lower bounds proves

$$\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L}{\sqrt{\alpha_{N_0+2j+1}}} \right), \quad r \geq 1. \quad (\text{TVB.36})$$

Every term is at the same fixed real tilt, packet and tensor degree. No relation between different k or a limiting value of either determinant is used. Under $\mathcal{V}$ the minimum representatives at all degrees retain jets η and their original quotient classes; each difference of two such lifts is the explicit boundary in (TVB.7)–(TVB.9).

34.8 The first admitted degree and the empty quotient

At $N = q - 1$, the first q monic remainder columns are a basis, so $b_i^* G_{q-1} b_j = \delta_{ij} \omega_j$ for $i, j < q$. Expand the original polynomial, with its coefficients retained,

$$\chi = p_q + \sum_{j=0}^{q-1} \gamma_j p_j, \quad \gamma_j = \frac{\langle p_j, \chi \rangle_\theta}{\omega_j}. \quad (\text{TVB.37})$$

Reduction gives $b_q = -\sum_{j<q} \gamma_j b_j$. Hence

$$a = \omega_{q-1}, \quad z_{q-1} = -\gamma_{q-1} \omega_{q-1}, \quad d_+ = \sum_{j<q} |\gamma_j|^2 \omega_j = \nu_0 - \omega_q, \quad \gamma_{q-1} = -i\psi_{q-1}.$$

The last equality follows from (TVB.26), which does not require K_{q-2}^{-1} . Substitution gives the exact CV.12-Toda endpoint

$$\epsilon_{q-1}^2 = \frac{\sum_{j=0}^{q-2} |\gamma_j|^2 \omega_j}{\omega_{q-1}} = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - \psi_{q-1}^2. \quad (\text{TVB.38})$$

There is no V_{q-2} here. The omitted determinant instead records $\tilde{D}_{q-1}/D_{q-1}^{\text{CV}} = \psi_{q-1}^2/\alpha_q$. The polynomial projection $\mathbf{P}_{q-2}\chi = \sum_{j<q-1} \gamma_j p_j$ has the actual jet $\eta \sum_{j<q-1} \gamma_j b_j$, whereas χ itself has zero jet and primitive (TVB.9). Its squared source norm in (TVB.6) is the numerator in (TVB.38). This is the exact map from the whole relation to its lower orthogonal projection, including the possibly nonzero resulting observation. For $q = 1$, the sum is empty, the lower source is $\mathcal{P}_{-1} = 0$, and $\epsilon_N = 0$ at every admitted degree because its one-dimensional relative control is trace zero.

The endpoint projection and the minor vector have a stronger exact typed correspondence. Assume $q \geq 2$, let $D = D_{q-1}^{\text{CV}} = \prod_{j<q} \omega_j^{-1}$, and set $I_j = \{0, \dots, q-2\} \setminus \{j\}$ for $0 \leq j \leq q-2$. The determinant of the first q monic remainder columns is 1. Substituting $b_q = -\sum_{l<q} \gamma_l b_l$ in the minor with columns $I_j \cup \{q-1, q\}$ leaves only $l = j$. Moving b_j from its last position to position j takes $q-1-j$ transpositions; with the preceding minus sign this proves

$$\det B_{I_j \cup \{q-1, q\}} = (-1)^{q-j} \gamma_j. \quad (\text{TVB.38a})$$

Write $e_j = (\bigwedge_{i \in I_j} p_i) \wedge p_{q-1} \wedge p_q$. The explicit conjugate-linear isomorphism

$$\mathcal{H} : \mathcal{P}_{q-2} \longrightarrow \mathcal{E}_{q-1}, \quad \mathcal{H} \left(\sum_{j<q-1} a_j p_j \right) = \sum_{j<q-1} (-1)^{q-j} \overline{a_j} \frac{\omega_j D}{\omega_q} e_j \quad (\text{TVB.38b})$$

has inverse $\sum t_j e_j \mapsto \sum (-1)^{q-j} \omega_q \overline{t_j} p_j / (\omega_j D)$. The source and target basis vectors are orthogonal and $\|e_j\|_\theta^2 = \omega_q / (D \omega_j)$. Consequently

$$\|\mathcal{H}P\|_\theta^2 = \frac{D}{\omega_q} \|P\|_\theta^2, \quad \zeta_{q-1} = \mathcal{H}(\mathbf{P}_{q-2}\chi), \quad \Delta_{q-1} = \frac{D}{\omega_q} \|\mathbf{P}_{q-2}\chi\|_\theta^2. \quad (\text{TVB.38c})$$

This follows term by term from (TVB.31) and (TVB.38a). The actual alternating tensor image has squared norm $q! D \|\mathbf{P}_{q-2}\chi\|_\theta^2 / \omega_q$ in the specified tilted tensor observation, and in the original source norm at $\theta = 0$. The top coefficient is retained separately: $\tilde{D}_{q-1} = D |\gamma_{q-1}|^2 \omega_{q-1} / \omega_q$. For $q = 1$ the map $\mathcal{H}$ is the unique map between zero spaces, with the same zero identities; the top-coefficient formula still records the entire omitted determinant.

For completeness the finite upper bound is strict when $q = 1$ and $N \geq 1$. Consecutive source polynomials have no common root: their three-term recurrence with its nonzero coefficient $\alpha_j > 0$ would propagate such a root backwards to $p_0 = 1$. For any nonzero quotient $q \geq 1$, both b_N, b_{N+1} therefore cannot vanish, since a root of χ would then be common to both polynomials. When $N \geq q$, K_{N-1} is positive definite and $K_{N+1} - K_{N-1}$ is a nonzero positive semidefinite matrix. Conjugating by $K_{N-1}^{-1/2}$ and multiplying its eigenvalues proves $D_{N+1}^{\text{CV}} > D_{N-1}^{\text{CV}}$, hence $\mathcal{R}_N > 1$ and $U_N > 0$. At $q = 1$ this is strictly larger than the already proved $\epsilon_N = 0$.

Finally let $h = 1$, so $\chi = 1$, $q = 0$, and $C = 0$. The analytic function g , w_1 and all source norms still exist. For every $N \geq 0$ the source sequence is multiplication by one from $\mathcal{P}_N$ onto itself followed by the zero quotient. Then $m = N+1$, $Z_\chi = Z$, $\mathfrak{B}_m = \mathfrak{D}_m$, and the determinant of the zero-dimensional Gram is the empty determinant $V_N = 1$. Its logarithm and derivatives are zero. The minimum lift, its curvature maps, and the relative control are the unique zero maps, with $\epsilon_N = L = 0$. The two Toda coefficients in their curvature difference are equal, so that difference is zero. No positive-dimensional inverse is introduced. The support lift retains $G(0) = \{\tau, e\}$: every represented polynomial, including each nonzero source relation, maps to e , while τ maps to τ .

34.9 The complete join with the original spectral coupling

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Its two inputs are the delivered Toda equation (28), proved here as (TVB.32), and the complete original-metric exterior identity E8. The latter source is retained in the cyclic companion at commit 161089942cce70a09fbc441d08ba4d36c1fd810c. We reproduce the finite identity and its typed coupling map here to make the join complete; no novelty or limiting estimate is claimed.

All quantities in this section are at the original observation $\theta = 0$, with $N \geq q$. Write $A^\# = G_N^{-1} A^* G_N$ and $\mathcal{H}_N = A^\# + A - kI$, acting on the original (C, G_N) . Let C_+ be the full generalized spectral subspace with $\Re \lambda > c$, let $C_- = C_+^{\perp_{G_N}}$, and let $P : C \rightarrow C$ be their orthogonal projection onto C_+ . Suppose first $C_+ \neq 0$. Its trace L from (TVB.35) is positive, so $\epsilon_N > 0$. Choose G_N -unit eigenvectors u_+, u_- of $\mathcal{H}_N$ at $\epsilon_N, -\epsilon_N$; selfadjointness makes them orthogonal. They specify only the coefficient injection

$$E : \mathbb{C}^2 \longrightarrow C, \quad E(z_+, z_-) = z_+ u_+ + z_- u_-, \quad E^\# E = I_2, \quad M = E^\# P E. \quad (\text{TVB.39})$$

Both M and $I_2 - M$ are positive semidefinite, since their quadratic forms are respectively the squared original norms of $P E z$ and $(1 - P) E z$. In the original orthogonal decomposition define

$$B = A|_{C_+}, \quad D^{\text{op}} = (1 - P)A|_{C_-}, \quad C = PA|_{C_-} : C_- \longrightarrow C_+.$$

Invariance of C_+ gives $A = \begin{pmatrix} B & C \\ 0 & D^{\text{op}} \end{pmatrix}$. Its adjoint has the conjugate-transpose blocks in the original restricted metrics, so $P\mathcal{H}_N|_{C_-} = C$ exactly.

Put $a_+ = \|Pu_+\|^2$, $a_- = \|Pu_-\|^2$, and $z = \langle Pu_+, Pu_- \rangle_{G_N}$. The compression M has these two diagonal entries and off-diagonal entries $z, \bar{z}$. The original rank-two spectral expansion gives $L = \epsilon_N(a_+ - a_-)$ and

$$\begin{aligned} \|C\|_{\text{HS}, G_N}^2 &= \text{Tr}(P\mathcal{H}_N^2 P) - \text{Tr}((P\mathcal{H}_N P)^2) \\ &= \epsilon_N^2(a_+ + a_- - a_+^2 - a_-^2 + 2|z|^2). \end{aligned}$$

The cross trace in the second equation is $|\langle v, w \rangle|^2$ for the two rank-one maps $v\langle v, \cdot \rangle$ and $w\langle w, \cdot \rangle$, as follows by composing them. Subtract the displayed norm and $\epsilon_N^2(a_+ - a_-)^2$ from ϵ_N^2 ; direct expansion proves

$$\epsilon_N^2 - L^2 = \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}. \quad (\text{TVB.40})$$

Both determinants are nonnegative by the quadratic forms after (TVB.39). Adding (TVB.40) to (TVB.32) cancels precisely the intermediate ϵ_N^2 and gives all four nonnegative losses

$$\boxed{\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1}(e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2\{\det M + \det(I_2 - M)\}.} \quad (\text{TVB.41})$$

Every adjoint, projection, norm and coefficient is for the same original minimum metric as the Toda volumes.

The spectral projector has an exact map to the coupling. Factor the minimal polynomial of A as $m_+ m_-$ over the selected and remaining full generalized blocks, with their original exponents. Bezout polynomials a, b with $am_+ + bm_- = 1$ give $Q = b(A)m_-(A)$: it is identity on each selected block and zero on each other block. Thus $Q^2 = Q$, $AQ = QA$, and in the preceding orthogonal decomposition $Q = \begin{pmatrix} I & Y \\ 0 & 0 \end{pmatrix}$ for a unique map $Y : C_- \rightarrow C_+$. Block multiplication proves

$$\mathcal{S} : \text{Hom}(C_-, C_+) \longrightarrow \text{Hom}(C_-, C_+), \quad \mathcal{S}(Y) = BY - YD^{\text{op}} = C, \quad Q - P = \begin{pmatrix} 0 & Y \\ 0 & 0 \end{pmatrix}. \quad (\text{TVB.42})$$

This morphism is invertible with its full original nilpotent parts:

$$\mathcal{S}^{-1}(Z) = \int_0^\infty e^{-tB} Z e^{tD^{\text{op}}} dt. \quad (\text{TVB.43})$$

Indeed every selected eigenvalue has real part greater than c and every remaining one has real part at most c . On a full block at λ the first exponential is exactly $e^{-t\lambda} \sum_{j < \ell_\lambda} (-t)^j (B - \lambda I)^j / j!$, with its corresponding projector, and the second has the same formula with positive t on its remaining blocks. Summing these finite expressions gives a polynomial times a strictly decaying exponential for their product in (TVB.43). This proves convergence in the original operator norm. Differentiating the integrand gives $-B e^{-tB} Z e^{tD^{\text{op}}} + e^{-tB} Z e^{tD^{\text{op}}} D^{\text{op}}$. Its integral has lower endpoint Z and zero upper endpoint, proving the inverse identity. If $BY - Y D^{\text{op}} = 0$, the same product with $Z = Y$ is constant and decays to zero, hence $Y = 0$. This proves uniqueness as well. The space C_- is nonzero: otherwise $C_+ = C$ and $L = \text{Tr } \mathcal{H}_N = 0$, contradicting $L > 0$. Together with $C_+ \neq 0$ this makes the unit sphere in $\text{Hom}(C_-, C_+)$ nonempty. The actual smallest singular value $s_{B,D} > 0$ of $\mathcal{S}$ in the original Hilbert-Schmidt pairing exists by compactness of its unit sphere and injectivity. Consequently $\|C\|_{\text{HS}, G_N} \geq s_{B,D} \|Q - P\|_{\text{HS}, G_N}$. No bound on $s_{B,D}$ independent of the arithmetic inputs is used.

Retaining the phase and coupling in (TVB.41) gives the finite strengthening

$$\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{\sqrt{L^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2}}{\sqrt{\alpha_{N+1}}} \right). \quad (\text{TVB.44})$$

It follows by subtracting only the nonnegative imbalance and compression-determinant terms and applying the same inverse sinh as in (TVB.35). Summation over the disjoint indices $N = N_0 + 2j$ has the same exact endpoints as (TVB.36), with this degree-dependent square root in each summand. If $C_+ = 0$, use (TVB.32) directly; no positive-eigenline compression is asserted when $\epsilon_N = 0$. At $N = q - 1$, the scalar term instead uses the separate endpoint expression (TVB.38). These identities supply no asymptotic estimate and change no arithmetic class.

34.10 Crosswalk and exact continuing arithmetic quantity

CV.1–CV.3 are (TVB.22)–(TVB.24), with the additional proved identification $D_N^{\text{CV}} = \mathfrak{B}_m / \mathfrak{D}_{N+1}$. CV.4–CV.5 become (TVB.27)–(TVB.29), where the whole complex cross term is $i\omega_N(\sigma - \ell'_N)$. CV.6 is the minimum lift (TVB.14) and its determinant-line map after (TVB.15). CV.10–CV.10a are (TVB.30)–(TVB.31), now with the exact two-loss identity (TVB.34). CV.11–CV.12 are precisely (TVB.37)–(TVB.38). The new operator refinement is (TVB.20)–(TVB.21): it computes both individual rank-one curvature costs, including their shared retained quantity $\alpha_{N+1} \delta_N$.

For an actual quartet polynomial the exterior multiplicities and their lower trace L remain those of that exact polynomial; for a larger packet they remain a lower contribution to its actual full trace. All admission thresholds here use its actual $q = \deg \chi$. The finite formulas prove an upper estimate by the exact source and relation determinants and prove the telescope (TVB.36). They do not estimate the growth of α_{N+1} or the joint determinant ratio as k and N grow. That continuing arithmetic question is the original expression (TVB.27), equivalently the positive-minor source norm in (TVB.28)–(TVB.31), with both losses (TVB.32) available without changing any of its inputs.

35 The original theta input and certified infinite-series tails

This continuation supplies the omitted-tail and ball-arithmetic certificate for the analytic input of Section 9 of *Source-boundary volume dynamics for the arithmetic control*. Every theta coefficient, Fourier factor, conjugation, and complex branch used below is specified. The original seed has total

mass $M_1(0)$; the measure is never divided by that mass. The result computes analytic input of the source determinant flow. Its exact connection with a finite zero packet is given below by Mellin division and the multiplier $h(1/2 + it)$.

35.1 The original theta function and its Mellin transform

Put $D = -z\partial_z$, let $\mathcal{S}_{\pi/4} = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \pi/4\}$, and use the restriction of the principal logarithm to this sector. Define

$$\vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{TI.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n=1}^{\infty} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{TI.2})$$

On a compact subset of the sector, each differentiated summand is bounded by a polynomial in n times e^{-an^2} with $a > 0$. Consequently both series define holomorphic functions there, and all the displayed derivatives are termwise valid.

Lemma 35.1 (The exact theta inversion). *On $\mathcal{S}_{\pi/4}$,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{TI.3})$$

For each $0 \leq \alpha < \pi/4$, every integer $j \geq 0$, and every real $A > 0$, $D^j f_0(z) = O(|z|^{-A})$ as $|z| \rightarrow \infty$ and $D^j f_0(z) = O(|z|^A)$ as $|z| \rightarrow 0$, uniformly for $|\arg z| \leq \alpha$.

Proof. For $x > 0$, the Fourier transform with kernel $e^{-2\pi i \xi t}$ of $e^{-\pi x^2 t^2}$ is $x^{-1}e^{-\pi \xi^2/x^2}$. For completeness, the value at $\xi = 0$ follows from $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$, whose square is evaluated in polar coordinates. Differentiation in ξ and integration by parts give the differential equation $I'(\xi) = -2\pi \xi I(\xi)/x^2$; the stated transform is its solution with the stated initial value. Periodize the Gaussian over integer translates. The resulting smooth periodic function has the absolutely convergent Fourier series with coefficients $x^{-1}e^{-\pi n^2/x^2}$. Evaluation at zero gives $\vartheta(x) = x^{-1}\vartheta(1/x)$. Both sides are holomorphic in the connected sector, so the identity theorem gives the first equation in (TI.3).

For the linear involution $(Rf)(z) = z^{-1}f(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this identity to $\vartheta = R\vartheta$ proves the second equation. The real coefficients in (TI.2) give the third. On a closed subsector, $\Re(z^2) \geq |z|^2 \cos(2\alpha) > 0$; the series and each derivative therefore have Gaussian decay at infinity, with a fixed polynomial factor. The second equation in (TI.3), differentiated using $DR = R(1 - D)$, transports that bound to zero and proves all the endpoint assertions. $\square$

For functions with these endpoint bounds set

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{TI.4})$$

The Mellin integral is entire. When $\Re s > 1$, the sum of the absolute integrals of the terms in (TI.2) is finite, because it is bounded by a constant depending on s times $\sum n^{-\Re s}$. Thus

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s) (4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)) \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s). \end{aligned} \quad (\text{TI.5})$$

The first line contains the factor 2 of (TI.2) and the factor $1/2$ from the substitution $u = \pi n^2 x^2$. The second follows from $\Gamma(z+1) = z\Gamma(z)$, which follows by integration by parts. Equation (TI.5) gives the entire continuation of its right side. In particular $g = 2\zeta$ with the original convention in the source note. Substituting $x \mapsto 1/x$ in (TI.4), using (TI.3), proves

$$g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s). \quad (\text{TI.6})$$

35.2 Mellin division, the dagger sign, and the tilted observation

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a monic finite product of actual zeros of g , with the selected orders at most their actual orders and with the selected multiset invariant under $\rho \mapsto 1 - \bar{\rho}$. The complete-order packets used in the source note satisfy these conditions. Write $d = \deg h$ and

$$v_h = g/h, \quad h^{\dagger}(s) = \overline{h(1 - \bar{s})}, \quad v_h^{\dagger}(s) = \overline{v_h(1 - \bar{s})}. \quad (\text{TI.7})$$

Removable singularities are filled in. The leading coefficient of $h^{\dagger}$ is $(-1)^d$, and its zero multiset is that of h ; therefore

$$h^{\dagger} = (-1)^d h, \quad v_h^{\dagger} = (-1)^d v_h. \quad (\text{TI.8})$$

Proposition 35.2 (Construction without an endpoint assumption). *For every $\sigma \in \mathbb{R}$ the integral*

$$F_h(z) = \frac{1}{2\pi i} \int_{\sigma - i\infty}^{\sigma + i\infty} v_h(s) z^{-s} ds \quad (\text{TI.9})$$

converges on the sector, is independent of σ , defines a holomorphic function there, and has the rapid endpoint bounds in the first lemma on each closed subsector. It satisfies

$$\mathcal{M}F_h = v_h, \quad h(D)F_h = f_0, \quad F_h(1/\bar{z}) = (-1)^d \bar{z} \overline{F_h(z)}. \quad (\text{TI.10})$$

Proof. Rotate the Mellin integral of f_0 to the ray $e^{i\beta}\mathbb{R}_{>0}$, where $|\beta| < \pi/4$. The endpoint estimates make both closing arcs vanish, and give

$$g(s) = e^{i\beta s} \int_0^{\infty} f_0(e^{i\beta}x) x^{s-1} dx.$$

For σ in any compact interval, the absolute integral on the right is uniformly bounded. Choosing the sign of β to be that of t proves $|g(\sigma + it)| \leq C e^{-|\beta||t|}$. For sufficiently large $|t|$, the explicit finite product h is bounded below by $c(1 + |t|)^d$ on the same strip. On the remaining compact rectangle, v_h is bounded because its singularities are removable. These facts give the same exponential estimate for v_h , with a different constant. It is valid for every $|\beta| < \pi/4$ and every compact interval of σ .

In (TI.9), choose $|\beta| > |\arg z|$. Absolute convergence and local uniform convergence of every differentiated integral follow. Horizontal contour segments tend to zero by the same estimate, proving independence of σ . Multiplication by s^j computes $D^j F_h$; moving σ arbitrarily far to the right or left gives respectively the asserted infinity or zero bounds.

The change $x = e^y$ turns (TI.9) on the line $\Re s = \sigma$ into Fourier inversion for $e^{\sigma y} F_h(e^y)$. The exponentially decreasing function $v_h(\sigma + it)$ and its derivatives have the required integrability, and inversion gives $\mathcal{M}F_h = v_h$. The Fourier inversion and Plancherel conventions here can be verified by inserting $e^{-\varepsilon t^2}$ in the inverse integral: Fubini gives convolution with the Gaussian approximate identity of integral one; letting $\varepsilon \downarrow 0$ gives inversion, and applying the same calculation to the inner product gives the factor $1/(2\pi)$ in Plancherel. Thus no change of Fourier normalization is implicit. Multiplication by $h(s)$ in (TI.9) is $h(D)$ on the source. Fourier inversion of (TI.5) proves $h(D)F_h = f_0$.

For the conjugate-linear involution $(\mathcal{R}F)(x) = x^{-1} \overline{F(1/\bar{x})}$, substitution gives $\mathcal{M}(\mathcal{R}F)(s) = \overline{\mathcal{M}F(1 - \bar{s})}$. Equations (TI.8) and Mellin uniqueness imply $\mathcal{R}F_h = (-1)^d F_h$ on the positive axis. Both sides extend holomorphically to the sector as $z^{-1} \overline{F_h(1/\bar{z})}$ and $(-1)^d F_h(z)$. Conjugating the resulting identity gives the last equation in (TI.10). $\square$

Here are the exact Hilbert-space maps. The change of variables

$$U : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}, dy), \quad (UF)(y) = e^{y/2} F(e^y) \quad (\text{TI.11})$$

is unitary, with inverse $u \mapsto [x \mapsto x^{-1/2} u(\log x)]$. On the rapid sector functions constructed above define, for real $|\theta| < \pi/2$,

$$(E_{\theta}F)(y) = e^{(y+i\theta/2)/2} F(e^{y+i\theta/2}). \quad (\text{TI.12})$$

This map has codomain $L^2(\mathbb{R}, dy)$; it is the analytic translation of UF on this specified domain. Its norm is not asserted to equal the untilted norm. Shifting the y contour by $i\theta/2$ in (TI.4), with the closing vertical edges vanishing by the proved endpoint bounds, gives

$$\mathcal{F}_+(E_\theta F_h)(t) = e^{\theta t/2} v_h(1/2 + it). \quad (\text{TI.13})$$

Indeed $y = z - i\theta/2$ contributes $e^{ity} = e^{\theta t/2} e^{itz}$. In the source expression the factor $e^{i\theta/4}$ remains present.

Define the original, unscaled density and its mass by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad \mu_h = M_h(0). \quad (\text{TI.14})$$

The exponential estimate above proves that the integral defining M_h and each of its derivatives converge locally uniformly on $|\Re\theta| < \pi/2$: for a compact substrip choose $2|\beta| > \sup |\Re\theta|$. By (TI.11)–(TI.13) and Plancherel,

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2} x)|^2 dx = 2 \int_1^\infty |F_h(e^{i\theta/2} x)|^2 dx \quad (\theta \text{ real}). \quad (\text{TI.15})$$

For the second equality the last identity in (TI.10) gives

$$F_h(e^{i\theta/2}/x) = (-1)^d e^{-i\theta/2} x \overline{F_h(e^{i\theta/2} x)}.$$

After taking absolute squares, the factor x^2 cancels the Jacobian of $x \mapsto 1/x$. The dagger sign is retained before that operation. Dagger stability alone gives this endpoint identity. If the selected multiset is additionally stable under ordinary conjugation, then $v_h(\bar{s}) = \overline{v_h(s)}$, so $w_h(-t) = w_h(t)$ and $M_h(-\theta) = M_h(\theta)$. This proves the precise extra symmetry needed for evenness.

The typed relation between seed and packet input is also explicit:

$$F_h \xrightarrow{h(D)} f_0 \quad v_h(1/2 + it) \xrightarrow{h(1/2 + it)} g(1/2 + it), \quad w_1(t) = |h(1/2 + it)|^2 w_h(t). \quad (\text{TI.16})$$

The arrow in the first display means the linear differential map on the rapid sector-function space; (TI.10) and (TI.13) prove its commutation with the transform. The scalar equality of densities holds even at selected line zeros by continuity; it does not require division of a measured zero value by zero.

35.3 The complete double series and an explicit omitted tail

For $h = 1$ we have $F_1 = f_0$. Set $c_1 = -6$, $c_2 = 4$ and, for real θ first,

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}), \quad \alpha_{rs} = r + s + \frac{1}{2}. \quad (\text{TI.17})$$

For $\Re C > 0$ define

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}. \quad (\text{TI.18})$$

The power uses the holomorphic logarithm of the right half-plane, real on its positive axis. To check (TI.18), first substitute $u = Cx^2$ for positive real C . Both sides extend holomorphically to the right half-plane: the integral by dominated convergence and the right side by the same continued incomplete gamma integral with its ray deformable inside that half-plane. The identity theorem then gives (TI.18) there with the stated branch.

Theorem 35.3 (The retained factor and the locally uniform tail). *For every θ with $|\Re\theta| < \pi/2$,*

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(\alpha_{rs}, C_{mn}(\theta))}{C_{mn}(\theta)^{\alpha_{rs}}}. \quad (\text{TI.19})$$

Let $M_{1,J}$ be exactly this sum with $1 \leq m, n \leq J$, for an integer $J \geq 0$. Fix $0 \leq \theta_0 < \pi/2$ and $B \geq 0$, and put

$$d_0 = \pi e^{-B} \cos \theta_0, \quad L = J + 1, \quad \rho_a = e^{-d_0(2a+1)}, \quad (\text{TI.20})$$

$$H_p(a, d_0) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{matrix} j \\ b \end{matrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{TI.21})$$

$$A_a(d_0, B) = 6\pi e^B H_2(a, d_0) + 4\pi^2 e^{2B} H_4(a, d_0). \quad (\text{TI.22})$$

Here $\left\{ \begin{matrix} j \\ b \end{matrix} \right\}$ is determined by $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$, $\left\{ \begin{matrix} j \\ 0 \end{matrix} \right\} = 0$ for $j > 0$, $\left\{ \begin{matrix} 0 \\ b \end{matrix} \right\} = 0$ for $b > 0$, and $\left\{ \begin{matrix} j+1 \\ b \end{matrix} \right\} = b \left\{ \begin{matrix} j \\ b \end{matrix} \right\} + \left\{ \begin{matrix} j \\ b-1 \end{matrix} \right\}$, with out-of-range indices zero. For every $|\Re\theta| \leq \theta_0$ and $|\Im\theta| \leq B$,

$$|M_1(\theta) - M_{1,J}(\theta)| \leq \mathcal{E}_J(\theta_0, B) := 16A_1(d_0, B)A_L(d_0, B)I_8(d_0(1+L^2)). \quad (\text{TI.23})$$

Every quantity on the right is explicit and positive. It tends to zero as $J \rightarrow \infty$. In particular (TI.23) supplies the requested locally uniform real-tilt bound by taking $B = 0$.

Proof. For real θ , substitute (TI.2) into the last expression in (TI.15). A product contributes $2 \cdot 2 \cdot 2 = 8$ before integration: the endpoint factor 2 and the two original theta coefficients 2. Equation (TI.18) contributes $1/2$. The remaining factor is precisely 4, and the powers of the two dilations give $e^{i\theta r} e^{-i\theta s}$. This establishes the coefficients and phases in (TI.19), subject to the absolute convergence now proved.

For integers $a \geq 1$, $p \geq 0$, and $x \geq 1$,

$$\begin{aligned} \sum_{n \geq a} n^p e^{-d_0 n^2 x^2} &\leq e^{-d_0 a^2 x^2} \sum_{j \geq 0} (a+j)^p e^{-d_0(2a+1)j} \\ &= e^{-d_0 a^2 x^2} H_p(a, d_0). \end{aligned}$$

The inequality uses $(a+j)^2 - a^2 = 2aj + j^2 \geq (2a+1)j$ and $x^2 \geq 1$. For the equality expand $(a+j)^p$ by the binomial theorem and use $j^r = \sum_b \left\{ \begin{matrix} r \\ b \end{matrix} \right\} (j)_b$. That last identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$. Thus (TI.21) is a finite positive rational function, not an unevaluated infinite sum.

For complex θ in the rectangle, let $f^\pm(x) = f_0(e^{\pm i\theta/2} x)$ and let $f_J^\pm$ contain only $1 \leq n \leq J$ in the original series. Since

$$\Re(\pi n^2 e^{\pm i\theta}) = \pi n^2 e^{\mp \Im\theta} \cos(\Re\theta) \geq d_0 n^2,$$

and $|e^{\pm i\theta}| \leq e^{rB}$, the positive bound just proved implies

$$\begin{aligned} |f^\pm(x)|, |f_J^\pm(x)| &\leq 2A_1(d_0, B) x^4 e^{-d_0 x^2}, \\ |f^\pm(x) - f_J^\pm(x)| &\leq 2A_L(d_0, B) x^4 e^{-d_0 L^2 x^2}. \end{aligned}$$

Here $x^2 \leq x^4$ retains a bound for the original -6 term; no coefficient of the exact series is changed. These inequalities prove absolute integrability of the full expanded double series, locally uniformly on the rectangle. They also prove

$$2 \int_1^\infty |f^+(x)f^-(x) - f_J^+(x)f_J^-(x)| dx \leq 16A_1(d_0, B)A_L(d_0, B) \int_1^\infty x^8 e^{-d_0(1+L^2)x^2} dx,$$

because the product difference is $(f^+ - f_J^+)f^- + f_J^+(f^- - f_J^-)$. The right side is (TI.23). For fixed $d_0 > 0$, formula (TI.21) bounds $H_p(L, d_0)$ by a fixed constant times $(1 + L)^p$: its denominators are bounded away from zero uniformly for $L \geq 1$. Moreover $I_8(b) \leq e^{-b} \int_1^\infty x^8 e^{-d_0(x^2-1)} dx$ for $b \geq d_0$. Hence (TI.23) tends to zero exponentially in L^2 .

For complex θ , $C_{mn}(\theta)$ stays in the right half-plane by the displayed inequality. The integral $2 \int_1^\infty f^+(x)f^-(x) dx$ is holomorphic on the whole strip. It equals M_1 on the real interval by (TI.15) and conjugation of f_0 . Holomorphic uniqueness proves equality on the strip. The same locally uniform bounds justify (TI.19) there and finish the proof. $\square$

The tail controls differentiated moments as well. If the closed disc $\{z : |z - \theta| \leq r\}$ lies in the rectangle of the theorem, Cauchy's integral formula gives, for every integer $j \geq 0$,

$$|M_1^{(j)}(\theta) - M_{1,J}^{(j)}(\theta)| \leq \frac{j!}{r^j} \mathcal{E}_J(\theta_0, B). \quad (\text{TI.24})$$

The bound includes arbitrary finite derivative orders used by a Hankel determinant. It does not assert a useful estimate when the derivative order or determinant dimension itself grows without bound.

35.4 The second moment and the first tensor coefficient

Let $H = (D - 1/2)f_0$. With $y = \pi n^2 x^2$ and $P(y) = 8y^2 - 12y$, literal differentiation gives

$$\begin{aligned} (D - 1/2)(P(y)e^{-y}) &= (-2yP'(y) + 2yP(y) - P(y)/2) e^{-y} \\ &= (16y^3 - 60y^2 + 30y)e^{-y}. \end{aligned} \quad (\text{TI.25})$$

Thus

$$H(x) = \sum_{n \geq 1} \sum_{r=1}^3 d_r (\pi n^2)^r x^{2r} e^{-\pi n^2 x^2}, \quad (d_1, d_2, d_3) = (30, -60, 16). \quad (\text{TI.26})$$

The unitary map (TI.11) satisfies $U(D - 1/2)F = -(UF)'$ by differentiation. Integration by parts in the Fourier transform therefore gives $\mathcal{F}_+(UH)(t) = it g(1/2 + it)$. It follows, with the original $1/(2\pi)$ factor, that

$$M_1''(0) = \frac{1}{2\pi} \int_{\mathbb{R}} t^2 |g(1/2 + it)|^2 dt = \int_0^\infty |H(x)|^2 dx = 2 \int_1^\infty H(x)^2 dx. \quad (\text{TI.27})$$

For the last equality, $DR = R(1 - D)$ and $Rf_0 = f_0$ give $RH = -H$, that is $H(1/x) = -xH(x)$. Its square gives precisely the same Jacobian cancellation as before, with the new minus sign retained before squaring.

Let H_J restrict (TI.26) to $n \leq J$ and put $T_J = 2 \int_1^\infty H_J(x)^2 dx$. Then

$$T_J = \sum_{m,n=1}^J \sum_{r,s=1}^3 d_r d_s (\pi m^2)^r (\pi n^2)^s \frac{\Gamma(r+s+1/2, \pi(m^2+n^2))}{(\pi(m^2+n^2))^{r+s+1/2}}. \quad (\text{TI.28})$$

There is no coefficient 4 in this last expression: (TI.26) already contains its fully differentiated coefficients, and the endpoint factor 2 cancels the $1/2$ in (TI.18). Set

$$B_a = 30\pi H_2(a, \pi) + 60\pi^2 H_4(a, \pi) + 16\pi^3 H_6(a, \pi). \quad (\text{TI.29})$$

The proof of (TI.23), now with degree six in x , gives

$$|M_1''(0) - T_J| \leq \mathcal{T}_J := 4B_1 B_{J+1} I_{12}(\pi(1 + (J+1)^2)). \quad (\text{TI.30})$$

In detail, $|H|, |H_J| \leq B_1 x^6 e^{-\pi x^2}$ and $|H - H_J| \leq B_{J+1} x^6 e^{-\pi(J+1)^2 x^2}$; apply $|H^2 - H_J^2| \leq |H - H_J|(|H| + |H_J|)$ and the factor 2 of (TI.27).

For tensor degree $k \geq 1$, let $m_k = w_1^{*k}$ with its total mass μ_1^k , where $\mu_1 = M_1(0)$. Absolute integrability and Fubini give $Z_k(\theta) = \int e^{\theta u} m_k(u) du = M_1(\theta)^k$. The function w_1 is even by (TI.6), so $Z'_k(0) = M'_1(0) = 0$. The degree-zero monic source polynomial is 1, of squared norm $h_0 = \mu_1^k$, and its monic degree-one orthogonal polynomial in the original sum variable u is u , of squared norm $h_1 = Z''_k(0) = k\mu_1^{k-1}M''_1(0)$. Therefore the first recurrence coefficient is exactly

$$a_1^{(k)} = \frac{h_1}{h_0} = k \frac{M''_1(0)}{M_1(0)}. \quad (\text{TI.31})$$

The quotient of norms in (TI.31) is the coefficient in the monic three-term recurrence. It leaves both norms and the measure mass specified, and performs no replacement of m_k by a probability law.

35.5 Certified numerical enclosures and their executable scope

The accompanying script `toda_theta_input_tail_check_20260912.py` evaluates (TI.19), (TI.28), (TI.23), and (TI.30) at $J = 8$ with `python-flint 0.9.0`, using Arb real balls and Acb rectangular complex balls at 256-bit precision and one arithmetic thread. All integer and rational input, including $\theta = 1/10$, is supplied exactly; π , exponentials, powers, and incomplete gamma functions are enclosed by those ball operations. The incomplete gamma call has argument C and parameter α , that is `C.gamma_upper(alpha)`. Its branch is the one in (TI.18). The documented ball and endpoint semantics used here are in [the Arb interface](#) and [the Acb interface](#).

The script adds a symmetric error ball whose radius is the upward rounded upper endpoint of the proved tail bound. Thus the result encloses the infinite sum, rather than only its finite arithmetic. The finite real-tilt sum is real by the proved conjugation identity; the script additionally checks that its imaginary ball contains zero. The positive real mass enclosure is then used in interval division for (TI.31). The following are rational decimal intervals:

$$\begin{aligned} M_1(0) &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ M_1(1/10) &\in [1.3459071784582497969682205923576298051604, \\ &\quad 1.3459071784582497969682205923576298051607], \\ M''_1(0) &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ a_1^{(k)}/k &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967]. \end{aligned}$$

Every width is $3 \cdot 10^{-40}$. The computed positive tail bounds satisfy, respectively,

$$\mathcal{E}_8(0,0) < 7 \cdot 10^{-107}, \quad \mathcal{E}_8(1/10,0) < 3 \cdot 10^{-106}, \quad \mathcal{T}_8 < 4 \cdot 10^{-103}. \quad (\text{TI.32})$$

The receipt preserves both the displayed decimal intervals and exact rational dyadic lower and upper endpoints of each resulting ball. Its decimal endpoints are generated by outward rounding at 40 decimal places, with one additional unit on each side, and are subsequently checked by rigorous ball comparisons. Normal and optimized execution produce identical mathematical records. The requested negative-control mode inserts the false inequality $a_1^{(k)}/k < 1$ and must raise an error; it uses an explicit checked exception rather than a Python assertion, so optimization cannot remove that check.

The numerical certificate uses the standard correctness contract of Arb/Acb ball arithmetic together with the written infinite-tail proof. It is not inferred from agreement of two numerical precisions. Its output concerns the three specified infinite analytic-seed quantities and their recurrence

ratio. For $h = 1$ the cyclic arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the actual quotient map sends each polynomial to zero, while the positive seed source norm and (TI.31) remain. For a selected packet h , the precise passage from the present input to w_h is (TI.16), obtained from the globally defined Mellin division g/h . No packet determinant or bound as k tends to infinity is certified by these four scalar intervals.

36 An exact comparison of equality, volume contraction, and its discarded square

This calculation extends the positive Gaussian calibration of E35–E36 through two further source degrees. Every measure, phase, coefficient and quotient basis is specified. It is a calibration of the same constructions; the roots below are declared coefficients, with no assertion that they are zeros of the arithmetic function.

Fix $\mu > 0, \sigma > 0, k = 1$ and $c = 1/2$. Put

$$d\nu(u) = \frac{\mu}{\sqrt{2\pi\sigma}} e^{-u^2/(2\sigma^2)} du, \quad x = S - c, \quad \chi(S) = x^2 - \sigma^2, \quad C = \mathbb{C}[S]/(\chi). \quad (\text{TC1})$$

The original source pairing is $\langle P, Q \rangle = \int \overline{P(c + iu)} Q(c + iu) d\nu(u)$. We use both the original quotient basis $(1, S)$ and the displayed basis $(1, x)$. The exact coordinate matrix, sending columns in the latter basis to columns in the former, is

$$L_{\text{coord}} = \begin{pmatrix} 1 & -c \\ 0 & 1 \end{pmatrix}, \quad A_S = \begin{pmatrix} 0 & \sigma^2 - c^2 \\ 1 & 2c \end{pmatrix}, \quad A_x = \begin{pmatrix} c & \sigma^2 \\ 1 & c \end{pmatrix}, \quad A_S L_{\text{coord}} = L_{\text{coord}} A_x. \quad (\text{TC2})$$

Here A_x represents multiplication by S in the basis $(1, x)$; multiplication by x is $A_x - cI$. Thus $G_{N,x} = L_{\text{coord}}^* G_{N,S} L_{\text{coord}}$ and $K_{N,S} = L_{\text{coord}} K_{N,x} L_{\text{coord}}^*$. In particular $\det L_{\text{coord}} = 1$ preserves the specified determinant, while the matrices and source polynomials retain their full coordinate transformations.

Integration by parts, with the Gaussian boundary term zero, gives $\int u^{2j} d\nu = \mu(2j-1)!!\sigma^{2j}$ and all odd moments zero: indeed the zeroth integral is μ , and differentiating $u^{2j-1}e^{-u^2/(2\sigma^2)}$ gives the recurrence $m_{2j} = (2j-1)\sigma^2 m_{2j-2}$. The first four monic source polynomials and their full squared norms are consequently

j	$p_j(S)$	ω_j
0	1	μ
1	x	$\mu\sigma^2$
2	$x^2 + \sigma^2$	$2\mu\sigma^4$
3	$x^3 + 3\sigma^2 x$	$6\mu\sigma^6$

(TC3)

For example $p_2(c + iu) = -(u^2 - \sigma^2)$ and $p_3(c + iu) = -i(u^3 - 3\sigma^2 u)$. Substitution of the moments proves every pairwise orthogonality and every displayed norm; the phases -1 and $-i$ are retained before taking the pairings.

In the basis $(1, x)$ the quotient columns $b_j = [p_j]_\chi$ are

$$b_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b_1 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 2\sigma^2 \\ 0 \end{pmatrix}, \quad b_3 = \begin{pmatrix} 0 \\ 4\sigma^2 \end{pmatrix}. \quad (\text{TC4})$$

For completeness, the minimum-source metric is obtained by writing a source polynomial as $\sum_{j=0}^N z_j p_j$. Its norm is $\sum \omega_j |z_j|^2$ and its quotient is $\sum b_j z_j$. The constrained minimizer is $z_j = \omega_j^{-1} b_j^* G_{N,x} v$, where $K_{N,x} = \sum b_j b_j^* / \omega_j$ and $G_{N,x} = K_{N,x}^{-1}$. Indeed these coefficients have quotient v , and their inner product with every quotient-kernel coefficient vector is zero. Pythagoras

then proves uniqueness and minimum norm $v^* G_{N,x} v$. Set $V_N = \det G_{N,x} = \det G_{N,S}$. Directly from (TC3)–(TC4),

$$\begin{aligned} G_{1,x} &= \text{diag}(\mu, \mu\sigma^2), \\ G_{2,x} &= \text{diag}(\mu/3, \mu\sigma^2), & (V_1, V_2, V_3) &= \mu^2\sigma^2(1, 1/3, 1/11). \\ G_{3,x} &= \text{diag}(\mu/3, 3\mu\sigma^2/11), \end{aligned} \quad (\text{TC5})$$

This is also exactly the source/relation determinant formula. On the original vertical line $\chi(c + iu) = -(u^2 + \sigma^2)$, so its zeroth relation norm and its next monic relation norm are

$$\nu_0 = \int (u^2 + \sigma^2)^2 d\nu = 6\mu\sigma^4, \quad \nu_1 = \int u^2(u^2 + \sigma^2)^2 d\nu = 22\mu\sigma^6. \quad (\text{TC6})$$

The two relation columns $\chi, \chi x$ are orthogonal by parity. For $n = 2, 3, 4$, the source Gram determinant is $\mathfrak{D}_n = \prod_{j < n} \omega_j$; the corresponding relation determinants are $\mathfrak{B}_0 = 1, \mathfrak{B}_1 = \nu_0, \mathfrak{B}_2 = \nu_0\nu_1$. These identities follow by the triangular monic changes of basis. They give $\mathfrak{D}_{N+1}/\mathfrak{B}_{N-1} = V_N$ for each $N = 1, 2, 3$, including the literal empty determinant at $N = 1$.

Let $H_N = A_x + G_{N,x}^{-1} A_x^* G_{N,x} - 2cI$. Multiplication of the displayed matrices yields

$$H_1 = \begin{pmatrix} 0 & 2\sigma^2 \\ 2 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 4\sigma^2 \\ 4/3 & 0 \end{pmatrix}, \quad \epsilon_1^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3. \quad (\text{TC7})$$

Each H_N is self-adjoint in its stated positive metric; its squared matrix is the corresponding scalar multiple of I . Thus these are the squares of its operator norm, or equivalently $\text{Tr}(H_N^2)/2$. The spectral aggregate for A_x is $L_{\text{spec}} = 2\sigma$, since its eigenvalues are $c + \sigma, c - \sigma$. Hence equality $\epsilon_1 = L_{\text{spec}}$ holds at the first admitted degree, whereas $\epsilon_2 > L_{\text{spec}}$ holds at the next. No freedom to choose another metric was used in either calculation.

To compute the auxiliary derivative at the original observation, replace $d\nu$ by $e^{\theta u} d\nu$ for real θ and set $\ell_N(\theta) = \log \det G_N(\theta)$ in either specified quotient basis. The precise real-coordinate moment Grams are $H_{rs}(\theta) = \int u^{r+s} e^{\theta u} d\nu$ and $H_{rs}^X(\theta) = \int u^{r+s} (u^2 + \sigma^2)^2 e^{\theta u} d\nu$. Their matrices at $-\theta$ are conjugated by $\text{diag}(1, -1, 1, \dots)$ from those at θ . The exact map to original monomials is $S^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r$. On dimension n its triangular determinant is $i^{n(n-1)/2}$, of modulus one. It therefore identifies these determinants with the original source and relation determinants, including the relation multiplier χ . Both determinants are even functions of θ . Their positive determinant ratio is therefore even, giving $\ell'_N(0) = 0$. Also $\sigma_T := \text{Tr}((A_x - cI)/i) = 0$. This proves the vanishing of the phase subtraction at zero tilt; it does not impose evenness at a nonzero tilt.

Finally set $\delta_N = V_N/V_{N-1}$ at the two degrees where both metrics exist, and set $\mathcal{R}_2 = V_1/V_3 = (\delta_2\delta_3)^{-1}$. Equations (TC3) and (TC5) give

$$\delta_2 = 1/3, \quad \delta_3 = 3/11, \quad a_3 = \omega_3/\omega_2 = 3\sigma^2, \quad \mathcal{R}_2 = 11. \quad (\text{TC8})$$

The complete Toda expression and its finite upper bound are respectively

$$a_3(1 - \delta_2)(\delta_3^{-1} - 1) = 16\sigma^2/3, \quad a_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} = 75\sigma^2/11. \quad (\text{TC9})$$

To identify the exact lost term, retain $s = (\log 3 + \log(11/3))/2 = \log(11)/2$ and $t = (\log(11/3) - \log 3)/2$. Then $e^t = \sqrt{11}/3$ and $\cosh s = 6/\sqrt{11}$, whence

$$a_3(e^t - \cosh s)^2 = 49\sigma^2/33 = 75\sigma^2/11 - 16\sigma^2/3. \quad (\text{TC10})$$

Thus the entire gap between the exact arithmetic-construction radius in this specified calibration and its Toda upper expression is precisely the retained imbalance square. The phase square is zero

for the proved parity reason. Both metric degrees, the quotient matrix, the source and relation volumes, and the original mass μ occur explicitly in this comparison.

There is also an explicit comparison with the spectral and orthogonal projectors in the exterior equality proof. In the $(1, x)$ coordinates let $v_+ = (\sigma, 1)^\top$ and $v_- = (-\sigma, 1)^\top$. They are eigenvectors of A_x for $c + \sigma$ and $c - \sigma$. Directly,

$$v_+^* G_{1,x} v_- = 0, \quad v_+^* G_{2,x} v_- = 2\mu\sigma^2/3. \quad (\text{TC11})$$

At $N = 2$ the orthogonal projector to $\mathbb{C}v_+$ and the algebraic spectral projector along $\mathbb{C}v_-$ are respectively

$$P = \frac{v_+ v_+^* G_{2,x}}{v_+^* G_{2,x} v_+} = \begin{pmatrix} 1/4 & 3\sigma/4 \\ 1/(4\sigma) & 3/4 \end{pmatrix}, \quad Q = \frac{A_x - (c - \sigma)I}{2\sigma} = \begin{pmatrix} 1/2 & \sigma/2 \\ 1/(2\sigma) & 1/2 \end{pmatrix}. \quad (\text{TC12})$$

These formulas prove $P^2 = P$, $P^{\dagger G_{2,x}} = P$, $Q^2 = Q$, and $\text{im } P = \text{im } Q = \mathbb{C}v_+$. Their exact kernels are $\ker P = \mathbb{C}(-3\sigma, 1)^\top = v_+^{\perp G_{2,x}}$ and $\ker Q = \mathbb{C}v_-$: direct multiplication proves the two zero values, while $Pv_+ = Qv_+ = v_+$ proves rank one. In particular $Pv_- = v_+/2$. They also retain the change to original coordinates: both matrices are sent to $L_{\text{coord}}(\cdot)L_{\text{coord}}^{-1}$ and paired with $G_{2,S} = L_{\text{coord}}^{-*} G_{2,x} L_{\text{coord}}^{-1}$. For $X = Q - P$, the exact Hilbert–Schmidt pairing gives

$$\|X\|_{\text{HS}, G_{2,x}}^2 = \text{Tr}(G_{2,x}^{-1} X^* G_{2,x} X) = 1/3, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|X\|_{\text{HS}, G_{2,x}}^2 = 4\sigma^2/3. \quad (\text{TC13})$$

For example the two diagonal entries of $G_{2,x}^{-1} X^* G_{2,x} X$ are $1/4$ and $1/12$, which proves the trace value without changing its pairing. The last equality follows also from (TC7). Thus the two explicit gaps have different exact source maps: (TC13) measures the spectral/orthogonal projector discrepancy, while (TC10) measures the square discarded by the Toda upper bound. Both maps and both positive quantities are calculated in the original specified metric.

37 The original gamma convolution and the exact arithmetic coefficient maps

37.1 Source, attribution, and original coordinates

This chapter proves the coefficient and cochain maps needed to integrate *Tau Gamma Convolution Descent*, whose original NOTE.tex has SHA256

7cf984d87de405b815250b44454c4ee27c8ab46574da245b377ebc6e49327fa2.

The analytic generating identities (GC.24)–(GC.26) were supplied by the *Zeta Function Literature & Local Corpus Foundation* task on 13 September 2026. The proofs here verify and integrate that supplied join; they do not claim its discovery. The original gamma generating integral and monic family are the existing TG construction, with the classical Meixner–Pollaczek parameters retained.

All scalar source coordinates remain

$$s = \frac{1}{2} + it, \quad S = s_1 + \cdots + s_k = \frac{k}{2} + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GC.1})$$

Here $k \geq 1$ is an integer. We use inner products conjugate-linear in the first argument. For a polynomial, an overline over its name means coefficientwise conjugation. In particular $\overline{P(c + iu)} = \overline{P(c - iu)}$ when c, u are real.

37.1.1 The analytic source and its actual theta boundary

Let $D = -x\partial_x$. Let V be the complex vector space of even Schwartz functions ϕ on $\mathbb{R}$ with $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi(x) dx = 0$. Let $\mathcal{B}$ be the smooth functions on $\mathbb{R}_{>0}$ such that for every $a > 0$ and every integer $j \geq 0$,

$$D^j F(x) = O(x^a) \quad (x \downarrow 0), \quad D^j F(x) = O(x^{-a}) \quad (x \rightarrow \infty).$$

The original theta map and cochain grading are

$$\Theta : V \longrightarrow \mathcal{B}, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1. \quad (\text{GC.2})$$

The sum and its differentiated sums converge at infinity by the Schwartz bounds. Poisson summation gives its corresponding bounds at zero: the zero Fourier coefficient vanishes by $\int \phi = 0$, and the original zero term vanishes by $\phi(0) = 0$. Differentiation preserves both conditions, because $D\phi(0) = 0$ and integration by parts gives $\int D\phi = \int \phi = 0$. Thus $D\Theta = \Theta D$ on these spaces. The rapid bounds make the Mellin integral

$$\mathcal{M}F(s) = \int_0^\infty F(x)x^{s-1} dx$$

entire for $F \in \mathcal{B}$ and give $\mathcal{M}(DF)(s) = s\mathcal{M}F(s)$, including the vanished endpoint terms.

For clarity the seed and all its constants can be constructed directly. Put

$$\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad f_0 = D(D-1)\vartheta = \Theta\phi_0. \quad (\text{GC.3})$$

Gaussian integration gives $\int 4\pi^2 x^4 e^{-\pi x^2} dx = 3$ and $\int 6\pi x^2 e^{-\pi x^2} dx = 3$ on $\mathbb{R}$, and $\phi_0(0) = 0$, so $\phi_0 \in V$. The Gaussian Fourier transform in the convention $\widehat{f}(y) = \int f(x)e^{-2\pi ixy} dx$ is $e^{-\pi y^2}$: differentiation under the integral and integration by parts give $\widehat{f}' = -2\pi y\widehat{f}$, and the integral at zero is one. Periodizing this Gaussian gives an absolutely convergent Fourier series with those computed coefficients. Evaluating it at zero proves $\vartheta(1/x) = x\vartheta(x)$ for $x > 0$. Holomorphic continuation gives the same identity for $|\arg z| < \pi/4$, where both theta series converge normally on compact subsets.

For $RF(x) = x^{-1}F(1/x)$, direct differentiation gives $DR = R(1-D)$. Hence $D(D-1)$ commutes with R and

$$f_0(1/z) = zf_0(z). \quad (\text{GC.4})$$

On every closed narrower sector, each derivative of f_0 at infinity is bounded by a polynomial times $e^{-c|z|^2}$ for some $c > 0$. Equation (GC.4), differentiated finitely many times, proves rapid decay at zero too. In particular $f_0 \in \mathcal{B}$.

For $\Re s > 1$, absolute convergence permits termwise Mellin integration of $2 \sum_{n \geq 1} \phi_0(nx)$. The change $y = \pi n^2 x^2$ gives

$$\begin{aligned} \mathcal{M}f_0(s) &= \pi^{-s/2} \zeta(s) \{4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)\} \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GC.5})$$

The first line retains the theta factor two and the integration Jacobian $1/2$. The gamma recurrence proves the second line. The Mellin integral gives its entire continuation.

Let $h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}$ be a finite monic packet of actual nontrivial zeros of g , each selected with its complete order. The empty packet is $h = 1$. Write $d = \deg h$ and $v_h = g/h$. At each selected centre, cancelling its exact local factor constructs the analytic value of v_h ; hence v_h is entire. It is nonzero at each selected centre and is not identically zero.

We record the bounds that justify the integrals below. For any closed real interval of σ , any $0 < \beta < \pi/4$, and any $j \geq 0$,

$$|v_h(\sigma + it)| \leq C_{j,\beta,h}(1 + |t|)^{-j} e^{-\beta|t|}. \quad (\text{GC.6})$$

To prove this first for g , rotate its Mellin contour to $z = e^{i\beta}x$ for $t \geq 0$ and to $z = e^{-i\beta}x$ for $t < 0$. The sector estimates after (GC.4) make the two closing arcs vanish, uniformly for the stated σ interval. The rotation contributes $e^{\mp\beta t}$ and a unit-modulus factor depending on σ . Repeated integration by parts along that ray gives every displayed power of $(1 + |t|)^{-1}$; all ray endpoint terms vanish by the same estimates. For $|t|$ larger than twice the moduli of all packet centres relative to that σ interval, monicity gives a lower bound for $|h(\sigma + it)|$ by a positive constant times $|t|^d$. On the remaining compact rectangle the cancelled entire quotient is bounded. This proves (GC.6) for v_h .

For any real σ define

$$F_h(x) = \frac{1}{2\pi} \int_{\mathbb{R}} v_h(\sigma + it) x^{-\sigma - it} dt. \quad (\text{GC.7})$$

Horizontal rectangle estimates from (GC.6) prove independence of σ . Differentiating in x multiplies the integrand by a power of $\sigma + it$ and remains absolutely convergent. Choosing σ arbitrarily large positive at infinity and arbitrarily large negative at zero proves $F_h \in \mathcal{B}$. Fourier inversion in $\log x$, with the displayed $1/(2\pi)$ constant, gives $\mathcal{M}F_h = v_h$. Consequently Mellin injectivity gives

$$h(D)F_h = f_0 = \Theta\phi_0. \quad (\text{GC.8})$$

Define the actual density, mass, and Laplace integral by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt. \quad (\text{GC.9})$$

Equation (GC.6) proves holomorphy of M_h on $|\Re\theta| < \pi/2$, with $M_h^{(j)}(\theta) = \int t^j e^{\theta t} w_h(t) dt$ there. Indeed any compact substrip is dominated using a β for which 2β is larger than its real width. The mass is positive because v_h is a nonzero analytic function; its real-line zeros are discrete. Every original spectral moment and its 2π factor are retained.

37.2 The gamma reference with its actual mass

Fix $\lambda > 0$ and write $\alpha = 2\lambda$. Define

$$d\sigma_\lambda(t) = r_\lambda(t) dt, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (\text{GC.10})$$

The product of two gamma integrals, changed by $(a, b) = (ry, (1 - r)y)$ with Jacobian y , proves the beta integral. Its substitution $r/(1 - r) = e^x$ gives

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{\Gamma(\lambda + iv)\Gamma(\lambda - iv)}{\Gamma(2\lambda)}.$$

The integrand before Fourier transformation and its second derivative are integrable. Two integrations by parts make its transform integrable, so Fourier inversion is valid. Alternatively, multiplying that transform by $e^{-\varepsilon v^2}$ gives an absolutely integrable Gaussian approximation and proves the same inversion by dominated convergence. The substitution $t = 2v$, with Jacobian two, therefore proves

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GC.11})$$

The beta integrand extends holomorphically to $|\Im x| < \pi$ using the branch positive on $\mathbb{R}$. On every closed narrower strip it is bounded by a constant times $e^{-\lambda|\Re x|}$. Shifting the contour up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_{\lambda,b} e^{-b|v|}$ for every $b < \pi$. Thus σ_λ has exponential moments of every order less than $\pi/2$ in the original t variable.

Fourier uniqueness for finite measures and multiplication of (GC.11) give the literal sum density

$$r_{\lambda,k} := r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}, \quad \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (\text{GC.12})$$

Its mass is c_λ^k . Analytic continuation justified by the proved exponential moments gives

$$Z_{\lambda,k}^\Gamma(\theta) = \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-k\alpha}, \quad |\Re \theta| < \pi/2. \quad (\text{GC.13})$$

The power uses the analytic logarithm of $\cos \theta$ equal to zero at zero; this cosine has no zeros on the stated strip.

For $|z| < 1$, let

$$\begin{aligned} \mathcal{G}_\lambda(z, t) &= (1 - iz)^{-\lambda+it/2} (1 + iz)^{-\lambda-it/2} \\ &= \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n. \end{aligned} \quad (\text{GC.14})$$

Both logarithms are the principal logarithms and vanish at zero, since $1 \pm iz$ have positive real part. For real z, w near zero, the Laplace transform and the elementary cosine addition identity give

$$\int \mathcal{G}_\lambda(z, t) \mathcal{G}_\lambda(w, t) d\sigma_\lambda(t) = c_\lambda (1 - zw)^{-\alpha}.$$

On a sufficiently small closed bidisc all derivatives of the integrands are bounded by a polynomial times $e^{a|t|}$ with $a < \pi/2$. The exponential moment proof permits their integration. Coefficient comparison then proves, with the monic leading coefficient obtained from $\arctan z = z + O(z^3)$,

$$\begin{aligned} \int b_n^{(\lambda)} b_j^{(\lambda)} d\sigma_\lambda &= c_\lambda n! (\alpha)_n \delta_{nj}, \\ b_0^{(\lambda)} &= 1, \quad b_1^{(\lambda)} = t, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \end{aligned} \quad (\text{GC.15})$$

The recurrence follows directly by differentiating (GC.14): $(1 + z^2) \partial_z \mathcal{G}_\lambda = (t - \alpha z) \mathcal{G}_\lambda$. The polynomials have real coefficients.

They are complete in $L^2(r_{\lambda,k} du)$ at parameter $k\lambda$. In fact, an orthogonal vector f defines the finite complex measure $f(u) r_{\lambda,k}(u) du$. Cauchy-Schwarz and exponential integrability give its Fourier transform on a nonempty complex strip. All derivatives at zero vanish by polynomial orthogonality, so analytic uniqueness and Fourier uniqueness imply $f = 0$ almost everywhere. This argument uses the full measure mass.

37.3 The sum projection and exact coefficient powers

On the actual spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du),$$

let $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$. Fubini and (GC.12) prove that it is an isometry. The affine fibre chart has Jacobian one, so its adjoint for $k \geq 2$ is

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GC.16})$$

The denominator is positive everywhere. Fibre Cauchy-Schwarz and Fubini give $\|C_\Sigma F\| \leq \|F\|$, so the formula defines the indicated adjoint on all of $\mathcal{H}_k$. When $k = 1$, both maps are the identity. In every case $C_\Sigma U_\Sigma = 1$ and $P_\Sigma = U_\Sigma C_\Sigma$ is the orthogonal projection. The maps

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n, \quad n \in \ker C_\Sigma \quad (\text{GC.17})$$

are inverse Hilbert isometries. This retains the entire relative fibre.

Multiplying the k generating functions gives a finite identity at each coefficient:

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1+\dots+n_k=n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (\text{GC.18})$$

Pairing it with a product polynomial and using (GC.15) leaves only its total degree $n = \sum_i n_i$. The pairing at that degree is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$ and the sum norm is $c_\lambda^k n! (k\alpha)_n$. Completeness therefore proves

$$C_\Sigma \prod_i b_{n_i}^{(\lambda)} = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}. \quad (\text{GC.19})$$

The squared norm of the retained complementary polynomial is exactly

$$c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right], \quad (\text{GC.20})$$

by the orthogonal decomposition (GC.17).

For the original arithmetic density define, for every $\lambda > 0$,

$$\begin{aligned} A_{h,\lambda}(t) &= \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda} = |A_{h,\lambda}|^2, \quad w_h = B_{h,\lambda} r_\lambda, \\ c_{h,n} &= \frac{\int b_n^{(\lambda)}(t) w_h(t) dt}{c_\lambda n! (\alpha)_n}, \quad \mathcal{A}_h(z) = \sum_{n \geq 0} (\alpha)_n c_{h,n} z^n, \\ d_{h,k,n} &= [z^n] \mathcal{A}_h(z)^k. \end{aligned} \quad (\text{GC.21})$$

The notation $\mathcal{A}_h$ retains the fixed parameter λ . The complex amplitude $A_{h,\lambda}$ and the generating function $\mathcal{A}_h$ are different specified maps. The integrals in (GC.21) converge by (GC.6), before any convergence claim about the power series. Each $d_{h,k,n}$ is the finite expression

$$d_{h,k,n} = \sum_{n_1+\dots+n_k=n} \prod_i (\alpha)_{n_i} c_{h,n_i}. \quad (\text{GC.22})$$

Let $m_{h,k} = w_h^{*k}$. Absolute moments and Fubini give, by (GC.18),

$$\int_{\mathbb{R}} b_n^{(k\lambda)}(u) m_{h,k}(u) du = c_\lambda^k n! d_{h,k,n}. \quad (\text{GC.23})$$

This finite identity holds for every $\lambda > 0$ and every n ; it does not require an infinite product to be interchanged with an integral. In particular $c_{h,0} = \mu_h/c_\lambda$ and the total mass is μ_h^k .

37.4 The supplied analytic generating join, with its branches

Theorem 37.1 (The exact analytic generating function). *For the original arithmetic measure and every $\lambda > 0$, the series in (GC.21) converges holomorphically on $|z| < 1$ and is exactly*

$$\mathcal{A}_h(z) = \frac{1}{c_\lambda} (1 + z^2)^{-\lambda} M_h(\arctan z). \quad (\text{GC.24})$$

Here $\arctan z = (\log(1 + iz) - \log(1 - iz))/(2i)$, and every logarithm is the branch inherited from zero. For real $|\theta| < \pi/4$,

$$M_h(\theta) = c_\lambda \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta). \quad (\text{GC.25})$$

With the original tensor mass and reference mass,

$$\frac{Z_{h,k}(\theta)}{Z_{\lambda,k}^\Gamma(\theta)} = \mathcal{A}_h(\tan \theta)^k, \quad Z_{h,k}(\theta) = \int e^{\theta u} m_{h,k}(u) du = M_h(\theta)^k. \quad (\text{GC.26})$$

Proof. For $|z| < 1$, both $1 \pm iz$ have positive real part. Their principal logs are analytic, and their sum is the analytic logarithm of $1 + z^2$ vanishing at zero. Direct substitution in (GC.14) gives

$$\mathcal{G}_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GC.27})$$

There is no omitted phase in this identity: the $it/2$ coefficient multiplies $\log(1 - iz) - \log(1 + iz) = -2i \arctan z$.

The ratio $w = (1 + iz)/(1 - iz)$ has

$$\Re w = \frac{1 - |z|^2}{|1 - iz|^2} > 0, \quad \tan(2\Re \arctan z) = \frac{2\Re z}{1 - |z|^2}.$$

The argument of w lies in $(-\pi/2, \pi/2)$ and equals $2\Re \arctan z$. Thus for every $0 \leq r < 1$,

$$|z| \leq r \implies |\Re \arctan z| \leq \arctan r < \pi/4, \quad |\mathcal{G}_\lambda(z, t)| \leq (1 - r^2)^{-\lambda} e^{\arctan(r)|t|}. \quad (\text{GC.28})$$

For the angle inequality use monotonicity of arctangent and $2|\Re z|/(1 - |z|^2) \leq 2r/(1 - r^2)$, together with $2\arctan r \in [0, \pi/2)$. For the modulus inequality use $|1 + z^2| \geq 1 - r^2$ and real $\lambda > 0$.

Equation (GC.6) makes the right side of (GC.28) integrable against $w_h(t) dt$, uniformly on every compact subdisc. Dominated integration therefore defines a holomorphic function $c_\lambda^{-1} \int \mathcal{G}_\lambda(z, t) w_h(t) dt$ on the whole unit disc. One explicit justification is Morera's theorem: domination permits exchanging the integral with each triangle contour, and the inner holomorphic contour integral is zero. For its derivatives on $|z| \leq r$, choose $r < R < 1$ and apply Cauchy's derivative formula on circles of radius $R - r$; the kernel derivative $|\partial_z^n \mathcal{G}_\lambda(z, t)|$ is bounded by $n!(R - r)^{-n}(1 - R^2)^{-\lambda} e^{\arctan(R)|t|}$. This integrable bound justifies every derivative interchange.

At zero, (GC.14) now gives

$$\mathcal{A}_h^{(n)}(0) = \frac{1}{c_\lambda} \int b_n^{(\lambda)}(t) w_h(t) dt = n!(\alpha)_n c_{h,n}. \quad (\text{GC.29})$$

Thus the Taylor series of this holomorphic integral is exactly the original coefficient series. Substitution of (GC.27) in the integral proves (GC.24). All its constants occur before the integration.

For real $|\theta| < \pi/4$, $|\tan \theta| < 1$, $\arctan(\tan \theta) = \theta$ on the specified branch, and $1 + \tan^2 \theta = \sec^2 \theta > 0$. Solving (GC.24) for $M_h(\theta)$ gives (GC.25). Finally the absolute integral of $\prod_i e^{\theta t_i} w_h(t_i)$ is finite throughout $|\Re \theta| < \pi/2$. Fubini gives $Z_{h,k} = M_h^k$ with no mass division. Divide the k th power of (GC.25) by (GC.13) to obtain (GC.26). At zero this ratio is explicitly $(\mu_h/c_\lambda)^k$. $\square$

The inverse also has a precise complex domain. If $|\Re \theta| < \pi/4$, then

$$|\tan \theta|^2 = \frac{\cosh(2\Im \theta) - \cos(2\Re \theta)}{\cosh(2\Im \theta) + \cos(2\Re \theta)} < 1.$$

The denominator is positive and the branch satisfies $\arctan(\tan \theta) = \theta$, first on a neighbourhood of zero and then on this connected strip by analytic uniqueness. Hence (GC.25)–(GC.26) extend to that strip with the analytic branch of the cosine power in (GC.13). No logarithm of $\mathcal{A}_h$ is asserted: it can have complex zeros, whereas all real-tilt integrals $M_h(\theta)$ are strictly positive.

For each finite derivative there is also an exact moment formula. Write $b_n^{(\lambda)}(t) = \sum_{j=0}^n b_{n,j} t^j$. Then

$$n!(\alpha)_n c_{h,n} = \frac{1}{c_\lambda} \sum_{j=0}^n b_{n,j} M_h^{(j)}(0), \quad d_{h,k,n} = \frac{1}{n!} \left. \frac{d^n}{dz^n} \mathcal{A}_h(z)^k \right|_{z=0}. \quad (\text{GC.30})$$

The first identity follows by the finite polynomial expansion and (GC.9); the second is Taylor's formula, whose Leibniz sum is exactly (GC.22). The triangular coefficients $b_{n,n} = 1$ give an invertible finite map between moments through degree n and coefficients through degree n , with the full diagonal factors $c_\lambda n!(\alpha)_n$ retained.

37.5 The complete density expansion for the original bounded references

The finite identities and analytic join above hold for every $\lambda > 0$. For the original full-quartet reference choose $\lambda = 1/4$. More generally choose the actual shifted reference

$$\lambda = \frac{1}{4} + L, \quad L \text{ an integer}, \quad L \geq \max(0, 3 - d). \quad (\text{GC.31})$$

These choices admit the following proved bounded multiplier and hence the complete L^2 density expansion.

Indeed at $s = 1/2 + it$ the defining g identity gives

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (\text{GC.32})$$

This expression is continued through each selected zero by the already defined entire quotient. Integrating the step function $\lfloor x \rfloor$, initially for $\Re s > 1$, gives $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$. The last integral is holomorphic for $\Re s > 0$, so the identity continues there away from 1. On the original line it gives $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For a nonempty packet put $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$; for $h = 1$ put $T_h = 1$. For $|t| \geq T_h$, every original packet factor has modulus at least $|t|/2$. Gamma recurrence then gives

$$\begin{aligned} A_{h,1/4+L}(t) &= \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}, \\ B_{h,1/4+L}(t) &\leq \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} |t|^{6-2d-2L}. \end{aligned} \quad (\text{GC.33})$$

For the constant 25, the two squared numerator estimates at $|t| \geq 1$ are $(t^2 + 1/4)^2 \leq 25t^4/16$ and $|\zeta(s)|^2 \leq 16t^2$. Each shift factor has modulus at least $|t|/2$. The exponent in (GC.33) is nonpositive for (GC.31). Thus a completely specified finite bound is

$$C_{h,L} = \max \left\{ \sup_{|t| \leq T_h} B_{h,1/4+L}(t), \frac{25 \cdot 2^{2(d+L)}}{\sqrt{\pi}} T_h^{6-2d-2L} \right\} > 0. \quad (\text{GC.34})$$

The compact term is finite by continuity of the cancelled multiplier. This is a mathematical bound with a specified supremum, not a claim that a numerical packet supremum has already been enclosed.

For these references $B \in L^2(d\sigma_\lambda)$, because the reference has finite mass and $0 \leq B \leq C_{h,L}$. Fibre integration of $w_h = Br_\lambda$ proves

$$B_{h,\lambda;k} = C_\Sigma(B^{\otimes k}), \quad m_{h,k} = r_{\lambda,k} B_{h,\lambda;k}, \quad 0 \leq B_{h,\lambda;k} \leq C_{h,L}^k. \quad (\text{GC.35})$$

For $k \geq 2$ it is positive at every sum coordinate: the zeros of B form a countable discrete set, and their forbidden fibre coordinates are a countable union of affine hyperplanes of measure zero. Every remaining reference factor is positive. For $k = 1$ the original isolated zeros remain.

Combining (GC.23), completeness, and the sum norm in (GC.15) gives the full, convergent $L^2(r_{\lambda,k} du)$ identity

$$B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (\text{GC.36})$$

In particular the norm of the retained relative density vector is

$$\|B^{\otimes k} - U_\Sigma B_{h,\lambda;k}\|^2 = \left(\int B^2 d\sigma_\lambda \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (\text{GC.37})$$

This is Pythagoras for (GC.17), so every norm and mass factor is retained. A truncation of (GC.36) is not asserted pointwise positive.

37.6 Original-coordinate finite Grams and every tilt derivative

Let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ for $N \geq 0$, with original ordered basis $1, S, \dots, S^N$, and set $c = k/2$. The arithmetic source Gram and its reference are

$$\begin{aligned} (\mathbf{M}_{h,k,N}(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} m_{h,k}(u) du, \\ (\mathbf{M}_{\lambda,k,N}^\Gamma(\theta))_{ij} &= \int (c - iu)^i (c + iu)^j e^{\theta u} r_{\lambda,k}(u) du, \quad 0 \leq i, j \leq N. \end{aligned} \quad (\text{GC.38})$$

They are holomorphic entrywise for $|\Re \theta| < \pi/2$ and positive definite at every real tilt in that strip: a nonzero polynomial has only finitely many real-line roots and each measure is positive almost everywhere. Let $\mathbf{H}_{h,k,N}$ denote the corresponding real- u monomial Hankel matrix. The exact matrix of the unchanged coordinate substitution is

$$(\mathbf{L}_{N+1})_{aj} = \begin{cases} \binom{j}{a} c^{j-a} i^a, & 0 \leq a \leq j \leq N, \\ 0, & a > j, \end{cases} \quad \det \mathbf{L}_{N+1} = i^{N(N+1)/2}. \quad (\text{GC.39})$$

Expanding each original S monomial proves

$$\mathbf{M}_{h,k,N} = \mathbf{L}_{N+1}^* \mathbf{H}_{h,k,N} \mathbf{L}_{N+1} \quad (\text{GC.40})$$

and the identical congruence for the reference. Since the displayed determinant has modulus one, it also proves exact equality of the two coordinate determinants. The congruence, rather than a relabeling of a matrix, preserves every complex phase in its entries.

For an integer $r \geq 0$, expand the specified polynomial uniquely as

$$u^r (c - iu)^i (c + iu)^j = \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} b_n^{(k,\lambda)}(u). \quad (\text{GC.41})$$

The monic triangular basis makes these coefficients finite, unique, and computable by polynomial subtraction in the written order. Differentiation of (GC.38) and (GC.23) now give the exact formula

$$\partial_\theta^r (\mathbf{M}_{h,k,N})_{ij} |_{\theta=0} = c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.42})$$

The derivative interchange is justified by the exponential bounds after (GC.9), or directly by Fubini on the k -fold product. Since $d_{h,k,n}$ uses exactly the finite coefficient power (GC.22), the entire r th derivative matrix is determined by $c_{h,0}, \dots, c_{h,2N+r}$. At zero tilt $r = 0$, the list ends at $2N$. There is no truncation error in this finite statement.

Equation (GC.42) supplies each derivative at zero from its stated finite coefficient list. Evaluation at a nonzero tilt is related to those derivatives by the following exact analytic Taylor identity, valid for $|\theta| < \pi/2$:

$$(\mathbf{M}_{h,k,N}(\theta))_{ij} = \sum_{r=0} \frac{\theta^r}{r!} c_\lambda^k \sum_{n=0}^{i+j+r} e_{ijr,n}^{(k,\lambda)} n! d_{h,k,n}. \quad (\text{GC.43})$$

Absolute convergence follows by bounding the absolute exponential series by $e^{|\theta||u|}$ times the original polynomial modulus. For larger imaginary components within the full holomorphic strip, analytic continuation of these entries is the defined original Laplace integral (GC.38). This specifies the relation between all finite coefficient jets and the full tilted metric.

37.7 The original cyclic quotient and the exact cochain correction

For a nonempty packet let $E_h = \mathbb{C}[s]/(h)$ and $\mathcal{E}_h = E_h^{\otimes k}$. Let a be the element $s_1 + \cdots + s_k$ of this finite algebra. The kernel of $\mathbb{C}[S] \rightarrow \mathcal{E}_h, P \mapsto P(a)$, is generated by its unique monic polynomial χ . It is nonzero since the powers of a lie in a finite-dimensional vector space. Put $q = \deg \chi \geq 1$, $C = \mathbb{C}[S]/(\chi)$, and $A = M_S : C \rightarrow C$. The induced map $C \rightarrow \mathcal{E}_h$ is injective by the definition of the kernel. For each selected zero, the complete Taylor value of g/h is nonzero at its centre; its residue $v_h = j_h(g/h) \in E_h$ is consequently a unit in every local factor and in E_h . Thus the original arithmetic injection

$$\eta_{h,k} : C \hookrightarrow \mathcal{E}_h, \quad [P] \mapsto v_h^{\otimes k} P(a) \quad (\text{GC.44})$$

retains the entire local unit and every selected multiplicity.

The original analytic polynomial map is

$$\mathcal{V}_{h,k} P = P(D_1 + \cdots + D_k) F_h^{\otimes k}. \quad (\text{GC.45})$$

Its tensor Mellin transform is $P(s_1 + \cdots + s_k) \prod_i v_h(s_i)$. Taking complete selected jets gives precisely (GC.44) after the quotient π_χ . No modulus operation has been applied to that arithmetic jet map.

Successive monic divisions, in the order $s_1, \dots, s_k$, provide specific polynomials $Q_i(s_1, \dots, s_k)$ with

$$\chi(s_1 + \cdots + s_k) = \sum_{i=1}^k h(s_i) Q_i(s_1, \dots, s_k). \quad (\text{GC.46})$$

To prove this, divide in s_1 , retain its remainder, divide that remainder in s_2 , and continue. The final remainder has degree less than d in every coordinate. Its residue in $\mathcal{E}_h$ is zero by the defining property of χ , and the tensor remainder monomials form a basis, so every coefficient is zero. This proves the displayed identity and fixes the division order.

The operators D_i commute on the tensor complex. For a polynomial P , define the actual cochain of degree $k - 1$ by

$$\begin{aligned} \mathcal{P}_{\chi,h,k}(P) = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) P(D_1 + \cdots + D_k) \\ \cdot (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \end{aligned} \quad (\text{GC.47})$$

Every summand belongs to the indicated tensor source, since D preserves V and $\mathcal{B}$. The first $i - 1$ factors have cochain degree one. The tensor differential on the i th factor contributes $(-1)^{i-1}$, cancelling the written sign. The other factors have zero differential. Equations (GC.8) and (GC.46) therefore give

$$d_{\text{tot}} \mathcal{P}_{\chi,h,k}(P) = \mathcal{V}_{h,k}(\chi P). \quad (\text{GC.48})$$

This identifies the cochain primitive, its sign, and its boundary in their different cochain degrees.

For the full quotient at finite source degree take $N \geq q - 1$ and $m = N - q + 1$. With $\mathcal{P}_{-1} = 0$, monic division gives the exact sequence in original coordinates

$$0 \longrightarrow \mathcal{P}_{m-1} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad J_N P = [P]_\chi. \quad (\text{GC.49})$$

The kernel assertion is divisibility, and the first q monomial columns prove surjectivity. Let B_χ be the $(N+1) \times m$ matrix of M_χ in the original ordered bases. Its entries are $[S^i](\chi(S)S^j)$; it is independent of the metric and tilt. The relation Gram and every zero-tilt derivative are therefore

$$\mathbf{B}_{h,k,N}(\theta) = B_\chi^* \mathbf{M}_{h,k,N}(\theta) B_\chi, \quad \mathbf{B}_{h,k,N}^{(r)}(0) = B_\chi^* \mathbf{M}_{h,k,N}^{(r)}(0) B_\chi. \quad (\text{GC.50})$$

Equivalently its polynomial integrand is $u^r |\chi(c+iu)|^2 (c-iu)^i (c+iu)^j$, where $0 \leq i, j < m$. Its degree is at most $r + 2q + 2(m-1) = r + 2N$, so the relation derivative uses the same list through $2N + r$

as the source derivative. Both complex polynomial factors, their conjugates, and all χ coefficients remain in this formula.

For a fixed real admitted tilt write $M_h = \mathbf{M}_{h,k,N}(\theta)$ and $M_\Gamma = \mathbf{M}_{\lambda,k,N}^\Gamma(\theta)$. Their canonical quotient Grams and minimum lifts are

$$G_h = (J_N M_h^{-1} J_N^*)^{-1}, \quad R_h = M_h^{-1} J_N^* G_h, \quad G_\Gamma = (J_N M_\Gamma^{-1} J_N^*)^{-1}, \quad R_\Gamma = M_\Gamma^{-1} J_N^* G_\Gamma. \quad (\text{GC.51})$$

Surjectivity of J and positivity of the source Grams prove every inverse exists. These formulas give $JR_h = JR_\Gamma = I_C$ and make each lift orthogonal to $\ker J$ in its own metric, proving its minimum property by the orthogonal Pythagorean identity. For $m > 0$ put

$$K = (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_\Gamma : C \longrightarrow \mathbb{C}^m. \quad (\text{GC.52})$$

The relation Gram is positive definite because B_χ is injective. The section $R_\Gamma - B_\chi K$ still has quotient identity and is orthogonal to the relation space in M_h . The preceding uniqueness gives the exact section map

$$\boxed{R_h = R_\Gamma - B_\chi K.} \quad (\text{GC.53})$$

At $N = q - 1$, $m = 0$, take K to be the zero map to the zero vector space and omit the inverse in (GC.52); (GC.53) remains the identity of the unique degree-less-than- q representatives.

For $u_0 \in C$, let $\kappa_{u_0}(S)$ be the polynomial whose coefficient vector is Ku_0 . Combining (GC.48) and (GC.53) proves the exact cochain correction

$$\boxed{\mathcal{V}_{h,k}(R_h u_0 - R_\Gamma u_0) = -\mathcal{V}_{h,k}(\chi \kappa_{u_0}) = d_{\text{tot}}(-\mathcal{P}_{\chi,h,k}(\kappa_{u_0})).} \quad (\text{GC.54})$$

Thus applying $\mathcal{V}$ gives the boundary in degree k ; the explicit negative cochain on the right is its primitive in degree $k - 1$. Passing to the original arithmetic quotient sends this boundary to zero. Explicitly, for the original set of labels $\mathcal{L}$ and vector spaces E_ℓ , write the supplied carrier as $\{\tau\} \sqcup \bigsqcup_{\ell \in \mathcal{L}} (\{\ell\} \times E_\ell)$. A family of linear maps $L_\ell : E_\ell \rightarrow F_\ell$ has the exact support lift $\tau \mapsto \tau$ and $(\ell, v) \mapsto (\ell, L_\ell v)$, with $e_\ell = (\ell, 0)$ in each target fibre. Substitution proves that these lifts preserve identities and composition. Applied to the original quotient, the relation boundary in (GC.54) therefore has image e_ℓ , while the external τ has image τ . The source vector and its primitive remain available before that quotient.

37.8 Determinant indices and the original Toda substitution

For real admitted θ , write $\mathfrak{D}_n$ for the original degree-less-than- n source Gram determinant and $\mathfrak{B}_m$ for the relation Gram determinant on m multiplication columns. Empty determinants are one. The analogous reference determinants carry the superscript Γ . For $N \geq q - 1$, set

$$V_N = \det G_h, \quad V_N^\Gamma = \det G_\Gamma, \quad X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_m = \mathfrak{B}_m / \mathfrak{B}_m^\Gamma.$$

The quotient-first coefficient matrix with columns $1, S, \dots, S^{q-1}, \chi, \chi S, \dots, \chi S^{m-1}$ has determinant one: its columns have successively distinct degrees and all their leading coefficients are one. Replacing its first q columns by the corresponding minimum sections only subtracts relation columns, so it still has determinant one. Orthogonality then gives the exact block determinant formula

$$V_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}} = V_N^\Gamma T_N, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}, \quad N \geq q - 1. \quad (\text{GC.55})$$

In particular the first full quotient has denominator $\mathfrak{B}_0 = 1$ and requires no relation inverse.

For completeness the reference source determinant is

$$\mathfrak{D}_n^\Gamma(\theta) = c_\chi^{kn} \prod_{j=0}^{n-1} j!(k\alpha)_j (\cos \theta)^{-n(k\alpha+n-1)}, \quad a_n^\Gamma(\theta) = n(n + k\alpha - 1) \sec^2 \theta. \quad (\text{GC.56})$$

Here a_n is the ratio of consecutive monic squared norms in the real- u coordinate; the norm-preserving phase of (GC.39) gives the same ratios in the original S coordinate. A direct proof of the Hankel identity used to obtain (GC.56) is as follows. Let p_j be the monic orthogonal polynomials for an $e^{\theta u}$ -tilted positive measure, let h_j be their squared norms, and write $up_j = p_{j+1} + b_j p_j + a_j p_{j-1}$, with $p_{-1} = 0$ and $a_0 = 0$. For $j \geq 1$, pairing with p_{j-1} gives $a_j = h_j/h_{j-1}$, and pairing with p_j gives $h'_j = b_j h_j$ for every $j \geq 0$. Differentiating orthogonality for $j \geq 1$ shows $\partial_\theta p_j = -a_j p_{j-1}$: its pairings with degrees at most $j-2$ vanish, and the remaining pairing is $-h_j$. For $j \geq 1$, write $p_j = u^j + c_j u^{j-1} + \dots$, and set $c_0 = 0$ for $p_0 = 1$. The derivative statement for p_0 is $\partial_\theta p_0 = 0$. Thus $c'_j = -a_j$ and $b_j = c_j - c_{j+1}$ hold for every $j \geq 0$, and consequently $b'_j = a_{j+1} - a_j$. Since $\mathfrak{D}_n = \prod_{j < n} h_j$,

$$(\log \mathfrak{D}_n)'' = \sum_{j < n} b'_j = a_n = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1} / \mathfrak{D}_n^2, \quad n \geq 1.$$

For the reference, the determinant values at $n=0$ and $n=1$ are 1 and (GC.13). The right side of the proposed formula (GC.56) has second logarithmic derivative $n(k\alpha + n - 1) \sec^2 \theta$. Its consecutive ratio has exactly the same value, with constants $n!(k\alpha)_n / ((n-1)!(k\alpha)_{n-1})$ and cosine exponent -2 . Positive induction in the displayed determinant identity proves (GC.56) for every n .

The reference relation determinant is the determinant of the moment matrix with entries

$$\partial_\theta^{i+j} \{ \bar{\chi}(c - i\partial_\theta) \chi(c + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \}, \quad 0 \leq i, j < m. \quad (\text{GC.57})$$

Differentiation under the integral gives the unchanged weight $|\chi(c + iu)|^2 r_{\lambda,k}(u)$ in this u -coordinate matrix. The relation version of (GC.39)–(GC.40) has determinant of modulus one, proving that (GC.57) is the original S -coordinate relation determinant as stated. It contains the actual χ with all coefficients.

At $\theta = 0$ put

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

The consecutive full-quotient formulas have domain $N \geq q$ and give, by literal substitution in (GC.55)–(GC.56),

$$a_{N+1} = (N+1)(N+k\alpha)Q_N, \quad \frac{V_{N-1}}{V_{N+1}} = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}. \quad (\text{GC.58})$$

The domain $N \geq q$ is necessary because V_{N-1} occurs. The separate original endpoint at $N = q - 1$ is not replaced by a q -dimensional metric at degree $q - 2$. At every fixed admissible real tilt the phase derivative remains the exact equality

$$\partial_\theta \log V_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N, \quad N \geq q - 1. \quad (\text{GC.59})$$

All these derivatives are of positive real-tilt determinants. Formula (GC.42) gives their underlying source and relation entry derivatives with the precise finite coefficient indices; their quotient and determinant operations use those original matrices.

For $h = 1$ the finite arithmetic quotient is zero and $\chi = 1$. Its relation and source spaces coincide at every degree N , so $\mathfrak{B}_{N+1} = \mathfrak{D}_{N+1}$, $V_N = V_N^\Gamma = T_N = 1$. The quotient lift has zero-dimensional domain and the cochain correction (GC.54) is the zero map on that domain. The nonzero analytic density, its mass μ_1^k , its coefficient series and (GC.24)–(GC.43) remain as constructed. The support lift is $G(0) = \{\tau, e\}$; it does not produce a nonzero finite spectral quotient from the analytic seed.

38 Certified theta moments and the original gamma arithmetic coefficients

This continuation proves and evaluates the fourth and sixth moments of the original theta seed, then transports the enclosures through the exact coefficient map for the reference parameter $\lambda = 13/4$

in *An exact gamma pushforward for the arithmetic Toda control*. The seed retains $g = 2\xi$, the density $|g(1/2 + it)|^2/(2\pi)$, the Fourier coordinate t , and its actual total mass. The reference retains the argument $\lambda + it/2$ and its different actual mass. We give the full analytic formulas, an explicit infinite-square tail for every finite derivative order, and the exact rational enclosures used by the executable certificate.

38.1 The theta input and its transform, with the original factors

On $\mathcal{S} = \{z \neq 0 : |\arg z| < \pi/4\}$ use the principal logarithm, and put

$$D = -z\partial_z, \quad L = D - \frac{1}{2}, \quad \vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{GS.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{GS.2})$$

The series and each differentiated series converge locally uniformly: on a compact subset of $\mathcal{S}$, every summand is bounded by a polynomial in n times e^{-an^2} for some $a > 0$.

Lemma 38.1 (Inversion and both endpoints). *On the specified sector,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{GS.3})$$

Every D -derivative of f_0 decays faster than every power both at zero and at infinity on a closed proper subsector.

Proof. For $x > 0$, Fourier transformation with kernel $e^{-2\pi i \eta u}$ maps $e^{-\pi x^2 u^2}$ to $x^{-1}e^{-\pi \eta^2/x^2}$. The value at $\eta = 0$ follows from the Gaussian integral, whose square in polar coordinates is one. Differentiation in η and integration by parts give $I'(\eta) = -2\pi \eta I(\eta)/x^2$, which proves the stated formula with that initial value. Periodize this Gaussian over integer translates. Its Fourier coefficients are the displayed rapidly decreasing transform values; its absolutely convergent Fourier series evaluated at zero proves $\vartheta(x) = x^{-1}\vartheta(1/x)$. Holomorphic uniqueness extends the identity to $\mathcal{S}$.

For $(RF)(z) = z^{-1}F(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this to $R\vartheta = \vartheta$ proves the second equality in (GS.3). The real series coefficients prove the third. On $|\arg z| \leq \beta < \pi/4$ one has $\Re z^2 \geq |z|^2 \cos(2\beta) > 0$; the series and each derivative therefore have Gaussian decay at infinity with a fixed polynomial factor. Inversion and $DR = R(1 - D)$ transport all these bounds to zero, with every derivative retained. $\square$

Define, on these rapid sector functions,

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (UF)(y) = e^{y/2}F(e^y), \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{GS.4})$$

Here $U : L^2(\mathbb{R}_{>0}, dx) \rightarrow L^2(\mathbb{R}, dy)$ is a surjective linear isometry: $x = e^y$ proves the norm identity, and its inverse is $u \mapsto [x \mapsto x^{-1/2}u(\log x)]$. The endpoint bounds prove that the Mellin integral is entire. For $\Re s > 1$, absolute integration of (GS.2) is justified by $\sum_{n \geq 1} n^{-\Re s} < \infty$. Substitution $v = \pi n^2 x^2$ then gives

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2}\zeta(s)\{4\Gamma(s/2+2) - 6\Gamma(s/2+1)\} \\ &= s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) =: g(s) = 2\xi(s). \end{aligned} \quad (\text{GS.5})$$

The factor 2 in (GS.2) and the substitution factor $1/2$ both occur before the first equality. Integration by parts proves the gamma recurrence used in the second equality. The entire Mellin integral continues the right side. Substitution $x \mapsto 1/x$ and real conjugation in (GS.4) prove

$$g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}. \quad (\text{GS.6})$$

In particular the following original density is even:

$$w(t) = \frac{|g(1/2 + it)|^2}{2\pi}, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt. \quad (\text{GS.7})$$

All moments exist. To see the exponential bound directly, rotate the Mellin contour to the ray $e^{i\beta}\mathbb{R}_{>0}$ with $|\beta| < \pi/4$. The endpoint bounds make the closing arcs vanish, and the rotated formula has the factor $e^{i\beta s}$. On the line $\Re s = 1/2$, choosing the sign of β to match $\Im s$ yields $|g(1/2 + it)| \leq C_\beta e^{-|\beta||t|}$. Thus $\int e^{a|t|} w(t) dt < \infty$ for every $0 < a < \pi/2$. Evenness gives $\mu_{2r+1} = 0$.

38.2 Every derivative polynomial and the exact higher-moment map

Let $H_r = L^r f_0$ for integers $r \geq 0$. Direct differentiation of (GS.4) gives $ULF = -(UF)'$ on the rapid sector functions. Integration by parts in $\mathcal{F}_+$, with the endpoint bounds already proved, therefore gives the commuting transform identity

$$\mathcal{F}_+(UH_r)(t) = (it)^r g(1/2 + it). \quad (\text{GS.8})$$

Every sign and power i^r belongs to this map.

Theorem 38.2 (Exact higher-moment formula). *Define polynomials $P_r \in \mathbb{Q}[y]$ by*

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y), \quad (\text{GS.9})$$

$$P_r(y) = \sum_{a=1}^{r+2} p_{r,a} y^a, \quad p_{r+1,a} = 2p_{r,a-1} - (2a + \tfrac{1}{2})p_{r,a}, \quad (\text{GS.10})$$

where coefficients outside their indicated range are zero. Then

$$H_r(x) = \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2}, \quad \mu_{2r} = \int_0^\infty |H_r(x)|^2 dx = 2 \int_1^\infty H_r(x)^2 dx. \quad (\text{GS.11})$$

The first four complete polynomials are

$$\begin{aligned} P_0 &= 8y^2 - 12y, \\ P_1 &= 16y^3 - 60y^2 + 30y, \\ P_2 &= 32y^4 - 224y^3 + 330y^2 - 75y, \\ P_3 &= 64y^5 - 720y^4 + 2116y^3 - 1635y^2 + \tfrac{375}{2}y. \end{aligned} \quad (\text{GS.12})$$

Proof. With $y = \pi n^2 x^2$, the operator D is $-2y\partial_y$. Consequently $L(P(y)e^{-y}) = \{-2yP'(y) + (2y - 1/2)P(y)\}e^{-y}$. This proves (GS.9), the series identity in (GS.11), and (GS.10) by collecting each specified coefficient. It also proves inductively that the degree is $r+2$, the leading coefficient is 2^{r+3} , and the constant coefficient is zero. Applying the recurrence three times gives the complete rational lists (GS.12).

Fourier Plancherel in the convention (GS.4) has factor $1/(2\pi)$: $\int |u|^2 dy = (2\pi)^{-1} \int |\mathcal{F}_+ u|^2 dt$. For the Schwartz functions here this follows by inserting $e^{-\varepsilon t^2}$ into Fourier inversion, applying Fubini, and evaluating the Gaussian kernel of integral one; letting $\varepsilon \downarrow 0$ gives inversion and the inner-product identity. Applying it to (GS.8) and the unitary U proves the first integral equality in (GS.11).

The relation $DR = R(1 - D)$ gives $RL = -LR$, while $Rf_0 = f_0$. Thus $RH_r = (-1)^r H_r$, or $H_r(1/x) = (-1)^r x H_r(x)$ for $x > 0$. In the integral over $(0, 1)$ substitute $x = 1/v$. The factor v^2 from the squared identity cancels the Jacobian v^{-2} and yields exactly the integral over $(1, \infty)$. The sign $(-1)^r$ is retained before squaring. All values $H_r(x)$ are real because their series has real coefficients, which finishes (GS.11). $\square$

The numerical driver in the delivered Gamma note writes $q_0 = 4y^2 - 6y$ and uses the same linear recurrence for q_r . Its exact coefficient map to this presentation is $q_r \mapsto P_r = 2q_r$, with inverse $P_r \mapsto P_r/2$. Thus $H_r = 2 \sum_n q_r(\pi n^2 x^2) e^{-\pi n^2 x^2}$. At the integral level (GS.11) consequently gives a factor 8 in front of the sum using $q_r q_r I_{2a+2b}$. At the gamma-function level below it gives a factor 4 in front of $q_r q_r \Gamma/C^\alpha$. The coefficient map accounts for both presentations without changing the theta input or its mass.

38.3 The incomplete-gamma square and a proved omitted tail

For real $C > 0$ and integers $p \geq 0$ put

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}, \quad \Gamma(a, C) = \int_C^\infty v^{a-1} e^{-v} dv. \quad (\text{GS.13})$$

Substitution $v = Cx^2$ proves the equality. Only positive real arguments and their positive real powers occur here. The companion recurrence, including its endpoint term, is

$$I_{p+2}(C) = \frac{p+1}{2C} I_p(C) + \frac{e^{-C}}{2C}. \quad (\text{GS.14})$$

It follows by integration by parts in (GS.13), whose upper endpoint vanishes exponentially.

Let $H_{r,J}$ restrict the original series (GS.11) to $1 \leq n \leq J$, where $J \geq 0$ is an integer. Put $T_{r,J} = 2 \int_1^\infty H_{r,J}^2 dx$. The entire finite expression is

$$T_{r,J} = \sum_{m,n=1}^J \sum_{a,b=1}^{r+2} p_{r,a} p_{r,b} (\pi m^2)^a (\pi n^2)^b \frac{\Gamma(a+b+1/2, \pi(m^2+n^2))}{[\pi(m^2+n^2)]^{a+b+1/2}}. \quad (\text{GS.15})$$

Here the endpoint factor 2 in (GS.11) cancels the factor $1/2$ in (GS.13); the coefficients $p_{r,a}$ already contain the complete original theta factor. Empty sums at $J = 0$ have value zero.

To specify the tail by finite positive expressions, define Stirling numbers of the second kind by $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$, zero when one index is zero and the other positive, zero outside $0 \leq b \leq j$, and

$$\left\{ \begin{smallmatrix} j+1 \\ b \end{smallmatrix} \right\} = b \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} + \left\{ \begin{smallmatrix} j \\ b-1 \end{smallmatrix} \right\}.$$

For $a \geq 1, d > 0$, integers $p \geq 0$, set

$$\rho_a = e^{-d(2a+1)}, \quad \mathbf{H}_p(a, d) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{smallmatrix} j \\ b \end{smallmatrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{GS.16})$$

$$\mathbf{A}_r(a) = \sum_{b=1}^{r+2} |p_{r,b}| \pi^b \mathbf{H}_{2b}(a, \pi). \quad (\text{GS.17})$$

Theorem 38.3 (Every finite derivative order has an explicit square tail). *For every integer $r \geq 0$ and $J \geq 0$, with $K = J + 1$,*

$$|\mu_{2r} - T_{r,J}| \leq E_{r,J} := 4\mathbf{A}_r(1)\mathbf{A}_r(K)I_{4r+8}(\pi(1+K^2)). \quad (\text{GS.18})$$

The right side is a finite positive expression and tends to zero as $J \rightarrow \infty$. In particular the infinite version of (GS.15) is absolutely convergent and equals μ_{2r} .

Proof. For $x \geq 1$, integers $a \geq 1$ and $p \geq 0$, write $n = a + j$. The inequality $(a + j)^2 - a^2 = 2aj + j^2 \geq (2a + 1)j$ proves

$$\sum_{n \geq a} n^p e^{-\pi n^2 x^2} \leq e^{-\pi a^2 x^2} \sum_{j \geq 0} (a + j)^p \rho_a^j = e^{-\pi a^2 x^2} H_p(a, \pi). \quad (\text{GS.19})$$

For the equality, expand $(a + j)^p$ and then use $j^v = \sum_b \left\{ \begin{smallmatrix} v \\ b \end{smallmatrix} \right\} (j)_b$, where $(j)_b = j(j-1) \cdots (j-b+1)$. This identity follows inductively from $j(j)_b = (j)_{b+1} + b(j)_b$ and the displayed Stirling recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1 - \rho)^{b+1}$, which proves the last equality in (GS.19).

Since $x^{2b} \leq x^{2r+4}$ for every original exponent $b \leq r+2$, (GS.19) gives the three explicit inequalities

$$\begin{aligned} |H_r(x)|, |H_{r,J}(x)| &\leq A_r(1) x^{2r+4} e^{-\pi x^2}, \\ |H_r(x) - H_{r,J}(x)| &\leq A_r(K) x^{2r+4} e^{-\pi K^2 x^2}. \end{aligned}$$

The same estimate for the sum of absolute values of all summands proves absolute integrability of the expanded double square. In particular termwise integration in the infinite formula is legitimate. Now use $|H_r^2 - H_{r,J}^2| \leq |H_r - H_{r,J}|(|H_r| + |H_{r,J}|)$ and the endpoint factor 2 in (GS.11). The resulting constant is exactly 4, the power is $4r + 8$, and integration gives (GS.18).

For fixed r , the denominators in (GS.16) are bounded away from zero uniformly for integers $a \geq 1$; thus $H_p(a, \pi)$ is at most a constant times $(1 + a)^p$, and $A_r(a)$ has at most polynomial growth. For $C \geq \pi$,

$$I_p(C) \leq e^{-C} \int_1^\infty x^p e^{-\pi(x^2-1)} dx.$$

The last integral is finite. Hence the factor with $C = \pi(1 + K^2)$ dominates that polynomial growth and forces $E_{r,J} \rightarrow 0$, proving the remaining assertions. $\square$

In the two new cases the coefficients in (GS.17) are completely specified by

$$\begin{aligned} A_2(a) &= 75\pi H_2 + 330\pi^2 H_4 + 224\pi^3 H_6 + 32\pi^4 H_8, \\ A_3(a) &= \frac{375}{2}\pi H_2 + 1635\pi^2 H_4 + 2116\pi^3 H_6 + 720\pi^4 H_8 + 64\pi^5 H_{10}, \end{aligned} \quad (\text{GS.20})$$

where each H_p on the right means $H_p(a, \pi)$. These absolute coefficients enter the error estimate only. Every original negative coefficient stays in the exact square (GS.15).

38.4 The coefficient morphism at the actual gamma reference

For $\lambda > 0$ put

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad \alpha = 2\lambda. \quad (\text{GS.21})$$

We prove the constants needed for the coefficient map. The beta integral follows from the product of two gamma integrals by the substitution $v = x_1 + x_2$, $u = x_1/(x_1 + x_2)$ with Jacobian v . After $u/(1-u) = e^x$ it reads

$$\int_{\mathbb{R}} (2 \cosh(x/2))^{-2\lambda} e^{ivx} dx = \frac{|\Gamma(\lambda + iv)|^2}{\Gamma(2\lambda)}. \quad (\text{GS.22})$$

The integrand before transformation and its first two derivatives decay exponentially at both ends. Two integrations by parts therefore give an integrable Fourier transform, so Fourier inversion applies with the same Gaussian justification as above. Using its evenness and substituting $t = 2v$ gives

$$\int_{\mathbb{R}} e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}. \quad (\text{GS.23})$$

The factor 2 from $t = 2v$ is present in c_λ . The function $(2 \cosh(x/2))^{-2\lambda}$ is holomorphic on $|\Im x| < \pi$ with the branch positive on the real line and decays exponentially on each closed narrower strip. Shifting the contour in (GS.22) up for $v > 0$ and down for $v < 0$ proves $|\Gamma(\lambda + iv)|^2 \leq C_c e^{-c|v|}$ for each $0 < c < \pi$. Thus the density in (GS.21) has exponential moments for every exponent of absolute value less than $\pi/2$. Analytic continuation of (GS.23) yields

$$\int e^{\theta t} r_\lambda(t) dt = c_\lambda (\cos \theta)^{-\alpha} \quad (|\theta| < \pi/2). \quad (\text{GS.24})$$

Define the monic polynomials by the local generating function

$$F_\lambda(z, t) = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = \sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n, \quad (\text{GS.25})$$

$$b_0 = 1, \quad b_1 = t, \quad b_{n+1} = tb_n - n(n + \alpha - 1)b_{n-1}. \quad (\text{GS.26})$$

The logarithms in (GS.25) vanish at zero. For real z close to zero, $F_\lambda(z, t) = (1 + z^2)^{-\lambda} e^{t \arctan z}$. Equation (GS.24) and $\cos(\arctan z + \arctan v) = (1 - zv)/\sqrt{(1 + z^2)(1 + v^2)}$ give

$$\int F_\lambda(z, t) F_\lambda(v, t) r_\lambda(t) dt = c_\lambda (1 - zv)^{-\alpha}. \quad (\text{GS.27})$$

Exponential integrability permits every derivative at $(0, 0)$ under the integral. Coefficient comparison proves

$$\int b_n^{(\lambda)}(t) b_j^{(\lambda)}(t) r_\lambda(t) dt = c_\lambda n! (\alpha)_n \delta_{nj}. \quad (\text{GS.28})$$

The highest coefficient of (GS.25) is $t^n/n!$, so these polynomials are monic. Finally differentiating (GS.25) gives $(1 + z^2) \partial_z F_\lambda = (t - \alpha z) F_\lambda$; coefficient comparison proves (GS.26), including all its constants.

The actual amplitude linking the two measures is

$$A_{1,\lambda}(t) = \frac{g(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{1,\lambda}(t) = |A_{1,\lambda}(t)|^2, \quad w(t) = B_{1,\lambda}(t) r_\lambda(t). \quad (\text{GS.29})$$

Gamma has no zeros and no poles at the displayed arguments. Here is a direct proof on their half-plane. For $\Re z > 0$, repeated integration by parts gives $B(z, n + 1) = n!/[z(z + 1) \cdots (z + n)]$. Substitution $u = v/n$ in the beta integral and $(1 - v/n)^n \leq e^{-v}$ prove $n^z B(z, n + 1) \rightarrow \Gamma(z)$ by dominated convergence. The reciprocal finite products converge to

$$ze^{\gamma z} \prod_{j \geq 1} (1 + z/j) e^{-z/j}, \quad \gamma = \lim_{n \rightarrow \infty} \left(\sum_{j=1}^n j^{-1} - \log n \right).$$

Integral comparison proves existence of the real limit. The logarithm of the tail product converges absolutely because $\log(1 + z/j) - z/j = O(j^{-2})$, and every factor is nonzero on $\Re z > 0$. The product is therefore finite and nonzero; by the previous limit it is $1/\Gamma(z)$. The original gamma integral is holomorphic on this half-plane, which also excludes poles there. The phase-preserving source map is multiplication by $A_{1,\lambda}$ from $L^2(w dt)$ to $L^2(r_\lambda dt)$; the exact norm identity follows by (GS.29). It is onto: the zero set of g on the line is discrete because g is entire and $g(2) > 0$ by (GS.5). Division by the amplitude off that null set gives the inverse and the same norm equality.

For completeness the full source phase at the chosen parameter is

$$A_{1,13/4}(t) = \frac{-\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it)}{(1/4 + it/2)(5/4 + it/2)(9/4 + it/2)}. \quad (\text{GS.30})$$

It follows from (GS.5), $s(s-1) = -(t^2 + 1/4)$, and three gamma recurrences. The elementary identity $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ first follows by summation by parts for $\Re s > 1$ and extends to $\Re s > 0$, $s \neq 1$, because its integral converges absolutely and locally uniformly there. On the line in (GS.30) it proves $|\zeta(s)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 4\sqrt{t^2 + 1/4}$. The squared modulus of the denominator in (GS.30) is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, so

$$0 \leq B_{1,13/4}(t) \leq \frac{1024}{\sqrt{\pi}} \frac{(t^2 + 1/4)^2}{(t^2 + 25/4)(t^2 + 81/4)} \leq \frac{1024}{\sqrt{\pi}}. \quad (\text{GS.31})$$

The last denominator exceeds the numerator by $26t^2 + 253/2 > 0$ after expansion. Thus $B_{1,13/4}$ belongs to $L^2(r_{13/4} dt)$, and its gamma coefficients have their literal Hilbert-space type.

Write $c = c_{13/4}$ and $c_j = c_{1,j}$. Equations (GS.28)–(GS.29) give the linear coefficient map on the moment functional:

$$c_j = \frac{\int b_j^{(13/4)}(t) w(t) dt}{c j! (13/2)_j}, \quad c = 2^{-11/2} \Gamma(13/2) = \frac{10395\sqrt{\pi}}{2048\sqrt{2}}. \quad (\text{GS.32})$$

The last identity follows by six gamma recurrences from $\Gamma(1/2) = \sqrt{\pi}$. At this exact parameter the complete polynomials needed are

$$\begin{aligned} b_0 &= 1, & b_1 &= t, & b_2 &= t^2 - \frac{13}{2}, \\ b_3 &= t^3 - \frac{43}{2}t, & b_4 &= t^4 - 47t^2 + \frac{663}{4}, \\ b_5 &= t^5 - 85t^3 + \frac{3931}{4}t, \\ b_6 &= t^6 - \frac{275}{2}t^4 + \frac{13801}{4}t^2 - \frac{69615}{8}. \end{aligned} \quad (\text{GS.33})$$

Substituting successively into (GS.26) proves every line. Evenness gives $c_1 = c_3 = c_5 = 0$, and the exact even coefficient map becomes

$$\begin{aligned} c_0 &= \mu_0/c, \\ c_2 &= (2\mu_2 - 13\mu_0)/(195c), \\ c_4 &= (4\mu_4 - 188\mu_2 + 663\mu_0)/(377910c), \\ c_6 &= (8\mu_6 - 1100\mu_4 + 27602\mu_2 - 69615\mu_0)/(2737957950c). \end{aligned} \quad (\text{GS.34})$$

Here $2!(13/2)_2 = 195/2$, $4!(13/2)_4 = 188955/2$, and $6!(13/2)_6 = 1368978975/4$. These factors show directly how every denominator in (GS.34) arises. The invertible triangular moment map has the complete inverse

$$\begin{aligned} \mu_0 &= cc_0, \\ \mu_2 &= (195cc_2 + 13\mu_0)/2, \\ \mu_4 &= (377910cc_4 + 188\mu_2 - 663\mu_0)/4, \\ \mu_6 &= (2737957950cc_6 + 1100\mu_4 - 27602\mu_2 + 69615\mu_0)/8. \end{aligned} \quad (\text{GS.35})$$

Substitution in (GS.34) proves that the displayed maps are mutual inverses. In particular the mass in (GS.7) remains μ_0 , while the reference mass in (GS.21) remains c .

38.5 Certified infinite quantities and exact rational error transport

The companion checker evaluates (GS.15) for $r = 0, 1, 2, 3$ and $J = 8$ in Arb real balls at 256-bit precision, with one arithmetic thread and python-flint 0.9.0. All polynomial input is exact rational arithmetic; π , exponentials, powers and gamma functions are enclosed by Arb. The upper-incomplete-gamma call is `C.gamma_upper(alpha)`, with argument C and order α , precisely as in (GS.13). Each finite

ball is enlarged by a symmetric ball with radius the upward-rounded upper endpoint of (GS.18). It therefore encloses the infinite quantity. The computed tails satisfy the rational inequalities

$$E_{0,8} < 7 \cdot 10^{-107}, \quad E_{1,8} < 4 \cdot 10^{-103}, \quad E_{2,8} < 2 \cdot 10^{-99}, \quad E_{3,8} < 14 \cdot 10^{-96}. \quad (\text{GS.36})$$

Each inequality is tested as a strict rigorous ball comparison.

The resulting exact rational decimal intervals are:

$$\begin{aligned} \mu_0 &\in [1.2790072478464851404795335922671932744916, \\ &\quad 1.2790072478464851404795335922671932744919], \\ \mu_2 &\in [13.0555493025705584353926846581231936824343, \\ &\quad 13.0555493025705584353926846581231936824346], \\ \mu_4 &\in [383.2743417173642425544072374791308324300288, \\ &\quad 383.2743417173642425544072374791308324300291], \\ \mu_6 &\in [17946.5273016052076235667646919834204772848931, \\ &\quad 17946.5273016052076235667646919834204772848934], \\ c &\in [6.3614260045872192926298535487331092297380, \\ &\quad 6.3614260045872192926298535487331092297383], \\ c_0 &\in [0.2010566886927858658099161632202187798614, \\ &\quad 0.2010566886927858658099161632202187798617], \\ c_2 &\in [0.0076454429994864011881338514807939967436, \\ &\quad 0.0076454429994864011881338514807939967439], \\ c_4 &\in [-0.0000305169208037620437467248395269085390, \\ &\quad -0.0000305169208037620437467248395269085387], \\ c_6 &\in [-0.0000003851554876709816618220743079695913, \\ &\quad -0.0000003851554876709816618220743079695910]. \end{aligned}$$

Each interval has width $3 \cdot 10^{-40}$. The coefficient balls are computed directly from the infinite-moment balls in (GS.34), before their conversion to the displayed decimal intervals. Interval dependency can enlarge enclosures; it cannot invalidate inclusion. The signs $c_4 < 0$ and $c_6 < 0$ follow from their negative upper endpoints and are preserved.

Here is the exact outward conversion used by the checker. If the resulting Arb ball is Q , let ℓ and u be its outward dyadic endpoints and set $d = 40$,

$$L_Q = 10^{-d}(\lfloor 10^d \ell \rfloor - 1), \quad U_Q = 10^{-d}(\lceil 10^d u \rceil + 1). \quad (\text{GS.37})$$

The checker then verifies $Q > L_Q$ and $Q < U_Q$ by rigorous ball comparison. The receipt stores L_Q, U_Q both as decimal strings and exact rational strings, and stores the dyadic endpoints as exact rationals. Thus every display can be checked with integer arithmetic after ball generation. The same receipt records the tail's exact upward dyadic rational.

The executable finite checks additionally verify all exact rational polynomial lists in (GS.12) and (GS.33) and the four exact rational denominator factors following (GS.34). They compare the ball evaluations of the polynomial and integer numerator forms in (GS.34) for overlap, and similarly compare the endpoint-containing recurrence (GS.14) at each argument pair used. These checks detect convention mistakes; the analytic certificate is the ball enclosure enlarged by the proved infinite tail, independently of any truncation agreement.

Four fault modes inject specific false mathematical assertions: discarding the inversion factor two preserves μ_4 ; assigning reference mass one preserves c_0 ; c_4 is nonnegative; and the cutoff-zero sum with zero error encloses the full positive seed mass. The already certified disjoint intervals reject each assertion. These checks use explicit conditionals and a nonzero exit status; Python optimization

cannot remove them. Normal and optimized runs produce the same mathematical results and each fault produces one recorded rejection.

The source note's two floating truncations remain its original numerical evidence. The intervals above independently certify their displayed values through the original theta series and (GS.18). The correctness contract for the elementary ball operations and outward endpoints is documented in the [python-flint Arb interface](#). No proof of that arithmetic library's implementation is asserted.

38.6 Exact tensor coefficient interface through degree six

The map from the certified one-factor data to every finite tensor degree can also be stated without changing the mass or sum variable. Let $k \geq 1$, $u = t_1 + \cdots + t_k$, and $m_k = w^{*k}$. Fubini gives $\int m_k = \mu_0^k$, and exponential integrability proves all the moments below are finite. Multiplying the generating functions (GS.25) gives the finite polynomial identity

$$b_n^{(k\lambda)}(u) = \sum_{n_1 + \cdots + n_k = n} \frac{n!}{n_1! \cdots n_k!} \prod_{j=1}^k b_{n_j}^{(\lambda)}(t_j). \quad (\text{GS.38})$$

Its proof is literal multiplication: the exponents of $1 \mp iz$ add to $-k\lambda \pm iu/2$, and the coefficient of degree n has only finitely many terms.

Use $\lambda = 13/4$, $\alpha = 13/2$, $a_j = (\alpha)_j c_j$, and the formal coefficient series $\mathcal{A}(z) = \sum_{j \geq 0} a_j z^j$. For any coefficient appearing below only finitely many c_j are needed. Integration of (GS.38), using (GS.32) separately in each factor, proves

$$\int b_n^{(k\lambda)}(u) m_k(u) du = c^k n! d_{k,n}, \quad d_{k,n} = [z^n] \mathcal{A}(z)^k. \quad (\text{GS.39})$$

No interchange of an infinite formal series with an integral is required. Evenness gives the complete first four nonzero expressions

$$\begin{aligned} d_{k,0} &= c_0^k, \\ d_{k,2} &= k c_0^{k-1} a_2, \\ d_{k,4} &= k c_0^{k-1} a_4 + \binom{k}{2} c_0^{k-2} a_2^2, \\ d_{k,6} &= k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3. \end{aligned} \quad (\text{GS.40})$$

For degree six the partitions into even positive degrees are 6, 4 + 2 and 2 + 2 + 2; their distinct-factor choices have respectively k , $k(k-1)$ and $\binom{k}{3}$ possibilities. This proves all three coefficients, including the mixed term and its sign. The analogous enumeration proves degree two and four. Since $c_0 > 0$, even the displayed negative powers at $k = 1, 2$ are defined; their accompanying zero combinatorial factors give the correct absent terms.

Equations (GS.34)–(GS.35), (GS.39)–(GS.40) are the exact map from the certified original moments to these all- k low-degree arithmetic gamma moments, retaining c^k and the coordinate u . For example their degree-zero identity is $c^k d_{k,0} = \mu_0^k$. They give interval evaluations at every specified finite integer k by rational and ball arithmetic with the already enclosed inputs. They assert no uniform estimate when polynomial degree grows. For this actual seed $h = 1$, the finite arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the quotient map sends every polynomial to zero because every polynomial lies in the ideal (1). The analytic source measure and all the coefficients above remain nonzero before that quotient. No nonzero packet quotient is inferred from these seed calculations.

39 The first three original source costs at every tensor degree

Retain the exact theta seed, its Fourier coordinate and its measure

$$w(t) dt = |g(1/2 + it)|^2 \frac{dt}{2\pi}, \quad g = 2\xi, \quad \mu_j = \int_{\mathbb{R}} t^j w(t) dt.$$

Equations (GS.1)–(GS.18) prove that w is even, has positive mass μ_0 , and has finite moments of every order. The certificate of (GS.36)–(GS.37) encloses the original $\mu_0, \mu_2, \mu_4, \mu_6$. This section transports those existing certified integrals by exact rational arithmetic. It performs no further quadrature.

For each integer $k \geq 1$ put $u = t_1 + \cdots + t_k$, $m_k = w^{*k}$ and $M_j(k) = \int u^j m_k(u) du$. Fubini and evenness give the full first four even sum moments

$$\begin{aligned} M_0(k) &= \mu_0^k, \\ M_2(k) &= k\mu_2\mu_0^{k-1}, \\ M_4(k) &= k\mu_4\mu_0^{k-1} + 3k(k-1)\mu_2^2\mu_0^{k-2}, \\ M_6(k) &= k\mu_6\mu_0^{k-1} + 15k(k-1)\mu_4\mu_2\mu_0^{k-2} + 15k(k-1)(k-2)\mu_2^3\mu_0^{k-3}. \end{aligned} \quad (\text{GS.41})$$

Here every missing term integrates to zero because at least one factor has an odd moment. For M_4 , the surviving exponent partitions are 4 and $2 + 2$. The second has $\binom{k}{2} 4!/(2!2!) = 3k(k-1)$ terms. For M_6 the partitions are 6, $4 + 2$ and $2 + 2 + 2$; their numbers of terms are respectively k , $k(k-1)6!/(4!2!) = 15k(k-1)$ and $\binom{k}{3} 6!/(2!2!2!) = 15k(k-1)(k-2)$. The second-moment expansion has precisely k surviving terms. These finite counts prove (GS.41) with its full mass factors. Because $\mu_0 > 0$, the displayed powers are defined even when a zero combinatorial coefficient accompanies a negative exponent.

All nonzero polynomial norms in $L^2(m_k du)$ are strictly positive. Indeed the nonnegative density m_k has positive mass μ_0^k , so its positivity set has positive Lebesgue measure. A nonzero polynomial has finitely many real roots, as repeated division by its distinct linear factors cannot exceed its degree. Its squared modulus times m_k is therefore positive on a set of positive measure, and its integral is positive. This proves every positivity and division below for the actual source.

Put $\nu = \mu_2/\mu_0$. With the norms taken in the unchanged measure $m_k du$, the first four monic orthogonal polynomials and all their squared norms are

$$\begin{aligned} p_0(u) &= 1, & h_0(k) &= M_0(k), \\ p_1(u) &= u, & h_1(k) &= M_2(k), \\ p_2(u) &= u^2 - k\nu, & h_2(k) &= M_4(k) - \frac{M_2(k)^2}{M_0(k)}, \\ p_3(u) &= u^3 - \frac{M_4(k)}{M_2(k)}u, & h_3(k) &= M_6(k) - \frac{M_4(k)^2}{M_2(k)}. \end{aligned} \quad (\text{GS.42})$$

To verify orthogonality of p_2 , its pairing with 1 is $M_2 - k\nu M_0 = 0$, and its pairing with u is zero by evenness. For p_3 , parity gives zero against 1 and p_2 ; its pairing with u is $M_4 - (M_4/M_2)M_2 = 0$. Expanding each squared polynomial gives the norms in (GS.42). Uniqueness follows because the difference of two monic orthogonal polynomials of the same degree lies in the lower-degree space, is orthogonal to itself, and hence is zero by the preceding positivity argument.

Define the positive original monic source costs $a_j^{(k)} = h_j(k)/h_{j-1}(k)$. These are exactly the lower coefficients in the real-coordinate three-term recurrence: in up_j , its pairing with p_{j-1} equals the pairing of p_j with up_{j-1} , which is h_j because $up_{j-1} - p_j$ has lower degree. Pairings with lower degrees vanish by the same argument. The degree- j coefficient vanishes by evenness. Therefore $up_j = p_{j+1} + a_j^{(k)} p_{j-1}$. For $j = 3$, the next monic polynomial p_4 exists by finite Gram orthogonalization, whose Gram is positive definite by the proved polynomial norm property; its full formula is not

needed to determine the already specified coefficient h_3/h_2 . Substituting (GS.41) into (GS.42) gives

$$\begin{aligned} a_1^{(k)} &= k\nu, \\ a_2^{(k)} &= \frac{\mu_4}{\mu_2} + (2k-3)\nu, \\ a_3^{(k)} &= \frac{M_6(k) - M_4(k)^2/M_2(k)}{M_4(k) - M_2(k)^2/M_0(k)}. \end{aligned} \quad (\text{GS.43})$$

For the second equality one obtains $M_4/M_2 = \mu_4/\mu_2 + 3(k-1)\nu$ and $M_2/M_0 = k\nu$, whose exact difference is the displayed cost. For an explicit expanded version of the third equality, write $r_4 = \mu_4/\mu_0$ and $r_6 = \mu_6/\mu_0$, solely as the indicated scalar quotients of the original moments. Then

$$a_3^{(k)} = \frac{r_6 - r_4^2/\nu + 9(k-1)r_4\nu + 3(k-1)(2k-7)\nu^3}{r_4 + (2k-3)\nu^2}. \quad (\text{GS.44})$$

To check every term, divide the numerator of (GS.43) by $k\mu_0^k$ using (GS.41). The square term contributes $r_4^2/\nu + 6(k-1)r_4\nu + 9(k-1)^2\nu^3$; subtracting it from $r_6 + 15(k-1)r_4\nu + 15(k-1)(k-2)\nu^3$ gives the numerator in (GS.44). The denominator of (GS.43), divided by the same $k\mu_0^k$, is $r_4 + (2k-3)\nu^2$. Thus both original norms and all mass factors are recovered from these scalar abbreviations.

The original spectral coordinate is $S = k/2 + iu$. Let $c_k = k/2$ and define the algebra isomorphism

$$\Psi_k : \mathbb{C}[S] \longrightarrow \mathbb{C}[u], \quad (\Psi_k P)(u) = P(c_k + iu), \quad \Psi_k^{-1} p(S) = p((S - c_k)/i). \quad (\text{GS.45})$$

Direct substitution proves the two inverse identities. Give $\mathbb{C}[S]$ the source norm $\int |P(c_k + iu)|^2 m_k(u) du$; then Ψ_k is a linear isometry onto the polynomial source in the real coordinate. On the degree- j monic affine space, its phase-preserving monic inverse is $p \mapsto P_j = i^j \Psi_k^{-1} p$. Since $|i^j| = 1$, this map preserves the original norm and has inverse $P \mapsto i^{-j} \Psi_k P$ on that affine space. In particular the complete first four spectral monic polynomials are

$$\begin{aligned} P_0 &= 1, \quad P_1 = S - c_k, \quad P_2 = (S - c_k)^2 + k\nu, \\ P_3 &= (S - c_k)^3 + \left\{ \frac{\mu_4}{\mu_2} + 3(k-1)\nu \right\} (S - c_k). \end{aligned} \quad (\text{GS.46})$$

Their squared norms are exactly h_0, h_1, h_2, h_3 from (GS.42). Multiplying the real recurrence by i^{j+1} proves the spectral recurrence with its actual lower sign,

$$SP_j = P_{j+1} + c_k P_j - a_j^{(k)} P_{j-1}. \quad (\text{GS.47})$$

Thus the positive norm ratio $a_j^{(k)}$ and the negative lower coefficient in the S presentation have their exact phase map.

To connect these costs to the existing Gamma determinant corrections, let D_n be the determinant of the source Gram on $1, u, \dots, u^{n-1}$, with $D_0 = 1$, and let D_n^Γ use the Gamma sum measure $r_{13/4}^{*k} du$. Its mass is $c_{13/4}^k$, and its monic norms are $c_{13/4}^k j!(13k/2)_j$, by (GS.23), Fourier uniqueness and (GS.28). Specifically Fourier transformation of the convolution gives $c_{13/4}^k \cosh^{-13k/2} y$; division by the mass constant in (GS.23) identifies its density as a positive scalar times $r_{13k/4}$, proving the asserted norms without any change to u . Monic orthogonalization is triangular with diagonal one, so $D_n = \prod_{j < n} h_j$ and the analogous Gamma product holds. Define $X_n = D_n/D_n^\Gamma$ and

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad a_{N+1}^{(k)} = (N+1)(N+13k/2)Q_N. \quad (\text{GS.48})$$

Dividing the two consecutive determinant products proves the second identity. In particular, with $\eta = \mu_4/\mu_2$,

$$\begin{aligned} Q_0 &= \frac{2\nu}{13}, \\ Q_1(k) &= \frac{\eta + (2k-3)\nu}{13k+2} = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \eta - \frac{43}{13}\nu, \\ Q_2(k) &= \frac{2a_3^{(k)}}{39k+12}. \end{aligned} \tag{GS.49}$$

Every equality is obtained by substituting (GS.43) into (GS.48). No volume of a nonzero packet quotient enters these source costs.

The pure rational checker reads the exact dyadic moment endpoints from the pinned GS.36 certificate and propagates them through (GS.43)–(GS.49) using `fractions.Fraction` exclusively. For intervals $[a, b]$ and $[c, d]$, it uses endpoint addition, $[a - d, b - c]$ for subtraction, the minimum and maximum of the four endpoint products for multiplication, and multiplication by $[1/d, 1/c]$ for division after checking $c > 0$. These rules follow from linearity in each variable and the decreasing map $x \mapsto 1/x$ on the positive axis. Thus they preserve inclusion without any floating-point operation. Applying (GS.37) with exact rational floor and ceiling yields, among the saved intervals,

$$\begin{aligned} \nu &\in [10.2075647534857216888657612522375551749964, \\ &\quad 10.2075647534857216888657612522375551749967], \\ Q_0 &\in [1.5703945774593417982870401926519315653839, \\ &\quad 1.5703945774593417982870401926519315653842], \\ a_2^{(1)} &\in [19.1496329281085602472067784658376980738791, \\ &\quad 19.1496329281085602472067784658376980738794], \\ \rho &\in [-4.4062857337815667270988244239412754068822, \\ &\quad -4.4062857337815667270988244239412754068819]. \end{aligned}$$

All four widths are $3 \cdot 10^{-40}$. The receipt also encloses μ_4/μ_2 , $\mu_4\mu_0 - \mu_2^2$, $a_2^{(2)}$, $Q_1(1)$, $a_3^{(1)}$ and $Q_2(1)$. In particular $\rho < 0$ and (GS.49) give the exact fixed-degree all- k comparison

$$0 < Q_1(1) \leq Q_1(k) < Q_0, \quad Q_1(k+1) - Q_1(k) = \frac{-13\rho}{(13k+2)(13k+15)} > 0. \tag{GS.50}$$

The positive lower bound follows either from $h_2/h_1 > 0$ or its certified $k = 1$ enclosure; the difference follows by literal subtraction in (GS.49). The computation concerns original source polynomial degrees at most three, at every specified tensor degree. The analytic seed still has quotient $\mathbb{C}[S]/(1) = 0$, its quotient determinant is the empty determinant one, and every nonzero source norm above remains before that zero quotient.

40 Finite arithmetic Gram transfer with the original minimum quotient metric

This continuation proves a finite coefficient-to-metric map, its exact boundary-resolvent remainder, and a two-degree determinant formula for the consecutive volume ratio in the arithmetic Toda control. The original Gamma mass, tensor sum, arithmetic amplitude, quotient coordinates and relation polynomial remain in every formula. A bounded positive-weight calibration at the end proves that the relative Gamma correction can increase with degree even while the actual minimum volume decreases. That calibration is explicitly an auxiliary measure, with no assertion that its polynomial is an annihilator of actual zeta zeros.

The sources read for this calculation are the complete `RESEARCH_NOTE.md` and `NOTE.tex` in the delivered Tau Gamma Convolution Descent, and the original arithmetic definitions and proofs TVB.1–TVB.38 in `toda_cv_exact_bridge_20260912.tex`. The finite transfer below is at the original observation $\theta = 0$. Its proof does not replace a nonzero exponential tilt by a polynomial test. All interval claims here are finite rational inequalities; no uniform estimate as packet, tensor degree or polynomial degree grow is asserted.

40.1 The original source, relation and coordinate maps

Let $h(s)$ be a monic nonempty packet of actual zeros of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s),$$

including the complete order at each selected zero. Put $v_h = g/h$, $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$, and $\mu_h = \int_{\mathbb{R}} w_h(t) dt$. The quotient is entire: local Taylor division removes exactly the stated vanishing order. Its value at each selected zero is nonzero. The full Taylor remainder $[v_h]_h$ is therefore a unit of $E_h = \mathbb{C}[s]/(h)$. Indeed in $\mathbb{C}[t]/(t^m)$ the inverse of $\sum_{j < m} a_j t^j$, $a_0 \neq 0$, has coefficients $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$; the Chinese remainder map from coprime factors is the exact algebra isomorphism obtained by Bezout's identity. This retains every jet and multiplicity.

The original tensor coordinate is

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GMT.1})$$

The source observation is the conjugate-linear-first pairing

$$\langle P, Q \rangle_h = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k. \quad (\text{GMT.2})$$

The change $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Thus Fubini gives (GMT.2) from the literal tensor Mellin observation without a divisor or an average coordinate. The complex Mellin amplitude remains

$$\prod_{i=1}^k \frac{v_h(1/2 + it_i)}{\sqrt{2\pi}} P(c + i \sum_i t_i).$$

Let $I = (h(s_1), \dots, h(s_k))$ and let χ be the monic generator of $I \cap \mathbb{C}[S]$, of degree q . Existence follows from an annihilating polynomial of multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$. Division by its least-degree monic member proves that every member of the intersection is a multiple of χ . For $N \geq q - 1$ define $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and the exact maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0, \quad M_\chi P = \chi P, \quad \mathcal{C} = \mathbb{C}[S]/(\chi), \quad J_N P = [P]_\chi. \quad (\text{GMT.3})$$

Here $\mathcal{P}_{-1} = 0$. All coordinate vectors of $\mathcal{C}$ are in the written basis $1, S, \dots, S^{q-1}$. Its map to the original tensor jet is

$$\eta[P] = [v_h]_h^{\otimes k} [P(S)]_I.$$

Its kernel is zero by the preceding intersection ideal and invertible unit. No Gamma coefficient calculation changes this map.

Write B_N for the coefficient matrix of the columns $\chi, \chi S, \dots, \chi S^{N-q}$ in $\mathcal{P}_N$. The matrix $[E_N, B_N]$, where E_N includes the first q monomials, is triangular with diagonal one. Thus its determinant is exactly one in this order; putting the relation block first contributes $(-1)^{q(N-q+1)}$.

The original theta map is $\Theta\phi(x) = \sum_{n \neq 0} \phi(nx)$, with $D = -x\partial_x$, and F_h satisfies $\mathcal{M}F_h = v_h$, $h(D)F_h = \Theta\phi_0$, where $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. The exact primitive attached to a column χP is

retained as follows. Successive monic division in $s_1, \dots, s_k$ gives $\chi(S) = \sum_i h(s_i) Q_i(\mathbf{s})$. Its tensor primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) P(\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GMT.4})$$

The differential at the i th position has sign $(-1)^{i-1}$, so differentiation gives precisely the original source vector $\mathcal{V}_{h,k}(\chi P)$. This proves the relation-to-boundary map, including its sign. The construction of F_h uses the actual zero-division operator $S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ at a vanishing Mellin value. The two integrals give respectively every large- and small- x power bound. Differentiation gives $(D - \rho) S_\rho F = F$, hence all Euler derivative bounds and the Mellin quotient by $s - \rho$. Starting from $\Theta \phi_0$ and successively dividing the complete root list constructs the displayed F_h . The seed identity follows by substituting $y = \pi x^2$ in its Mellin integral, which gives $\mathcal{M}\phi_0(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2)$ for $\Re s > -2$, followed by the absolutely convergent theta sum for $\Re s > 1$ and analytic continuation.

On each original support label ℓ , a boundary still maps to $(\ell, 0) = e_\ell$; external absence τ still maps to τ . The linear polynomial maps (GMT.3)–(GMT.4) precede the quadratic map $R \mapsto R^* H R$ from source representatives to Hermitian forms. For the empty packet $h = 1$, the analytic source remains, but $\chi = 1, q = 0, \mathcal{C} = 0, G_N$ is the zero-dimensional form and its determinant is one. Every quotient formula below then has its empty-block interpretation.

40.2 Full Gamma mass and the finite coefficient evaluation

For $\lambda > 0$ set

$$\alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda} \Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{GMT.5})$$

The beta integral with its coordinate $x = 2y$ gives

$$|\Gamma(\lambda + it/2)|^2 = 2\Gamma(2\lambda) \int_{\mathbb{R}} e^{ity} (2 \cosh y)^{-2\lambda} dy.$$

Fourier inversion therefore gives $\int e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}$. Both functions are integrable: the right integrand and all its derivatives decay exponentially, so integration by parts bounds its Fourier transform by every inverse power. Exponential integrability of r_λ on $|a| < \pi/2$ follows by rotating the gamma integral by any φ with $|a| < \varphi < \pi/2$; the squared modulus has a bound $\Gamma(\lambda)^2 (\cos \varphi)^{-2\lambda} e^{-\varphi|t|}$. Analytic continuation now proves

$$\int e^{au} r_{\lambda,k}(u) du = C_k (\cos a)^{-\beta}, \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{C_k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GMT.6})$$

The last identity is Fourier uniqueness with the exact product mass.

Let $b_j^{(\lambda)}$ denote the monic polynomials with

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GMT.7})$$

These agree with $j! P_j^{(\lambda)}(t/2; \pi/2)$ in the classical generating function at [DLMF 18.23.7](#). The next proof fixes their scale without relying on that identification. Differentiating (GMT.7) gives

$$b_{j+1}^{(\lambda)} = t b_j^{(\lambda)} - j(j + \alpha - 1) b_{j-1}^{(\lambda)}.$$

For small real z, w , integration of the product of generating functions by (GMT.6) gives $c_\lambda(1-zw)^{-\alpha}$, since $\cos(\arctan z + \arctan w) = (1-zw)/\sqrt{(1+z^2)(1+w^2)}$. Local exponential domination justifies every coefficient derivative. Thus

$$\int b_i^{(\lambda)} b_j^{(\lambda)} r_\lambda = \delta_{ij} c_\lambda j! (\alpha)_j, \quad \int b_i^{(k\lambda)} b_j^{(k\lambda)} r_{\lambda,k} = \delta_{ij} C_k j! (\beta)_j. \quad (\text{GMT.8})$$

Put $B(t) = |v_h(1/2 + it)/\Gamma(\lambda + it/2)|^2$, so that $w_h = Br_\lambda$ with its full complex amplitude still present in the Mellin source. One may use the bounded reference constructed in the Gamma descent: $\lambda = 1/4$ for degree at least three, or $\lambda = 1/4 + L$, $L \geq \max(0, 3 - \deg h)$ otherwise. The relevant boundedness follows directly from the compact entire quotient, $|h(1/2 + it)| \geq (|t|/2)^{\deg h}$ beyond twice the root radius, $|\zeta(1/2 + it)| \leq 2(1 + |t|)$, and the nonvanishing factors $\prod_{j < L} (j + 1/4 + it/2)$ in the gamma shift. In particular $B \in L^2(r_\lambda dt)$. All statements of this section also hold for any specified nonnegative nonzero $B \in L^2(r_\lambda dt)$. For that domain write $w_B = Br_\lambda$ and $m_{B,k} = w_B^{*k}$; at the arithmetic input these equal the original $w_h, m_{h,k}$. Cauchy-Schwarz with (GMT.6) gives all polynomial moments. The calibration below uses this latter domain before it is explicitly capped to a bounded function.

The sum adjoint is the actual conditional integral

$$C_\Sigma F(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GMT.9})$$

Its adjoint inclusion $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$ is isometric by Fubini; hence $C_\Sigma U_\Sigma = 1$ and $F \mapsto (C_\Sigma F, (1 - U_\Sigma C_\Sigma)F)$ has inverse $(f, n) \mapsto U_\Sigma f + n$ on $L^2(r_{\lambda,k}) \oplus \ker C_\Sigma$. At $k = 1$ both maps are the identity. The normal component and every support label remain in this paired map. The specified sum density is $m_{B,k} = r_{\lambda,k} B_k$, $B_k = C_\Sigma B^{\otimes k}$; in particular it is $m_{h,k}$ at the arithmetic input.

Define the real coefficients $c_{B,j}$ (the original $c_{h,j}$ for the arithmetic multiplier $B = B_{h,\lambda}$) and formal coefficient power

$$c_{B,j} = \frac{\int b_j^{(\lambda)} w_B}{c_\lambda j! (\alpha)_j}, \quad A(z) = \sum_{j \geq 0} (\alpha)_j c_{B,j} z^j, \quad d_n = [z^n] A(z)^k. \quad (\text{GMT.10})$$

Multiplying the finite generating coefficients in (GMT.7) gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing with $B^{\otimes k}$ gives the exact finite identity

$$\int b_n^{(k\lambda)}(u) m_{B,k}(u) du = C_k n! d_n. \quad (\text{GMT.11})$$

Every term is integrable by Cauchy-Schwarz; no infinite formal-series product was interchanged with an integral.

For a specified finite list of polynomials $p_0, \dots, p_r$ in $\mathcal{P}_N$, expand the actual tests

$$\overline{p_a(c + iu)} p_b(c + iu) = \sum_{n=0}^{2N} L_{ab,n} b_n^{(k\lambda)}(u).$$

This expansion is a finite algorithm: subtract the top coefficient times the monic polynomial of the same degree, then continue in descending degree through zero. The resulting remainder is exactly zero, by the decreasing degree. Equations (GMT.10)–(GMT.11) prove

$$\boxed{H(p)_{ab} = C_k \sum_{n=0}^{2N} n! L_{ab,n} d_n.} \quad (\text{GMT.12})$$

The coefficient-to-form map is homogeneous of degree k in $c_{B,0}, \dots, c_{B,2N}$. Its full coordinate type is $\mathbb{R}^{2N+1} \rightarrow \text{Herm}(\mathcal{P}_N)$ on the actual moment vectors, followed by congruence $H \mapsto P^*HP$ for a polynomial coefficient matrix P . The nonnegative density has positive mass. Its positivity set has positive Lebesgue measure, and a nonzero polynomial has only finitely many real roots. Thus the integral of its squared modulus is strictly positive, proving positivity of every finite source Gram and of every injective relation Gram. Choosing $p_a = S^a$ gives H_N ; choosing the columns of B_N gives exactly $B_N^* H_N B_N$. Choosing actual representatives gives their actual Gram. Every relation column has degree at most N , including its complete χ factor.

One may instead evaluate in the monic Gamma columns $p_j^\Gamma(S) = i^j b_j^{(k\lambda)}((S - c)/i)$. Their coefficient matrix U_N is triangular with diagonal one, and their reference Gram is

$$U_N^* H_N^\Gamma U_N = \text{diag}\{C_k j! (\beta)_j : 0 \leq j \leq N\}. \quad (\text{GMT.13})$$

This is a stated invertible coordinate map, with all squared norms retained. In particular it divides neither the measure nor the polynomials by their mass or norms. The original reduction matrix becomes $J_N U_N$; the original source form is recovered by $U_N^{-*}(\cdot) U_N^{-1}$.

40.3 Finite coefficient errors and polynomial-test bounds

Finite errors propagate before any inverse is taken. For real central coefficients $\widehat{c}_{B,j}$ and radii $e_j \geq 0$ enclosing the actual $c_{B,j}$ for $0 \leq j \leq 2N$, define the finite polynomials

$$\widehat{A} = \sum_{j=0}^{2N} (\alpha)_j \widehat{c}_{B,j} z^j, \quad A_+ = \sum_{j=0}^{2N} (\alpha)_j |\widehat{c}_{B,j}| z^j, \quad E_+ = \sum_{j=0}^{2N} (\alpha)_j e_j z^j,$$

and set

$$\widehat{d}_n = [z^n] \widehat{A}^k, \quad e_n^{(k)} = [z^n] \{(A_+ + E_+)^k - A_+^k\}. \quad (\text{GMT.14})$$

The multinomial expansion of each difference of products groups terms by the nonempty set of factors containing a coefficient error. Taking absolute values term by term proves $|d_n - \widehat{d}_n| \leq e_n^{(k)}$ for $n \leq 2N$. Thus the Hermitian central matrix $\widehat{H}$ from (GMT.12) satisfies

$$|(H - \widehat{H})_{ab}| \leq \varepsilon_{ab} := C_k \sum_{n=0}^{2N} n! |L_{ab,n}| e_n^{(k)}. \quad (\text{GMT.15})$$

The masses in this formula are exact constants. An interval evaluation of those constants is propagated by the same product rule; it cannot be replaced by setting $C_k = 1$. Exact polynomial coefficients of χ remain fixed in $L_{ab,n}$; any numerical enclosure of such coefficients must likewise be propagated before reporting a matrix certificate.

There is also a strictly polynomial-test refinement of the L^2 tail bound in the supplied descent. Write $B_L = \sum_{j \leq L} c_{B,j} b_j^{(\lambda)}$ and $E_k = B_k - C_\Sigma(B_L^{\otimes k})$. Orthogonality proves $\|B_L\| \leq \|B\|$. Expanding the tensor difference into k terms and using contraction of C_Σ proves $\|E_k\| \leq k \|B\|^{k-1} \|B - B_L\|$ with the original measures. Equation (GMT.11) applied degree by degree shows that E_k is orthogonal to all polynomials of degree at most L . Therefore

$$\left| \int \overline{p_a(c + iu)} p_b(c + iu) E_k(u) r_{\lambda,k}(u) du \right| \leq \|E_k\| \left(C_k \sum_{n > L} n! (\beta)_n |L_{ab,n}|^2 \right)^{1/2}. \quad (\text{GMT.16})$$

This follows by discarding only the orthogonal test components whose pairings are proved zero, and then applying Cauchy–Schwarz and (GMT.8). It is exactly zero at $L \geq 2N$. In particular a signed truncated density has the same finite positive Gram at that degree; pointwise positivity of the truncation is not required by this map.

An intrinsic finite polynomial-test bound is also available. The reference evaluation kernel is

$$\mathcal{K}_N^\Gamma(u) = \sum_{j=0}^N \frac{b_j^{(k\lambda)}(u)^2}{C_k j! (\beta)_j}.$$

Expansion in the original monic basis and Cauchy-Schwarz give $|P(c + iu)|^2 \leq \|P\|_\Gamma^2 \mathcal{K}_N^\Gamma(u)$. It follows that the form induced by a real error E_k obeys

$$|\langle P, P \rangle_{E_k}| \leq \|P\|_\Gamma^2 \|E_k\|_{L^2(r_{\lambda,k})} \|\mathcal{K}_N^\Gamma\|_{L^2(r_{\lambda,k})}. \quad (\text{GMT.17})$$

The last squared norm is an exact Gamma integral of a degree- $4N$ polynomial, evaluable by (GMT.8) or (GMT.12) with the reference weight. This is a finite evaluation-functional estimate, retaining every Christoffel-kernel denominator and the full mass.

40.4 Exact least lifts and the boundary-resolvent remainder

For a positive definite source matrix H in the fixed source coordinates, define

$$K(H) = J_N H^{-1} J_N^*, \quad G(H) = K(H)^{-1}, \quad R(H) = H^{-1} J_N^* G(H). \quad (\text{GMT.18})$$

Surjectivity of J_N makes $K(H)$ positive definite. Multiplication gives $J_N R(H) = 1$ and $B_N^* H R(H) = 0$. Every other lift is $R(H)x + B_N y$, whose squared norm is $x^* G(H)x + y^* B_N^* H B_N y$. This proves minimum, uniqueness and $R(H)^* H R(H) = G(H)$.

Let H_0 and H_1 be two actual positive finite source forms on the same polynomial space. In particular they may be the Gamma and arithmetic forms already constructed. Put

$$H_t = H_0 + tE, \quad E = H_1 - H_0, \quad 0 \leq t \leq 1, \quad A_0 = B_N^* H_0 B_N, \quad F = B_N^* E B_N, \quad C = B_N^* E R_0,$$

where $R_0 = R(H_0)$ and $G_0 = G(H_0)$. The whole interval of forms is positive because it is the convex combination of the two actual positive forms. Thus every nonempty $A_t = A_0 + tF$ is positive. The complete, signed resolvent formulas are

$$\boxed{R_t = R_0 - tB_N A_t^{-1} C, \quad G_t = G_0 + tR_0^* E R_0 - t^2 C^* A_t^{-1} C.} \quad (\text{GMT.19})$$

To prove the first equation, its reduction is the identity and $B_N^* H_t R_0 = tC$, while $B_N^* H_t B_N = A_t$. Consequently its boundary pairing is zero, so (GMT.18) identifies it. Expanding its full H_t norm gives $R_0^* H_t R_0 - 2t^2 C^* A_t^{-1} C + t^2 C^* A_t^{-1} A_t A_t^{-1} C$. The two negative terms and one positive term yield exactly (GMT.19). In particular the unscaled matrix $C^* A_t^{-1} C$ is positive semidefinite and vanishes precisely when $C = 0$. Pairing $A_t^{-1} > 0$ with every column of C proves this criterion. The full quadratic loss $t^2 C^* A_t^{-1} C$ vanishes precisely when $t = 0$ or $C = 0$; for an empty boundary block it is the zero form throughout.

The representative difference is a literal boundary with coefficient vector $-tA_t^{-1} Cx$; inserting that polynomial into (GMT.4) gives its full original theta primitive. The same jets ηx persist. No statement of metric nonidentity leaves its comparison map unspecified.

For completeness the exact derivative identities are

$$R'_t = -B_N A_t^{-1} B_N^* E R_t, \quad G'_t = R_t^* E R_t, \quad G''_t = -2(B_N^* E R_t)^* A_t^{-1} (B_N^* E R_t). \quad (\text{GMT.20})$$

Differentiate $J_N R_t = 1$ and $B_N^* H_t R_t = 0$ to obtain the first formula. The derivative R'_t is a boundary, hence its two pairings with R_t in the derivative of $R_t^* H_t R_t$ vanish. This proves the second formula. Differentiating it and substituting the first proves the third. This is a proved concavity statement along the actual finite metric segment, with the entire boundary coupling retained.

The determinant-line calculation has no hidden source-size exponent. The change $[E_N, B_N] \mapsto [R(H), B_N]$ adds boundary columns to the first q columns and has determinant one. Its Gram is block diagonal. Therefore

$$V_N(H) = \det G(H) = \frac{\det H}{\det(B_N^* H B_N)}. \quad (\text{GMT.21})$$

For the original arithmetic and Gamma forms this is exactly $V_N = V_N^\Gamma T_N$, with the same $T_N = X_{N+1}/Y_{N-q+1}$ as the supplied descent. Formula (GMT.19) gives its complete matrix-level map and its nonlinear remainder.

40.5 A two-degree map for the actual consecutive ratio

Fix $N \geq q$. In original source monomials split degree $N + 1$ into degrees $0, \dots, N - 1$ followed by S^N, S^{N+1} , and write

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_-^{-1} C_D, \quad J_+ = [J_-, J_D], \quad F_D = J_D - J_- H_-^{-1} C_D. \quad (\text{GMT.22})$$

Here $H_- = H_{N-1}$, $H_+ = H_{N+1}$, W is 2×2 , and $F_D : \mathbb{C}^2 \rightarrow \mathcal{C}$. Its columns are the quotient classes of the two specified polynomials after subtracting their actual orthogonal projections onto $\mathcal{P}_{N-1}$. The block map

$$(x, y) \mapsto (x - H_-^{-1} C_D y, y)$$

has determinant one and carries the source Gram to $\text{diag}(H_-, W)$. Positivity proves $W > 0$ and, by applying the reduction map to this exact orthogonal direct sum,

$$K_+ = K_- + F_D W^{-1} F_D^*, \quad Z_D = F_D^* G_- F_D \succeq 0, \quad G_- = K_-^{-1}. \quad (\text{GMT.23})$$

Block elimination of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in its two orders proves $\det(I + AB) = \det(I + BA)$. Applying this identity to (GMT.23) gives the exact scalar maps

$$\boxed{\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W} = 1 + \text{Tr}(W^{-1} Z_D) + \frac{\det Z_D}{\det W}.} \quad (\text{GMT.24})$$

The last equality is the complete characteristic polynomial of a 2×2 matrix. All denominators are positive. This gives the actual monotonicity because both eigenvalues of $W^{-1/2} Z_D W^{-1/2}$ are non-negative.

The same block also gives the adjacent source recurrence coefficient. In the order in (GMT.22), the monic residual of S^N has squared norm W_{11} , and eliminating it from the residual of S^{N+1} gives squared norm $\det W / W_{11}$. Therefore

$$\boxed{a_{N+1} = \frac{\omega_{N+1}}{\omega_N} = \frac{\det W}{W_{11}^2}.} \quad (\text{GMT.25})$$

The first residual is monic because the subtracted projection has smaller degree; so is the second after its additional elimination. The norms have not been rescaled. The coefficient map $S^j = \sum_{l \leq j} \binom{j}{l} c^{j-l} u^l$ has determinant $i^{j(j-1)/2}$ in dimension j and hence Gram determinant multiplier one, which proves that these are also the source Hankel-Toda norms.

Apply (GMT.22)–(GMT.25) separately to the Gamma reference and the arithmetic source, retaining their same J_+ and χ . Then

$$\boxed{\frac{T_{N-1}}{T_{N+1}} = \frac{\det(W + Z_D) / \det W}{\det(W^\Gamma + Z_D^\Gamma) / \det W^\Gamma}.} \quad (\text{GMT.26})$$

Every entry is a finite polynomial in the one-factor coefficients through degree $2N + 2$, followed by explicitly stated inverses of positive matrices. In the supplied Toda upper estimate, substitute the two positive scalars $a_{N+1} = \det W / W_{11}^2$ and $\mathcal{R}_N = \det(W + Z_D) / \det W$. Its full right side becomes

$$\sqrt{\frac{\det W}{W_{11}^2} \frac{\det(W + Z_D) / \det W - 1}{2\sqrt{\det(W + Z_D) / \det W}}}. \quad (\text{GMT.27})$$

This is an exact substitution into that existing inequality; its discarded phase and volume-imbalance squares remain the ones proved in TVB.32–TVB.34a. Formula (GMT.27) alone does not remove them or assert equality. The two-degree determinant formula itself also works at $q = 0$, where F_D and Z_D are zero and the ratio is one.

40.6 Certified finite envelopes and quotient-dimensional bounds

Suppose a finite coefficient calculation supplies the entry bounds (GMT.15), or the proved polynomial-test bounds (GMT.16). Let $v_i > 0$ be any specified positive rational weights and set

$$\Delta = \text{diag}(\delta_i), \quad \delta_i = \sum_j \varepsilon_{ij} \frac{v_j}{v_i}, \quad H^{\text{lo}} = \hat{H} - \Delta, \quad H^{\text{hi}} = \hat{H} + \Delta. \quad (\text{GMT.28})$$

The radii may be replaced by symmetric rational upper radii. For each pair i, j , the inequality $2|x_i||x_j| \leq (v_j/v_i)|x_i|^2 + (v_i/v_j)|x_j|^2$ gives $|x^*(H - \hat{H})x| \leq x^*\Delta x$ after summation. This proves $H^{\text{lo}} \preceq H \preceq H^{\text{hi}}$ without using a pointwise lower bound on B .

The positive-pivot certificate for H^{lo} is entirely finite. Eliminate successively its first diagonal pivot, recording

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \mapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \mapsto (x - a^{-1}b^*y, y).$$

The displayed congruence is invertible. Positive recorded rational pivots prove positivity by induction; nonpositive pivots reject that particular enclosure. Actual matrix positivity is already proved by the source integral. For any sequence of valid entry intervals tending to their actual entries, the lower envelopes tend to that positive matrix. Its least unit-sphere quadratic value is positive by compactness, and convergence in the finite sum of entry errors eventually makes every lower envelope positive. Hence the certificate process succeeds at every fixed finite degree when its actual input intervals are refined; no success or interval refinement is asserted for absent arithmetic data.

For a positive certified lower envelope, calculate the actual finite quotient forms $G^{\text{lo}}, G^{\text{hi}}$ by (GMT.18), or by the equivalent Schur formula

$$G(H) = E_N^* H E_N - E_N^* H B_N (B_N^* H B_N)^{-1} B_N^* H E_N. \quad (\text{GMT.29})$$

The formula follows by substituting $E_N - B_N (B_N^* H B_N)^{-1} B_N^* H E_N$ for the unique least lift. The same radii certify the representative coefficients themselves. Write $\hat{R} = R(\hat{H})$, $A^{\text{lo}} = B_N^* H^{\text{lo}} B_N$ and define, using any rational upper bounds for the displayed complex moduli,

$$\begin{aligned} s_{ai} &= \sum_{r,l} |(B_N)_{ra}| \varepsilon_{rl} |\hat{R}_{li}|, \\ k_{ab} &= \frac{1}{2} ((A^{\text{lo}})_{aa}^{-1} + (A^{\text{lo}})_{bb}^{-1}), \\ e_{ri}^R &= \sum_{a,b} |(B_N)_{ra}| k_{ab} s_{bi}. \end{aligned}$$

Then the full original-coordinate error and its quadratic loss obey

$$\boxed{|(R(H) - \hat{R})_{ri}| \leq e_{ri}^R, \quad \hat{R}^* H \hat{R} - G(H) = (R(H) - \hat{R})^* H (R(H) - \hat{R})}. \quad (\text{GMT.29a})$$

To prove the entry bound set $A = B_N^* H B_N$ and $S_E = B_N^* (H - \widehat{H}) \widehat{R}$. Formula (GMT.19) gives $R(H) - \widehat{R} = -B_N A^{-1} S_E$. The triangle inequality gives $|(S_E)_{ai}| \leq s_{ai}$. Inversion reverses positive order: conjugating $A \succeq A^{\text{lo}}$ by $(A^{\text{lo}})^{-1/2}$, inverting its eigenvalues, and conjugating back gives $A^{-1} \preceq (A^{\text{lo}})^{-1}$. Positivity of each two-coordinate Gram minor of A^{-1} gives $|A_{ab}^{-1}| \leq \sqrt{A_{aa}^{-1} A_{bb}^{-1}} \leq k_{ab}$, where the last step is the arithmetic-geometric mean inequality and the proved diagonal bounds. Matrix multiplication now proves the coefficientwise bound in (GMT.29a). For its second identity, $\widehat{R} - R(H)$ is an actual boundary, orthogonal to $R(H)$ in H . Expanding $\widehat{R}^* H \widehat{R}$ proves the equality.

This loss has an explicit q -dimensional rational upper form. Choose positive rational weights v_i in the original quotient coordinates and set

$$\tau_{ij} = \sum_{a,b} s_{ai} |(A^{\text{lo}})^{-1}|_{ab} s_{bj}, \quad \Delta_Q = \text{diag} \left\{ \sum_j \tau_{ij} \frac{v_j}{v_i} \right\}.$$

The same entrywise quadratic inequality as (GMT.28) proves

$$0 \preceq \widehat{R}^* H \widehat{R} - G(H) \preceq \Delta_Q. \quad (\text{GMT.29b})$$

Indeed the middle matrix is $S_E^* A^{-1} S_E \preceq S_E^* (A^{\text{lo}})^{-1} S_E$. Its value on x is at most $\sum_{i,j} \tau_{ij} |x_i| |x_j| \leq x^* \Delta_Q x$. Every source row, relation coefficient and quotient coordinate is retained in these sums. At an empty boundary block both error forms are zero. Applying $\mathcal{V}_{h,k}$ to each representative difference gives the theta primitive (GMT.4), so the certified coefficient error also has its original source and quotient interpretation.

Minimizing the two proven source inequalities over the same affine space of representatives gives

$$G^{\text{lo}} \preceq G(H) \preceq G^{\text{hi}}, \quad \det G^{\text{lo}} \leq V_N(H) \leq \det G^{\text{hi}}. \quad (\text{GMT.30})$$

For the determinant assertion, conjugate by the positive square root of the lower form and multiply the eigenvalues, all of which are at least one; apply the same argument for the upper form.

Construct the degree $N - 1$ envelope by taking the corresponding principal submatrix of the same degree $N + 1$ envelope. Then

$$\frac{\det G_-^{\text{lo}}}{\det G_+^{\text{hi}}} \leq \frac{V_{N-1}}{V_{N+1}} \leq \frac{\det G_-^{\text{hi}}}{\det G_+^{\text{lo}}}. \quad (\text{GMT.31})$$

Division by the exact positive Gamma ratio from (GMT.26) gives the certified interval for T_{N-1}/T_{N+1} . These are determinants of size q , with no independent numerator/denominator error estimate on the larger source and relation determinants.

For example a relative certificate $(1 - \eta)\widehat{H} \preceq H \preceq (1 + \eta)\widehat{H}$, with a recorded rational $0 \leq \eta < 1$ and $\widehat{H} > 0$, gives exactly

$$(1 - \eta)^q \det G(\widehat{H}) \leq V_N(H) \leq (1 + \eta)^q \det G(\widehat{H}),$$

and a common such certificate for the two nested degrees gives ratio multipliers $((1 - \eta)/(1 + \eta))^q$ and its reciprocal. This follows because $R(aH) = R(H)$ and $G(aH) = aG(H)$ for $a > 0$, as direct substitution in (GMT.18) proves. The mass is scaled only in this explicitly declared comparison, never assigned a new value in the original arithmetic source.

40.7 Exact calibration and a bounded relative-volume counterexample

Take the explicitly auxiliary one-factor weight

$$\lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = \frac{1}{2}, \quad C_2 = \frac{1}{4}, \quad B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = (S - 1)^2 - 2. \quad (\text{GMT.32})$$

The positive polynomial B lies in $L^2(r_1)$ by (GMT.6). It is not the function $|g/h\Gamma|^2$ of a claimed zeta packet. Since $b_2^{(1)} = t^2 - 2$, its coefficients are $c_{B,0} = 3/2$, $c_{B,2} = 1/4$ and all others zero. Therefore

$$A(z) = \frac{3}{2} + \frac{3}{2}z^2, \quad (d_0, d_2, d_4) = \left(\frac{9}{4}, \frac{9}{2}, \frac{9}{4}\right), \quad B_2(u) = \frac{54}{35} + \frac{39}{280}u^2 + \frac{3}{1120}u^4. \quad (\text{GMT.33})$$

To verify the last equality, use $b_2^{(2)} = u^2 - 4$, $b_4^{(2)} = u^4 - 32u^2 + 72$ in $\sum_n d_n b_n^{(2)}/(4)_n$ and expand all three terms. The one-factor mass is $3/4$ and the sum mass is $9/16$. Repeated multiplication by t in the recurrence (GMT.8) gives the one-factor reference even moments through degree eight as $(1/2, 1, 8, 136, 3968)$. Inserting $B = 1 + t^2/4$ and expanding $(t_1 + t_2)^n$ with the full binomial coefficients gives sum moments

$$(\mu_0, \mu_1, \dots, \mu_8) = (9/16, 0, 9/2, 0, 117, 0, 5472, 0, 385272). \quad (\text{GMT.34})$$

This is also (GMT.11)–(GMT.12); the two calculations independently fix its mass and tensor cross term.

The original degree-three source Gram and reference Gram are

$$H_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \quad H_3^\Gamma = \frac{1}{4} \begin{pmatrix} 1 & 1 & -3 & -11 \\ 1 & 5 & 5 & -55 \\ -3 & 5 & 65 & 65 \\ -11 & -55 & 65 & 1685 \end{pmatrix}. \quad (\text{GMT.35})$$

Every entry follows by expanding $(1 - iu)^a(1 + iu)^b$ against (GMT.34), or against the reference moments. The complete original relation and reduction matrices are

$$B_3 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \\ 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}. \quad (\text{GMT.36})$$

These follow from $S^2 = 2S + 1$ and $S^3 = 5S + 2$ modulo χ ; multiplication verifies $J_3 B_3 = 0$ and $\det[E_3, B_3] = 1$. Direct substitution into (GMT.18) gives

$$G_1 = \frac{1}{16} \begin{pmatrix} 9 & 9 \\ 9 & 81 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 81/244 & 81/244 \\ 81/244 & 175041/80764 \end{pmatrix},$$

$$R_3 = \begin{pmatrix} 117/122 & -801/40382 \\ -5/61 & 18963/20191 \\ 5/122 & -907/40382 \\ 0 & 7/331 \end{pmatrix}, \quad G_3^\Gamma = \begin{pmatrix} 5/38 & 5/38 \\ 5/38 & 2255/4142 \end{pmatrix}. \quad (\text{GMT.37})$$

These values can be checked without trusting an inverse algorithm: multiply to obtain $J_3 R_3 = I$, $B_3^* H_3 R_3 = 0$ and $R_3^* H_3 R_3 = G_3$, then use the uniqueness proved in (GMT.18).

For the degree-one to degree-three split, the full two-degree data are

$$W = \begin{pmatrix} 81 & 243 \\ 243 & 3159 \end{pmatrix}, \quad F_D = \begin{pmatrix} 10 & 2 \\ 0 & 28 \end{pmatrix}, \quad Z_D = \frac{1}{4} \begin{pmatrix} 225 & 675 \\ 675 & 16137 \end{pmatrix}. \quad (\text{GMT.38})$$

Equation (GMT.22), evaluated with (GMT.35)–(GMT.36), proves these entries by ordinary rational multiplication. Consequently

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad a_3 = 30, \quad \frac{V_1^\Gamma}{V_3^\Gamma} = \frac{2071}{450}, \quad \boxed{\frac{T_1}{T_3} = \frac{100955}{111834} < 1}. \quad (\text{GMT.39})$$

Here $V_1^\Gamma = 1/4$, $V_3^\Gamma = 225/4142$, obtained by taking the two displayed 2×2 determinants. Thus no unrecorded cancellation of reference masses is used.

The phenomenon has a completely specified bounded positive version. Define $B^{[200]}(t) = \min(1 + t^2/4, 10001)$ and $w^{[200]} = B^{[200]}r_1$. This changes the auxiliary weight only; it leaves $S = 1 + iu$, χ , the reference Gamma mass and the unscaled convolution operation fixed. For $0 \leq j \leq 6$,

$$\int |t|^j (w - w^{[200]})(t) dt \leq \frac{1}{4} \int_{|t|>200} |t|^{j+2} r_1(t) dt < \frac{8!}{2^{92}} =: \epsilon_*. \quad (\text{GMT.40})$$

Here is a proof with no floating evaluation. For every integer $r \geq 0$, $x^r \leq r!2^r e^{x/2}$; on $x > 200$ multiply by $e^{(x-200)/2} \geq 1$. Equation (GMT.6) and evenness give

$$\int_{|t|>200} |t|^r r_1(t) dt \leq 2c_1 r! 2^r e^{-100} (\cos 1)^{-2} \leq 4r! 2^r e^{-100}.$$

The estimate $\cos 1 \geq 1/2$ follows from $1 - \cos 1 = \int_0^1 \sin x dx \leq \int_0^1 x dx = 1/2$, using $|\sin x| \leq x$ on this interval. Since $e > 2$, $e^{-100} < 2^{-100}$. Use $r = j + 2 \leq 8$ and the factor $1/4$ to obtain (GMT.40).

For $0 \leq j \leq 6$, the absolute j th one-factor moment of w is at most $A_* = \mu_0(w) + \mu_6(w) = 3/4 + 1128 = 4515/4$, because $|t|^j \leq 1 + |t|^6$. The exact tensor difference is $(w - w^{[200]}) \otimes w + w^{[200]} \otimes (w - w^{[200]})$. Expanding $|t_1 + t_2|^n \leq \sum_j \binom{n}{j} |t_1|^j |t_2|^{n-j}$ proves a sum-moment difference at degree $n \leq 6$ of at most $2^{n+1} A_* \epsilon_* \leq 128 A_* \epsilon_*$. The coefficient absolute sum in $(1 - iu)^a (1 + iu)^b$ is at most $2^{a+b} \leq 64$ for $a, b \leq 3$. Hence every source entry changes by at most $8192 A_* \epsilon_* < 1/40000$. The last inequality is an integer comparison after substitution of A_*, ϵ_* . Equation (GMT.28) with four weights equal to one now proves

$$H_3 - 10^{-4}I \prec H_3^{[200]} \prec H_3 + 10^{-4}I. \quad (\text{GMT.41})$$

Strictness follows because the error in each row sum is strictly less than 10^{-4} . The four leading principal minors of the lower matrix are

$$\frac{703}{1250}, \quad \frac{253068751}{100000000}, \quad \frac{102481960035937}{500000000000}, \quad \frac{4980299350501627807501}{10000000000000000}.$$

They are positive; their successive ratios are exactly the positive pivots in the elimination proving (GMT.28). Strict source inequalities give strict quotient inequalities: evaluate the lower form on the actual minimum lift, and the actual form on the upper minimum lift. Each lift of a nonzero quotient vector is nonzero, so the positive source difference has a strictly positive quadratic value. Computing (GMT.29) for these two rational endpoint matrices, and the degree-one principal submatrices, gives the strict ratio enclosure

$$\frac{20694424798404596925036}{4984219548001747192501} < \frac{V_1^{[200]}}{V_3^{[200]}} < \frac{20703435010591353075036}{4980299350501627807501} < \frac{2071}{450}. \quad (\text{GMT.42})$$

The first two strict comparisons follow from that strict minimization and (GMT.31); the final comparison follows by positive-denominator cross multiplication. The rational envelope computation is reproduced by the exact checker accompanying this proof. Division by the fixed positive Gamma ratio therefore proves $T_1^{[200]}/T_3^{[200]} < 1$ for a function bounded between 1 and 10001. This is an explicit obstruction to deriving relative volume monotonicity from positivity and boundedness of a multiplier alone. Its exact relation to the arithmetic problem is the same coefficient, relation, least-lift and determinant maps (GMT.12), (GMT.18), (GMT.24) and (GMT.26). It does not assert that the actual arithmetic multiplier has this behavior.

40.8 Verification scope

The accompanying exact checker constructs moments in two ways: Jacobi recurrence and the literal two-factor binomial expansion, then compares them with the Gamma coefficient formula. It verifies the original S -coordinate Gram, every displayed lift and relation identity, the full boundary-resolvent loss, five points on its actual positive matrix segment, both two-degree formulas and the rational envelope calculation. Its deliberate formula changes remove the mass, omit the tensor cross term, reverse the boundary-loss sign, remove the conjugation, or remove one power in the Schur denominator. Each change is rejected by the corresponding exact check in ordinary and optimized Python. The written proofs establish the all-degree identities; these finite checks provide reproducible regression evidence. There is no Lean execution or analytic interval claim for a nonempty zeta packet in this verification.

41 The complete complex amplitude in gamma sum descent

41.1 Original objects and the question answered

The delivered Gamma Convolution Descent note constructs the conditional expectation of the squared arithmetic amplitude. We construct the exact comparison for its full complex amplitude. The two resulting source Grams add to the original arithmetic Gram. We prove a strict positive contribution of the retained relative component for every full quartet, and give its exact map into the original minimum quotient metric. All measures below retain their masses. The variable is the original sum $u = t_1 + \cdots + t_k$, and the arithmetic variable remains $S = k/2 + iu$.

Fix $\lambda > 0$ and an integer $k \geq 1$. Retain

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad r_{\lambda,k} = r_\lambda^{*k} = \frac{c_\lambda^k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GP.1})$$

Thus $\int r_{\lambda,k} = c_\lambda^k$. The beta-integral proof of this identity, with its original Jacobian $t = 2v$, is retained in the cumulative reader, TG.7–TG.8, and the delivered note, (5)–(9). In particular

$$\int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}, \quad |\theta| < \pi/2. \quad (\text{GP.2})$$

Let h be the original finite polynomial packet of complete orders of actual zeros of $g = 2\xi$, and include $h = 1$ when considering the analytic seed. Define the entire quotient $v_h = g/h$ and keep

$$A(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B(t) = |A(t)|^2, \quad w(t) = B(t)r_\lambda(t), \quad \mu_h = \int w(t) dt. \quad (\text{GP.3})$$

Use one of the references proved in the delivered note for which $B \leq C_h < \infty$. In particular a full quartet of degree $d \geq 4$ uses its original $\lambda = 1/4$. The seed uses $\lambda = 13/4$ and $C_1 = 1024/\sqrt{\pi}$. The statements involving decay explicitly use the quartet reference. The function A is nonzero and analytic on the real line: v_h is a nonzero entire function and reciprocal gamma is entire and nonzero on this line. Its real zeros form a discrete set. No value at a removed packet zero is assigned using an uncanceled fraction.

Put $d\nu(\mathbf{t}) = \prod_i r_\lambda(t_i) d\mathbf{t}$ and $d\nu_\Sigma(u) = r_{\lambda,k}(u) du$. In $\mathcal{H} = L^2(\nu)$ and $\mathcal{H}_\Sigma = L^2(\nu_\Sigma)$ write

$$Uf(\mathbf{t}) = f(\sum_i t_i), \quad C = U^*, \quad P = UC, \quad a(\mathbf{t}) = \prod_i A(t_i), \quad (\text{GP.4})$$

$$b_k = C(|a|^2), \quad a_k = C(a), \quad d_k = b_k - |a_k|^2.$$

For $k = 1$, $C = U = I$, $a_1 = A$, $b_1 = B$, $d_1 = 0$. For $k \geq 2$ the conditional integral CF is precisely

$$(CF)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GP.5})$$

These are probability measures on individual fibres, derived by dividing by their stated fibre mass; neither original measure is replaced by a probability measure. Fubini proves $CU = I$ and the isometry of U . Jensen's inequality in (GP.5) proves $d_k \geq 0$ and $0 \leq b_k \leq C_h^k$. For $k \geq 2$, $b_k(u) > 0$ at every real u : the discrete zeros of each factor exclude only countably many hyperplanes in this fibre, which have zero Lebesgue measure.

41.2 Unitary transport of the actual conditional projection

Theorem 41.1 (Complete amplitude comparison). *Use the actual arithmetic spaces*

$$\mathcal{H}^h = L^2(\mathbb{R}^k, |a|^2 d\nu), \quad \mathcal{H}_\Sigma^h = L^2(\mathbb{R}, b_k d\nu_\Sigma).$$

The maps

$$T : \mathcal{H}^h \longrightarrow \mathcal{H}, \quad F \longmapsto aF, \quad T_\Sigma : \mathcal{H}_\Sigma^h \longrightarrow \mathcal{H}_\Sigma, \quad f \longmapsto \sqrt{b_k}f \quad (\text{GP.6})$$

are surjective isometries. The positive square root in T_Σ is a specified comparison map; the original amplitude in T remains complex. The actual arithmetic sum inclusion $U_h f = f(\sum t_i)$ and its adjoint C_h^{sum} satisfy

$$C_h^{\text{sum}} F = \frac{C(|a|^2 F)}{b_k}, \quad J = TU_h T_\Sigma^{-1} = M_a U M_{b_k^{-1/2}}, \quad J^* = M_{b_k^{-1/2}} C M_{\bar{a}}, \quad J^* J = I. \quad (\text{GP.7})$$

All quotients here are defined almost everywhere. In particular their operator extensions are bounded even when the individual inverse multiplier is unbounded. The transported arithmetic projection is

$$Q = TP_h T^{-1} = J J^*, \quad QF = a U \left(\frac{C(\bar{a}F)}{b_k} \right), \quad P_h = U_h C_h^{\text{sum}}. \quad (\text{GP.8})$$

Thus P and Q are related by the displayed exact transport from the original arithmetic spaces.

Proof. The squared norms in (GP.6) agree by direct substitution. The zero set of a is a union of null hyperplanes; b_k is positive almost everywhere, including $k = 1$. Division off those null sets constructs the two inverses with their exact squared norms. Applying (GP.5) to the arithmetic fibre density gives C_h^{sum} . Pairing $M_a U M_{b_k^{-1/2}} f$ with F proves the stated adjoint. The identity $C(|a|^2 U f) = b_k f$ then gives $J^* J = I$. Conjugating P_h gives (GP.8). These norm identities first hold for bounded functions supported on finite-measure sets on which the denominators are bounded below; truncation and the isometries extend them to the indicated complete Hilbert spaces. This also proves that no unbounded multiplier composition is being asserted on a larger domain. $\square$

Theorem 41.2 (Exact fibre angle and retained defect). *Define $\gamma_k = a_k / \sqrt{b_k}$ almost everywhere. Then*

$$U^* J = M_{\gamma_k}, \quad J = U M_{\gamma_k} + N, \quad U^* N = 0, \quad N^* N = M_{1-|\gamma_k|^2} = M_{d_k/b_k}. \quad (\text{GP.9})$$

For every real u at which $b_k > 0$, the two fibre projections onto 1 and $a/\sqrt{b_k}$ obey

$$\text{Tr}_u(P_u - Q_u)^2 = 2(1 - |\gamma_k(u)|^2), \quad \|[P_u, Q_u]\|_{\text{HS},u}^2 = 2|\gamma_k(u)|^2(1 - |\gamma_k(u)|^2). \quad (\text{GP.10})$$

These are finite-rank traces in the specified fibre Hilbert space, not a claim that the global decomposable operators are trace class. The complete unscaled variance also has the formula

$$d_k(u) = \frac{1}{2} \iint |a(\mathbf{t}) - a(\mathbf{s})|^2 d\nu_u(\mathbf{t}) d\nu_u(\mathbf{s}), \quad (\text{GP.11})$$

where both ν_u are the exact conditional measures in (GP.5).

Proof. Applying C to Jf gives $\gamma_k f$. Orthogonal decomposition with P gives (GP.9), and taking squared norms uses $J^*J = I$. On a fibre let $e_0 = 1$ and $v = a/\sqrt{b_k}$, so both have norm one and $\langle e_0, v \rangle = \gamma_k$, using the inner product antilinear in its first argument. If $|\gamma_k| < 1$, put $e_1 = (v - \gamma_k e_0)/\sqrt{1 - |\gamma_k|^2}$. In the displayed orthonormal comparison frame, without altering either original vector, the two projections are

$$P_u = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q_u = \begin{pmatrix} |\gamma_k|^2 & \gamma_k \sqrt{1 - |\gamma_k|^2} \\ \overline{\gamma_k} \sqrt{1 - |\gamma_k|^2} & 1 - |\gamma_k|^2 \end{pmatrix}.$$

Matrix multiplication proves (GP.10); both vanish when $|\gamma_k| = 1$. Expanding the square in (GP.11), and using the unit total mass of each conditional measure, gives $C(|a|^2) - |Ca|^2$. No phase is discarded in this calculation. $\square$

41.3 Strictness for an actual quartet, with its exact constant

For a full quartet retain $\lambda = 1/4$, degree $d \geq 4$ and $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. The delivered proof gives

$$|A(t)|^2 \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d} \quad (|t| \geq T_h). \quad (\text{GP.12})$$

Here is its direct calculation. In the unchanged amplitude,

$$A(t) = -\pi^{-1/4} (t^2 + 1/4) e^{-it \log \pi/2} \zeta(1/2 + it) / h(1/2 + it).$$

The continuation formula $\zeta(s) = s/(s-1) - s \int_1^\infty \{x\} x^{-s-1} dx$ gives $|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|)$. For $|t| \geq T_h$, every retained root factor has modulus at least $|t|/2$, so $|h| \geq (|t|/2)^d$. Also $(t^2 + 1/4)^2 \leq 25t^4/16$ and $4(1 + |t|)^2 \leq 16t^2$ for $|t| \geq 1$. Multiplication proves (GP.12). Consequently $A(t) \rightarrow 0$ at both real ends.

Theorem 41.3 (Strict relative contribution on every sum fibre). *For every such actual quartet, every $k \geq 2$, and every real u ,*

$$d_k(u) > 0, \quad |\gamma_k(u)| < 1. \quad (\text{GP.13})$$

Thus the relative component is nonzero at every fibre. It is given exactly by (GP.9) and (GP.11); strictness does not assign a uniform positive lower bound to it.

Proof. Fix u . If $d_k(u) = 0$, (GP.11), or equality in the variance identity, says that the continuous function a is constant almost everywhere on the fibre. Its conditional density is strictly positive everywhere, so continuity makes it constant everywhere. For $k = 2$, this would make $A(t)A(u - t)$ constant in t . Its limit is zero by (GP.12) and boundedness of A , whereas some value is nonzero: avoid the two discrete sets of zeros of $A(t)$ and $A(u - t)$. For $k > 2$, choose fixed $t_2, \dots, t_{k-1}$ at nonzeros of A and vary $t_1 = t$, $t_k = u - t - \sum_{2 \leq i < k} t_i$. The identical argument makes the product both zero and nonzero, a contradiction. The assertion for γ_k follows by dividing by the strictly positive b_k already proved. This proof uses only the explicit quartet tail, not a conjecture about positions of unselected zeros. $\square$

41.4 Full complex coefficient transform and exact variance norm

Put $\alpha = 2\lambda$ and keep the original monic polynomials

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}, \quad \|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n. \quad (\text{GP.14})$$

The specified gamma measure has exponential moments. Its polynomials are complete in L^2 : for a vector orthogonal to them, multiplying it by the measure gives a finite complex measure by Cauchy-Schwarz; the same inequality with exponential moments makes its Fourier transform holomorphic

on a strip about the real axis. All derivatives at zero vanish. Analytic uniqueness and Fourier uniqueness make the measure, hence the vector, zero. The same proof applies to the sum measure with mass c_λ^k .

Define complex coefficients, retaining conjugation conventions,

$$a_j = \frac{\int b_j^{(\lambda)}(t)A(t)r_\lambda(t)dt}{c_\lambda j!(\alpha)_j}, \quad \mathfrak{A}(z) = \sum_{j \geq 0} (\alpha)_j a_j z^j, \quad e_{k,n} = [z^n] \mathfrak{A}(z)^k. \quad (\text{GP.15})$$

The square-integrable vector A has these orthogonal coefficients. The series in (GP.15) may be used coefficientwise; no interchange of an infinite product with an integral is required below.

Theorem 41.4 (Complex convolution and its exact lost norm). *With all original masses,*

$$a_k(u) = \sum_{n \geq 0} \frac{e_{k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u) \quad \text{in } L^2(r_{\lambda,k} du),$$

$$\int d_k r_{\lambda,k} = \mu_h^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |e_{k,n}|^2. \quad (\text{GP.16})$$

For every polynomial F in the original S coordinate and every real $|\theta| < \pi/2$, the exact weighted version is

$$\|a U(e^{\theta u/2} F(k/2 + iu))\|^2 = \int e^{\theta u} |F(k/2 + iu)|^2 |a_k(u)|^2 r_{\lambda,k}(u) du$$

$$+ \int e^{\theta u} |F(k/2 + iu)|^2 d_k(u) r_{\lambda,k}(u) du. \quad (\text{GP.17})$$

Proof. Multiplication of the finite-degree coefficients of the generating functions gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing this finite identity with a gives $c_\lambda^k n! e_{k,n}$. Division by the original sum norm $c_\lambda^k n! (k\alpha)_n$, followed by completeness, gives the first identity. Pythagoras for $P = UC$ gives the second since $\|a\|^2 = \mu_h^k$. Finally conditional expectation commutes with multiplication by a function of the sum. Apply the same orthogonal decomposition to $aU(e^{\theta u/2} F)$ to obtain (GP.17). Boundedness by C_h^k , (GP.2), and polynomial growth prove all integrals finite, including any fixed number of theta derivatives. $\square$

41.5 The exact induced change of the original finite quotient metric

Fix the actual annihilator $\chi = \chi_{h,k}$, degree $q > 0$, and $N \geq q - 1$. Use original polynomial spaces $E = \mathcal{P}_N(S)$, $E_\partial = \mathcal{P}_{N-q}(S)$, where a negative degree means the zero space, and $C_\chi = \mathbb{C}[S]/(\chi)$. Let $B_\chi : E_\partial \rightarrow E$ be multiplication by χ and $\pi : E \rightarrow C_\chi$ the quotient. Use the original monomial bases, or any explicitly transported fixed bases, in every matrix below. For a coefficient column x denote its polynomial by F_x . Define three source matrices by

$$x^* M_h x = \int e^{\theta u} |F_x(k/2 + iu)|^2 b_k r_{\lambda,k} du,$$

$$x^* M_{\text{coh}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 |a_k|^2 r_{\lambda,k} du,$$

$$x^* M_{\text{rel}} x = \int e^{\theta u} |F_x(k/2 + iu)|^2 d_k r_{\lambda,k} du. \quad (\text{GP.18})$$

These matrices are elements of $\text{Herm}(E)$, equivalently linear maps $E \rightarrow E^\vee$ with $E^\vee$ the conjugate dual under $x \mapsto [y \mapsto y^* M x]$. The associated linear observation maps are $F \mapsto aU(e^{\theta u/2} F)$, $F \mapsto$

$U(e^{\theta u/2} a_k F)$, and $F \mapsto (I - P)aU(e^{\theta u/2} F)$. The Gram operation itself has type $\text{Hom}_{\mathbb{C}}(E, \mathcal{H}) \rightarrow \text{Herm}(E)$, $R \mapsto R^* R$. Thus

$$M_h = M_{\text{coh}} + M_{\text{rel}}. \quad (\text{GP.19})$$

The original arithmetic matrix M_h is unchanged.

Lemma 41.5. M_{coh} is positive definite for every packet/reference admitted above, every $k \geq 1$, every finite N , and real $|\theta| < \pi/2$. For the quartets of (GP.12) and $k \geq 2$, M_{rel} is also positive definite.

Proof. Put $f = Ar_{\lambda}$. It is a nonzero L^1 function, since A is bounded and nonzero almost everywhere. The numerator of a_k is f^{*k} . If this convolution were zero almost everywhere, its Fourier transform $(\hat{f})^k$ would vanish everywhere, whence $\hat{f} = 0$ and $f = 0$ by Fourier uniqueness, a contradiction. Consequently $|a_k|^2 r_{\lambda,k}$ is positive on a set of positive Lebesgue measure. A nonzero polynomial on the indicated affine line has only finitely many real zeros, so its squared norm is strictly positive. This proves the first assertion. The second follows from $d_k(u) > 0$ for every u , proved in (GP.13). $\square$

Let $R_h, R_{\text{coh}} : C_{\chi} \rightarrow E$ be the least-norm sections for M_h, M_{coh} , and write $G_h = R_h^* M_h R_h$, $G_{\text{coh}} = R_{\text{coh}}^* M_{\text{coh}} R_{\text{coh}}$. They exist uniquely: for any section R_0 , subtract $B_{\chi}(B_{\chi}^* M B_{\chi})^{-1} B_{\chi}^* M R_0$; the boundary Gram is invertible because B_{χ} is injective and M positive definite. An empty boundary contributes the zero map.

Theorem 41.6 (Exact two positive terms in the quotient change). *In these original coordinates put*

$$K = (B_{\chi}^* M_h B_{\chi})^{-1} B_{\chi}^* M_{\text{rel}} R_{\text{coh}}. \quad (\text{GP.20})$$

It is a map $C_{\chi} \rightarrow E_{\partial}$, or the unique zero map when the boundary is empty. Then

$$R_h = R_{\text{coh}} - B_{\chi} K, \quad \boxed{G_h = G_{\text{coh}} + K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h.} \quad (\text{GP.21})$$

For an actual full quartet, $k \geq 2$, and $q > 0$, this gives $G_h - G_{\text{coh}} > 0$ for every admitted N and theta. More precisely, with the actual positive matrix

$$E_q = G_{\text{coh}}^{-1/2} (K^* B_{\chi}^* M_{\text{coh}} B_{\chi} K + R_h^* M_{\text{rel}} R_h) G_{\text{coh}}^{-1/2}, \quad (\text{GP.22})$$

the original determinants satisfy

$$\det G_h = \det G_{\text{coh}} \det(I + E_q). \quad (\text{GP.23})$$

These comparison conjugations do not replace either quotient metric.

Proof. $B_{\chi}^* M_{\text{coh}} R_{\text{coh}} = 0$, so the formula for the M_h section using R_{coh} as initial section gives (GP.20) and the first identity. Expand $R_h^* M_{\text{coh}} R_h$ about R_{coh} . The two cross terms vanish by the stated orthogonality. Add $R_h^* M_{\text{rel}} R_h$ using (GP.19); this proves (GP.21) with both terms retained. R_h is injective because $\pi R_h = I$. Thus strict positivity of M_{rel} makes its displayed pullback positive definite on the nonzero quotient. Conjugation by the actual positive square root of G_{coh} proves (GP.23). For $N = q - 1$ the first positive term is zero and the second remains as stated. For $q = 0$ the original quotient is zero: both section maps are zero, and the empty determinant is one. The nonzero analytic source does not supply an extra quotient dimension. $\square$

41.6 Support, the original unit, and the domain of arithmetic observation

The original cyclic source is $\mathcal{V}P = P(D_1 + \cdots + D_k)F_h^{\otimes k}$. In its Mellin realization its full image is

$$\prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} a(\mathbf{t}) P(k/2 + i \sum_i t_i). \quad (\text{GP.24})$$

Multiplication by the displayed gamma factors is an isometry $\mathcal{H} \rightarrow L^2(d\mathbf{t})$. Equations (GP.9) and (GP.17) split the vector before this multiplication and reconstruct it by addition. On the image graph

$$\mathcal{G}_N = \{(U(a_k F), (I - P)aUF) : F \in \mathcal{P}_N(S)\}$$

addition $\text{add}_N : \mathcal{G}_N \rightarrow aU(\mathcal{P}_N)$ is injective, since the squared norm of the reconstructed vector is the strictly positive polynomial norm M_h . It is onto the displayed image by construction. Denote the multiplication in (GP.24) by $M_\Gamma : \mathcal{H} \rightarrow L^2(d\mathbf{t})$ and the original unitary Mellin map by $\mathcal{M}_{\text{unit}}$. The graph observation is exactly $J^{(k)}\mathcal{M}_{\text{unit}}^{-1}M_\Gamma \text{add}_N$, and likewise with $q^{(k)}$. These are well-defined linear maps because the reconstructed image is precisely the original admitted Mellin source. Their values on the graph of P are exactly

$$J^{(k)}\mathcal{V}P = \eta_{h,k}\pi P, \quad q^{(k)}\mathcal{V}P = \sigma_h^{\otimes k}\eta_{h,k}\pi P, \quad \eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1. \quad (\text{GP.25})$$

This specifies the domain of the observation; no arithmetic jet is assigned to an arbitrary independent measurable relative vector. The graph in this display uses theta zero. At a nonzero admitted theta, multiply both graph components by the same $e^{\theta u/2}$; its inverse multiplies by $e^{-\theta u/2}$ before the original inverse Mellin map. Both maps are isometries between the declared tilted and untilted spaces.

Here is the full cochain map for the relation. In the original tensor algebra

$$\mathbb{C}[s_1, \dots, s_k]/(h(s_1), \dots, h(s_k)),$$

the cyclic annihilator satisfies $\chi(s_1 + \dots + s_k) = 0$. Successive monic division in the fixed order $s_1, \dots, s_k$ gives unique quotients at each division step and polynomials Q_i with

$$\chi(s_1 + \dots + s_k) = \sum_{i=1}^k h(s_i)Q_i(s_1, \dots, s_k).$$

The final remainder is zero because the monomials of degree less than $\deg h$ in each variable are a basis of that tensor algebra. In $[V \xrightarrow{\Theta} \mathcal{B}]^{\otimes k}$ put $D = -x\partial_x$ and $\phi_0(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. It is even, $\phi_0(0) = 0$ and $\int_{\mathbb{R}} \phi_0 = 0$: the two even Gaussian integrals are $4\pi^2 \cdot 3/(4\pi^2)$ and $6\pi \cdot 1/(2\pi)$, both equal to three. For $\Re s > 1$, termwise Mellin integration of $\Theta\phi_0(x) = \sum_{n \neq 0} \phi_0(nx)$ yields $2\zeta(s)\mathcal{M}\phi_0(s)$. Direct integration with $y = \pi x^2$ gives

$$\mathcal{M}\phi_0(s) = \pi^{-s/2}(2\Gamma(s/2 + 2) - 3\Gamma(s/2 + 1)) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2).$$

Thus $\mathcal{M}(\Theta\phi_0) = g$ there and by continuation on the original Mellin domain. The original inverse Mellin definition of F_h then gives $h(D)F_h = \Theta\phi_0$. The precise primitive is

$$\mathcal{P}_\chi F = \sum_{i=1}^k (-1)^{i-1} Q_i(D_1, \dots, D_k) F(D_1 + \dots + D_k) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GP.26})$$

Euler differentiation preserves the even Schwartz source and its two conditions: its value at zero vanishes and its integral is the original integral by integration by parts. It also preserves rapid decay at both multiplicative ends and commutes with Θ . Every term therefore lies in the asserted cochain domain. The preceding $i - 1$ degree-one factors give the tensor differential sign $(-1)^{i-1}$, cancelling the written sign. Hence, with no sign omitted,

$$d\mathcal{P}_\chi F = \mathcal{V}(\chi F), \quad d[-\mathcal{P}_\chi K] = \mathcal{V}(R_h - R_{\text{coh}}). \quad (\text{GP.27})$$

This proves the cochain correction map on C_χ in (GP.21), including its zero value when the boundary is empty.

Every linear map on this graph and every original source or quotient has its original fixed-support lift: $(\ell, x) \mapsto (\ell, Lx)$, and external absence maps to external absence. The maps preserve addition and scalar action because the underlying maps are linear. For an actual relation the quotient value is $(\ell, 0)$, its supported zero, with the source and the two graph components still reconstructible. The norm has type $\mathcal{H} \rightarrow \mathbb{R}_{\geq 0}$ and is positively homogeneous; its square is quadratic. Trace has type $\text{End}_{\mathbb{C}}(E) \rightarrow \mathbb{C}$ and is linear; its restriction to Hermitian endomorphisms is real-linear. The Gram map and squared modulus are quadratic on the underlying real vector spaces, and trace composed with a Gram map is quadratic. The exact linear observation maps producing the source Grams and the quotient values are (GP.18), (GP.21), and (GP.24)–(GP.25).

41.7 Exact polynomial calibration with the original gamma mass

The following auxiliary calibration tests the complete complex-phase identities by rational arithmetic. It is a declared polynomial amplitude, with all fixed moments finite in the original gamma measure. It does not invoke the bounded packet estimate (GP.12). Take $\lambda = 1$, $k = 2$, $c_1 = 1/2$, $c_1^2 = 1/4$, and $A(t) = 1 + it/2$. The original sum coordinate is $S = 1 + iu$. Direct conditional polynomial calculation gives

$$\begin{aligned} a_2(u) &= \frac{6}{5} + \frac{iu}{2} - \frac{u^2}{20}, \\ b_2(u) &= \frac{54}{35} + \frac{39u^2}{280} + \frac{3u^4}{1120}, \\ d_2(u) &= \frac{18}{175} + \frac{13u^2}{1400} + \frac{u^4}{5600} > 0. \end{aligned} \tag{GP.28}$$

Here are enough exact polynomial identities to prove the calculation. For the one-factor parameter $\alpha = 2$, $b_2^{(1)}(t) = t^2 - 2$. For the sum parameter 4, the second and fourth monic polynomials are $u^2 - 4$ and $u^4 - 32u^2 + 72$. The conditional addition formula yields

$$\begin{aligned} C(t_1 t_2) &= (u^2 - 4)/5, \\ C(t_1^2) &= C(t_2^2) = 3(u^2 - 4)/10 + 2, \\ C(t_1^2 t_2^2) &= \frac{3}{70}(u^4 - 32u^2 + 72) + \frac{6}{5}(u^2 - 4) + 4. \end{aligned}$$

Indeed the respective coefficients are $2^2/(4)_2$, $(2)_2/(4)_2$, and $(2)_2^2/(4)_4$. Expanding $(1 + it_1/2)(1 + it_2/2)$ and $(1 + t_1^2/4)(1 + t_2^2/4)$ using these identities gives the first two lines of (GP.28). Subtracting the squared modulus gives its last line. This is the complete complex subtraction, including its imaginary linear coefficient.

For the auxiliary annihilator $\chi(S) = (S - 1)^2 - 2$, use the original S monomials at $N = 1$, where the boundary is empty. The sum gamma moments of orders zero, two, four, six are respectively $1/4, 1, 14, 376$, obtained by differentiating $\frac{1}{4} \cos^{-4} \theta$. Substitution in (GP.18) gives

$$\begin{aligned} G_h &= \begin{pmatrix} 9/16 & 9/16 \\ 9/16 & 81/16 \end{pmatrix}, \quad G_{\text{coh}} = \begin{pmatrix} 21/40 & 21/40 \\ 21/40 & 189/40 \end{pmatrix}, \\ G_h - G_{\text{coh}} &= \begin{pmatrix} 3/80 & 3/80 \\ 3/80 & 27/80 \end{pmatrix}, \\ \det G_h &= 81/32, \quad \det G_{\text{coh}} = 441/200, \quad \det(G_h - G_{\text{coh}}) = 9/800. \end{aligned} \tag{GP.29}$$

For example $\int b_2 r_{1,2} = 9/16$ and $\int u^2 b_2 r_{1,2} = 9/2$, while the corresponding coherent moments are $21/40$ and $21/5$. Odd moments vanish, so the original off-diagonal S entries equal the masses. These calculations prove every entry above. They exhibit the strictly positive contribution in the unchanged quotient coordinates.

The accompanying checker additionally derives moments from the Laplace function, solves the independent product-moment equations, and checks all moments through degree eight and the section/quotient identities through $N = 4$. It uses a separate two-atom conditional fibre of original masses two and three to test the complex adjoint and both projection traces. Its mathematical controls change a conjugate transpose, omit an original mass factor, or omit the relative quotient term. Those finite checks supplement the proofs above and do not estimate an actual growing-degree zeta quotient.

41.8 Provenance and consequence

The source constants, packet, conditional density, and coefficient addition identity come from the complete delivered Gamma Convolution Descent note, equations (2), (5)–(19), (24)–(27), and (32). The original unit and theta relation primitive come from the retained cyclic source construction and TVB in the cumulative reader. The classical generating function is [NIST DLMF 18.23.7](#), with $b_n^{(\lambda)}(t) = n!P_n^{(\lambda)}(t/2; \pi/2)$. The exact specialization and norms are proved in TG.7–TG.14. The bounded amplitude is used here to prove the full phase transport, fibre projection formulas, strict quartet variance, complex coefficient transform, and two-term positive quotient identity. These equations provide additional exact data for the same Toda quotient. They do not give a growing-degree estimate for its consecutive volume ratio. That ratio remains the actual determinant expression of the retained metric, now with the explicit contribution (GP.21)–(GP.23).

42 Which completion retains the arithmetic jets?

This section extends the Hilbert/jet calculation of the September 12 orthogonal-boundary note. All changes of topology are specified maps on the same arithmetic functions. The theta relation space, the operator $D = -x\partial_x$, and $g = 2\xi$ are retained.

42.1 Logarithmic coordinates and the original measure

Let $\mathcal{B}, V, \Theta$ have the definitions in Section 6. Put $\mathcal{H} = L^2(\mathbb{R}_+, dx)$. The map

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{L}^{-1}f(x) = x^{-1/2}f(\log x)$$

is unitary from $\mathcal{H}$ to $L^2(\mathbb{R}, dy)$, since $dx = e^y dy$. Direct differentiation gives

$$\mathcal{L}D\mathcal{L}^{-1} = \frac{1}{2} - \partial_y, \quad \mathcal{M}F(\rho) = \int_{\mathbb{R}} (\mathcal{L}F)(y)e^{(\rho-1/2)y} dy. \quad (\text{JT1})$$

On $\mathcal{H}$, D denotes the closed maximal distributional operator which $\mathcal{L}$ identifies with $1/2 - \partial_y$ on the Sobolev domain $H^1(\mathbb{R})$. For an integer $r \geq 0$, let $\mathcal{H}^{(r)}$ be the domain of the first r powers of D in $\mathcal{H}$, with norm $\|F\|_r^2 = \sum_{k=0}^r \|D^k F\|_{\mathcal{H}}^2$. The Mellin-line transform makes this the norm with density $w_r(t) = \sum_{k=0}^r |1/2 + it|^{2k}$ relative to $dt/(2\pi)$.

Theorem 42.1 (Finite differential graph norms and the complete jet graph). *Let Z be a nonempty finite packet of actual nontrivial zeros with full multiplicities, $E_Z = \mathbb{C}[t]/h_Z$, and $J_Z : \mathcal{B} \rightarrow E_Z$ the complete Mellin-jet map. For every finite r ,*

$$\overline{\text{span}\{D^j f_0 : j \geq 0\}}^{\mathcal{H}^{(r)}} = \mathcal{H}^{(r)}, \quad f_0 = \Theta\phi_*, \quad (\text{JT2})$$

$$\overline{\{(F, J_Z F) : F \in \mathcal{B}\}}^{\mathcal{H}^{(r)} \oplus E_Z} = \mathcal{H}^{(r)} \oplus E_Z. \quad (\text{JT3})$$

Thus J_Z is not closable in this norm. The completed theta-boundary graph is $\mathcal{H}^{(r)} \oplus 0$; its quotient in (JT3) is E_Z through $[(F, u)] \mapsto u$.

Proof. For each $0 < c < \pi/4$, the actual source gives $|g(1/2 + it)| \leq C_c e^{-c|t|}$. Indeed $e^{z/2} f_0(e^z)$ is holomorphic in $|\operatorname{Im} z| < \pi/4$: its Gaussian series and derivatives converge uniformly on compact subsets since $\operatorname{Re}(e^{2z}) > 0$. On smaller horizontal strips it decays faster than every exponential at the positive end. Poisson gives $Jf_0 = f_0$, so analytic continuation gives $e^{z/2} f_0(e^z) = e^{-z/2} f_0(e^{-z})$; this controls the negative end. Shift the Fourier contour to height c or $-c$. The vertical integrals vanish by that decay, proving the bound.

The measure $d\mu_r = w_r(t)|g(1/2 + it)|^2 dt/(2\pi)$ has a positive exponential moment. If $u \in L^2(\mu_r)$ is orthogonal to every polynomial, $d\nu = \bar{u} d\mu_r$ is a finite complex measure with a smaller positive exponential moment, by Cauchy–Schwarz. Its Fourier transform is analytic in a strip, and every derivative at zero vanishes. Analytic continuation makes it zero there. Fourier uniqueness gives $\nu = 0$, hence $u = 0$. For this uniqueness step, convolve with Gaussian approximate identities: each convolution is in L^2 , since $\|\nu * \gamma_\epsilon\|_2 \leq \|\nu\|_{\text{TV}} \|\gamma_\epsilon\|_2$; its transform is zero, so Plancherel makes it zero, and weak convergence then gives $\nu = 0$. Polynomials are therefore dense in $L^2(\mu_r)$. Multiplication by $g(1/2 + it)$ is an isometry onto $L^2(w_r dt/(2\pi))$: divide almost everywhere, since this nonzero analytic function vanishes on a discrete, measure-zero set. The Mellin transform of $D^j f_0$ is $(1/2 + it)^j g(1/2 + it)$, proving (JT2).

Every theta polynomial has zero J_Z -jet. Thus the closure in (JT3) contains $(h, 0)$ for every h . The actual packet section $R_{\text{ref}} : E_Z \rightarrow \mathcal{B}$ satisfies $J_Z R_{\text{ref}} = 1$. Approximate $R_{\text{ref}} u$ in $\mathcal{H}^{(r)}$ by theta polynomials $p_n(D)f_0$. Then $(R_{\text{ref}} u - p_n(D)f_0, u) \rightarrow (0, u)$. This proves (JT3). For $u \neq 0$ it prevents closability. The boundary and quotient statements follow from the coordinate projections. $\square$

Corollary 42.2 (Differential on the completed graph). *The continuous map*

$$(D, A) : \mathcal{H}^{(r+1)} \oplus E_Z \longrightarrow \mathcal{H}^{(r)} \oplus E_Z, \quad (F, u) \mapsto (DF, Au) \quad (\text{JT4})$$

extends the original $(F, J_Z F) \mapsto (DF, J_Z DF)$. On the quotient by the boundary closure it is exactly $A = M_t$.

Proof. The norm definitions bound $D : \mathcal{H}^{(r+1)} \rightarrow \mathcal{H}^{(r)}$; the jet identity $J_Z D = A J_Z$ follows by retaining the complete Taylor polynomial of $s\mathcal{M}F(s)$. Apply (JT3). $\square$

Its split observation records $(e_{\mathcal{H}}, u^\bullet)$ when the Hilbert component tends to its supported zero and the jet survives. The projections compare this point to $e_{\mathcal{H}}$ and to $u^\bullet$; they do not identify it with (τ, τ) .

42.2 Exponential weights and exact jet norms

For $a \geq 0$, retain the original functions and introduce

$$\mathcal{H}_a = L^2(\mathbb{R}_+, e^{2a|\log x|} dx), \quad \|F\|_a^2 = \int_{\mathbb{R}} |\mathcal{L}F(y)|^2 e^{2a|y|} dy. \quad (\text{JT5})$$

The identity inclusions $\mathcal{B} \hookrightarrow \mathcal{H}_a \hookrightarrow \mathcal{H}_0$ are continuous. Source seminorms with powers exceeding $a + 1/2$ control both ends. Smooth functions compactly supported in log coordinates prove density.

Theorem 42.3 (Exact threshold, including the boundary). *Put $\ell_{\rho,k}(F) = (\mathcal{M}F)^{(k)}(\rho)/k!$ and $b = \operatorname{Re} \rho - 1/2$, with $k \geq 0$. The functional on $\mathcal{B}$ extends continuously to $\mathcal{H}_a$ if and only if $|b| < a$. Its exact squared norm is*

$$\|\ell_{\rho,k}\|^2 = \frac{(2k)!}{(k!)^2} \left(\frac{1}{[2(a-b)]^{2k+1}} + \frac{1}{[2(a+b)]^{2k+1}} \right). \quad (\text{JT6})$$

Proof. Inner products are conjugate-linear in their first argument. The Riesz vector in log coordinates must be

$$r_{\rho,k}(y) = \frac{y^k}{k!} e^{(\bar{\rho}-1/2)y} e^{-2a|y|}.$$

Its norm squared is $(k!)^{-2} \int y^{2k} e^{2by-2a|y|} dy$. The two half-lines converge exactly when $|b| < a$; their gamma integrals give (JT6). If $|b| \geq a$, testing any putative Riesz representative against compactly supported smooth functions forces it to be this same non-square-integrable density, a contradiction. The equality case is included: one end has no exponential decay. $\square$

Theorem 42.4 (Complete confluent Cauchy Gram). *For distinct centres ρ_i with $|\operatorname{Re} \rho_i - 1/2| < a$ and orders $0 \leq k < m_i$, let $J_a : \mathcal{H}_a \rightarrow \mathbb{C}^d$ be their Taylor-jet map. In the original Taylor coordinates its Gram $C_a = J_a J_a^*$ is*

$$(C_a)_{(i,k),(j,l)} = \frac{(k+l)!}{k!l!} \left(\frac{1}{(2a - z_{ij})^{k+l+1}} + \frac{(-1)^{k+l}}{(2a + z_{ij})^{k+l+1}} \right), \quad z_{ij} = \rho_i + \bar{\rho}_j - 1. \quad (\text{JT7})$$

It is positive definite. The exact right inverse $S_a = J_a^* C_a^{-1}$ satisfies

$$J_a S_a = 1, \quad \|S_a u\|_a^2 = u^* C_a^{-1} u, \quad \|F\|_a^2 = \|(1 - S_a J_a)F\|_a^2 + (J_a F)^* C_a^{-1} (J_a F). \quad (\text{JT8})$$

Proof. The entry is $\langle r_{\rho_i,k}, r_{\rho_j,l} \rangle_a$; integrating $y^{k+l} e^{z_{ij}y-2a|y|} / (k!l!)$ on both half-lines proves (JT7) with its stated sign. The vectors are independent. After removing their common nonzero factor, a dependence is an exponential polynomial $\sum_i P_i(y) e^{\bar{\rho}_i y} = 0$. For each i , apply $\prod_{j \neq i} (\partial_y - \bar{\rho}_j)^{m_j}$. On the i -th polynomial factor every nonzero constant plus derivative is invertible on the finite-degree polynomial space; hence $P_i = 0$. Thus $C_a > 0$. Now $S_a J_a$ is the orthogonal projection onto $\operatorname{range} J_a^* = (\ker J_a)^\perp$. Multiplying the specified matrices proves all of (JT8). $\square$

For an actual packet the unchanged theta subspace lies in $\ker J_a$. Consequently there is an onto continuous quotient map

$$\mathcal{H}_a / \overline{\Theta V}^{\mathcal{H}_a} \longrightarrow \mathbb{C}^d, \quad [F] \mapsto J_a F, \quad \ker = (\ker J_a) / \overline{\Theta V}^{\mathcal{H}_a}. \quad (\text{JT9})$$

Its surjectivity follows from (JT8). Define the retained coordinate map

$$T : E_Z \longrightarrow \mathbb{C}^d, \quad T[p] = (p^{(k)}(\rho_i)/k!)_{i,k}.$$

Then $J_a|_{\mathcal{B}} = T J_Z$, and the exact representative difference is $R_{\text{ref}} T^{-1} - S_a : \mathbb{C}^d \rightarrow \ker J_a$. Here R_{ref} is the original source map; S_a is a Hilbert map, with no claim it lies in $\mathcal{B}$. In E_Z coordinates the observation Gram is $C_E = T^{-1} C_a (T^{-1})^*$ and the right inverse is $S_a T$.

42.3 The energy term carried by the weighted topology

Theorem 42.5 (Weighted boundary identity on the actual representatives). *On $\mathcal{B}$, with $(\Sigma H)(x) = \operatorname{sgn}(\log x) H(x)$,*

$$\langle DF, H \rangle_a + \langle F, DH \rangle_a - \langle F, H \rangle_a = 2a \langle F, \Sigma H \rangle_a. \quad (\text{JT10})$$

For an actual packet representative $R : E_Z \rightarrow \mathcal{B}$ with $DR - RA = B = \Theta k$, put $G_a = R^ R$, $T_a = R^* \Sigma R$. Adjoints use exactly $\mathcal{H}_a$. Then*

$$A^* G_a + G_a A - G_a = 2a T_a - (B^* R + R^* B), \quad -G_a \leq T_a \leq G_a. \quad (\text{JT11})$$

For $Au = \rho u \neq 0$,

$$2 \operatorname{Re} \rho - 1 = 2a \frac{u^* T_a u}{u^* G_a u} - 2 \operatorname{Re} \frac{\langle Ru, Bu \rangle_a}{\|Ru\|_a^2}. \quad (\text{JT12})$$

Proof. In log coordinates integrate $-(\bar{f}h)' e^{2a|y|}$ on the two half-lines. Source decay kills the terms at infinity; those at zero cancel because the weight is continuous there. The weak derivative of the weight is $2a \operatorname{sgn}(y) e^{2a|y|}$. This proves (JT10) with its positive sign. Substitute $DR = RA + B$ and move both boundary terms to obtain (JT11). Multiplication by sign is a self-adjoint contraction, proving the form bounds. Evaluation on u proves (JT12). $\square$

At $a = 0$, (JT11) is the source boundary-control formula. At $a > 0$, the topology making a jet continuous has its explicitly retained energy term $2aT_a$. Off-line jets cannot remain continuous as $a \downarrow 0$. For line-centred jets, all $a > 0$ work but (JT6) diverges as $a \downarrow 0$. These are exact comparisons on the same Mellin map and differential, without a weight theorem assumed.

42.4 The tessarine coordinate in the actual weight form

Retain $c_{-+} = 4(\beta - 1/2)$, $c_{--} = 4i\gamma$ for $\rho = \beta + i\gamma$. For $p > 1$ and $|\psi| = 1$,

$$z = \psi \exp\left(\frac{\log p}{4}(c_{-+} + c_{--})\right), \quad \frac{u^*(A^*G + GA - G)u}{u^*Gu} = \frac{c_{-+}}{2}, \quad |b| < a \iff |c_{-+}| < 4a. \quad (\text{JT13})$$

The first equality is exactly $z = \psi p^{\rho-1/2}$; the second is evaluation on $Au = \rho u$. Thus the original sector, local holonomy radius, arithmetic weight form and jet-continuity strip have specified comparison maps. The scalar evaluations do not replace the original matrix or its support-labelled source.

43 The current arithmetic route and its next calculation

The source chain now supplies an actual $\mathbb{C}[t]$ -module cohomology Q , a finite invariant packet inside that cohomology, a specified nonzero extension class, an original theta boundary for its representative defect, and an exact arithmetic formula for the relative weight form. The local continuation makes that last formula an explicit confluent interpolation problem. These constructions are logically prior to any purity conclusion. The subsequent continuation now supplies certified entries of the original arithmetic moment matrix, the exact finite-compression and infinite-tail resolvent comparison, and a convergence rate for the initial resolvent including its complete off-line jets.

43.1 A single chain of exact comparison maps

For each actual finite packet, retaining every full order, the maps proved in this reader are

$$E = \mathbb{C}[t]/h \xrightarrow{R_{\text{ref}}} \mathcal{B} \xrightarrow{q} Q, \quad J_Z R_{\text{ref}} = 1, \quad D R_{\text{ref}} - R_{\text{ref}} A = \Theta \phi_* \ell.$$

Every admitted polynomial correction keeps these same quotient and jet maps. At its distinguished orthogonal level let $B_{N,u}(t) = (p_{N,u}(t) - R_Z(u)(t))/h(t)$, a polynomial of degree at most m . The complete calculation is

$$\begin{aligned} d\nu_Z(t) &= \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}, \quad N = d + m, \quad C_N P = j_h P, \quad K_N = C_N (H_N^\nu)^{-1} C_N^*, \\ p_{N,u} &= (H_N^\nu)^{-1} C_N^* K_N^{-1} U_\varepsilon u, \quad R_m u = R_{\text{ref}} u + \Theta(B_{N,u}(D) \phi_*), \\ G_m &= U_\varepsilon^* K_N^{-1} U_\varepsilon, \quad W_m = U_\varepsilon^* K_N^{-1} (A K_N + K_N A^* - K_N) K_N^{-1} U_\varepsilon. \end{aligned} \quad (\text{R1})$$

The middle line evaluates the specified polynomial at the original D ; every source vector lies in V .

Write $v_n = j_h q_n$ and $\kappa_n = \|q_n\|_{\nu_Z}^2$. The original arithmetic measure now determines the exact quantity

$$\epsilon_m = \frac{|\operatorname{Re}(v_N^* K_N^{-1} v_{N+1})| + \sqrt{(v_{N+1}^* K_N^{-1} v_{N+1})(v_N^* K_N^{-1} v_N) - (\operatorname{Im}(v_N^* K_N^{-1} v_{N+1}))^2}}{\kappa_N}. \quad (\text{R2})$$

The proofs in Section 9 establish (R1)–(R2) before passing to cohomology. For a reflection-stable packet the first absolute-value term is zero by the exact trace identity. The same sequence obeys

$$\epsilon_m \geq \sum_{\rho \in Z} m_\rho \left| \operatorname{Re} \rho - \frac{1}{2} \right|. \quad (\text{R3})$$

Thus the rank-two boundary detects the combined displacement of the entire packet, not only its largest individual displacement. The estimate concerns the distinguished R_m sequence. The larger family of all polynomial source corrections has the separately proved metric image and sharp infimum in Section 8; those results do not identify its arbitrary member with R_m .

The full two-row source observation in Section 9 is injective after the original A -iterates are retained. It therefore offers a finite arithmetic calculation preserving every nilpotent jet: compute the source moments, the adjacent monic polynomials and their complete residue vectors, rather than discard a component whose scalar trace vanishes.

43.2 Topology, duality and the two sides of a weight estimate

The actual source moments now have the absolutely convergent incomplete-gamma expression and explicit double-tail bound of Section 7. Its certified six-by-six Gram matrix concerns the original $|2\xi|^2 dt/(2\pi)$ measure. The exact identity (AM.19) supplies the original monic recurrence norms through degree five, with interval enclosures.

For the $|g/h|^2$ kernel, Section 10 proves a universal polynomial quadrature formula, including centres that coincide with a quadrature node. On the precise nonintersection domain its alternate expression is

$$K_{n-1} = \kappa_{n-1}^{-1} q_n(A) S_n q_n(A)^*, \quad AS_n + S_n A^* - S_n = cr_n^* + r_n c^*, \quad r_n = F_n(A)c.$$

The exact Schur comparison (KR.15)–(KR.20) then joins the initial and terminal observations to the actual infinite-dimensional tail. Section 11 proves the explicit initial-resolvent rate (GC.8) and its complete finite-algebra bound (GC.23). The projection to that off-line domain retains the critical summand as its stated kernel. All factors in the Schur identity remain in converting the initial estimate to the terminal form that determines ϵ_m .

In particular, Section 13 identifies exactly the two contributions in a reflection-stable packet:

$$\epsilon_m^2 = 2 \sum_{\rho \in Z} m_\rho (\operatorname{Re} \rho - 1/2)^2 + \mathfrak{d}_{G_m}(A).$$

Its full-jet inequalities also quantify the required metric ratios along each retained nilpotent chain. These identities do not infer a relative estimate from absolute matrix shrinkage.

The theta-polynomial span is dense in every finite differential graph norm. The completed jet graph, however, is $\mathcal{H}^{(r)} \oplus E$, with quotient E , not a quotient of $\mathcal{H}^{(r)}$ in which the arithmetic component has vanished. This supplies the exact comparison between the shrinking Hilbert representative and its unchanged cohomological class.

The exponential observation norm supplies a second exact comparison. A Mellin jet at ρ is bounded precisely in the strip

$$|\operatorname{Re} \rho - \tfrac{1}{2}| < a,$$

with the factorial norm in Theorem 42.3. The original differential then has the energy term $2a \operatorname{sgn}(\log x)$ in Theorem 42.5. Consequently stronger topology and weight control are related by that explicit term; choosing a positive observation norm does not remove it.

For any packet Z , reflection gives the conjugate-linear ring isomorphism

$$\dagger_Z : E_Z \longrightarrow E_{Z^\dagger}, \quad f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad Z^\dagger = \{1 - \bar{\rho} : \rho \in Z\},$$

with all multiplicities retained. In fact $h_Z^\dagger = (-1)^d h_{Z^\dagger}$; this proves well-definedness, and applying the same operation twice is the identity. Restrict the following internal residue pairing to reflection-stable Z , so that this map is an involution on E_Z . The arithmetic residue form is the source form

$$\mathcal{R}_Z(f, u) = \sum_{\rho \in Z} \operatorname{Res}_\rho \frac{f^\dagger u}{g} ds.$$

Its Jacobian contraction is

$$\mathcal{R}_Z(f, g'u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho). \quad (\text{R4})$$

Indeed locally $g = z^{m_\rho} w(z)$ gives $g'/g = m_\rho/z + w'/w$, proving (R4) with multiplicities. The trace form pairs reflected centres and can retain nontrivial null directions from nilpotents. In the retained basis $1, t, \dots, t^{d-1}$, define

$$S_{ij} = \sum_{\rho \in Z} \text{Res}_\rho \frac{(s^i)^\dagger s^j}{g(s)} ds, \quad 0 \leq i, j < d, \quad J_g = M_{j_h g'}.$$

Changing representatives by h makes the local quotient by g holomorphic, proving that this matrix represents the intrinsic form. By (R4) the trace pairing has matrix SJ_g . Its exact comparison to the positive source metric is therefore $G_m^{-1}SJ_g$, since $f^*G_m(G_m^{-1}SJ_g)u = f^*SJ_gu$. Its positivity is an arithmetic question about this operator, not a consequence of the positive Gram construction.

The finite-field two-sided weight theorem remains a source for methods, with its compact-support comparison, duality and tensor hypotheses. This reader has not constructed a realization meeting those geometric hypotheses for the entire arithmetic quotient. It has instead calculated the current characteristic-zero objects and their original maps, including the local-cohomology spectral extraction and meromorphic complement documented in the complete finite-comparison appendix.

43.3 Provenance and mathematical status

The coherent interpolation now has the full coefficient identity $J_{h,N} = U_{v_h} C_N$, with $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$. Equations (CT.3)–(CT.7) prove equality with (R1), including the original representative, and compatibility with the complete CRT injections between finite packets. The actual reflection preserves the unique minimum and proves the same-metric signed pairing of (CT.10) and (CT.16a).

On the original tensor complex, the complete calculation is

$$\begin{aligned} E^{(k)} &= E_h^{\otimes k}, \quad A^{(k)} = \sum_{j=1}^k A_{h,j}, \quad \mathcal{I}_{h,k,M} = \mathcal{P}_M^{(k)} \cap (h(s_1), \dots, h(s_k)), \\ R_M &= (I - P_{\mathcal{B}_M}) s_h^{\otimes k}, \quad G_M = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}, \\ W_M &= -(Y_M^* C_M + C_M^* Y_M), \quad G_M - G_{M+1} = Y_M^* Y_M, \quad Y_M, C_M : E^{(k)} \longrightarrow \mathcal{E}_M, \\ \dim \mathcal{E}_M &= \binom{M+k}{k-1}. \end{aligned} \quad (\text{R5})$$

Section 14 proves the original ordered division, both cochain signs, every layer map, and the extra orthogonal projection required when tensor packets are enlarged. Its formula (CT.18) calculates the resulting change in both G and W .

For a reflection-stable packet, let p_M be either signed inertia of W_M , and let $\mathfrak{d}_M^{(k)}$ be its complete tensor departure from (CT.20). The same source layers give

$$\mathcal{D}_k \leq p_M \epsilon_M, \quad p_M \epsilon_M^2 \geq 2kd^{k-1} \sum_{\rho \in Z} m_\rho (\text{Re } \rho - 1/2)^2 + \mathfrak{d}_M^{(k)}, \quad (\text{R6})$$

$$p_M \leq \min\{\text{rank } Y_M, \text{rank } C_M, \lfloor d^k/2 \rfloor\}, \quad \mathcal{D}_k^2 \leq \text{Tr}(G_M^{-1}(G_M - G_{M+1})) \text{Tr}(G_M^{-1} C_M^* C_M).$$

The zero-inertia case is proved directly in (CT.25). The full ordered sum $\mathcal{D}_k$ and all multiplicities are retained in (CT.20). The joint density and exact graph closure in Section 15 show $G_M \downarrow 0$ for this actual total-degree sequence and identify every retained jet in the vertical fibre of (JD.11). The same section proves the coercive inverse-interpolation limit with its arithmetic unit, the supported-zero limit morphism, and the accumulated determinant cost (JD.14)–(JD.15). These formulas connect the relative layer energy to the exact rate of determinant loss through the same original matrices.

The September 11 support, Rees, comparison and audit packages are retained in the companion source directory. The current analytic baseline is the September 12 sequence: finite operator comparison (PR 8), chain descent (PR 9), global retraction (PR 10), homotopy control (PR 11), formal tau recovery (PR 12) orthogonal boundary control (PR 13), and coherent interpolation with joint tensor boundaries (PR 14). Part II includes the complete selected source proof notes with their exact revisions. Historical statements that a map remains unconstructed belong to their stated source dates; a later explicit construction in this reader or the source chain supersedes that particular statement.

The new local proofs are the full support-index equivalence, the corrected carrier and sector comparisons, the arithmetic confluent-kernel minimum and complete two-row source observation, the classification of source-metric images and aggregate control, and the graph/weighted-jet comparison. The next tranche adds the certified original theta-moment series and its full truncation bound, the confluent quadrature and finite/terminal/tail resolvent identities, the explicit initial convergence rate, and the metric-departure and nilpotent-chain identities. The recovered June 24 historical packet calculation is reproduced with a full-jet comparison in Section 2. Its exact A -invariant saturation is every off-axis local factor; the quotient, involutions, source-metric transport and split-zero endomorphism descent are proved there. The original transcript excerpt and its response locators are retained in the source package. This is an explicitly recorded historical recovery, with its new arithmetic saturation proof distinguished from the original permutation calculation. The subsequent historical reconstruction in Section 3 proves the inverse map from a complete local Hurwitz zero-cluster germ to every moment of its original compact probability measure, followed by the explicit Bernstein recovery of that measure. The monic cluster polynomial retains arbitrary zero multiplicity. The same section proves the complete Laurent period maps and their collision kernels. For two distinct primes, the joint Laurent map is surjective onto the full finite spectral algebra; Theorem 3.1 gives every primary kernel and the inverse original spectral coordinate, including its truncated logarithm on each nilpotent factor. The factorwise Split-Zero lifts and their exact comparison with the synchronized global lift are proved in (S4)–(S5). Their elementary algebraic and analytic steps are written here. No claim of global mathematical priority follows from finding or proving an identity in this research programme.

The finite validation records distinguish exact symbolic identities, rational calibration models, numerical quadratures and intentionally rejected negative controls. They do not count test methods as theorems or imply analytic certification from an L^2 numerical experiment. The supplied formalization records are also kept separate: successful exact-head CI metadata does not turn the unformalized analytic theta arguments into Lean proofs. No Lean run was performed for this continuation.

The current research calculation includes (R2) on the actual $|g/h|^2$ -moments, with its determinant update and two-row recurrence, together with the topology constants of Theorem 42.3. No uniform arithmetic estimate forcing (R2) to vanish has been established in this reader. RH, GRH and a global purity theorem are not conclusions of it. The exact maps and source constraints above specify the live route on which tandem work can now continue. The next estimate must propagate (GC.8) through (KR.16), retaining q_n^2 , κ_n , a_n , the terminal ratio F_n and the actual tail denominator. The arithmetic moment formulas give an executable source for those finite data. The exact departure identity and the full-jet metric bounds specify what a resulting relative estimate would control in the original packet; no component is removed by changing observation.

The next step of that propagation is now calculated in Section 12. For $u_n = q_n(A)c$, $w_n = q_{n-1}(A)c$, and $M_n = \kappa_{n-1}^{-1}K_{n-1}^{-1}$, let $\beta_n = u_n^* M_n u_n$, and let P_n be the exact M_n -orthogonal projection with kernel $\mathcal{C}u_n$ when $\beta_n > 0$. Put $p_n = P_n w_n$, $z_n = P_n A u_n$. The universal step is

$$\begin{aligned} M_{n+1} &= a_n^{-1} \left(M_n - \frac{M_n u_n u_n^* M_n}{a_n + \beta_n} \right), & \frac{\det G_{m+1}}{\det G_m} &= \frac{a_n}{a_n + \beta_n}, \\ \Delta_{n+1} &= \frac{\beta_n \|z_n + a_n p_n\|_n^2}{a_n(a_n + \beta_n)}, & \epsilon^{(n)} &= |gz| + \sqrt{g_Z^2 + \Delta_n}, \quad gz = \operatorname{Re} \operatorname{tr} A - d/2. \end{aligned} \tag{R7}$$

Here $m = n - d - 1$. The zero-vector exceptional step is given completely in (KT.14), so critical centres

and every polynomial coincidence remain in (R7)'s source calculation. The alternate terminal Gram follows by the exact isometry (KT.15) and recurrence (KT.16). The full scalar-to-matrix extraction uses the doubled off-line algebra and the explicit sequence (KT.23); at a reflected pair its matrix coefficient is exactly $(-1)^l F_n^{[k+l+1]}(\rho)$. Finally (KT.24)–(KT.25) recover these coefficients from the original initial error and tail, with both reciprocal series and all polynomial/norm factors displayed. The remaining energy estimate is therefore attached to the three explicit terms of (KT.11), including their actual cross pairing, on the same arithmetic moment sequence.

For a packet closed under both conjugation and reflection, the original real-coefficient calculation in (KT.28) closes this step in terms of consecutive determinant ratios. With the fixed section metric G_{-1} retained, put $r_m = \det G_m / \det G_{m-1}$ for $m \geq 0$. Then the actual arithmetic allowance obeys exactly

$$\epsilon_m^2 = a_{d+m+1} \frac{(1-r_m)(1-r_{m+1})}{r_{m+1}}. \quad (\text{R8})$$

The coefficient reality and full reflection trace prove $\gamma_n = 0$ in the original coordinates; (KT.7)–(KT.8) then prove (R8), including zero-step cases. This places the relative quartet control directly on the same monic norm ratios and consecutive Gram determinants as the certified moment problem.

This also gives a direct arithmetic bound for the original tesserine violation channel. Retain the original quartet element

$$Q(\rho) = \rho 1 + (1-\rho)\sigma + \bar{\rho}\kappa + (1-\bar{\rho})\sigma\kappa, \quad \rho = \beta + i\gamma,$$

and its character $c_{-+}(\rho) = 2(2\beta - 1)$, as calculated in the carrier section. If $\gamma \neq 0$, $\beta \neq 1/2$, and Z is this entire four-point zero orbit with common full order μ , then $d = 4\mu$ and for every $m \geq 0$,

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = a_{4\mu+m+1} \frac{(1-r_m)(1-r_{m+1})}{r_{m+1}}. \quad (\text{R9})$$

To prove every coefficient, conjugation and the functional equation preserve the full order, and the four real displacements are $\beta - 1/2, \beta - 1/2, -(\beta - 1/2), -(\beta - 1/2)$. The proved aggregate bound (R3) is therefore $\epsilon_m \geq 4\mu|\beta - 1/2| = \mu|2(2\beta - 1)|$. Squaring and using (R8) gives (R9). The sector morphism on the original supported quartet is precisely

$$G(\chi_{-+}) : G(\mathbb{C}[K_4]) \longrightarrow G(\mathbb{C}), \quad Q(\rho)^\bullet \longmapsto c_{-+}(\rho)^\bullet, \quad \tau \longmapsto \tau.$$

Here $\chi_{-+} = \text{ev}_{-1,+1}$ is the original group character extended linearly; it preserves multiplication because the character does, and hence its displayed G -lift is a semiring morphism. At $\beta = 1/2$ this morphism takes the supported value e ; the same original quartet has phase channel $c_{--} = 4i\gamma$. The orbit then has two distinct points when $\gamma \neq 0$, so its degree is 2μ and (R8) applies with that exact degree; the left-hand side of the sector bound is zero. Thus the arithmetic volume comparison and the supported critical value use the same typed character map, with the phase coordinate retained.

43.4 The original theta norm and the positive arithmetic frontier

The two newly delivered source notes, reproduced completely in the appendices, admit the following exact common formula with (R8). Keep $u_j = (I - P_{L_{j-1}})D^j f_0$, $\mathfrak{h}_j = \|u_j\|^2$, and the original measure $|g|^2 dt / (2\pi)$. The polynomial division and original minimum in (KL.23)–(KL.27) prove, at every full packet,

$$\kappa_{d+j} = \mathfrak{h}_j r_j \quad (j \geq 0), \quad \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}) - \mathfrak{h}_{m+1}^2 |c_m|^2 \quad (\text{R10})$$

for reflection-stable packets and $m \geq 0$, where $c_m = d_m G_m^{-1} d_{m+1}^*$ uses the original theta projection rows. The first identity includes $j = 0$ and the fixed section G_{-1} : the actual new-layer vector at that

endpoint is exactly f_0 . The row dictionary (KL.30) retains its separate domain $m \geq 0$; its cross term is $\mathfrak{h}_{m+1}c_m = \overline{\gamma}_n$, with $n = d + m + 1$. Both signs and the conjugation are obtained by multiplying those rows against the original inverse metric, including both local-unit matrices. Equation (KL.25) substitutes the first identity in (R10) into the complete terminal energy, proving its second identity. Under conjugation closure the original coefficients are real, so γ_n is real; the reflection trace makes its real part zero. Thus $c_m = 0$ in this case, by precisely the same calculation, with no discarded cross term.

For the full quartet in (R9), this gives the proved chain

$$\mu^2 |c_{-+}(\rho)|^2 \leq \epsilon_m^2 = \frac{\mathfrak{h}_{m+1}}{\mathfrak{h}_m} (r_m^{-1} - 1)(1 - r_{m+1}). \quad (\text{R11})$$

Indeed the order-preserving symmetries establish the hypotheses of (R10) and its vanishing cross term; the lower bound was proved with every multiplicity in (R9). The norms $\mathfrak{h}_j$ are independent of the packet. For $h \mid H$, the original polynomial map $P \mapsto (H/h)P$ is an isometry because $(g/H)(H/h)P = (g/h)P$. Its jets are the old complete jets and zero jets at each added centre. Its fixed representative and all its corrected representatives agree under this insertion, as proved in (KL.28). Thus the common theta sequence in (R11) connects the actual nested-packet maps; the packet dependence remains in the two displayed determinant ratios.

For the tensor frontier, retain the original full column maps $F, E_+ : \mathbb{C}^r \rightarrow E_h^{\otimes k}$ and its original norm diagonal Ω in (AP.4). With G_M the actual total-degree minimum, define

$$\begin{aligned} \mathbf{A} &= \Omega^{-1/2} F^* G_M F \Omega^{-1/2}, & \mathbf{B} &= \Omega^{-1/2} E_+^* G_M E_+ \Omega^{-1/2}, \\ \Gamma_{k,M} &= \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}), & \pi_M &= \frac{\det G_{M+1}}{\det G_M}. \end{aligned} \quad (\text{R12})$$

All complete conjugation-and-reflection packets satisfy

$$\epsilon_{k,M}^2 = \lambda_{\max}(\mathbf{A}^{1/2} \mathbf{B} \mathbf{A}^{1/2}) \leq \Gamma_{k,M} \lambda_{\max}(\mathbf{A}) \leq \Gamma_{k,M} (\pi_M^{-1} - 1). \quad (\text{R13})$$

Here the square roots are the explicit metric isometries, with their inverses in (AP.12); the original matrices in (R12) remain part of the formula. The full proof in (AP.7)–(AP.21) starts from the actual linear reflection of the original Mellin source. It places the two frontiers in opposite orthogonal eigenspaces and gives $U^*V = 0$. The relative weight is consequently the two off-diagonal blocks UV^* and its adjoint. Its complete singular-vector decomposition identifies its norm squared with the positive operator in (R13), including all zero vectors. The explicit incidence row sum gives $VV^* \preceq \Gamma I$; the inverse-metric update gives $\pi_M^{-1} = \det(I + UU^*)$. Multiplication of its nonnegative eigenvalue factors proves the two inequalities. For $k = 1, M = d + m$, (R13), (R11), and (AP.23) calculate the same original tesserine bound through its norm, volume, and positive-operator presentations.

43.5 The symmetric summand, the actual relation layer, and support

Every factorial in the symmetric comparison is retained. The literal orbit-sum injection I , its diagonal $D_o = I^*I$, and $L = D_o^{-1}I^*$ in (AP.24) give $G_M^s = I^*G_M I$ and the exact identity $(G_M^s)^{-1} = L G_M^{-1} L^*$. The proof uses permutation invariance of the actual minimum, so that its metric commutes with the projection IL . Hence the original redundant frontier has symmetric maps $F_s = LF, E_s = LE_+$ with the unchanged Ω , and

$$\begin{aligned} \mathbf{A}_s &= \Omega^{-1/2} F_s^* G_M^s F_s \Omega^{-1/2}, & \mathbf{B}_s &= \Omega^{-1/2} E_s^* G_M^s E_s \Omega^{-1/2}, \\ (\epsilon_M^s)^2 &= \lambda_{\max}(\mathbf{A}_s^{1/2} \mathbf{B}_s \mathbf{A}_s^{1/2}) \leq \Gamma_{k,M} ((\pi_M^s)^{-1} - 1), & \pi_M^s &= \frac{\det G_{M+1}^s}{\det G_M^s}. \end{aligned} \quad (\text{R14})$$

Equations (AP.25)–(AP.36) prove all these maps, the full positive and zero spectra, the opposite reflection parities, and the orbit-factorial grouping. In particular the isometry $Q_M = G_M^{1/2} I(G_M^s)^{-1/2}$ satisfies $Q_M^* Q_M = I$ and $Q_M Q_M^* = IL$; it carries the symmetric relative operator to the invariant restriction of the original one, retaining the complementary operator on $\ker L$. The continuous cochain projector in (SS.1)–(SS.9) has exactly this cohomology image. Its finite packet version uses the original section and full jet, and all Koszul signs in that construction remain in its primitive.

The actual next relation layer underlying these matrices is proved in (KL.7)–(KL.20). Its complete graph frame, Gram, fixed-order division primitive, derivative image, and derivative cokernel are given there. The comparison with a homogeneous shell is the exact polynomial identity (KPC.7), with every lower coefficient in (KPC.6). Its full graph is the fibre product (KPC.12), and its two projection kernels are specified in (KPC.11). For two tensor coordinates the original ratio algebra embeds by (KPC.16), with exponent $\max_{\rho/\sigma=r}(m_\rho + m_\sigma - 1)$ at each coincident ratio. Its adjacent-shell generator action is (KPC.22a); the missing second-coordinate multiplier remains in that formula. The finite example (KPC.23)–(KPC.28a) calculates both original shell ranks, all nine node values, and the actual minimum, establishing the exact scope of that comparison.

For every linear map in these graph, kernel, quotient and cochain comparisons, the fixed-support lift at the end of the projective-comparison section maps (λ, v) to (λ, Tv) and maps external absence to absence. Its scalar and additive identities are proved there, including supported zero. For measured support, the historical branch chapter proves the separate full connection from ordinary branches to frame points through $\text{Idl}(L) \rightarrow L$, with its exact kernel congruence and descent criterion. Its L^∞ correspondence identifies every Boolean branch with a multiplicative state and its normal branches with actual measure atoms. The factor $\mu(a)^{-1} \int_a f d\mu$ is retained. These explicit maps specify the support observations available to the analytic construction and preserve the nonatomic support component alongside its atomic quotient.

43.6 The original derivative cost and both positive-half-line kernels

The same original derivative layer has the stronger arithmetic identity proved in (QD.6)–(QD.15). In its exact metric coordinates, $-\hat{Y}^* \hat{C} = UV^*$ occupies one off-diagonal reflection block. Thus its norm is the full relative allowance, and the original trace cost χ_M , operator cost ζ_M and largest update eigenvalue λ_M satisfy

$$\epsilon_{k,M}^2 \leq \frac{\lambda_M}{1 + \lambda_M} \zeta_M \leq \frac{\lambda_M}{1 + \lambda_M} \chi_M, \quad \epsilon_{1,M}^2 = (1 - \pi_M) \chi_M. \quad (\text{R15})$$

The first two inequalities follow from the norm of the product of the two actual layer maps and the eigenvalues of $UU^*(I + UU^*)^{-1}$. For one factor, the layer is one-dimensional, so that product is the outer product of its two original rows and its norm is exactly their norm product. Its rank-one determinant update is $\pi_M = (1 + \lambda_M)^{-1}$, proving the last equality. This argument includes either row zero. The full lower bounds $4k^2\delta_*^2$ and all finite accumulated volume constraints follow in (QD.10)–(QD.13), with their positive denominators established before division.

For one factor, put $z = s - 1/2$, $x_h = -z^2$, and retain the monic polynomial H defined by $h(s) = (-1)^D H(x_h)$, $d = 2D$. The exact algebra is $A_H \oplus zA_H = \mathbb{C}[x_h, z]/(H, z^2 + x_h)$, with its two local length- m_ρ branches explicitly inverted in (SP.8)–(SP.9). The original source is the Hilbert direct sum for $d\lambda_h = |V(x_h)|^2 dx_h / (2\pi\sqrt{x_h})$ and $x_h d\lambda_h$. Here $v_h(s) = V(x_h)$ is the entire original quotient and its full unit modulo H enters both kernels. With those kernels and their inverse metrics, (SP.23)–(SP.35) give

$$\begin{aligned} \epsilon_{N-d} &= \left\| G_e^{-1/2} (G_o - G_e X) G_o^{-1/2} \right\|, \\ K_e - X K_o &= \begin{cases} f_n e_n^* / p_n, & N = 2n, \\ -f_{n+1} e_n^* / u_n, & N = 2n + 1. \end{cases} \end{aligned} \quad (\text{R16})$$

The first line follows by substituting the exact multiplication matrix $\frac{1}{2}I + \begin{pmatrix} 0 & -X \\ I & 0 \end{pmatrix}$ in the original weight form. The second is its full inverse-metric recurrence: all interior adjacent terms cancel with their original norm ratios, and the last pair has the two displayed parity signs. The finite kernels have indices $(n, n-1)$ and (n, n) , respectively. Hence the norm is given by the full quadratic contractions in (SP.33), with at most one nonzero singular value. Every nilpotent jet remains inside f_n, e_n, X and both metrics. The exact derivative cost (SP.34) identifies (R16) with (R15), including the two zero-frontier cases.

43.7 The gamma determinant and the retained spectral-sum fibre

The original completed function has the exact factorization $g(1/2 + it) = \Gamma(1/4 + it/2)a(t)$, with $a(t) = -\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)$. The beta integral and generating-function calculation (TG.7)–(TG.14) prove the reference monic norms $\gamma_n = \sqrt{2} n! \Gamma(n + 1/2)$. Keeping the actual multiplier in the full positive arithmetic Gram matrix gives its determinant D_n , with $D_{-1} = 1$. Monic triangular Gram congruence proves $\mathfrak{h}_n = \gamma_n D_n / D_{n-1}$, so the actual quartet formula is exactly

$$\epsilon_m^2 = (m+1)(m + \frac{1}{2}) \frac{D_{m+1} D_{m-1}}{D_m^2} (r_m^{-1} - 1)(1 - r_{m+1}), \quad m \geq 0. \quad (\text{R17})$$

This follows by substituting the consecutive norm ratio into (R10), whose cross term vanishes by the proved quartet symmetry. The full packet determinants are independently expressed in (TG.17)–(TG.18), and the original confluent kernel is the complete Gram congruence (TG.20)–(TG.22). The explicit triangular maps to the parameter $3/4$ family and their inverses retain every phase and degree. The theorem-hypothesis analysis (TG.25)–(TG.28) proves the obstruction from the infinitely many retained critical zeros on positive-measure intervals; it also supplies these exact Gram maps for continuing beyond that obstruction.

The web spectral-sum construction retains an additional relative coordinate $\Delta = (s_1 - s_2)^2$. Write $S = s_1 + s_2$, $Z = S - 1$ and $x_\Sigma = -Z^2$. Its complete reflection components, proved in Section 24, are

$$B_e = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, x_\Sigma \Delta A_1)}, \quad B_o = \frac{\mathbb{C}[x_\Sigma, \Delta]}{(A_0, \Delta A_1)}. \quad (\text{R18})$$

Here the original polynomial identities are $H_0(S, \Delta) = A_0(x_\Sigma, \Delta)$ and $H_1(S, \Delta) = Z A_1(x_\Sigma, \Delta)$; all coefficients come from $h((S+r)/2) = H_0 + r H_1$, $r^2 = \Delta$. The map $\alpha : B_o \rightarrow B_e$ multiplies by x_Σ , and $\beta : B_e \rightarrow B_o$ is the quotient. They satisfy $\alpha\beta = x_\Sigma$ and $\beta\alpha = x_\Sigma$. The original generator is $S = I + \begin{pmatrix} 0 & -\alpha \\ \beta & 0 \end{pmatrix}$. Their dimensions are $D(D+1)$ and D^2 ; both values follow by taking the symmetric square of the original two D -dimensional reflection eigenspaces. The exact maps, rather than an identification of these unequal spaces, enter the weight.

The full kernel and cokernel of $S - 1$ both recover A_H . Indeed $A_0(0, \Delta) = (-1)^D H(-\Delta/4) \neq 0$ makes α injective, while β is onto. Its cokernel map evaluates $x_\Sigma = 0$ and sends Δ to $-4x_h$. The kernel injection sends $q(x_h)$ to $\Delta A_1 q(-\Delta/4)$ in B_e , with the full polynomial ΔA_1 retained before taking the quotient. The ideal calculation and the dimension count prove both exact sequences in the cited section, including all orders. On both defect modules the actual tensor unit acts as $V(x_h)^2$, because at $S = 1$ the two original arguments are s and $1 - s$. Thus the complete paired packet, its nilpotents and its arithmetic unit remain in the relative fibre.

The exact matrix-weight pushforward retains all powers of Δ , the Jacobian $dx_\Sigma / \sqrt{x_\Sigma}$, and both vector Hilbert measures. Its actual least-norm quotient gives the two metrics in (R18); the upper block of the original weight is $\beta^* G_o - G_e \alpha$, with its full rectangular singular spectrum. All maps preserve external absence and carry a vanishing amplitude to its supported zero through the displayed fixed-support lifts. These finite identities and complete source maps are established without an asymptotic estimate for (R17).

43.8 The sum derivative, its strict normal energy and the full conormal algebra

For the actual sum amplitude retain $a_h^{\text{SC}}(t) = v_h(1/2 + it)/\sqrt{2\pi}$, $w = |a_h^{\text{SC}}|^2$, $\mu_h = \int w$, and $I_h = 4 \int |(a_h^{\text{SC}})'|^2$. In the original coordinates $t_i = u/k + y_i$, $y_k = -\sum_{i < k} y_i$, the full product amplitude is $\Psi = \prod_i a_h^{\text{SC}}(t_i)$ and its squared fibre norm is $m = w^{*k}$. The exact Jacobian has absolute value one. Differentiating before integration gives $\partial_u \Psi = k^{-1} \sum_i a_i' \prod_{j \neq i} a_j$. The mixed integrated terms vanish because $\int \bar{a} a' = 0$. Orthogonal projection onto the actual product-amplitude line gives

$$\int_{\mathbb{R}} \frac{|m'|^2}{m} du + 4 \int_{\mathbb{R}} \|n(u)\|^2 du = \frac{\mu_h^{k-1}}{k} I_h, \quad n = \partial_u \Psi - \frac{m'}{2m} \Psi. \quad (\text{R19})$$

The proof in (FC.1)–(FC.4) retains the original mass and phase, and establishes every required integrability statement. For every finite independent relative polynomial frame, $j_u c = \Psi \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}$, the corresponding exact derivative is

$$\partial_u(j_u c) = j_u(c' + \Gamma_u c) + N_u c, \quad \Gamma = W^{-1} j^* j', \quad N = (I - j W^{-1} j^*) j', \quad j^* N = 0. \quad (\text{R20})$$

This identity follows by inserting the orthogonal projection onto $\text{im } j_u$, with $W = j^* j > 0$. Its full normal Gram is $\mathcal{N} = N^* N$.

Every finite packet leaves infinitely many critical-line zeros of the original amplitude. At such a zero T of complete order ℓ_T , the transverse expansion in (FC.25) has coefficients $c_T A_b \delta^{\ell_T}$ for Ψ and $(\ell_T/k) c_T A_b \delta^{\ell_T-1}$ for $\partial_u \Psi$, with $c_T A_b \neq 0$. Thus an equality $N_u c = 0$ forces the relative polynomial to vanish on the entire hyperplane $y_1 = T - u/k$ whenever the other factors can be chosen nonzero. For $k \geq 3$ this holds at every u ; for $k = 2$ it holds outside the countable sumset $\mathcal{Z}_h + \mathcal{Z}_h$. Infinitely many such hyperplanes force the finite polynomial to vanish. Consequently (FC.21)–(FC.27) prove

$$\begin{aligned} \mathcal{N}(u) &> 0 \text{ everywhere for } k \geq 3, \\ \mathcal{N}(u) &> 0 \text{ almost everywhere for } k = 2, \quad \int \frac{|m'|^2}{m} < \frac{\mu_h^{k-1}}{k} I_h. \end{aligned} \quad (\text{R21})$$

The full proof retains the orders, coefficient domains and the quartet two-factor zero fibre $N_0 = 0$.

The two-branch transformation is equally exact for a matrix weight which is not even. At $x > 0$, put $v = \sqrt{x}$ and

$$M = \begin{pmatrix} I & ivI \\ I & -ivI \end{pmatrix}, \quad H = \frac{M^* \text{diag}(W(v), W(-v)) M}{2v}.$$

The original coefficient is $c(\pm v) = a(x) \pm ivb(x)$. Equations (FC.10)–(FC.18) prove the full unitary correspondence, the covariant derivative $\nabla_x = \partial_x + \mathcal{A}$, and its energy

$$\begin{aligned} \mathcal{A} &= M^{-1} \text{diag}\left(\frac{\Gamma(v)}{2v}, -\frac{\Gamma(-v)}{2v}\right) M + \text{diag}(0, I/(2x)), \\ \int \|\partial_u(jc)\|^2 du &= 4 \int_0^\infty x (\nabla_x q)^* H \nabla_x q dx + \int_0^\infty q^* Q q dx, \quad q = (a, b). \end{aligned} \quad (\text{R22})$$

Here $Q = M^* \text{diag}(\mathcal{N}(v), \mathcal{N}(-v)) M / (2v)$. The retained Jacobian drift is $\mathcal{A}^* H + H \mathcal{A} = H' + H/(2x)$. The two endpoint traces $a + i\sqrt{x}b$ and $a - i\sqrt{x}b$ must agree at $x = 0$; this is the proved joining condition for the original derivative domain.

The arithmetic derivative has the unital algebra bridge

$$\Phi : P/I^2 \longrightarrow (P/I)[\epsilon]/(\epsilon^2), \quad [p] \longmapsto [p] + \epsilon[\partial_{\Sigma} p], \quad I = (h(s_1), \dots, h(s_k)), \quad \partial_{\Sigma} = k^{-1} \sum_i \partial_{s_i}. \quad (\text{R23})$$

The product rule modulo I proves multiplicativity. Its conormal restriction is the row $(a_i) \mapsto k^{-1} \sum_i h_i' a_i$ on $I/I^2 \cong (P/I)^k$. On an ordered local block of lengths m_i , multiplication by h_i' sends

the input constant in variable i to $m_i c_i(0) z_i^{m_i-1}$ times that constant. Counting the monomials having at least one top exponent proves its full rank $d^k - (d - r_0)^k$, r_0 the number of distinct packet roots. The kernel and image of the algebra map, including its linear cokernel, are proved in (CW.6)–(CW.11).

For the original monic remainder section $j : P/I \rightarrow P/I^2$, $Z = S - k/2$ and $D_0 = \delta j$, the exact section defect and correction are

$$[Z, D_0] = -I + J, \quad J = \frac{1}{k} \sum_i M_{h'_i} c_i \ell_i = \frac{d}{k} \sum_i P_i, \quad P_i = \frac{1}{d} M_{h'_i} c_i \ell_i. \quad (R24)$$

Here ℓ_i extracts the original s_i^{d-1} coefficient and c_i inserts the constant polynomial in that variable. The identity $\ell_i M_{h'_i} c_i = dI$ proves that the P_i are commuting idempotents. Their complete tensor decomposition gives eigenvalues dj/k with multiplicities $\binom{k}{j} (d-1)^{k-j}$ for $d > 1$, and $J = I$ for $d = 1$. Its trace is d^k and its rank $d^k - (d-1)^k$. The explicit factorization (CW.22) through the full conormal row proves the relationship between these two rank formulas.

The complete arithmetic unit $\widehat{U} = [\prod_i v_h(s_i)]_{I^2}$ gives $j_U = \widehat{U} j U^{-1}$, $D_U = M_{U^{-1} \delta \widehat{U}} + M_U D_0 M_U^{-1}$, and $[Z, D_U] = -I + U J U^{-1}$. For the actual minimum section j_M , the difference $a_M = j_M - j_U$ is its admitted original relation in I/I^2 ; the correction is exactly $U J U^{-1} + [Z, \delta a_M]$, with trace d^k . These are the full source maps proved in (CW.23)–(CW.30). In one variable they also give

$$j_U u = (j_h g') \mathcal{R}_Z(1, u), \quad \mathcal{R}_Z(1, u) = \ell(U^{-1} u), \quad \mathcal{R}_Z(1, j_h g') = \sum_{\rho} m_{\rho} = d. \quad (R25)$$

The partial-fraction proof in (CW.32)–(CW.36) retains the full poles before taking residues. Finally $G(\Phi)$ is the proved semiring lift; the derivative is its infinitesimal coefficient map. Thus every relation killed by the quotient retains its supported zero and its explicit derivative observation.

43.9 The full relative limit and the original arithmetic observation

Retain $k \geq 2$ and the nonempty original packet $d = \deg h \geq 1$ throughout this subsection. For two nested relative frames, keep the literal coefficient inclusion E_{LM} and the original matrices of (NT.1)–(NT.37). Their exact connection difference $K_{LM} = \Gamma_L E_{LM} - E_{LM} \Gamma_M$ obeys

$$N_M = j_L K_{LM} + N_L E_{LM}, \quad N_M^* N_M = K_{LM}^* W_L K_{LM} + E_{LM}^* N_L^* N_L E_{LM}. \quad (R26)$$

The cross terms vanish because $j_L^* N_L = 0$. Thus enlarging a frame transfers the first displayed component into its tangent part, with its complete positive Gram retained. The arithmetic exponential bounds and weighted polynomial density proved in (ND.1)–(ND.34) give $\Pi_M \rightarrow I$ strongly on every original relative fibre. For each fixed original source polynomial P , (NN.1)–(NN.22) prove $r_M(P) = (I - \Pi_M) \partial_u(\Psi P) \rightarrow 0$ in the original global L^2 norm. Their normal-image transitions are the explicit contractions $r_M(P) \mapsto (I - \Pi_L) r_M(P) = r_L(P)$; their full derivative-graph transitions keep $\partial_u(\Psi P)$ unchanged and are isometric.

The same proofs retain the actual logarithmic arithmetic observation. The invariant relation $P_{\text{rel}} = \sum_i h(s_i)$ has zero ordinary jet but derivative jet $(U_k/k) \sum_i [h'(s_i)] \neq 0$. For $d = 1$ the sum is k ; for $d > 1$ its distinct highest standard monomials have coefficients d , so it cannot vanish. Its normal seminorm tends to zero, while its algebraic observation remains this specified nonzero vector. The exact split quotient of that zero limiting seminorm sends every supported source to e and external absence to τ ; the nonzero observation therefore cannot factor through that quotient. Its original finite maps and full derivative graph remain. For the changing canonical target metrics the complete comparison is

$$G_M \preceq G_*, \quad B_M = G_*^{-1/2} G_M G_*^{-1/2}, \quad x^* G_M x = (G_*^{1/2} x)^* B_M (G_*^{1/2} x). \quad (R27)$$

The fixed-source limit has not estimated this changing B_M . For $h = 1$ the finite coefficient quotient is zero, so the arithmetic observation above is its zero map; the original analytic source and the fixed-source density and derivative-graph maps remain as in (NN.22). The cyclic/full-frame derivative split and canonical-section difference are proved with all four metric terms in (CM.1)–(CM.19).

43.10 Every relation depth and its full cyclic complement

For the same monic h and $I = (h(s_1), \dots, h(s_k))$, the complete depth calculation (CC.1)–(CC.52) proves

$$\begin{aligned} I^r \cap \mathbb{C}[S] &= (\chi_r), \quad \chi_r(T) = \prod_{\lambda} (T - \lambda)^{\ell_{\lambda,r}}, \\ \ell_{\lambda,r} &= \max_{\sum_i \rho_i = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) + (r - 1) \max_i m_{\rho_i} \right), \\ C_r &= \mathbb{C}[T]/\chi_r \xrightarrow{\eta_r = U_r \alpha_r} A_r = (P/I^r)^{\mathfrak{S}_k}, \quad \Pi_r \eta_r = I, \quad A_r \cong C_r \oplus \ker \Pi_r. \end{aligned} \quad (\text{R28})$$

The coefficient-extraction formula (CC.13) constructs this retraction, and (CC.10) counts every invariant orbit before taking its trace. The quotient derivative $\delta_r : P/I^{r+1} \rightarrow P/I^r$ and its scalar counterpart satisfy the commuting squares (CC.17). Their full unit term has the proved decomposition $\delta_r \eta_{r+1} = \eta_r(\partial_r + M_{b_r} \rho_r) + \mathcal{N}_r \rho_r$ of (CC.23); its complementary quotient map vanishes exactly when $U_r^{-1} \delta_r U_{r+1} \in \alpha_r C_r$. The first-jet maps, their full kernels, the product-Jacobian factorization, and the exact two-factor parity ideals at every depth are proved in (CC.29)–(CC.51b). An arbitrary weighted multiplier has all four blocks (CC.45); its lower diagonal block is a genuine quotient operator precisely when the lower left block is zero. Thus the earlier weighted-trace language has the exact operator interpretation (CC.45)–(CC.49), with no complementary direction lost.

43.11 Exterior amplification and the arithmetic volume to estimate

For the actual cyclic quotient, keep $A = M_S$, its canonical G_N and $H = G_N^{-1}(A^* G_N + G_N A - k G_N)$. The complete exterior source note proves that H has spectrum $\epsilon, -\epsilon, 0, \dots, 0$. The additive exterior action $A^{[p]}$ and the original unscaled representative $R_N^{\otimes p} \text{Alt}_p$ have source Gram $p!(\det(G_N)_{I,J})_{I,J}$; their control has the same literal factorial and allowance ϵ . On the determinant of the full positive generalized spectral subspace this gives

$$L_{h,k} = \sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (\text{R29})$$

The full cochain primitive carries the sign $(-1)^{k(j-1)}$ in block j , which the tensor differential repeats; their product is one. Hence this exterior comparison uses the original theta boundary. The cochain idempotent splits full cohomology, while the ordinary trace in (R29) is on the specified finite arithmetic packet.

For the exact complete nonreal quartet polynomial of (E33), with common order m , let $\ell_k = 1 + k(m-1)$. The counted sum grid yields

$$q = \ell_k(k+1)^2, \quad L_{h,k} = 2\delta \ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq q-1. \quad (\text{R30})$$

The explicit four occupation counts in (E33)–(E34) prove this formula without a noncollision assumption on ordered tuples. In a larger packet it remains a lower bound and the threshold is its actual $\deg \chi_{h,k} - 1$. The equality analysis (E1)–(E36) further proves, in the original metric, with positive spectral projector Q , orthogonal range projection P , and $T = PA(1-P)$,

$$\begin{aligned} \epsilon^2 - L^2 &= \|T\|_{\text{HS}, G_N}^2 + \epsilon^2 \{\det M + \det(I_2 - M)\}, \\ BX - XD &= T, \quad X = (Q - P)|_{\ker P}, \quad X = \int_0^\infty e^{-tB} T e^{tD} dt. \end{aligned} \quad (\text{R31})$$

Here M is the explicit positive compression in (E5)–(E7), and B, D are the original diagonal blocks of A in the orthogonal sum. The finite nilpotent exponential formulas (E15) prove convergence of the integral and that this Sylvester map is invertible. For $L > 0$, equality $L = \epsilon$ is equivalent to $P = Q$, or to $T = 0$. A retained critical Jordan block forces strict inequality. The canonical Gaussian equality calibration (E35)–(E36) proves why positivity of a moment measure alone supplies no further strictness.

Finally the arithmetic matrices themselves give the exact calculation (CV.1)–(CV.12). Set $D_N = \det K_N$ and $\tilde{K}_N = K_{N-1} + b_{N+1}b_{N+1}^*/\omega_{N+1}$. Then

$$\begin{aligned} \epsilon^2 &= \frac{\omega_{N+1}}{\omega_N D_N} \Delta_N, & \Delta_N &= D_{N+1} + D_{N-1} - D_N - \det \tilde{K}_N, \\ \Delta_N &= \sum_{\substack{I \subset \{0, \dots, N-1\} \\ |I|=q-2}} \frac{|\det[b_I, b_N, b_{N+1}]|^2}{\omega_N \omega_{N+1} \prod_{i \in I} \omega_i} \quad (q \geq 2). \end{aligned} \quad (\text{R32})$$

Determinant multilinearity proves the positive sum, including the singular first-degree boundary, and (CV.10a) constructs its original source vector with norm squared Δ_N and tensor norm squared $q! \Delta_N$. At $N = q - 1$, (CV.11)–(CV.12) give the same control as $\sum_{j < q-1} |\gamma_j|^2 \omega_j / \omega_{q-1}$ in the unchanged orthogonal expansion of χ . Thus the current quantitative expression is $\omega_{N+1} \Delta_N / (k^6 \omega_N D_N)$ along admitted growing degrees. Its limit has not been estimated. The complete lower bound (R30), the positive exterior data (R32), and the changing metric (R27) are retained together for that calculation.

43.12 The original Toda volumes and the complete four-term loss

The Toda continuation now calculates this same quantity from the original source and relation volumes. In the notation proved in (TVB.1)–(TVB.38c), use $k \geq 1$, a nonempty dagger-stable packet with its full zero orders, $q \geq 1$, real $|\theta| < \pi/2$, and $N \geq q$, except at the separately stated $N = q - 1$ endpoint, throughout (R33)–(R34). We retain $c = k/2$, $Z(\theta) = M_h(\theta)^k$ and $Z_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta)Z$. The Hankel determinants $\mathfrak{D}_n = \det[Z^{(a+b)}]_{a,b < n}$ and $\mathfrak{B}_m = \det[Z_\chi^{(a+b)}]_{a,b < m}$ are the actual source and relation Gram determinants. With $m = N - q + 1$ their exact determinant-line map and the rank-one update give

$$\begin{aligned} V_N &= \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}, & \delta_N &= \frac{V_N}{V_{N-1}}, & \alpha_{N+1} &= \frac{\omega_{N+1}}{\omega_N}, \\ \psi_N &= \frac{\text{Tr } A - cq}{i} - \partial_\theta \log V_N, & \frac{b_N^* G_N b_{N+1}}{\omega_N} &= i\psi_N, \\ \epsilon_N^2 &= \alpha_{N+1}(1 - \delta_N)(\delta_{N+1}^{-1} - 1) - \psi_N^2 = \alpha_{N+1} \frac{\Delta_N}{\det K_N}, & N &\geq q. \end{aligned} \quad (\text{R33})$$

Every equality has its complete source, phase and determinant proof in the cited bridge. At $N = q - 1$, the exact replacement remains $\epsilon_{q-1}^2 = (\nu_0 - \omega_q)/\omega_{q-1} - \psi_{q-1}^2$, with $\nu_0 = \|\chi\|^2$; no preceding inverse occurs. The explicit conjugate-linear map (TVB.38b) sends the lower orthogonal projection of χ to the actual positive-minor exterior source vector, with the full squared-norm factor $\det K_{q-1}/\omega_q$.

The two scalar Toda curvatures have full source maps. For $X = (S - c)/i$, the actual least-norm lift satisfies $R'_N = -I_N$, where $I_N = P_{\mathcal{D}_N} X R_N$ and $O_N = (1 - P_N) X R_N$ in the current tilted pairing, where P_N is the tilted source projection onto $\mathcal{P}_N$. With $\beta_m = \nu_m/\nu_{m-1}$ for $m \geq 1$, the proofs of (TVB.17)–(TVB.21) give

$$\begin{aligned} G_N'' - G_N' G_N^{-1} G_N' &= O_N^* O_N - I_N^* I_N, \\ \text{Tr}(G_N^{-1} O_N^* O_N) &= \alpha_{N+1}(1 - \delta_N), & \text{Tr}(G_N^{-1} I_N^* I_N) &= \beta_m - \alpha_{N+1} \delta_N. \end{aligned} \quad (\text{R34})$$

The common subtracted term is present in both positive costs. At the first admitted degree the two costs are α_q and zero, with no negative-index relation norm. Every derivative boundary has the literal cochain primitive (TVB.9), including its tensor signs.

The coordinating Zeta contribution joins the two volume losses with the original spectral coupling. At $\theta = 0$, $N \geq q$ and $L > 0$, let P be the G_N -orthogonal projection onto the full positive spectral subspace, $C = PA(1 - P)$ its restricted coupling, and $M = E^\sharp P E$, where E embeds the two G_N -unit eigenvectors of $A^\sharp + A - kI$ at $\pm \epsilon_N$. Set $v_N = -\log \delta_N$, $s_N = (v_N + v_{N+1})/2$ and $t_N = (v_{N+1} - v_N)/2$. The complete derivation in (TVB.39)–(TVB.43), using the same metric on both sides, proves

$$\alpha_{N+1} \sinh^2 s_N - L^2 = \alpha_{N+1} (e^{t_N} - \cosh s_N)^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2 + \epsilon_N^2 \{\det M + \det(I_2 - M)\}. \quad (\text{R35})$$

All four terms are nonnegative. In particular the two-step volume lower bound retains $L^2 + \psi_N^2 + \|C\|_{\text{HS}, G_N}^2$ inside its inverse-hyperbolic-sine expression. The exact Sylvester inverse maps C to the difference between spectral and orthogonal projectors; its full original nilpotent factors and singular value remain specified. The complete supplied join is also reproduced as a source appendix.

43.13 Certified theta input and a calculated comparison of the two gaps

The theta continuation proves the sector construction and the exact maps

$$h(D)F_h = f_0, \quad w_1(t) = |h(1/2 + it)|^2 w_h(t), \quad M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2} x)|^2 dx. \quad (\text{R36})$$

The absolute-square integral is for real $|\theta| < \pi/2$; its holomorphic strip continuation is proved in the theta input chapter. The dagger sign, complex Fourier phase, endpoint Jacobian and source factor $g = 2\xi$ are all retained in (TI.1)–(TI.16). The exact incomplete-gamma double series and its explicit infinite-square-tail bounds (TI.19)–(TI.30) supply rigorous interval enclosures for $M_1(0)$, $M_1(1/10)$, $M_1''(0)$ and $a_1^{(k)}/k = M_1''(0)/M_1(0)$. Each displayed rational interval in that proof has width $3 \cdot 10^{-40}$. The certificate adds the proved infinite tail to outward-rounded ball arithmetic; it does not infer an enclosure from precision agreement. The source norm is positive for $h = 1$ while its arithmetic cyclic quotient is zero. The density identity in (R36), and the globally constructed Mellin division, specify the exact passage to packet input without division of a measured zero value by zero. No packet determinant estimate follows from these four seed intervals alone. For the empty packet $h = 1$, the arithmetic quotient is zero and $V_N = 1$ by the empty-determinant convention. For $q = 1$, $\epsilon_N = 0$ at every admitted degree, whereas the upper Toda expression is strictly positive for $N \geq 1$, as proved in the bridge.

Finally (TC1)–(TC13) calculate both gaps in the original Gaussian calibration with mass μ , variance σ^2 , $c = 1/2$ and $\chi = (S - c)^2 - \sigma^2$. The source and quotient coordinate maps are fully displayed there. They give

$$(V_1, V_2, V_3) = \mu^2 \sigma^2 (1, 1/3, 1/11), \quad \epsilon_1^2 = L_{\text{spec}}^2 = 4\sigma^2, \quad \epsilon_2^2 = 16\sigma^2/3, \\ \alpha_3 \frac{(\mathcal{R}_2 - 1)^2}{4\mathcal{R}_2} - \epsilon_2^2 = 49\sigma^2/33, \quad \epsilon_2^2 - L_{\text{spec}}^2 = L_{\text{spec}}^2 \|Q - P\|_{\text{HS}, G_2}^2 = 4\sigma^2/3. \quad (\text{R37})$$

The first gap is precisely the Toda imbalance square; the second has the explicitly calculated spectral/orthogonal projector map. This calibration uses declared polynomial coefficients. The actual growing packet calculation continues with (R33)–(R36), retaining its arithmetic measure and all four terms of (R35).

43.14 Gamma convolution, certified higher input, and the complete quotient comparison

Result R38: an exact analytic and finite coefficient interface. Keep the original $w_h(t) = |g(1/2 + it)/h(1/2 + it)|^2/(2\pi)$ and $r_\lambda(t) = |\Gamma(\lambda + it/2)|^2/(2\pi)$, with $c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda)$. The full

Gamma source appendix and the coefficient-join proof construct

$$r_{\lambda,k} = \frac{c_{\lambda}^k}{c_{k\lambda}} r_{k\lambda}, \quad m_{h,k} = w_h^{*k} = r_{\lambda,k} C_{\Sigma}(B_{h,\lambda}^{\otimes k}), \quad B_{h,\lambda} = |v_h/\Gamma(\lambda + it/2)|^2.$$

The Hilbert maps $U_{\Sigma}f = f(\sum t_i)$ and its conditional adjoint C_{Σ} have the inverse paired decomposition $F \leftrightarrow (C_{\Sigma}F, (I - U_{\Sigma}C_{\Sigma})F)$. All masses and the relative vector remain. With $\alpha = 2\lambda$,

$$c_{h,j} = \frac{\int b_j^{(\lambda)} w_h}{c_{\lambda} j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j, \quad d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k,$$

the exact finite moment is $\int b_n^{(k\lambda)} m_{h,k} = c_{\lambda}^k n! d_{h,k,n}$. The original degree- N source and relation Grams at theta zero use $c_{h,0}, \dots, c_{h,2N}$; their derivative of order r uses the list through $2N+r$, through the fully written original- S polynomial test. The analytic join supplied by the Zeta task and proved in the coefficient chapter is

$$\mathcal{A}_h(z) = c_{\lambda}^{-1} (1+z^2)^{-\lambda} M_h(\arctan z), \quad |z| < 1, \quad M_h(\theta) = \int e^{\theta t} w_h(t) dt.$$

The specified branches at zero and exponential domination justify all coefficient derivatives. Its inverse uses $M_h(\theta) = c_{\lambda} \sec^{2\lambda} \theta \mathcal{A}_h(\tan \theta)$ for $|\operatorname{Re} \theta| < \pi/4$, with the analytic branches continued from zero. The complete source appendix is preserved alongside the explicit cochain correction; applying $\mathcal{V}$ to a section difference gives the boundary, and (GP.26)–(GP.27) construct its primitive.

Result R39: certified original moments and signed coefficients. Equations (GS.1)–(GS.40) prove the original theta derivative recurrence

$$P_0(y) = 8y^2 - 12y, \quad P_{r+1}(y) = -2yP'_r(y) + (2y - \tfrac{1}{2})P_r(y),$$

$$\mu_{2r} = 2 \int_1^{\infty} \left| \sum_{n \geq 1} P_r(\pi n^2 x^2) e^{-\pi n^2 x^2} \right|^2 dx.$$

Their fully specified positive square tail (GS.18) bounds the omitted infinite integer sum at every fixed order. At order two and three, the rigorous rational enclosures are

$$\mu_4 \in [383.2743417173642425544072374791308324300288, \\ 383.2743417173642425544072374791308324300291],$$

$$\mu_6 \in [17946.5273016052076235667646919834204772848931, \\ 17946.5273016052076235667646919834204772848934].$$

The original reference $c_{13/4} = 10395\sqrt{\pi}/(2048\sqrt{2})$ is enclosed separately from the actual mass μ_0 . The complete invertible moment/coefficient map (GS.34)–(GS.35) certifies c_0, c_2, c_4, c_6 and proves $c_4 < 0, c_6 < 0$. Its tensor coefficient of degree six retains all three terms

$$d_{k,6} = k c_0^{k-1} a_6 + k(k-1) c_0^{k-2} a_2 a_4 + \binom{k}{3} c_0^{k-3} a_2^3, \quad a_j = (13/2)_j c_j.$$

The accompanying source-cost continuation (GS.41)–(GS.50) proves the first four monic norms h_0, h_1, h_2, h_3 and all three positive ratios $a_j^{(k)} = h_j/h_{j-1}$ from these same moments. In particular, with $\nu = \mu_2/\mu_0$ and the original sum moments $M_j(k)$,

$$a_1^{(k)} = k\nu, \quad a_2^{(k)} = \frac{\mu_4}{\mu_2} + (2k-3)\nu, \quad a_3^{(k)} = \frac{M_6 - M_4^2/M_2}{M_4 - M_2^2/M_0}.$$

The exact polynomial isometry $P(S) \mapsto P(k/2 + iu)$ and its monic phase i^j preserve these norms and prove the original spectral recurrence $SP_j = P_{j+1} + (k/2)P_j - a_j^{(k)}P_{j-1}$. For the source determinant corrections, put

$$Q_N = \frac{X_{N+2}X_N}{X_{N+1}^2}, \quad Q_0 = \frac{2\nu}{13}, \quad Q_1(k) = Q_0 + \frac{\rho}{13k+2}, \quad \rho = \frac{\mu_4}{\mu_2} - \frac{43\nu}{13}.$$

The certified interval for ρ is strictly negative; hence $0 < Q_1(1) \leq Q_1(k) < Q_0$ and $Q_1(k+1) - Q_1(k) = -13\rho/((13k+2)(13k+15)) > 0$. These are exact statements at every tensor degree for the specified source polynomial degrees. The analytic seed has the original zero arithmetic quotient $\mathbb{C}[S]/(1)$ throughout.

Result R40: finite error propagation to the actual consecutive ratio. The exact polynomial coefficient power and test expansion in (GMT.12)–(GMT.17) propagate finite coefficient enclosures into Hermitian source envelopes. The full boundary-resolvent formula (GMT.19)–(GMT.20) carries those actual forms to their least lifts, including the signed quadratic loss and its original theta primitive. Equations (GMT.29a)–(GMT.29b) also certify every coefficient of the least lift and the full representative loss $\hat{R}^*H\hat{R} - G(H)$ by an explicit positive q -dimensional rational upper form. The positive finite envelopes induce quotient-dimensional determinant enclosures by minimizing on the same affine space of representatives. For $N \geq q$, retain the exact two-degree source block

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^*H_-^{-1}C_D, \quad F_D = J_D - J_-H_-^{-1}C_D, \quad Z_D = F_D^*G_-F_D.$$

Equations (GMT.22)–(GMT.27) prove, in the original coordinates,

$$\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W}, \quad a_{N+1} = \frac{\det W}{W_{11}^2}.$$

Applying the same maps to the Gamma reference gives the exact relative correction T_{N-1}/T_{N+1} as the quotient of these two positive ratios. The fully proved auxiliary bounded multiplier $\min(1 + t^2/4, 10001)$ with $\lambda = 1$, $k = 2$, and the declared $\chi(S) = (S - 1)^2 - 2$ has $T_1/T_3 < 1$. Its exact finite tail, rational envelopes, relation matrices, and source/quotient maps are (GMT.32)–(GMT.42). Thus positivity and boundedness alone do not imply relative correction monotonicity. This explicit counterexample is connected to the arithmetic problem by those same coefficient, least-lift, and determinant maps; its auxiliary weight is not assigned the value of an actual zeta packet weight.

Result R41: the complete phase and a strict relative contribution. For the actual complex product $a = \prod_i A_{h,\lambda}(t_i)$, put $a_k = C_{\Sigma}a$, $b_k = C_{\Sigma}|a|^2$, and $d_k = b_k - |a_k|^2$. Equations (GP.6)–(GP.11) prove the exact unitary transport of the arithmetic sum projection and its fibre defect. Equations (GP.12)–(GP.13) use the original quartet tail, with its full constant $25 \cdot 2^{2d}/\sqrt{\pi}$, to prove $d_k(u) > 0$ at every real u for $\lambda = 1/4$, $d \geq 4$, and $k \geq 2$. The full complex coefficient transform and its Parseval identity preserve the factors c_{λ}^k and $n!$. The source matrices and the original quotient satisfy

$$\begin{aligned} M_h &= M_{\text{coh}} + M_{\text{rel}}, & R_h &= R_{\text{coh}} - B_{\chi}K, \\ K &= (B_{\chi}^*M_hB_{\chi})^{-1}B_{\chi}^*M_{\text{rel}}R_{\text{coh}}, \\ G_h &= G_{\text{coh}} + K^*B_{\chi}^*M_{\text{coh}}B_{\chi}K + R_h^*M_{\text{rel}}R_h. \end{aligned}$$

For every admitted nonzero quartet quotient the last difference is positive definite. The empty boundary at $N = q - 1$ and zero quotient at $q = 0$ retain their explicit separate values. The graph-to-source addition, gamma multiplication, inverse Mellin map, and full jet unit in (GP.24)–(GP.27) reconstruct the original arithmetic observation and its exact supported zero.

The new finite coefficients, strict fibre contribution, and interval certificates therefore enter the same original Toda and exterior quantities in R33–R37. In particular the lower spectral cost and the upper expression are compared in the same G_N , with its volume-imbalance square, phase square, spectral coupling, and projection terms retained. The displayed finite maps do not yet bound these arithmetic determinant corrections when N grows with the full amplified packet. That degree-dependent calculation remains the active research continuation, with the exact finite input and its certified error maps now constructed.

Part II**Complete source proof notes and chronology**

A PR 8: Finite operator comparison

Source attribution. Web contribution, pull request #8, [Zeta research repository](#).

Exact revision. 25689c5376166df37801229bc275ff5a896007d9

Source file. workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 57502c7fe6a7e1de239c94c3479abbb6a70b4718b551ad1bb1e26e2d3e4f4c22

A.1 Exact finite operator quotients of the tau-base theta cohomology

Owner-directed continuation, 12 September 2026. Mathematical derivation and research draft; no new analytic theorem below is claimed to have been checked by Lean.

A.1.1 0. Inputs, provenance, and the new result

This note continues TAU_BASE_MODEL.md, rather than moving the programme back to a finite-field variety. Its arithmetic coefficient object is the original split semiring

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e = 0_R^\bullet.$$

The geometric base remains the one-point absolute pointed base described there. The quotient

$$p_{\mathbb{Z}} : G(\mathbb{Z}) \longrightarrow \mathbb{Z}$$

and its square with $G(\mathbb{C}) \rightarrow \mathbb{C}$ remain part of the marked object. No ideal norm, Euler factor, scaling parameter, Gaussian, or completion factor is changed here.

The supplied derived formalization (PR #6, 5f6ee6575550d8a20c6fc3990359cb4f67855f80) provides the actual reconstruction, internal quotient, homology-kernel equivalence, and compatible kernel actions. Its finite-jet module supplies the balanced tensor-quotient comparison with the split-reflection square. Those results are reused; this note does not duplicate their Lean implementation.

The new analytic step computes the previously only surjective finite Mellin-jet observation as an exact operator quotient. For every polynomial h whose roots lie in the open critical strip, it constructs

$$Q/h(D)Q \xrightarrow{\sim} E_h/g_h E_h, \quad Q[h(D)] \xrightarrow{\sim} \ker(g_h : E_h \rightarrow E_h),$$

with $E_h = \mathbb{C}[t]/(h)$ and $g = 2\xi$. It then constructs finite spectral projectors on the original Q , not only quotients of Q , and identifies the torsion subsheaf of the balanced analytic realization with $\mathcal{O}/(2\xi)$. Their kernels, original supported zeros, nilpotent actions, and trace-producing residue pairing are all retained.

The general existence of spectral realizations of zeta zeros is prior literature (not a novelty claim here): see R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. The contribution of this note is the explicit comparison in the specified workbench, together with its original SplitZero quotients and its complete finite kernel calculation.

A.1.2 1. The exact arithmetic complex and its operator

Retain

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for all } b \in \mathbb{Z}, k \geq 0\}.$$

The source maps, with their original factors, are

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

The Fourier transform is $\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx$. Poisson and the two moment conditions put ΘV in $\mathcal{B}$. Also $D\Theta = \Theta D$, and D preserves V .

Set

$$A = \mathbb{C}[t], \quad t \cdot F = DF, \quad t \cdot \phi = D\phi,$$

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}] \text{ in degrees } 0, 1, \quad Q = \mathcal{B}/\Theta V.$$

The existing injectivity proof of Θ gives $H^0(C_+) = 0$ and $H^1(C_+) = Q$.

This is the one-leg subcomplex of the actual tau-base chart complex

$$C_{+-} = [V \oplus V \xrightarrow{\Theta\phi - \Theta\widehat{\psi}} \mathcal{B}].$$

Its inclusion is $\phi \mapsto (\phi, 0)$ in degree zero and the identity in degree one. The quotient complex is the second copy of V in degree zero. The explicit section is $\psi \mapsto (\psi, \psi)$. The operator on that second copy is $1 - D$, because $D\widehat{\psi} = (1 - D)\psi$. Thus the additional joint-support degree-zero cohomology is retained in this split exact sequence; it is not included in Q and then silently discarded.

The actual source multiplier is

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s),$$

$$H_\phi(s) = c(s)M_+\phi(s), \quad c(s) = \frac{2\pi^{s/2}}{s(s-1)\Gamma(s/2)}, \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}.$$

On the open strip $\mathcal{S} = \{0 < \Re s < 1\}$, $c(s)$ is holomorphic and nowhere zero. The source identity is

$$\mathcal{M}\Theta\phi = gH_\phi.$$

The specific function

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}$$

has $H_{\phi_*} = 1$. Consequently

$$H_{P(D)\phi_*}(s) = P(s), \quad \mathcal{M}\Theta(P(D)\phi_*) = g(s)P(s).$$

These are identities with the stated Gaussian and $g = 2\xi$; no rescaling is performed.

A.1.3 2. A continuous inverse to the actual Euler differential

For any $\rho \in \mathbb{C}$, define on the subspace $\{H \in \mathcal{B} : \mathcal{M}H(\rho) = 0\}$

$$S_\rho H(x) = x^{-\rho} \int_x^\infty H(t) t^{\rho-1} dt.$$

All powers use the real logarithm on $x, t > 0$. Direct differentiation gives

$$(D - \rho)S_\rho H = H.$$

Endpoint control. Write $x = e^y$, $h(y) = H(e^y)$, and $\sigma = \Re \rho$. At $y \geq 0$, the displayed tail and a bound $|h(t)| \leq C_N e^{-Nt}$ give

$$|S_\rho H(e^y)| \leq \frac{C_N}{N - \sigma} e^{-Ny} \quad (N > \sigma).$$

At $y \leq 0$, the zero moment changes the same expression into the negative lower tail. A bound $|h(t)| \leq C'_N e^{Nt}$ gives

$$|S_\rho H(e^y)| \leq \frac{C'_N}{N + \sigma} e^{Ny} \quad (N + \sigma > 0).$$

Differentiating uses $(S_\rho H)'_y = -\rho S_\rho H - H(e^y)$, so every derivative has the same family of estimates. This proves membership in $\mathcal{B}$ and continuity in its stated seminorms. The homogeneous solution $x^{-\rho}$ fails the required rapid decrease unless its coefficient is zero. Thus $D - \rho$ is injective, and

$$(D - \rho)\mathcal{B} = \ker(F \mapsto \mathcal{M}F(\rho)).$$

Mellin intertwining gives

$$\mathcal{M}(S_\rho H)(s) = \frac{\mathcal{M}H(s)}{s - \rho},$$

with the removable value filled at $s = \rho$.

There is also a source inverse. For $\rho \in \mathcal{S}$ and $\phi \in V$ with $H_\phi(\rho) = 0$, use the same tail for $x > 0$ and extend evenly. The nonvanishing of $c(\rho)$ gives $M_+\phi(\rho) = 0$. Near zero the formula is

$$S_\rho \phi(x) = - \int_0^1 u^{\rho-1} \phi(xu) du.$$

Differentiation under this integral proves smoothness at zero in every order. The function is even and its value at zero is zero. The upper-tail formula proves Schwartz decay at infinity. Finally

$$(1 - \rho) \int_{\mathbb{R}} S_\rho \phi(x) dx = \int_{\mathbb{R}} \phi(x) dx = 0,$$

so the second moment condition is preserved. This proves the source range identity

$$(D - \rho)V = \ker(\phi \mapsto H_\phi(\rho)) \quad (\rho \in \mathcal{S}).$$

At the level of Taylor coefficients $\phi(x) = \sum_{j \geq 1} a_{2j} x^{2j}$, the local expression is $-\sum_{j \geq 1} a_{2j} x^{2j} / (\rho + 2j)$, interpreted to each finite Taylor order with its smooth remainder. All denominators and the minus sign are retained. Endpoints 0, 1 are not included in this source range theorem.

A.1.4 3. Every finite operator test, without choosing zeros first

Choose any finite set of distinct centres $\rho_1, \dots, \rho_r \in \mathcal{S}$ and positive integers m_i . Define the particular polynomial

$$h(t) = \prod_{i=1}^r (t - \rho_i)^{m_i}, \quad d = \deg h, \quad E_h = A/(h).$$

This defines the test; it does not presume that any centre is a zeta zero. The Chinese-remainder map is

$$\theta_h : E_h \xrightarrow{\sim} \prod_i \mathbb{C}[z_i]/(z_i^{m_i}), \quad [P(t)] \mapsto [P(\rho_i + z_i)].$$

For a holomorphic germ family f at the centres, $j_h f$ denotes its Taylor jets transported back by θ_h^{-1} . In particular $g_h = j_h g$ retains the original local units of g .

Iterate Section 2 through the factors of h . Uniqueness makes the inverse independent of their chosen order. This gives the continuous inverse

$$S_h : \ker(j_h \mathcal{M}) \longrightarrow \mathcal{B}, \quad h(D)S_h = \text{id}.$$

The exact range identities are

$$h(D)\mathcal{B} = \ker(j_h \mathcal{M}), \quad h(D)V = \ker(j_h H).$$

Surjectivity on the source follows from $H_{P(D)\phi_*} = P$. Surjectivity of the target jet map follows directly from finite Mellin interpolation: on an interval in $y = \log x$, the functions $e^{\rho_i y} y^j / j!$ are linearly independent. For a nonnegative smooth bump η positive on a subinterval, the Gram matrix

$$G_{(i,j),(k,l)} = \int \eta(y) e^{(\rho_i + \bar{\rho}_k)y} \frac{y^{j+l}}{j!l!} dy$$

is positive definite. Multiplying its inverse into the prescribed jet vector produces a compactly supported smooth preimage. This is only an interpolation device; it does not define or alter the arithmetic Weil pairing.

Consequently the vertical arrows in

$$\begin{array}{ccc} V/h(D)V & \xrightarrow{\bar{\theta}} & \mathcal{B}/h(D)\mathcal{B} \\ \alpha_h \downarrow \wr & & \wr \downarrow \beta_h \\ E_h & \xrightarrow{\times g_h} & E_h \end{array}$$

are actual A -module isomorphisms:

$$\alpha_h[\phi] = j_h H_\phi, \quad \beta_h[F] = j_h \mathcal{M}F.$$

An inverse to α_h sends the unique polynomial representative P of degree less than d to $[P(D)\phi_*]$. An inverse to β_h is any of the above explicit interpolation lifts, taken modulo $h(D)\mathcal{B}$; the class is independent of that choice by the range theorem.

A.1.5 4. The kernel and quotient of the global arithmetic operator

Both V and $\mathcal{B}$ are torsion-free A -modules: every factor $D - \rho$ is injective, since its homogeneous solution cannot be Schwartz at infinity. Over the PID A , they are flat. The bounded complex C_+ is therefore a valid flat model for derived tensoring. The displayed isomorphism computes

$$C_+ \otimes_A^{\mathbf{L}} E_h \simeq [E_h \xrightarrow{\times g_h} E_h]$$

in degrees zero and one. It is the base-change of the actual theta complex, not a newly declared spectral complex with an assigned differential.

The original exact sequence $0 \rightarrow V \xrightarrow{\Theta} \mathcal{B} \rightarrow Q \rightarrow 0$ also gives

$$0 \rightarrow Q[h(D)] \xrightarrow{\kappa_h} E_h \xrightarrow{\times g_h} E_h \rightarrow Q/h(D)Q \rightarrow 0.$$

Here $Q[h(D)] = \ker(h(D) : Q \rightarrow Q)$. The maps are explicit:

- Given $[F] \in Q[h(D)]$, uniquely write $h(D)F = \Theta\phi$. Then $\kappa_h[F] = j_h H_\phi$.
- The quotient observation sends $[F]$ to $j_h \mathcal{M}F$ modulo $g_h E_h$.
- For $P \in \ker(g_h : E_h \rightarrow E_h)$, choose its polynomial representative of degree less than d and set

$$\partial_h(P) = [S_h \Theta(P(D)\phi_*)] \in Q[h(D)].$$

The last expression is defined because its theta input has zero jets modulo h . Replacing P by $P + hR$ changes the result by $\Theta(R(D)\phi_*)$. Thus it is independent of representatives and is the inverse of κ_h onto its kernel. The two injectivity and surjectivity assertions also follow directly by lifting representatives in the preceding exact sequence; no dimension-only argument is needed.

All maps commute with D , with $U_a F(x) = F(x/a)$, and with compactly supported multiplicative convolution. On E_h , the scaling action is multiplication by $j_h(a^s)$, including its entire finite nilpotent Taylor polynomial.

At one centre, retain $g(\rho + z) = u(z)z^r$, where $u(0) \neq 0$ (and $r = 0$ is allowed). For test order m , the two cohomology modules have dimension $\min(m, r)$. When $r = 0$, multiplication by g_h is invertible. The derived amplitude complex is then acyclic, while its reconstructed supported homology still has each labelled internal zero.

The degree-zero term is $Q[h(D)]$; the degree-one term is $Q/h(D)Q$. Their finite alternating traces cancel. The arithmetic packet trace below is the degree-one action, or equivalently its explicitly constructed direct summand in the original Q . The canonical truncation triangle retains the degree-zero term rather than relabelling it as zero.

A.1.6 5. Exact finite spectral projectors on the original cohomology

Now let Z be any finite set of actual nontrivial zeros and retain their full multiplicities r_ρ . Put

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{r_\rho}, \quad E_Z = A/(h_Z), \quad v_Z(s) = g(s)/h_Z(s).$$

The quotient v_Z is entire and its germs are nonzero at every point of Z . Hence $v_{Z,h} = j_{h_Z} v_Z$ is an invertible element of E_Z . This inverse retains the actual Taylor coefficients of g/h_Z ; it is not set to one.

Here $g_{h_Z} = 0$, and the exact sequence becomes

$$Q[h_Z(D)] \xrightarrow[\kappa_Z]{\sim} E_Z, \quad Q/h_Z(D)Q \xrightarrow[\beta_Z]{\sim} E_Z.$$

There is a further map that must be calculated, not assumed: the inclusion of the kernel into Q , followed by the quotient. For $P \in E_Z$,

$$\mathcal{M}(S_{h_Z}\Theta(P(D)\phi_*)) = \frac{gP}{h_Z} = v_Z P.$$

Therefore this kernel-to-quotient comparison is precisely

$$\boxed{\beta_Z q_Z \partial_Z = M_{v_{Z,h}} : E_Z \xrightarrow{\sim} E_Z.}$$

Define

$$\sigma_Z = \partial_Z \circ M_{v_{Z,h}}^{-1} : E_Z \longrightarrow Q,$$

$$\boxed{\Pi_Z = \sigma_Z J_Z : Q \longrightarrow Q, \quad J_Z[F] = j_{h_Z} \mathcal{M}F.}$$

Then

$$\boxed{J_Z \sigma_Z = \text{id}, \quad \Pi_Z^2 = \Pi_Z, \quad \text{im } \Pi_Z = Q[h_Z(D)], \quad \ker \Pi_Z = h_Z(D)Q.}$$

This gives the actual equivariant direct sum

$$\boxed{Q = Q[h_Z(D)] \oplus h_Z(D)Q.}$$

A representative formula is completely determined. For $F \in \mathcal{B}$, let $P_{Z,F}$ be the unique polynomial of degree less than $\deg h_Z$ representing $v_{Z,h}^{-1} j_{h_Z} \mathcal{M}F$. Then

$$\boxed{\Pi_Z[F] = [S_{h_Z}\Theta(P_{Z,F}(D)\phi_*)].}$$

For a simple actual zero ρ , this specializes to

$$\Pi_{\{\rho\}}[F] = \frac{\mathcal{M}F(\rho)}{g'(\rho)} \left[x^{-\rho} \int_x^\infty \Theta\phi_*(t) t^{\rho-1} dt \right].$$

The displayed denominator is the actual $g'(\rho) = 2\xi'(\rho)$; it is not omitted. The bracketed class is an actual nonzero eigenclass in Q , proved by the exact sequence, even though its representative is only an eigenvector modulo the theta relation.

For finite packets the projectors satisfy

$$\Pi_Z \Pi_W = \Pi_{Z \cap W}, \quad \Pi_{Z \cup W} = \Pi_Z + \Pi_W \quad (Z \cap W = \emptyset).$$

These follow from the direct sums, coprimality of disjoint packet polynomials, and commutation with D . Inclusions of finite packets therefore give compatible invariant subspaces and quotient observations. No convergence of an infinite sum of projectors is asserted.

A.1.7 6. Carry every projector through the original split quotient

For any vector space M the single-active-fibre lift is $M^\tau = \{\tau\} \sqcup \{m^\bullet : m \in M\}$. For a submodule N use exactly

$$q_N^{\text{sp}} : M^\tau \rightarrow (M/N)^\tau, \quad m^\bullet \mapsto [m]^\bullet, \quad \tau \mapsto \tau.$$

This is the original coequalizer of the relation inclusion and its internal-zero companion. Its universal property is the one implemented in PR #6, including arbitrary noncancellative target semi-modules.

Consequently the finite observation has the split-linear equivalence

$$(Q/h(D)Q)^\tau \xrightarrow{\sim} G(E_h/g_h E_h), \quad [F]^\bullet \mapsto (j_h \mathcal{M} F \bmod g_h E_h)^\bullet.$$

The domain is being used as a split semimodule, not as an unconstructed ring. A class in $h(D)Q$ maps to the supported zero, and external absence alone maps to external absence.

Lift the projector as

$$\tilde{\Pi}_Z(\lambda, [F]) = (\lambda, \Pi_Z[F]).$$

This formula applies to the actual nonempty support faces of the tau-base theta model; their degree-one transports are the stated identities of Q . At bottom it is the original zero map of the bottom module. The full reconstruction interface permits a nonzero bottom fibre when a subsequent diagram requires one; it is not silently specialized away on an opposite-index dual diagram.

For disjoint packets,

$$\tilde{\Pi}_Z \tilde{\Pi}_W = \varepsilon, \quad \varepsilon(\lambda, x) = (\lambda, 0).$$

This is the supported lift of the zero linear operator, not the constant external-zero map. More uniformly, take $\Pi_\emptyset = 0_Q$ and lift it to ε . The scalar representation

$$G(\text{End}_{\mathbb{C}} Q) \longrightarrow \text{End}_{G(\mathbb{C})}(Q^\tau)$$

sends $f^\bullet$ to its split lift and sends τ to the constant- τ map. It is a unital semiring morphism and sends the supported scalar zero to ε .

Retain smaller admitted source labels For a subspace $W \subset V$, first use its specified enlargement $W_D = \mathbb{C}[D]W$ and keep the map

$$\mathcal{B}/\Theta W \longrightarrow Q_{W_D} := \mathcal{B}/\Theta W_D, \quad \ker = \Theta W_D/\Theta W.$$

For a D -stable W , define the ideal

$$J_{W,h} = \{j_h H_\phi : \phi \in W\} \subseteq E_h.$$

It is an ideal because $H_{P(D)\phi} = PH_\phi$ and every element of E_h has a polynomial representative. The same target range calculation gives

$$Q_W/h(D)Q_W \cong E_h/(g_h J_{W,h}).$$

Enlarging to $W + \mathbb{C}[D]\phi_*$ makes $J_{W,h} = E_h$. The exact finite comparison kernel is

$$\boxed{g_h E_h / (g_h J_{W,h}).}$$

Thus the finite equality for the full arithmetic relation space has a precise smaller-label comparison. It is not asserted for every original W without this kernel.

A.1.8 7. The Hausdorff observation and the remaining global kernel

The projectors just constructed have continuous finite-rank representatives on $\mathcal{B}$: J_Z is continuous, E_Z is finite dimensional, and S_{h_Z} is continuous on its zero-jet domain. They annihilate ΘV , hence also its closure. They therefore descend to

$$Q^H = \mathcal{B} / \overline{\Theta V}.$$

The finite observation still has kernel exactly $h_Z(D)Q^H$. Indeed

$$\ker J_Z = h_Z(D)\mathcal{B} + \Theta V,$$

and continuity gives $\overline{\Theta V} \subset \ker J_Z$. Hence adding the closure to the right side leaves it unchanged. The same finite section yields a continuous direct sum

$$Q^H = \sigma_Z(E_Z) \oplus h_Z(D)Q^H.$$

The algebraic packet $Q[h_Z(D)]$ injects into this Hausdorff quotient: its composite with J_Z is already an isomorphism onto E_Z . No conclusion about vanishing of the entire algebraic-to-Hausdorff kernel follows from this finite statement.

Let

$$\mathcal{R}_\infty = \bigcap_{Z \text{ finite}} \ker \Pi_Z = \bigcap_{Z \text{ finite}} h_Z(D)Q.$$

The map $Q \rightarrow \prod_\rho \mathcal{O}_\rho / (g)$ has exactly this kernel. Every nonzero polynomial whose roots are in $\mathcal{S}$ acts bijectively on $\mathcal{R}_\infty$. To verify this, its kernel on Q lies in the finite packet at its zero centres; its cokernel is supported there as calculated in Section 4. On the complementary summand it is bijective, and its inverse preserves all other commuting finite projectors. This proves injectivity and surjectivity on the intersection.

The continuation retains $\mathcal{R}_\infty$ and its scaling action. Finite packets do not prove it zero. A topological spectral-synthesis theorem or a global cohomological trace argument must address that specific invariant module rather than infer its disappearance from finite-jet isomorphisms.

7.1 The global balanced kernel has a computed local-cohomology complement Work now on the analytic strip $\mathcal{S}$, with its sheaf $\mathcal{O}$ and sheaf $\mathcal{K}$ of meromorphic functions. Form the sheafified balanced realization

$$\mathcal{M}_Q = \mathcal{O} \otimes_A Q, \quad \beta : \mathcal{M}_Q \longrightarrow \mathcal{O}/(g), \quad b \otimes [F] \longmapsto [bMF].$$

Let $\text{tors}_{\mathcal{O}}(\mathcal{M}_Q)$ denote its torsion subsheaf: a germ is in it when some nonzero holomorphic germ annihilates it. This is torsion over the specified analytic coefficient ring; its connection to the original split support is the reconstruction of the inclusion and quotient maps below.

Theorem. The actual Mellin map restricts to an isomorphism

$$\boxed{\beta|_{\text{tors}} : \text{tors}_{\mathcal{O}}(\mathcal{M}_Q) \xrightarrow{\sim} \mathcal{O}/(g).}$$

Moreover, $\mathcal{N} = \ker \beta$ has a canonical $\mathcal{K}$ -module structure, and the displayed map canonically splits the sequence:

$$\boxed{\mathcal{M}_Q \cong \mathcal{O}/(g) \oplus \mathcal{N}, \quad \mathcal{K} \otimes_{\mathcal{O}} \mathcal{N} \cong \mathcal{N}.}$$

Proof at every stalk. Fix $\rho \in \mathcal{S}$ and set $x = s - \rho$. If $g(\rho) \neq 0$, Section 4 with $h = t - \rho$ proves that $D - \rho$ is bijective on Q . Consequently x is invertible on $\mathcal{M}_{Q,\rho}$, and this module is a module over the meromorphic field $\mathcal{O}_\rho[1/x]$. Its torsion is zero, as is $\mathcal{O}_\rho/(g)$.

If $r = \text{ord}_\rho g > 0$, the finite projector for this one full packet gives

$$Q = Q[(D - \rho)^r] \oplus (D - \rho)^r Q.$$

On the second summand, $(D - \rho)^r$ is bijective: it is injective by the direct-sum intersection, and

$$(D - \rho)^r Q = (D - \rho)^{2r} Q$$

follows by applying $(D - \rho)^r$ to the direct sum. Hence $D - \rho$ itself is bijective there. Tensoring the direct sum with $\mathcal{O}_\rho$ gives a first summand $\mathcal{O}_\rho/(x^r)$, on which β is the isomorphism calculated in Section 5, and a second summand on which x is invertible. The map β kills the second summand because it is in $x^r \mathcal{M}_{Q,\rho}$, while x^r annihilates its target. Every nonzero germ of $\mathcal{O}_\rho$ is a unit times a power of x , so that second summand is a module over $\mathcal{K}_\rho$ and is torsion-free.

This proves the stated torsion isomorphism at every stalk. It glues without choosing neighbourhood projectors: both the torsion subsheaf and the restricted map β are intrinsic. The inverse of that isomorphism, followed by the torsion inclusion, is the global section of β . Local inverses to nonzero holomorphic scalars on $\mathcal{N}$ are unique, so they glue to its $\mathcal{K}$ -module structure. This proves the theorem.

7.2 The extraction is an actual local-cohomology map At the same stalk the principal-ideal local-cohomology complex is the explicit localization complex

$$R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) = [\mathcal{M}_{Q,\rho} \longrightarrow \mathcal{M}_{Q,\rho}[1/x]], \quad \text{degrees } 0, 1.$$

The decomposition just proved computes it:

$$\boxed{R\Gamma_{(x)}(\mathcal{M}_{Q,\rho}) \xrightarrow{\sim} (\mathcal{O}_\rho/(g))[0].}$$

The finite torsion summand maps to zero upon localization; the meromorphic summand maps isomorphically to itself. The map to the displayed target is β in degree zero and zero in degree one. Thus the supported arithmetic divisor is the actual cohomological image of the localization kernel, not a prescribed deletion of a list of spectral points.

Before analytic extension there is the corresponding operator-localization map

$$Q \longrightarrow A[h_Z^{-1}] \otimes_A Q,$$

with kernel $Q[h_Z(D)]$ and cokernel zero, because $h_Z(D)$ is invertible on its complementary summand. Its supported lift sends that kernel to the internal supported zero, not to external absence.

At every original label λ , reconstruct the localization complex using

$$d(\lambda, m) = (\lambda, m/1), \quad \varepsilon(\lambda, m) = (\lambda, 0).$$

Its internal degree-zero cycles are exactly the reconstructed torsion module. Its internal first cohomology is the support skeleton, since the ordinary first cohomology vanishes. The map $p_{\mathbb{Z}}$ remains

in the structural coefficient square throughout; x is the analytic image of the operator $t - \rho$, not a replacement for the scalar e or τ .

The theorem does not assert $\mathcal{N} = 0$. It computes what local cohomology does to it: localization is an isomorphism on that module, so it contributes no cohomology to the stated local-support complex. Its possible contribution to a different global trace functor has not been declared absent.

A.1.9 8. The residue duality is now inside an actual derived finite comparison

Assume the centres and their orders in h are stable under $\iota(s) = 1 - \bar{s}$. For germs define

$$f^\dagger(s) = \overline{f(1 - \bar{s})}.$$

The chosen monic polynomial satisfies $h^\dagger = (-1)^d h$. Retain that factor. Define the perfect finite algebra pairing

$$R_h(f, u) = \sum_i \text{Res}_{s=\rho_i} \frac{f^\dagger(s)u(s)}{h(s)} ds.$$

It satisfies $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$. No phase is inserted to change this sign. Since $g^\dagger = g$, multiplication by g_h is self-adjoint for this pairing. Therefore it induces a perfect pairing

$$\overline{\text{coker } g_h} \otimes \ker g_h \longrightarrow \mathbb{C}.$$

Well-definedness is the identity $R_h(ga, u) = R_h(a, gu)$; nondegeneracy follows by taking the annihilator of the image in the finite perfect algebra pairing. Through Section 4 this pairs the two actual cohomology groups of $C_+ \otimes_A^{\mathbf{L}} E_h$.

For an exact packet h_Z , set $v_{Z,h} = j_h(g/h_Z)$ as above and

$$j_Z : A_Z \rightarrow \ker g_h = E_Z, \quad [f] \mapsto [j_h(h_Z/g)f].$$

Here $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g) \cong E_Z$ by the specified local quotient maps; h_Z/g is holomorphic at every centre. The local units are not replaced by one. The exact contraction is

$$R_h(f, j_Z k) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger k}{g} ds =: R_Z(f, k).$$

Thus the previously constructed arithmetic residue form is the duality pairing of the actual finite operator comparison after the displayed unit morphism.

The Jacobian map is still $J_g[k] = [g'k]$. Its contraction is

$$R_h(f, j_Z J_g k) = R_Z(f, J_g k) = \sum_{\rho \in Z} r_\rho \overline{f(1 - \bar{\rho})} k(\rho) = \text{Tr}(M_{f^\dagger k} | A_Z).$$

In particular the factor 2 in $g = 2\xi$ is retained in both the denominator and the Jacobian. The trace contraction is invariant under their combined explicitly specified transformation, not by deleting that factor in only one place.

Every map lifts to the original support fibres. Dual support arrows are precomposition maps on the opposite index category. A vanishing pairing value is the supported scalar $e_{\mathbb{C}}$; an absent argument gives τ .

A.1.10 9. Cohomological placement and arithmetic traces

On the tau-base chart use the one-leg sheaf $\mathcal{T}_+$ and its A -action. The bounded projective resolution from TAU_BASE_MODEL.md gives the canonical map

$$\mathbf{A}_\tau(\mathcal{T}_+ \otimes_A^{\mathbf{L}} E_h) \xrightarrow{\sim} \mathbf{A}_\tau(\mathcal{T}_+) \otimes_A^{\mathbf{L}} E_h \xrightarrow{\sim} [E_h \xrightarrow{g_h} E_h].$$

These isomorphisms use finite evaluation sums and the displayed free/flat resolutions. They are not compact-support base-change assertions for a different arithmetic scheme.

The packet projector is now an actual finite-rank endomorphism of the original $H_\tau^1 = Q$ (and of each of its retained support fibres). For an admissible smooth compactly supported multiplicative test k ,

$$H_k F(x) = \int_0^\infty k(a) F(x/a) \frac{da}{a}$$

commutes with the projector. Its ordinary finite-rank trace is

$$\boxed{\mathrm{Tr}(H_k \Pi_Z | Q) = \sum_{\rho \in Z} r_\rho \mathcal{M}k(\rho).}$$

This is a finite-rank trace on the original cohomology, using its explicit finite invariant image. It is stronger than declaring that a surjective finite observation has a trace. Infinite trace summation still requires the specified topology and admissible test class.

On r -fold products over the tau-point, the already calculated external tensor map gives $H^r = Q^{\otimes r}$. The operator $\Pi_Z^{\otimes r}$ is a finite projector onto $Q[h_Z]^{\otimes r}$. The full nilpotent scaling action there is

$$a^{\rho_1 + \dots + \rho_r} \exp \left((\log a) \sum_{j=1}^r N_{\rho_j, j} \right).$$

For a repeated exponent the exact modulus comparison remains

$$\log \frac{|a^{r\rho}|^2}{a^r} = r(2\Re\rho - 1) \log a.$$

This supplies actual finite invariant objects on which a Deligne-style tensor estimate would have to act. It does not prove that estimate. The source's opposite weight bounds, its global dualizing theory, and the admissibility of its geometric hypotheses have not been inferred from these finite isomorphisms.

A.1.11 10. Status and publication boundary

Established in this note, with written proofs: the differential range identities on the stated spaces; the exact finite operator quotient and torsion kernel; the derived finite comparison; the analytic torsion-subsheaf and local-cohomology identification with $\mathcal{O}/(2\xi)$; finite spectral direct summands with explicit projectors; the original split lifts and smaller-label kernels; the residue/Jacobian contraction on that comparison; and compatible finite-rank arithmetic traces and tensor actions.

Not claimed: a global injectivity theorem for the full balanced analytic realization; vanishing of $\mathcal{R}_\infty$; identification of the entire four-point chart category with an arithmetic curve; a full compact-support/ordinary-cohomology realization satisfying the hypotheses of Weil II; positivity of the unshifted Weil form; RH or GRH. The finite tests are not a Lean or analytic proof certificate.

The prior base note and this continuation are published as additive research material beside PR #6. The original scalar core, formal modules, frozen reader and all existing branches remain unchanged. New derivations are AI-assisted work under the owner's programme; the structural source and its Lean implementation retain their own attribution. No copyrighted source corpus or private conversation is included.

References and source interfaces

- Owner's SplitZero programme: the original scalar, support fibre and quotient constructions; PR #6 at 5f6ee6575550d8a20c6fc3990359cb4f67855f80, especially SplitZeroHomology.lean, SplitZeroComplex.lean, SplitZeroInternalQuotient.lean, and SplitZeroFiniteJets.lean.
- TAU_BASE_MODEL.md: the actual characteristic-zero chart, cohomology, right-adjoint and tensor constructions used here. It remains an authored research model, not an accepted geometric RH realization.
- Deligne, P., *La conjecture de Weil. II* (1980), sections 3.3.4-3.3.6 and 6.2.1-6.2.7. Selected supplied TeX passages were reread for the intended comparison/duality architecture; no source proof is republished or claimed fully audited.
- Meyer, R., *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277. Prior spectral-realization context; no theorem from it is silently used to identify the test spaces here.
- The Stacks Project, Derived tensor product, Tag 06XY. The general bounded-flat-complex construction used in Section 4.

B PR 9: Chain descent

Source attribution. Web contribution, pull request #9, [Zeta research repository](#).

Exact revision. 8f99b94d306c3b1aaa817d52c57fa6fd298d511c

Source file. workbenches/tau-chain-descent/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 1ab370cf2db9fce70020bebbab76d3dbd3bd37d39e28c83114eeb0eccfa2f0b3

B.1 Chain-level arithmetic descent over the tau base

Owner-directed research continuation, 12 September 2026. Draft proofs for review; no new Lean certification or RH claim.

B.1.1 0. Fixed inputs and publication scope

This is an additive continuation of Zeta PR #8 at 25689c5376166df37801229bc275ff5a896007d9, particularly workbenches/tau-base-cohomology/FINITE_OPERATOR_COMPARISON.md (Git blob 68f59d098d60708f8e899bb960abbd1135ebb3b4). Its sections 1–5 supply the stated analytic range theorem and the finite packet maps. The new result here is their chain-level descent, its actual extension class, and the connection of that class to residue duality and the arithmetic trace.

The source programme retains

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$p_R(\tau) = 0_R, \quad p_R(r^\bullet) = r.$$

The absolute pointed base and the coefficient square remain

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the first vertical map remains the infinite ring of integers. Every new arrow below is a coefficient or cochain construction over this marked base. No base field, ideal norm, Euler factor, or completion constant is changed.

The supplied derived note and PR #6 provide `Recovery.recoverEquiv`, `Relations.coequalizer_universal`, the internal-homology coequalizer, `ChainMap.homologyKernelEquiv`, and `homology_int_ertwines`. Their algebraic conclusions are used at their stated types. No existing formal module is modified. Analytic inputs from PR #8 remain written research arguments; they are not promoted to Lean theorems.

B.1.2 1. The actual arithmetic source and packet

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |(-x\partial_x)^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}.$$

Retain

$$D = -x\partial_x, \quad \Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x},$$

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad A = \mathbb{C}[t], \quad t \cdot F = DF.$$

The one-leg tau-chart complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad |V| = 0, \quad |\mathcal{B}| = 1, \quad Q = H^1(C_+).$$

The injectivity of theta gives $H^0(C_+) = 0$. The two-leg complex is

$$C_{+-} = [V \oplus V \longrightarrow \mathcal{B}], \quad (\phi, \psi) \mapsto \Theta\phi - \Theta\widehat{\psi}.$$

Its source operator is $(D, 1 - D)$; the inclusion $C_+ \rightarrow C_{+-}$ is $\phi \mapsto (\phi, 0)$ and the identity in degree one. It leaves the extra degree-zero copy of V in C_{+-} present.

Use the unchanged arithmetic multiplier

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

The input function is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2},$$

and the established identities are

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

Fix a nonempty finite packet Z of actual nontrivial zeros, with full orders m_ρ . Set

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = A/(h), \quad v(s) = g(s)/h(s).$$

The specific monic polynomial defines this finite test; it does not rescale g . Write j_h for the complete Taylor-jet/Chinese-remainder map at these centres, and retain the invertible element

$$v_h = j_h v \in E^\times, \quad \varepsilon_Z = v_h^{-1}.$$

For $u \in E$, let $R_Z(u) \in \mathbb{C}[t]_{<d}$ be the unique polynomial representing $\varepsilon_Z u$.

The input range theorem supplies the inverse $S_h^{\mathcal{B}}$ of $h(D)$ on the zero-jet range, and the analogous source inverse S_h^V . For one factor,

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy.$$

The stated source inverse uses $0 < \Re \rho < 1$ and the zero moment. Iteration supplies S_h . Define the actual maps

$$J : \mathcal{B} \rightarrow E, \quad JF = j_h \mathcal{M}F,$$

$$s : E \rightarrow \mathcal{B}, \quad s(u) = S_h^{\mathcal{B}} \Theta(R_Z(u)(D)\phi_*).$$

The inverse is defined because $gR_Z(u)$ has all required zero jets. Directly,

$$\mathcal{M}s(u) = vR_Z(u), \quad Js = 1_E, \quad J\Theta = 0, \quad h(D)s(u) = \Theta(R_Z(u)(D)\phi_*). \quad (1)$$

These equations fix the representative map, including all derivatives of g/h . On Q , the existing projector is $\Pi_Z = \bar{s}\bar{J}$, with image $Q[h(D)]$ and kernel $h(D)Q$.

B.1.3 2. The exact equivariant extension and its class

Define an actual subspace of the original test-function space:

$$\mathcal{B}_Z = \{F \in \mathcal{B} : [F] \in Q[h(D)]\}.$$

Then

$$\mathcal{B}_Z = \Theta V \oplus s(E) \quad \text{as complex vector spaces,}$$

with maps

$$(\phi, u) \mapsto \Theta\phi + s(u), \quad F \mapsto (\Theta^{-1}(F - sJF), JF). \quad (2)$$

For the inverse, $F - sJF$ has zero class in Q by the packet projector theorem; injectivity of theta makes its source unique. The intersection is zero by $Js = 1$ and $J\Theta = 0$.

The original operator D preserves $\mathcal{B}_Z$, and there is a short exact sequence of A -modules

$$\boxed{0 \longrightarrow V \xrightarrow{\Theta} \mathcal{B}_Z \xrightarrow{J} E \longrightarrow 0.} \quad (3)$$

The arrows in (3), not just the three modules, specify its extension class. Compute it using the resolution

$$0 \longrightarrow A \xrightarrow{\times h} A \longrightarrow E \longrightarrow 0.$$

The resulting isomorphism $\text{Ext}_A^1(E, V) \cong V/h(D)V$ sends an extension to $h(D)b$ in the source, where b is any lift of $1 \in E$. Fixing this explicit rule fixes its sign. Under the existing map $[\phi] \mapsto j_h H_\phi$, the class of (3) is

$$\boxed{[\mathcal{B}_Z] = \varepsilon_Z = j_h(h/g) \in E^\times.} \quad (4)$$

Proof: use the lift $s(1)$ and (1). Its source boundary is $R_Z(1)(D)\phi_*$, whose H -jet is ε_Z .

In particular, (3) has no A -linear section. This also follows immediately from the injectivity of $h(D)$ on $\mathcal{B}$: an A -linear map $E \rightarrow \mathcal{B}$ has image killed by $h(D)$ and is therefore zero. The map s in (2) is the explicit complex-linear section, and the next section computes exactly how it fails to intertwine the arithmetic action. This non-splitting occurs for every nonempty packet, including critical-line packets; it is not an RH-failure criterion.

The general use of extension classes is standard homological algebra (Stacks, Tags 010I and 06XP). Formula (4) calculates the specific extension arising from this theta complex.

B.1.4 3. The scaling defect is an original theta boundary

For $P \in A$ and $u \in E$, define

$$q_P(u) = \frac{P(t)R_Z(u) - R_Z(Pu)}{h(t)} \in A, \quad k_P(u) = q_P(u)(D)\phi_* \in V. \quad (5)$$

The numerator is divisible by h , since both of its residues are $P\varepsilon_Z u$. Using (1) and the injectivity of $h(D)$ on $\mathcal{B}$ gives

$$\boxed{P(D)s - sM_P = \Theta k_P.} \quad (6)$$

No quotient has yet been applied in (6). The right side is an actual theta function. The composition law is

$$\boxed{k_{PQ} = P(D)k_Q + k_P M_Q.} \quad (7)$$

It follows either by expanding the numerator of (5), or by expanding the commutator in (6) and using injectivity of theta.

For $P = t$, the numerator of (5) has degree at most d . Hence

$$\ell_Z(u) = [t^{d-1}]R_Z(u), \quad q_t(u) = \ell_Z(u), \quad \boxed{Ds(u) - s(tu) = \ell_Z(u)\Theta\phi_*} \quad (8)$$

Thus, in the exact coordinates (2), the original operator is

$$\boxed{D_{\mathcal{B}_Z} = \begin{pmatrix} D_V & \phi_*\ell_Z \\ 0 & M_t \end{pmatrix}.} \quad (9)$$

Its off-diagonal map is rank one and its coefficient is determined by the retained local arithmetic unit. It is not chosen to improve a spectral estimate.

For one simple zero,

$$h = t - \rho, \quad \varepsilon_Z = 1/g'(\rho),$$

$$s(1) = \frac{1}{g'(\rho)} x^{-\rho} \int_x^\infty \Theta\phi_*(y) y^{\rho-1} dy,$$

$$\boxed{(D - \rho)s(1) = \frac{1}{g'(\rho)} \Theta\phi_*} \quad (10)$$

The coefficient is $1/(2\xi'(\rho))$, with the original factor two retained.

The full dilation group For $a > 0$, retain $U_a F(x) = F(x/a)$ and $A_a = M_{j_h(a^s)}$ on E . Define

$$k_a(u) = S_h^V (U_a(R_Z(u)(D)\phi_*) - R_Z(A_a u)(D)\phi_*). \quad (11)$$

The source of S_h^V has zero H -jets because $a^s \varepsilon_Z u$ and $\varepsilon_Z A_a u$ have equal jets. All powers use the real logarithm of a . The exact identities are

$$U_a s - s A_a = \Theta k_a, \quad k_{ab} = U_a k_b + k_a A_b, \quad k_1 = 0. \quad (12)$$

The same construction applies to a smooth compactly supported multiplicative test f : replace U_a by $H_f = \int f(a) U_a da/a$ and A_a by $M_{j_h \mathcal{M} f}$. The source integral belongs to V ; its H -jets give the same cancellation. These are actual linear maps, not a choice of a formal matrix on E alone.

B.1.5 4. A derived equivariant inclusion with its full homotopy

Put $P_Z = sJ : \mathcal{B} \rightarrow \mathcal{B}$. It is a finite-rank idempotent, and $P_Z\Theta = 0$. Consequently

$$\mathfrak{p}_Z = (0, P_Z) : C_+ \rightarrow C_+ \quad (13)$$

is a cochain idempotent. Its image is the two-term complex $[0 \rightarrow s(E)]$, complex-linearly isomorphic to $E[-1]$.

For the operator $P(D)$ on C_+ , define a degree-minus-one map with sole component

$$\mathbf{H}_P^1 = k_P J : \mathcal{B} \rightarrow V.$$

Equation (6) gives

$$\boxed{P(D)\mathfrak{p}_Z - \mathfrak{p}_Z P(D) = d\mathbf{H}_P + \mathbf{H}_P d.} \quad (14)$$

In degree one this is $\Theta k_P J$; in degree zero it is zero because $J\Theta = 0$. For dilation use $\mathbf{H}_a^1 = k_a J$. Equation (12) is the coherence law of those homotopies.

There is also a strictly A -linear derived presentation. Set

$$C_Z = [V \xrightarrow{\Theta} \mathcal{B}_Z].$$

In the coordinates (2) its differential is $\phi \mapsto (\phi, 0)$ and its action is (9). The maps

$$\iota : C_Z \hookrightarrow C_+, \quad \pi : C_Z \rightarrow E[-1], \quad \pi^0 = 0, \pi^1 = J \quad (15)$$

are strictly A -linear cochain maps, and π is a quasi-isomorphism. Thus the actual equivariant inclusion is the roof

$$\boxed{E[-1] \xleftarrow[\simeq]{\pi} C_Z \xrightarrow{\iota} C_+.} \quad (16)$$

The complex-linear deformation retraction also has an exact contracting homotopy: on $\mathcal{B}_Z$, send $\Theta\phi + s(u)$ to $\phi \in V$. Its equation is $1 - s\pi = dH + Hd$ (with s interpreted as the degree-one cochain section). This homotopy is not claimed A -linear; the A -linear replacement is the roof (16).

The strictly A -linear map $j : C_+ \rightarrow E[-1]$ is J in degree one and zero in degree zero. It satisfies $j\iota = \pi$. This gives a split retract in $D(A)$ without falsely asserting an A -linear section into $\mathcal{B}$.

On the two-leg complex use $\mathfrak{p}_Z^0 = 0$ on $V \oplus V$, $\mathfrak{p}_Z^1 = P_Z$, and $\mathbf{H}_a^1(F) = (k_a JF, 0)$. The source dilation is $(U_a, aU_{1/a})$, so (14) holds with its original two-leg differential. The inclusion $C_+ \rightarrow C_{+-}$ commutes with these cochain projectors. The extra degree-zero cohomology of C_{+-} is killed by this projector; it remains in the complementary complex.

B.1.6 5. The mixed-support homotopies require the actual synchronization map

Let $L = \mathcal{P}(\{+, -\})$ be the original chart-support lattice. Its three active complexes are

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad C_- = [V \xrightarrow{-\Theta\mathcal{F}} \mathcal{B}], \quad C_{+-} = [V \oplus V \xrightarrow{\Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Here $\mathcal{F}$ is the specified Fourier transform, with $\mathcal{F}^2 = 1$ on even functions. The source inclusions into the joint fibre are $v \mapsto (v, 0)$ and $v \mapsto (0, v)$; degree-one transports are identities. The bottom complex is zero in this particular model. The new formalization allows a nonzero bottom fibre in the general reconstruction; no assertion about all diagrams is obtained by specializing this model.

The reconstructed operations are

$$(\lambda, x) + (\mu, y) = (\lambda \vee \mu, \rho_{\lambda, \lambda \vee \mu} x + \rho_{\mu, \lambda \vee \mu} y),$$

$$e(\lambda, x) = (\lambda, 0), \quad \tau(\lambda, x) = (\perp, 0). \quad (17)$$

The cochain projector $\tilde{p}_Z$ is zero-in-its-fibre in degree zero and P_Z in degree one. It commutes with the support transports. So does the actual action: U_a on the plus source, $aU_{1/a}$ on the minus source, their direct sum on the joint source, and U_a at every active overlap.

The objectwise homotopies are, however, different maps. With $b = k_a JF$, they are

$$H_{a,+}^1(F) = b, \quad H_{a,-}^1(F) = -\hat{b}.$$

Their images in the joint source are $(b, 0)$ and $(0, -\hat{b})$. The difference is the explicit cycle

$$(b, \hat{b}) \in \ker d_{+-} = H^0(C_{+-}). \quad (18)$$

For nonzero k_a , there is no label-preserving family of homotopies compatible with both source inclusions: the joint component would have to equal both $(b, 0)$ and $(0, -\hat{b})$. This says exactly which naturality square fails. It does not discard the cohomology or the arithmetic action.

The replacement is the programme's actual synchronization. Define the support map

$$r_L(\perp) = \perp, \quad r_L(\lambda) = \{+, -\} \quad (\lambda \neq \perp).$$

In degree zero use literal multiplication by $E = (1, e)$ on the original two-coordinate support module:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea).$$

In degree one define $\mathfrak{r}^1(\lambda, F) = (r_L \lambda, F)$ for active labels and preserve global zero. An absent source coordinate becomes its supported vector zero, so the source differential has unchanged amplitude. Hence

$$d\mathfrak{r}^0 = \mathfrak{r}^1 d, \quad \mathfrak{r}^2 = \mathfrak{r}. \quad (19)$$

This uses the operation verified in the synchronization attachment, now on the actual complex. The fixed active fibre is exactly C_{+-} . On degree-one homology, the map is

$$(\lambda, [F]) \mapsto (\{+, -\}, [F]).$$

It has no nonzero amplitude kernel, because all degree-one transports are the identity of Q . The original mask is retained through

$$x \mapsto (\text{supp}(x), \mathfrak{r}x).$$

This map is injective in each degree of this chart complex: in degree zero the coordinate inclusions are injective, and in degree one the transport is the identity. This injectivity is a proved property of these maps, not an assumption for general diagrams with noninjective transports.

There are now two actual split-linear homotopies with the same support map r_L . Define

$$\tilde{H}_a^{\text{left},1}(\lambda, F) = (\{+, -\}, (k_a JF, 0)),$$

$$\tilde{H}_a^{\text{right},1}(\lambda, F) = (\{+, -\}, (0, -\mathcal{F}k_a JF)),$$

and send global zero to global zero. Both satisfy

$$\tau(\tilde{U}_a \tilde{p}_Z + (-1)^{\bullet} \tilde{p}_Z \tilde{U}_a) = \tilde{d} \tilde{H}_a + \tilde{H}_a \tilde{d}. \quad (20)$$

In degree one this is (6); in degree zero it follows from $Jd = 0$. The two choices differ by the retained cycle (18). No averaging or removal of that cycle is performed. This is the precise role of synchronization here: it permits a common supported comparison fibre for the homotopy, while the original mask and the difference of homotopies remain recorded.

In the simple-zero case (10), the defect in any source support is the nonzero chain amplitude $\Theta(\phi_*/g'(\rho))$. Its internal homology class is the zero of that support, not external absence.

For every chain comparison $f : C \rightarrow C'$, the supplied formalized kernel equivalence remains

$$\{z \in Z(C) : f(z) \in B(C')\} / B(C) \xrightarrow{\sim} \ker H(f).$$

Applied to π in (15), it computes the zero kernel in degree one from its actual exact sequence. Applied to support synchronization, it yields the degree-one assertion just proved and the separately retained degree-zero data. It is not applied to a different global comparison without carrying that comparison's maps.

For any smaller D -stable admitted source $W \subset V$, the map $Q_W \rightarrow Q$ has kernel $\Theta V / \Theta W$. A homotopy value may not already lie in W . The explicit replacement is enlargement by the actual source values, with this quotient map and its kernel retained. The present chart uses full V , where every displayed homotopy is defined.

B.1.7 6. The same extension class produces the arithmetic duality factor

Assume Z is stable under $\rho \mapsto 1 - \bar{\rho}$. Then

$$f^\dagger(s) = \overline{f(1 - \bar{s})}, \quad h^\dagger = (-1)^d h.$$

The perfect finite pairing is

$$R_h(f, u) = \sum_{\rho \in Z} \text{Res}_{s=\rho} \frac{f^\dagger(s) u(s)}{h(s)} ds,$$

with exact symmetry $\overline{R_h(u, f)} = (-1)^{d+1} R_h(f, u)$.

Identify E with the actual divisor algebra $A_Z = \prod_{\rho \in Z} \mathcal{O}_\rho / (g)$ by polynomial evaluation and the complete Chinese-remainder jets. The nonzero local units in g/h make this an isomorphism. The operator represented by the extension class is

$$M_{\varepsilon_Z} : E \rightarrow E.$$

Substituting (4) into the residue formula gives

$$R_h(f, M_{\varepsilon_Z} u) = \sum_{\rho \in Z} \text{Res}_\rho \frac{f^\dagger u}{g} ds = R_Z(f, u). \quad (21)$$

This is independent of representatives: changing ε_Z by h times a germ adds a holomorphic differential. Thus the actual extension class is precisely the unit morphism converting the polynomial-resolution residue form into the arithmetic residue form.

Let $J_g = M_{j_h g'}$. Then

$$R_h(f, M_{\varepsilon_Z} J_g u) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} u(\rho) = \text{Tr}(M_{f^\dagger u} | E). \quad (22)$$

Locally $g = z^m w(z)$ gives $g'/g = m/z + w'/w$. This proves the formula and retains every multiplicity. The product $\varepsilon_Z J_g$ carries both factors associated to $g = 2\xi$.

The pairing-valued support map sends a zero value to $e_{\mathbb{C}}$ and an absent argument to τ . Its original input support and relation fibre remain available upstream. Formula (22) is a calculated trace, not a positivity assertion.

Reflection is strict for this specified section Define conjugate-linear involutions

$$j_B F(x) = x^{-1} \overline{F(1/x)}, \quad j_V \phi = \widehat{\phi}, \quad j_E u = u^\dagger.$$

They satisfy $\Theta j_V = j_B \Theta$ and $J j_B = j_E J$. The actual source function is Fourier-fixed: writing $\phi_* = D(D-1)e^{-\pi x^2}$, Fourier sends the two factors to $(1-D)(-D) = D(D-1)$ and fixes the Gaussian. It is real, so $j_V \phi_* = \phi_*$.

Since $\varepsilon_Z^\dagger = (-1)^d \varepsilon_Z$ and reflection preserves polynomial degree,

$$R_Z(j_E u) = (-1)^d (R_Z(u))^\dagger.$$

Uniqueness of the inverse of $h(D)$ on its range gives

$$j_B S_h^{\mathcal{B}} = (-1)^d S_h^{\mathcal{B}} j_B.$$

Both factors $(-1)^d$ are retained in the substitution, and their product is one. Therefore

$$j_B s = s j_E, \quad j_B P_Z = P_Z j_B. \quad (23)$$

This is an actual equality for this section; it would not follow for an arbitrary choice of representatives. Its compatibility with the nonzero dilation homotopy is

$$j_V k_a = a k_{1/a} j_E. \quad (24)$$

Indeed, apply j_B to (12), use $j_B U_a = a U_{1/a} j_B$ and (23), then use injectivity of theta. The split lift is semilinear through $r^\bullet \mapsto \bar{r}^\bullet$, which fixes both e and τ .

B.1.8 7. A finite-rank Lefschetz trace on the actual chain complex

For $f \in C_c^\infty(\mathbb{R}_{>0})$, let U_f be its convolution action on C_+ . The cochain endomorphism

$$p_Z U_f = (0, s J H_f)$$

has finite-rank components. Its cochain supertrace is therefore defined without an infinite-dimensional regularization:

$$\text{Str}(p_Z U_f | C_+) = -\text{Tr}(s J H_f | \mathcal{B}) = -\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (25)$$

The finite-rank identity $\text{Tr}(AB) = \text{Tr}(BA)$ reduces the middle expression to $\text{Tr}(J H_f s | E)$; this is multiplication by $j_h \mathcal{M}f$. Equation (25) equals the supertrace of the induced operator on homology.

On C_{+-} its degree-zero contribution is still zero, since the same cochain projector has zero degree-zero component.

The boundary representatives in (6), (14), and (20) are retained even though their contribution to homology is zero. The finite-rank trace does not delete their complex or assign its zero to external absence.

For the n -fold tensor complex over the tau-base product, use the original coefficient tensor over $\mathbb{C}$ and its differential

$$d(x_1 \otimes \cdots \otimes x_n) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_n.$$

The projector $\mathfrak{p}_Z^{\otimes n}$ has image $E^{\otimes n}[-n]$. For independently specified tests $f_1, \dots, f_n$,

$$\text{Str} \left(\bigotimes_j (\mathfrak{p}_Z \cup_{f_j}) \right) = (-1)^n \prod_j \sum_{\rho \in Z} m_\rho \mathcal{M} f_j(\rho). \quad (26)$$

Let $A = U_a \mathfrak{p}_Z$ and $B = \mathfrak{p}_Z U_a$ as cochain maps. A homotopy for $A^{\otimes n} - B^{\otimes n}$ is

$$\sum_{j=1}^n B^{\otimes(j-1)} \otimes H_a \otimes A^{\otimes(n-j)},$$

with the displayed Koszul sign on homogeneous inputs when the degree-minus-one map crosses the preceding factors. The telescoping identity and (14) prove it. The full support mask maps to the product of its input masks; vanishing products of represented amplitudes stay supported.

B.1.9 8. What now enters the Weil II programme

The structural base, the infinite quotient, and the actual theta complex are unchanged. The new objects are the extension (3), its unit class (4), its synchronized homotopy (20), its strict equivariant derived roof (16), and the chain trace (25). The arithmetic residue unit in (21) is now supplied by that very extension rather than appended without its source.

Deligne's inspected comparison-image argument (Weil II 3.3.4–3.3.6) requires two weight bounds on the same cohomological image. The inspected derived formulation (6.2.1–6.2.7) retains the dualizing operation, its degree and Tate shifts, and its exchange of direct-image functors. This continuation supplies coherent arithmetic maps and finite trace operations for the tau model; it does not transfer the finite-field hypotheses or assert those two bounds.

The retained error is explicit: it is the original boundary Θ_{k_a} on representatives, with the nonzero Ext class (4). It is removed at homology by the original supported quotient, not by making its input absent. A geometric comparison can now be tested for compatibility with this class and the residue map (21). A complex-linear section that ignores (6) fails that test.

No infinite sum of packet projectors is asserted to converge. The existing meromorphic/kernel summand from PR #8 remains in its own exact sequence, and a different global trace functor must retain or analyze its contribution. The source's finite-jet isomorphisms and the new finite-rank chain trace do not by themselves set that summand to zero.

B.1.10 9. Verification and references

The new finite checker tests polynomial remainder identities, the extension class, rank-one scaling defects, cochain homotopies, dilation cocycles, residue contraction, tensor signs, trace identities, and

original support operations. These finite algebraic models are not numerical tests of zeta zeros and not certificates for the analytic range theorem. No Lean execution is claimed for this continuation.

Read source interfaces:

- PR #8 at the pinned revision above, `FINITE_OPERATOR_COMPARISON.md`, sections 1–10; `TAU_BASE_MODEL.md` is its already-published model.
- Owner’s `SplitZero_Derived_Mathematics_2026-09-12(1).md`, sections 1–7. In particular its allowance of a nonzero bottom fibre and the exact homology-kernel/action theorems are retained.
- `formal/splitzero/SplitZeroHomology.lean`, lines 175–end, read at the same revision; blob `7b9e5c49a494dd0c751133e0234b3e01048f7b67`.
- Supplied Deligne TeX excerpts: French segment 2, original lines 882–950; segment 3, original lines 675–756. The used equations are the image bounds and duality operations, not every assertion in the historical transcription. No source corpus is republished with this contribution.
- The Stacks Project, Tags 010I and 06XP, for extension classes and their projective-resolution interpretation.
- R. Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277, for prior spectral-realization context. No global priority claim is made here.

New calculations are AI-assisted work under the owner’s programme. Existing mathematical files, formalizations, licensing, and source attribution remain untouched.

C PR 10: Global retraction

Source attribution. Web contribution, pull request #10, [Zeta research repository](#).

Exact revision. d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7

Source file. workbenches/tau-global-retraction/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 496f3ec752e3e67ddafaae7423feb300b1faa789ca6907872f8d539bdf6e1859

C.1 A global retraction of the original theta complex over the tau base

Owner-directed continuation of Zeta PR #9, 12 September 2026. New derivations are submitted as research proofs for review, not as Lean-certified analytic theorems.

C.1.1 0. Fixed source, result, and provenance

The base is PR #9 at 8f99b94d306c3b1aaa817d52c57fa6fd298d511c; its chain-descent note has Git blob 86db838e79d10a8fe7d4376303706c4ea64570a1. Its source and the owner's derived SplitZero attachment remain the inputs. This continuation constructs an inverse on the entire original theta image, not only on a chosen finite spectral packet.

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\mathfrak{b}_\tau = \operatorname{Spec} \mathbf{F}_{1,\tau}, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The first vertical target is still infinite. The geometric base, the original additive laws, and the arithmetic quotient are unchanged. The auxiliary cutoffs below are parameters of an explicit continuous representative map; they do not change an Euler factor, the theta relation space, or the completed zeta function.

Result. For every pair of the cutoffs specified in section 3 there is an explicitly constructed continuous complex-linear map

$$\Lambda_{\alpha,\beta} : \mathcal{B} \longrightarrow V, \quad \boxed{\Lambda_{\alpha,\beta}\Theta = \operatorname{id}_V}. \quad (2)$$

Consequently the actual image ΘV is closed and complemented in the stated strong-Schwartz topology. The original quotient $Q = \mathcal{B}/\Theta V$ is Hausdorff. The note constructs its global section, the full supported cochain homotopy, the exact change of the finite sections in #9, its continuous dual sequence, and the completed tensor powers. The proof does not assume RH, simplicity of any zero, or vanishing of the separate balanced-Mellin kernel.

Spectral realizations on nuclear Frechet spaces are prior work, notably R. Meyer, arXiv:math/0412277. That paper's abstract was consulted for scope, but no theorem from an unread body of that paper is imported. The proof below is given for the workbench's precise spaces. No global novelty claim is made.

C.1.2 1. Spaces and all transforms

Let

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : p_{b,k}(F) := \sup_{x>0} x^b |D^k F(x)| < \infty \text{ for every } b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x. \quad (3)$$

These seminorms give $\mathcal{B}$ its Frechet topology; under $x = e^y$ they are the weighted uniform derivative seminorms on smooth functions of y . Completeness follows by uniform convergence on compact sets of every derivative, followed by the weighted bounds. V has its closed-subspace Schwartz topology.

The original maps are

$$\begin{aligned} \Theta\phi(x) &= \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \\ \mathcal{F}\phi(\xi) &= \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad \mathcal{F}^{-1}u(x) = \int_{\mathbb{R}} u(\xi) e^{2\pi i x \xi} d\xi, \\ JF(x) &= x^{-1}F(1/x), \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (4)$$

Poisson summation, with both source moments zero, gives

$$\Theta\mathcal{F} = J\Theta, \quad J^2 = 1, \quad DJ = J(1 - D). \quad (5)$$

The theta sum and each derivative converge absolutely on closed subsets of $x > 0$. At infinity, Schwartz estimates give all $p_{b,k}$ bounds. At zero use (5) and the same estimates on $\hat{\phi}$. This proves continuity $\Theta : V \rightarrow \mathcal{B}$. Also J is continuous on $\mathcal{B}$: $p_{b,k}(JF)$ is bounded by the binomial sum of $p_{1-b,j}(F)$, $0 \leq j \leq k$.

The original complex is $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, over $A = \mathbb{C}[t]$ with t acting as D . The two-leg tau chart retains $C_{+-} = [V^2 \rightarrow \mathcal{B}]$ with differential $\Theta(\phi - \hat{\psi})$ and source action $(D, 1 - D)$.

C.1.3 2. Arithmetic inversion on the two exterior regions

Let μ be the ordinary Mobius function, characterized by

$$\sum_{d|n} \mu(d) = \begin{cases} 1, & n = 1, \\ 0, & n > 1. \end{cases}$$

For $F \in \mathcal{B}$ define, for $x \neq 0$,

$$\boxed{\mathcal{U}F(x) = \frac{1}{2} \sum_{n \geq 1} \mu(n) F(n|x|)}. \quad (6)$$

The factor $1/2$ is the inverse of the full nonzero-integer theta sum in (4). The map is complex-linear into smooth even functions on $\mathbb{R} \setminus \{0\}$. It is not asserted to be smooth at zero for arbitrary F .

Here are the required estimates. For $j \geq 0$, let $P_j(T) = (-1)^j T(T+1) \cdots (T+j-1)$, with $P_0 = 1$. Then $x^j \partial_x^j = P_j(D)$. If $C_{N,j}(F)$ is the sum of the $p_{N,k}(F)$ weighted by the absolute coefficients of P_j , then, for $x > 0$ and integer $N > 1$,

$$|\partial_x^j \mathcal{U}F(x)| \leq \frac{1}{2} \zeta(N) C_{N,j}(F) x^{-N-j}. \quad (7)$$

Indeed each summand is bounded by $C_{N,j}(F) n^{-N} x^{-N-j}/2$. This proves convergence of every derivative uniformly for $|x| \geq \delta > 0$, and gives rapid decay there. It also proves continuity of the cut-off exterior maps used below.

For every $\phi \in V$,

$$\boxed{\mathcal{U}\Theta\phi(x) = \phi(x), \quad \mathcal{U}J\Theta\phi(x) = \widehat{\phi}(x) \quad (x \neq 0).} \quad (8)$$

For the first equality the double series is absolutely convergent for each $x > 0$: its absolute value is bounded by $\sum_{k \geq 1} d(k) |\phi(kx)|$, which is finite by Schwartz decay. Rearrange by $k = mn$ and use the displayed divisor identity. Evenness handles negative x . The second equality follows from (5) and the first equality applied to $\widehat{\phi} \in V$.

In the split lift, every $\mu(n)$ is the supported integer $\mu(n)^\bullet$, including $\mu(n) = 0$. A convergent sum within a specified active fibre begins at that fibre's zero and is taken in that fibre's vector-space topology. Formula (8) thus lifts to a support-preserving equality. A computed cancellation is not external absence.

C.1.4 3. The reconstruction operator, with the strict contraction proved

Choose and retain real, even $\alpha, \beta \in C_c^\infty(\mathbb{R})$ satisfying $0 \leq \alpha, \beta \leq 1$ and equal to 1 on neighborhoods of zero. Their supports and all values are part of the auxiliary data. Work on $\mathcal{H} = L^2(\mathbb{R}, dx)$ with the Fourier convention in (4).

Define bounded operators

$$A_\alpha = M_\alpha, \quad B_\beta = \mathcal{F}^{-1} M_\beta \mathcal{F}, \quad T_{\alpha,\beta} = A_\alpha B_\beta. \quad (9)$$

Lemma. $T_{\alpha,\beta}$ is compact and $c_{\alpha,\beta} := \|T_{\alpha,\beta}\| < 1$.

Proof. Its kernel is $\alpha(x)\check{\beta}(x-y)$, whose squared integral is $\|\alpha\|_2^2 \|\beta\|_2^2$. Hence it is Hilbert-Schmidt. Both factors are contractions. If the norm were 1, compactness would supply a unit vector f attaining it. Equality in

$$\|A_\alpha B_\beta f\| \leq \|B_\beta f\| \leq \|f\|$$

forces $\widehat{f}$ to be supported where $\beta = 1$; there $B_\beta f = f$. The first equality forces f to be supported where $\alpha = 1$. Both supports are compact. The inverse Fourier transform of compactly supported $\widehat{f} \in L^2$ is entire, by differentiation under its finite-interval integral. Its continuous restriction agrees with the compactly supported f , so it vanishes on an open real interval and hence identically. This contradicts $\|f\| = 1$. Thus $c_{\alpha,\beta} < 1$.

Consequently the exact Neumann inverse exists:

$$\boxed{(1 - T_{\alpha,\beta})^{-1} = \sum_{k \geq 0} T_{\alpha,\beta}^k, \quad \|(1 - T_{\alpha,\beta})^{-1}\| \leq \frac{1}{1 - c_{\alpha,\beta}}.} \quad (10)$$

No uniform bound as the cutoffs vary is claimed. In particular the gap $1 - c_{\alpha,\beta}$ is retained, not replaced by a positive universal constant.

The operator $T_{\alpha,\beta}$ maps L^2 continuously to the Schwartz space. For each derivative, compact Fourier support and Cauchy-Schwarz bound $\partial^j B_\beta f$ uniformly by a constant times $\|f\|_2$; multiplication by the smooth compactly supported α gives every Schwartz seminorm.

Define the two exterior data and their combination:

$$a_F = (1 - \alpha)\mathcal{U}F, \quad b_F = (1 - \beta)\mathcal{U}JF,$$

$$Y_{\alpha,\beta}F = a_F + \alpha\mathcal{F}^{-1}b_F. \quad (11)$$

The exterior terms are extended by zero near the origin, where their cutoff is identically zero. Estimates (7) prove that both are even Schwartz functions and depend continuously on F . Therefore $Y : \mathcal{B} \rightarrow \mathcal{S}(\mathbb{R})^{\text{even}}$ is continuous.

Put

$$\Phi_{\alpha,\beta}F = (1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta}F. \quad (12)$$

This initially defines an L^2 vector, but the identity $\Phi F = YF + T\Phi F$ proves it is Schwartz. If q is any Schwartz seminorm and $q(Tu) \leq C_q\|u\|_2$, then

$$q(\Phi F) \leq q(YF) + \frac{C_q}{1 - c_{\alpha,\beta}}\|YF\|_2. \quad (13)$$

This is an explicit continuity estimate. Both sides are even because the cutoff operators preserve parity.

For $F = \Theta\phi$, equations (8) give

$$Y\Theta\phi = (1 - A_\alpha)\phi + A_\alpha(1 - B_\beta)\phi = (1 - T_{\alpha,\beta})\phi,$$

hence

$$\boxed{\Phi_{\alpha,\beta}\Theta\phi = \phi.} \quad (14)$$

C.1.5 4. Keep both original moment conditions

For arbitrary F , (12) need not satisfy the two source moments. The correction is the following specified projection, not an alteration of V .

Retain

$$\gamma_0(x) = e^{-\pi x^2}, \quad \gamma_2(x) = 2\pi x^2 e^{-\pi x^2}.$$

Their moment columns $(\phi(0), \int \phi)$ are $(1, 1)$ and $(0, 1)$. Define

$$\mathbf{m}(\phi) = (\phi(0), \int_{\mathbb{R}} \phi), \quad j_0(a, b) = a\gamma_0 + (b - a)\gamma_2,$$

$$P_V\phi = \phi - j_0\mathbf{m}(\phi) = \phi - \phi(0)\gamma_0 - \left(\int \phi - \phi(0)\right)\gamma_2. \quad (15)$$

Then $\mathbf{m}j_0 = 1$, $P_V^2 = P_V$, $\text{im } P_V = V$, and $P_V|_V = 1$. All maps are continuous.

The promised map is

$$\boxed{\Lambda_{\alpha,\beta} = P_V(1 - T_{\alpha,\beta})^{-1}Y_{\alpha,\beta} : \mathcal{B} \rightarrow V.} \quad (16)$$

Equation (14) and $P_V|_V = 1$ prove (2). The discarded moment vector has the recorded map $m\Phi : \mathcal{B} \rightarrow \mathbb{C}^2$ and the explicit correction $j_0 m\Phi$. It vanishes on the actual theta image. No endpoint constant was absorbed into the zeta function.

C.1.6 5. The full quotient is closed and has a continuous section

Write $\Lambda = \Lambda_{\alpha,\beta}$, and define

$$P_\Theta = \Theta\Lambda, \quad K = 1_{\mathcal{B}} - \Theta\Lambda. \quad (17)$$

Direct substitution gives

$$P_\Theta^2 = P_\Theta, \quad K^2 = K, \quad \ker K = \Theta V, \quad \text{im } K = \ker \Lambda. \quad (18)$$

Since K is continuous, its kernel is closed. Thus $Q = \mathcal{B}/\Theta V$ with its actual quotient topology is Frechet and Hausdorff. If $q : \mathcal{B} \rightarrow Q$ is the original quotient map, the continuous section is

$$\boxed{s : Q \rightarrow \mathcal{B}, \quad s[F] = KF, \quad qs = 1, \quad \Lambda s = 0.} \quad (19)$$

The explicit topological direct-sum isomorphism is

$$\boxed{V \oplus Q \xrightarrow{\sim} \mathcal{B}, \quad (\phi, u) \mapsto \Theta\phi + su, \quad F \mapsto (\Lambda F, qF).} \quad (20)$$

For a convergent sequence $\Theta\phi_n \rightarrow F$ in $\mathcal{B}$, one can check the closedness without any abstract theorem: $\phi_n = \Lambda\Theta\phi_n \rightarrow \Lambda F$ in V , so $F = \Theta\Lambda F$ by continuity of theta.

This computes the earlier Hausdorff arrow:

$$Q^{\text{alg}} = \mathcal{B}/\Theta V \longrightarrow Q^{\text{H}} = \mathcal{B}/\overline{\Theta V}, \quad [F] \mapsto [F], \quad (21)$$

as an isomorphism with zero amplitude kernel. It does not say the map from Q into its product of spectral jets is injective. The latter map and the balanced analytic map still have their own named kernels:

$$Q \rightarrow \prod_{\rho} \mathcal{O}_{\rho}/(g), \quad \mathcal{O} \otimes_{\mathbb{C}[t]} Q \xrightarrow{\beta} \mathcal{O}/(g).$$

No claim that $\mathcal{R}_{\infty}$ or $\ker \beta$ vanishes is inferred from (21).

Smaller admitted relations For any specified subspace $W \subset V$, retain

$$Q_W = \mathcal{B}/\Theta W \longrightarrow Q, \quad \ker = \Theta V/\Theta W \cong V/W. \quad (22)$$

The isomorphism sends $[\phi] \mapsto [\Theta\phi]$. It is algebraic for arbitrary W ; when W is closed, (20) makes it a topological isomorphism onto the kernel. The pullback of the global section has the representative relation kernel in (22); the full-source result is not silently asserted with W in place of V .

C.1.7 6. Perform the original internal quotient and synchronization

On the single-leg map, or a fixed-index diagram of that same map, the split lifts are

$$\tilde{\Theta}(\lambda, \phi) = (\lambda, \Theta\phi), \quad \tilde{\Lambda}(\lambda, F) = (\lambda, \Lambda F), \quad \tilde{K}(\lambda, F) = (\lambda, KF). \quad (23)$$

They preserve external bottom and commute with scalar reflection. In particular

$$\tilde{K}\tilde{\Theta}(\lambda, \phi) = (\lambda, 0_{\mathcal{B}}) = z_{V, \mathcal{B}}(\lambda, \phi).$$

Here $z_{V, \mathcal{B}}$ is the label-preserving zero map from the source fibre to the target fibre. The mixed chart, whose minus restriction is $-\Theta\mathcal{F}$, is treated separately by the explicit maps (25)-(28), without changing that restriction. The actual chart has zero bottom fibre; a general nonzero bottom fibre is acted on by its stated linear map, rather than identified with the single global zero.

The supported image projection is idempotent, and the quotient map is exactly the coequalizer of the theta relation inclusion and its supported-zero companion. The implementation's universal property applies against all split-linear targets. A theta boundary is sent to $e_\lambda = (\lambda, 0)$; it is not sent to external absence.

For the single-leg complex there is an actual continuous contraction

$$C_+ \xrightleftharpoons[s]{q} Q[-1], \quad H^1 = \Lambda,$$

$$\boxed{qs = 1, \quad 1_{C_+} - sq = dH + Hd.} \quad (24)$$

In degree zero this is $\Lambda\Theta = 1_V$; in degree one it is $\Theta\Lambda = 1 - K$. In split notation the subtraction means addition of the supported scalar $(-1)^\bullet$ times the second map. It yields the correct labelled zero wherever the amplitude vanishes.

For the two-leg chart set $H_0 = V[0] \oplus Q[-1]$ with zero differential, and define

$$p_L^0(\phi, \psi) = \psi, \quad p_L^1 = q, \quad i^0(\psi) = (\widehat{\psi}, \psi), \quad i^1 = s,$$

$$H_L^1(F) = (\Lambda F, 0). \quad (25)$$

Then $p_L i = 1$, and $1 - ip_L = dH_L + H_L d$. The degree-zero identity is $(\phi, \psi) - (\widehat{\psi}, \psi) = (\phi - \widehat{\psi}, 0)$. The other leg gives

$$p_R^0(\phi, \psi) = \widehat{\phi}, \quad p_R^1 = q, \quad H_R^1(F) = (0, -\widehat{\Lambda F}). \quad (26)$$

It obeys $1 - ip_R = dH_R + H_R d$. Their homotopies differ by

$$\boxed{H_L^1 F - H_R^1 F = (\Lambda F, \widehat{\Lambda F}),} \quad (27)$$

the actual joint degree-zero cycle, which is retained.

On the complete original support diagram, use literal multiplication by $E_{\text{sync}} = (1, e)$ in degree zero and the source's active-label-to-joint map in degree one:

$$\mathfrak{r}^0(a, b) = (a + eb, b + ea), \quad \mathfrak{r}^1(\lambda, F) = (\{+, -\}, F) \quad (\lambda \neq \perp).$$

The absent input remains absent. The original differential obeys $d\mathfrak{r} = \mathfrak{r}d$ and $\mathfrak{r}^2 = \mathfrak{r}$. Consequently (25) lifts to the fully typed equation

$$\boxed{\mathfrak{r} - \tilde{i}\tilde{p}_L\mathfrak{r} = \tilde{d}\tilde{H}_L + \tilde{H}_L\tilde{d},} \quad (28)$$

where $\tilde{H}_L(\lambda, F) = (\{+, -\}, (\Lambda F, 0))$ for each active label. The right-leg equation is analogous with (26). The original mask is recorded with the injection $x \mapsto (\text{supp } x, \mathfrak{r}x)$ already proved for this chart. Neither synchronization nor the contraction identifies the three supported origins with τ .

C.1.8 7. Global scaling defects, and their agreement with the finite extensions

Let $U_a F(x) = F(x/a)$, $a > 0$, and denote its induced continuous action on Q by $\overline{U}_a$. The section (19) gives a global continuous boundary map

$$\boxed{k_a = \Lambda U_a s : Q \rightarrow V, \quad U_a s - s\overline{U}_a = \Theta k_a.} \quad (29)$$

The proof applies $K + \Theta\Lambda = 1$ to $U_a s$, then uses $qU_a s = \overline{U}_a$. Therefore

$$k_{ab} = U_a k_b + k_a \overline{U}_b, \quad k_1 = 0. \quad (30)$$

The same calculation works for D and compactly supported multiplicative convolution, with $k_D = \Lambda D s$. In the explicit coordinates (20),

$$\boxed{U_a|_{\mathcal{B}} = \begin{pmatrix} U_a|_V & k_a \\ 0 & \overline{U}_a \end{pmatrix}, \quad D|_{\mathcal{B}} = \begin{pmatrix} D|_V & k_D \\ 0 & D_Q \end{pmatrix}.} \quad (31)$$

Thus the topological splitting is not substituted for an equivariant splitting. Equation (29) is the precise comparison between them.

For two cutoff choices, let s_0, s_1 be their sections and set $b_{10} = \Lambda_0 s_1 : Q \rightarrow V$. Then

$$s_1 - s_0 = \Theta b_{10}, \quad k_a^{(1)} = k_a^{(0)} + U_a b_{10} - b_{10} \overline{U}_a. \quad (32)$$

All dependence on the chosen representative map is carried by this displayed coboundary. No canonical cutoff or average over choices is declared.

Restrict back to PR #9 without changing its extension class Use its existing maps $s_Z : E_Z \rightarrow \mathcal{B}$, $\sigma_Z = q s_Z : E_Z \rightarrow Q$, and $J_Z : \mathcal{B} \rightarrow E_Z$. Put $b_Z = \Lambda s_Z$. The new continuous representative of the same finite arithmetic class is

$$s_Z^{\text{new}} = s \sigma_Z = K s_Z = s_Z - \Theta b_Z. \quad (33)$$

If $P(D)s_Z - s_Z M_P = \Theta k_{P,Z}$, direct substitution gives

$$k_{P,Z}^{\text{new}} = k_{P,Z} - P(D)b_Z + b_Z M_P. \quad (34)$$

For $P = h_Z$, this changes the source boundary by $-h_Z(D)b_Z$. It is zero in $V/h_Z(D)V$, so the extension class remains exactly

$$\varepsilon_Z = j_{h_Z}(h_Z/g), \quad g = 2\xi. \quad (35)$$

The polynomial residue form still becomes the arithmetic one by M_{ε_Z} ; its Jacobian contraction remains $M_{\varepsilon_Z} M_{j_h g'}$. No unit, derivative, sign, or factor of 2 is changed.

The two finite cochain projectors differ by the explicit homotopy $-b_Z J_Z$ in degree minus one. Their finite-rank trace difference is zero because $J_Z \Theta = 0$. Hence their cochain Lefschetz trace is the unchanged

$$-\sum_{\rho \in Z} m_\rho \mathcal{M}f(\rho). \quad (36)$$

C.1.9 8. Continuous duality and completed tensor powers of the full object

Use continuous duals, denoted E'_b with the strong topology. Transpose the actual topological splitting (20). It gives

$$\boxed{0 \rightarrow Q'_b \xrightarrow{q'} \mathcal{B}'_b \xrightarrow{\Theta'} V'_b \rightarrow 0, \quad \Theta' \Lambda' = 1.} \quad (37)$$

The continuous cochain-dual complex is $[\mathcal{B}'_b \xrightarrow{\Theta'} V'_b]$ in degrees $-1, 0$. Its cohomology is Q'_b in degree -1 and zero in degree 0 . The transpose contraction proves this directly; no general exactness theorem for arbitrary locally convex quotients is assumed.

The actual finite residue map into that continuous dual is

$$\overline{E_Z} \longrightarrow Q'_b, \quad \bar{f} \longmapsto (u \mapsto R_Z(f, j_Z u)), \quad (38)$$

where $j_Z q = J_Z$. Continuity follows from the finite Mellin jets and finite dimensionality of E_Z . It is injective by their surjectivity and perfect residue duality. Multiplication by g' then gives the arithmetic finite trace pairing, with its previously calculated radical retained.

Let $\widehat{\otimes}_\pi$ denote completed projective tensor product. The comparison from the previous algebraic tensor is its canonical completion map. All chain operators in (24) are continuous. Thus their finite tensor products and homotopies extend to the completed products. With $P = sq$ on C_+ , the homotopy

$$H_n = \sum_{j=1}^n P^{\widehat{\otimes}(j-1)} \widehat{\otimes} H \widehat{\otimes} 1^{\widehat{\otimes}(n-j)} \quad (39)$$

includes $(-1)^{\sum_{i < j} |x_i|}$ on homogeneous inputs. It satisfies

$$dH_n + H_n d = 1 - P^{\widehat{\otimes} n}.$$

Consequently, for the completed external coefficient complex on the same tau-base chart product,

$$\boxed{H^k(C_+^{\widehat{\otimes} n}) = \begin{cases} Q^{\widehat{\otimes} n}, & k = n, \\ 0, & k \neq n. \end{cases}} \quad (40)$$

This follows from a split continuous contraction, not from a blanket claim that completed tensor product preserves every exact sequence.

For the joint chart use (25). Its full cohomology in degree k is the direct sum, indexed by subsets $J \subseteq \{1, \dots, n\}$ with $|J| = k$, of the ordered completed tensors having Q in the slots of J and V in the other slots. In those V slots the scaling action is $aU_{1/a}$, exactly as in the original minus-coordinate representative. The degree-zero contribution is not discarded. Split reconstruction retains each product mask and each internal zero.

C.1.10 9. The pairing calculation is now a specified global boundary problem

The section $s : Q \rightarrow \mathcal{B} \subset L^2(dx)$ supplies a positive definite auxiliary form

$$\mathcal{H}_s(u, v) = \langle su, sv \rangle_{L^2(dx)}.$$

This is not assigned to the arithmetic Weil form. The explicit weight-covariance defect is calculated using (29) and $\langle U_a f, U_a g \rangle = a \langle f, g \rangle$:

$$\boxed{\mathcal{H}_s(\overline{U}_a u, \overline{U}_a v) - a \mathcal{H}_s(u, v) = -\langle U_a s u, \Theta k_a v \rangle - \langle \Theta k_a u, U_a s v \rangle + \langle \Theta k_a u, \Theta k_a v \rangle.} \quad (41)$$

The original source boundary and both of its cross-pairings remain. Being a boundary in Q does not assert these pairings are zero in the representative Hilbert space. In a split lift each computed zero is its supported value e .

For a reduced eigenline of the actual dilation representation with exponent ρ , the left side at $u = v$ is $(a^{2\rho} - a) \mathcal{H}_s(u, u)$. Formula (41) locates that defect in the full global boundary cocycle, with the original operator retained.

A concrete comparison explaining which topology must be kept In fact ΘV is dense in $L^2(\mathbb{R}_{>0}, dx)$, although it is closed in $\mathcal{B}$. Here is a proof retaining the map between them. Set $b_* = \Theta \phi_*$ and

$$W : L^2(dx) \xrightarrow{\sim} L^2(dy), \quad W F(y) = e^{y/2} F(e^y).$$

The Fourier transform in y , with kernel $e^{-2\pi i \nu y}$, gives

$$\mathcal{F}_y W b_*(\nu) = g(1/2 - 2\pi i \nu).$$

This is nonzero almost everywhere, because g is a nonzero entire function. The span of all translates of $W b_*$ is dense in $L^2(dy)$: a vector orthogonal to all translates has Fourier transform whose product with $\overline{\mathcal{F}_y W b_*}$ is an L^1 function with zero Fourier transform, hence vanishes almost everywhere. The nonvanishing just stated then makes that vector zero. Since $W U_a b_* = a^{1/2} (W b_*)(y - \log a)$ and $U_a b_* \in \Theta V$, density follows.

The exact quotient comparison is therefore

$$\boxed{Q = \mathcal{B}/\Theta V \longrightarrow L^2(dx)/\overline{\Theta V}^{L^2} = 0.} \quad (42)$$

Its kernel is all of Q . Its split lift sends every supported input to the supported zero in $G(0) = \mathbb{B}$, while keeping external absence separate. The constructive replacement is the Frechet quotient and the explicit global section (19), not a claim that the ordinary Hilbert quotient secretly retains those classes.

C.1.11 10. Scope of the progress toward the Deligne comparison

The selected supplied Weil II passages 3.3.4-3.3.6 put two bounds on the same compact-to-ordinary image; 6.2 retains the dualizing operation and its shifts. This continuation does not apply those bounds over a new base without their geometric proof. It makes the existing characteristic-zero arithmetic complex globally split in its precise topological category, gives its continuous dual and completed products, and computes the operator defects on those same objects.

The formerly unresolved algebraic-to-Hausdorff kernel is now zero by (16)-(21). The finite extension class and its residue/Jacobian trace are carried by (33)-(36). The full scaling boundary is (29), its dependence on the auxiliary cutoffs is (32), and its metric defect is (41). These are concrete maps available for a global comparison.

Still retained: the common kernel of all finite spectral observations, the meromorphic summand of the balanced analytic realization, and the requirement to identify the correct arithmetic polarization/weight control on the global geometric image. The density map (42) explains why discarding the stronger topology would remove the entire arithmetic cohomology, not why the supported programme should be abandoned.

The reconstruction estimate contains $1 - c_{\alpha,\beta}$; no asymptotic uniform estimate for it is proved. No global positivity, RH, GRH, or infinite trace-class theorem is claimed. No theorem from the earlier finite-range work is upgraded to a Lean certificate. The new global retraction proof itself uses only the original theta/Poisson identities, absolute Mobius inversion, the displayed Fourier compactness argument, and the two retained moment maps.

C.1.12 11. Verification and references

The written analytic proofs are the evidence for the new analytic statements. The accompanying unittest script tests exact finite identities: divisor cancellation with supported zeros, cutoff-resolvent identities in a finite unitary model, moment corrections, contractions, synchronization masks, operator cocycles, changes of sections, preservation of extension classes, finite trace changes, and completed-tensor algebra before completion. It does not numerically certify a Fourier operator norm, an analytic convergence statement, or a Riemann zero.

Source interfaces read:

- Owner's SplitZero_Derived_Mathematics_2026-09-12(1).md, sections 1-7; the source permits nonzero bottom fibres in its general interface.
- PR #9, workbenches/tau-chain-descent/RESEARCH_NOTE.md, at the commit above; local bytes match its Git blob.
- PR #8's finite comparison as retained and invoked in #9; the sections using s_Z and J_Z are explicitly dependencies of (33)-(38), not inputs to the proof of (2).
- Supplied Weil II TeX: the actual 3.3.1-3.3.6 excerpt, including its finite-type base and duality assumptions. It was read as TeX, not obtained by OCR. No full source-proof audit is claimed.
- Ralf Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, arXiv:math/0412277: prior-context abstract only in this continuation. A source-archive retrieval failed; no unread detailed theorem is cited as a premise.

The contribution is AI-assisted under the owner's mathematical direction. Original mathematical files, formal sources, attribution and licenses are unchanged. Only authored exposition, test code and validation records are proposed for public addition; no private transcript or source-literature corpus is published.

D PR 11: Homotopy control

Source attribution. Web contribution, pull request #11, [Zeta research repository](#).

Exact revision. d37eade4f9e6a82c1a88aeff710a3b851a34eb49

Source file. workbenches/tau-homotopy-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9924b358c0365a8cbdcddb3a7496f610c65d64617cd1860679b42e371d2fe88d

D.1 Homotopy choices and source-preserving arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #10, d26c283c2a58da57bb5a5a4d4bc8669019f6ebb7. This integration note records the new derivations; the conversation delivery also contains a longer equation-by-equation exposition. No original scalar definitions or formal modules are changed.

D.1.1 1. Reviewed theorem and analytic specialization

The supplied other-session proof classifies all homotopies for

$$C = [V \oplus V \xrightarrow{d} B], \quad d(v, w) = \Theta(v - \mathcal{F}w), \quad Q = B/\Theta V,$$

where Θ is injective and $\mathcal{F}$ is an R -linear automorphism. For a fixed $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, every solution of BOTH equations

$$dH = \Theta \kappa, \quad Hd = 0$$

has the unique form

$$H_\alpha(b) = (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b), \quad \alpha : Q \rightarrow_R V.$$

Proof: writing $H=(u,v)$, injectivity gives $u-\mathcal{F}v=\kappa b$. Since $d(x,0)=\Theta x$, $Hd=0$ gives $H \Theta = 0$. Hence $\mathcal{F}v$ factors uniquely through q , and supplies α . Conversely substitution proves both equations. The reference solution is $H_0=(\kappa, 0)$; the family is an affine space under $\text{Hom}_R(Q, V)$, relative to that reference.

Let $\gamma(v)=(v, \mathcal{F}^{-1} v)$. Then

$$H_\alpha - H_\beta = \gamma(\alpha - \beta)q.$$

The difference is a map $H_1(C) \rightarrow H_0(C)$. There are no degree-minus-two endomorphisms of this two-term coefficient complex. Thus it is a genuine degree-minus-one Hom-complex class, not an erased difference.

The supplied theorem is mathematically correct. Its Lean draft explicitly remains uncompiled, and its earlier failed chart build is not relabelled successful. No Lean or Lake executable is installed in the review environment.

For the actual analytic spaces instantiate $R=\mathbb{C}$. With the quotient topology on Q , the continuous version follows: $\alpha q = \mathcal{F} \text{pr}_2 H$ is continuous, so the quotient property makes α continuous. Conversely the displayed formula sends continuous α to continuous H .

For the polynomial action $A_0 = \mathbb{C}[t]$, Fourier has the exact type

$$\mathcal{F} : V^j \xrightarrow{\sim} V, \quad t \cdot_{V^j} v = (1 - D)v, \quad j(t) = 1 - t,$$

because $F D = (1 - D)F$. Thus d is A_0 -linear from $V + V^\wedge j$ to B . A complex-linear homotopy is not automatically A_0 -linear. The reflected module above is the actual comparison, not a change to the Fourier definition.

D.1.2 2. Original split support on every homotopy

Retain the base b_τ and the square $G(Z) \rightarrow G(C)$ over $Z > C$. The quotient $G(Z) \rightarrow Z$ remains infinite. Both lifted coefficient maps preserve τ and the supported scalar e separately.

Use the original support lattice $L = \{\text{bottom}, +, -, J\}$, $J = \{+, -\}$. Degree-zero fibres are $0, V, V, V + V$ with the coordinate inclusions. Degree-one active fibres are all B , with identity transports. Their differentials are Θ , $-\Theta$, F and d . Retain zero modules in the other degrees; their reconstructed zero map is $z(\lambda, x) = (\lambda, 0)$.

Literal multiplication by $E_{\text{sync}} = (1, e)$ induces

$$r^0(a, b) = (a + eb, b + ea), \quad r_L(\perp) = \perp, \quad r_L(\lambda) = J(\lambda \neq \perp).$$

Every homotopy lifts by

$$\tilde{H}_\alpha(\lambda, b) = (J, (\kappa b + \alpha q b, \mathcal{F}^{-1} \alpha q b)), \quad \lambda \neq \perp; \quad \tilde{H}_\alpha(\tau) = \tau.$$

This is split-linear: two active inputs add at their joined label, both outputs add at J , and their amplitudes obey the two displayed linear formulas. A supported scalar zero gives $(J, (0, 0))$; external τ gives the global zero. Therefore

$$\tilde{d} \tilde{H}_\alpha + \tilde{H}_\alpha \tilde{d} = r \tilde{\Delta}.$$

The difference is $(J, \gamma(\alpha - \beta)qb)$, retaining its degree-zero class. For this chart, the pair $(\text{support}(x), r(x))$ is injective. No such injectivity is assumed for arbitrary diagrams with noninjective transports. For a smaller relation space W , keep $Q \rightarrow W$ with kernel $\Theta V / \Theta W$; q in the formulas means the displayed quotient by the full admitted V .

D.1.3 3. Coherence for all dilations

Let $S_a = U_a$ on V , $T_a = U_a$ on B , $U_a F(x) = F(x/a)$, and T_{bar}_a the quotient action. The actual cochain action is

$$\mathsf{T}_a^0 = (S_a, \mathcal{F}^{-1} S_a \mathcal{F}) = (U_a, a U_{1/a}), \quad \mathsf{T}_a^1 = T_a.$$

It satisfies $T_a^0 \gamma = \gamma S_a$ and $q T_a = T_{\text{bar}}_a q$. With the PR #10 section s , take $P = (0, sq)$, and write

$$[\mathsf{T}_a, P]^1 = \Theta \kappa_a, \quad \kappa_a = k_a q, \quad T_a s - s \bar{T}_a = \Theta k_a.$$

The commutator product identity gives $\kappa_{ab} = S_a \kappa_b + \kappa_a T_b$. The family of homotopies is composition-coherent,

$$H_{ab} = \mathsf{T}_a H_b + H_a \mathsf{T}_b,$$

exactly when

$$\alpha_{ab} = S_a \alpha_b + \alpha_a \bar{T}_b.$$

Proof: the reference family $(\kappa_a, 0)$ is coherent. Subtract it and use injectivity of γ and surjectivity of q . No action or boundary is omitted.

For a smoothly parameterized continuous-linear family with $L = (d/dt)|_0 \alpha(e^t)$, the exact formula is

$$\alpha_{e^t} = \int_0^t S_{e^{t-r}} L \bar{T}_{e^r} dr.$$

This is a vectorwise integral in the complete Frechet source, with the original oriented integral for negative t . Differentiating the composition equation gives the formula; splitting the integral proves its converse. In particular $\alpha_a = S_a b - b \bar{T}_a$ has infinitesimal parameter $D_V b - b D_Q$.

D.1.4 4. A homotopy parameter leaves the control form fixed

Fix an actual finite arithmetic packet with full multiplicities,

$$h_Z(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad E_Z = \mathbb{C}[t]/(h_Z), \quad A = M_t.$$

Use the existing $R: E_Z \rightarrow B$ and $J_Z: B \rightarrow E_Z$ with

$$qR = \sigma_Z, \quad J_Z R = 1, \quad J_Z \Theta = 0, \quad J_Z D = A J_Z, \quad DR - RA = \Theta k.$$

The cochain projector $P_R = (0, R|_Z)$ has commutator $(0, \Theta k|_Z)$, so the reviewed theorem applies with $\kappa = k|_Z$. The quotient $\text{jet } J_{\text{bar}}_Z$ satisfies $J_{\text{bar}}_Z \sigma_Z = 1$, hence any $\alpha_E: E_Z \rightarrow V$ extends to $\alpha_E J_{\text{bar}}_Z$ on Q . Restriction to R gives

$$H_{\alpha_E} R = (k + \alpha_E, \mathcal{F}^{-1} \alpha_E), \quad dH_{\alpha_E} R = \Theta k.$$

Take $H = L^2(R_+, dx)$ with the original inner product, conjugate-linear in the first variable. Define

$$G_R = R^* R, \quad B_R = \Theta k, \quad W_R = -(R^* B_R + B_R^* R).$$

The composite of actual maps

$$\text{Hom}_{\mathbb{C}}(E_Z, V) \longrightarrow \text{Hom}(E_Z, V^2) \xrightarrow{d \circ (-)} \text{Hom}(E_Z, H) \xrightarrow{B \mapsto -(R^* B + B^* R)} \text{Herm}(E_Z)$$

is constant at W_R . Its last map is real-linear. The retained H^0 ambiguity does not change this boundary contraction.

Integration by parts on the actual source functions gives

$$A^* G_R + G_R A - G_R = W_R.$$

The boundary term $[x \text{ conjugate}(F)H]_0^\infty$ vanishes by the stated decay. No claim that a θ boundary is orthogonal to a representative is used.

D.1.5 5. The actual representative-change family

Every representative of the same inclusion σ_Z is uniquely

$$R_b = R + \Theta b, \quad b : E_Z \rightarrow_{\mathbb{C}} V.$$

Indeed $q(R'-R)=0$ gives values in ΘV , and the PR #10 retraction gives the unique $b=\Lambda(R'-R)$. Finite source dimension gives continuity. Global continuous sections have the identical formula with $b:Q \rightarrow V$ continuous.

The complete changes are

$$k_b = k + D_V b - bA,$$

$$\delta G_b = R^* \Theta b + (\Theta b)^* R + (\Theta b)^* \Theta b,$$

$$\boxed{W_{R_b} - W_R = A^* \delta G_b + \delta G_b A - \delta G_b.}$$

Their extension-coordinate map is $U_b(v,u)=(v+bu,u)$, and

$$U_b^{-1} \begin{pmatrix} D_V & k \\ 0 & A \end{pmatrix} U_b = \begin{pmatrix} D_V & k + D_V b - bA \\ 0 & A \end{pmatrix}.$$

Lift that automorphism on the common active fibre $(V+E_Z)^{\tau}$. For the original masks, first keep the pair $(\text{support}(x), r(x))$; apply the shear on the joint fibre, and recover the original amplitudes by U_b^{-1} and the original mask by its recorded component. No inverse to the unrecorded support collapse is asserted.

The projectors obey $P_{(R_b)} - P_R = \text{partial } L_b$ with $L_b^{-1}(F) = (b|_Z F, 0)$. Their finite cochain traces agree because

$$\text{Tr}(\Theta b J_Z H_f) = \text{Tr}(J_Z H_f \Theta b) = 0.$$

The cochain supertrace sign remains -1 in degree one. The arithmetic extension class changes by $h_Z(D)b(1)$, so its class in $V/h_Z(D)V$ remains $\epsilon_Z = j_h(h_Z/(2\pi))$. The residue/Jacobian trace form is consequently unchanged.

D.1.6 6. Finite actual source ansatz and one master Gram matrix

Retain $\phi = (4\pi^2 x^4 - 6\pi x^2) \exp(-\pi x^2)$, $\phi_j = D^j \phi$, and

$$\Phi_m : \mathbb{C}^{m+1} \rightarrow V, \quad (c_j) \mapsto \sum_{j=0}^m c_j \phi_j.$$

Their actual Mellin identities are $M \Theta \phi_j(s) = 2\pi i(s)^j$. They remain in the original two-moment source and are linearly independent. The source section $R_0 = s_Z$ has the exact rank-one defect $DR_0 - R_0 A = \Theta \phi^* \ell_Z$, with the unchanged arithmetic remainder functional ℓ_Z .

Define maps, including the target enlargement,

$$I_m e_j = e_j, \quad S_m e_j = e_{j+1}, \quad I_m, S_m : \mathbb{C}^{m+1} \rightarrow \mathbb{C}^{m+2}, \quad D\Phi_m = \Phi_{m+1} S_m.$$

For B in $\text{Mat}_{((m+1) \times d)}(\mathbb{C})$, $d = \dim E_Z$, put

$$R_B = R_0 + \Theta \Phi_m B, \quad K_B = e_0 \ell_Z + S_m B - I_m B A.$$

Then

$$DR_B - R_B A = \Theta \Phi_{m+1} K_B, \quad q R_B = \sigma_Z, \quad J_Z R_B = 1.$$

Take the actual column map and its Gram matrix

$$\mathcal{T}_m : E_Z \oplus \mathbb{C}^{m+2} \rightarrow L^2(\mathbb{R}_+, dx), \quad (u, c) \mapsto R_0 u + \Theta \Phi_{m+1} c, \quad M_m = \mathcal{T}_m^* \mathcal{T}_m.$$

M_m is positive definite. Applying J_Z to a vanishing combination gives $u=0$; Mellin then gives $c=0$. Its entries are the convergent integrals of the displayed arithmetic functions, with the original measure.

Define

$$X_B = \begin{pmatrix} I_d \\ I_m B \end{pmatrix}, \quad Y_B = \begin{pmatrix} 0 \\ K_B \end{pmatrix}.$$

The complete control calculation is

$$\boxed{G_B = X_B^* M_m X_B, \quad W_B = -(X_B^* M_m Y_B + Y_B^* M_m X_B) = A^* G_B + G_B A - G_B.}$$

This restricts representative choice to an actual finite family of theta boundaries. It is not an arbitrary positive matrix ansatz, and it is not asserted to exhaust all continuous source corrections. All packet coordinates, multiplicities, local units and finite source-degree changes are retained.

D.1.7 7. Error transport, duality, and the remaining estimate

With actual enclosures $||M\text{-}\hat{M}|| \leq \eta_M$, $||X\text{-}\hat{X}|| \leq \eta_X$, $||Y\text{-}\hat{Y}|| \leq \eta_Y$, put $x = ||\hat{X}||$, $y = ||\hat{Y}||$, $m_0 = ||\hat{M}||$. Product estimates give

$$\eta_G = \eta_M (x + \eta_X)^2 + m_0 (2x\eta_X + \eta_X^2),$$

$$\eta_W = 2[\eta_M (x + \eta_X)(y + \eta_Y) + m_0 (x\eta_Y + y\eta_X + \eta_X \eta_Y)].$$

These include coefficient errors, not just quadrature errors. Exact certificates for $\hat{G}\text{-}\eta_G \geq 0$ and

$$\epsilon \hat{G} \pm \hat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0$$

imply the actual two-sided control. This is an error-transfer theorem; arithmetic enclosures have not been computed here.

The unchanged residue matrix S satisfies $S^* = -S$ and $A^* S + S A = S$. Its exact dual transport is

$$G_B^D = S^* G_B^{-1} S, \quad W_B^D = -S^* G_B^{-1} W_B G_B^{-1} S.$$

Thus an upper certificate gives a lower certificate on the same arithmetic packet. The Weil form stays $S J_g$, with metric comparison $G_B^{-1} S J_g$; no positivity is assigned by choosing G_B .

For tensor degree n , retain $A_n = \sum_j 1$ tensor A tensor 1 and $G_{(B,n)} = G_B^{\wedge}(\text{tensor } n)$. The product control is $\sum_j G_B$ tensor W_B tensor G_B . Changing a homotopy by $\gamma \alpha q$ changes

the tensor homotopy by its Koszul-signed insertions; its differential is zero, hence it does not change that control form.

The existing orbit construction on this actual tensor image gives

$$G_{B,n,T} = \int_0^T e^{-nt} e^{tA_n^*} G_{B,n} e^{tA_n} dt,$$

$$A_n^* G_{B,n,T} + G_{B,n,T} A_n - n G_{B,n,T} = e^{-nT} e^{TA_n^*} G_{B,n} e^{TA_n} - G_{B,n}.$$

Every occurrence of $G_{B,n}$ now has the explicitly constrained source in section 6.

A nonzero boundary need not have nonzero control. The exact calibration $D_{\text{amb}} = \text{diag}(1/2+i, 1/2+2i)$, $\Theta(v) = (v, 0)$, $A = 1/2+2i$, $R(u) = (u, u)$ has boundary $\Theta(-i)$, Gram 2, and control zero. It is an algebraic calibration, not a zeta packet. The real-linear map $B \mapsto (R^*B + B^*R)$ explains which nonzero boundaries can have zero control.

Conversely every eigenvector $Av = \rho v$ obeys

$$\frac{v^* W_B v}{v^* G_B v} = 2\Re \rho - 1.$$

This ensures that changing representatives has not changed a spectral coordinate. The analytic target is now to bound the actual matrices from section 6, or their tensor-orbit endpoints, with all packet and tensor dependence retained. No uniform purity estimate, RH, GRH, or global spectral-kernel vanishing is proved here.

D.1.8 8. Validation and attribution

The attachment supplies the homotopy-classification theorem and the uncompiled draft. This note verifies its proof and adds the continuous/twisted interface, supported reconstruction, coherent-family condition, and the connection to the actual control matrices. The original owner-directed SplitZero construction remains the source.

The independent checker has sixteen exact unittest methods, including finite parameter families. They passed normally and under python -O with identical successful JSON records. Intentional negative controls failed in both modes. The previous control checker also passed its sixteen methods. These finite checks cover algebraic models and polynomial-times-exponential integral calibrations, not evaluated zeta packets or analytic/Lean certification.

General references: The Stacks Project, Tags 0A8H and 0115 for the Hom-complex differential and homotopy, and 010I and 06XP for extension classes. The formulas above are proved on the specified programme objects; those references are not imported as a number-field weight theorem. Only authored notes, test code and validation records are published; no private transcript, source-literature corpus, or font files are included.

E PR 13: Orthogonal boundary control

Source attribution. Web contribution, pull request #13, [Zeta research repository](#).

Exact revision. e7ba4c29d4503bf20d06cb70c93c3347f199f1b4

Source file. workbenches/tau-orthogonal-boundary-control/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 9d364d9cc2f6840af8eb39555e4b89622acd7714fbcab7ca135e8ec0ce25ab0b

E.1 Orthogonal theta boundaries and rank-two arithmetic control

Owner-directed SplitZero continuation, 12 September 2026. Base: PR #11, d37eade4f9e6a82c1a88aeff710a3b851a34eb49. This is an additive research contribution with written proofs for review, not a new Lean certificate for the analytic results. The conversation delivery contains a longer numbered LaTeX/HTML version, the full reading record, and exploratory arithmetic moments.

E.1.1 1. Reviewed input and exact types

The supplied other-session theorem is correct: for injective $\Theta: V \rightarrow B$, an R -linear automorphism F , $Q = B/\Theta V$, and $\kappa: B \rightarrow V$ with $\kappa \Theta = 0$, all homotopies satisfying BOTH $dH = \Theta \kappa$ and $Hd = 0$, for $d(v, w) = \Theta(v - Fw)$, are

$$H_\alpha(b) = (\kappa b + \alpha \kappa b, F^{-1} \alpha \kappa b), \quad \alpha: Q \rightarrow R V.$$

Writing $H = (u, v)$ gives $u - Fv = \kappa$. Since $d(x, 0) = \Theta x$, $Hd = 0$ gives $H \Theta = 0$; hence Fv descends uniquely to α . Substitution proves the converse. The difference is $\gamma(\alpha - \beta)q$ with $\gamma(v) = (v, F^{-1}v)$. The solution set is affine relative to H_0 , not generally a vector space with the zero homotopy as origin.

The attachment's uncompiled-draft status has now been superseded by the other session's successful PR #12 at 5d2772ea0a16d177ac3e01ff70394b90ba95a232. The actual SplitZeroTauHomotopy.lean, completed job 103535438942, and its full log were read. Both cochain equations and the inverse equivalence passed the specified kernel checks. Retraction and quotient-topology interfaces also passed with their explicit hypotheses. This is the other session's formalization; neither the actual analytic theta construction nor the new Gram/control results below are certified by it. No local Lean/Lake execution is claimed. See COORDINATION_UPDATE.md for exact scope. PR #11 already supplies the continuous specialization, the Fourier twist over $C[t]$, support-changing lifts, coherent dilations and representative-change formulas. Those are retained inputs, not new results here.

Keep the original tau base and the square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$, with $p: G(Z) \rightarrow Z$ still infinite, $e = \text{ofR}(0)$, and tau the external scalar zero. No Boolean localization replaces the base quotient. All quotient relations below use the original internal coequalizer and preserve the support label.

Retain V, B and the actual operators of PRs #8-#11:

```
V={even Schwartz phi on R: phi(0)=0 and integral(phi)=0};
B={smooth F on R_+: sup x^b |D^k F(x)| < infinity for all integer b and k>=0};
D=-x d/dx; Theta phi(x)=2 sum_(n>=1) phi(nx);
M F(s)=integral_0^infinity F(x)x^s dx/x;
phi_*= (4 pi^2 x^4 - 6 pi x^2)exp(-pi x^2);
phi_j=D^j phi_*; f_j=Theta phi_j=D^j f_0;
M f_j(s)=s^j g(s), g=2 xi.
```


The Hilbert observation is $H=L^2(\mathbb{R}_+, dx)$, conjugate-linear in the first argument. Polynomial packet coordinates are retained, not changed to orthonormal coordinates.

For an actual finite packet with all multiplicities, let $E=C[t]/h_Z$, $A=M_t$. The previous analytic constructions supply

$R_\theta: E \rightarrow B$, $J_Z: B \rightarrow E$, $qR_\theta = \sigma_Z$, $J_{ZR_\theta} = 1$, $J_Z \theta = 0$,
 $DR_\theta - R_\theta A = f_\theta \in \ell_Z$.

The exact ℓ_Z comes from $\epsilonpsilon_Z = j_h(h/(2\xi))$; at a simple zero its value on 1 is $1/(2\xi'(\rho))$. These analytic constructions remain the cited research-proof dependencies, not theorems certified by the finite tests here.

E.1.2 2. An actual minimum over the admitted theta relations

For $m \geq 0$ define

$W_m = \text{span}(\phi_0, \dots, \phi_m)$;
 $\phi_m: C^{(m+1)} \rightarrow V$, $c \mapsto \sum c_j \phi_j$;
 $F_m = \theta \phi_m: C^{(m+1)} \rightarrow H$;
 $H_m = F_m^* F_m$; $C_m = F_m^* R_\theta$; $G_\theta = R_\theta^* R_\theta$.

The ϕ_j are independent: Mellin sends a relation to $g(s)P(s)=0$. Thus H_m is positive definite. Here C_m is a cross-Gram matrix $E \rightarrow C^{(m+1)}$, not the complex of that name in the endpoint workbench.

Set

$B_m = -H_m^{-1} C_m$;
 $\pi_m = F_m H_m^{-1} F_m^*$;
 $R_m = R_\theta + \theta \phi_m B_m = (1 - \pi_m) R_\theta$.

The correction is a finite original theta boundary, so R_m lands in B , $qR_m = \sigma_Z$, and $J_{ZR_m} = 1$ at every m . Direct multiplication proves

$F_m^* R_m = 0$;
 $G_m = R_m^* R_m = G_\theta - C_m^* H_m^{-1} C_m \succ 0$.

Positive definiteness follows from $J_{ZR_m} = 1$. For any $B: E \rightarrow C^{(m+1)}$,

$(R_\theta + F_m B)^* (R_\theta + F_m B)$
 $= G_\theta + (B - B_m)^* H_m (B - B_m)$.

This is the minimum of the Hilbert Gram within the stated source family, not yet a minimum of its relative weight error. Also $(1 - \pi_m)(R_\theta + F_m B') = R_m$: changing the initial representative by a relation already in W_m leaves the result unchanged.

E.1.3 3. The complete control form has rank at most two

Let $I_m, S_m: C^{(m+1)} \rightarrow C^{(m+2)}$ be $I_m e_j = e_j$ and $S_m e_j = e_{j+1}$. Define

$K_m = e_\theta \in \ell_Z + S_m B_m - I_m B_m A$.

Then

$DR_m - R_m A = F_{(m+1)} K_m = \theta (\phi_{(m+1)} K_m)$.

No boundary component is discarded in this identity. Put $b_m = e_m^* B_m: E \rightarrow C$, and $z_m = R_m^* f_{(m+1)}: C \rightarrow E$. Integration by parts on the stated source functions gives

$$\langle DF, H \rangle + \langle F, DH \rangle = \langle F, H \rangle,$$

with vanishing boundary term $[x \text{ conjugate}(F)H]_0^\infty$. Hence

$$\begin{aligned} W_m &= A^* G_m + G_m A - G_m \\ &= -(R_m^* F_{(m+1)} K_m + K_m^* F_{(m+1)} R_m) \\ &= -z_m b_m - b_m^* z_m^*. \end{aligned}$$

The last equality uses $F_m^* R_m = 0$ and the exact last row of K_m , which is b_m . Thus $\text{rank}(W_m) \leq 2$ for every packet dimension and every finite m . The pairings with the first $m+1$ theta directions vanish; those supported vectors themselves are retained.

Define the actual scalar quantities

$$\begin{aligned} u_m &= z_m^* G_m^{-1} z_m \geq 0; \\ v_m &= b_m^* G_m^{-1} b_m \geq 0; \\ c_m &= b_m^* G_m^{-1} z_m. \end{aligned}$$

The nonzero spectrum of $G_m^{-1} W_m$ is the nonzero spectrum of

$$-\begin{bmatrix} c_m & v_m \\ u_m & \text{conjugate}(c_m) \end{bmatrix}.$$

This follows by factoring through the two-column map $[z_m, b_m^*]$ and using the equality of nonzero spectra of the two compositions. Its trace is $-2 \text{Re}(c_m)$, determinant $|c_m|^2 - u_m v_m$. Therefore the least symmetric control allowance is

$$\begin{aligned} \epsilon_m &= |\text{Re}(c_m)| + \sqrt{u_m v_m - (\text{Im}(c_m))^2}, \\ -\epsilon_m G_m &\leq W_m \leq \epsilon_m G_m. \end{aligned}$$

Cauchy-Schwarz makes the radicand nonnegative. The formula also covers rank one. A reflection-stable full packet has

$$\text{trace}(G_m^{-1} W_m) = 2 \text{Re}(\text{trace } A) - \dim E = 0,$$

so $\text{Re}(c_m) = 0$. This is an exact trace identity, not a choice of coordinates. For every eigenvector $Av = \rho v$,

$$(v^* W_m v) / (v^* G_m v) = 2 \text{Re}(\rho) - 1.$$

Thus the relative control still detects precisely the original spectral weight.

E.1.4 4. The next relation is a rank-one update with its own retained coordinate

Set

$$\begin{aligned} a_{(m+1)} &= (1 - \Pi_m) f_{(m+1)} = \Theta \psi_{(m+1)}, \\ \psi_{(m+1)} &= \phi_{(m+1)} - \Pi_m H_m^{-1} F_m^* f_{(m+1)}, \\ \eta_{(m+1)} &= \|a_{(m+1)}\|^2 > 0. \end{aligned}$$

Then $R_m^* a_{(m+1)} = z_m$, and orthogonal decomposition gives

$$\begin{aligned} \Pi_{(m+1)} &= \Pi_m + a_{(m+1)} \eta_{(m+1)}^{-1} a_{(m+1)}^*; \\ R_{(m+1)} &= R_m - a_{(m+1)} \eta_{(m+1)}^{-1} z_m^*; \\ G_{(m+1)} &= G_m - \eta_{(m+1)}^{-1} z_m z_m^*. \end{aligned}$$

The control update is obtained by applying the same real-linear operator $X \mapsto A^*X + XA - X$ to the last identity.

The original quotient transport is

$$Q_W = B/\Theta W_m \rightarrow Q_{W(m+1)} = B/\Theta W_{(m+1)},$$

$$\text{kernel} = \Theta W_{(m+1)} / \Theta W_m \simeq W_{(m+1)} / W_m.$$

That kernel is one-dimensional. The square $E \rightarrow Q_W$ by R_m and $E \rightarrow Q_{W(m+1)}$ by $R_{(m+1)}$ commutes. The new relation becomes the supported zero only through the actual next-level quotient; it is not another scalar adjunction at the bottom.

There is a form on the entire quotient:

$$\text{Hcal}_m([F], [G]) = \langle (1 - \Pi_m)F, (1 - \Pi_m)G \rangle.$$

It is positive definite since the kernel of $1 - \Pi_m$ in B is exactly ΘW_m . Let $\lambda_{(m+1)}[F] = \langle a_{(m+1)}, F \rangle$. This is well defined on Q_W and

$$\text{Hcal}_m(x, y) = \text{Hcal}_{(m+1)}(qx, qy) + \eta_{(m+1)}^{-1} \text{conjugate}(\lambda_{(m+1)}(x)) \lambda_{(m+1)}(y).$$

The map $(q, \lambda_{(m+1)}): Q_W \rightarrow Q_{W(m+1)}$ plus C is a linear isomorphism. Its inverse is

$$([F], c) \mapsto [F - a \eta_{(m+1)}^{-1} \langle a, F \rangle + a \eta_{(m+1)}^{-1} c]_W.$$

Changing F by a relation at level $m+1$ changes this expression only by a relation at level m . This supplies the exact metric comparison between levels instead of treating the quotient transports as isometries.

Every displayed linear map has its original fixed-support lift. On the full source chain, reconstructed addition uses the join $\max(m, n)$ and the above transports. The scalar e sends $(m, [F])$ to $(m, [0])$; τ sends it to external absence. The two-leg primitives are $(k_m u, 0)$ and $(0, \text{Fourier}^{-1} k_m u)$, where $k_m = \Phi_{(m+1)} K_m$. Synchronization by the original $E_{\text{sync}} = (1, e)$ supplies their common joint support. Their difference remains the H_0 cycle $(k_m u, \text{Fourier}^{-1} k_m u)$. The other session's $\text{Hom}(Q, H_0)$ freedom leaves their differential and hence W_m unchanged.

E.1.5 5. Actual arithmetic moments and their truncation allowance

The matrices H_m come from the original arithmetic functions. Since $JF(x) = x^{-1} F(1/x)$ is unitary on $L_2(dx)$, $Jf_0 = f_0$ and $JD = (1 - D)J$,

$$H_{ij} = \int_{-\infty}^{\infty} [\text{conjugate}(f_i) f_j + \text{conjugate}(f_i^{\text{ref}}) f_j^{\text{ref}}] dx,$$

$$f_j^{\text{ref}} = (1 - D)^j f_0.$$

This retains both multiplicative ends and permits absolutely convergent Gaussian sums on $x \geq 1$. Write

$$\phi_j(x) = P_j(x) \exp(-\pi x^2),$$

$$P_0 = 4\pi^2 x^4 - 6\pi x^2,$$

$$P_{(j+1)} = 2\pi x^2 P_j - x P_j',$$

$$P_j = \sum_r c_{(j,r)} x^{(2r)},$$

$$c_{(j+1,r)} = 2\pi c_{(j,r-1)} - 2r c_{(j,r)}.$$

For $f_j^N = 2 \sum_{(n=1)}^N P_j(nx) \exp(-\pi n^2 x^2)$, put

$$C_j = \sum_r |c(j, r)| \sqrt{\Gamma(2r+1/2) / (2 \pi^{(2r+1/2)})}.$$

Then

$$\begin{aligned} & \|f_j - f_j^N\|_{L^2([1, \infty))} \\ & \leq 2 C_j (N+1)^{-1/2} \exp(-\pi(N+1)^2/2) \\ & \quad / [1 - \exp(-\pi(2N+3)/2)]. \end{aligned}$$

Proof: on $x \geq 1$, $2 \pi n^2 x^2 \geq \pi n^2 + \pi n^2 x^2$. Integrating each monomial's squared bound gives $n^{-1/2} \exp(-\pi n^2/2)$ times the displayed gamma constant. Minkowski and a geometric bound on consecutive n terms give the formula. The reflected derivative retains the absolute binomial sum of these allowances.

The finite integral entries use exactly

$$\begin{aligned} & \text{integral}_1^{\infty} x^{(2k)} \exp(-c x^2) dx \\ & = \Gamma(k+1/2, c) / (2 c^{(k+1/2)}), \quad c = \pi(n^2 + n'^2). \end{aligned}$$

For truncation tail norms δ_i, δ_j , each inner-product error is bounded by $\delta_i \|f_j^N\| + \|f_i^N\| \delta_j + \delta_i \delta_j$, plus its reflected contribution. Numerical evaluation errors for the finite expression still require their own enclosure. The optional conversation computation is explicitly exploratory, not interval-certified.

The exact consistency check is $H_{(i+1, j)} + H_{(i, j+1)} = H_{ij}$. Cross-Gram columns $F_m \cdot R_0$ remain their actual source integrals and are not replaced by arbitrary matrices.

E.1.6 6. The same source proves a strong topology comparison

For this actual f_0 ,

$$\text{closure span}\{D^j f_0 : j \geq 0\} \text{ in } L^2(\mathbb{R}_+, dx) = H.$$

Proof. The explicit unitary Mellin-line map sends F to $M F(1/2+it)$ in $L^2(\mathbb{R}, dt/(2\pi))$. It sends D to $1/2+it$ and f_j to $(1/2+it)^j g(1/2+it)$. The log-space function $h(z) = \exp(z/2) f_0(\exp z)$ is holomorphic in $|\text{Im } z| < \pi/4$ by its Gaussian series. For $|\text{Im } z| \leq a < \pi/4$ it decays superexponentially as $\text{Re } z$ tends to positive infinity. Poisson gives $h(z) = h(-z)$, so the same holds at the other end. Fourier contour displacement therefore gives $|g(1/2+it)| \leq C_a \exp(-a|t|)$.

The measure $d\mu = |g(1/2+it)|^2 dt/(2\pi)$ has a positive exponential moment. If an $L^2(\mu)$ function is orthogonal to all polynomials, multiplying μ by its conjugate gives a finite complex measure with a smaller positive exponential moment. Its Fourier transform is analytic in a strip and all derivatives at zero vanish. Analyticity and uniqueness of the Fourier transform imply that measure is zero. Thus polynomials are dense in $L^2(\mu)$. Multiplication by g is unitary from this weighted space onto $L^2(dt/(2\pi))$, because the nonzero analytic function g on the line is nonzero almost everywhere. This proves the assertion without RH.

As a result, for a fixed finite packet,

$$\begin{aligned} & R_m \rightarrow 0 \text{ in } \text{Hom}(E, H), \quad G_m \rightarrow 0 \text{ and } W_m \rightarrow 0, \\ & \text{while } q_{R_m} = \sigma_Z \text{ and } J_{ZR_m} = 1 \text{ for every } m. \end{aligned}$$

This specifies why unscaled smallness of W_m cannot be used alone as weight control: its denominator G_m is also shrinking. The exact relative formula in section 3 must be retained.

The corrective map is the Hilbert/jet graph:

$$\begin{aligned} & \text{closure}\{(F, J_{ZF}) : F \text{ in } B\} = H \text{ plus } E; \\ & \text{closure}\{(\Theta \phi, 0) : \phi \text{ in } V\} = H \text{ plus } 0. \end{aligned}$$

Indeed finite theta polynomials give every $(h,0)$ in the closure, and the sequence $(R_m u, u)$ gives $(0, u)$. For a specified positive metric on E , completion in $\|F\|_2^2 + \|J_Z F\|_E^2$ is this graph closure, and the quotient by the boundary closure is E by $[(h, u)] \rightarrow u$. The added metric is retained supplied data, not an implied positivity theorem for the Weil pairing.

Its split observation is

$$((R_m u)^{\bullet}, u^{\bullet}) \rightarrow (e_H, u^{\bullet})$$

in the product of the stated support-fibre topologies. The arithmetic component survives above a supported Hilbert zero. External absence is (τ, τ) .

E.1.7 7. Duality, trace and scope

The original residue map has matrix S with $S^* = -S$ and $A^*S + SA = S$. Its pullback of the dual metric is

$$\begin{aligned} G_m^* D &= S^* G_m^{-1} S, \\ W_m^* D &= -S^* G_m^{-1} W_m G_m^{-1} S. \end{aligned}$$

Thus the upper-to-lower comparison is retained on the same arithmetic packet. The Weil pairing stays $S J_g$, with explicit metric comparison $G_m^{-1} S J_g$. Its positivity is not imposed by choosing a positive Gram matrix.

All representative changes above are original theta boundaries. Their extension class $j_h(h/(2\xi))$ is unchanged; the induced finite cochain projectors differ by the homotopy $(b)_Z, 0$. The trace difference is zero because $J_Z \Theta = 0$. Degree-one and tensor Koszul signs remain those of the source complex. On a raw tensor metric, the inserted W_m terms give the copied allowance $n \epsilon_m$; tensoring alone does not improve it.

The new results are the actual orthogonal source choice, rank-two boundary formula, exact relative certificate, rank-one source-kernel recurrence, arithmetic moment tail bound, and the full Hilbert/jet topology comparison. No uniform arithmetic purity estimate, global spectral-kernel vanishing, RH or GRH is asserted.

The thirteen finite exact tests check the encoded algebra and declared calibration models, not infinite analytic arguments or actual zeta zeros. The complete cochain-homotopy theorem remains the other session's result, now supported by the independently inspected successful PR #12 run. Its certificate does not extend to the new analytic control results. Existing Lean modules and earlier validation receipts are untouched.

References: live PR #11 at the pinned head; successful PR #12 and source/job references in COORDINATION_UPDATE.md; the supplied `Tau_Deligne_Control` and `Tau_Global_Comparison` source notes; the supplied other-session homotopy proof; the previously retained Weil II 3.3.4-3.3.6 and 6.2 duality excerpts. Standard orthogonal-polynomial context is NIST DLMF 18.2; the density argument needed here is proved above for the actual arithmetic measure.

F PR 17: Spectral-sum descent and the complete relative arithmetic fibre

Source attribution. Web contribution, pull request #17, [Zeta research repository](#).

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. workbenches/tau-spectral-sum/RESEARCH_NOTE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 551e3fe89b4409d5ed1eeacfaac4cb7d9c1291978127de1b4f989b272bd24a47

F.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation, 12 September 2026. Draft for review; no purity or RH theorem is claimed. Base: PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc.

F.1.1 1. Original source and scope

Keep $G(R) = \{\tau\} \text{ disjoint-union } \{r^\bullet\}$, $e_R = 0_R^\bullet$, and the original square $G(Z) \rightarrow G(C)$ over $Z \rightarrow C$. The first arithmetic quotient remains infinite. Fixed-label linear maps lift by $(\lambda, v) \rightarrow (\lambda, Tv)$ and fix external absence; a vanishing value remains the original labelled supported zero.

Use the existing $C_+ = [V \rightarrow \Theta_B]$, $Q = B/\Theta \setminus V$, $D = -x/d/dx$, $g = 2/x$, and the actual source function $\phi_* = (4\pi^2 x^4 - 6\pi x^2)\exp(-\pi x^2)$. For a finite packet h of actual zeros, including full orders, the prior analytic construction supplies F_h with Mellin transform g/h . Retain

$$T_h(P) = P(D)F_h, \quad T_h(hP) = \Theta(P(D)\phi_*), \quad J_h T_h(P) = v_h[P]_h,$$

where $\text{upsilon}_h = j_h(g/h)$ is the full invertible jet, not the scalar e . These analytic range statements are inputs from the previous work, not newly Lean-certified here.

For k factors, the actual numerator is $P_M = C[s_1, \dots, s_k]_{(\text{degree} \leq M)}$, the relation space is its intersection with $(h(s_1), \dots, h(s_k))$, and its theta realization is $B_-(k, M)$. The canonical representative $R_-(k, M)$ is the unique least-norm lift with exactly these relations allowed. Its full jets, arithmetic class, Gram G and control W remain those of #14.

The supplied Kernel Layer Integration supplement agrees with the Symmetric Frontier note under $K = C_{\text{inverse}}$, $U = F$, $D = \Omega$, $E_{\text{layer}} = b_M$, and $F_{\text{derivative}} = \Omega^{(-1)}E_{\text{plus}}$. Its sharper rank and original-layer identities are retained as source results, not counted again as new work.

F.1.2 2. An actual one-parameter pushforward and its resolution

Define

$$\iota_k : \mathbb{C}[S] \rightarrow E_h^{\otimes k}, \quad S \mapsto [s_1 + \dots + s_k].$$

Restriction of modules through this map is the exact quasi-coherent pushforward along the finite scheme morphism. Its adjunction maps are $f \rightarrow (1 \rightarrow f(1 \otimes 1))$ and $a \otimes 1 \rightarrow a f(1)$. Applying G gives $p \setminus G(\iota) = \iota \setminus p$, with e sent to e and τ to τ ; the morphism is over the same absolute pointed base. No finite field is substituted.

For the actual finite image E , with sum operator A , there is the free resolution

$$0 \rightarrow \mathbb{C}[S] \otimes E \xrightarrow{SI-A} \mathbb{C}[S] \otimes E \xrightarrow{\text{ev}_A} E_A \rightarrow 0.$$

Its augmentation sends $P \backslash \text{tensor} \backslash v$ to $P(A)v$. Injectivity follows from the highest polynomial coefficient. Exactness follows from

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Dualizing with $\mathbb{C}[S] dS$ gives $SI - A^\vee$ in degrees 0,1, where $A^\vee(\lambda) = \lambda$ compose A is the algebraic transpose, not a Hermitian adjoint. Thus

$$\text{Ext}_{\mathbb{C}[S]}^1(E_A, \mathbb{C}[S]dS) \cong E^\vee,$$

with $[\sum S^j \lambda_j dS] \rightarrow \sum (A^\vee)^j \lambda_j$ and inverse the constant class. With $\mathbb{C}[b]^{i=b} = \mathbb{C}^{i=b}$, the exact shifted statement is

$$R\text{Hom}(E_A[-k], \mathbb{C}[S]dS[1]) \simeq E^\vee[k].$$

The right adjoint is the specified module functor $R\text{Hom}_{\mathbb{C}[S]}(E_h^{\text{tensor } k}, -)$. This is not an assertion about pushforward on all sheaves of modules. See Stacks Tags 0AVV and 0AWZ.

The same duality has the concrete pairing

$$\langle v, [P(S)dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S) ((SI - A)^{-1} v) dS.$$

Expansion at infinity gives $\sum \lambda_j (A^j v)$, proving the formula and perfectness. On a length- l block,

$$(SI - A)^{-1} = \sum_{j=0}^{l-1} \frac{N^j}{(S - \lambda)^{j+1}}.$$

All poles are retained before taking the trace. The further observation is

$$\text{Tr}(\psi(A)) = \sum_{\lambda} \text{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS.$$

Here $\chi_A = \det(SI - A)$ is the finite characteristic polynomial, not a replacement for $g=2x$. With $\psi(S) = a^S$ this is the finite trace of the same diagonal scaling correspondence; its cochain sign is $(-1)^k$.

F.1.3 3. Exact symmetric-square relation algebra

Retain the coordinate morphisms

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2, \quad s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and negates r . Even/odd polynomial expansion gives the filtered algebra isomorphism $\Phi: \mathbb{C}[S, \Delta] \rightarrow \mathbb{C}[s_1, s_2]^{\text{swap}}$, with $\deg(S)=1$, $\deg(\Delta)=2$.

Define H_0, H_1 without localizing away from $r=0$:

$$h((S+r)/2) = H_0(S, r^2) + rH_1(S, r^2), \quad h((S-r)/2) = H_0(S, r^2) - rH_1(S, r^2).$$

Then

$$(E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).$$

Proof: the original ideal in $\mathbb{C}[S, r]$ is (H_0, rH_1) . Averaging an expression $H_0 \setminus a + r \setminus H_1 \setminus b$ under $r \rightarrow -r$ gives exactly $H_0 \setminus a_{\text{even}} + \Delta \setminus H_1 \setminus b_{\text{odd}}$. Conversely both generators lie in the original ideal. Reynolds averaging is exact over $\mathbb{C}$, so the invariant quotient has precisely this ideal, including all nilpotents.

The unscaled orbit sums of $s_1^{i_1} \dots s_d^{i_d}$, $0 \leq i_1 \leq \dots \leq i_d \leq d$, give dimension $d(d+1)/2$. The associated graded Hilbert series for the original degree filtration is

$$\sum_{0 \leq i_1 \leq \dots \leq i_d \leq d} z^{i_1 + \dots + i_d} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}.$$

No grading of a nonhomogeneous h is asserted.

The original primitive map is explicit:

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}.$$

Its two cochain primitives use $\phi_* \setminus \text{tensor} \setminus F_h$ and $F_h \setminus \text{tensor} \setminus \phi_*$, respectively. The second primitive has a minus sign, matched by the second tensor-differential sign. For a relation of degree M , ordered division followed by swap averaging gives $\deg A_0 \leq M-d$, $\deg A_1 \leq M-d-1$.

The unit $\epsilon_h \setminus \text{tensor} \setminus \epsilon_h$ descends to the invariant algebra and remains in its full jet map. All finite relation quotients and the transitions $M \rightarrow M+1$ commute with Φ . A class killed by the next relation maps to the receiving e -zero, never to external absence.

F.1.4 4. Push the actual norm to a matrix weight, not a scalar observation

Set

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}.$$

The exact real coordinate map is

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad dt_1 dt_2 = \frac{1}{2} du dv.$$

It gives $S=1+iu$, $\Delta=-v^2$. For

$$P(S, \Delta) = \sum_{a=0}^{\lfloor M/2 \rfloor} \Delta^a f_a(S), \quad \deg f_a \leq M - 2a,$$

define

$$W_{ac}(u) = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv.$$

Then the original arithmetic norm is exactly

$$\|T_h^{(2)} \Phi P\|^2 = \int_{\mathbb{R}} f(1+iu)^* \mathbf{W}(u) f(1+iu) du.$$

Proof: substitute the coordinate map and expand the finite polynomial. Exponential moments justify Fubini. Each finite matrix $W(u)$ is strictly positive: the integrand is the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. The entire nonzero g/h has only discrete zeros on these real lines. The source exponential bound also gives convergence for each fixed u .

Its scalar entry is $(w_h^* w_h)(u)$. Its other entries retain the relative-coordinate moments; they are not replaced by that scalar entry.

For general k retain $S = \sum s_i$, $x_i = s_i$ for $i < k$ and $s_k = S - \sum s_i$. This is a polynomial isomorphism and the real Jacobian is one. Writing $P = \sum x^\alpha f_\alpha(S)$ gives

$$W_{\alpha\beta}(u) = \int_{\mathbb{R}^{k-1}} \prod_{i < k} (1/2 + it_i)^{\alpha_i} \prod_{i < k} (1/2 + it_i)^{\beta_i} \prod_{i < k} w_h(t_i) w_h(u - \sum_{i < k} t_i) d^{k-1}t.$$

The outer coordinate remains $S = k/2 + iu$, and $\deg f_\alpha \leq M - |\alpha|$. This supplies an all- k isometry retaining the entire filtered source. Constants in exponential-moment bounds depend on h, k and the degrees; no uniform tensor bound is inferred.

F.1.5 5. Canonical control commutes with the descent

For the explicit source-coordinate matrix C and quotient-coordinate matrix U , keep

$$M' = C^* M C, \quad J' = U^{-1} J C, \quad K' = U^{-1} K (U^{-1})^*, \quad G' = U^* G U,$$

$$A' = U^{-1} A U, \quad W' = U^* W U, \quad R' = R U.$$

These follow by substituting into $K = J M^{\wedge}(-1) J^*$ and the exact constrained-minimum formula. Thus no positive metric has been freely selected. The same isometry carries the actual next relation layer and both boundary maps Y, C_{rel} ; their cross-pairing is unchanged.

There is a further exact base/relative map. Write $w(u) = [m_0, b], [b^*, c_{\text{rel}}]$, with $m_0 > 0$. Then

$$f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}})$$

is an isometry into the direct sum weighted by m_0 and $c_{\text{rel}} - b^* m_0^{\wedge}(-1) b$. Its inverse is $(a, z) \mapsto (a - m_0^{\wedge}(-1) b z, z)$. The precise image of the polynomial source is retained; the ratio need not be polynomial. Multiplication by S commutes with this map. Neither the scalar mass nor positivity of this representative norm is substituted for the arithmetic Weil pairing.

F.1.6 6. Full nilpotent fibres for every tensor degree

For an ordered tuple of roots with orders m_i , the local algebra is

$$L = \mathbb{C}[z_1, \dots, z_k] / (z_i^{m_i}), \quad A = \rho_{\Sigma} I + N, \quad N = M_{\sum z_i}, \quad D_0 = \sum_i (m_i - 1).$$

Let $c_r = [t^{\wedge r}] \prod_i (1 + t + \dots + t^{\wedge(m_i-1)})$, $c_{-1} = 0$. Then the complete module pushforward is

$$\pi_* L \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{c_r - c_{r-1}}.$$

Proof and maps: on z_{-i}^j define $F_{-i} = j(m_{-i} - j)z_{-i}^{j-1}$ and $H_{-i} = (2j - (m_{-i} - 1))z_{-i}^j$. Direct calculation gives $[N, F] = H$, $[H, N] = 2N$, $[H, F] = -2F$. The explicit auxiliary diagonal Gram

$$\|z^\alpha\|_0^2 = \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}$$

makes $F = N^*$. A degree- r primitive p with $Fp = 0$ has ladder length $w+1$, $w = D_0 - 2r$, since

$$FN^j p = j(w - j + 1)N^{j-1}p, \quad \|N^j p\|_0^2 = \frac{j!w!}{(w - j)!} \|p\|_0^2.$$

Orthogonal invariant complements construct all ladders; degree counts give $c_r - c_{(r-1)}$. The actual module map is $[P(S)] \rightarrow P(\rho_\Sigma + N)p$ on each chosen primitive. This auxiliary Gram is related to the arithmetic one by the explicit operator $G_{\text{arithmetic}}^{(-1)} G_0$; it is used for the algebraic decomposition only. All cross-ladder entries $U^* G_{\text{arithmetic}} U$ remain in control estimates.

For an occupation list n_ρ with sum k , the symmetric component has graded dimensions given by

$$\prod_{\rho} \prod_{j=1}^{n_{\rho}} \frac{1 - t^{m_{\rho} + j - 1}}{1 - t^j}.$$

These are polynomial orbit-count identities, not evaluations of a rational function at roots of unity. Apply the same primitive construction to invariants, with $D_0 = \sum n_\rho (m_\rho - 1)$ and $\rho_\Sigma a = \sum n_\rho \rho$. Central occupation idempotents are retained even when different occupations have the same spectral sum.

For an order-three local input taken twice, the symmetric fibre has ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad X = z_1 + z_2, \quad Xp = 0.$$

It is $\mathbb{C}[S]/(S - 2\rho)^5$ direct-sum $\mathbb{C}[S]/(S - 2\rho)$, dimension six. This is a local algebra calculation, not an assertion that ζ has a triple zero. The full arithmetic unit commutes with A and carries the ladder coordinates to the original jets.

Each occupation trace is its full dimension times $a^{(\sum n_\rho \rho)}$, with dimension product $\text{binomial}(n_\rho + m_\rho - 1, n_\rho)$. The cochain trace retains $(-1)^k$. Quotienting a top ladder vector, when explicitly requested as a module quotient, makes it the supported zero; the full pushforward itself keeps it.

F.1.7 7. Exact calibration and scope

For the declared measure $\exp(-t^2/2)dt$, not the arithmetic measure, the three relative directions $1, \Delta, \Delta^2$ give

$$W(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix}.$$

The unscaled coordinate columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ turn this into $\sqrt{\pi} \exp(-u^2/4) \text{diag}(1, 8, 384)$. The full two-variable mass is still 2π . The published core

checker verifies these signs, constants, algebraic primitives, sum resolution and nilpotent blocks. The longer conversation package has a separate 24-method suite, plus independent reruns of the source 18- and 22-method suites; those are not all claimed to be tests in this smaller repository checker.

The new reduction isolates the convolution mass, the relative covariance, and full-jet interpolation on one explicit filtered image. It does not prove their required uniform sublinear tensor estimate. The finite pushforward and matrix-weight model are not asserted to be lisse sheaves on Deligne's Lefschetz pencil. They supply a precisely defined product-to-one-parameter operation to study next, with its adjoint, cohomology, arithmetic trace and original supported relations retained.

References: Stacks Tags 0AVV and 0AWZ; NIST DLMF 18.2; Deligne, La conjecture de Weil II, IHES 52 (1980), 137-252. A bounded supplied French TeX window around 3.2.11–3.2.15 was reread. Its historical line 872 contains an inconsistent numerical implication and is not used as a theorem. No complete new audit of Weil II, analytic integral certificate, Lean execution, or global novelty claim is made.

G Derived Split-Zero mathematics

Source attribution. Web contribution retained on the pre-integration main revision.

Exact revision. 14c69d604044b848d64318751a7bee19edfd9aba

Source file. formal/splitzero/DERIVED_MATHEMATICS.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

SHA-256. 6a290e193161eb09f49b6faba578193fdaf28b5ae3a97340d1c41a40aa99e695

G.1 SplitZero: reconstruction, internal homology, and balanced finite jets

Owner-directed formalization continuation, 12 September 2026.

G.1.1 Scope and provenance

This contribution extends formal/splitzero on the owner's Zeta workbench. Its base is the synchronization PR #5, commit 236f0a8370477991e2d0cf334ad1ebfe4f3f8b17, itself based on the structural PR #4. The original scalar core and all five preceding extension sources remain unchanged. There is no Sneed-side submission or integration in this continuation.

The mathematical source is the owner's SplitZero programme, including the supplied continuation's internal homology and balanced-Mellin sections. The implementation below distinguishes source-derived structure, ordinary reusable algebra, and the new explicit comparison example. It makes no global priority claim and does not treat the number of Lean declarations as a number of mathematical discoveries. No private conversation or source-literature corpus is republished.

The exact build certificate for a revision is the completed SplitZero\derived\constructions workflow on that commit, including direct source checks and its generated AuditDerived.lean. The PR records the accepted execution revision and run. A failed earlier development run is not a certificate for later source.

G.1.2 1. Reverse reconstruction, with actual scalar action

Let R be a commutative ring, L a join-semilattice with bottom, and let D consist of R -modules V_i with linear transition maps

$$\begin{aligned} \text{rho_ij} &: V_i \rightarrow V_j \quad (i \leq j), \\ \text{rho_ii} &= \text{id}, \quad \text{rho_jk} \circ \text{rho_ij} = \text{rho_ik}. \end{aligned}$$

No finiteness, field, injectivity, or nontriviality assumption is imposed. Construct

$$\text{Total}(D) = \text{disjoint union of } V_i \text{ over } i \text{ in } L.$$

The global zero is $(\text{bottom}, 0)$, and

$$\begin{aligned} (i, x) + (j, y) &= (i \text{ join } j, \text{rho_i, i-join-j}(x) + \text{rho_j, i-join-j}(y)). \\ \text{tau} \cdot (i, x) &= (\text{bottom}, 0). \\ \text{ofR}(r) \cdot (i, x) &= (i, r \cdot x). \end{aligned}$$

`LinearDiagram.totalAddCommMonoid` and `LinearDiagram.totalModule` construct the actual instances and prove every law. For associativity, transport both sides to $i \text{ join } j \text{ join } k$. Linearity and transition composition reduce both to the same three-term sum. The scalar laws follow in the supported case from the R -module laws and in the absent cases from the chosen global zero. The supported-zero scalar sends (i,x) to $(i,0)$.

The bottom fibre is allowed to be nonzero. Specializing it to the zero module recovers the source's carrier consisting of one absent point and the non-bottom fibres. This is an explicit generalization, not an assumption that the bottom fibre in every later dual diagram vanishes. Opposite-index duality itself is not constructed here.

Fixed-index natural families of linear maps produce actual $G(R)$ -linear maps between reconstructed totals. Identity and composition are proved. More substantially, `Recovery.recoverEquiv` reconstructs the intrinsic support fibres of **any** existing $G(R)$ -semimodule M and gives a full $G(R)$ -linear equivalence back to M :

$$(1,x) \mapsto x, \quad m \mapsto (e\ m, m).$$

Its addition proof uses the original absorption identity $m + e\ m = m$. It is not merely a bijection of carriers. A bundled equivalence of the two full categories, including arbitrary changes of support index and both natural inverse comparisons, is a separate library interface not claimed by this implementation.

G.1.3 2. The internal quotient is characterized by a universal property

Let B_i be transition-stable submodules of V_i . The quotient diagram has fibres V_i/B_i and the genuinely induced quotient transition maps. The projection

$$q : \text{Total}(D) \rightarrow \text{Total}(D/B), \quad q(i,x) = (i,[x])$$

is a surjective $G(R)$ -linear map. The code proves the exact equality criterion

$$q(i,x)=q(i,y) \text{ iff } x-y \text{ is in } B_i.$$

Different labels cannot become equal under q . In particular a relation maps to its own fibre zero.

The main point is stronger than that observation. Let N be **any** $G(R)$ -semimodule, not assumed additively cancellative, and let $f:\text{Total}(D)\rightarrow N$ be split-linear. If

$$f(i,b)=f(i,0) \text{ for every } b \text{ in } B_i,$$

then there exists a unique split-linear $g:\text{Total}(D/B)\rightarrow N$ with $g\ q=f$. The proof does not subtract elements in N . If $x-y$ is in B_i , use the identity $y+(x-y)=x$ inside V_i and then apply additivity of f . This proves constancy on q -fibres. Surjectivity constructs the descended function; linearity and uniqueness follow by lifting representatives.

`Relations.coequalizer_universal` states this against the actual inclusion of the relation diagram and its supported-zero companion. Thus the universal property is proved against arbitrary semimodule targets, not only against a selected class of fixed-support maps. The implementation gives the complete unbundled universal property; it does not merely define the desired quotient and label it a coequalizer.

G.1.4 3. Actual homology and the exact transported-class kernel

At a degree i of a cochain complex, use the window

$C^{(i-1)} \xrightarrow{d_{\text{Prev}}} C^i \xrightarrow{d_{\text{Next}}} C^{(i+1)}, \quad d_{\text{Next}} d_{\text{Prev}} = 0.$

Window.boundaryToCycles constructs the incoming map into $\ker(d_{\text{Next}})$. Boundaries are its actual linear range, and H is the module quotient of cycles by that range. Window.ofCochain imports these data at every integer degree of a Mathlib CochainComplex; ChainMap.ofCochain imports an actual cochain map without replacing its differentials.

For a chain map $f:C \rightarrow D$, the code constructs

$f_Z : Z(C) \rightarrow Z(D),$
 $H(f) : Z(C)/B(C) \rightarrow Z(D)/B(D).$

It proves identity, composition, and

$H(f)([z]) = 0 \iff \text{there exists } y \text{ with } d_{\text{Prev}_D}(y) = f_i(z).$

Define the module of killed representatives and its original relation submodule:

$\text{Krep} = \{z \text{ in } Z(C) : f_Z(z) \text{ is in } B(D)\},$
 $\text{Bold} = B(C), \text{ viewed as a submodule of Krep.}$

There is a constructed linear equivalence

$\text{Krep}/\text{Bold} \simeq \ker H(f).$

Forward map: the class of z goes to the kernel element $[z]$. It is well-defined because Bold consists exactly of original source boundaries. It is injective because equality of source classes is exactly a difference in Bold. It is surjective because every kernel class has a cycle representative whose image is a boundary. This is ChainMap.homologyKernelEquiv, obtained from a general quotient-kernel equivalence proved in the same file.

The calculation retains actions as well. Given chain endomorphisms a of C and b of D satisfying $f \circ a = b \circ f$, homology_intertwines proves $H(f) \circ H(a) = H(b) \circ H(f)$. kernelAction constructs the restriction of $H(a)$ to $\ker H(f)$. Thus passing to the comparison kernel does not discard a compatible operator or require asserting that the kernel is zero.

G.1.5 4. Internal homology on the complete support diagram

For coherent support-indexed chain windows, the code constructs the diagrams of preceding terms, middle terms, following terms, cycles, boundaries, and homology. Let R_L denote the reverse reconstruction.

The differential identity is typed as

$R_L(d_{\text{Next}}) \circ R_L(d_{\text{Prev}}) = z_L : R_L(C^{(i-1)}) \rightarrow R_L(C^{(i+1)}),$
 $z_L(1, x) = (1, 0).$

The source and target degrees are explicit. A theorem shows that z_L differs from the constant global-zero map whenever L has a non-bottom element. Cycles satisfy the actual equalizer equation $d(x) = e.d(x)$.

The internal homology projection

$q : R_L(Z(C)) \rightarrow R_L(H(C))$

is the coequalizer of the **original incoming differential into cycles** and its supported-zero companion. `homology_coequalizes` and `homology_coequalizer` establish both parts of that universal property against every split-linear target. This is not an assumed homology compatibility of reconstruction.

For every transition $i \leq j$,

$\text{rho}^{\wedge} H_{ij}([z]) = 0$ iff $\text{rho}_{ij}(z)$ is a boundary at j .

If z is not a boundary at i , its retained class $(i, [z])$ is not $(i, 0)$, regardless of whether that transition kills it. The quotient-kernel equivalence of section 3 computes all such classes as a module, not just a yes/no predicate.

G.1.6 5. A concrete comparison that detects more than the support labels

Fix any nontrivial commutative ring k . In the displayed degree let

$C: k \xrightarrow{-\theta} k \xrightarrow{-\theta} k,$
 $D: k \xrightarrow{-a} (a, \theta) \xrightarrow{-} k^2 \xrightarrow{-\theta} k.$

Use zero maps in the outside degrees. The two middle chain maps are

$f(a) = (a, \theta),$ $g(a) = (\theta, a).$

The code proves that the source class $[1]$ is nonzero. Its f -image is the **nonzero vector** $(1, 0)$, but that vector is a boundary in D , so $H(f)([1]) = 0$. In fact $H(f) = 0$ on every source class. The g -image $(0, 1)$ cannot be a boundary, because every boundary has second coordinate zero. Hence $H(g)([1])$ is nonzero and $H(f)$ is not $H(g)$.

The ordinary calculation gives $H(C) = k$ and $H(D) = k$ (the latter through second-coordinate projection). These identifications explain the example; the Lean acceptance tests directly prove the nonzero source class, the nonzero chain image, the zero induced map, the surviving class, and the distinction of induced maps rather than relying on a dimension computation.

Consequently the source and target complexes, individual fibre homology, and terminal homology can be identical while the transition kernel differs. This example is parametric over all nontrivial commutative rings, not a finite numerical experiment. Together with the general assembly theorem it supplies an explicit test of the information retained by support-indexed homology. It is an algebraic test case, not a claimed model of the global theta comparison.

G.1.7 6. Balanced finite jets, including the original split quotient square

Let k and B be commutative rings and fix a $k[t]$ -algebra structure on B . Let $u: k[t] \rightarrow B$ denote that map and

$I_{-(\text{rho}, m)} = ((t - \text{rho})^{\wedge} m), \quad m \geq 0.$

There is an explicit B -algebra equivalence

$B \otimes_{k[t]} (k[t]/I_{-(\text{rho}, m)}) \simeq B / ((u(t) - u(\text{rho}))^{\wedge} m).$

It maps $b \otimes [p]$ to $[u(p)b]$, with inverse $[b]$ to $b \otimes 1$. No flatness assumption is used for this quotient base-change theorem. This is a specialization of the existing Mathlib theorem `Algebra.tensorProduct.quotIdealMapEquivTensorQuot`, not a claim to have independently developed general tensor-product quotient theory.

splitBaseChange lifts this equivalence to the original G-construction on both sides and proves the commutative square with reflect. It preserves τ and e separately and is a genuine semiring equivalence, not just an amplitude comparison. Thus every element of the finite-jet quotient has an inverse image before forgetting split support.

For any $k[t]$ -module V and v in V , the exact balancing calculation is

$$(1-u(t)) \text{ tensor } (t.v) - u(t) \text{ tensor } (v-t.v) = 0.$$

Expanding gives $1 \text{ tensor } (t.v) - u(t) \text{ tensor } v$, which vanishes by the tensor balancing relation. This is the algebraic equation of the source's Mellin repair. The choice $u(t)=s$ and $t.v=Dv$ is an application once those analytic module structures and intertwining maps have been constructed. It does not assert that the entire global balanced kernel vanishes, or that analytic scalar extension is flat without proof.

G.1.8 7. Where Deligne enters the intended RH route

None of sections 1-6 imports a Deligne purity theorem. They provide the algebraic maps with which a proposed realization can now be specified and its kernel retained.

For this programme, the subsequent dependency diagram should distinguish the following mathematical obligations:

1. Construct the actual operator-balanced theta complex at the relevant support labels, and its action-compatible analytic comparison β to the spectral sheaf $O/(2\pi i)$. Identify its kernel or cone using the same maps; a finite-jet isomorphism is not a global injectivity argument.
2. Construct the geometric compact-support and ordinary complexes, their actual comparison c , their correspondence action, and a compatible realization identifying the required spectral sheaf with the image of $H^1(c)$. Any additional kernel/cone contributions to the trace must be kept or proved to vanish.
3. Establish two-sided weight control on that **same image**. In Deligne's geometric setting the compact-support upper estimate and the duality-derived ordinary-cohomology lower estimate meet on the image. The dualizing operation, degree and Tate twists, tensor-power maps, and hypotheses of the estimates are part of this step, not consequences stipulated in the definition of support.
4. Transport the spectral normalization without changing the arithmetic function. For a dilation parameter $a>1$, an eigenvalue a^ρ has modulus $a^{(\text{Re } \rho)}$. A proved weight-one modulus $a^{(1/2)}$, for the representation identified with every relevant spectral zero, then forces $\text{Re } \rho=1/2$. This last calculation does not supply the missing weight estimate; it states the precise final implication of this route.

These are dependencies of the specified route, not a claim that every possible proof of RH must follow it. The newly implemented kernel action is useful in steps 1-2 precisely because it keeps the part killed by a comparison available to the operator and trace calculation.

The supplied Connes-Consani article has an actual quotient-sheaf realization with a specified closure and scaling action. Its Theorem 1.2 realizes **critical zeros**. That theorem alone cannot be substituted for an identification representing all nontrivial zeros in the desired purity argument. This is a concrete source/target distinction for a future comparison map, not a dismissal of the construction.

G.1.9 8. Reading record and references

The six supplied S20 chunks were checked against their individual declared hashes and concatenated in order. The resulting archive has 216,580,466 bytes, 312 members, and SHA-256

e2005bf31e1fcf362765f73c315c855e0f504f24f34d3a0ab188049da522b7c8.

The current top-level readers/english_standalone.html and readers/source_language.html were used for the selected readings, not the nested zero-accepted salvage corpus. The inspected windows include the tensor-square improvement in 3.2.13, the upper-bound statement in 3.3.1, and the duality/image argument in 3.3.4-3.3.6. This is not an independent audit of every page or of the entire historical proof. The Connes-Consani introduction and Theorem 1.2 were read in the supplied volume, printed pages 83-85 (physical PDF pages 94-96). No copyrighted source corpus is included in this contribution.

Deligne, P. (1980). La conjecture de Weil. II. *Publications Mathematiques de l'IHES*, 52, 137-252. <https://doi.org/10.1007/BF02684780>

Connes, A., & Consani, C. (2023). Hochschild homology, trace map and zeta-cycles. In A. Connes, C. Consani, B. I. Dundas, M. Khalkhali, & H. Moscovici (Eds.), *Cyclic cohomology at 40: Achievements and future prospects* (Proceedings of Symposia in Pure Mathematics, Vol. 105, pp. 83-102). American Mathematical Society.

Mathlib contributors. (2026). *Mathlib4*, revision fabf563a7c95a166b8d7b6efca11c8b4dc9d911f. In particular, LinearAlgebra/Quotient/{Defs,Basic}.lean, RingTheory/TensorProduct/Quotient.lean, Algebra/Homology/HomologicalComplex.lean, and Algebra/Category/ModuleCat/Basic.lean.

G.1.10 9. Reproduction

From formal/splitzero on this branch, with elan and Python installed:

```
python3 prepare.py
python3 check_derived.py --prepare
python3 -m unittest -v test_derived_checker.py
lake update
# Verify the resolved Mathlib revision before fetching its cache.
test "$(git -C .lake/packages/mathlib rev-parse HEAD)" = fabf563a7c95a166b8d7b6efca11c8b4dc9d911f
lake exe cache get
lake -KmaxJobs=1 build
for name in SplitZeroReconstruction SplitZeroRecovery SplitZeroHomology SplitZeroInternalQuotient
↪ SplitZeroComplex SplitZeroHomologyExample SplitZeroFiniteJets; do
  lake env lean --trust=0 -DwarningAsError=true "$name.lean"
done
lake env lean --trust=0 -DwarningAsError=true AuditDerived.lean > AuditDerived.log
python3 check_derived.py AuditDerived.log
```

The manifest enumerates the selected declarations, including the principal structures, universal properties, equivalences, and example proofs. Missing, duplicate, unexpected, or nonstandard axiom reports fail. Source checks reject unfinished proofs and native-reduction escapes. The Python tests validate this checking machinery; the mathematical certificate is the successful Lean execution. Build-job totals include dependencies and are not theorem counts.

H PR 14: Coherent arithmetic interpolation and joint tensor boundary control

Source attribution. Complete TeX witness delivered with [web pull request #14](#); the associated Git revision is recorded in its delivery receipt.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Coherent_Interpolation_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 01a4fa76072ca4f3d8266c9fdbf61c816e53d09c673f8b8776b981b736c09e26

Original archive. Tau_Coherent_Interpolation_Control_2026-09-12.zip

Archive SHA-256. a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26

H.1 Coherent arithmetic interpolation and joint tensor boundary control

Owner-directed continuation, 12 September 2026. Written mathematical arguments for review; the new analytic and matrix results in this note have not been Lean-checked.

H.1.1 0. Accepted inputs and the change in the continuation

The new formalization receipt reports—and the repository metadata and workflow step summaries confirm—a successful check at 5d2772ea0a16d177ac3e01ff70394b90ba95a232 in PR #12. That certificate covers the specified retraction’s algebraic/topological consequences, the complete coefficient homotopy family, and the stated finite-chart dual maps. It does not certify the analytic construction of the theta retraction, support-changing synchronization homotopies, or the matrix estimates developed here. Its branch is left untouched.

The second attachment proves, for an unrestricted finite correction matrix B , the exact formula

$$G_B = G_{\perp,m} + (B + H_m^{-1}C_m)^* H_m (B + H_m^{-1}C_m). \quad (0.1)$$

Once $m+1 \geq \dim E_Z$, every positive metric ray is realized by an actual theta-boundary correction, with the arithmetic jets unchanged. Its further conclusion

$$\inf_B \epsilon_A(G_B) = \max_{\rho \in Z} |2\Re \rho - 1| \quad (0.2)$$

therefore has to be retained. This note does not count unrestricted metric optimization as a new arithmetic estimate. It instead uses the unique least-norm representative for a fixed, increasing space of the original arithmetic boundaries. It proves compatibility of these representatives for all finite packets, an exact full-jet moment-matrix formula, and a joint total-degree tensor construction.

All conclusions are derived on the existing characteristic-zero theta complex over the marked tau-base. None is asserted to be an application of finite-field purity to a space merely renamed tau.

H.1.2 1. Fixed scalar, cochain, and arithmetic maps

Retain the original scalar construction

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the entire square

$$\boxed{\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}} \quad (1.1)$$

The target of $p_{\mathbb{Z}}$ is infinite. This quotient remains part of the marked geometry over $\mathfrak{b}_\tau$. Coefficient maps below fix external absence and carry a represented scalar zero to the represented scalar zero of the target.

Use the unchanged spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^k F(x)| < \infty \forall b \in \mathbb{Z}, k \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{m \neq 0} \phi(mx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Write $q : \mathcal{B} \rightarrow Q$. Its action is by the original D , not a shifted operator. Let

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad f_0 = \Theta\phi_*,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad \mathcal{M}f_0 = g. \quad (1.2)$$

The established range identities are used with these exact constants. They give $\mathcal{M}\Theta\phi = gH_\phi$, with H_ϕ entire. In particular every finite jet map used below annihilates ΘV .

For $n \geq 0$, retain

$$V_n = \text{span}\{\phi_*, D\phi_*, \dots, D^n\phi_*\}, \quad L_n = \Theta V_n,$$

$$Q_n = \mathcal{B}/L_n \longrightarrow Q_{n+1}, \quad \ker = L_{n+1}/L_n. \quad (1.3)$$

Their split reconstructions have labels V_n . The transport of a class in L_{n+1}/L_n is the supported zero at V_{n+1} ; it is not external absence. The derivative has type $Q_n \rightarrow Q_{n+1}$ because $DL_n \subseteq L_{n+1}$.

H.1.3 2. A polynomial presentation of the actual finite representatives

Take a nonempty finite set Z of actual zeros of g , with their complete orders m_ρ , and put

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[t]/(h), \quad v_h(s) = g(s)/h(s). \quad (2.1)$$

The polynomial variable t acts on the source by D and is evaluated at s by Mellin transformation. Full orders make v_h nonzero at all selected centres. Let j_h be the complete Hermite-jet/Chinese-remainder map into E_h and retain

$$v_h = j_h v_h \in E_h^\times, \quad \varepsilon_h = v_h^{-1}. \quad (2.2)$$

The unit ε_h is not the scalar e . Its split scalar is $\varepsilon_h^\bullet$.

There is an actual $F_h \in \mathcal{B}$ with $h(D)F_h = f_0$ and $\mathcal{M}F_h = v_h$. It can be obtained successively from

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy, \quad (D - \rho)S_\rho F = F, \quad (2.3)$$

whenever the required Mellin value vanishes. At zero, replace the integral by its negative from 0 to x , using that vanishing; at infinity use the displayed integral. The defining estimates of $\mathcal{B}$ prove rapid decay of every scaling derivative at both ends. Repeated zero jets permit iteration. Uniqueness follows from injectivity of Mellin and $h\mathcal{M}F = 0 \Rightarrow F = 0$.

Define the linear map

$$\boxed{\mathcal{T}_h : \mathbb{C}[t] \longrightarrow \mathcal{B}, \quad P \longmapsto P(D)F_h.} \quad (2.4)$$

It is injective and satisfies

$$\mathcal{M}\mathcal{T}_h(P) = \frac{g}{h}P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h\mathcal{T}_h(P) = v_h[P]_h. \quad (2.5)$$

Here $J_h F = j_h \mathcal{M}F$. The first equation retains the whole completed function, not its values only on the critical line.

For $N = n + d$, let $\mathcal{P}_N = \mathbb{C}[t]_{\leq N}$, and define

$$\mathcal{J}_{h,N} : \mathcal{P}_N \twoheadrightarrow E_h, \quad P \longmapsto v_h[P]_h.$$

Its exact sequence is

$$\boxed{0 \longrightarrow \mathcal{P}_n \xrightarrow{\times h} \mathcal{P}_{n+d} \xrightarrow{\mathcal{J}_{h,n+d}} E_h \longrightarrow 0.} \quad (2.6)$$

Applying $\mathcal{T}_h$ sends the left map to the original theta relations L_n . Thus its split internal quotient has exactly the relation $P - P' \in h\mathcal{P}_n$, with the support label retained.

The original source section is

$$s_h(u) = \mathcal{T}_h(\text{rem}_h(\varepsilon_h u)), \quad J_h s_h = 1, \quad q s_h = \sigma_h. \quad (2.7)$$

No arbitrary metric or section has entered (2.6).

H.1.4 3. Exact arithmetic moment matrices and the full-jet interpolation formula

The Hilbert observation is $\mathcal{H} = L^2((0, \infty), dx)$. The precise transform map is

$$\mathcal{H} \xrightarrow{\mathcal{L}} L^2(\mathbb{R}, dy) \xrightarrow{\mathcal{F}_+} L^2(\mathbb{R}, dt/(2\pi)),$$

$$\mathcal{L}F(y) = e^{y/2}F(e^y), \quad \mathcal{F}_+\psi(t) = \int_{\mathbb{R}} \psi(y)e^{ity}dy, \quad (\mathcal{F}_+\mathcal{L}F)(t) = \mathcal{M}F(\tfrac{1}{2} + it). \quad (3.1)$$

Both arrows are unitary with these measures. The inverse Fourier factor is $1/(2\pi)$; no mass of a measure below is set equal to one. The same map carries D to multiplication by $s_t = 1/2 + it$.

Define the finite positive measure

$$d\nu_h(t) = \left| \frac{g(\frac{1}{2} + it)}{h(\frac{1}{2} + it)} \right|^2 \frac{dt}{2\pi}. \quad (3.2)$$

Any apparent singularity at a selected critical-line zero is removed by the entire function v_h . All moments are finite. In the fixed basis $1, t, \dots, t^N$ of $\mathcal{P}_N$, put

$$(M_{h,N})_{ij} = \int_{\mathbb{R}} \overline{s_t^i} s_t^j d\nu_h(t), \quad 0 \leq i, j \leq N. \quad (3.3)$$

In the integrand t is the real integration variable; the polynomial basis is evaluated at s_t . This prevents the polynomial variable from being confused with the spectral-height coordinate.

The matrix $M_{h,N}$ is positive definite. A polynomial with zero norm vanishes almost everywhere on a line on which v_h is nonzero almost everywhere, hence is zero. Write $J_{h,N}$ for the matrix of $\mathcal{J}_{h,N}$ in the fixed quotient basis $1, t, \dots, t^{d-1}$. It includes multiplication by the actual unit v_h .

Set

$$\mathcal{C}_{h,N} = J_{h,N} M_{h,N}^{-1} J_{h,N}^*, \quad C_{h,N} = M_{h,N}^{-1} J_{h,N}^* \mathcal{C}_{h,N}^{-1}. \quad (3.4)$$

These are finite matrices on declared coordinate spaces. $J_{h,N} C_{h,N} = 1$. The actual test-function map is

$$R_{h,n} = \mathcal{T}_h C_{h,n+d} : E_h \longrightarrow \mathcal{B}. \quad (3.5)$$

Theorem. This is precisely the preceding orthogonal representative $(1 - P_n)s_h$, where P_n is the $L^2(dx)$ orthogonal projection onto L_n . In particular,

$$qR_{h,n} = \sigma_h, \quad J_h R_{h,n} = 1, \quad G_{h,n} := R_{h,n}^* R_{h,n} = \mathcal{C}_{h,n+d}^{-1}. \quad (3.6)$$

Proof. Every polynomial with jet u is $C_{h,N}u + z$ with $J_{h,N}z = 0$. The cross term vanishes because

$$z^* M_{h,N} C_{h,N} u = z^* J_{h,N}^* \mathcal{C}_{h,N}^{-1} u = 0.$$

Its norm is therefore $u^* \mathcal{C}_{h,N}^{-1} u + z^* M_{h,N} z$. This proves the unique minimizer. By (2.6), its difference from the degree-below- d lift (2.7) is exactly a member of L_n . It must be $(1 - P_n)s_h$. This also proves (3.6).

Thus the subtraction-of-nearby-Grams formula of the previous note has the exact alternative (3.6): invert the full-jet interpolation matrix. This is a matrix-valued Christoffel construction with the arithmetic local unit retained. The finite least-norm identity itself is standard linear algebra; its application here identifies the actual original representative, source relation, and completed multiplier.

For one simple zero ρ , the same formula is

$$G_{\rho,n} = \frac{1}{|g'(\rho)|^2 \mathcal{K}_{\nu_{\rho,n+1}}(\rho, \rho)}, \quad \mathcal{K}_{\nu,N}(z, w) = (1, z, \dots, z^N) M_N^{-1} (1, w, \dots, w^N)^*. \quad (3.7)$$

The complex conjugate in the second evaluation is part of the displayed kernel. Higher orders use (3.4), not just point evaluations. The factor $g' = 2\xi'$ is retained.

H.1.5 4. All finite packets fit the same source filtration

Let $h \mid H$ be polynomials for nested packets with full orders and set $m = H/h$. Since the newly added centres are disjoint, h and m are coprime. There is a $\mathbb{C}[t]$ -linear CRT injection

$$\iota_{hH} : E_h \hookrightarrow E_H,$$

which inserts zero jets at the new centres. It is multiplication by the appropriate CRT idempotent after lifting a residue; it sends the unit of E_h to that idempotent, so it is not being asserted to be a unital ring map.

The polynomial map $P \mapsto mP$ has the exact properties

$$\mathcal{T}_H(mP) = \mathcal{T}_h(P), \quad J_H \mathcal{T}_h(P) = \iota_{hH} \mathcal{J}_h(P), \quad (4.1)$$

and is isometric from ν_h to ν_H because $|v_H mP| = |v_h P|$. The CRT equations and degree bounds prove

$$s_H \iota_{hH} = s_h.$$

The original L_n is independent of the packet, so

$$\boxed{R_{H,n} \iota_{hH} = R_{h,n}, \quad \iota_{hH}^* G_{H,n} \iota_{hH} = G_{h,n}, \quad \iota_{hH}^* W_{H,n} \iota_{hH} = W_{h,n}.} \quad (4.2)$$

The action also intertwines: $A_H \iota_{hH} = \iota_{hH} A_h$, where $A_h = M_t$. Thus a certificate on the larger packet restricts by this actual congruence to the smaller packet. The inverse interpolation matrices satisfy the corresponding inverse-compression formula; inverse and compression are not commuted.

The finite sections consequently glue on the algebraic direct limit of the actual packet injections. This statement does not assert convergence of an infinite sum of packet projectors or identify the whole Q with that direct limit.

At every support label, lift these maps as $(\lambda, u) \mapsto (\lambda, \iota_{hH} u)$ and $(\lambda, u) \mapsto (\lambda, R_{h,n} u)$. Added zero jets remain represented zeros. The label map in (1.3) is still the map that admits one more source relation.

H.1.6 5. A global density result, with its precise cohomological consequence

The single derivative family $\{D^n f_0 : n \geq 0\}$ is already dense in $\mathcal{H}$.

Proof. Put $d\nu(t) = |g(1/2 + it)|^2 dt / (2\pi)$. It has an exponential moment of every order $c < \pi/2$. To verify this, the alternating-series formula and partial summation give

$$|\zeta(\tfrac{1}{2} + it)| \leq \frac{2|\tfrac{1}{2} + it|}{\sqrt{2} - 1}.$$

The gamma vertical-line asymptotic gives, for sufficiently large $|t|$,

$$|g(\tfrac{1}{2} + it)|^2 \leq C(1 + |t|)^{11/2} e^{-\pi|t|/2}.$$

The constant C is a fixed bound, not a numerical enclosure claimed here. The gamma fact used is NIST DLMF 5.11.9. The same exponential-moment property holds for ν_h , since v_h is entire at the finitely many possible denominator zeros and has polynomially modified tails.

If $f \in L^2(\nu)$ is orthogonal to every polynomial, the finite measure $\bar{f} d\nu$ has a holomorphic Fourier transform in a nonempty horizontal strip, by Cauchy–Schwarz and the exponential moment. All

derivatives at the origin vanish. Analytic uniqueness makes its transform zero; Fourier uniqueness for finite measures makes the measure zero. Thus polynomials are dense in $L^2(\nu)$.

Multiplication by the actual function $g(1/2 + it)$ is a unitary map $L^2(\nu) \rightarrow L^2(dt/(2\pi))$: its inverse divides by g away from its measure-zero set of real-line zeros. The inverse is a map of weighted L^2 spaces, not holomorphic division through an arithmetic zero. Combining it with (3.1) sends polynomials to $P(D)f_0$. This proves the assertion.

It follows for each fixed packet that

$$\boxed{G_{h,n} \longrightarrow 0 \quad \text{in every matrix entry and in operator norm.}} \quad (5.1)$$

Indeed $P_n \rightarrow 1$ strongly on $\mathcal{H}$, and the finite number of columns of s_h is fixed. Since each $G_{h,n}$ is positive definite, the least eigenvalue of its inverse tends to infinity. Every nonzero arithmetic row functional $\ell : E_h \rightarrow \mathbb{C}$ therefore has

$$\boxed{\|\ell\|_{G_{h,n}^\vee}^2 = \ell \mathcal{C}_{h,n+d} \ell^* \longrightarrow \infty.} \quad (5.2)$$

These facts do not kill a cohomology class. The actual maps are

$$\mathcal{B}/L_n \longrightarrow \mathcal{H}/L_n, \quad Q = \mathcal{B}/\Theta V \longrightarrow \mathcal{H}/\overline{\Theta V}^{\mathcal{H}} = 0. \quad (5.3)$$

The first map is injective because L_n is finite-dimensional and consists of the same smooth functions in both spaces. The second map forgets the stronger topology and has the whole Q as kernel. Its split lift lands at the represented zero in $G(0)$, not at τ . $G(0)$ retains both elements. An active class may therefore tend to zero under the L^2 observation while remaining fixed by $qR_{h,n} = \sigma_h$ and $J_h R_{h,n} = 1$.

The constructive consequence is to estimate relative boundary flux against the complete interpolation cost (3.4), not to use the absolute smallness of $G_{h,n}$ as a weight bound.

H.1.7 6. Tail representation and same-metric reflection control

Retain the unscaled orthogonal vectors from the previous calculation:

$$u_0 = f_0, \quad u_n = (1 - P_{n-1})D^n f_0, \quad \mathfrak{h}_n = \|u_n\|^2, \quad (6.1)$$

$$Du_n = u_{n+1} + \frac{1}{2}u_n - \frac{\mathfrak{h}_n}{\mathfrak{h}_{n-1}}u_{n-1} \quad (n \geq 1).$$

The diagonal $1/2$ and the negative last coefficient follow from $D^* = 1 - D$ on these functions. Neither the u_n nor their norms are rescaled. Define $r_n = \mathfrak{h}_n^{-1}u_n^* s_h$. The new density result makes the entire tail formula exact:

$$\boxed{G_{h,n} = \sum_{j>n} \mathfrak{h}_j r_j^* r_j.} \quad (6.2)$$

This series converges in finite matrix norm. Its source map is $u \mapsto (r_j u)_{j>n}$ into the weighted sequence space with norm $\sum_{j>n} \mathfrak{h}_j |c_j|^2$. The map retains every coefficient; it is an isometry for $G_{h,n}$.

The previous boundary calculation, or telescoping (6.1), gives

$$\boxed{W_{h,n} = A_h^* G_{h,n} + G_{h,n} A_h - G_{h,n} = \mathfrak{h}_{n+1}(r_{n+1}^* r_n + r_n^* r_{n+1}).} \quad (6.3)$$

It pairs two explicit maps into L_{n+1}/L_n . At the original support V_n the derivative defect is $-[u_{n+1}]r_n$; at V_{n+1} it becomes the supported zero. This is also the input to the now-checked coefficient homotopy family, with $\kappa_n = K_n J_h$ and $\kappa_n \Theta = 0$.

For a reflection-stable packet define the actual antilinear involution

$$j_h u(s) = \overline{u(1 - \bar{s})} \pmod{h}.$$

On functions use $j_B F(x) = x^{-1} \overline{F(1/x)}$. It is antiunitary on $\mathcal{H}$ and satisfies $Dj_B = j_B(1 - D)$. The specific section (2.7) satisfies $j_B s_h = s_h j_h$: since $h^\dagger = (-1)^d h$, both the unit and the numerator have the corresponding $(-1)^d$ factor. Also $j_B L_n = L_n$, so uniqueness of orthogonal projection gives $j_B R_{h,n} = R_{h,n} j_h$.

Write $j_h u = C_h \bar{u}$ in the retained coordinates. Then

$$\boxed{C_h^* G_{h,n} C_h = \overline{G_{h,n}}, \quad A_h C_h = C_h (I - \overline{A_h}), \quad C_h^* W_{h,n} C_h = -\overline{W_{h,n}}.} \quad (6.4)$$

Consequently the generalized spectrum of $(W_{h,n}, G_{h,n})$ is symmetric under $\lambda \mapsto -\lambda$ **for this same canonical metric**, with full multiplicities retained. An upper inequality $W_{h,n} \preceq \epsilon G_{h,n}$ therefore implies the lower inequality $W_{h,n} \succeq -\epsilon G_{h,n}$ by applying j_h to the input vector.

The residue duality is still the separate explicit morphism with matrix S_h , and its pulled-back metric is $S_h^* G_{h,n}^{-1} S_h$. The relation to the current metric is the linear map $G_{h,n}^{-1} S_h$ into its antilinear dual coordinates. The arithmetic Weil form remains $S_h J_g$; no positivity is supplied merely by (6.4).

For completeness, retain the full rank-two formula

$$a_n = r_n G_{h,n}^{-1} r_n^*, \quad b_n = r_{n+1} G_{h,n}^{-1} r_{n+1}^*, \quad c_n = r_n G_{h,n}^{-1} r_{n+1}^*.$$

The least control excess is

$$\epsilon_{h,n} = \mathfrak{h}_{n+1} \left(|\Re c_n| + \sqrt{a_n b_n - (\Im c_n)^2} \right). \quad (6.5)$$

For reflection-stable Z , the additional proved identity is $\Re c_n = 0$: take the trace of $G_{h,n}^{-1} W_{h,n}$, which is $2\Re \operatorname{Tr} A_h - d = 0$. Thus (6.5) also equals $\mathfrak{h}_{n+1} \sqrt{a_n b_n - (\Im c_n)^2}$. Both equations are retained. Equation (3.6) supplies every occurrence of $G_{h,n}^{-1}$ without a difference of nearly equal Gram matrices.

H.1.8 7. Joint total-degree control on the actual product complex

The tensor degree will be denoted k , independently of the relation degree n and the packet multiplicities. Put

$$E_h^{(k)} = E_h^{\otimes k}, \quad A_h^{(k)} = \sum_{j=1}^k I^{\otimes(j-1)} \otimes A_h \otimes I^{\otimes(k-j)}.$$

The existing product complex over $\mathfrak{b}_\tau$ has differential

$$d(x_1 \otimes \cdots \otimes x_k) = \sum_j (-1)^{\sum_{i < j} |x_i|} x_1 \otimes \cdots \otimes dx_j \otimes \cdots \otimes x_k.$$

Define $\mathcal{P}_M^{(k)} = \mathbb{C}[t_1, \dots, t_k]_{\text{total degree} \leq M}$ for $M \geq k(d-1)$ and

$$\mathcal{T}_h^{(k)} P = P(D_1, \dots, D_k)(F_h^{\otimes k}). \quad (7.1)$$

Its Mellin transform is $P(s_1, \dots, s_k) \prod_j g(s_j)/h(s_j)$. Its full jet map $J_{k,M}$ is reduction modulo each $h(t_j)$ followed by multiplication by $v_h^{\otimes k}$. The target is $E_h^{(k)}$ and this map is surjective. The tensor remainder basis has total degree at most $k(d-1)$.

The exact relation space is

$$\mathcal{I}_{k,M} = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)). \quad (7.2)$$

Polynomial division in the fixed order $t_1, \dots, t_k$ represents each member as $\sum_j h(t_j)P_j$, with $\deg P_j \leq M-d$. It terminates because every h is monic and replacing a leading power lowers total degree. Under (7.1), each term is an actual top cochain boundary: in the j -th source factor use ϕ_* in place of F_h and give its primitive the factor $(-1)^{j-1}$. The differential contributes the second $(-1)^{j-1}$. The resulting boundary is $+h(t_j)P_j$ with no removed sign.

Thus these joint corrections are maps in the original product complex, not freely chosen positive tensor metrics. Set $\mathcal{B}_{k,M} = \mathcal{T}_h^{(k)} \mathcal{I}_{k,M}$.

The moment matrix on the monomials of total degree at most M is

$$(M_{k,M})_{\alpha\beta} = \prod_{j=1}^k \int \overline{s_t^{\alpha_j}} s_t^{\beta_j} d\nu_h(t). \quad (7.3)$$

Its positive definiteness follows because a polynomial cannot vanish almost everywhere on the full real product unless it is zero. It is a principal submatrix of the stated product moment matrix, not a declaration that the total-degree space is a tensor product of smaller univariate spaces.

The actual joint least-norm representative and its metric are

$$R_{k,M} = \mathcal{T}_h^{(k)} M_{k,M}^{-1} J_{k,M}^* (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1},$$

$$G_{k,M} = (J_{k,M} M_{k,M}^{-1} J_{k,M}^*)^{-1}. \quad (7.4)$$

The proof is the same explicit constrained-minimum calculation as (3.6), now with the relation ideal (7.2). In particular its arithmetic class is exactly $\sigma_h^{\otimes k}$ and every jet is fixed.

For a rectangular degree budget N one has the actual inclusion

$$\mathcal{P}_N^{\otimes k} \hookrightarrow \mathcal{P}_{kN}^{(k)}.$$

It carries the old product representative into the feasible set of the new problem. Therefore

$$G_{k,kN} \preceq G_{h,N-d}^{\otimes k} \quad (N \geq d). \quad (7.5)$$

This is an inequality from an inclusion of admissible original boundaries. It is not asserted to imply the same ordering of control forms, whose dependence on A is retained separately.

H.1.9 8. The product error is one new total-degree layer

Let $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, and let $P_{\mathcal{B}_{k,M}}$ be its orthogonal projection onto the finite-dimensional actual boundary space. With the fixed low-degree section $s_h^{\otimes k}$,

$$R_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) s_h^{\otimes k}.$$

Put

$$\mathcal{E}_{k,M} = (1 - P_{\mathcal{B}_{k,M}})\mathcal{B}_{k,M+1}.$$

Because the jet map is already onto at both degrees,

$$\dim \mathcal{E}_{k,M} = \binom{M+k}{k-1}. \quad (8.1)$$

This is $\dim \mathcal{P}_{M+1}^{(k)} - \dim \mathcal{P}_M^{(k)}$; the d^k jet constraints cancel from the dimension difference.

Let $D^{(k)} = \sum_j D_j$. Its induced action on jets is $A_h^{(k)}$. The actual derivative defect lies in $\mathcal{B}_{k,M+1}$:

$$B_{k,M} = D^{(k)} R_{k,M} - R_{k,M} A_h^{(k)}.$$

Define the two explicitly typed maps into the same new relation layer

$$Y_{k,M} = P_{\mathcal{E}_{k,M}} R_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M},$$

$$C_{k,M} = (1 - P_{\mathcal{B}_{k,M}}) B_{k,M} : E_h^{(k)} \rightarrow \mathcal{E}_{k,M}.$$

Integration by parts gives $(D^{(k)})^* + D^{(k)} = kI$ on these functions. Orthogonality then proves

$$W_{k,M} := (A_h^{(k)})^* G_{k,M} + G_{k,M} A_h^{(k)} - k G_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad (8.2)$$

$$G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}, \quad \text{rank } W_{k,M} \leq 2 \binom{M+k}{k-1}. \quad (8.3)$$

For (8.3), use the orthogonal decomposition $\mathcal{B}_{k,M+1} = \mathcal{B}_{k,M} \oplus \mathcal{E}_{k,M}$. Its projection sends $R_{k,M}$ to $Y_{k,M}$ and the new representative is $R_{k,M} - Y_{k,M}$.

The source quotient map is the same degree transport as before:

$$\mathcal{B}_{k,M+1}/\mathcal{B}_{k,M} \xrightarrow{\sim} \mathcal{E}_{k,M}, \quad [z] \mapsto (1 - P_{\mathcal{B}_{k,M}})z.$$

A derivative class in this kernel becomes the supported zero only at the next total-degree label. The primitive retains which tensor factor supplied the theta boundary and its cochain sign. Synchronization by $(1, e)$ in each required two-leg factor supplies the common joint support, while the original mask remains recorded.

Thus the product estimate need not simply copy k identical one-factor errors: (8.2) gives a new jointly optimized boundary-layer expression with completely specified moment data and maps. Its rank bound alone does not bound its generalized eigenvalues.

H.1.10 9. Same-metric reflection and residue duality on the product

For a reflection-stable packet, $j_h^{\otimes k}$ is the antilinear involution on $E_h^{(k)}$, defined on pure tensors by applying j_h to every factor. The corresponding function reflection is $F(x_1, \dots, x_k) \mapsto \prod_j x_j^{-1} \overline{F(x_1^{-1}, \dots, x_k^{-1})}$.

It preserves both the total-degree numerator space and its relation ideal. Since $(g/h)^\dagger = (-1)^d (g/h)$, its action on a numerator is $(-1)^{kd} P^\dagger$; this factor is retained. Uniqueness of the minimum in (7.4) proves reflection compatibility of $R_{k,M}$. Therefore the analogue of (6.4) gives symmetry of the generalized spectrum of $(W_{k,M}, G_{k,M})$ on the same metric.

Residue duality remains the explicitly specified map with matrix $S_h^{\otimes k}$ and target line $\mathcal{L}_k$, on which dilation acts by a^k . It satisfies

$$(A_h^{(k)})^* S_h^{\otimes k} + S_h^{\otimes k} A_h^{(k)} = k S_h^{\otimes k}.$$

Any fixed degree-orientation scalar in the cochain evaluation is retained separately; the ordered product of the scalar residue pairings here has matrix exactly $S_h^{\otimes k}$.

Its dual metric and control are

$$\begin{aligned} G_{k,M}^D &= (S_h^{\otimes k})^* G_{k,M}^{-1} S_h^{\otimes k}, \\ W_{k,M}^D &= -(S_h^{\otimes k})^* G_{k,M}^{-1} W_{k,M} G_{k,M}^{-1} S_h^{\otimes k}. \end{aligned} \quad (9.1)$$

The Jacobian contraction is still the original multiplication trace, with every multiplicity, and is unchanged by the source boundaries used to construct (7.4). Neither (9.1) nor reflection alone establishes an arithmetic upper bound.

H.1.11 10. Quantitative certificate and the Deligne comparison

For one or several variables write M for the positive moment matrix, J for its fixed full-jet map, and $\mathcal{C} = JM^{-1}J^*$. These actual matrices now replace an unrestricted metric search.

A useful deterministic enclosure is as follows. Suppose a Hermitian approximation $\widehat{M}$ has a verified bound $\|M - \widehat{M}\| \leq \eta$ and $\widehat{M} \succeq \mu I$ for a retained $\mu > \eta$. Then

$$\|M^{-1} - \widehat{M}^{-1}\| \leq \frac{\eta}{\mu(\mu - \eta)}, \quad \|\mathcal{C} - J\widehat{M}^{-1}J^*\| \leq \|J\|^2 \frac{\eta}{\mu(\mu - \eta)}. \quad (10.1)$$

This is the resolvent identity for inverses. The positive lower bound on $\mathcal{C}$ must also be retained before a second inversion. With $\widehat{G}$ and $\widehat{W}$ and verified errors η_G, η_W , the sufficient matrix certificate is

$$\widehat{G} - \eta_G I \succ 0, \quad \epsilon \widehat{G} \pm \widehat{W} - (\epsilon \eta_G + \eta_W) I \succeq 0. \quad (10.2)$$

No numerical interval enclosure for the actual infinite arithmetic moments is claimed merely from a floating-point calculation. The previous exterior-theta sums give another exact representation of their one-variable inputs; all tails and roundoff must be supplied to (10.1).

Deligne's retained Corollaries 3.3.4–3.3.6 place upper and dual lower bounds on the same compact-to-ordinary image; the tensor step in 3.2.13 improves an estimate only after additional geometric control. This note supplies a canonical coherent family, its exact source relation ideal on products, and the same-metric reflection/dual operations with which such an upper estimate can now be tested. It does not import finite-field hypotheses into the tau chart.

For an actual eigenvector $A_h v = \rho v$, every constructed tensor metric still has the exact diagnostic

$$\frac{(v^{\otimes k})^* W_{k,M} v^{\otimes k}}{(v^{\otimes k})^* G_{k,M} v^{\otimes k}} = k(2\Re \rho - 1). \quad (10.3)$$

The quantity to bound is therefore the generalized boundary form (8.2) relative to (7.4). A dimension/rank estimate, a decrease of a Gram matrix, or unrestricted metric realizability is not substituted for that relative estimate. The new constructive route is the exact finite full-jet moment problem together with its next-layer flux, not another homotopy choice.

H.1.12 11. Reading, provenance, and validation

Newly read: both supplied formalization/control notes in full; the preceding local `Tau_Homotopy_Boundary_Control/NOTE.tex`, especially sections 4–10; the retained Deligne English TeX excerpt 3.3.4–3.3.6 and French 6.2.4–6.2.7. This is not a new audit of every page of Weil II. No OCR or PDF extraction was used. Source hashes are in the package reading record.

External context checked: NIST DLMF 5.11.9 for the gamma bound; Connes–Consani–Moscovici, arXiv:2403.01247, abstract, for prior moment/Jacobi constructions from squared absolute local factors; Romik, arXiv:1902.06330, abstract, for prior orthogonal-polynomial expansions of Ξ ; Lasserre–Pauwels, arXiv:1606.03858, abstract, for inverse-moment/Christoffel context. Only the gamma asymptotic is used as an external estimate; the other papers are priority/context references, not sources of an unread theorem. The arguments above prove the specific maps used here.

The accompanying exact checker uses declared finite discrete calibration measures and rational complex polynomial data. It checks the general finite matrix identities, joint relation ranks, signs, full repeated-root reductions, reflection, naturality, and comparison metrics. Its model polynomial is never labelled an arithmetic zeta counterexample. Analytic density, asymptotic gamma bounds, and the original theta range identities are written arguments, not certified by those finite tests. No new Lean execution is claimed. No RH or GRH conclusion is claimed.

I Full-jet interpolation kernels and the actual next relation layer

Source attribution. Complete TeX witness from the dated kernel-layer supplement, delivered 12 September 2026. Its handoff identifies PR 14 as the arithmetic source and the existing PR 15 revision as its formal baseline. The supplement’s add-only patch was verified locally; its receipt records no remote branch modification. These NOTE bytes are identified by the source and archive hashes below; neither baseline revision is represented as their Git head.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Kernel_Layer_Integration/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Arithmetic source baseline PR 14

Cited existing PR 15 formal baseline. 21970bbf4760d0bbca512a4a7996a2e38f968947

Source SHA-256. be8485471fc7dac41d97e4a9fd52413da14e83fbc42c6c55c01a5f0908caa06c

Original archive. Tau_Kernel_Layer_Integration_2026-09-12(1).zip

Archive SHA-256. ebd927c8a9afa90600f5320bd4189ecb48085889835205057b1bfafdb68d23d2

I.1 Full-jet interpolation kernels and the actual next relation layer

I.1.1 Integration supplement — 12 September 2026

This supplement reviews the supplied **Coherent arithmetic interpolation and joint tensor boundary control** package and derives an additional finite-stage bridge. The source’s canonical representative, original theta boundaries, full multiplicities, and support transport are retained. The new formulas below have written proofs and exact finite regression tests; they are **not new Lean certificates**.

The supplied ZIP has SHA-256 a753df8341d05b2fccb6487b1dabc0044153a6c1abcf35092504aadb95c51d26. All 29 entries of its manifest verified. Its note, handoff, and checker match the published Git blobs at PR #14, commit 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc. The source checker was independently rerun: nineteen tests passed normally and under optimized Python, with identical result records; the intentionally false twentieth test failed in both modes. See SOURCE_REVIEW_RECEIPT.json.

The comparison with PR #15 is limited precisely: its existing five-module receipt certifies its stated support-changing, relation-layer, finite-Krylov and rank-two interfaces. This supplement does not extend that receipt to the source’s theta analytics, all-degree orthogonal-polynomial construction, or the new tensor-kernel calculations.

I.1.2 1. Source results retained, rather than reconstructed differently

Write the source polynomial degree as N , to distinguish it from a moment matrix. Fix tensor degree $k \geq 1$, a nonempty packet polynomial h of degree d , and $N \geq k(d - 1)$. The packet includes the complete orders of the selected actual zeros. Retain

$$g = 2\xi, \quad E = E_h^{\otimes k}, \quad A = A_h^{(k)}, \quad \mathcal{J}_N : \mathcal{P}_N^{(k)} \twoheadrightarrow E. \quad (1)$$

The last map is reduction modulo the original ideal $(h(t_1), \dots, h(t_k))$, followed by multiplication by the original local unit $v_h^{\otimes k}$. Its kernel is $\mathcal{I}_{k,N}$. The source map is

$$\mathcal{T}(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{MT}(P) = P(s_1, \dots, s_k) \prod_i \frac{g(s_i)}{h(s_i)}. \quad (2)$$

Let $\mathcal{L}_N = \mathcal{TI}_{k,N}$, $P_N = P_{\mathcal{L}_N}$, and let $\sigma : E \hookrightarrow Q^{\otimes k}$ be the source arithmetic inclusion. Its canonical representative and forms are

$$R_N = (I - P_N)s_h^{\otimes k}, \quad G_N = R_N^* R_N \succ 0, \quad q^{(k)} R_N = \sigma, \quad J^{(k)} R_N = I, \quad (3)$$

$$B_N = D^{(k)} R_N - R_N A, \quad W_N = A^* G_N + G_N A - k G_N. \quad (4)$$

The supplied note proves $B_N(E) \subseteq \mathcal{L}_{N+1}$ and, with $\mathcal{E}_N = \mathcal{L}_{N+1} \cap \mathcal{L}_N^\perp$,

$$Y_N = P_{\mathcal{E}_N} R_N, \quad C_N = (I - P_N) B_N, \quad W_N = -(Y_N^* C_N + C_N^* Y_N). \quad (5)$$

These are source results (source note, §§2–4 and 7–9), not claimed as discoveries of this supplement. Their finite-dimensional derivations check: the constrained minimum is over precisely the original ideal, the derivative increases total degree by one, and orthogonal projection removes only the admitted relation space. The source's exact sequence and packet naturality require full orders. Its total-degree statements require the threshold in (1). Its analytic function-space identities remain the stated analytic inputs, not consequences of a finite fixture.

The original coefficient square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, with projections $p_{\mathbb{Z}}, p_{\mathbb{C}}$, is unchanged. A vector in an admitted relation maps to the receiving fibre's supported zero, never to external absence τ .

1.1.3 2. Calculate the inverse interpolation metric by full-jet columns

Use exactly the source measure

$$d\nu_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{dt}{2\pi}. \quad (6)$$

Let $p_j^{(h)}(s)$ be its monic orthogonal polynomials, with the polynomial variable evaluated at $s = 1/2 + it$, and retain their positive squared norms

$$\kappa_j^{(h)} = \int |p_j^{(h)}(1/2 + it)|^2 d\nu_h(t) > 0. \quad (7)$$

These are **not** the earlier theta-Krylov vectors u_j , nor their norms $\mathfrak{h}_j$. In particular they depend on h ; they are not assigned norm one. Existence through any finite degree follows from the source measure's positive-definite moment matrices. No limiting assertion is needed here.

Suppress h in notation. For a multi-index α , define

$$p_\alpha = \prod_i p_{\alpha_i}, \quad \kappa_\alpha = \prod_i \kappa_{\alpha_i}, \quad a_j = v_h[p_j]_h \in E_h, \quad a_\alpha = \bigotimes_i a_{\alpha_i} \in E. \quad (8)$$

Every a_j is the full jet, including the local unit. Product orthogonality gives $\langle \mathcal{T}p_\alpha, \mathcal{T}p_\gamma \rangle = \kappa_\alpha \delta_{\alpha\gamma}$. These polynomials form a basis of $\mathcal{P}_N^{(k)}$, because the monic triangular basis change preserves total degree.

Set

$$K_N := G_N^{-1} = \sum_{|\alpha| \leq N} \frac{a_\alpha a_\alpha^*}{\kappa_\alpha}. \quad (9)$$

Proof. In this basis the source moment matrix is $\text{diag}(\kappa_\alpha)$ and the full-jet matrix has columns a_α . Substitute into the source identity $G_N^{-1} = J_N M_N^{-1} J_N^*$. A change of polynomial basis does not change that product. This proves (9) for the same canonical metric, not a newly selected metric.

Its representative has the explicit coefficient formula

$$R_N u = \sum_{|\alpha| \leq N} \mathcal{T}(p_\alpha) \frac{a_\alpha^* G_N u}{\kappa_\alpha}. \quad (10)$$

In particular $R_N K_N$ is an actual map from finite antidual coordinates into the original test-function space. It satisfies $J^{(k)} R_N K_N = K_N$ and $q^{(k)} R_N K_N = \sigma K_N$. The invertible coordinate change K_N is retained, not identified with the identity.

1.1.4 3. An explicit basis of the original next relation layer

Let $\Sigma_{N+1} = \{\beta \in \mathbb{N}^k : |\beta| = N + 1\}$. Define actual test functions

$$e_\beta = \mathcal{T}(p_\beta) - R_N a_\beta, \quad \beta \in \Sigma_{N+1}. \quad (11)$$

They are in $\mathcal{E}_N$. First, their full jets vanish by (3), so their numerator polynomials lie in the original ideal at degree at most $N + 1$; thus $e_\beta \in \mathcal{L}_{N+1}$. Second, $\mathcal{T}p_\beta$ is orthogonal to all degree-at-most- N polynomials and $R_N a_\beta$ is orthogonal to $\mathcal{L}_N$. Hence $e_\beta \perp \mathcal{L}_N$.

Define

$$\mathcal{E}_N : \mathbb{C}^{\Sigma_{N+1}} \longrightarrow \mathcal{E}_N, \quad c \longmapsto \sum_{\beta} c_\beta e_\beta, \quad (12)$$

$$U_N = (a_\beta)_{\beta \in \Sigma_{N+1}}, \quad D_N = \text{diag}(\kappa_\beta)_{\beta \in \Sigma_{N+1}}. \quad (13)$$

The map $\mathcal{E}_N$ is an isomorphism. Independence follows by taking the degree- $N + 1$ polynomial part of (11). For spanning, subtract from a relation its degree- $N + 1$ combination of the e_β . The remainder is a relation of degree at most N , hence belongs to $\mathcal{L}_N$. An element of $\mathcal{E}_N$ has no such remaining component.

Its Gram matrix and pairing with the representative are exactly

$$H_N := \mathcal{E}_N^* \mathcal{E}_N = D_N + U_N^* G_N U_N, \quad \mathcal{E}_N^* R_N = -U_N^* G_N. \quad (14)$$

The cross terms vanish because the new-degree polynomials are orthogonal to the old polynomial space. In particular $H_N \succ 0$, and no original norm has been suppressed.

This identifies the existing quotient equivalence in coordinates:

$$\mathbb{C}^{\Sigma_{N+1}} \xrightarrow{\mathcal{E}_N} \mathcal{E}_N \xrightarrow{e \mapsto [e]} \mathcal{L}_{N+1} / \mathcal{L}_N. \quad (15)$$

The inverse quotient-to-layer map is the source's $[z] \mapsto (I - P_N)z$. For each nonzero coordinate vector, its class in this relative quotient is nonzero, and its image in the next admitted quotient is zero. At the split level this remains the next supported zero.

For an explicit theta primitive, form $p_\beta - \sum_{|\alpha| \leq N} p_\alpha a_\alpha^* G_N a_\beta / \kappa_\alpha$. Its full jet is zero. Fixed-order monic division expresses it as $\sum_i h(t_i) Q_{i,\beta}$, with $\deg Q_{i,\beta} \leq N + 1 - d$. In the i -th cochain factor use the original ϕ_* in place of F_h , and multiply its primitive by $(-1)^{i-1}$. The differential supplies the same sign, producing the displayed relation with positive sign. This supplies the actual primitive, not just a dimension count.

I.1.5 4. Both boundary maps now have explicit coordinates

Define the matrix $F_N : E \rightarrow \mathbb{C}^{\Sigma_{N+1}}$ by its rows

$$(F_N)_\beta = \sum_{i: \beta_i > 0} \frac{a_{\beta - \mathbf{e}_i}^*}{\kappa_{\beta - \mathbf{e}_i}}. \quad (16)$$

Then

$$\boxed{Y_N = -\mathcal{E}_N H_N^{-1} U_N^* G_N, \quad C_N = \mathcal{E}_N F_N G_N.} \quad (17)$$

The first equation is the actual orthogonal projection formula using (14). For the second, multiply the finite expansion (10) by $s_1 + \cdots + s_k$. Monicity implies that its degree- $N + 1$ coefficient at p_β is exactly $(F_N)_\beta G_N u$. The term $R_N A u$ has degree at most N . Subtract $\mathcal{E}_N F_N G_N u$ from $B_N u$: the result has degree at most N and zero full jets, hence is in $\mathcal{L}_N$. Projecting by $I - P_N$ proves the equation.

Consequently the old quotient records

$$\boxed{[B_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N} = [\mathcal{E}_N F_N G_N u]_{\mathcal{B}^{(k)}/\mathcal{L}_N}.} \quad (18)$$

It becomes the supported zero under the next relation transition. The symbol $\mathcal{B}^{(k)}$ denotes the original product test-function space used by the source; no derivative on all of its L^2 completion is defined here.

I.1.6 5. The control form is a top-degree full-jet endpoint

Define

$$\boxed{V_N := A K_N + K_N A^* - k K_N.} \quad (19)$$

Substituting (17) into the source boundary identity (5), and using $\mathcal{E}_N^* \mathcal{E}_N = H_N$, gives

$$\boxed{W_N = G_N (U_N F_N + F_N^* U_N^*) G_N.} \quad (20)$$

Meanwhile direct multiplication of (4) by K_N on both sides gives $K_N W_N K_N = V_N$. Thus

$$\boxed{V_N = U_N F_N + F_N^* U_N^* = \sum_{|\alpha|=N} \sum_{i=1}^k \frac{a_{\alpha + \mathbf{e}_i} a_\alpha^* + a_\alpha a_{\alpha + \mathbf{e}_i}^*}{\kappa_\alpha}.} \quad (21)$$

This is a finite endpoint identity derived through the actual theta relation layer. It does not assume a small boundary, discard mixed terms, or replace the packet operator. It can also be obtained by finite Christoffel–Darboux telescoping. The standard scalar orthogonal-polynomial kernel and telescoping identities are recorded by NIST (n.d., §18.2(v)); the full-jet, original-layer identification (11)–(21) is proved here rather than imported from that scalar formula.

For this alternative telescoping proof, multiplication satisfies $s p_j = p_{j+1} + \beta_j p_j - (\kappa_j / \kappa_{j-1}) p_{j-1}$, with $\operatorname{Re} \beta_j = 1/2$. The imaginary part need not vanish for the packet-dependent measure. Only $\beta_j + \overline{\beta_j} = 1$ is needed. Interior adjacent terms cancel; (21) is the uncanceled total-degree boundary. The new tests include an asymmetric measure with a nonzero imaginary diagonal coefficient.

Regroup by α and put $b_\alpha = \sum_i a_{\alpha + \mathbf{e}_i}$. Equation (21) factors through the two sets of columns $(a_\alpha)_{|\alpha|=N}$ and $(b_\alpha)_{|\alpha|=N}$. Therefore

$$\boxed{\text{rank } W_N = \text{rank } V_N \leq \min \left\{ d^k, 2 \binom{N+k-1}{k-1} \right\}.} \quad (22)$$

This sharpens the source's general next-layer bound $2 \binom{N+k}{k-1}$ for this product-polynomial construction. It does **not** change the actual dimension of $\mathcal{E}_N$, which remains $\binom{N+k}{k-1}$. The sharpening comes from the scalar summed derivative and regrouping its top-degree columns. For $k = 2, d = 4, N = 6$, the two structural bounds are 14 and 16 on a 16-dimensional packet; the exact regression tests cover this nonvacuous case. Neither rank bound is a small-eigenvalue estimate.

1.1.7 6. Estimate the same form without a second inversion

Since K_N is Hermitian positive definite,

$$\boxed{-\epsilon G_N \preceq W_N \preceq \epsilon G_N \iff -\epsilon K_N \preceq V_N \preceq \epsilon K_N.} \quad (23)$$

This is congruence by the specified invertible map K_N . More explicitly, for $u = K_N \lambda$, $u^* G_N u = \lambda^* K_N \lambda$ and $u^* W_N u = \lambda^* V_N \lambda$. This identifies the generalized Rayleigh quotients, including multiplicities of generalized eigenvalues. The original generator A is not replaced.

The source's same-metric reflection therefore still transfers an upper estimate to a lower estimate for precisely the original G_N . Under the natural weight- k antidual action $A^D = kI - A^*$, the defect of the antidual metric K_N is $-V_N$, not V_N . The dual sign and weight are both retained.

The reflection map itself can be retained explicitly. If the source involution is $j(u) = C\bar{u}$, its antidual involution is $j^\#(\lambda) = C^T \bar{\lambda}$. From $C\bar{C} = I$ and $C^* G_N C = \overline{G_N}$, inversion gives $(C^T)^* K_N C^T = \overline{K_N}$. The control identity transports to $(C^T)^* V_N C^T = -\overline{V_N}$. Thus the two-sided equivalence retains the actual involution, not only spectral symmetry.

Equation (23) avoids the *second* inversion $G_N = K_N^{-1}$ in the final inequality test. It does not eliminate the need to enclose moments, full-jet coefficients, or the conditioning of the first construction of K_N .

A deterministic finite certificate can therefore work directly with $\hat{K}$, $\hat{A}$ and error bounds $\|K_N - \hat{K}\| \leq \eta_K$, $\|A - \hat{A}\| \leq \eta_A$, where $\hat{K}$ is Hermitian. Put

$$\hat{V} = \hat{A}\hat{K} + \hat{K}\hat{A}^* - k\hat{K}, \quad \eta_V = (2\|\hat{A}\| + k)\eta_K + 2\eta_A(\|\hat{K}\| + \eta_K). \quad (24)$$

Expanding both perturbations proves $\|V_N - \hat{V}\| \leq \eta_V$. Sufficient conditions, for $\epsilon \geq 0$, are

$$\hat{K} - \eta_K I \succ 0, \quad \epsilon \hat{K} \pm \hat{V} - (\epsilon \eta_K + \eta_V) I \succeq 0. \quad (25)$$

No enclosure of the actual infinite arithmetic moments is asserted in this supplement.

1.1.8 7. The metric update and the univariate connection

The positive update from (9) and its inverse are

$$\boxed{K_{N+1} = K_N + U_N D_N^{-1} U_N^*,} \quad (26)$$

$$\boxed{G_{N+1} = G_N - G_N U_N (D_N + U_N^* G_N U_N)^{-1} U_N^* G_N.} \quad (27)$$

Equation (27) follows either by direct multiplication with (26), or from $R_{N+1} = R_N - Y_N$ and the orthogonal decomposition. The middle matrix is exactly the actual layer Gram matrix H_N in (14), not an unrelated auxiliary matrix. The lost Gram form is $Y_N^* Y_N$.

For $k = 1$ and $N = n + d$, (11) recovers the earlier monic theta vector:

$$\boxed{e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1}.} \quad (28)$$

Indeed the numerator of (11) is monic of degree $n+d+1$, divisible by monic h , and its quotient has degree $n+1$ and leading coefficient one. Applying $\mathcal{T}_h$ gives f_{n+1} plus an element of L_n , orthogonal to L_n . This uniquely identifies u_{n+1} . Thus the packet-dependent polynomial construction is connected to the packet-independent original theta norm by an exact identity; those two norm families were not conflated.

I.1.9 8. Degree recursion and the remaining estimate

For $A_j = a_j a_j^* / \kappa_j$, define finite matrix sums

$$Q_{k,m} = \sum_{|\alpha|=m} \bigotimes_i A_{\alpha_i}, \quad Q_{k,m} = \sum_{j=0}^m A_j \otimes Q_{k-1,m-j}, \quad K_{k,N} = \sum_{m=0}^N Q_{k,m}. \quad (29)$$

These are finite degree convolutions; they do not assert convergence of an infinite generating series. They compute the product full-jet interpolation matrix using one-variable polynomial norms and jets, without inverting a total-degree moment matrix of dimension $\binom{N+k}{k}$. The target dimension remains d^k ; no polynomial-time complexity or unconditional numerical stability is claimed.

In univariate packet inclusions, the source's exact equality $R_{H,n} \iota_{hH} = R_{h,n}$ remains in force. The packet-dependent polynomial bases used here need not themselves commute under inclusion. No extension of this univariate equality to different joint total-degree budgets is presumed.

Finally, on $Av = k\rho v$, with $v = v_0^{\otimes k}$, and $\lambda = G_N v$,

$$\frac{\lambda^* V_N \lambda}{\lambda^* K_N \lambda} = k(2 \operatorname{Re} \rho - 1). \quad (30)$$

The additional work is therefore the explicit layer basis, derivative-coordinate map, endpoint identity, and direct relative certificate for the same source-controlled object. It is not a proof of a uniform sublinear excess. No arithmetic block or nilpotent jet has been removed.

I.1.10 Verification and literature scope

The independent new checker contains eighteen exact test methods. They cover the kernel basis change, repeated jets, two-variable layers, actual derivative-class coordinates, Gram updates, the original univariate theta vector, endpoint factorization, a nonvacuous improved rank bound, antidual signs, generalized characteristic polynomials, nonzero off-line *polynomial model* diagnostics, nonreal recurrence diagonals, fixed-order theta primitives, and deterministic matrix perturbation certificates. Normal and optimized result records agree. Both intentional-failure modes return failure.

These are finite regression tests, not proofs of the general analytic inputs and not Lean execution. The general finite identities have the written proofs above. The source's nineteen tests and this supplement's eighteen tests are separate suites, not counts of novel theorems.

The matrix-kernel and Christoffel–Darboux methods have established antecedents. NIST's general orthogonal-polynomial formulas supply the standard scalar comparison. Lasserre and Pauwels (2016) connect inverse moment matrices with orthogonal polynomials and the Christoffel function; only their abstract was examined here, so no theorem from that paper is used as a premise. No global priority claim is made for standard kernel or inverse-update identities.

I.1.11 References

Coherent arithmetic interpolation and joint tensor boundary control. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #14, commit 3bff1b2bd87dfa2592f65c5cc231f6ee9541dfc. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/14>

Formalize support-changing relation layers and the Krylov boundary-control calculation. (2026, September 12). Zeta-function-research-reader research workbench, draft PR #15, verified source commit 21970bbf4760d0bbca512a4a7996a2e38f968947. <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/15>

Lasserre, J.-B., & Pauwels, E. (2016). *Sorting out typicality with the inverse moment matrix SOS polynomial* [Preprint]. arXiv. <https://arxiv.org/abs/1606.03858>

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, §18.2(v), Christoffel–Darboux formula. <https://dlmf.nist.gov/18.2#v>

J Arithmetic frontier estimates and symmetric tensor control

Source attribution. Complete TeX witness from the dated symmetric-frontier continuation, delivered 12 September 2026. Its delivery receipt records an add-only local patch based on PR 14, with no remote write. The exact revision below is that patch base; it is not a new remote head for this NOTE. The source and archive hashes identify the complete delivered witness.

Exact revision. 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc

Source file. Tau_Symmetric_Frontier_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Recorded add-only patch base, PR 14

Source SHA-256. 734b6c935892ebd9d18b506f6b592f1ca3489afcb95d5c669a489f4ca04656cf

Original archive. Tau_Symmetric_Frontier_Control_2026-09-12(1).zip

Archive SHA-256. a024ee4bbb4df9555bc6f94c2d74a2e90c077533ad2ce312863342b758539796

J.1 Arithmetic frontier estimates and symmetric tensor control

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. New written deductions from the precise arithmetic source of PR #14, submitted for mathematical review. The supplied Lean receipt certifies its stated coefficient, quotient, homotopy, and topology interfaces; it does not certify the analytic inputs or the results in this note. No purity estimate is assumed. No claim of global novelty is made for the underlying orthogonal-polynomial, finite-group, or matrix identities.

J.1.1 0. Source and purpose

The source is PR #14 at 3bff1b2bd87dfa2592f65c5ccc231f6ee9541dfc, research-note Git blob 4ec40c15397128552bc276ec903fe59c89657d4e. Its actual total-degree representative, full-jet map, and theta relation ideal are used without changing them.

The source obtained two maps into the next relation layer, with

$$W_{k,M} = -(Y_{k,M}^* C_{k,M} + C_{k,M}^* Y_{k,M}), \quad G_{k,M} - G_{k,M+1} = Y_{k,M}^* Y_{k,M}. \quad (0.1)$$

This note constructs coordinates on that *actual relation layer*, computes the relative control in them, proves a finite arithmetic upper bound, and constructs a smaller symmetric tensor cohomology summand that still contains every repeated eigenline used in amplification. Its exact complement and its trace are retained.

The decisive new coordinate map is

$$b_M(q) = \mathcal{T}_h^{(k)} q - R_{k,M} \mathcal{J} q. \quad (0.2)$$

It converts the highest orthogonal polynomial degree into an original theta boundary and has a calculated inverse and Gram matrix. A highest-degree polynomial alone need not be a boundary; its full-jet value is subtracted by the specific source representative in (0.2).

J.1.2 1. Unchanged split base, arithmetic source, and quotients

Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.1)$$

The base is the marked absolute point $\mathfrak{b}_\tau$. The first arithmetic target in (1.1) is still infinite. Every fixed-label linear map below is reconstructed by $(\lambda, x) \mapsto (\lambda, Tx)$; its value on external absence is external absence. A vanishing amplitude at an active label remains that label's supported zero.

Use the original $C_+ = [V \xrightarrow{\Theta} \mathcal{B}]$ in degrees 0, 1, where

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ \Theta\phi(x) &= \sum_{n \neq 0} \phi(nx), \quad q : \mathcal{B} \rightarrow Q = \mathcal{B}/\Theta V. \end{aligned}$$

The input is

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_*(s) = g(s) = 2\xi(s).$$

Fix a nonempty finite packet of actual nontrivial zeros of g , with their full orders, and define

$$h(t) = \prod_{\rho \in Z} (t - \rho)^{m_\rho}, \quad d = \deg h, \quad E = \mathbb{C}[t]/(h), \quad v_h = g/h, \quad v = j_h v_h \in E^\times. \quad (1.2)$$

No squarefree assumption is made. The source constructs $F_h \in \mathcal{B}$ with $h(D)F_h = \Theta\phi_*$ and $\mathcal{M}F_h = v_h$. It supplies

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{M}\mathcal{T}_h(P) = v_h P, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v[P]_h. \quad (1.3)$$

For tensor degree k , put $E_k = E^{\otimes k}$ and $A_k = \sum_{j=1}^k A_j$, where $A = M_t$. Let

$$\begin{aligned} \mathcal{P}_M^{(k)} &= \mathbb{C}[t_1, \dots, t_k]_{\deg \leq M}, \quad M \geq k(d-1), \\ \mathcal{T}^{(k)}P &= P(D_1, \dots, D_k)F_h^{\otimes k}, \quad \mathcal{J}P = v^{\otimes k}[P]_{(h(t_1), \dots, h(t_k))}. \end{aligned} \quad (1.4)$$

The relation space and its realization are

$$\mathcal{I}_M = \mathcal{P}_M^{(k)} \cap (h(t_1), \dots, h(t_k)), \quad \mathcal{B}_M = \mathcal{T}^{(k)}\mathcal{I}_M. \quad (1.5)$$

The source's least-norm representative $R_M : E_k \rightarrow \mathcal{B}^{\otimes k} \subset \mathcal{H}_k$, with $\mathcal{H}_k = L^2(\mathbb{R}_{>0}^k, d^k x)$, satisfies

$$J^{(k)}R_M = 1, \quad q^{\otimes k}R_M = \sigma_h^{\otimes k}, \quad R_M(E_k) \perp \mathcal{B}_M. \quad (1.6)$$

Its finite Gram is $G_M = R_M^* R_M \succ 0$. Every adjoint here is of a finite-domain map into the declared Hilbert observation; no bounded Hilbert-space extension of the full theta map is assumed.

J.1.3 2. Unscaled polynomial coordinates with every moment retained

The arithmetic measure is exactly

$$d\nu_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 \frac{dy}{2\pi}. \quad (2.1)$$

The source's log-coordinate and Fourier maps give $\|\mathcal{T}_h P\|_{L^2(dx)}^2 = \int |P(\tfrac{1}{2} + iy)|^2 d\nu_h(y)$. The mass of this measure is not set to one.

Starting from $1, t, t^2, \dots$, subtract its orthogonal projections onto the preceding span:

$$p_0 = 1, \quad p_n = t^n - \text{proj}_{\mathcal{P}_{n-1}} t^n, \quad \omega_n = \int |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y) > 0. \quad (2.2)$$

No polynomial is divided by its norm. Let T_N be the triangular matrix with columns $p_0, \dots, p_N$ in the original power basis. The exact coordinate relation is

$$T_N^* M_{h,N} T_N = \text{diag}(\omega_0, \dots, \omega_N). \quad (2.3)$$

The earlier seed-boundary vectors $u_n = P_n(D)\Theta\phi_*$ are related to these coordinates by $\mathcal{T}_h(hP_n) = u_n$. Their measure is $|g|^2 dy/(2\pi)$; equation (1.3) is the map to the present $|g/h|^2$ measure. The two orthogonalizations have not been interchanged.

Recurrence. Define $a_0 = 0$ and $a_n = \omega_n/\omega_{n-1}$ for $n \geq 1$. Then

$$tp_n = p_{n+1} + b_n p_n - a_n p_{n-1}, \quad b_n = \frac{\int (\tfrac{1}{2} + iy) |p_n(\tfrac{1}{2} + iy)|^2 d\nu_h(y)}{\omega_n}, \quad \Re b_n = \tfrac{1}{2}. \quad (2.4)$$

Here $p_{-1} = 0$. The possibly nonzero $\Im b_n$ remains. The packet measure need not be even in the height coordinate y .

Proof. Multiplication by t has adjoint multiplication by $1 - t$ for this polynomial inner product. Pairing tp_n with $p_j, j < n-1$, gives zero. Its coefficient of p_{n+1} is one by its leading term; pairing with p_{n-1} gives $-\omega_n/\omega_{n-1}$. The diagonal is the displayed integral. This proves every coefficient. These are the usual orthogonal-polynomial operations, applied through the specified coordinate map rather than changing the arithmetic generator.

For $\alpha \in \mathbb{N}^k$, define

$$p_\alpha = \prod_j p_{\alpha_j}(t_j), \quad \omega_\alpha = \prod_j \omega_{\alpha_j}, \quad z_\alpha = \mathcal{J}p_\alpha \in E_k. \quad (2.5)$$

The columns z_α contain the complete unit $v^{\otimes k}$ and all nilpotent jets.

Inverse-metric formula. In the fixed original basis of E_k ,

$$\mathcal{C}_M = G_M^{-1} = \sum_{|\alpha| \leq M} \frac{z_\alpha z_\alpha^*}{\omega_\alpha}. \quad (2.6)$$

Indeed (2.3) applied to the tensor polynomial basis transforms $JM^{-1}J^*$ exactly into this sum. The representative coefficient on p_α is

$$c_\alpha(u) = \omega_\alpha^{-1} z_\alpha^* G_M u. \quad (2.7)$$

These formulas avoid inversion of a large raw monomial moment matrix, without imposing a new metric.

J.1.4 3. The highest polynomial degree is isomorphic to the actual new relation layer

Let $\mathcal{U}_M = \text{span}\{p_\beta : |\beta| = M + 1\}$ and retain its ordered basis. Put

$$\Omega = \text{diag}(\omega_\beta)_{|\beta|=M+1}, \quad F = [z_\beta]_{|\beta|=M+1} : \mathbb{C}^r \rightarrow E_k, \quad r = \binom{M+k}{k-1}. \quad (3.1)$$

Write $T_+ : \mathbb{C}^r \rightarrow \mathcal{H}_k$ for the map with columns $\mathcal{T}^{(k)} p_\beta$. Its image is orthogonal to the entire degree- M polynomial image, including $R_M(E_k)$.

The prior actual new relation layer is

$$\mathcal{E}_M = (1 - P_{\mathcal{B}_M})\mathcal{B}_{M+1}.$$

Theorem: relation-graph isomorphism. The map

$$b_M = T_+ - R_M F : \mathbb{C}^r \xrightarrow{\sim} \mathcal{E}_M \quad (3.2)$$

is an isomorphism and has

$$H_M = b_M^* b_M = \Omega + F^* G_M F, \quad b_M^* R_M = -F^* G_M. \quad (3.3)$$

Proof. Its full jet is $F - F = 0$, so (1.5) makes it an actual boundary of degree at most $M + 1$. Both terms are orthogonal to $\mathcal{B}_M$. If $b_M u = 0$, its component in the highest orthogonal polynomial degree is $T_+ u = 0$, so $u = 0$. The source calculation gives $\dim \mathcal{E}_M = r$, proving surjectivity. Alternatively every degree- $M + 1$ boundary has a unique highest orthogonal component; subtract its b_M image and the result lies in $\mathcal{B}_M$. This also constructs the inverse. Orthogonality $T_+^* R_M = 0$ proves (3.3).

A primitive is fully specified: form the numerator $p_\beta - c_M(z_\beta)$, where c_M is (2.7), divide it in the fixed variable order by $h(t_1), \dots, h(t_k)$, and use the original theta primitive in the factor where $h(t_j)$ appears. Give that primitive coefficient $(-1)^{j-1}$; the tensor differential supplies the same sign. No relation is inferred solely from a dimension count.

The comparison with the original internal quotient is

$$\mathbb{C}^r \xrightarrow{b_M} \mathcal{E}_M \xrightarrow{v \mapsto [v]} \mathcal{B}_{M+1}/\mathcal{B}_M \hookrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_M \longrightarrow \mathcal{B}^{\otimes k}/\mathcal{B}_{M+1}. \quad (3.4)$$

The last composite is zero. Its split lift sends (λ_M, u) to $(\lambda_{M+1}, 0)$, not to τ . At λ_M , $u \neq 0$ represents a nonzero kernel class. The map b_M describes the entire class that is about to become a supported zero.

Representative update. Orthogonal projection onto $b_M(\mathbb{C}^r)$ gives

$$Y_M = -b_M H_M^{-1} F^* G_M, \quad R_{M+1} = R_M + b_M H_M^{-1} F^* G_M. \quad (3.5)$$

Consequently

$$G_{M+1} = G_M - G_M F (\Omega + F^* G_M F)^{-1} F^* G_M, \quad (3.6)$$

$$\mathcal{C}_{M+1} = \mathcal{C}_M + F \Omega^{-1} F^*. \quad (3.7)$$

The complete volume identity is

$$\frac{\det \mathcal{C}_{M+1}}{\det \mathcal{C}_M} = \frac{\det(\Omega + F^* G_M F)}{\det \Omega}. \quad (3.8)$$

The full matrices in (3.6)–(3.7) remain available; their determinant in (3.8) is an additional observation, not their replacement.

J.1.5 4. Relative control is a highest-degree Christoffel–Darboux identity

For each $|\beta| = M + 1$, put

$$w_\beta = \sum_{j: \beta_j > 0} a_{\beta_j} z_{\beta - e_j}, \quad E_+ = [w_\beta]_{|\beta|=M+1}. \quad (4.1)$$

The name E_+ denotes this finite matrix, not the scalar e or the arithmetic quotient E_h ; its type is $\mathbb{C}^r \rightarrow E_k$.

Theorem: exact displacement.

$$\boxed{Z_M := A_k C_M + C_M A_k^* - k C_M = F \Omega^{-1} E_+^* + E_+ \Omega^{-1} F^*}. \quad (4.2)$$

Proof. Apply (2.4) in each factor of (2.6). A pair of adjacent indices inside $|\alpha| \leq M$ cancels because $a_{\alpha_j+1} = \omega_{\alpha+e_j}/\omega_\alpha$. The diagonal contributes $\sum_j (b_{\alpha_j} + \bar{b}_{\alpha_j}) = k$, precisely the term subtracted in (4.2). The remaining ordered pairs have $|\beta| = M + 1$ and predecessor $\beta - e_j$; collecting them gives (4.1)–(4.2). The imaginary diagonal coefficients cancel with their conjugates here; they were retained in the recurrence and are not assigned zero.

This identity is the tensor total-degree version of the Christoffel–Darboux cancellation. The standard one-variable formula is background; the maps to the full arithmetic quotient and its source boundaries are specified above.

The derivative defect also has an exact expression in the same relation-graph coordinates:

$$B_M = D^{(k)} R_M - R_M A_k, \quad \boxed{C_M^{\text{rel}} := (1 - P_{\mathcal{B}_M}) B_M = b_M \Omega^{-1} E_+^* G_M}. \quad (4.3)$$

Proof. The coefficient of p_β in $D^{(k)} R_M u$ is $\sum_j \omega_{\beta-e_j}^{-1} z_{\beta-e_j}^* G_M u = \omega_\beta^{-1} w_\beta^* G_M u$. The defect is an original boundary. The inverse of (3.2), followed by projection away from $\mathcal{B}_M$, therefore gives (4.3).

Substituting (3.5) and (4.3) into the previous cross-pairing formula proves

$$\boxed{W_M = A_k^* G_M + G_M A_k - k G_M = G_M Z_M G_M = -(Y_M^* C_M^{\text{rel}} + (C_M^{\text{rel}})^* Y_M)}. \quad (4.4)$$

Thus the new highest-degree computation is connected to both maps in the preceding note, including their minus sign and their support-valued boundary primitive.

J.1.6 4.1 Exact finite certificate and a proved upper estimate

For every real ϵ , congruence by the actual invertible G_M gives

$$\boxed{-\epsilon G_M \preceq W_M \preceq \epsilon G_M \iff -\epsilon C_M \preceq Z_M \preceq \epsilon C_M}. \quad (4.5)$$

This transports the inequality; no eigenvalue or generator has been rescaled.

Put $\mathcal{B} = [F, E_+]$ and

$$J = \begin{pmatrix} 0 & \Omega^{-1} \\ \Omega^{-1} & 0 \end{pmatrix}.$$

The nonzero generalized eigenvalues of (Z_M, C_M) are the nonzero eigenvalues of

$$\boxed{J \mathcal{B}^* G_M \mathcal{B}}. \quad (4.6)$$

The proof is the pair of maps $\mathcal{B} : \mathbb{C}^{2r} \rightarrow E_k$ and $J\mathcal{B}^* G_M : E_k \rightarrow \mathbb{C}^{2r}$ and the equality of nonzero spectra of their two composites. These eigenvalues are real because the first composite is the operator of the Hermitian pencil in (4.5). The $2r$ -matrix itself need not be Hermitian in its displayed coordinates.

There is also a direct upper bound. Let

$$\lambda_{k,M} = \max_{v \neq 0} \frac{v^* F \Omega^{-1} F^* v}{v^* \mathcal{C}_M v},$$

$$\Gamma_{k,M} = \max_{|\alpha|=M} \sum_{j=1}^k (a_{\alpha_j+1} + (k-1)a_{\alpha_j}). \quad (4.7)$$

Then the actual arithmetic control excess satisfies

$$\boxed{\epsilon_{k,M} \leq 2\sqrt{\Gamma_{k,M} \lambda_{k,M}}}. \quad (4.8)$$

Proof. Let Z_M be the columns z_α , $|\alpha| = M$, and $\Omega_M = \text{diag}(\omega_\alpha)$. The lowering-incidence matrix L has $L_{\alpha\beta} = a_{\beta_j}$ when $\alpha = \beta - e_j$, summed over matching j . Then $E_+ = Z_M L$. The matrix $\Omega_M L \Omega^{-1} L^*$ is similar to a positive semidefinite matrix and has nonnegative entries. Its row sum at α is exactly the sum in (4.7), so its spectral radius is at most $\Gamma_{k,M}$. This proves

$$E_+ \Omega^{-1} E_+^* \preceq \Gamma_{k,M} Z_M \Omega_M^{-1} Z_M^* \preceq \Gamma_{k,M} \mathcal{C}_M.$$

Cauchy–Schwarz in the same Ω^{-1} pairing now gives

$$|v^* \mathcal{Z}_M v| \leq 2\sqrt{v^* F \Omega^{-1} F^* v} \sqrt{v^* E_+ \Omega^{-1} E_+^* v} \leq 2\sqrt{\lambda_{k,M} \Gamma_{k,M}} v^* \mathcal{C}_M v.$$

This proves (4.8). The phase-sensitive formula (4.6) is retained for a sharper estimate. Equation (4.8) is not claimed sublinear in k ; its two arithmetic factors are now fully specified.

1.1.7 5. A symmetric cohomology summand that retains the amplification witnesses

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$ with its original tensor differential. For $\pi \in S_k$, the chain permutation is

$$T_\pi(x_1 \otimes \cdots \otimes x_k) = (-1)^{\sum_{i < j, \pi(i) > \pi(j)} |x_i| |x_j|} x_{\pi^{-1}(1)} \otimes \cdots \otimes x_{\pi^{-1}(k)}. \quad (5.1)$$

The exponent uses the original inputs crossed by π . Adjacent transpositions verify the cochain equation and the permutation relations. In top degree every input has degree one, so $T_\pi = \text{sgn}(\pi) P_\pi$, where P_π is ordinary factor permutation.

Define the actual idempotent

$$\boxed{S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi}. \quad (5.2)$$

The unscaled numerator squares to $k!$ times itself. The displayed $1/k!$ is necessary to make the specified morphism idempotent and is not absorbed into a pairing or an arithmetic function. In degree k the two signs multiply to $+1$, hence

$$\boxed{H^k(S_k \mathcal{C}_k) \cong (Q^{\widehat{\otimes} k})^{S_k}}. \quad (5.3)$$

On a finite packet this contains $(E^{\otimes k})^{S_k}$ with its entire nilpotent action. If $Av = \rho v$, then $v^{\otimes k}$ is nonzero, lies in this summand, and has generator eigenvalue $k\rho$. Using the unsigned chain average instead would give the alternating top-degree summand and would kill a repeated vector for $k > 1$; equation (5.2) is the precise replacement.

The complement is retained through the actual cochain decomposition

$$\mathcal{C}_k \xrightarrow{x \mapsto (S_k x, (1-S_k)x)} \text{im } S_k \oplus \text{im}(1 - S_k), \quad (u, v) \mapsto u + v. \quad (5.4)$$

At an invariant joint support this is reconstructed fibrewise. On all support masks the lift of (5.2) has the specified support map $\lambda \mapsto \bigvee_{\pi} \pi \lambda$. Its formula is the sum of the actual supported scalars $(\text{sgn } \pi / k!) \bullet T_{\pi}$. A vanishing projected amplitude lands at the orbit-joined supported zero, not at τ . Original transports into that joint fibre remain recorded. No injectivity of arbitrary support transports is assumed.

J.1.8 5.1 The canonical representatives commute with this operation

Permutation of variables preserves the total-degree numerator, its relation ideal, the product arithmetic measure, and the full unit $v^{\otimes k}$. Uniqueness of the constrained minimum gives

$$R_M P_{\pi} = P_{\pi} R_M. \quad (5.5)$$

Consequently R_M restricts to the invariant arithmetic module

$$E_k^{\text{sym}} = (E^{\otimes k})^{S_k}, \quad \boxed{\dim E_k^{\text{sym}} = \binom{k+d-1}{d-1}}. \quad (5.6)$$

This is not a change of the spectral packet. The inclusion $I_k : E_k^{\text{sym}} \hookrightarrow E_k$ gives

$$G_M^{\text{sym}} = I_k^* G_M I_k, \quad W_M^{\text{sym}} = I_k^* W_M I_k, \quad A_k I_k = I_k A_k^{\text{sym}}. \quad (5.7)$$

All repeated eigenlines remain, so this smaller family still contains every witness needed by the tensor-amplification upper estimate. For example, $d = 3, k = 8$ gives dimension 45 rather than $3^8 = 6561$. This is a dimension reduction, not a bound on W_M^{sym} .

J.1.9 6. Orbit sums give the symmetric matrices with all factorials retained

For a nondecreasing tuple $\lambda \in \mathbb{N}^k$, let $\text{Orb } \lambda$ denote its distinct permutations and

$$n_{\lambda} = \frac{k!}{\prod_{j \geq 0} m_j(\lambda)!}, \quad p_{[\lambda]} = \sum_{\alpha \in \text{Orb } \lambda} p_{\alpha}, \quad \omega_{[\lambda]} = n_{\lambda} \prod_j \omega_{\lambda_j}. \quad (6.1)$$

Distinct product basis vectors are orthogonal, proving the last formula directly. The orbit sum is not divided by its norm or its orbit size.

The target basis is likewise the unscaled orbit sums of $e_{a_1} \otimes \cdots \otimes e_{a_k}$, $0 \leq a_j < d$, where $e_i = [t^i]$. Let I_k be their column inclusion and

$$D_k = I_k^* I_k = \text{diag} \left(\frac{k!}{\prod_i n_i!} \right), \quad L_k = D_k^{-1} I_k^*. \quad (6.2)$$

The explicit L_k extracts the common coefficient on an orbit. For each orbit polynomial, define

$$z_{[\lambda]} = L_k \mathcal{J} p_{[\lambda]}.$$

Then

$$\mathcal{C}_M^{\text{sym}} = \sum_{|\lambda| \leq M} \frac{z^{[\lambda]} z_{[\lambda]}^*}{\omega_{[\lambda]}}, \quad G_M^{\text{sym}} = (\mathcal{C}_M^{\text{sym}})^{-1}. \quad (6.3)$$

The comparison with full coordinates is

$$\mathcal{C}_M^{\text{sym}} = L_k \mathcal{C}_M L_k^*, \quad G_M^{\text{sym}} = I_k^* G_M I_k. \quad (6.4)$$

Inversion and compression have not been commuted without the displayed D_k factors. Equation (6.4) follows because the full form commutes with the orthogonal permutation projection and by substituting $L_k = D_k^{-1} I_k^*$.

The symmetric polynomial numerator has one basis vector per partition of its degree into at most k parts. For $M \geq k(d-1)$ the jet map is onto, because a tensor remainder has total degree at most $k(d-1)$ and averaging makes a symmetric lift of every invariant jet. Thus

$$\dim(\mathcal{E}_M^{S_k}) = p_{\leq k}(M+1), \quad (6.5)$$

where $p_{\leq k}(n)$ is the number of partitions of n into at most k parts. The maps $b_M, Y_M, C_M^{\text{rel}}$ are permutation-equivariant and therefore restrict to that very subspace. Fixed-order polynomial division yields a primitive; applying S_k to it gives a symmetric cochain primitive because $dS_k = S_k d$.

The highest-layer formulas in sections 3–4 hold on the invariant coordinates by grouping F, E_+ into orbit sums, applying L_k , and using the diagonal entries $\omega_{[\lambda]}$. This computes the symmetric bound directly, instead of building a d^k -dimensional matrix first. The upper-bound row-sum estimate can still use the fully retained $\Gamma_{k,M}$; the exact grouped incidence matrix can give a smaller constant.

J.1.10 6.1 Reflection and the original trace

For a reflection-stable packet the source involution j_h induces $j_h^{\otimes k}$ on both summands in (5.4), commutes with permutations, and preserves the canonical representatives. Hence on the symmetric metric an upper control implies the opposite lower control by the same involution. The residue pairing and its Jacobian map are still transported through their specified inclusions; no auxiliary positive Gram is declared to be the Weil form.

For a finite packet action $U(a) = \exp((\log a)A)$, the exact trace on the invariant summand is

$$\text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) = \frac{1}{k!} \sum_{\pi \in S_k} \prod_{c \in \text{Cycles}(\pi)} \text{Tr}(U(a)^{|c|} | E). \quad (6.6)$$

The proof expands the permutation operator in the tensor basis; each cycle forces the corresponding contracted indices and gives the indicated trace. It holds for nonsemisimple U . The cohomological supertrace is $(-1)^k$ times (6.6). The complement trace remains

$$(-1)^k \left(\text{Tr}(U(a) | E)^k - \text{Tr}(U(a)^{\otimes k} | E_k^{\text{sym}}) \right). \quad (6.7)$$

These are traces of finite-rank cochain projectors on the original complex. An infinite packet sum and its trace topology are not supplied by these formulas.

J.1.11 7. An exact joint-control calculation with the nilpotent direction retained

This section is a declared algebraic calibration, not a packet of zeta zeros. Take the polynomial quotient

$$E = \mathbb{C}[s]/((s - \frac{1}{2})^2), \quad d\nu(y) = e^{-y^2/2} dy, \quad c = \sqrt{2\pi}.$$

The measure has its literal mass c , and the unit in the calibration jet map is 1. The polynomial basis comparison $x = s - 1/2$ is given by

$$B : \text{span}(1, x) \xrightarrow{\sim} \text{span}(1, s), \quad B = \begin{pmatrix} 1 & -1/2 \\ 0 & 1 \end{pmatrix}. \quad (7.1)$$

The operator remains multiplication by $s = 1/2 + x$. In the symmetric basis

$$1 \otimes 1, \quad 1 \otimes x + x \otimes 1, \quad x \otimes x,$$

its two-fold generator is

$$A_2^{\text{sym}} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}. \quad (7.2)$$

The two-fold total-degree-4 calculation gives

$$\boxed{G_{2,4}^{\text{sym}} = c^2 \text{diag}(1/3, 2/3, 1/4),} \quad (7.3)$$

$$\boxed{W_{2,4}^{\text{sym}} = c^2 \begin{pmatrix} 0 & 2/3 & 0 \\ 2/3 & 0 & 1/2 \\ 0 & 1/2 & 0 \end{pmatrix}.} \quad (7.4)$$

Its generalized characteristic polynomial is

$$\frac{\det(zG_{2,4}^{\text{sym}} - W_{2,4}^{\text{sym}})}{\det G_{2,4}^{\text{sym}}} = z(z^2 - 7/2),$$

so the exact two-sided excess is $\sqrt{7/2}$. The independent rectangular degree-2-per-factor construction gives $\sqrt{6}$ at the same containing total degree. This proves an actual improvement for the specified joint construction in this calibration; the nilpotent operator in (7.2) is unchanged.

For comparison, the smaller joint total degree $M = 2$ has excess $\sqrt{6}$, whereas the rectangular degree-1 construction contained in it has excess 2. Consequently a larger correction space alone does not monotonically improve the control constant. The typed comparison is inclusion of the two polynomial numerators and the resulting Gram compression; the direct formula (4.2) supplies the actual control value. Both calibrations are retained so that the first improvement is not promoted into a false monotonicity theorem.

J.1.12 8. Exact target, counterfactuals, and handoff

The next estimate is now an arithmetic highest-layer estimate, either by the phase-sensitive pencil (4.5)–(4.6), or by bounding the two factors in (4.8). On the symmetric cohomology family it concerns the same G_M^{sym} and W_M^{sym} , with all relation primitives retained. The source of every matrix entry is the actual measure $|2\xi/h|^2 dy/(2\pi)$, not a fitted metric.

A sublinear excess in tensor degree has not been proved. For every actual eigenvector $Av = \rho v$, the retained witness in the symmetric summand satisfies

$$\frac{(v^{\otimes k})^* W_M(v^{\otimes k})}{(v^{\otimes k})^* G_M(v^{\otimes k})} = k(2\Re\rho - 1). \quad (8.1)$$

The symmetric reduction therefore retains the witness needed by Deligne-style amplification; it does not assume its disappearance. The input Deligne passages use tensoring *and* a separate geometric estimate. Our new work supplies a computable smaller arithmetic target and a proved finite bound, not that geometric estimate by renaming a category.

Constructive continuations are recorded with their exact maps:

- Raw moment inversion: use (2.3)–(2.7), retaining every ω_j and the full jet unit.
- Highest polynomials that are not relations: use $b_M = T_+ - R_M F$, not an assertion that high degree is zero.
- Absolute Gram decay: use the congruence (4.5), retaining $\mathcal{C}_M = G_M^{-1}$ and the whole highest-layer update.
- Excessive tensor dimension: use the chain projector (5.2) and the symmetric numerator/quotient (6.1)–(6.5), while retaining the complementary cohomology and trace.
- A Cauchy–Schwarz bound too large: use the exact signed matrix (4.6); its phases and cross terms have not been replaced by their absolute values in the stored data.
- Ordinary zero observations: reconstruct the maps and kernels in (3.4). A new relation is *e*-zero at the next label; the original class and the full kernel remain upstream.

The supplied quotient/homotopy Lean results can formalize the finite algebra of (3.2)–(4.6) and the chain-permutation calculation. The original analytic range theorem, arithmetic moment bounds, and any uniform tensor estimate remain separate written-analysis inputs.

J.1.13 References and evidence scope

The current input was read in the mounted NOTE.tex and checked against the pinned public mark-down (especially sections 7–9). The prior formalization receipt was read at its recorded scope: its section-change and internal-quotient statements are used, without treating its analytic assumptions as proved Lean theorems.

The classical orthogonal-polynomial recurrence and Christoffel–Darboux formula are documented in NIST DLMF sections 18.2(iv)–(v), <https://dlmf.nist.gov/18.2>. The tensor complex and its signed commutativity are documented in the Stacks Project, Tag 0GWN, <https://stacks.math.columbia.edu/tag/0GWN>. These are background mechanisms; sections 3–6 above prove their specific maps on this arithmetic source.

The retained Deligne reading for the intended amplification is the supplied account of Weil II 3.2.13 and 3.3.4–3.3.6. No new full rereading of Weil II is claimed in this turn. No external source corpus is redistributed in this package.

The checker uses exact finite Gaussian moment fixtures, including repeated-root quotients, non-constant units, a nonzero imaginary recurrence diagonal, and both signed chain degrees. They are not arithmetic measurements. All derived infinite-dimensional statements rest on the written arguments and the explicitly named prior analytic inputs, not on the finite tests.

K Complete spectral-sum descent: relative arithmetic fibres, dual resolution and all nilpotents

Source attribution. Complete spectral-sum TeX witness retained inside the sum-connection delivery archive. The earlier public PR 17 Markdown note is separately preserved in this reader. This longer witness includes the original sum-action resolution, linear dual and antidual comparison, Ext and resolvent calculations, full matrix weight and constrained minimum. The associated public revision below is not a Git identity assigned to these longer TeX bytes; the source and archive hashes identify them.

Exact revision. 33b29f706008124886614ba4bd55bffc489df9e2

Source file. Tau_Sum_Connection_Control/sources/Tau_Spectral_Sum_Descent/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Associated public PR 17 note revision; full delivered TeX identified by its separate hash

Source SHA-256. a3becf8455ddf96e8e382762acbc0e9d0548ea7342c01d70aecdad744ae79552

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

K.1 Spectral-sum descent with the relative arithmetic fibre retained

Owner-directed continuation of the split-zero programme. 12 September 2026.

Status. Written deductions for review. These results use the actual theta range and representative constructions stated in the preceding packages. Their analytic inputs are not newly Lean-certified. The finite polynomial, quotient, and matrix statements below have independent proofs and executable finite tests. No RH, GRH, or sublinear arithmetic weight bound is asserted.

K.1.1 1. Integration and purpose

The new Kernel-Layer Integration supplement and the preceding Symmetric Frontier note overlap in their explicit relation basis and inverse-metric control. We retain the dictionary

$$K_N = \mathcal{C}_M, \quad U_N = F, \quad D_N = \Omega, \quad \mathcal{E}_N = b_M, \quad F_N = \Omega^{-1}E_+^*, \quad V_N = \mathcal{Z}_M, \quad N = M. \quad (1.1)$$

The supplement's sharper rank bound follows by grouping the summed derivative at degree M rather than counting all degree- $M+1$ coordinates. Its identities are not counted again as new work. In particular its one-variable map

$$e_{n+d+1} = u_{n+1}, \quad \mathfrak{h}_{n+1} = \kappa_{n+d+1}^{(h)} + a_{n+d+1}^* G_{h,n} a_{n+d+1} \quad (1.2)$$

is retained: it identifies the packet-dependent interpolation layer with the original packet-independent theta relation by an actual polynomial-divisibility map.

The new step is to push the arithmetic tensor system along its **sum coordinate**, retaining the relative coordinates in a matrix-valued arithmetic measure. We calculate the entire symmetric-square relation algebra, its original theta primitives, the exact norm under the pushforward, and the full

nilpotent fibres of the sum action. This is a product-to-one-parameter construction on the existing characteristic-zero object over the tau base. It is not a use of finite-field purity with relabelled coefficients.

K.1.2 2. Unchanged source, split quotient, and tensor cohomology

Keep

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (2.1)$$

The first quotient remains infinite. All spaces and morphisms are marked over the existing absolute pointed base $\mathfrak{b}_\tau$. At an active coefficient label λ , the lift of a linear map T is $(\lambda, x) \mapsto (\lambda, Tx)$; it fixes external absence. Its amplitude-zero value is $(\lambda, 0)$, the image of the original supported-zero action, not τ .

Use precisely

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \quad \mathcal{B} = \{F : \sup_{x>0} x^b |D^j F(x)| < \infty (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g, \quad g(s) = 2\xi(s). \quad (2.2)$$

For a finite nonempty packet of actual zeros with their full orders, put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad d = \deg h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h v_h \in E_h^\times.$$

The prior analytic construction supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. Retain its maps

$$\mathcal{T}_h(P) = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h. \quad (2.3)$$

For tensor degree k , use $\mathcal{P}_M = \mathbb{C}[s_1, \dots, s_k]_{\deg \leq M}$ and

$$E_k = E_h^{\otimes k}, \quad A_k = M_{s_1 + \dots + s_k}, \quad \mathcal{T}_k(P) = P(D_1, \dots, D_k)F_h^{\otimes k}, \\ I_{k,M} = \mathcal{P}_M \cap (h(s_1), \dots, h(s_k)), \quad B_{k,M} = \mathcal{T}_k I_{k,M}. \quad (2.4)$$

The jet map $\mathcal{J}_k P = v_h^{\otimes k}[P]$ has this exact kernel and is onto when $M \geq k(d-1)$. This map is linear, with an invertible unit multiplication after the algebra quotient; it is not called a unital algebra homomorphism. The source representative $R_{k,M}$ is its unique least-norm lift with the specified theta relations allowed. Thus

$$J^{(k)} R_{k,M} = 1, \quad q^{(k)} R_{k,M} = \sigma_h^{\otimes k}, \quad G_{k,M} = R_{k,M}^* R_{k,M}, \quad W_{k,M} = A_k^* G_{k,M} + G_{k,M} A_k - k G_{k,M}. \quad (2.5)$$

The tensor differential retains $(-1)^{j-1}$ when it applies the differential in the j th factor to a degree- $(k-1)$ primitive with all other factors of degree one. A primitive for the positive relation $h(s_j)P_j$ has the matching sign $(-1)^{j-1}$.

In top degree the signed chain projector

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi \quad (2.6)$$

selects ordinary symmetric tensors: the Koszul sign in T_π supplies the second permutation sign. The factor $1/k!$ is retained. We work on this image while retaining the complementary chain projector $1 - S_k$ and its trace. A source mask is transported to the union of its permutation orbit before averaging; its cancelled coordinates remain supported at that joined label.

K.1.3 3. The spectral-sum direct image is an exact functor with its action fixed

Define the actual ring map

$$\iota_k : \mathbb{C}[S] \longrightarrow E_k, \quad S \longmapsto [s_1 + \cdots + s_k]. \quad (3.1)$$

For an E_k -module M , let $\pi_{k*}M$ be that same module with the $\mathbb{C}[S]$ action restricted through ι_k . The functor is exact because its maps, kernels, and underlying abelian groups are unchanged. It has the explicit adjunction

$$\text{Hom}_{E_k}(E_k \otimes_{\mathbb{C}[S]} L, M) \cong \text{Hom}_{\mathbb{C}[S]}(L, \pi_{k*}M), \quad (3.2)$$

with forward map $f \mapsto (l \mapsto f(1 \otimes l))$ and inverse $a \otimes l \mapsto af(l)$. On quasi-coherent sheaves this is pushforward along the finite morphism $\text{Spec } E_k \rightarrow \text{Spec } \mathbb{C}[S]$. The affine pushforward statement concerns quasi-coherent complexes, not all sheaves of modules (Stacks, Tag 0AVV).

The split map is $G(\iota_k)$, with

$$p_{E_k} G(\iota_k) = \iota_k p_{\mathbb{C}[S]}, \quad G(\iota_k)(e_{\mathbb{C}[S]}) = e_{E_k}, \quad G(\iota_k)(\tau) = \tau. \quad (3.3)$$

Consequently the geometric arrow is over the same $\mathfrak{b}_\tau$. It does not replace that base by a finite field. The additional line $\text{Spec } \mathbb{C}[S]$ is the recipient of a specified arithmetic correspondence.

At the original cochain level, S acts by $D_1 + \cdots + D_k$. Its finite representative equation remains

$$D^{(k)} R_{k,M} - R_{k,M} A_k = dK_{k,M} \in B_{k,M+1}. \quad (3.4)$$

The jet and cohomology maps intertwine that action exactly. Restriction of scalars preserves their kernels and this boundary equation. It does not declare the chosen representative $R_{k,M}$ equivariant.

On the symmetric summand the ring is $B_k = (E_k)^{S_k}$, its inclusion is the original invariant inclusion, and S acts by the same sum. Exactness of taking invariants here follows from the specified Reynolds projector $k!^{-1} \sum P_\pi$ over $\mathbb{C}$.

K.1.4 3.1 The pushforward has a computed free resolution and dualizing comparison

Let E denote the actual finite ordered or symmetric image, with its exact sum operator A . Put $R = \mathbb{C}[S]$ and $n = \dim_{\mathbb{C}} E$. There is the actual free resolution

$$0 \longrightarrow R \otimes_{\mathbb{C}} E \xrightarrow{SI-A} R \otimes_{\mathbb{C}} E \xrightarrow{\text{ev}_A} \pi_* E \longrightarrow 0, \quad \text{ev}_A(P \otimes v) = P(A)v. \quad (3.5)$$

Injectivity of $SI - A$ follows either from its monic nonzero determinant or by inspecting the highest coefficient of a polynomial vector. To identify the kernel of ev_A , use

$$S^j v - A^j v = (SI - A) \sum_{i=0}^{j-1} S^{j-1-i} A^i v.$$

Subtracting these displayed relations leaves precisely the constant vector $\text{ev}_A(f)$. This proves exactness with both original arrows retained.

Let $E^\vee = \text{Hom}_{\mathbb{C}}(E, \mathbb{C})$ and $A^\vee \lambda = \lambda \circ A$. This is the algebraic dual; its matrix is a transpose. The coefficient conjugation $\overline{E^\vee} \rightarrow E^\#$, $\bar{\lambda} \mapsto (v \mapsto \overline{\lambda(v)})$, connects it explicitly to the antidual used in the control matrices.

Applying $\text{Hom}_R(-, R dS)$ to (3.5) gives

$$R \text{Hom}_R(\pi_* E, R dS) \simeq [R \otimes E^\vee dS \xrightarrow{SI - A^\vee} R \otimes E^\vee dS], \quad (3.6)$$

in degrees 0, 1. Its degree-zero group is zero, and

$$\boxed{\text{Ext}_R^1(\pi_* E, R dS) \cong E^\vee, \quad [P(S) dS] \mapsto P(A^\vee).} \quad (3.7)$$

Here $P(A^\vee)$ means $\sum_j (A^\vee)^j \lambda_j$ for a polynomial dual vector $P(S) = \sum_j S^j \lambda_j$. The inverse is the constant class $\lambda \mapsto [\lambda dS]$.

For the explicit cohomological shift, $C[b]^i = C^{i+b}$. Thus

$$R \text{Hom}_R(\pi_* E[-k], R dS[1]) \simeq E^\vee[k]. \quad (3.8)$$

The degree- k arithmetic image becomes degree $-k$ under the stated dualizing object of the sum line. These shifts are obtained from (3.6), not inferred by calling the finite packet a curve.

The adjoint behind this calculation is the actual module-category map

$$\text{Hom}_R(\pi_* M, L) \cong \text{Hom}_{B_k}(M, \text{Hom}_R(B_k, L)), \quad f \mapsto (m \mapsto (b \mapsto f(bm))). \quad (3.9)$$

The inverse evaluates the inner map at 1. Passing to derived Hom gives the corresponding derived right adjoint. This concerns the exact restriction functor on module categories; no exactness of pushforward on all sheaves is asserted (Stacks, Tag 0AWZ).

K.1.5 3.2 Resolve the finite arithmetic trace through that same duality

The pairing in (3.7) is equivalently the concrete resolvent-residue pairing

$$\boxed{\langle v, [P(S) dS] \rangle = \sum_{\lambda \in \text{Spec } A} \text{Res}_{S=\lambda} P(S) ((SI - A)^{-1} v) dS.} \quad (3.10)$$

Relations from $SI - A^\vee$ produce polynomial expressions with no finite residue. Expanding the resolvent at infinity proves that the sum is $P(A^\vee)(v)$: the coefficient of S^{-1} is $\sum_j \lambda_j (A^j v)$. This proves perfectness without deleting nilpotents.

On a block $A = \lambda I + N$ the retained resolvent is

$$(SI - A)^{-1} = \sum_{j=0}^{\ell-1} \frac{N^j}{(S - \lambda)^{j+1}}, \quad N^\ell = 0. \quad (3.11)$$

The trace observation of that resolvent is a further specified map. For a function ψ holomorphic at the finite spectrum,

$$\boxed{\mathrm{Tr}(\psi(A)) = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \mathrm{Tr}((SI - A)^{-1}) dS = \sum_{\lambda} \mathrm{Res}_{S=\lambda} \psi(S) \frac{\chi'_A(S)}{\chi_A(S)} dS,} \quad (3.12)$$

where $\chi_A(S) = \det(SI - A)$ is the exact finite characteristic polynomial. It is not substituted for the original $g = 2\xi$. With $\psi(S) = a^S$, (3.12) is the finite Lefschetz contribution of the same diagonal scaling correspondence. The cochain image is in degree k , so its supertrace includes $(-1)^k$.

The constant-class maps in (3.5)–(3.7), and the linear functional in (3.10), lift fibrewise to the original split objects. Every zero functional or zero residue remains supported. At fixed label the equations forming the resolution have composite the supported zero map. The residue pairing is an ordinary bilinear pairing in that coefficient fibre, with its split lift separately additive, not a linear map on its Cartesian product.

K.1.6 4. The symmetric tensor-square has an explicit two-generator relation algebra

Use three distinguished polynomials

$$S = s_1 + s_2, \quad r = s_1 - s_2, \quad \Delta = r^2. \quad (4.1)$$

The inverse map before taking invariants is

$$s_1 = (S + r)/2, \quad s_2 = (S - r)/2.$$

Swap fixes S and sends r to $-r$. Therefore the algebra map

$$\Phi : \mathbb{C}[S, \Delta] \xrightarrow{\sim} \mathbb{C}[s_1, s_2]^{S_2}, \quad S \mapsto s_1 + s_2, \quad \Delta \mapsto (s_1 - s_2)^2 \quad (4.2)$$

is an isomorphism. Its proof is even/odd expansion in r , using the displayed inverse with both factors $1/2$. Give S degree one and Δ degree two; then Φ preserves the entire original total-degree filtration.

Define actual polynomials H_0, H_1 by

$$\begin{aligned} h((S + r)/2) &= H_0(S, r^2) + rH_1(S, r^2), \\ h((S - r)/2) &= H_0(S, r^2) - rH_1(S, r^2). \end{aligned} \quad (4.3)$$

In particular H_0 is the half-sum and H_1 is the difference divided by $2r$. That difference is an odd polynomial, so this is polynomial division, not localization away from $r = 0$.

Theorem (complete symmetric relation ideal).

$$\boxed{B_2 = (E_h \otimes E_h)^{S_2} \cong \mathbb{C}[S, \Delta]/(H_0, \Delta H_1).} \quad (4.4)$$

Proof. In $\mathbb{C}[S, r]$, the original ideal $(h(s_1), h(s_2))$ is (H_0, rH_1) . Its even part is exactly (H_0, r^2H_1) : average an expression $H_0a + rH_1b$ under $r \mapsto -r$. Writing $a = a_0 + ra_1$, $b = b_0 + rb_1$ gives the even expression $H_0a_0 + \Delta H_1b_1$. Conversely both generators are in the original ideal. Invariants of the quotient equal the quotient of invariants because Reynolds averaging is an exact projector. This proves (4.4), including coincident roots and all nilpotents.

The quotient has dimension $d(d+1)/2$. Indeed the unscaled orbit sums of the basis $s_1^i s_2^j$, $0 \leq i, j < d$, give that many independent invariant vectors. The associated graded Hilbert series is

$$\sum_{0 \leq i \leq j < d} z^{i+j} = \frac{(1-z^d)(1-z^{d+1})}{(1-z)(1-z^2)}. \quad (4.5)$$

Here associated graded refers to the original degree filtration; (4.5) is not a claim that a nonhomogeneous h defines a graded algebra. Polynomial remainder lowers degree, and Reynolds averaging respects the filtration, proving the formula for every monic h of degree d .

K.1.7 4.1 All original primitives and their degrees survive

Let a symmetric relation of degree at most M have, after original ordered division, the expression $h(s_1)Q_1 + h(s_2)Q_2$ with $\deg Q_i \leq M-d$. Averaging yields paired coefficients

$$C_1 = (Q_1 + Q_2 \circ \text{swap})/2, \quad C_2 = C_1 \circ \text{swap}.$$

Put $A_0 = C_1 + C_2$ and $A_1 = (C_1 - C_2)/r$. Then A_0, A_1 are polynomials in S, Δ and

$$H_0 A_0 + \Delta H_1 A_1 = h(s_1) \frac{A_0 + r A_1}{2} + h(s_2) \frac{A_0 - r A_1}{2}. \quad (4.6)$$

Their weighted degree bounds are $\deg A_0 \leq M-d$ and $\deg A_1 \leq M-d-1$. The two source primitives are respectively

$$\frac{A_0 + r A_1}{2} (D_1, D_2) (\phi_* \otimes F_h), \quad -\frac{A_0 - r A_1}{2} (D_1, D_2) (F_h \otimes \phi_*). \quad (4.7)$$

The negative sign in the second primitive meets the negative cochain sign of the second differential. Applying d gives exactly (4.6) through $\mathcal{T}_2$.

The unit $v_h \otimes v_h$ is symmetric. Under (4.4) denote its full class by $v_{2,h}$. The jet map in these coordinates is exactly

$$\mathcal{J}'_{2,M} : \mathbb{C}[S, \Delta]_{\deg S + 2 \deg \Delta \leq M} \longrightarrow B_2, \quad P \longmapsto v_{2,h}[P]. \quad (4.8)$$

Its kernel is the full weighted-degree part of $(H_0, \Delta H_1)$. It is onto for $M \geq 2(d-1)$. This is the original full-jet map conjugated by Φ , not a scalar evaluation at the sum coordinate.

On the split totals, $\tilde{\Phi}(\lambda_M, P) = (\lambda_M, \Phi P)$ and $\tilde{\Phi}(\tau) = \tau$. The induced isomorphism of relation quotients commutes with every $M \rightarrow M+1$. A nonzero new relation therefore becomes $(\lambda_{M+1}, 0)$ under exactly the same degree transition as before.

K.1.8 5. The arithmetic norm becomes a one-variable matrix weight

Let

$$w_h(t) = \left| \frac{g(1/2 + it)}{h(1/2 + it)} \right|^2 \frac{1}{2\pi}, \quad d\nu_h(t) = w_h(t) dt. \quad (5.1)$$

No mass is assigned the value one. The source norm of a two-variable polynomial is

$$\|\mathcal{T}_2 P\|^2 = \int_{\mathbb{R}^2} |P(1/2 + it_1, 1/2 + it_2)|^2 w_h(t_1) w_h(t_2) dt_1 dt_2.$$

Make the exact real coordinate change

$$u = t_1 + t_2, \quad v = t_1 - t_2, \quad (t_1, t_2) = ((u+v)/2, (u-v)/2), \quad dt_1 dt_2 = \frac{1}{2} du dv. \quad (5.2)$$

Then $S = 1 + iu$, $r = iv$, and $\Delta = -v^2$. If

$$P(S, \Delta) = \sum_{a=0}^b \Delta^a f_a(S), \quad b = \lfloor M/2 \rfloor, \quad \deg f_a \leq M - 2a,$$

define the full matrix

$$\mathbf{W}_{h,b}(u)_{ac} = \frac{(-1)^{a+c}}{2} \int_{\mathbb{R}} v^{2(a+c)} w_h((u+v)/2) w_h((u-v)/2) dv. \quad (5.3)$$

Theorem (isometric spectral-sum realization).

$$\|\mathcal{T}_2 \Phi P\|^2 = \int_{\mathbb{R}} f(1 + iu)^* \mathbf{W}_{h,b}(u) f(1 + iu) du. \quad (5.4)$$

This follows by (5.2), finite polynomial expansion, and Fubini. All integrals converge by the source's exponential moments. For every real u , the matrix $\mathbf{W}_{h,b}(u)$ is positive definite: its quadratic form is the integral of the squared modulus of a nonzero polynomial in $-v^2$ against a density positive almost everywhere. Zeros of the nonzero entire g/h are discrete and cannot remove a positive-measure set of the v -line. The exponential bounds prove convergence also for each fixed u .

For $b = 0$ the scalar coefficient is the convolution

$$\mathbf{W}_{h,0}(u) = (w_h * w_h)(u). \quad (5.5)$$

For $b > 0$, its additional rows are the actual relative-coordinate moments. They have not been replaced by (5.5).

The map in (5.4) sends a polynomial to its vector of coefficient functions; its inverse is $f \mapsto \sum \Delta^a f_a(S)$ on the specified filtered image. Multiplication by $s_1 + s_2$ is exactly multiplication by S on every component. Hence its generator is still A_2 , its degree increase is still one, and its boundary is still (3.4).

K.1.9 5.1 General tensor degree

For any k , set $S = \sum s_i$, retain $x_i = s_i$ for $i < k$, and put $s_k = S - \sum_{i < k} x_i$. This is a polynomial algebra isomorphism with determinant one on the real variables $(t_1, \dots, t_k) \leftrightarrow (t_1, \dots, t_{k-1}, u = \sum t_i)$.

Write $P = \sum_{\alpha} x^{\alpha} f_{\alpha}(S)$, with $\deg f_{\alpha} \leq M - |\alpha|$. Its matrix weight is

$$\mathbf{W}_{k,M}(u)_{\alpha\beta} = \int_{\mathbb{R}^{k-1}} \prod_{i < k} \overline{(1/2 + it_i)^{\alpha_i}} \prod_{i < k} (1/2 + it_i)^{\beta_i} \prod_{i < k} w_h(t_i) w_h(u - \sum_{i < k} t_i) d^{k-1}t. \quad (5.6)$$

The outer coordinate is $S = k/2 + iu$. The same proof gives an isometry on the actual degree- M source, with every original relation substituted through this algebra isomorphism. Its entries have exponential moments in u for every exponent below the source threshold $\pi/2$: use $|\sum t_i| \leq \sum |t_i|$, the polynomial factors, and the one-variable exponential moment. Constants depend on the retained packet, k , and degrees; no uniform-in- k estimate is inferred.

K.1.10 6. The canonical representative, kernel control and relation layers commute with the descent

Let C_M be the explicit coefficient matrix of Φ from $S^a \Delta^b$ to the original symmetric polynomial basis. It is invertible between those two filtered spaces. Let U be the exact coordinate map of (4.4) to the original unscaled invariant quotient basis. Retain its orbit coefficients.

For the source moment and full-jet matrices,

$$M'_M = C_M^* M_M C_M, \quad J'_M = U^{-1} J_M C_M. \quad (6.1)$$

The matrix in (5.3) gives exactly M'_M by one-dimensional integration of its entries against the retained powers of S .

The inverse interpolation kernel and its metric satisfy

$$K'_M = J'_M (M'_M)^{-1} (J'_M)^* = U^{-1} K_M (U^{-1})^*, \quad G'_M = U^* G_M U. \quad (6.2)$$

These identities follow by multiplying the stated inverses, not by selecting a new metric. The least-norm representatives obey $R'_M = R_M U$ after their polynomial-source identification. Their action and control are

$$A'_2 = U^{-1} A_2 U, \quad W'_M = U^* W_M U, \quad \mathcal{Z}'_M = U^{-1} \mathcal{Z}_M (U^{-1})^*. \quad (6.3)$$

Thus every control inequality is exactly transported. The same isometry sends the actual next boundary layer, its Gram matrix, and its two maps Y_M, C_M^{rel} to their images in (5.4). It preserves the finer rank bound supplied by the new attachment.

The arithmetic representative defect remains a relation with an explicit primitive (4.7). Its cross-pairing is evaluated by (5.3); it is not erased when the relation becomes a supported zero in the next internal quotient.

K.1.11 6.1 Retain the base and relative parts of the norm separately through a triangular map

Write the finite matrix weight as

$$W(u) = \begin{pmatrix} m_0(u) & b(u) \\ b(u)^* & C(u) \end{pmatrix}, \quad m_0(u) > 0.$$

There is the exact isometric map into the stated direct-sum weighted space

$$\mathcal{A} : f \mapsto (f_0 + m_0^{-1} b f_{\text{rel}}, f_{\text{rel}}), \quad (6.4)$$

with inverse $(a, z) \mapsto (a - m_0^{-1} b z, z)$ and weights

$$m_0(u), \quad C(u) - b(u)^* m_0(u)^{-1} b(u). \quad (6.5)$$

Completing the square proves the isometry and positivity of the second weight. No measure is renormalized: m_0 and its reciprocal are both in the formulas. The image $\mathcal{A}(\mathcal{P}_M^{\text{sym}})$ is retained, not enlarged to arbitrary functions in the target. The ratio $m_0^{-1} b$ need not be polynomial. Multiplication by $S = 1 + iu$ commutes with this pointwise map, so the derivative action is still specified without a new connection term.

This supplies two concrete analytic quantities for the next estimates: the convolution mass and the full relative covariance matrix. Neither positivity of this representative norm nor the scalar mass alone is being substituted for the arithmetic Weil pairing.

K.1.12 7. Calculate every nilpotent fibre of the sum map

The pushforward must not replace a multiple jet by its residue. We now give its complete Jordan lengths.

For an ordered root tuple $(\rho_1, \dots, \rho_k)$ with retained orders m_i , its local algebra is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_1}, \dots, z_k^{m_k}), \quad A_k = \rho_\Sigma I + N, \quad \rho_\Sigma = \sum \rho_i, \quad N = M_{\sum z_i}. \quad (7.1)$$

Let $D_0 = \sum (m_i - 1)$ and $c_r = [t^r] \prod_i (1 + t + \dots + t^{m_i-1})$, with $c_{-1} = 0$.

Theorem (full ordered sum-fibre decomposition). As a $\mathbb{C}[S]$ -module,

$$\pi_{k*} L_\rho \cong \bigoplus_{r=0}^{\lfloor D_0/2 \rfloor} (\mathbb{C}[S]/(S - \rho_\Sigma)^{D_0-2r+1})^{\oplus (c_r - c_{r-1})}. \quad (7.2)$$

Here the isomorphism is constructed by primitive vectors, as follows. On z_i^j set

$$F_i z_i^j = j(m_i - j) z_i^{j-1}, \quad H_i z_i^j = (2j - (m_i - 1)) z_i^j.$$

For $F = \sum F_i$ and $H = \sum H_i$, direct substitution gives

$$[N, F] = H, \quad [H, N] = 2N, \quad [H, F] = -2F. \quad (7.3)$$

An auxiliary positive inner product making $F = N^*$ is explicitly diagonal:

$$\langle z^\alpha, z^\beta \rangle_0 = \delta_{\alpha\beta} \prod_i \frac{\alpha_i! (m_i - 1)!}{(m_i - 1 - \alpha_i)!}. \quad (7.4)$$

These are specified positive numbers. For the actual arithmetic Gram $G_{k,M}$ and the displayed diagonal Gram G_0 , their exact comparison is the invertible operator $C_0 = G_{k,M}^{-1} G_0$, with inverse $G_0^{-1} G_{k,M}$:

$$\langle v, w \rangle_0 = \langle v, C_0 w \rangle_{G_{k,M}}. \quad (7.4a)$$

Equation (7.4) is used only to prove the algebraic decomposition. All arithmetic control remains with $G_{k,M}$. In the primitive-ladder basis its entries are $(N^i p)^* G_{k,M} (N^j q)$, including all cross-ladder terms. They are not discarded because the auxiliary Gram admits orthogonal ladders.

A primitive vector p has degree r and satisfies $Fp = 0$. Put $w = D_0 - 2r$. The commutators prove

$$F N^j p = j(w - j + 1) N^{j-1} p, \quad \|N^j p\|_0^2 = \frac{j! w!}{(w - j)!} \|p\|_0^2 \quad (0 \leq j \leq w). \quad (7.5)$$

Thus its ladder has length $w + 1$ and $N^{w+1} p = 0$. Conversely a nonzero lowest vector has $r \leq D_0/2$, since $\|Np\|_0^2 = (D_0 - 2r) \|p\|_0^2$. Orthogonal complements of the constructed ladders are invariant under N, F, H , so induction decomposes the full finite space. Counting degree- r vectors before the middle gives $c_r - c_{r-1}$ new primitive ladders. The isomorphism in (7.2) maps $[q(S)]$ in a chosen primitive copy to $q(\rho_\Sigma + N)p$.

K.1.13 7.1 All symmetric packets, including collisions of sums

For an occupation list $\mathbf{n} = (n_\rho)_\rho$, $\sum n_\rho = k$, the corresponding invariant local component is

$$B_{\mathbf{n}} = \bigotimes_{\rho} \left(\mathbb{C}[z_1, \dots, z_{n_\rho}] / (z_j^{m_\rho}) \right)^{S_{n_\rho}}.$$

It retains its own canonical central idempotent even when two different occupation lists have the same sum $\sum n_\rho \rho$. Its graded dimensions c_r are counted by unscaled orbit monomials; equivalently their generating polynomial is

$$\prod_{\rho} \prod_{j=1}^{n_\rho} \frac{1 - t^{m_\rho + j - 1}}{1 - t^j}. \quad (7.6)$$

For $n_\rho = 0$ the factor is one. This is a polynomial identity for the orbit counts; no division at a root of unity is needed. The same operators preserve invariants and the same primitive proof gives (7.2) with

$$D_0 = \sum n_\rho (m_\rho - 1), \quad \rho_\Sigma = \sum n_\rho \rho$$

and these invariant c_r . It includes all k and all finite multiplicities.

For one order-three zero taken twice, write $z_i = s_i - \rho$, $X = z_1 + z_2$. The symmetric component has the exact ladders

$$1, X, X^2, X^3, X^4, \quad p = z_1^2 - z_1 z_2 + z_2^2, \quad Xp = 0, \quad (7.7)$$

with $X^3 = 3(z_1^2 z_2 + z_1 z_2^2)$ and $X^4 = 6z_1^2 z_2^2$. Its pushforward is

$$\mathbb{C}[S]/(S - 2\rho)^5 \oplus \mathbb{C}[S]/(S - 2\rho). \quad (7.8)$$

The actual total dimension is six. Merely retaining one evaluation at 2ρ would lose five dimensions; the displayed injective primitive maps and direct sum retain all six.

The original $v_h^{\otimes k}$ acts by an invertible multiplication map on every component and commutes with A_k . It carries these coordinates to the original full-jet coordinates. Its coefficients are not assigned one.

K.1.14 8. Pairings, traces, and the support-layer meaning of the ladders

The algebraic direct image in (3.1) keeps the full underlying vector space. Therefore its trace of a^{A_k} is unchanged. On the occupation component,

$$\mathrm{Tr}(a^{A_k} \mid B_{\mathbf{n}}) = \left[\prod_{\rho} \binom{n_\rho + m_\rho - 1}{n_\rho} \right] a^{\sum n_\rho \rho}. \quad (8.1)$$

This is valid with all nonzero nilpotents, since their exponential is triangular with diagonal one. The cochain summand has degree k , so its supertrace is $(-1)^k$ times (8.1). The complementary symmetry summands retain their previously calculated traces. No packet is deleted by pushing to the sum line.

Each ladder map in (7.2) has a split lift fixing τ . In the module $\mathbb{C}[S]/(S - \rho_\Sigma)^\ell$, the transition to the quotient of length $\ell - 1$ takes the nonzero top jet to the supported zero. On its original side the quotient map is the corresponding quotient by $\mathbb{C}N^{\ell-1}p$; it is a $\mathbb{C}[S]$ -module quotient, not a declaration

that this line is an algebra ideal. The full direct-image theorem retains the line before that optional observation.

The residue pairing and arithmetic Jacobian trace form are transported by the same full coordinate matrices. For an invertible coordinate map U , retain

$$S' = U^* S U, \quad J'_g = U^{-1} J_g U, \quad G' = U^* G U, \quad S' J'_g = U^* S J_g U. \quad (8.2)$$

On product cohomology one additionally retains the original tensor-duality signs. No auxiliary inner product from (7.4) is substituted in (8.2). A supported zero of the trace and a full vanishing jet are connected only through the stated Jacobian/trace map.

K.1.15 9. A complete unscaled arithmetic-moment calibration

For a declared Gaussian calibration, not an asserted zeta packet, take the one-variable measure $e^{-t^2/2} dt$. The change (5.2) gives

$$\mathbf{W}_2(u) = \sqrt{\pi} e^{-u^2/4} \begin{pmatrix} 1 & -2 & 12 \\ -2 & 12 & -120 \\ 12 & -120 & 1680 \end{pmatrix} \quad (9.1)$$

on coefficient functions of $1, \Delta, \Delta^2$. This uses

$$\frac{1}{2} \int v^{2j} e^{-v^2/4} dv = 2^j (2j-1)!! \sqrt{\pi},$$

with $(-1)!! = 1$. The actual triangular coordinate matrix with columns $1, \Delta + 2, \Delta^2 + 12\Delta + 12$ is

$$T = \begin{pmatrix} 1 & 2 & 12 \\ 0 & 1 & 12 \\ 0 & 0 & 1 \end{pmatrix}, \quad T^* \mathbf{W}_2(u) T = \sqrt{\pi} e^{-u^2/4} \text{diag}(1, 8, 384). \quad (9.2)$$

All three fibre directions and their norms survive. For the first two directions, the exact base/relative decomposition is

$$\|f_0 + \Delta f_1\|^2 = \sqrt{\pi} \int e^{-u^2/4} (|f_0 - 2f_1|^2 + 8|f_1|^2) du. \quad (9.3)$$

The source metric has total two-variable mass 2π , retained by these formulas. Exact symbolic tests compare the moments of $S^a \Delta^b$ computed directly in (t_1, t_2) and through (9.1), and compare the resulting constrained representatives, jet maps, Grams and controls.

For the actual arithmetic measure, (5.3) supplies the corresponding nonconstant matrix. Its relative covariance need not have the constant Gaussian form. Neither the calibration's triangular coefficients nor its control are imported into the arithmetic case.

K.1.16 10. Position in the Deligne-style programme

The source's tensor-square step is followed by geometric pushforward and control on that pushforward, not by tensoring alone. The construction here now supplies a **specified sum-coordinate pushforward** of the actual finite arithmetic image and its original filtered representatives. Its metric realization is the exact one-dimensional matrix weight (5.3) or (5.6), with the relative directions retained. The symmetric relation presentation (4.4) and original primitives (4.7) say exactly which zero-amplitude relations are present in that fibre.

The result does not identify this finite spectral scheme or matrix-weight model with a lisse sheaf on a Lefschetz pencil. The map actually constructed is the quasi-coherent finite pushforward (3.1), connected to the original analytic representatives by (2.3), (4.8), and (5.4). Deligne’s geometric hypotheses and estimates have not thereby been imported.

The immediate control task now has an explicit base/relative split: estimate the actual convolution mass, the full relative covariance, and the full-jet interpolation on the **image** of the filtered polynomial family in (6.4). These determine the same $K_M, \mathcal{Z}_M$ as the preceding work. The sum map retains A_k and every repeated eigenline, so a bound sublinear in k would still have the intended force. No such bound is proved in this note.

The concrete advances are the exact filtered symmetric relation algebra, an all- k spectral-sum norm realization, and a full nilpotent direct-image decomposition that retains multiplicities and colliding occupations. They are new applications to this specified programme; the reusable algebra and integration tools are standard and are not asserted to be globally novel.

K.1.17 References and reading scope

1. Supplied **Tau Kernel Layer Integration**, RESEARCH_NOTE.md and NOTE.tex; read in full in this continuation. Its finite-layer identifications and sharper rank are retained as source results.
2. Supplied **Tau Symmetric Frontier Control**, RESEARCH_NOTE.md; read in full. Its signed projector, orbit factors, next-layer map, and finite bound are retained.
3. Supplied **Tau Coherent Interpolation Control**, especially polynomial/jet definitions and joint total-degree sections; used as stated analytic input.
4. The Stacks Project, Tag **0AVV**, affine pushforward of quasi-coherent complexes; Tag **0AWZ**, the precise categories for the finite-pushforward adjoint. <https://stacks.math.columbia.edu/tag/0AVV> and <https://stacks.math.columbia.edu/tag/0AWZ>.
5. NIST DLMF, §18.2, polynomial moment and kernel background. <https://dlmf.nist.gov/18.2>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. The retained French TeX window around §§3.2.11–3.2.15 was reread. Its line 872 contains an inconsistent numerical implication in the historical transcription; that literal line is not used as a proof step here. The source-based tensor/pushforward architecture is used, not a certification of the entire 116-page proof. Original bibliographic record: <https://archive.numdam.org/articles/10.1007/BF02684780/>.

L PR 20: Arithmetic sum density, the full relative connection and conormal derivatives

Source attribution. Complete TeX proof witness delivered in the sum-connection archive on 12 September 2026. The verified PR 20 head contains its shorter public note and checker. This full NOTE is identified by its own source and archive hashes, and is not assigned the public note's Git blob. The local conormal continuation proves the input-constant to socle map with its full multiplicities. The local connection continuation supplies the exact projected derivative pair for the graph map. The reproduced source witness is preserved with its original statements.

Exact revision. 2144c358c2b41aa235b15cf0fa472289aec8673f

Source file. Tau_Sum_Connection_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Verified public PR 20 head; the complete delivered witness has the separate hash below

Source SHA-256. 11870208e3a08df186e42629ae1a6c956faaf28f79dfce0dc2f2d44dc64b4687

Original archive. Tau_Sum_Connection_Control_2026-09-12.zip

Archive SHA-256. f2de513041b7336b495a338e90325e84c5ce53cfa98799e1b89c53af308356de

L.1 Quantitative sum-fibre connection and the conormal arithmetic trace

Owner-directed continuation of the split-zero programme, 12 September 2026.

Status. Written analytic and algebraic deductions for review. The original theta source, its asserted range theorem, and the canonical finite representatives are inputs from the preceding work. No new Lean execution or arithmetic purity theorem is claimed. The new analytic estimate is a mass-retaining contraction for the actual spectral-sum density. The full matrix connection, its normal component, and the conormal derivative of the original theta relations are constructed as well; the contraction is not assigned to the full Weil control form without a comparison.

L.1.1 1. Source intake and the unchanged marked object

The newly uploaded Kernel Layer Integration archive is byte-identical to its earlier uploaded copy. Its thirty-one manifest entries and the fifty-three entries in the Spectral Sum Descent archive were independently verified. The kernel-layer note's identities are used as source results, not counted as new deductions:

$$K_M = G_M^{-1}, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \quad (1.1)$$

In particular the same control problem is transported by the actual map $v = K_M \lambda$. The source's next theta-relation layer, its full-jet columns, and its map into B_{M+1}/B_M remain attached. Its formalization interface is the existing internal quotient, not a replacement quotient.

Retain the marked absolute base $\mathfrak{b}_\tau$, the scalar semiring, and the original infinite quotient:

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet, \quad \begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1.2)$$

The target of $p_{\mathbb{Z}}$ is infinite. Every ring map below is lifted through G and this square. A complex-linear map $T : V \rightarrow W$ is lifted instead by the explicitly defined split-linear map $T^{\tau}(v^{\bullet}) = (Tv)^{\bullet}$, $T^{\tau}(\tau) = \tau$. These are different types of construction, connected by their respective arithmetic projection squares. In the full support diagram the same formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_{\lambda}v)$, with the displayed support transport φ . In particular amplitude zero is the receiving fibre's zero, not external absence.

The original test spaces and operators are

$$\begin{aligned} V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^{\infty}(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\ C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \\ \phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_{\phi}, \quad H_{\phi_*} = 1. \end{aligned} \quad (1.3)$$

Fix a finite packet of actual zeros with full multiplicities, stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_{\rho}}, \quad d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h v_h \in E_h^{\times}. \quad (1.4)$$

The source supplies $F_h \in \mathcal{B}$ with $\mathcal{M}F_h = v_h$ and $h(D)F_h = \Theta\phi_*$. It also supplies

$$\mathcal{T}_h P = P(D)F_h, \quad \mathcal{T}_h(hP) = \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h P = v_h[P]_h. \quad (1.5)$$

The statements below also apply to $h = 1$ as the actual theta-seed calculation, but that case has zero-dimensional arithmetic quotient and is not substituted for a nonempty packet.

L.1.2 2. The sum coordinate and its conjugate logarithmic operator

Let $k \geq 2$, $S = \sum_{i=1}^k s_i$, and use the explicit centered relative coordinates

$$z_i = s_i - S/k \quad (i < k), \quad z_k = -\sum_{i < k} z_i, \quad s_i = S/k + z_i. \quad (2.1)$$

On the original integration lines set

$$s_i = \frac{1}{2} + it_i, \quad S = \frac{k}{2} + iu, \quad u = \sum_i t_i, \quad y_i = t_i - u/k \ (i < k), \quad y_k = -\sum_{i < k} y_i. \quad (2.2)$$

The inverse is $t_i = u/k + y_i$. The real Jacobian has absolute value one. For $k = 2$, the earlier relative coordinate was $v = t_1 - t_2 = 2y_1$, so $dt_1 dt_2 = du dy_1 = \frac{1}{2} du dv$. Thus the earlier factor $1/2$ remains exactly in that coordinate comparison. Its Δ is $4z_1^2 = -4y_1^2$.

Write

$$a_h(t) = \frac{v_h(\frac{1}{2} + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int_{\mathbb{R}} w_h(t) dt = \|F_h\|_{L^2(dx)}^2. \quad (2.3)$$

The factor $\sqrt{2\pi}$ is the stated Mellin–Fourier Plancherel factor. No mass has been set to one. Define the actual fibre amplitude

$$\Psi_k(u, \mathbf{y}) = \prod_{i=1}^k a_h(u/k + y_i). \quad (2.4)$$

Mellin–Fourier followed by (2.2) is the unitary map

$$\mathcal{U}_k : L^2(\mathbb{R}_{>0}^k, d^k x) \longrightarrow L^2(\mathbb{R}_u \times \mathbb{R}_{\mathbf{y}}^{k-1}, du d\mathbf{y}).$$

On the given test functions it satisfies

$$\begin{aligned} \mathcal{U}_k(D^{(k)}F) &= (\tfrac{k}{2} + iu)\mathcal{U}_kF, \quad D^{(k)} = \sum_i D_i, \\ \partial_u \mathcal{U}_kF &= i\mathcal{U}_k(\mathcal{L}_kF), \quad \mathcal{L}_k = \frac{1}{k} \sum_{i=1}^k \log x_i, \quad [D^{(k)}, \mathcal{L}_k] = -1. \end{aligned} \quad (2.5)$$

Here $\mathcal{L}_k$ is multiplication by the displayed function; $1/k$ comes from the inverse coordinate map keeping the relative z_i fixed. It does not change the scaling generator $D^{(k)}$. The identities follow by differentiating the Mellin integral. Logarithmic multiplication preserves $\mathcal{B}$ and the corresponding tensor test space.

For the polynomial source,

$$\mathcal{U}_k \mathcal{T}_h^{(k)} P = \Psi_k(u, \mathbf{y}) P(S/k + iy_1, \dots, S/k + iy_k). \quad (2.6)$$

Thus the operator whose new estimate is computed is related to the original scaling operator by the precise commutator (2.5), not equated with the generator or its weight defect.

L.1.3 3. A quantitative contraction on the actual arithmetic density

Define the convolution mass and its logarithmic-derivative energy by

$$\begin{aligned} m_k(u) &= \int_{\mathbb{R}^{k-1}} |\Psi_k(u, \mathbf{y})|^2 d\mathbf{y} = w_h^{*k}(u), \quad \int m_k = \mu_h^k, \\ \mathcal{I}_h &= \int_{\mathbb{R}} \frac{|w'_h(t)|^2}{w_h(t)} dt, \quad \mathcal{I}_{h,k} = \int_{\mathbb{R}} \frac{|m'_k(u)|^2}{m_k(u)} du. \end{aligned} \quad (3.1)$$

At isolated zeros of w_h , the integrand is assigned its limiting value where applicable, or any value on that null set. The construction and proofs use the smooth amplitude a_h , so no division at a zero is required. For $k \geq 2$, $m_k(u) > 0$ for every real u : each factor w_h is positive almost everywhere, and its convolution integral has a positive integrand almost everywhere on its fibre.

The original functional equation gives

$$g^\dagger = g, \quad h^\dagger = (-1)^d h, \quad v_h^\dagger = (-1)^d v_h, \quad \overline{a_h(t)} = (-1)^d a_h(t). \quad (3.2)$$

Accordingly a_h and a'_h have the same fixed real/imaginary phase and

$$\boxed{\mathcal{I}_h = 4 \int_{\mathbb{R}} |a'_h(t)|^2 dt = 4 \int_0^\infty (\log x)^2 |F_h(x)|^2 dx < \infty, \quad \int \overline{a_h} a'_h = 0.} \quad (3.3)$$

Proof: outside the discrete zero set, $w'_h = 2\overline{a_h}a'_h$ is real, giving $w_h'^2/w_h = 4|a'_h|^2$. Mellin Plancherel and $\partial_t v_h(\frac{1}{2} + it) = i v'_h(\frac{1}{2} + it)$ give the second equality. The zero integral is the integral of $w'_h/2$. The decay needed at both endpoints follows from $F_h \in \mathcal{B}$: $e^{y/2} F_h(e^y)$ is Schwartz, as are its polynomially weighted versions. Reflection (3.2) is essential for the equality rather than just an upper bound.

Let $\mathcal{H}_u = L^2(\mathbb{R}^{k-1}, d\mathbf{y})$. The line inclusion and projection are

$$j_{0,u} : \mathbb{C} \rightarrow \mathcal{H}_u, \quad c \mapsto c\Psi_k(u, \cdot), \quad \Pi_{0,u}Z = \Psi_k \frac{\langle \Psi_k, Z \rangle}{m_k(u)}.$$

Put

$$n_k(u) = (1 - \Pi_{0,u})\partial_u \Psi_k = \partial_u \Psi_k - \frac{m'_k}{2m_k} \Psi_k. \quad (3.4)$$

The inner product is antilinear in its first argument. Equation (3.2) makes $\langle \Psi_k, \partial_u \Psi_k \rangle = m'_k/2$ real. Therefore (3.4) is precisely an orthogonal decomposition, including at zeros of individual amplitude factors.

Theorem 3.1 (mass-retaining sum contraction and exact relative term).

$$\mathcal{I}_{h,k} + 4 \int_{\mathbb{R}} \|n_k(u)\|_{\mathcal{H}_u}^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.5)$$

In particular

$$\mathcal{I}_{h,k} \leq \frac{\mu_h^{k-1}}{k} \mathcal{I}_h. \quad (3.6)$$

Proof. Differentiate the actual product amplitude in the centered coordinates:

$$\partial_u \Psi_k = \frac{1}{k} \sum_{i=1}^k a'_h(t_i) \prod_{j \neq i} a_h(t_j).$$

Integrating its squared modulus over all original t_i , the k diagonal terms each give $\mu_h^{k-1} \|a'_h\|^2$. Every distinct-index cross term is $\mu_h^{k-2} |\int \overline{a_h} a'_h|^2 = 0$. Thus

$$\int \|\partial_u \Psi_k\|^2 du = \frac{\mu_h^{k-1}}{4k} \mathcal{I}_h.$$

The pointwise orthogonal decomposition (3.4) gives

$$\|\partial_u \Psi_k\|^2 = \frac{m_k'^2}{4m_k} + \|n_k\|^2.$$

Integrate and multiply by four. This proves both statements and the finiteness of $\mathcal{I}_{h,k}$. All masses and the factor four remain.

This is an application of the projection mechanism underlying Fisher-information convolution inequalities (Johnson–Barron), with the actual arithmetic amplitude, its phase, and its residual relative vector kept explicitly. It does not rely on log-concavity, a Gaussian approximation, or locations of the zeros beyond the stipulated reflection-stable packet and the source construction of F_h .

The pair $(\Pi_{0,u}\partial_u \Psi_k, n_k)$ retains the full derivative by addition. Its split lift can have an active zero in either component without converting the other component or its source to τ . Dropping n_k would replace equality (3.5) by an inequality and lose the quantified relative contribution; the actual projection and its complementary map remain available.

L.1.4 4. The full relative matrix has a computed connection and normal variation

Take any finite linearly independent family of relative polynomials $\theta_\alpha(\mathbf{z})$, independent of S . For the original degree- M source use the actual coefficient functions

$$P(\mathbf{s}) = \sum_{\alpha} \theta_{\alpha}(\mathbf{z}) f_{\alpha}(S), \quad \deg f_{\alpha} \leq M - \deg \theta_{\alpha}. \quad (4.1)$$

Choose a homogeneous relative monomial basis, or the stated unscaled invariant basis on a symmetric summand, so that these degree conditions specify the exact original filtered image. This is the centered-coordinate version of the preceding matrix weight; substitution $s_i = S/k + z_i$ is triangular by total degree and invertible.

Define the actual fibre map, its derivative, and its three Gram matrices:

$$\begin{aligned} j_u : \mathbb{C}^r &\rightarrow \mathcal{H}_u, & j_u c &= \Psi_k(u, \mathbf{y}) \sum_{\alpha} \theta_{\alpha}(i\mathbf{y}) c_{\alpha}, \\ W(u) &= j_u^* j_u, & B(u) &= j_u^* j'_u, & T(u) &= (j'_u)^* j'_u. \end{aligned} \quad (4.2)$$

These are integrals in the same arithmetic measure. $W(u)$ is strictly positive because the relative polynomials are independent and Ψ_k is nonzero almost everywhere. Every entry is smooth by differentiated Schwartz estimates on compact u intervals. Its domain has its stated finite size r ; no assertion is made that r remains fixed as the source degree grows.

In the fixed-phase case (3.2),

$$B(u) = \frac{1}{2} W'(u). \quad (4.3)$$

Indeed $\overline{\Psi_k} \partial_u \Psi_k$ is real almost everywhere, hence B is Hermitian; differentiate W to get $W' = B + B^*$. The integrals remain valid at the zero set because the smooth product amplitude, not its logarithm, is used.

Let

$$\Pi_u = j_u W^{-1} j_u^*, \quad \Gamma_u = W^{-1} B = \frac{1}{2} W^{-1} W', \quad N_u = (1 - \Pi_u) j'_u. \quad (4.4)$$

The exact differential map is

$$\boxed{\partial_u(j_u c(u)) = j_u(c'(u) + \Gamma_u c(u)) + N_u c(u), \quad j_u^* N_u = 0.} \quad (4.5)$$

Thus $\nabla_u c = c' + \Gamma_u c$ is a metric-compatible connection on this finite coefficient bundle:

$$\Gamma_u^* W + W \Gamma_u = W'. \quad (4.6)$$

The complete normal form is

$$\boxed{\mathcal{N}(u) := N_u^* N_u = T - B^* W^{-1} B = T - \frac{1}{4} W' W^{-1} W' \succeq 0.} \quad (4.7)$$

In particular no commutation of W and W' is assumed. Multiplication proves the identity and positivity follows from $1 - \Pi_u$ being an orthogonal projection. The norm decomposition is

$$\boxed{\|\partial_u(j_u c)\|^2 = (\nabla_u c)^* W (\nabla_u c) + c^* \mathcal{N} c.} \quad (4.8)$$

For the one-column constant family, (4.8) is exactly (3.5) before integration. For the full relative family it retains every matrix entry in both terms. Scalar inequality (3.6) is not inserted as a bound

on this larger matrix: its precise connection is the inclusion of the constant column and the pair of projections $\Pi_{0,u}, \Pi_u$ with $\text{im } \Pi_{0,u} \subset \text{im } \Pi_u$.

L.1.5 Change of relative frame

For an explicitly given invertible smooth matrix $C(u)$, replace the fibre map by $j_u^C = j_u C(u)$. The exact transformed connection and normal map are

$$W^C = C^* W C, \quad \Gamma^C = C^{-1} \Gamma C + C^{-1} C', \quad N^C = N C. \quad (4.12)$$

These follow by differentiating j^C and applying the same orthogonal projection. This is the additional derivative term needed when using the earlier pointwise triangular square-completion map, whose coefficients can depend on u . Multiplication by S still intertwines without such a term; the derivative's relationship to it is the explicit commutator (2.5). The exact filtered image of the original coefficient functions is retained after this change of frame.

L.1.6 4.1 The differentiated filtered image is retained

For the actual source coefficient function $c_P(u) = (f_\alpha(k/2 + iu))_\alpha$, define the linear graph map

$$\mathcal{D}_M : P \mapsto (j_u \nabla_u c_P, N_u c_P). \quad (4.9)$$

Its values lie in the specified graph image $\mathcal{G}_M$, not in two freely chosen function spaces. Addition of the two components recovers

$$\text{Add } \mathcal{D}_M(P) = \partial_u \mathcal{U}_k \mathcal{T}_h^{(k)} P = i \mathcal{U}_k \mathcal{L}_k \mathcal{T}_h^{(k)} P. \quad (4.10)$$

The two summands are pointwise orthogonal, so this addition is an isometric injection on the graph into the derivative's actual image. This supplies a precise way to recover its arithmetic jet:

$$\mathcal{G}_M \xrightarrow{\text{Add}} i \mathcal{U}_k (\mathcal{B}^{\widehat{\otimes} k}) \xrightarrow{-i \mathcal{U}_k^{-1}} \mathcal{B}^{\widehat{\otimes} k} \xrightarrow{J^{(k)}} E_h^{\otimes k}. \quad (4.11)$$

No separate arithmetic-jet map on an arbitrary measurable normal section is asserted. The combined graph is exactly the domain on which the formula is defined. The original support label is retained in both components and in (4.11). The cohomological control still uses A_k, R_M, G_M, W_M and (1.1); (2.5), (4.10), and (4.11) are the actual comparison to this new analytic derivative.

L.1.7 5. Differentiating a supported relation requires its first infinitesimal layer

The logarithmic observable in (2.5) has a calculated effect on the original theta relation, even in one variable:

$$\boxed{J_h(\log x \Theta \phi) = j_h(g') j_h(H_\phi)}. \quad (5.1)$$

Proof: logarithmic multiplication differentiates the actual Mellin transform, so

$$\mathcal{M}(\log x \Theta \phi) = g' H_\phi + g H'_\phi.$$

Taking full jets modulo h removes the second term, but not the first. For $\phi = P(D)\phi_*$, $H_\phi = P$; hence all E_h values of $j_h H_\phi$ are supplied by actual original source functions.

This is the map connecting the original supported boundary to the Jacobian trace contraction. In the original quotient $q\Theta = 0$, while $J_h(\log x \Theta)$ has the explicitly displayed image. The typed resolution of that effect is a first thickening of the *same* relation ideal.

Put

$$\mathcal{P} = \mathbb{C}[s_1, \dots, s_k], \quad I = (h(s_1), \dots, h(s_k)), \quad E = \mathcal{P}/I, \quad E^{[2]} = \mathcal{P}/I^2.$$

There is the actual exact sequence

$$0 \longrightarrow I/I^2 \longrightarrow E^{[2]} \xrightarrow{\pi} E \longrightarrow 0. \quad (5.2)$$

The coefficient conormal map is the isomorphism

$$\iota_N : E^k \xrightarrow{\sim} I/I^2, \quad (a_1, \dots, a_k) \mapsto \left[\sum_i h(s_i) \tilde{a}_i \right]. \quad (5.3)$$

A direct proof uses monic division in each variable. The polynomial ring is free over $\mathbb{C}[h(s_1), \dots, h(s_k)]$ with basis $\prod s_i^{a_i}$, $0 \leq a_i < d$. Modulo the square of the ideal of these latter variables, only coefficient degrees zero and one survive. This proves uniqueness and surjectivity in (5.3), retaining all nilpotents in E .

Define the sum derivative at fixed relative coordinates

$$\partial_S^{\text{rel}} = \frac{1}{k} \sum_i \partial_{s_i}. \quad (5.4)$$

It maps I^2 into I , so it induces a *complex-linear* map

$$\delta_S : E^{[2]} \longrightarrow E, \quad [P]_{I^2} \mapsto [\partial_S^{\text{rel}} P]_I. \quad (5.5)$$

It is a derivation along π :

$$\delta_S(ab) = \pi(a)\delta_S(b) + \pi(b)\delta_S(a), \quad \delta_S(S) = 1.$$

Its restriction to the retained conormal fibre is the E -linear row

$$\delta_S \iota_N(a_1, \dots, a_k) = \frac{1}{k} \sum_i h'(s_i) a_i. \quad (5.6)$$

The full conormal differential, before contraction by ∂_S^{rel} , is

$$I/I^2 \xrightarrow{d_I} \bigoplus_i E \, ds_i, \quad (a_i)_i \mapsto \sum_i h'(s_i) a_i \, ds_i. \quad (5.7)$$

This is the two-term cotangent presentation for the complete intersection, in cohomological degrees $-1, 0$. Its universal differential and (5.5) are related by contraction $\sum b_i ds_i \mapsto k^{-1} \sum b_i$. This is the established conormal/cotangent construction applied to this exact packet ideal (Stacks, Tags 00S0 and 08SH).

The original split diagram is

$$G(E^{[2]}) \xrightarrow{G(\pi)} G(E), \quad \delta_S^\tau : G(E^{[2]}) \longrightarrow G(E) \quad \text{as } G(\mathbb{C})\text{-semimodules.} \quad (5.8)$$

The first is a semiring morphism; the second is a split-linear map of coefficient modules, not a semiring morphism. Both send external absence to external absence and supported zero to supported zero. A supported conormal class has image e_E under the first map, while its image under the second is the supported scalar specified in (5.6). The larger domain retains the datum needed to differentiate that relation.

For one local block $h = (s - \rho)^m$ times a retained unit, (5.6) detects the z^{m-1} coefficient with its factor m . In an ordered product of local block lengths m_i , the image of (5.6) has dimension

$$\prod_i m_i - \prod_i (m_i - 1), \quad (5.9)$$

because it is the ideal generated by the unit multiples of $z_i^{m_i-1}$. Each unit is invertible in its actual local algebra. Monomials missing all top exponents form the complementary vector count; they are not discarded from E .

L.1.8 5.1 Carry the arithmetic jet unit through the derivative

Let $U(\mathbf{s}) = \prod_i v_h(s_i)$, and retain its full Taylor class $\hat{U} \in E^{[2]}$. This class is invertible. Define

$$\hat{\mathcal{J}}P = \hat{U}[P]_{I^2}, \quad \mathcal{J}P = U[P]_I, \quad \beta_h = U^{-1}\delta_S\hat{U} \in E. \quad (5.10)$$

The finite Taylor realization here uses the entire functions on neighbourhoods of the finite zero set; polynomial Hermite interpolation supplies their classes. $E^{[2]}$ is *not* being called a new zeta-zero packet: it also retains first-order source relations not annihilated by g .

The exact commuting equation is

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \delta_S\hat{\mathcal{J}}P = \mathcal{J}(\partial_S^{\text{rel}}P) + \beta_h\mathcal{J}P.} \quad (5.11)$$

Proof: its Mellin transform is $\partial_S^{\text{rel}}(UP)$. Apply the product rule before taking jets. Thus the coefficient-connection term is $\beta_h = k^{-1}\sum_i j(v'_h/v_h)(s_i)$, with no discarded local unit.

For an original relation $P = \sum_i h(s_i)P_i$, the last term vanishes and

$$\boxed{J^{(k)}\mathcal{L}_k\mathcal{T}_h^{(k)}P = \frac{1}{k}U\sum_i h'(s_i)[P_i]_I.} \quad (5.12)$$

Modulo I , $v_h(s_i)h'(s_i) = g'(s_i)$, so the same equation can be written with $g'(s_i)$ and the other retained v_h factors. This is the precise relationship between the analytic sum derivative in (4.10) and the conormal row (5.6).

In one variable the source's residue pairing therefore obeys

$$\boxed{\mathcal{R}_Z(f, J_h(\log x \Theta(P(D)\phi_*))) = \sum_{\rho \in Z} m_\rho \overline{f(1 - \bar{\rho})} P(\rho).} \quad (5.13)$$

Indeed (5.1) makes the second argument $g'P$, and g'/g has residue m_ρ . The residue pairing, its Jacobian contraction, and the logarithmic derivative of an actual original theta relation have now been connected by a single formula. No new positivity assertion is inferred from that equality.

L.1.9 6. All higher derivative layers and the old cochain signs remain available

For $r \geq 1$, the derivation gives actual maps

$$\partial_S^{\text{rel}} : \mathcal{P}/I^{r+1} \longrightarrow \mathcal{P}/I^r. \quad (6.1)$$

They commute with the quotient maps wherever their sources and targets match. The associated-graded algebra is explicitly

$$\text{gr}_I \mathcal{P} \cong E[\eta_1, \dots, \eta_k], \quad \eta_i \longleftrightarrow [h(s_i)]. \quad (6.2)$$

On it, the degree-minus-one derivative is

$$\boxed{\text{gr}(\partial_S^{\text{rel}}) = \frac{1}{k} \sum_i h'(s_i) \partial_{\eta_i}.} \quad (6.3)$$

The proof expands $\partial(h^\alpha P)$. The term differentiating P stays in $I^{|\alpha|}$ and disappears only in the explicitly specified quotient $I^{|\alpha|-1}/I^{|\alpha|}$. The other terms give (6.3) with every factor α_i retained. Consequently this layer has rank $\binom{r+k-1}{k-1}$ over E , not one new scalar support.

There is also a full mixed derivative formula retaining all coordinate directions:

$$\boxed{\left[\partial_{s_1} \cdots \partial_{s_k} \left(UP \prod_i h(s_i) \right) \right]_I = \left[UP \prod_i h'(s_i) \right]_I = \left[P \prod_i g'(s_i) \right]_I.} \quad (6.4)$$

Every other product-rule term contains an undifferentiated $h(s_i)$ and maps to the supported zero in E . At the test-function level these derivatives are multiplication by $\prod_i \log x_i$. At the residue level (6.4) is the product Jacobian in the existing tensor arithmetic trace. The ordered volume form $ds_1 \wedge \cdots \wedge ds_k$ and the previously specified tensor-duality sign are retained; no new identification of the iterated residue with an unsigned cochain supertrace is made.

The original primitive of $h(s_i)P_i$ still has the sign $(-1)^{i-1}$ from the prior product complex. Its cochain differential supplies the second $(-1)^{i-1}$. Conormal reduction records the resulting coefficient $[P_i]_I$. It does not replace that source primitive or the higher syzygies of its Koszul complex.

For a finite total-degree relation space, (6.1) is applied to the image of that exact space in $\mathcal{P}/I^{r+1}$. It does not enlarge the set of admissible representatives. The degree- M next-relation map from the attachment feeds (5.12) by its actual polynomial numerator. This supplies a differentiated observation of those very same e -relations.

L.1.10 7. Exact calibrations and a bounded arithmetic seed evaluation

L.1.11 7.1 Gaussian fibre: the relative connection can be flat without deleting it

Take the declared calibration $a(t) = e^{-t^2/4}$, $w(t) = e^{-t^2/2}$, $\mu = \sqrt{2\pi}$. This is a test of the formula, not the arithmetic input. For two factors,

$$\Psi_2(u, y) = e^{-u^2/8} e^{-y^2/2}, \quad m_2(u) = \sqrt{\pi} e^{-u^2/4}.$$

For every fixed relative polynomial frame, $j'_u = -(u/4)j_u$, so

$$\Gamma_u = -\frac{u}{4}I, \quad \mathcal{N}(u) = 0, \quad \mathcal{I}_2 = \frac{\mu \mathcal{I}_1}{2} = \pi. \quad (7.1)$$

All relative directions and their Gram entries remain; zero normal variation is a proved consequence of this particular product, not an instruction to erase the relative fibre in the arithmetic case.

L.1.12 7.2 A zero-bearing calibration has nonzero retained normal variation

Take instead $a(t) = te^{-t^2/4}$ and $w(t) = t^2e^{-t^2/2}$. Then

$$\begin{aligned}\mu &= \sqrt{2\pi}, & \mathcal{I}_1 &= 3\sqrt{2\pi}, \\ \Psi_2(u, y) &= (u^2/4 - y^2)e^{-u^2/8 - y^2/2}, \\ m_2(u) &= \frac{\sqrt{\pi}}{16}e^{-u^2/4}(u^4 - 4u^2 + 12).\end{aligned}\tag{7.2}$$

These identities follow from the complete Gaussian even moments; the polynomial has positive minimum 8, so m_2 has no zero. For $u \neq 0$, $\partial_u \Psi_2$ is not a multiple of Ψ_2 as a polynomial in y , so $n_2(u) \neq 0$. The exact identity is

$$\mathcal{I}_2 + 4 \int \|n_2\|^2 = 3\pi, \quad \mathcal{I}_2 < 3\pi.\tag{7.3}$$

The checker calculates the two-column matrix connection and nonzero positive normal Gram at $u = 1$, retaining the relative basis and its mass factors. This also tests the handling of zeros without assuming logarithmic scores are bounded.

L.1.13 7.3 The actual arithmetic seed, with no selected zero coordinates

For $h = 1$, $F_h = f_0 = \Theta\phi_*$. Its exact inversion is $f_0(x) = x^{-1}f_0(1/x)$. Hence

$$\mu_1 = 2 \int_1^\infty f_0(x)^2 dx, \quad \mathcal{I}_1 = 8 \int_1^\infty (\log x)^2 f_0(x)^2 dx.\tag{7.4}$$

On that half-line use the actual absolutely convergent sum

$$f_0(x) = 2 \sum_{n \geq 1} (4\pi^2 n^4 x^4 - 6\pi n^2 x^2) e^{-\pi n^2 x^2}.$$

The numerical evaluations at 45 digits with six integer terms and at 65 digits with eight integer terms agree on the displayed 35-digit values:

$$\mu_1 \approx 1.2790072478464851404795335922671933, \quad \mathcal{I}_1 \approx 0.12575246821348846545509759756793817.$$

The optional computation evaluates (7.4) at two stated decimal precisions and two integer cutoffs. Its output is labeled numerical, not an interval certificate. The theorem uses the exact integrals; no numerical value or cutoff is imported into a purity assertion.

L.1.14 8. What the new estimate controls in the programme

The earlier arithmetic defect remains

$$W_M = A_k^* G_M + G_M A_k - k G_M, \quad V_M = A_k K_M + K_M A_k^* - k K_M, \quad W_M = G_M V_M G_M. \tag{8.1}$$

The new operations retain the path

$$\mathcal{T}_k(P) \xrightarrow{\mathcal{U}_k} \Psi_{kCP} \xrightarrow{\partial_u} \mathcal{G}_M \xrightarrow{\text{Add}, -i\mathcal{U}_k^{-1}} \mathcal{L}_k \mathcal{T}_k(P) \xrightarrow{J^{(k)}} \delta_S \hat{\mathcal{J}} P. \tag{8.2}$$

This is the explicit connection between the positive differential estimate, the matrix-valued relative fibre, and the first conormal layer of the original arithmetic quotient. It does not replace the scaling generator A_k with differentiation. Their relation is the commutator (2.5), and the source's original representative relation $D^{(k)}R_M - R_MA_k \in B_{M+1}$ continues to determine (8.1).

The result of this continuation is a genuine $1/k$ factor in (3.5), with the literal mass μ_h^{k-1} , and an exact decomposition of the full matrix derivative. Uniform relative interpolation control is still needed to transfer that integrated smoothing to the extremal finite-packet estimate (8.1). The required comparison cannot be inferred from the scalar density alone; (4.9)–(4.11) and (5.11) provide the actual objects on which to work. All normal terms and the conormal Jacobian remain in that comparison.

Deligne's source architecture requires a further estimate after the product-to-one-parameter operation, on the same arithmetic image and with its duality. This note constructs and estimates a differential of that new-base pushforward, and identifies its exact interaction with the trace-producing Jacobian. It does not assert that a Fisher inequality is already Deligne purity, or that the analytic matrix bundle is a finite-field lisse sheaf. The finite-point cohomology, spectral-sum restriction functor, kernel-layer calculation, and first infinitesimal quotient remain connected by the typed arrows above.

L.1.15 References and reading record

1. Supplied *Tau Kernel Layer Integration*, NOTE.tex and RESEARCH_NOTE.md, read for the original relation-layer and antidual comparisons; the archive matched the earlier copy exactly and all manifest entries verified. The source's eighteen-method exact suite was rerun normally and with optimized Python. This is not a new formal certificate.
2. Supplied *Tau Spectral Sum Descent*, NOTE.tex, read in full, especially §§2, 5, 6, 8 and 10. The same sum coordinate, matrix density, relative polynomial image, full jets, and residue comparison are used.
3. Supplied *SplitZero: reconstruction, internal homology, and balanced finite jets*, §§1–4, for the existing original internal quotient and supported cohomology interfaces. No new scalar implementation is introduced.
4. Oliver Johnson and Andrew Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020, published in Probability Theory and Related Fields 129 (2004), 391–409. Their projection method is established background; Theorem 3.1 is proved here directly with the unaltered arithmetic mass and phase. <https://arxiv.org/abs/math/0111020>.
5. The Stacks Project, Tags 00S0 and 08SH, the conormal presentation and cotangent complex of a local complete intersection. These are reusable algebraic constructions; the explicit arithmetic conormal row and the trace connection are computed in §§5–6. <https://stacks.math.columbia.edu/tag/00S0> and <https://stacks.math.columbia.edu/tag/08SH>.
6. Deligne, *La conjecture de Weil II*, IHES 52 (1980), 137–252. This continuation relies on the previously identified tensor/pushforward/dual-control architecture, not a fresh reading or independent audit of all pages. No finite-field theorem is asserted on the new base merely from terminology.

M Cyclic arithmetic sum control and the complete contracted relation tower

Source attribution. Complete TeX witness supplied in the Cyclic Sum Control archive. Its exact source and archive hashes identify the delivered text. The seven retained prior source files are identified separately; the earlier 48-entry validation claim concerns the separate Sum Connection archive. The new 24-method finite checker has a separate independent replay record. No Lean execution, remote CI or uniform arithmetic estimate is inferred from this source witness.

Exact revision. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Source file. Tau_Cyclic_Sum_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Source witness SHA-256; no remote Git revision is assigned to this delivery

Source SHA-256. 2cdfd5b25631464f9500dc5fd07c4c8f5c927c8877eb1bf076307ec5dcbc2e8c

Original archive. Tau_Cyclic_Sum_Control_2026-09-12.zip

Archive SHA-256. e0e4d0ecda69e4738dcb4cdd3b750e623d863eac8b44f353737393e720a840a7

M.1 Sum-generated arithmetic cohomology, scalar control, and exact conormal depth

Owner-directed continuation of the split-zero programme — 12 September 2026.

Status. Written deductions for mathematical review. This note uses the original theta source and its stated finite-packet realization from the preceding work. It does not claim a new Lean certificate, a global comparison-kernel theorem, or a uniform arithmetic purity estimate. The new constructions are (i) an explicitly injected sum-generated arithmetic submodule containing every amplified spectral value, (ii) its scalar convolution norm and rank-two control formula, (iii) raw-moment formulas for the complete differentiated polynomial image, and (iv) the exact contraction of every original conormal thickening to that sum coordinate. The full relative quotient, its action, and its trace are retained.

M.1.1 1. Source intake and fixed objects

The two new transcript files were searched for changes to the arithmetic/control and formalization interfaces. They preserve the earlier full-jet kernel/layer calculations and the formalized internal quotient; they do not supply a later analytic proof of the remaining estimate. The previous Tau_Sum_Connection_Control archive was opened directly, its LaTeX and mathematical note were read, and all 48 entries of its supplied manifest verified. Its scalar contraction and matrix connection are inputs [S1], not attributed to the present continuation. The previous transcript's checked algebraic interfaces remain within their reported scope [S2–S4].

The original base is still b_τ . Retain

$$G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

$$\begin{array}{ccc}
G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\
p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\
\mathbb{Z} & \xrightarrow{j} & \mathbb{C}.
\end{array} \tag{1.1}$$

The target of $p\mathbb{Z}$ is infinite. A ring homomorphism is lifted through G . A complex-linear map $T : V \rightarrow W$ has instead the explicitly defined $G(\mathbb{C})$ -linear lift $T^\tau(v^\bullet) = (Tv)^\bullet$, $T^\tau(\tau) = \tau$. In a support diagram the formula is $(\lambda, v) \mapsto (\varphi(\lambda), T_\lambda v)$, with its stated support map φ . These formulas commute with the respective amplitude observations.

Use exactly

$$\begin{aligned}
V &= \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\}, \\
\mathcal{B} &= \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0)\}, \quad D = -x\partial_x, \\
C_+ &= [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \\
\phi_* &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, \quad g(s) = 2\xi(s), \quad \mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1.
\end{aligned} \tag{1.2}$$

Let $h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}$ be a **nonempty** finite packet of actual zeros of g , with their full orders. Assume it is stable under $\rho \mapsto 1 - \bar{\rho}$. Put

$$d = \deg h, \quad v_h = g/h, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = j_h(v_h) \in E_h^\times. \tag{1.3}$$

The preceding source provides the actual test function F_h and map

$$\begin{aligned}
\mathcal{M}F_h &= v_h, \quad h(D)F_h = \Theta\phi_*, \quad \mathcal{T}_h(P) = P(D)F_h, \\
\mathcal{T}_h(hP) &= \Theta(P(D)\phi_*), \quad J_h \mathcal{T}_h(P) = v_h[P]_h, \quad q\mathcal{T}_h(P) = \sigma_h(v_h[P]_h).
\end{aligned} \tag{1.4}$$

For $k \geq 2$, define

$$\begin{aligned}
\mathcal{P}_k &= \mathbb{C}[s_1, \dots, s_k], \quad I_k = (h(s_1), \dots, h(s_k)), \quad B_k = \mathcal{P}_k/I_k = E_h^{\otimes k}, \\
S &= s_1 + \dots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k} \in B_k^\times.
\end{aligned} \tag{1.5}$$

The ordinary invariant submodule is $B_k^{\text{sym}} = B_k^{S_k}$. Its cochain realization is the existing signed chain projector $\frac{1}{k!} \sum_{\pi} \text{sgn}(\pi) T_\pi$: in top degree the Koszul sign in T_π cancels the displayed sign. The complementary cochain summand and both traces remain in the programme. No change to that projector is made here.

M.1.2 2. The sum-generated submodule and its exact spectrum

Index the distinct roots by $\rho_1, \dots, \rho_a$, with orders $m_1, \dots, m_a$. For an occupation vector $\mathbf{n} = (n_1, \dots, n_a)$ with $\sum n_i = k$, set

$$\lambda_{\mathbf{n}} = \sum_i n_i \rho_i, \quad L_{\mathbf{n}} = 1 + \sum_i n_i (m_i - 1). \tag{2.1}$$

Let Λ_k be the set of distinct values of $\lambda_{\mathbf{n}}$, and put

$$\ell_\lambda = \max_{\mathbf{n}: \lambda_{\mathbf{n}} = \lambda} L_{\mathbf{n}}, \quad \chi_{h,k}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_\lambda}. \tag{2.2}$$

Equal sums are grouped by their exact equality; no assumption of linear independence or of absent collisions is used. The distinct occupation idempotents stay in B_k^{sym} even when their sums agree.

Theorem 2.1. The polynomial (2.2) is the minimal polynomial of A_k both on B_k and on B_k^{sym} . There are an injective unital algebra map and a separate injective module map

$$\begin{aligned} \alpha_{h,k} : C_{h,k} := \mathbb{C}[S]/(\chi_{h,k}) &\hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto P(A_k)1, \\ \eta_{h,k} = M_{U_k} \alpha_{h,k} : C_{h,k} &\hookrightarrow B_k^{\text{sym}}, \quad [P] \mapsto U_k P(A_k)1. \end{aligned} \quad (2.3)$$

The map M_{U_k} is an invertible B_k^{sym} -module map on its carrier. It is not asserted to be a unital algebra homomorphism; its inverse is $M_{U_k^{-1}}$. The full jet unit is retained by this pair of maps rather than assigned the value one.

Proof. At an ordered root tuple the local factor is

$$L_\rho = \mathbb{C}[z_1, \dots, z_k]/(z_1^{m_{i_1}}, \dots, z_k^{m_{i_k}}), \quad A_k = \lambda I + N, \quad N = M_{z_1 + \dots + z_k}.$$

Every monomial of degree greater than $\sum_j (m_{i_j} - 1)$ vanishes. In precisely that degree, $N^D 1$ contains the nonzero monomial $\prod z_j^{m_{i_j} - 1}$ with coefficient $D! / \prod (m_{i_j} - 1)!$. Thus its nilpotence index is $D + 1$. The minimal polynomial on the product takes the maximum index over factors at the same eigenvalue, which is (2.2). For the invariant algebra, the vector 1 is invariant and its annihilator under $\mathbb{C}[S]$ is already the same polynomial; indeed $P(A_k)1 = 0$ is equivalent to the algebra element $P(S)$ being zero. Thus restriction to invariants does not reduce this minimal polynomial. Multiplication by the actual unit U_k has zero kernel and commutes with A_k . This proves all assertions.

In particular, for every $\rho \in Z$, the value $k\rho$ occurs in Λ_k . The cyclic module has a nonzero eigenvector at every $\lambda \in \Lambda_k$: take the local class $(S - \lambda)^{\ell_\lambda - 1}$ in its Chinese-remainder factor, and zero in its other factors. Its image under $\eta_{h,k}$ is still nonzero and has eigenvalue λ .

This retains every exponent required for tensor amplification. It does not claim that a chosen repeated pure tensor is literally the same vector as this cyclic eigenvector when different occupations have the same sum. The original repeated-local-factor projection is the explicit map

$$C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \longrightarrow L_{(\rho, \dots, \rho)}; \quad (2.4)$$

its complete nilpotent image is generated by the local unit times $1, N, \dots, N^{k(m_\rho - 1)}$. A cyclic eigenvector supported at a collided sum can have zero projection to a shorter local block. The spectral value is nevertheless present by the proved injection (2.3).

M.1.3 The remainder of the arithmetic module remains as a quotient

Let

$$T_{h,k} = B_k^{\text{sym}} / \eta_{h,k} C_{h,k}.$$

Then

$$0 \rightarrow C_{h,k} \xrightarrow{\eta_{h,k}} B_k^{\text{sym}} \rightarrow T_{h,k} \rightarrow 0 \quad (2.5)$$

is an exact sequence of $\mathbb{C}[S]$ -modules. This is a module quotient, not a new algebra quotient of B_k . Its split lift is the original $G(\mathbb{C})$ -linear internal quotient. An element of the retained submodule maps to the supported zero; τ remains τ .

The sequence also has a constructed $\mathbb{C}[S]$ -linear retraction. On the generalized λ -part let $N_\lambda = A_k - \lambda I$, $w_\lambda = e_\lambda(A_k)U_k 1$, and $\ell = \ell_\lambda$. Choose the fixed-order root tuple attaining ℓ and let ϑ_λ extract

one of the top monomial coefficients of $N_\lambda^{\ell-1}w_\lambda$ there. That coefficient is its nonzero local arithmetic unit times the corresponding multinomial coefficient. In particular

$$\vartheta_\lambda(N_\lambda^{\ell-1}w_\lambda) \neq 0.$$

Define

$$\pi_\lambda^0(v) = \sum_{j=0}^{\ell-1} \vartheta_\lambda(N_\lambda^{\ell-1-j}v)X^j \pmod{X^\ell}, \quad p_\lambda(X) = \pi_\lambda^0(w_\lambda).$$

Its constant term is nonzero, so p_λ has a unique multiplicative inverse modulo X^ℓ . The actual retraction is

$$\boxed{\pi_\lambda(v) = p_\lambda(X)^{-1}\pi_\lambda^0(v)}. \quad (2.5a)$$

Indeed $\pi_\lambda^0(N_\lambda v) = X\pi_\lambda^0(v)$, and $\pi_\lambda(w_\lambda) = 1$. Combining these maps with the CRT idempotents gives $\pi : B_k^{\text{sym}} \rightarrow C_{h,k}$ with $\pi\eta_{h,k} = 1$. Thus the inverse pair is

$$\begin{aligned} C_{h,k} \oplus \ker \pi &\longrightarrow B_k^{\text{sym}}, & (c, t) &\mapsto \eta_{h,k}c + t, \\ v &\longmapsto (\pi v, v - \eta_{h,k}\pi v). \end{aligned} \quad (2.5b)$$

The quotient $T_{h,k}$ is linked to $\ker \pi$ by $[v] \mapsto v - \eta\pi v$, with inverse given by inclusion followed by the quotient. This retains every remaining nilpotent block; at each sum precisely one maximal-length block is assigned to the cyclic summand. The choice of top coefficient is recorded. This splitting is not asserted orthogonal: an arithmetic Gram H pulls back to the full block matrix containing $\eta^*H\eta$, $\eta^*H|_{\ker \pi}$, and its adjoint cross term, not just their diagonal blocks.

On a fixed sum λ , the dimension of the symmetric generalized eigenspace is

$$a_\lambda = \sum_{\mathbf{n}:\lambda_{\mathbf{n}}=\lambda} \prod_i \binom{n_i + m_i - 1}{m_i - 1}. \quad (2.6)$$

Thus $T_{h,k}$ has dimension $a_\lambda - \ell_\lambda$ there. For every entire f , trace additivity in (2.5) proves

$$\text{Tr}(f(A_k)|B_k^{\text{sym}}) = \sum_\lambda \ell_\lambda f(\lambda) + \sum_\lambda (a_\lambda - \ell_\lambda) f(\lambda). \quad (2.7)$$

The first term is the cyclic trace; the second is the retained quotient trace. All terms get the same cohomological factor $(-1)^k$ in top-degree supertrace. This calculation is valid with nilpotents; it does not replace their modules by their residue values.

For example, the algebraic fixture with simple roots $1/4, 1/2, 3/4$ and $k = 2$ has a six-dimensional symmetric module but a five-dimensional cyclic module. The two occupations with sum 1 give a one-dimensional quotient at that sum. The fixture with one root of order three and $k = 2$ has symmetric dimension six, cyclic length five, and quotient length one at 2ρ . These are fixtures of the formulas, not assertions of zeta zeros at those locations or multiplicities.

M.1.4 3. An original source realization whose norm is the scalar sum density

Define the linear map

$$\boxed{\mathcal{V}_{h,k} : \mathbb{C}[S] \rightarrow \mathcal{B}^{\otimes k}, \quad P \mapsto P(D_1 + \cdots + D_k)(F_h^{\otimes k})}. \quad (3.1)$$

Its Mellin transform is

$$\mathcal{M}^{(k)}\mathcal{V}_{h,k}(P) = \left(\prod_i v_h(s_i) \right) P(s_1 + \cdots + s_k).$$

It is injective: if this analytic function vanishes, it vanishes on an open set where all v_h are nonzero, forcing the polynomial P to vanish identically. Its actual arithmetic map is

$$\boxed{J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,} \quad (3.2)$$

where $\pi_\chi(P) = [P]_\chi$. Thus its arithmetic kernel is exactly (χ) and not a selected subset of the original relations.

Let $q_k = \deg \chi_{h,k}$ and $N \geq q_k - 1$. At finite source degree,

$$\boxed{0 \rightarrow \mathbb{C}[S]_{\leq N-q_k} \xrightarrow{\times \chi} \mathbb{C}[S]_{\leq N} \xrightarrow{\pi_\chi} C_{h,k} \rightarrow 0.} \quad (3.3)$$

A negative bound means the zero space. Define the actual boundary space

$$\mathcal{D}_{h,k,N} = \mathcal{V}_{h,k}(\chi \mathbb{C}[S]_{\leq N-q_k}).$$

Because $\chi(s_1 + \cdots + s_k) \in I_k$, fixed-order monic division gives

$$\chi(S)Q(S) = \sum_i h(s_i)Q_i(\mathbf{s}). \quad (3.4)$$

Each term is an original top cochain boundary: replace F_h in factor i by ϕ_* , apply the polynomial $Q_i(D_1, \dots, D_k)$, and give the primitive $(-1)^{i-1}$. The tensor differential gives the second $(-1)^{i-1}$. Their product is $+1$. This supplies the original primitive rather than defining a new vanishing relation by fiat.

The split quotient of (3.3) has equality criterion $P \sim P'$ precisely when $P - P' \in \chi \mathbb{C}[S]_{\leq N-q_k}$. A relation maps to the supported zero at label λ_N . Degree transport retains the preceding label and its killed-representative kernel.

M.1.5 The exact norm

Retain from [S1]

$$a_h(t) = \frac{v_h(1/2 + it)}{\sqrt{2\pi}}, \quad w_h(t) = |a_h(t)|^2, \quad \mu_h = \int w_h, \quad m_{h,k} = w_h^{*k}.$$

In centered sum coordinates $u = \sum t_i$, $y_i = t_i - u/k$, the amplitude is $\Psi_{h,k}(u, \mathbf{y}) = \prod_i a_h(u/k + y_i)$. Its relative integral is $m_{h,k}(u)$. Therefore

$$\boxed{\|\mathcal{V}_{h,k}P\|_{L^2(d^k x)}^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du.} \quad (3.5)$$

This is the restriction of the preceding full matrix-weight realization to the **specific source family** $P(S)$, through (3.1) and (3.2). It does not identify every relative coefficient with the constant relative coefficient.

The full relative module remains in (2.5). For comparison with the previous least-norm representative, at the same ambient degree N the maps obey

$$\mathcal{V}_{h,k}(\mathbb{C}[S]_{\leq N}) \hookrightarrow \mathcal{T}_h^{(k)}(\mathcal{P}_{k,N}), \quad \mathcal{D}_{h,k,N} \hookrightarrow \mathcal{B}_{k,N}.$$

Let $R_{k,N}^{\text{full}}$ be the old joint minimum, and $\mathcal{R}_N$ the cyclic minimum constructed next. Their exact relationship is

$$\Delta_N = \mathcal{R}_N - R_{k,N}^{\text{full}} \eta_{h,k} \in \text{Hom}(C_{h,k}, \mathcal{B}_{k,N}),$$

$$\boxed{\mathcal{G}_N = \eta_{h,k}^* G_{k,N}^{\text{full}} \eta_{h,k} + \Delta_N^* \Delta_N.} \quad (3.6)$$

Both maps represent the same full jets; their difference is in the original relation ideal, and the full minimum is orthogonal to it. This proves (3.6). The corresponding control difference is

$$\mathcal{W}_N - \eta_{h,k}^* W_{k,N}^{\text{full}} \eta_{h,k} = A_\chi^* \Delta_N^* \Delta_N + \Delta_N^* \Delta_N A_\chi - k \Delta_N^* \Delta_N. \quad (3.7)$$

This explicitly retains why the Gram inequality does not itself order the control forms.

The scalar source must reach degree $q_k - 1$ to realize all cyclic jets. This can exceed $k(d - 1)$. The reduction in relative rank has a degree cost which remains in every estimate.

M.1.6 4. Canonical scalar representatives and rank-two arithmetic control

Apply Gram-Schmidt by subtracting lower-degree projections from $1, S, S^2, \dots$ for the exact measure in (3.5). Let $p_j^{[h,k]}$ be the resulting polynomial with leading coefficient one, and retain

$$\omega_j^{[h,k]} = \int |p_j^{[h,k]}(k/2 + iu)|^2 m_{h,k}(u) du > 0. \quad (4.1)$$

No polynomial is divided by its norm. In particular $\omega_0^{[h,k]} = \mu_h^k$. For this section suppress the indices h, k and write

$$b_j = [p_j]_\chi \in C_{h,k}, \quad \mathcal{K}_N = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad \mathcal{G}_N = \mathcal{K}_N^{-1}. \quad (4.2)$$

The coordinates on $C_{h,k}$ are the fixed remainder basis $1, S, \dots, S^{q_k-1}$. Its inclusion into the original full jets is $\eta_{h,k}$, carrying U_k ; b_j is not an assertion that $U_k = 1$.

The unique least-norm representative modulo the specified $\mathcal{D}_{h,k,N}$ is

$$\boxed{\mathcal{R}_N u = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* \mathcal{G}_N u}{\omega_j} \right).} \quad (4.3)$$

Its identities are

$$J^{(k)} \mathcal{R}_N = \eta_{h,k}, \quad q^{(k)} \mathcal{R}_N = \sigma_h^{\otimes k} \eta_{h,k}, \quad \mathcal{R}_N^* \mathcal{R}_N = \mathcal{G}_N. \quad (4.4)$$

Proof. The coefficient map in (4.3) is the weighted adjoint of the full remainder map followed by $(JJ^*)^{-1}$ in the displayed orthogonal coordinates. Multiplication verifies its right-inverse identity. If Z is another polynomial with the same remainder, their difference lies in the exact kernel (3.3); the cross term with (4.3) is zero by the adjoint formula. The norm is therefore the displayed minimum plus the squared norm of that difference. This proves uniqueness and all equations, before any completion of the arithmetic quotient.

Let $A_\chi = M_S$ on $C_{h,k}$. Multiplication by S has adjoint multiplication by $k - S$ on the integration line. Orthogonality and leading coefficients give

$$Sp_j = p_{j+1} + \beta_j p_j - \frac{\omega_j}{\omega_{j-1}} p_{j-1}, \quad \Re \beta_j = k/2, \quad (4.5)$$

with the last term absent at $j = 0$. This is the ordinary orthogonal-polynomial recurrence in the retained complex coordinate; its general framework is recorded in DLMF §18.2 [E1].

Theorem 4.1 (two-row control on a spectrum-complete submodule).

$$\mathcal{Z}_N := A_\chi \mathcal{K}_N + \mathcal{K}_N A_\chi^* - k \mathcal{K}_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N}. \quad (4.6)$$

Consequently

$$\mathcal{W}_N := A_\chi^* \mathcal{G}_N + \mathcal{G}_N A_\chi - k \mathcal{G}_N = \mathcal{G}_N \mathcal{Z}_N \mathcal{G}_N, \quad \text{rank } \mathcal{W}_N \leq 2. \quad (4.7)$$

Proof. Substitute (4.5) into the finite sum (4.2). Each adjacent pair internal to $0, \dots, N$ cancels because the negative lower coefficient is exactly $-\omega_j/\omega_{j-1}$. Each diagonal pair is $k b_j b_j^*/\omega_j$. The only remaining adjacent pair is the one in (4.6). Congruence gives (4.7). This is a Christoffel–Darboux cancellation on the actual cyclic source, not a statement that the full relative module’s control has rank two.

M.1.7 The original next relation is retained

Define

$$\mathfrak{b}_N = \mathcal{V}_{h,k}(p_{N+1}) - \mathcal{R}_N b_{N+1}.$$

Its full jets vanish and its polynomial degree is $N + 1$. It gives the explicit isomorphism

$$\mathbb{C} \xrightarrow{c \mapsto c \mathfrak{b}_N} \mathcal{D}_{h,k,N+1} \cap \mathcal{D}_{h,k,N}^\perp \xrightarrow{z \mapsto [z]} \mathcal{D}_{h,k,N+1} / \mathcal{D}_{h,k,N}. \quad (4.8)$$

The leading term proves injectivity; the one-dimensional degree increment proves surjectivity. Its norm and pairing are

$$\|\mathfrak{b}_N\|^2 = \omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}, \quad \mathfrak{b}_N^* \mathcal{R}_N = -b_{N+1}^* \mathcal{G}_N. \quad (4.9)$$

The class is nonzero at degree N and becomes the receiving supported zero at degree $N + 1$. Its theta primitive is obtained from the original division (3.4). The scalar support itself is not adjoined again.

The exact update is

$$\begin{aligned} \mathcal{K}_{N+1} &= \mathcal{K}_N + b_{N+1} b_{N+1}^* / \omega_{N+1}, \\ \mathcal{G}_{N+1} &= \mathcal{G}_N - \frac{\mathcal{G}_N b_{N+1} b_{N+1}^* \mathcal{G}_N}{\omega_{N+1} + b_{N+1}^* \mathcal{G}_N b_{N+1}}. \end{aligned} \quad (4.10)$$

M.1.8 Same-metric reflection and exact excess

Let $P^{\dagger k}(S) = \overline{P(k - \bar{S})}$. The polynomial χ obeys $\chi^{\dagger k} = (-1)^{q_k} \chi$. The induced antilinear involution on the **weighted** cyclic module is

$$j_C[P] = (-1)^{kd} [P^{\dagger k}]. \quad (4.11)$$

This sign is forced by $U_k^\dagger = (-1)^{kd}U_k$ in the original embedding. It intertwines with the actual function reflection and preserves the source degree and norm. Hence the unique minimum obeys reflection on the same $\mathcal{G}_N$.

Uniqueness of the monic orthogonal polynomial gives $p_j^{\dagger k} = (-1)^j p_j$. In particular

$$c_N = b_N^* \mathcal{G}_N b_{N+1} \quad \text{satisfies} \quad \bar{c}_N = -c_N.$$

Write $a_N = b_N^* \mathcal{G}_N b_N$ and $d_N = b_{N+1}^* \mathcal{G}_N b_{N+1}$. The nonzero generalized control eigenvalues are the eigenvalues of

$$\frac{1}{\omega_N} \begin{pmatrix} \bar{c}_N & d_N \\ a_N & c_N \end{pmatrix}.$$

Therefore the exact least two-sided excess is

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}. \quad (4.12)$$

The determinant is nonnegative by the Gram inequality. Zero and rank-one degeneracies are covered by the same formula; in the reflection-stable case a nonzero rank-one signed defect cannot occur. No phase is dropped.

For the actual eigenvector at $k\rho$ constructed in Section 2,

$$k|2\Re\rho - 1| \leq \epsilon_{h,k,N}^{\text{cyc}}. \quad (4.13)$$

This is a proved diagnostic of this explicit family. A sublinear bound on the calculated right side as k grows is still to be established from the arithmetic input. The rank-two identity is not such a bound by itself.

M.1.9 5. Keep packet changes through their actual source seeds

Let $h \mid H$ include complete multiplicities, and let $m = H/h$. Then

$$F_h = m(D)F_H, \quad \mathcal{V}_{h,k}P = \mathcal{T}_H^{(k)} \left(\prod_i m(s_i) P(S) \right). \quad (5.1)$$

The full CRT inclusion on jets is the $\mathbb{C}[S]$ -linear map which inserts zero jets at the added centres. Its old unit maps to the associated idempotent, so it is not described as a unital algebra inclusion into the entire larger packet.

The right side of (5.1) has the fixed seed $a_{hH}(\mathbf{s}) = \prod_i m(s_i)$. Its source norm is exactly the old norm, since $(g/H)m = g/h$ in every factor. It therefore defines an isometric map of the marked polynomial sources and their internal quotients, with the old cyclic annihilator $\chi_{h,k}$.

This is the correct packet-coherence comparison. The new seed is generally not a polynomial in S alone. For example, if $m(s) = s - 1/2$ and $k = 2$, then

$$m(s_1)m(s_2) = \frac{(S-1)^2 - (s_1 - s_2)^2}{4}. \quad (5.2)$$

The relative square and its coefficient are retained. The explicit source-seed map (5.1), followed by the larger packet's full relative source inclusion, replaces any unjustified identification with that larger packet's unmarked constant-seed cyclic submodule. No original packet, supported relation, or canonical scalar norm is changed by this marked comparison.

M.1.10 6. The scalar Fisher estimate now acts on this source, and all polynomial derivative moments are calculable

From [S1],

$$\mathcal{I}_{h,k} + 4 \int \|n_{h,k}(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h, \quad \mathcal{I}_h = 4 \int |a'_h(t)|^2 dt. \quad (6.1)$$

The cyclic source (3.5) uses **that very density** $m_{h,k}$, not another scalar replacement for the full matrix norm. The full derivative still has the specified two components:

$$\partial_u(\Psi_{h,k}P(k/2 + iu)) = \Psi_{h,k} \left(iP'(k/2 + iu) + \frac{m'_{h,k}}{2m_{h,k}} P(k/2 + iu) \right) + n_{h,k}P(k/2 + iu). \quad (6.2)$$

They are pointwise orthogonal. The second component is retained even though the original coefficient polynomial depends only on the sum. Its arithmetic interpretation is the graph map and logarithmic observable from [S1]. The generator remains A_k ; the derivative corresponds to $i\mathcal{L}_k$, with $[D^{(k)}, \mathcal{L}_k] = -1$.

M.1.11 Raw moments give the complete differentiated finite source without multidimensional quadrature

Define the finite moments

$$\mu_j = \int t^j |a_h(t)|^2 dt, \quad \nu_j = \int t^j |a'_h(t)|^2 dt, \quad \theta_j = \int t^j \overline{a_h(t)} a'_h(t) dt.$$

Fixed arithmetic phase and integration by parts give

$$\theta_0 = 0, \quad \theta_j = -\frac{j}{2} \mu_{j-1} \quad (j \geq 1). \quad (6.3)$$

Use the **formal** exponential generating series

$$\mathcal{M}(z) = \sum_{j \geq 0} \mu_j z^j / j!, \quad \mathcal{N}(z) = \sum_{j \geq 0} \nu_j z^j / j!.$$

No convergence of these series is needed for the finite coefficient identities. Their constant terms remain the actual mass and derivative energy.

Let

$$M_{j,k} = \int u^j m_{h,k}(u) du, \quad B_{j,k} = \int u^j \|\partial_u \Psi_{h,k}(u, \cdot)\|^2 du.$$

Theorem 6.1 (all polynomial derivative moments).

$$M_{j,k} = j! [z^j] \mathcal{M}(z)^k, \quad (6.4)$$

$$B_{j,k} = j! [z^j] \left[\frac{1}{k} \mathcal{N}(z) \mathcal{M}(z)^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}(z)^k \right]. \quad (6.5)$$

Proof. Expand $u^j = (t_1 + \dots + t_k)^j$ and use Fubini; all integrals are absolutely finite. For (6.5), the diagonal terms of $|k^{-1} \sum_i a'_i \prod_{r \neq i} a_r|^2$ give $k^{-1} \mathcal{N} \mathcal{M}^{k-1}$. The distinct-index terms give

$(k-1)\Theta(z)^2\mathcal{M}^{k-2}/k$, where $\Theta(z) = \sum \theta_j z^j/j! = -z\mathcal{M}(z)/2$ by (6.3). Substitution gives the displayed expression. The second term is not discarded: the cross terms vanish only for the unweighted zeroth moment, not for arbitrary polynomial moments.

In particular,

$$\begin{aligned} M_{0,k} &= \mu_0^k, & M_{1,k} &= k\mu_1\mu_0^{k-1}, \\ M_{2,k} &= k\mu_2\mu_0^{k-1} + k(k-1)\mu_1^2\mu_0^{k-2}, \\ B_{0,k} &= \frac{1}{k}\nu_0\mu_0^{k-1} = \frac{\mu_h^{k-1}}{4k}\mathcal{I}_h. \end{aligned} \quad (6.6)$$

The first orthogonal norm and diagonal coefficient on the sum line are accordingly

$$\omega_0 = \mu_0^k, \quad \beta_0 = \frac{k}{2} + i\frac{k\mu_1}{\mu_0}, \quad \omega_1 = k\mu_0^{k-2}(\mu_0\mu_2 - \mu_1^2). \quad (6.7)$$

No symmetry assumption setting $\mu_1 = 0$ is made.

For any two polynomials P, Q , let $c_P(u) = P(k/2 + iu)$ and $c_Q(u) = Q(k/2 + iu)$. Their complete graph-energy form is

$$\begin{aligned} \mathfrak{D}_k(P, Q) &= \int \overline{c_P}c_Q \|\Psi'_k\|^2 du + \int \overline{c'_P}c'_Q m_k du \\ &\quad + \frac{1}{2} \int \left(\overline{c_P}c'_Q + \overline{c'_P}c_Q \right) m'_k du. \end{aligned} \quad (6.8)$$

All four terms remain. Equations (6.4)–(6.5) and $\frac{1}{2} \int u^j m'_k = -jM_{j-1,k}/2$ evaluate every matrix entry on a fixed polynomial degree using only μ_j, ν_j . This is the complete derivative graph norm from the existing sum-fibre construction, not merely the scalar score norm. It is positive definite on every nonzero polynomial source: a zero derivative would make $\Psi_k P$ constant in u for almost every relative coordinate, while its decay forces that constant and then P to be zero.

For a finite canonical representative, its graph-energy matrix is obtained by applying its actual coefficient matrix from (4.3) on both sides of (6.8). Thus there are now explicit moment matrices for both the original metric and the differentiated representative. No continuous jet functional on arbitrary L^2 classes is assumed.

M.1.12 7. Exact sum-coordinate conormal depth at every original thickening

The derivative graph still needs the class **before** an original relation is sent to e . The required domains can now be computed exactly on the sum line.

For $r \geq 1$, retain the original thickened ring

$$B_{k,r} = \mathcal{P}_k/I_k^r.$$

Define, for an ordered root tuple $\rho = (\rho_{i_1}, \dots, \rho_{i_k})$,

$$\begin{aligned} L_{\rho,r} &= 1 + \sum_{j=1}^k (m_{i_j} - 1) + (r-1) \max_j m_{i_j}, \\ \ell_{\lambda,r} &= \max_{\rho: \sum \rho_{i_j} = \lambda} L_{\rho,r}, \end{aligned} \quad \boxed{\chi_{h,k,r}(S) = \prod_{\lambda \in \Lambda_k} (S - \lambda)^{\ell_{\lambda,r}}.} \quad (7.1)$$

Theorem 7.1. The exact contracted ideal is

$$\boxed{I_k^r \cap \mathbb{C}[S] = (\chi_{h,k,r}).} \quad (7.2)$$

Consequently there is an injective unital algebra map

$$\boxed{\alpha_r : C_{h,k,r} := \mathbb{C}[S]/(\chi_{h,k,r}) \hookrightarrow B_{k,r}^{S_k}, \quad [P] \mapsto [P(S)].} \quad (7.3)$$

Proof. At a root tuple, the units in $h(\rho_i + z_i) = u_i(z_i)z_i^{m_i}$ make the original local ideal equal to $(z_1^{m_1}, \dots, z_k^{m_k})$, with its r th power retained. A monomial z^α survives precisely when

$$\sum_i \left\lfloor \frac{\alpha_i}{m_i} \right\rfloor \leq r - 1. \quad (7.4)$$

The largest surviving total degree is $\sum_i (m_i - 1) + (r - 1) \max_i m_i$. It is attained by assigning all $r - 1$ full blocks to an index of maximal m_i , and taking remainders $m_i - 1$ in every index. The corresponding term of $(\sum z_i)^D$ has coefficient $D! / \prod \alpha_i! \neq 0$. All terms of degree $D + 1$ vanish. Thus the exact nilpotence index is (7.1). Take the maximum over collided sums and then the product over different sums. This gives precisely the annihilator of the algebra element S , proving (7.2). The same cyclic image is in the invariants because S and 1 are invariant. No local unit has been discarded: it enters the specified invertible change of generators of the ideal, not an alteration of the physical coordinate S .

M.1.13 Quotient and derivative maps on the same tower

There are commuting quotient squares

$$\begin{array}{ccc} C_{h,k,r+1} & \xrightarrow{\alpha_{r+1}} & B_{k,r+1} \\ \downarrow & & \downarrow \\ C_{h,k,r} & \xrightarrow{\alpha_r} & B_{k,r}. \end{array} \quad (7.5)$$

Each quotient is a ring map, hence has the original split lift. Its kernel is retained. On the sum side,

$$0 \rightarrow \mathbb{C}[S]/(\chi_{r+1}/\chi_r) \xrightarrow{\times \chi_r} C_{r+1} \rightarrow C_r \rightarrow 0. \quad (7.6)$$

The first arrow is explicitly a module map. Under α_{r+1} its image lies in I_k^r/I_k^{r+1} ; this is the actual relation fibre sent to the supported zero by the right vertical quotient.

The exact exponent satisfies $\ell_{\lambda,r+1} \geq \ell_{\lambda,r} + 1$. Hence χ_r divides χ'_{r+1} , and the linear derivative map is well-defined:

$$\boxed{\partial_r : C_{r+1} \rightarrow C_r, \quad [P] \mapsto [P']}. \quad (7.7)$$

It commutes with the original relative-sum derivative

$$\delta_r : B_{k,r+1} \rightarrow B_{k,r}, \quad [P] \mapsto \left[\frac{1}{k} \sum_i \partial_{s_i} P \right],$$

$$\boxed{\delta_r \alpha_{r+1} = \alpha_r \partial_r.} \quad (7.8)$$

The derivatives are coefficient-linear maps and are lifted as such, not called ring homomorphisms. For example,

$$h(s) = (s - \rho)^2, \quad k = 2, \quad X = S - 2\rho \implies \chi_r = X^{2r+1}. \quad (7.9)$$

The class X^3 is nonzero modulo X^5 , maps to the supported zero modulo X^3 , and its derivative maps to $3X^2$, which is nonzero there.

This also calculates the relationship to a naive scalar power thickening:

$$\mathbb{C}[S]/(\chi_1^r) \twoheadrightarrow C_r \hookrightarrow B_{k,r}. \quad (7.10)$$

For the fixture in (7.9) at $r = 2$, the first map is $\mathbb{C}[X]/(X^6) \twoheadrightarrow \mathbb{C}[X]/(X^5)$, with kernel the retained class of X^5 . The original joint thickening therefore carries a different sum-coordinate depth, and the precise quotient above gives their relationship. This is a calculation about powers of the original relation ideal; it is not iteration of the scalar zero adjunction.

M.1.14 The arithmetic Taylor unit remains through differentiation

Let $U_{k,r}$ be the full Taylor class of $\prod v_h(s_i)$ in $B_{k,r}$. It is invertible and compatible under the quotient tower. Define the coefficient-linear inclusion

$$\eta_r = M_{U_{k,r}} \alpha_r.$$

The full equation is

$$\boxed{\delta_r \eta_{r+1}(P) = U_{k,r} \alpha_r(P') + (\delta_r U_{k,r+1}) \alpha_r(P).} \quad (7.11)$$

The second term remains in the relative ring $B_{k,r}$. It is not presumed to be in the cyclic image. Its exact map is $M_{\delta_r U_{k,r+1}} \alpha_r$, and the complementary quotient $B_{k,r}/\eta_r C_r$ records its noncyclic part. This is the algebraic counterpart of the retained normal derivative in (6.2).

At $r = 1$, the quotient and derivative thus give two actual maps out of the same lifted thickening. A conormal relation can map to e under the quotient and to a nonzero supported value under the derivative. Differentiating the original theta relation, not the scalar e itself, is what produces that value.

M.1.15 8. The conormal derivative reaches both cyclic and full arithmetic traces, with their difference retained

Choose an original division

$$\chi_1(S) = \sum_i h(s_i) B_i(\mathbf{s}).$$

Applying the actual derivative and then the original quotient yields

$$\boxed{\alpha_1(\chi_1') = \frac{1}{k} \sum_i h'(s_i) [B_i]_{I_k}.} \quad (8.1)$$

The right side is the source's conormal row. For the weighted relation $U \chi_1 P$, the term containing δU has an undifferentiated χ_1 and vanishes in B_k . Therefore

$$\boxed{\delta_1[U \chi_1 P]_{I_k^2} = U_k \alpha_1(\chi_1' P).} \quad (8.2)$$

It has the actual analytic representative

$$J^{(k)} \mathcal{L}_k \mathcal{V}_{h,k}(\chi_1 P) = U_k \alpha_1(\chi'_1 P), \quad \mathcal{L}_k = \frac{1}{k} \sum_i \log x_i. \quad (8.3)$$

Here the left source is an original theta boundary, supplied by (3.4). Its first differentiated arithmetic value is read before it is collapsed to its supported zero.

For $C_1 = \mathbb{C}[S]/(\chi_1)$, the explicitly chosen finite residue pairing is

$$R_\chi(f, P) = \sum_\lambda \text{Res}_{S=\lambda} \frac{f^{\dagger k}(S) P(S)}{\chi_1(S)} dS.$$

Its Jacobian contraction is

$$R_\chi(f, \chi'_1 P) = \sum_\lambda \ell_\lambda \overline{f(k - \bar{\lambda})} P(\lambda) = \text{Tr}(M_{f^{\dagger k} P} | C_1). \quad (8.4)$$

This is the cyclic trace. The original full symmetric trace has a_λ instead of ℓ_λ and is their sum with the quotient trace (2.7). Equivalently, let $\chi_{\text{full}}(S) = \prod_\lambda (S - \lambda)^{a_\lambda}$. Its difference is the explicit residue functional

$$\text{Tr}(f(A_k) | T_{h,k}) = \sum_\lambda \text{Res}_{S=\lambda} f(S) \left(\frac{\chi'_{\text{full}}}{\chi_{\text{full}}} - \frac{\chi'_1}{\chi_1} \right) dS. \quad (8.5)$$

There is a further explicit comparison for the original arithmetic vector pairing. Its pullback through the **weighted** inclusion is

$$\text{Tr}_{B_k^{\text{sym}}} M_{(\eta f)^{\dagger} \eta P} = \text{Tr}_{B_k^{\text{sym}}} M_{U_k^{\dagger} U_k \alpha_1(f^{\dagger k} P)}. \quad (8.6)$$

The factor $U_k^{\dagger} U_k$ remains. It need not lie in the sum-generated subalgebra. The trace decomposition (2.7) applies to the invariant operator $f(A_k)$; it is not an assertion that this weighted vector pairing is automatically the cyclic residue form (8.4).

Neither polynomial is substituted for the original $g = 2\xi$. Their finite trace comparison is (8.5), while $U_{k,r}$ and the map (8.3) retain their arithmetic source. The original product residue/Jacobian trace can still be evaluated on the full module and its retained quotient. No trace contribution is declared absent.

M.1.16 9. Checks, counterfactual tests, and the next estimate

The exact checker implements finite polynomial, matrix, and Gaussian-amplitude fixtures. Its tests cover exact thickening dimensions and nilpotence; collision grouping; cyclic injections and trace complements; the non-naive thickening map; the derivative square; supported-zero versus absence; the original conormal row; full unit differentiation; scalar minimum and updates; rank-two control; reflection with a nonzero imaginary cross term; amplified off-line test exponents; raw convolution and complete polynomial derivative moments; and the marked source-seed transition. These are regression checks, not tests asserting that a fixture's roots are zeta zeros.

In particular an off-line reflected polynomial fixture remains off-line under the new construction: its exact eigenline identity is still $2\Re\lambda - k$. The scalar Fisher contraction does not annul this value. This guards against an inference from convolution smoothing alone to purity. The constructive object supplied instead is (4.12): two explicitly computed full-jet rows relative to the exact cyclic interpolation metric, with every scalar moment obtainable from the original one-factor moments by (6.4).

The degree cost $N \geq \deg \chi_{h,k} - 1$ and the reciprocal metric in (4.12) are retained as k grows. The derivative matrix (6.8) likewise keeps the polynomially weighted normal contribution. Controlling those quantities on the actual arithmetic inputs, rather than just the constant polynomial, is the remaining quantitative step of this branch.

The connection to the Deligne amplification programme is now a smaller, exactly included cohomological image with every amplified exponent represented; its source norm is the scalar sum-density norm for which (6.1) was proved. The original multi-direction module, its complementary trace, and all powers of its conormal ideal remain as explicit surrounding objects. This is not a transfer of finite-field purity by notation.

No remote repository is modified in this delivery. The user has assigned GitHub integration and Lean work to the parallel session. The authored note, checker, and a bounded handoff are ready for that session without changes to its existing files.

M.1.17 Source record and references

- **[S1]** Owner programme, `Tau_Sum_Connection_Control/NOTE.tex` and `RESEARCH_NOTE.md`, archive `Tau_Sum_Connection_Control_2026-09-12.zip`. Selected analytic source, scalar information identity, full matrix connection, original first and higher conormal derivative. Exact hashes and source-manifest verification are in `checks/source_manifest.json` and `SOURCE_READING.json`.
- **[S2]** User upload `Untitled\1765(1).md`: retained progression from globalization to coherent interpolation, highest relation layer and spectral-sum descent. Read as a research transcript, not as an independent certificate for every embedded claim.
- **[S3]** User upload `Untitled\1766(1).md`: formalization progression, retained bounded scopes, and the kernel-layer integration. No new Lean run is claimed in this note.
- **[S4]** `SplitZero_Derived_Mathematics_2026-09-12(1).md` and retained source-derived interfaces: internal quotient, transported-class kernel and tensor base-change. Used within their stated hypotheses.
- **[E1]** NIST, *Digital Library of Mathematical Functions*, §18.2, particularly recurrence, Christoffel–Darboux, and moments. <https://dlmf.nist.gov/18.2>. General framework only; the arithmetic source maps and cyclic calculation are proved here.
- **[E2]** The Stacks Project, Tag 00S0, *The naive cotangent complex*. <https://stacks.math.columbia.edu/tag/00S0>. Background for the conormal map; (7.1)–(7.11) compute the exact present ideals and coordinate derivative.
- **[E3]** O. Johnson and A. Barron, *Fisher Information inequalities and the Central Limit Theorem*, arXiv:math/0111020. <https://arxiv.org/abs/math/0111020>. General convolution-projection mechanism; the mass-retaining arithmetic identity is the earlier source [S1].
- **[D]** P. Deligne, *La conjecture de Weil. II*, supplied TeX/source excerpts in the programme, especially §3.2.13 and §3.3.4–3.3.6. This turn uses the previously read amplification target; it does not claim to have reread or independently verified the whole paper.

N Exterior trace amplification with the original arithmetic Gram and full quartet multiplicities

Source attribution. Complete TeX witness supplied in the Exterior Trace Amplification archive. The original Gram, additive exterior action, factorial tensor inclusion, full generalized eigenspaces and quartet multiplicities are retained. The source supplies an amplification lower bound and explicitly leaves the actual subcubic arithmetic upper estimate unproved. Its finite fixtures and source-reported verification have their recorded scopes; the separate PR 21 formalization is identified independently.

Exact revision. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Source file. Tau_Exterior_Trace_Amplification/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete source witness SHA-256; cited repository main is a source baseline

Source SHA-256. c7050e1c2f76b31b72af1d237efd29f656fd34a92492e5a11f76d8bc55b2acd9

Original archive. Tau_Exterior_Trace_Amplification_2026-09-12.zip

Archive SHA-256. 168f977a9b54cf461f564e099fe55dc7b8feffbc09c563f541e0be2bb5746a94

N.1 Result and scope

The previous cyclic-sum construction has a Hermitian control form of rank at most two. The continuation here uses that specific property, rather than changing its metric. Passing to an exterior power of the same arithmetic module does **not** multiply the control allowance by the exterior degree. It makes it possible to control a determinant line containing the sum of the positive real-part defects.

For the actual cyclic packet and its specified least-norm representative, the new inequality is

$$\sum_{\Re \lambda > k/2} \ell_{\lambda}(2\Re \lambda - k) \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

For a complete nonreal off-line quartet with real displacement $\delta > 0$ and common multiplicity m , it gives the exact lower bound

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta [1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor, \quad N \geq [1 + k(m-1)](k+1)^2 - 1.$$

For simple roots this grows as $\delta k^3/2$. The corresponding sufficient analytic target is therefore a **subcubic** upper bound on this same explicitly defined allowance, rather than the earlier sublinear target obtained from one repeated eigenvector. This note proves the lower bound and constructs the exterior/cohomological maps. It does not prove that analytic upper bound, and it does not claim RH.

The finite-dimensional extremal-trace ingredient is classical: it is a rank-two case of the Ky Fan principle. The contribution here is its application through the existing arithmetic representatives, their original theta boundaries, their split-support lifts, and the exact sum-set and multiplicity calculation. General finite-dimensional linear algebra is not presented as a new classical theorem.

N.2 1. GitHub corrections integrated at a pinned revision

The repository read for this continuation is KokunoYumeto/zeta-function-research-reader, with pinned main revision

a4494fba4968db837a8af6fcb952c8688cd5d1da.

The inspection was bounded: the latest PR metadata, the main revision, the current sum-connection note, selected parts of the new Stieltjes/conormal continuation, and the formal frontier note. It was not an independent audit of the entire 292-page edition.

N.2.1 1.1 The empty packet

The current PR #20 source explicitly corrects the scope of its monic-freeness argument. For a nonempty packet $d = \deg h \geq 1$, the original conormal calculation applies. For $h = 1$ and $k \geq 2$,

$$I = (h(s_1), \dots, h(s_k)) = \mathbb{C}[s_1, \dots, s_k], \quad \mathbb{C}[\mathbf{s}]/I^r = 0 \quad (r \geq 1).$$

The algebraic spectral module is then zero. Its split lift $G(0)$ still contains the external absence τ and the supported zero e . The analytic amplitude g and its mass do not vanish. No Gram inverse on a nonzero spectral space is inferred from that analytic amplitude.

Every inverse matrix below is on a **nonempty** actual spectral packet. The empty case has the unique maps between zero coefficient modules and the corresponding original split lifts.

The reviewed public note blob is 797d8bc298aec06b112b53a03654a9664323e748. Its historical validation file records an earlier note blob; that historical receipt is not relabelled as a new proof check.

N.2.2 1.2 Metric restriction retains its hypotheses

The current formal frontier note requires, for rectangular inclusion and extraction matrices,

$$LI = 1, \quad P = IL = P^*, \quad PK = KP, \quad GK = 1$$

before concluding

$$(I^*GI)(LKL^*) = 1.$$

The spectral projection used later need not be orthogonal for the arithmetic metric. Accordingly this note uses the actual compressed Gram $C_+^*GC_+$ and its inverse. It does not replace that inverse by $C_+^*G^{-1}C_+$.

N.2.3 1.3 Formalization and source boundaries

The source library already constructs the original internal quotient, cohomology maps, comparison kernels, support-changing morphisms, and homology of a cochain-idempotent image. These are inputs. This continuation neither reruns Lean nor extends the scope of their published certificates to the analytic theta construction.

The new tandem note also constructs the conormal algebra map into dual numbers and keeps the variable-frame connection term. Those results remain compatible inputs; they are not recounted as new results here.

N.3 2. The original scalar and arithmetic source

Retain the original split scalar

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet.$$

Its original quotient and coefficient maps remain

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

Here j is the specified unital embedding. The target of $p_{\mathbb{Z}}$ is infinite. Every lifted coefficient map sends e to e and τ to τ before the arithmetic observation.

Keep the source spaces

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = 0, \int_{\mathbb{R}} \phi = 0\},$$

$$\mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \text{ for all } b \in \mathbb{Z}, j \geq 0 \right\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = H^1(C_+) = \mathcal{B}/\Theta V.$$

The complex is in degrees 0, 1. Retain

$$\phi_*(x) = (4\pi^2 x^4 - 6\pi x^2) e^{-\pi x^2}, \quad \mathcal{M}\Theta\phi_* = g,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

These definitions and the continuous/finite source comparisons are inherited from the supplied cyclic-sum note. Their analytic proofs are not newly re-established in this note.

For a nonempty finite reflection-stable actual zero packet with complete orders, set

$$h(s) = \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad E_h = \mathbb{C}[s]/(h), \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times.$$

For $k \geq 1$, the full tensor arithmetic algebra, generator and unit are

$$B_k = E_h^{\otimes k}, \quad S = s_1 + \cdots + s_k, \quad A_k = M_S, \quad U_k = v_h^{\otimes k}.$$

At a sum λ , its maximal local nilpotent length is

$$\ell_\lambda = \max_{\rho_1 + \cdots + \rho_k = \lambda} \left(1 + \sum_i (m_{\rho_i} - 1) \right).$$

Use the original cyclic algebra and its arithmetic inclusion

$$\chi(S) = \prod_{\lambda} (S - \lambda)^{\ell_\lambda}, \quad C = \mathbb{C}[S]/(\chi), \quad q_k = \deg \chi,$$

$$\alpha : C \hookrightarrow B_k^{S_k}, \quad [P] \mapsto P(A_k)1, \quad \eta = M_{U_k} \alpha : C \hookrightarrow B_k^{S_k}.$$

The first arrow is a unital algebra map. The map M_{U_k} and its inverse $M_{U_k}^{-1}$ are explicit module automorphisms. Thus η retains the arithmetic unit without declaring it to be a unital algebra inclusion.

The source already provides an equivariant retraction $\pi : B_k^{S_k} \rightarrow C$, $\pi\eta = 1$, and retains $\ker \pi$. Together with the full finite jet map and the original signed symmetrization this gives a left inverse to the cyclic inclusion in the top cohomology of $C_+^{\widehat{\otimes} k}$. No cyclic complement is discarded in this continuation.

N.4 3. The existing canonical metric and its rank-two operator

The actual source map is

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad \mathcal{M}F_h = g/h.$$

It obeys

$$J^{(k)}\mathcal{V}_{h,k} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta\pi_\chi.$$

The norm remains

$$\|\mathcal{V}_{h,k}P\|^2 = \int_{\mathbb{R}} |P(k/2 + iu)|^2 m_{h,k}(u) du,$$

where

$$a_h(t) = \frac{(g/h)(1/2 + it)}{\sqrt{2\pi}}, \quad w_h = |a_h|^2, \quad \mu_h = \int_{\mathbb{R}} w_h, \quad m_{h,k} = w_h^{*k}.$$

The total mass is μ_h^k . Construct the monic orthogonal polynomials $p_j(S)$ by subtracting lower-degree projections from S^j ; retain their squared norms ω_j , including $\omega_0 = \mu_h^k$.

For $N \geq q_k - 1$, in the fixed remainder coordinates on C , put

$$b_j = [p_j]_\chi, \quad K = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}, \quad G = K^{-1},$$

$$Ru = \mathcal{V}_{h,k} \left(\sum_{j=0}^N p_j \frac{b_j^* G u}{\omega_j} \right).$$

These are the actual canonical representative and metric:

$$R^*R = G, \quad J^{(k)}R = \eta, \quad q^{(k)}R = \sigma_h^{\otimes k}\eta.$$

For $A = M_S$ on C , the source's exact boundary control is

$$W = A^*G + GA - kG = G \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N} G.$$

Introduce the metric raising map $G : C \rightarrow C^\#$, $v \mapsto (w \mapsto w^*Gv)$, where $C^\#$ is the space of conjugate-linear functionals. Its inverse is K . Define the corresponding Gram adjoint and control endomorphism by the typed composites

$$A^{\sharp G} = G^{-1} A^* G : C \rightarrow C, \quad H = G^{-1} W = A^{\sharp G} + A - kI_C.$$

Then $H^{\sharp G} = H$ and $\text{rank } H \leq 2$. Reflection pairs λ with $k - \bar{\lambda}$, with equal full length. Hence

$$\text{Tr } H = 2\Re \text{Tr } A - kq_k = 0.$$

Thus the nonzero spectrum of H is $\{+\epsilon, -\epsilon\}$ for the actual nonnegative number

$$\epsilon = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}, \quad a_N = b_N^* G b_N, \quad d_N = b_{N+1}^* G b_{N+1}, \quad c_N = b_N^* G b_{N+1}.$$

The source reflection gives $\bar{c}_N = -c_N$. Equivalently $\epsilon^2 = \text{Tr}(H^2)/2$. The case $\epsilon = 0$ means $H = 0$ by its G -self-adjointness. No orthonormal replacement of the retained polynomial coordinates is made.

N.5 4. Exterior control: the allowance does not multiply by exterior degree

Let $1 \leq p \leq q_k$. On $\bigwedge^p C$, define the **additive exterior generator**

$$A^{[p]}(v_1 \wedge \cdots \wedge v_p) = \sum_{j=1}^p v_1 \wedge \cdots \wedge A v_j \wedge \cdots \wedge v_p.$$

Its relation to the multiplicative exterior functor is

$$A^{[p]} = \left. \frac{d}{dt} \right|_{t=0} \bigwedge^p (e^{tA}), \quad \bigwedge^p (e^{tA}) = e^{tA^{[p]}}.$$

In the unscaled increasing-index wedge basis, the Gram matrix is

$$G_{I,J}^{[p]} = \det G_{I,J}.$$

It is positive definite and is the determinant inner product induced from the original G .

Theorem 4.1. The exact exterior control identity is

$$(A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} = G^{[p]} H^{[p]}.$$

For $1 \leq p \leq q_k - 1$,

$$-\epsilon G^{[p]} \preceq (A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \preceq \epsilon G^{[p]}.$$

If $\epsilon > 0$, this is the exact optimal allowance. For $p = q_k$ the control form is zero.

Proof. The induced Gram adjoint satisfies

$$(A^{[p]})^{\sharp G^{[p]}} = (A^{\sharp G})^{[p]}.$$

This follows by applying the product rule to the determinant pairing of decomposable vectors. Since $(kI_C)^{[p]} = kpI$, add the two terms to obtain the identity.

For the order bound, keep the G -orthogonal eigenspace decomposition

$$C = L_+ \oplus Z_0 \oplus L_-, \quad H|_{L_+} = \epsilon, \quad H|_{Z_0} = 0, \quad H|_{L_-} = -\epsilon.$$

Both L_+ and L_- have dimension one, and $\dim Z_0 = q_k - 2$. The exterior product of each direct-summand inclusion gives the complete decomposition

$$\begin{aligned} \bigwedge^p C = \bigwedge^p Z_0 \oplus (L_+ \wedge \bigwedge^{p-1} Z_0) \\ \oplus (L_- \wedge \bigwedge^{p-1} Z_0) \oplus (L_+ \wedge L_- \wedge \bigwedge^{p-2} Z_0). \end{aligned}$$

The eigenvalues of $H^{[p]}$ on these four summands are respectively $0, \epsilon, -\epsilon, 0$. The multiplicity of each nonzero eigenvalue is $\binom{q_k-2}{p-1}$; summands with impossible degree are zero modules. This proves every assertion, including the top-degree case. When $\epsilon = 0$, the identity gives the conclusion directly. $\square$

This is the mechanism that makes a determinant-line estimate stronger than testing one eigenvector: a wedge cannot contain the same one-dimensional positive control direction twice.

N.6 5. The exterior maps live on the original cochain source

Define the unscaled alternating map

$$\text{Alt}_p : \bigwedge^p C \hookrightarrow C^{\otimes p}, \quad v_1 \wedge \cdots \wedge v_p \mapsto \sum_{\pi \in S_p} \text{sgn}(\pi) v_{\pi(1)} \otimes \cdots \otimes v_{\pi(p)}.$$

If $\text{wed} : C^{\otimes p} \rightarrow \bigwedge^p C$ is the exterior quotient, then

$$\text{wed} \text{Alt}_p = p! I.$$

The actual source representative is

$$R_p = R^{\otimes p} \text{Alt}_p : \bigwedge^p C \longrightarrow \mathcal{B}^{\widehat{\otimes} kp}.$$

It retains

$$R_p^* R_p = p! G^{[p]}, \quad J^{(kp)} R_p = \eta^{\otimes p} \text{Alt}_p,$$

$$q^{(kp)} R_p = (\sigma_h^{\otimes k} \eta)^{\otimes p} \text{Alt}_p.$$

The last map is injective: compose with the existing finite-jet/retraction map in each block, then with $\text{wed} / p!$. The factorial is part of this inverse composite; the representative itself is not divided by its norm.

The actual source control is therefore

$$W_p^{\text{source}} = p! \left((A^{[p]})^* G^{[p]} + G^{[p]} A^{[p]} - kp G^{[p]} \right).$$

Both its Gram and its control retain the same literal $p!$. Its relative allowance is the original ϵ .

N.6.1 5.1 The cochain idempotent and both permutation signs

Let $\mathcal{C}_k = C_+^{\widehat{\otimes} k}$. Let T_π be the actual Koszul permutation on $\mathcal{C}_k^{\widehat{\otimes} p}$. Its top-degree action is $(\text{sgn } \pi)^k P_\pi$, with P_π ordinary block permutation. The chain idempotent for the exterior top-degree component is

$$A_{p,k} = \frac{1}{p!} \sum_{\pi \in S_p} (\text{sgn } \pi)^{k+1} T_\pi.$$

The two top-degree signs multiply to $\text{sgn } \pi$. Its square equals itself by the group-algebra calculation, with the factor $1/p!$ retained. The complementary idempotent $1 - A_{p,k}$ remains, with its own cohomology and trace. The existing projected-homology theorem supplies inclusion and retraction on the original cycle/boundary quotient.

These sign rules are the standard symmetric-monoidal structure on complexes, instantiated here on the retained arithmetic degree k ; they are not an assumption that degree-one vectors commute without sign.

N.6.2 5.2 Propagate the actual theta primitive

The previous source construction supplies a map

$$K_N^{\text{prim}} : C \rightarrow \mathcal{C}_k^{k-1}, \quad D^{(k)} R - RA = dK_N^{\text{prim}}.$$

Define

$$K_{N,p}^{\text{prim}} = \sum_{j=1}^p (-1)^{k(j-1)} R^{\otimes(j-1)} \otimes K_N^{\text{prim}} \otimes R^{\otimes(p-j)} \text{Alt}_p.$$

The source and target are $\bigwedge^p C$ and $\mathcal{C}_k^{\widehat{\otimes} p, kp-1}$. Applying the tensor differential contributes the same $(-1)^{k(j-1)}$ on the displayed summand. Differentials on the top-degree R factors vanish. Consequently

$$D^{(kp)} R_p - R_p A^{[p]} = dK_{N,p}^{\text{prim}}.$$

This is the exact boundary producing the exterior control form on the original source.

N.6.3 5.3 Original support and the alternative full-product minimum

Every coefficient map above has its original $G(\mathbb{C})$ -linear reconstructed lift. At a chosen active support label, a term with zero amplitude remains at that label's zero. The sums of differently located primitives use the joins of their actual tensor support masks; the character average uses the corresponding orbit join. External scalar τ maps every input to external absence, while e maps it to its fibre zero. In particular the boundary in the preceding equation is sent by the next internal quotient to the receiving supported zero, not to τ .

There is also an exact comparison with the earlier full-product minimum. Set

$$I_{k,p} = \eta^{\otimes p} \text{Alt}_p, \quad M \geq \max\{pN, kp(\deg h - 1)\}.$$

Let $R_{kp,M}^{\text{full}}$ be the already-constructed canonical full-jet representative. Both it and R_p lie in the stated degree- M source after applying $I_{k,p}$. Hence

$$\Delta = R_p - R_{kp,M}^{\text{full}} I_{k,p}$$

has zero full jets, and its numerator belongs to the original ideal $(h(s_1), \dots, h(s_{kp}))$. Thus it is an original theta boundary with the source's polynomial-division primitive. Orthogonality of the full minimum gives

$$p!G^{[p]} = I_{k,p}^* G_{kp,M}^{\text{full}} I_{k,p} + \Delta^* \Delta.$$

The complete control comparison is

$$W_p^{\text{source}} = I_{k,p}^* W_{kp,M}^{\text{full}} I_{k,p} + (A^{[p]})^* \Delta^* \Delta + \Delta^* \Delta A^{[p]} - kp \Delta^* \Delta.$$

This connects the two representative choices without identifying their metrics or discarding their boundary difference.

N.7 6. A determinant line measures the sum of the positive defects

Let

$$C_{>} = \bigoplus_{\Re \lambda > k/2} C_{\lambda} \subseteq C$$

be the sum of the **full generalized eigenspaces**, of total dimension p . Here $C_{\lambda} \cong \mathbb{C}[S]/(S - \lambda)^{\ell_{\lambda}}$. Write its inclusion as $I_{>} : C_{>} \hookrightarrow C$. Its determinant line has the actual injection

$$\det(C_{>}) = \bigwedge^p C_{>} \xrightarrow{\bigwedge^p I_{>}} \bigwedge^p C \xrightarrow{R_p} \mathcal{B}^{\widehat{\otimes} kp}.$$

The additive exterior generator acts on this line by

$$\zeta_{>} = \text{Tr}(A|_{C_{>}}) = \sum_{\Re \lambda > k/2} \ell_{\lambda} \lambda.$$

Indeed expansion of A on a wedge of a basis keeps only its diagonal coefficients: every off-diagonal replacement repeats one basis vector and has zero wedge. The full nilpotent modules are retained before this determinant map; their dimensions appear in the trace. No simplicity assertion follows: a critical-line Jordan block can have nonzero numerical control while contributing zero real-part defect.

Define

$$L_{h,k} := 2\Re \zeta_{>} - kp = \sum_{\Re \lambda > k/2} \ell_{\lambda} (2\Re \lambda - k).$$

Theorem 6.1. For every admitted N ,

$$0 \leq L_{h,k} \leq \epsilon_{h,k,N}^{\text{cyc}}.$$

Proof. If $p = 0$, the left side is zero. Otherwise reflection implies $p < q_k$. Let v be the nonzero determinant vector of an unscaled basis of $C_{>}$. On this actual eigenline,

$$\frac{v^* W_p^{\text{source}} v}{v^* (p!G^{[p]}) v} = 2\Re \zeta_{>} - kp = L_{h,k}.$$

Theorem 4.1 bounds the same quotient by ϵ . $\square$

The old eigenline argument gave the maximum individual defect. This theorem controls their entire positive sum, with their full lengths. Reflection also gives

$$L_{h,k} = \frac{1}{2} \sum_{\lambda} \ell_{\lambda} |2\Re \lambda - k| = \sum_{\lambda} \ell_{\lambda} |\Re \lambda - k/2|.$$

N.7.1 6.2 The exact arithmetic volume equation

Let $B_{>} : \mathbb{C}^p \rightarrow C$ be the column map of a fixed unscaled basis of $C_{>}$, and set $G_{>} = B_{>}^* G B_{>}$. Its determinant vector $v_{>}$ has squared source norm

$$\|R_p v_{>}\|^2 = p! \det G_{>}.$$

For the original parameter $a > 0$, let $U_{>}(a) = a^{A|C_{>}}$. Then

$$\det(U_{>}(a)^* G_{>} U_{>}(a)) = a^{kp+L_{h,k}} \det G_{>}.$$

The entire exterior control integrates for $a \geq 1$ to

$$a^{kp-\epsilon} p! \det G_{>} \leq \|R_p \bigwedge^p U_{>}(a) v_{>}\|^2 \leq a^{kp+\epsilon} p! \det G_{>}.$$

Differentiating the norm along $a = e^t$ and applying the exact quadratic inequality proves the bound. Comparing with the exact determinant equality again gives $L_{h,k} \leq \epsilon$. The determinant, its norm, and its weight are all produced by maps from the original arithmetic module.

N.8 7. Spectral and metric projections are linked explicitly

The determinant argument has an equivalent trace calculation useful for computation. Let $Q_{>} = e_{>}(A)$ be the actual CRT spectral idempotent, where $e_{>}$ is one modulo each positive-defect factor and zero modulo every other factor. It commutes with A .

Choose the original CRT column map $B_{>} : \mathbb{C}^p \rightarrow C$, including every jet. Its G -orthogonal projection is

$$P_{>} = B_{>} (B_{>}^* G B_{>})^{-1} B_{>}^* G.$$

Their exact relationship is

$$Q_{>} P_{>} = P_{>}, \quad P_{>} Q_{>} = Q_{>}, \quad P_{>}^{\sharp G} = P_{>}, \quad (P_{>} - Q_{>})^2 = 0.$$

The first two identities follow from their common range and identity restrictions there. Expanding the square gives the last identity. Invariance of that range gives $P_{>} A P_{>} = A P_{>}$ and

$$\mathrm{Tr}(P_{>} A) = \mathrm{Tr}(Q_{>} A) = \zeta_{>}.$$

Since $\mathrm{Tr}(P_{>} A^{\sharp G})$ is the conjugate of $\mathrm{Tr}(P_{>} A)$,

$$\mathrm{Tr}(P_{>} H) = L_{h,k}.$$

For $\epsilon > 0$, the two G -orthogonal rank-one control projections are the exact polynomials

$$F_+ = \frac{H^2 + \epsilon H}{2\epsilon^2}, \quad F_- = \frac{H^2 - \epsilon H}{2\epsilon^2}.$$

They give the full slack formula

$$L_{h,k} = \epsilon(1 - \eta_{h,k,N}), \quad \eta_{h,k,N} = 1 - \text{Tr}(P_{>}F_+) + \text{Tr}(P_{>}F_-) \geq 0.$$

With $\|T\|_{\text{HS},G}^2 = \text{Tr}(T^\sharp G T)$, this is

$$\eta_{h,k,N} = \|(1 - P_{>})F_+\|_{\text{HS},G}^2 + \|P_{>}F_-\|_{\text{HS},G}^2.$$

Thus the gap has a specified projection meaning rather than an omitted term. This is the elementary rank-two instance of the classical extremal-trace principle.

The retained finite trace has its resolvent description as well:

$$\zeta_{>} = \frac{1}{2\pi i} \int_{\Gamma_{>}} S \text{Tr}((SI - A)^{-1}) dS,$$

where the positively oriented contour encloses exactly the indicated finite blocks. The complementary trace remains

$$\text{Tr } f(A) = \text{Tr}(Q_{>}f(A)) + \text{Tr}((1 - Q_{>})f(A)).$$

Multiplication by $(-1)^k$ gives the inherited cohomological sign. The determinant line occurs in degree kp , retaining $(-1)^{kp}$ for its trace. Split lifts of either projector send an existing class with zero projected amplitude to a supported zero, while the paired projector retains the complementary component.

N.9 8. Exact sum counting for a nonreal quartet

Take a complete actual nontrivial quartet

$$\rho_{\varepsilon,\eta} = \frac{1}{2} + \varepsilon\delta + i\eta\gamma, \quad \varepsilon, \eta \in \{+1, -1\}, \quad \delta > 0, \quad \gamma > 0,$$

with common full multiplicity m . This is a specification for the following theorem, not an assertion that an off-line zero has been found. The two standard symmetries of $g = 2\xi$ give the full quartet and equal orders from any such actual zero.

The distinct k -fold sums are precisely

$$\lambda_{a,b} = \frac{k}{2} + \delta(2a - k) + i\gamma(2b - k), \quad 0 \leq a, b \leq k.$$

Proof of existence. Let a count positive real signs and b positive imaginary signs. Choose an integer t with

$$\max(0, a + b - k) \leq t \leq \min(a, b).$$

The four occupation counts

$$(t, a - t, b - t, k - a - b + t)$$

are nonnegative and sum to k , with the required two margins. Conversely every tuple has such margins. Since δ and γ are nonzero real numbers, equality of two sums forces equality of both margins. No hypothesis of algebraic independence of zero ordinates is used.

All ordered tuple blocks have the same maximal length

$$\ell_k = 1 + k(m-1).$$

The nilpotency proof is the original one: the top monomial in $(z_1 + \cdots + z_k)^{k(m-1)}$ has coefficient $(k(m-1))!/((m-1)!)^k \neq 0$. It survives before the next power vanishes. Hence

$$\chi_{h,k}(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^{\ell_k}, \quad q_k = \ell_k(k+1)^2.$$

For the positive-defect subspace,

$$p = \ell_k(k+1) \left\lceil \frac{k}{2} \right\rceil.$$

Summing its real defects gives

$$\begin{aligned} L_{h,k} &= 2\delta\ell_k(k+1) \sum_{a=\lfloor k/2 \rfloor+1}^k (2a-k) \\ &= 2\delta\ell_k(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor. \end{aligned}$$

For $k = 2n$, the last sum is $n(n+1)$; for $k = 2n+1$, it is $(n+1)^2$. The imaginary parts cancel in the determinant trace because $\sum_{b=0}^k (2b-k) = 0$.

Combining this calculation with Theorem 6.1 proves

$$2\delta[1+k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_{h,k,N}^{\text{cyc}}, \quad N \geq [1+k(m-1)](k+1)^2 - 1.$$

Every multiplicity is retained. For simple roots the leading term is $\delta k^3/2$; for $m > 1$ it is $\delta(m-1)k^4/2$.

For the two-point reflected pair alone, the analogous exact lower bound is

$$\epsilon_{h,k,N}^{\text{cyc}} \geq 2\delta[1+k(m-1)] \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

The quartet supplies the additional, rigorously counted factor $k+1$ through its distinct imaginary sums.

N.10 9. The quantitative target after this construction

The current source formula is still

$$\epsilon_{h,k,N}^{\text{cyc}} = \frac{\sqrt{a_N d_N - |c_N|^2}}{\omega_N}.$$

A contradiction to any actual off-line quartet now follows from a bound

$$\frac{\sqrt{a_N d_N - |c_N|^2}}{k^3 \omega_N} \rightarrow 0$$

along a permitted choice $N = N(k) \geq [1+k(m-1)](k+1)^2 - 1$, proved for its actual arithmetic measure and complete jet map. For this contradiction the packet h is fixed: constants may depend

on h , and uniformity over all zero locations is not required. The expression itself has not been shown to tend to zero here. The result is a stronger amplification theorem and a weaker sufficient growth target for the next analytic estimate.

For a simple quartet, the source degree is at least quadratic in k . This cost is retained; a fixed-degree central-limit estimate cannot simply be inserted at that growing degree. The previously proved scalar Fisher contraction, with its factor μ_h^{k-1}/k , is not assigned to every polynomial moment.

The existing polynomial derivative formulas supply those actual moments:

$$\begin{aligned}\mathcal{M}_h(z) &= \sum_{j \geq 0} \mu_j \frac{z^j}{j!}, & \mathcal{N}_h(z) &= \sum_{j \geq 0} \nu_j \frac{z^j}{j!}, \\ \int u^j m_{h,k}(u) du &= j! [z^j] \mathcal{M}_h(z)^k, \\ \int u^j \|\partial_u \Psi_{h,k}\|^2 du &= j! [z^j] \left[\frac{1}{k} \mathcal{N}_h \mathcal{M}_h^{k-1} + \frac{k-1}{4k} z^2 \mathcal{M}_h^k \right].\end{aligned}$$

All masses, polynomial degrees and cross terms remain. The new determinant bound uses their already-defined Gram and control matrices rather than changing the source in order to obtain a favourable value.

The real interval $0 < s < 1$ causes no missing quartet case: the alternating zeta series there is positive when grouped into consecutive pairs, while $1 - 2^{1-s} < 0$, so $\zeta(s) < 0$. An actual non-trivial off-line zero is therefore nonreal and is covered by its complete quartet. This argument only identifies the case split; it supplies no upper estimate.

N.11 10. Deligne architecture and the boundary retained in this step

The inherited Weil-II reading identifies an amplification pattern: carry the same arithmetic spectral information into a cohomological tensor image, obtain a further estimate there, and transport dual control without changing the representation. The finite-field geometric theorem is not being applied on a renamed base.

Here the new characteristic-zero operation is

$$\begin{aligned}& \text{original theta complex over } \mathfrak{b}_\tau \\ & \downarrow \text{existing } k\text{-fold cyclic sum image} \\ & (C, A, R_N, G_N, W_N) \\ & \downarrow \text{signed exterior cochain map} \\ & \left(\bigwedge^p C, A^{[p]}, R_p, p! G^{[p]} \right) \\ & \uparrow \det(C_{>}) \\ & \text{exact sum of positive arithmetic defects.}\end{aligned}$$

The exterior operation controls that determinant line with the **same** allowance as the cyclic image. The original e -relation primitives are carried by $K_{N,p}^{\text{prim}}$; the scalar τ remains external absence. The spectral and metric projectors, their difference, the cyclic complement, the exterior complement and their finite traces are all explicit.

No direct implication from the cardinality of the infinite quotient to positivity is used. Its role is to remain the structural arithmetic quotient throughout the construction, not to be replaced by a finite-field scalar or by the Boolean support observation.

N.12 11. Verification and reproducibility

The supplied cyclic-sum archive was safely extracted and all 42 entries of its SHA-256 manifest verified. Its 24-method checker was rerun normally and under Python optimization, with matching success records. The new checker contains 22 exact finite test methods; its normal, optimized and deliberately failing results are recorded in checks/.

The Gaussian calculations in the checker are declared algebraic calibrations. Their literal measure is

$$\frac{7^k}{\sqrt{2\pi k}} e^{-u^2/(2k)} du.$$

They test companion matrices, full repeated-root modules, the canonical inverse-moment Gram, exterior factorials, the additive generator, the trace bound and the quartet counting. The mass is 7^k , not one. These examples do not claim that the calibration roots are zeta zeros.

The written arguments are general. Finite tests are regression evidence, not a Lean certificate, interval arithmetic for the actual spectral integrals, or independent certification of the whole analytic source corpus. No repository or other-session branch is changed by this continuation.

N.13 References and provenance

1. Owner programme, **Tau Cyclic Sum Control**, delivered NOTE.tex, 12 September 2026. The original packet, norm, canonical section, rank-two identity, reflection and thickened relation maps are source inputs. Archive and member hashes are in checks/source_manifest.json.
2. Owner repository, pinned main a4494fba4968db837a8af6fcb952c8688cd5d1da. Reviewed workbenches/tau-sum-connection/RESEARCH_NOTE.md, current blob 797d8bc298aec06b112b53a03654a9664323e748; workbenches/tau-frontier-formal/RESEARCH_NOTE.md; selected sum_connection_stieltjes.tex and conormal_finite_weyl.tex. Exact reading scope is in SOURCE_REVIEW.md.
3. Original SplitZero derived/reconstruction note and the actual universal property of its internal quotient; retained in the owner's supplied formalization material.
4. Overton, M. L., and Womersley, R. S. (1992), **On the Sum of the Largest Eigenvalues of a Symmetric Matrix**, SIAM Journal on Matrix Analysis and Applications 13(1), 41–45, doi:10.1137/0613006. Used only as attribution for the classical extremal-trace ingredient. The rank-two proof used here is fully written above.
5. The Stacks Project, Tag **OGWN**, tensor products of complexes. Used for the standard tensor differential and Koszul permutation types; the specific arithmetic signs and primitive are calculated above.
6. NIST DLMF §25.4, equations 25.4.3–25.4.4, the functional equation and the literal definition of ξ . The original arithmetic function in this program remains 2ξ .
7. Deligne, P. (1980), **La conjecture de Weil II**, Publications Mathématiques de l'IHÉS 52, 137–252. The supplied TeX/reading record at §3.2.13 motivates the amplification architecture. This note does not claim a new full reading or a finite-field purity theorem on the τ -base.

0 Source and relation volume dynamics for the original arithmetic control

Source attribution. Complete source note from the original Toda Volume Control archive. The source and relation Hankel determinants, tilt derivatives, imaginary cross term, first-degree endpoint, both loss squares and original theta input are retained. Its endpoint proof uses a vector symbol for a scalar norm; the explicit typed correction is proved in the integrated Toda bridge, while these original source bytes remain unchanged. The delivered floating evaluations are historical computations without interval or tail certification. The new local theta-tail proof and its separately scoped interval certificate supply that additional evidence for the analytic seed. Neither source asserts the growing-degree arithmetic estimate, and no local Lean execution is attributed to this edition.

Exact revision. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Source file. Tau_Toda_Volume_Control/NOTE.tex

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Revision scope. Complete delivered source witness SHA-256; no new remote revision is assigned

Source SHA-256. 027cd27a5fc104c5ebdf0102ee43aafd7b6218da4929ceb1b8913e3448cd2375

Original archive. Tau_Toda_Volume_Control_2026-09-12.zip

Archive SHA-256. 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41

0.1 Results and scope

The preceding exterior construction proves an aggregate lower bound on the canonical arithmetic control. This continuation constructs a scalar route to its upper estimate without selecting another metric on the arithmetic quotient.

The actual canonical quotient determinant is

$$V_N(\theta) = \det G_N(\theta) = \frac{\mathfrak{D}_{N+1}(\theta)}{\mathfrak{B}_{N-q+1}(\theta)}. \quad (1)$$

The numerator is the Gram determinant of the original polynomial source. The denominator is the Gram determinant of the original theta relations in that source. Both determinants come from one retained arithmetic Laplace transform and an explicitly given differential filter. Both obey the classical Hankel-Toda identity. No new general integrable-system theorem is claimed.

For $N \geq q$, put

$$v_N = \log \frac{V_{N-1}}{V_N}, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}, \quad \ell_N = \log V_N, \quad \sigma = \frac{\text{Tr } A - (k/2)q}{i} \in \mathbb{R}.$$

The following is an exact formula, at every real tilt in the stated domain:

$$\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \partial_\theta \ell_N)^2. \quad (2)$$

At $\theta = 0$, ϵ_N is the original allowance in the exterior-amplification note. The new formula retains the cross-term phase through a derivative of a scalar determinant. It is not an estimate that forgets that term.

In particular, with the actual two-step volume ratio

$$\mathcal{R}_N = \frac{V_{N-1}}{V_{N+1}},$$

one obtains the proved finite upper bound

$$\epsilon_N \leq \sqrt{a_{N+1}} \frac{\mathcal{R}_N - 1}{2\sqrt{\mathcal{R}_N}}. \quad (3)$$

Both nonnegative losses in this upper estimate are computed below. The previous aggregate spectral lower bound therefore forces an explicit minimum two-step contraction of the source/boundary volume ratio. This turns the remaining asymptotic question into a comparison between two specified scalar determinant flows. It does not prove that this comparison has the required subcubic bound.

0.2 1. Integration of the new formalization report

The supplied report identifies verified source commit 2abc351424ba87aeda948a5cfb846e15ed9373d1 in draft PR #21. The current PR metadata and its complete mathematical note were read. The report's successful execution is used with its stated scope; this session did not run Lean or replay the workflow log.

The checked coefficient interfaces are the original quotient derivative

$$\delta_r : A_0/I^{r+1} \longrightarrow A_0/I^r,$$

its square with quotient transitions, the cyclic injection obtained from the **literal** pullback

$$\mathcal{K}_r = \{P : P(S) \in I^r\},$$

and the full multiplier identity

$$\delta_r([uP(S)]) = [uP'(S)] + [(\partial u)P(S)].$$

The report separately identifies the local quotient-basis and CRT interpretation as written proofs. That separation is maintained. Its relation-depth maximum includes attainment and permits the tuple maximizing a collided sum to change with depth.

Here the scalar relation determinant uses the first cyclic relation $\chi = \chi_{h,k,1}$. The weight $|\chi|^2$ is the **squared Hilbert norm of the multiplication map** M_χ ; it is not a declaration that the next contracted ideal is (χ^2) . The map used is displayed in (11). At higher original depth the relevant polynomial remains $\chi_{h,k,r}$, through the checked pullback $\mathcal{K}_r$, not an assumed power of χ .

The empty-packet correction is also kept. For $h = 1$, I is the whole coefficient ring and every positive-depth quotient is zero. The analytic seed g and its mass remain. The split lift $G(0)$ retains two elements. In that case the determinant quotient below is the determinant of the zero-dimensional module, namely 1, and no positive-dimensional inverse or spectral estimate is inferred.

0.3 2. Original base, coefficient square, and arithmetic source

Keep

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

and the original square

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (4)$$

The target of $p\mathbb{Z}$ remains infinite. The base remains $\mathfrak{b}_\tau$. Every lifted coefficient map fixes external absence and carries represented zero to represented zero.

The analytic source remains

$$V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = \int \phi = 0\},$$

$$\mathcal{B} = \{F \in C^\infty(\mathbb{R}_{>0}) : \sup_{x>0} x^b |D^j F(x)| < \infty \quad \forall b \in \mathbb{Z}, j \geq 0\}, \quad D = -x\partial_x,$$

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V,$$

$$g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Let h be a nonempty reflection-stable packet of actual zeros with complete orders. Set

$$E_h = \mathbb{C}[s]/h, \quad v_h = g/h, \quad v_h = j_h(v_h) \in E_h^\times, \quad \mathcal{M}F_h = v_h.$$

Fix tensor degree $k \geq 1$. Use the cyclic algebra already constructed:

$$C = \mathbb{C}[S]/\chi, \quad q = \deg \chi \geq 1, \quad c = k/2, \quad A = M_S,$$

$$\eta : C \hookrightarrow (E_h^{\otimes k})^{S_k}, \quad \eta[P] = v_h^{\otimes k} P \left(\sum_i M_{s_i} \right) 1.$$

The source map is

$$\mathcal{V}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}).$$

It retains

$$J^{(k)}\mathcal{V} = \eta\pi_\chi, \quad q^{(k)}\mathcal{V} = \sigma_h^{\otimes k}\eta\pi_\chi. \quad (5)$$

Here $q^{(k)}$ denotes the original top cohomology quotient; $q = \deg \chi$ denotes a dimension. The full arithmetic unit is still in η .

The scalar norm is

$$\|\mathcal{V}P\|^2 = \int_{\mathbb{R}} |P(c+iu)|^2 m_k(u) du, \quad m_k = w_h^{*k}, \quad w_h(t) = \left| \frac{v_h(1/2+it)}{\sqrt{2\pi}} \right|^2. \quad (6)$$

Its mass is μ_h^k , with $\mu_h = \int w_h$. No measure is divided by that mass.

0.4 3. One arithmetic function generates both determinant flows

0.4.1 3.1 Analytic domain

Define

$$M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad Z_{h,k}(\theta) = M_h(\theta)^k. \quad (7)$$

These are analytic on $|\Re \theta| < \pi/2$. Here and below positivity statements concern real θ in that interval.

A direct bound suffices. At $\Re s = 1/2$, bounded partial sums of the alternating series and summation by parts give $|\eta(s)| \leq C(1 + |s|)$. The denominator $|1 - 2^{1-s}|$ is bounded below by $\sqrt{2} - 1$. Stirling's vertical-strip estimate then gives, beyond a fixed compact set,

$$w_h(t) \leq C_h(1 + |t|)^{11/2 - 2 \deg h} e^{-\pi|t|/2}.$$

On the compact set, g/h is entire because the selected zero orders are complete. Thus every differentiated integral is locally uniformly dominated on the displayed strip. This is a sufficient domain, not a claim that it is the maximal analytic continuation of M_h .

Fubini and the exact sum coordinate give

$$Z_{h,k}(\theta) = \int e^{\theta u} m_k(u) du, \quad Z_{h,k}^{(j)}(\theta) = \int u^j e^{\theta u} m_k(u) du. \quad (8)$$

In particular $Z_{h,k}(0) = \mu_h^k$.

0.4.2 3.2 The source and the relation filter

Let $\bar{\chi}$ mean coefficientwise conjugation, and define the differential operator and its output

$$\mathcal{F}_\chi = \bar{\chi}(c - i\partial_\theta)\chi(c + i\partial_\theta), \quad Z_\chi(\theta) = \mathcal{F}_\chi Z_{h,k}(\theta). \quad (9)$$

Differentiation under the integral proves

$$\boxed{Z_\chi(\theta) = \int |\chi(c + iu)|^2 e^{\theta u} m_k(u) du > 0.} \quad (10)$$

Both factors, their signs, and the coefficient $c = k/2$ remain. This filter is multiplication by a source polynomial under the Laplace transform. It is not the quotient-level derivation δ_r .

For $\mathcal{P}_n = \mathbb{C}[S]_{\leq n}$ and $N \geq q - 1$, the actual source sequence is

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{\pi_\chi} C \longrightarrow 0, \quad M_\chi P = \chi P, \quad (11)$$

with $\mathcal{P}_{-1} = 0$. Applying $\mathcal{V}$ identifies its first image with the original admitted theta boundaries.

The Laplace observation gives the typed identity

$$\mathcal{L}_\theta(SP) = (c + i\partial_\theta)\mathcal{L}_\theta(P), \quad \mathcal{L}_\theta(P) = \int P(c + iu) e^{\theta u} m_k(u) du.$$

Equation (10) is the Gram observation of M_χ . A source relation has nonzero positive norm before the quotient sends it to a fibre zero.

0.4.3 3.3 The two determinants

Put

$$\mathfrak{D}_n(\theta) = \det[Z_{h,k}^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad \mathfrak{B}_n(\theta) = \det[Z_\chi^{(i+j)}(\theta)]_{i,j=0}^{n-1}, \quad (12)$$

with the actual empty determinants $\mathfrak{D}_0 = \mathfrak{B}_0 = 1$. Every positive-order determinant is positive. A zero norm polynomial would vanish on a full-measure subset of the real line.

The use of u -moments is linked to the original S -basis by

$$S^j = (c + iu)^j = \sum_{r=0}^j \binom{j}{r} c^{j-r} i^r u^r.$$

This triangular coordinate map has determinant $i^{n(n-1)/2}$ on dimension n . Its Gram determinant multiplier is exactly 1, because that determinant has modulus one. No polynomial coefficient or factorial has been dropped.

The determinant has the exact integral representation

$$\mathfrak{D}_n(\theta) = \frac{1}{n!} \int_{\mathbb{R}^n} \prod_{i < j} (u_i - u_j)^2 \prod_{j=1}^n e^{\theta u_j} m_k(u_j) d^n u. \quad (13)$$

For $\mathfrak{B}_n$, insert $\prod_j |\chi(c + iu_j)|^2$. Expanding both Vandermonde determinants and integrating proves the formula, including $1/n!$. The total mass in the first expression scales as $(\mu_h^k)^n$, not as one.

0.5 4. The quotient determinant is the source/boundary volume ratio

At real θ equip the same polynomial source with

$$\langle P, Q \rangle_\theta = \int \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_k(u) du. \quad (14)$$

Let $H_\theta = L^2(\mathbb{R}, e^{\theta u} m_k(u) du)$. The map $f \mapsto e^{\theta u/2} f$ is an isometry $H_\theta \rightarrow H_0$ with inverse $f \mapsto e^{-\theta u/2} f$. The source in each space is exactly the image of $\mathcal{P}_N$ under evaluation. This is an observation of the original source, not an enlargement of its admitted polynomials.

Let J_N be its quotient matrix and $M_N(\theta)$ its source Gram. Define

$$K_N = J_N M_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad \hat{R}_N = M_N^{-1} J_N^* G_N, \quad R_N = \mathcal{V} \hat{R}_N. \quad (15)$$

The adjoints in this calculation use (14). At $\theta = 0$ these are the preceding canonical arithmetic representatives and their original norm. At every θ ,

$$J^{(k)} R_N = \eta, \quad q^{(k)} R_N = \sigma_h^{\otimes k} \eta.$$

Use the source basis

$$1, S, \dots, S^{q-1}, \quad \chi, \chi S, \dots, \chi S^{N-q}.$$

Its change from $1, S, \dots, S^N$ is triangular with diagonal one because the original χ is monic. The Schur complement of its boundary Gram is G_N . This proves

$$V_N := \det G_N = \frac{\mathfrak{D}_{N+1}}{\mathfrak{B}_{N-q+1}}. \quad (16)$$

Equivalently, the determinant line of (11) is mapped by the wedge of the **specified** kernel columns and quotient lifts. Orthogonal quotient representatives supply the Schur complement. This is the exact determinant-line map behind the ratio.

The original support lift is

$$(\lambda_N, P) \mapsto (\lambda_N, [P]_\chi), \quad (\lambda_N, \chi Q) \mapsto (\lambda_N, 0), \quad \tau \mapsto \tau. \quad (17)$$

The determinant $\mathfrak{B}_{N-q+1}$ retains the norm-volume of those nonzero source relations. Replacing that source by its quotient value would incorrectly replace the positive denominator by information-free zeros. The construction evaluates its Gram before applying (17).

For $h = 1$, $\chi = 1$ and $C = 0$. Then $Z_\chi = Z_{h,k}$ and (16) is 1 identically. This is the empty determinant, not an inverse on a nonzero spectral module.

0.6 5. The two classical Toda identities have their actual arithmetic input

The source determinants satisfy

$$\mathfrak{D}_n \mathfrak{D}_n'' - (\mathfrak{D}_n')^2 = \mathfrak{D}_{n+1} \mathfrak{D}_{n-1}, \quad n \geq 1. \quad (18)$$

The relation determinants satisfy the same equation with $\mathfrak{D}$ replaced by $\mathfrak{B}$.

Here is a proof with the constants retained. For a real positive weight $e^{\theta u} d\mu(u)$, let Q_j be the monic orthogonal polynomial obtained by subtracting lower-degree projections, with norm h_j . Multiplication has the recurrence

$$uQ_j = Q_{j+1} + b_j Q_j + a_j Q_{j-1}, \quad a_j = h_j/h_{j-1}, \quad a_0 = 0,$$

The variable remains u . Differentiating orthogonality gives

$$\partial_\theta Q_j = -a_j Q_{j-1}, \quad \partial_\theta \log h_j = b_j, \quad b_j' = a_{j+1} - a_j.$$

For the last identity, differentiate $h_j b_j = \int u Q_j^2 e^{\theta u} d\mu$, substitute the derivative of Q_j , and use the recurrence. Since $\mathfrak{D}_n = \prod_{j < n} h_j$, the sum telescopes:

$$\partial_\theta^2 \log \mathfrak{D}_n = a_n = \frac{\mathfrak{D}_{n+1} \mathfrak{D}_{n-1}}{\mathfrak{D}_n^2}.$$

This proves (18). The relation weight is the same positive weight multiplied by $|\chi(c+iu)|^2$, which is independent of θ , so the proof applies without alteration to its determinants.

The original S -monic polynomial norms are exactly

$$\omega_j = \frac{\mathfrak{D}_{j+1}}{\mathfrak{D}_j}, \quad \nu_j = \frac{\mathfrak{B}_{j+1}}{\mathfrak{B}_j}. \quad (19)$$

The phases linking monic polynomials in S and u are i^j and i^{-j} ; their norms agree because the phase has modulus one.

For $m = N - q + 1 \geq 0$ this gives the exact quotient-volume curvature

$$\partial_\theta^2 \log V_N = \frac{\mathfrak{D}_{N+2} \mathfrak{D}_N}{\mathfrak{D}_{N+1}^2} - \frac{\mathfrak{B}_{m+1} \mathfrak{B}_{m-1}}{\mathfrak{B}_m^2}. \quad (20)$$

The second term is defined to be zero for $m = 0$, because $\log \mathfrak{B}_0 = 0$. It is not a quotient involving $\mathfrak{B}_{-1}$.

0.6.1 5.1 The curvature comes from two actual source maps

Let X be multiplication by u on the polynomial realization in H_θ . Let P_N project to $\mathcal{P}_N$ and $P_{\mathcal{D}_N}$ project to the original boundary subspace $\mathcal{D}_N = \chi \mathcal{P}_{N-q}$. Write R for its least-norm polynomial lift.

Differentiating the quotient condition and its orthogonality proves

$$\partial_\theta R = -P_{\mathcal{D}_N} X R. \quad (21)$$

Thus its derivative is an original boundary, with a primitive obtained by literal monic division by χ and the existing theta division maps. Its quotient class and arithmetic jets stay fixed.

Define

$$O_N = (1 - P_N) X R : C \rightarrow \mathcal{P}_{N+1} \cap \mathcal{P}_N^\perp, \quad I_N = P_{\mathcal{D}_N} X R : C \rightarrow \mathcal{D}_N.$$

Direct differentiation, with the derivative of the measure included, yields

$$\boxed{G_N'' - G_N' G_N^{-1} G_N' = O_N^* O_N - I_N^* I_N.} \quad (22)$$

Indeed $G_N' = R^* X R$, $G_N'' = R^* X^2 R - 2R^* X P_{\mathcal{D}_N} X R$, and $R G_N^{-1} R^*$ is the projection to the orthogonal quotient representatives. Their orthogonal direct sum with $\mathcal{D}_N$ is $\mathcal{P}_N$, which proves (22).

Both maps have rank at most one. For O_N this follows from the highest polynomial degree. For I_N^* , multiplication sends every boundary of degree below its highest degree back into $\mathcal{D}_N$, orthogonal to R ; only the highest boundary direction can contribute.

Taking $\text{Tr}(G_N^{-1} \cdot)$ gives (20). The positive source contribution and the negative boundary contribution are retained. Positivity of each Gram separately is not assigned to their difference.

0.7 6. The phase in the rank-two control is a scalar volume derivative

Put $T = (A - cI)/i$ on C , with $c = k/2$. This is the coefficient operator corresponding to multiplication by u under the explicit ring map $S \mapsto c + iu$. The actual scaling generator is still $A = cI + iT$.

Reflection makes the polynomial $i^{-q} \chi(c + iu)$ real. This assertion follows by substituting $S = c + iu$ into $\chi^\dagger = (-1)^q \chi$. The retained leading phase of $\chi(c + iu)$ is i^q . Its remainder ideal and coefficient extraction are transported through that scalar unit, rather than altering χ in the arithmetic source.

In those real coefficient coordinates let $Q_j(u)$ be the monic source polynomials and $d_j = [Q_j]$. Their norm is ω_j . The metric and representatives have the same quotient definition. Define

$$\ell_N = \log V_N, \quad \sigma = \text{Tr } T \in \mathbb{R}.$$

The exact identity is

$$\boxed{\frac{d_N^* G_N d_{N+1}}{\omega_N} = \sigma - \ell_N'.} \quad (23)$$

To prove it, let $\mathfrak{b}_N = Q_{N+1} - R_N d_{N+1}$ in the polynomial source. It is the actual next boundary and

$$R_N^* \mathfrak{b}_N = -G_N d_{N+1}.$$

Multiplying the least-norm polynomial representative by u , its highest coefficient is $d_N^* G_N / \omega_N$. Therefore, modulo old boundaries,

$$X R_N - R_N T = \mathfrak{b}_N \frac{d_N^* G_N}{\omega_N}.$$

Multiplication by R_N^* eliminates precisely those old boundary terms. Consequently

$$\mathrm{Tr}(G_N^{-1} R_N^* X R_N) = \mathrm{Tr} T - \frac{d_N^* G_N d_{N+1}}{\omega_N}.$$

The left side is ℓ'_N by (21) and differentiation of the norm. This proves (23).

In the original S -polynomial coordinates, $p_j(S)$ maps to $i^j Q_j(u)$. Hence the preceding source cross term satisfies

$$\boxed{\frac{b_N^* G_N b_{N+1}}{\omega_N} = i(\sigma - \ell'_N).} \quad (24)$$

Both the sign and the factor i are retained. In particular an auxiliary real tilt generally creates a nonzero imaginary cross term even from an initially even measure.

0.8 7. The exact scalar formula and a two-step upper bound

For $N \geq q$, define

$$\delta_N = \frac{V_N}{V_{N-1}}, \quad v_N = -\log \delta_N \geq 0, \quad a_{N+1} = \frac{\omega_{N+1}}{\omega_N}.$$

The nonnegativity follows from the same minimum-norm comparison at nested source degrees. The rank-one kernel update and the matrix determinant lemma give

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (25)$$

Substitute (24)–(25) into the inherited exact rank-two radius. This proves

$$\boxed{\epsilon_N^2 = a_{N+1}(1 - e^{-v_N})(e^{v_{N+1}} - 1) - (\sigma - \ell'_N)^2.} \quad (26)$$

All scalar inputs come from (7), (9), and (12). In particular

$$\delta_N = \frac{\omega_N}{\nu_{N-q}}, \quad \delta_{N+1} = \frac{\omega_{N+1}}{\nu_{N-q+1}}.$$

This does not assert that Hankel determinant evaluation is well-conditioned; the exact formula still requires error enclosures for numerical certification.

0.8.1 7.1 The first admitted degree

At $N = q - 1$, K_{N-1} is not invertible. The displayed formula is replaced by its direct endpoint calculation:

$$\boxed{\epsilon_{q-1}^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}} - (\sigma - \ell'_{q-1})^2.} \quad (27)$$

Here $\nu_0 = Z_\chi$ is the actual norm of the first source relation. To prove (27), the first q remainder vectors are a basis, so the final included column has leverage one; the next determinant update gives $d_N = \nu_0 - \omega_q$. Formula (24) supplies the phase. No V_{q-2}^{-1} is introduced.

0.8.2 7.2 Both losses in the finite upper bound

For $N \geq q$, put

$$s_N = \frac{v_N + v_{N+1}}{2}, \quad t_N = \frac{v_{N+1} - v_N}{2}.$$

Direct scalar multiplication gives

$$(1 - e^{-v_N})(e^{v_{N+1}} - 1) = \sinh^2 s_N - (e^{t_N} - \cosh s_N)^2.$$

Thus

$$\boxed{\epsilon_N^2 = a_{N+1} \sinh^2 s_N - a_{N+1} (e^{t_N} - \cosh s_N)^2 - (\sigma - \ell'_N)^2.} \quad (28)$$

In particular, with $\mathcal{R}_N = e^{2s_N} = V_{N-1}/V_{N+1} \geq 1$,

$$\boxed{\epsilon_N^2 \leq a_{N+1} \frac{(\mathcal{R}_N - 1)^2}{4\mathcal{R}_N}.} \quad (29)$$

This upper estimate retains two computable losses: imbalance between consecutive volume contractions and the original cross-term phase. When the bound is too coarse, (26) or (28) is the exact replacement within the same construction.

0.9 8. What the exterior lower bound forces on these arithmetic volumes

The preceding exterior theorem gives

$$L_{h,k} := \sum_{\Re \lambda > k/2} \ell_\lambda (2\Re \lambda - k) \leq \epsilon_{h,k,N}.$$

It uses the original representative on its determinant line, not a newly chosen scalar metric. Combining it with (29), at $\theta = 0$, proves

$$\boxed{\log \frac{V_{N-1}}{V_{N+1}} \geq 2 \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N+1}}} \right), \quad N \geq q.} \quad (30)$$

The equivalent inequality without logarithms is

$$4\mathcal{R}_N L_{h,k}^2 \leq a_{N+1} (\mathcal{R}_N - 1)^2.$$

For disjoint successive two-degree steps the logarithms telescope. For every $N_0 \geq q$ and integer $r \geq 1$,

$$\boxed{\log \frac{V_{N_0-1}}{V_{N_0+2r-1}} \geq 2 \sum_{j=0}^{r-1} \operatorname{arsinh} \left(\frac{L_{h,k}}{\sqrt{a_{N_0+2j+1}}} \right).} \quad (31)$$

This is a source-volume consequence for the actual nested relation quotients, not a claim that arithmetic classes disappear when their representative norms decrease. At every degree their jets and original classes remain fixed by (5) and (15).

For a full off-line quartet with displacement $\delta > 0$, height $\gamma > 0$, and common multiplicity m , the previous count is

$$q = [1 + k(m - 1)](k + 1)^2,$$

$$L_{h,k} = 2\delta[1 + k(m - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor. \quad (32)$$

The quotient (16) and its derivative therefore give an entirely scalar test of the exterior obstruction. For this full quartet m_k and $|\chi(c + iu)|^2$ are even, so $\sigma = 0$ and $\ell'_N(0) = 0$. The phase term vanishes at zero tilt by this **additional** symmetry; it has not been removed from the general formula.

The sufficient subcubic target can now be tested from the quantities

$$\frac{\sqrt{a_{N(k)+1}}}{k^3} \frac{\mathcal{R}_{N(k)} - 1}{2\sqrt{\mathcal{R}_{N(k)}}}, \quad N(k) \geq q, \quad (33)$$

or, more sharply, from the full expression (26). An upper limit of zero for (33) would contradict (32) for that fixed quartet. This continuation proves the formula, the finite upper inequality, and the forced volume lower inequalities. It does not prove that limit.

0.10 9. The Laplace input can be evaluated from the original theta source

The inverse Mellin representation makes F_h holomorphic in $|\arg z| < \pi/4$. Uniform vertical-strip bounds for v_h permit differentiation and shifts on closed subsectors, and shifting the Mellin contour gives the required rapid endpoint bounds there.

The exact analytic-continuation map is

$$E_\theta F_h(y) = e^{(y+i\theta/2)/2} F_h(e^{y+i\theta/2}).$$

For $|\theta| < \pi/2$, its Fourier transform, with kernel e^{ity} , is

$$\mathcal{F}_+ E_\theta F_h(t) = e^{\theta t/2} v_h(1/2 + it).$$

Contour shift proves the equality; Plancherel yields

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (34)$$

The phase $e^{i\theta/4}$ in E_θ has modulus one and is retained before taking the norm.

Dagger stability gives the analytic identity

$$F_h(1/\bar{z}) = (-1)^{\deg h} \bar{z} \overline{F_h(z)}.$$

Substitute $z = e^{i\theta/2}x$ and change variables on $(0, 1)$. This proves

$$M_h(\theta) = 2 \int_1^\infty |F_h(e^{i\theta/2}x)|^2 dx. \quad (35)$$

Evenness in θ requires conjugation stability as well; it is not inferred just from this endpoint identity.

For the **analytic seed** $h = 1$, keep

$$f_0(z) = 2 \sum_{n \geq 1} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}.$$

Let $c_1 = -6, c_2 = 4$ and

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}).$$

Substitution into (35), where $\Re C_{mn} > 0$, gives

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(r+s+1/2, C_{mn}(\theta))}{C_{mn}(\theta)^{r+s+1/2}}. \quad (36)$$

The complex power is the analytic branch on the open right half-plane agreeing with the positive real power on its positive axis. The sum converges locally uniformly for $|\theta| < \pi/2$, by the exterior Gaussian estimates. Thus all derivatives used by (12) can be taken there. The factor 4 includes both the original two-sided theta sum and the change from the integral to the incomplete gamma function.

The accompanying numerical script evaluated (36) at 50 and 70 digits, with integer cutoffs 6 and 8. Their printed values agree:

$$M_1(0) \approx 1.279007247846485140479533592267193274492,$$

$$M_1(0.1) \approx 1.345907178458249796968220592357629805161.$$

An independent source integral gives

$$M_1''(0) = \|(D - 1/2)f_0\|^2 \approx 13.05554930257055843539268465812319368243.$$

Here $M_1'(0) = 0$. Consequently the first source Toda coefficient for tensor degree k is exactly

$$a_1^{(k)} = k \frac{M_1''(0)}{M_1(0)} \approx 10.207564753485721688865761252237555175 k. \quad (37)$$

This evaluates initial analytic data of the determinant flow. It is not a finite-packet control estimate: for $h = 1$ the arithmetic cyclic quotient is zero. The numerical results have neither directed-rounding enclosures nor a numerical omitted-tail certificate.

0.11 10. Exact finite checks, an essential phase example, and current endpoint

The checker independently constructs source Gram matrices, quotient maps, representatives, action matrices, and scalar determinants in rational Gaussian fixtures. It compares the scalar result against $\text{Tr}((G^{-1}W)^2)/2$.

For the declared coefficient fixture with weight $7e^{-u^2/2} du / \sqrt{2\pi}$, relation $u^2 + u + 1$, and degree $N = 3$, it obtains

$$\ell'_N(0) = -52/111, \quad \sigma = -1, \quad \epsilon_N^2 = 1156/333.$$

The phase subtraction is $(59/111)^2$, which is nonzero. Omitting it changes the answer. A tilted even-relation fixture gives the same warning: parity at zero tilt is not imposed at another tilt.

Twenty-four exact test methods passed in normal and optimized Python, with identical successful records. Both deliberate-failure controls failed. Repeated roots, source-coordinate phases, mass factors, the minimal-degree endpoint, both Toda equations, the curvature difference, the complete scalar radius, and the supported-zero relation are included. These are finite regression tests, not Lean certificates or verification of infinite arithmetic integrals.

All 34 members named in the preceding exterior package's manifest verified. Its 22-method checker passed a fresh normal run. A combined command attempting a subsequent optimized rerun timed out; no completed optimized rerun is claimed for that inherited suite. The new suite passed in both modes.

The present advance is an upper-estimate mechanism on the exact original family: its input is the one-factor arithmetic Laplace transform; its relation filter is the original cyclic annihilator; its output is the canonical quotient volume and complete control radius. The source-boundary curvature and the positive volume retained before an e -quotient are explicit maps. No uniform subcubic upper estimate, RH/GRH proof, additional Lean execution, or remote repository modification is claimed.

0.12 References and source boundary

The original source, support, and cohomological conventions are those in the supplied `Tau_Exterior_Trace_Amplification` package and its predecessor `Tau_Cyclic_Sum_Control`. The new formalization report and pinned PR #21 note supply the checked coefficient interfaces, not the analytic estimates in this note.

The general determinant, norm, and Christoffel–Darboux identities are classical. NIST DLMF §18.2(iv), (v), and (ix) records them, including the caution that raw Hankel determinant formulas are often poorly conditioned numerically: <https://dlmf.nist.gov/18.2>. The proof in §5 derives the needed Toda identity directly for the actual exponential deformation, without an isomonodromic or rational-logarithmic-derivative hypothesis. Gamma vertical-strip estimates used in §3 are recorded in DLMF §5.11: <https://dlmf.nist.gov/5.11>.

The finite exterior trace bound remains the theorem proved in the supplied preceding note; this continuation changes its analytic upper-estimate realization, not its algebraic proof or the scope of the parallel formalization.

P Supplied conormal derivative, symmetric restriction and uncompiled tower draft

Source attribution. This appendix reproduces the complete supplied mathematical continuation. Its first-person reports about PR 16, PR 18, combined CI, Mathlib inspection and 161 Wolfram checks are historical source statements, not additional verification performed for this edition. Both Lean blocks are retained; the proposed continuation is explicitly uncompiled. The monic-division coordinates and displayed dimension formulas have positive packet degree and tensor degree at least one; the empty packet is handled separately in the integrated proof. Full arithmetic unit and reflection formulas retain the cited source packet hypotheses.

Exact revision. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Source file. sources/web_conormal_tower_paste/CONORMAL_TOWER_SOURCE.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 18f86ad411d0dfafc003aabddf25bccae2d38dcc4006c4d27e5be8aece9347b8

Formal status. Both Lean blocks are source text; the draft has not been compiled for this edition.

Codex's integration has advanced: PR #16 is merged, and the combined frontier/Codex run has now passed. I checked the current repository state and the completed job: it rebuilt the inherited sources, checked the frontier files, imported both developments into one Lean environment, and completed their selected axiom audits. The “combined run pending” sentence in PR #18's description is therefore out of date.

I continued with the new sum-connection note rather than duplicating that work. **The additional result is a complete calculation of what its conormal derivative detects, what it misses, and how that calculation restricts to the symmetric tensor summand.** It also packages the quotient and derivative together as a genuine algebra homomorphism.

The new results below have complete mathematical proofs and exact finite algebra checks. The Lean continuation at the end is an **uncompiled draft**: the local execution service is timing out, and the currently exposed GitHub actions provide no write operation. No remote files were changed in this turn.

P.0.1 1. The derivative now has a complete quotient construction

Retain the source's notation, with tensor degree $k \geq 1$:

$$P = \mathbb{C}[s_1, \dots, s_k], \quad h_i = h(s_i), \quad I = (h_1, \dots, h_k),$$

$$E = P/I, \quad E^{[2]} = P/I^2, \quad \partial_S = \frac{1}{k} \sum_{i=1}^k \partial_{s_i}.$$

Here h is the original monic packet polynomial, including all multiplicities. These are precisely the relation ideal and derivative in the new attachment.

For any derivation D of a commutative algebra and any ideal I ,

$$\boxed{D(I^{r+1}) \subseteq I^r \quad (r \geq 0).}$$

Proof. An element of I^{r+1} is a sum of multiples of products of $r+1$ elements of I . The product rule differentiates one factor at a time, leaving at least r factors in I . Differentiating an exterior coefficient leaves all $r+1$ factors. Additivity completes the proof.

Consequently the source's maps exist at every order:

$$\delta_r : P/I^{r+1} \longrightarrow P/I^r, \quad [F] \longmapsto [\partial_S F].$$

They commute with the quotient transitions. In particular,

$$\delta_1 : E^{[2]} \longrightarrow E$$

is a derivation **along** the quotient $\pi : E^{[2]} \rightarrow E$:

$$\boxed{\delta_1(xy) = \pi(x)\delta_1(y) + \pi(y)\delta_1(x).}$$

This is the appropriate product rule; δ_1 is not an endomorphism of the arithmetic algebra E .

Monic division constructs all the retained layers There is a unique finite expansion

$$F = \sum_{\alpha \in \mathbb{N}^k} h_1^{\alpha_1} \cdots h_k^{\alpha_k} r_\alpha, \quad \deg_{s_i} r_\alpha < d := \deg h.$$

Existence follows by repeated monic division in the separate variables. For uniqueness, the leading monomials of

$$h^\alpha s^\beta, \quad 0 \leq \beta_i < d,$$

have exponent vectors $d\alpha + \beta$, which are distinct. A finite dependence therefore has a unique largest leading monomial and cannot cancel.

This gives explicit coordinates—not just an existence statement—for

$$I^r/I^{r+1} \cong E[\eta_1, \dots, \eta_k]_r, \quad \eta_i \longleftrightarrow [h_i].$$

In particular,

$$I/I^2 \cong E^k, \quad (a_i) \longmapsto \left[\sum_i h_i \tilde{a}_i \right],$$

and

$$\dim_{\mathbb{C}} P/I^{r+1} = d^k \binom{k+r}{k}.$$

On the associated graded, the derivative is exactly

$$\boxed{\text{gr}(\partial_S) = \frac{1}{k} \sum_i [h'_i] \partial_{\eta_i}.$$

Indeed, differentiating a coefficient of $h^\alpha r_\alpha$ stays in $I^{|\alpha|}$, while differentiating one h_i -factor lowers the ideal order and contributes the literal exponent α_i . This proves the higher-layer formula in the attachment without deleting those exponent factors.

The standard conormal construction supplies the map

$$I/I^2 \longrightarrow \Omega_{P/\mathbb{C}} \otimes_P E.$$

The calculation here identifies its coordinates for this particular ideal and then contracts it by the specified sum direction; those are distinct operations (The Stacks Project Authors, n.d., Tag 00S0). (The Stacks Project)

P.0.2 2. The exact information detected by the sum derivative

Let

$$A_h = \mathbb{C}[s]/(h), \quad E = A_h^{\otimes k},$$

and write

$$c_i = [h'(s_i)] \in E.$$

Under the preceding conormal coordinates, the restriction of δ_1 is the E -linear map

$$\partial_N : E^k \longrightarrow E, \quad (a_1, \dots, a_k) \longmapsto \frac{1}{k} \sum_i c_i a_i.$$

The attachment gives this row. The following kernel, cokernel, and rank calculations are additional deductions.

Let

$$b = \deg \gcd(h, h').$$

Over $\mathbb{C}$, if h has q distinct roots, then

$$b = d - q = \sum_{\rho} (m_{\rho} - 1).$$

The image and cokernel are

$$\mathrm{im} \partial_N = (c_1, \dots, c_k) \subseteq E,$$

$$\mathrm{coker} \partial_N \cong (\mathbb{C}[s]/(h, h'))^{\otimes k}.$$

Proof. The image is the ideal generated by the row entries because $1/k$ is invertible. Quotienting E by those entries imposes $h(s_i) = h'(s_i) = 0$ in each separate variable, giving the displayed tensor product.

Therefore

$$\mathrm{rank}_{\mathbb{C}} \partial_N = d^k - b^k, \quad \dim_{\mathbb{C}} \ker \partial_N = (k-1)d^k + b^k.$$

These formulas separate two sources of information lost by this particular observation.

When h is squarefree, $b = 0$, so the contracted derivative is onto. Nevertheless, for $k > 1$ its kernel still has dimension $(k-1)d^k$: **contracting to the sum direction does not observe every relative conormal direction, even without repeated roots.**

When roots are repeated, the additional term b^k records the failure of the Jacobian entries to generate the unit ideal.

For example, take the declared algebraic calibration

$$h(s) = s^2, \quad k = 2, \quad E = \mathbb{C}[x, y]/(x^2, y^2).$$

Then

$$\partial_N(a, b) = xa + yb.$$

Its image is

$$\text{span}\{x, y, xy\},$$

so its rank is three and its kernel has dimension five. The conormal vector $(x, 0)$ is nonzero but lies in that kernel. Its representative is x^3 modulo

$$I^2 = (x^4, x^2y^2, y^4),$$

so the retained relation is genuinely nonzero before observation.

These are ranks of the **conormal restriction**, not of the whole map $\delta_1 : E^{[2]} \rightarrow E$. The latter is onto: in coordinates

$$S = \sum_i s_i, \quad z_i = s_i - S/k,$$

one has $\partial_S = \partial/\partial S$, and a polynomial primitive is obtained by integrating its powers of S . The same argument makes every δ_r onto.

Package the quotient and derivative as one algebra map Introduce an ordinary formal square-zero parameter ϵ , distinct from the supported scalar e . Define

$$\boxed{\mathcal{J}_S^1 : E^{[2]} \longrightarrow E[\epsilon]/(\epsilon^2), \quad x \longmapsto \pi(x) + \epsilon \delta_1(x).}$$

This is a unital $\mathbb{C}$ -algebra homomorphism. Multiplication follows from the relative product rule:

$$(\pi x + \epsilon \delta x)(\pi y + \epsilon \delta y) = \pi(xy) + \epsilon \delta(xy).$$

Thus the source's two differently typed maps can be retained together in a single multiplicative observation. Its kernel is exactly

$$\ker \mathcal{J}_S^1 = \{n \in I/I^2 : \partial_N n = 0\},$$

and consequently

$$\boxed{\dim \ker \mathcal{J}_S^1 = (k-1)d^k + b^k, \quad \dim \text{im } \mathcal{J}_S^1 = 2d^k - b^k.}$$

The proof of the image dimension is the exact sequence

$$0 \longrightarrow \epsilon \text{ im } \partial_N \longrightarrow \text{im } \mathcal{J}_S^1 \longrightarrow E \longrightarrow 0.$$

This also describes the precise quotient of the first thickening seen by the sum derivative. It does not silently identify that quotient with the entire first thickening.

Retaining every first derivative is a different observation If all coordinate derivatives are retained, the map is

$$\mathcal{J}_{\text{all}}^1 : E^{[2]} \longrightarrow E \oplus E^k, \quad [F] \longmapsto ([F], ([\partial_i F])_i),$$

where E^k is a square-zero ideal in the target.

On $I/I^2 \cong E^k$, its derivative component is

$$(a_i) \longmapsto (c_i a_i)_i.$$

Each multiplication map M_{c_i} has kernel dimension $b d^{k-1}$. Hence

$$\dim \ker \mathcal{J}_{\text{all}}^1 = k b d^{k-1}.$$

In the squarefree case this full first-jet map is an algebra isomorphism onto the square-zero extension $E \oplus E^k$. With repeated roots it still loses the explicitly computed Jacobian-annihilator directions, which remain in the higher source layers.

The split lifts retain their original meaning throughout:

$$G(\pi)(n^\bullet) = e_E, \quad G(\mathcal{J}_S^1)(n^\bullet) = (\epsilon \partial_N n)^\bullet.$$

External absence maps to external absence. This is a calculation of a retained relation before quotienting, not a rule that differentiates e into a nonzero scalar. The attachment makes that distinction explicitly.

P.0.3 3. This calculation restricts to the symmetric tensor construction

The preceding maps commute with simultaneous permutation of variables and conormal indices. Let

$$N = I/I^2 \cong E^k, \quad T = \mathbb{C}[s]/(h, h').$$

The cokernel sequence is equivariant:

$$N \xrightarrow{\partial_N} E \longrightarrow T^{\otimes k} \longrightarrow 0.$$

Taking S_k -invariants remains exact: given an invariant element in an image, average any preimage with the literal factor $1/k!$. Thus

$$\text{coker}(\partial_N : N^{S_k} \rightarrow E^{S_k}) \cong (T^{\otimes k})^{S_k}.$$

The invariant conormal domain has the explicit description

$$N^{S_k} \cong A_h \otimes (A_h^{\otimes(k-1)})^{S_{k-1}}.$$

An invariant tuple is determined by its first component, which is invariant under permutations fixing the first index; the other components are its prescribed translates. This constructs both directions of the identification.

Consequently,

$$\dim N^{S_k} = d \binom{d+k-2}{k-1},$$

$$\text{rank } \partial_N^{S_k} = \binom{d+k-1}{k} - \binom{b+k-1}{k},$$

and

$$\dim \ker \partial_N^{S_k} = d \binom{d+k-2}{k-1} - \binom{d+k-1}{k} + \binom{b+k-1}{k}.$$

For $b = 0$, the final binomial is zero.

For $h = s^2$ and $k = 2$, the symmetric conormal domain has dimension four, the symmetric arithmetic target has dimension three, and the restricted derivative has rank two. Its kernel still has dimension two.

These are ordinary variable-permutation invariants of the relation algebra. To place them in the original cochain construction, use the source's **signed chain idempotent**

$$S_k = \frac{1}{k!} \sum_{\pi \in S_k} \text{sgn}(\pi) T_\pi.$$

Its two top-degree signs produce precisely the ordinary symmetric tensors used here. Replacing it by the unsigned chain average would change the selected summand.

This gives the smaller cohomological calculation its own exact derivative kernel; selecting the symmetric summand does not amount to presuming that the remaining relative information vanishes.

P.0.4 4. The arithmetic unit and trace remain attached

The new note retains

$$U = \prod_i (g/h)(s_i)$$

and its Taylor class $\widehat{U} \in E^{[2]}$. The relative Leibniz rule proves

$$\delta_1(\widehat{U}x) = U \delta_1(x) + \delta_1(\widehat{U})\pi(x).$$

On a conormal class n , the second term vanishes:

$$\delta_1(\widehat{U}n) = U \partial_N n.$$

Because U is a unit, this arithmetic multiplication changes neither the kernel nor the ranks calculated above. It is nevertheless retained in the actual map, rather than assigned the value one. This is the source's specified unit correction.

There is also an important distinction between the **sum-Jacobian response** and the **product Jacobian used by the full product trace**.

Multiplication by

$$\prod_i h'(s_i)$$

on $E = A_h^{\otimes k}$ has rank

$$\boxed{q^k},$$

where q is the number of distinct roots of h . Its kernel has dimension

$$d^k - q^k$$

and is the nilradical of E .

Proof. At a root ρ of multiplicity m , write

$$h(\rho + z) = z^m u_\rho(z), \quad u_\rho(0) \neq 0.$$

Modulo z^m ,

$$h'(\rho + z) = m u_\rho(0) z^{m-1}.$$

Multiplication by this element has one-dimensional image and kernel (z) . At a k -tuple of roots, the product Jacobian similarly sends the constant coefficient to a nonzero multiple of

$$z_1^{m_1-1} \cdots z_k^{m_k-1}$$

and annihilates the maximal ideal. There is one image dimension per root tuple, giving q^k .

This is not the rank $d^k - b^k$ of the contracted conormal row. The two operators use the same derivative factors but perform different observations.

The one-variable trace identity remains exact. Let

$$\lambda_h([P]) = \sum_{\rho} \operatorname{Res}_{s=\rho} \frac{P(s)}{h(s)} ds.$$

Then

$$\boxed{\lambda_h(h'P) = \operatorname{Tr}(M_P \mid A_h).}$$

Locally, $h'/h = m_\rho/(s - \rho) + u'_\rho/u_\rho$, so its residue against P is $m_\rho P(\rho)$. Multiplication by P on the local algebra has that same trace because its nilpotent terms have zero diagonal.

Writing $v_h = j_h(g/h)$ and $\varepsilon_h = v_h^{-1}$, the actual arithmetic pairing gives

$$\begin{aligned} \mathcal{R}_Z(f, j_h(g')P) &= \lambda_h(\varepsilon_h f^\dagger v_h h'P) \\ &= \lambda_h(f^\dagger h'P) \\ &= \operatorname{Tr}(M_{f^\dagger P} \mid A_h). \end{aligned}$$

Both unit factors are present before cancellation. This is the algebraic form of the attachment's observation of

$$(\log x)\Theta(P(D)\phi_*),$$

with the original $g = 2\xi$, reflection, and multiplicities retained.

P.0.5 5. Lean continuation against the integrated library

The merged Codex source already supplies

`SplitZero.RelationLayer.derivative`

as the quotient map induced by a linear map that carries the source relations into the target relations. I read its implementation and reused that interface below rather than introducing another quotient.

The following proposed `SplitZeroConormalTower.lean` proves the ideal-power containment from the derivation law, uses it to construct every descended derivative, proves compatibility between levels, and computes the derivative of an original relation. Its proof scripts are complete, but **compilation has not been checked**.

```
import SplitZeroRelationLayerIntegration
import Mathlib.RingTheory.Derivation.Basic
import Mathlib.RingTheory.Ideal.Operations

/-!
Derivative maps between the original ideal-power quotients.

This draft reuses RelationLayer.derivative. It does not assume that
the ideal is derivative-stable or that the packet is squarefree.
-/

noncomputable section

namespace SplitZero.ConormalTower

variable {R A : Type*}
[CommRing R] [CommRing A] [Algebra R A]

/-- The original ideal-power relation, viewed over the coefficient ring. -/
def level (I : Ideal A) (r : ℕ) : Submodule R A :=
  (I ^ r).restrictScalars R

/-- A derivation lowers ideal order by at most one. -/
theorem deriv_mem_pow
  (D : Derivation R A A) (I : Ideal A) (r : ℕ)
  {x : A} (hx : x ∈ I ^ (r + 1)) :
  D x ∈ I ^ r := by
  induction r generalizing x with
  | zero =>
    rw [Submodule.pow_zero, Ideal.one_eq_top]
    exact Submodule.mem_top
  | succ r ih =>
    rw [Submodule.pow_succ] at hx
    refine Submodule.mul_induction_on hx ?_ ?_
    · intro a ha b hb
      rw [D.leibniz]
      change a * D b + b * D a ∈ I ^ (r + 1)
      have hd : D a * b ∈ I ^ (r + 1) := by
        rw [Submodule.pow_succ]
        exact Ideal.mul_mem_mul (ih ha) hb
      exact (I ^ (r + 1)).add_mem
        ((I ^ (r + 1)).mul_mem_right (D b) ha)
        (by simp only [mul_comm] using hd)
    · intro a b ha hb
      rw [map_add]
      exact (I ^ (r + 1)).add_mem ha hb

/-- The derivative on  $A/I^{(r+1)}$ , with values in  $A/I^{(r)}$ . -/
```

```

def descended
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.derivative
  (level I (r + 1)) (level I r) D.toLinearMap
  (fun _ hx => deriv_mem_pow D I r hx)

@[simp]
theorem descended_mk
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) (x : A) :
  descended D I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ (D x) :=
  rfl

/-- Forget one ideal-adic layer without differentiating. -/
def projection (I : Ideal A) (r : ℕ) :
  (A / level (R := R) I (r + 1)) →1[R]
  (A / level (R := R) I r) :=
  SplitZero.RelationLayer.transport
  (level I (r + 1)) (level I r)
  (fun _ hx => Ideal.pow_le_pow_right (Nat.le_succ r) hx)

@[simp]
theorem projection_mk (I : Ideal A) (r : ℕ) (x : A) :
  projection (R := R) I r ((level I (r + 1)).mkQ x) =
  (level I r).mkQ x :=
  rfl

/-- Derivative and quotient transition form the actual commuting square. -/
theorem tower_square
  (D : Derivation R A A) (I : Ideal A) (r : ℕ) :
  (projection I r).comp (descended D I (r + 1)) =
  (descended D I r).comp (projection I (r + 1)) := by
  apply LinearMap.ext
  intro z
  obtain (x, rfl) :=
    Submodule.Quotient.mk_surjective (level I (r + 2)) z
  rfl

/-- On a retained relation z*a, only a*D(z) survives modulo I. -/
theorem conormal_product
  (D : Derivation R A A) (I : Ideal A)
  (z a : A) (hz : z ∈ I) :
  descended D I 1 ((level I 2).mkQ (z * a)) =
  (level I 1).mkQ (a * D z) := by
  rw [descended_mk, D.leibniz]
  change
    (level I 1).mkQ (z * D a + a * D z) =
    (level I 1).mkQ (a * D z)
  rw [map_add]
  have hzero : (level (R := R) I 1).mkQ (z * D a) = 0 := by
    apply (Submodule.Quotient.mk_eq_zero _).mpr
    change z * D a ∈ I ^ 1
    rw [Submodule.pow_one]
    exact I.mul_mem_right (D a) hz
  rw [hzero, zero_add]

end SplitZero.ConormalTower

```

The derivation and ideal-product APIs used here were checked against the repository's existing Mathlib revision, rather than assumed from a different release. That source inspection does not substitute for running Lean.

P.0.6 Verification and the remaining quantitative connection

The fresh Wolfram calculations passed **161 exact checks**: conormal ranks and invariant restrictions, full first-jet ranks, ideal-square annihilation, residue traces, product-Jacobian ranks, higher derivative identities, and retained-unit trace identities. The fixtures include squarefree polynomials, repeated roots, mixed multiplicities, and tensor degrees through three. An initial variable-binding error in my checker was corrected before the successful runs. These are finite algebra checks, not a new Lean certificate or evaluations of arithmetic zero locations.

The new result does not identify the Hilbert-space normal map N_u with the algebraic conormal module I/I^2 . The attachment's filtered graph supplies the stated route from the full derivative to arithmetic jets. Its exact scalar identity

$$\mathcal{I}_{h,k} + 4 \int \|n_k(u)\|^2 du = \frac{\mu_h^{k-1}}{k} \mathcal{I}_h$$

retains the Hilbert normal contribution; the conormal calculations above now identify the algebraic kernel encountered by the subsequent sum-direction observation. Those two pieces must remain separately typed when estimating the original weight-control form.

The continuation now provides an exact first-jet algebra map, its full retained kernel, the corresponding symmetric restriction, and all-order quotient derivatives. Codex can build those on the merged quotient library without recreating the scalar, homology, or boundary-control implementations.

Reference The Stacks Project Authors. (n.d.). *The naive cotangent complex* (Tag 00S0). *The Stacks Project*. ([The Stacks Project](#))

Q PR 21: conormal tower and cyclic depth formalization with its complete mathematical note

Source attribution. Complete mathematical note at PR 21 implementation commit 2abc351424ba87aeda948a5cfb846e15ed9373d1. The staged source provenance identifies all eight added Git blobs. The historical pasted draft, these repository modules and the observed remote CI evidence retain their distinct revisions and verification scopes.

Exact revision. 2abc351424ba87aeda948a5cfb846e15ed9373d1

Source file. `sources/web_conormal_cyclic_formal_delivery/repository/workbenches/tau-conormal-cyclic-formal/RESEARCH_NOTE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 202d7e9111c0a88fdd30c15554f3dd1e0907b2fc4fb2e11da129c97b10a053fd

Formal status. The associated two Lean modules and observed remote CI evidence are archived separately. No local Lean execution is attributed to this edition.

Q.1 Original conormal and cyclic relation towers

Q.1.1 Scope and source reconciliation

This add-only continuation starts from main at a4494fba4968db837a8af6fcb952c8688cd5d1da. It retains the merged Codex quotient/homotopy/boundary implementation, the merged finite-frontier implementation, and the reviewed sum-connection note. It uses the owner's attached **Handoff: sum-generated arithmetic control**, especially targets 1, 2, 4 and 8. The current upload is an analytic/algebraic continuation with a separately reported 24-method finite checker; it is not a Lean certificate. The local execution service failed before its ZIP could be opened, so neither the ZIP manifest nor that checker has been independently rerun here.

Two actual GitHub corrections are incorporated, not merely acknowledged:

- Commit 600ab7a527f7a626c162bf0199d093d454b86c14 makes the nonempty hypothesis explicit in the monic-freeness argument and calculates $h=1$ separately. Every positive-depth relation ideal for $h=1$ is the whole polynomial ring. The corresponding underlying quotients are zero, while the split lift still retains tau and supported zero. The analytic theta seed and its mass do not disappear. `ConormalTower.top_level_zero` checks the full-ideal quotient at every level.
- Commit 7aec03a93b66d3d97fbf6f7042f58bb2c1019543 corrects the completed integration audit to 68 selected declarations and enables the existing frontier workflow on main and relevant pull requests. This branch inherits those files unchanged. The current 68-target audit, not the historical 66-target description, is rerun by the added workflow.

The original scalar G , e , tau, arithmetic projection to the infinite ring, reconstructed carriers, quotient transports and theta normalization $g=2$ xi are unchanged. No claim about a global arithmetic small-control estimate follows from the present algebraic results.

Q.1.2 1. An actual derivative between consecutive original quotients

Let R and A be commutative rings, A an R -algebra, I an ideal of A , and D an R -derivation of A . Then

$D(I^{(r+1)})$ is contained in I^r , $r \geq 0$.

Proof: use induction on r and the additive generation of an ideal product by products. For a in $I^{(r+1)}$, b in I , the Leibniz terms $a D(b)$ and $b D(a)$ both lie in $I^{(r+1)}$: the first by the ideal property, the second by the induction hypothesis. At $r=0$ the target ideal is the whole ring.

SplitZeroConormalTower.lean implements that proof as `deriv_mem_pow`. It then uses the merged `SplitZero.RelationLayer.derivative` to construct

$\text{delta}_r : A/I^{(r+1)} \rightarrow_R A/I^r, [x] \mapsto [D x]$.

It separately constructs the quotient projection p_r between the same types and proves

$p_r \text{delta}_{(r+1)} = \text{delta}_r p_{(r+1)}$.

The derivative is not an endomorphism of A/I . These are coefficient-linear maps on the actual original quotient types, not new quotient carriers. At a retained relation z in I ,

$\text{delta}_1([z a]) = [a D(z)]$.

For the full arithmetic multiplier u , the implemented formula is

$\text{delta}_r([u x]) = [u D(x)] + [x D(u)]$.

It applies in particular to the retained Taylor unit, without assigning the unit the value one or deleting its derivative. The definitions work for arbitrary commutative rings and do not assume characteristic zero or squarefreeness.

Q.1.3 2. The exact cyclic pullback, before a polynomial is used to describe it

Fix S in A . At each depth define the actual coefficient submodule

$K_r = \{P \text{ in } R[X] : P(S) \text{ belongs to } I^r\}$.

The implementation calls this `cyclicLevel`; it is literally the comap of the original relation submodule along `Polynomial.aeval\S`. Its constructed quotient map is

$\text{alpha}_r : R[X]/K_r \rightarrow_R A/I^r, [P] \mapsto [P(S)]$.

`evaluation_injective` proves injectivity by the two original quotient equality criteria. No assumed injectivity of polynomial evaluation is required. This implementation is a linear map; its usual algebra-homomorphism packaging is not claimed as a new bundled declaration.

If $D(S)=1$, Mathlib's formal polynomial chain rule gives

$D(P(S)) = P'(S)$.

Thus P in $K_{(r+1)}$ implies P' in K_r . The code constructs the cyclic derivative and proves the full square

$$\begin{array}{ccc} R[X]/K_{(r+1)} & \xrightarrow{-d/dX-} & R[X]/K_r \\ \downarrow \text{alpha}_{(r+1)} & & \downarrow \text{alpha}_r \\ A/I^{(r+1)} & \xrightarrow{\text{D}} & A/I^r. \end{array}$$

The actual unit comparison is also proved:

$\text{delta}_r([u P(S)]) = [u P'(S)] + [D(u) P(S)]$.

The final term need not be in the cyclic image. The equation retains its larger target. Most importantly, K_r is not replaced by K_1^r . The explicit annihilator calculation below explains the difference.

Q.1.4 3. Exact local maximum, including an attaining monomial

Here the base field is $\mathbb{C}$. Let $k \geq 1$, $d \geq 1$, $r \geq 1$ and write the complete packet polynomial as

$$h(s) = \prod_{\rho} (s - \rho)^{m_{\rho}}.$$

For a root tuple $\mathbf{\rho} = (\rho_1, \dots, \rho_k)$, put $m_i = m_{(\rho_i)}$ and $y_i = s_i - \rho_i$. Locally the ideal $(h(s_i))$ is $(y_i^{m_i})$, because each remaining factor is a unit. In the quotient by its r -th power the monomial basis is precisely

$$y^{\mathbf{a}} \text{ such that } \sum_i \lfloor a_i / m_i \rfloor < r.$$

Write $a_i = m_i q_i + t_i$ with $0 \leq t_i < m_i$. Its total degree satisfies

$$\sum_i a_i \leq \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

The proof uses $\sum_i q_i \leq r - 1$ and bounds each m_i by its maximum. This bound is attained: choose j with $m_j = \max_i m_i$, let every $t_i = m_i - 1$, let $q_j = r - 1$, and let every other $q_i = 0$.

SplitZeroCyclicDepth.lean formalizes the quotient/remainder predicate, the degree bound, that explicit exponent vector, and attainment. The floor-sum formulation is linked through Euclidean division. This is an all-index, all-depth result, not a finite calibration. The additional algebraic identification of these monomials with the polynomial quotient and the characteristic-zero multinomial step below have written proofs, not an additional Lean certificate in these two modules.

Let $N = y_1 + \dots + y_k$. Its exact nilpotency index is

$$L_r(\mathbf{\rho}) = 1 + \sum_i (m_i - 1) + (r - 1) \max_i m_i.$$

Indeed every monomial of total degree L_r is killed. The attaining monomial at degree $L_r - 1$ has coefficient

$$(L_r - 1)! / \prod_i a_i!$$

in $\mathbb{N}^{L_r - 1}$, a nonzero positive integer in characteristic zero. Distinct surviving monomials cannot cancel. This proves both vanishing and minimality.

The proof also counts the local dimension as

$$(\prod_i m_i) * \text{binomial}(k + r - 1, k).$$

The q_i with sum less than r give the binomial factor, and their remainders give the product factor.

Q.1.5 4. Collisions require a maximum at every depth

Set $S = \sum_i s_i$. At a collided sum λ , take the maximum over all root tuples with that sum:

$$\begin{aligned} \text{ell}(\lambda, r) &= \max_{\sum \rho_i = \lambda} L_r(\mathbf{\rho}). \\ \chi_r(X) &= \prod_{\lambda} (X - \lambda)^{(\text{ell}(\lambda, r))}. \end{aligned}$$

The Chinese-remainder decomposition and the exact local nilpotency prove

$$I^{\wedge r} \text{ intersect } \mathbb{C}[S] = (\chi_r(S)).$$

The same annihilator is obtained in the variable-permutation invariant subalgebra because every polynomial in S is already invariant. Thus α_r identifies the cyclic algebra $C[X]/(\chi_r)$ with its actual image, with no squarefree simplification.

The maximizing tuple can change with depth. In the declared calibration packet with roots 0,1,2 of multiplicities 5,4,1, respectively, the sum $\lambda=2$ in tensor degree two has

$$\text{ell}_r(2, r) = \max(5r, 4r+3).$$

At depths 1,...,5 the orders are 7,11,15,20,25. A tuple chosen only at depth one is insufficient for the whole tower. This is an algebraic test packet, not a statement about actual zeta zeros.

For every λ ,

$$\begin{aligned} \text{ell}_r(\lambda, r+1) &\geq \text{ell}_r(\lambda, r)+1, \\ \text{ell}_r(\lambda, r) &\leq r \text{ell}_r(\lambda, 1). \end{aligned}$$

The first follows by retaining a maximizing tuple from depth r ; its next slope is a positive multiplicity. For the second, note $\max_i m_i \leq 1 + \sum_i (m_i - 1)$ for every tuple. Hence

$$\begin{aligned} \chi_r &\text{ divides } \chi_{r+1}, \\ \chi_r &\text{ divides } \chi_{r+1}', \\ \chi_r &\text{ divides } \chi_1^r. \end{aligned}$$

The last divisibility supplies the stated surjection $C[X]/(\chi_1^r) \rightarrow C[X]/(\chi_r)$, not an equality. For $h(s)=(s-\rho)^2$ and $k=2$,

$$\begin{aligned} \chi_r(X) &= (X-2\rho)^{(2r+1)}, \\ \chi_1(X)^r &= (X-2\rho)^{(3r)}. \end{aligned}$$

They differ as soon as $r>1$. The derivative maps $C[X]/\chi_{r+1} \rightarrow C[X]/\chi_r$ are onto over C : integrate any polynomial representative, then take its source class. Their kernel dimensions are $\deg(\chi_{r+1}) - \deg(\chi_r)$.

For the empty packet $h=1$, handled separately as required by the GitHub correction, $I=P$, $K_r=C[X]$, and $\chi_r=1$ for all positive depths. The cyclic arithmetic quotient is zero. No maximum over an empty tuple set or inverse of a zero-dimensional Gram matrix is used to define it. The original nonzero analytic seed remains available as an analytic object.

Q.1.6 5. Explicit equivariant retraction of the cyclic summand (written proof)

This checks the handoff's retraction formula, including its inverse factor. On a λ -primary part M of the symmetric finite algebra, let $N=S-\lambda$, $N^{\text{ell}}=0$, and let w be the λ -component of the retained arithmetic unit U . Exact maximality gives $N^{(\text{ell}-1)}w \neq 0$. Choose the stated top-monomial coefficient functional θ with $\theta(N^{(\text{ell}-1)}w) \neq 0$. If it is realized first on an ordered root component, it can be restricted to the invariant subspace; the component of the symmetric vector is still detected.

Define

$$\pi_0(v) = \sum_{(j=0)}^{(\text{ell}-1)} \theta(N^{(\text{ell}-1-j)}v) X^j \bmod X^{\text{ell}}.$$

Coefficient comparison gives $\pi_0(Nv)=X \pi_0(v)$: the constant term on the left is $\theta(N^{\text{ell}}v)=0$ and the other coefficients shift exactly. Put $c=\pi_0(w)$. Its constant coefficient is nonzero, so it is invertible modulo X^{ell} . Explicitly, if $c=c_0(1+n)$, the inverse is $c_0^{-1} \sum_{(j=0)}^{(\text{ell}-1)} (-n)^j$.

Then $\pi=c^{-1}\pi_0$ satisfies

$\pi(P(N)w) = P(X) \bmod X^{e11}.$

Therefore the unit-weighted cyclic injection η has an S -equivariant left inverse. Summing over λ constructs the full retraction, and $\ker(\pi)$ is an invariant complement. This is a module splitting, not an assertion that π is an algebra map or that the summands are orthogonal for the canonical arithmetic Gram. All cross terms under that splitting remain.

This cyclic summand retains each amplified value $k \rho$; collided repeated pure tensors need not equal the particular cyclic eigenvector. The cyclic residue trace and the full symmetric trace have their distinct multiplicities. No part of the present calculation identifies those traces or drops the U -dagger U factor in the ambient arithmetic pairing.

Q.1.7 6. Formal versus analytic scope

The new formal outputs are the ideal-power derivative, actual quotient transitions, exact cyclic pull-back/injection, derivative square, unit rule, full-ideal case, and the all-degree attained monomial ceiling. Existing SplitZero support maps lift their coefficient maps at the original labels. No duplicate scalar construction or alternative homology core is introduced.

The exact χ_r annihilator, local basis/multinomial identification and the retraction in Sections 3-5 are supplied here as complete written arguments. Their finite calibrations are separate Python tests. The new upload's analytic cyclic convolution measure, canonical Gram, differentiated moments, rank-two control and full trace comparison remain its separately stated inputs and targets. This work does not independently certify that archive, an analytic moment enclosure, or a tensor-uniform small-control bound.

Q.1.8 Reproduction

Run the added SplitZero\conormal\cyclic\tower workflow on this branch. It installs unchanged Lean 4.31.0 and Mathlib fabf563a7c95a166b8d7b6efca11c8b4dc9d911f, rebuilds all inherited modules, checks every new source with `- \/-trust=0 \-DwarningAsError=true`, imports the combined sources, and rejects any selected transitive axiom outside `propext`, `Classical.choice` and `Quot.sound`. It reuses the previously tested axiom parser. The original core's byte pin is checked by the inherited prepare script.

The new finite checker directly multiplies the nilpotent sum in explicit local monomial quotients. It tests attainment and its literal multinomial coefficient, collided maxima, derivative divisibility, local dimensions and the $h=1$ branch, in normal and optimized Python with deliberate-failure controls. These finite tests do not replace the general Lean proofs or the written polynomial arguments. The final PR record identifies the exact completed run; no pending run is a certificate.

R The coordinating Zeta contribution: determinant losses and spectral coupling

Source attribution. Complete mathematical note supplied by the coordinating Zeta task on 13 September 2026. The direct sum of the two exact identities retains the original quotient metric, phase, spectral coupling and compression remainder. Its full formulas, domains, endpoint scope and Sylvester comparison are reproduced as supplied. The integrated bridge gives the same join with its complete derivation. No asymptotic estimate or novelty claim is inferred.

Exact revision. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Source file. sources/web_toda_parent_join/PARENT_EXACT_JOIN.md

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 71e275a8543448e57bcfafc5ead8cf3c7f98d594ae994582d1180999d98c6370

Formal status. Complete written join of Toda (28) and frozen exterior identity E8; no new Lean execution.

R.1 Exact join: determinant losses and spectral coupling

This short continuation joins two already written identities on the same finite arithmetic packet and its original least-norm metric. It does not replace that metric or claim an asymptotic estimate. It is a direct algebraic consequence of the supplied Toda note (28) and the frozen companion reader's exterior equality calculation (E8); no novelty claim or Lean execution.

R.1.1 Original objects and notation

Fix the existing nonempty packet, its tensor degree k , cyclic algebra $C = C[S]/\chi$, $q = \deg \chi$, $A = M_S$, and an admitted degree $N \geq q$. All quantities below are at the original observation $\theta = 0$. Retain G_N , the least-norm quotient Gram, and the adjoint $A^\sharp = G_N^{-1} A^* G_N$. Write $H_N = A^\sharp + A - kI$. Its eigenvalues are $+\epsilon_N$, $-\epsilon_N$, and zero, with the multiplicities from the rank-two calculation.

Let V_+ be the full generalized right-half-plane spectral subspace of A , and P_N its G_N -orthogonal projector. Its positive excess is

$$L = \sum_{\text{Re } \lambda > k/2} \ell_\lambda (2 \text{Re } \lambda - k).$$

Consider the nonempty V_+ case, so $L > 0$ and $\epsilon_N > 0$ by the proved exterior trace bound. Let u_+ , u_- be G_N -unit eigenvectors of H_N at $+\epsilon_N$ and $-\epsilon_N$. The coordinate injection $E_N : C^2 \rightarrow C$ sends (z_+, z_-) to $z_+ u_+ + z_- u_-$. Set $M_N = E_N^\sharp P_N E_N$. Thus $0 \leq M_N \leq I_2$. These eigen-coordinates describe a compression only; they do not replace the arithmetic metric.

Use $C_N = P_N A (1 - P_N) : V_+^\perp \rightarrow V_+$ for the actual off-diagonal coupling. The symbol C_N is used here to avoid confusing this coupling with the Toda note's operator $T = (A - kI/2)/i$. Every Hilbert-Schmidt norm below uses the original restricted G_N metrics.

For the scalar quantities retain

$$V_N = \det G_N = D_{(N+1)}/B_{(N-q+1)}, \quad v_N = \log(V_{(N-1)}/V_N), \quad s_N = (v_N + v_{(N+1)})/2, \quad t_N = (v_{(N+1)} - v_N)/2, \quad a_{(N+1)} = \omega_{(N+1)}/\omega_N, \quad \phi_N = \sigma - \partial_\theta \log V_N|_{\theta=0}, \quad \sigma = \text{Tr}((A - kI/2)/i).$$

Here D and B are exactly the source and relation Hankel determinant sequences in the supplied note, not alternative free sequences.

R.1.2 Complete loss identity

The Toda identity gives

$$a_{(N+1)} \sinh(s_N)^2 - \epsilon_N^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2.$$

The companion's E8 gives, on the same original metric,

$$\epsilon_N^2 - L^2 = \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

Adding these equalities cancels precisely ϵ_N^2 and proves

$$a_{(N+1)} \sinh(s_N)^2 - L^2 = a_{(N+1)} (\exp(t_N) - \cosh(s_N))^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2 + \epsilon_N^2 (\det M_N + \det(I_2 - M_N)).$$

All four terms on the right are nonnegative. The first two are the successive-volume imbalance and the retained phase. The last two are the original spectral-subspace coupling and the compression remainder. There is no cancellation between them.

The exact Sylvester morphism is also retained. In the G_N -orthogonal decomposition $C = V_+ \text{ plus } V_+^{\perp}$, write $A = [[B_N, C_N], [0, D_N^{\text{op}}]]$, and the CRT spectral idempotent as $Q = [[I, X_N], [0, 0]]$. The identity $AQ = QA$ is exactly

$$B_N X_N - X_N D_N^{\text{op}} = C_N.$$

Its inverse is $\int_0^\infty \exp(-t B_N) Y \exp(t D_N^{\text{op}}) dt$; the strict real spectral separation and the complete finite nilpotent polynomial factors give convergence. Consequently the coupling term controls $\|Q - P_N\|$ by the actual smallest singular value of this Sylvester map, as in E14-E18. No uniform bound for that singular value is inserted here.

R.1.3 What this adds to the two-step inequality

Let $R_N = V_{(N-1)}/V_{(N+1)} = \exp(2s_N)$. Dropping only the imbalance square and the nonnegative compression remainder gives the stronger finite inequality

$$a_{(N+1)} (R_N - 1)^2 / (4 R_N) \geq L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2.$$

Since $s_N \geq 0$, this is equivalently

$$\log R_N \geq 2 \operatorname{arsinh}(\sqrt{L^2 + \phi_N^2 + \|C_N\|_-(HS, G_N)^2} / \sqrt{a_{(N+1)}}).$$

For a full quartet at zero tilt, the supplied symmetry proves $\phi_N = 0$; the coupling term is still present. If that coupling is nonzero, the original lower bound with L alone is strictly sharpened at that N . Nothing here supplies uniform strictness or large- k decay. The retained endpoint $N = q-1$ uses the source note's separate endpoint formula rather than these preceding-volume ratios.

The empty selected subspace has $L = 0$ and should use the Toda identity directly; no positive-eigenline construction is asserted when $\epsilon_N = 0$.

R.1.4 Evidence and reading scope

The parent read the complete supplied NOTE.tex, including its sections 1-10, references and explicit first-degree calculation, directly from `Tau_Toda_Volume_Control_2026-09-12.zip` (SHA-256 80b365233ba5ebaf1b52aeea369a7771d74bf774dbb4416f10a56503321b8f41). The companion E8/E14-E18 source is pinned by GitHub commit 161089942cce70a09fbe441d08ba4d36c1fd810c, under `workbenches/splitzero-tandem/continuations/20260912-cyclic/tex/` `exterior_trace_equality_continuation.tex`. The displayed join was checked by literal addition of the two equalities. The independent delivery/checker review is separate in `toda_review/`.

Primary-source context checked in this turn: DLMF 5.11.9 for gamma vertical-strip decay, and DLMF 18.2(ix), especially 18.2.30 and the following numerical-conditioning warning. The supplied 1997 Toda paper was verified at publisher abstract/metadata scope; its full proof was not fetched. These references identify the classical ingredients, not a prior source for this arithmetic specialization.

S Gamma convolution descent and the original arithmetic coefficient correction

Source attribution. Complete original Gamma Convolution Descent Markdown note delivered on 13 September 2026. Its archive and alternate complete TeX note are retained byte-for-byte. The full local coefficient chapter proves the exact coordinate maps, finite derivative indices, cochain primitive and transform domains alongside this unmodified source witness. Source reports about remote revisions and computations retain the verification scope of their original receipts.

Exact revision. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Source file. `sources/web_gamma_convolution_delivery/Tau_Gamma_Convolution_Descent/RESEARCH_NO TE.md`

This appendix reproduces the complete proof note as a dated source witness. Descriptions of what is new, unavailable, or left for continuation refer to that source revision. The current local continuations and their proofs are in Part I. The appendix is attributed to its originating web contribution; it is not presented as a newly discovered local proof.

Source SHA-256. 886c82e2562d9b4846adb44640d28a34552e38eb2518a3cd47f90dbbedd95e5b

Formal status. Complete source witness. Finite checker records are separately retained. No new local Lean execution is claimed.

S.1 1. Scope and the inspected collaboration state

This is a mathematical continuation of **Tau Toda Volume Control** and **Tau Exterior Trace Amplification**. It retains their arithmetic source, quotient, scaling action, full packet multiplicities, and canonical minimum-norm metric. The new calculation links those objects to the gamma reference that is already in the repository. The gamma orthogonal polynomials and their general convolution identities are classical. The contribution here is the explicit arithmetic coefficient transform, its source and quotient maps, its quantitative bounds, and the resulting exact substitution into the existing Toda/exterior estimate.

The GitHub read was pinned to main fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f. The separate formalization was read at PR #22 head 811210d24b80813a08972ca23f919db015383137. At that inspected revision the PR explicitly said strict verification was still in progress. No successful certificate for that revision is inferred here. Its invariant-filtration trace budget is retained as the other session's result, not presented as a new deduction of this note.

The main-branch gamma source read was `workbenches/splitzero-tandem/continuations/20260912-stieltjes/tex/theta_gamma_reference.tex`, equations TG.2–TG.14. It already constructs the exact one-factor gamma measure, its constants, its monic polynomials and its arithmetic multiplier. We extend that specified reference through the sum pushforward at every tensor degree.

There is no new RH proof or arithmetic asymptotic assertion here. Theorems below have written proofs. The checker supplies finite exact regression evidence, not analytic interval certification and not Lean execution. No inherited file or other-session branch is modified.

S.2 2. The original arithmetic source and its split maps

Retain

$$G(R) = \{\tau_R\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet,$$

with the original maps

$$\begin{array}{ccc} G(\mathbb{Z}) & \xrightarrow{G(j)} & G(\mathbb{C}) \\ p\mathbb{Z} \downarrow & & \downarrow p\mathbb{C} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array} \quad (1)$$

The target of the first quotient is infinite. Its structural base is the same absolute pointed base $\mathfrak{b}_\tau$. The supported scalar and the external absorber have not changed.

The source complex is

$$C_+ = [V \xrightarrow{\Theta} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V, \quad D = -x\partial_x,$$

where V consists of the original even Schwartz functions with both original moments zero, and $\mathcal{B}$ has the original rapid decay at both multiplicative ends with all D -derivatives. Keep

$$\Theta\phi(x) = \sum_{n \neq 0} \phi(nx), \quad g(s) = 2\xi(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Fix a nonempty finite packet $h(s) = \prod_\rho (s - \rho)^{m_\rho}$ of actual zeros, with every selected order complete. For reflection statements require the same dagger stability as the source. Write

$$v_h = g/h, \quad \mathcal{M}F_h = v_h, \quad w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad \mu_h = \int w_h(t) dt. \quad (2)$$

The cyclic tensor source and its maps remain

$$\mathcal{V}_{h,k}P = P(D_1 + \cdots + D_k)(F_h^{\otimes k}), \quad C_{h,k} = \mathbb{C}[S]/\chi_{h,k}, \quad A = M_S, \quad (3)$$

$$J^{(k)}\mathcal{V}_{h,k} = \eta_{h,k}\pi_\chi, \quad q^{(k)}\mathcal{V}_{h,k} = \sigma_h^{\otimes k}\eta_{h,k}\pi_\chi,$$

where $S = s_1 + \cdots + s_k$ and $\eta_{h,k}[P] = v_h^{\otimes k}P(A_k)1$, with $v_h = j_h(g/h)$. All local units remain in this map.

At polynomial degree N , the original relation map is

$$M_\chi : \mathcal{P}_{N-q} \longrightarrow \mathcal{P}_N, \quad P \mapsto \chi P, \quad q = \deg \chi, \quad (4)$$

and its image under $\mathcal{V}_{h,k}$ consists of the already constructed theta boundaries. Every norm construction below concerns this same map and its quotient. Its squared-norm weight $|\chi|^2$ does not mean that the next conormal ideal is (χ^2) : that ideal remains the exact pullback of the original I^r .

For $h = 1$, the analytic source (2) still exists, but the finite arithmetic quotient is the zero module. Its split lift has e and τ . No nonzero packet or Gram inverse is inferred from that analytic branch.

S.3 3. Carry the repository's gamma reference through every tensor power

For a fixed real $\lambda > 0$ define exactly

$$r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad d\sigma_\lambda(t) = r_\lambda(t)dt, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda). \quad (5)$$

The repository proves, by the beta integral and Fourier inversion,

$$\int e^{iyt} d\sigma_\lambda(t) = c_\lambda (\cosh y)^{-2\lambda}. \quad (6)$$

Its factor 2 comes from $t = 2u$ in the beta integral. Thus c_λ , rather than 1, is the mass. Fourier uniqueness for finite measures gives the all-tensor identity

$$r_{\lambda,k} := r_\lambda^{*k} = \kappa_{\lambda,k} r_{k\lambda}, \quad \kappa_{\lambda,k} = \frac{c_\lambda^k}{c_{k\lambda}} = 2^{k-1} \frac{\Gamma(2\lambda)^k}{\Gamma(2k\lambda)}. \quad (7)$$

In particular the mass of the sum measure is c_λ^k . For $\lambda = 1/4$ this reads

$$r_{1/4,k}(u) = \frac{2^{k-1} \pi^{k/2}}{\Gamma(k/2)} \frac{|\Gamma(k/4 + iu/2)|^2}{2\pi}, \quad \int r_{1/4,k} = (2\pi)^{k/2}. \quad (8)$$

The literal coordinate map is $(t_1, \dots, t_k) \mapsto u = \sum_i t_i$; u is not replaced by an average or a rescaled variable. The original arithmetic scaling coordinate is still $S = k/2 + iu$.

For real $|\theta| < \pi/2$, analytic continuation of (6) gives

$$Z_{\lambda,k}^\Gamma(\theta) := \int e^{\theta u} r_{\lambda,k}(u) du = c_\lambda^k (\cos \theta)^{-2k\lambda}. \quad (9)$$

The exponential integrability follows either from the original beta-contour estimate or the gamma vertical-strip bound. This justifies all finite differentiated moment integrals used below.

S.4 4. The sum projection and its complete polynomial complement

Define actual Hilbert spaces

$$\mathcal{H}_k = L^2(\mathbb{R}^k, \prod_i d\sigma_\lambda(t_i)), \quad \mathcal{H}_\Sigma = L^2(\mathbb{R}, r_{\lambda,k}(u) du).$$

Their isometric inclusion is

$$U_\Sigma : \mathcal{H}_\Sigma \rightarrow \mathcal{H}_k, \quad (U_\Sigma f)(\mathbf{t}) = f(t_1 + \dots + t_k). \quad (10)$$

For $k \geq 2$, its adjoint has the explicit disintegration formula

$$(C_\Sigma F)(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (11)$$

Its denominator is the actual fibre mass. Fubini proves $C_\Sigma U_\Sigma = 1$ and $C_\Sigma = U_\Sigma^*$. For $k = 1$ both maps are the identity. Put $P_\Sigma = U_\Sigma C_\Sigma$ and retain

$$F \mapsto (C_\Sigma F, (1 - P_\Sigma)F), \quad (f, n) \mapsto U_\Sigma f + n. \quad (12)$$

These are inverse Hilbert isometries onto $\mathcal{H}_\Sigma \oplus \ker C_\Sigma$. The second component is not discarded. At each original support label ℓ , their reconstructed maps are $(\ell, F) \mapsto (\ell, C_\Sigma F)$ and $(\ell, F) \mapsto (\ell, (1 - P_\Sigma)F)$. A nonzero normal vector has first image $(\ell, 0)$, while its second image retains that vector. Both maps send external absence to external absence; their paired inverse is still (12). Norm and Gram observations have their actual quadratic types, not the type of a linear quotient: $\text{Gram} : \text{Hom}_\mathbb{C}(E, \mathcal{H}) \rightarrow \text{Herm}(E), R \mapsto R^* R$.

S.4.1 4.1 Exact action on every product-polynomial degree

Let $b_n^{(\lambda)}(t)$ be the repository's monic gamma polynomial, with

$$\sum_{n \geq 0} \frac{b_n^{(\lambda)}(t)}{n!} z^n = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2}.$$

Set $\alpha = 2\lambda$. Its complete norm and recurrence are

$$\|b_n^{(\lambda)}\|^2 = c_\lambda n! (\alpha)_n, \quad b_{n+1}^{(\lambda)} = t b_n^{(\lambda)} - n(n + \alpha - 1) b_{n-1}^{(\lambda)}. \quad (13)$$

The polynomials are not divided by their norms. Multiplying the k generating functions proves the literal addition formula

$$b_n^{(k\lambda)}\left(\sum_i t_i\right) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i). \quad (14)$$

For a multi-index $\mathbf{n}$, put $n = \sum_i n_i$. The exact sum projection is

$$C_\Sigma \left(\prod_i b_{n_i}^{(\lambda)}(t_i) \right) = \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}(u). \quad (15)$$

Proof. Pair the product against (14), in the actual product measure. Orthogonality makes the answer zero unless the sum degree is n , and at degree n the answer is $c_\lambda^k n! \prod_i (\alpha)_{n_i}$. The squared norm of the sum polynomial is $c_\lambda^k n! (k\alpha)_n$. Their quotient gives (15).

These tests identify the whole conditional projection, not merely its finite polynomial moments: the gamma sum measure has exponential moments, so its polynomials are dense in L^2 . To verify that last statement, an L^2 vector orthogonal to all polynomials defines a finite measure whose Fourier transform is analytic on a strip by Cauchy-Schwarz and exponential integrability. All derivatives at zero vanish; analytic uniqueness and uniqueness of Fourier transforms give the zero vector.

The retained normal polynomial is

$$n_{\mathbf{n}} = \prod_i b_{n_i}^{(\lambda)}(t_i) - \frac{\prod_i (\alpha)_{n_i}}{(k\alpha)_n} b_n^{(k\lambda)}\left(\sum_i t_i\right).$$

Its exact squared norm is

$$\|n_{\mathbf{n}}\|^2 = c_\lambda^k \left[\prod_i n_i! (\alpha)_{n_i} - \frac{n! \prod_i (\alpha)_{n_i}^2}{(k\alpha)_n} \right]. \quad (16)$$

Thus every degree, every factorial, and the relative component have an explicit map and norm. This is the classical Meixner-Pollaczek addition mechanism, applied with the source's exact scale and masses; it is not a claim to have discovered a new general family of orthogonal polynomials.

S.5 5. Insert the complete arithmetic multiplier, not a substituted gamma measure

Define the complex amplitude and its squared modulus by

$$A_{h,\lambda}(t) = \frac{v_h(1/2 + it)}{\Gamma(\lambda + it/2)}, \quad B_{h,\lambda}(t) = |A_{h,\lambda}(t)|^2, \quad w_h = B_{h,\lambda} r_\lambda. \quad (17)$$

The entire quotient g/h supplies values at the selected zeros. No pole is inserted into the multiplier at those locations. The full complex amplitude, not only its modulus, remains in the source comparison

$$P \mapsto \prod_i \frac{\Gamma(\lambda + it_i/2)}{\sqrt{2\pi}} A_{h,\lambda}(t_i) P(k/2 + i \sum_i t_i). \quad (18)$$

This is the original Mellin image of $\mathcal{V}_{h,k}P$.

The scalar multiplier on the sum line is

$$B_{h,\lambda;k} = C_\Sigma(B_{h,\lambda}^{\otimes k}).$$

The exact pushforward identity is

$$\boxed{m_{h,k}(u) = w_h^{*k}(u) = r_{\lambda,k}(u) B_{h,\lambda;k}(u)}. \quad (19)$$

This is a norm observation of the original source; it does not project the original finite arithmetic module or change its trace. The maps in (3), including the full unit in η , remain attached to every polynomial. The full field $B_{h,\lambda}^{\otimes k}$ is recoverable from (12), so its relative complement is retained too.

S.5.1 5.1 A bounded multiplier for every full quartet

For $\lambda = 1/4$, the original formula is

$$A_{h,1/4}(t) = \frac{-\pi^{-1/4}(t^2 + 1/4)e^{-it \log \pi/2} \zeta(1/2 + it)}{h(1/2 + it)}. \quad (20)$$

The minus sign and phase are still in (18). The elementary continuation formula

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} dx$$

gives, on this line,

$$|\zeta(1/2 + it)| \leq 1 + 2\sqrt{t^2 + 1/4} \leq 2(1 + |t|).$$

Let $d = \deg h \geq 3$ and take $T_h = \max(1, 2 \max_\rho |\rho - 1/2|)$. For $|t| \geq T_h$, $|h(1/2 + it)| \geq (|t|/2)^d$, and direct substitution gives

$$B_{h,1/4}(t) \leq \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} |t|^{6-2d}.$$

Consequently the finite, specified constant

$$C_h = \max \left\{ \sup_{|t| \leq T_h} |A_{h,1/4}(t)|^2, \frac{25 \cdot 2^{2d}}{\sqrt{\pi}} T_h^{6-2d} \right\}$$

satisfies $0 \leq B_{h,1/4} \leq C_h$. The compact supremum uses the analytic quotient, not the expression with a removable denominator left untreated. A full nonreal quartet has $d = 4m \geq 4$, so this applies to the entire packet used in the exterior amplification, with constants allowed to depend on that fixed packet.

The positive integral in (11) now proves

$$\boxed{0 < B_{h,1/4;k}(u) \leq C_h^k \quad (k \geq 2), \quad 0 \leq m_{h,k}(u) \leq C_h^k r_{1/4,k}(u).} \quad (21)$$

Strict positivity for $k \geq 2$ follows because the one-factor multiplier only vanishes at a discrete set; its excluded hyperplanes have zero fibre measure. For $k = 1$ retain its actual isolated zeros.

S.5.2 5.2 The other packet sizes and the empty analytic source

For a chosen reference below, C_h always denotes an explicitly specified upper bound for that reference's $B_{h,\lambda}$. An explicit alternate reference is available without changing the arithmetic source. For integer $L \geq 0$,

$$A_{h,1/4+L}(t) = \frac{A_{h,1/4}(t)}{\prod_{j=0}^{L-1} (j + 1/4 + it/2)}. \quad (22)$$

Its product is a known nonvanishing gamma-shift factor on the integration line. Taking $L \geq \max(0, 3 - d)$ makes its squared modulus bounded by the same compact/tail argument. Equations (17)–(19) connect both references to the identical original w_h ; no coordinate or original mass is reassigned.

For the actual seed $h = 1$, choosing $L = 3$ gives the explicit all-real estimate

$$\boxed{B_{1,13/4}(t) \leq 1024/\sqrt{\pi}.} \quad (23)$$

Indeed the gamma-shift denominator is $4^{-3}(t^2 + 1/4)(t^2 + 25/4)(t^2 + 81/4)$, and $|\zeta(1/2 + it)| \leq 4\sqrt{t^2 + 1/4}$. The remaining fraction is bounded by one since $(t^2 + 1/4)^2 \leq (t^2 + 25/4)(t^2 + 81/4)$.

This is an analytic estimate for a nonzero seed. Its finite spectral quotient is still zero, as required by the merged correction.

S.6 6. Exact arithmetic coefficients at every tensor degree

Use a reference for which $B = B_{h,\lambda}$ is bounded, as constructed above. Define its actual coefficients

$$c_{h,j} = \frac{\int b_j^{(\lambda)}(t) w_h(t) dt}{c_\lambda j! (\alpha)_j}, \quad \mathcal{A}_h(z) = \sum_{j \geq 0} (\alpha)_j c_{h,j} z^j. \quad (24)$$

The series is used coefficientwise; no convergence radius for this formal series is stipulated. Each coefficient is an actual one-factor arithmetic integral. Define

$$d_{h,k,n} = [z^n] \mathcal{A}_h(z)^k.$$

Then the complete $L^2(r_{\lambda,k} du)$ expansion is

$$\boxed{B_{h,\lambda;k}(u) = \sum_{n \geq 0} \frac{d_{h,k,n}}{(k\alpha)_n} b_n^{(k\lambda)}(u).} \quad (25)$$

Proof. Pair $B^{\otimes k}$ with the finite addition formula (14). Each one-factor pairing is $c_\lambda j! (\alpha)_j c_{h,j}$. The multinomial coefficient cancels the displayed factorials and gives

$$\int B_{h,\lambda;k} b_n^{(k\lambda)} r_{\lambda,k} = c_\lambda^n n! [z^n] \mathcal{A}_h(z)^k.$$

Divide by the exact norm $c_\lambda^k n! (k\alpha)_n$. Completeness gives (25). No exchange of an infinite product of formal series with an integral was needed: every coefficient used a finite polynomial identity.

At degree zero,

$$c_{h,0} = \mu_h / c_\lambda, \quad d_{h,k,0} = (\mu_h / c_\lambda)^k,$$

so (25) recovers exactly $\int m_{h,k} = \mu_h^k$.

The retained relative residual is

$$\mathcal{N}_{h,k} = B^{\otimes k} - U_\Sigma B_{h,\lambda;k}.$$

Its whole norm is calculated by Pythagoras:

$$\|\mathcal{N}_{h,k}\|^2 = \left(\int B(t)^2 d\sigma_\lambda(t) \right)^k - c_\lambda^k \sum_{n \geq 0} \frac{n!}{(k\alpha)_n} |d_{h,k,n}|^2. \quad (26)$$

The difference is nonnegative because it is the norm of the explicitly retained vector. It is not set to zero by calling the sum map a pushforward.

S.6.1 6.1 Finite exactness and approximation errors

Let $B_L = \sum_{j=0}^L c_{h,j} b_j^{(\lambda)}$ and $B_{k,L} = C_\Sigma(B_L^{\otimes k})$. Projection contraction and the tensor telescoping identity give

$$\|B_{h,\lambda;k} - B_{k,L}\| \leq k \|B\|^{k-1} \|B - B_L\|. \quad (27)$$

These norms have the respective original reference measures. The proof uses $\|B_L\| \leq \|B\|$ and expands the tensor difference into k terms. Both sides retain the full mass factors.

More strongly, the moments of $B_{k,L} r_{\lambda,k}$ against every polynomial of degree at most L are **exactly** those of $m_{h,k}$, since coefficient $n \leq L$ in (25) uses only $c_{h,j}$ with $j \leq n$.

It follows that the entire degree- N source Gram, and its degree- N relation Gram, are determined exactly by $c_{h,0}, \dots, c_{h,2N}$. The relation integrands are also degree at most $2N$. Their tensor dependence is the explicit finite coefficient power in (24), not a fresh k -dimensional integral.

A truncated $B_{k,L}$ need not be nonnegative pointwise. Its relationship to the true positive metric is the exact finite-moment equality just proved when $L \geq 2N$, or the error estimate (27) and Cauchy-Schwarz for the specified polynomial tests when $L < 2N$. Positivity is not inferred from a plotted truncated density.

S.6.2 6.2 Actual seed coefficients, and their tensor cross term

The supplied numerical driver evaluates the original theta seed on $x \geq 1$, using its inversion symmetry. With $L = D - 1/2$, the even spectral moments are $\mu_{2r} = 2 \int_1^\infty |L^r f_0(x)|^2 dx$. It constructs their source polynomials by $q_0(z) = 4z^2 - 6z$ and $q_{r+1} = 2z(q_r - q'_r) - q_r/2$, and integrates the integer double sums using upper incomplete gamma integrals. This is a numerical check, not a validated interval computation.

For $h = 1$ and the bounded-multiplier reference $\lambda = 13/4$, the 50-digit, cutoff-6 and 70-digit, cutoff-8 calculations agree in the displayed values:

Quantity	Numerical value
μ_0	1.279007247846485140479533592267
μ_2	13.055549302570558435392684658123
μ_4	383.274341717364242554407237479131
μ_6	17946.5273016052076235667646919834
$c_{h,0}$	0.201056688692785865809916163220
$c_{h,2}$	0.007645442999486401188133851481
$c_{h,4}$	-0.000030516920803762043746724840

The reference mass is $c_{13/4} = 6.361426004587219292629853548733 \dots$, so it has not been assigned the actual mass μ_0 or the value one. The negative fourth expansion coefficient is retained; a positive density need not have positive coefficients in this orthogonal basis.

For every even arithmetic multiplier, the first tensor terms from (25) are exactly

$$d_{h,k,2} = k c_{h,0}^{k-1}(\alpha)_2 c_{h,2},$$

$$d_{h,k,4} = k c_{h,0}^{k-1}(\alpha)_4 c_{h,4} + \binom{k}{2} c_{h,0}^{k-2}((\alpha)_2 c_{h,2})^2.$$

Both terms in the fourth coefficient remain. These are coefficients of the exact arithmetic source norm; the empty packet still supplies no nonzero finite spectral quotient. Agreement of two numerical truncations is not a bound on their shared truncation or rounding error.

S.7 7. Feed the result into the two original Toda determinants

Put $\alpha_k = 2k\lambda$. The full gamma source determinants are explicit:

$$\mathfrak{D}_n^\Gamma(\theta) = c_\lambda^{kn} \prod_{j=0}^{n-1} j! (\alpha_k)_j (\cos \theta)^{-n(\alpha_k+n-1)}. \quad (28)$$

At $\theta = 0$ this is the product of the monic norms from (13) and (7). One direct proof for all real θ starts with $\mathfrak{D}_0^\Gamma = 1$ and $\mathfrak{D}_1^\Gamma = c_\lambda^k \cos^{-\alpha_k} \theta$. Apply the already-proved Hankel-Toda identity recursively; the second derivative of the logarithm of the right side is $n(\alpha_k + n - 1) \sec^2 \theta$. Substitution proves the next determinant with all constants. Strict positivity makes the recurrence division valid.

In particular

$$\omega_n^\Gamma(\theta) = c_\lambda^k n! (\alpha_k)_n (\cos \theta)^{-(\alpha_k+2n)}, \quad a_n^\Gamma(\theta) = n(n + \alpha_k - 1) \sec^2 \theta. \quad (29)$$

The arithmetic source determinant is the previous $\mathfrak{D}_n$. The reference relation determinant is

$$\mathfrak{B}_n^\Gamma(\theta) = \det \left[\partial_\theta^{i+j} \{ \bar{\chi}(k/2 - i\partial_\theta) \chi(k/2 + i\partial_\theta) Z_{\lambda,k}^\Gamma(\theta) \} \right]_{i,j < n}. \quad (30)$$

It is the Gram of the **same** multiplication map M_χ . This is an explicit finite expression in gamma constants, $\cos \theta$, $\tan \theta$, and the original coefficients of χ .

Define positive scalar corrections, including the empty-index value one,

$$X_n = \mathfrak{D}_n / \mathfrak{D}_n^\Gamma, \quad Y_n = \mathfrak{B}_n / \mathfrak{B}_n^\Gamma.$$

The source-boundary determinant theorem now factors as

$$\boxed{V_N = V_N^\Gamma T_N, \quad V_N^\Gamma = \frac{\mathfrak{D}_{N+1}^\Gamma}{\mathfrak{B}_{N-q+1}^\Gamma}, \quad T_N = \frac{X_{N+1}}{Y_{N-q+1}}.} \quad (31)$$

This retains the entire arithmetic correction in both numerator and denominator. It is not an assertion that the arithmetic quotient metric equals the reference metric.

S.7.1 7.1 The representative map between the two metrics

Use the same finite polynomial source. Let B_χ be the injective matrix of its original relation columns, and let R_N^Γ be the least-norm lift for the gamma observation. Let M_h be the actual arithmetic source Gram. Then the actual lift is

$$\boxed{R_N = R_N^\Gamma - B_\chi (B_\chi^* M_h B_\chi)^{-1} B_\chi^* M_h R_N^\Gamma.} \quad (32)$$

For an empty boundary block the correction is the zero map. The proof is direct: the correction is an old relation, so the quotient remains the identity; multiplication by $B_\chi^* M_h$ gives zero. These two properties characterize the arithmetic least-norm lift.

Applying $\mathcal{V}_{h,k}$ to (32) supplies the original theta primitive of the difference. In the reconstructed support carrier, the difference maps to the supported zero in its fibre. It does not disappear from the source metric or from its boundary primitive. The comparison keeps the complete unit in (3).

S.7.2 7.2 A genuine source-side upper bound

From (21), for every admitted polynomial,

$$\|P\|_h^2 \leq C_h^k \|P\|_\Gamma^2.$$

Evaluate this inequality on $R_N^\Gamma u$ and then minimize on the arithmetic side. This proves

$$\boxed{G_N \preceq C_h^k G_N^\Gamma, \quad 0 < T_N \leq C_h^{kq}.} \quad (33)$$

The second assertion is the determinant inequality in dimension q . It has the quotient dimension, rather than the full polynomial-source dimension, in its exponent. The corresponding source estimate is $0 < X_n \leq C_h^{kn}$, and similarly $0 < Y_n \leq C_h^{kn}$.

These are one-sided estimates. They do not bound the consecutive ratio T_{N-1}/T_{N+1} , which is the quantity required by the Toda upper bound. Its exact map to that estimate is the following.

S.8 8. The current upper-control expression, with all reference costs exposed

Define

$$Q_N = \frac{X_{N+2} X_N}{X_{N+1}^2}, \quad \mathcal{R}_N^\Gamma = \frac{V_{N-1}^\Gamma}{V_{N+1}^\Gamma}.$$

At the original observation $\theta = 0$, (29)–(31) give

$$\boxed{a_{N+1} = (N+1)(N+\alpha_k)Q_N, \quad \mathcal{R}_N = \mathcal{R}_N^\Gamma \frac{T_{N-1}}{T_{N+1}}.} \quad (34)$$

The preceding exact phase identity is also retained through

$$\ell'_N = \partial_\theta \log V_N^\Gamma + \partial_\theta \log T_N.$$

Thus its finite upper bound becomes exactly

$$\epsilon_{h,k,N} \leq \sqrt{(N+1)(N+\alpha_k)Q_N} \frac{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1} - 1}{2\sqrt{\mathcal{R}_N^\Gamma T_{N-1}/T_{N+1}}}. \quad (35)$$

The nonnegative numerator uses the original monotonicity of the actual quotient minimum, not a presumed monotonicity of T_N . When this inequality is coarse, the inherited equality with the volume-imbalance square and the phase square remains the sharper calculation.

For the full quartet one can keep $\lambda = 1/4$, so $\alpha_k = k/2$. The gamma source recurrence is now explicit at every degree; the remaining arithmetic dependence is in Q_N and the relative source-boundary correction T_{N-1}/T_{N+1} . The reference relation term still contains the full spectral annihilator χ . It is not bounded by pretending its roots are on the integration line.

The exterior theorem remains

$$L_{h,k} \leq \epsilon_{h,k,N}, \quad L_{h,k} = 2\delta[1 + k(m-1)](k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor.$$

Neither (33) nor (35) proves a subcubic bound. Equation (35) identifies exactly which degree-to-degree arithmetic corrections must be controlled next. All entries of those finite determinants can now be generated by the all- k coefficient transform (25), with the source-boundary comparison map (32) retained.

S.9 9. A proved all-degree tail estimate for the actual arithmetic moments

The multiplier bound gives more than a change of coordinates. For $a, b > 0$ with $a + b < \pi/2$, every integer $r \geq 0$ and $T \geq 0$,

$$\int_{|u|>T} |u|^r m_{h,k}(u) du \leq 2C_h^k c_\lambda^k r! a^{-r} e^{-bT} (\cos(a+b))^{-2k\lambda}. \quad (36)$$

Proof. For $x \geq 0$, $x^r \leq r! a^{-r} e^{ax}$. On the indicated tail, $1 \leq e^{b(x-T)}$. Apply (21) and integrate. Since the reference density is even,

$$\int e^{(a+b)|u|} r_{\lambda,k}(u) du \leq Z_{\lambda,k}^\Gamma(a+b) + Z_{\lambda,k}^\Gamma(-a-b).$$

Equation (9) gives (36). This estimate retains the original mass and the complete tensor cost. It is a theorem with specified constants, not a claim that a floating-point quadrature has validated them numerically.

For a source or relation moment whose integrand is a polynomial $P(u) = \sum_j p_j u^j$, sum $|p_j|$ times (36). In particular this controls every entry of a degree- N source and relation Gram, including the full $|\chi|^2$ factor. It is then combined with an independently certified compact-interval quadrature and the existing matrix inverse/error inequalities; it does not certify that quadrature by itself.

This is an explicit tail interface for the actual arithmetic input at all degrees and tensor powers. For the empty analytic seed, (23) gives a completely stated bound without locating any zeta zero. For a nonempty packet the compact constant in C_h must be enclosed using the actual analytic quotient.

S.10 10. What has and has not been concluded

The constructed chain is

- original split-supported polynomial source
- full gamma product fibre with arithmetic amplitude
- sum density and retained relative component
- exact all-tensor arithmetic coefficients
- the same source–boundary Toda determinant ratio.

The finite coefficient projection, polynomial degree, all masses, all gamma factors, the arithmetic unit, and the original theta primitives are explicit. The same finite quotient is used throughout; an existing relation becomes e only under its original quotient map, while the source norm and all derivative data remain available before that map. No new scalar-zero adjunction is performed at a tensor or relation level.

There is now a proved bound for the full arithmetic sum density, explicit closed-form reference determinants at every tensor degree, an exact finite coefficient algorithm, and a rigorous moment-tail bound. There is not yet a uniform small bound for the consecutive arithmetic determinant correction. The upper-estimate problem is sharpened to that specific comparison rather than being claimed solved by gamma convolution or by positivity of individual source Grams.

S.11 References and exact source roles

- Repository main at `fa4be32f87a9a90f3c02a07f7790dea95a1e7b0f`, `tex/theta_gamma_reference.tex` in the 20260912-stieltjes continuation, TG.2–TG.14: original one-factor gamma reference, arithmetic amplitude, constants and polynomial norms. Only the stated TeX window was read in this review; no complete reader audit is claimed.
- PR #22 head `811210d24b80813a08972ca23f919db015383137`, `workbenches/tau-exterior-trace-formal/RESEARCH_NOTE.md`: current exterior and filtration formalization work, with its separate verification scope. Its reported in-progress status is not promoted to a completed certificate.
- Delivered `Tau_Toda_Volume_Control/NOTE.tex`: original source and relation determinants, quotient metric, phase derivative and finite upper inequality. Its supplied manifest was independently checked here.
- NIST Digital Library of Mathematical Functions, §§18.22 and 18.23: classical Meixner–Pollaczek recurrence and generating function. The formulas here retain the repository’s original $t/2$ scale and all factorials.
- Koelink, H. T., and Van der Jeugt, J. (1998). Convolutions for orthogonal polynomials from Lie and quantum algebra representations. *SIAM Journal on Mathematical Analysis*, 29(3), 794–822. arXiv:q-alg/9607010. Background for the classical addition/convolution mechanism, not a claim that this article treats the present arithmetic quotient.

The proofs in this note establish the stated specialized maps directly. The regression checker tests finite exact polynomial and matrix consequences; it does not replace the analytic proofs or establish the outstanding uniform estimate.